



Vulkan[®] 1.0.300 - A Specification

The Khronos[®] Vulkan Working Group

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Chapter 1. Preamble

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This Specification contains substantially unmodified functionality from, and is a successor to, Khronos specifications including OpenGL, OpenGL ES and OpenCL.

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Some parts of this Specification are purely informative and so are EXCLUDED the Scope of this Specification. The [Document Conventions](#) section of the [Introduction](#) defines how these parts of the Specification are identified.

Where this Specification uses [technical terminology](#), defined in the [Glossary](#) or otherwise, that refer to enabling technologies that are not expressly set forth in this Specification, those enabling technologies are EXCLUDED from the Scope of this Specification. For clarity, enabling technologies not disclosed with particularity in this Specification (e.g. semiconductor manufacturing technology, hardware architecture, processor architecture or microarchitecture, memory architecture, compiler technology, object oriented technology, basic operating system technology, compression technology, algorithms, and so on) are NOT to be considered expressly set forth; only those application program interfaces and data structures disclosed with particularity are included in the Scope of this Specification.

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Chapter 2. Introduction

This document, referred to as the “Vulkan Specification” or just the “Specification” hereafter, describes the Vulkan Application Programming Interface (API). Vulkan is a [C99](#) API designed for explicit control of low-level graphics and compute functionality.

The canonical version of the Specification is available in the official [Vulkan Registry](https://registry.khronos.org/vulkan/) (<https://registry.khronos.org/vulkan/>). The source files used to generate the Vulkan specification are stored in the [Vulkan Documentation Repository](https://github.com/KhronosGroup/Vulkan-Docs) (<https://github.com/KhronosGroup/Vulkan-Docs>).

The source repository additionally has a public issue tracker and allows the submission of pull requests that improve the specification.

2.1. Document Conventions

The Vulkan specification is intended for use by both implementors of the API and application developers seeking to make use of the API, forming a contract between these parties. Specification text may address either party; typically the intended audience can be inferred from context, though some sections are defined to address only one of these parties. (For example, [Valid Usage](#) sections only address application developers). Any requirements, prohibitions, recommendations or options defined by [normative terminology](#) are imposed only on the audience of that text.

NOTE

Structure and enumerated types defined in extensions that were promoted to core in a later version of Vulkan are now defined in terms of the equivalent Vulkan core interfaces. This affects the Vulkan Specification, the Vulkan header files, and the corresponding XML Registry.

2.1.1. Ratification

Ratification of a Vulkan core version or extension is a status conferred by vote of the Khronos Board of Promoters, bringing that core version or extension under the umbrella of the Khronos IP Policy.

All Vulkan core versions and [KHR](#) extensions (including provisional specifications) are ratified, as are some multi-vendor [EXT](#) extensions. Ratification status of extensions is described in the [Layers & Extensions \(Informative\)](#) appendix.

NOTE

Ratification status is primarily of interest to IHVs developing GPU hardware and Vulkan implementations

For developers, ratification does not necessarily mean that an extension is “better”; has a more stable API; or is more widely supported than alternative ways of achieving that functionality.

Interactions between ratified and non-ratified extensions are not themselves ratified.

2.1.2. Informative Language

Some language in the specification is purely informative, intended to give background or suggestions to implementors or developers.

If an entire chapter or section contains only informative language, its title will be suffixed with “(Informative)”.

All NOTES are implicitly informative.

2.1.3. Normative Terminology

Within this specification, the key words **must**, **required**, **should**, **recommended**, **may**, and **optional** are to be interpreted as described in [RFC 2119 - Key words for use in RFCs to Indicate Requirement Levels](https://www.ietf.org/rfc/rfc2119.txt) (<https://www.ietf.org/rfc/rfc2119.txt>). The additional key word **optionally** is an alternate form of **optional**, for use where grammatically appropriate.

These key words are highlighted in the specification for clarity. In text addressing application developers, their use expresses requirements that apply to application behavior. In text addressing implementors, their use expresses requirements that apply to implementations.

In text addressing application developers, the additional key words **can** and **cannot** are to be interpreted as describing the capabilities of an application, as follows:

can

This word means that the application is able to perform the action described.

cannot

This word means that the API and/or the execution environment provide no mechanism through which the application can express or accomplish the action described.

These key words are never used in text addressing implementors.

NOTE

There is an important distinction between **cannot** and **must not**, as used in this Specification. **Cannot** means something the application literally is unable to express or accomplish through the API, while **must not** means something that the application is capable of expressing through the API, but that the consequences of doing so are undefined and potentially unrecoverable for the implementation (see [Valid Usage](#)).

Unless otherwise noted in the section heading, all sections and appendices in this document are normative.

2.1.4. Technical Terminology

The Vulkan Specification makes use of common engineering and graphics terms such as **Pipeline**, **Shader**, and **Host** to identify and describe Vulkan API constructs and their attributes, states, and behaviors. The [Glossary](#) defines the basic meanings of these terms in the context of the Specification. The Specification text provides fuller definitions of the terms and may elaborate,

extend, or clarify the [Glossary](#) definitions. When a term defined in the [Glossary](#) is used in normative language within the Specification, the definitions within the Specification govern and supersede any meanings the terms may have in other technical contexts (i.e. outside the Specification).

2.1.5. Normative References

References to external documents are considered normative references if the Specification uses any of the normative terms defined in [Normative Terminology](#) to refer to them or their requirements, either as a whole or in part.

The following documents are referenced by normative sections of the specification:

IEEE. August, 2008. *IEEE Standard for Floating-Point Arithmetic*. IEEE Std 754-2008. <https://dx.doi.org/10.1109/IEEESTD.2008.4610935> .

Andrew Garrard. *Khronos Data Format Specification, version 1.3*. <https://registry.khronos.org/DataFormat/specs/1.3/dataformat.1.3.html> .

John Kessenich. *SPIR-V Extended Instructions for GLSL, Version 1.00* (February 10, 2016). <https://registry.khronos.org/spir-v/> .

John Kessenich, Boaz Ouriel, and Raun Krisch. *SPIR-V Specification, Version 1.5, Revision 3, Unified* (April 24, 2020). <https://registry.khronos.org/spir-v/> .

ITU-T. *H.264 Advanced Video Coding for Generic Audiovisual Services* (August, 2021). <https://www.itu.int/rec/T-REC-H.264-202108-I/> .

ITU-T. *H.265 High Efficiency Video Coding* (August, 2021). <https://www.itu.int/rec/T-REC-H.265-202108-S/> .

Alliance for Open Media. *AV1 Bitstream & Decoding Process Specification* (January 8, 2019). <https://aomediacodec.github.io/av1-spec/av1-spec.pdf> .

Jon Leech. *The Khronos Vulkan API Registry* (February 26, 2023). <https://registry.khronos.org/vulkan/specs/1.3/registry.html> .

Jon Leech and Tobias Hector. *Vulkan Documentation and Extensions: Procedures and Conventions* (February 26, 2023). <https://registry.khronos.org/vulkan/specs/1.3/styleguide.html> .

Architecture of the Vulkan Loader Interfaces (October, 2021). <https://github.com/KhronosGroup/Vulkan-Loader/blob/main/docs/LoaderInterfaceArchitecture.md> .

Chapter 3. Fundamentals

This chapter introduces fundamental concepts including the Vulkan architecture and execution model, API syntax, queues, pipeline configurations, numeric representation, state and state queries, and the different types of objects and shaders. It provides a framework for interpreting more specific descriptions of commands and behavior in the remainder of the Specification.

3.1. Host and Device Environment

The Vulkan Specification assumes and requires: the following properties of the host environment with respect to Vulkan implementations:

- The host **must** have runtime support for 8, 16, 32 and 64-bit signed and unsigned twos-complement integers, all addressable at the granularity of their size in bytes.
- The host **must** have runtime support for 32- and 64-bit floating-point types satisfying the range and precision constraints in the [Floating-Point Computation](#) section.
- The representation and endianness of these types on the host **must** match the representation and endianness of the same types on every physical device supported.

NOTE

Since a variety of data types and structures in Vulkan **may** be accessible by both host and physical device operations, the implementation **should** be able to access such data efficiently in both paths in order to facilitate writing portable and performant applications.

3.2. Execution Model

This section outlines the execution model of a Vulkan system.

Vulkan exposes one or more *devices*, each of which exposes one or more *queues* which **may** process work asynchronously to one another. The set of queues supported by a device is partitioned into *families*. Each family supports one or more types of functionality and **may** contain multiple queues with similar characteristics. Queues within a single family are considered *compatible* with one another, and work produced for a family of queues **can** be executed on any queue within that family. This specification defines the following types of functionality that queues **may** support: graphics, compute, sparse memory management, and transfer.

NOTE

A single device **may** report multiple similar queue families rather than, or as well as, reporting multiple members of one or more of those families. This indicates that while members of those families have similar capabilities, they are *not* directly compatible with one another.

Device memory is explicitly managed by the application. Each device **may** advertise one or more heaps, representing different areas of memory. Memory heaps are either device-local or host-local, but are always visible to the device. Further detail about memory heaps is exposed via memory types available on that heap. Examples of memory areas that **may** be available on an implementation include:

- *device-local* is memory that is physically connected to the device.
- *device-local, host visible* is device-local memory that is visible to the host.
- *host-local, host visible* is memory that is local to the host and visible to the device and host.

On other architectures, there **may** only be a single heap that **can** be used for any purpose.

3.2.1. Queue Operation

Vulkan queues provide an interface to the execution engines of a device. Commands for these execution engines are recorded into command buffers ahead of execution time, and then submitted to a queue for execution. Once submitted to a queue, command buffers will begin and complete execution without further application intervention, though the order of this execution is dependent on a number of [implicit and explicit ordering constraints](#).

Work is submitted to queues using *queue submission commands* that typically take the form `vkQueue*` (e.g. `vkQueueSubmit`, `vkQueueBindSparse`), and **can** take a list of semaphores upon which to wait before work begins and a list of semaphores to signal once work has completed. The work itself, as well as signaling and waiting on the semaphores are all *queue operations*. Queue submission commands return control to the application once queue operations have been submitted - they do not wait for completion.

There are no implicit ordering constraints between queue operations on different queues, or between queues and the host, so these **may** operate in any order with respect to each other. Explicit ordering constraints between different queues or with the host **can** be expressed with [semaphores](#) and [fences](#).

Command buffer submissions to a single queue respect [submission order](#) and other [implicit ordering guarantees](#), but otherwise **may** overlap or execute out of order. Other types of batches and queue submissions against a single queue (e.g. [sparse memory binding](#)) have no implicit ordering constraints with any other queue submission or batch. Additional explicit ordering constraints between queue submissions and individual batches can be expressed with [semaphores](#) and [fences](#).

Before a fence or semaphore is signaled, it is guaranteed that any previously submitted queue operations have completed execution, and that memory writes from those queue operations are [available](#) to future queue operations. Waiting on a signaled semaphore or fence guarantees that previous writes that are available are also [visible](#) to subsequent commands.

Command buffer boundaries, both between primary command buffers of the same or different batches or submissions as well as between primary and secondary command buffers, do not introduce any additional ordering constraints. In other words, submitting the set of command buffers (which **can** include executing secondary command buffers) between any semaphore or fence operations execute the recorded commands as if they had all been recorded into a single primary command buffer, except that the current state is [reset](#) on each boundary. Explicit ordering constraints **can** be expressed with [explicit synchronization primitives](#).

There are a few [implicit ordering guarantees](#) between commands within a command buffer, but only covering a subset of execution. Additional explicit ordering constraints can be expressed with the various [explicit synchronization primitives](#).

NOTE

Implementations have significant freedom to overlap execution of work submitted to a queue, and this is common due to deep pipelining and parallelism in Vulkan devices.

Commands recorded in command buffers can perform actions, set state that persists across commands, synchronize other commands, or indirectly launch other commands, with some commands fulfilling several of these roles. The “Command Properties” section for each such command lists which of these roles the command takes:

Action

Action commands perform operations that can update values in memory. E.g. [draw commands](#), [dispatch commands](#).

State

State setting commands update the current state of a command buffer, affecting the operation of future action commands.

Synchronization

Synchronization commands impose ordering constraints on action commands, by introducing explicit [execution and memory dependencies](#).

Indirection

Indirection commands execute other commands which were not directly recorded in the same command buffer.

NOTE

In the absence of explicit synchronization or [implicit ordering guarantees](#), action commands may overlap execution or execute out of order, potentially leading to data races. However, such reordering does not affect the current state observed by any action command. Each action command uses the state in effect at the point where the command occurs in the command buffer, regardless of when it is executed.

3.3. Object Model

The devices, queues, and other entities in Vulkan are represented by Vulkan objects. At the API level, all objects are referred to by handles. There are two classes of handles, dispatchable and non-dispatchable. *Dispatchable* handle types are a pointer to an opaque type. This pointer **may** be used by layers as part of intercepting API commands, and thus each API command takes a dispatchable type as its first parameter. Each object of a dispatchable type **must** have a unique handle value during its lifetime.

Non-dispatchable handle types are a 64-bit integer type whose meaning is implementation-dependent. Non-dispatchable handles **may** encode object information directly in the handle rather than acting as a reference to an underlying object, and thus **may** not have unique handle values. If handle values are not unique, then destroying one such handle **must** not cause identical handles of other types to become invalid, and **must** not cause identical handles of the same type to become invalid if that handle value has been created more times than it has been destroyed.

All objects created or allocated from a `VkDevice` (i.e. with a `VkDevice` as the first parameter) are private to that device, and **must** not be used on other devices.

3.3.1. Object Lifetime

Objects are created or allocated by `vkCreate*` and `vkAllocate*` commands, respectively. Once an object is created or allocated, its “structure” is considered to be immutable, though the contents of certain object types is still free to change. Objects are destroyed or freed by `vkDestroy*` and `vkFree*` commands, respectively.

Objects that are allocated (rather than created) take resources from an existing pool object or memory heap, and when freed return resources to that pool or heap. While object creation and destruction are generally expected to be low-frequency occurrences during runtime, allocating and freeing objects **can** occur at high frequency. Pool objects help accommodate improved performance of the allocations and frees.

It is an application’s responsibility to track the lifetime of Vulkan objects, and not to destroy them while they are still in use.

The ownership of application-owned memory is immediately acquired by any Vulkan command it is passed into. Ownership of such memory **must** be released back to the application at the end of the duration of the command, so that the application **can** alter or free this memory as soon as all the commands that acquired it have returned.

The following object types are consumed when they are passed into a Vulkan command and not further accessed by the objects they are used to create. They **must** not be destroyed in the duration of any API command they are passed into:

- `VkShaderModule`
- `VkPipelineCache`

A `VkRenderPass` object passed as a parameter to create another object is not further accessed by that object after the duration of the command it is passed into. A `VkRenderPass` used in a command buffer follows the rules described below.

A `VkPipelineLayout` object **must** not be destroyed while any command buffer that uses it is in the recording state.

`VkDescriptorSetLayout` objects **may** be accessed by commands that operate on descriptor sets allocated using that layout, and those descriptor sets **must** not be updated with `vkUpdateDescriptorSets` after the descriptor set layout has been destroyed. Otherwise, a `VkDescriptorSetLayout` object is no longer referenced by an API command it is passed into once host execution of that command completes.

The application **must** not destroy any other type of Vulkan object until all uses of that object by the device (such as via command buffer execution) have completed.

The following Vulkan objects **must** not be destroyed while any command buffers using the object are in the `pending state`:

- `VkEvent`
- `VkQueryPool`
- `VkBuffer`
- `VkBufferView`
- `VkImage`
- `VkImageView`
- `VkPipeline`
- `VkSampler`
- `VkDescriptorPool`
- `VkFramebuffer`
- `VkRenderPass`
- `VkCommandBuffer`
- `VkCommandPool`
- `VkDeviceMemory`
- `VkDescriptorSet`

Destroying these objects will move any command buffers that are in the [recording or executable state](#), and are using those objects, to the [invalid state](#).

The following Vulkan objects **must** not be destroyed while any queue is executing commands that use the object:

- `VkFence`
- `VkSemaphore`
- `VkCommandBuffer`
- `VkCommandPool`

In general, objects **can** be destroyed or freed in any order, even if the object being freed is involved in the use of another object (e.g. use of a resource in a view, use of a view in a descriptor set, use of an object in a command buffer, binding of a memory allocation to a resource), as long as any object that uses the freed object is not further used in any way except to be destroyed or to be reset in such a way that it no longer uses the other object (such as resetting a command buffer). If the object has been reset, then it **can** be used as if it never used the freed object. An exception to this is when there is a parent/child relationship between objects. In this case, the application **must** not destroy a parent object before its children, except when the parent is explicitly defined to free its children when it is destroyed (e.g. for pool objects, as defined below).

`VkCommandPool` objects are parents of `VkCommandBuffer` objects. `VkDescriptorPool` objects are parents of `VkDescriptorSet` objects. `VkDevice` objects are parents of many object types (all that take a `VkDevice` as a parameter to their creation).

The following Vulkan objects have specific restrictions for when they **can** be destroyed:

- **VkQueue** objects **cannot** be explicitly destroyed. Instead, they are implicitly destroyed when the **VkDevice** object they are retrieved from is destroyed.
- Destroying a pool object implicitly frees all objects allocated from that pool. Specifically, destroying **VkCommandPool** frees all **VkCommandBuffer** objects that were allocated from it, and destroying **VkDescriptorPool** frees all **VkDescriptorSet** objects that were allocated from it.
- **VkDevice** objects **can** be destroyed when all **VkQueue** objects retrieved from them are idle, and all objects created from them have been destroyed.
 - This includes the following objects:
 - **VkFence**
 - **VkSemaphore**
 - **VkEvent**
 - **VkQueryPool**
 - **VkBuffer**
 - **VkBufferView**
 - **VkImage**
 - **VkImageView**
 - **VkShaderModule**
 - **VkPipelineCache**
 - **VkPipeline**
 - **VkPipelineLayout**
 - **VkSampler**
 - **VkDescriptorSetLayout**
 - **VkDescriptorPool**
 - **VkFramebuffer**
 - **VkRenderPass**
 - **VkCommandPool**
 - **VkCommandBuffer**
 - **VkDeviceMemory**
- **VkPhysicalDevice** objects **cannot** be explicitly destroyed. Instead, they are implicitly destroyed when the **VkInstance** object they are retrieved from is destroyed.
- **VkInstance** objects **can** be destroyed once all **VkDevice** objects created from any of its **VkPhysicalDevice** objects have been destroyed.

3.4. Application Binary Interface

The mechanism by which Vulkan is made available to applications is platform- or implementation-defined. On many platforms the C interface described in this Specification is provided by a shared

library. Since shared libraries can be changed independently of the applications that use them, they present particular compatibility challenges, and this Specification places some requirements on them.

Shared library implementations **must** use the default Application Binary Interface (ABI) of the standard C compiler for the platform, or provide customized API headers that cause application code to use the implementation's non-default ABI. An ABI in this context means the size, alignment, and layout of C data types; the procedure calling convention; and the naming convention for shared library symbols corresponding to C functions. Customizing the calling convention for a platform is usually accomplished by defining [calling convention macros](#) appropriately in `vk_platform.h`.

On platforms where Vulkan is provided as a shared library, library symbols beginning with “vk” and followed by a digit or uppercase letter are reserved for use by the implementation. Applications which use Vulkan **must** not provide definitions of these symbols. This allows the Vulkan shared library to be updated with additional symbols for new API versions or extensions without causing symbol conflicts with existing applications.

Shared library implementations **should** provide library symbols for commands in the highest version of this Specification they support, and for Window System Integration extensions relevant to the platform. They **may** also provide library symbols for commands defined by additional extensions.

NOTE

These requirements and recommendations are intended to allow implementors to take advantage of platform-specific conventions for SDKs, ABIs, library versioning mechanisms, etc. while still minimizing the code changes necessary to port applications or libraries between platforms. Platform vendors, or providers of the *de facto* standard Vulkan shared library for a platform, are encouraged to document what symbols the shared library provides and how it will be versioned when new symbols are added.

Applications **should** only rely on shared library symbols for commands in the minimum core version required by the application. [vkGetInstanceProcAddr](#) and [vkGetDeviceProcAddr](#) **should** be used to obtain function pointers for commands in core versions beyond the application's minimum required version.

3.5. Command Syntax and Duration

The Specification describes Vulkan commands as functions or procedures using C99 syntax. Language bindings for other languages such as C++ and JavaScript **may** allow for stricter parameter passing, or object-oriented interfaces.

Vulkan uses the standard C types for the base type of scalar parameters (e.g. types from `<stdint.h>`), with exceptions described below, or elsewhere in the text when appropriate:

[VkBool32](#) represents boolean `True` and `False` values, since C does not have a sufficiently portable built-in boolean type:

```
// Provided by VK_VERSION_1_0
typedef uint32_t VkBool32;
```

`VK_TRUE` represents a boolean **True** (unsigned integer 1) value, and `VK_FALSE` a boolean **False** (unsigned integer 0) value.

All values returned from a Vulkan implementation in a `VkBool32` will be either `VK_TRUE` or `VK_FALSE`.

Applications **must** not pass any other values than `VK_TRUE` or `VK_FALSE` into a Vulkan implementation where a `VkBool32` is expected.

`VK_TRUE` is a constant representing a `VkBool32` **True** value.

```
#define VK_TRUE 1U
```

`VK_FALSE` is a constant representing a `VkBool32` **False** value.

```
#define VK_FALSE 0U
```

`VkDeviceSize` represents device memory size and offset values:

```
// Provided by VK_VERSION_1_0
typedef uint64_t VkDeviceSize;
```

`VkDeviceAddress` represents device buffer address values:

```
// Provided by VK_VERSION_1_0
typedef uint64_t VkDeviceAddress;
```

Commands that create Vulkan objects are of the form `vkCreate*` and take `Vk*CreateInfo` structures with the parameters needed to create the object. These Vulkan objects are destroyed with commands of the form `vkDestroy*`.

The last in-parameter to each command that creates or destroys a Vulkan object is `pAllocator`. The `pAllocator` parameter **can** be a non-NULL value, in which case allocations for the given object are delegated to an application provided callback. Refer to the [Memory Allocation](#) chapter for further details.

Commands that allocate Vulkan objects owned by pool objects are of the form `vkAllocate*`, and take `Vk*AllocateInfo` structures. These Vulkan objects are freed with commands of the form `vkFree*`. These objects do not take allocators; if host memory is needed, they will use the allocator that was specified when their parent pool was created.

Commands are recorded into a command buffer by calling API commands of the form `vkCmd*`. Each such command **may** have different restrictions on where it **can** be used: in a primary and/or

secondary command buffer, inside and/or outside a render pass, and in one or more of the supported queue types. These restrictions are documented together with the definition of each such command.

The *duration* of a Vulkan command refers to the interval between calling the command and its return to the caller.

3.5.1. Lifetime of Retrieved Results

Information is retrieved from the implementation with commands of the form `vkGet*` and `vkEnumerate*`.

Unless otherwise specified for an individual command, the results are *invariant*; that is, they will remain unchanged when retrieved again by calling the same command with the same parameters, so long as those parameters themselves all remain valid.

3.5.2. Array Results

Some query commands of the form `vkGet*` and `vkEnumerate*` enable retrieving multiple results in the form of a return array. Such commands typically have two pointer arguments as follows:

- An element count pointer pointing to an integer variable, conventionally named as `p*Count` where `*` is the capitalized singular form of the name of the retrieved values.
- A pointer to an array where the result array is retrieved, conventionally named as `p*` where `*` is the capitalized plural form of the name of the retrieved values.

If such commands are called with the array pointer set to `NULL`, then the number of retrievable elements is returned in the variable pointed to by the element count pointer. Otherwise, the element count pointer **must** point to a variable set by the application to the number of elements in the return array, and on return the variable is overwritten with the number of elements actually written to the return array. If the input element count is less than the number of retrievable array elements, the query will write only as many elements to the return array as specified by the element count variable set by the application, and the command will return `VK_INCOMPLETE` instead of `VK_SUCCESS`, to indicate that not all retrievable array elements were returned.

NOTE

In practice, this means that applications will typically call such query commands twice:

- First, with the array pointer set to `NULL`, to retrieve the number of retrievable elements.
- Second, with the array pointer pointing to an application allocated storage for at least as many elements as indicated by the variable pointed to by the element count pointer, to retrieve at most as many of the retrievable elements.

Query commands that return one or more structures, regardless of whether they return a single or an array of structures with or without a `pNext` chain, **may** also contain arrays within those structures. Such return arrays are typically defined in the form of two members as follows:

- An integer value specifying the element count, conventionally named as `*Count` where `*` is the singular form of the name of the retrieved values.
- A pointer to an array where the result array is retrieved, conventionally named as `p*` where `*` is the capitalized plural form of the name of the retrieved values.

Analogously to query commands that return multiple results, if the command is called with the array pointer member of the output structure in question set to `NULL`, then the number of retrievable elements is returned in the element count member of that output structure. Otherwise, the element count **must** specify the number of elements in the return array, and on return the element count member is overwritten with the number of elements actually written to the return array. If the input element count is less than the number of retrievable array elements, the query will write only as many elements to the return array as specified by the input element count, and the command will return `VK_INCOMPLETE` instead of `VK_SUCCESS`, if the query command has a `VkResult` return type, to indicate that not all retrievable array elements were returned.

Applications need to separately track the value they provided as the input element count member for such arrays and compare those with the returned element counts in order to determine whether the actually returned element count is smaller than the size of the return array. Another side effect of this is that it is impossible for the application to determine if the number of retrievable elements has increased beyond the provided input element count so using return arrays in output structures **should** be limited to *invariant* array results. In practice, this means that applications will typically call such query commands multiple times:

NOTE

- First, with the array pointer member(s) set to `NULL`, to retrieve the number(s) of retrievable elements.
- Second, with the array pointer(s) pointing to an application allocated storage for at least as many elements as indicated by the element count member(s), to retrieve at most as many of the retrievable elements.
- Then the process **may** need to be repeated for all other newly introduced return arrays in any nested output structures indirectly specified through the previously retrieved result arrays.

Regardless of the type of query command, any array pointer member of an output structure **must** either be `NULL`, or point to an application-allocated array. Query commands **must** not return a pointer to implementation allocated storage in any output structure.

3.6. Threading Behavior

Vulkan is intended to provide scalable performance when used on multiple host threads. All commands support being called concurrently from multiple threads, but certain parameters, or components of parameters are defined to be *externally synchronized*. This means that the caller **must** guarantee that no more than one thread is using such a parameter at a given time.

More precisely, Vulkan commands use simple stores to update the state of Vulkan objects. A parameter declared as externally synchronized **may** have its contents updated at any time during the host execution of the command. If two commands operate on the same object and at least one of

the commands declares the object to be externally synchronized, then the caller **must** guarantee not only that the commands do not execute simultaneously, but also that the two commands are separated by an appropriate memory barrier (if needed).

NOTE

Memory barriers are particularly relevant for hosts based on the ARM CPU architecture, which is more weakly ordered than many developers are accustomed to from x86/x64 programming. Fortunately, most higher-level synchronization primitives (like the pthread library) perform memory barriers as a part of mutual exclusion, so mutexing Vulkan objects via these primitives will have the desired effect.

Similarly the application **must** avoid any potential data hazard of application-owned memory that has its [ownership temporarily acquired](#) by a Vulkan command. While the ownership of application-owned memory remains acquired by a command the implementation **may** read the memory at any point, and it **may** write non-**const** qualified memory at any point. Parameters referring to non-**const** qualified application-owned memory are not marked explicitly as *externally synchronized* in the Specification.

Many object types are *immutable*, meaning the objects **cannot** change once they have been created. These types of objects never need external synchronization, except that they **must** not be destroyed while they are in use on another thread. In certain special cases mutable object parameters are internally synchronized, making external synchronization unnecessary. Any command parameters that are not labeled as externally synchronized are either not mutated by the command or are internally synchronized. Additionally, certain objects related to a command's parameters (e.g. command pools and descriptor pools) **may** be affected by a command, and **must** also be externally synchronized. These implicit parameters are documented as described below.

Parameters of commands that are externally synchronized are listed below.

Externally Synchronized Parameters

- The **instance** parameter in [vkDestroyInstance](#)
- The **device** parameter in [vkDestroyDevice](#)
- The **queue** parameter in [vkQueueSubmit](#)
- The **fence** parameter in [vkQueueSubmit](#)
- The **queue** parameter in [vkQueueWaitIdle](#)
- The **memory** parameter in [vkFreeMemory](#)
- The **memory** parameter in [vkMapMemory](#)
- The **memory** parameter in [vkUnmapMemory](#)
- The **buffer** parameter in [vkBindBufferMemory](#)
- The **image** parameter in [vkBindImageMemory](#)
- The **queue** parameter in [vkQueueBindSparse](#)
- The **fence** parameter in [vkQueueBindSparse](#)

- The `fence` parameter in `vkDestroyFence`
- The `semaphore` parameter in `vkDestroySemaphore`
- The `event` parameter in `vkDestroyEvent`
- The `event` parameter in `vkSetEvent`
- The `event` parameter in `vkResetEvent`
- The `queryPool` parameter in `vkDestroyQueryPool`
- The `buffer` parameter in `vkDestroyBuffer`
- The `bufferView` parameter in `vkDestroyBufferView`
- The `image` parameter in `vkDestroyImage`
- The `imageView` parameter in `vkDestroyImageView`
- The `shaderModule` parameter in `vkDestroyShaderModule`
- The `pipelineCache` parameter in `vkDestroyPipelineCache`
- The `dstCache` parameter in `vkMergePipelineCaches`
- The `pipeline` parameter in `vkDestroyPipeline`
- The `pipelineLayout` parameter in `vkDestroyPipelineLayout`
- The `sampler` parameter in `vkDestroySampler`
- The `descriptorSetLayout` parameter in `vkDestroyDescriptorSetLayout`
- The `descriptorPool` parameter in `vkDestroyDescriptorPool`
- The `descriptorPool` parameter in `vkResetDescriptorPool`
- The `descriptorPool` member of the `pAllocateInfo` parameter in `vkAllocateDescriptorSets`
- The `descriptorPool` parameter in `vkFreeDescriptorSets`
- The `framebuffer` parameter in `vkDestroyFramebuffer`
- The `renderPass` parameter in `vkDestroyRenderPass`
- The `commandPool` parameter in `vkDestroyCommandPool`
- The `commandPool` parameter in `vkResetCommandPool`
- The `commandPool` member of the `pAllocateInfo` parameter in `vkAllocateCommandBuffers`
- The `commandPool` parameter in `vkFreeCommandBuffers`
- The `commandBuffer` parameter in `vkBeginCommandBuffer`
- The `commandBuffer` parameter in `vkEndCommandBuffer`
- The `commandBuffer` parameter in `vkResetCommandBuffer`
- The `commandBuffer` parameter in `vkCmdBindPipeline`
- The `commandBuffer` parameter in `vkCmdSetViewport`
- The `commandBuffer` parameter in `vkCmdSetScissor`
- The `commandBuffer` parameter in `vkCmdSetLineWidth`
- The `commandBuffer` parameter in `vkCmdSetDepthBias`

- The `commandBuffer` parameter in `vkCmdSetBlendConstants`
- The `commandBuffer` parameter in `vkCmdSetDepthBounds`
- The `commandBuffer` parameter in `vkCmdSetStencilCompareMask`
- The `commandBuffer` parameter in `vkCmdSetStencilWriteMask`
- The `commandBuffer` parameter in `vkCmdSetStencilReference`
- The `commandBuffer` parameter in `vkCmdBindDescriptorSets`
- The `commandBuffer` parameter in `vkCmdBindIndexBuffer`
- The `commandBuffer` parameter in `vkCmdBindVertexBuffers`
- The `commandBuffer` parameter in `vkCmdDraw`
- The `commandBuffer` parameter in `vkCmdDrawIndexed`
- The `commandBuffer` parameter in `vkCmdDrawIndirect`
- The `commandBuffer` parameter in `vkCmdDrawIndexedIndirect`
- The `commandBuffer` parameter in `vkCmdDispatch`
- The `commandBuffer` parameter in `vkCmdDispatchIndirect`
- The `commandBuffer` parameter in `vkCmdCopyBuffer`
- The `commandBuffer` parameter in `vkCmdCopyImage`
- The `commandBuffer` parameter in `vkCmdBlitImage`
- The `commandBuffer` parameter in `vkCmdCopyBufferToImage`
- The `commandBuffer` parameter in `vkCmdCopyImageToBuffer`
- The `commandBuffer` parameter in `vkCmdUpdateBuffer`
- The `commandBuffer` parameter in `vkCmdFillBuffer`
- The `commandBuffer` parameter in `vkCmdClearColorImage`
- The `commandBuffer` parameter in `vkCmdClearDepthStencilImage`
- The `commandBuffer` parameter in `vkCmdClearAttachments`
- The `commandBuffer` parameter in `vkCmdResolveImage`
- The `commandBuffer` parameter in `vkCmdSetEvent`
- The `commandBuffer` parameter in `vkCmdResetEvent`
- The `commandBuffer` parameter in `vkCmdWaitEvents`
- The `commandBuffer` parameter in `vkCmdPipelineBarrier`
- The `commandBuffer` parameter in `vkCmdBeginQuery`
- The `commandBuffer` parameter in `vkCmdEndQuery`
- The `commandBuffer` parameter in `vkCmdResetQueryPool`
- The `commandBuffer` parameter in `vkCmdWriteTimestamp`
- The `commandBuffer` parameter in `vkCmdCopyQueryPoolResults`
- The `commandBuffer` parameter in `vkCmdPushConstants`

- The `commandBuffer` parameter in `vkCmdBeginRenderPass`
- The `commandBuffer` parameter in `vkCmdNextSubpass`
- The `commandBuffer` parameter in `vkCmdEndRenderPass`
- The `commandBuffer` parameter in `vkCmdExecuteCommands`

There are also a few instances where a command **can** take in an application-allocated list whose contents are externally synchronized parameters. In these cases, the caller **must** guarantee that at most one thread is using a given element within the list at a given time. These parameters are listed below.

Externally Synchronized Parameter Lists

- Each element of the `pFences` parameter in `vkResetFences`
- Each element of the `pDescriptorSets` parameter in `vkFreeDescriptorSets`
- Each element of the `pCommandBuffers` parameter in `vkFreeCommandBuffers`

In addition, there are some implicit parameters that need to be externally synchronized. For example, when a `commandBuffer` parameter needs to be externally synchronized, it implies that the `commandPool` from which that command buffer was allocated also needs to be externally synchronized. The implicit parameters and their associated object are listed below.

Implicit Externally Synchronized Parameters

- All `VkPhysicalDevice` objects enumerated from `instance` in `vkDestroyInstance`
- All `VkQueue` objects created from `device` in `vkDestroyDevice`
- All `VkQueue` objects created from `device` in `vkDeviceWaitIdle`
- Any `VkDescriptorSet` objects allocated from `descriptorPool` in `vkResetDescriptorPool`
- The `VkCommandPool` that `commandBuffer` was allocated from in `vkBeginCommandBuffer`
- The `VkCommandPool` that `commandBuffer` was allocated from in `vkEndCommandBuffer`
- The `VkCommandPool` that `commandBuffer` was allocated from in `vkResetCommandBuffer`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBindPipeline`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetViewport`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetScissor`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetLineWidth`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetDepthBias`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetBlendConstants`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetDepthBounds`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetStencilCompareMask`

- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetStencilWriteMask`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetStencilReference`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBindDescriptorSets`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBindIndexBuffer`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBindVertexBuffers`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdDraw`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdDrawIndexed`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdDrawIndirect`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdDrawIndexedIndirect`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdDispatch`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdDispatchIndirect`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdCopyBuffer`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdCopyImage`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBlitImage`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdCopyBufferToImage`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdCopyImageToBuffer`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdUpdateBuffer`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdFillBuffer`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdClearColorImage`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdClearDepthStencilImage`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdClearAttachments`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdResolveImage`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetEvent`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdResetEvent`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdWaitEvents`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdPipelineBarrier`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBeginQuery`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdEndQuery`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdResetQueryPool`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdWriteTimestamp`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdCopyQueryPoolResults`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdPushConstants`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBeginRenderPass`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdNextSubpass`

- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdEndRenderPass`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdExecuteCommands`

3.7. Valid Usage

Valid usage defines a set of conditions which **must** be met in order to achieve well-defined runtime behavior in an application. These conditions depend only on Vulkan state, and the parameters or objects whose usage is constrained by the condition.

The core layer assumes applications are using the API correctly. Except as documented elsewhere in the Specification, the behavior of the core layer to an application using the API incorrectly is undefined, and **may** include program termination. However, implementations **must** ensure that incorrect usage by an application does not affect the integrity of the operating system, the Vulkan implementation, or other applications in the system using Vulkan. In particular, any guarantees made by an operating system about whether memory from one process **can** be visible to another process or not **must** not be violated by a Vulkan implementation for **any memory allocation**. Vulkan implementations are not **required** to make additional security or integrity guarantees beyond those provided by the OS unless explicitly directed by the application's use of a particular feature or extension.

NOTE

For instance, if an operating system guarantees that data in all its memory allocations are set to zero when newly allocated, the Vulkan implementation **must** make the same guarantees for any allocations it controls (e.g. `VkDeviceMemory`).

Similarly, if an operating system guarantees that use-after-free of host allocations will not result in values written by another process becoming visible, the same guarantees **must** be made by the Vulkan implementation for device memory.

Some valid usage conditions have dependencies on runtime limits or feature availability. It is possible to validate these conditions against Vulkan's minimum supported values for these limits and features, or some subset of other known values.

Valid usage conditions do not cover conditions where well-defined behavior (including returning an error code) exists.

Valid usage conditions **should** apply to the command or structure where complete information about the condition would be known during execution of an application. This is such that a validation layer or linter **can** be written directly against these statements at the point they are specified.

NOTE

This does lead to some non-obvious places for valid usage statements. For instance, the valid values for a structure might depend on a separate value in the calling command. In this case, the structure itself will not reference this valid usage as it is impossible to determine validity from the structure that it is invalid - instead this valid usage would be attached to the calling command.

Another example is draw state - the state setters are independent, and can cause a

legitimately invalid state configuration between draw calls; so the valid usage statements are attached to the place where all state needs to be valid - at the drawing command.

Valid usage conditions are described in a block labeled “Valid Usage” following each command or structure they apply to.

3.7.1. Usage Validation

Vulkan is a layered API. The lowest layer is the core Vulkan layer, as defined by this Specification. The application **can** use additional layers above the core for debugging, validation, and other purposes.

One of the core principles of Vulkan is that building and submitting command buffers **should** be highly efficient. Thus error checking and validation of state in the core layer is minimal, although more rigorous validation **can** be enabled through the use of layers.

Validation of correct API usage is left to validation layers. Applications **should** be developed with validation layers enabled, to help catch and eliminate errors. Once validated, released applications **should** not enable validation layers by default.

3.7.2. Implicit Valid Usage

Some valid usage conditions apply to all commands and structures in the API, unless explicitly denoted otherwise for a specific command or structure. These conditions are considered *implicit*, and are described in a block labeled “Valid Usage (Implicit)” following each command or structure they apply to. Implicit valid usage conditions are described in detail below.

Valid Usage for Object Handles

Any input parameter to a command that is an object handle **must** be a valid object handle, unless otherwise specified. An object handle is valid if:

- It has been created or allocated by a previous, successful call to the API. Such calls are noted in the Specification.
- It has not been deleted or freed by a previous call to the API. Such calls are noted in the Specification.
- Any objects used by that object, either as part of creation or execution, **must** also be valid.

The reserved values `VK_NULL_HANDLE` and `NULL` **can** be used in place of valid non-dispatchable handles and dispatchable handles, respectively, when *explicitly called out in the Specification*. Any command that creates an object successfully **must** not return these values. It is valid to pass these values to `vkDestroy*` or `vkFree*` commands, which will silently ignore these values.

Valid Usage for Pointers

Any parameter that is a pointer **must** be a *valid pointer* only if it is explicitly called out by a Valid Usage statement.

A pointer is “valid” if it points at memory containing values of the number and type(s) expected by the command, and all fundamental types accessed through the pointer (e.g. as elements of an array or as members of a structure) satisfy the alignment requirements of the host processor.

Valid Usage for Strings

Any parameter that is a pointer to **char** **must** be a finite sequence of values terminated by a null character, or if *explicitly called out in the Specification*, **can** be **NULL**.

Strings specified as UTF-8 encoded **must** not contain invalid UTF-8 sequences. See [String Representation](#) for additional information about strings.

Valid Usage for Enumerated Types

Any parameter of an enumerated type **must** be a valid enumerant for that type. Use of an enumerant is valid if the following conditions are true:

- The enumerant is defined as part of the enumerated type.
- The enumerant is not a value suffixed with **_MAX_ENUM**.
 - This value exists only to ensure that C **enum** types are 32 bits in size and **must** not be used by applications.
- If the enumerant is used in a function that has a [VkInstance](#) as its first parameter and either:
 - it was added by a core version that is supported and the value of [VkApplicationInfo::apiVersion](#) is greater than or equal to the version that added it; or
 - it was added by an [instance extension](#) that was enabled for the instance.
- If the enumerant is used in a function that has a [VkPhysicalDevice](#) object as its first parameter and either:
 - it was added by a core version that is supported by that device (as reported by [VkPhysicalDeviceProperties::apiVersion](#));
 - it was added by an [instance extension](#) that was enabled for the instance; or
 - it was added by a [device extension](#) that is supported by that device.
- If the enumerant is used in a function that has any other dispatchable object as its first parameter and either:
 - it was added by a core version that is supported for the device (as reported by [VkPhysicalDeviceProperties::apiVersion](#)); or
 - it was added by a [device extension](#) that was enabled for the device.

Any enumerated type returned from a query command or otherwise output from Vulkan to the application **must** not have a reserved value. Reserved values are values not defined by any extension for that enumerated type.

NOTE

In some special cases, an enumerant is only meaningful if a feature defined by an extension is also enabled, as well as the extension itself. The global “valid enumerant” rule described here does not address such cases.

NOTE

This language is intended to accommodate cases such as “hidden” extensions known only to driver internals, or layers enabling extensions without knowledge of the application, without allowing return of values not defined by any extension.

NOTE

Application developers are encouraged to be careful when using `switch` statements with Vulkan API enums. This is because new extensions can add new values to existing enums. Using a `default:` statement within a `switch` may avoid future compilation issues.

Valid Usage for Flags

A collection of flags is represented by a bitmask using the type `VkFlags`:

```
// Provided by VK_VERSION_1_0
typedef uint32_t VkFlags;
```

Bitmasks are passed to many commands and structures to compactly represent options, but `VkFlags` is not used directly in the API. Instead, a `Vk*Flags` type which is an alias of `VkFlags`, and whose name matches the corresponding `Vk*FlagBits` that are valid for that type, is used.

Any `Vk*Flags` member or parameter used in the API as an input **must** be a valid combination of bit flags. A valid combination is either zero or the bitwise OR of valid bit flags.

An individual bit flag is valid for a `Vk*Flags` type if it would be a [valid enumerant](#) when used with the equivalent `Vk*FlagBits` type, where the bits type is obtained by taking the flag type and replacing the trailing `Flags` with `FlagBits`. For example, a flag value of type `VkColorComponentFlags` **must** contain only bit flags defined by `VkColorComponentFlagBits`.

Any `Vk*Flags` member or parameter returned from a query command or otherwise output from Vulkan to the application **may** contain bit flags undefined in its corresponding `Vk*FlagBits` type. An application **cannot** rely on the state of these unspecified bits.

Only the low-order 31 bits (bit positions zero through 30) are available for use as flag bits.

NOTE

This restriction is due to poorly defined behavior by C compilers given a C enumerant value of `0x80000000`. In some cases adding this enumerant value may increase the size of the underlying `Vk*FlagBits` type, breaking the ABI.

Valid Usage for Structure Types

Any parameter that is a structure containing a `sType` member **must** have a value of `sType` which is a valid `VkStructureType` value matching the type of the structure.

Valid Usage for Structure Pointer Chains

Any parameter that is a structure containing a `void* pNext` member **must** have a value of `pNext` that is either `NULL`, or is a pointer to a valid *extending structure*, containing `sType` and `pNext` members as described in the [Vulkan Documentation and Extensions](#) document in the section “Extending

Structures”. The set of structures connected by **pNext** pointers is referred to as a **pNext chain**.

Each structure included in the **pNext** chain **must** be defined at runtime by either:

- a core version which is supported
- an extension which is enabled

Each type of extending structure **must** not appear more than once in a **pNext** chain, including any [aliases](#). This general rule may be explicitly overridden for specific structures.

Any component of the implementation (the loader, any enabled layers, and drivers) **must** skip over, without processing (other than reading the **sType** and **pNext** members) any extending structures in the chain not defined by core versions or extensions supported by that component.

As a convenience to implementations and layers needing to iterate through a structure pointer chain, the Vulkan API provides two *base structures*. These structures allow for some type safety, and can be used by Vulkan API functions that operate on generic inputs and outputs.

The **VkBaseInStructure** structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkBaseInStructure {
    VkStructureType      sType;
    const struct VkBaseInStructure* pNext;
} VkBaseInStructure;
```

- **sType** is the structure type of the structure being iterated through.
- **pNext** is **NULL** or a pointer to the next structure in a structure chain.

VkBaseInStructure can be used to facilitate iterating through a read-only structure pointer chain.

The **VkBaseOutStructure** structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkBaseOutStructure {
    VkStructureType      sType;
    struct VkBaseOutStructure* pNext;
} VkBaseOutStructure;
```

- **sType** is the structure type of the structure being iterated through.
- **pNext** is **NULL** or a pointer to the next structure in a structure chain.

VkBaseOutStructure can be used to facilitate iterating through a structure pointer chain that returns data back to the application.

Valid Usage for Nested Structures

The above conditions also apply recursively to members of structures provided as input to a

command, either as a direct argument to the command, or themselves a member of another structure.

Specifics on valid usage of each command are covered in their individual sections.

Valid Usage for Extensions

Instance-level functionality or behavior added by an [instance extension](#) to the API **must** not be used unless that extension is supported by the instance as determined by [vkEnumerateInstanceExtensionProperties](#), and that extension is enabled in [VkInstanceCreateInfo](#).

Physical-device-level functionality or behavior added by an [instance extension](#) to the API **must** not be used unless that extension is supported by the instance as determined by [vkEnumerateInstanceExtensionProperties](#), and that extension is enabled in [VkInstanceCreateInfo](#).

Device-level functionality added by a [device extension](#) that is dispatched from a [VkDevice](#), or from a child object of a [VkDevice](#) **must** not be used unless that extension is supported by the device as determined by [vkEnumerateDeviceExtensionProperties](#), and that extension is enabled in [VkDeviceCreateInfo](#).

Valid Usage for Newer Core Versions

Physical-device-level functionality or behavior added by a [new core version](#) of the API **must** not be used unless it is supported by the physical device as determined by [VkPhysicalDeviceProperties::apiVersion](#) and the specified version of [VkApplicationInfo::apiVersion](#).

Device-level functionality or behavior added by a [new core version](#) of the API **must** not be used unless it is supported by the device as determined by [VkPhysicalDeviceProperties::apiVersion](#) and the specified version of [VkApplicationInfo::apiVersion](#).

3.8. [VkResult](#) Return Codes

While the core Vulkan API is not designed to capture incorrect usage, some circumstances still require return codes. Commands in Vulkan return their status via return codes that are in one of two categories:

- Successful completion codes are returned when a command needs to communicate success or status information. All successful completion codes are non-negative values.
- Runtime error codes are returned when a command needs to communicate a failure that could only be detected at runtime. All runtime error codes are negative values.

All return codes in Vulkan are reported via [VkResult](#) return values. The possible codes are:

```
// Provided by VK_VERSION_1_0
typedef enum VkResult {
    VK_SUCCESS = 0,
    VK_NOT_READY = 1,
    VK_TIMEOUT = 2,
    VK_EVENT_SET = 3,
    VK_EVENT_RESET = 4,
    VK_INCOMPLETE = 5,
    VK_ERROR_OUT_OF_HOST_MEMORY = -1,
    VK_ERROR_OUT_OF_DEVICE_MEMORY = -2,
    VK_ERROR_INITIALIZATION_FAILED = -3,
    VK_ERROR_DEVICE_LOST = -4,
    VK_ERROR_MEMORY_MAP_FAILED = -5,
    VK_ERROR_LAYER_NOT_PRESENT = -6,
    VK_ERROR_EXTENSION_NOT_PRESENT = -7,
    VK_ERROR_FEATURE_NOT_PRESENT = -8,
    VK_ERROR_INCOMPATIBLE_DRIVER = -9,
    VK_ERROR_TOO_MANY_OBJECTS = -10,
    VK_ERROR_FORMAT_NOT_SUPPORTED = -11,
    VK_ERROR_FRAGMENTED_POOL = -12,
    VK_ERROR_UNKNOWN = -13,
} VkResult;
```

Success Codes

- **VK_SUCCESS** Command successfully completed
- **VK_NOT_READY** A fence or query has not yet completed
- **VK_TIMEOUT** A wait operation has not completed in the specified time
- **VK_EVENT_SET** An event is signaled
- **VK_EVENT_RESET** An event is unsignaled
- **VK_INCOMPLETE** A return array was too small for the result

Error Codes

- **VK_ERROR_OUT_OF_HOST_MEMORY** A host memory allocation has failed.
- **VK_ERROR_OUT_OF_DEVICE_MEMORY** A device memory allocation has failed.
- **VK_ERROR_INITIALIZATION_FAILED** Initialization of an object could not be completed for implementation-specific reasons.
- **VK_ERROR_DEVICE_LOST** The logical or physical device has been lost. See [Lost Device](#)
- **VK_ERROR_MEMORY_MAP_FAILED** Mapping of a memory object has failed.
- **VK_ERROR_LAYER_NOT_PRESENT** A requested layer is not present or could not be loaded.
- **VK_ERROR_EXTENSION_NOT_PRESENT** A requested extension is not supported.
- **VK_ERROR_FEATURE_NOT_PRESENT** A requested feature is not supported.
- **VK_ERROR_INCOMPATIBLE_DRIVER** The requested version of Vulkan is not supported by the driver or is otherwise incompatible for implementation-specific reasons.

- **VK_ERROR_TOO_MANY_OBJECTS** Too many objects of the type have already been created.
- **VK_ERROR_FORMAT_NOT_SUPPORTED** A requested format is not supported on this device.
- **VK_ERROR_FRAGMENTED_POOL** A pool allocation has failed due to fragmentation of the pool's memory. This **must** only be returned if no attempt to allocate host or device memory was made to accommodate the new allocation.
- **VK_ERROR_UNKNOWN** An unknown error has occurred; either the application has provided invalid input, or an implementation failure has occurred.

If a command returns a runtime error, unless otherwise specified any output parameters will have undefined contents, except that if the output parameter is a structure with **sType** and **pNext** fields, those fields will be unmodified. Any structures chained from **pNext** will also have undefined contents, except that **sType** and **pNext** will be unmodified.

VK_ERROR_OUT_OF_*_MEMORY errors do not modify any currently existing Vulkan objects. Objects that have already been successfully created **can** still be used by the application.

NOTE

As a general rule, **Free**, **Release**, and **Reset** commands do not return **VK_ERROR_OUT_OF_HOST_MEMORY**, while any other command with a return code **may** return it. Any exceptions from this rule are described for those commands.

VK_ERROR_UNKNOWN will be returned by an implementation when an unexpected error occurs that cannot be attributed to valid behavior of the application and implementation. Under these conditions, it **may** be returned from any command returning a **VkResult**.

NOTE

VK_ERROR_UNKNOWN is not expected to ever be returned if the application behavior is valid, and if the implementation is bug-free. If **VK_ERROR_UNKNOWN** is received, the application should be checked against the latest validation layers to verify correct behavior as much as possible. If no issues are identified it could be an implementation issue, and the implementor should be contacted for support.

Performance-critical commands generally do not have return codes. If a runtime error occurs in such commands, the implementation will defer reporting the error until a specified point. For commands that record into command buffers (**vkCmd***) runtime errors are reported by **vkEndCommandBuffer**.

3.9. Numeric Representation and Computation

Implementations normally perform computations in floating-point, and **must** meet the range and precision requirements defined under “Floating-Point Computation” below.

These requirements only apply to computations performed in Vulkan operations outside of shader execution, such as texture image specification and sampling, and per-fragment operations. Range and precision requirements during shader execution differ and are specified by the [Precision and Operation of SPIR-V Instructions](#) section.

In some cases, the representation and/or precision of operations is implicitly limited by the specified format of vertex or texel data consumed by Vulkan. Specific floating-point formats are

described later in this section.

3.9.1. Floating-Point Computation

Most floating-point computation is performed in SPIR-V shader modules. The properties of computation within shaders are constrained as defined by the [Precision and Operation of SPIR-V Instructions](#) section.

Some floating-point computation is performed outside of shaders, such as viewport and depth range calculations. For these computations, we do not specify how floating-point numbers are to be represented, or the details of how operations on them are performed, but only place minimal requirements on representation and precision as described in the remainder of this section.

We require simply that numbers' floating-point parts contain enough bits and that their exponent fields are large enough so that individual results of floating-point operations are accurate to about 1 part in 10^5 . The maximum representable magnitude for all floating-point values **must** be at least 2^{32} .

$$x \times 0 = 0 \times x = 0 \text{ for any non-infinite and non-NaN } x.$$

$$1 \times x = x \times 1 = x.$$

$$x + 0 = 0 + x = x.$$

$$0^0 = 1.$$

Occasionally, further requirements will be specified. Most single-precision floating-point formats meet these requirements.

The special values Inf and -Inf encode values with magnitudes too large to be represented; the special value NaN encodes “Not A Number” values resulting from undefined arithmetic operations such as $0 / 0$. Implementations **may** support Inf and NaN in their floating-point computations. Any computation which does not support either Inf or NaN, for which that value is an input or output will yield an undefined value.

3.9.2. Floating-Point Format Conversions

When a value is converted to a defined floating-point representation, finite values falling between two representable finite values are rounded to one or the other. The rounding mode is not defined. Finite values whose magnitude is larger than that of any representable finite value may be rounded either to the closest representable finite value or to the appropriately signed infinity. For unsigned destination formats any negative values are converted to zero. Positive infinity is converted to positive infinity; negative infinity is converted to negative infinity in signed formats and to zero in unsigned formats; and any NaN is converted to a NaN.

3.9.3. 16-Bit Floating-Point Numbers

16-bit floating-point numbers are defined in the “16-bit floating-point numbers” section of the [Khronos Data Format Specification](#).

3.9.4. Unsigned 11-Bit Floating-Point Numbers

Unsigned 11-bit floating-point numbers are defined in the “Unsigned 11-bit floating-point numbers” section of the [Khronos Data Format Specification](#).

3.9.5. Unsigned 10-Bit Floating-Point Numbers

Unsigned 10-bit floating-point numbers are defined in the “Unsigned 10-bit floating-point numbers” section of the [Khronos Data Format Specification](#).

3.9.6. General Requirements

Any representable floating-point value in the appropriate format is legal as input to a Vulkan command that requires floating-point data. The result of providing a value that is not a floating-point number to such a command is unspecified, but **must** not lead to Vulkan interruption or termination. For example, providing a negative zero (where applicable) or a denormalized number to a Vulkan command **must** yield deterministic results, while providing a NaN or Inf yields unspecified results.

Some calculations require division. In such cases (including implied divisions performed by vector normalization), division by zero produces an unspecified result but **must** not lead to Vulkan interruption or termination.

3.10. Fixed-Point Data Conversions

When generic vertex attributes and pixel color or depth *components* are represented as integers, they are often (but not always) considered to be *normalized*. Normalized integer values are treated specially when being converted to and from floating-point values, and are usually referred to as *normalized fixed-point*.

In the remainder of this section, b denotes the bit width of the fixed-point integer representation. When the integer is one of the types defined by the API, b is the bit width of that type. When the integer comes from an [image](#) containing color or depth component texels, b is the number of bits allocated to that component in its [specified image format](#).

The signed and unsigned fixed-point representations are assumed to be b -bit binary two's-complement integers and binary unsigned integers, respectively.

3.10.1. Conversion From Normalized Fixed-Point to Floating-Point

Unsigned normalized fixed-point integers represent numbers in the range $[0,1]$. The conversion from an unsigned normalized fixed-point value c to the corresponding floating-point value f is defined as

$$f = \frac{c}{2^b - 1}$$

Signed normalized fixed-point integers represent numbers in the range [-1,1]. The conversion from a signed normalized fixed-point value c to the corresponding floating-point value f is performed using

$$f = \max\left(\frac{c}{2^b - 1}, -1.0\right)$$

Only the range $[-2^{b-1} + 1, 2^{b-1} - 1]$ is used to represent signed fixed-point values in the range [-1,1]. For example, if $b = 8$, then the integer value -127 corresponds to -1.0 and the value 127 corresponds to 1.0. This equation is used everywhere that signed normalized fixed-point values are converted to floating-point.

Note that while zero is exactly expressible in this representation, one value (-128 in the example) is outside the representable range, and implementations **must** clamp it to -1.0. Where the value is subject to further processing by the implementation, e.g. during texture filtering, values less than -1.0 **may** be used but the result **must** be clamped before the value is returned to shaders.

3.10.2. Conversion From Floating-Point to Normalized Fixed-Point

The conversion from a floating-point value f to the corresponding unsigned normalized fixed-point value c is defined by first clamping f to the range [0,1], then computing

$$c = \text{convertFloatToUint}(f \times (2^b - 1), b)$$

where $\text{convertFloatToUint}(r,b)$ returns one of the two unsigned binary integer values with exactly b bits which are closest to the floating-point value r . Implementations **should** round to nearest. If r is equal to an integer, then that integer value **must** be returned. In particular, if f is equal to 0.0 or 1.0, then c **must** be assigned 0 or $2^b - 1$, respectively.

The conversion from a floating-point value f to the corresponding signed normalized fixed-point value c is performed by clamping f to the range [-1,1], then computing

$$c = \text{convertFloatToInt}(f \times (2^{b-1} - 1), b)$$

where $\text{convertFloatToInt}(r,b)$ returns one of the two signed two's-complement binary integer values with exactly b bits which are closest to the floating-point value r . Implementations **should** round to nearest. If r is equal to an integer, then that integer value **must** be returned. In particular, if f is equal to -1.0, 0.0, or 1.0, then c **must** be assigned $-(2^{b-1} - 1)$, 0, or $2^{b-1} - 1$, respectively.

This equation is used everywhere that floating-point values are converted to signed normalized fixed-point.

3.11. String Representation

Strings passed into and returned from Vulkan API commands are usually defined to be null-

terminated and UTF-8 encoded.

NOTE

Exceptions to this rule exist only when strings are defined or used by operating system APIs where that OS has a different convention. For example, `VkExportMemoryWin32HandleInfoKHR::name` is a null-terminated UTF-16 encoded string used in conjunction with Windows handles.

When a UTF-8 string is **returned from** a Vulkan API query, it is returned in a fixed-length buffer of C `char`. For example, a string returned in `VkPhysicalDeviceProperties::deviceName` has maximum length `VK_MAX_PHYSICAL_DEVICE_NAME_SIZE`, and a string returned in `VkExtensionProperties::extensionName` has maximum length `VK_MAX_EXTENSION_NAME_SIZE`. The string, **including** its null terminator, will always fit completely within this buffer. If the string is shorter than the buffer size, the contents of `char` in the buffer following the null terminator are undefined.

When a UTF-8 string is **passed into** a Vulkan API, such as `VkDeviceCreateInfo::ppEnabledExtensionNames`, there is no explicit limit on the length of that string. However, the string **must** contain a valid UTF-8 encoded string and **must** be null-terminated.

3.12. Common Object Types

Some types of Vulkan objects are used in many different structures and command parameters, and are described here. These types include *offsets*, *extents*, and *rectangles*.

3.12.1. Offsets

Offsets are used to describe a pixel location within an image or framebuffer, as an (x,y) location for two-dimensional images, or an (x,y,z) location for three-dimensional images.

A two-dimensional offset is defined by the structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkOffset2D {
    int32_t    x;
    int32_t    y;
} VkOffset2D;
```

- `x` is the x offset.
- `y` is the y offset.

A three-dimensional offset is defined by the structure:


```
// Provided by VK_VERSION_1_0
typedef struct VkOffset3D {
    int32_t    x;
    int32_t    y;
    int32_t    z;
} VkOffset3D;
```

- **x** is the x offset.
- **y** is the y offset.
- **z** is the z offset.

3.12.2. Extents

Extents are used to describe the size of a rectangular region of pixels within an image or framebuffer, as (width,height) for two-dimensional images, or as (width,height,depth) for three-dimensional images.

A two-dimensional extent is defined by the structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkExtent2D {
    uint32_t    width;
    uint32_t    height;
} VkExtent2D;
```

- **width** is the width of the extent.
- **height** is the height of the extent.

A three-dimensional extent is defined by the structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkExtent3D {
    uint32_t    width;
    uint32_t    height;
    uint32_t    depth;
} VkExtent3D;
```

- **width** is the width of the extent.
- **height** is the height of the extent.
- **depth** is the depth of the extent.

3.12.3. Rectangles

Rectangles are used to describe a specified rectangular region of pixels within an image or framebuffer. Rectangles include both an offset and an extent of the same dimensionality, as

described above. Two-dimensional rectangles are defined by the structure

```
// Provided by VK_VERSION_1_0
typedef struct VkRect2D {
    VkOffset2D    offset;
    VkExtent2D    extent;
} VkRect2D;
```

- `offset` is a `VkOffset2D` specifying the rectangle offset.
- `extent` is a `VkExtent2D` specifying the rectangle extent.

3.12.4. Structure Types

Each value corresponds to a particular structure with a `sType` member with a matching name. As a general rule, the name of each `VkStructureType` value is obtained by taking the name of the structure, stripping the leading `Vk`, prefixing each capital letter with `_`, converting the entire resulting string to upper case, and prefixing it with `VK_STRUCTURE_TYPE_`. For example, structures of type `VkImageCreateInfo` correspond to a `VkStructureType` value of `VK_STRUCTURE_TYPE_IMAGE_CREATE_INFO`, and thus a structure of this type **must** have its `sType` member set to this value before it is passed to the API.

The values `VK_STRUCTURE_TYPE_LOADER_INSTANCE_CREATE_INFO` and `VK_STRUCTURE_TYPE_LOADER_DEVICE_CREATE_INFO` are reserved for internal use by the loader, and do not have corresponding Vulkan structures in this Specification.

Structure types supported by the Vulkan API include:

```
// Provided by VK_VERSION_1_0
typedef enum VkStructureType {
    VK_STRUCTURE_TYPE_APPLICATION_INFO = 0,
    VK_STRUCTURE_TYPE_INSTANCE_CREATE_INFO = 1,
    VK_STRUCTURE_TYPE_DEVICE_QUEUE_CREATE_INFO = 2,
    VK_STRUCTURE_TYPE_DEVICE_CREATE_INFO = 3,
    VK_STRUCTURE_TYPE_SUBMIT_INFO = 4,
    VK_STRUCTURE_TYPE_MEMORY_ALLOCATE_INFO = 5,
    VK_STRUCTURE_TYPE_MAPPED_MEMORY_RANGE = 6,
    VK_STRUCTURE_TYPE_BIND_SPARSE_INFO = 7,
    VK_STRUCTURE_TYPE_FENCE_CREATE_INFO = 8,
    VK_STRUCTURE_TYPE_SEMAPHORE_CREATE_INFO = 9,
    VK_STRUCTURE_TYPE_EVENT_CREATE_INFO = 10,
    VK_STRUCTURE_TYPE_QUERY_POOL_CREATE_INFO = 11,
    VK_STRUCTURE_TYPE_BUFFER_CREATE_INFO = 12,
    VK_STRUCTURE_TYPE_BUFFER_VIEW_CREATE_INFO = 13,
    VK_STRUCTURE_TYPE_IMAGE_CREATE_INFO = 14,
    VK_STRUCTURE_TYPE_IMAGE_VIEW_CREATE_INFO = 15,
    VK_STRUCTURE_TYPE_SHADER_MODULE_CREATE_INFO = 16,
    VK_STRUCTURE_TYPE_PIPELINE_CACHE_CREATE_INFO = 17,
    VK_STRUCTURE_TYPE_PIPELINE_SHADER_STAGE_CREATE_INFO = 18,
```

```

VK_STRUCTURE_TYPE_PIPELINE_VERTEX_INPUT_STATE_CREATE_INFO = 19,
VK_STRUCTURE_TYPE_PIPELINE_INPUT_ASSEMBLY_STATE_CREATE_INFO = 20,
VK_STRUCTURE_TYPE_PIPELINE_TESSELLATION_STATE_CREATE_INFO = 21,
VK_STRUCTURE_TYPE_PIPELINE_VIEWPORT_STATE_CREATE_INFO = 22,
VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_STATE_CREATE_INFO = 23,
VK_STRUCTURE_TYPE_PIPELINE_MULTISAMPLE_STATE_CREATE_INFO = 24,
VK_STRUCTURE_TYPE_PIPELINE_DEPTH_STENCIL_STATE_CREATE_INFO = 25,
VK_STRUCTURE_TYPE_PIPELINE_COLOR_BLEND_STATE_CREATE_INFO = 26,
VK_STRUCTURE_TYPE_PIPELINE_DYNAMIC_STATE_CREATE_INFO = 27,
VK_STRUCTURE_TYPE_GRAPHICS_PIPELINE_CREATE_INFO = 28,
VK_STRUCTURE_TYPE_COMPUTE_PIPELINE_CREATE_INFO = 29,
VK_STRUCTURE_TYPE_PIPELINE_LAYOUT_CREATE_INFO = 30,
VK_STRUCTURE_TYPE_SAMPLER_CREATE_INFO = 31,
VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_CREATE_INFO = 32,
VK_STRUCTURE_TYPE_DESCRIPTOR_POOL_CREATE_INFO = 33,
VK_STRUCTURE_TYPE_DESCRIPTOR_SET_ALLOCATE_INFO = 34,
VK_STRUCTURE_TYPE_WRITE_DESCRIPTOR_SET = 35,
VK_STRUCTURE_TYPE_COPY_DESCRIPTOR_SET = 36,
VK_STRUCTURE_TYPE_FRAMEBUFFER_CREATE_INFO = 37,
VK_STRUCTURE_TYPE_RENDER_PASS_CREATE_INFO = 38,
VK_STRUCTURE_TYPE_COMMAND_POOL_CREATE_INFO = 39,
VK_STRUCTURE_TYPE_COMMAND_BUFFER_ALLOCATE_INFO = 40,
VK_STRUCTURE_TYPE_COMMAND_BUFFER_INHERITANCE_INFO = 41,
VK_STRUCTURE_TYPE_COMMAND_BUFFER_BEGIN_INFO = 42,
VK_STRUCTURE_TYPE_RENDER_PASS_BEGIN_INFO = 43,
VK_STRUCTURE_TYPE_BUFFER_MEMORY_BARRIER = 44,
VK_STRUCTURE_TYPE_IMAGE_MEMORY_BARRIER = 45,
VK_STRUCTURE_TYPE_MEMORY_BARRIER = 46,
VK_STRUCTURE_TYPE_LOADER_INSTANCE_CREATE_INFO = 47,
VK_STRUCTURE_TYPE_LOADER_DEVICE_CREATE_INFO = 48,
} VkStructureType;

```

3.13. API Name Aliases

A small number of APIs did not follow the [naming conventions](#) when initially defined. For consistency, when we discover an API name that violates the naming conventions, we rename it in the Specification, XML, and header files. For backwards compatibility, the original (incorrect) name is retained as a “typo alias”. The alias is deprecated and should not be used, but will be retained indefinitely.

NOTE

VK_STENCIL_FRONT_AND_BACK is an example of a *typo alias*. It was initially defined as part of **VkStencilFaceFlagBits**. Once the naming inconsistency was noticed, it was renamed to **VK_STENCIL_FACE_FRONT_AND_BACK**, and the old name was aliased to the correct name.

Chapter 4. Initialization

Before using Vulkan, an application **must** initialize it by loading the Vulkan commands, and creating a `VkInstance` object.

4.1. Command Function Pointers

Vulkan commands are not necessarily exposed by static linking on a platform. Commands to query function pointers for Vulkan commands are described below.

NOTE

When extensions are [promoted](#) or otherwise incorporated into another extension or Vulkan core version, command [aliases](#) may be included. Whilst the behavior of each command alias is identical, the behavior of retrieving each alias's function pointer is not. A function pointer for a given alias can only be retrieved if the extension or version that introduced that alias is supported and enabled, irrespective of whether any other alias is available.

Function pointers for all Vulkan commands **can** be obtained by calling:

```
// Provided by VK_VERSION_1_0
PFN_vkVoidFunction vkGetInstanceProcAddr(
    VkInstance          instance,
    const char*         pName);
```

- `instance` is the instance that the function pointer will be compatible with, or `NULL` for commands not dependent on any instance.
- `pName` is the name of the command to obtain.

`vkGetInstanceProcAddr` itself is obtained in a platform- and loader- specific manner. Typically, the loader library will export this command as a function symbol, so applications **can** link against the loader library, or load it dynamically and look up the symbol using platform-specific APIs.

The table below defines the various use cases for `vkGetInstanceProcAddr` and expected return value (“fp” is “function pointer”) for each case. A valid returned function pointer (“fp”) **must** not be `NULL`.

The returned function pointer is of type `PFN_vkVoidFunction`, and **must** be cast to the type of the command being queried before use.

Table 1. `vkGetInstanceProcAddr` behavior

<code>instance</code>	<code>pName</code>	return value
* ¹	<code>NULL</code>	undefined
invalid non- <code>NULL</code> instance	* ¹	undefined
<code>NULL</code>	<i>global command</i> ²	fp
instance	<code>vkGetInstanceProcAddr</code>	fp

<i>instance</i>	<i>pName</i>	return value
instance	core <i>dispatchable command</i>	fp ³
instance	enabled instance extension dispatchable command for <i>instance</i>	fp ³
instance	available device extension ⁴ dispatchable command for <i>instance</i>	fp ³
any other case, not covered above		NULL

1

"*" means any representable value for the parameter (including valid values, invalid values, and NULL).

2

The global commands are: [vkEnumerateInstanceExtensionProperties](#), [vkEnumerateInstanceLayerProperties](#), and [vkCreateInstance](#). Dispatchable commands are all other commands which are not global.

3

The returned function pointer **must** only be called with a dispatchable object (the first parameter) that is *instance* or a child of *instance*, e.g. [VkInstance](#), [VkPhysicalDevice](#), [VkDevice](#), [VkQueue](#), or [VkCommandBuffer](#).

4

An “available device extension” is a device extension supported by any physical device enumerated by *instance*.

Valid Usage (Implicit)

- VUID-vkGetInstanceProcAddr-instance-parameter
If *instance* is not NULL, *instance* **must** be a valid [VkInstance](#) handle
- VUID-vkGetInstanceProcAddr-pName-parameter
pName **must** be a null-terminated UTF-8 string

In order to support systems with multiple Vulkan implementations, the function pointers returned by [vkGetInstanceProcAddr](#) **may** point to dispatch code that calls a different real implementation for different [VkDevice](#) objects or their child objects. The overhead of the internal dispatch for [VkDevice](#) objects can be avoided by obtaining device-specific function pointers for any commands that use a device or device-child object as their dispatchable object. Such function pointers **can** be obtained by calling:

```
// Provided by VK_VERSION_1_0
PFN_vkVoidFunction vkGetDeviceProcAddr(
    VkDevice          device,
    const char*       pName);
```

The table below defines the various use cases for `vkGetDeviceProcAddr` and expected return value (“fp” is “function pointer”) for each case. A valid returned function pointer (“fp”) **must** not be `NULL`.

The returned function pointer is of type `PFN_vkVoidFunction`, and **must** be cast to the type of the command being queried before use. The function pointer **must** only be called with a dispatchable object (the first parameter) that is `device` or a child of `device`.

Table 2. `vkGetDeviceProcAddr` behavior

device	pName	return value
<code>NULL</code>	* ¹	undefined
invalid device	* ¹	undefined
device	<code>NULL</code>	undefined
device	requested core version ² device-level dispatchable command ³	fp ⁴
device	enabled extension device-level dispatchable command ³	fp ⁴
any other case, not covered above		<code>NULL</code>

1

“*” means any representable value for the parameter (including valid values, invalid values, and `NULL`).

2

Device-level commands which are part of the core version specified by `VkApplicationInfo::apiVersion` when creating the instance will always return a valid function pointer. Core commands beyond that version which are supported by the implementation **may** either return `NULL` or a function pointer. If a function pointer is returned, it **must** not be called.

3

In this function, device-level excludes all physical-device-level commands.

4

The returned function pointer **must** only be called with a dispatchable object (the first parameter) that is `device` or a child of `device` e.g. `VkDevice`, `VkQueue`, or `VkCommandBuffer`.

Valid Usage (Implicit)

- VUID-vkGetDeviceProcAddr-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkGetDeviceProcAddr-pName-parameter
pName **must** be a null-terminated UTF-8 string

The definition of [PFN_vkVoidFunction](#) is:

```
// Provided by VK_VERSION_1_0
typedef void (VKAPI_PTR *PFN_vkVoidFunction)(void);
```

This type is returned from command function pointer queries, and **must** be cast to an actual command function pointer before use.

4.2. Instances

There is no global state in Vulkan and all per-application state is stored in a [VkInstance](#) object. Creating a [VkInstance](#) object initializes the Vulkan library and allows the application to pass information about itself to the implementation.

Instances are represented by [VkInstance](#) handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_HANDLE(VkInstance)
```

To create an instance object, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateInstance(
    const VkInstanceCreateInfo*      pCreateInfo,
    const VkAllocationCallbacks*    pAllocator,
    VkInstance*                      pInstance);
```

- **pCreateInfo** is a pointer to a [VkInstanceCreateInfo](#) structure controlling creation of the instance.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.
- **pInstance** points a [VkInstance](#) handle in which the resulting instance is returned.

[vkCreateInstance](#) verifies that the requested layers exist. If not, [vkCreateInstance](#) will return [VK_ERROR_LAYER_NOT_PRESENT](#). Next [vkCreateInstance](#) verifies that the requested extensions are supported (e.g. in the implementation or in any enabled instance layer) and if any requested extension is not supported, [vkCreateInstance](#) **must** return [VK_ERROR_EXTENSION_NOT_PRESENT](#). After verifying and enabling the instance layers and extensions the [VkInstance](#) object is created and

returned to the application. If a requested extension is only supported by a layer, both the layer and the extension need to be specified at `vkCreateInstance` time for the creation to succeed.

Valid Usage

- VUID-vkCreateInstance-ppEnabledExtensionNames-01388

All [required extensions](#) for each extension in the [VkInstanceCreateInfo::ppEnabledExtensionNames](#) list **must** also be present in that list

Valid Usage (Implicit)

- VUID-vkCreateInstance-pCreateInfo-parameter
`pCreateInfo` **must** be a valid pointer to a valid [VkInstanceCreateInfo](#) structure
- VUID-vkCreateInstance-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkCreateInstance-pInstance-parameter
`pInstance` **must** be a valid pointer to a [VkInstance](#) handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_INITIALIZATION_FAILED`
- `VK_ERROR_LAYER_NOT_PRESENT`
- `VK_ERROR_EXTENSION_NOT_PRESENT`
- `VK_ERROR_INCOMPATIBLE_DRIVER`

The [VkInstanceCreateInfo](#) structure is defined as:


```
// Provided by VK_VERSION_1_0
typedef struct VkInstanceCreateInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkInstanceCreateFlags flags;
    const VkApplicationInfo* pApplicationInfo;
    uint32_t             enabledLayerCount;
    const char* const*   ppEnabledLayerNames;
    uint32_t             enabledExtensionCount;
    const char* const*   ppEnabledExtensionNames;
} VkInstanceCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkInstanceCreateFlagBits` indicating the behavior of the instance.
- `pApplicationInfo` is `NULL` or a pointer to a `VkApplicationInfo` structure. If not `NULL`, this information helps implementations recognize behavior inherent to classes of applications. `VkApplicationInfo` is defined in detail below.
- `enabledLayerCount` is the number of global layers to enable.
- `ppEnabledLayerNames` is a pointer to an array of `enabledLayerCount` null-terminated UTF-8 strings containing the names of layers to enable for the created instance. The layers are loaded in the order they are listed in this array, with the first array element being the closest to the application, and the last array element being the closest to the driver. See the [Layers](#) section for further details.
- `enabledExtensionCount` is the number of global extensions to enable.
- `ppEnabledExtensionNames` is a pointer to an array of `enabledExtensionCount` null-terminated UTF-8 strings containing the names of extensions to enable.

Valid Usage (Implicit)

- VUID-VkInstanceCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_INSTANCE_CREATE_INFO`
- VUID-VkInstanceCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkInstanceCreateInfo-flags-zero bitmask
`flags` **must** be `0`
- VUID-VkInstanceCreateInfo-pApplicationInfo-parameter
If `pApplicationInfo` is not `NULL`, `pApplicationInfo` **must** be a valid pointer to a valid `VkApplicationInfo` structure
- VUID-VkInstanceCreateInfo-ppEnabledLayerNames-parameter
If `enabledLayerCount` is not `0`, `ppEnabledLayerNames` **must** be a valid pointer to an array of `enabledLayerCount` null-terminated UTF-8 strings

- `VUID-VkInstanceCreateInfo-ppEnabledExtensionNames-parameter`
If `enabledExtensionCount` is not 0, `ppEnabledExtensionNames` **must** be a valid pointer to an array of `enabledExtensionCount` null-terminated UTF-8 strings

```
// Provided by VK_VERSION_1_0
typedef enum VkInstanceCreateFlagBits {
} VkInstanceCreateFlagBits;
```

NOTE

All bits for this type are defined by extensions, and none of those extensions are enabled in this build of the specification.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkInstanceCreateFlags;
```

`VkInstanceCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

The `VkApplicationInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkApplicationInfo {
    VkStructureType    sType;
    const void*        pNext;
    const char*        pApplicationName;
    uint32_t           applicationVersion;
    const char*        pEngineName;
    uint32_t           engineVersion;
    uint32_t           apiVersion;
} VkApplicationInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pApplicationName` is `NULL` or is a pointer to a null-terminated UTF-8 string containing the name of the application.
- `applicationVersion` is an unsigned integer variable containing the developer-supplied version number of the application.
- `pEngineName` is `NULL` or is a pointer to a null-terminated UTF-8 string containing the name of the engine (if any) used to create the application.
- `engineVersion` is an unsigned integer variable containing the developer-supplied version number of the engine used to create the application.
- `apiVersion` is the version of the Vulkan API against which the application expects to run, encoded as described in [Version Numbers](#). If `apiVersion` is 0 the implementation **must** ignore it, otherwise if the implementation does not support the requested `apiVersion`, or an effective

substitute for `apiVersion`, it **must** return `VK_ERROR_INCOMPATIBLE_DRIVER`. The patch version number specified in `apiVersion` is ignored when creating an instance object. The variant version of the instance **must** match that requested in `apiVersion`.

Valid Usage

- VUID-VkApplicationInfo-apiVersion-04010
If `apiVersion` is not `0`, then it **must** be greater than or equal to `VK_API_VERSION_1_0`

Valid Usage (Implicit)

- VUID-VkApplicationInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_APPLICATION_INFO`
- VUID-VkApplicationInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkApplicationInfo-pApplicationName-parameter
If `pApplicationName` is not `NULL`, `pApplicationName` **must** be a null-terminated UTF-8 string
- VUID-VkApplicationInfo-pEngineName-parameter
If `pEngineName` is not `NULL`, `pEngineName` **must** be a null-terminated UTF-8 string

To destroy an instance, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyInstance(
    VkInstance          instance,
    const VkAllocationCallbacks* pAllocator);
```

- `instance` is the handle of the instance to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyInstance-instance-00629
All child objects created using `instance` **must** have been destroyed prior to destroying `instance`
- VUID-vkDestroyInstance-instance-00630
If `VkAllocationCallbacks` were provided when `instance` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyInstance-instance-00631
If no `VkAllocationCallbacks` were provided when `instance` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyInstance-instance-parameter
If `instance` is not `NULL`, `instance` **must** be a valid `VkInstance` handle
- VUID-vkDestroyInstance-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure

Host Synchronization

- Host access to `instance` **must** be externally synchronized
- Host access to all `VkPhysicalDevice` objects enumerated from `instance` **must** be externally synchronized

Chapter 5. Devices and Queues

Once Vulkan is initialized, devices and queues are the primary objects used to interact with a Vulkan implementation.

Vulkan separates the concept of *physical* and *logical* devices. A physical device usually represents a single complete implementation of Vulkan (excluding instance-level functionality) available to the host, of which there are a finite number. A logical device represents an instance of that implementation with its own state and resources independent of other logical devices.

Physical devices are represented by `VkPhysicalDevice` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_HANDLE(VkPhysicalDevice)
```

5.1. Physical Devices

To retrieve a list of physical device objects representing the physical devices installed in the system, call:

```
// Provided by VK_VERSION_1_0
VkResult vkEnumeratePhysicalDevices(
    VkInstance          instance,
    uint32_t*           pPhysicalDeviceCount,
    VkPhysicalDevice*   pPhysicalDevices);
```

- `instance` is a handle to a Vulkan instance previously created with `vkCreateInstance`.
- `pPhysicalDeviceCount` is a pointer to an integer related to the number of physical devices available or queried, as described below.
- `pPhysicalDevices` is either `NULL` or a pointer to an array of `VkPhysicalDevice` handles.

If `pPhysicalDevices` is `NULL`, then the number of physical devices available is returned in `pPhysicalDeviceCount`. Otherwise, `pPhysicalDeviceCount` **must** point to a variable set by the application to the number of elements in the `pPhysicalDevices` array, and on return the variable is overwritten with the number of handles actually written to `pPhysicalDevices`. If `pPhysicalDeviceCount` is less than the number of physical devices available, at most `pPhysicalDeviceCount` structures will be written, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that not all the available physical devices were returned.

Valid Usage (Implicit)

- VUID-vkEnumeratePhysicalDevices-instance-parameter `instance` **must** be a valid `VkInstance` handle
- VUID-vkEnumeratePhysicalDevices-pPhysicalDeviceCount-parameter

`pPhysicalDeviceCount` **must** be a valid pointer to a `uint32_t` value

- VUID-vkEnumeratePhysicalDevices-pPhysicalDevices-parameter

If the value referenced by `pPhysicalDeviceCount` is not 0, and `pPhysicalDevices` is not NULL, `pPhysicalDevices` **must** be a valid pointer to an array of `pPhysicalDeviceCount` `VkPhysicalDevice` handles

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_INITIALIZATION_FAILED`

To query general properties of physical devices once enumerated, call:

```
// Provided by VK_VERSION_1_0
void vkGetPhysicalDeviceProperties(
    VkPhysicalDevice          physicalDevice,
    VkPhysicalDeviceProperties* pProperties);
```

- `physicalDevice` is the handle to the physical device whose properties will be queried.
- `pProperties` is a pointer to a `VkPhysicalDeviceProperties` structure in which properties are returned.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceProperties-physicalDevice-parameter
`physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceProperties-pProperties-parameter
`pProperties` **must** be a valid pointer to a `VkPhysicalDeviceProperties` structure

The `VkPhysicalDeviceProperties` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPhysicalDeviceProperties {
    uint32_t          apiVersion;
    uint32_t          driverVersion;
    uint32_t          vendorID;
    uint32_t          deviceID;
    VkPhysicalDeviceType deviceType;
    char              deviceName[VK_MAX_PHYSICAL_DEVICE_NAME_SIZE];
    uint8_t           pipelineCacheUUID[VK_UUID_SIZE];
    VkPhysicalDeviceLimits limits;
    VkPhysicalDeviceSparseProperties sparseProperties;
} VkPhysicalDeviceProperties;
```

- **apiVersion** is the version of Vulkan supported by the device, encoded as described in [Version Numbers](#).
- **driverVersion** is the vendor-specified version of the driver.
- **vendorID** is a unique identifier for the *vendor* (see below) of the physical device.
- **deviceID** is a unique identifier for the physical device among devices available from the vendor.
- **deviceType** is a [VkPhysicalDeviceType](#) specifying the type of device.
- **deviceName** is an array of `VK_MAX_PHYSICAL_DEVICE_NAME_SIZE` `char` containing a null-terminated UTF-8 string which is the name of the device.
- **pipelineCacheUUID** is an array of `VK_UUID_SIZE` `uint8_t` values representing a universally unique identifier for the device.
- **limits** is the [VkPhysicalDeviceLimits](#) structure specifying device-specific limits of the physical device. See [Limits](#) for details.
- **sparseProperties** is the [VkPhysicalDeviceSparseProperties](#) structure specifying various sparse related properties of the physical device. See [Sparse Properties](#) for details.

NOTE

The encoding of **driverVersion** is implementation-defined. It **may** not use the same encoding as **apiVersion**. Applications should follow information from the *vendor* on how to extract the version information from **driverVersion**.

The **vendorID** and **deviceID** fields are provided to allow applications to adapt to device characteristics that are not adequately exposed by other Vulkan queries.

NOTE

These **may** include performance profiles, hardware errata, or other characteristics.

The *vendor* identified by **vendorID** is the entity responsible for the most salient characteristics of the underlying implementation of the [VkPhysicalDevice](#) being queried.

NOTE

For example, in the case of a discrete GPU implementation, this **should** be the GPU chipset vendor. In the case of a hardware accelerator integrated into a system-on-chip (SoC), this **should** be the supplier of the silicon IP used to create the accelerator.

If the vendor has a [PCI vendor ID](#), the low 16 bits of **vendorID** **must** contain that PCI vendor ID, and the remaining bits **must** be zero. Otherwise, the value returned **must** be a valid Khronos vendor ID, obtained as described in the [Vulkan Documentation and Extensions: Procedures and Conventions](#) document in the section “Registering a Vendor ID with Khronos”. Khronos vendor IDs are allocated starting at 0x10000, to distinguish them from the PCI vendor ID namespace. Khronos vendor IDs are symbolically defined in the [VkVendorId](#) type.

The vendor is also responsible for the value returned in **deviceID**. If the implementation is driven primarily by a [PCI device](#) with a [PCI device ID](#), the low 16 bits of **deviceID** **must** contain that PCI device ID, and the remaining bits **must** be zero. Otherwise, the choice of what values to return **may** be dictated by operating system or platform policies - but **should** uniquely identify both the device version and any major configuration options (for example, core count in the case of multicore devices).

NOTE

The same device ID **should** be used for all physical implementations of that device version and configuration. For example, all uses of a specific silicon IP GPU version and configuration **should** use the same device ID, even if those uses occur in different SoCs.

Khronos vendor IDs which **may** be returned in [VkPhysicalDeviceProperties::vendorID](#) are:

```
// Provided by VK_VERSION_1_0
typedef enum VkVendorId {
    VK_VENDOR_ID_KHRONOS = 0x10000,
    VK_VENDOR_ID_VIV = 0x10001,
    VK_VENDOR_ID_VSI = 0x10002,
    VK_VENDOR_ID_KAZAN = 0x10003,
    VK_VENDOR_ID_CODEPLAY = 0x10004,
    VK_VENDOR_ID_MESA = 0x10005,
    VK_VENDOR_ID_POCL = 0x10006,
    VK_VENDOR_ID_MOBILEYE = 0x10007,
} VkVendorId;
```

NOTE

Khronos vendor IDs may be allocated by vendors at any time. Only the latest canonical versions of this Specification, of the corresponding [vk.xml](#) API Registry, and of the corresponding [vulkan_core.h](#) header file **must** contain all reserved Khronos vendor IDs.

Only Khronos vendor IDs are given symbolic names at present. PCI vendor IDs returned by the implementation can be looked up in the PCI-SIG database.

VK_MAX_PHYSICAL_DEVICE_NAME_SIZE is the length in **char** values of an array containing a physical device name string, as returned in [VkPhysicalDeviceProperties::deviceName](#).

```
#define VK_MAX_PHYSICAL_DEVICE_NAME_SIZE 256U
```

The physical device types which **may** be returned in [VkPhysicalDeviceProperties::deviceType](#) are:


```
// Provided by VK_VERSION_1_0
typedef enum VkPhysicalDeviceType {
    VK_PHYSICAL_DEVICE_TYPE_OTHER = 0,
    VK_PHYSICAL_DEVICE_TYPE_INTEGRATED_GPU = 1,
    VK_PHYSICAL_DEVICE_TYPE_DISCRETE_GPU = 2,
    VK_PHYSICAL_DEVICE_TYPE_VIRTUAL_GPU = 3,
    VK_PHYSICAL_DEVICE_TYPE_CPU = 4,
} VkPhysicalDeviceType;
```

- **VK_PHYSICAL_DEVICE_TYPE_OTHER** - the device does not match any other available types.
- **VK_PHYSICAL_DEVICE_TYPE_INTEGRATED_GPU** - the device is typically one embedded in or tightly coupled with the host.
- **VK_PHYSICAL_DEVICE_TYPE_DISCRETE_GPU** - the device is typically a separate processor connected to the host via an interlink.
- **VK_PHYSICAL_DEVICE_TYPE_VIRTUAL_GPU** - the device is typically a virtual node in a virtualization environment.
- **VK_PHYSICAL_DEVICE_TYPE_CPU** - the device is typically running on the same processors as the host.

The physical device type is advertised for informational purposes only, and does not directly affect the operation of the system. However, the device type **may** correlate with other advertised properties or capabilities of the system, such as how many memory heaps there are.

To query properties of queues available on a physical device, call:

```
// Provided by VK_VERSION_1_0
void vkGetPhysicalDeviceQueueFamilyProperties(
    VkPhysicalDevice          physicalDevice,
    uint32_t*                 pQueueFamilyPropertyCount,
    VkQueueFamilyProperties*   pQueueFamilyProperties);
```

- **physicalDevice** is the handle to the physical device whose properties will be queried.
- **pQueueFamilyPropertyCount** is a pointer to an integer related to the number of queue families available or queried, as described below.
- **pQueueFamilyProperties** is either **NULL** or a pointer to an array of **VkQueueFamilyProperties** structures.

If **pQueueFamilyProperties** is **NULL**, then the number of queue families available is returned in **pQueueFamilyPropertyCount**. Implementations **must** support at least one queue family. Otherwise, **pQueueFamilyPropertyCount** **must** point to a variable set by the application to the number of elements in the **pQueueFamilyProperties** array, and on return the variable is overwritten with the number of structures actually written to **pQueueFamilyProperties**. If **pQueueFamilyPropertyCount** is less than the number of queue families available, at most **pQueueFamilyPropertyCount** structures will be written.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceQueueFamilyProperties-physicalDevice-parameter `physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceQueueFamilyProperties-pQueueFamilyPropertyCount-parameter `pQueueFamilyPropertyCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDeviceQueueFamilyProperties-pQueueFamilyProperties-parameter
If the value referenced by `pQueueFamilyPropertyCount` is not 0, and `pQueueFamilyProperties` is not `NULL`, `pQueueFamilyProperties` **must** be a valid pointer to an array of `pQueueFamilyPropertyCount` `VkQueueFamilyProperties` structures

The `VkQueueFamilyProperties` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkQueueFamilyProperties {
    VkQueueFlags    queueFlags;
    uint32_t        queueCount;
    uint32_t        timestampValidBits;
    VkExtent3D      minImageTransferGranularity;
} VkQueueFamilyProperties;
```

- `queueFlags` is a bitmask of `VkQueueFlagBits` indicating capabilities of the queues in this queue family.
- `queueCount` is the unsigned integer count of queues in this queue family. Each queue family **must** support at least one queue.
- `timestampValidBits` is the unsigned integer count of meaningful bits in the timestamps written via `vkCmdWriteTimestamp`. The valid range for the count is 36 to 64 bits, or a value of 0, indicating no support for timestamps. Bits outside the valid range are guaranteed to be zeros.
- `minImageTransferGranularity` is the minimum granularity supported for image transfer operations on the queues in this queue family.

The value returned in `minImageTransferGranularity` has a unit of compressed texel blocks for images having a block-compressed format, and a unit of texels otherwise.

Possible values of `minImageTransferGranularity` are:

- (0,0,0) specifies that only whole mip levels **must** be transferred using the image transfer operations on the corresponding queues. In this case, the following restrictions apply to all offset and extent parameters of image transfer operations:
 - The `x`, `y`, and `z` members of a `VkOffset3D` parameter **must** always be zero.
 - The `width`, `height`, and `depth` members of a `VkExtent3D` parameter **must** always match the width, height, and depth of the image subresource corresponding to the parameter, respectively.

- (A_x, A_y, A_z) where A_x , A_y , and A_z are all integer powers of two. In this case the following restrictions apply to all image transfer operations:
 - x , y , and z of a `VkOffset3D` parameter **must** be integer multiples of A_x , A_y , and A_z , respectively.
 - `width` of a `VkExtent3D` parameter **must** be an integer multiple of A_x , or else $x + \text{width}$ **must** equal the width of the image subresource corresponding to the parameter.
 - `height` of a `VkExtent3D` parameter **must** be an integer multiple of A_y , or else $y + \text{height}$ **must** equal the height of the image subresource corresponding to the parameter.
 - `depth` of a `VkExtent3D` parameter **must** be an integer multiple of A_z , or else $z + \text{depth}$ **must** equal the depth of the image subresource corresponding to the parameter.
 - If the format of the image corresponding to the parameters is one of the block-compressed formats then for the purposes of the above calculations the granularity **must** be scaled up by the compressed texel block dimensions.

Queues supporting graphics and/or compute operations **must** report (1,1,1) in `minImageTransferGranularity`, meaning that there are no additional restrictions on the granularity of image transfer operations for these queues. Other queues supporting image transfer operations are only **required** to support whole mip level transfers, thus `minImageTransferGranularity` for queues belonging to such queue families **may** be (0,0,0).

The [Device Memory](#) section describes memory properties queried from the physical device.

For physical device feature queries see the [Features](#) chapter.

Bits which **may** be set in `VkQueueFamilyProperties::queueFlags`, indicating capabilities of queues in a queue family are:

```
// Provided by VK_VERSION_1_0
typedef enum VkQueueFlagBits {
    VK_QUEUE_GRAPHICS_BIT = 0x00000001,
    VK_QUEUE_COMPUTE_BIT = 0x00000002,
    VK_QUEUE_TRANSFER_BIT = 0x00000004,
    VK_QUEUE_SPARSE_BINDING_BIT = 0x00000008,
} VkQueueFlagBits;
```

- `VK_QUEUE_GRAPHICS_BIT` specifies that queues in this queue family support graphics operations.
- `VK_QUEUE_COMPUTE_BIT` specifies that queues in this queue family support compute operations.
- `VK_QUEUE_TRANSFER_BIT` specifies that queues in this queue family support transfer operations.
- `VK_QUEUE_SPARSE_BINDING_BIT` specifies that queues in this queue family support sparse memory management operations (see [Sparse Resources](#)). If any of the sparse resource features are enabled, then at least one queue family **must** support this bit.

If an implementation exposes any queue family that supports graphics operations, at least one queue family of at least one physical device exposed by the implementation **must** support both graphics and compute operations.

NOTE

All commands that are allowed on a queue that supports transfer operations are also allowed on a queue that supports either graphics or compute operations. Thus, if the capabilities of a queue family include `VK_QUEUE_GRAPHICS_BIT` or `VK_QUEUE_COMPUTE_BIT`, then reporting the `VK_QUEUE_TRANSFER_BIT` capability separately for that queue family is **optional**.

For further details see [Queues](#).

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkQueueFlags;
```

`VkQueueFlags` is a bitmask type for setting a mask of zero or more [VkQueueFlagBits](#).

5.2. Devices

Device objects represent logical connections to physical devices. Each device exposes a number of *queue families* each having one or more *queues*. All queues in a queue family support the same operations.

As described in [Physical Devices](#), a Vulkan application will first query for all physical devices in a system. Each physical device **can** then be queried for its capabilities, including its queue and queue family properties. Once an acceptable physical device is identified, an application will create a corresponding logical device. The created logical device is then the primary interface to the physical device.

How to enumerate the physical devices in a system and query those physical devices for their queue family properties is described in the [Physical Device Enumeration](#) section above.

5.2.1. Device Creation

Logical devices are represented by `VkDevice` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_HANDLE(VkDevice)
```

A logical device is created as a *connection* to a physical device. To create a logical device, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateDevice(
    VkPhysicalDevice          physicalDevice,
    const VkDeviceCreateInfo* pCreateInfo,
    const VkAllocationCallbacks* pAllocator,
    VkDevice*                 pDevice);
```

- `physicalDevice` **must** be one of the device handles returned from a call to `vkEnumeratePhysicalDevices` (see [Physical Device Enumeration](#)).

- `pCreateInfo` is a pointer to a `VkDeviceCreateInfo` structure containing information about how to create the device.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.
- `pDevice` is a pointer to a handle in which the created `VkDevice` is returned.

`vkCreateDevice` verifies that extensions and features requested in the `ppEnabledExtensionNames` and `pEnabledFeatures` members of `pCreateInfo`, respectively, are supported by the implementation. If any requested extension is not supported, `vkCreateDevice` **must** return `VK_ERROR_EXTENSION_NOT_PRESENT`. If any requested feature is not supported, `vkCreateDevice` **must** return `VK_ERROR_FEATURE_NOT_PRESENT`. Support for extensions **can** be checked before creating a device by querying `vkEnumerateDeviceExtensionProperties`. Support for features **can** similarly be checked by querying `vkGetPhysicalDeviceFeatures`.

After verifying and enabling the extensions the `VkDevice` object is created and returned to the application.

Multiple logical devices **can** be created from the same physical device. Logical device creation **may** fail due to lack of device-specific resources (in addition to other errors). If that occurs, `vkCreateDevice` will return `VK_ERROR_TOO_MANY_OBJECTS`.

Valid Usage

- VUID-vkCreateDevice-ppEnabledExtensionNames-01387
All [required device extensions](#) for each extension in the `VkDeviceCreateInfo::ppEnabledExtensionNames` list **must** also be present in that list

Valid Usage (Implicit)

- VUID-vkCreateDevice-physicalDevice-parameter
`physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkCreateDevice-pCreateInfo-parameter
`pCreateInfo` **must** be a valid pointer to a valid `VkDeviceCreateInfo` structure
- VUID-vkCreateDevice-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkCreateDevice-pDevice-parameter
`pDevice` **must** be a valid pointer to a `VkDevice` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_INITIALIZATION_FAILED`
- `VK_ERROR_EXTENSION_NOT_PRESENT`
- `VK_ERROR_FEATURE_NOT_PRESENT`
- `VK_ERROR_TOO_MANY_OBJECTS`
- `VK_ERROR_DEVICE_LOST`

The `VkDeviceCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDeviceCreateInfo {
    VkStructureType           sType;
    const void*               pNext;
    VkDeviceCreateFlags       flags;
    uint32_t                  queueCreateInfoCount;
    const VkDeviceQueueCreateInfo* pQueueCreateInfos;
    // enabledLayerCount is deprecated and should not be used
    uint32_t                  enabledLayerCount;
    // ppEnabledLayerNames is deprecated and should not be used
    const char* const*        ppEnabledLayerNames;
    uint32_t                  enabledExtensionCount;
    const char* const*        ppEnabledExtensionNames;
    const VkPhysicalDeviceFeatures* pEnabledFeatures;
} VkDeviceCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `queueCreateInfoCount` is the unsigned integer size of the `pQueueCreateInfos` array. Refer to the [Queue Creation](#) section below for further details.
- `pQueueCreateInfos` is a pointer to an array of `VkDeviceQueueCreateInfo` structures describing the queues that are requested to be created along with the logical device. Refer to the [Queue Creation](#) section below for further details.
- `enabledLayerCount` is deprecated and ignored.
- `ppEnabledLayerNames` is deprecated and ignored. See [Device Layer Deprecation](#).
- `enabledExtensionCount` is the number of device extensions to enable.
- `ppEnabledExtensionNames` is a pointer to an array of `enabledExtensionCount` null-terminated UTF-8 strings containing the names of extensions to enable for the created device. See the [Extensions](#) section for further details.

- `pEnabledFeatures` is `NULL` or a pointer to a `VkPhysicalDeviceFeatures` structure containing boolean indicators of all the features to be enabled. Refer to the [Features](#) section for further details.

Valid Usage

- VUID-VkDeviceCreateInfo-queueFamilyIndex-02802
The `queueFamilyIndex` member of each element of `pQueueCreateInfos` **must** be unique within `pQueueCreateInfos`

Valid Usage (Implicit)

- VUID-VkDeviceCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DEVICE_CREATE_INFO`
- VUID-VkDeviceCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkDeviceCreateInfo-flags-zeroBitmask
`flags` **must** be `0`
- VUID-VkDeviceCreateInfo-pQueueCreateInfos-parameter
`pQueueCreateInfos` **must** be a valid pointer to an array of `queueCreateInfoCount` valid `VkDeviceQueueCreateInfo` structures
- VUID-VkDeviceCreateInfo-ppEnabledLayerNames-parameter
If `enabledLayerCount` is not `0`, `ppEnabledLayerNames` **must** be a valid pointer to an array of `enabledLayerCount` null-terminated UTF-8 strings
- VUID-VkDeviceCreateInfo-ppEnabledExtensionNames-parameter
If `enabledExtensionCount` is not `0`, `ppEnabledExtensionNames` **must** be a valid pointer to an array of `enabledExtensionCount` null-terminated UTF-8 strings
- VUID-VkDeviceCreateInfo-pEnabledFeatures-parameter
If `pEnabledFeatures` is not `NULL`, `pEnabledFeatures` **must** be a valid pointer to a valid `VkPhysicalDeviceFeatures` structure
- VUID-VkDeviceCreateInfo-queueCreateInfoCount-arraylength
`queueCreateInfoCount` **must** be greater than `0`

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkDeviceCreateFlags;
```

`VkDeviceCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

5.2.2. Device Use

The following is a high-level list of `VkDevice` uses along with references on where to find more information:

- Creation of queues. See the [Queues](#) section below for further details.
- Creation and tracking of various synchronization constructs. See [Synchronization and Cache Control](#) for further details.
- Allocating, freeing, and managing memory. See [Memory Allocation](#) and [Resource Creation](#) for further details.
- Creation and destruction of command buffers and command buffer pools. See [Command Buffers](#) for further details.
- Creation, destruction, and management of graphics state. See [Pipelines](#) and [Resource Descriptors](#), among others, for further details.

5.2.3. Lost Device

A logical device **may** become *lost* for a number of implementation-specific reasons, indicating that pending and future command execution **may** fail and cause resources and backing memory to become undefined.

NOTE

Typical reasons for device loss will include things like execution timing out (to prevent denial of service), power management events, platform resource management, implementation errors.

Applications not adhering to [valid usage](#) may also result in device loss being reported, however this is not guaranteed. Even if device loss is reported, the system may be in an unrecoverable state, and further usage of the API is still considered invalid.

When this happens, certain commands will return `VK_ERROR_DEVICE_LOST`. After any such event, the logical device is considered *lost*. It is not possible to reset the logical device to a non-lost state, however the lost state is specific to a logical device (`VkDevice`), and the corresponding physical device (`VkPhysicalDevice`) **may** be otherwise unaffected.

In some cases, the physical device **may** also be lost, and attempting to create a new logical device will fail, returning `VK_ERROR_DEVICE_LOST`. This is usually indicative of a problem with the underlying implementation, or its connection to the host. If the physical device has not been lost, and a new logical device is successfully created from that physical device, it **must** be in the non-lost state.

NOTE

Whilst logical device loss **may** be recoverable, in the case of physical device loss, it is unlikely that an application will be able to recover unless additional, unaffected physical devices exist on the system. The error is largely informational and intended only to inform the application that a platform issue has occurred, and **should** be investigated further. For example, underlying hardware **may** have developed a fault or become physically disconnected from the rest of the system. In many cases, physical device loss **may** cause other more serious issues such as the operating system crashing; in which case it **may** not be reported via the Vulkan API.

When a device is lost, its child objects are not implicitly destroyed and their handles are still valid. Those objects **must** still be destroyed before their parents or the device **can** be destroyed (see the [Object Lifetime](#) section). The host address space corresponding to device memory mapped using

`vkMapMemory` is still valid, and host memory accesses to these mapped regions are still valid, but the contents are undefined. It is still legal to call any API command on the device and child objects.

Once a device is lost, command execution **may** fail, and certain commands that return a `VkResult` **may** return `VK_ERROR_DEVICE_LOST`. These commands can be identified by the inclusion of `VK_ERROR_DEVICE_LOST` in the Return Codes section for each command. Commands that do not allow runtime errors **must** still operate correctly for valid usage and, if applicable, return valid data.

Commands that wait indefinitely for device execution (namely `vkDeviceWaitIdle`, `vkQueueWaitIdle`, `vkWaitForFences` with a maximum `timeout`, and `vkGetQueryPoolResults` with the `VK_QUERY_RESULT_WAIT_BIT` bit set in `flags`) **must** return in finite time even in the case of a lost device, and return either `VK_SUCCESS` or `VK_ERROR_DEVICE_LOST`. For any command that **may** return `VK_ERROR_DEVICE_LOST`, for the purpose of determining whether a command buffer is in the `pending` state, or whether resources are considered in-use by the device, a return value of `VK_ERROR_DEVICE_LOST` is equivalent to `VK_SUCCESS`.

5.2.4. Device Destruction

To destroy a device, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyDevice(
    VkDevice device,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

To ensure that no work is active on the device, `vkDeviceWaitIdle` **can** be used to gate the destruction of the device. Prior to destroying a device, an application is responsible for destroying/freeing any Vulkan objects that were created using that device as the first parameter of the corresponding `vkCreate*` or `vkAllocate*` command.

NOTE

The lifetime of each of these objects is bound by the lifetime of the `VkDevice` object. Therefore, to avoid resource leaks, it is critical that an application explicitly free all of these resources prior to calling `vkDestroyDevice`.

Valid Usage

- VUID-vkDestroyDevice-device-05137
All child objects created on `device` **must** have been destroyed prior to destroying `device`
- VUID-vkDestroyDevice-device-00379
If `VkAllocationCallbacks` were provided when `device` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyDevice-device-00380
If no `VkAllocationCallbacks` were provided when `device` was created, `pAllocator` **must** be

NULL

Valid Usage (Implicit)

- VUID-vkDestroyDevice-device-parameter
If **device** is not NULL, **device** must be a valid [VkDevice](#) handle
- VUID-vkDestroyDevice-pAllocator-parameter
If **pAllocator** is not NULL, **pAllocator** must be a valid pointer to a valid [VkAllocationCallbacks](#) structure

Host Synchronization

- Host access to **device** must be externally synchronized
- Host access to all [VkQueue](#) objects created from **device** must be externally synchronized

5.3. Queues

5.3.1. Queue Family Properties

As discussed in the [Physical Device Enumeration](#) section above, the [vkGetPhysicalDeviceQueueFamilyProperties](#) command is used to retrieve details about the queue families and queues supported by a device.

Each index in the [pQueueFamilyProperties](#) array returned by [vkGetPhysicalDeviceQueueFamilyProperties](#) describes a unique queue family on that physical device. These indices are used when creating queues, and they correspond directly with the [queueFamilyIndex](#) that is passed to the [vkCreateDevice](#) command via the [VkDeviceQueueCreateInfo](#) structure as described in the [Queue Creation](#) section below.

Grouping of queue families within a physical device is implementation-dependent.

NOTE

The general expectation is that a physical device groups all queues of matching capabilities into a single family. However, while implementations **should** do this, it is possible that a physical device **may** return two separate queue families with the same capabilities.

Once an application has identified a physical device with the queue(s) that it desires to use, it will create those queues in conjunction with a logical device. This is described in the following section.

5.3.2. Queue Creation

Creating a logical device also creates the queues associated with that device. The queues to create are described by a set of [VkDeviceQueueCreateInfo](#) structures that are passed to [vkCreateDevice](#) in [pQueueCreateInfos](#).

Queues are represented by `VkQueue` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_HANDLE(VkQueue)
```

The `VkDeviceQueueCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDeviceQueueCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkDeviceQueueCreateFlags flags;
    uint32_t           queueFamilyIndex;
    uint32_t           queueCount;
    const float*       pQueuePriorities;
} VkDeviceQueueCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `queueFamilyIndex` is an unsigned integer indicating the index of the queue family in which to create the queues on this device. This index corresponds to the index of an element of the `pQueueFamilyProperties` array that was returned by `vkGetPhysicalDeviceQueueFamilyProperties`.
- `queueCount` is an unsigned integer specifying the number of queues to create in the queue family indicated by `queueFamilyIndex`, and with the behavior specified by `flags`.
- `pQueuePriorities` is a pointer to an array of `queueCount` normalized floating-point values, specifying priorities of work that will be submitted to each created queue. See [Queue Priority](#) for more information.

Valid Usage

- VUID-VkDeviceQueueCreateInfo-queueFamilyIndex-00381
`queueFamilyIndex` **must** be less than `pQueueFamilyPropertyCount` returned by `vkGetPhysicalDeviceQueueFamilyProperties`
- VUID-VkDeviceQueueCreateInfo-queueCount-00382
`queueCount` **must** be less than or equal to the `queueCount` member of the `VkQueueFamilyProperties` structure, as returned by `vkGetPhysicalDeviceQueueFamilyProperties` in the `pQueueFamilyProperties[queueFamilyIndex]`
- VUID-VkDeviceQueueCreateInfo-pQueuePriorities-00383
Each element of `pQueuePriorities` **must** be between `0.0` and `1.0` inclusive

Valid Usage (Implicit)

- VUID-VkDeviceQueueCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DEVICE_QUEUE_CREATE_INFO`
- VUID-VkDeviceQueueCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkDeviceQueueCreateInfo-flags-zeroBitmask
`flags` **must** be `0`
- VUID-VkDeviceQueueCreateInfo-pQueuePriorities-parameter
`pQueuePriorities` **must** be a valid pointer to an array of `queueCount` `float` values
- VUID-VkDeviceQueueCreateInfo-queueCount-arraylength
`queueCount` **must** be greater than `0`

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkDeviceQueueCreateFlags;
```

`VkDeviceQueueCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

To retrieve a handle to a `VkQueue` object, call:

```
// Provided by VK_VERSION_1_0
void vkGetDeviceQueue(
    VkDevice          device,
    uint32_t          queueFamilyIndex,
    uint32_t          queueIndex,
    VkQueue*          pQueue);
```

- `device` is the logical device that owns the queue.
- `queueFamilyIndex` is the index of the queue family to which the queue belongs.
- `queueIndex` is the index within this queue family of the queue to retrieve.
- `pQueue` is a pointer to a `VkQueue` object that will be filled with the handle for the requested queue.

Valid Usage

- VUID-vkGetDeviceQueue-queueFamilyIndex-00384
`queueFamilyIndex` **must** be one of the queue family indices specified when `device` was created, via the `VkDeviceQueueCreateInfo` structure
- VUID-vkGetDeviceQueue-queueIndex-00385
`queueIndex` **must** be less than the value of `VkDeviceQueueCreateInfo::queueCount` for the queue family indicated by `queueFamilyIndex` when `device` was created

- VUID-vkGetDeviceQueue-flags-01841
`VkDeviceQueueCreateInfo::flags` **must** have been zero when `device` was created

Valid Usage (Implicit)

- VUID-vkGetDeviceQueue-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetDeviceQueue-pQueue-parameter
`pQueue` **must** be a valid pointer to a `VkQueue` handle

5.3.3. Queue Family Index

The queue family index is used in multiple places in Vulkan in order to tie operations to a specific family of queues.

When retrieving a handle to the queue via `vkGetDeviceQueue`, the queue family index is used to select which queue family to retrieve the `VkQueue` handle from as described in the previous section.

When creating a `VkCommandPool` object (see [Command Pools](#)), a queue family index is specified in the `VkCommandPoolCreateInfo` structure. Command buffers from this pool **can** only be submitted on queues corresponding to this queue family.

When creating `VkImage` (see [Images](#)) and `VkBuffer` (see [Buffers](#)) resources, a set of queue families is included in the `VkImageCreateInfo` and `VkBufferCreateInfo` structures to specify the queue families that **can** access the resource.

When inserting a `VkBufferMemoryBarrier` or `VkImageMemoryBarrier` (see [Pipeline Barriers](#)), a source and destination queue family index is specified to allow the ownership of a buffer or image to be transferred from one queue family to another. See the [Resource Sharing](#) section for details.

5.3.4. Queue Priority

Each queue is assigned a priority, as set in the `VkDeviceQueueCreateInfo` structures when creating the device. The priority of each queue is a normalized floating-point value between 0.0 and 1.0, which is then translated to a discrete priority level by the implementation. Higher values indicate a higher priority, with 0.0 being the lowest priority and 1.0 being the highest.

Within the same device, queues with higher priority **may** be allotted more processing time than queues with lower priority. The implementation makes no guarantees with regards to ordering or scheduling among queues with the same priority, other than the constraints defined by any [explicit synchronization primitives](#). The implementation makes no guarantees with regards to queues across different devices.

An implementation **may** allow a higher-priority queue to starve a lower-priority queue on the same `VkDevice` until the higher-priority queue has no further commands to execute. The relationship of queue priorities **must** not cause queues on one `VkDevice` to starve queues on another `VkDevice`.

No specific guarantees are made about higher priority queues receiving more processing time or better quality of service than lower priority queues.

5.3.5. Queue Submission

Work is submitted to a queue via *queue submission* commands such as [vkQueueSubmit](#). Queue submission commands define a set of *queue operations* to be executed by the underlying physical device, including synchronization with semaphores and fences.

Submission commands take as parameters a target queue, zero or more *batches* of work, and an **optional** fence to signal upon completion. Each batch consists of three distinct parts:

1. Zero or more semaphores to wait on before execution of the rest of the batch.
 - If present, these describe a [semaphore wait operation](#).
2. Zero or more work items to execute.
 - If present, these describe a *queue operation* matching the work described.
3. Zero or more semaphores to signal upon completion of the work items.
 - If present, these describe a [semaphore signal operation](#).

If a fence is present in a queue submission, it describes a [fence signal operation](#).

All work described by a queue submission command **must** be submitted to the queue before the command returns.

Sparse Memory Binding

In Vulkan it is possible to sparsely bind memory to buffers and images as described in the [Sparse Resource](#) chapter. Sparse memory binding is a queue operation. A queue whose flags include the [VK_QUEUE_SPARSE_BINDING_BIT](#) **must** be able to support the mapping of a virtual address to a physical address on the device. This causes an update to the page table mappings on the device. This update **must** be synchronized on a queue to avoid corrupting page table mappings during execution of graphics commands. By binding the sparse memory resources on queues, all commands that are dependent on the updated bindings are synchronized to only execute after the binding is updated. See the [Synchronization and Cache Control](#) chapter for how this synchronization is accomplished.

5.3.6. Queue Destruction

Queues are created along with a logical device during [vkCreateDevice](#). All queues associated with a logical device are destroyed when [vkDestroyDevice](#) is called on that device.

Chapter 6. Command Buffers

Command buffers are objects used to record commands which **can** be subsequently submitted to a device queue for execution. There are two levels of command buffers - *primary command buffers*, which **can** execute secondary command buffers, and which are submitted to queues, and *secondary command buffers*, which **can** be executed by primary command buffers, and which are not directly submitted to queues.

Command buffers are represented by `VkCommandBuffer` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_HANDLE(VkCommandBuffer)
```

Recorded commands include commands to bind pipelines and descriptor sets to the command buffer, commands to modify dynamic state, commands to draw (for graphics rendering), commands to dispatch (for compute), commands to execute secondary command buffers (for primary command buffers only), commands to copy buffers and images, and other commands.

Each command buffer manages state independently of other command buffers. There is no inheritance of state across primary and secondary command buffers, or between secondary command buffers. When a command buffer begins recording, all state in that command buffer is undefined. When secondary command buffer(s) are recorded to execute on a primary command buffer, the secondary command buffer inherits no state from the primary command buffer, and all state of the primary command buffer is undefined after an execute secondary command buffer command is recorded. There is one exception to this rule - if the primary command buffer is inside a render pass instance, then the render pass and subpass state is not disturbed by executing secondary command buffers. For state dependent commands (such as draws and dispatches), any state consumed by those commands **must** not be undefined.

Unless otherwise specified, and without explicit synchronization, the various commands submitted to a queue via command buffers **may** execute in arbitrary order relative to each other, and/or concurrently. Also, the memory side effects of those commands **may** not be directly visible to other commands without explicit memory dependencies. This is true within a command buffer, and across command buffers submitted to a given queue. See [the synchronization chapter](#) for information on [implicit](#) and explicit synchronization between commands.

6.1. Command Buffer Lifecycle

Each command buffer is always in one of the following states:

Initial

When a command buffer is [allocated](#), it is in the *initial state*. Some commands are able to *reset* a command buffer (or a set of command buffers) back to this state from any of the executable, recording or invalid state. Command buffers in the initial state **can** only be moved to the recording state, or freed.

Recording

`vkBeginCommandBuffer` changes the state of a command buffer from the initial state to the *recording state*. Once a command buffer is in the recording state, `vkCmd*` commands **can** be used to record to the command buffer.

Executable

`vkEndCommandBuffer` ends the recording of a command buffer, and moves it from the recording state to the *executable state*. Executable command buffers **can** be *submitted*, reset, or *recorded to another command buffer*.

Pending

Queue submission of a command buffer changes the state of a command buffer from the executable state to the *pending state*. Whilst in the pending state, applications **must** not attempt to modify the command buffer in any way - as the device **may** be processing the commands recorded to it. Once execution of a command buffer completes, the command buffer either reverts back to the *executable state*, or if it was recorded with `VK_COMMAND_BUFFER_USAGE_ONE_TIME_SUBMIT_BIT`, it moves to the *invalid state*. A *synchronization* command **should** be used to detect when this occurs.

Invalid

Some operations, such as *modifying or deleting a resource* that was used in a command recorded to a command buffer, will transition the state of that command buffer into the *invalid state*. Command buffers in the invalid state **can** only be reset or freed.

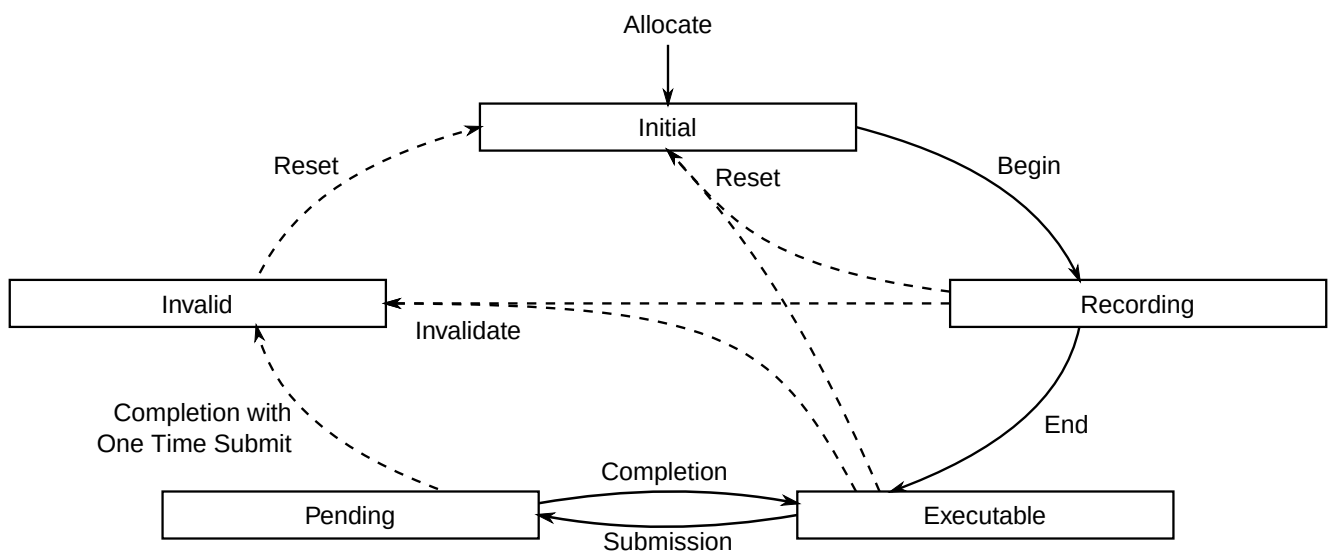


Figure 1. Lifecycle of a command buffer

Any given command that operates on a command buffer has its own requirements on what state a command buffer **must** be in, which are detailed in the valid usage constraints for that command.

Resetting a command buffer is an operation that discards any previously recorded commands and puts a command buffer in the *initial state*. Resetting occurs as a result of `vkResetCommandBuffer` or `vkResetCommandPool`, or as part of `vkBeginCommandBuffer` (which additionally puts the command buffer in the *recording state*).

Secondary command buffers **can** be recorded to a primary command buffer via

[vkCmdExecuteCommands](#). This partially ties the lifecycle of the two command buffers together - if the primary is submitted to a queue, both the primary and any secondaries recorded to it move to the *pending state*. Once execution of the primary completes, so it does for any secondary recorded within it. After all executions of each command buffer complete, they each move to their appropriate completion state (either to the *executable state* or the *invalid state*, as specified above).

If a secondary moves to the *invalid state* or the *initial state*, then all primary buffers it is recorded in move to the *invalid state*. A primary moving to any other state does not affect the state of a secondary recorded in it.

NOTE

Resetting or freeing a primary command buffer removes the lifecycle linkage to all secondary command buffers that were recorded into it.

6.2. Command Pools

Command pools are opaque objects that command buffer memory is allocated from, and which allow the implementation to amortize the cost of resource creation across multiple command buffers. Command pools are externally synchronized, meaning that a command pool **must** not be used concurrently in multiple threads. That includes use via recording commands on any command buffers allocated from the pool, as well as operations that allocate, free, and reset command buffers or the pool itself.

Command pools are represented by [VkCommandPool](#) handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkCommandPool)
```

To create a command pool, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateCommandPool(
    VkDevice                                device,
    const VkCommandPoolCreateInfo*          pCreateInfo,
    const VkAllocationCallbacks*            pAllocator,
    VkCommandPool*                          pCommandPool);
```

- [device](#) is the logical device that creates the command pool.
- [pCreateInfo](#) is a pointer to a [VkCommandPoolCreateInfo](#) structure specifying the state of the command pool object.
- [pAllocator](#) controls host memory allocation as described in the [Memory Allocation](#) chapter.
- [pCommandPool](#) is a pointer to a [VkCommandPool](#) handle in which the created pool is returned.

Valid Usage

- VUID-vkCreateCommandPool-queueFamilyIndex-01937

`pCreateInfo->queueFamilyIndex` **must** be the index of a queue family available in the logical device `device`

Valid Usage (Implicit)

- VUID-vkCreateCommandPool-device-parameter `device` **must** be a valid `VkDevice` handle
- VUID-vkCreateCommandPool-pCreateInfo-parameter `pCreateInfo` **must** be a valid pointer to a valid `VkCommandPoolCreateInfo` structure
- VUID-vkCreateCommandPool-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkCreateCommandPool-pCommandPool-parameter
`pCommandPool` **must** be a valid pointer to a `VkCommandPool` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkCommandPoolCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkCommandPoolCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkCommandPoolCreateFlags flags;
    uint32_t           queueFamilyIndex;
} VkCommandPoolCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkCommandPoolCreateFlagBits` indicating usage behavior for the pool and command buffers allocated from it.
- `queueFamilyIndex` designates a queue family as described in section [Queue Family Properties](#). All command buffers allocated from this command pool **must** be submitted on queues from the same queue family.

Valid Usage

Valid Usage (Implicit)

- VUID-VkCommandPoolCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_COMMAND_POOL_CREATE_INFO`
- VUID-VkCommandPoolCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkCommandPoolCreateInfo-flags-parameter
`flags` **must** be a valid combination of `VkCommandPoolCreateFlagBits` values

Bits which **can** be set in `VkCommandPoolCreateInfo::flags`, specifying usage behavior for a command pool, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkCommandPoolCreateFlagBits {
    VK_COMMAND_POOL_CREATE_TRANSIENT_BIT = 0x00000001,
    VK_COMMAND_POOL_CREATE_RESET_COMMAND_BUFFER_BIT = 0x00000002,
} VkCommandPoolCreateFlagBits;
```

- `VK_COMMAND_POOL_CREATE_TRANSIENT_BIT` specifies that command buffers allocated from the pool will be short-lived, meaning that they will be reset or freed in a relatively short timeframe. This flag **may** be used by the implementation to control memory allocation behavior within the pool.
- `VK_COMMAND_POOL_CREATE_RESET_COMMAND_BUFFER_BIT` allows any command buffer allocated from a pool to be individually reset to the [initial state](#); either by calling `vkResetCommandBuffer`, or via the implicit reset when calling `vkBeginCommandBuffer`. If this flag is not set on a pool, then `vkResetCommandBuffer` **must** not be called for any command buffer allocated from that pool.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkCommandPoolCreateFlags;
```

`VkCommandPoolCreateFlags` is a bitmask type for setting a mask of zero or more `VkCommandPoolCreateFlagBits`.

To reset a command pool, call:

```
// Provided by VK_VERSION_1_0
VkResult vkResetCommandPool(
    VkDevice          device,
    VkCommandPool     commandPool,
    VkCommandPoolResetFlags flags);
```

- **device** is the logical device that owns the command pool.
- **commandPool** is the command pool to reset.
- **flags** is a bitmask of **VkCommandPoolResetFlagBits** controlling the reset operation.

Resetting a command pool recycles all of the resources from all of the command buffers allocated from the command pool back to the command pool. All command buffers that have been allocated from the command pool are put in the **initial state**.

Any primary command buffer allocated from another **VkCommandPool** that is in the **recording or executable state** and has a secondary command buffer allocated from **commandPool** recorded into it, becomes **invalid**.

Valid Usage

- VUID-vkResetCommandPool-commandPool-00040
All **VkCommandBuffer** objects allocated from **commandPool** **must** not be in the **pending state**

Valid Usage (Implicit)

- VUID-vkResetCommandPool-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkResetCommandPool-commandPool-parameter
commandPool **must** be a valid **VkCommandPool** handle
- VUID-vkResetCommandPool-flags-parameter
flags **must** be a valid combination of **VkCommandPoolResetFlagBits** values
- VUID-vkResetCommandPool-commandPool-parent
commandPool **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **commandPool** **must** be externally synchronized

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_DEVICE_MEMORY**

Bits which **can** be set in **vkResetCommandPool::flags**, controlling the reset operation, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkCommandPoolResetFlagBits {
    VK_COMMAND_POOL_RESET_RELEASE_RESOURCES_BIT = 0x00000001,
} VkCommandPoolResetFlagBits;
```

- **VK_COMMAND_POOL_RESET_RELEASE_RESOURCES_BIT** specifies that resetting a command pool recycles all of the resources from the command pool back to the system.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkCommandPoolResetFlags;
```

VkCommandPoolResetFlags is a bitmask type for setting a mask of zero or more **VkCommandPoolResetFlagBits**.

To destroy a command pool, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyCommandPool(
    VkDevice                device,
    VkCommandPool           commandPool,
    const VkAllocationCallbacks* pAllocator);
```

- **device** is the logical device that destroys the command pool.
- **commandPool** is the handle of the command pool to destroy.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.

When a pool is destroyed, all command buffers allocated from the pool are **freed**.

Any primary command buffer allocated from another **VkCommandPool** that is in the **recording or executable state** and has a secondary command buffer allocated from **commandPool** recorded into it, becomes **invalid**.

Valid Usage

- VUID-vkDestroyCommandPool-commandPool-00041
All **VkCommandBuffer** objects allocated from **commandPool** **must** not be in the **pending state**
- VUID-vkDestroyCommandPool-commandPool-00042
If **VkAllocationCallbacks** were provided when **commandPool** was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyCommandPool-commandPool-00043
If no **VkAllocationCallbacks** were provided when **commandPool** was created, **pAllocator** **must** be **NULL**

Valid Usage (Implicit)

- VUID-vkDestroyCommandPool-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkDestroyCommandPool-commandPool-parameter
If `commandPool` is not `VK_NULL_HANDLE`, `commandPool` **must** be a valid `VkCommandPool` handle
- VUID-vkDestroyCommandPool-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkDestroyCommandPool-commandPool-parent
If `commandPool` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `commandPool` **must** be externally synchronized

6.3. Command Buffer Allocation and Management

To allocate command buffers, call:

```
// Provided by VK_VERSION_1_0
VkResult vkAllocateCommandBuffers(
    VkDevice device,
    const VkCommandBufferAllocateInfo* pAllocateInfo,
    VkCommandBuffer* pCommandBuffers);
```

- `device` is the logical device that owns the command pool.
- `pAllocateInfo` is a pointer to a `VkCommandBufferAllocateInfo` structure describing parameters of the allocation.
- `pCommandBuffers` is a pointer to an array of `VkCommandBuffer` handles in which the resulting command buffer objects are returned. The array **must** be at least the length specified by the `commandBufferCount` member of `pAllocateInfo`. Each allocated command buffer begins in the initial state.

When command buffers are first allocated, they are in the `initial state`.

Valid Usage (Implicit)

- VUID-vkAllocateCommandBuffers-device-parameter
`device` **must** be a valid `VkDevice` handle

- VUID-vkAllocateCommandBuffers-pAllocateInfo-parameter
`pAllocateInfo` **must** be a valid pointer to a valid `VkCommandBufferAllocateInfo` structure
- VUID-vkAllocateCommandBuffers-pCommandBuffers-parameter
`pCommandBuffers` **must** be a valid pointer to an array of `pAllocateInfo->commandBufferCount` `VkCommandBuffer` handles
- VUID-vkAllocateCommandBuffers-pAllocateInfo::commandBufferCount-arraylength
`pAllocateInfo->commandBufferCount` **must** be greater than 0

Host Synchronization

- Host access to `pAllocateInfo->commandPool` **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkCommandBufferAllocateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkCommandBufferAllocateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkCommandPool      commandPool;
    VkCommandBufferLevel level;
    uint32_t           commandBufferCount;
} VkCommandBufferAllocateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `commandPool` is the command pool from which the command buffers are allocated.
- `level` is a `VkCommandBufferLevel` value specifying the command buffer level.
- `commandBufferCount` is the number of command buffers to allocate from the pool.

Valid Usage (Implicit)

- VUID-VkCommandBufferAllocateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_COMMAND_BUFFER_ALLOCATE_INFO`

- VUID-VkCommandBufferAllocateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkCommandBufferAllocateInfo-commandPool-parameter
`commandPool` **must** be a valid `VkCommandPool` handle
- VUID-VkCommandBufferAllocateInfo-level-parameter
`level` **must** be a valid `VkCommandBufferLevel` value

Possible values of `VkCommandBufferAllocateInfo::level`, specifying the command buffer level, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkCommandBufferLevel {
    VK_COMMAND_BUFFER_LEVEL_PRIMARY = 0,
    VK_COMMAND_BUFFER_LEVEL_SECONDARY = 1,
} VkCommandBufferLevel;
```

- `VK_COMMAND_BUFFER_LEVEL_PRIMARY` specifies a primary command buffer.
- `VK_COMMAND_BUFFER_LEVEL_SECONDARY` specifies a secondary command buffer.

To reset a command buffer, call:

```
// Provided by VK_VERSION_1_0
VkResult vkResetCommandBuffer(
    VkCommandBuffer          commandBuffer,
    VkCommandBufferResetFlags flags);
```

- `commandBuffer` is the command buffer to reset. The command buffer **can** be in any state other than `pending`, and is moved into the `initial state`.
- `flags` is a bitmask of `VkCommandBufferResetFlagBits` controlling the reset operation.

Any primary command buffer that is in the `recording or executable state` and has `commandBuffer` recorded into it, becomes `invalid`.

Valid Usage

- VUID-vkResetCommandBuffer-commandBuffer-00045
`commandBuffer` **must** not be in the `pending state`
- VUID-vkResetCommandBuffer-commandBuffer-00046
`commandBuffer` **must** have been allocated from a pool that was created with the `VK_COMMAND_POOL_CREATE_RESET_COMMAND_BUFFER_BIT`

Valid Usage (Implicit)

- VUID-vkResetCommandBuffer-commandBuffer-parameter

`commandBuffer` **must** be a valid `VkCommandBuffer` handle

- VUID-vkResetCommandBuffer-flags-parameter

`flags` **must** be a valid combination of `VkCommandBufferResetFlagBits` values

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

Bits which **can** be set in `vkResetCommandBuffer::flags`, controlling the reset operation, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkCommandBufferResetFlagBits {
    VK_COMMAND_BUFFER_RESET_RELEASE_RESOURCES_BIT = 0x00000001,
} VkCommandBufferResetFlagBits;
```

- `VK_COMMAND_BUFFER_RESET_RELEASE_RESOURCES_BIT` specifies that most or all memory resources currently owned by the command buffer **should** be returned to the parent command pool. If this flag is not set, then the command buffer **may** hold onto memory resources and reuse them when recording commands. `commandBuffer` is moved to the `initial state`.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkCommandBufferResetFlags;
```

`VkCommandBufferResetFlags` is a bitmask type for setting a mask of zero or more `VkCommandBufferResetFlagBits`.

To free command buffers, call:

```
// Provided by VK_VERSION_1_0
void vkFreeCommandBuffers(
    VkDevice          device,
    VkCommandPool     commandPool,
    uint32_t          commandBufferCount,
    const VkCommandBuffer* pCommandBuffers);
```

- **device** is the logical device that owns the command pool.
- **commandPool** is the command pool from which the command buffers were allocated.
- **commandBufferCount** is the length of the **pCommandBuffers** array.
- **pCommandBuffers** is a pointer to an array of handles of command buffers to free.

Any primary command buffer that is in the **recording or executable state** and has any element of **pCommandBuffers** recorded into it, becomes **invalid**.

Valid Usage

- VUID-vkFreeCommandBuffers-pCommandBuffers-00047
All elements of **pCommandBuffers** **must** not be in the **pending state**
- VUID-vkFreeCommandBuffers-pCommandBuffers-00048
pCommandBuffers **must** be a valid pointer to an array of **commandBufferCount** **VkCommandBuffer** handles, each element of which **must** either be a valid handle or **NULL**

Valid Usage (Implicit)

- VUID-vkFreeCommandBuffers-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkFreeCommandBuffers-commandPool-parameter
commandPool **must** be a valid **VkCommandPool** handle
- VUID-vkFreeCommandBuffers-commandBufferCount-arraylength
commandBufferCount **must** be greater than **0**
- VUID-vkFreeCommandBuffers-commandPool-parent
commandPool **must** have been created, allocated, or retrieved from **device**
- VUID-vkFreeCommandBuffers-pCommandBuffers-parent
Each element of **pCommandBuffers** that is a valid handle **must** have been created, allocated, or retrieved from **commandPool**

Host Synchronization

- Host access to **commandPool** **must** be externally synchronized
- Host access to each member of **pCommandBuffers** **must** be externally synchronized

6.4. Command Buffer Recording

To begin recording a command buffer, call:

```
// Provided by VK_VERSION_1_0
VkResult vkBeginCommandBuffer(
    VkCommandBuffer          commandBuffer,
    const VkCommandBufferBeginInfo* pBeginInfo);
```

- `commandBuffer` is the handle of the command buffer which is to be put in the recording state.
- `pBeginInfo` is a pointer to a `VkCommandBufferBeginInfo` structure defining additional information about how the command buffer begins recording.

Valid Usage

- VUID-vkBeginCommandBuffer-commandBuffer-00049
`commandBuffer` **must** not be in the `recording or pending` state
- VUID-vkBeginCommandBuffer-commandBuffer-00050
If `commandBuffer` was allocated from a `VkCommandPool` which did not have the `VK_COMMAND_POOL_CREATE_RESET_COMMAND_BUFFER_BIT` flag set, `commandBuffer` **must** be in the `initial` state
- VUID-vkBeginCommandBuffer-commandBuffer-00051
If `commandBuffer` is a secondary command buffer, the `pInheritanceInfo` member of `pBeginInfo` **must** be a valid `VkCommandBufferInheritanceInfo` structure
- VUID-vkBeginCommandBuffer-commandBuffer-00052
If `commandBuffer` is a secondary command buffer and either the `occlusionQueryEnable` member of the `pInheritanceInfo` member of `pBeginInfo` is `VK_FALSE`, or the `occlusionQueryPrecise` feature is not enabled, then `pBeginInfo->pInheritanceInfo->queryFlags` **must** not contain `VK_QUERY_CONTROL_PRECISE_BIT`
- VUID-vkBeginCommandBuffer-commandBuffer-02840
If `commandBuffer` is a primary command buffer, then `pBeginInfo->flags` **must** not set both the `VK_COMMAND_BUFFER_USAGE_ONE_TIME_SUBMIT_BIT` and the `VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT` flags

Valid Usage (Implicit)

- VUID-vkBeginCommandBuffer-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkBeginCommandBuffer-pBeginInfo-parameter
`pBeginInfo` **must** be a valid pointer to a valid `VkCommandBufferBeginInfo` structure

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkCommandBufferBeginInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkCommandBufferBeginInfo {
    VkStructureType           sType;
    const void*               pNext;
    VkCommandBufferUsageFlags flags;
    const VkCommandBufferInheritanceInfo* pInheritanceInfo;
} VkCommandBufferBeginInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkCommandBufferUsageFlagBits` specifying usage behavior for the command buffer.
- `pInheritanceInfo` is a pointer to a `VkCommandBufferInheritanceInfo` structure, used if `commandBuffer` is a secondary command buffer. If this is a primary command buffer, then this value is ignored.

Valid Usage

- VUID-VkCommandBufferBeginInfo-flags-09123
If `flags` contains `VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT`, the `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-VkCommandBufferBeginInfo-flags-00055
If `flags` contains `VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT`, the `framebuffer` member of `pInheritanceInfo` **must** be either `VK_NULL_HANDLE`, or a valid `VkFramebuffer` that is compatible with the `renderPass` member of `pInheritanceInfo`

- VUID-VkCommandBufferBeginInfo-flags-06000
If `flags` contains `VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT` the `renderPass` member of `pInheritanceInfo` **must** be a valid `VkRenderPass`
- VUID-VkCommandBufferBeginInfo-flags-06001
If `flags` contains `VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT` the `subpass` member of `pInheritanceInfo` **must** be a valid subpass index within the `renderPass` member of `pInheritanceInfo`

Valid Usage (Implicit)

- VUID-VkCommandBufferBeginInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_COMMAND_BUFFER_BEGIN_INFO`
- VUID-VkCommandBufferBeginInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkCommandBufferBeginInfo-flags-parameter
`flags` **must** be a valid combination of `VkCommandBufferUsageFlagBits` values

Bits which **can** be set in `VkCommandBufferBeginInfo::flags`, specifying usage behavior for a command buffer, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkCommandBufferUsageFlagBits {
    VK_COMMAND_BUFFER_USAGE_ONE_TIME_SUBMIT_BIT = 0x00000001,
    VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT = 0x00000002,
    VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT = 0x00000004,
} VkCommandBufferUsageFlagBits;
```

- `VK_COMMAND_BUFFER_USAGE_ONE_TIME_SUBMIT_BIT` specifies that each recording of the command buffer will only be submitted once, and the command buffer will be reset and recorded again between each submission.
- `VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT` specifies that a secondary command buffer is considered to be entirely inside a render pass. If this is a primary command buffer, then this bit is ignored.
- `VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT` specifies that a command buffer **can** be resubmitted to any queue of the same queue family while it is in the *pending state*, and recorded into multiple primary command buffers.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkCommandBufferUsageFlags;
```

`VkCommandBufferUsageFlags` is a bitmask type for setting a mask of zero or more `VkCommandBufferUsageFlagBits`.

If the command buffer is a secondary command buffer, then the `VkCommandBufferInheritanceInfo` structure defines any state that will be inherited from the primary command buffer:

```
// Provided by VK_VERSION_1_0
typedef struct VkCommandBufferInheritanceInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkRenderPass          renderPass;
    uint32_t              subpass;
    VkFramebuffer         framebuffer;
    VkBool32              occlusionQueryEnable;
    VkQueryControlFlags   queryFlags;
    VkQueryPipelineStatisticFlags pipelineStatistics;
} VkCommandBufferInheritanceInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `renderPass` is a `VkRenderPass` object defining which render passes the `VkCommandBuffer` will be compatible with and **can** be executed within.
- `subpass` is the index of the subpass within the render pass instance that the `VkCommandBuffer` will be executed within.
- `framebuffer` **can** refer to the `VkFramebuffer` object that the `VkCommandBuffer` will be rendering to if it is executed within a render pass instance. It **can** be `VK_NULL_HANDLE` if the framebuffer is not known.

NOTE

Specifying the exact framebuffer that the secondary command buffer will be executed with **may** result in better performance at command buffer execution time.

- `occlusionQueryEnable` specifies whether the command buffer **can** be executed while an occlusion query is active in the primary command buffer. If this is `VK_TRUE`, then this command buffer **can** be executed whether the primary command buffer has an occlusion query active or not. If this is `VK_FALSE`, then the primary command buffer **must** not have an occlusion query active.
- `queryFlags` specifies the query flags that **can** be used by an active occlusion query in the primary command buffer when this secondary command buffer is executed. If this value includes the `VK_QUERY_CONTROL_PRECISE_BIT` bit, then the active query **can** return boolean results or actual sample counts. If this bit is not set, then the active query **must** not use the `VK_QUERY_CONTROL_PRECISE_BIT` bit.
- `pipelineStatistics` is a bitmask of `VkQueryPipelineStatisticFlagBits` specifying the set of pipeline statistics that **can** be counted by an active query in the primary command buffer when this secondary command buffer is executed. If this value includes a given bit, then this command buffer **can** be executed whether the primary command buffer has a pipeline statistics query active that includes this bit or not. If this value excludes a given bit, then the active pipeline statistics query **must** not be from a query pool that counts that statistic.

If the `VkCommandBuffer` will not be executed within a render pass instance, `renderPass`, `subpass`, and `framebuffer` are ignored.

Valid Usage

- VUID-VkCommandBufferInheritanceInfo-occlusionQueryEnable-00056
If the `inheritedQueries` feature is not enabled, `occlusionQueryEnable` **must** be `VK_FALSE`
- VUID-VkCommandBufferInheritanceInfo-queryFlags-00057
If the `inheritedQueries` feature is enabled, `queryFlags` **must** be a valid combination of `VkQueryControlFlagBits` values
- VUID-VkCommandBufferInheritanceInfo-queryFlags-02788
If the `inheritedQueries` feature is not enabled, `queryFlags` **must** be `0`
- VUID-VkCommandBufferInheritanceInfo-pipelineStatistics-02789
If the `pipelineStatisticsQuery` feature is enabled, `pipelineStatistics` **must** be a valid combination of `VkQueryPipelineStatisticFlagBits` values
- VUID-VkCommandBufferInheritanceInfo-pipelineStatistics-00058
If the `pipelineStatisticsQuery` feature is not enabled, `pipelineStatistics` **must** be `0`

Valid Usage (Implicit)

- VUID-VkCommandBufferInheritanceInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_COMMAND_BUFFER_INHERITANCE_INFO`
- VUID-VkCommandBufferInheritanceInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkCommandBufferInheritanceInfo-commonparent
Both of `framebuffer`, and `renderPass` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

NOTE

On some implementations, not using the `VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT` bit enables command buffers to be patched in-place if needed, rather than creating a copy of the command buffer.

If a command buffer is in the `invalid, or executable state`, and the command buffer was allocated from a command pool with the `VK_COMMAND_POOL_CREATE_RESET_COMMAND_BUFFER_BIT` flag set, then `vkBeginCommandBuffer` implicitly resets the command buffer, behaving as if `vkResetCommandBuffer` had been called with `VK_COMMAND_BUFFER_RESET_RELEASE_RESOURCES_BIT` not set. After the implicit reset, `commandBuffer` is moved to the `recording state`.

Once recording starts, an application records a sequence of commands (`vkCmd*`) to set state in the command buffer, draw, dispatch, and other commands.

To complete recording of a command buffer, call:

```
// Provided by VK_VERSION_1_0
VkResult vkEndCommandBuffer(
    VkCommandBuffer                                commandBuffer);
```

- `commandBuffer` is the command buffer to complete recording.

The command buffer **must** have been in the `recording state`, and, if successful, is moved to the `executable state`.

If there was an error during recording, the application will be notified by an unsuccessful return code returned by `vkEndCommandBuffer`, and the command buffer will be moved to the `invalid state`.

Valid Usage

- VUID-vkEndCommandBuffer-commandBuffer-00059
`commandBuffer` **must** be in the `recording state`
- VUID-vkEndCommandBuffer-commandBuffer-00060
If `commandBuffer` is a primary command buffer, there **must** not be an active render pass instance
- VUID-vkEndCommandBuffer-commandBuffer-00061
All queries made `active` during the recording of `commandBuffer` **must** have been made inactive

Valid Usage (Implicit)

- VUID-vkEndCommandBuffer-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

When a command buffer is in the executable state, it **can** be submitted to a queue for execution.

6.5. Command Buffer Submission

NOTE

Submission can be a high overhead operation, and applications **should** attempt to batch work together into as few calls to `vkQueueSubmit` as possible.

To submit command buffers to a queue, call:

```
// Provided by VK_VERSION_1_0
VkResult vkQueueSubmit(
    VkQueue          queue,
    uint32_t         submitCount,
    const VkSubmitInfo* pSubmits,
    VkFence          fence);
```

- `queue` is the queue that the command buffers will be submitted to.
- `submitCount` is the number of elements in the `pSubmits` array.
- `pSubmits` is a pointer to an array of `VkSubmitInfo` structures, each specifying a command buffer submission batch.
- `fence` is an **optional** handle to a fence to be signaled once all submitted command buffers have completed execution. If `fence` is not `VK_NULL_HANDLE`, it defines a [fence signal operation](#).

`vkQueueSubmit` is a [queue submission command](#), with each batch defined by an element of `pSubmits`. Batches begin execution in the order they appear in `pSubmits`, but **may** complete out of order.

Fence and semaphore operations submitted with `vkQueueSubmit` have additional ordering constraints compared to other submission commands, with dependencies involving previous and subsequent queue operations. Information about these additional constraints can be found in the [semaphore](#) and [fence](#) sections of [the synchronization chapter](#).

Details on the interaction of `pWaitDstStageMask` with synchronization are described in the [semaphore wait operation](#) section of [the synchronization chapter](#).

The order that batches appear in `pSubmits` is used to determine [submission order](#), and thus all the [implicit ordering guarantees](#) that respect it. Other than these implicit ordering guarantees and any [explicit synchronization primitives](#), these batches **may** overlap or otherwise execute out of order.

If any command buffer submitted to this queue is in the [executable state](#), it is moved to the [pending state](#). Once execution of all submissions of a command buffer complete, it moves from the [pending state](#), back to the [executable state](#). If a command buffer was recorded with the `VK_COMMAND_BUFFER_USAGE_ONE_TIME_SUBMIT_BIT` flag, it instead moves to the [invalid state](#).

If `vkQueueSubmit` fails, it **may** return `VK_ERROR_OUT_OF_HOST_MEMORY` or `VK_ERROR_OUT_OF_DEVICE_MEMORY`. If it does, the implementation **must** ensure that the state and contents of any resources or synchronization primitives referenced by the submitted command buffers and any semaphores referenced by `pSubmits` is unaffected by the call or its failure. If `vkQueueSubmit` fails in such a way

that the implementation is unable to make that guarantee, the implementation **must** return `VK_ERROR_DEVICE_LOST`. See [Lost Device](#).

Valid Usage

- VUID-vkQueueSubmit-fence-00063
If `fence` is not `VK_NULL_HANDLE`, `fence` **must** be unsignaled
- VUID-vkQueueSubmit-fence-00064
If `fence` is not `VK_NULL_HANDLE`, `fence` **must** not be associated with any other queue command that has not yet completed execution on that queue
- VUID-vkQueueSubmit-pCommandBuffers-00065
Any calls to `vkCmdSetEvent`, `vkCmdResetEvent` or `vkCmdWaitEvents` that have been recorded into any of the command buffer elements of the `pCommandBuffers` member of any element of `pSubmits`, **must** not reference any `VkEvent` that is referenced by any of those commands in a command buffer that has been submitted to another queue and is still in the *pending state*
- VUID-vkQueueSubmit-pWaitDstStageMask-00066
Any stage flag included in any element of the `pWaitDstStageMask` member of any element of `pSubmits` **must** be a pipeline stage supported by one of the capabilities of `queue`, as specified in the [table of supported pipeline stages](#)
- VUID-vkQueueSubmit-pSignalSemaphores-00067
Each binary semaphore element of the `pSignalSemaphores` member of any element of `pSubmits` **must** be unsignaled when the semaphore signal operation it defines is executed on the device
- VUID-vkQueueSubmit-pWaitSemaphores-00068
When a semaphore wait operation referring to a binary semaphore defined by any element of the `pWaitSemaphores` member of any element of `pSubmits` executes on `queue`, there **must** be no other queues waiting on the same semaphore
- VUID-vkQueueSubmit-pWaitSemaphores-03238
All elements of the `pWaitSemaphores` member of all elements of `pSubmits` **must** reference a semaphore signal operation that has been submitted for execution and any [semaphore signal operations](#) on which it depends **must** have also been submitted for execution
- VUID-vkQueueSubmit-pCommandBuffers-00070
Each element of the `pCommandBuffers` member of each element of `pSubmits` **must** be in the [pending or executable state](#)
- VUID-vkQueueSubmit-pCommandBuffers-00071
If any element of the `pCommandBuffers` member of any element of `pSubmits` was not recorded with the `VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT`, it **must** not be in the [pending state](#)
- VUID-vkQueueSubmit-pCommandBuffers-00072
Any [secondary command buffers recorded](#) into any element of the `pCommandBuffers` member of any element of `pSubmits` **must** be in the [pending or executable state](#)
- VUID-vkQueueSubmit-pCommandBuffers-00073

If any **secondary command buffers recorded** into any element of the **pCommandBuffers** member of any element of **pSubmits** was not recorded with the **VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT**, it **must** not be in the **pending state**

- VUID-vkQueueSubmit-pCommandBuffers-00074

Each element of the **pCommandBuffers** member of each element of **pSubmits** **must** have been allocated from a **VkCommandPool** that was created for the same queue family **queue** belongs to

- VUID-vkQueueSubmit-pSubmits-02207

If any element of **pSubmits**→**pCommandBuffers** includes a **Queue Family Ownership Transfer Acquire Operation**, there **must** exist a previously submitted **Queue Family Ownership Transfer Release Operation** on a queue in the queue family identified by the acquire operation, with parameters matching the acquire operation as defined in the definition of such **acquire operations**, and which happens-before the acquire operation

- VUID-vkQueueSubmit-pSubmits-02808

Any resource created with **VK_SHARING_MODE_EXCLUSIVE** that is read by an operation specified by **pSubmits** **must** not be owned by any queue family other than the one which **queue** belongs to, at the time it is executed

- VUID-vkQueueSubmit-pSubmits-04626

Any resource created with **VK_SHARING_MODE_CONCURRENT** that is accessed by an operation specified by **pSubmits** **must** have included the queue family of **queue** at resource creation time

Valid Usage (Implicit)

- VUID-vkQueueSubmit-queue-parameter

queue **must** be a valid **VkQueue** handle

- VUID-vkQueueSubmit-pSubmits-parameter

If **submitCount** is not 0, **pSubmits** **must** be a valid pointer to an array of **submitCount** valid **VkSubmitInfo** structures

- VUID-vkQueueSubmit-fence-parameter

If **fence** is not **VK_NULL_HANDLE**, **fence** **must** be a valid **VkFence** handle

- VUID-vkQueueSubmit-commonparent

Both of **fence**, and **queue** that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same **VkDevice**

Host Synchronization

- Host access to **queue** **must** be externally synchronized
- Host access to **fence** **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
-	-	Any	-

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_DEVICE_LOST`

The `VkSubmitInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSubmitInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           waitSemaphoreCount;
    const VkSemaphore* pWaitSemaphores;
    const VkPipelineStageFlags* pWaitDstStageMask;
    uint32_t           commandBufferCount;
    const VkCommandBuffer* pCommandBuffers;
    uint32_t           signalSemaphoreCount;
    const VkSemaphore* pSignalSemaphores;
} VkSubmitInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `waitSemaphoreCount` is the number of semaphores upon which to wait before executing the command buffers for the batch.
- `pWaitSemaphores` is a pointer to an array of `VkSemaphore` handles upon which to wait before the command buffers for this batch begin execution. If semaphores to wait on are provided, they define a [semaphore wait operation](#).
- `pWaitDstStageMask` is a pointer to an array of pipeline stages at which each corresponding semaphore wait will occur.
- `commandBufferCount` is the number of command buffers to execute in the batch.
- `pCommandBuffers` is a pointer to an array of `VkCommandBuffer` handles to execute in the batch.

- `signalSemaphoreCount` is the number of semaphores to be signaled once the commands specified in `pCommandBuffers` have completed execution.
- `pSignalSemaphores` is a pointer to an array of `VkSemaphore` handles which will be signaled when the command buffers for this batch have completed execution. If semaphores to be signaled are provided, they define a `semaphore signal operation`.

The order that command buffers appear in `pCommandBuffers` is used to determine `submission order`, and thus all the `implicit ordering guarantees` that respect it. Other than these implicit ordering guarantees and any `explicit synchronization primitives`, these command buffers **may** overlap or otherwise execute out of order.

Valid Usage

- VUID-VkSubmitInfo-pWaitDstStageMask-04090
If the `geometryShader` feature is not enabled, `pWaitDstStageMask` **must** not contain `VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT`
- VUID-VkSubmitInfo-pWaitDstStageMask-04091
If the `tessellationShader` feature is not enabled, `pWaitDstStageMask` **must** not contain `VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT` or `VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT`
- VUID-VkSubmitInfo-pWaitDstStageMask-04996
`pWaitDstStageMask` **must** not be 0
- VUID-VkSubmitInfo-pCommandBuffers-00075
Each element of `pCommandBuffers` **must** not have been allocated with `VK_COMMAND_BUFFER_LEVEL_SECONDARY`
- VUID-VkSubmitInfo-pWaitDstStageMask-00078
Each element of `pWaitDstStageMask` **must** not include `VK_PIPELINE_STAGE_HOST_BIT`

Valid Usage (Implicit)

- VUID-VkSubmitInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_SUBMIT_INFO`
- VUID-VkSubmitInfo-pNext-pNext
`pNext` **must** be NULL
- VUID-VkSubmitInfo-pWaitSemaphores-parameter
If `waitSemaphoreCount` is not 0, `pWaitSemaphores` **must** be a valid pointer to an array of `waitSemaphoreCount` valid `VkSemaphore` handles
- VUID-VkSubmitInfo-pWaitDstStageMask-parameter
If `waitSemaphoreCount` is not 0, `pWaitDstStageMask` **must** be a valid pointer to an array of `waitSemaphoreCount` valid combinations of `VkPipelineStageFlagBits` values
- VUID-VkSubmitInfo-pCommandBuffers-parameter
If `commandBufferCount` is not 0, `pCommandBuffers` **must** be a valid pointer to an array of `commandBufferCount` valid `VkCommandBuffer` handles

- VUID-VkSubmitInfo-pSignalSemaphores-parameter
If `signalSemaphoreCount` is not 0, `pSignalSemaphores` **must** be a valid pointer to an array of `signalSemaphoreCount` valid `VkSemaphore` handles
- VUID-VkSubmitInfo-commonparent
Each of the elements of `pCommandBuffers`, the elements of `pSignalSemaphores`, and the elements of `pWaitSemaphores` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

6.6. Queue Forward Progress

When using binary semaphores, the application **must** ensure that command buffer submissions will be able to complete without any subsequent operations by the application on any queue. After any call to `vkQueueSubmit` (or other queue operation), for every queued wait on a semaphore there **must** be a prior signal of that semaphore that will not be consumed by a different wait on the semaphore.

If a command buffer submission waits for any events to be signaled, the application **must** ensure that command buffer submissions will be able to complete without any subsequent operations by the application. Events signaled by the host **must** be signaled before the command buffer waits on those events.

NOTE

The ability for commands to wait on the host to set an events was originally added to allow low-latency updates to resources between host and device. However, to ensure quality of service, implementations would necessarily detect extended stalls in execution and timeout after a short period. As this period is not defined in the Vulkan specification, it is impossible to correctly validate any application with any wait period. Since the original users of this functionality were highly limited and platform-specific, this functionality is now considered defunct and should not be used.

6.7. Secondary Command Buffer Execution

Secondary command buffers **must** not be directly submitted to a queue. To record a secondary command buffer to execute as part of a primary command buffer, call:

```
// Provided by VK_VERSION_1_0
void vkCmdExecuteCommands(
    VkCommandBuffer          commandBuffer,
    uint32_t                 commandBufferCount,
    const VkCommandBuffer*   pCommandBuffers);
```

- `commandBuffer` is a handle to a primary command buffer that the secondary command buffers are executed in.
- `commandBufferCount` is the length of the `pCommandBuffers` array.

- `pCommandBuffers` is a pointer to an array of `commandBufferCount` secondary command buffer handles, which are recorded to execute in the primary command buffer in the order they are listed in the array.

If any element of `pCommandBuffers` was not recorded with the `VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT` flag, and it was recorded into any other primary command buffer which is currently in the `executable` or `recording` state, that primary command buffer becomes `invalid`.

Valid Usage

- VUID-vkCmdExecuteCommands-pCommandBuffers-00088
Each element of `pCommandBuffers` **must** have been allocated with a `level` of `VK_COMMAND_BUFFER_LEVEL_SECONDARY`
- VUID-vkCmdExecuteCommands-pCommandBuffers-00089
Each element of `pCommandBuffers` **must** be in the `pending` or `executable` state
- VUID-vkCmdExecuteCommands-pCommandBuffers-00091
If any element of `pCommandBuffers` was not recorded with the `VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT` flag, it **must** not be in the `pending` state
- VUID-vkCmdExecuteCommands-pCommandBuffers-00092
If any element of `pCommandBuffers` was not recorded with the `VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT` flag, it **must** not have already been recorded to `commandBuffer`
- VUID-vkCmdExecuteCommands-pCommandBuffers-00093
If any element of `pCommandBuffers` was not recorded with the `VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT` flag, it **must** not appear more than once in `pCommandBuffers`
- VUID-vkCmdExecuteCommands-pCommandBuffers-00094
Each element of `pCommandBuffers` **must** have been allocated from a `VkCommandPool` that was created for the same queue family as the `VkCommandPool` from which `commandBuffer` was allocated
- VUID-vkCmdExecuteCommands-pCommandBuffers-00096
If `vkCmdExecuteCommands` is being called within a render pass instance, each element of `pCommandBuffers` **must** have been recorded with the `VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT`
- VUID-vkCmdExecuteCommands-pCommandBuffers-00099
If `vkCmdExecuteCommands` is being called within a render pass instance, and any element of `pCommandBuffers` was recorded with `VkCommandBufferInheritanceInfo::framebuffer` not equal to `VK_NULL_HANDLE`, that `VkFramebuffer` **must** match the `VkFramebuffer` used in the current render pass instance
- VUID-vkCmdExecuteCommands-contents-09680
If `vkCmdExecuteCommands` is being called within a render pass instance begun with `vkCmdBeginRenderPass`, and `vkCmdNextSubpass` has not been called in the current render pass instance, the `contents` parameter of `vkCmdBeginRenderPass` **must** have been

VK_SUBPASS_CONTENTS_SECONDARY_COMMAND_BUFFERS

- VUID-vkCmdExecuteCommands-None-09681

If `vkCmdExecuteCommands` is being called within a render pass instance begun with `vkCmdBeginRenderPass`, and `vkCmdNextSubpass` has been called in the current render pass instance, the `contents` parameter of the last call to `vkCmdNextSubpass` **must** have been `VK_SUBPASS_CONTENTS_SECONDARY_COMMAND_BUFFERS`

- VUID-vkCmdExecuteCommands-pCommandBuffers-06019

If `vkCmdExecuteCommands` is being called within a render pass instance begun with `vkCmdBeginRenderPass`, each element of `pCommandBuffers` **must** have been recorded with `VkCommandBufferInheritanceInfo::subpass` set to the index of the subpass which the given command buffer will be executed in

- VUID-vkCmdExecuteCommands-pBeginInfo-06020

If `vkCmdExecuteCommands` is being called within a render pass instance begun with `vkCmdBeginRenderPass`, the render passes specified in the `pBeginInfo->pInheritanceInfo->renderPass` members of the `vkBeginCommandBuffer` commands used to begin recording each element of `pCommandBuffers` **must** be `compatible` with the current render pass

- VUID-vkCmdExecuteCommands-pCommandBuffers-00100

If `vkCmdExecuteCommands` is not being called within a render pass instance, each element of `pCommandBuffers` **must** not have been recorded with the `VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT`

- VUID-vkCmdExecuteCommands-commandBuffer-00101

If the `inheritedQueries` feature is not enabled, `commandBuffer` **must** not have any queries `active`

- VUID-vkCmdExecuteCommands-commandBuffer-00102

If `commandBuffer` has a `VK_QUERY_TYPE_OCCLUSION` query `active`, then each element of `pCommandBuffers` **must** have been recorded with `VkCommandBufferInheritanceInfo::occlusionQueryEnable` set to `VK_TRUE`

- VUID-vkCmdExecuteCommands-commandBuffer-00103

If `commandBuffer` has a `VK_QUERY_TYPE_OCCLUSION` query `active`, then each element of `pCommandBuffers` **must** have been recorded with `VkCommandBufferInheritanceInfo::queryFlags` having all bits set that are set for the query

- VUID-vkCmdExecuteCommands-commandBuffer-00104

If `commandBuffer` has a `VK_QUERY_TYPE_PIPELINE_STATISTICS` query `active`, then each element of `pCommandBuffers` **must** have been recorded with `VkCommandBufferInheritanceInfo::pipelineStatistics` having all bits set that are set in the `VkQueryPool` the query uses

- VUID-vkCmdExecuteCommands-pCommandBuffers-00105

Each element of `pCommandBuffers` **must** not begin any query types that are `active` in `commandBuffer`

- VUID-vkCmdExecuteCommands-commandBuffer-07594

`commandBuffer` **must** not have any queries other than `VK_QUERY_TYPE_OCCLUSION` and `VK_QUERY_TYPE_PIPELINE_STATISTICS` `active`

- VUID-vkCmdExecuteCommands-commandBuffer-06533

If `vkCmdExecuteCommands` is being called within a render pass instance and any recorded

command in `commandBuffer` in the current subpass will write to an image subresource as an attachment, commands recorded in elements of `pCommandBuffers` **must** not read from the memory backing that image subresource in any other way

- VUID-vkCmdExecuteCommands-commandBuffer-06534

If `vkCmdExecuteCommands` is being called within a render pass instance and any recorded command in `commandBuffer` in the current subpass will read from an image subresource used as an attachment in any way other than as an attachment, commands recorded in elements of `pCommandBuffers` **must** not write to that image subresource as an attachment

- VUID-vkCmdExecuteCommands-pCommandBuffers-06535

If `vkCmdExecuteCommands` is being called within a render pass instance and any recorded command in a given element of `pCommandBuffers` will write to an image subresource as an attachment, commands recorded in elements of `pCommandBuffers` at a higher index **must** not read from the memory backing that image subresource in any other way

- VUID-vkCmdExecuteCommands-pCommandBuffers-06536

If `vkCmdExecuteCommands` is being called within a render pass instance and any recorded command in a given element of `pCommandBuffers` will read from an image subresource used as an attachment in any way other than as an attachment, commands recorded in elements of `pCommandBuffers` at a higher index **must** not write to that image subresource as an attachment

- VUID-vkCmdExecuteCommands-commandBuffer-09375

`commandBuffer` **must** not be a [secondary command buffer](#)

Valid Usage (Implicit)

- VUID-vkCmdExecuteCommands-commandBuffer-parameter

`commandBuffer` **must** be a valid [VkCommandBuffer](#) handle

- VUID-vkCmdExecuteCommands-pCommandBuffers-parameter

`pCommandBuffers` **must** be a valid pointer to an array of `commandBufferCount` valid [VkCommandBuffer](#) handles

- VUID-vkCmdExecuteCommands-commandBuffer-recording

`commandBuffer` **must** be in the [recording state](#)

- VUID-vkCmdExecuteCommands-commandBuffer-cmdpool

The [VkCommandPool](#) that `commandBuffer` was allocated from **must** support transfer, graphics, or compute operations

- VUID-vkCmdExecuteCommands-commandBufferCount-arraylength

`commandBufferCount` **must** be greater than 0

- VUID-vkCmdExecuteCommands-commonparent

Both of `commandBuffer`, and the elements of `pCommandBuffers` **must** have been created, allocated, or retrieved from the same [VkDevice](#)

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Transfer Graphics Compute	Indirection

Chapter 7. Synchronization and Cache Control

Synchronization of access to resources is primarily the responsibility of the application in Vulkan. The order of execution of commands with respect to the host and other commands on the device has few implicit guarantees, and needs to be explicitly specified. Memory caches and other optimizations are also explicitly managed, requiring that the flow of data through the system is largely under application control.

Whilst some implicit guarantees exist between commands, five explicit synchronization mechanisms are exposed by Vulkan:

Fences

Fences **can** be used to communicate to the host that execution of some task on the device has completed, controlling resource access between host and device.

Semaphores

Semaphores **can** be used to control resource access across multiple queues.

Events

Events provide a fine-grained synchronization primitive which **can** be signaled either within a command buffer or by the host, and **can** be waited upon within a command buffer or queried on the host. Events **can** be used to control resource access within a single queue.

Pipeline Barriers

Pipeline barriers also provide synchronization control within a command buffer, but at a single point, rather than with separate signal and wait operations. Pipeline barriers **can** be used to control resource access within a single queue.

Render Pass Objects

Render pass objects provide a synchronization framework for rendering tasks, built upon the concepts in this chapter. Many cases that would otherwise need an application to use other synchronization primitives **can** be expressed more efficiently as part of a render pass. Render pass objects **can** be used to control resource access within a single queue.

7.1. Execution and Memory Dependencies

An *operation* is an arbitrary amount of work to be executed on the host, a device, or an external entity such as a presentation engine. Synchronization commands introduce explicit *execution dependencies*, and *memory dependencies* between two sets of operations defined by the command's two *synchronization scopes*.

The synchronization scopes define which other operations a synchronization command is able to create execution dependencies with. Any type of operation that is not in a synchronization command's synchronization scopes will not be included in the resulting dependency. For example, for many synchronization commands, the synchronization scopes **can** be limited to just operations executing in specific [pipeline stages](#), which allows other pipeline stages to be excluded from a

dependency. Other scoping options are possible, depending on the particular command.

An *execution dependency* is a guarantee that for two sets of operations, the first set **must happen-before** the second set. If an operation happens-before another operation, then the first operation **must** complete before the second operation is initiated. More precisely:

- Let **Ops₁** and **Ops₂** be separate sets of operations.
- Let **Sync** be a synchronization command.
- Let **Scope_{1st}** and **Scope_{2nd}** be the synchronization scopes of **Sync**.
- Let **ScopedOps₁** be the intersection of sets **Ops₁** and **Scope_{1st}**.
- Let **ScopedOps₂** be the intersection of sets **Ops₂** and **Scope_{2nd}**.
- Submitting **Ops₁**, **Sync** and **Ops₂** for execution, in that order, will result in execution dependency **ExeDep** between **ScopedOps₁** and **ScopedOps₂**.
- Execution dependency **ExeDep** guarantees that **ScopedOps₁** happen-before **ScopedOps₂**.

An *execution dependency chain* is a sequence of execution dependencies that form a happens-before relation between the first dependency's **ScopedOps₁** and the final dependency's **ScopedOps₂**. For each consecutive pair of execution dependencies, a chain exists if the intersection of **Scope_{2nd}** in the first dependency and **Scope_{1st}** in the second dependency is not an empty set. The formation of a single execution dependency from an execution dependency chain can be described by substituting the following in the description of execution dependencies:

- Let **Sync** be a set of synchronization commands that generate an execution dependency chain.
- Let **Scope_{1st}** be the first synchronization scope of the first command in **Sync**.
- Let **Scope_{2nd}** be the second synchronization scope of the last command in **Sync**.

Execution dependencies alone are not sufficient to guarantee that values resulting from writes in one set of operations **can** be read from another set of operations.

Three additional types of operations are used to control memory access. *Availability operations* cause the values generated by specified memory write accesses to become *available* to a memory domain for future access. Any available value remains available until a subsequent write to the same memory location occurs (whether it is made available or not) or the memory is freed. *Memory domain operations* cause writes that are available to a source memory domain to become available to a destination memory domain (an example of this is making writes available to the host domain available to the device domain). *Visibility operations* cause values available to a memory domain to become *visible* to specified memory accesses.

A *memory dependency* is an execution dependency which includes availability and visibility operations such that:

- The first set of operations happens-before the availability operation.
- The availability operation happens-before the visibility operation.
- The visibility operation happens-before the second set of operations.

Once written values are made visible to a particular type of memory access, they **can** be read or

written by that type of memory access. Most synchronization commands in Vulkan define a memory dependency.

The specific memory accesses that are made available and visible are defined by the *access scopes* of a memory dependency. Any type of access that is in a memory dependency's first access scope and occurs in **ScopedOps₁** is made available. Any type of access that is in a memory dependency's second access scope and occurs in **ScopedOps₂** has any available writes made visible to it. Any type of operation that is not in a synchronization command's access scopes will not be included in the resulting dependency.

A memory dependency enforces availability and visibility of memory accesses and execution order between two sets of operations. Adding to the description of [execution dependency chains](#):

- Let **MemOps₁** be the set of memory accesses performed by **ScopedOps₁**.
- Let **MemOps₂** be the set of memory accesses performed by **ScopedOps₂**.
- Let **AccessScope_{1st}** be the first access scope of the first command in the **Sync** chain.
- Let **AccessScope_{2nd}** be the second access scope of the last command in the **Sync** chain.
- Let **ScopedMemOps₁** be the intersection of sets **MemOps₁** and **AccessScope_{1st}**.
- Let **ScopedMemOps₂** be the intersection of sets **MemOps₂** and **AccessScope_{2nd}**.
- Submitting **Ops₁**, **Sync**, and **Ops₂** for execution, in that order, will result in a memory dependency **MemDep** between **ScopedOps₁** and **ScopedOps₂**.
- Memory dependency **MemDep** guarantees that:
 - Memory writes in **ScopedMemOps₁** are made available.
 - Available memory writes, including those from **ScopedMemOps₁**, are made visible to **ScopedMemOps₂**.

NOTE

Execution and memory dependencies are used to solve data hazards, i.e. to ensure that read and write operations occur in a well-defined order. Write-after-read hazards can be solved with just an execution dependency, but read-after-write and write-after-write hazards need appropriate memory dependencies to be included between them. If an application does not include dependencies to solve these hazards, the results and execution orders of memory accesses are undefined.

7.1.1. Image Layout Transitions

Image subresources **can** be transitioned from one [layout](#) to another as part of a [memory dependency](#) (e.g. by using an [image memory barrier](#)). When a layout transition is specified in a memory dependency, it happens-after the availability operations in the memory dependency, and happens-before the visibility operations. Image layout transitions **may** perform read and write accesses on all memory bound to the image subresource range, so applications **must** ensure that all memory writes have been made [available](#) before a layout transition is executed. Available memory is automatically made visible to a layout transition, and writes performed by a layout transition are automatically made available.

Layout transitions always apply to a particular image subresource range, and specify both an old

layout and new layout. The old layout **must** either be `VK_IMAGE_LAYOUT_UNDEFINED`, or match the current layout of the image subresource range. If the old layout matches the current layout of the image subresource range, the transition preserves the contents of that range. If the old layout is `VK_IMAGE_LAYOUT_UNDEFINED`, the contents of that range **may** be discarded.

NOTE

Image layout transitions with `VK_IMAGE_LAYOUT_UNDEFINED` allow the implementation to discard the image subresource range, which can provide performance or power benefits. Tile-based architectures may be able to avoid flushing tile data to memory, and immediate style renderers may be able to achieve fast metadata clears to reinitialize frame buffer compression state, or similar.

If the contents of an attachment are not needed after a render pass completes, then applications **should** use `VK_ATTACHMENT_STORE_OP_DONT_CARE`.

NOTE

Applications **must** ensure that layout transitions happen-after all operations accessing the image with the old layout, and happen-before any operations that will access the image with the new layout. Layout transitions are potentially read/write operations, so not defining appropriate memory dependencies to guarantee this will result in a data race.

Image layout transitions interact with [memory aliasing](#).

Layout transitions that are performed via image memory barriers execute in their entirety in [submission order](#), relative to other image layout transitions submitted to the same queue, including those performed by [render passes](#). In effect there is an implicit execution dependency from each such layout transition to all layout transitions previously submitted to the same queue.

7.1.2. Pipeline Stages

The work performed by an [action command](#) consists of multiple operations, which are performed as a sequence of logically independent steps known as *pipeline stages*. The exact pipeline stages executed depend on the particular command that is used, and current command buffer state when the command was recorded.

NOTE

Operations performed by synchronization commands (e.g. [availability and visibility operations](#)) are not executed by a defined pipeline stage. However other commands can still synchronize with them by using the [synchronization scopes](#) to create a [dependency chain](#).

Execution of operations across pipeline stages **must** adhere to [implicit ordering guarantees](#), particularly including [pipeline stage order](#). Otherwise, execution across pipeline stages **may** overlap or execute out of order with regards to other stages, unless otherwise enforced by an execution dependency.

Several of the synchronization commands include pipeline stage parameters, restricting the [synchronization scopes](#) for that command to just those stages. This allows fine grained control over the exact execution dependencies and accesses performed by action commands. Implementations **should** use these pipeline stages to avoid unnecessary stalls or cache flushing.

Bits which **can** be set in a [VkPipelineStageFlags](#) mask, specifying stages of execution, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkPipelineStageFlagBits {
    VK_PIPELINE_STAGE_TOP_OF_PIPE_BIT = 0x00000001,
    VK_PIPELINE_STAGE_DRAW_INDIRECT_BIT = 0x00000002,
    VK_PIPELINE_STAGE_VERTEX_INPUT_BIT = 0x00000004,
    VK_PIPELINE_STAGE_VERTEX_SHADER_BIT = 0x00000008,
    VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT = 0x00000010,
    VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT = 0x00000020,
    VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT = 0x00000040,
    VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT = 0x00000080,
    VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT = 0x00000100,
    VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT = 0x00000200,
    VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT = 0x00000400,
    VK_PIPELINE_STAGE_COMPUTE_SHADER_BIT = 0x00000800,
    VK_PIPELINE_STAGE_TRANSFER_BIT = 0x00001000,
    VK_PIPELINE_STAGE_BOTTOM_OF_PIPE_BIT = 0x00002000,
    VK_PIPELINE_STAGE_HOST_BIT = 0x00004000,
    VK_PIPELINE_STAGE_ALL_GRAPHICS_BIT = 0x00008000,
    VK_PIPELINE_STAGE_ALL_COMMANDS_BIT = 0x00010000,
} VkPipelineStageFlagBits;
```

- [VK_PIPELINE_STAGE_DRAW_INDIRECT_BIT](#) specifies the stage of the pipeline where [VkDrawIndirect*](#) / [VkDispatchIndirect*](#) / [VkTraceRaysIndirect*](#) data structures are consumed.
- [VK_PIPELINE_STAGE_VERTEX_INPUT_BIT](#) specifies the stage of the pipeline where vertex and index buffers are consumed.
- [VK_PIPELINE_STAGE_VERTEX_SHADER_BIT](#) specifies the vertex shader stage.
- [VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT](#) specifies the tessellation control shader stage.
- [VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT](#) specifies the tessellation evaluation shader stage.
- [VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT](#) specifies the geometry shader stage.
- [VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT](#) specifies the fragment shader stage.
- [VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT](#) specifies the stage of the pipeline where early fragment tests (depth and stencil tests before fragment shading) are performed. This stage also includes [render pass load operations](#) for framebuffer attachments with a depth/stencil format.
- [VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT](#) specifies the stage of the pipeline where late fragment tests (depth and stencil tests after fragment shading) are performed. This stage also includes [render pass store operations](#) for framebuffer attachments with a depth/stencil format.
- [VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT](#) specifies the stage of the pipeline after blending where the final color values are output from the pipeline. This stage includes [blending](#), [logic operations](#), render pass [load](#) and [store](#) operations for color attachments, [render pass multisample resolve operations](#), and [vkCmdClearAttachments](#).

- `VK_PIPELINE_STAGE_COMPUTE_SHADER_BIT` specifies the execution of a compute shader.
- `VK_PIPELINE_STAGE_TRANSFER_BIT` specifies the following commands:
 - All [copy commands](#), including [vkCmdCopyQueryPoolResults](#)
 - [vkCmdBlitImage](#)
 - [vkCmdResolveImage](#)
 - All [clear commands](#), with the exception of [vkCmdClearAttachments](#)
- `VK_PIPELINE_STAGE_HOST_BIT` specifies a pseudo-stage indicating execution on the host of reads/writes of device memory. This stage is not invoked by any commands recorded in a command buffer.
- `VK_PIPELINE_STAGE_ALL_GRAPHICS_BIT` specifies the execution of all graphics pipeline stages, and is equivalent to the logical OR of:
 - `VK_PIPELINE_STAGE_DRAW_INDIRECT_BIT`
 - `VK_PIPELINE_STAGE_VERTEX_INPUT_BIT`
 - `VK_PIPELINE_STAGE_VERTEX_SHADER_BIT`
 - `VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT`
 - `VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT`
 - `VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT`
 - `VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT`
 - `VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT`
 - `VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT`
 - `VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT`
- `VK_PIPELINE_STAGE_ALL_COMMANDS_BIT` specifies all operations performed by all commands supported on the queue it is used with.
- `VK_PIPELINE_STAGE_TOP_OF_PIPE_BIT` is equivalent to `VK_PIPELINE_STAGE_ALL_COMMANDS_BIT` with [VkAccessFlags](#) set to `0` when specified in the second synchronization scope, but specifies no stage of execution when specified in the first scope.
- `VK_PIPELINE_STAGE_BOTTOM_OF_PIPE_BIT` is equivalent to `VK_PIPELINE_STAGE_ALL_COMMANDS_BIT` with [VkAccessFlags](#) set to `0` when specified in the first synchronization scope, but specifies no stage of execution when specified in the second scope.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineStageFlags;
```

`VkPipelineStageFlags` is a bitmask type for setting a mask of zero or more [VkPipelineStageFlagBits](#).

If a synchronization command includes a source stage mask, its first [synchronization scope](#) only includes execution of the pipeline stages specified in that mask and any [logically earlier](#) stages. Its first [access scope](#) only includes memory accesses performed by pipeline stages explicitly specified in the source stage mask.

If a synchronization command includes a destination stage mask, its second [synchronization scope](#) only includes execution of the pipeline stages specified in that mask and any [logically later](#) stages. Its second [access scope](#) only includes memory accesses performed by pipeline stages explicitly specified in the destination stage mask.

NOTE

Note that [access scopes](#) do not interact with the logically earlier or later stages for either scope - only the stages the application specifies are considered part of each access scope.

Certain pipeline stages are only available on queues that support a particular set of operations. The following table lists, for each pipeline stage flag, which queue capability flag **must** be supported by the queue. When multiple flags are enumerated in the second column of the table, it means that the pipeline stage is supported on the queue if it supports any of the listed capability flags. For further details on queue capabilities see [Physical Device Enumeration](#) and [Queues](#).

Table 3. Supported Pipeline Stage Flags

Pipeline stage flag	Required queue capability flag
VK_PIPELINE_STAGE_2_NONE	None required
VK_PIPELINE_STAGE_2_TOP_OF_PIPE_BIT	None required
VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT
VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT	VK_QUEUE_COMPUTE_BIT
VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT or VK_QUEUE_TRANSFER_BIT
VK_PIPELINE_STAGE_2_BOTTOM_OF_PIPE_BIT	None required
VK_PIPELINE_STAGE_2_HOST_BIT	None required
VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT	None required
VK_PIPELINE_STAGE_2_COPY_BIT	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT or VK_QUEUE_TRANSFER_BIT

Pipeline stage flag	Required queue capability flag
VK_PIPELINE_STAGE_2_RESOLVE_BIT	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT or VK_QUEUE_TRANSFER_BIT
VK_PIPELINE_STAGE_2_BLIT_BIT	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT or VK_QUEUE_TRANSFER_BIT
VK_PIPELINE_STAGE_2_CLEAR_BIT	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT or VK_QUEUE_TRANSFER_BIT
VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_PRE_RASTERIZATION_SHADERS_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_EXT	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT

Pipeline stages that execute as a result of a command logically complete execution in a specific order, such that completion of a logically later pipeline stage **must** not happen-before completion of a logically earlier stage. This means that including any stage in the source stage mask for a particular synchronization command also implies that any logically earlier stages are included in **Scope_{1st}** for that command.

Similarly, initiation of a logically earlier pipeline stage **must** not happen-after initiation of a logically later pipeline stage. Including any given stage in the destination stage mask for a particular synchronization command also implies that any logically later stages are included in **Scope_{2nd}** for that command.

NOTE

Implementations **may** not support synchronization at every pipeline stage for every synchronization operation. If a pipeline stage that an implementation does not support synchronization for appears in a source stage mask, it **may** substitute any logically later stage in its place for the first synchronization scope. If a pipeline stage that an implementation does not support synchronization for appears in a destination stage mask, it **may** substitute any logically earlier stage in its place for the second synchronization scope.

For example, if an implementation is unable to signal an event immediately after vertex shader execution is complete, it **may** instead signal the event after color attachment output has completed.

If an implementation makes such a substitution, it **must** not affect the semantics of execution or memory dependencies or image and buffer memory barriers.

Graphics pipelines are executable on queues supporting **VK_QUEUE_GRAPHICS_BIT**. Stages executed by graphics pipelines **can** only be specified in commands recorded for queues supporting **VK_QUEUE_GRAPHICS_BIT**.

The graphics pipeline executes the following stages, with the logical ordering of the stages matching the order specified here:

- `VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT`
- `VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT`
- `VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT`
- `VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT`
- `VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT`
- `VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT`
- `VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT`
- `VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT`
- `VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT`
- `VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT`
- `VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT`

For the compute pipeline, the following stages occur in this order:

- `VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT`
- `VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT`

For the transfer pipeline, the following stages occur in this order:

- `VK_PIPELINE_STAGE_2_TRANSFER_BIT`

For host operations, only one pipeline stage occurs, so no order is guaranteed:

- `VK_PIPELINE_STAGE_2_HOST_BIT`

7.1.3. Access Types

Memory in Vulkan **can** be accessed from within shader invocations and via some fixed-function stages of the pipeline. The *access type* is a function of the [descriptor type](#) used, or how a fixed-function stage accesses memory.

Some synchronization commands take sets of access types as parameters to define the [access scopes](#) of a memory dependency. If a synchronization command includes a *source access mask*, its first [access scope](#) only includes accesses via the access types specified in that mask. Similarly, if a synchronization command includes a *destination access mask*, its second [access scope](#) only includes accesses via the access types specified in that mask.

Bits which **can** be set in the `srcAccessMask` and `dstAccessMask` members of [VkSubpassDependency](#), [VkMemoryBarrier](#), [VkBufferMemoryBarrier](#), and [VkImageMemoryBarrier](#), specifying access behavior, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkAccessFlagBits {
    VK_ACCESS_INDIRECT_COMMAND_READ_BIT = 0x00000001,
    VK_ACCESS_INDEX_READ_BIT = 0x00000002,
    VK_ACCESS_VERTEX_ATTRIBUTE_READ_BIT = 0x00000004,
    VK_ACCESS_UNIFORM_READ_BIT = 0x00000008,
    VK_ACCESS_INPUT_ATTACHMENT_READ_BIT = 0x00000010,
    VK_ACCESS_SHADER_READ_BIT = 0x00000020,
    VK_ACCESS_SHADER_WRITE_BIT = 0x00000040,
    VK_ACCESS_COLOR_ATTACHMENT_READ_BIT = 0x00000080,
    VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT = 0x00000100,
    VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_READ_BIT = 0x00000200,
    VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT = 0x00000400,
    VK_ACCESS_TRANSFER_READ_BIT = 0x00000800,
    VK_ACCESS_TRANSFER_WRITE_BIT = 0x00001000,
    VK_ACCESS_HOST_READ_BIT = 0x00002000,
    VK_ACCESS_HOST_WRITE_BIT = 0x00004000,
    VK_ACCESS_MEMORY_READ_BIT = 0x00008000,
    VK_ACCESS_MEMORY_WRITE_BIT = 0x00010000,
} VkAccessFlagBits;
```

- **VK_ACCESS_MEMORY_READ_BIT** specifies all read accesses. It is always valid in any access mask, and is treated as equivalent to setting all **READ** access flags that are valid where it is used.
- **VK_ACCESS_MEMORY_WRITE_BIT** specifies all write accesses. It is always valid in any access mask, and is treated as equivalent to setting all **WRITE** access flags that are valid where it is used.
- **VK_ACCESS_INDIRECT_COMMAND_READ_BIT** specifies read access to indirect command data read as part of an indirect drawing or dispatching command. Such access occurs in the **VK_PIPELINE_STAGE_DRAW_INDIRECT_BIT** pipeline stage.
- **VK_ACCESS_INDEX_READ_BIT** specifies read access to an index buffer as part of an indexed drawing command, bound by **vkCmdBindIndexBuffer**. Such access occurs in the **VK_PIPELINE_STAGE_VERTEX_INPUT_BIT** pipeline stage.
- **VK_ACCESS_VERTEX_ATTRIBUTE_READ_BIT** specifies read access to a vertex buffer as part of a drawing command, bound by **vkCmdBindVertexBuffers**. Such access occurs in the **VK_PIPELINE_STAGE_VERTEX_INPUT_BIT** pipeline stage.
- **VK_ACCESS_UNIFORM_READ_BIT** specifies read access to a **uniform buffer** in any shader pipeline stage.
- **VK_ACCESS_INPUT_ATTACHMENT_READ_BIT** specifies read access to an **input attachment** within a render pass during fragment shading. Such access occurs in the **VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT** pipeline stage.
- **VK_ACCESS_SHADER_READ_BIT** specifies read access to a **uniform texel buffer**, **sampled image**, **storage buffer**, **storage texel buffer**, or **storage image** in any shader pipeline stage.
- **VK_ACCESS_SHADER_WRITE_BIT** specifies write access to a **storage buffer**, **storage texel buffer**, or **storage image** in any shader pipeline stage.
- **VK_ACCESS_COLOR_ATTACHMENT_READ_BIT** specifies read access to a **color attachment**, such as via

blending, logic operations or certain render pass load operations. Such access occurs in the `VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT` pipeline stage.

- `VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT` specifies write access to a color or resolve attachment during a render pass or via certain render pass load and store operations. Such access occurs in the `VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT` pipeline stage.
- `VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_READ_BIT` specifies read access to a depth/stencil attachment, via depth or stencil operations or certain render pass load operations. Such access occurs in the `VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT` or `VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT` pipeline stages.
- `VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT` specifies write access to a depth/stencil attachment, via depth or stencil operations or certain render pass load and store operations. Such access occurs in the `VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT` or `VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT` pipeline stages.
- `VK_ACCESS_TRANSFER_READ_BIT` specifies read access to an image or buffer in a copy operation.
- `VK_ACCESS_TRANSFER_WRITE_BIT` specifies write access to an image or buffer in a clear or copy operation.
- `VK_ACCESS_HOST_READ_BIT` specifies read access by a host operation. Accesses of this type are not performed through a resource, but directly on memory. Such access occurs in the `VK_PIPELINE_STAGE_HOST_BIT` pipeline stage.
- `VK_ACCESS_HOST_WRITE_BIT` specifies write access by a host operation. Accesses of this type are not performed through a resource, but directly on memory. Such access occurs in the `VK_PIPELINE_STAGE_HOST_BIT` pipeline stage.

Certain access types are only performed by a subset of pipeline stages. Any synchronization command that takes both stage masks and access masks uses both to define the access scopes - only the specified access types performed by the specified stages are included in the access scope. An application **must** not specify an access flag in a synchronization command if it does not include a pipeline stage in the corresponding stage mask that is able to perform accesses of that type. The following table lists, for each access flag, which pipeline stages **can** perform that type of access.

Table 4. Supported Access Types

Access flag	Supported pipeline stages
<code>VK_ACCESS_2_NONE</code>	Any
<code>VK_ACCESS_2_INDIRECT_COMMAND_READ_BIT</code>	<code>VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT</code> ,
<code>VK_ACCESS_2_INDEX_READ_BIT</code>	<code>VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT</code> , <code>VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT</code>
<code>VK_ACCESS_2_VERTEX_ATTRIBUTE_READ_BIT</code>	<code>VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT</code> , <code>VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT</code>

Access flag	Supported pipeline stages
VK_ACCESS_2_UNIFORM_READ_BIT	VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT, VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT, VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT,
VK_ACCESS_2_INPUT_ATTACHMENT_READ_BIT	VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT,
VK_ACCESS_2_SHADER_READ_BIT	VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT, VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT, VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT,
VK_ACCESS_2_SHADER_WRITE_BIT	VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT, VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT, VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT,
VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT	VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT
VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT	VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT
VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT	VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT
VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT	VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT
VK_ACCESS_2_TRANSFER_READ_BIT	VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT, VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_RESOLVE_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT,
VK_ACCESS_2_TRANSFER_WRITE_BIT	VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT, VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_RESOLVE_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT, VK_PIPELINE_STAGE_2_CLEAR_BIT,

Access flag	Supported pipeline stages
VK_ACCESS_2_HOST_READ_BIT	VK_PIPELINE_STAGE_2_HOST_BIT
VK_ACCESS_2_HOST_WRITE_BIT	VK_PIPELINE_STAGE_2_HOST_BIT
VK_ACCESS_2_MEMORY_READ_BIT	Any
VK_ACCESS_2_MEMORY_WRITE_BIT	Any
VK_ACCESS_2_SHADER_SAMPLED_READ_BIT	VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT, VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT, VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT,
VK_ACCESS_2_SHADER_STORAGE_READ_BIT	VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT, VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT, VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT,
VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT	VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT, VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT, VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT,
VK_ACCESS_2_COMMAND_PREPROCESS_READ_BIT_EXT	VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_EXT
VK_ACCESS_2_COMMAND_PREPROCESS_WRITE_BIT_EXT	VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_EXT

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkAccessFlags;
```

VkAccessFlags is a bitmask type for setting a mask of zero or more [VkAccessFlagBits](#).

If a memory object does not have the **VK_MEMORY_PROPERTY_HOST_COHERENT_BIT** property, then [vkFlushMappedMemoryRanges](#) **must** be called in order to guarantee that writes to the memory object from the host are made available to the host domain, where they **can** be further made available to the device domain via a domain operation. Similarly, [vkInvalidateMappedMemoryRanges](#) **must** be called to guarantee that writes which are available to the host domain are made visible to host operations.

If the memory object does have the **VK_MEMORY_PROPERTY_HOST_COHERENT_BIT** property flag, writes to

the memory object from the host are automatically made available to the host domain. Similarly, writes made available to the host domain are automatically made visible to the host.

NOTE

Queue submission commands automatically perform a domain operation from host to device for all writes performed before the command executes, so in most cases an explicit memory barrier is not needed for this case. In the few circumstances where a submit does not occur between the host write and the device read access, writes **can** be made available by using an explicit memory barrier.

7.1.4. Framebuffer Region Dependencies

Pipeline stages that operate on, or with respect to, the framebuffer are collectively the *framebuffer-space* pipeline stages. These stages are:

- `VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT`
- `VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT`
- `VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT`
- `VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT`

For these pipeline stages, an execution or memory dependency from the first set of operations to the second set **can** either be a single *framebuffer-global* dependency, or split into multiple *framebuffer-local* dependencies. A dependency with non-framebuffer-space pipeline stages is neither framebuffer-global nor framebuffer-local.

A *framebuffer region* is a set of sample (x, y, layer, sample) coordinates that is a subset of the entire framebuffer.

Both *synchronization scopes* of a framebuffer-local dependency include only the operations performed within corresponding framebuffer regions (as defined below). No ordering guarantees are made between different framebuffer regions for a framebuffer-local dependency.

Both *synchronization scopes* of a framebuffer-global dependency include operations on all framebuffer-regions.

If the first synchronization scope includes operations on pixels/fragments with N samples and the second synchronization scope includes operations on pixels/fragments with M samples, where N does not equal M, then a framebuffer region containing all samples at a given (x, y, layer) coordinate in the first synchronization scope corresponds to a region containing all samples at the same coordinate in the second synchronization scope. In other words, it is a pixel granularity dependency. If N equals M, then a framebuffer region containing a single (x, y, layer, sample) coordinate in the first synchronization scope corresponds to a region containing the same sample at the same coordinate in the second synchronization scope. In other words, it is a sample granularity dependency.

NOTE

Since fragment shader invocations are not specified to run in any particular groupings, the size of a framebuffer region is implementation-dependent, not known to the application, and **must** be assumed to be no larger than specified

above.

NOTE

Practically, the pixel vs. sample granularity dependency means that if an input attachment has a different number of samples than the pipeline's `rasterizationSamples`, then a fragment **can** access any sample in the input attachment's pixel even if it only uses framebuffer-local dependencies. If the input attachment has the same number of samples, then the fragment **can** only access the covered samples in its input `SampleMask` (i.e. the fragment operations happen-after a framebuffer-local dependency for each sample the fragment covers). To access samples that are not covered, a framebuffer-global dependency is required.

If a synchronization command includes a `dependencyFlags` parameter, and specifies the `VK_DEPENDENCY_BY_REGION_BIT` flag, then it defines framebuffer-local dependencies for the framebuffer-space pipeline stages in that synchronization command, for all framebuffer regions. If no `dependencyFlags` parameter is included, or the `VK_DEPENDENCY_BY_REGION_BIT` flag is not specified, then a framebuffer-global dependency is specified for those stages. The `VK_DEPENDENCY_BY_REGION_BIT` flag does not affect the dependencies between non-framebuffer-space pipeline stages, nor does it affect the dependencies between framebuffer-space and non-framebuffer-space pipeline stages.

NOTE

Framebuffer-local dependencies are more efficient for most architectures; particularly tile-based architectures - which can keep framebuffer-regions entirely in on-chip registers and thus avoid external bandwidth across such a dependency. Including a framebuffer-global dependency in your rendering will usually force all implementations to flush data to memory, or to a higher level cache, breaking any potential locality optimizations.

7.2. Implicit Synchronization Guarantees

A small number of implicit ordering guarantees are provided by Vulkan, ensuring that the order in which commands are submitted is meaningful, and avoiding unnecessary complexity in common operations.

Submission order is a fundamental ordering in Vulkan, giving meaning to the order in which [action and synchronization commands](#) are recorded and submitted to a single queue. Explicit and implicit ordering guarantees between commands in Vulkan all work on the premise that this ordering is meaningful. This order does not itself define any execution or memory dependencies; synchronization commands and other orderings within the API use this ordering to define their scopes.

Submission order for any given set of commands is based on the order in which they were recorded to command buffers and then submitted. This order is determined as follows:

1. The initial order is determined by the order in which `vkQueueSubmit` commands are executed on the host, for a single queue, from first to last.
2. The order in which `VkSubmitInfo` structures are specified in the `pSubmits` parameter of `vkQueueSubmit`, from lowest index to highest.

3. The order in which command buffers are specified in the `pCommandBuffers` member of `VkSubmitInfo` from lowest index to highest.
4. The order in which commands outside of a render pass were recorded to a command buffer on the host, from first to last.
5. The order in which commands inside a single subpass were recorded to a command buffer on the host, from first to last.

NOTE

When using a `render pass object` with multiple subpasses, commands in different subpasses have no defined submission order relative to each other, regardless of the order in which the subpasses were recorded. Commands within a subpass are still ordered relative to other commands in the same subpass, and those outside of the render pass.

`State commands` do not execute any operations on the device, instead they set the state of the command buffer when they execute on the host, in the order that they are recorded. `Action commands` consume the current state of the command buffer when they are recorded, and will execute state changes on the device as required to match the recorded state.

The order of primitives passing through the graphics pipeline and image layout transitions as part of an image memory barrier provide additional guarantees based on submission order.

Execution of `pipeline stages` within a given command also has a loose ordering, dependent only on a single command.

Signal operation order is a fundamental ordering in Vulkan, giving meaning to the order in which semaphore and fence signal operations occur when submitted to a single queue. The signal operation order for queue operations is determined as follows:

1. The initial order is determined by the order in which `vkQueueSubmit` commands are executed on the host, for a single queue, from first to last.
2. The order in which `VkSubmitInfo` structures are specified in the `pSubmits` parameter of `vkQueueSubmit`, from lowest index to highest.
3. The fence signal operation defined by the `fence` parameter of a `vkQueueSubmit` or `vkQueueBindSparse` command is ordered after all semaphore signal operations defined by that command.

Semaphore signal operations defined by a single `VkSubmitInfo` or `VkBindSparseInfo` structure are unordered with respect to other semaphore signal operations defined within the same structure.

7.3. Fences

Fences are a synchronization primitive that **can** be used to insert a dependency from a queue to the host. Fences have two states - signaled and unsignaled. A fence **can** be signaled as part of the execution of a `queue submission` command. Fences **can** be unsignaled on the host with `vkResetFences`. Fences **can** be waited on by the host with the `vkWaitForFences` command, and the current state **can** be queried with `vkGetFenceStatus`.

Fences are represented by **VkFence** handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkFence)
```

To create a fence, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateFence(
    VkDevice                                device,
    const VkFenceCreateInfo*                pCreateInfo,
    const VkAllocationCallbacks*            pAllocator,
    VkFence*                                pFence);
```

- **device** is the logical device that creates the fence.
- **pCreateInfo** is a pointer to a **VkFenceCreateInfo** structure containing information about how the fence is to be created.
- **pAllocator** controls host memory allocation as described in the **Memory Allocation** chapter.
- **pFence** is a pointer to a handle in which the resulting fence object is returned.

Valid Usage (Implicit)

- VUID-vkCreateFence-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkCreateFence-pCreateInfo-parameter
pCreateInfo **must** be a valid pointer to a valid **VkFenceCreateInfo** structure
- VUID-vkCreateFence-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid **VkAllocationCallbacks** structure
- VUID-vkCreateFence-pFence-parameter
pFence **must** be a valid pointer to a **VkFence** handle

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_OUT_OF_DEVICE_MEMORY**

The **VkFenceCreateInfo** structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkFenceCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkFenceCreateFlags flags;
} VkFenceCreateInfo;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **flags** is a bitmask of [VkFenceCreateFlagBits](#) specifying the initial state and behavior of the fence.

Valid Usage (Implicit)

- VUID-VkFenceCreateInfo-sType-sType
sType must be **VK_STRUCTURE_TYPE_FENCE_CREATE_INFO**
- VUID-VkFenceCreateInfo-pNext-pNext
pNext must be **NULL**
- VUID-VkFenceCreateInfo-flags-parameter
flags must be a valid combination of [VkFenceCreateFlagBits](#) values

```
// Provided by VK_VERSION_1_0
typedef enum VkFenceCreateFlagBits {
    VK_FENCE_CREATE_SIGNALED_BIT = 0x00000001,
} VkFenceCreateFlagBits;
```

- **VK_FENCE_CREATE_SIGNALED_BIT** specifies that the fence object is created in the signaled state. Otherwise, it is created in the unsignaled state.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkFenceCreateFlags;
```

VkFenceCreateFlags is a bitmask type for setting a mask of zero or more [VkFenceCreateFlagBits](#).

To destroy a fence, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyFence(
    VkDevice          device,
    VkFence           fence,
    const VkAllocationCallbacks* pAllocator);
```

- **device** is the logical device that destroys the fence.
- **fence** is the handle of the fence to destroy.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyFence-fence-01120
All [queue submission](#) commands that refer to **fence** **must** have completed execution
- VUID-vkDestroyFence-fence-01121
If [VkAllocationCallbacks](#) were provided when **fence** was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyFence-fence-01122
If no [VkAllocationCallbacks](#) were provided when **fence** was created, **pAllocator** **must** be **NULL**

Valid Usage (Implicit)

- VUID-vkDestroyFence-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkDestroyFence-fence-parameter
If **fence** is not [VK_NULL_HANDLE](#), **fence** **must** be a valid [VkFence](#) handle
- VUID-vkDestroyFence-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkDestroyFence-fence-parent
If **fence** is a valid handle, it **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **fence** **must** be externally synchronized

To query the status of a fence from the host, call:

```
// Provided by VK_VERSION_1_0
VkResult vkGetFenceStatus(
    VkDevice      device,
    VkFence       fence);
```

- **device** is the logical device that owns the fence.
- **fence** is the handle of the fence to query.

Upon success, `vkGetFenceStatus` returns the status of the fence object, with the following return codes:

Table 5. Fence Object Status Codes

Status	Meaning
<code>VK_SUCCESS</code>	The fence specified by <code>fence</code> is signaled.
<code>VK_NOT_READY</code>	The fence specified by <code>fence</code> is unsignaled.
<code>VK_ERROR_DEVICE_LOST</code>	The device has been lost. See Lost Device .

If a [queue submission](#) command is pending execution, then the value returned by this command **may** immediately be out of date.

If the device has been lost (see [Lost Device](#)), `vkGetFenceStatus` **may** return any of the above status codes. If the device has been lost and `vkGetFenceStatus` is called repeatedly, it will eventually return either `VK_SUCCESS` or `VK_ERROR_DEVICE_LOST`.

Valid Usage (Implicit)

- VUID-vkGetFenceStatus-device-parameter
`device` **must** be a valid [VkDevice](#) handle
- VUID-vkGetFenceStatus-fence-parameter
`fence` **must** be a valid [VkFence](#) handle
- VUID-vkGetFenceStatus-fence-parent
`fence` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_SUCCESS`
- `VK_NOT_READY`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_DEVICE_LOST`

To set the state of fences to unsignaled from the host, call:

```
// Provided by VK_VERSION_1_0
VkResult vkResetFences(
    VkDevice          device,
    uint32_t          fenceCount,
    const VkFence*    pFences);
```

- **device** is the logical device that owns the fences.
- **fenceCount** is the number of fences to reset.
- **pFences** is a pointer to an array of fence handles to reset.

When **vkResetFences** is executed on the host, it defines a *fence unsignal operation* for each fence, which resets the fence to the unsignaled state.

If any member of **pFences** is already in the unsignaled state when **vkResetFences** is executed, then **vkResetFences** has no effect on that fence.

Valid Usage

- VUID-vkResetFences-pFences-01123
Each element of **pFences** **must** not be currently associated with any queue command that has not yet completed execution on that queue

Valid Usage (Implicit)

- VUID-vkResetFences-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkResetFences-pFences-parameter
pFences **must** be a valid pointer to an array of **fenceCount** valid **VkFence** handles
- VUID-vkResetFences-fenceCount-arraylength
fenceCount **must** be greater than 0
- VUID-vkResetFences-pFences-parent
Each element of **pFences** **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to each member of **pFences** **must** be externally synchronized

Return Codes

Success

- **VK_SUCCESS**

Failure

- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

When a fence is submitted to a queue as part of a [queue submission](#) command, it defines a memory dependency on the batches that were submitted as part of that command, and defines a *fence signal operation* which sets the fence to the signaled state.

The first [synchronization scope](#) includes every batch submitted in the same [queue submission](#) command. Fence signal operations that are defined by `vkQueueSubmit` additionally include in the first synchronization scope all commands that occur earlier in [submission order](#). Fence signal operations that are defined by `vkQueueSubmit` or `vkQueueBindSparse` additionally include in the first synchronization scope any semaphore and fence signal operations that occur earlier in [signal operation order](#).

The second [synchronization scope](#) only includes the fence signal operation.

The first [access scope](#) includes all memory access performed by the device.

The second [access scope](#) is empty.

To wait for one or more fences to enter the signaled state on the host, call:

```
// Provided by VK_VERSION_1_0
VkResult vkWaitForFences(
    VkDevice          device,
    uint32_t          fenceCount,
    const VkFence*     pFences,
    VkBool32           waitAll,
    uint64_t           timeout);
```

- `device` is the logical device that owns the fences.
- `fenceCount` is the number of fences to wait on.
- `pFences` is a pointer to an array of `fenceCount` fence handles.
- `waitAll` is the condition that **must** be satisfied to successfully unblock the wait. If `waitAll` is `VK_TRUE`, then the condition is that all fences in `pFences` are signaled. Otherwise, the condition is that at least one fence in `pFences` is signaled.
- `timeout` is the timeout period in units of nanoseconds. `timeout` is adjusted to the closest value allowed by the implementation-dependent timeout accuracy, which **may** be substantially longer than one nanosecond, and **may** be longer than the requested period.

If the condition is satisfied when `vkWaitForFences` is called, then `vkWaitForFences` returns immediately. If the condition is not satisfied at the time `vkWaitForFences` is called, then `vkWaitForFences` will block and wait until the condition is satisfied or the `timeout` has expired, whichever is sooner.

If `timeout` is zero, then `vkWaitForFences` does not wait, but simply returns the current state of the fences. `VK_TIMEOUT` will be returned in this case if the condition is not satisfied, even though no

actual wait was performed.

If the condition is satisfied before the `timeout` has expired, `vkWaitForFences` returns `VK_SUCCESS`. Otherwise, `vkWaitForFences` returns `VK_TIMEOUT` after the `timeout` has expired.

If device loss occurs (see [Lost Device](#)) before the timeout has expired, `vkWaitForFences` **must** return in finite time with either `VK_SUCCESS` or `VK_ERROR_DEVICE_LOST`.

NOTE

While we guarantee that `vkWaitForFences` **must** return in finite time, no guarantees are made that it returns immediately upon device loss. However, the application can reasonably expect that the delay will be on the order of seconds and that calling `vkWaitForFences` will not result in a permanently (or seemingly permanently) dead process.

Valid Usage (Implicit)

- VUID-vkWaitForFences-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkWaitForFences-pFences-parameter
`pFences` **must** be a valid pointer to an array of `fenceCount` valid `VkFence` handles
- VUID-vkWaitForFences-fenceCount-arraylength
`fenceCount` **must** be greater than 0
- VUID-vkWaitForFences-pFences-parent
Each element of `pFences` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_SUCCESS`
- `VK_TIMEOUT`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_DEVICE_LOST`

An execution dependency is defined by waiting for a fence to become signaled, either via `vkWaitForFences` or by polling on `vkGetFenceStatus`.

The first [synchronization scope](#) includes only the fence signal operation.

The second [synchronization scope](#) includes the host operations of `vkWaitForFences` or `vkGetFenceStatus` indicating that the fence has become signaled.

NOTE

Signaling a fence and waiting on the host does not guarantee that the results of memory accesses will be visible to the host, as the access scope of a memory dependency defined by a fence only includes device access. A [memory barrier](#) or other memory dependency **must** be used to guarantee this. See the description of [host access types](#) for more information.

7.4. Semaphores

Semaphores are a synchronization primitive that **can** be used to insert a dependency between queue operations. Semaphores have two states - signaled and unsignaled. A semaphore **can** be signaled after execution of a queue operation is completed, and a queue operation **can** wait for a semaphore to become signaled before it begins execution.

Semaphores are represented by `VkSemaphore` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkSemaphore)
```

To create a semaphore, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateSemaphore(
    VkDevice                device,
    const VkSemaphoreCreateInfo* pCreateInfo,
    const VkAllocationCallbacks* pAllocator,
    VkSemaphore*             pSemaphore);
```

- `device` is the logical device that creates the semaphore.
- `pCreateInfo` is a pointer to a [VkSemaphoreCreateInfo](#) structure containing information about how the semaphore is to be created.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.
- `pSemaphore` is a pointer to a handle in which the resulting semaphore object is returned.

This command creates a *binary semaphore* that has a boolean payload indicating whether the semaphore is currently signaled or unsignaled. When created, the semaphore is in the unsignaled state.

Valid Usage (Implicit)

- VUID-vkCreateSemaphore-device-parameter
`device` **must** be a valid [VkDevice](#) handle
- VUID-vkCreateSemaphore-pCreateInfo-parameter
`pCreateInfo` **must** be a valid pointer to a valid [VkSemaphoreCreateInfo](#) structure
- VUID-vkCreateSemaphore-pAllocator-parameter

If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure

- `VUID-vkCreateSemaphore-pSemaphore-parameter`
`pSemaphore` **must** be a valid pointer to a `VkSemaphore` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkSemaphoreCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSemaphoreCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkSemaphoreCreateFlags flags;
} VkSemaphoreCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.

Valid Usage (Implicit)

- `VUID-VkSemaphoreCreateInfo-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_SEMAPHORE_CREATE_INFO`
- `VUID-VkSemaphoreCreateInfo-pNext-pNext`
`pNext` **must** be `NULL`
- `VUID-VkSemaphoreCreateInfo-flags-zero bitmask`
`flags` **must** be 0

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkSemaphoreCreateFlags;
```

`VkSemaphoreCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

To destroy a semaphore, call:

```
// Provided by VK_VERSION_1_0
void vkDestroySemaphore(
    VkDevice          device,
    VkSemaphore       semaphore,
    const VkAllocationCallbacks* pAllocator);
```

- **device** is the logical device that destroys the semaphore.
- **semaphore** is the handle of the semaphore to destroy.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroySemaphore-semaphore-05149
All submitted batches that refer to **semaphore** **must** have completed execution
- VUID-vkDestroySemaphore-semaphore-01138
If **VkAllocationCallbacks** were provided when **semaphore** was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroySemaphore-semaphore-01139
If no **VkAllocationCallbacks** were provided when **semaphore** was created, **pAllocator** **must** be **NULL**

Valid Usage (Implicit)

- VUID-vkDestroySemaphore-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkDestroySemaphore-semaphore-parameter
If **semaphore** is not **VK_NULL_HANDLE**, **semaphore** **must** be a valid **VkSemaphore** handle
- VUID-vkDestroySemaphore-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid **VkAllocationCallbacks** structure
- VUID-vkDestroySemaphore-semaphore-parent
If **semaphore** is a valid handle, it **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **semaphore** **must** be externally synchronized

7.4.1. Semaphore Signaling

When a batch is submitted to a queue via a [queue submission](#), and it includes semaphores to be

signaled, it defines a memory dependency on the batch, and defines *semaphore signal operations* which set the semaphores to the signaled state.

The first [synchronization scope](#) includes every command submitted in the same batch. Semaphore signal operations that are defined by [vkQueueSubmit](#) additionally include all commands that occur earlier in [submission order](#). Semaphore signal operations that are defined by [vkQueueSubmit](#) or [vkQueueBindSparse](#) additionally include in the first synchronization scope any semaphore and fence signal operations that occur earlier in [signal operation order](#).

The second [synchronization scope](#) includes only the semaphore signal operation.

The first [access scope](#) includes all memory access performed by the device.

The second [access scope](#) is empty.

7.4.2. Semaphore Waiting

When a batch is submitted to a queue via a [queue submission](#), and it includes semaphores to be waited on, it defines a memory dependency between prior semaphore signal operations and the batch, and defines *semaphore wait operations*.

Such semaphore wait operations set the semaphores to the unsignaled state.

The first [synchronization scope](#) includes one [semaphore signal operation](#) for each semaphore waited on by this batch. The specific signal operation waited on for each semaphore **must** meet the following criteria:

- the signal operation is either earlier in [submission order](#) on the same queue, or is submitted by a command whose host operation happens-before this batch is submitted on the host
- no wait operation exists that happens-after the signal operation and happens-before this wait operation
- the signal operation is not guaranteed to happen-after the semaphore wait operation in this batch

If multiple semaphore signal operations meet these criteria, any of those operations **may** be included in the first [synchronization scope](#). When waiting on a binary semaphore, applications **must** ensure that exactly one semaphore signal operation meets these criteria.

The second [synchronization scope](#) includes every command submitted in the same batch. In the case of [vkQueueSubmit](#), the second synchronization scope is limited to operations on the pipeline stages determined by the [destination stage mask](#) specified by the corresponding element of [pWaitDstStageMask](#). Also, in the case of [vkQueueSubmit](#), the second synchronization scope additionally includes all commands that occur later in [submission order](#).

The first [access scope](#) is empty.

The second [access scope](#) includes all memory access performed by the device.

The semaphore wait operation happens-after the first set of operations in the execution dependency, and happens-before the second set of operations in the execution dependency.

NOTE

Unlike fences or events, waiting for a binary semaphore also unsignals that semaphore when the wait completes. Applications **must** ensure that between two such wait operations, the semaphore is signaled again, with execution dependencies used to ensure these occur in order. Binary semaphore waits and signals should thus occur in discrete 1:1 pairs.

7.4.3. Semaphore State Requirements for Wait Operations

Before waiting on a semaphore, the application **must** ensure the semaphore is in a valid state for a wait operation. Specifically, when a [semaphore wait operation](#) is submitted to a queue:

- A binary semaphore **must** be signaled, or have an associated [semaphore signal operation](#) that is pending execution.
- Any [semaphore signal operations](#) on which the pending binary semaphore signal operation depends **must** also be completed or pending execution.
- There **must** be no other queue waiting on the same binary semaphore when the operation executes.

7.5. Events

Events are a synchronization primitive that **can** be used to insert a fine-grained dependency between commands submitted to the same queue, or between the host and a queue. Events **must** not be used to insert a dependency between commands submitted to different queues. Events have two states - signaled and unsignaled. An application **can** signal or unsignal an event either on the host or on the device. A device **can** be made to wait for an event to become signaled before executing further operations. No command exists to wait for an event to become signaled on the host, but the current state of an event **can** be queried.

Events are represented by [VkEvent](#) handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkEvent)
```

To create an event, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateEvent(
    VkDevice                                device,
    const VkEventCreateInfo*                pCreateInfo,
    const VkAllocationCallbacks*            pAllocator,
    VkEvent*                                 pEvent);
```

- [device](#) is the logical device that creates the event.
- [pCreateInfo](#) is a pointer to a [VkEventCreateInfo](#) structure containing information about how the event is to be created.

- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.
- `pEvent` is a pointer to a handle in which the resulting event object is returned.

When created, the event object is in the unsigned state.

Valid Usage

- VUID-vkCreateEvent-device-09672
`device` **must** support at least one queue family with one of the `VK_QUEUE_COMPUTE_BIT`, or `VK_QUEUE_GRAPHICS_BIT` capabilities

Valid Usage (Implicit)

- VUID-vkCreateEvent-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkCreateEvent-pCreateInfo-parameter
`pCreateInfo` **must** be a valid pointer to a valid `VkEventCreateInfo` structure
- VUID-vkCreateEvent-pAllocator-parameter
 If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkCreateEvent-pEvent-parameter
`pEvent` **must** be a valid pointer to a `VkEvent` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkEventCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkEventCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkEventCreateFlags flags;
} VkEventCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkEventCreateFlagBits` defining additional creation parameters.

Valid Usage (Implicit)

- VUID-VkEventCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_EVENT_CREATE_INFO`
- VUID-VkEventCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkEventCreateInfo-flags-zero-bitmask
`flags` **must** be `0`

```
// Provided by VK_VERSION_1_0
typedef enum VkEventCreateFlagBits {
} VkEventCreateFlagBits;
```

NOTE

All bits for this type are defined by extensions, and none of those extensions are enabled in this build of the specification.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkEventCreateFlags;
```

`VkEventCreateFlags` is a bitmask type for setting a mask of `VkEventCreateFlagBits`.

To destroy an event, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyEvent(
    VkDevice          device,
    VkEvent           event,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that destroys the event.
- `event` is the handle of the event to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyEvent-event-01145
All submitted commands that refer to `event` **must** have completed execution
- VUID-vkDestroyEvent-event-01146
If `VkAllocationCallbacks` were provided when `event` was created, a compatible set of

callbacks **must** be provided here

- VUID-vkDestroyEvent-event-01147

If no **VkAllocationCallbacks** were provided when **event** was created, **pAllocator** **must** be **NULL**

Valid Usage (Implicit)

- VUID-vkDestroyEvent-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkDestroyEvent-event-parameter
If **event** is not **VK_NULL_HANDLE**, **event** **must** be a valid **VkEvent** handle
- VUID-vkDestroyEvent-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid **VkAllocationCallbacks** structure
- VUID-vkDestroyEvent-event-parent
If **event** is a valid handle, it **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **event** **must** be externally synchronized

To query the state of an event from the host, call:

```
// Provided by VK_VERSION_1_0
VkResult vkGetEventStatus(
    VkDevice          device,
    VkEvent           event);
```

- **device** is the logical device that owns the event.
- **event** is the handle of the event to query.

Upon success, **vkGetEventStatus** returns the state of the event object with the following return codes:

Table 6. Event Object Status Codes

Status	Meaning
VK_EVENT_SET	The event specified by event is signaled.
VK_EVENT_RESET	The event specified by event is unsignaled.

If a **vkCmdSetEvent** or **vkCmdResetEvent** command is in a command buffer that is in the **pending state**, then the value returned by this command **may** immediately be out of date.

The state of an event **can** be updated by the host. The state of the event is immediately changed, and subsequent calls to `vkGetEventStatus` will return the new state. If an event is already in the requested state, then updating it to the same state has no effect.

Valid Usage (Implicit)

- VUID-vkGetEventStatus-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetEventStatus-event-parameter
`event` **must** be a valid `VkEvent` handle
- VUID-vkGetEventStatus-event-parent
`event` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_EVENT_SET`
- `VK_EVENT_RESET`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_DEVICE_LOST`

To set the state of an event to signaled from the host, call:

```
// Provided by VK_VERSION_1_0
VkResult vkSetEvent(
    VkDevice          device,
    VkEvent           event);
```

- `device` is the logical device that owns the event.
- `event` is the event to set.

When `vkSetEvent` is executed on the host, it defines an *event signal operation* which sets the event to the signaled state.

If `event` is already in the signaled state when `vkSetEvent` is executed, then `vkSetEvent` has no effect, and no event signal operation occurs.

NOTE

If a command buffer is waiting for an event to be signaled from the host, the application must signal the event before submitting the command buffer, as described in the [queue forward progress](#) section.

Valid Usage (Implicit)

- VUID-vkSetEvent-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkSetEvent-event-parameter
event **must** be a valid [VkEvent](#) handle
- VUID-vkSetEvent-event-parent
event **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **event** **must** be externally synchronized

Return Codes

Success

- [VK_SUCCESS](#)

Failure

- [VK_ERROR_OUT_OF_HOST_MEMORY](#)
- [VK_ERROR_OUT_OF_DEVICE_MEMORY](#)

To set the state of an event to unsignaled from the host, call:

```
// Provided by VK_VERSION_1_0
VkResult vkResetEvent(
    VkDevice          device,
    VkEvent           event);
```

- **device** is the logical device that owns the event.
- **event** is the event to reset.

When [vkResetEvent](#) is executed on the host, it defines an *event unsignal operation* which resets the event to the unsignaled state.

If **event** is already in the unsignaled state when [vkResetEvent](#) is executed, then [vkResetEvent](#) has no effect, and no event unsignal operation occurs.

Valid Usage

- VUID-vkResetEvent-event-03821
There **must** be an execution dependency between [vkResetEvent](#) and the execution of any [vkCmdWaitEvents](#) that includes **event** in its [pEvents](#) parameter

Valid Usage (Implicit)

- VUID-vkResetEvent-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkResetEvent-event-parameter
event **must** be a valid [VkEvent](#) handle
- VUID-vkResetEvent-event-parent
event **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **event** **must** be externally synchronized

Return Codes

Success

- [VK_SUCCESS](#)

Failure

- [VK_ERROR_OUT_OF_DEVICE_MEMORY](#)

The state of an event **can** also be updated on the device by commands inserted in command buffers.

To set the state of an event to signaled from a device, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetEvent(
    VkCommandBuffer          commandBuffer,
    VkEvent                  event,
    VkPipelineStageFlags     stageMask);
```

- **commandBuffer** is the command buffer into which the command is recorded.
- **event** is the event that will be signaled.
- **stageMask** specifies the [source stage mask](#) used to determine the first [synchronization scope](#).

When [vkCmdSetEvent](#) is submitted to a queue, it defines an execution dependency on commands that were submitted before it, and defines an event signal operation which sets the event to the signaled state.

The first [synchronization scope](#) includes all commands that occur earlier in [submission order](#). The synchronization scope is limited to operations on the pipeline stages determined by the [source stage mask](#) specified by **stageMask**.

The second [synchronization scope](#) includes only the event signal operation.

If **event** is already in the signaled state when [vkCmdSetEvent](#) is executed on the device, then [vkCmdSetEvent](#) has no effect, no event signal operation occurs, and no execution dependency is generated.

Valid Usage

- VUID-vkCmdSetEvent-stageMask-04090
If the **geometryShader** feature is not enabled, **stageMask** **must** not contain **VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT**
- VUID-vkCmdSetEvent-stageMask-04091
If the **tessellationShader** feature is not enabled, **stageMask** **must** not contain **VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT** or **VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT**
- VUID-vkCmdSetEvent-stageMask-04996
stageMask **must** not be 0
- VUID-vkCmdSetEvent-stageMask-06457
Any pipeline stage included in **stageMask** **must** be supported by the capabilities of the queue family specified by the **queueFamilyIndex** member of the [VkCommandPoolCreateInfo](#) structure that was used to create the [VkCommandPool](#) that **commandBuffer** was allocated from, as specified in the [table of supported pipeline stages](#)
- VUID-vkCmdSetEvent-stageMask-01149
stageMask **must** not include **VK_PIPELINE_STAGE_HOST_BIT**

Valid Usage (Implicit)

- VUID-vkCmdSetEvent-commandBuffer-parameter
commandBuffer **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdSetEvent-event-parameter
event **must** be a valid [VkEvent](#) handle
- VUID-vkCmdSetEvent-stageMask-parameter
stageMask **must** be a valid combination of [VkPipelineStageFlagBits](#) values
- VUID-vkCmdSetEvent-commandBuffer-recording
commandBuffer **must** be in the [recording state](#)
- VUID-vkCmdSetEvent-commandBuffer-cmdpool
The [VkCommandPool](#) that **commandBuffer** was allocated from **must** support graphics, or compute operations
- VUID-vkCmdSetEvent-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdSetEvent-commonparent
Both of **commandBuffer**, and **event** **must** have been created, allocated, or retrieved from the same [VkDevice](#)

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Graphics Compute	Synchronization

To set the state of an event to unsignaled from a device, call:

```
// Provided by VK_VERSION_1_0
void vkCmdResetEvent(
    VkCommandBuffer          commandBuffer,
    VkEvent                  event,
    VkPipelineStageFlags     stageMask);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `event` is the event that will be unsignaled.
- `stageMask` is a bitmask of `VkPipelineStageFlagBits` specifying the `source stage mask` used to determine when the `event` is unsignaled.

When `vkCmdResetEvent` is submitted to a queue, it defines an execution dependency on commands that were submitted before it, and defines an event unsignal operation which resets the event to the unsignaled state.

The first `synchronization scope` includes all commands that occur earlier in `submission order`. The synchronization scope is limited to operations on the pipeline stages determined by the `source stage mask` specified by `stageMask`.

The second `synchronization scope` includes only the event unsignal operation.

If `event` is already in the unsignaled state when `vkCmdResetEvent` is executed on the device, then `vkCmdResetEvent` has no effect, no event unsignal operation occurs, and no execution dependency is generated.

Valid Usage

- VUID-vkCmdResetEvent-stageMask-04090
If the `geometryShader` feature is not enabled, `stageMask` **must** not contain

VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT

- VUID-vkCmdResetEvent-stageMask-04091

If the `tessellationShader` feature is not enabled, `stageMask` **must** not contain `VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT` or `VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT`

- VUID-vkCmdResetEvent-stageMask-04996

`stageMask` **must** not be 0

- VUID-vkCmdResetEvent-stageMask-06458

Any pipeline stage included in `stageMask` **must** be supported by the capabilities of the queue family specified by the `queueFamilyIndex` member of the `VkCommandPoolCreateInfo` structure that was used to create the `VkCommandPool` that `commandBuffer` was allocated from, as specified in the [table of supported pipeline stages](#)

- VUID-vkCmdResetEvent-stageMask-01153

`stageMask` **must** not include `VK_PIPELINE_STAGE_HOST_BIT`

- VUID-vkCmdResetEvent-event-03834

There **must** be an execution dependency between `vkCmdResetEvent` and the execution of any `vkCmdWaitEvents` that includes `event` in its `pEvents` parameter

Valid Usage (Implicit)

- VUID-vkCmdResetEvent-commandBuffer-parameter

`commandBuffer` **must** be a valid `VkCommandBuffer` handle

- VUID-vkCmdResetEvent-event-parameter

`event` **must** be a valid `VkEvent` handle

- VUID-vkCmdResetEvent-stageMask-parameter

`stageMask` **must** be a valid combination of `VkPipelineStageFlagBits` values

- VUID-vkCmdResetEvent-commandBuffer-recording

`commandBuffer` **must** be in the `recording` state

- VUID-vkCmdResetEvent-commandBuffer-cmdpool

The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, or compute operations

- VUID-vkCmdResetEvent-renderpass

This command **must** only be called outside of a render pass instance

- VUID-vkCmdResetEvent-commonparent

Both of `commandBuffer`, and `event` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized

- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally

synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Graphics Compute	Synchronization

To wait for one or more events to enter the signaled state on a device, call:

```
// Provided by VK_VERSION_1_0
void vkCmdWaitEvents(
    VkCommandBuffer                commandBuffer,
    uint32_t                       eventCount,
    const VkEvent*                 pEvents,
    VkPipelineStageFlags           srcStageMask,
    VkPipelineStageFlags           dstStageMask,
    uint32_t                       memoryBarrierCount,
    const VkMemoryBarrier*         pMemoryBarriers,
    uint32_t                       bufferMemoryBarrierCount,
    const VkBufferMemoryBarrier*   pBufferMemoryBarriers,
    uint32_t                       imageMemoryBarrierCount,
    const VkImageMemoryBarrier*    pImageMemoryBarriers);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `eventCount` is the length of the `pEvents` array.
- `pEvents` is a pointer to an array of event object handles to wait on.
- `srcStageMask` is a bitmask of `VkPipelineStageFlagBits` specifying the [source stage mask](#).
- `dstStageMask` is a bitmask of `VkPipelineStageFlagBits` specifying the [destination stage mask](#).
- `memoryBarrierCount` is the length of the `pMemoryBarriers` array.
- `pMemoryBarriers` is a pointer to an array of `VkMemoryBarrier` structures.
- `bufferMemoryBarrierCount` is the length of the `pBufferMemoryBarriers` array.
- `pBufferMemoryBarriers` is a pointer to an array of `VkBufferMemoryBarrier` structures.
- `imageMemoryBarrierCount` is the length of the `pImageMemoryBarriers` array.
- `pImageMemoryBarriers` is a pointer to an array of `VkImageMemoryBarrier` structures.

When `vkCmdWaitEvents` is submitted to a queue, it defines a memory dependency between prior event signal operations on the same queue or the host, and subsequent commands. `vkCmdWaitEvents` **must** not be used to wait on event signal operations occurring on other queues.

The first synchronization scope only includes event signal operations that operate on members of

pEvents, and the operations that happened-before the event signal operations. Event signal operations performed by **vkCmdSetEvent** that occur earlier in **submission order** are included in the first synchronization scope, if the **logically latest** pipeline stage in their **stageMask** parameter is **logically earlier** than or equal to the **logically latest** pipeline stage in **srcStageMask**. Event signal operations performed by **vkSetEvent** are only included in the first synchronization scope if **VK_PIPELINE_STAGE_HOST_BIT** is included in **srcStageMask**.

The second **synchronization scope** includes all commands that occur later in **submission order**. The second synchronization scope is limited to operations on the pipeline stages determined by the **destination stage mask** specified by **dstStageMask**.

The first **access scope** is limited to accesses in the pipeline stages determined by the **source stage mask** specified by **srcStageMask**. Within that, the first access scope only includes the first access scopes defined by elements of the **pMemoryBarriers**, **pBufferMemoryBarriers** and **pImageMemoryBarriers** arrays, which each define a set of **memory barriers**. If no memory barriers are specified, then the first access scope includes no accesses.

The second **access scope** is limited to accesses in the pipeline stages determined by the **destination stage mask** specified by **dstStageMask**. Within that, the second access scope only includes the second access scopes defined by elements of the **pMemoryBarriers**, **pBufferMemoryBarriers** and **pImageMemoryBarriers** arrays, which each define a set of **memory barriers**. If no memory barriers are specified, then the second access scope includes no accesses.

NOTE

vkCmdWaitEvents is used with **vkCmdSetEvent** to define a memory dependency between two sets of action commands, roughly in the same way as pipeline barriers, but split into two commands such that work between the two **may** execute unhindered.

Unlike **vkCmdPipelineBarrier**, a **queue family ownership transfer cannot** be performed using **vkCmdWaitEvents**.

NOTE

Applications should be careful to avoid race conditions when using events. There is no direct ordering guarantee between **vkCmdWaitEvents** and **vkCmdResetEvent**, or **vkCmdSetEvent**. Another execution dependency (e.g. a pipeline barrier or semaphore with **VK_PIPELINE_STAGE_ALL_COMMANDS_BIT**) is needed to prevent such a race condition.

Valid Usage

- VUID-vkCmdWaitEvents-srcStageMask-04090
If the **geometryShader** feature is not enabled, **srcStageMask** **must** not contain **VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT**
- VUID-vkCmdWaitEvents-srcStageMask-04091
If the **tessellationShader** feature is not enabled, **srcStageMask** **must** not contain **VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT** or **VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT**
- VUID-vkCmdWaitEvents-srcStageMask-04996

`srcStageMask` **must** not be 0

- VUID-vkCmdWaitEvents-dstStageMask-04090

If the `geometryShader` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT`

- VUID-vkCmdWaitEvents-dstStageMask-04091

If the `tessellationShader` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT` or `VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT`

- VUID-vkCmdWaitEvents-dstStageMask-04996

`dstStageMask` **must** not be 0

- VUID-vkCmdWaitEvents-srcAccessMask-02815

The `srcAccessMask` member of each element of `pMemoryBarriers` **must** only include access flags that are supported by one or more of the pipeline stages in `srcStageMask`, as specified in the [table of supported access types](#)

- VUID-vkCmdWaitEvents-dstAccessMask-02816

The `dstAccessMask` member of each element of `pMemoryBarriers` **must** only include access flags that are supported by one or more of the pipeline stages in `dstStageMask`, as specified in the [table of supported access types](#)

- VUID-vkCmdWaitEvents-pBufferMemoryBarriers-02817

For any element of `pBufferMemoryBarriers`, if its `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members are equal, or if its `srcQueueFamilyIndex` is the queue family index that was used to create the command pool that `commandBuffer` was allocated from, then its `srcAccessMask` member **must** only contain access flags that are supported by one or more of the pipeline stages in `srcStageMask`, as specified in the [table of supported access types](#)

- VUID-vkCmdWaitEvents-pBufferMemoryBarriers-02818

For any element of `pBufferMemoryBarriers`, if its `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members are equal, or if its `dstQueueFamilyIndex` is the queue family index that was used to create the command pool that `commandBuffer` was allocated from, then its `dstAccessMask` member **must** only contain access flags that are supported by one or more of the pipeline stages in `dstStageMask`, as specified in the [table of supported access types](#)

- VUID-vkCmdWaitEvents-pImageMemoryBarriers-02819

For any element of `pImageMemoryBarriers`, if its `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members are equal, or if its `srcQueueFamilyIndex` is the queue family index that was used to create the command pool that `commandBuffer` was allocated from, then its `srcAccessMask` member **must** only contain access flags that are supported by one or more of the pipeline stages in `srcStageMask`, as specified in the [table of supported access types](#)

- VUID-vkCmdWaitEvents-pImageMemoryBarriers-02820

For any element of `pImageMemoryBarriers`, if its `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members are equal, or if its `dstQueueFamilyIndex` is the queue family index that was used to create the command pool that `commandBuffer` was allocated from,

then its `dstAccessMask` member **must** only contain access flags that are supported by one or more of the pipeline stages in `dstStageMask`, as specified in the [table of supported access types](#)

- VUID-vkCmdWaitEvents-srcStageMask-06459

Any pipeline stage included in `srcStageMask` **must** be supported by the capabilities of the queue family specified by the `queueFamilyIndex` member of the `VkCommandPoolCreateInfo` structure that was used to create the `VkCommandPool` that `commandBuffer` was allocated from, as specified in the [table of supported pipeline stages](#)

- VUID-vkCmdWaitEvents-dstStageMask-06460

Any pipeline stage included in `dstStageMask` **must** be supported by the capabilities of the queue family specified by the `queueFamilyIndex` member of the `VkCommandPoolCreateInfo` structure that was used to create the `VkCommandPool` that `commandBuffer` was allocated from, as specified in the [table of supported pipeline stages](#)

- VUID-vkCmdWaitEvents-srcStageMask-01158

`srcStageMask` **must** be the bitwise OR of the `stageMask` parameter used in previous calls to `vkCmdSetEvent` with any of the elements of `pEvents` and `VK_PIPELINE_STAGE_HOST_BIT` if any of the elements of `pEvents` was set using `vkSetEvent`

- VUID-vkCmdWaitEvents-srcStageMask-07308

If `vkCmdWaitEvents` is being called inside a render pass instance, `srcStageMask` **must** not include `VK_PIPELINE_STAGE_HOST_BIT`

- VUID-vkCmdWaitEvents-srcQueueFamilyIndex-02803

The `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members of any element of `pBufferMemoryBarriers` or `pImageMemoryBarriers` **must** be equal

Valid Usage (Implicit)

- VUID-vkCmdWaitEvents-commandBuffer-parameter

`commandBuffer` **must** be a valid `VkCommandBuffer` handle

- VUID-vkCmdWaitEvents-pEvents-parameter

`pEvents` **must** be a valid pointer to an array of `eventCount` valid `VkEvent` handles

- VUID-vkCmdWaitEvents-srcStageMask-parameter

`srcStageMask` **must** be a valid combination of `VkPipelineStageFlagBits` values

- VUID-vkCmdWaitEvents-dstStageMask-parameter

`dstStageMask` **must** be a valid combination of `VkPipelineStageFlagBits` values

- VUID-vkCmdWaitEvents-pMemoryBarriers-parameter

If `memoryBarrierCount` is not 0, `pMemoryBarriers` **must** be a valid pointer to an array of `memoryBarrierCount` valid `VkMemoryBarrier` structures

- VUID-vkCmdWaitEvents-pBufferMemoryBarriers-parameter

If `bufferMemoryBarrierCount` is not 0, `pBufferMemoryBarriers` **must** be a valid pointer to an array of `bufferMemoryBarrierCount` valid `VkBufferMemoryBarrier` structures

- VUID-vkCmdWaitEvents-pImageMemoryBarriers-parameter

If `imageMemoryBarrierCount` is not 0, `pImageMemoryBarriers` **must** be a valid pointer to an

array of `imageMemoryBarrierCount` valid `VkImageMemoryBarrier` structures

- VUID-vkCmdWaitEvents-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdWaitEvents-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, or compute operations
- VUID-vkCmdWaitEvents-eventCount-arraylength
`eventCount` **must** be greater than 0
- VUID-vkCmdWaitEvents-commonparent
Both of `commandBuffer`, and the elements of `pEvents` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Graphics Compute	Synchronization

7.6. Pipeline Barriers

To record a pipeline barrier, call:

```
// Provided by VK_VERSION_1_0
void vkCmdPipelineBarrier(
    VkCommandBuffer          commandBuffer,
    VkPipelineStageFlags     srcStageMask,
    VkPipelineStageFlags     dstStageMask,
    VkDependencyFlags        dependencyFlags,
    uint32_t                 memoryBarrierCount,
    const VkMemoryBarrier*   pMemoryBarriers,
    uint32_t                 bufferMemoryBarrierCount,
    const VkBufferMemoryBarrier* pBufferMemoryBarriers,
    uint32_t                 imageMemoryBarrierCount,
    const VkImageMemoryBarrier* pImageMemoryBarriers);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `srcStageMask` is a bitmask of `VkPipelineStageFlagBits` specifying the [source stages](#).
- `dstStageMask` is a bitmask of `VkPipelineStageFlagBits` specifying the [destination stages](#).
- `dependencyFlags` is a bitmask of `VkDependencyFlagBits` specifying how execution and memory dependencies are formed.
- `memoryBarrierCount` is the length of the `pMemoryBarriers` array.
- `pMemoryBarriers` is a pointer to an array of `VkMemoryBarrier` structures.
- `bufferMemoryBarrierCount` is the length of the `pBufferMemoryBarriers` array.
- `pBufferMemoryBarriers` is a pointer to an array of `VkBufferMemoryBarrier` structures.
- `imageMemoryBarrierCount` is the length of the `pImageMemoryBarriers` array.
- `pImageMemoryBarriers` is a pointer to an array of `VkImageMemoryBarrier` structures.

When `vkCmdPipelineBarrier` is submitted to a queue, it defines a memory dependency between commands that were submitted to the same queue before it, and those submitted to the same queue after it.

If `vkCmdPipelineBarrier` was recorded outside a render pass instance, the first [synchronization scope](#) includes all commands that occur earlier in [submission order](#). If `vkCmdPipelineBarrier` was recorded inside a render pass instance, the first synchronization scope includes only commands that occur earlier in [submission order](#) within the same subpass. In either case, the first synchronization scope is limited to operations on the pipeline stages determined by the [source stage mask](#) specified by `srcStageMask`.

If `vkCmdPipelineBarrier` was recorded outside a render pass instance, the second [synchronization scope](#) includes all commands that occur later in [submission order](#). If `vkCmdPipelineBarrier` was recorded inside a render pass instance, the second synchronization scope includes only commands that occur later in [submission order](#) within the same subpass. In either case, the second synchronization scope is limited to operations on the pipeline stages determined by the [destination stage mask](#) specified by `dstStageMask`.

The first [access scope](#) is limited to accesses in the pipeline stages determined by the [source stage mask](#) specified by `srcStageMask`. Within that, the first access scope only includes the first access scopes defined by elements of the `pMemoryBarriers`, `pBufferMemoryBarriers` and `pImageMemoryBarriers` arrays, which each define a set of [memory barriers](#). If no memory barriers are specified, then the first access scope includes no accesses.

The second [access scope](#) is limited to accesses in the pipeline stages determined by the [destination stage mask](#) specified by `dstStageMask`. Within that, the second access scope only includes the second access scopes defined by elements of the `pMemoryBarriers`, `pBufferMemoryBarriers` and `pImageMemoryBarriers` arrays, which each define a set of [memory barriers](#). If no memory barriers are specified, then the second access scope includes no accesses.

If `dependencyFlags` includes `VK_DEPENDENCY_BY_REGION_BIT`, then any dependency between [framebuffer-space](#) pipeline stages is [framebuffer-local](#) - otherwise it is [framebuffer-global](#).

Valid Usage

- VUID-vkCmdPipelineBarrier-srcStageMask-04090
If the `geometryShader` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT`
- VUID-vkCmdPipelineBarrier-srcStageMask-04091
If the `tessellationShader` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT` or `VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT`
- VUID-vkCmdPipelineBarrier-srcStageMask-04996
`srcStageMask` **must** not be 0
- VUID-vkCmdPipelineBarrier-dstStageMask-04090
If the `geometryShader` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT`
- VUID-vkCmdPipelineBarrier-dstStageMask-04091
If the `tessellationShader` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT` or `VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT`
- VUID-vkCmdPipelineBarrier-dstStageMask-04996
`dstStageMask` **must** not be 0
- VUID-vkCmdPipelineBarrier-srcAccessMask-02815
The `srcAccessMask` member of each element of `pMemoryBarriers` **must** only include access flags that are supported by one or more of the pipeline stages in `srcStageMask`, as specified in the [table of supported access types](#)
- VUID-vkCmdPipelineBarrier-dstAccessMask-02816
The `dstAccessMask` member of each element of `pMemoryBarriers` **must** only include access flags that are supported by one or more of the pipeline stages in `dstStageMask`, as specified in the [table of supported access types](#)
- VUID-vkCmdPipelineBarrier-pBufferMemoryBarriers-02817
For any element of `pBufferMemoryBarriers`, if its `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members are equal, or if its `srcQueueFamilyIndex` is the queue family index that was used to create the command pool that `commandBuffer` was allocated from, then its `srcAccessMask` member **must** only contain access flags that are supported by one or more of the pipeline stages in `srcStageMask`, as specified in the [table of supported access types](#)
- VUID-vkCmdPipelineBarrier-pBufferMemoryBarriers-02818
For any element of `pBufferMemoryBarriers`, if its `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members are equal, or if its `dstQueueFamilyIndex` is the queue family index that was used to create the command pool that `commandBuffer` was allocated from, then its `dstAccessMask` member **must** only contain access flags that are supported by one or more of the pipeline stages in `dstStageMask`, as specified in the [table of supported access types](#)

- VUID-vkCmdPipelineBarrier-pImageMemoryBarriers-02819
For any element of `pImageMemoryBarriers`, if its `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members are equal, or if its `srcQueueFamilyIndex` is the queue family index that was used to create the command pool that `commandBuffer` was allocated from, then its `srcAccessMask` member **must** only contain access flags that are supported by one or more of the pipeline stages in `srcStageMask`, as specified in the [table of supported access types](#)
- VUID-vkCmdPipelineBarrier-pImageMemoryBarriers-02820
For any element of `pImageMemoryBarriers`, if its `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members are equal, or if its `dstQueueFamilyIndex` is the queue family index that was used to create the command pool that `commandBuffer` was allocated from, then its `dstAccessMask` member **must** only contain access flags that are supported by one or more of the pipeline stages in `dstStageMask`, as specified in the [table of supported access types](#)
- VUID-vkCmdPipelineBarrier-pDependencies-02285
If `vkCmdPipelineBarrier` is called within a render pass instance using a `VkRenderPass` object, the render pass **must** have been created with at least one subpass dependency that expresses a dependency from the current subpass to itself, does not include `VK_DEPENDENCY_BY_REGION_BIT` if this command does not, and has [synchronization scopes](#) and [access scopes](#) that are all supersets of the scopes defined in this command
- VUID-vkCmdPipelineBarrier-bufferMemoryBarrierCount-01178
If `vkCmdPipelineBarrier` is called within a render pass instance using a `VkRenderPass` object, it **must** not include any buffer memory barriers
- VUID-vkCmdPipelineBarrier-image-04073
If `vkCmdPipelineBarrier` is called within a render pass instance using a `VkRenderPass` object, the `image` member of any image memory barrier included in this command **must** be an attachment used in the current subpass both as an input attachment, and as either a color, or depth/stencil attachment
- VUID-vkCmdPipelineBarrier-oldLayout-01181
If `vkCmdPipelineBarrier` is called within a render pass instance, the `oldLayout` and `newLayout` members of any image memory barrier included in this command **must** be equal
- VUID-vkCmdPipelineBarrier-srcQueueFamilyIndex-01182
If `vkCmdPipelineBarrier` is called within a render pass instance, the `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members of any memory barrier included in this command **must** be equal
- VUID-vkCmdPipelineBarrier-None-07890
If `vkCmdPipelineBarrier` is called within a render pass instance, and the source stage masks of any memory barriers include [framebuffer-space stages](#), destination stage masks of all memory barriers **must** only include [framebuffer-space stages](#)
- VUID-vkCmdPipelineBarrier-dependencyFlags-07891
If `vkCmdPipelineBarrier` is called within a render pass instance, and the source stage masks of any memory barriers include [framebuffer-space stages](#), then `dependencyFlags` **must** include `VK_DEPENDENCY_BY_REGION_BIT`

- VUID-vkCmdPipelineBarrier-None-07892
If `vkCmdPipelineBarrier` is called within a render pass instance, the source and destination stage masks of any memory barriers **must** only include graphics pipeline stages
- VUID-vkCmdPipelineBarrier-srcStageMask-06461
Any pipeline stage included in `srcStageMask` **must** be supported by the capabilities of the queue family specified by the `queueFamilyIndex` member of the `VkCommandPoolCreateInfo` structure that was used to create the `VkCommandPool` that `commandBuffer` was allocated from, as specified in the [table of supported pipeline stages](#)
- VUID-vkCmdPipelineBarrier-dstStageMask-06462
Any pipeline stage included in `dstStageMask` **must** be supported by the capabilities of the queue family specified by the `queueFamilyIndex` member of the `VkCommandPoolCreateInfo` structure that was used to create the `VkCommandPool` that `commandBuffer` was allocated from, as specified in the [table of supported pipeline stages](#)
- VUID-vkCmdPipelineBarrier-srcStageMask-09633
If either `srcStageMask` or `dstStageMask` includes `VK_PIPELINE_STAGE_HOST_BIT`, for any element of `pImageMemoryBarriers`, `srcQueueFamilyIndex` and `dstQueueFamilyIndex` **must** be equal
- VUID-vkCmdPipelineBarrier-srcStageMask-09634
If either `srcStageMask` or `dstStageMask` includes `VK_PIPELINE_STAGE_HOST_BIT`, for any element of `pBufferMemoryBarriers`, `srcQueueFamilyIndex` and `dstQueueFamilyIndex` **must** be equal

Valid Usage (Implicit)

- VUID-vkCmdPipelineBarrier-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdPipelineBarrier-srcStageMask-parameter
`srcStageMask` **must** be a valid combination of `VkPipelineStageFlagBits` values
- VUID-vkCmdPipelineBarrier-dstStageMask-parameter
`dstStageMask` **must** be a valid combination of `VkPipelineStageFlagBits` values
- VUID-vkCmdPipelineBarrier-dependencyFlags-parameter
`dependencyFlags` **must** be a valid combination of `VkDependencyFlagBits` values
- VUID-vkCmdPipelineBarrier-pMemoryBarriers-parameter
If `memoryBarrierCount` is not 0, `pMemoryBarriers` **must** be a valid pointer to an array of `memoryBarrierCount` valid `VkMemoryBarrier` structures
- VUID-vkCmdPipelineBarrier-pBufferMemoryBarriers-parameter
If `bufferMemoryBarrierCount` is not 0, `pBufferMemoryBarriers` **must** be a valid pointer to an array of `bufferMemoryBarrierCount` valid `VkBufferMemoryBarrier` structures
- VUID-vkCmdPipelineBarrier-pImageMemoryBarriers-parameter
If `imageMemoryBarrierCount` is not 0, `pImageMemoryBarriers` **must** be a valid pointer to an array of `imageMemoryBarrierCount` valid `VkImageMemoryBarrier` structures
- VUID-vkCmdPipelineBarrier-commandBuffer-recording

`commandBuffer` **must** be in the `recording` state

- VUID-vkCmdPipelineBarrier-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support transfer, graphics, or compute operations

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Transfer Graphics Compute	Synchronization

Bits which **can** be set in `vkCmdPipelineBarrier::dependencyFlags`, specifying how execution and memory dependencies are formed, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkDependencyFlagBits {
    VK_DEPENDENCY_BY_REGION_BIT = 0x00000001,
} VkDependencyFlagBits;
```

- `VK_DEPENDENCY_BY_REGION_BIT` specifies that dependencies will be `framebuffer-local`.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkDependencyFlags;
```

`VkDependencyFlags` is a bitmask type for setting a mask of zero or more `VkDependencyFlagBits`.

7.7. Memory Barriers

Memory barriers are used to explicitly control access to buffer and image subresource ranges. Memory barriers are used to `transfer ownership between queue families`, `change image layouts`, and define `availability and visibility operations`. They explicitly define the `access types` and buffer and image subresource ranges that are included in the `access scopes` of a memory dependency that is created by a synchronization command that includes them.

7.7.1. Global Memory Barriers

Global memory barriers apply to memory accesses involving all memory objects that exist at the time of its execution.

The `VkMemoryBarrier` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkMemoryBarrier {
    VkStructureType    sType;
    const void*        pNext;
    VkAccessFlags       srcAccessMask;
    VkAccessFlags       dstAccessMask;
} VkMemoryBarrier;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcAccessMask` is a bitmask of `VkAccessFlagBits` specifying a [source access mask](#).
- `dstAccessMask` is a bitmask of `VkAccessFlagBits` specifying a [destination access mask](#).

The first [access scope](#) is limited to access types in the [source access mask](#) specified by `srcAccessMask`.

The second [access scope](#) is limited to access types in the [destination access mask](#) specified by `dstAccessMask`.

Valid Usage (Implicit)

- VUID-VkMemoryBarrier-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_MEMORY_BARRIER`
- VUID-VkMemoryBarrier-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkMemoryBarrier-srcAccessMask-parameter
`srcAccessMask` **must** be a valid combination of `VkAccessFlagBits` values
- VUID-VkMemoryBarrier-dstAccessMask-parameter
`dstAccessMask` **must** be a valid combination of `VkAccessFlagBits` values

7.7.2. Buffer Memory Barriers

Buffer memory barriers only apply to memory accesses involving a specific buffer range. That is, a memory dependency formed from a buffer memory barrier is [scoped](#) to access via the specified buffer range. Buffer memory barriers **can** also be used to define a [queue family ownership transfer](#) for the specified buffer range.

The `VkBufferMemoryBarrier` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkBufferMemoryBarrier {
    VkStructureType    sType;
    const void*        pNext;
    VkAccessFlags       srcAccessMask;
    VkAccessFlags       dstAccessMask;
    uint32_t            srcQueueFamilyIndex;
    uint32_t            dstQueueFamilyIndex;
    VkBuffer            buffer;
    VkDeviceSize        offset;
    VkDeviceSize        size;
} VkBufferMemoryBarrier;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcAccessMask` is a bitmask of `VkAccessFlagBits` specifying a [source access mask](#).
- `dstAccessMask` is a bitmask of `VkAccessFlagBits` specifying a [destination access mask](#).
- `srcQueueFamilyIndex` is the source queue family for a [queue family ownership transfer](#).
- `dstQueueFamilyIndex` is the destination queue family for a [queue family ownership transfer](#).
- `buffer` is a handle to the buffer whose backing memory is affected by the barrier.
- `offset` is an offset in bytes into the backing memory for `buffer`; this is relative to the base offset as bound to the buffer (see [vkBindBufferMemory](#)).
- `size` is a size in bytes of the affected area of backing memory for `buffer`, or `VK_WHOLE_SIZE` to use the range from `offset` to the end of the buffer.

The first [access scope](#) is limited to access to memory through the specified buffer range, via access types in the [source access mask](#) specified by `srcAccessMask`. If `srcAccessMask` includes `VK_ACCESS_HOST_WRITE_BIT`, a [memory domain operation](#) is performed where available memory in the host domain is also made available to the device domain.

The second [access scope](#) is limited to access to memory through the specified buffer range, via access types in the [destination access mask](#) specified by `dstAccessMask`. If `dstAccessMask` includes `VK_ACCESS_HOST_WRITE_BIT` or `VK_ACCESS_HOST_READ_BIT`, a [memory domain operation](#) is performed where available memory in the device domain is also made available to the host domain.

NOTE When `VK_MEMORY_PROPERTY_HOST_COHERENT_BIT` is used, available memory in host domain is automatically made visible to host domain, and any host write is automatically made available to host domain.

If `srcQueueFamilyIndex` is not equal to `dstQueueFamilyIndex`, and `srcQueueFamilyIndex` is equal to the current queue family, then the memory barrier defines a [queue family release operation](#) for the specified buffer range, and the second synchronization scope of the calling command does not apply to this operation.

If `dstQueueFamilyIndex` is not equal to `srcQueueFamilyIndex`, and `dstQueueFamilyIndex` is equal to the

current queue family, then the memory barrier defines a [queue family acquire operation](#) for the specified buffer range, and the first synchronization scope of the calling command does not apply to this operation.

Valid Usage

- VUID-VkBufferMemoryBarrier-offset-01187
`offset` **must** be less than the size of `buffer`
- VUID-VkBufferMemoryBarrier-size-01188
If `size` is not equal to `VK_WHOLE_SIZE`, `size` **must** be greater than 0
- VUID-VkBufferMemoryBarrier-size-01189
If `size` is not equal to `VK_WHOLE_SIZE`, `size` **must** be less than or equal to the size of `buffer` minus `offset`
- VUID-VkBufferMemoryBarrier-buffer-01931
If `buffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkBufferMemoryBarrier-buffer-09095
If `buffer` was created with a sharing mode of `VK_SHARING_MODE_EXCLUSIVE`, and `srcQueueFamilyIndex` and `dstQueueFamilyIndex` are not equal, `srcQueueFamilyIndex` **must** be a valid queue family
- VUID-VkBufferMemoryBarrier-buffer-09096
If `buffer` was created with a sharing mode of `VK_SHARING_MODE_EXCLUSIVE`, and `srcQueueFamilyIndex` and `dstQueueFamilyIndex` are not equal, `dstQueueFamilyIndex` **must** be a valid queue family
- VUID-VkBufferMemoryBarrier-None-09050
If `buffer` was created with a sharing mode of `VK_SHARING_MODE_CONCURRENT`, `srcQueueFamilyIndex` **must** be `VK_QUEUE_FAMILY_IGNORED`
- VUID-VkBufferMemoryBarrier-None-09051
If `buffer` was created with a sharing mode of `VK_SHARING_MODE_CONCURRENT`, `dstQueueFamilyIndex` **must** be `VK_QUEUE_FAMILY_IGNORED`

Valid Usage (Implicit)

- VUID-VkBufferMemoryBarrier-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_BUFFER_MEMORY_BARRIER`
- VUID-VkBufferMemoryBarrier-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkBufferMemoryBarrier-buffer-parameter
`buffer` **must** be a valid [VkBuffer](#) handle

`VK_WHOLE_SIZE` is a special value indicating that the entire remaining length of a buffer following a given `offset` should be used. It **can** be specified for `VkBufferMemoryBarrier::size` and other

structures.

```
#define VK_WHOLE_SIZE          (~0ULL)
```

7.7.3. Image Memory Barriers

Image memory barriers only apply to memory accesses involving a specific image subresource range. That is, a memory dependency formed from an image memory barrier is [scoped](#) to access via the specified image subresource range. Image memory barriers **can** also be used to define [image layout transitions](#) or a [queue family ownership transfer](#) for the specified image subresource range.

The `VkImageMemoryBarrier` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageMemoryBarrier {
    VkStructureType    sType;
    const void*        pNext;
    VkAccessFlags       srcAccessMask;
    VkAccessFlags       dstAccessMask;
    VkImageLayout       oldLayout;
    VkImageLayout       newLayout;
    uint32_t            srcQueueFamilyIndex;
    uint32_t            dstQueueFamilyIndex;
    VkImage             image;
    VkImageSubresourceRange subresourceRange;
} VkImageMemoryBarrier;
```

- `sType` is a [VkStructureType](#) value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcAccessMask` is a bitmask of [VkAccessFlagBits](#) specifying a [source access mask](#).
- `dstAccessMask` is a bitmask of [VkAccessFlagBits](#) specifying a [destination access mask](#).
- `oldLayout` is the old layout in an [image layout transition](#).
- `newLayout` is the new layout in an [image layout transition](#).
- `srcQueueFamilyIndex` is the source queue family for a [queue family ownership transfer](#).
- `dstQueueFamilyIndex` is the destination queue family for a [queue family ownership transfer](#).
- `image` is a handle to the image affected by this barrier.
- `subresourceRange` describes the [image subresource range](#) within `image` that is affected by this barrier.

The first [access scope](#) is limited to access to memory through the specified image subresource range, via access types in the [source access mask](#) specified by `srcAccessMask`. If `srcAccessMask` includes `VK_ACCESS_HOST_WRITE_BIT`, memory writes performed by that access type are also made visible, as that access type is not performed through a resource.

The second [access scope](#) is limited to access to memory through the specified image subresource range, via access types in the [destination access mask](#) specified by `dstAccessMask`. If `dstAccessMask` includes `VK_ACCESS_HOST_WRITE_BIT` or `VK_ACCESS_HOST_READ_BIT`, available memory writes are also made visible to accesses of those types, as those access types are not performed through a resource.

If `srcQueueFamilyIndex` is not equal to `dstQueueFamilyIndex`, and `srcQueueFamilyIndex` is equal to the current queue family, then the memory barrier defines a [queue family release operation](#) for the specified image subresource range, and the second synchronization scope of the calling command does not apply to this operation.

If `dstQueueFamilyIndex` is not equal to `srcQueueFamilyIndex`, and `dstQueueFamilyIndex` is equal to the current queue family, then the memory barrier defines a [queue family acquire operation](#) for the specified image subresource range, and the first synchronization scope of the calling command does not apply to this operation.

`oldLayout` and `newLayout` define an [image layout transition](#) for the specified image subresource range.

Valid Usage

- VUID-VkImageMemoryBarrier-oldLayout-01208
If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL` then `image` **must** have been created with `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT`
- VUID-VkImageMemoryBarrier-oldLayout-01209
If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL` then `image` **must** have been created with `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`
- VUID-VkImageMemoryBarrier-oldLayout-01210
If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL` then `image` **must** have been created with `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`
- VUID-VkImageMemoryBarrier-oldLayout-01211
If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL` then `image` **must** have been created with `VK_IMAGE_USAGE_SAMPLED_BIT` or `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT`
- VUID-VkImageMemoryBarrier-oldLayout-01212
If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL` then `image` **must** have been created with `VK_IMAGE_USAGE_TRANSFER_SRC_BIT`
- VUID-VkImageMemoryBarrier-oldLayout-01213

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL` then `image` **must** have been created with `VK_IMAGE_USAGE_TRANSFER_DST_BIT`

- VUID-VkImageMemoryBarrier-oldLayout-01197

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), `oldLayout` **must** be `VK_IMAGE_LAYOUT_UNDEFINED` or the current layout of the image subresources affected by the barrier

- VUID-VkImageMemoryBarrier-newLayout-01198

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), `newLayout` **must** not be `VK_IMAGE_LAYOUT_UNDEFINED` or `VK_IMAGE_LAYOUT_PREINITIALIZED`

- VUID-VkImageMemoryBarrier-image-09117

If `image` was created with a sharing mode of `VK_SHARING_MODE_EXCLUSIVE`, and `srcQueueFamilyIndex` and `dstQueueFamilyIndex` are not equal, `srcQueueFamilyIndex` **must** be a valid queue family

- VUID-VkImageMemoryBarrier-image-09118

If `image` was created with a sharing mode of `VK_SHARING_MODE_EXCLUSIVE`, and `srcQueueFamilyIndex` and `dstQueueFamilyIndex` are not equal, `dstQueueFamilyIndex` **must** be a valid queue family

- VUID-VkImageMemoryBarrier-subresourceRange-01486

`subresourceRange.baseMipLevel` **must** be less than the `mipLevels` specified in [VkImageCreateInfo](#) when `image` was created

- VUID-VkImageMemoryBarrier-subresourceRange-01724

If `subresourceRange.levelCount` is not `VK_REMAINING_MIP_LEVELS`, `subresourceRange.baseMipLevel + subresourceRange.levelCount` **must** be less than or equal to the `mipLevels` specified in [VkImageCreateInfo](#) when `image` was created

- VUID-VkImageMemoryBarrier-subresourceRange-01488

`subresourceRange.baseArrayLayer` **must** be less than the `arrayLayers` specified in [VkImageCreateInfo](#) when `image` was created

- VUID-VkImageMemoryBarrier-subresourceRange-01725

If `subresourceRange.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, `subresourceRange.baseArrayLayer + subresourceRange.layerCount` **must** be less than or equal to the `arrayLayers` specified in [VkImageCreateInfo](#) when `image` was created

- VUID-VkImageMemoryBarrier-image-01932

If `image` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-VkImageMemoryBarrier-image-09241

If `image` has a color format then the `aspectMask` member of `subresourceRange` **must** be `VK_IMAGE_ASPECT_COLOR_BIT`

- VUID-VkImageMemoryBarrier-image-03320

If `image` has a depth/stencil format with both depth and stencil then the `aspectMask`

member of `subresourceRange` **must** include both `VK_IMAGE_ASPECT_DEPTH_BIT` and `VK_IMAGE_ASPECT_STENCIL_BIT`

- VUID-VkImageMemoryBarrier-subresourceRange-09601
`subresourceRange.aspectMask` **must** be valid for the `format` the `image` was created with
- VUID-VkImageMemoryBarrier-None-09053
If `image` was created with a sharing mode of `VK_SHARING_MODE_CONCURRENT`, `srcQueueFamilyIndex` **must** be `VK_QUEUE_FAMILY_IGNORED`
- VUID-VkImageMemoryBarrier-None-09054
If `image` was created with a sharing mode of `VK_SHARING_MODE_CONCURRENT`, `dstQueueFamilyIndex` **must** be `VK_QUEUE_FAMILY_IGNORED`

Valid Usage (Implicit)

- VUID-VkImageMemoryBarrier-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_IMAGE_MEMORY_BARRIER`
- VUID-VkImageMemoryBarrier-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkImageMemoryBarrier-oldLayout-parameter
`oldLayout` **must** be a valid `VkImageLayout` value
- VUID-VkImageMemoryBarrier-newLayout-parameter
`newLayout` **must** be a valid `VkImageLayout` value
- VUID-VkImageMemoryBarrier-image-parameter
`image` **must** be a valid `VkImage` handle
- VUID-VkImageMemoryBarrier-subresourceRange-parameter
`subresourceRange` **must** be a valid `VkImageSubresourceRange` structure

7.7.4. Queue Family Ownership Transfer

Resources created with a `VkSharingMode` of `VK_SHARING_MODE_EXCLUSIVE` **must** have their ownership explicitly transferred from one queue family to another in order to access their content in a well-defined manner on a queue in a different queue family.

The special queue family index `VK_QUEUE_FAMILY_IGNORED` indicates that a queue family parameter or member is ignored.

```
#define VK_QUEUE_FAMILY_IGNORED        (~0U)
```

If memory dependencies are correctly expressed between uses of such a resource between two queues in different families, but no ownership transfer is defined, the contents of that resource are undefined for any read accesses performed by the second queue family.

NOTE If an application does not need the contents of a resource to remain valid when

transferring from one queue family to another, then the ownership transfer **should** be skipped.

A queue family ownership transfer consists of two distinct parts:

1. Release exclusive ownership from the source queue family
2. Acquire exclusive ownership for the destination queue family

An application **must** ensure that these operations occur in the correct order by defining an execution dependency between them, e.g. using a semaphore.

A *release operation* is used to release exclusive ownership of a range of a buffer or image subresource range. A release operation is defined by executing a [buffer memory barrier](#) (for a buffer range) or an [image memory barrier](#) (for an image subresource range) using a pipeline barrier command, on a queue from the source queue family. The [srcQueueFamilyIndex](#) parameter of the barrier **must** be the source queue family index, and the [dstQueueFamilyIndex](#) parameter to the destination queue family index. [dstAccessMask](#) is ignored for such a barrier, such that no visibility operation is executed - the value of this mask does not affect the validity of the barrier. The release operation happens-after the availability operation. [dstStageMask](#) is also ignored for such a barrier as defined by [buffer memory ownership transfer](#) and [image memory ownership transfer](#).

An *acquire operation* is used to acquire exclusive ownership of a range of a buffer or image subresource range. An acquire operation is defined by executing a [buffer memory barrier](#) (for a buffer range) or an [image memory barrier](#) (for an image subresource range) using a pipeline barrier command, on a queue from the destination queue family. The buffer range or image subresource range specified in an acquire operation **must** match exactly that of a previous release operation. The [srcQueueFamilyIndex](#) parameter of the barrier **must** be the source queue family index, and the [dstQueueFamilyIndex](#) parameter to the destination queue family index. [srcAccessMask](#) is ignored for such a barrier, such that no availability operation is executed - the value of this mask does not affect the validity of the barrier. The acquire operation happens-before the visibility operation. [srcStageMask](#) is also ignored for such a barrier as defined by [buffer memory ownership transfer](#) and [image memory ownership transfer](#). As the first synchronization scope for an acquire operation is empty there is no happens-before dependency. Such a dependency **can** be introduced by using [VK_PIPELINE_STAGE_ALL_COMMANDS_BIT](#).

NOTE

Whilst it is not invalid to provide destination or source access masks for memory barriers used for release or acquire operations, respectively, they have no practical effect. Access after a release operation has undefined results, and so visibility for those accesses has no practical effect. Similarly, write access before an acquire operation will produce undefined results for future access, so availability of those writes has no practical use. In an earlier version of the specification, these were required to match on both sides - but this was subsequently relaxed. These masks **should** be set to 0.

NOTE

Since a release and acquire operation does not synchronize with second and first scopes respectively, the [VK_PIPELINE_STAGE_ALL_COMMANDS_BIT](#) stage must be used to wait for a release operation to complete. Typically, a release and acquire pair is performed by a [VkSemaphore](#) signal and wait in their respective queues. Signaling

a semaphore with `vkQueueSubmit` waits for `VK_PIPELINE_STAGE_ALL_COMMANDS_BIT`. Similarly, for the `acquire` operation, waiting for a semaphore must use `VK_PIPELINE_STAGE_ALL_COMMANDS_BIT` to make sure the `acquire` operation is synchronized.

If the transfer is via an image memory barrier, and an `image layout transition` is desired, then the values of `oldLayout` and `newLayout` in the *release operation*'s memory barrier **must** be equal to values of `oldLayout` and `newLayout` in the *acquire operation*'s memory barrier. Although the image layout transition is submitted twice, it will only be executed once. A layout transition specified in this way happens-after the *release operation* and happens-before the *acquire operation*.

If the values of `srcQueueFamilyIndex` and `dstQueueFamilyIndex` are equal, no ownership transfer is performed, and the barrier operates as if they were both set to `VK_QUEUE_FAMILY_IGNORED`.

Queue family ownership transfers **may** perform read and write accesses on all memory bound to the image subresource or buffer range, so applications **must** ensure that all memory writes have been made `available` before a queue family ownership transfer is executed. Available memory is automatically made visible to queue family release and acquire operations, and writes performed by those operations are automatically made available.

Once a queue family has acquired ownership of a buffer range or image subresource range of a `VK_SHARING_MODE_EXCLUSIVE` resource, its contents are undefined to other queue families unless ownership is transferred. The contents of any portion of another resource which aliases memory that is bound to the transferred buffer or image subresource range are undefined after a release or acquire operation.

NOTE

Because `events` **cannot** be used directly for inter-queue synchronization, and because `vkCmdSetEvent` does not have the queue family index or memory barrier parameters needed by a *release operation*, the release and acquire operations of a queue family ownership transfer **can** only be performed using `vkCmdPipelineBarrier`.

7.8. Wait Idle Operations

To wait on the host for the completion of outstanding queue operations for a given queue, call:

```
// Provided by VK_VERSION_1_0
VkResult vkQueueWaitIdle(
    VkQueue queue);
```

- `queue` is the queue on which to wait.

`vkQueueWaitIdle` is equivalent to having submitted a valid fence to every previously executed `queue submission command` that accepts a fence, then waiting for all of those fences to signal using `vkWaitForFences` with an infinite timeout and `waitAll` set to `VK_TRUE`.

Valid Usage (Implicit)

- VUID-vkQueueWaitIdle-queue-parameter
queue must be a valid **VkQueue** handle

Host Synchronization

- Host access to **queue** must be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
-	-	Any	-

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_OUT_OF_DEVICE_MEMORY**
- **VK_ERROR_DEVICE_LOST**

To wait on the host for the completion of outstanding queue operations for all queues on a given logical device, call:

```
// Provided by VK_VERSION_1_0
VkResult vkDeviceWaitIdle(
    VkDevice device);
```

- **device** is the logical device to idle.

vkDeviceWaitIdle is equivalent to calling **vkQueueWaitIdle** for all queues owned by **device**.

Valid Usage (Implicit)

- VUID-vkDeviceWaitIdle-device-parameter
device must be a valid **VkDevice** handle

Host Synchronization

- Host access to all `VkQueue` objects created from `device` **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_DEVICE_LOST`

7.9. Host Write Ordering Guarantees

When batches of command buffers are submitted to a queue via a [queue submission command](#), it defines a memory dependency with prior host operations, and execution of command buffers submitted to the queue.

The first [synchronization scope](#) includes execution of `vkQueueSubmit` on the host and anything that happened-before it, as defined by the host memory model.

NOTE

Some systems allow writes that do not directly integrate with the host memory model; these have to be synchronized by the application manually. One example of this is non-temporal store instructions on x86; to ensure these happen-before submission, applications should call `_mm_sfence()`.

The second [synchronization scope](#) includes all commands submitted in the same [queue submission](#), and all commands that occur later in [submission order](#).

The first [access scope](#) includes all host writes to mappable device memory that are available to the host memory domain.

The second [access scope](#) includes all memory access performed by the device.

Chapter 8. Render Pass

Draw commands **must** be recorded within a *render pass instance*. Each render pass instance defines a set of image resources, referred to as *attachments*, used during rendering.

A render pass object represents a collection of attachments, subpasses, and dependencies between the subpasses, and describes how the attachments are used over the course of the subpasses.

Render passes are represented by **VkRenderPass** handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkRenderPass)
```

An *attachment description* describes the properties of an attachment including its format, sample count, and how its contents are treated at the beginning and end of each render pass instance.

A *subpass* represents a phase of rendering that reads and writes a subset of the attachments in a render pass. Rendering commands are recorded into a particular subpass of a render pass instance.

A *subpass description* describes the subset of attachments that is involved in the execution of a subpass. Each subpass **can** read from some attachments as *input attachments*, write to some as *color attachments* or *depth/stencil attachments*, and perform *multisample resolve operations* to *resolve attachments*. A subpass description **can** also include a set of *preserve attachments*, which are attachments that are not read or written by the subpass but whose contents **must** be preserved throughout the subpass.

A subpass *uses an attachment* if the attachment is a color, depth/stencil, resolve, or input attachment for that subpass (as determined by the **pColorAttachments**, **pDepthStencilAttachment**, **pResolveAttachments**, and **pInputAttachments** members of **VkSubpassDescription**, respectively). A subpass does not use an attachment if that attachment is preserved by the subpass. The *first use of an attachment* is in the lowest numbered subpass that uses that attachment. Similarly, the *last use of an attachment* is in the highest numbered subpass that uses that attachment.

The subpasses in a render pass all render to the same dimensions, and fragments for pixel (x,y,layer) in one subpass **can** only read attachment contents written by previous subpasses at that same (x,y,layer) location.

NOTE

By describing a complete set of subpasses in advance, render passes provide the implementation an opportunity to optimize the storage and transfer of attachment data between subpasses.

In practice, this means that subpasses with a simple framebuffer-space dependency **may** be merged into a single tiled rendering pass, keeping the attachment data on-chip for the duration of a render pass instance. However, it is also quite common for a render pass to only contain a single subpass.

Subpass dependencies describe **execution and memory dependencies** between subpasses.

A *subpass dependency chain* is a sequence of subpass dependencies in a render pass, where the source subpass of each subpass dependency (after the first) equals the destination subpass of the previous dependency.

Execution of subpasses **may** overlap or execute out of order with regards to other subpasses, unless otherwise enforced by an execution dependency. Each subpass only respects [submission order](#) for commands recorded in the same subpass, and the [vkCmdBeginRenderPass](#) and [vkCmdEndRenderPass](#) commands that delimit the render pass - commands within other subpasses are not included. This affects most other [implicit ordering guarantees](#).

A render pass describes the structure of subpasses and attachments independent of any specific image views for the attachments. The specific image views that will be used for the attachments, and their dimensions, are specified in [VkFramebuffer](#) objects. Framebuffers are created with respect to a specific render pass that the framebuffer is compatible with (see [Render Pass Compatibility](#)). Collectively, a render pass and a framebuffer define the complete render target state for one or more subpasses as well as the algorithmic dependencies between the subpasses.

The various pipeline stages of the drawing commands for a given subpass **may** execute concurrently and/or out of order, both within and across drawing commands, whilst still respecting [pipeline order](#). However for a given (x,y,layer,sample) sample location, certain per-sample operations are performed in [rasterization order](#).

[VK_ATTACHMENT_UNUSED](#) is a constant indicating that a render pass attachment is not used.

```
#define VK_ATTACHMENT_UNUSED (~0U)
```

8.1. Render Pass Creation

To create a render pass, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateRenderPass(
    VkDevice device,
    const VkRenderPassCreateInfo* pCreateInfo,
    const VkAllocationCallbacks* pAllocator,
    VkRenderPass* pRenderPass);
```

- [device](#) is the logical device that creates the render pass.
- [pCreateInfo](#) is a pointer to a [VkRenderPassCreateInfo](#) structure describing the parameters of the render pass.
- [pAllocator](#) controls host memory allocation as described in the [Memory Allocation](#) chapter.
- [pRenderPass](#) is a pointer to a [VkRenderPass](#) handle in which the resulting render pass object is returned.

Valid Usage

- VUID-vkCreateRenderPass-device-10000
device **must** support at least one queue family with the `VK_QUEUE_GRAPHICS_BIT` capability

Valid Usage (Implicit)

- VUID-vkCreateRenderPass-device-parameter
device **must** be a valid `VkDevice` handle
- VUID-vkCreateRenderPass-pCreateInfo-parameter
pCreateInfo **must** be a valid pointer to a valid `VkRenderPassCreateInfo` structure
- VUID-vkCreateRenderPass-pAllocator-parameter
If **pAllocator** is not `NULL`, **pAllocator** **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkCreateRenderPass-pRenderPass-parameter
pRenderPass **must** be a valid pointer to a `VkRenderPass` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkRenderPassCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkRenderPassCreateInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkRenderPassCreateFlags flags;
    uint32_t             attachmentCount;
    const VkAttachmentDescription* pAttachments;
    uint32_t             subpassCount;
    const VkSubpassDescription* pSubpasses;
    uint32_t             dependencyCount;
    const VkSubpassDependency* pDependencies;
} VkRenderPassCreateInfo;
```

- **sType** is a `VkStructureType` value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.

- `flags` is reserved for future use.
- `attachmentCount` is the number of attachments used by this render pass.
- `pAttachments` is a pointer to an array of `attachmentCount` `VkAttachmentDescription` structures describing the attachments used by the render pass.
- `subpassCount` is the number of subpasses to create.
- `pSubpasses` is a pointer to an array of `subpassCount` `VkSubpassDescription` structures describing each subpass.
- `dependencyCount` is the number of memory dependencies between pairs of subpasses.
- `pDependencies` is a pointer to an array of `dependencyCount` `VkSubpassDependency` structures describing dependencies between pairs of subpasses.

NOTE

Care should be taken to avoid a data race here; if any subpasses access attachments with overlapping memory locations, and one of those accesses is a write, a subpass dependency needs to be included between them.

Valid Usage

- VUID-VkRenderPassCreateInfo-attachment-00834
If the `attachment` member of any element of `pInputAttachments`, `pColorAttachments`, `pResolveAttachments` or `pDepthStencilAttachment`, or any element of `pPreserveAttachments` in any element of `pSubpasses` is not `VK_ATTACHMENT_UNUSED`, then it **must** be less than `attachmentCount`
- VUID-VkRenderPassCreateInfo-pAttachments-00836
For any member of `pAttachments` with a `loadOp` equal to `VK_ATTACHMENT_LOAD_OP_CLEAR`, the first use of that attachment **must** not specify a `layout` equal to `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`
- VUID-VkRenderPassCreateInfo-pAttachments-02511
For any member of `pAttachments` with a `stencilLoadOp` equal to `VK_ATTACHMENT_LOAD_OP_CLEAR`, the first use of that attachment **must** not specify a `layout` equal to `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`
- VUID-VkRenderPassCreateInfo-pDependencies-00837
For any element of `pDependencies`, if the `srcSubpass` is not `VK_SUBPASS_EXTERNAL`, all stage flags included in the `srcStageMask` member of that dependency **must** be a pipeline stage supported by the `pipeline` identified by the `pipelineBindPoint` member of the source subpass
- VUID-VkRenderPassCreateInfo-pDependencies-00838
For any element of `pDependencies`, if the `dstSubpass` is not `VK_SUBPASS_EXTERNAL`, all stage flags included in the `dstStageMask` member of that dependency **must** be a pipeline stage supported by the `pipeline` identified by the `pipelineBindPoint` member of the destination subpass
- VUID-VkRenderPassCreateInfo-pDependencies-06866

For any element of `pDependencies`, if its `srcSubpass` is not `VK_SUBPASS_EXTERNAL`, it **must** be less than `subpassCount`

- VUID-VkRenderPassCreateInfo-pDependencies-06867

For any element of `pDependencies`, if its `dstSubpass` is not `VK_SUBPASS_EXTERNAL`, it **must** be less than `subpassCount`

Valid Usage (Implicit)

- VUID-VkRenderPassCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_RENDER_PASS_CREATE_INFO`
- VUID-VkRenderPassCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkRenderPassCreateInfo-flags-zeroBitmask
`flags` **must** be `0`
- VUID-VkRenderPassCreateInfo-pAttachments-parameter
If `attachmentCount` is not `0`, `pAttachments` **must** be a valid pointer to an array of `attachmentCount` valid `VkAttachmentDescription` structures
- VUID-VkRenderPassCreateInfo-pSubpasses-parameter
`pSubpasses` **must** be a valid pointer to an array of `subpassCount` valid `VkSubpassDescription` structures
- VUID-VkRenderPassCreateInfo-pDependencies-parameter
If `dependencyCount` is not `0`, `pDependencies` **must** be a valid pointer to an array of `dependencyCount` valid `VkSubpassDependency` structures
- VUID-VkRenderPassCreateInfo-subpassCount-arraylength
`subpassCount` **must** be greater than `0`

Bits which **can** be set in `VkRenderPassCreateInfo::flags`, describing additional properties of the render pass, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkRenderPassCreateFlagBits {
} VkRenderPassCreateFlagBits;
```

NOTE

All bits for this type are defined by extensions, and none of those extensions are enabled in this build of the specification.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkRenderPassCreateFlags;
```

`VkRenderPassCreateFlags` is a bitmask type for setting a mask of zero or more `VkRenderPassCreateFlagBits`.

The `VkAttachmentDescription` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkAttachmentDescription {
    VkAttachmentDescriptionFlags    flags;
    VkFormat                       format;
    VkSampleCountFlagBits          samples;
    VkAttachmentLoadOp              loadOp;
    VkAttachmentStoreOp             storeOp;
    VkAttachmentLoadOp              stencilLoadOp;
    VkAttachmentStoreOp             stencilStoreOp;
    VkImageLayout                   initialLayout;
    VkImageLayout                   finalLayout;
} VkAttachmentDescription;
```

- `flags` is a bitmask of `VkAttachmentDescriptionFlagBits` specifying additional properties of the attachment.
- `format` is a `VkFormat` value specifying the format of the image view that will be used for the attachment.
- `samples` is a `VkSampleCountFlagBits` value specifying the number of samples of the image.
- `loadOp` is a `VkAttachmentLoadOp` value specifying how the contents of color and depth components of the attachment are treated at the beginning of the subpass where it is first used.
- `storeOp` is a `VkAttachmentStoreOp` value specifying how the contents of color and depth components of the attachment are treated at the end of the subpass where it is last used.
- `stencilLoadOp` is a `VkAttachmentLoadOp` value specifying how the contents of stencil components of the attachment are treated at the beginning of the subpass where it is first used.
- `stencilStoreOp` is a `VkAttachmentStoreOp` value specifying how the contents of stencil components of the attachment are treated at the end of the last subpass where it is used.
- `initialLayout` is the layout the attachment image subresource will be in when a render pass instance begins.
- `finalLayout` is the layout the attachment image subresource will be transitioned to when a render pass instance ends.

If the attachment uses a color format, then `loadOp` and `storeOp` are used, and `stencilLoadOp` and `stencilStoreOp` are ignored. If the format has depth and/or stencil components, `loadOp` and `storeOp` apply only to the depth data, while `stencilLoadOp` and `stencilStoreOp` define how the stencil data is handled. `loadOp` and `stencilLoadOp` define the `load operations` for the attachment. `storeOp` and `stencilStoreOp` define the `store operations` for the attachment. If an attachment is not used by any subpass, `loadOp`, `storeOp`, `stencilStoreOp`, and `stencilLoadOp` will be ignored for that attachment, and no load or store ops will be performed. However, any transition specified by `initialLayout` and `finalLayout` will still be executed.

If `flags` includes `VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT`, then the attachment is treated as if it shares physical memory with another attachment in the same render pass. This information limits the ability of the implementation to reorder certain operations (like layout transitions and the

loadOp) such that it is not improperly reordered against other uses of the same physical memory via a different attachment. This is described in more detail below.

If a render pass uses multiple attachments that alias the same device memory, those attachments **must** each include the `VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT` bit in their attachment description flags. Attachments aliasing the same memory occurs in multiple ways:

- Multiple attachments being assigned the same image view as part of framebuffer creation.
- Attachments using distinct image views that correspond to the same image subresource of an image.
- Attachments using views of distinct image subresources which are bound to overlapping memory ranges.

NOTE

Render passes **must** include subpass dependencies (either directly or via a subpass dependency chain) between any two subpasses that operate on the same attachment or aliasing attachments and those subpass dependencies **must** include execution and memory dependencies separating uses of the aliases, if at least one of those subpasses writes to one of the aliases. These dependencies **must** not include the `VK_DEPENDENCY_BY_REGION_BIT` if the aliases are views of distinct image subresources which overlap in memory.

Multiple attachments that alias the same memory **must** not be used in a single subpass. A given attachment index **must** not be used multiple times in a single subpass, with one exception: two subpass attachments **can** use the same attachment index if at least one use is as an input attachment and neither use is as a resolve or preserve attachment. In other words, the same view **can** be used simultaneously as an input and color or depth/stencil attachment, but **must** not be used as multiple color or depth/stencil attachments nor as resolve or preserve attachments.

If a set of attachments alias each other, then all except the first to be used in the render pass **must** use an `initialLayout` of `VK_IMAGE_LAYOUT_UNDEFINED`, since the earlier uses of the other aliases make their contents undefined. Once an alias has been used and a different alias has been used after it, the first alias **must** not be used in any later subpasses. However, an application **can** assign the same image view to multiple aliasing attachment indices, which allows that image view to be used multiple times even if other aliases are used in between.

NOTE

Once an attachment needs the `VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT` bit, there **should** be no additional cost of introducing additional aliases, and using these additional aliases **may** allow more efficient clearing of the attachments on multiple uses via `VK_ATTACHMENT_LOAD_OP_CLEAR`.

Valid Usage

- VUID-VkAttachmentDescription-format-06699
If `format` includes a color or depth component and `loadOp` is `VK_ATTACHMENT_LOAD_OP_LOAD`, then `initialLayout` **must** not be `VK_IMAGE_LAYOUT_UNDEFINED`
- VUID-VkAttachmentDescription-finalLayout-00843
`finalLayout` **must** not be `VK_IMAGE_LAYOUT_UNDEFINED` or `VK_IMAGE_LAYOUT_PREINITIALIZED`

- VUID-VkAttachmentDescription-format-03280
If `format` is a color format, `initialLayout` **must** not be `VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`
- VUID-VkAttachmentDescription-format-03281
If `format` is a depth/stencil format, `initialLayout` **must** not be `VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL`
- VUID-VkAttachmentDescription-format-03282
If `format` is a color format, `finalLayout` **must** not be `VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`
- VUID-VkAttachmentDescription-format-03283
If `format` is a depth/stencil format, `finalLayout` **must** not be `VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL`
- VUID-VkAttachmentDescription-samples-08745
`samples` **must** be a valid `VkSampleCountFlagBits` value that is set in `imageCreateSampleCounts` (as defined in [Image Creation Limits](#)) for the given `format`
- VUID-VkAttachmentDescription-format-06698
`format` **must** not be `VK_FORMAT_UNDEFINED`
- VUID-VkAttachmentDescription-format-06700
If `format` includes a stencil component and `stencilLoadOp` is `VK_ATTACHMENT_LOAD_OP_LOAD`, then `initialLayout` **must** not be `VK_IMAGE_LAYOUT_UNDEFINED`
- VUID-VkAttachmentDescription-format-03292
If `format` is a depth/stencil format which includes only the stencil component, `initialLayout` **must** not be `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`
- VUID-VkAttachmentDescription-format-03293
If `format` is a depth/stencil format which includes only the stencil component, `finalLayout` **must** not be `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`
- VUID-VkAttachmentDescription-format-06242
If `format` is a depth/stencil format which includes both depth and stencil components, `initialLayout` **must** not be `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`
- VUID-VkAttachmentDescription-format-06243
If `format` is a depth/stencil format which includes both depth and stencil components, `finalLayout` **must** not be `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`

Valid Usage (Implicit)

- VUID-VkAttachmentDescription-flags-parameter
`flags` **must** be a valid combination of `VkAttachmentDescriptionFlagBits` values

- VUID-VkAttachmentDescription-format-parameter **format** must be a valid [VkFormat](#) value
- VUID-VkAttachmentDescription-samples-parameter **samples** must be a valid [VkSampleCountFlagBits](#) value
- VUID-VkAttachmentDescription-loadOp-parameter **loadOp** must be a valid [VkAttachmentLoadOp](#) value
- VUID-VkAttachmentDescription-storeOp-parameter **storeOp** must be a valid [VkAttachmentStoreOp](#) value
- VUID-VkAttachmentDescription-stencilLoadOp-parameter **stencilLoadOp** must be a valid [VkAttachmentLoadOp](#) value
- VUID-VkAttachmentDescription-stencilStoreOp-parameter **stencilStoreOp** must be a valid [VkAttachmentStoreOp](#) value
- VUID-VkAttachmentDescription-initialLayout-parameter **initialLayout** must be a valid [VkImageLayout](#) value
- VUID-VkAttachmentDescription-finalLayout-parameter **finalLayout** must be a valid [VkImageLayout](#) value

Bits which **can** be set in [VkAttachmentDescription::flags](#), describing additional properties of the attachment, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkAttachmentDescriptionFlagBits {
    VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT = 0x00000001,
} VkAttachmentDescriptionFlagBits;
```

- **VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT** specifies that the attachment aliases the same device memory as other attachments.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkAttachmentDescriptionFlags;
```

[VkAttachmentDescriptionFlags](#) is a bitmask type for setting a mask of zero or more [VkAttachmentDescriptionFlagBits](#).

The [VkSubpassDescription](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSubpassDescription {
    VkSubpassDescriptionFlags    flags;
    VkPipelineBindPoint          pipelineBindPoint;
    uint32_t                     inputAttachmentCount;
    const VkAttachmentReference* pInputAttachments;
    uint32_t                     colorAttachmentCount;
    const VkAttachmentReference* pColorAttachments;
    const VkAttachmentReference* pResolveAttachments;
    const VkAttachmentReference* pDepthStencilAttachment;
    uint32_t                     preserveAttachmentCount;
    const uint32_t*              pPreserveAttachments;
} VkSubpassDescription;
```

- `flags` is a bitmask of `VkSubpassDescriptionFlagBits` specifying usage of the subpass.
- `pipelineBindPoint` is a `VkPipelineBindPoint` value specifying the pipeline type supported for this subpass.
- `inputAttachmentCount` is the number of input attachments.
- `pInputAttachments` is a pointer to an array of `VkAttachmentReference` structures defining the input attachments for this subpass and their layouts.
- `colorAttachmentCount` is the number of color attachments.
- `pColorAttachments` is a pointer to an array of `colorAttachmentCount` `VkAttachmentReference` structures defining the color attachments for this subpass and their layouts.
- `pResolveAttachments` is `NULL` or a pointer to an array of `colorAttachmentCount` `VkAttachmentReference` structures defining the resolve attachments for this subpass and their layouts.
- `pDepthStencilAttachment` is a pointer to a `VkAttachmentReference` structure specifying the depth/stencil attachment for this subpass and its layout.
- `preserveAttachmentCount` is the number of preserved attachments.
- `pPreserveAttachments` is a pointer to an array of `preserveAttachmentCount` render pass attachment indices identifying attachments that are not used by this subpass, but whose contents **must** be preserved throughout the subpass.

Each element of the `pInputAttachments` array corresponds to an input attachment index in a fragment shader, i.e. if a shader declares an image variable decorated with a `InputAttachmentIndex` value of `X`, then it uses the attachment provided in `pInputAttachments[X]`. Input attachments **must** also be bound to the pipeline in a descriptor set. If the `attachment` member of any element of `pInputAttachments` is `VK_ATTACHMENT_UNUSED`, the application **must** not read from the corresponding input attachment index. Fragment shaders **can** use subpass input variables to access the contents of an input attachment at the fragment's (x, y, layer) framebuffer coordinates.

Each element of the `pColorAttachments` array corresponds to an output location in the shader, i.e. if the shader declares an output variable decorated with a `Location` value of `X`, then it uses the attachment provided in `pColorAttachments[X]`. If the `attachment` member of any element of

`pColorAttachments` is `VK_ATTACHMENT_UNUSED`, then writes to the corresponding location by a fragment shader are discarded.

If `pResolveAttachments` is not `NULL`, each of its elements corresponds to a color attachment (the element in `pColorAttachments` at the same index), and a [multisample resolve operation](#) is defined for each attachment unless the resolve attachment index is `VK_ATTACHMENT_UNUSED`.

If `pDepthStencilAttachment` is `NULL`, or if its attachment index is `VK_ATTACHMENT_UNUSED`, it indicates that no depth/stencil attachment will be used in the subpass.

The contents of an attachment within the render area become undefined at the start of a subpass `S` if all of the following conditions are true:

- The attachment is used as a color, depth/stencil, or resolve attachment in any subpass in the render pass.
- There is a subpass `S1` that uses or preserves the attachment, and a subpass dependency from `S1` to `S`.
- The attachment is not used or preserved in subpass `S`.

Once the contents of an attachment become undefined in subpass `S`, they remain undefined for subpasses in subpass dependency chains starting with subpass `S` until they are written again. However, they remain valid for subpasses in other subpass dependency chains starting with subpass `S1` if those subpasses use or preserve the attachment.

Valid Usage

- VUID-VkSubpassDescription-attachment-06912
If the `attachment` member of an element of `pInputAttachments` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL`
- VUID-VkSubpassDescription-attachment-06913
If the `attachment` member of an element of `pColorAttachments` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL`
- VUID-VkSubpassDescription-attachment-06914
If the `attachment` member of an element of `pResolveAttachments` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL`
- VUID-VkSubpassDescription-attachment-06915
If the `attachment` member of `pDepthStencilAttachment` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL`
- VUID-VkSubpassDescription-pipelineBindPoint-04952
`pipelineBindPoint` **must** be `VK_PIPELINE_BIND_POINT_GRAPHICS`
- VUID-VkSubpassDescription-colorAttachmentCount-00845

`colorAttachmentCount` **must** be less than or equal to `VkPhysicalDeviceLimits::maxColorAttachments`

- VUID-VkSubpassDescription-loadOp-00846
If the first use of an attachment in this render pass is as an input attachment, and the attachment is not also used as a color or depth/stencil attachment in the same subpass, then `loadOp` **must** not be `VK_ATTACHMENT_LOAD_OP_CLEAR`
- VUID-VkSubpassDescription-pResolveAttachments-00847
If `pResolveAttachments` is not `NULL`, for each resolve attachment that is not `VK_ATTACHMENT_UNUSED`, the corresponding color attachment **must** not be `VK_ATTACHMENT_UNUSED`
- VUID-VkSubpassDescription-pResolveAttachments-00848
If `pResolveAttachments` is not `NULL`, for each resolve attachment that is not `VK_ATTACHMENT_UNUSED`, the corresponding color attachment **must** not have a sample count of `VK_SAMPLE_COUNT_1_BIT`
- VUID-VkSubpassDescription-pResolveAttachments-00849
If `pResolveAttachments` is not `NULL`, each resolve attachment that is not `VK_ATTACHMENT_UNUSED` **must** have a sample count of `VK_SAMPLE_COUNT_1_BIT`
- VUID-VkSubpassDescription-pResolveAttachments-00850
If `pResolveAttachments` is not `NULL`, each resolve attachment that is not `VK_ATTACHMENT_UNUSED` **must** have the same `VkFormat` as its corresponding color attachment
- VUID-VkSubpassDescription-pColorAttachments-09430
All attachments in `pColorAttachments` that are not `VK_ATTACHMENT_UNUSED` **must** have the same sample count
- VUID-VkSubpassDescription-pInputAttachments-02647
All attachments in `pInputAttachments` that are not `VK_ATTACHMENT_UNUSED` **must** have image formats whose `potential format features` contain at least `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT` or `VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT`
- VUID-VkSubpassDescription-pColorAttachments-02648
All attachments in `pColorAttachments` that are not `VK_ATTACHMENT_UNUSED` **must** have image formats whose `potential format features` contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT`
- VUID-VkSubpassDescription-pResolveAttachments-02649
All attachments in `pResolveAttachments` that are not `VK_ATTACHMENT_UNUSED` **must** have image formats whose `potential format features` contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT`
- VUID-VkSubpassDescription-pDepthStencilAttachment-02650
If `pDepthStencilAttachment` is not `NULL` and the attachment is not `VK_ATTACHMENT_UNUSED` then it **must** have an image format whose `potential format features` contain `VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT`
- VUID-VkSubpassDescription-pDepthStencilAttachment-01418
If `pDepthStencilAttachment` is not `VK_ATTACHMENT_UNUSED` and any attachments in `pColorAttachments` are not `VK_ATTACHMENT_UNUSED`, they **must** have the same sample count

- VUID-VkSubpassDescription-attachment-00853
Each element of `pPreserveAttachments` **must** not be `VK_ATTACHMENT_UNUSED`
- VUID-VkSubpassDescription-pPreserveAttachments-00854
Each element of `pPreserveAttachments` **must** not also be an element of any other member of the subpass description
- VUID-VkSubpassDescription-layout-02519
If any attachment is used by more than one `VkAttachmentReference` member, then each use **must** use the same `layout`
- VUID-VkSubpassDescription-pDepthStencilAttachment-04438
`pDepthStencilAttachment` and `pColorAttachments` **must** not contain references to the same attachment

Valid Usage (Implicit)

- VUID-VkSubpassDescription-flags-zeroBitmask
`flags` **must** be 0
- VUID-VkSubpassDescription-pipelineBindPoint-parameter
`pipelineBindPoint` **must** be a valid `VkPipelineBindPoint` value
- VUID-VkSubpassDescription-pInputAttachments-parameter
If `inputAttachmentCount` is not 0, `pInputAttachments` **must** be a valid pointer to an array of `inputAttachmentCount` valid `VkAttachmentReference` structures
- VUID-VkSubpassDescription-pColorAttachments-parameter
If `colorAttachmentCount` is not 0, `pColorAttachments` **must** be a valid pointer to an array of `colorAttachmentCount` valid `VkAttachmentReference` structures
- VUID-VkSubpassDescription-pResolveAttachments-parameter
If `colorAttachmentCount` is not 0, and `pResolveAttachments` is not `NULL`, `pResolveAttachments` **must** be a valid pointer to an array of `colorAttachmentCount` valid `VkAttachmentReference` structures
- VUID-VkSubpassDescription-pDepthStencilAttachment-parameter
If `pDepthStencilAttachment` is not `NULL`, `pDepthStencilAttachment` **must** be a valid pointer to a valid `VkAttachmentReference` structure
- VUID-VkSubpassDescription-pPreserveAttachments-parameter
If `preserveAttachmentCount` is not 0, `pPreserveAttachments` **must** be a valid pointer to an array of `preserveAttachmentCount` `uint32_t` values

Bits which **can** be set in `VkSubpassDescription::flags`, specifying usage of the subpass, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSubpassDescriptionFlagBits {
} VkSubpassDescriptionFlagBits;
```

NOTE

All bits for this type are defined by extensions, and none of those extensions are

enabled in this build of the specification.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkSubpassDescriptionFlags;
```

`VkSubpassDescriptionFlags` is a bitmask type for setting a mask of zero or more `VkSubpassDescriptionFlagBits`.

The `VkAttachmentReference` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkAttachmentReference {
    uint32_t      attachment;
    VkImageLayout  layout;
} VkAttachmentReference;
```

- `attachment` is either an integer value identifying an attachment at the corresponding index in `VkRenderPassCreateInfo::pAttachments`, or `VK_ATTACHMENT_UNUSED` to signify that this attachment is not used.
- `layout` is a `VkImageLayout` value specifying the layout the attachment uses during the subpass.

Valid Usage

- VUID-VkAttachmentReference-layout-03077

If `attachment` is not `VK_ATTACHMENT_UNUSED`, `layout` **must** not be `VK_IMAGE_LAYOUT_UNDEFINED`, `VK_IMAGE_LAYOUT_PREINITIALIZED`, or `VK_IMAGE_LAYOUT_PRESENT_SRC_KHR`

Valid Usage (Implicit)

- VUID-VkAttachmentReference-layout-parameter

`layout` **must** be a valid `VkImageLayout` value

`VK_SUBPASS_EXTERNAL` is a special subpass index value expanding synchronization scope outside a subpass. It is described in more detail by `VkSubpassDependency`.

```
#define VK_SUBPASS_EXTERNAL          (~0U)
```

The `VkSubpassDependency` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSubpassDependency {
    uint32_t          srcSubpass;
    uint32_t          dstSubpass;
    VkPipelineStageFlags srcStageMask;
    VkPipelineStageFlags dstStageMask;
    VkAccessFlags      srcAccessMask;
    VkAccessFlags      dstAccessMask;
    VkDependencyFlags  dependencyFlags;
} VkSubpassDependency;
```

- `srcSubpass` is the subpass index of the first subpass in the dependency, or `VK_SUBPASS_EXTERNAL`.
- `dstSubpass` is the subpass index of the second subpass in the dependency, or `VK_SUBPASS_EXTERNAL`.
- `srcStageMask` is a bitmask of `VkPipelineStageFlagBits` specifying the [source stage mask](#).
- `dstStageMask` is a bitmask of `VkPipelineStageFlagBits` specifying the [destination stage mask](#).
- `srcAccessMask` is a bitmask of `VkAccessFlagBits` specifying a [source access mask](#).
- `dstAccessMask` is a bitmask of `VkAccessFlagBits` specifying a [destination access mask](#).
- `dependencyFlags` is a bitmask of `VkDependencyFlagBits`.

If `srcSubpass` is equal to `dstSubpass` then the `VkSubpassDependency` does not directly define a [dependency](#). Instead, it enables pipeline barriers to be used in a render pass instance within the identified subpass, where the scopes of one pipeline barrier **must** be a subset of those described by one subpass dependency. Subpass dependencies specified in this way that include [framebuffer-space stages](#) in the `srcStageMask` **must** only include [framebuffer-space stages](#) in `dstStageMask`, and **must** include `VK_DEPENDENCY_BY_REGION_BIT`.

If `srcSubpass` and `dstSubpass` are not equal, when a render pass instance which includes a subpass dependency is submitted to a queue, it defines a [dependency](#) between the subpasses identified by `srcSubpass` and `dstSubpass`.

If `srcSubpass` is equal to `VK_SUBPASS_EXTERNAL`, the first [synchronization scope](#) includes commands that occur earlier in [submission order](#) than the `vkCmdBeginRenderPass` used to begin the render pass instance. Otherwise, the first set of commands includes all commands submitted as part of the subpass instance identified by `srcSubpass` and any [load](#), [store](#), or [multisample resolve](#) operations on attachments used in `srcSubpass`. In either case, the first synchronization scope is limited to operations on the pipeline stages determined by the [source stage mask](#) specified by `srcStageMask`.

If `dstSubpass` is equal to `VK_SUBPASS_EXTERNAL`, the second [synchronization scope](#) includes commands that occur later in [submission order](#) than the `vkCmdEndRenderPass` used to end the render pass instance. Otherwise, the second set of commands includes all commands submitted as part of the subpass instance identified by `dstSubpass` and any [load](#), [store](#), and [multisample resolve](#) operations on attachments used in `dstSubpass`. In either case, the second synchronization scope is limited to operations on the pipeline stages determined by the [destination stage mask](#) specified by `dstStageMask`.

The first [access scope](#) is limited to accesses in the pipeline stages determined by the [source stage mask](#) specified by `srcStageMask`. It is also limited to access types in the [source access mask](#) specified by `srcAccessMask`.

The second [access scope](#) is limited to accesses in the pipeline stages determined by the [destination stage mask](#) specified by `dstStageMask`. It is also limited to access types in the [destination access mask](#) specified by `dstAccessMask`.

The [availability and visibility operations](#) defined by a subpass dependency affect the execution of [image layout transitions](#) within the render pass.

NOTE

For non-attachment resources, the memory dependency expressed by subpass dependency is nearly identical to that of a [VkMemoryBarrier](#) (with matching `srcAccessMask` and `dstAccessMask` parameters) submitted as a part of a [vkCmdPipelineBarrier](#) (with matching `srcStageMask` and `dstStageMask` parameters). The only difference being that its scopes are limited to the identified subpasses rather than potentially affecting everything before and after.

For attachments however, subpass dependencies work more like a [VkImageMemoryBarrier](#) defined similarly to the [VkMemoryBarrier](#) above, the queue family indices set to `VK_QUEUE_FAMILY_IGNORED`, and layouts as follows:

- The equivalent to `oldLayout` is the attachment's layout according to the subpass description for `srcSubpass`.
- The equivalent to `newLayout` is the attachment's layout according to the subpass description for `dstSubpass`.

Valid Usage

- VUID-VkSubpassDependency-srcStageMask-04090
If the `geometryShader` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT`
- VUID-VkSubpassDependency-srcStageMask-04091
If the `tessellationShader` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT` or `VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT`
- VUID-VkSubpassDependency-srcStageMask-04996
`srcStageMask` **must** not be 0
- VUID-VkSubpassDependency-dstStageMask-04090
If the `geometryShader` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT`
- VUID-VkSubpassDependency-dstStageMask-04091
If the `tessellationShader` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT` or `VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT`

- VUID-VkSubpassDependency-dstStageMask-04996
`dstStageMask` **must** not be 0
- VUID-VkSubpassDependency-srcSubpass-00864
`srcSubpass` **must** be less than or equal to `dstSubpass`, unless one of them is `VK_SUBPASS_EXTERNAL`, to avoid cyclic dependencies and ensure a valid execution order
- VUID-VkSubpassDependency-srcSubpass-00865
`srcSubpass` and `dstSubpass` **must** not both be equal to `VK_SUBPASS_EXTERNAL`
- VUID-VkSubpassDependency-srcSubpass-06809
If `srcSubpass` is equal to `dstSubpass` and `srcStageMask` includes a `framebuffer-space` stage, `dstStageMask` **must** only contain `framebuffer-space` stages
- VUID-VkSubpassDependency-srcAccessMask-00868
Any access flag included in `srcAccessMask` **must** be supported by one of the pipeline stages in `srcStageMask`, as specified in the [table of supported access types](#)
- VUID-VkSubpassDependency-dstAccessMask-00869
Any access flag included in `dstAccessMask` **must** be supported by one of the pipeline stages in `dstStageMask`, as specified in the [table of supported access types](#)
- VUID-VkSubpassDependency-srcSubpass-02243
If `srcSubpass` equals `dstSubpass`, and `srcStageMask` and `dstStageMask` both include a `framebuffer-space` stage, then `dependencyFlags` **must** include `VK_DEPENDENCY_BY_REGION_BIT`

Valid Usage (Implicit)

- VUID-VkSubpassDependency-srcStageMask-parameter
`srcStageMask` **must** be a valid combination of `VkPipelineStageFlagBits` values
- VUID-VkSubpassDependency-dstStageMask-parameter
`dstStageMask` **must** be a valid combination of `VkPipelineStageFlagBits` values
- VUID-VkSubpassDependency-srcAccessMask-parameter
`srcAccessMask` **must** be a valid combination of `VkAccessFlagBits` values
- VUID-VkSubpassDependency-dstAccessMask-parameter
`dstAccessMask` **must** be a valid combination of `VkAccessFlagBits` values
- VUID-VkSubpassDependency-dependencyFlags-parameter
`dependencyFlags` **must** be a valid combination of `VkDependencyFlagBits` values

If there is no subpass dependency from `VK_SUBPASS_EXTERNAL` to the first subpass that uses an attachment, then an implicit subpass dependency exists from `VK_SUBPASS_EXTERNAL` to the first subpass it is used in. The implicit subpass dependency only exists if there exists an automatic layout transition away from `initialLayout`. The subpass dependency operates as if defined with the following parameters:

```
VkSubpassDependency implicitDependency = {
    .srcSubpass = VK_SUBPASS_EXTERNAL,
    .dstSubpass = firstSubpass, // First subpass attachment is used in
```

```

        .srcStageMask = VK_PIPELINE_STAGE_TOP_OF_PIPE_BIT,
        .dstStageMask = VK_PIPELINE_STAGE_ALL_COMMANDS_BIT,
        .srcAccessMask = 0,
        .dstAccessMask = VK_ACCESS_INPUT_ATTACHMENT_READ_BIT |
                        VK_ACCESS_COLOR_ATTACHMENT_READ_BIT |
                        VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT |
                        VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_READ_BIT |
                        VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT,
        .dependencyFlags = 0
    };

```

Similarly, if there is no subpass dependency from the last subpass that uses an attachment to `VK_SUBPASS_EXTERNAL`, then an implicit subpass dependency exists from the last subpass it is used in to `VK_SUBPASS_EXTERNAL`. The implicit subpass dependency only exists if there exists an automatic layout transition into `finalLayout`. The subpass dependency operates as if defined with the following parameters:

```

VkSubpassDependency implicitDependency = {
    .srcSubpass = lastSubpass, // Last subpass attachment is used in
    .dstSubpass = VK_SUBPASS_EXTERNAL,
    .srcStageMask = VK_PIPELINE_STAGE_ALL_COMMANDS_BIT,
    .dstStageMask = VK_PIPELINE_STAGE_BOTTOM_OF_PIPE_BIT,
    .srcAccessMask = VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT |
                    VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT,
    .dstAccessMask = 0,
    .dependencyFlags = 0
};

```

As subpasses **may** overlap or execute out of order with regards to other subpasses unless a subpass dependency chain describes otherwise, the layout transitions required between subpasses **cannot** be known to an application. Instead, an application provides the layout that each attachment **must** be in at the start and end of a render pass, and the layout it **must** be in during each subpass it is used in. The implementation then **must** execute layout transitions between subpasses in order to guarantee that the images are in the layouts required by each subpass, and in the final layout at the end of the render pass.

Automatic layout transitions apply to the entire image subresource attached to the framebuffer. If the attachment is a view of a 1D or 2D image, the automatic layout transitions apply to the number of layers specified by `VkFramebufferCreateInfo::layers`.

Automatic layout transitions away from the layout used in a subpass happen-after the availability operations for all dependencies with that subpass as the `srcSubpass`.

Automatic layout transitions into the layout used in a subpass happen-before the visibility operations for all dependencies with that subpass as the `dstSubpass`.

Automatic layout transitions away from `initialLayout` happen-after the availability operations for all dependencies with a `srcSubpass` equal to `VK_SUBPASS_EXTERNAL`, where `dstSubpass` uses the

attachment that will be transitioned. For attachments created with `VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT`, automatic layout transitions away from `initialLayout` happen-after the availability operations for all dependencies with a `srcSubpass` equal to `VK_SUBPASS_EXTERNAL`, where `dstSubpass` uses any aliased attachment.

Automatic layout transitions into `finalLayout` happen-before the visibility operations for all dependencies with a `dstSubpass` equal to `VK_SUBPASS_EXTERNAL`, where `srcSubpass` uses the attachment that will be transitioned. For attachments created with `VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT`, automatic layout transitions into `finalLayout` happen-before the visibility operations for all dependencies with a `dstSubpass` equal to `VK_SUBPASS_EXTERNAL`, where `srcSubpass` uses any aliased attachment.

If two subpasses use the same attachment, and both subpasses use the attachment in a read-only layout, no subpass dependency needs to be specified between those subpasses. If an implementation treats those layouts separately, it **must** insert an implicit subpass dependency between those subpasses to separate the uses in each layout. The subpass dependency operates as if defined with the following parameters:

```
// Used for input attachments
VkPipelineStageFlags inputAttachmentStages = VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT;
VkAccessFlags inputAttachmentDstAccess = VK_ACCESS_INPUT_ATTACHMENT_READ_BIT;

// Used for depth/stencil attachments
VkPipelineStageFlags depthStencilAttachmentStages =
VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT |
VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT;
VkAccessFlags depthStencilAttachmentDstAccess =
VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_READ_BIT;

VkSubpassDependency implicitDependency = {
    .srcSubpass = firstSubpass;
    .dstSubpass = secondSubpass;
    .srcStageMask = inputAttachmentStages | depthStencilAttachmentStages;
    .dstStageMask = inputAttachmentStages | depthStencilAttachmentStages;
    .srcAccessMask = 0;
    .dstAccessMask = inputAttachmentDstAccess | depthStencilAttachmentDstAccess;
    .dependencyFlags = 0;
};
```

To destroy a render pass, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyRenderPass(
    VkDevice                device,
    VkRenderPass            renderPass,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that destroys the render pass.

- `renderPass` is the handle of the render pass to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyRenderPass-renderPass-00873
All submitted commands that refer to `renderPass` **must** have completed execution
- VUID-vkDestroyRenderPass-renderPass-00874
If `VkAllocationCallbacks` were provided when `renderPass` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyRenderPass-renderPass-00875
If no `VkAllocationCallbacks` were provided when `renderPass` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyRenderPass-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkDestroyRenderPass-renderPass-parameter
If `renderPass` is not `VK_NULL_HANDLE`, `renderPass` **must** be a valid `VkRenderPass` handle
- VUID-vkDestroyRenderPass-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkDestroyRenderPass-renderPass-parent
If `renderPass` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `renderPass` **must** be externally synchronized

8.2. Render Pass Compatibility

Framebuffers and graphics pipelines are created based on a specific render pass object. They **must** only be used with that render pass object, or one compatible with it.

Two attachment references are compatible if they have matching format and sample count, or are both `VK_ATTACHMENT_UNUSED` or the pointer that would contain the reference is `NULL`.

Two arrays of attachment references are compatible if all corresponding pairs of attachments are compatible. If the arrays are of different lengths, attachment references not present in the smaller array are treated as `VK_ATTACHMENT_UNUSED`.

Two render passes are compatible if their corresponding color, input, resolve, and depth/stencil attachment references are compatible and if they are otherwise identical except for:

- Initial and final image layout in attachment descriptions
- Load and store operations in attachment descriptions
- Image layout in attachment references

As an additional special case, if two render passes have a single subpass, the resolve attachment reference compatibility requirements are ignored.

A framebuffer is compatible with a render pass if it was created using the same render pass or a compatible render pass.

8.3. Framebuffers

Render passes operate in conjunction with *framebuffers*. Framebuffers represent a collection of specific memory attachments that a render pass instance uses.

Framebuffers are represented by `VkFramebuffer` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkFramebuffer)
```

To create a framebuffer, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateFramebuffer(
    VkDevice                                device,
    const VkFramebufferCreateInfo*          pCreateInfo,
    const VkAllocationCallbacks*           pAllocator,
    VkFramebuffer*                          pFramebuffer);
```

- `device` is the logical device that creates the framebuffer.
- `pCreateInfo` is a pointer to a `VkFramebufferCreateInfo` structure describing additional information about framebuffer creation.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.
- `pFramebuffer` is a pointer to a `VkFramebuffer` handle in which the resulting framebuffer object is returned.

Valid Usage

- VUID-vkCreateFramebuffer-device-10002
`device` **must** support at least one queue family with the `VK_QUEUE_GRAPHICS_BIT` capability
- VUID-vkCreateFramebuffer-pCreateInfo-02777

If `pCreateInfo->flags` does not include `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, and `attachmentCount` is not `0`, each element of `pCreateInfo->pAttachments` **must** have been created on device

Valid Usage (Implicit)

- VUID-vkCreateFramebuffer-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkCreateFramebuffer-pCreateInfo-parameter
`pCreateInfo` **must** be a valid pointer to a valid `VkFramebufferCreateInfo` structure
- VUID-vkCreateFramebuffer-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkCreateFramebuffer-pFramebuffer-parameter
`pFramebuffer` **must** be a valid pointer to a `VkFramebuffer` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkFramebufferCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkFramebufferCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkFramebufferCreateFlags flags;
    VkRenderPass        renderPass;
    uint32_t            attachmentCount;
    const VkImageView*  pAttachments;
    uint32_t            width;
    uint32_t            height;
    uint32_t            layers;
} VkFramebufferCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkFramebufferCreateFlagBits`

- **renderPass** is a render pass defining what render passes the framebuffer will be compatible with. See [Render Pass Compatibility](#) for details.
- **attachmentCount** is the number of attachments.
- **pAttachments** is a pointer to an array of [VkImageView](#) handles, each of which will be used as the corresponding attachment in a render pass instance.
- **width**, **height** and **layers** define the dimensions of the framebuffer.

It is legal for a subpass to use no color or depth/stencil attachments, either because it has no attachment references or because all of them are **VK_ATTACHMENT_UNUSED**. This kind of subpass **can** use shader side effects such as image stores and atomics to produce an output. In this case, the subpass continues to use the **width**, **height**, and **layers** of the framebuffer to define the dimensions of the rendering area, and the **rasterizationSamples** from each pipeline's [VkPipelineMultisampleStateCreateInfo](#) to define the number of samples used in rasterization; however, if [VkPhysicalDeviceFeatures::variableMultisampleRate](#) is **VK_FALSE**, then all pipelines to be bound with the subpass **must** have the same value for [VkPipelineMultisampleStateCreateInfo::rasterizationSamples](#). In all such cases, **rasterizationSamples** **must** be a valid [VkSampleCountFlagBits](#) value that is set in [VkPhysicalDeviceLimits::framebufferNoAttachmentsSampleCounts](#).

Valid Usage

- VUID-VkFramebufferCreateInfo-attachmentCount-00876
attachmentCount **must** be equal to the attachment count specified in **renderPass**
- VUID-VkFramebufferCreateInfo-flags-02778
If **flags** does not include **VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT** and **attachmentCount** is not 0, **pAttachments** **must** be a valid pointer to an array of **attachmentCount** valid [VkImageView](#) handles
- VUID-VkFramebufferCreateInfo-pAttachments-00877
If **flags** does not include **VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT**, each element of **pAttachments** that is used as a color attachment or resolve attachment by **renderPass** **must** have been created with a **usage** value including **VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT**
- VUID-VkFramebufferCreateInfo-pAttachments-02633
If **flags** does not include **VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT**, each element of **pAttachments** that is used as a depth/stencil attachment by **renderPass** **must** have been created with a **usage** value including **VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT**
- VUID-VkFramebufferCreateInfo-pAttachments-00879
If **renderpass** is not **VK_NULL_HANDLE**, **flags** does not include **VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT**, each element of **pAttachments** that is used as an input attachment by **renderPass** **must** have been created with a **usage** value including **VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT**
- VUID-VkFramebufferCreateInfo-pAttachments-00880
If **flags** does not include **VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT**, each element of **pAttachments** **must** have been created with a **VkFormat** value that matches the **VkFormat** specified by the corresponding **VkAttachmentDescription** in **renderPass**

- VUID-VkFramebufferCreateInfo-pAttachments-00881
If **flags** does not include **VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT**, each element of **pAttachments** **must** have been created with a **samples** value that matches the **samples** value specified by the corresponding **VkAttachmentDescription** in **renderPass**
- VUID-VkFramebufferCreateInfo-flags-04533
If **flags** does not include **VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT**, each element of **pAttachments** that is used as an input, color, resolve, or depth/stencil attachment by **renderPass** **must** have been created with a **VkImageCreateInfo::extent.width** greater than or equal to **width**
- VUID-VkFramebufferCreateInfo-flags-04534
If **flags** does not include **VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT**, each element of **pAttachments** that is used as an input, color, resolve, or depth/stencil attachment by **renderPass** **must** have been created with a **VkImageCreateInfo::extent.height** greater than or equal to **height**
- VUID-VkFramebufferCreateInfo-flags-04535
If **flags** does not include **VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT**, each element of **pAttachments** that is used as an input, color, resolve, or depth/stencil attachment by **renderPass** **must** have been created with a **VkImageViewCreateInfo::subresourceRange.layerCount** greater than or equal to **layers**
- VUID-VkFramebufferCreateInfo-pAttachments-00883
If **flags** does not include **VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT**, each element of **pAttachments** **must** only specify a single mip level
- VUID-VkFramebufferCreateInfo-pAttachments-00884
If **flags** does not include **VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT**, each element of **pAttachments** **must** have been created with the identity swizzle
- VUID-VkFramebufferCreateInfo-width-00885
width **must** be greater than 0
- VUID-VkFramebufferCreateInfo-width-00886
width **must** be less than or equal to **maxFramebufferWidth**
- VUID-VkFramebufferCreateInfo-height-00887
height **must** be greater than 0
- VUID-VkFramebufferCreateInfo-height-00888
height **must** be less than or equal to **maxFramebufferHeight**
- VUID-VkFramebufferCreateInfo-layers-00889
layers **must** be greater than 0
- VUID-VkFramebufferCreateInfo-layers-00890
layers **must** be less than or equal to **maxFramebufferLayers**
- VUID-VkFramebufferCreateInfo-flags-04113
If **flags** does not include **VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT**, each element of **pAttachments** **must** have been created with **VkImageViewCreateInfo::viewType** not equal to **VK_IMAGE_VIEW_TYPE_3D**

Valid Usage (Implicit)

- VUID-VkFramebufferCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_FRAMEBUFFER_CREATE_INFO`
- VUID-VkFramebufferCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkFramebufferCreateInfo-flags-zeroBitmask
`flags` **must** be `0`
- VUID-VkFramebufferCreateInfo-renderPass-parameter
`renderPass` **must** be a valid `VkRenderPass` handle
- VUID-VkFramebufferCreateInfo-commonParent
Both of `renderPass`, and the elements of `pAttachments` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

Bits which **can** be set in `VkFramebufferCreateInfo::flags`, specifying options for framebuffers, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkFramebufferCreateFlagBits {
} VkFramebufferCreateFlagBits;
```

NOTE

All bits for this type are defined by extensions, and none of those extensions are enabled in this build of the specification.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkFramebufferCreateFlags;
```

`VkFramebufferCreateFlags` is a bitmask type for setting a mask of zero or more `VkFramebufferCreateFlagBits`.

To destroy a framebuffer, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyFramebuffer(
    VkDevice          device,
    VkFramebuffer     framebuffer,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that destroys the framebuffer.
- `framebuffer` is the handle of the framebuffer to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyFramebuffer-framebuffer-00892
All submitted commands that refer to **framebuffer** **must** have completed execution
- VUID-vkDestroyFramebuffer-framebuffer-00893
If **VkAllocationCallbacks** were provided when **framebuffer** was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyFramebuffer-framebuffer-00894
If no **VkAllocationCallbacks** were provided when **framebuffer** was created, **pAllocator** **must** be **NULL**

Valid Usage (Implicit)

- VUID-vkDestroyFramebuffer-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkDestroyFramebuffer-framebuffer-parameter
If **framebuffer** is not **VK_NULL_HANDLE**, **framebuffer** **must** be a valid **VkFramebuffer** handle
- VUID-vkDestroyFramebuffer-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid **VkAllocationCallbacks** structure
- VUID-vkDestroyFramebuffer-framebuffer-parent
If **framebuffer** is a valid handle, it **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **framebuffer** **must** be externally synchronized

8.4. Render Pass Load Operations

Render pass load operations define the initial values of an attachment during a render pass instance.

Load operations for attachments with a depth/stencil format execute in the **VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT** pipeline stage. Load operations for attachments with a color format execute in the **VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT** pipeline stage. The load operation for each sample in an attachment happens-before any recorded command which accesses the sample in that render pass instance via that attachment or an alias.

NOTE

Because load operations always happen first, external synchronization with attachment access only needs to synchronize the load operations with previous

commands; not the operations within the render pass instance.

Load operations only update values within the defined render area for the render pass instance. However, any writes performed by a load operation (as defined by its access masks) to a given attachment **may** read and write back any memory locations within the image subresource bound for that attachment. For depth/stencil images, writes to one aspect **may** also result in read-modify-write operations for the other aspect.

NOTE

As entire subresources could be accessed by load operations, applications cannot safely access values outside of the render area during a render pass instance when a load operation that modifies values is used.

Load operations that **can** be used for a render pass are:

```
// Provided by VK_VERSION_1_0
typedef enum VkAttachmentLoadOp {
    VK_ATTACHMENT_LOAD_OP_LOAD = 0,
    VK_ATTACHMENT_LOAD_OP_CLEAR = 1,
    VK_ATTACHMENT_LOAD_OP_DONT_CARE = 2,
} VkAttachmentLoadOp;
```

- **VK_ATTACHMENT_LOAD_OP_LOAD** specifies that the previous contents of the image within the render area will be preserved as the initial values. For attachments with a depth/stencil format, this uses the access type **VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_READ_BIT**. For attachments with a color format, this uses the access type **VK_ACCESS_COLOR_ATTACHMENT_READ_BIT**.
- **VK_ATTACHMENT_LOAD_OP_CLEAR** specifies that the contents within the render area will be cleared to a uniform value, which is specified when a render pass instance is begun. For attachments with a depth/stencil format, this uses the access type **VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT**. For attachments with a color format, this uses the access type **VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT**.
- **VK_ATTACHMENT_LOAD_OP_DONT_CARE** specifies that the previous contents within the area need not be preserved; the contents of the attachment will be undefined inside the render area. For attachments with a depth/stencil format, this uses the access type **VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT**. For attachments with a color format, this uses the access type **VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT**.

During a render pass instance, input and color attachments with color formats that have a component size of 8, 16, or 32 bits **must** be represented in the attachment's format throughout the instance. Attachments with other floating- or fixed-point color formats, or with depth components **may** be represented in a format with a precision higher than the attachment format, but **must** be represented with the same range. When such a component is loaded via the **loadOp**, it will be converted into an implementation-dependent format used by the render pass. Such components **must** be converted from the render pass format, to the format of the attachment, before they are resolved or stored at the end of a render pass instance via **storeOp**. Conversions occur as described in [Numeric Representation and Computation](#) and [Fixed-Point Data Conversions](#).

8.5. Render Pass Store Operations

Render pass store operations define how values written to an attachment during a render pass instance are stored to memory.

Store operations for attachments with a depth/stencil format execute in the `VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT` pipeline stage. Store operations for attachments with a color format execute in the `VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT` pipeline stage. The store operation for each sample in an attachment happens-after any recorded command which accesses the sample via that attachment or an alias.

NOTE

Because store operations always happen after other accesses in a render pass instance, external synchronization with attachment access in an earlier render pass only needs to synchronize with the store operations; not the operations within the render pass instance.

Store operations only update values within the defined render area for the render pass instance. However, any writes performed by a store operation (as defined by its access masks) to a given attachment **may** read and write back any memory locations within the image subresource bound for that attachment. For depth/stencil images, writes to one aspect **may** also result in read-modify-write operations for the other aspect.

NOTE

As entire subresources could be accessed by store operations, applications cannot safely access values outside of the render area via aliased resources during a render pass instance when a store operation that modifies values is used.

Possible values of `VkAttachmentDescription::storeOp` and `stencilStoreOp`, specifying how the contents of the attachment are treated, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkAttachmentStoreOp {
    VK_ATTACHMENT_STORE_OP_STORE = 0,
    VK_ATTACHMENT_STORE_OP_DONT_CARE = 1,
} VkAttachmentStoreOp;
```

- `VK_ATTACHMENT_STORE_OP_STORE` specifies the contents generated during the render pass and within the render area are written to memory. For attachments with a depth/stencil format, this uses the access type `VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT`. For attachments with a color format, this uses the access type `VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT`.
- `VK_ATTACHMENT_STORE_OP_DONT_CARE` specifies the contents within the render area are not needed after rendering, and **may** be discarded; the contents of the attachment will be undefined inside the render area. For attachments with a depth/stencil format, this uses the access type `VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT`. For attachments with a color format, this uses the access type `VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT`.

NOTE

`VK_ATTACHMENT_STORE_OP_DONT_CARE` **can** cause contents generated during previous render passes to be discarded before reaching memory, even if no write to the

attachment occurs during the current render pass.

8.6. Render Pass Multisample Resolve Operations

Render pass multisample resolve operations combine sample values from a single pixel in a multisample attachment and store the result to the corresponding pixel in a single sample attachment.

Multisample resolve operations for attachments execute in the `VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT` pipeline stage. A final resolve operation for all pixels in the render area happens after any recorded command which writes a pixel via the multisample attachment to be resolved or an explicit alias of it in the subpass that it is specified. Any single sample attachment specified for use in a multisample resolve operation **may** have its contents modified at any point once rendering begins for the render pass instance. Reads from the multisample attachment can be synchronized with `VK_ACCESS_COLOR_ATTACHMENT_READ_BIT`. Access to the single sample attachment can be synchronized with `VK_ACCESS_COLOR_ATTACHMENT_READ_BIT` and `VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT`.

When using render pass objects, a subpass dependency specified with the above pipeline stages and access flags will ensure synchronization with multisample resolve operations for any attachments that were last accessed by that subpass. This allows later subpasses to read resolved values as input attachments.

Resolve operations only update values within the defined render area for the render pass instance. However, any writes performed by a resolve operation (as defined by its access masks) to a given attachment **may** read and write back any memory locations within the image subresource bound for that attachment. For depth/stencil images, writes to one aspect **may** also result in read-modify-write operations for the other aspect.

NOTE

As entire subresources could be accessed by multisample resolve operations, applications cannot safely access values outside of the render area via aliased resources during a render pass instance when a multisample resolve operation is performed.

The average value of samples for a given pixel in the multisample attachment is taken and written to the single sample attachment.

8.7. Render Pass Commands

An application records the commands for a render pass instance one subpass at a time, by beginning a render pass instance, iterating over the subpasses to record commands for that subpass, and then ending the render pass instance.

To begin a render pass instance, call:

```
// Provided by VK_VERSION_1_0
void vkCmdBeginRenderPass(
    VkCommandBuffer                      commandBuffer,
    const VkRenderPassBeginInfo*        pRenderPassBegin,
    VkSubpassContents                    contents);
```

- `commandBuffer` is the command buffer in which to record the command.
- `pRenderPassBegin` is a pointer to a `VkRenderPassBeginInfo` structure specifying the render pass to begin an instance of, and the framebuffer the instance uses.
- `contents` is a `VkSubpassContents` value specifying how the commands in the first subpass will be provided.

After beginning a render pass instance, the command buffer is ready to record the commands for the first subpass of that render pass.

Valid Usage

- VUID-vkCmdBeginRenderPass-initialLayout-00895
If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is `VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL` then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with a `usage` value including `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT`
- VUID-vkCmdBeginRenderPass-initialLayout-01758
If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is `VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL`, or `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL` then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with a `usage` value including `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`
- VUID-vkCmdBeginRenderPass-initialLayout-00897
If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL` then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with a `usage` value including `VK_IMAGE_USAGE_SAMPLED_BIT` or `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT`
- VUID-vkCmdBeginRenderPass-initialLayout-00898
If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is

`VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL` then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with a `usage` value including `VK_IMAGE_USAGE_TRANSFER_SRC_BIT`

- VUID-vkCmdBeginRenderPass-initialLayout-00899

If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL` then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with a `usage` value including `VK_IMAGE_USAGE_TRANSFER_DST_BIT`

- VUID-vkCmdBeginRenderPass-initialLayout-00900

If the `initialLayout` member of any of the `VkAttachmentDescription` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is not `VK_IMAGE_LAYOUT_UNDEFINED`, then each such `initialLayout` **must** be equal to the current layout of the corresponding attachment image subresource of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin`

- VUID-vkCmdBeginRenderPass-srcStageMask-06451

The `srcStageMask` members of any element of the `pDependencies` member of `VkRenderPassCreateInfo` used to create `renderPass` **must** be supported by the capabilities of the queue family identified by the `queueFamilyIndex` member of the `VkCommandPoolCreateInfo` used to create the command pool which `commandBuffer` was allocated from

- VUID-vkCmdBeginRenderPass-dstStageMask-06452

The `dstStageMask` members of any element of the `pDependencies` member of `VkRenderPassCreateInfo` used to create `renderPass` **must** be supported by the capabilities of the queue family identified by the `queueFamilyIndex` member of the `VkCommandPoolCreateInfo` used to create the command pool which `commandBuffer` was allocated from

- VUID-vkCmdBeginRenderPass-framebuffer-02532

For any attachment in `framebuffer` that is used by `renderPass` and is bound to memory locations that are also bound to another attachment used by `renderPass`, and if at least one of those uses causes either attachment to be written to, both attachments **must** have had the `VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT` set

- VUID-vkCmdBeginRenderPass-framebuffer-09045

If any attachments specified in `framebuffer` are used by `renderPass` and are bound to overlapping memory locations, there **must** be only one that is used as a color attachment, depth/stencil, or resolve attachment in any subpass

Valid Usage (Implicit)

- VUID-vkCmdBeginRenderPass-commandBuffer-parameter

`commandBuffer` **must** be a valid `VkCommandBuffer` handle

- VUID-vkCmdBeginRenderPass-pRenderPassBegin-parameter

`pRenderPassBegin` **must** be a valid pointer to a valid `VkRenderPassBeginInfo` structure

- VUID-vkCmdBeginRenderPass-contents-parameter
contents **must** be a valid [VkSubpassContents](#) value
- VUID-vkCmdBeginRenderPass-commandBuffer-recording
commandBuffer **must** be in the [recording state](#)
- VUID-vkCmdBeginRenderPass-commandBuffer-cmdpool
The [VkCommandPool](#) that **commandBuffer** was allocated from **must** support graphics operations
- VUID-vkCmdBeginRenderPass-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdBeginRenderPass-bufferlevel
commandBuffer **must** be a primary [VkCommandBuffer](#)

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the [VkCommandPool](#) that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary	Outside	Graphics	Action State Synchronization

The [VkRenderPassBeginInfo](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkRenderPassBeginInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkRenderPass        renderPass;
    VkFramebuffer        framebuffer;
    VkRect2D            renderArea;
    uint32_t            clearValueCount;
    const VkClearValue* pClearValues;
} VkRenderPassBeginInfo;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is [NULL](#) or a pointer to a structure extending this structure.

- **renderPass** is the render pass to begin an instance of.
- **framebuffer** is the framebuffer containing the attachments that are used with the render pass.
- **renderArea** is the render area that is affected by the render pass instance, and is described in more detail below.
- **clearValueCount** is the number of elements in **pClearValues**.
- **pClearValues** is a pointer to an array of **clearValueCount** **VkClearColorValue** structures containing clear values for each attachment, if the attachment uses a **loadOp** value of **VK_ATTACHMENT_LOAD_OP_CLEAR** or if the attachment has a depth/stencil format and uses a **stencilLoadOp** value of **VK_ATTACHMENT_LOAD_OP_CLEAR**. The array is indexed by attachment number. Only elements corresponding to cleared attachments are used. Other elements of **pClearValues** are ignored.

renderArea is the render area that is affected by the render pass instance. The effects of attachment load, store and multisample resolve operations are restricted to the pixels whose x and y coordinates fall within the render area on all attachments. The render area extends to all layers of **framebuffer**. The application **must** ensure (using scissor if necessary) that all rendering is contained within the render area. The render area **must** be contained within the framebuffer dimensions.

NOTE

There **may** be a performance cost for using a render area smaller than the framebuffer, unless it matches the render area granularity for the render pass.

Valid Usage

- VUID-VkRenderPassBeginInfo-clearValueCount-00902
clearValueCount **must** be greater than the largest attachment index in **renderPass** specifying a **loadOp** (or **stencilLoadOp**, if the attachment has a depth/stencil format) of **VK_ATTACHMENT_LOAD_OP_CLEAR**
- VUID-VkRenderPassBeginInfo-clearValueCount-04962
If **clearValueCount** is not 0, **pClearValues** **must** be a valid pointer to an array of **clearValueCount** **VkClearColorValue** unions
- VUID-VkRenderPassBeginInfo-renderPass-00904
renderPass **must** be **compatible** with the **renderPass** member of the **VkFramebufferCreateInfo** structure specified when creating **framebuffer**
- VUID-VkRenderPassBeginInfo-None-08996
renderArea.extent.width **must** be greater than 0
- VUID-VkRenderPassBeginInfo-None-08997
renderArea.extent.height **must** be greater than 0
- VUID-VkRenderPassBeginInfo-pNext-02850
renderArea.offset.x **must** be greater than or equal to 0
- VUID-VkRenderPassBeginInfo-pNext-02851
renderArea.offset.y **must** be greater than or equal to 0
- VUID-VkRenderPassBeginInfo-pNext-02852
renderArea.offset.x + **renderArea.extent.width** **must** be less than or equal to

`VkFramebufferCreateInfo::width` the `framebuffer` was created with

- VUID-VkRenderPassBeginInfo-pNext-02853
`renderArea.offset.y + renderArea.extent.height` **must** be less than or equal to `VkFramebufferCreateInfo::height` the `framebuffer` was created with

Valid Usage (Implicit)

- VUID-VkRenderPassBeginInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_RENDER_PASS_BEGIN_INFO`
- VUID-VkRenderPassBeginInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkRenderPassBeginInfo-renderPass-parameter
`renderPass` **must** be a valid `VkRenderPass` handle
- VUID-VkRenderPassBeginInfo-framebuffer-parameter
`framebuffer` **must** be a valid `VkFramebuffer` handle
- VUID-VkRenderPassBeginInfo-commonparent
Both of `framebuffer`, and `renderPass` **must** have been created, allocated, or retrieved from the same `VkDevice`

Possible values of `vkCmdBeginRenderPass::contents`, specifying how the commands in the first subpass will be provided, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSubpassContents {
    VK_SUBPASS_CONTENTS_INLINE = 0,
    VK_SUBPASS_CONTENTS_SECONDARY_COMMAND_BUFFERS = 1,
} VkSubpassContents;
```

- `VK_SUBPASS_CONTENTS_INLINE` specifies that the contents of the subpass will be recorded inline in the primary command buffer, and secondary command buffers **must** not be executed within the subpass.
- `VK_SUBPASS_CONTENTS_SECONDARY_COMMAND_BUFFERS` specifies that the contents are recorded in secondary command buffers that will be called from the primary command buffer, and `vkCmdExecuteCommands` is the only valid command in the command buffer until `vkCmdNextSubpass` or `vkCmdEndRenderPass`.

To query the render area granularity, call:

```
// Provided by VK_VERSION_1_0
void vkGetRenderAreaGranularity(
    VkDevice          device,
    VkRenderPass      renderPass,
    VkExtent2D*       pGranularity);
```

- **device** is the logical device that owns the render pass.
- **renderPass** is a handle to a render pass.
- **pGranularity** is a pointer to a [VkExtent2D](#) structure in which the granularity is returned.

The conditions leading to an optimal **renderArea** are:

- the **offset.x** member in **renderArea** is a multiple of the **width** member of the returned [VkExtent2D](#) (the horizontal granularity).
- the **offset.y** member in **renderArea** is a multiple of the **height** member of the returned [VkExtent2D](#) (the vertical granularity).
- either the **extent.width** member in **renderArea** is a multiple of the horizontal granularity or **offset.x+extent.width** is equal to the **width** of the **framebuffer** in the [VkRenderPassBeginInfo](#).
- either the **extent.height** member in **renderArea** is a multiple of the vertical granularity or **offset.y+extent.height** is equal to the **height** of the **framebuffer** in the [VkRenderPassBeginInfo](#).

Subpass dependencies are not affected by the render area, and apply to the entire image subresources attached to the framebuffer as specified in the description of [automatic layout transitions](#). Similarly, pipeline barriers are valid even if their effect extends outside the render area.

Valid Usage (Implicit)

- VUID-vkGetRenderAreaGranularity-device-parameter **device** **must** be a valid [VkDevice](#) handle
- VUID-vkGetRenderAreaGranularity-renderPass-parameter **renderPass** **must** be a valid [VkRenderPass](#) handle
- VUID-vkGetRenderAreaGranularity-pGranularity-parameter **pGranularity** **must** be a valid pointer to a [VkExtent2D](#) structure
- VUID-vkGetRenderAreaGranularity-renderPass-parent **renderPass** **must** have been created, allocated, or retrieved from **device**

To transition to the next subpass in the render pass instance after recording the commands for a subpass, call:


```
// Provided by VK_VERSION_1_0
void vkCmdNextSubpass(
    VkCommandBuffer                commandBuffer,
    VkSubpassContents               contents);
```

- **commandBuffer** is the command buffer in which to record the command.
- **contents** specifies how the commands in the next subpass will be provided, in the same fashion as the corresponding parameter of [vkCmdBeginRenderPass](#).

The subpass index for a render pass begins at zero when [vkCmdBeginRenderPass](#) is recorded, and increments each time [vkCmdNextSubpass](#) is recorded.

After transitioning to the next subpass, the application **can** record the commands for that subpass.

Valid Usage

- VUID-vkCmdNextSubpass-None-00909
The current subpass index **must** be less than the number of subpasses in the render pass minus one

Valid Usage (Implicit)

- VUID-vkCmdNextSubpass-commandBuffer-parameter
commandBuffer **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdNextSubpass-contents-parameter
contents **must** be a valid [VkSubpassContents](#) value
- VUID-vkCmdNextSubpass-commandBuffer-recording
commandBuffer **must** be in the [recording state](#)
- VUID-vkCmdNextSubpass-commandBuffer-cmdpool
The [VkCommandPool](#) that **commandBuffer** was allocated from **must** support graphics operations
- VUID-vkCmdNextSubpass-renderpass
This command **must** only be called inside of a render pass instance
- VUID-vkCmdNextSubpass-bufferlevel
commandBuffer **must** be a primary [VkCommandBuffer](#)

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the [VkCommandPool](#) that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary	Inside	Graphics	Action State Synchronization

To record a command to end a render pass instance after recording the commands for the last subpass, call:

```
// Provided by VK_VERSION_1_0
void vkCmdEndRenderPass(
    VkCommandBuffer                      commandBuffer);
```

- `commandBuffer` is the command buffer in which to end the current render pass instance.

Ending a render pass instance performs any multisample resolve operations on the final subpass.

Valid Usage

- VUID-vkCmdEndRenderPass-None-00910
The current subpass index **must** be equal to the number of subpasses in the render pass minus one
- VUID-vkCmdEndRenderPass-None-07004
If `vkCmdBeginQuery*` was called within a subpass of the render pass, the corresponding `vkCmdEndQuery*` **must** have been called subsequently within the same subpass

Valid Usage (Implicit)

- VUID-vkCmdEndRenderPass-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdEndRenderPass-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdEndRenderPass-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdEndRenderPass-renderpass
This command **must** only be called inside of a render pass instance
- VUID-vkCmdEndRenderPass-bufferlevel
`commandBuffer` **must** be a primary `VkCommandBuffer`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary	Inside	Graphics	Action State Synchronization

8.8. Common Render Pass Data Races (Informative)

Due to the complexity of how rendering is performed, there are several ways an application can accidentally introduce a data race, usually by doing something that may seem benign but actually cannot be supported. This section indicates a number of the more common cases as guidelines to help avoid them.

8.8.1. Sampling From a Read-only Attachment

Vulkan includes read-only layouts for depth/stencil images, that allow the images to be both read during a render pass for the purposes of depth/stencil tests, and read as a non-attachment.

However, because `VK_ATTACHMENT_STORE_OP_STORE` and `VK_ATTACHMENT_STORE_OP_DONT_CARE` may perform write operations, even if no recorded command writes to an attachment, reading from an image while also using it as an attachment with these store operations can result in a data race. If the reads from the non-attachment are performed in a fragment shader where the accessed samples match those covered by the fragment shader, no data race will occur as store operations are guaranteed to operate after fragment shader execution for the set of samples the fragment covers. Notably, input attachments can also be used for this case. Reading other samples or in any other shader stage can result in unexpected behavior due to the potential for a data race, and validation errors should be generated for doing so. In practice, many applications have shipped reading samples outside of the covered fragment without any observable issue, but there is no guarantee that this will always work, and it is not advisable to rely on this in new or re-worked code bases.

8.8.2. Non-overlapping Access Between Resources

When relying on non-overlapping accesses between attachments and other resources, it is important to note that `load` and `store` operations have fairly wide alignment requirements - potentially affecting entire subresources and adjacent depth/stencil aspects. This makes it invalid to

access a non-attachment subresource that is simultaneously being used as an attachment where either access performs a write operation.

8.8.3. Depth/Stencil and Input Attachments

When rendering to only the depth OR stencil aspect of an image, an input attachment accessing the other aspect will always result in a data race.

8.8.4. Synchronization Options

There are several synchronization options available to synchronize between accesses to resources within a render pass. Some of the options are outlined below:

- A [VkSubpassDependency](#) in a render pass object can synchronize attachment writes and [multisample resolve operations](#) from a prior subpass for subsequent input attachment reads.
- A [vkCmdPipelineBarrier](#) inside a subpass can synchronize prior attachment writes in the subpass with subsequent input attachment reads.

Chapter 9. Shaders

A shader specifies programmable operations that execute for each vertex, control point, tessellated vertex, primitive, fragment, or workgroup in the corresponding stage(s) of the graphics and compute pipelines.

Graphics pipelines include vertex shader execution as a result of [primitive assembly](#), followed, if enabled, by tessellation control and evaluation shaders operating on [patches](#), geometry shaders, if enabled, operating on primitives, and fragment shaders, if present, operating on fragments generated by [Rasterization](#). In this specification, vertex, tessellation control, tessellation evaluation and geometry shaders are collectively referred to as [pre-rasterization shader stages](#) and occur in the logical pipeline before rasterization. The fragment shader occurs logically after rasterization.

Only the compute shader stage is included in a compute pipeline. Compute shaders operate on compute invocations in a workgroup.

Shaders **can** read from input variables, and read from and write to output variables. Input and output variables **can** be used to transfer data between shader stages, or to allow the shader to interact with values that exist in the execution environment. Similarly, the execution environment provides constants describing capabilities.

Shader variables are associated with execution environment-provided inputs and outputs using *built-in* decorations in the shader. The available decorations for each stage are documented in the following subsections.

9.1. Shader Modules

Shader modules contain *shader code* and one or more entry points. Shaders are selected from a shader module by specifying an entry point as part of [pipeline](#) creation. The stages of a pipeline **can** use shaders that come from different modules. The shader code defining a shader module **must** be in the SPIR-V format, as described by the [Vulkan Environment for SPIR-V](#) appendix.

Shader modules are represented by `VkShaderModule` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkShaderModule)
```

To create a shader module, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateShaderModule(
    VkDevice                                device,
    const VkShaderModuleCreateInfo*         pCreateInfo,
    const VkAllocationCallbacks*           pAllocator,
    VkShaderModule*                         pShaderModule);
```

- `device` is the logical device that creates the shader module.

- `pCreateInfo` is a pointer to a `VkShaderModuleCreateInfo` structure.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.
- `pShaderModule` is a pointer to a `VkShaderModule` handle in which the resulting shader module object is returned.

Once a shader module has been created, any entry points it contains **can** be used in pipeline shader stages as described in [Compute Pipelines](#) and [Graphics Pipelines](#).

Valid Usage

- VUID-vkCreateShaderModule-pCreateInfo-06904
If `pCreateInfo` is not `NULL`, `pCreateInfo->pNext` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkCreateShaderModule-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkCreateShaderModule-pCreateInfo-parameter
`pCreateInfo` **must** be a valid pointer to a valid `VkShaderModuleCreateInfo` structure
- VUID-vkCreateShaderModule-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkCreateShaderModule-pShaderModule-parameter
`pShaderModule` **must** be a valid pointer to a `VkShaderModule` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkShaderModuleCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkShaderModuleCreateInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkShaderModuleCreateFlags flags;
    size_t               codeSize;
    const uint32_t*       pCode;
} VkShaderModuleCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `codeSize` is the size, in bytes, of the code pointed to by `pCode`.
- `pCode` is a pointer to code that is used to create the shader module. The type and format of the code is determined from the content of the memory addressed by `pCode`.

Valid Usage

- VUID-VkShaderModuleCreateInfo-codeSize-08735
`codeSize` **must** be a multiple of 4
- VUID-VkShaderModuleCreateInfo-pCode-08736
`pCode` **must** point to valid SPIR-V code, formatted and packed as described by the [Khronos SPIR-V Specification](#)
- VUID-VkShaderModuleCreateInfo-pCode-08737
`pCode` **must** adhere to the validation rules described by the [Validation Rules within a Module](#) section of the [SPIR-V Environment](#) appendix
- VUID-VkShaderModuleCreateInfo-pCode-08738
`pCode` **must** declare the `Shader` capability for SPIR-V code
- VUID-VkShaderModuleCreateInfo-pCode-08739
`pCode` **must** not declare any capability that is not supported by the API, as described by the [Capabilities](#) section of the [SPIR-V Environment](#) appendix
- VUID-VkShaderModuleCreateInfo-pCode-08740
and `pCode` declares any of the capabilities listed in the [SPIR-V Environment](#) appendix, one of the corresponding requirements **must** be satisfied
- VUID-VkShaderModuleCreateInfo-pCode-08741
`pCode` **must** not declare any SPIR-V extension that is not supported by the API, as described by the [Extension](#) section of the [SPIR-V Environment](#) appendix
- VUID-VkShaderModuleCreateInfo-pCode-08742
and `pCode` declares any of the SPIR-V extensions listed in the [SPIR-V Environment](#) appendix, one of the corresponding requirements **must** be satisfied
- VUID-VkShaderModuleCreateInfo-codeSize-01085
`codeSize` **must** be greater than 0

Valid Usage (Implicit)

- VUID-VkShaderModuleCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_SHADER_MODULE_CREATE_INFO`
- VUID-VkShaderModuleCreateInfo-flags-zeroBitmask
`flags` **must** be `0`
- VUID-VkShaderModuleCreateInfo-pCode-parameter
`pCode` **must** be a valid pointer to an array of $\frac{\text{codeSize}}{4}$ `uint32_t` values

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkShaderModuleCreateFlags;
```

`VkShaderModuleCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

To destroy a shader module, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyShaderModule(
    VkDevice          device,
    VkShaderModule    shaderModule,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that destroys the shader module.
- `shaderModule` is the handle of the shader module to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

A shader module **can** be destroyed while pipelines created using its shaders are still in use.

Valid Usage

- VUID-vkDestroyShaderModule-shaderModule-01092
If `VkAllocationCallbacks` were provided when `shaderModule` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyShaderModule-shaderModule-01093
If no `VkAllocationCallbacks` were provided when `shaderModule` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyShaderModule-device-parameter
`device` **must** be a valid `VkDevice` handle

- VUID-vkDestroyShaderModule-shaderModule-parameter
If `shaderModule` is not `VK_NULL_HANDLE`, `shaderModule` **must** be a valid `VkShaderModule` handle
- VUID-vkDestroyShaderModule-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkDestroyShaderModule-shaderModule-parent
If `shaderModule` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `shaderModule` **must** be externally synchronized

9.2. Binding Shaders

Before a shader can be used it **must** be first bound to the command buffer.

Calling `vkCmdBindPipeline` binds all stages corresponding to the `VkPipelineBindPoint`.

The following table describes the relationship between shader stages and pipeline bind points:

Shader stage	Pipeline bind point	behavior controlled
• <code>VK_SHADER_STAGE_VERTEX_BIT</code> • <code>VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT</code> • <code>VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT</code> • <code>VK_SHADER_STAGE_GEOMETRY_BIT</code> • <code>VK_SHADER_STAGE_FRAGMENT_BIT</code>	<code>VK_PIPELINE_BIND_POINT_GRAPHICS</code>	all <code>drawing commands</code>
• <code>VK_SHADER_STAGE_COMPUTE_BIT</code>	<code>VK_PIPELINE_BIND_POINT_COMPUTE</code>	all <code>dispatch commands</code>

9.3. Shader Execution

At each stage of the pipeline, multiple invocations of a shader **may** execute simultaneously. Further, invocations of a single shader produced as the result of different commands **may** execute simultaneously. The relative execution order of invocations of the same shader type is undefined. Shader invocations **may** complete in a different order than that in which the primitives they originated from were drawn or dispatched by the application. However, fragment shader outputs are written to attachments in `rasterization order`.

The relative execution order of invocations of different shader types is largely undefined. However, when invoking a shader whose inputs are generated from a previous pipeline stage, the shader invocations from the previous stage are guaranteed to have executed far enough to generate input values for all required inputs.

9.3.1. Shader Termination

A shader invocation that is *terminated* has finished executing instructions.

Executing `OpReturn` in the entry point, or executing `OpTerminateInvocation` in any function will terminate an invocation. Implementations **may** also terminate a shader invocation when `OpKill` is executed in any function; otherwise it becomes a [helper invocation](#).

In addition to the above conditions, [helper invocations](#) **may** be terminated when all non-helper invocations in the same [derivative group](#) either terminate or become [helper invocations](#).

A shader stage for a given command completes execution when all invocations for that stage have terminated.

NOTE

`OpKill` will behave the same as either `OpTerminateInvocation` or `OpDemoteToHelperInvocation` depending on the implementation. It is recommended that shader authors use `OpTerminateInvocation` or `OpDemoteToHelperInvocation` instead of `OpKill` whenever possible to produce more predictable behavior.

9.4. Shader Memory Access Ordering

The order in which image or buffer memory is read or written by shaders is largely undefined. For some shader types (vertex, tessellation evaluation, and in some cases, fragment), even the number of shader invocations that **may** perform loads and stores is undefined.

In particular, the following rules apply:

- [Vertex](#) and [tessellation evaluation](#) shaders will be invoked at least once for each unique vertex, as defined in those sections.
- [Fragment](#) shaders will be invoked zero or more times, as defined in that section.
- The relative execution order of invocations of the same shader type is undefined. A store issued by a shader when working on primitive B might complete prior to a store for primitive A, even if primitive A is specified prior to primitive B. This applies even to fragment shaders; while fragment shader outputs are always written to the framebuffer in [rasterization order](#), stores executed by fragment shader invocations are not.
- The relative execution order of invocations of different shader types is largely undefined.

NOTE

The above limitations on shader invocation order make some forms of synchronization between shader invocations within a single set of primitives unimplementable. For example, having one invocation poll memory written by another invocation assumes that the other invocation has been launched and will complete its writes in finite time.

Stores issued to different memory locations within a single shader invocation **may** not be visible to other invocations, or **may** not become visible in the order they were performed.

The **OpMemoryBarrier** instruction **can** be used to provide stronger ordering of reads and writes performed by a single invocation. **OpMemoryBarrier** guarantees that any memory transactions issued by the shader invocation prior to the instruction complete prior to the memory transactions issued after the instruction. Memory barriers are needed for algorithms that require multiple invocations to access the same memory and require the operations to be performed in a partially-defined relative order. For example, if one shader invocation does a series of writes, followed by an **OpMemoryBarrier** instruction, followed by another write, then the results of the series of writes before the barrier become visible to other shader invocations at a time earlier or equal to when the results of the final write become visible to those invocations. In practice it means that another invocation that sees the results of the final write would also see the previous writes. Without the memory barrier, the final write **may** be visible before the previous writes.

Writes that are the result of shader stores through a variable decorated with **Coherent** automatically have available writes to the same buffer, buffer view, or image view made visible to them, and are themselves automatically made available to access by the same buffer, buffer view, or image view. Reads that are the result of shader loads through a variable decorated with **Coherent** automatically have available writes to the same buffer, buffer view, or image view made visible to them. The order that coherent writes to different locations become available is undefined, unless enforced by a memory barrier instruction or other memory dependency.

NOTE	Explicit memory dependencies must still be used to guarantee availability and visibility for access via other buffers, buffer views, or image views.
-------------	---

The built-in atomic memory transaction instructions **can** be used to read and write a given memory address atomically. While built-in atomic functions issued by multiple shader invocations are executed in undefined order relative to each other, these functions perform both a read and a write of a memory address and guarantee that no other memory transaction will write to the underlying memory between the read and write. Atomic operations ensure automatic availability and visibility for writes and reads in the same way as those to **Coherent** variables.

NOTE	Memory accesses performed on different resource descriptors with the same memory backing may not be well-defined even with the Coherent decoration or via atomics, due to things such as image layouts or ownership of the resource - as described in the Synchronization and Cache Control chapter.
-------------	--

NOTE	Atomics allow shaders to use shared global addresses for mutual exclusion or as counters, among other uses.
-------------	---

The SPIR-V **SubgroupMemory**, **CrossWorkgroupMemory**, and **AtomicCounterMemory** memory semantics are ignored. Sequentially consistent atomics and barriers are not supported and **SequentiallyConsistent** is treated as **AcquireRelease**. **SequentiallyConsistent** **should** not be used.

9.5. Shader Inputs and Outputs

Data is passed into and out of shaders using variables with input or output storage class, respectively. User-defined inputs and outputs are connected between stages by matching their **Location** decorations. Additionally, data **can** be provided by or communicated to special functions provided by the execution environment using **BuiltIn** decorations.

In many cases, the same **BuiltIn** decoration **can** be used in multiple shader stages with similar meaning. The specific behavior of variables decorated as **BuiltIn** is documented in the following sections.

9.6. Vertex Shaders

Each vertex shader invocation operates on one vertex and its associated **vertex attribute** data, and outputs one vertex and associated data. Graphics pipelines **must** include a vertex shader, and the vertex shader stage is always the first shader stage in the graphics pipeline.

9.6.1. Vertex Shader Execution

A vertex shader **must** be executed at least once for each vertex specified by a drawing command. During execution, the shader is presented with the index of the vertex and instance for which it has been invoked. Input variables declared in the vertex shader are filled by the implementation with the values of vertex attributes associated with the invocation being executed.

If the same vertex is specified multiple times in a drawing command (e.g. by including the same index value multiple times in an index buffer) the implementation **may** reuse the results of vertex shading if it can statically determine that the vertex shader invocations will produce identical results.

NOTE

It is implementation-dependent when and if results of vertex shading are reused, and thus how many times the vertex shader will be executed. This is true also if the vertex shader contains stores or atomic operations (see **vertexPipelineStoresAndAtomics**).

9.7. Tessellation Control Shaders

The tessellation control shader is used to read an input patch provided by the application and to produce an output patch. Each tessellation control shader invocation operates on an input patch (after all control points in the patch are processed by a vertex shader) and its associated data, and outputs a single control point of the output patch and its associated data, and **can** also output additional per-patch data. The input patch is sized according to the **patchControlPoints** member of **VkPipelineTessellationStateCreateInfo**, as part of input assembly.

The size of the output patch is controlled by the **OpExecutionMode OutputVertices** specified in the tessellation control or tessellation evaluation shaders, which **must** be specified in at least one of the shaders. The size of the input and output patches **must** each be greater than zero and less than or equal to **VkPhysicalDeviceLimits::maxTessellationPatchSize**.

9.7.1. Tessellation Control Shader Execution

A tessellation control shader is invoked at least once for each *output* vertex in a patch.

Inputs to the tessellation control shader are generated by the vertex shader. Each invocation of the tessellation control shader **can** read the attributes of any incoming vertices and their associated data. The invocations corresponding to a given patch execute logically in parallel, with undefined relative execution order. However, the `OpControlBarrier` instruction **can** be used to provide limited control of the execution order by synchronizing invocations within a patch, effectively dividing tessellation control shader execution into a set of phases. Tessellation control shaders will read undefined values if one invocation reads a per-vertex or per-patch output written by another invocation at any point during the same phase, or if two invocations attempt to write different values to the same per-patch output in a single phase.

9.8. Tessellation Evaluation Shaders

The Tessellation Evaluation Shader operates on an input patch of control points and their associated data, and a single input barycentric coordinate indicating the invocation's relative position within the subdivided patch, and outputs a single vertex and its associated data.

9.8.1. Tessellation Evaluation Shader Execution

A tessellation evaluation shader is invoked at least once for each unique vertex generated by the tessellator.

9.9. Geometry Shaders

The geometry shader operates on a group of vertices and their associated data assembled from a single input primitive, and emits zero or more output primitives and the group of vertices and their associated data required for each output primitive.

9.9.1. Geometry Shader Execution

A geometry shader is invoked at least once for each primitive produced by the tessellation stages, or at least once for each primitive generated by [primitive assembly](#) when tessellation is not in use. A shader can request that the geometry shader runs multiple [instances](#). A geometry shader is invoked at least once for each instance.

9.10. Fragment Shaders

Fragment shaders are invoked as a [fragment operation](#) in a graphics pipeline. Each fragment shader invocation operates on a single fragment and its associated data. With few exceptions, fragment shaders do not have access to any data associated with other fragments and are considered to execute in isolation of fragment shader invocations associated with other fragments.

9.11. Compute Shaders

Compute shaders are invoked via `vkCmdDispatch` and `vkCmdDispatchIndirect` commands. In general, they have access to similar resources as shader stages executing as part of a graphics pipeline.

Compute workloads are formed from groups of work items called workgroups and processed by the compute shader in the current compute pipeline. A workgroup is a collection of shader invocations that execute the same shader, potentially in parallel. Compute shaders execute in *global workgroups* which are divided into a number of *local workgroups* with a size that **can** be set by assigning a value to the `LocalSize` execution mode or via an object decorated by the `WorkgroupSize` decoration. An invocation within a local workgroup **can** share data with other members of the local workgroup through shared variables and issue memory and control flow barriers to synchronize with other members of the local workgroup.

9.12. Interpolation Decorations

Variables in the `Input` storage class in a fragment shader's interface are interpolated from the values specified by the primitive being rasterized.

NOTE

Interpolation decorations can be present on input and output variables in pre-rasterization shaders but have no effect on the interpolation performed.

An undecorated input variable will be interpolated with perspective-correct interpolation according to the primitive type being rasterized. `Lines` and `polygons` are interpolated in the same way as the primitive's clip coordinates. If the `NoPerspective` decoration is present, linear interpolation is instead used for `lines` and `polygons`. For points, as there is only a single vertex, input values are never interpolated and instead take the value written for the single vertex.

If the `Flat` decoration is present on an input variable, the value is not interpolated, and instead takes its value directly from the `provoking vertex`. Fragment shader inputs that are signed or unsigned integers, integer vectors, or any double-precision floating-point type **must** be decorated with `Flat`.

Interpolation of input variables is performed at an implementation-defined position within the fragment area being shaded. The position is further constrained as follows:

- If the `Centroid` decoration is used, the interpolation position used for the variable **must** also fall within the bounds of the primitive being rasterized.
- If the `Sample` decoration is used, the interpolation position used for the variable **must** be at the position of the sample being shaded by the current fragment shader invocation.
- If a sample count of 1 is used, the interpolation position **must** be at the center of the fragment area.

NOTE

As `Centroid` restricts the possible interpolation position to the covered area of the primitive, the position can be forced to vary between neighboring fragments when it otherwise would not. Derivatives calculated based on these differing locations can

produce inconsistent results compared to undecorated inputs. It is recommended that input variables used in derivative calculations are not decorated with **Centroid**.

9.13. Static Use

A SPIR-V module declares a global object in memory using the **OpVariable** instruction, which results in a pointer **x** to that object. A specific entry point in a SPIR-V module is said to *statically use* that object if that entry point's call tree contains a function containing a instruction with **x** as an **id** operand. A shader entry point also *statically uses* any variables explicitly declared in its interface.

9.14. Scope

A *scope* describes a set of shader invocations, where each such set is a *scope instance*. Each invocation belongs to one or more scope instances, but belongs to no more than one scope instance for each scope.

The operations available between invocations in a given scope instance vary, with smaller scopes generally able to perform more operations, and with greater efficiency.

9.14.1. Cross Device

All invocations executed in a Vulkan instance fall into a single *cross device scope instance*.

Whilst the **CrossDevice** scope is defined in SPIR-V, it is disallowed in Vulkan. API [synchronization](#) commands **can** be used to communicate between devices.

9.14.2. Device

All invocations executed on a single device form a *device scope instance*.

There is no method to synchronize the execution of these invocations within SPIR-V, and this **can** only be done with API synchronization primitives.

The scope only extends to the queue family, not the whole device.

9.14.3. Queue Family

Invocations executed by queues in a given queue family form a *queue family scope instance*.

This scope is identified in SPIR-V as the **Device Scope**, which **can** be used as a **Memory Scope** for barrier and atomic operations.

There is no method to synchronize the execution of these invocations within SPIR-V, and this **can** only be done with API synchronization primitives.

Each invocation in a queue family scope instance **must** be in the same [device scope instance](#).

9.14.4. Command

Any shader invocations executed as the result of a single command such as `vkCmdDispatch` or `vkCmdDraw` form a *command scope instance*. For indirect drawing commands with `drawCount` greater than one, invocations from separate draws are in separate command scope instances.

There is no specific `Scope` for communication across invocations in a command scope instance. As this has a clear boundary at the API level, coordination here **can** be performed in the API, rather than in SPIR-V.

Each invocation in a command scope instance **must** be in the same *queue-family scope instance*.

For shaders without defined `workgroups`, this set of invocations forms an *invocation group* as defined in the *SPIR-V specification*.

9.14.5. Primitive

Any fragment shader invocations executed as the result of rasterization of a single primitive form a *primitive scope instance*.

There is no specific `Scope` for communication across invocations in a primitive scope instance.

Any generated `helper invocations` are included in this scope instance.

Each invocation in a primitive scope instance **must** be in the same *command scope instance*.

Any input variables decorated with `Flat` are uniform within a primitive scope instance.

9.14.6. Workgroup

A *local workgroup* is a set of invocations that can synchronize and share data with each other using memory in the `Workgroup` storage class.

The `Workgroup Scope` can be used as both an `Execution Scope` and `Memory Scope` for barrier and atomic operations.

Each invocation in a local workgroup **must** be in the same *command scope instance*.

Only compute shaders have defined workgroups - other shader types **cannot** use workgroup functionality. For shaders that have defined workgroups, this set of invocations forms an *invocation group* as defined in the *SPIR-V specification*.

The amount of storage consumed by the variables declared with the `Workgroup` storage class is implementation-dependent. However, the amount of storage consumed may not exceed the largest block size that would be obtained if all active variables declared with `Workgroup` storage class were assigned offsets in an arbitrary order by successively taking the smallest valid offset according to the *Standard Storage Buffer Layout* rules, and with `Boolean` values considered as 32-bit integer values for the purpose of this calculation. (This is equivalent to using the GLSL std430 layout rules.)

9.14.7. Quad

A *quad scope instance* is formed of four shader invocations.

In a fragment shader, each invocation in a quad scope instance is formed of invocations in neighboring framebuffer locations (x_i, y_i) , where:

- i is the index of the invocation within the scope instance.
- w and h are the number of pixels the fragment covers in the x and y axes.
- w and h are identical for all participating invocations.
- $(x_0) = (x_1 - w) = (x_2) = (x_3 - w)$
- $(y_0) = (y_1) = (y_2 - h) = (y_3 - h)$
- Each invocation has the same layer and sample indices.

The specific set of invocations that make up a quad scope instance in other shader stages is undefined.

In a fragment shader, each invocation in a quad scope instance **must** be in the same [primitive scope instance](#).

For [shaders that have defined workgroups](#), each invocation in a quad scope instance **must** be in the same [local workgroup](#).

In other shader stages, each invocation in a quad scope instance **must** be in the same [device scope instance](#).

Fragment shaders have defined quad scope instances.

9.14.8. Invocation

The smallest *scope* is a single invocation; this is represented by the [Invocation Scope](#) in SPIR-V.

Fragment shader invocations **must** be in a [primitive scope instance](#).

Invocations in [shaders that have defined workgroups](#) **must** be in a [local workgroup](#).

Invocations in [shaders that have a defined quad scope](#) **must** be in a [quad scope instance](#).

All invocations in all stages **must** be in a [command scope instance](#).

9.15. Derivative Operations

Derivative operations calculate the partial derivative for an expression P as a function of an invocation's x and y coordinates.

Derivative operations operate on a set of invocations known as a *derivative group* as defined in the [SPIR-V specification](#).

A derivative group in a fragment shader is equivalent to the [primitive scope instance](#).

Derivatives are calculated assuming that P is piecewise linear and continuous within the derivative group.

The following control-flow restrictions apply to derivative operations:

- dynamic instances of explicit derivative instructions (**OpDPdx***, **OpDPdy***, and **OpFwidth***) **must** be executed in control flow that is uniform within a derivative group.
- dynamic instances of implicit derivative operations **can** be executed in control flow that is not uniform within the derivative group, but results are undefined.

Fragment shaders that statically execute derivative operations **must** launch sufficient invocations to ensure their correct operation; additional **helper invocations** are launched for framebuffer locations not covered by rasterized fragments if necessary.

Derivative operations calculate their results as the difference between the result of P across invocations in the quad. For fine derivative operations (**OpDPdxFine** and **OpDPdyFine**), the values of $DPdx(P_i)$ are calculated as

$$DPdx(P_0) = DPdx(P_1) = P_1 - P_0$$

$$DPdx(P_2) = DPdx(P_3) = P_3 - P_2$$

and the values of $DPdy(P_i)$ are calculated as

$$DPdy(P_0) = DPdy(P_2) = P_2 - P_0$$

$$DPdy(P_1) = DPdy(P_3) = P_3 - P_1$$

where i is the index of each invocation as described in [Quad](#).

Coarse derivative operations (**OpDPdxCoarse** and **OpDPdyCoarse**), calculate their results in roughly the same manner, but **may** only calculate two values instead of four (one for each of $DPdx$ and $DPdy$), reusing the same result no matter the originating invocation. If an implementation does this, it **should** use the fine derivative calculations described for P_0 .

NOTE

Derivative values are calculated between fragments rather than pixels. If the fragment shader invocations involved in the calculation cover multiple pixels, these operations cover a wider area, resulting in larger derivative values. This in turn will result in a coarser LOD being selected for image sampling operations using derivatives.

Applications may want to account for this when using multi-pixel fragments; if pixel derivatives are desired, applications should use explicit derivative operations and divide the results by the size of the fragment in each dimension as follows:

$$DPdx(P_n)' = DPdx(P_n) / w$$

$$DPdy(P_n)' = DPdy(P_n) / h$$

where w and h are the size of the fragments in the quad, and $DPdx(P_n)'$ and $DPdy(P_n)'$ are the pixel derivatives.

The results for `OpDPdx` and `OpDPdy` **may** be calculated as either fine or coarse derivatives, with implementations favoring the most efficient approach. Implementations **must** choose coarse or fine consistently between the two.

Executing `OpFwidthFine`, `OpFwidthCoarse`, or `OpFwidth` is equivalent to executing the corresponding `OpDPdx*` and `OpDPdy*` instructions, taking the absolute value of the results, and summing them.

Executing an `OpImage*Sample*ImplicitLod` instruction is equivalent to executing `OpDPdx(Coordinate)` and `OpDPdy(Coordinate)`, and passing the results as the `Grad` operands `dx` and `dy`.

NOTE

It is expected that using the `ImplicitLod` variants of sampling functions will be substantially more efficient than using the `ExplicitLod` variants with explicitly generated derivatives.

9.16. Helper Invocations

When performing `derivative` operations in a fragment shader, additional invocations **may** be spawned in order to ensure correct results. These additional invocations are known as *helper invocations* and **can** be identified by a non-zero value in the `HelperInvocation` built-in. Stores and atomics performed by helper invocations **must** not have any effect on memory except for the `Function`, `Private` and `Output` storage classes, and values returned by atomic instructions in helper invocations are undefined.

NOTE

While storage to `Output` storage class has an effect even in helper invocations, it does not mean that helper invocations have an effect on the framebuffer. `Output` variables in fragment shaders can be read from as well, and they behave more like `Private` variables for the duration of the shader invocation.

Chapter 10. Pipelines

The following [figure](#) shows a block diagram of the Vulkan pipelines. Some Vulkan commands specify geometric objects to be drawn or computational work to be performed, while others specify state controlling how objects are handled by the various pipeline stages, or control data transfer between memory organized as images and buffers. Commands are effectively sent through a processing pipeline, either a *graphics pipeline*, or a *compute pipeline*.

The first stage of the [graphics pipeline](#) ([Input Assembler](#)) assembles vertices to form geometric primitives such as points, lines, and triangles, based on a requested primitive topology. In the next stage ([Vertex Shader](#)) vertices **can** be transformed, computing positions and attributes for each vertex. If [tessellation](#) and/or [geometry](#) shaders are supported, they **can** then generate multiple primitives from a single input primitive, possibly changing the primitive topology or generating additional attribute data in the process.

The final resulting primitives are [clipped](#) to a clip volume in preparation for the next stage, [Rasterization](#). The rasterizer produces a series of *fragments* associated with a region of the framebuffer, from a two-dimensional description of a point, line segment, or triangle. These fragments are processed by [fragment operations](#) to determine whether generated values will be written to the framebuffer. [Fragment shading](#) determines the values to be written to the framebuffer attachments. Framebuffer operations then read and write the color and depth/stencil attachments of the framebuffer for a given subpass of a [render pass instance](#). The attachments **can** be used as input attachments in the fragment shader in a later subpass of the same render pass.

The [compute pipeline](#) is a separate pipeline from the graphics pipeline, which operates on one-, two-, or three-dimensional workgroups which **can** read from and write to buffer and image memory.

This ordering is meant only as a tool for describing Vulkan, not as a strict rule of how Vulkan is implemented, and we present it only as a means to organize the various operations of the pipelines. Actual ordering guarantees between pipeline stages are explained in detail in the [synchronization chapter](#).

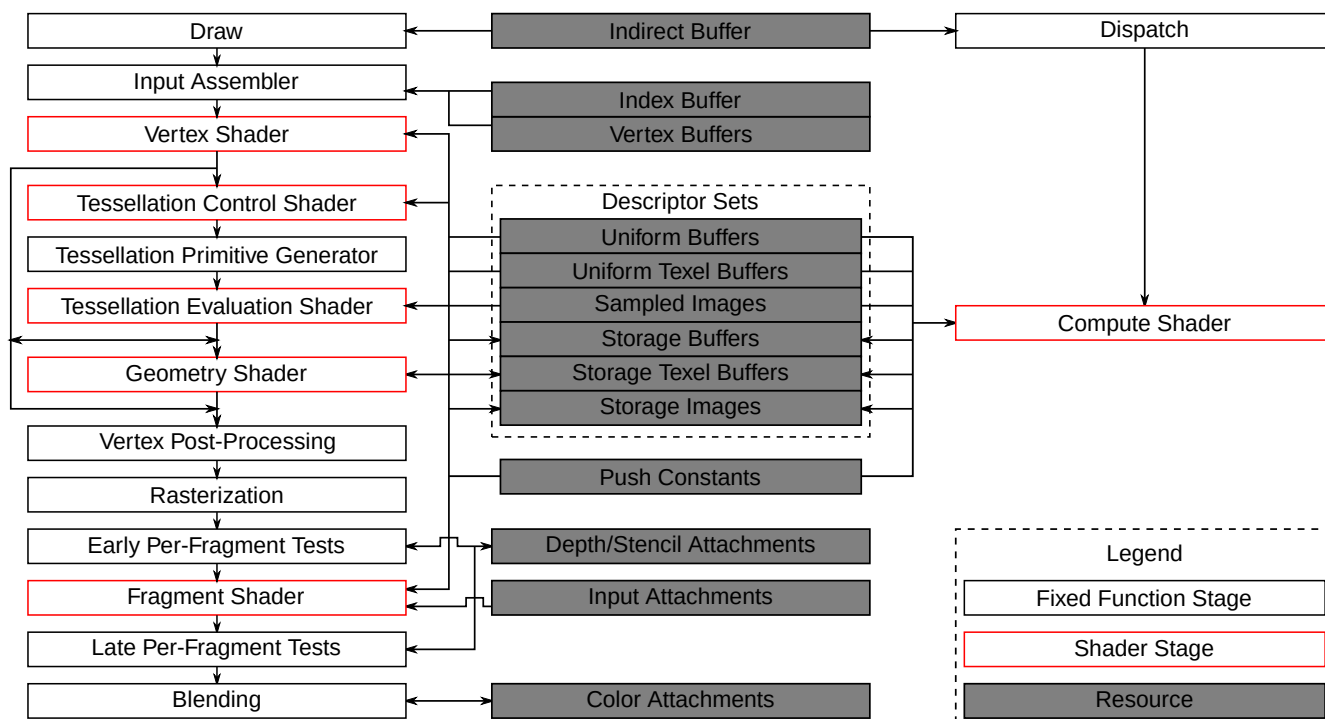


Figure 2. Block diagram of the Vulkan pipeline

Each pipeline is controlled by a monolithic object created from a description of all of the shader stages and any relevant fixed-function stages. [Linking](#) the whole pipeline together allows the optimization of shaders based on their input/outputs and eliminates expensive draw time state validation.

A pipeline object is bound to the current state using [vkCmdBindPipeline](#). Any pipeline object state that is specified as [dynamic](#) is not applied to the current state when the pipeline object is bound, but is instead set by dynamic state setting commands.

No state, including dynamic state, is inherited from one command buffer to another.

Compute, and graphics pipelines are each represented by [VkPipeline](#) handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkPipeline)
```

10.1. Multiple Pipeline Creation

Multiple pipelines **can** be created in a single call by commands such as [vkCreateComputePipelines](#), and [vkCreateGraphicsPipelines](#).

The creation commands are passed an array [pCreateInfos](#) of [Vk*PipelineCreateInfo](#) structures specifying parameters of each pipeline to be created, and return a corresponding array of handles in [pPipelines](#). Each element index i of [pPipelines](#) is created based on the corresponding element i of [pCreateInfos](#).

Applications **can** group together similar pipelines to be created in a single call, and implementations are encouraged to look for reuse opportunities when creating a group.

When attempting to create many pipelines in a single command, it is possible that creation **may** fail for a subset of them. In this case, the corresponding elements of `pPipelines` will be `VK_NULL_HANDLE`. If creation fails for a pipeline despite valid arguments (for example, due to out of memory errors), the `VkResult` code returned by the pipeline creation command will indicate why. The implementation will attempt to create all pipelines, and only return `VK_NULL_HANDLE` values for those that actually failed.

If creation fails for multiple pipelines, the returned `VkResult` **must** be the return value of any one of the pipelines which did not succeed. An application **can** reliably clean up from a failed call by iterating over the `pPipelines` array and destroying every element that is not `VK_NULL_HANDLE`.

If the entire command fails and no pipelines are created, all elements of `pPipelines` will be `VK_NULL_HANDLE`.

10.2. Compute Pipelines

Compute pipelines consist of a single static compute shader stage and the pipeline layout.

The compute pipeline represents a compute shader and is created by calling `vkCreateComputePipelines` with `module` and `pName` selecting an entry point from a shader module, where that entry point defines a valid compute shader, in the `VkPipelineShaderStageCreateInfo` structure contained within the `VkComputePipelineCreateInfo` structure.

To create compute pipelines, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateComputePipelines(
    VkDevice                device,
    VkPipelineCache          pipelineCache,
    uint32_t                 createInfoCount,
    const VkComputePipelineCreateInfo* pCreateInfos,
    const VkAllocationCallbacks* pAllocator,
    VkPipeline*              pPipelines);
```

- `device` is the logical device that creates the compute pipelines.
- `pipelineCache` is either `VK_NULL_HANDLE`, indicating that pipeline caching is disabled; or the handle of a valid `pipeline cache` object, in which case use of that cache is enabled for the duration of the command.
- `createInfoCount` is the length of the `pCreateInfos` and `pPipelines` arrays.
- `pCreateInfos` is a pointer to an array of `VkComputePipelineCreateInfo` structures.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.
- `pPipelines` is a pointer to an array of `VkPipeline` handles in which the resulting compute pipeline objects are returned.

Pipelines are created and returned as described for [Multiple Pipeline Creation](#).

Valid Usage

- VUID-vkCreateComputePipelines-device-09661
device **must** support at least one queue family with the `VK_QUEUE_COMPUTE_BIT` capability
- VUID-vkCreateComputePipelines-flags-00695
If the **flags** member of any element of **pCreateInfo**s contains the `VK_PIPELINE_CREATE_DERIVATIVE_BIT` flag, and the **basePipelineIndex** member of that same element is not `-1`, **basePipelineIndex** **must** be less than the index into **pCreateInfo**s that corresponds to that element
- VUID-vkCreateComputePipelines-flags-00696
If the **flags** member of any element of **pCreateInfo**s contains the `VK_PIPELINE_CREATE_DERIVATIVE_BIT` flag, the base pipeline **must** have been created with the `VK_PIPELINE_CREATE_ALLOW_DERIVATIVES_BIT` flag set

Valid Usage (Implicit)

- VUID-vkCreateComputePipelines-device-parameter
device **must** be a valid `VkDevice` handle
- VUID-vkCreateComputePipelines-pipelineCache-parameter
If **pipelineCache** is not `VK_NULL_HANDLE`, **pipelineCache** **must** be a valid `VkPipelineCache` handle
- VUID-vkCreateComputePipelines-pCreateInfo-parameter
pCreateInfos **must** be a valid pointer to an array of `createInfoCount` valid `VkComputePipelineCreateInfo` structures
- VUID-vkCreateComputePipelines-pAllocator-parameter
If **pAllocator** is not `NULL`, **pAllocator** **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkCreateComputePipelines-pPipelines-parameter
pPipelines **must** be a valid pointer to an array of `createInfoCount` `VkPipeline` handles
- VUID-vkCreateComputePipelines-createInfoCount-arraylength
createInfoCount **must** be greater than `0`
- VUID-vkCreateComputePipelines-pipelineCache-parent
If **pipelineCache** is a valid handle, it **must** have been created, allocated, or retrieved from **device**

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`

- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkComputePipelineCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkComputePipelineCreateInfo {
    VkStructureType          sType;
    const void*              pNext;
    VkPipelineCreateFlags     flags;
    VkPipelineShaderStageCreateInfo stage;
    VkPipelineLayout          layout;
    VkPipeline                basePipelineHandle;
    int32_t                  basePipelineIndex;
} VkComputePipelineCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkPipelineCreateFlagBits` specifying how the pipeline will be generated.
- `stage` is a `VkPipelineShaderStageCreateInfo` structure describing the compute shader.
- `layout` is the description of binding locations used by both the pipeline and descriptor sets used with the pipeline.
- `basePipelineHandle` is a pipeline to derive from.
- `basePipelineIndex` is an index into the `pCreateInfos` parameter to use as a pipeline to derive from.

The parameters `basePipelineHandle` and `basePipelineIndex` are described in more detail in [Pipeline Derivatives](#).

Valid Usage

- VUID-VkComputePipelineCreateInfo-None-09497
`flags` **must** be a valid combination of `VkPipelineCreateFlagBits` values
- VUID-VkComputePipelineCreateInfo-flags-07984
If `flags` contains the `VK_PIPELINE_CREATE_DERIVATIVE_BIT` flag, and `basePipelineIndex` is -1, `basePipelineHandle` **must** be a valid compute `VkPipeline` handle
- VUID-VkComputePipelineCreateInfo-flags-07985
If `flags` contains the `VK_PIPELINE_CREATE_DERIVATIVE_BIT` flag, and `basePipelineHandle` is `VK_NULL_HANDLE`, `basePipelineIndex` **must** be a valid index into the calling command's `pCreateInfos` parameter
- VUID-VkComputePipelineCreateInfo-flags-07986
If `flags` contains the `VK_PIPELINE_CREATE_DERIVATIVE_BIT` flag, `basePipelineIndex` **must** be -1 or `basePipelineHandle` **must** be `VK_NULL_HANDLE`
- VUID-VkComputePipelineCreateInfo-layout-07987

If a push constant block is declared in a shader, a push constant range in **layout** **must** match the shader stage

- VUID-VkComputePipelineCreateInfo-layout-10069

If a push constant block is declared in a shader, the block must be contained inside the push constant range in **layout** that matches the stage

- VUID-VkComputePipelineCreateInfo-layout-07988

If a **resource variables** is declared in a shader, a descriptor slot in **layout** **must** match the shader stage

- VUID-VkComputePipelineCreateInfo-layout-07990

If a **resource variables** is declared in a shader, a descriptor slot in **layout** **must** match the descriptor type

- VUID-VkComputePipelineCreateInfo-layout-07991

If a **resource variables** is declared in a shader as an array, a descriptor slot in **layout** **must** match the descriptor count

- VUID-VkComputePipelineCreateInfo-stage-00701

The **stage** member of **stage** **must** be **VK_SHADER_STAGE_COMPUTE_BIT**

- VUID-VkComputePipelineCreateInfo-stage-00702

The shader code for the entry point identified by **stage** and the rest of the state identified by this structure **must** adhere to the pipeline linking rules described in the **Shader Interfaces** chapter

- VUID-VkComputePipelineCreateInfo-layout-01687

The number of resources in **layout** accessible to the compute shader stage **must** be less than or equal to **VkPhysicalDeviceLimits::maxPerStageResources**

Valid Usage (Implicit)

- VUID-VkComputePipelineCreateInfo-sType-sType

sType **must** be **VK_STRUCTURE_TYPE_COMPUTE_PIPELINE_CREATE_INFO**

- VUID-VkComputePipelineCreateInfo-pNext-pNext

pNext **must** be **NULL**

- VUID-VkComputePipelineCreateInfo-stage-parameter

stage **must** be a valid **VkPipelineShaderStageCreateInfo** structure

- VUID-VkComputePipelineCreateInfo-layout-parameter

layout **must** be a valid **VkPipelineLayout** handle

- VUID-VkComputePipelineCreateInfo-commonparent

Both of **basePipelineHandle**, and **layout** that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same **VkDevice**

The **VkPipelineShaderStageCreateInfo** structure is defined as:


```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineShaderStageCreateInfo {
    VkStructureType          sType;
    const void*              pNext;
    VkPipelineShaderStageCreateFlags flags;
    VkShaderStageFlagBits    stage;
    VkShaderModule            module;
    const char*              pName;
    const VkSpecializationInfo* pSpecializationInfo;
} VkPipelineShaderStageCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkPipelineShaderStageCreateFlagBits` specifying how the pipeline shader stage will be generated.
- `stage` is a `VkShaderStageFlagBits` value specifying a single pipeline stage.
- `module` is a `VkShaderModule` object containing the shader code for this stage.
- `pName` is a pointer to a null-terminated UTF-8 string specifying the entry point name of the shader for this stage.
- `pSpecializationInfo` is a pointer to a `VkSpecializationInfo` structure, as described in [Specialization Constants](#), or `NULL`.

The shader code used by the pipeline is defined by `module`.

Valid Usage

- VUID-VkPipelineShaderStageCreateInfo-stage-00704
If the `geometryShader` feature is not enabled, `stage` **must** not be `VK_SHADER_STAGE_GEOMETRY_BIT`
- VUID-VkPipelineShaderStageCreateInfo-stage-00705
If the `tessellationShader` feature is not enabled, `stage` **must** not be `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT` or `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`
- VUID-VkPipelineShaderStageCreateInfo-stage-00706
`stage` **must** not be `VK_SHADER_STAGE_ALL_GRAPHICS`, or `VK_SHADER_STAGE_ALL`
- VUID-VkPipelineShaderStageCreateInfo-pName-00707
`pName` **must** be the name of an `OpEntryPoint` in `module` with an execution model that matches `stage`
- VUID-VkPipelineShaderStageCreateInfo-maxClipDistances-00708
If the identified entry point includes any variable in its interface that is declared with the `ClipDistance BuiltIn` decoration, that variable **must** not have an array size greater than `VkPhysicalDeviceLimits::maxClipDistances`
- VUID-VkPipelineShaderStageCreateInfo-maxCullDistances-00709

If the identified entry point includes any variable in its interface that is declared with the `CullDistance BuiltIn` decoration, that variable **must** not have an array size greater than `VkPhysicalDeviceLimits::maxCullDistances`

- VUID-VkPipelineShaderStageCreateInfo-maxCombinedClipAndCullDistances-00710

If the identified entry point includes variables in its interface that are declared with the `ClipDistance BuiltIn` decoration and variables in its interface that are declared with the `CullDistance BuiltIn` decoration, those variables **must** not have array sizes which sum to more than `VkPhysicalDeviceLimits::maxCombinedClipAndCullDistances`

- VUID-VkPipelineShaderStageCreateInfo-maxSampleMaskWords-00711

If the identified entry point includes any variable in its interface that is declared with the `SampleMask BuiltIn` decoration, that variable **must** not have an array size greater than `VkPhysicalDeviceLimits::maxSampleMaskWords`

- VUID-VkPipelineShaderStageCreateInfo-stage-00713

If `stage` is `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT` or `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`, and the identified entry point has an `OpExecutionMode` instruction specifying a patch size with `OutputVertices`, the patch size **must** be greater than `0` and less than or equal to `VkPhysicalDeviceLimits::maxTessellationPatchSize`

- VUID-VkPipelineShaderStageCreateInfo-stage-00714

If `stage` is `VK_SHADER_STAGE_GEOMETRY_BIT`, the identified entry point **must** have an `OpExecutionMode` instruction specifying a maximum output vertex count that is greater than `0` and less than or equal to `VkPhysicalDeviceLimits::maxGeometryOutputVertices`

- VUID-VkPipelineShaderStageCreateInfo-stage-00715

If `stage` is `VK_SHADER_STAGE_GEOMETRY_BIT`, the identified entry point **must** have an `OpExecutionMode` instruction specifying an invocation count that is greater than `0` and less than or equal to `VkPhysicalDeviceLimits::maxGeometryShaderInvocations`

- VUID-VkPipelineShaderStageCreateInfo-stage-02596

If `stage` is either `VK_SHADER_STAGE_VERTEX_BIT`, `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT`, `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`, or `VK_SHADER_STAGE_GEOMETRY_BIT`, and the identified entry point writes to `Layer` for any primitive, it **must** write the same value to `Layer` for all vertices of a given primitive

- VUID-VkPipelineShaderStageCreateInfo-stage-02597

If `stage` is either `VK_SHADER_STAGE_VERTEX_BIT`, `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT`, `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`, or `VK_SHADER_STAGE_GEOMETRY_BIT`, and the identified entry point writes to `ViewportIndex` for any primitive, it **must** write the same value to `ViewportIndex` for all vertices of a given primitive

- VUID-VkPipelineShaderStageCreateInfo-stage-06685

If `stage` is `VK_SHADER_STAGE_FRAGMENT_BIT`, and the identified entry point writes to `FragDepth` in any execution path, all execution paths that are not exclusive to helper invocations **must** either discard the fragment, or write or initialize the value of `FragDepth`

- VUID-VkPipelineShaderStageCreateInfo-stage-08771

`module` **must** be a valid `VkShaderModule`

- VUID-VkPipelineShaderStageCreateInfo-pSpecializationInfo-06849

The shader code used by the pipeline **must** be valid as described by the [Khronos SPIR-V](#)

[Specification](#) after applying the specializations provided in [pSpecializationInfo](#), if any, and then converting all specialization constants into fixed constants

Valid Usage (Implicit)

- VUID-VkPipelineShaderStageCreateInfo-sType-sType
[sType](#) **must** be [VK_STRUCTURE_TYPE_PIPELINE_SHADER_STAGE_CREATE_INFO](#)
- VUID-VkPipelineShaderStageCreateInfo-pNext-pNext
[pNext](#) **must** be [NULL](#) or a pointer to a valid instance of [VkShaderModuleCreateInfo](#)
- VUID-VkPipelineShaderStageCreateInfo-sType-unique
The [sType](#) value of each struct in the [pNext](#) chain **must** be unique
- VUID-VkPipelineShaderStageCreateInfo-flags-zero bitmask
[flags](#) **must** be 0
- VUID-VkPipelineShaderStageCreateInfo-stage-parameter
[stage](#) **must** be a valid [VkShaderStageFlagBits](#) value
- VUID-VkPipelineShaderStageCreateInfo-module-parameter
If [module](#) is not [VK_NULL_HANDLE](#), [module](#) **must** be a valid [VkShaderModule](#) handle
- VUID-VkPipelineShaderStageCreateInfo-pName-parameter
[pName](#) **must** be a null-terminated UTF-8 string
- VUID-VkPipelineShaderStageCreateInfo-pSpecializationInfo-parameter
If [pSpecializationInfo](#) is not [NULL](#), [pSpecializationInfo](#) **must** be a valid pointer to a valid [VkSpecializationInfo](#) structure

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineShaderStageCreateFlags;
```

[VkPipelineShaderStageCreateFlags](#) is a bitmask type for setting a mask of zero or more [VkPipelineShaderStageCreateFlagBits](#).

Possible values of the [flags](#) member of [VkPipelineShaderStageCreateInfo](#) specifying how a pipeline shader stage is created, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkPipelineShaderStageCreateFlagBits {
} VkPipelineShaderStageCreateFlagBits;
```

Bits which **can** be set by commands and structures, specifying one or more shader stages, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkShaderStageFlagBits {
    VK_SHADER_STAGE_VERTEX_BIT = 0x00000001,
    VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT = 0x00000002,
    VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT = 0x00000004,
    VK_SHADER_STAGE_GEOMETRY_BIT = 0x00000008,
    VK_SHADER_STAGE_FRAGMENT_BIT = 0x00000010,
    VK_SHADER_STAGE_COMPUTE_BIT = 0x00000020,
    VK_SHADER_STAGE_ALL_GRAPHICS = 0x0000001F,
    VK_SHADER_STAGE_ALL = 0x7FFFFFFF,
} VkShaderStageFlagBits;
```

- **VK_SHADER_STAGE_VERTEX_BIT** specifies the vertex stage.
- **VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT** specifies the tessellation control stage.
- **VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT** specifies the tessellation evaluation stage.
- **VK_SHADER_STAGE_GEOMETRY_BIT** specifies the geometry stage.
- **VK_SHADER_STAGE_FRAGMENT_BIT** specifies the fragment stage.
- **VK_SHADER_STAGE_COMPUTE_BIT** specifies the compute stage.
- **VK_SHADER_STAGE_ALL_GRAPHICS** is a combination of bits used as shorthand to specify all graphics stages defined above (excluding the compute stage).
- **VK_SHADER_STAGE_ALL** is a combination of bits used as shorthand to specify all shader stages supported by the device, including all additional stages which are introduced by extensions.

NOTE

VK_SHADER_STAGE_ALL_GRAPHICS only includes the original five graphics stages included in Vulkan 1.0, and not any stages added by extensions. Thus, it may not have the desired effect in all cases.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkShaderStageFlags;
```

VkShaderStageFlags is a bitmask type for setting a mask of zero or more **VkShaderStageFlagBits**.

10.3. Graphics Pipelines

Graphics pipelines consist of multiple shader stages, multiple fixed-function pipeline stages, and a pipeline layout.

To create graphics pipelines, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateGraphicsPipelines(
    VkDevice                                device,
    VkPipelineCache                        pipelineCache,
    uint32_t                              createInfoCount,
    const VkGraphicsPipelineCreateInfo*    pCreateInfos,
    const VkAllocationCallbacks*          pAllocator,
    VkPipeline*                            pPipelines);
```

- **device** is the logical device that creates the graphics pipelines.
- **pipelineCache** is either [VK_NULL_HANDLE](#), indicating that pipeline caching is disabled; or the handle of a valid [pipeline cache](#) object, in which case use of that cache is enabled for the duration of the command.
- **createInfoCount** is the length of the **pCreateInfos** and **pPipelines** arrays.
- **pCreateInfos** is a pointer to an array of [VkGraphicsPipelineCreateInfo](#) structures.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.
- **pPipelines** is a pointer to an array of [VkPipeline](#) handles in which the resulting graphics pipeline objects are returned.

The [VkGraphicsPipelineCreateInfo](#) structure includes an array of [VkPipelineShaderStageCreateInfo](#) structures for each of the desired active shader stages, as well as creation information for all relevant fixed-function stages, and a pipeline layout.

Pipelines are created and returned as described for [Multiple Pipeline Creation](#).

Valid Usage

- VUID-vkCreateGraphicsPipelines-device-09662
device **must** support at least one queue family with the [VK_QUEUE_GRAPHICS_BIT](#) capability
- VUID-vkCreateGraphicsPipelines-flags-00720
If the **flags** member of any element of **pCreateInfos** contains the [VK_PIPELINE_CREATE_DERIVATIVE_BIT](#) flag, and the **basePipelineIndex** member of that same element is not -1, **basePipelineIndex** **must** be less than the index into **pCreateInfos** that corresponds to that element
- VUID-vkCreateGraphicsPipelines-flags-00721
If the **flags** member of any element of **pCreateInfos** contains the [VK_PIPELINE_CREATE_DERIVATIVE_BIT](#) flag, the base pipeline **must** have been created with the [VK_PIPELINE_CREATE_ALLOW_DERIVATIVES_BIT](#) flag set

Valid Usage (Implicit)

- VUID-vkCreateGraphicsPipelines-device-parameter
device **must** be a valid [VkDevice](#) handle

- VUID-vkCreateGraphicsPipelines-pipelineCache-parameter
If `pipelineCache` is not `VK_NULL_HANDLE`, `pipelineCache` **must** be a valid `VkPipelineCache` handle
- VUID-vkCreateGraphicsPipelines-pCreateInfos-parameter
`pCreateInfos` **must** be a valid pointer to an array of `createInfoCount` valid `VkGraphicsPipelineCreateInfo` structures
- VUID-vkCreateGraphicsPipelines-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkCreateGraphicsPipelines-pPipelines-parameter
`pPipelines` **must** be a valid pointer to an array of `createInfoCount` `VkPipeline` handles
- VUID-vkCreateGraphicsPipelines-createInfoCount-arraylength
`createInfoCount` **must** be greater than `0`
- VUID-vkCreateGraphicsPipelines-pipelineCache-parent
If `pipelineCache` is a valid handle, it **must** have been created, allocated, or retrieved from device

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkGraphicsPipelineCreateInfo` structure is defined as:

```

// Provided by VK_VERSION_1_0
typedef struct VkGraphicsPipelineCreateInfo {
    VkStructureType           sType;
    const void*               pNext;
    VkPipelineCreateFlags     flags;
    uint32_t                  stageCount;
    const VkPipelineShaderStageCreateInfo* pStages;
    const VkPipelineVertexInputStateCreateInfo* pVertexInputState;
    const VkPipelineInputAssemblyStateCreateInfo* pInputAssemblyState;
    const VkPipelineTessellationStateCreateInfo* pTessellationState;
    const VkPipelineViewportStateCreateInfo* pViewportState;
    const VkPipelineRasterizationStateCreateInfo* pRasterizationState;
    const VkPipelineMultisampleStateCreateInfo* pMultisampleState;
    const VkPipelineDepthStencilStateCreateInfo* pDepthStencilState;
    const VkPipelineColorBlendStateCreateInfo* pColorBlendState;
    const VkPipelineDynamicStateCreateInfo* pDynamicState;
    VkPipelineLayout          layout;
    VkRenderPass              renderPass;
    uint32_t                  subpass;
    VkPipeline                basePipelineHandle;
    int32_t                   basePipelineIndex;
} VkGraphicsPipelineCreateInfo;

```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkPipelineCreateFlagBits` specifying how the pipeline will be generated.
- `stageCount` is the number of entries in the `pStages` array.
- `pStages` is a pointer to an array of `stageCount` `VkPipelineShaderStageCreateInfo` structures describing the set of the shader stages to be included in the graphics pipeline.
- `pVertexInputState` is a pointer to a `VkPipelineVertexInputStateCreateInfo` structure.
- `pInputAssemblyState` is a pointer to a `VkPipelineInputAssemblyStateCreateInfo` structure which determines input assembly behavior for vertex shading, as described in [Drawing Commands](#).
- `pTessellationState` is a pointer to a `VkPipelineTessellationStateCreateInfo` structure defining tessellation state used by tessellation shaders.
- `pViewportState` is a pointer to a `VkPipelineViewportStateCreateInfo` structure defining viewport state used when rasterization is enabled.
- `pRasterizationState` is a pointer to a `VkPipelineRasterizationStateCreateInfo` structure defining rasterization state.
- `pMultisampleState` is a pointer to a `VkPipelineMultisampleStateCreateInfo` structure defining multisample state used when rasterization is enabled.
- `pDepthStencilState` is a pointer to a `VkPipelineDepthStencilStateCreateInfo` structure defining depth/stencil state used when rasterization is enabled for depth or stencil attachments accessed during rendering.

- `pColorBlendState` is a pointer to a [VkPipelineColorBlendStateCreateInfo](#) structure defining color blend state used when rasterization is enabled for any color attachments accessed during rendering.
- `pDynamicState` is a pointer to a [VkPipelineDynamicStateCreateInfo](#) structure defining which properties of the pipeline state object are dynamic and **can** be changed independently of the pipeline state. This **can** be `NULL`, which means no state in the pipeline is considered dynamic.
- `layout` is the description of binding locations used by both the pipeline and descriptor sets used with the pipeline.
- `renderPass` is a handle to a render pass object describing the environment in which the pipeline will be used. The pipeline **must** only be used with a render pass instance compatible with the one provided. See [Render Pass Compatibility](#) for more information.
- `subpass` is the index of the subpass in the render pass where this pipeline will be used.
- `basePipelineHandle` is a pipeline to derive from.
- `basePipelineIndex` is an index into the `pCreateInfos` parameter to use as a pipeline to derive from.

The parameters `basePipelineHandle` and `basePipelineIndex` are described in more detail in [Pipeline Derivatives](#).

The state required for a graphics pipeline is divided into [vertex input state](#), [pre-rasterization shader state](#), [fragment shader state](#), and [fragment output state](#).

Vertex Input State

Vertex input state is defined by:

- [VkPipelineVertexInputStateCreateInfo](#)
- [VkPipelineInputAssemblyStateCreateInfo](#)

This state **must** be specified to create a [complete graphics pipeline](#).

Pre-Rasterization Shader State

Pre-rasterization shader state is defined by:

- [VkPipelineShaderStageCreateInfo](#) entries for:
 - Vertex shaders
 - Tessellation control shaders
 - Tessellation evaluation shaders
 - Geometry shaders
- Within the [VkPipelineLayout](#), the full pipeline layout **must** be specified.
- [VkPipelineViewportStateCreateInfo](#)
- [VkPipelineRasterizationStateCreateInfo](#)
- [VkPipelineTessellationStateCreateInfo](#)
- [VkRenderPass](#) and `subpass` parameter

This state **must** be specified to create a [complete graphics pipeline](#).

Fragment Shader State

Fragment shader state is defined by:

- A [VkPipelineShaderStageCreateInfo](#) entry for the fragment shader
- Within the [VkPipelineLayout](#), the full pipeline layout **must** be specified.
- [VkPipelineMultisampleStateCreateInfo](#)
- [VkPipelineDepthStencilStateCreateInfo](#)
- [VkRenderPass](#) and [subpass](#) parameter

If [rasterizerDiscardEnable](#) is [VK_FALSE](#) this state **must** be specified to create a [complete graphics pipeline](#).

Fragment Output State

Fragment output state is defined by:

- [VkPipelineColorBlendStateCreateInfo](#)
- [VkRenderPass](#) and [subpass](#) parameter
- [VkPipelineMultisampleStateCreateInfo](#)

If [rasterizerDiscardEnable](#) is [VK_FALSE](#) this state **must** be specified to create a [complete graphics pipeline](#).

Dynamic State

Dynamic state values set via [pDynamicState](#) **must** be ignored if the state they correspond to is not otherwise statically set by one of the state subsets used to create the pipeline. For example, if a pipeline only included [pre-rasterization shader state](#), then any dynamic state value corresponding to depth or stencil testing has no effect.

Complete Graphics Pipelines

A complete graphics pipeline always includes [pre-rasterization shader state](#), with other subsets included depending on that state as specified in the above sections.

Valid Usage

- VUID-VkGraphicsPipelineCreateInfo-None-09497
[flags](#) **must** be a valid combination of [VkPipelineCreateFlagBits](#) values
- VUID-VkGraphicsPipelineCreateInfo-flags-07984
If [flags](#) contains the [VK_PIPELINE_CREATE_DERIVATIVE_BIT](#) flag, and [basePipelineIndex](#) is -1, [basePipelineHandle](#) **must** be a valid graphics [VkPipeline](#) handle
- VUID-VkGraphicsPipelineCreateInfo-flags-07985
If [flags](#) contains the [VK_PIPELINE_CREATE_DERIVATIVE_BIT](#) flag, and [basePipelineHandle](#) is [VK_NULL_HANDLE](#), [basePipelineIndex](#) **must** be a valid index into the calling command's [pCreateInfos](#) parameter

- VUID-VkGraphicsPipelineCreateInfo-flags-07986
If **flags** contains the **VK_PIPELINE_CREATE_DERIVATIVE_BIT** flag, **basePipelineIndex** **must** be -1 or **basePipelineHandle** **must** be **VK_NULL_HANDLE**
- VUID-VkGraphicsPipelineCreateInfo-layout-07987
If a push constant block is declared in a shader, a push constant range in **layout** **must** match the shader stage
- VUID-VkGraphicsPipelineCreateInfo-layout-10069
If a push constant block is declared in a shader, the block must be contained inside the push constant range in **layout** that matches the stage
- VUID-VkGraphicsPipelineCreateInfo-layout-07988
If a **resource variables** is declared in a shader, a descriptor slot in **layout** **must** match the shader stage
- VUID-VkGraphicsPipelineCreateInfo-layout-07990
If a **resource variables** is declared in a shader, a descriptor slot in **layout** **must** match the descriptor type
- VUID-VkGraphicsPipelineCreateInfo-layout-07991
If a **resource variables** is declared in a shader as an array, a descriptor slot in **layout** **must** match the descriptor count
- VUID-VkGraphicsPipelineCreateInfo-stage-02096
If the pipeline requires **pre-rasterization shader state** the **stage** member of one element of **pStages** **must** be **VK_SHADER_STAGE_VERTEX_BIT**
- VUID-VkGraphicsPipelineCreateInfo-pStages-00729
If the pipeline requires **pre-rasterization shader state** and **pStages** includes a tessellation control shader stage, it **must** include a tessellation evaluation shader stage
- VUID-VkGraphicsPipelineCreateInfo-pStages-00730
If the pipeline requires **pre-rasterization shader state** and **pStages** includes a tessellation evaluation shader stage, it **must** include a tessellation control shader stage
- VUID-VkGraphicsPipelineCreateInfo-pStages-09022
If the pipeline requires **pre-rasterization shader state** and **pStages** includes a tessellation control shader stage, **pTessellationState** **must** be a valid pointer to a valid **VkPipelineTessellationStateCreateInfo** structure
- VUID-VkGraphicsPipelineCreateInfo-pStages-00732
If the pipeline requires **pre-rasterization shader state** and **pStages** includes tessellation shader stages, the shader code of at least one stage **must** contain an **OpExecutionMode** instruction specifying the type of subdivision in the pipeline
- VUID-VkGraphicsPipelineCreateInfo-pStages-00733
If the pipeline requires **pre-rasterization shader state** and **pStages** includes tessellation shader stages, and the shader code of both stages contain an **OpExecutionMode** instruction specifying the type of subdivision in the pipeline, they **must** both specify the same subdivision mode
- VUID-VkGraphicsPipelineCreateInfo-pStages-00734
If the pipeline requires **pre-rasterization shader state** and **pStages** includes tessellation shader stages, the shader code of at least one stage **must** contain an **OpExecutionMode**

instruction specifying the output patch size in the pipeline

- VUID-VkGraphicsPipelineCreateInfo-pStages-00735

If the pipeline requires [pre-rasterization shader state](#) and `pStages` includes tessellation shader stages, and the shader code of both contain an `OpExecutionMode` instruction specifying the out patch size in the pipeline, they **must** both specify the same patch size

- VUID-VkGraphicsPipelineCreateInfo-pStages-08888

If the pipeline is being created with [pre-rasterization shader state](#) and [vertex input state](#) and `pStages` includes tessellation shader stages, the `topology` member of `pInputAssembly` **must** be `VK_PRIMITIVE_TOPOLOGY_PATCH_LIST`

- VUID-VkGraphicsPipelineCreateInfo-topology-08889

If the pipeline is being created with [pre-rasterization shader state](#) and [vertex input state](#) and the `topology` member of `pInputAssembly` is `VK_PRIMITIVE_TOPOLOGY_PATCH_LIST`, then `pStages` **must** include tessellation shader stages

- VUID-VkGraphicsPipelineCreateInfo-TessellationEvaluation-07723

If the pipeline is being created with a `TessellationEvaluation Execution Model`, no `Geometry Execution Model`, uses the `PointMode Execution Mode`, and `shaderTessellationAndGeometryPointSize` is enabled, a `PointSize` decorated variable **must** be written to

- VUID-VkGraphicsPipelineCreateInfo-topology-08773

If the pipeline is being created with a `Vertex Execution Model` and no `TessellationEvaluation` or `Geometry Execution Model`, and the `topology` member of `pInputAssembly` is `VK_PRIMITIVE_TOPOLOGY_POINT_LIST`, a `PointSize` decorated variable **must** be written to

- VUID-VkGraphicsPipelineCreateInfo-TessellationEvaluation-07724

If the pipeline is being created with a `TessellationEvaluation Execution Model`, no `Geometry Execution Model`, uses the `PointMode Execution Mode`, and `shaderTessellationAndGeometryPointSize` is not enabled, a `PointSize` decorated variable **must** not be written to

- VUID-VkGraphicsPipelineCreateInfo-shaderTessellationAndGeometryPointSize-08776

If the pipeline is being created with a `Geometry Execution Model`, uses the `OutputPoints Execution Mode`, and `shaderTessellationAndGeometryPointSize` is enabled, a `PointSize` decorated variable **must** be written to for every vertex emitted

- VUID-VkGraphicsPipelineCreateInfo-Geometry-07726

If the pipeline is being created with a `Geometry Execution Model`, uses the `OutputPoints Execution Mode`, and `shaderTessellationAndGeometryPointSize` is not enabled, a `PointSize` decorated variable **must** not be written to

- VUID-VkGraphicsPipelineCreateInfo-pStages-00738

If the pipeline requires [pre-rasterization shader state](#) and `pStages` includes a geometry shader stage, and does not include any tessellation shader stages, its shader code **must** contain an `OpExecutionMode` instruction specifying an input primitive type that is [compatible](#) with the primitive topology specified in `pInputAssembly`

- VUID-VkGraphicsPipelineCreateInfo-pStages-00739

If the pipeline requires [pre-rasterization shader state](#) and `pStages` includes a geometry shader stage, and also includes tessellation shader stages, its shader code **must** contain an

`OpExecutionMode` instruction specifying an input primitive type that is `compatible` with the primitive topology that is output by the tessellation stages

- VUID-VkGraphicsPipelineCreateInfo-pStages-00740
If the pipeline requires `pre-rasterization shader state` and `fragment shader state`, it includes both a fragment shader and a geometry shader, and the fragment shader code reads from an input variable that is decorated with `PrimitiveId`, then the geometry shader code **must** write to a matching output variable, decorated with `PrimitiveId`, in all execution paths
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06038
If `renderPass` is not `VK_NULL_HANDLE` and the pipeline is being created with `fragment shader state` the fragment shader **must** not read from any input attachment that is defined as `VK_ATTACHMENT_UNUSED` in `subpass`
- VUID-VkGraphicsPipelineCreateInfo-pStages-00742
If the pipeline requires `pre-rasterization shader state` and multiple pre-rasterization shader stages are included in `pStages`, the shader code for the entry points identified by those `pStages` and the rest of the state identified by this structure **must** adhere to the pipeline linking rules described in the `Shader Interfaces` chapter
- VUID-VkGraphicsPipelineCreateInfo-None-04889
If the pipeline requires `pre-rasterization shader state` and `fragment shader state`, the fragment shader and last `pre-rasterization shader stage` and any relevant state **must** adhere to the pipeline linking rules described in the `Shader Interfaces` chapter
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06041
If `renderPass` is not `VK_NULL_HANDLE`, and the pipeline is being created with `fragment output interface state`, then for each color attachment in the subpass, if the `potential format features` of the format of the corresponding attachment description do not contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`, then the `blendEnable` member of the corresponding element of the `pAttachments` member of `pColorBlendState` **must** be `VK_FALSE`
- VUID-VkGraphicsPipelineCreateInfo-renderPass-07609
If `renderPass` is not `VK_NULL_HANDLE`, the pipeline is being created with `fragment output interface state`, the `pColorBlendState` pointer is not `NULL`, the `attachmentCount` member of `pColorBlendState` is not ignored, and the subpass uses color attachments, the `attachmentCount` member of `pColorBlendState` **must** be equal to the `colorAttachmentCount` used to create `subpass`
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-04130
If the pipeline requires `pre-rasterization shader state`, and `pViewportState->pViewports` is not dynamic, then `pViewportState->pViewports` **must** be a valid pointer to an array of `pViewportState->viewportCount` valid `VkViewport` structures
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-04131
If the pipeline requires `pre-rasterization shader state`, and `pViewportState->pScissors` is not dynamic, then `pViewportState->pScissors` **must** be a valid pointer to an array of `pViewportState->scissorCount` `VkRect2D` structures
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-00749
If the pipeline requires `pre-rasterization shader state`, and the `widelines` feature is not enabled, and no element of the `pDynamicStates` member of `pDynamicState` is

VK_DYNAMIC_STATE_LINE_WIDTH, the lineWidth member of pRasterizationState **must** be 1.0

- VUID-VkGraphicsPipelineCreateInfo-rasterizerDiscardEnable-09024

If the pipeline requires [pre-rasterization shader state](#), and the rasterizerDiscardEnable member of pRasterizationState is VK_FALSE, pViewportState **must** be a valid pointer to a valid [VkPipelineViewportStateCreateInfo](#) structure

- VUID-VkGraphicsPipelineCreateInfo-pMultisampleState-09026

If the pipeline requires [fragment output interface state](#), pMultisampleState **must** be a valid pointer to a valid [VkPipelineMultisampleStateCreateInfo](#) structure

- VUID-VkGraphicsPipelineCreateInfo-pMultisampleState-09027

If pMultisampleState is not NULL it **must** be a valid pointer to a valid [VkPipelineMultisampleStateCreateInfo](#) structure

- VUID-VkGraphicsPipelineCreateInfo-alphaToCoverageEnable-08891

If the pipeline is being created with [fragment shader state](#), the [VkPipelineMultisampleStateCreateInfo::alphaToCoverageEnable](#) is not ignored and is VK_TRUE, then the [Fragment Output Interface](#) **must** contain a variable for the alpha Component word in Location 0 at Index 0

- VUID-VkGraphicsPipelineCreateInfo-renderPass-09028

If renderPass is not VK_NULL_HANDLE, the pipeline is being created with [fragment shader state](#), and subpass uses a depth/stencil attachment, pDepthStencilState **must** be a valid pointer to a valid [VkPipelineDepthStencilStateCreateInfo](#) structure

- VUID-VkGraphicsPipelineCreateInfo-pDepthStencilState-09029

If pDepthStencilState is not NULL it **must** be a valid pointer to a valid [VkPipelineDepthStencilStateCreateInfo](#) structure

- VUID-VkGraphicsPipelineCreateInfo-renderPass-09030

If renderPass is not VK_NULL_HANDLE, the pipeline is being created with [fragment output interface state](#), and subpass uses color attachments, pColorBlendState **must** be a valid pointer to a valid [VkPipelineColorBlendStateCreateInfo](#) structure

- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-00754

If the pipeline requires [pre-rasterization shader state](#), the depthBiasClamp feature is not enabled, no element of the pDynamicStates member of pDynamicState is VK_DYNAMIC_STATE_DEPTH_BIAS, and the depthBiasEnable member of pRasterizationState is VK_TRUE, the depthBiasClamp member of pRasterizationState **must** be 0.0

- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-02510

If the pipeline requires [fragment shader state](#), and no element of the pDynamicStates member of pDynamicState is VK_DYNAMIC_STATE_DEPTH_BOUNDS, and the depthBoundsTestEnable member of pDepthStencilState is VK_TRUE, the minDepthBounds and maxDepthBounds members of pDepthStencilState **must** be between 0.0 and 1.0, inclusive

- VUID-VkGraphicsPipelineCreateInfo-subpass-00758

If the pipeline requires [fragment output interface state](#), rasterizationSamples is not dynamic, and subpass does not use any color and/or depth/stencil attachments, then the rasterizationSamples member of pMultisampleState **must** follow the rules for a [zero-attachment subpass](#)

- VUID-VkGraphicsPipelineCreateInfo-renderPass-06046

If `renderPass` is not `VK_NULL_HANDLE`, `subpass` **must** be a valid subpass within `renderPass`

- VUID-VkGraphicsPipelineCreateInfo-layout-01688

The number of resources in `layout` accessible to each shader stage that is used by the pipeline **must** be less than or equal to `VkPhysicalDeviceLimits::maxPerStageResources`

- VUID-VkGraphicsPipelineCreateInfo-pStages-02097

If the pipeline requires `vertex input state`, and `pVertexInputState` is not dynamic, then `pVertexInputState` **must** be a valid pointer to a valid `VkPipelineVertexInputStateCreateInfo` structure

- VUID-VkGraphicsPipelineCreateInfo-Input-07904

If the pipeline is being created with `vertex input state` and `pVertexInputState` is not dynamic, then all variables with the `Input` storage class decorated with `Location` in the `Vertex Execution Model OpEntryPoint` **must** contain a location in `VkVertexInputAttributeDescription::location`

- VUID-VkGraphicsPipelineCreateInfo-Input-08733

If the pipeline requires `vertex input state` and `pVertexInputState` is not dynamic, then the numeric type associated with all `Input` variables of the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` **must** be the same as `VkVertexInputAttributeDescription::format`

- VUID-VkGraphicsPipelineCreateInfo-pVertexInputState-08929

If the pipeline is being created with `vertex input state` and `pVertexInputState` is not dynamic, and `VkVertexInputAttributeDescription::format` has a 64-bit component, then the scalar width associated with all `Input` variables of the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` **must** be 64-bit

- VUID-VkGraphicsPipelineCreateInfo-pVertexInputState-08930

If the pipeline is being created with `vertex input state` and `pVertexInputState` is not dynamic, and the scalar width associated with a `Location` decorated `Input` variable in the `Vertex Execution Model OpEntryPoint` is 64-bit, then the corresponding `VkVertexInputAttributeDescription::format` **must** have a 64-bit component

- VUID-VkGraphicsPipelineCreateInfo-pVertexInputState-09198

If the pipeline is being created with `vertex input state` and `pVertexInputState` is not dynamic, and `VkVertexInputAttributeDescription::format` has a 64-bit component, then all `Input` variables at the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` **must** not use components that are not present in the format

- VUID-VkGraphicsPipelineCreateInfo-dynamicPrimitiveTopologyUnrestricted-09031

If the pipeline requires `vertex input state`, `pInputAssemblyState` **must** be a valid pointer to a valid `VkPipelineInputAssemblyStateCreateInfo` structure

- VUID-VkGraphicsPipelineCreateInfo-pInputAssemblyState-09032

If `pInputAssemblyState` is not `NULL` it **must** be a valid pointer to a valid `VkPipelineInputAssemblyStateCreateInfo` structure

- VUID-VkGraphicsPipelineCreateInfo-pStages-06600

If the pipeline requires `pre-rasterization shader state` or `fragment shader state`, `pStages` **must** be a valid pointer to an array of `stageCount` valid `VkPipelineShaderStageCreateInfo` structures

- VUID-VkGraphicsPipelineCreateInfo-stageCount-09587
If the pipeline does not require [pre-rasterization shader state](#) or [fragment shader state](#), **stageCount** **must** be zero
- VUID-VkGraphicsPipelineCreateInfo-pRasterizationState-06601
If the pipeline requires [pre-rasterization shader state](#), **pRasterizationState** **must** be a valid pointer to a valid [VkPipelineRasterizationStateCreateInfo](#) structure
- VUID-VkGraphicsPipelineCreateInfo-layout-06602
If the pipeline requires [fragment shader state](#) or [pre-rasterization shader state](#), **layout** **must** be a valid [VkPipelineLayout](#) handle
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06603
If the pipeline requires [pre-rasterization shader state](#), [fragment shader state](#), or [fragment output state](#), **renderPass** **must** be a valid [VkRenderPass](#) handle
- VUID-VkGraphicsPipelineCreateInfo-stageCount-09530
If the pipeline requires [pre-rasterization shader state](#), **stageCount** **must** be greater than 0
- VUID-VkGraphicsPipelineCreateInfo-pStages-06894
If the pipeline requires [pre-rasterization shader state](#) but not [fragment shader state](#), elements of **pStages** **must** not have **stage** set to [VK_SHADER_STAGE_FRAGMENT_BIT](#)
- VUID-VkGraphicsPipelineCreateInfo-pStages-06895
If the pipeline requires [fragment shader state](#) but not [pre-rasterization shader state](#), elements of **pStages** **must** not have **stage** set to a shader stage which participates in pre-rasterization
- VUID-VkGraphicsPipelineCreateInfo-pStages-06896
If the pipeline requires [pre-rasterization shader state](#), all elements of **pStages** **must** have a **stage** set to a shader stage which participates in [fragment shader state](#) or [pre-rasterization shader state](#)
- VUID-VkGraphicsPipelineCreateInfo-stage-06897
If the pipeline requires [fragment shader state](#) and/or [pre-rasterization shader state](#), any value of **stage** **must** not be set in more than one element of **pStages**
- VUID-VkGraphicsPipelineCreateInfo-None-08893
The pipeline **must** be created with [pre-rasterization shader state](#)
- VUID-VkGraphicsPipelineCreateInfo-pStages-08894
If **pStages** includes a vertex shader stage, the pipeline **must** be created with [vertex input state](#)
- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-08896
If **pRasterizationState->rasterizerDiscardEnable** is [VK_FALSE](#), the pipeline **must** be created with [fragment shader state](#) and [fragment output interface state](#)
- VUID-VkGraphicsPipelineCreateInfo-None-09043
If the format of any color attachment is [VK_FORMAT_E5B9G9R9_UFLOAT_PACK32](#), the **colorWriteMask** member of the corresponding element of **pColorBlendState->pAttachments** **must** either include all of [VK_COLOR_COMPONENT_R_BIT](#), [VK_COLOR_COMPONENT_G_BIT](#), and [VK_COLOR_COMPONENT_B_BIT](#), or none of them

Valid Usage (Implicit)

- VUID-VkGraphicsPipelineCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_GRAPHICS_PIPELINE_CREATE_INFO`
- VUID-VkGraphicsPipelineCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-parameter
If `pDynamicState` is not `NULL`, `pDynamicState` **must** be a valid pointer to a valid `VkPipelineDynamicStateCreateInfo` structure
- VUID-VkGraphicsPipelineCreateInfo-commonparent
Each of `basePipelineHandle`, `layout`, and `renderPass` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

Bits which **can** be set in

- `VkGraphicsPipelineCreateInfo::flags`
- `VkComputePipelineCreateInfo::flags`

specify how a pipeline is created, and are:

```
// Provided by VK_VERSION_1_0
typedef enum VkPipelineCreateFlagBits {
    VK_PIPELINE_CREATE_DISABLE_OPTIMIZATION_BIT = 0x00000001,
    VK_PIPELINE_CREATE_ALLOW_DERIVATIVES_BIT = 0x00000002,
    VK_PIPELINE_CREATE_DERIVATIVE_BIT = 0x00000004,
} VkPipelineCreateFlagBits;
```

- `VK_PIPELINE_CREATE_DISABLE_OPTIMIZATION_BIT` specifies that the created pipeline will not be optimized. Using this flag **may** reduce the time taken to create the pipeline.
- `VK_PIPELINE_CREATE_ALLOW_DERIVATIVES_BIT` specifies that the pipeline to be created is allowed to be the parent of a pipeline that will be created in a subsequent pipeline creation call.
- `VK_PIPELINE_CREATE_DERIVATIVE_BIT` specifies that the pipeline to be created will be a child of a previously created parent pipeline.

It is valid to set both `VK_PIPELINE_CREATE_ALLOW_DERIVATIVES_BIT` and `VK_PIPELINE_CREATE_DERIVATIVE_BIT`. This allows a pipeline to be both a parent and possibly a child in a pipeline hierarchy. See [Pipeline Derivatives](#) for more information.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineCreateFlags;
```

`VkPipelineCreateFlags` is a bitmask type for setting a mask of zero or more `VkPipelineCreateFlagBits`.

The `VkPipelineDynamicStateCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineDynamicStateCreateInfo {
    VkStructureType          sType;
    const void*              pNext;
    VkPipelineDynamicStateCreateFlags flags;
    uint32_t                 dynamicStateCount;
    const VkDynamicState*     pDynamicStates;
} VkPipelineDynamicStateCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `dynamicStateCount` is the number of elements in the `pDynamicStates` array.
- `pDynamicStates` is a pointer to an array of `VkDynamicState` values specifying which pieces of pipeline state will use the values from dynamic state commands rather than from pipeline state creation information.

Valid Usage

- VUID-VkPipelineDynamicStateCreateInfo-pDynamicStates-01442
Each element of `pDynamicStates` **must** be unique

Valid Usage (Implicit)

- VUID-VkPipelineDynamicStateCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_DYNAMIC_STATE_CREATE_INFO`
- VUID-VkPipelineDynamicStateCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkPipelineDynamicStateCreateInfo-flags-zero bitmask
`flags` **must** be 0
- VUID-VkPipelineDynamicStateCreateInfo-pDynamicStates-parameter
If `dynamicStateCount` is not 0, `pDynamicStates` **must** be a valid pointer to an array of `dynamicStateCount` valid `VkDynamicState` values

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineDynamicStateCreateFlags;
```

`VkPipelineDynamicStateCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

The source of different pieces of dynamic state is specified by the `VkPipelineDynamicStateCreateInfo::pDynamicStates` property of the currently active pipeline, each of whose elements **must** be one of the values:

```
// Provided by VK_VERSION_1_0
typedef enum VkDynamicState {
    VK_DYNAMIC_STATE_VIEWPORT = 0,
    VK_DYNAMIC_STATE_SCISSOR = 1,
    VK_DYNAMIC_STATE_LINE_WIDTH = 2,
    VK_DYNAMIC_STATE_DEPTH_BIAS = 3,
    VK_DYNAMIC_STATE_BLEND_CONSTANTS = 4,
    VK_DYNAMIC_STATE_DEPTH_BOUNDS = 5,
    VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK = 6,
    VK_DYNAMIC_STATE_STENCIL_WRITE_MASK = 7,
    VK_DYNAMIC_STATE_STENCIL_REFERENCE = 8,
} VkDynamicState;
```

- `VK_DYNAMIC_STATE_VIEWPORT` specifies that the `pViewports` state in `VkPipelineViewportStateCreateInfo` will be ignored and **must** be set dynamically with `vkCmdSetViewport` before any drawing commands. The number of viewports used by a pipeline is still specified by the `viewportCount` member of `VkPipelineViewportStateCreateInfo`.
- `VK_DYNAMIC_STATE_SCISSOR` specifies that the `pScissors` state in `VkPipelineViewportStateCreateInfo` will be ignored and **must** be set dynamically with `vkCmdSetScissor` before any drawing commands. The number of scissor rectangles used by a pipeline is still specified by the `scissorCount` member of `VkPipelineViewportStateCreateInfo`.
- `VK_DYNAMIC_STATE_LINE_WIDTH` specifies that the `lineWidth` state in `VkPipelineRasterizationStateCreateInfo` will be ignored and **must** be set dynamically with `vkCmdSetLineWidth` before any drawing commands that generate line primitives for the rasterizer.
- `VK_DYNAMIC_STATE_DEPTH_BIAS` specifies that the `depthBiasConstantFactor`, `depthBiasClamp` and `depthBiasSlopeFactor` states in `VkPipelineRasterizationStateCreateInfo` will be ignored and **must** be set dynamically with `vkCmdSetDepthBias` before any draws are performed with `depth bias enabled`.
- `VK_DYNAMIC_STATE_BLEND_CONSTANTS` specifies that the `blendConstants` state in `VkPipelineColorBlendStateCreateInfo` will be ignored and **must** be set dynamically with `vkCmdSetBlendConstants` before any draws are performed with a pipeline state with `VkPipelineColorBlendAttachmentState` member `blendEnable` set to `VK_TRUE` and any of the blend functions using a constant blend color.
- `VK_DYNAMIC_STATE_DEPTH_BOUNDS` specifies that the `minDepthBounds` and `maxDepthBounds` states of `VkPipelineDepthStencilStateCreateInfo` will be ignored and **must** be set dynamically with `vkCmdSetDepthBounds` before any draws are performed with a pipeline state with `VkPipelineDepthStencilStateCreateInfo` member `depthBoundsTestEnable` set to `VK_TRUE`.
- `VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK` specifies that the `compareMask` state in `VkPipelineDepthStencilStateCreateInfo` for both `front` and `back` will be ignored and **must** be set dynamically with `vkCmdSetStencilCompareMask` before any draws are performed with a

pipeline state with `VkPipelineDepthStencilStateCreateInfo` member `stencilTestEnable` set to `VK_TRUE`

- `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK` specifies that the `writeMask` state in `VkPipelineDepthStencilStateCreateInfo` for both `front` and `back` will be ignored and **must** be set dynamically with `vkCmdSetStencilWriteMask` before any draws are performed with a pipeline state with `VkPipelineDepthStencilStateCreateInfo` member `stencilTestEnable` set to `VK_TRUE`
- `VK_DYNAMIC_STATE_STENCIL_REFERENCE` specifies that the `reference` state in `VkPipelineDepthStencilStateCreateInfo` for both `front` and `back` will be ignored and **must** be set dynamically with `vkCmdSetStencilReference` before any draws are performed with a pipeline state with `VkPipelineDepthStencilStateCreateInfo` member `stencilTestEnable` set to `VK_TRUE`

10.3.1. Valid Combinations of Stages for Graphics Pipelines

If tessellation shader stages are omitted, the tessellation shading and fixed-function stages of the pipeline are skipped.

If a geometry shader is omitted, the geometry shading stage is skipped.

If a fragment shader is omitted, fragment color outputs have undefined values, and the fragment depth value is determined by `Fragment Operations` state. This **can** be useful for depth-only rendering.

Presence of a shader stage in a pipeline is indicated by including a valid `VkPipelineShaderStageCreateInfo` with `module` and `pName` selecting an entry point from a shader module, where that entry point is valid for the stage specified by `stage`.

Presence of some of the fixed-function stages in the pipeline is implicitly derived from enabled shaders and provided state. For example, the fixed-function tessellator is always present when the pipeline has valid Tessellation Control and Tessellation Evaluation shaders.

For example:

- Depth/stencil-only rendering in a subpass with no color attachments
 - Active Pipeline Shader Stages
 - Vertex Shader
 - Required: Fixed-Function Pipeline Stages
 - `VkPipelineVertexInputStateCreateInfo`
 - `VkPipelineInputAssemblyStateCreateInfo`
 - `VkPipelineViewportStateCreateInfo`
 - `VkPipelineRasterizationStateCreateInfo`
 - `VkPipelineMultisampleStateCreateInfo`
 - `VkPipelineDepthStencilStateCreateInfo`
- Color-only rendering in a subpass with no depth/stencil attachment
 - Active Pipeline Shader Stages

- Vertex Shader
- Fragment Shader
- Required: Fixed-Function Pipeline Stages
 - [VkPipelineVertexInputStateCreateInfo](#)
 - [VkPipelineInputAssemblyStateCreateInfo](#)
 - [VkPipelineViewportStateCreateInfo](#)
 - [VkPipelineRasterizationStateCreateInfo](#)
 - [VkPipelineMultisampleStateCreateInfo](#)
 - [VkPipelineColorBlendStateCreateInfo](#)
- Rendering pipeline with tessellation and geometry shaders
 - Active Pipeline Shader Stages
 - Vertex Shader
 - Tessellation Control Shader
 - Tessellation Evaluation Shader
 - Geometry Shader
 - Fragment Shader
 - Required: Fixed-Function Pipeline Stages
 - [VkPipelineVertexInputStateCreateInfo](#)
 - [VkPipelineInputAssemblyStateCreateInfo](#)
 - [VkPipelineTessellationStateCreateInfo](#)
 - [VkPipelineViewportStateCreateInfo](#)
 - [VkPipelineRasterizationStateCreateInfo](#)
 - [VkPipelineMultisampleStateCreateInfo](#)
 - [VkPipelineDepthStencilStateCreateInfo](#)
 - [VkPipelineColorBlendStateCreateInfo](#)

10.4. Pipeline Destruction

To destroy a pipeline, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyPipeline(
    VkDevice          device,
    VkPipeline        pipeline,
    const VkAllocationCallbacks* pAllocator);
```

- **device** is the logical device that destroys the pipeline.

- `pipeline` is the handle of the pipeline to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyPipeline-pipeline-00765
All submitted commands that refer to `pipeline` **must** have completed execution
- VUID-vkDestroyPipeline-pipeline-00766
If `VkAllocationCallbacks` were provided when `pipeline` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyPipeline-pipeline-00767
If no `VkAllocationCallbacks` were provided when `pipeline` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyPipeline-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkDestroyPipeline-pipeline-parameter
If `pipeline` is not `VK_NULL_HANDLE`, `pipeline` **must** be a valid `VkPipeline` handle
- VUID-vkDestroyPipeline-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkDestroyPipeline-pipeline-parent
If `pipeline` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `pipeline` **must** be externally synchronized

10.5. Pipeline Derivatives

A pipeline derivative is a child pipeline created from a parent pipeline, where the child and parent are expected to have much commonality.

The goal of derivative pipelines is that they be cheaper to create using the parent as a starting point, and that it be more efficient (on either host or device) to switch/bind between children of the same parent.

A derivative pipeline is created by setting the `VK_PIPELINE_CREATE_DERIVATIVE_BIT` flag in the `Vk*PipelineCreateInfo` structure. If this is set, then exactly one of `basePipelineHandle` or `basePipelineIndex` members of the structure **must** have a valid handle/index, and specifies the

parent pipeline. If `basePipelineHandle` is used, the parent pipeline **must** have already been created. If `basePipelineIndex` is used, then the parent is being created in the same command. `VK_NULL_HANDLE` acts as the invalid handle for `basePipelineHandle`, and -1 is the invalid index for `basePipelineIndex`. If `basePipelineIndex` is used, the base pipeline **must** appear earlier in the array. The base pipeline **must** have been created with the `VK_PIPELINE_CREATE_ALLOW_DERIVATIVES_BIT` flag set.

10.6. Pipeline Cache

Pipeline cache objects allow the result of pipeline construction to be reused between pipelines and between runs of an application. Reuse between pipelines is achieved by passing the same pipeline cache object when creating multiple related pipelines. Reuse across runs of an application is achieved by retrieving pipeline cache contents in one run of an application, saving the contents, and using them to preinitialize a pipeline cache on a subsequent run. The contents of the pipeline cache objects are managed by the implementation. Applications **can** manage the host memory consumed by a pipeline cache object and control the amount of data retrieved from a pipeline cache object.

Pipeline cache objects are represented by `VkPipelineCache` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkPipelineCache)
```

10.6.1. Creating a Pipeline Cache

To create pipeline cache objects, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreatePipelineCache(
    VkDevice                                device,
    const VkPipelineCacheCreateInfo*        pCreateInfo,
    const VkAllocationCallbacks*           pAllocator,
    VkPipelineCache*                        pPipelineCache);
```

- `device` is the logical device that creates the pipeline cache object.
- `pCreateInfo` is a pointer to a `VkPipelineCacheCreateInfo` structure containing initial parameters for the pipeline cache object.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.
- `pPipelineCache` is a pointer to a `VkPipelineCache` handle in which the resulting pipeline cache object is returned.

NOTE

Applications **can** track and manage the total host memory size of a pipeline cache object using the `pAllocator`. Applications **can** limit the amount of data retrieved from a pipeline cache object in `vkGetPipelineCacheData`. Implementations **should** not internally limit the total number of entries added to a pipeline cache object or

the total host memory consumed.

Once created, a pipeline cache **can** be passed to the [vkCreateGraphicsPipelines](#) and [vkCreateComputePipelines](#) commands. If the pipeline cache passed into these commands is not [VK_NULL_HANDLE](#), the implementation will query it for possible reuse opportunities and update it with new content. The use of the pipeline cache object in these commands is internally synchronized, and the same pipeline cache object **can** be used in multiple threads simultaneously.

NOTE

Implementations **should** make every effort to limit any critical sections to the actual accesses to the cache, which is expected to be significantly shorter than the duration of the [vkCreate*Pipelines](#) commands.

Valid Usage (Implicit)

- VUID-vkCreatePipelineCache-device-parameter
[device](#) **must** be a valid [VkDevice](#) handle
- VUID-vkCreatePipelineCache-pCreateInfo-parameter
[pCreateInfo](#) **must** be a valid pointer to a valid [VkPipelineCacheCreateInfo](#) structure
- VUID-vkCreatePipelineCache-pAllocator-parameter
If [pAllocator](#) is not [NULL](#), [pAllocator](#) **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkCreatePipelineCache-pPipelineCache-parameter
[pPipelineCache](#) **must** be a valid pointer to a [VkPipelineCache](#) handle

Return Codes

Success

- [VK_SUCCESS](#)

Failure

- [VK_ERROR_OUT_OF_HOST_MEMORY](#)
- [VK_ERROR_OUT_OF_DEVICE_MEMORY](#)

The [VkPipelineCacheCreateInfo](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineCacheCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkPipelineCacheCreateFlags flags;
    size_t              initialDataSize;
    const void*        pInitialData;
} VkPipelineCacheCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `initialDataSize` is the number of bytes in `pInitialData`. If `initialDataSize` is zero, the pipeline cache will initially be empty.
- `pInitialData` is a pointer to previously retrieved pipeline cache data. If the pipeline cache data is incompatible (as defined below) with the device, the pipeline cache will be initially empty. If `initialDataSize` is zero, `pInitialData` is ignored.

Valid Usage

- VUID-VkPipelineCacheCreateInfo-initialDataSize-00768
If `initialDataSize` is not 0, it **must** be equal to the size of `pInitialData`, as returned by `vkGetPipelineCacheData` when `pInitialData` was originally retrieved
- VUID-VkPipelineCacheCreateInfo-initialDataSize-00769
If `initialDataSize` is not 0, `pInitialData` **must** have been retrieved from a previous call to `vkGetPipelineCacheData`

Valid Usage (Implicit)

- VUID-VkPipelineCacheCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_CACHE_CREATE_INFO`
- VUID-VkPipelineCacheCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkPipelineCacheCreateInfo-flags-zeroBitmask
`flags` **must** be 0
- VUID-VkPipelineCacheCreateInfo-pInitialData-parameter
If `initialDataSize` is not 0, `pInitialData` **must** be a valid pointer to an array of `initialDataSize` bytes

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineCacheCreateFlags;
```

`VkPipelineCacheCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

10.6.2. Merging Pipeline Caches

Pipeline cache objects **can** be merged using the command:


```
// Provided by VK_VERSION_1_0
VkResult vkMergePipelineCaches(
    VkDevice          device,
    VkPipelineCache   dstCache,
    uint32_t          srcCacheCount,
    const VkPipelineCache* pSrcCaches);
```

- **device** is the logical device that owns the pipeline cache objects.
- **dstCache** is the handle of the pipeline cache to merge results into.
- **srcCacheCount** is the length of the **pSrcCaches** array.
- **pSrcCaches** is a pointer to an array of pipeline cache handles, which will be merged into **dstCache**. The previous contents of **dstCache** are included after the merge.

NOTE

The details of the merge operation are implementation-dependent, but implementations **should** merge the contents of the specified pipelines and prune duplicate entries.

Valid Usage

- VUID-vkMergePipelineCaches-dstCache-00770
dstCache must not appear in the list of source caches

Valid Usage (Implicit)

- VUID-vkMergePipelineCaches-device-parameter
device must be a valid **VkDevice** handle
- VUID-vkMergePipelineCaches-dstCache-parameter
dstCache must be a valid **VkPipelineCache** handle
- VUID-vkMergePipelineCaches-pSrcCaches-parameter
pSrcCaches must be a valid pointer to an array of **srcCacheCount** valid **VkPipelineCache** handles
- VUID-vkMergePipelineCaches-srcCacheCount-arraylength
srcCacheCount must be greater than 0
- VUID-vkMergePipelineCaches-dstCache-parent
dstCache must have been created, allocated, or retrieved from **device**
- VUID-vkMergePipelineCaches-pSrcCaches-parent
Each element of **pSrcCaches must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **dstCache must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

10.6.3. Retrieving Pipeline Cache Data

Data **can** be retrieved from a pipeline cache object using the command:

```
// Provided by VK_VERSION_1_0
VkResult vkGetPipelineCacheData(
    VkDevice          device,
    VkPipelineCache    pipelineCache,
    size_t*           pDataSize,
    void*             pData);
```

- `device` is the logical device that owns the pipeline cache.
- `pipelineCache` is the pipeline cache to retrieve data from.
- `pDataSize` is a pointer to a `size_t` value related to the amount of data in the pipeline cache, as described below.
- `pData` is either `NULL` or a pointer to a buffer.

If `pData` is `NULL`, then the maximum size of the data that **can** be retrieved from the pipeline cache, in bytes, is returned in `pDataSize`. Otherwise, `pDataSize` **must** point to a variable set by the application to the size of the buffer, in bytes, pointed to by `pData`, and on return the variable is overwritten with the amount of data actually written to `pData`. If `pDataSize` is less than the maximum size that **can** be retrieved by the pipeline cache, at most `pDataSize` bytes will be written to `pData`, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that not all of the pipeline cache was returned.

Any data written to `pData` is valid and **can** be provided as the `pInitialData` member of the `VkPipelineCacheCreateInfo` structure passed to `vkCreatePipelineCache`.

Two calls to `vkGetPipelineCacheData` with the same parameters **must** retrieve the same data unless a command that modifies the contents of the cache is called between them.

The initial bytes written to `pData` **must** be a header as described in the [Pipeline Cache Header](#) section.

If `pDataSize` is less than what is necessary to store this header, nothing will be written to `pData` and zero will be written to `pDataSize`.

Valid Usage (Implicit)

- VUID-vkGetPipelineCacheData-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkGetPipelineCacheData-pipelineCache-parameter
pipelineCache **must** be a valid [VkPipelineCache](#) handle
- VUID-vkGetPipelineCacheData-pDataSize-parameter
pDataSize **must** be a valid pointer to a [size_t](#) value
- VUID-vkGetPipelineCacheData-pData-parameter
If the value referenced by **pDataSize** is not 0, and **pData** is not **NULL**, **pData** **must** be a valid pointer to an array of **pDataSize** bytes
- VUID-vkGetPipelineCacheData-pipelineCache-parent
pipelineCache **must** have been created, allocated, or retrieved from **device**

Return Codes

Success

- **VK_SUCCESS**
- **VK_INCOMPLETE**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_OUT_OF_DEVICE_MEMORY**

10.6.4. Pipeline Cache Header

Applications **can** store the data retrieved from the pipeline cache, and use these data, possibly in a future run of the application, to populate new pipeline cache objects. The results of pipeline compiles, however, **may** depend on the vendor ID, device ID, driver version, and other details of the device. To enable applications to detect when previously retrieved data is incompatible with the device, the pipeline cache data **must** begin with a valid pipeline cache header.

NOTE

Structures described in this section are not part of the Vulkan API and are only used to describe the representation of data elements in pipeline cache data. Accordingly, the valid usage clauses defined for structures defined in this section do not define valid usage conditions for APIs accepting pipeline cache data as input, as providing invalid pipeline cache data as input to any Vulkan API commands will result in the provided pipeline cache data being ignored.

Version one of the pipeline cache header is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineCacheHeaderVersionOne {
    uint32_t          headerSize;
    VkPipelineCacheHeaderVersion headerVersion;
    uint32_t          vendorID;
    uint32_t          deviceID;
    uint8_t           pipelineCacheUUID[VK_UUID_SIZE];
} VkPipelineCacheHeaderVersionOne;
```

- **headerSize** is the length in bytes of the pipeline cache header.
- **headerVersion** is a [VkPipelineCacheHeaderVersion](#) value specifying the version of the header. A consumer of the pipeline cache **should** use the cache version to interpret the remainder of the cache header.
- **vendorID** is the [VkPhysicalDeviceProperties::vendorID](#) of the implementation.
- **deviceID** is the [VkPhysicalDeviceProperties::deviceID](#) of the implementation.
- **pipelineCacheUUID** is the [VkPhysicalDeviceProperties::pipelineCacheUUID](#) of the implementation.

Unlike most structures declared by the Vulkan API, all fields of this structure are written with the least significant byte first, regardless of host byte-order.

The C language specification does not define the packing of structure members. This layout assumes tight structure member packing, with members laid out in the order listed in the structure, and the intended size of the structure is 32 bytes. If a compiler produces code that diverges from that pattern, applications **must** employ another method to set values at the correct offsets.

Valid Usage

- VUID-VkPipelineCacheHeaderVersionOne-headerSize-04967
headerSize **must** be 32
- VUID-VkPipelineCacheHeaderVersionOne-headerVersion-04968
headerVersion **must** be [VK_PIPELINE_CACHE_HEADER_VERSION_ONE](#)
- VUID-VkPipelineCacheHeaderVersionOne-headerSize-08990
headerSize **must** not exceed the size of the pipeline cache

Valid Usage (Implicit)

- VUID-VkPipelineCacheHeaderVersionOne-headerVersion-parameter
headerVersion **must** be a valid [VkPipelineCacheHeaderVersion](#) value

Possible values of the **headerVersion** value of the pipeline cache header are:

```
// Provided by VK_VERSION_1_0
typedef enum VkPipelineCacheHeaderVersion {
    VK_PIPELINE_CACHE_HEADER_VERSION_ONE = 1,
} VkPipelineCacheHeaderVersion;
```

- `VK_PIPELINE_CACHE_HEADER_VERSION_ONE` specifies version one of the pipeline cache, described by [VkPipelineCacheHeaderVersionOne](#).

10.6.5. Destroying a Pipeline Cache

To destroy a pipeline cache, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyPipelineCache(
    VkDevice                device,
    VkPipelineCache         pipelineCache,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that destroys the pipeline cache object.
- `pipelineCache` is the handle of the pipeline cache to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyPipelineCache-pipelineCache-00771
If `VkAllocationCallbacks` were provided when `pipelineCache` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyPipelineCache-pipelineCache-00772
If no `VkAllocationCallbacks` were provided when `pipelineCache` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyPipelineCache-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkDestroyPipelineCache-pipelineCache-parameter
If `pipelineCache` is not `VK_NULL_HANDLE`, `pipelineCache` **must** be a valid `VkPipelineCache` handle
- VUID-vkDestroyPipelineCache-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkDestroyPipelineCache-pipelineCache-parent
If `pipelineCache` is a valid handle, it **must** have been created, allocated, or retrieved from

device

Host Synchronization

- Host access to `pipelineCache` **must** be externally synchronized

10.7. Specialization Constants

Specialization constants are a mechanism whereby constants in a SPIR-V module **can** have their constant value specified at the time the `VkPipeline` is created. This allows a SPIR-V module to have constants that **can** be modified while executing an application that uses the Vulkan API.

NOTE

Specialization constants are useful to allow a compute shader to have its local workgroup size changed at runtime by the user, for example.

Each `VkPipelineShaderStageCreateInfo` structure contains a `pSpecializationInfo` member, which **can** be `NULL` to indicate no specialization constants, or point to a `VkSpecializationInfo` structure.

The `VkSpecializationInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSpecializationInfo {
    uint32_t          mapEntryCount;
    const VkSpecializationMapEntry* pMapEntries;
    size_t            dataSize;
    const void*        pData;
} VkSpecializationInfo;
```

- `mapEntryCount` is the number of entries in the `pMapEntries` array.
- `pMapEntries` is a pointer to an array of `VkSpecializationMapEntry` structures, which map constant IDs to offsets in `pData`.
- `dataSize` is the byte size of the `pData` buffer.
- `pData` contains the actual constant values to specialize with.

Valid Usage

- VUID-VkSpecializationInfo-offset-00773
The `offset` member of each element of `pMapEntries` **must** be less than `dataSize`
- VUID-VkSpecializationInfo-pMapEntries-00774
The `size` member of each element of `pMapEntries` **must** be less than or equal to `dataSize` minus `offset`
- VUID-VkSpecializationInfo-constantID-04911
The `constantID` value of each element of `pMapEntries` **must** be unique within `pMapEntries`

Valid Usage (Implicit)

- VUID-VkSpecializationInfo-pMapEntries-parameter
If `mapEntryCount` is not 0, `pMapEntries` **must** be a valid pointer to an array of `mapEntryCount` valid `VkSpecializationMapEntry` structures
- VUID-VkSpecializationInfo-pData-parameter
If `dataSize` is not 0, `pData` **must** be a valid pointer to an array of `dataSize` bytes

The `VkSpecializationMapEntry` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSpecializationMapEntry {
    uint32_t    constantID;
    uint32_t    offset;
    size_t      size;
} VkSpecializationMapEntry;
```

- `constantID` is the ID of the specialization constant in SPIR-V.
- `offset` is the byte offset of the specialization constant value within the supplied data buffer.
- `size` is the byte size of the specialization constant value within the supplied data buffer.

If a `constantID` value is not a specialization constant ID used in the shader, that map entry does not affect the behavior of the pipeline.

Valid Usage

- VUID-VkSpecializationMapEntry-constantID-00776
For a `constantID` specialization constant declared in a shader, `size` **must** match the byte size of the `constantID`. If the specialization constant is of type `boolean`, `size` **must** be the byte size of `VkBool32`

In human readable SPIR-V:

```
OpDecorate %x SpecId 13 ; decorate .x component of WorkgroupSize with ID 13
OpDecorate %y SpecId 42 ; decorate .y component of WorkgroupSize with ID 42
OpDecorate %z SpecId 3 ; decorate .z component of WorkgroupSize with ID 3
OpDecorate %wgsz BuiltIn WorkgroupSize ; decorate WorkgroupSize onto constant
%i32 = OpTypeInt 32 0 ; declare an unsigned 32-bit type
%uvec3 = OpTypeVector %i32 3 ; declare a 3 element vector type of unsigned 32-bit
%x = OpSpecConstant %i32 1 ; declare the .x component of WorkgroupSize
%y = OpSpecConstant %i32 1 ; declare the .y component of WorkgroupSize
%z = OpSpecConstant %i32 1 ; declare the .z component of WorkgroupSize
%wgsz = OpSpecConstantComposite %uvec3 %x %y %z ; declare WorkgroupSize
```

From the above we have three specialization constants, one for each of the x, y & z elements of the WorkgroupSize vector.

Now to specialize the above via the specialization constants mechanism:

```
const VkSpecializationMapEntry entries[] =
{
    {
        .constantID = 13,
        .offset = 0 * sizeof(uint32_t),
        .size = sizeof(uint32_t)
    },
    {
        .constantID = 42,
        .offset = 1 * sizeof(uint32_t),
        .size = sizeof(uint32_t)
    },
    {
        .constantID = 3,
        .offset = 2 * sizeof(uint32_t),
        .size = sizeof(uint32_t)
    }
};

const uint32_t data[] = { 16, 8, 4 }; // our workgroup size is 16x8x4

const VkSpecializationInfo info =
{
    .mapEntryCount = 3,
    .pMapEntries = entries,
    .dataSize = 3 * sizeof(uint32_t),
    .pData = data,
};
```

Then when calling `vkCreateComputePipelines`, and passing the `VkSpecializationInfo` we defined as the `pSpecializationInfo` parameter of `VkPipelineShaderStageCreateInfo`, we will create a compute pipeline with the runtime specified local workgroup size.

Another example would be that an application has a SPIR-V module that has some platform-dependent constants they wish to use.

In human readable SPIR-V:

```
OpDecorate %1 SpecId 0 ; decorate our signed 32-bit integer constant
OpDecorate %2 SpecId 12 ; decorate our 32-bit floating-point constant
%i32 = OpTypeInt 32 1 ; declare a signed 32-bit type
%float = OpTypeFloat 32 ; declare a 32-bit floating-point type
%1 = OpSpecConstant %i32 -1 ; some signed 32-bit integer constant
```



```
%2 = OpSpecConstant %float 0.5 ; some 32-bit floating-point constant
```

From the above we have two specialization constants, one is a signed 32-bit integer and the second is a 32-bit floating-point value.

Now to specialize the above via the specialization constants mechanism:

```
struct SpecializationData {
    int32_t data0;
    float data1;
};

const VkSpecializationMapEntry entries[] =
{
    {
        .constantID = 0,
        .offset = offsetof(SpecializationData, data0),
        .size = sizeof(SpecializationData::data0)
    },
    {
        .constantID = 12,
        .offset = offsetof(SpecializationData, data1),
        .size = sizeof(SpecializationData::data1)
    }
};

SpecializationData data;
data.data0 = -42;    // set the data for the 32-bit integer
data.data1 = 42.0f; // set the data for the 32-bit floating-point

const VkSpecializationInfo info =
{
    .mapEntryCount = 2,
    .pMapEntries = entries,
    .dataSize = sizeof(data),
    .pdata = &data,
};
```

It is legal for a SPIR-V module with specializations to be compiled into a pipeline where no specialization information was provided. SPIR-V specialization constants contain default values such that if a specialization is not provided, the default value will be used. In the examples above, it would be valid for an application to only specialize some of the specialization constants within the SPIR-V module, and let the other constants use their default values encoded within the OpSpecConstant declarations.

10.8. Pipeline Binding

Once a pipeline has been created, it **can** be bound to the command buffer using the command:

```
// Provided by VK_VERSION_1_0
void vkCmdBindPipeline(
    VkCommandBuffer          commandBuffer,
    VkPipelineBindPoint      pipelineBindPoint,
    VkPipeline                pipeline);
```

- `commandBuffer` is the command buffer that the pipeline will be bound to.
- `pipelineBindPoint` is a `VkPipelineBindPoint` value specifying to which bind point the pipeline is bound. Binding one does not disturb the others.
- `pipeline` is the pipeline to be bound.

Once bound, a pipeline binding affects subsequent commands that interact with the given pipeline type in the command buffer until a different pipeline of the same type is bound to the bind point. Commands that do not interact with the `given pipeline` type **must** not be affected by the pipeline state.

Valid Usage

- VUID-vkCmdBindPipeline-pipelineBindPoint-00777
If `pipelineBindPoint` is `VK_PIPELINE_BIND_POINT_COMPUTE`, the `VkCommandPool` that `commandBuffer` was allocated from **must** support compute operations
- VUID-vkCmdBindPipeline-pipelineBindPoint-00778
If `pipelineBindPoint` is `VK_PIPELINE_BIND_POINT_GRAPHICS`, the `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdBindPipeline-pipelineBindPoint-00779
If `pipelineBindPoint` is `VK_PIPELINE_BIND_POINT_COMPUTE`, `pipeline` **must** be a compute pipeline
- VUID-vkCmdBindPipeline-pipelineBindPoint-00780
If `pipelineBindPoint` is `VK_PIPELINE_BIND_POINT_GRAPHICS`, `pipeline` **must** be a graphics pipeline
- VUID-vkCmdBindPipeline-pipeline-00781
If the `variableMultisampleRate` feature is not supported, `pipeline` is a graphics pipeline, the current subpass `uses no attachments`, and this is not the first call to this function with a graphics pipeline after transitioning to the current subpass, then the sample count specified by this pipeline **must** match that set in the previous pipeline

Valid Usage (Implicit)

- VUID-vkCmdBindPipeline-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdBindPipeline-pipelineBindPoint-parameter
`pipelineBindPoint` **must** be a valid `VkPipelineBindPoint` value
- VUID-vkCmdBindPipeline-pipeline-parameter

`pipeline` **must** be a valid `VkPipeline` handle

- VUID-vkCmdBindPipeline-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdBindPipeline-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, or compute operations
- VUID-vkCmdBindPipeline-commonparent
Both of `commandBuffer`, and `pipeline` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Graphics Compute	State

Possible values of `vkCmdBindPipeline::pipelineBindPoint`, specifying the bind point of a pipeline object, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkPipelineBindPoint {
    VK_PIPELINE_BIND_POINT_GRAPHICS = 0,
    VK_PIPELINE_BIND_POINT_COMPUTE = 1,
} VkPipelineBindPoint;
```

- `VK_PIPELINE_BIND_POINT_COMPUTE` specifies binding as a compute pipeline.
- `VK_PIPELINE_BIND_POINT_GRAPHICS` specifies binding as a graphics pipeline.

10.9. Dynamic State

When a pipeline object is bound, any pipeline object state that is not specified as dynamic is applied to the command buffer state. Pipeline object state that is specified as dynamic is not applied to the command buffer state at this time.

Instead, dynamic state **can** be modified at any time and persists for the lifetime of the command

buffer, or until modified by another dynamic state setting command, or made invalid by binding a pipeline in which that state is statically specified.

When a pipeline object is bound, the following applies to each state parameter:

- If the state is not specified as dynamic in the new pipeline object, then that command buffer state is overwritten by the state in the new pipeline object. Before any draw or dispatch call with this pipeline there **must** not have been any calls to any of the corresponding dynamic state setting commands after this pipeline was bound.
- If the state is specified as dynamic in the new pipeline object, then that command buffer state is not disturbed. Before any draw or dispatch call with this pipeline there **must** have been at least one call to each of the corresponding dynamic state setting commands. The state-setting commands **must** be recorded after command buffer recording was begun, or after the last command binding a pipeline object with that state specified as static, whichever was the latter.
- If the state is not included (corresponding pointer in [VkGraphicsPipelineCreateInfo](#) was **NULL** or was ignored) in the new pipeline object, then that command buffer state is not disturbed.

Dynamic state that does not affect the result of operations **can** be left undefined.

NOTE

For example, if blending is disabled by the pipeline object state then the dynamic color blend constants do not need to be specified in the command buffer, even if this state is specified as dynamic in the pipeline object.

NOTE

Applications running on Vulkan implementations advertising an [VkPhysicalDeviceDriverProperties::conformanceVersion](#) less than 1.3.8.0 should be aware that rebinding the currently bound pipeline object may not reapply static state.

Chapter 11. Memory Allocation

Vulkan memory is broken up into two categories, *host memory* and *device memory*.

11.1. Host Memory

Host memory is memory needed by the Vulkan implementation for non-device-visible storage.

NOTE This memory **may** be used to store the implementation's representation and state of Vulkan objects.

Vulkan provides applications the opportunity to perform host memory allocations on behalf of the Vulkan implementation. If this feature is not used, the implementation will perform its own memory allocations. Since most memory allocations are off the critical path, this is not meant as a performance feature. Rather, this **can** be useful for certain embedded systems, for debugging purposes (e.g. putting a guard page after all host allocations), or for memory allocation logging.

Allocators are provided by the application as a pointer to a `VkAllocationCallbacks` structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkAllocationCallbacks {
    void*                                pUserData;
    PFN_vkAllocationFunction              pfnAllocation;
    PFN_vkReallocationFunction            pfnReallocation;
    PFN_vkFreeFunction                   pfnFree;
    PFN_vkInternalAllocationNotification pfnInternalAllocation;
    PFN_vkInternalFreeNotification       pfnInternalFree;
} VkAllocationCallbacks;
```

- `pUserData` is a value to be interpreted by the implementation of the callbacks. When any of the callbacks in `VkAllocationCallbacks` are called, the Vulkan implementation will pass this value as the first parameter to the callback. This value **can** vary each time an allocator is passed into a command, even when the same object takes an allocator in multiple commands.
- `pfnAllocation` is a `PFN_vkAllocationFunction` pointer to an application-defined memory allocation function.
- `pfnReallocation` is a `PFN_vkReallocationFunction` pointer to an application-defined memory reallocation function.
- `pfnFree` is a `PFN_vkFreeFunction` pointer to an application-defined memory free function.
- `pfnInternalAllocation` is a `PFN_vkInternalAllocationNotification` pointer to an application-defined function that is called by the implementation when the implementation makes internal allocations.
- `pfnInternalFree` is a `PFN_vkInternalFreeNotification` pointer to an application-defined function that is called by the implementation when the implementation frees internal allocations.

Valid Usage

- VUID-VkAllocationCallbacks-pfnAllocation-00632
`pfnAllocation` **must** be a valid pointer to a valid application-defined [PFN_vkAllocationFunction](#)
- VUID-VkAllocationCallbacks-pfnReallocation-00633
`pfnReallocation` **must** be a valid pointer to a valid application-defined [PFN_vkReallocationFunction](#)
- VUID-VkAllocationCallbacks-pfnFree-00634
`pfnFree` **must** be a valid pointer to a valid application-defined [PFN_vkFreeFunction](#)
- VUID-VkAllocationCallbacks-pfnInternalAllocation-00635
If either of `pfnInternalAllocation` or `pfnInternalFree` is not `NULL`, both **must** be valid callbacks

The type of `pfnAllocation` is:

```
// Provided by VK_VERSION_1_0
typedef void* (VKAPI_PTR *PFN_vkAllocationFunction)(
    void*                pUserData,
    size_t               size,
    size_t               alignment,
    VkSystemAllocationScope allocationScope);
```

- `pUserData` is the value specified for [VkAllocationCallbacks::pUserData](#) in the allocator specified by the application.
- `size` is the size in bytes of the requested allocation.
- `alignment` is the requested alignment of the allocation in bytes and **must** be a power of two.
- `allocationScope` is a [VkSystemAllocationScope](#) value specifying the allocation scope of the lifetime of the allocation, as described [here](#).

If `pfnAllocation` is unable to allocate the requested memory, it **must** return `NULL`. If the allocation was successful, it **must** return a valid pointer to memory allocation containing at least `size` bytes, and with the pointer value being a multiple of `alignment`.

NOTE

Correct Vulkan operation **cannot** be assumed if the application does not follow these rules.

For example, `pfnAllocation` (or `pfnReallocation`) could cause termination of running Vulkan instance(s) on a failed allocation for debugging purposes, either directly or indirectly. In these circumstances, it **cannot** be assumed that any part of any affected [VkInstance](#) objects are going to operate correctly (even [vkDestroyInstance](#)), and the application **must** ensure it cleans up properly via other means (e.g. process termination).

If `pfnAllocation` returns `NULL`, and if the implementation is unable to continue correct processing of the current command without the requested allocation, it **must** treat this as a runtime error, and generate `VK_ERROR_OUT_OF_HOST_MEMORY` at the appropriate time for the command in which the condition was detected, as described in [Return Codes](#).

If the implementation is able to continue correct processing of the current command without the requested allocation, then it **may** do so, and **must** not generate `VK_ERROR_OUT_OF_HOST_MEMORY` as a result of this failed allocation.

The type of `pfnReallocation` is:

```
// Provided by VK_VERSION_1_0
typedef void* (VKAPI_PTR *PFN_vkReallocationFunction)(
    void*                pUserData,
    void*                pOriginal,
    size_t               size,
    size_t               alignment,
    VkSystemAllocationScope allocationScope);
```

- `pUserData` is the value specified for `VkAllocationCallbacks::pUserData` in the allocator specified by the application.
- `pOriginal` **must** be either `NULL` or a pointer previously returned by `pfnReallocation` or `pfnAllocation` of a compatible allocator.
- `size` is the size in bytes of the requested allocation.
- `alignment` is the requested alignment of the allocation in bytes and **must** be a power of two.
- `allocationScope` is a `VkSystemAllocationScope` value specifying the allocation scope of the lifetime of the allocation, as described [here](#).

If the reallocation was successful, `pfnReallocation` **must** return an allocation with enough space for `size` bytes, and the contents of the original allocation from bytes zero to `min(original size, new size) - 1` **must** be preserved in the returned allocation. If `size` is larger than the old size, the contents of the additional space are undefined. If satisfying these requirements involves creating a new allocation, then the old allocation **should** be freed.

If `pOriginal` is `NULL`, then `pfnReallocation` **must** behave equivalently to a call to `PFN_vkAllocationFunction` with the same parameter values (without `pOriginal`).

If `size` is zero, then `pfnReallocation` **must** behave equivalently to a call to `PFN_vkFreeFunction` with the same `pUserData` parameter value, and `pMemory` equal to `pOriginal`.

If `pOriginal` is non-`NULL`, the implementation **must** ensure that `alignment` is equal to the `alignment` used to originally allocate `pOriginal`.

If this function fails and `pOriginal` is non-`NULL` the application **must** not free the old allocation.

`pfnReallocation` **must** follow the same [rules for return values](#) as `PFN_vkAllocationFunction`.

The type of `pfnFree` is:

```
// Provided by VK_VERSION_1_0
typedef void (VKAPI_PTR *PFN_vkFreeFunction)(
    void*                                pUserData,
    void*                                pMemory);
```

- `pUserData` is the value specified for `VkAllocationCallbacks::pUserData` in the allocator specified by the application.
- `pMemory` is the allocation to be freed.

`pMemory` **may** be `NULL`, which the callback **must** handle safely. If `pMemory` is non-`NULL`, it **must** be a pointer previously allocated by `pfnAllocation` or `pfnReallocation`. The application **should** free this memory.

The type of `pfnInternalAllocation` is:

```
// Provided by VK_VERSION_1_0
typedef void (VKAPI_PTR *PFN_vkInternalAllocationNotification)(
    void*                                pUserData,
    size_t                                size,
    VkInternalAllocationType             allocationType,
    VkSystemAllocationScope              allocationScope);
```

- `pUserData` is the value specified for `VkAllocationCallbacks::pUserData` in the allocator specified by the application.
- `size` is the requested size of an allocation.
- `allocationType` is a `VkInternalAllocationType` value specifying the requested type of an allocation.
- `allocationScope` is a `VkSystemAllocationScope` value specifying the allocation scope of the lifetime of the allocation, as described [here](#).

This is a purely informational callback.

The type of `pfnInternalFree` is:

```
// Provided by VK_VERSION_1_0
typedef void (VKAPI_PTR *PFN_vkInternalFreeNotification)(
    void*                                pUserData,
    size_t                                size,
    VkInternalAllocationType             allocationType,
    VkSystemAllocationScope              allocationScope);
```

- `pUserData` is the value specified for `VkAllocationCallbacks::pUserData` in the allocator specified by the application.
- `size` is the requested size of an allocation.

- `allocationType` is a `VkInternalAllocationType` value specifying the requested type of an allocation.
- `allocationScope` is a `VkSystemAllocationScope` value specifying the allocation scope of the lifetime of the allocation, as described [here](#).

Each allocation has an *allocation scope* defining its lifetime and which object it is associated with. Possible values passed to the `allocationScope` parameter of the callback functions specified by `VkAllocationCallbacks`, indicating the allocation scope, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSystemAllocationScope {
    VK_SYSTEM_ALLOCATION_SCOPE_COMMAND = 0,
    VK_SYSTEM_ALLOCATION_SCOPE_OBJECT = 1,
    VK_SYSTEM_ALLOCATION_SCOPE_CACHE = 2,
    VK_SYSTEM_ALLOCATION_SCOPE_DEVICE = 3,
    VK_SYSTEM_ALLOCATION_SCOPE_INSTANCE = 4,
} VkSystemAllocationScope;
```

- `VK_SYSTEM_ALLOCATION_SCOPE_COMMAND` specifies that the allocation is scoped to the duration of the Vulkan command.
- `VK_SYSTEM_ALLOCATION_SCOPE_OBJECT` specifies that the allocation is scoped to the lifetime of the Vulkan object that is being created or used.
- `VK_SYSTEM_ALLOCATION_SCOPE_CACHE` specifies that the allocation is scoped to the lifetime of a `VkPipelineCache` object.
- `VK_SYSTEM_ALLOCATION_SCOPE_DEVICE` specifies that the allocation is scoped to the lifetime of the Vulkan device.
- `VK_SYSTEM_ALLOCATION_SCOPE_INSTANCE` specifies that the allocation is scoped to the lifetime of the Vulkan instance.

Most Vulkan commands operate on a single object, or there is a sole object that is being created or manipulated. When an allocation uses an allocation scope of `VK_SYSTEM_ALLOCATION_SCOPE_OBJECT` or `VK_SYSTEM_ALLOCATION_SCOPE_CACHE`, the allocation is scoped to the object being created or manipulated.

When an implementation requires host memory, it will make callbacks to the application using the most specific allocator and allocation scope available:

- If an allocation is scoped to the duration of a command, the allocator will use the `VK_SYSTEM_ALLOCATION_SCOPE_COMMAND` allocation scope. The most specific allocator available is used: if the object being created or manipulated has an allocator, that object's allocator will be used, else if the parent `VkDevice` has an allocator it will be used, else if the parent `VkInstance` has an allocator it will be used. Else,
- If an allocation is associated with a `VkPipelineCache` object, the allocator will use the `VK_SYSTEM_ALLOCATION_SCOPE_CACHE` allocation scope. The most specific allocator available is used (cache, else device, else instance). Else,

- If an allocation is scoped to the lifetime of an object, that object is being created or manipulated by the command, and that object's type is not `VkDevice` or `VkInstance`, the allocator will use an allocation scope of `VK_SYSTEM_ALLOCATION_SCOPE_OBJECT`. The most specific allocator available is used (object, else device, else instance). Else,
- If an allocation is scoped to the lifetime of a device, the allocator will use an allocation scope of `VK_SYSTEM_ALLOCATION_SCOPE_DEVICE`. The most specific allocator available is used (device, else instance). Else,
- If the allocation is scoped to the lifetime of an instance and the instance has an allocator, its allocator will be used with an allocation scope of `VK_SYSTEM_ALLOCATION_SCOPE_INSTANCE`.
- Otherwise an implementation will allocate memory through an alternative mechanism that is unspecified.

Objects that are allocated from pools do not specify their own allocator. When an implementation requires host memory for such an object, that memory is sourced from the object's parent pool's allocator.

The application is not expected to handle allocating memory that is intended for execution by the host due to the complexities of differing security implementations across multiple platforms. The implementation will allocate such memory internally and invoke an application provided informational callback when these *internal allocations* are allocated and freed. Upon allocation of executable memory, `pfnInternalAllocation` will be called. Upon freeing executable memory, `pfnInternalFree` will be called. An implementation will only call an informational callback for executable memory allocations and frees.

The `allocationType` parameter to the `pfnInternalAllocation` and `pfnInternalFree` functions **may** be one of the following values:

```
// Provided by VK_VERSION_1_0
typedef enum VkInternalAllocationType {
    VK_INTERNAL_ALLOCATION_TYPE_EXECUTABLE = 0,
} VkInternalAllocationType;
```

- `VK_INTERNAL_ALLOCATION_TYPE_EXECUTABLE` specifies that the allocation is intended for execution by the host.

An implementation **must** only make calls into an application-provided allocator during the execution of an API command. An implementation **must** only make calls into an application-provided allocator from the same thread that called the provoking API command. The implementation **should** not synchronize calls to any of the callbacks. If synchronization is needed, the callbacks **must** provide it themselves. The informational callbacks are subject to the same restrictions as the allocation callbacks.

If an implementation intends to make calls through a `VkAllocationCallbacks` structure between the time a `vkCreate*` command returns and the time a corresponding `vkDestroy*` command begins, that implementation **must** save a copy of the allocator before the `vkCreate*` command returns. The callback functions and any data structures they rely upon **must** remain valid for the lifetime of the object they are associated with.

If an allocator is provided to a `vkCreate*` command, a *compatible* allocator **must** be provided to the corresponding `vkDestroy*` command. Two `VkAllocationCallbacks` structures are compatible if memory allocated with `pfnAllocation` or `pfnReallocation` in each **can** be freed with `pfnReallocation` or `pfnFree` in the other. An allocator **must** not be provided to a `vkDestroy*` command if an allocator was not provided to the corresponding `vkCreate*` command.

If a non-NULL allocator is used, the `pfnAllocation`, `pfnReallocation` and `pfnFree` members **must** be non-NULL and point to valid implementations of the callbacks. An application **can** choose to not provide informational callbacks by setting both `pfnInternalAllocation` and `pfnInternalFree` to NULL. `pfnInternalAllocation` and `pfnInternalFree` **must** either both be NULL or both be non-NULL.

If `pfnAllocation` or `pfnReallocation` fail, the implementation **may** fail object creation and/or generate a `VK_ERROR_OUT_OF_HOST_MEMORY` error, as appropriate.

Allocation callbacks **must** not call any Vulkan commands.

The following sets of rules define when an implementation is permitted to call the allocator callbacks.

`pfnAllocation` or `pfnReallocation` **may** be called in the following situations:

- Allocations scoped to a `VkDevice` or `VkInstance` **may** be allocated from any API command.
- Allocations scoped to a command **may** be allocated from any API command.
- Allocations scoped to a `VkPipelineCache` **may** only be allocated from:
 - `vkCreatePipelineCache`
 - `vkMergePipelineCaches` for `dstCache`
 - `vkCreateGraphicsPipelines` for `pipelineCache`
 - `vkCreateComputePipelines` for `pipelineCache`
- Allocations scoped to a `VkDescriptorPool` **may** only be allocated from:
 - any command that takes the pool as a direct argument
 - `vkAllocateDescriptorSets` for the `descriptorPool` member of its `pAllocateInfo` parameter
 - `vkCreateDescriptorPool`
- Allocations scoped to a `VkCommandPool` **may** only be allocated from:
 - any command that takes the pool as a direct argument
 - `vkCreateCommandPool`
 - `vkAllocateCommandBuffers` for the `commandPool` member of its `pAllocateInfo` parameter
 - any `vkCmd*` command whose `commandBuffer` was allocated from that `VkCommandPool`
- Allocations scoped to any other object **may** only be allocated in that object's `vkCreate*` command.

`pfnFree`, or `pfnReallocation` with zero `size`, **may** be called in the following situations:

- Allocations scoped to a `VkDevice` or `VkInstance` **may** be freed from any API command.

- Allocations scoped to a command **must** be freed by any API command which allocates such memory.
- Allocations scoped to a `VkPipelineCache` **may** be freed from `vkDestroyPipelineCache`.
- Allocations scoped to a `VkDescriptorPool` **may** be freed from
 - any command that takes the pool as a direct argument
- Allocations scoped to a `VkCommandPool` **may** be freed from:
 - any command that takes the pool as a direct argument
 - `vkResetCommandBuffer` whose `commandBuffer` was allocated from that `VkCommandPool`
- Allocations scoped to any other object **may** be freed in that object's `vkDestroy*` command.
- Any command that allocates host memory **may** also free host memory of the same scope.

11.2. Device Memory

Device memory is memory that is visible to the device—for example the contents of the image or buffer objects, which **can** be natively used by the device.

11.2.1. Device Memory Properties

Memory properties of a physical device describe the memory heaps and memory types available.

To query memory properties, call:

```
// Provided by VK_VERSION_1_0
void vkGetPhysicalDeviceMemoryProperties(
    VkPhysicalDevice          physicalDevice,
    VkPhysicalDeviceMemoryProperties* pMemoryProperties);
```

- `physicalDevice` is the handle to the device to query.
- `pMemoryProperties` is a pointer to a `VkPhysicalDeviceMemoryProperties` structure in which the properties are returned.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceMemoryProperties-physicalDevice-parameter `physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceMemoryProperties-pMemoryProperties-parameter `pMemoryProperties` **must** be a valid pointer to a `VkPhysicalDeviceMemoryProperties` structure

The `VkPhysicalDeviceMemoryProperties` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPhysicalDeviceMemoryProperties {
    uint32_t      memoryTypeCount;
    VkMemoryType  memoryTypes[VK_MAX_MEMORY_TYPES];
    uint32_t      memoryHeapCount;
    VkMemoryHeap  memoryHeaps[VK_MAX_MEMORY_HEAPS];
} VkPhysicalDeviceMemoryProperties;
```

- `memoryTypeCount` is the number of valid elements in the `memoryTypes` array.
- `memoryTypes` is an array of `VK_MAX_MEMORY_TYPES` `VkMemoryType` structures describing the *memory types* that **can** be used to access memory allocated from the heaps specified by `memoryHeaps`.
- `memoryHeapCount` is the number of valid elements in the `memoryHeaps` array.
- `memoryHeaps` is an array of `VK_MAX_MEMORY_HEAPS` `VkMemoryHeap` structures describing the *memory heaps* from which memory **can** be allocated.

The `VkPhysicalDeviceMemoryProperties` structure describes a number of *memory heaps* as well as a number of *memory types* that **can** be used to access memory allocated in those heaps. Each heap describes a memory resource of a particular size, and each memory type describes a set of memory properties (e.g. host cached vs. uncached) that **can** be used with a given memory heap. Allocations using a particular memory type will consume resources from the heap indicated by that memory type's heap index. More than one memory type **may** share each heap, and the heaps and memory types provide a mechanism to advertise an accurate size of the physical memory resources while allowing the memory to be used with a variety of different properties.

The number of memory heaps is given by `memoryHeapCount` and is less than or equal to `VK_MAX_MEMORY_HEAPS`. Each heap is described by an element of the `memoryHeaps` array as a `VkMemoryHeap` structure. The number of memory types available across all memory heaps is given by `memoryTypeCount` and is less than or equal to `VK_MAX_MEMORY_TYPES`. Each memory type is described by an element of the `memoryTypes` array as a `VkMemoryType` structure.

At least one heap **must** include `VK_MEMORY_HEAP_DEVICE_LOCAL_BIT` in `VkMemoryHeap::flags`. If there are multiple heaps that all have similar performance characteristics, they **may** all include `VK_MEMORY_HEAP_DEVICE_LOCAL_BIT`. In a unified memory architecture (UMA) system there is often only a single memory heap which is considered to be equally “local” to the host and to the device, and such an implementation **must** advertise the heap as device-local.

Each memory type returned by `vkGetPhysicalDeviceMemoryProperties` **must** have its `propertyFlags` set to one of the following values:

- 0
- `VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT` | `VK_MEMORY_PROPERTY_HOST_COHERENT_BIT`
- `VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT` | `VK_MEMORY_PROPERTY_HOST_CACHED_BIT`
- `VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT` |

- VK_MEMORY_PROPERTY_HOST_CACHED_BIT |
 VK_MEMORY_PROPERTY_HOST_COHERENT_BIT
- VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT
- VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT |
 VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT |
 VK_MEMORY_PROPERTY_HOST_COHERENT_BIT
- VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT |
 VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT |
 VK_MEMORY_PROPERTY_HOST_CACHED_BIT
- VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT |
 VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT |
 VK_MEMORY_PROPERTY_HOST_CACHED_BIT |
 VK_MEMORY_PROPERTY_HOST_COHERENT_BIT
- VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT |
 VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT

There **must** be at least one memory type with both the `VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT` and `VK_MEMORY_PROPERTY_HOST_COHERENT_BIT` bits set in its `propertyFlags`. There **must** be at least one memory type with the `VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT` bit set in its `propertyFlags`.

For each pair of elements **X** and **Y** returned in `memoryTypes`, **X** **must** be placed at a lower index position than **Y** if:

- the set of bit flags returned in the `propertyFlags` member of **X** is a strict subset of the set of bit flags returned in the `propertyFlags` member of **Y**; or
- the `propertyFlags` members of **X** and **Y** are equal, and **X** belongs to a memory heap with greater performance (as determined in an implementation-specific manner)

NOTE

There is no ordering requirement between **X** and **Y** elements for the case their `propertyFlags` members are not in a subset relation. That potentially allows more than one possible way to order the same set of memory types. Notice that the [list of all allowed memory property flag combinations](#) is written in a valid order. But if instead `VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT` was before `VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT | VK_MEMORY_PROPERTY_HOST_COHERENT_BIT`, the list would still be in a valid order.

This ordering requirement enables applications to use a simple search loop to select the desired memory type along the lines of:

```
// Find a memory in 'memoryTypeBitsRequirement' that includes all of
// 'requiredProperties'
int32_t findProperties(const VkPhysicalDeviceMemoryProperties* pMemoryProperties,
                     uint32_t memoryTypeBitsRequirement,
                     VkMemoryPropertyFlags requiredProperties) {
    const uint32_t memoryCount = pMemoryProperties->memoryTypeCount;
    for (uint32_t memoryIndex = 0; memoryIndex < memoryCount; ++memoryIndex) {
```

```

const uint32_t memoryTypeBits = (1 << memoryIndex);
const bool isRequiredMemoryType = memoryTypeBitsRequirement & memoryTypeBits;

const VkMemoryPropertyFlags properties =
    pMemoryProperties->memoryTypes[memoryIndex].propertyFlags;
const bool hasRequiredProperties =
    (properties & requiredProperties) == requiredProperties;

if (isRequiredMemoryType && hasRequiredProperties)
    return static_cast<int32_t>(memoryIndex);
}

// failed to find memory type
return -1;
}

// Try to find an optimal memory type, or if it does not exist try fallback memory
type
// `device` is the VkDevice
// `image` is the VkImage that requires memory to be bound
// `memoryProperties` properties as returned by vkGetPhysicalDeviceMemoryProperties
// `requiredProperties` are the property flags that must be present
// `optimalProperties` are the property flags that are preferred by the application
VkMemoryRequirements memoryRequirements;
vkGetImageMemoryRequirements(device, image, &memoryRequirements);
int32_t memoryType =
    findProperties(&memoryProperties, memoryRequirements.memoryTypeBits,
optimalProperties);
if (memoryType == -1) // not found; try fallback properties
    memoryType =
        findProperties(&memoryProperties, memoryRequirements.memoryTypeBits,
requiredProperties);

```

`VK_MAX_MEMORY_TYPES` is the length of an array of `VkMemoryType` structures describing memory types, as returned in `VkPhysicalDeviceMemoryProperties::memoryTypes`.

```
#define VK_MAX_MEMORY_TYPES 32U
```

`VK_MAX_MEMORY_HEAPS` is the length of an array of `VkMemoryHeap` structures describing memory heaps, as returned in `VkPhysicalDeviceMemoryProperties::memoryHeaps`.

```
#define VK_MAX_MEMORY_HEAPS 16U
```

The `VkMemoryHeap` structure is defined as:


```
// Provided by VK_VERSION_1_0
typedef struct VkMemoryHeap {
    VkDeviceSize      size;
    VkMemoryHeapFlags flags;
} VkMemoryHeap;
```

- **size** is the total memory size in bytes in the heap.
- **flags** is a bitmask of [VkMemoryHeapFlagBits](#) specifying attribute flags for the heap.

Bits which **may** be set in [VkMemoryHeap::flags](#), indicating attribute flags for the heap, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkMemoryHeapFlagBits {
    VK_MEMORY_HEAP_DEVICE_LOCAL_BIT = 0x00000001,
} VkMemoryHeapFlagBits;
```

- **VK_MEMORY_HEAP_DEVICE_LOCAL_BIT** specifies that the heap corresponds to device-local memory. Device-local memory **may** have different performance characteristics than host-local memory, and **may** support different memory property flags.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkMemoryHeapFlags;
```

VkMemoryHeapFlags is a bitmask type for setting a mask of zero or more [VkMemoryHeapFlagBits](#).

The **VkMemoryType** structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkMemoryType {
    VkMemoryPropertyFlags propertyFlags;
    uint32_t              heapIndex;
} VkMemoryType;
```

- **heapIndex** describes which memory heap this memory type corresponds to, and **must** be less than **memoryHeapCount** from the [VkPhysicalDeviceMemoryProperties](#) structure.
- **propertyFlags** is a bitmask of [VkMemoryPropertyFlagBits](#) of properties for this memory type.

Bits which **may** be set in [VkMemoryType::propertyFlags](#), indicating properties of a memory type, are:


```
// Provided by VK_VERSION_1_0
typedef enum VkMemoryPropertyFlagBits {
    VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT = 0x00000001,
    VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT = 0x00000002,
    VK_MEMORY_PROPERTY_HOST_COHERENT_BIT = 0x00000004,
    VK_MEMORY_PROPERTY_HOST_CACHED_BIT = 0x00000008,
    VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT = 0x00000010,
} VkMemoryPropertyFlagBits;
```

- **VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT** bit specifies that memory allocated with this type is the most efficient for device access. This property will be set if and only if the memory type belongs to a heap with the **VK_MEMORY_HEAP_DEVICE_LOCAL_BIT** set.
- **VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT** bit specifies that memory allocated with this type **can** be mapped for host access using [vkMapMemory](#).
- **VK_MEMORY_PROPERTY_HOST_COHERENT_BIT** bit specifies that the host cache management commands [vkFlushMappedMemoryRanges](#) and [vkInvalidateMappedMemoryRanges](#) are not needed to manage [availability and visibility](#) on the host.
- **VK_MEMORY_PROPERTY_HOST_CACHED_BIT** bit specifies that memory allocated with this type is cached on the host. Host memory accesses to uncached memory are slower than to cached memory, however uncached memory is always host coherent.
- **VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT** bit specifies that the memory type only allows device access to the memory. Memory types **must** not have both **VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT** and **VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT** set. Additionally, the object's backing memory **may** be provided by the implementation lazily as specified in [Lazily Allocated Memory](#).

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkMemoryPropertyFlags;
```

VkMemoryPropertyFlags is a bitmask type for setting a mask of zero or more [VkMemoryPropertyFlagBits](#).

11.2.2. Device Memory Objects

A Vulkan device operates on data in device memory via memory objects that are represented in the API by a **VkDeviceMemory** handle:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkDeviceMemory)
```

11.2.3. Device Memory Allocation

To allocate memory objects, call:

```
// Provided by VK_VERSION_1_0
VkResult vkAllocateMemory(
    VkDevice                                device,
    const VkMemoryAllocateInfo*            pAllocateInfo,
    const VkAllocationCallbacks*           pAllocator,
    VkDeviceMemory*                         pMemory);
```

- **device** is the logical device that owns the memory.
- **pAllocateInfo** is a pointer to a **VkMemoryAllocateInfo** structure describing parameters of the allocation. A successfully returned allocation **must** use the requested parameters—no substitution is permitted by the implementation.
- **pAllocator** controls host memory allocation as described in the **Memory Allocation** chapter.
- **pMemory** is a pointer to a **VkDeviceMemory** handle in which information about the allocated memory is returned.

Allocations returned by **vkAllocateMemory** are guaranteed to meet any alignment requirement of the implementation. For example, if an implementation requires 128 byte alignment for images and 64 byte alignment for buffers, the device memory returned through this mechanism would be 128-byte aligned. This ensures that applications **can** correctly suballocate objects of different types (with potentially different alignment requirements) in the same memory object.

When memory is allocated, its contents are undefined.

The maximum number of valid memory allocations that **can** exist simultaneously within a **VkDevice** **may** be restricted by implementation- or platform-dependent limits. The **maxMemoryAllocationCount** feature describes the number of allocations that **can** exist simultaneously before encountering these internal limits.

NOTE

For historical reasons, if **maxMemoryAllocationCount** is exceeded, some implementations may return **VK_ERROR_TOO_MANY_OBJECTS**. Exceeding this limit will result in undefined behavior, and an application should not rely on the use of the returned error code in order to identify when the limit is reached.

NOTE

Many protected memory implementations involve complex hardware and system software support, and often have additional and much lower limits on the number of simultaneous protected memory allocations (from memory types with the **VK_MEMORY_PROPERTY_PROTECTED_BIT** property) than for non-protected memory allocations. These limits can be system-wide, and depend on a variety of factors outside of the Vulkan implementation, so they cannot be queried in Vulkan. Applications **should** use as few allocations as possible from such memory types by suballocating aggressively, and be prepared for allocation failure even when there is apparently plenty of capacity remaining in the memory heap. As a guideline, the Vulkan conformance test suite requires that at least 80 minimum-size allocations can exist concurrently when no other uses of protected memory are active in the system.

Some platforms **may** have a limit on the maximum size of a single allocation. For example, certain systems **may** fail to create allocations with a size greater than or equal to 4GB. Such a limit is implementation-dependent, and if such a failure occurs then the error `VK_ERROR_OUT_OF_DEVICE_MEMORY` **must** be returned.

Valid Usage

- VUID-vkAllocateMemory-pAllocateInfo-01713
`pAllocateInfo->allocationSize` **must** be less than or equal to `VkPhysicalDeviceMemoryProperties::memoryHeaps[memindex].size` where `memindex` = `VkPhysicalDeviceMemoryProperties::memoryTypes[pAllocateInfo->memoryTypeIndex].heapIndex` as returned by `vkGetPhysicalDeviceMemoryProperties` for the `VkPhysicalDevice` that `device` was created from
- VUID-vkAllocateMemory-pAllocateInfo-01714
`pAllocateInfo->memoryTypeIndex` **must** be less than `VkPhysicalDeviceMemoryProperties::memoryTypeCount` as returned by `vkGetPhysicalDeviceMemoryProperties` for the `VkPhysicalDevice` that `device` was created from
- VUID-vkAllocateMemory-maxMemoryAllocationCount-04101
There **must** be less than `VkPhysicalDeviceLimits::maxMemoryAllocationCount` device memory allocations currently allocated on the device

Valid Usage (Implicit)

- VUID-vkAllocateMemory-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkAllocateMemory-pAllocateInfo-parameter
`pAllocateInfo` **must** be a valid pointer to a valid `VkMemoryAllocateInfo` structure
- VUID-vkAllocateMemory-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkAllocateMemory-pMemory-parameter
`pMemory` **must** be a valid pointer to a `VkDeviceMemory` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkMemoryAllocateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkMemoryAllocateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkDeviceSize        allocationSize;
    uint32_t           memoryTypeIndex;
} VkMemoryAllocateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `allocationSize` is the size of the allocation in bytes.
- `memoryTypeIndex` is an index identifying a memory type from the `memoryTypes` array of the `VkPhysicalDeviceMemoryProperties` structure.

The internal data of an allocated device memory object **must** include a reference to implementation-specific resources, referred to as the memory object's *payload*.

Valid Usage

- VUID-VkMemoryAllocateInfo-allocationSize-07897
`allocationSize` **must** be greater than 0

Valid Usage (Implicit)

- VUID-VkMemoryAllocateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_MEMORY_ALLOCATE_INFO`
- VUID-VkMemoryAllocateInfo-pNext-pNext
`pNext` **must** be `NULL`

11.2.4. Freeing Device Memory

To free a memory object, call:

```
// Provided by VK_VERSION_1_0
void vkFreeMemory(
    VkDevice                device,
    VkDeviceMemory          memory,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that owns the memory.
- `memory` is the `VkDeviceMemory` object to be freed.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Before freeing a memory object, an application **must** ensure the memory object is no longer in use by the device—for example by command buffers in the *pending state*. Memory **can** be freed whilst still bound to resources, but those resources **must** not be used afterwards. Freeing a memory object releases the reference it held, if any, to its payload. If there are still any bound images or buffers, the memory object’s payload **may** not be immediately released by the implementation, but **must** be released by the time all bound images and buffers have been destroyed. Once all references to a payload are released, it is returned to the heap from which it was allocated.

How memory objects are bound to Images and Buffers is described in detail in the [Resource Memory Association](#) section.

If a memory object is mapped at the time it is freed, it is implicitly unmapped.

NOTE

As described [below](#), host writes are not implicitly flushed when the memory object is unmapped, but the implementation **must** guarantee that writes that have not been flushed do not affect any other memory.

Valid Usage

- VUID-vkFreeMemory-memory-00677
All submitted commands that refer to **memory** (via images or buffers) **must** have completed execution

Valid Usage (Implicit)

- VUID-vkFreeMemory-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkFreeMemory-memory-parameter
If **memory** is not [VK_NULL_HANDLE](#), **memory** **must** be a valid [VkDeviceMemory](#) handle
- VUID-vkFreeMemory-pAllocator-parameter
If **pAllocator** is not [NULL](#), **pAllocator** **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkFreeMemory-memory-parent
If **memory** is a valid handle, it **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **memory** **must** be externally synchronized

11.2.5. Host Access to Device Memory Objects

Memory objects created with [vkAllocateMemory](#) are not directly host accessible.

Memory objects created with the memory property [VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT](#) are

considered *mappable*. Memory objects **must** be mappable in order to be successfully mapped on the host.

To retrieve a host virtual address pointer to a region of a mappable memory object, call:

```
// Provided by VK_VERSION_1_0
VkResult vkMapMemory(
    VkDevice          device,
    VkDeviceMemory    memory,
    VkDeviceSize      offset,
    VkDeviceSize      size,
    VkMemoryMapFlags   flags,
    void**            ppData);
```

- **device** is the logical device that owns the memory.
- **memory** is the `VkDeviceMemory` object to be mapped.
- **offset** is a zero-based byte offset from the beginning of the memory object.
- **size** is the size of the memory range to map, or `VK_WHOLE_SIZE` to map from **offset** to the end of the allocation.
- **flags** is reserved for future use.
- **ppData** is a pointer to a `void*` variable in which a host-accessible pointer to the beginning of the mapped range is returned. The value of the returned pointer minus **offset** **must** be aligned to `VkPhysicalDeviceLimits::minMemoryMapAlignment`.

After a successful call to `vkMapMemory` the memory object **memory** is considered to be currently *host mapped*.

NOTE	It is an application error to call <code>vkMapMemory</code> on a memory object that is already <i>host mapped</i> .
-------------	---

NOTE	<code>vkMapMemory</code> will fail if the implementation is unable to allocate an appropriately sized contiguous virtual address range, e.g. due to virtual address space fragmentation or platform limits. In such cases, <code>vkMapMemory</code> must return <code>VK_ERROR_MEMORY_MAP_FAILED</code> . The application can improve the likelihood of success by reducing the size of the mapped range and/or removing unneeded mappings using <code>vkUnmapMemory</code> .
-------------	---

`vkMapMemory` does not check whether the device memory is currently in use before returning the host-accessible pointer. The application **must** guarantee that any previously submitted command that writes to this range has completed before the host reads from or writes to that range, and that any previously submitted command that reads from that range has completed before the host writes to that region (see [here](#) for details on fulfilling such a guarantee). If the device memory was allocated without the `VK_MEMORY_PROPERTY_HOST_COHERENT_BIT` set, these guarantees **must** be made for an extended range: the application **must** round down the start of the range to the nearest multiple of `VkPhysicalDeviceLimits::nonCoherentAtomSize`, and round the end of the range up to the nearest

multiple of `VkPhysicalDeviceLimits::nonCoherentAtomSize`.

While a range of device memory is host mapped, the application is responsible for synchronizing both device and host access to that memory range.

NOTE

It is important for the application developer to become meticulously familiar with all of the mechanisms described in the chapter on [Synchronization and Cache Control](#) as they are crucial to maintaining memory access ordering.

Valid Usage

- VUID-vkMapMemory-memory-00678
`memory` **must** not be currently host mapped
- VUID-vkMapMemory-offset-00679
`offset` **must** be less than the size of `memory`
- VUID-vkMapMemory-size-00680
If `size` is not equal to `VK_WHOLE_SIZE`, `size` **must** be greater than 0
- VUID-vkMapMemory-size-00681
If `size` is not equal to `VK_WHOLE_SIZE`, `size` **must** be less than or equal to the size of the `memory` minus `offset`
- VUID-vkMapMemory-memory-00682
`memory` **must** have been created with a memory type that reports `VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT`

Valid Usage (Implicit)

- VUID-vkMapMemory-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkMapMemory-memory-parameter
`memory` **must** be a valid `VkDeviceMemory` handle
- VUID-vkMapMemory-flags-zerobitmask
`flags` **must** be 0
- VUID-vkMapMemory-ppData-parameter
`ppData` **must** be a valid pointer to a pointer value
- VUID-vkMapMemory-memory-parent
`memory` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `memory` **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_MEMORY_MAP_FAILED`

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkMemoryMapFlags;
```

`VkMemoryMapFlags` is a bitmask type for setting a mask of zero or more `VkMemoryMapFlagBits`.

Two commands are provided to enable applications to work with non-coherent memory allocations: `vkFlushMappedMemoryRanges` and `vkInvalidateMappedMemoryRanges`.

NOTE

If the memory object was created with the `VK_MEMORY_PROPERTY_HOST_COHERENT_BIT` set, `vkFlushMappedMemoryRanges` and `vkInvalidateMappedMemoryRanges` are unnecessary and **may** have a performance cost. However, [availability and visibility operations](#) still need to be managed on the device. See the description of [host access types](#) for more information.

After a successful call to `vkMapMemory` the memory object `memory` is considered to be currently *host mapped*.

To flush ranges of non-coherent memory from the host caches, call:

```
// Provided by VK_VERSION_1_0
VkResult vkFlushMappedMemoryRanges(
    VkDevice          device,
    uint32_t          memoryRangeCount,
    const VkMappedMemoryRange* pMemoryRanges);
```

- `device` is the logical device that owns the memory ranges.
- `memoryRangeCount` is the length of the `pMemoryRanges` array.
- `pMemoryRanges` is a pointer to an array of `VkMappedMemoryRange` structures describing the memory ranges to flush.

`vkFlushMappedMemoryRanges` guarantees that host writes to the memory ranges described by `pMemoryRanges` are made available to the host memory domain, such that they **can** be made available to the device memory domain via [memory domain operations](#) using the `VK_ACCESS_HOST_WRITE_BIT` [access type](#).

The first [synchronization scope](#) includes all host operations that happened-before it, as defined by the host memory model.

NOTE	<i>Note</i> Some systems allow writes that do not directly integrate with the host memory model; these have to be synchronized by the application manually. One example of this is non-temporal store instructions on x86; to ensure these happen-before submission, applications should call <code>_mm_sfence()</code> .
-------------	--

The second [synchronization scope](#) is empty.

The first [access scope](#) includes host writes to the specified memory ranges.

NOTE	<i>Note</i> When a host write to a memory location is made available in this way, each whole aligned set of <code>nonCoherentAtomSize</code> bytes that the memory location exists in will also be made available as if they were written by the host. For example, with a <code>nonCoherentAtomSize</code> of 128, if an application writes to the first byte of a memory object via a host mapping, the first 128 bytes of the memory object will be made available by this command. While the value of the following 127 bytes will be unchanged, this does count as an access for the purpose of synchronization, so care must be taken to avoid data races.
-------------	---

The second [access scope](#) is empty.

Unmapping non-coherent memory does not implicitly flush the host mapped memory, and host writes that have not been flushed **may** not ever be visible to the device. However, implementations **must** ensure that writes that have not been flushed do not become visible to any other memory.

NOTE	The above guarantee avoids a potential memory corruption in scenarios where host writes to a mapped memory object have not been flushed before the memory is unmapped (or freed), and the virtual address range is subsequently reused for a different mapping (or memory allocation).
-------------	--

Valid Usage (Implicit)

- VUID-vkFlushMappedMemoryRanges-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkFlushMappedMemoryRanges-pMemoryRanges-parameter
`pMemoryRanges` **must** be a valid pointer to an array of `memoryRangeCount` valid `VkMappedMemoryRange` structures
- VUID-vkFlushMappedMemoryRanges-memoryRangeCount-arraylength
`memoryRangeCount` **must** be greater than 0

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

To invalidate ranges of non-coherent memory from the host caches, call:

```
// Provided by VK_VERSION_1_0
VkResult vkInvalidateMappedMemoryRanges(
    VkDevice          device,
    uint32_t          memoryRangeCount,
    const VkMappedMemoryRange* pMemoryRanges);
```

- `device` is the logical device that owns the memory ranges.
- `memoryRangeCount` is the length of the `pMemoryRanges` array.
- `pMemoryRanges` is a pointer to an array of `VkMappedMemoryRange` structures describing the memory ranges to invalidate.

`vkInvalidateMappedMemoryRanges` guarantees that device writes to the memory ranges described by `pMemoryRanges`, which have been made available to the host memory domain using the `VK_ACCESS_HOST_WRITE_BIT` and `VK_ACCESS_HOST_READ_BIT` access types, are made visible to the host. If a range of non-coherent memory is written by the host and then invalidated without first being flushed, its contents are undefined.

The first [synchronization scope](#) includes all host operations that happened-before it, as defined by the host memory model.

Note

This function does not synchronize with device operations directly - other host [synchronization operations](#) that depend on device operations such as [vkWaitForFences](#) must be executed beforehand. So for any non-coherent device write to be made visible to the host, there has to be a dependency chain along the following lines:

NOTE

1. Device write
2. Device memory barrier including host reads in its second scope
3. Signal on the device (e.g. a [fence signal operation](#))
4. Wait on the host (e.g. [vkWaitForFences](#))
5. [vkInvalidateMappedMemoryRanges](#)

The second [synchronization scope](#) includes all host operations that happen-after it, as defined by the host memory model.

The first [access scope](#) is empty.

The second [access scope](#) includes host reads to the specified memory ranges.

NOTE	<i>Note</i>
	When a device write to a memory location is made visible to the host in this way, each whole aligned set of <code>nonCoherentAtomSize</code> bytes that the memory location exists in will also be made visible as if they were written by the device. For example, with a <code>nonCoherentAtomSize</code> of 128, if an application writes to the first byte of a memory object on the device, the first 128 bytes of the memory object will be made visible by this command. While the value of the following 127 bytes will be unchanged, this does count as an access for the purpose of synchronization, so care must be taken to avoid data races.
NOTE	Mapping non-coherent memory does not implicitly invalidate that memory.

Valid Usage (Implicit)

- VUID-vkInvalidateMappedMemoryRanges-device-parameter
`device` **must** be a valid [VkDevice](#) handle
- VUID-vkInvalidateMappedMemoryRanges-pMemoryRanges-parameter
`pMemoryRanges` **must** be a valid pointer to an array of `memoryRangeCount` valid [VkMappedMemoryRange](#) structures
- VUID-vkInvalidateMappedMemoryRanges-memoryRangeCount-arraylength
`memoryRangeCount` **must** be greater than 0

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkMappedMemoryRange` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkMappedMemoryRange {
    VkStructureType    sType;
    const void*        pNext;
    VkDeviceMemory      memory;
    VkDeviceSize        offset;
    VkDeviceSize        size;
} VkMappedMemoryRange;
```

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **memory** is the memory object to which this range belongs.
- **offset** is the zero-based byte offset from the beginning of the memory object.
- **size** is either the size of range, or **VK_WHOLE_SIZE** to affect the range from **offset** to the end of the current mapping of the allocation.

Valid Usage

- VUID-VkMappedMemoryRange-memory-00684
memory must be currently host mapped
- VUID-VkMappedMemoryRange-size-00685
If **size** is not equal to **VK_WHOLE_SIZE**, **offset** and **size must** specify a range contained within the currently mapped range of **memory**
- VUID-VkMappedMemoryRange-size-00686
If **size** is equal to **VK_WHOLE_SIZE**, **offset must** be within the currently mapped range of **memory**
- VUID-VkMappedMemoryRange-offset-00687
offset must be a multiple of **VkPhysicalDeviceLimits::nonCoherentAtomSize**
- VUID-VkMappedMemoryRange-size-01389
If **size** is equal to **VK_WHOLE_SIZE**, the end of the current mapping of **memory must** either be a multiple of **VkPhysicalDeviceLimits::nonCoherentAtomSize** bytes from the beginning of the memory object, or be equal to the end of the memory object
- VUID-VkMappedMemoryRange-size-01390
If **size** is not equal to **VK_WHOLE_SIZE**, **size must** either be a multiple of **VkPhysicalDeviceLimits::nonCoherentAtomSize**, or **offset** plus **size must** equal the size of **memory**

Valid Usage (Implicit)

- VUID-VkMappedMemoryRange-sType-sType
sType must be **VK_STRUCTURE_TYPE_MAPPED_MEMORY_RANGE**
- VUID-VkMappedMemoryRange-pNext-pNext

pNext must be **NULL**

- VUID-VkMappedMemoryRange-memory-parameter
memory must be a valid **VkDeviceMemory** handle

To unmap a memory object once host access to it is no longer needed by the application, call:

```
// Provided by VK_VERSION_1_0
void vkUnmapMemory(
    VkDevice          device,
    VkDeviceMemory    memory);
```

- **device** is the logical device that owns the memory.
- **memory** is the memory object to be unmapped.

Valid Usage

- VUID-vkUnmapMemory-memory-00689
memory must be currently host mapped

Valid Usage (Implicit)

- VUID-vkUnmapMemory-device-parameter
device must be a valid **VkDevice** handle
- VUID-vkUnmapMemory-memory-parameter
memory must be a valid **VkDeviceMemory** handle
- VUID-vkUnmapMemory-memory-parent
memory must have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **memory** must be externally synchronized

11.2.6. Lazily Allocated Memory

If the memory object is allocated from a heap with the **VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT** bit set, that object's backing memory **may** be provided by the implementation lazily. The actual committed size of the memory **may** initially be as small as zero (or as large as the requested size), and monotonically increases as additional memory is needed.

A memory type with this flag set is only allowed to be bound to a **VkImage** whose usage flags include **VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT**.

NOTE

Using lazily allocated memory objects for framebuffer attachments that are not needed once a render pass instance has completed **may** allow some implementations to never allocate memory for such attachments.

To determine the amount of lazily-allocated memory that is currently committed for a memory object, call:

```
// Provided by VK_VERSION_1_0
void vkGetDeviceMemoryCommitment(
    VkDevice          device,
    VkDeviceMemory    memory,
    VkDeviceSize*     pCommittedMemoryInBytes);
```

- **device** is the logical device that owns the memory.
- **memory** is the memory object being queried.
- **pCommittedMemoryInBytes** is a pointer to a **VkDeviceSize** value in which the number of bytes currently committed is returned, on success.

The implementation **may** update the commitment at any time, and the value returned by this query **may** be out of date.

The implementation guarantees to allocate any committed memory from the **heapIndex** indicated by the memory type that the memory object was created with.

Valid Usage

- VUID-vkGetDeviceMemoryCommitment-memory-00690
memory must have been created with a memory type that reports **VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT**

Valid Usage (Implicit)

- VUID-vkGetDeviceMemoryCommitment-device-parameter
device must be a valid **VkDevice** handle
- VUID-vkGetDeviceMemoryCommitment-memory-parameter
memory must be a valid **VkDeviceMemory** handle
- VUID-vkGetDeviceMemoryCommitment-pCommittedMemoryInBytes-parameter
pCommittedMemoryInBytes must be a valid pointer to a **VkDeviceSize** value
- VUID-vkGetDeviceMemoryCommitment-memory-parent
memory must have been created, allocated, or retrieved from **device**

Chapter 12. Resource Creation

Vulkan supports two primary resource types: *buffers* and *images*. Resources are views of memory with associated formatting and dimensionality. Buffers provide access to raw arrays of bytes, whereas images **can** be multidimensional and **may** have associated metadata.

12.1. Buffers

Buffers represent linear arrays of data which are used for various purposes by binding them to a graphics or compute pipeline via descriptor sets or certain commands, or by directly specifying them as parameters to certain commands.

Buffers are represented by `VkBuffer` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkBuffer)
```

To create buffers, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateBuffer(
    VkDevice                                device,
    const VkBufferCreateInfo*               pCreateInfo,
    const VkAllocationCallbacks*           pAllocator,
    VkBuffer*                               pBuffer);
```

- `device` is the logical device that creates the buffer object.
- `pCreateInfo` is a pointer to a `VkBufferCreateInfo` structure containing parameters affecting creation of the buffer.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.
- `pBuffer` is a pointer to a `VkBuffer` handle in which the resulting buffer object is returned.

Valid Usage

- VUID-vkCreateBuffer-device-09664
`device` **must** support at least one queue family with one of the `VK_QUEUE_SPARSE_BINDING_BIT`, `VK_QUEUE_TRANSFER_BIT`, `VK_QUEUE_COMPUTE_BIT`, or `VK_QUEUE_GRAPHICS_BIT` capabilities
- VUID-vkCreateBuffer-flags-00911
If the `flags` member of `pCreateInfo` includes `VK_BUFFER_CREATE_SPARSE_BINDING_BIT`, creating this `VkBuffer` **must** not cause the total required sparse memory for all currently valid sparse resources on the device to exceed `VkPhysicalDeviceLimits::sparseAddressSpaceSize`

Valid Usage (Implicit)

- VUID-vkCreateBuffer-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkCreateBuffer-pCreateInfo-parameter
pCreateInfo **must** be a valid pointer to a valid [VkBufferCreateInfo](#) structure
- VUID-vkCreateBuffer-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkCreateBuffer-pBuffer-parameter
pBuffer **must** be a valid pointer to a [VkBuffer](#) handle

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_OUT_OF_DEVICE_MEMORY**

The [VkBufferCreateInfo](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkBufferCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkBufferCreateFlags flags;
    VkDeviceSize        size;
    VkBufferUsageFlags  usage;
    VkSharingMode        sharingMode;
    uint32_t            queueFamilyIndexCount;
    const uint32_t*      pQueueFamilyIndices;
} VkBufferCreateInfo;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **flags** is a bitmask of [VkBufferCreateFlagBits](#) specifying additional parameters of the buffer.
- **size** is the size in bytes of the buffer to be created.
- **usage** is a bitmask of [VkBufferUsageFlagBits](#) specifying allowed usages of the buffer.
- **sharingMode** is a [VkSharingMode](#) value specifying the sharing mode of the buffer when it will be accessed by multiple queue families.

- `queueFamilyIndexCount` is the number of entries in the `pQueueFamilyIndices` array.
- `pQueueFamilyIndices` is a pointer to an array of queue families that will access this buffer. It is ignored if `sharingMode` is not `VK_SHARING_MODE_CONCURRENT`.

Valid Usage

- VUID-VkBufferCreateInfo-None-09499
`usage` **must** be a valid combination of `VkBufferUsageFlagBits` values
- VUID-VkBufferCreateInfo-None-09500
`usage` **must** not be 0
- VUID-VkBufferCreateInfo-size-00912
`size` **must** be greater than 0
- VUID-VkBufferCreateInfo-sharingMode-00913
If `sharingMode` is `VK_SHARING_MODE_CONCURRENT`, `pQueueFamilyIndices` **must** be a valid pointer to an array of `queueFamilyIndexCount` `uint32_t` values
- VUID-VkBufferCreateInfo-sharingMode-00914
If `sharingMode` is `VK_SHARING_MODE_CONCURRENT`, `queueFamilyIndexCount` **must** be greater than 1
- VUID-VkBufferCreateInfo-sharingMode-01419
If `sharingMode` is `VK_SHARING_MODE_CONCURRENT`, each element of `pQueueFamilyIndices` **must** be unique and **must** be less than `pQueueFamilyPropertyCount` returned by `vkGetPhysicalDeviceQueueFamilyProperties` for the `physicalDevice` that was used to create `device`
- VUID-VkBufferCreateInfo-flags-00915
If the `sparseBinding` feature is not enabled, `flags` **must** not contain `VK_BUFFER_CREATE_SPARSE_BINDING_BIT`
- VUID-VkBufferCreateInfo-flags-00916
If the `sparseResidencyBuffer` feature is not enabled, `flags` **must** not contain `VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT`
- VUID-VkBufferCreateInfo-flags-00917
If the `sparseResidencyAliased` feature is not enabled, `flags` **must** not contain `VK_BUFFER_CREATE_SPARSE_ALIASED_BIT`
- VUID-VkBufferCreateInfo-flags-00918
If `flags` contains `VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT` or `VK_BUFFER_CREATE_SPARSE_ALIASED_BIT`, it **must** also contain `VK_BUFFER_CREATE_SPARSE_BINDING_BIT`
- VUID-VkBufferCreateInfo-flags-09641
If `flags` includes `VK_BUFFER_CREATE_PROTECTED_BIT`, then `usage` **must** not contain any of the following bits

Valid Usage (Implicit)

- VUID-VkBufferCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_BUFFER_CREATE_INFO`
- VUID-VkBufferCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkBufferCreateInfo-flags-parameter
`flags` **must** be a valid combination of `VkBufferCreateFlagBits` values
- VUID-VkBufferCreateInfo-sharingMode-parameter
`sharingMode` **must** be a valid `VkSharingMode` value

Bits which **can** be set in `VkBufferCreateInfo::usage`, specifying usage behavior of a buffer, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkBufferUsageFlagBits {
    VK_BUFFER_USAGE_TRANSFER_SRC_BIT = 0x00000001,
    VK_BUFFER_USAGE_TRANSFER_DST_BIT = 0x00000002,
    VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT = 0x00000004,
    VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT = 0x00000008,
    VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT = 0x00000010,
    VK_BUFFER_USAGE_STORAGE_BUFFER_BIT = 0x00000020,
    VK_BUFFER_USAGE_INDEX_BUFFER_BIT = 0x00000040,
    VK_BUFFER_USAGE_VERTEX_BUFFER_BIT = 0x00000080,
    VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT = 0x00000100,
} VkBufferUsageFlagBits;
```

- `VK_BUFFER_USAGE_TRANSFER_SRC_BIT` specifies that the buffer **can** be used as the source of a *transfer command* (see the definition of `VK_PIPELINE_STAGE_TRANSFER_BIT`).
- `VK_BUFFER_USAGE_TRANSFER_DST_BIT` specifies that the buffer **can** be used as the destination of a transfer command.
- `VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT` specifies that the buffer **can** be used to create a `VkBufferView` suitable for occupying a `VkDescriptorSet` slot of type `VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER`.
- `VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT` specifies that the buffer **can** be used to create a `VkBufferView` suitable for occupying a `VkDescriptorSet` slot of type `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER`.
- `VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT` specifies that the buffer **can** be used in a `VkDescriptorBufferInfo` suitable for occupying a `VkDescriptorSet` slot either of type `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER` or `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC`.
- `VK_BUFFER_USAGE_STORAGE_BUFFER_BIT` specifies that the buffer **can** be used in a `VkDescriptorBufferInfo` suitable for occupying a `VkDescriptorSet` slot either of type `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER` or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC`.
- `VK_BUFFER_USAGE_INDEX_BUFFER_BIT` specifies that the buffer is suitable for passing as the *buffer*

parameter to [vkCmdBindIndexBuffer](#).

- [VK_BUFFER_USAGE_VERTEX_BUFFER_BIT](#) specifies that the buffer is suitable for passing as an element of the `pBuffers` array to [vkCmdBindVertexBuffers](#).
- [VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT](#) specifies that the buffer is suitable for passing as the `buffer` parameter to [vkCmdDrawIndirect](#), [vkCmdDrawIndexedIndirect](#), or [vkCmdDispatchIndirect](#).

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkBufferUsageFlags;
```

`VkBufferUsageFlags` is a bitmask type for setting a mask of zero or more [VkBufferUsageFlagBits](#).

Bits which **can** be set in [VkBufferCreateInfo::flags](#), specifying additional parameters of a buffer, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkBufferCreateFlagBits {
    VK_BUFFER_CREATE_SPARSE_BINDING_BIT = 0x00000001,
    VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT = 0x00000002,
    VK_BUFFER_CREATE_SPARSE_ALIASED_BIT = 0x00000004,
} VkBufferCreateFlagBits;
```

- [VK_BUFFER_CREATE_SPARSE_BINDING_BIT](#) specifies that the buffer will be backed using sparse memory binding.
- [VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT](#) specifies that the buffer **can** be partially backed using sparse memory binding. Buffers created with this flag **must** also be created with the [VK_BUFFER_CREATE_SPARSE_BINDING_BIT](#) flag.
- [VK_BUFFER_CREATE_SPARSE_ALIASED_BIT](#) specifies that the buffer will be backed using sparse memory binding with memory ranges that might also simultaneously be backing another buffer (or another portion of the same buffer). Buffers created with this flag **must** also be created with the [VK_BUFFER_CREATE_SPARSE_BINDING_BIT](#) flag.

See [Sparse Resource Features](#) and [Physical Device Features](#) for details of the sparse memory features supported on a device.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkBufferCreateFlags;
```

`VkBufferCreateFlags` is a bitmask type for setting a mask of zero or more [VkBufferCreateFlagBits](#).

To destroy a buffer, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyBuffer(
    VkDevice          device,
    VkBuffer          buffer,
    const VkAllocationCallbacks* pAllocator);
```

- **device** is the logical device that destroys the buffer.
- **buffer** is the buffer to destroy.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyBuffer-buffer-00922
All submitted commands that refer to **buffer**, either directly or via a **VkBufferView**, **must** have completed execution
- VUID-vkDestroyBuffer-buffer-00923
If **VkAllocationCallbacks** were provided when **buffer** was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyBuffer-buffer-00924
If no **VkAllocationCallbacks** were provided when **buffer** was created, **pAllocator** **must** be **NULL**

Valid Usage (Implicit)

- VUID-vkDestroyBuffer-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkDestroyBuffer-buffer-parameter
If **buffer** is not **VK_NULL_HANDLE**, **buffer** **must** be a valid **VkBuffer** handle
- VUID-vkDestroyBuffer-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid **VkAllocationCallbacks** structure
- VUID-vkDestroyBuffer-buffer-parent
If **buffer** is a valid handle, it **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **buffer** **must** be externally synchronized

12.2. Buffer Views

A *buffer view* represents a contiguous range of a buffer and a specific format to be used to interpret the data. Buffer views are used to enable shaders to access buffer contents using [image operations](#). In order to create a valid buffer view, the buffer **must** have been created with at least one of the following usage flags:

- `VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT`
- `VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT`

Buffer views are represented by `VkBufferView` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkBufferView)
```

To create a buffer view, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateBufferView(
    VkDevice                                device,
    const VkBufferViewCreateInfo*           pCreateInfo,
    const VkAllocationCallbacks*           pAllocator,
    VkBufferView*                           pView);
```

- `device` is the logical device that creates the buffer view.
- `pCreateInfo` is a pointer to a `VkBufferViewCreateInfo` structure containing parameters to be used to create the buffer view.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.
- `pView` is a pointer to a `VkBufferView` handle in which the resulting buffer view object is returned.

Valid Usage

- VUID-vkCreateBufferView-device-09665
`device` **must** support at least one queue family with one of the `VK_QUEUE_COMPUTE_BIT` or `VK_QUEUE_GRAPHICS_BIT` capabilities

Valid Usage (Implicit)

- VUID-vkCreateBufferView-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkCreateBufferView-pCreateInfo-parameter
`pCreateInfo` **must** be a valid pointer to a valid `VkBufferViewCreateInfo` structure

- VUID-vkCreateBufferView-pAllocator-parameter

If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure

- VUID-vkCreateBufferView-pView-parameter

`pView` **must** be a valid pointer to a `VkBufferView` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkBufferViewCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkBufferViewCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkBufferViewCreateFlags flags;
    VkBuffer            buffer;
    VkFormat            format;
    VkDeviceSize        offset;
    VkDeviceSize        range;
} VkBufferViewCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `buffer` is a `VkBuffer` on which the view will be created.
- `format` is a `VkFormat` describing the format of the data elements in the buffer.
- `offset` is an offset in bytes from the base address of the buffer. Accesses to the buffer view from shaders use addressing that is relative to this starting offset.
- `range` is a size in bytes of the buffer view. If `range` is equal to `VK_WHOLE_SIZE`, the range from `offset` to the end of the buffer is used. If `VK_WHOLE_SIZE` is used and the remaining size of the buffer is not a multiple of the `texel block size` of `format`, the nearest smaller multiple is used.

The buffer view has a *buffer view usage* identifying which descriptor types can be created from it. This usage is equal to the `VkBufferCreateInfo::usage` value used to create `buffer`.

Valid Usage

- VUID-VkBufferViewCreateInfo-offset-00925
offset must be less than the size of **buffer**
- VUID-VkBufferViewCreateInfo-range-00928
If **range** is not equal to **VK_WHOLE_SIZE**, **range must** be greater than 0
- VUID-VkBufferViewCreateInfo-range-00929
If **range** is not equal to **VK_WHOLE_SIZE**, **range must** be an integer multiple of the texel block size of **format**
- VUID-VkBufferViewCreateInfo-range-00930
If **range** is not equal to **VK_WHOLE_SIZE**, the number of texel buffer elements given by $(\lceil \text{range} / (\text{texel block size}) \rceil \times (\text{texels per block}))$ where texel block size and texels per block are as defined in the [Compatible Formats](#) table for **format**, **must** be less than or equal to **VkPhysicalDeviceLimits::maxTexelBufferElements**
- VUID-VkBufferViewCreateInfo-offset-00931
If **range** is not equal to **VK_WHOLE_SIZE**, the sum of **offset** and **range must** be less than or equal to the size of **buffer**
- VUID-VkBufferViewCreateInfo-range-04059
If **range** is equal to **VK_WHOLE_SIZE**, the number of texel buffer elements given by $(\lceil (\text{size} - \text{offset}) / (\text{texel block size}) \rceil \times (\text{texels per block}))$ where size is the size of **buffer**, and texel block size and texels per block are as defined in the [Compatible Formats](#) table for **format**, **must** be less than or equal to **VkPhysicalDeviceLimits::maxTexelBufferElements**
- VUID-VkBufferViewCreateInfo-buffer-00932
buffer must have been created with a **usage** value containing at least one of **VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT** or **VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT**
- VUID-VkBufferViewCreateInfo-format-08778
If the **buffer view usage** contains **VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT**, then **format features** of **format must** contain **VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT**
- VUID-VkBufferViewCreateInfo-format-08779
If the **buffer view usage** contains **VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT**, then **format features** of **format must** contain **VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT**
- VUID-VkBufferViewCreateInfo-buffer-00935
If **buffer** is non-sparse then it **must** be bound completely and contiguously to a single **VkDeviceMemory** object
- VUID-VkBufferViewCreateInfo-offset-02749
offset must be a multiple of **VkPhysicalDeviceLimits::minTexelBufferOffsetAlignment**

Valid Usage (Implicit)

- VUID-VkBufferViewCreateInfo-sType-sType
sType must be **VK_STRUCTURE_TYPE_BUFFER_VIEW_CREATE_INFO**
- VUID-VkBufferViewCreateInfo-pNext-pNext

pNext must be **NULL**

- VUID-VkBufferViewCreateInfo-flags-zero-bitmask
flags must be 0
- VUID-VkBufferViewCreateInfo-buffer-parameter
buffer must be a valid **VkBuffer** handle
- VUID-VkBufferViewCreateInfo-format-parameter
format must be a valid **VkFormat** value

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkBufferViewCreateFlags;
```

VkBufferViewCreateFlags is a bitmask type for setting a mask, but is currently reserved for future use.

To destroy a buffer view, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyBufferView(
    VkDevice          device,
    VkBufferView      bufferView,
    const VkAllocationCallbacks* pAllocator);
```

- **device** is the logical device that destroys the buffer view.
- **bufferView** is the buffer view to destroy.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyBufferView-bufferView-00936
All submitted commands that refer to **bufferView** must have completed execution
- VUID-vkDestroyBufferView-bufferView-00937
If **VkAllocationCallbacks** were provided when **bufferView** was created, a compatible set of callbacks must be provided here
- VUID-vkDestroyBufferView-bufferView-00938
If no **VkAllocationCallbacks** were provided when **bufferView** was created, **pAllocator** must be **NULL**

Valid Usage (Implicit)

- VUID-vkDestroyBufferView-device-parameter
device must be a valid **VkDevice** handle

- VUID-vkDestroyBufferView-bufferView-parameter
If `bufferView` is not `VK_NULL_HANDLE`, `bufferView` **must** be a valid `VkBufferView` handle
- VUID-vkDestroyBufferView-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkDestroyBufferView-bufferView-parent
If `bufferView` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `bufferView` **must** be externally synchronized

12.2.1. Buffer View Format Features

Valid uses of a `VkBufferView` **may** depend on the buffer view's *format features*, defined below. Such constraints are documented in the affected valid usage statement.

- The buffer view's set of *format features* is the value of `VkFormatProperties::bufferFeatures` found by calling `vkGetPhysicalDeviceFormatProperties` on the same `format` as `VkBufferViewCreateInfo::format`.

12.3. Images

Images represent multidimensional - up to 3 - arrays of data which **can** be used for various purposes (e.g. attachments, textures), by binding them to a graphics or compute pipeline via descriptor sets, or by directly specifying them as parameters to certain commands.

Images are represented by `VkImage` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkImage)
```

To create images, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateImage(
    VkDevice                device,
    const VkImageCreateInfo* pCreateInfo,
    const VkAllocationCallbacks* pAllocator,
    VkImage*                pImage);
```

- `device` is the logical device that creates the image.
- `pCreateInfo` is a pointer to a `VkImageCreateInfo` structure containing parameters to be used to

create the image.

- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.
- **pImage** is a pointer to a [VkImage](#) handle in which the resulting image object is returned.

Valid Usage

- VUID-vkCreateImage-device-09666
device must support at least one queue family with one of the `VK_QUEUE_SPARSE_BINDING_BIT`, `VK_QUEUE_TRANSFER_BIT`, `VK_QUEUE_COMPUTE_BIT`, or `VK_QUEUE_GRAPHICS_BIT` capabilities
- VUID-vkCreateImage-flags-00939
If the **flags** member of **pCreateInfo** includes `VK_IMAGE_CREATE_SPARSE_BINDING_BIT`, creating this **VkImage must** not cause the total required sparse memory for all currently valid sparse resources on the device to exceed `VkPhysicalDeviceLimits::sparseAddressSpaceSize`

Valid Usage (Implicit)

- VUID-vkCreateImage-device-parameter
device must be a valid [VkDevice](#) handle
- VUID-vkCreateImage-pCreateInfo-parameter
pCreateInfo must be a valid pointer to a valid [VkImageCreateInfo](#) structure
- VUID-vkCreateImage-pAllocator-parameter
If **pAllocator** is not `NULL`, **pAllocator must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkCreateImage-pImage-parameter
pImage must be a valid pointer to a [VkImage](#) handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The [VkImageCreateInfo](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageCreateInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkImageCreateFlags    flags;
    VkImageType           imageType;
    VkFormat              format;
    VkExtent3D            extent;
    uint32_t              mipLevels;
    uint32_t              arrayLayers;
    VkSampleCountFlagBits samples;
    VkImageTiling          tiling;
    VkImageUsageFlags      usage;
    VkSharingMode          sharingMode;
    uint32_t              queueFamilyIndexCount;
    const uint32_t*        pQueueFamilyIndices;
    VkImageLayout          initialLayout;
} VkImageCreateInfo;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **flags** is a bitmask of [VkImageCreateFlagBits](#) describing additional parameters of the image.
- **imageType** is a [VkImageType](#) value specifying the basic dimensionality of the image. Layers in array textures do not count as a dimension for the purposes of the image type.
- **format** is a [VkFormat](#) describing the format and type of the texel blocks that will be contained in the image.
- **extent** is a [VkExtent3D](#) describing the number of data elements in each dimension of the base level.
- **mipLevels** describes the number of levels of detail available for minified sampling of the image.
- **arrayLayers** is the number of layers in the image.
- **samples** is a [VkSampleCountFlagBits](#) value specifying the number of [samples per texel](#).
- **tiling** is a [VkImageTiling](#) value specifying the tiling arrangement of the texel blocks in memory.
- **usage** is a bitmask of [VkImageUsageFlagBits](#) describing the intended usage of the image.
- **sharingMode** is a [VkSharingMode](#) value specifying the sharing mode of the image when it will be accessed by multiple queue families.
- **queueFamilyIndexCount** is the number of entries in the **pQueueFamilyIndices** array.
- **pQueueFamilyIndices** is a pointer to an array of queue families that will access this image. It is ignored if **sharingMode** is not `VK_SHARING_MODE_CONCURRENT`.
- **initialLayout** is a [VkImageLayout](#) value specifying the initial [VkImageLayout](#) of all image subresources of the image. See [Image Layouts](#).

Images created with **tiling** equal to `VK_IMAGE_TILING_LINEAR` have further restrictions on their limits and capabilities compared to images created with **tiling** equal to `VK_IMAGE_TILING_OPTIMAL`. Creation

of images with tiling `VK_IMAGE_TILING_LINEAR` **may** not be supported unless other parameters meet all of the constraints:

- `imageType` is `VK_IMAGE_TYPE_2D`
- `format` is not a depth/stencil format
- `mipLevels` is 1
- `arrayLayers` is 1
- `samples` is `VK_SAMPLE_COUNT_1_BIT`
- `usage` only includes `VK_IMAGE_USAGE_TRANSFER_SRC_BIT` and/or `VK_IMAGE_USAGE_TRANSFER_DST_BIT`

Implementations **may** support additional limits and capabilities beyond those listed above.

To determine the set of valid `usage` bits for a given format, call `vkGetPhysicalDeviceFormatProperties`.

If the size of the resultant image would exceed `maxResourceSize`, then `vkCreateImage` **must** fail and return `VK_ERROR_OUT_OF_DEVICE_MEMORY`. This failure **may** occur even when all image creation parameters satisfy their valid usage requirements.

Image Creation Limits

Valid values for some image creation parameters are limited by a numerical upper bound or by inclusion in a bitset. For example, `VkImageCreateInfo::arrayLayers` is limited by `imageCreateMaxArrayLayers`, defined below; and `VkImageCreateInfo::samples` is limited by `imageCreateSampleCounts`, also defined below.

Several limiting values are defined below, as well as assisting values from which the limiting values are derived. The limiting values are referenced by the relevant valid usage statements of `VkImageCreateInfo`.

- Let `VkBool32 imageCreateMaybeLinear` indicate if the resultant image may be `linear`. (The definition below is trivial because certain extensions are disabled in this build of the specification).
 - If `tiling` is `VK_IMAGE_TILING_LINEAR`, then `imageCreateMaybeLinear` is `VK_TRUE`.
 - If `tiling` is `VK_IMAGE_TILING_OPTIMAL`, then `imageCreateMaybeLinear` is `VK_FALSE`.
- Let `VkFormatFeatureFlags imageCreateFormatFeatures` be the set of valid *format features* available during image creation.
 - If `tiling` is `VK_IMAGE_TILING_LINEAR`, then `imageCreateFormatFeatures` is the value of `VkFormatProperties::linearTilingFeatures` found by calling `vkGetPhysicalDeviceFormatProperties` with parameter `format` equal to `VkImageCreateInfo::format`.
 - If `tiling` is `VK_IMAGE_TILING_OPTIMAL`, then `imageCreateFormatFeatures` is the value of `VkFormatProperties::optimalTilingFeatures` found by calling `vkGetPhysicalDeviceFormatProperties` with parameter `format` equal to `VkImageCreateInfo::format`.

- Let `uint32_t imageCreateMaxMipLevels` be the value of `VkImageFormatProperties::maxMipLevels` found by calling `vkGetPhysicalDeviceImageFormatProperties` with parameters `format`, `imageType`, `tiling`, `usage`, and `flags` equal to those in `VkImageCreateInfo`. If `vkGetPhysicalDeviceImageFormatProperties` returns an error, then the value of `imageCreateMaxMipLevels` is undefined.
- Let `uint32_t imageCreateMaxArrayLayers` be defined analogously to `imageCreateMaxMipLevels`.
- Let `VkExtent3D imageCreateMaxExtent` be defined analogously to `imageCreateMaxMipLevels`.
- Let `VkSampleCountFlags imageCreateSampleCounts` be defined analogously to `imageCreateMaxMipLevels`.

Valid Usage

- VUID-VkImageCreateInfo-imageCreateMaxMipLevels-02251
Each of the following values (as described in [Image Creation Limits](#)) **must** not be undefined : `imageCreateMaxMipLevels`, `imageCreateMaxArrayLayers`, `imageCreateMaxExtent`, and `imageCreateSampleCounts`
- VUID-VkImageCreateInfo-sharingMode-00941
If `sharingMode` is `VK_SHARING_MODE_CONCURRENT`, `pQueueFamilyIndices` **must** be a valid pointer to an array of `queueFamilyIndexCount` `uint32_t` values
- VUID-VkImageCreateInfo-sharingMode-00942
If `sharingMode` is `VK_SHARING_MODE_CONCURRENT`, `queueFamilyIndexCount` **must** be greater than 1
- VUID-VkImageCreateInfo-sharingMode-01392
If `sharingMode` is `VK_SHARING_MODE_CONCURRENT`, each element of `pQueueFamilyIndices` **must** be unique and **must** be less than `pQueueFamilyPropertyCount` returned by `vkGetPhysicalDeviceQueueFamilyProperties` for the `physicalDevice` that was used to create `device`
- VUID-VkImageCreateInfo-format-00943
`format` **must** not be `VK_FORMAT_UNDEFINED`
- VUID-VkImageCreateInfo-extent-00944
`extent.width` **must** be greater than 0
- VUID-VkImageCreateInfo-extent-00945
`extent.height` **must** be greater than 0
- VUID-VkImageCreateInfo-extent-00946
`extent.depth` **must** be greater than 0
- VUID-VkImageCreateInfo-mipLevels-00947
`mipLevels` **must** be greater than 0
- VUID-VkImageCreateInfo-arrayLayers-00948
`arrayLayers` **must** be greater than 0
- VUID-VkImageCreateInfo-flags-00949
If `flags` contains `VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT`, `imageType` **must** be

VK_IMAGE_TYPE_2D

- VUID-VkImageCreateInfo-flags-08865
If `flags` contains `VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT`, `extent.width` and `extent.height` **must** be equal
- VUID-VkImageCreateInfo-flags-08866
If `flags` contains `VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT`, `arrayLayers` **must** be greater than or equal to 6
- VUID-VkImageCreateInfo-imageType-10196
If `imageType` is `VK_IMAGE_TYPE_3D` and `flags` contains `VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT` then `flags` **must** not contain any of `VK_IMAGE_CREATE_SPARSE_BINDING_BIT`, `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`, and `VK_IMAGE_CREATE_SPARSE_ALIASED_BIT`
- VUID-VkImageCreateInfo-extent-02252
`extent.width` **must** be less than or equal to `imageCreateMaxExtent.width` (as defined in [Image Creation Limits](#))
- VUID-VkImageCreateInfo-extent-02253
`extent.height` **must** be less than or equal to `imageCreateMaxExtent.height` (as defined in [Image Creation Limits](#))
- VUID-VkImageCreateInfo-extent-02254
`extent.depth` **must** be less than or equal to `imageCreateMaxExtent.depth` (as defined in [Image Creation Limits](#))
- VUID-VkImageCreateInfo-imageType-00956
If `imageType` is `VK_IMAGE_TYPE_1D`, both `extent.height` and `extent.depth` **must** be 1
- VUID-VkImageCreateInfo-imageType-00957
If `imageType` is `VK_IMAGE_TYPE_2D`, `extent.depth` **must** be 1
- VUID-VkImageCreateInfo-mipLevels-00958
`mipLevels` **must** be less than or equal to the number of levels in the complete mipmap chain based on `extent.width`, `extent.height`, and `extent.depth`
- VUID-VkImageCreateInfo-mipLevels-02255
`mipLevels` **must** be less than or equal to `imageCreateMaxMipLevels` (as defined in [Image Creation Limits](#))
- VUID-VkImageCreateInfo-arrayLayers-02256
`arrayLayers` **must** be less than or equal to `imageCreateMaxArrayLayers` (as defined in [Image Creation Limits](#))
- VUID-VkImageCreateInfo-imageType-00961
If `imageType` is `VK_IMAGE_TYPE_3D`, `arrayLayers` **must** be 1
- VUID-VkImageCreateInfo-samples-02257
If `samples` is not `VK_SAMPLE_COUNT_1_BIT`, then `imageType` **must** be `VK_IMAGE_TYPE_2D`, `flags` **must** not contain `VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT`, `mipLevels` **must** be equal to 1, and `imageCreateMayBeLinear` (as defined in [Image Creation Limits](#)) **must** be `VK_FALSE`,
- VUID-VkImageCreateInfo-usage-00963
If `usage` includes `VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT`, then bits other than

VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT, VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT, and VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT **must** not be set

- VUID-VkImageCreateInfo-usage-00964
If `usage` includes `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT`, `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`, `VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT`, or `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT`, `extent.width` **must** be less than or equal to `VkPhysicalDeviceLimits::maxFramebufferWidth`
- VUID-VkImageCreateInfo-usage-00965
If `usage` includes `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT`, `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`, `VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT`, or `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT`, `extent.height` **must** be less than or equal to `VkPhysicalDeviceLimits::maxFramebufferHeight`
- VUID-VkImageCreateInfo-usage-00966
If `usage` includes `VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT`, `usage` **must** also contain at least one of `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT`, `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`, or `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT`
- VUID-VkImageCreateInfo-samples-02258
`samples` **must** be a valid `VkSampleCountFlagBits` value that is set in `imageCreateSampleCounts` (as defined in [Image Creation Limits](#))
- VUID-VkImageCreateInfo-usage-00968
If the `shaderStorageImageMultisample` feature is not enabled, and `usage` contains `VK_IMAGE_USAGE_STORAGE_BIT`, `samples` **must** be `VK_SAMPLE_COUNT_1_BIT`
- VUID-VkImageCreateInfo-flags-00969
If the `sparseBinding` feature is not enabled, `flags` **must** not contain `VK_IMAGE_CREATE_SPARSE_BINDING_BIT`
- VUID-VkImageCreateInfo-flags-01924
If the `sparseResidencyAliased` feature is not enabled, `flags` **must** not contain `VK_IMAGE_CREATE_SPARSE_ALIASED_BIT`
- VUID-VkImageCreateInfo-tiling-04121
If `tiling` is `VK_IMAGE_TILING_LINEAR`, `flags` **must** not contain `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`
- VUID-VkImageCreateInfo-imageType-00970
If `imageType` is `VK_IMAGE_TYPE_1D`, `flags` **must** not contain `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`
- VUID-VkImageCreateInfo-imageType-00971
If the `sparseResidencyImage2D` feature is not enabled, and `imageType` is `VK_IMAGE_TYPE_2D`, `flags` **must** not contain `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`
- VUID-VkImageCreateInfo-imageType-00972
If the `sparseResidencyImage3D` feature is not enabled, and `imageType` is `VK_IMAGE_TYPE_3D`, `flags` **must** not contain `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`
- VUID-VkImageCreateInfo-imageType-00973
If the `sparseResidency2Samples` feature is not enabled, `imageType` is `VK_IMAGE_TYPE_2D`, and `samples` is `VK_SAMPLE_COUNT_2_BIT`, `flags` **must** not contain

VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT

- VUID-VkImageCreateInfo-imageType-00974

If the `sparseResidency4Samples` feature is not enabled, `imageType` is `VK_IMAGE_TYPE_2D`, and `samples` is `VK_SAMPLE_COUNT_4_BIT`, `flags` **must** not contain `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`

- VUID-VkImageCreateInfo-imageType-00975

If the `sparseResidency8Samples` feature is not enabled, `imageType` is `VK_IMAGE_TYPE_2D`, and `samples` is `VK_SAMPLE_COUNT_8_BIT`, `flags` **must** not contain `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`

- VUID-VkImageCreateInfo-imageType-00976

If the `sparseResidency16Samples` feature is not enabled, `imageType` is `VK_IMAGE_TYPE_2D`, and `samples` is `VK_SAMPLE_COUNT_16_BIT`, `flags` **must** not contain `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`

- VUID-VkImageCreateInfo-flags-00987

If `flags` contains `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` or `VK_IMAGE_CREATE_SPARSE_ALIASED_BIT`, it **must** also contain `VK_IMAGE_CREATE_SPARSE_BINDING_BIT`

- VUID-VkImageCreateInfo-None-01925

If any of the bits `VK_IMAGE_CREATE_SPARSE_BINDING_BIT`, `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`, or `VK_IMAGE_CREATE_SPARSE_ALIASED_BIT` are set, `VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT` **must** not also be set

- VUID-VkImageCreateInfo-initialLayout-00993

`initialLayout` **must** be `VK_IMAGE_LAYOUT_UNDEFINED` or `VK_IMAGE_LAYOUT_PREINITIALIZED`

Valid Usage (Implicit)

- VUID-VkImageCreateInfo-sType-sType

`sType` **must** be `VK_STRUCTURE_TYPE_IMAGE_CREATE_INFO`

- VUID-VkImageCreateInfo-pNext-pNext

`pNext` **must** be `NULL`

- VUID-VkImageCreateInfo-flags-parameter

`flags` **must** be a valid combination of `VkImageCreateFlagBits` values

- VUID-VkImageCreateInfo-imageType-parameter

`imageType` **must** be a valid `VkImageType` value

- VUID-VkImageCreateInfo-format-parameter

`format` **must** be a valid `VkFormat` value

- VUID-VkImageCreateInfo-samples-parameter

`samples` **must** be a valid `VkSampleCountFlagBits` value

- VUID-VkImageCreateInfo-tiling-parameter

`tiling` **must** be a valid `VkImageTiling` value

- VUID-VkImageCreateInfo-usage-parameter

`usage` **must** be a valid combination of `VkImageUsageFlagBits` values

- VUID-VkImageCreateInfo-usage-requiredbitmask **usage** **must** not be 0
- VUID-VkImageCreateInfo-sharingMode-parameter **sharingMode** **must** be a valid `VkSharingMode` value
- VUID-VkImageCreateInfo-initialLayout-parameter **initialLayout** **must** be a valid `VkImageLayout` value

Bits which **can** be set in

- `VkImageCreateInfo::usage`

specify intended usage of an image, and are:

```
// Provided by VK_VERSION_1_0
typedef enum VkImageUsageFlagBits {
    VK_IMAGE_USAGE_TRANSFER_SRC_BIT = 0x00000001,
    VK_IMAGE_USAGE_TRANSFER_DST_BIT = 0x00000002,
    VK_IMAGE_USAGE_SAMPLED_BIT = 0x00000004,
    VK_IMAGE_USAGE_STORAGE_BIT = 0x00000008,
    VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT = 0x00000010,
    VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT = 0x00000020,
    VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT = 0x00000040,
    VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT = 0x00000080,
} VkImageUsageFlagBits;
```

- `VK_IMAGE_USAGE_TRANSFER_SRC_BIT` specifies that the image **can** be used as the source of a transfer command.
- `VK_IMAGE_USAGE_TRANSFER_DST_BIT` specifies that the image **can** be used as the destination of a transfer command.
- `VK_IMAGE_USAGE_SAMPLED_BIT` specifies that the image **can** be used to create a `VkImageView` suitable for occupying a `VkDescriptorSet` slot either of type `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE` or `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, and be sampled by a shader.
- `VK_IMAGE_USAGE_STORAGE_BIT` specifies that the image **can** be used to create a `VkImageView` suitable for occupying a `VkDescriptorSet` slot of type `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`.
- `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT` specifies that the image **can** be used to create a `VkImageView` suitable for use as a color or resolve attachment in a `VkFramebuffer`.
- `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT` specifies that the image **can** be used to create a `VkImageView` suitable for use as a depth/stencil attachment in a `VkFramebuffer`.
- `VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT` specifies that implementations **may** support using `memory allocations` with the `VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT` to back an image with this usage. This bit **can** be set for any image that **can** be used to create a `VkImageView` suitable for use as a color, resolve, depth/stencil, or input attachment.
- `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT` specifies that the image **can** be used to create a `VkImageView` suitable for occupying `VkDescriptorSet` slot of type

`VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT`; be read from a shader as an input attachment; and be used as an input attachment in a framebuffer.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkImageUsageFlags;
```

`VkImageUsageFlags` is a bitmask type for setting a mask of zero or more `VkImageUsageFlagBits`.

When creating a `VkImageView` one of the following `VkImageUsageFlagBits` **must** be set:

- `VK_IMAGE_USAGE_SAMPLED_BIT`
- `VK_IMAGE_USAGE_STORAGE_BIT`
- `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT`
- `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`
- `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT`
- `VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT`

Bits which **can** be set in `VkImageCreateInfo::flags`, specifying additional parameters of an image, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkImageCreateFlagBits {
    VK_IMAGE_CREATE_SPARSE_BINDING_BIT = 0x00000001,
    VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT = 0x00000002,
    VK_IMAGE_CREATE_SPARSE_ALIASED_BIT = 0x00000004,
    VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT = 0x00000008,
    VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT = 0x00000010,
} VkImageCreateFlagBits;
```

- `VK_IMAGE_CREATE_SPARSE_BINDING_BIT` specifies that the image will be backed using sparse memory binding.
- `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` specifies that the image **can** be partially backed using sparse memory binding. Images created with this flag **must** also be created with the `VK_IMAGE_CREATE_SPARSE_BINDING_BIT` flag.
- `VK_IMAGE_CREATE_SPARSE_ALIASED_BIT` specifies that the image will be backed using sparse memory binding with memory ranges that might also simultaneously be backing another image (or another portion of the same image). Images created with this flag **must** also be created with the `VK_IMAGE_CREATE_SPARSE_BINDING_BIT` flag.
- `VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT` specifies that the image **can** be used to create a `VkImageView` with a different format from the image.
- `VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT` specifies that the image **can** be used to create a `VkImageView` of type `VK_IMAGE_VIEW_TYPE_CUBE` or `VK_IMAGE_VIEW_TYPE_CUBE_ARRAY`.

See [Sparse Resource Features](#) and [Sparse Physical Device Features](#) for more details.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkImageCreateFlags;
```

`VkImageCreateFlags` is a bitmask type for setting a mask of zero or more `VkImageCreateFlagBits`.

Possible values of `VkImageCreateInfo::imageType`, specifying the basic dimensionality of an image, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkImageType {
    VK_IMAGE_TYPE_1D = 0,
    VK_IMAGE_TYPE_2D = 1,
    VK_IMAGE_TYPE_3D = 2,
} VkImageType;
```

- `VK_IMAGE_TYPE_1D` specifies a one-dimensional image.
- `VK_IMAGE_TYPE_2D` specifies a two-dimensional image.
- `VK_IMAGE_TYPE_3D` specifies a three-dimensional image.

Possible values of `VkImageCreateInfo::tiling`, specifying the tiling arrangement of texel blocks in an image, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkImageTiling {
    VK_IMAGE_TILING_OPTIMAL = 0,
    VK_IMAGE_TILING_LINEAR = 1,
} VkImageTiling;
```

- `VK_IMAGE_TILING_OPTIMAL` specifies optimal tiling (texels are laid out in an implementation-dependent arrangement, for more efficient memory access).
- `VK_IMAGE_TILING_LINEAR` specifies linear tiling (texels are laid out in memory in row-major order, possibly with some padding on each row).

To query the memory layout of an image subresource, call:

```
// Provided by VK_VERSION_1_0
void vkGetImageSubresourceLayout(
    VkDevice          device,
    VkImage           image,
    const VkImageSubresource* pSubresource,
    VkSubresourceLayout* pLayout);
```

- `device` is the logical device that owns the image.
- `image` is the image whose layout is being queried.

- **pSubresource** is a pointer to a [VkImageSubresource](#) structure selecting a specific image subresource from the image.
- **pLayout** is a pointer to a [VkSubresourceLayout](#) structure in which the layout is returned.

The image **must** be [linear](#). The returned layout is valid for [host access](#).

vkGetImageSubresourceLayout is invariant for the lifetime of a single image.

Valid Usage

- VUID-vkGetImageSubresourceLayout-image-07789
image must have been created with **tiling** equal to **VK_IMAGE_TILING_LINEAR**
- VUID-vkGetImageSubresourceLayout-aspectMask-00997
The **aspectMask** member of **pSubresource must** only have a single bit set
- VUID-vkGetImageSubresourceLayout-mipLevel-01716
The **mipLevel** member of **pSubresource must** be less than the **mipLevels** specified in **image**
- VUID-vkGetImageSubresourceLayout-arrayLayer-01717
The **arrayLayer** member of **pSubresource must** be less than the **arrayLayers** specified in **image**
- VUID-vkGetImageSubresourceLayout-format-08886
If **format** of the **image** is a color format and **tiling** of the **image** is **VK_IMAGE_TILING_LINEAR** or **VK_IMAGE_TILING_OPTIMAL**, the **aspectMask** member of **pSubresource must** be **VK_IMAGE_ASPECT_COLOR_BIT**
- VUID-vkGetImageSubresourceLayout-format-04462
If **format** of the **image** has a depth component, the **aspectMask** member of **pSubresource must** contain **VK_IMAGE_ASPECT_DEPTH_BIT**
- VUID-vkGetImageSubresourceLayout-format-04463
If **format** of the **image** has a stencil component, the **aspectMask** member of **pSubresource must** contain **VK_IMAGE_ASPECT_STENCIL_BIT**
- VUID-vkGetImageSubresourceLayout-format-04464
If **format** of the **image** does not contain a stencil or depth component, the **aspectMask** member of **pSubresource must** not contain **VK_IMAGE_ASPECT_DEPTH_BIT** or **VK_IMAGE_ASPECT_STENCIL_BIT**

Valid Usage (Implicit)

- VUID-vkGetImageSubresourceLayout-device-parameter
device must be a valid [VkDevice](#) handle
- VUID-vkGetImageSubresourceLayout-image-parameter
image must be a valid [VkImage](#) handle
- VUID-vkGetImageSubresourceLayout-pSubresource-parameter
pSubresource must be a valid pointer to a valid [VkImageSubresource](#) structure

- VUID-vkGetImageSubresourceLayout-pLayout-parameter
pLayout must be a valid pointer to a [VkSubresourceLayout](#) structure
- VUID-vkGetImageSubresourceLayout-image-parent
image must have been created, allocated, or retrieved from **device**

The [VkImageSubresource](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageSubresource {
    VkImageAspectFlags    aspectMask;
    uint32_t              mipLevel;
    uint32_t              arrayLayer;
} VkImageSubresource;
```

- **aspectMask** is a [VkImageAspectFlags](#) value selecting the image *aspect*.
- **mipLevel** selects the mipmap level.
- **arrayLayer** selects the array layer.

Valid Usage (Implicit)

- VUID-VkImageSubresource-aspectMask-parameter
aspectMask must be a valid combination of [VkImageAspectFlagBits](#) values
- VUID-VkImageSubresource-aspectMask-requiredbitmask
aspectMask must not be 0

Information about the layout of the image subresource is returned in a [VkSubresourceLayout](#) structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkSubresourceLayout {
    VkDeviceSize    offset;
    VkDeviceSize    size;
    VkDeviceSize    rowPitch;
    VkDeviceSize    arrayPitch;
    VkDeviceSize    depthPitch;
} VkSubresourceLayout;
```

- **offset** is the byte offset from the start of the image where the image subresource begins.
- **size** is the size in bytes of the image subresource. **size** includes any extra memory that is required based on **rowPitch**.
- **rowPitch** describes the number of bytes between each row of texels in an image.
- **arrayPitch** describes the number of bytes between each array layer of an image.

- **depthPitch** describes the number of bytes between each slice of 3D image.

If the image is **linear**, then **rowPitch**, **arrayPitch** and **depthPitch** describe the layout of the image subresource in linear memory. For uncompressed formats, **rowPitch** is the number of bytes between texels with the same x coordinate in adjacent rows (y coordinates differ by one). **arrayPitch** is the number of bytes between texels with the same x and y coordinate in adjacent array layers of the image (array layer values differ by one). **depthPitch** is the number of bytes between texels with the same x and y coordinate in adjacent slices of a 3D image (z coordinates differ by one). Expressed as an addressing formula, the starting byte of a texel in the image subresource has address:

```
// (x,y,z,layer) are in texel coordinates
address(x,y,z,layer) = layer*arrayPitch + z*depthPitch + y*rowPitch + x*elementSize +
offset
```

For compressed formats, the **rowPitch** is the number of bytes between compressed texel blocks in adjacent rows. **arrayPitch** is the number of bytes between compressed texel blocks in adjacent array layers. **depthPitch** is the number of bytes between compressed texel blocks in adjacent slices of a 3D image.

```
// (x,y,z,layer) are in compressed texel block coordinates
address(x,y,z,layer) = layer*arrayPitch + z*depthPitch + y*rowPitch + x
*compressedTexelBlockSize + offset;
```

The value of **arrayPitch** is undefined for images that were not created as arrays. **depthPitch** is defined only for 3D images.

If the image has a color format , then the **aspectMask** member of **VkImageSubresource** **must** be **VK_IMAGE_ASPECT_COLOR_BIT**.

If the image has a depth/stencil format , then **aspectMask** **must** be either **VK_IMAGE_ASPECT_DEPTH_BIT** or **VK_IMAGE_ASPECT_STENCIL_BIT**. On implementations that store depth and stencil aspects separately, querying each of these image subresource layouts will return a different **offset** and **size** representing the region of memory used for that aspect. On implementations that store depth and stencil aspects interleaved, the same **offset** and **size** are returned and represent the interleaved memory allocation.

To destroy an image, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyImage(
    VkDevice          device,
    VkImage           image,
    const VkAllocationCallbacks* pAllocator);
```

- **device** is the logical device that destroys the image.
- **image** is the image to destroy.

- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyImage-image-01000
All submitted commands that refer to **image**, either directly or via a **VkImageView**, **must** have completed execution
- VUID-vkDestroyImage-image-01001
If **VkAllocationCallbacks** were provided when **image** was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyImage-image-01002
If no **VkAllocationCallbacks** were provided when **image** was created, **pAllocator** **must** be **NULL**

Valid Usage (Implicit)

- VUID-vkDestroyImage-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkDestroyImage-image-parameter
If **image** is not **VK_NULL_HANDLE**, **image** **must** be a valid **VkImage** handle
- VUID-vkDestroyImage-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid **VkAllocationCallbacks** structure
- VUID-vkDestroyImage-image-parent
If **image** is a valid handle, it **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **image** **must** be externally synchronized

12.3.1. Image Format Features

Valid uses of a **VkImage** **may** depend on the image's *format features*, defined below. Such constraints are documented in the affected valid usage statement.

- If the image was created with **VK_IMAGE_TILING_LINEAR**, then its set of *format features* is the value of **VkFormatProperties::linearTilingFeatures** found by calling **vkGetPhysicalDeviceFormatProperties** on the same **format** as **VkImageCreateInfo::format**.
- If the image was created with **VK_IMAGE_TILING_OPTIMAL**, then its set of *format features* is the value of **VkFormatProperties::optimalTilingFeatures** found by calling **vkGetPhysicalDeviceFormatProperties** on the same **format** as **VkImageCreateInfo::format**.

12.3.2. Image Mip Level Sizing

A *complete mipmap chain* is the full set of mip levels, from the largest mip level provided, down to the *minimum mip level size*.

Conventional Images

For conventional images, the dimensions of each successive mip level, $n+1$, are:

$$\text{width}_{n+1} = \max(\lceil \text{width}_n / 2 \rceil, 1)$$

$$\text{height}_{n+1} = \max(\lceil \text{height}_n / 2 \rceil, 1)$$

$$\text{depth}_{n+1} = \max(\lceil \text{depth}_n / 2 \rceil, 1)$$

where width_n , height_n , and depth_n are the dimensions of the next larger mip level, n .

The minimum mip level size is:

- 1 for one-dimensional images,
- 1x1 for two-dimensional images, and
- 1x1x1 for three-dimensional images.

The number of levels in a complete mipmap chain is:

$$\lceil \log_2(\max(\text{width}_0, \text{height}_0, \text{depth}_0)) \rceil + 1$$

where width_0 , height_0 , and depth_0 are the dimensions of the largest (most detailed) mip level, 0.

12.4. Image Layouts

Images are stored in implementation-dependent opaque layouts in memory. Each layout has limitations on what kinds of operations are supported for image subresources using the layout. At any given time, the data representing an image subresource in memory exists in a particular layout which is determined by the most recent layout transition that was performed on that image subresource. Applications have control over which layout each image subresource uses, and **can** transition an image subresource from one layout to another. Transitions **can** happen with an image memory barrier, included as part of a [vkCmdPipelineBarrier](#) or a [vkCmdWaitEvents](#) command buffer command (see [Image Memory Barriers](#)), or as part of a subpass dependency within a render pass (see [VkSubpassDependency](#)).

Image layout is per-image subresource. Separate image subresources of the same image **can** be in different layouts at the same time, with the exception that depth and stencil aspects of a given image subresource **can** only be in different layouts if the [separateDepthStencilLayouts](#) feature is enabled.

NOTE

Each layout **may** offer optimal performance for a specific usage of image memory. For example, an image with a layout of `VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL` **may** provide optimal performance for use as a color attachment, but be unsupported for use in transfer commands. Applications **can** transition an image subresource from one layout to another in order to achieve optimal performance when the image subresource is used for multiple kinds of operations. After initialization, applications need not use any layout other than the general layout, though this **may** produce suboptimal performance on some implementations.

Upon creation, all image subresources of an image are initially in the same layout, where that layout is selected by the `VkImageCreateInfo::initialLayout` member. The `initialLayout` **must** be either `VK_IMAGE_LAYOUT_UNDEFINED` or `VK_IMAGE_LAYOUT_PREINITIALIZED`. If it is `VK_IMAGE_LAYOUT_PREINITIALIZED`, then the image data **can** be preinitialized by the host while using this layout, and the transition away from this layout will preserve that data. If it is `VK_IMAGE_LAYOUT_UNDEFINED`, then the contents of the data are considered to be undefined, and the transition away from this layout is not guaranteed to preserve that data. For either of these initial layouts, any image subresources **must** be transitioned to another layout before they are accessed by the device.

Host access to image memory is only well-defined for [linear](#) images and for image subresources of those images which are currently in either the `VK_IMAGE_LAYOUT_PREINITIALIZED` or `VK_IMAGE_LAYOUT_GENERAL` layout. Calling `vkGetImageSubresourceLayout` for a linear image returns a subresource layout mapping that is valid for either of those image layouts.

The set of image layouts consists of:

```
// Provided by VK_VERSION_1_0
typedef enum VkImageLayout {
    VK_IMAGE_LAYOUT_UNDEFINED = 0,
    VK_IMAGE_LAYOUT_GENERAL = 1,
    VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL = 2,
    VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL = 3,
    VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL = 4,
    VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL = 5,
    VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL = 6,
    VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL = 7,
    VK_IMAGE_LAYOUT_PREINITIALIZED = 8,
} VkImageLayout;
```

The type(s) of device access supported by each layout are:

- `VK_IMAGE_LAYOUT_UNDEFINED` specifies that the layout is unknown. Image memory **cannot** be transitioned into this layout. This layout **can** be used as the `initialLayout` member of `VkImageCreateInfo`. This layout **can** be used in place of the current image layout in a layout transition, but doing so will cause the contents of the image's memory to be undefined.
- `VK_IMAGE_LAYOUT_PREINITIALIZED` specifies that an image's memory is in a defined layout and **can** be populated by data, but that it has not yet been initialized by the driver. Image memory **cannot** be transitioned into this layout. This layout **can** be used as the `initialLayout` member of

[VkImageCreateInfo](#). This layout is intended to be used as the initial layout for an image whose contents are written by the host, and hence the data **can** be written to memory immediately, without first executing a layout transition. Currently, [VK_IMAGE_LAYOUT_PREINITIALIZED](#) is only useful with [linear](#) images because there is not a standard layout defined for [VK_IMAGE_TILING_OPTIMAL](#) images.

- [VK_IMAGE_LAYOUT_GENERAL](#) supports all types of device access.
- [VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL](#) **must** only be used as a color or resolve attachment in a [VkFramebuffer](#). This layout is valid only for image subresources of images created with the [VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT](#) usage bit enabled.
- [VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL](#) specifies a layout for both the depth and stencil aspects of a depth/stencil format image allowing read and write access as a depth/stencil attachment.
- [VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL](#) specifies a layout for both the depth and stencil aspects of a depth/stencil format image allowing read only access as a depth/stencil attachment or in shaders as a sampled image, combined image/sampler, or input attachment.
- [VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL](#) specifies a layout allowing read-only access in a shader as a sampled image, combined image/sampler, or input attachment. This layout is valid only for image subresources of images created with the [VK_IMAGE_USAGE_SAMPLED_BIT](#) or [VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT](#) usage bits enabled.
- [VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL](#) **must** only be used as a source image of a transfer command (see the definition of [VK_PIPELINE_STAGE_TRANSFER_BIT](#)). This layout is valid only for image subresources of images created with the [VK_IMAGE_USAGE_TRANSFER_SRC_BIT](#) usage bit enabled.
- [VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL](#) **must** only be used as a destination image of a transfer command. This layout is valid only for image subresources of images created with the [VK_IMAGE_USAGE_TRANSFER_DST_BIT](#) usage bit enabled.

The layout of each image subresource is not a state of the image subresource itself, but is rather a property of how the data in memory is organized, and thus for each mechanism of accessing an image in the API the application **must** specify a parameter or structure member that indicates which image layout the image subresource(s) are considered to be in when the image will be accessed. For transfer commands, this is a parameter to the command (see [Clear Commands](#) and [Copy Commands](#)). For use as a framebuffer attachment, this is a member in the substructures of the [VkRenderPassCreateInfo](#) (see [Render Pass](#)). For use in a descriptor set, this is a member in the [VkDescriptorImageInfo](#) structure (see [Descriptor Set Updates](#)).

12.4.1. Image Layout Matching Rules

At the time that any command buffer command accessing an image executes on any queue, the layouts of the image subresources that are accessed **must** all match exactly the layout specified via the API controlling those accesses.

When performing a layout transition on an image subresource, the old layout value **must** either equal the current layout of the image subresource (at the time the transition executes), or else be [VK_IMAGE_LAYOUT_UNDEFINED](#) (implying that the contents of the image subresource need not be preserved). The new layout used in a transition **must** not be [VK_IMAGE_LAYOUT_UNDEFINED](#) or

VK_IMAGE_LAYOUT_PREINITIALIZED.

12.5. Image Views

Image objects are not directly accessed by pipeline shaders for reading or writing image data. Instead, *image views* representing contiguous ranges of the image subresources and containing additional metadata are used for that purpose. Views **must** be created on images of compatible types, and **must** represent a valid subset of image subresources.

Image views are represented by `VkImageView` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkImageView)
```

`VK_REMAINING_ARRAY_LAYERS` is a special constant value used for image views to indicate that all remaining array layers in an image after the base layer should be included in the view.

```
#define VK_REMAINING_ARRAY_LAYERS        (~0U)
```

`VK_REMAINING_MIP_LEVELS` is a special constant value used for image views to indicate that all remaining mipmap levels in an image after the base level should be included in the view.

```
#define VK_REMAINING_MIP_LEVELS          (~0U)
```

The types of image views that **can** be created are:

```
// Provided by VK_VERSION_1_0
typedef enum VkImageViewType {
    VK_IMAGE_VIEW_TYPE_1D = 0,
    VK_IMAGE_VIEW_TYPE_2D = 1,
    VK_IMAGE_VIEW_TYPE_3D = 2,
    VK_IMAGE_VIEW_TYPE_CUBE = 3,
    VK_IMAGE_VIEW_TYPE_1D_ARRAY = 4,
    VK_IMAGE_VIEW_TYPE_2D_ARRAY = 5,
    VK_IMAGE_VIEW_TYPE_CUBE_ARRAY = 6,
} VkImageViewType;
```

To create an image view, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateImageView(
    VkDevice                                device,
    const VkImageViewCreateInfo*            pCreateInfo,
    const VkAllocationCallbacks*           pAllocator,
    VkImageView*                            pView);
```

- **device** is the logical device that creates the image view.
- **pCreateInfo** is a pointer to a **VkImageViewCreateInfo** structure containing parameters to be used to create the image view.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.
- **pView** is a pointer to a **VkImageView** handle in which the resulting image view object is returned.

Valid Usage

- VUID-vkCreateImageView-device-09667
device must support at least one queue family with one of the **VK_QUEUE_COMPUTE_BIT**, or **VK_QUEUE_GRAPHICS_BIT** capabilities
- VUID-vkCreateImageView-image-09179
VkImageViewCreateInfo::image must have been created from **device**

Valid Usage (Implicit)

- VUID-vkCreateImageView-device-parameter
device must be a valid **VkDevice** handle
- VUID-vkCreateImageView-pCreateInfo-parameter
pCreateInfo must be a valid pointer to a valid **VkImageViewCreateInfo** structure
- VUID-vkCreateImageView-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator must** be a valid pointer to a valid **VkAllocationCallbacks** structure
- VUID-vkCreateImageView-pView-parameter
pView must be a valid pointer to a **VkImageView** handle

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_OUT_OF_DEVICE_MEMORY**

The `VkImageViewCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageViewCreateInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkImageViewCreateFlags flags;
    VkImage              image;
    VkImageViewType       viewType;
    VkFormat              format;
    VkComponentMapping     components;
    VkImageSubresourceRange subresourceRange;
} VkImageViewCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkImageViewCreateFlagBits` specifying additional parameters of the image view.
- `image` is a `VkImage` on which the view will be created.
- `viewType` is a `VkImageViewType` value specifying the type of the image view.
- `format` is a `VkFormat` specifying the format and type used to interpret texel blocks of the image.
- `components` is a `VkComponentMapping` structure specifying a remapping of color components (or of depth or stencil components after they have been converted into color components).
- `subresourceRange` is a `VkImageSubresourceRange` structure selecting the set of mipmap levels and array layers to be accessible to the view.

Some of the `image` creation parameters are inherited by the view. In particular, image view creation inherits the implicit parameter `usage` specifying the allowed usages of the image view that, by default, takes the value of the corresponding `usage` parameter specified in `VkImageCreateInfo` at image creation time.

If `image` was created with the `VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT` flag, `format` **can** be different from the image's format, but if they are not equal they **must** be *compatible*. Image format compatibility is defined in the [Format Compatibility Classes](#) section. Views of compatible formats will have the same mapping between texel coordinates and memory locations irrespective of the `format`, with only the interpretation of the bit pattern changing.

NOTE

Values intended to be used with one view format **may** not be exactly preserved when written or read through a different format. For example, an integer value that happens to have the bit pattern of a floating-point denorm or NaN **may** be flushed or canonicalized when written or read through a view with a floating-point format. Similarly, a value written through a signed normalized format that has a bit pattern exactly equal to -2^b **may** be changed to $-2^b + 1$ as described in [Conversion from Normalized Fixed-Point to Floating-Point](#).

The `VkComponentMapping` `components` member describes a remapping from components of the image to components of the vector returned by shader image instructions. This remapping **must** be the identity swizzle for storage image descriptors, input attachment descriptors, and framebuffer attachments.

Table 7. Image Type and Image View Type Compatibility Requirements

Image View Type	Compatible Image Types
VK_IMAGE_VIEW_TYPE_1D	VK_IMAGE_TYPE_1D
VK_IMAGE_VIEW_TYPE_1D_ARRAY	VK_IMAGE_TYPE_1D
VK_IMAGE_VIEW_TYPE_2D	VK_IMAGE_TYPE_2D
VK_IMAGE_VIEW_TYPE_2D_ARRAY	VK_IMAGE_TYPE_2D
VK_IMAGE_VIEW_TYPE_CUBE	VK_IMAGE_TYPE_2D
VK_IMAGE_VIEW_TYPE_CUBE_ARRAY	VK_IMAGE_TYPE_2D
VK_IMAGE_VIEW_TYPE_3D	VK_IMAGE_TYPE_3D

Valid Usage

- VUID-VkImageViewCreateInfo-image-01003
If `image` was not created with `VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT` then `viewType` **must** not be `VK_IMAGE_VIEW_TYPE_CUBE` or `VK_IMAGE_VIEW_TYPE_CUBE_ARRAY`
- VUID-VkImageViewCreateInfo-viewType-01004
If the `imageCubeArray` feature is not enabled, `viewType` **must** not be `VK_IMAGE_VIEW_TYPE_CUBE_ARRAY`
- VUID-VkImageViewCreateInfo-image-04441
`image` **must** have been created with a `usage` value containing at least one of the usages defined in the [valid image usage](#) list for image views
- VUID-VkImageViewCreateInfo-None-02273
The [format features](#) of the resultant image view **must** contain at least one bit
- VUID-VkImageViewCreateInfo-usage-02274
If `usage` contains `VK_IMAGE_USAGE_SAMPLED_BIT`, then the [format features](#) of the resultant image view **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT`
- VUID-VkImageViewCreateInfo-usage-02275
If `usage` contains `VK_IMAGE_USAGE_STORAGE_BIT`, then the image view's [format features](#) **must** contain `VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT`
- VUID-VkImageViewCreateInfo-usage-02276
If `usage` contains `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT`, then the image view's [format features](#) **must** contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT`
- VUID-VkImageViewCreateInfo-usage-02277
If `usage` contains `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`, then the image view's [format features](#) **must** contain `VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT`
- VUID-VkImageViewCreateInfo-usage-08932
If `usage` contains `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT`,

then the image view's **format features** **must** contain at least one of
VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT or
VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT

- VUID-VkImageViewCreateInfo-subresourceRange-01478
subresourceRange.baseMipLevel **must** be less than the **mipLevels** specified in **VkImageCreateInfo** when **image** was created
- VUID-VkImageViewCreateInfo-subresourceRange-01718
If **subresourceRange.levelCount** is not **VK_REMAINING_MIP_LEVELS**, **subresourceRange.baseMipLevel + subresourceRange.levelCount** **must** be less than or equal to the **mipLevels** specified in **VkImageCreateInfo** when **image** was created
- VUID-VkImageViewCreateInfo-subresourceRange-01480
subresourceRange.baseArrayLayer **must** be less than the **arrayLayers** specified in **VkImageCreateInfo** when **image** was created
- VUID-VkImageViewCreateInfo-subresourceRange-01719
If **subresourceRange.layerCount** is not **VK_REMAINING_ARRAY_LAYERS**, **subresourceRange.baseArrayLayer + subresourceRange.layerCount** **must** be less than or equal to the **arrayLayers** specified in **VkImageCreateInfo** when **image** was created
- VUID-VkImageViewCreateInfo-image-01018
If **image** was created with the **VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT** flag, **format** **must** be compatible with the **format** used to create **image**, as defined in **Format Compatibility Classes**
- VUID-VkImageViewCreateInfo-image-01019
If **image** was not created with the **VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT** flag, **format** **must** be identical to the **format** used to create **image**
- VUID-VkImageViewCreateInfo-image-01020
If **image** is non-sparse then it **must** be bound completely and contiguously to a single **VkDeviceMemory** object
- VUID-VkImageViewCreateInfo-subResourceRange-01021
viewType **must** be compatible with the type of **image** as shown in the **view type compatibility table**
- VUID-VkImageViewCreateInfo-imageViewType-04973
If **viewType** is **VK_IMAGE_VIEW_TYPE_1D**, **VK_IMAGE_VIEW_TYPE_2D**, or **VK_IMAGE_VIEW_TYPE_3D**; and **subresourceRange.layerCount** is not **VK_REMAINING_ARRAY_LAYERS**, then **subresourceRange.layerCount** **must** be 1
- VUID-VkImageViewCreateInfo-imageViewType-04974
If **viewType** is **VK_IMAGE_VIEW_TYPE_1D**, **VK_IMAGE_VIEW_TYPE_2D**, or **VK_IMAGE_VIEW_TYPE_3D**; and **subresourceRange.layerCount** is **VK_REMAINING_ARRAY_LAYERS**, then the remaining number of layers **must** be 1
- VUID-VkImageViewCreateInfo-viewType-02960
If **viewType** is **VK_IMAGE_VIEW_TYPE_CUBE** and **subresourceRange.layerCount** is not **VK_REMAINING_ARRAY_LAYERS**, **subresourceRange.layerCount** **must** be 6
- VUID-VkImageViewCreateInfo-viewType-02961
If **viewType** is **VK_IMAGE_VIEW_TYPE_CUBE_ARRAY** and **subresourceRange.layerCount** is not

`VK_REMAINING_ARRAY_LAYERS`, `subresourceRange.layerCount` **must** be a multiple of 6

- VUID-VkImageViewCreateInfo-viewType-02962
If `viewType` is `VK_IMAGE_VIEW_TYPE_CUBE` and `subresourceRange.layerCount` is `VK_REMAINING_ARRAY_LAYERS`, the remaining number of layers **must** be 6
- VUID-VkImageViewCreateInfo-viewType-02963
If `viewType` is `VK_IMAGE_VIEW_TYPE_CUBE_ARRAY` and `subresourceRange.layerCount` is `VK_REMAINING_ARRAY_LAYERS`, the remaining number of layers **must** be a multiple of 6
- VUID-VkImageViewCreateInfo-subresourceRange-09594
`subresourceRange.aspectMask` **must** be valid for the `format` the `image` was created with

Valid Usage (Implicit)

- VUID-VkImageViewCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_IMAGE_VIEW_CREATE_INFO`
- VUID-VkImageViewCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkImageViewCreateInfo-flags-zerobitmask
`flags` **must** be 0
- VUID-VkImageViewCreateInfo-image-parameter
`image` **must** be a valid `VkImage` handle
- VUID-VkImageViewCreateInfo-viewType-parameter
`viewType` **must** be a valid `VkImageViewType` value
- VUID-VkImageViewCreateInfo-format-parameter
`format` **must** be a valid `VkFormat` value
- VUID-VkImageViewCreateInfo-components-parameter
`components` **must** be a valid `VkComponentMapping` structure
- VUID-VkImageViewCreateInfo-subresourceRange-parameter
`subresourceRange` **must** be a valid `VkImageSubresourceRange` structure

Bits which **can** be set in `VkImageViewCreateInfo::flags`, specifying additional parameters of an image view, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkImageViewCreateFlagBits {
} VkImageViewCreateFlagBits;
```

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkImageViewCreateFlags;
```

`VkImageViewCreateFlags` is a bitmask type for setting a mask of zero or more

VkImageViewCreateFlagBits.

The `VkImageSubresourceRange` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageSubresourceRange {
    VkImageAspectFlags    aspectMask;
    uint32_t              baseMipLevel;
    uint32_t              levelCount;
    uint32_t              baseArrayLayer;
    uint32_t              layerCount;
} VkImageSubresourceRange;
```

- `aspectMask` is a bitmask of `VkImageAspectFlagBits` specifying which aspect(s) of the image are included in the view.
- `baseMipLevel` is the first mipmap level accessible to the view.
- `levelCount` is the number of mipmap levels (starting from `baseMipLevel`) accessible to the view.
- `baseArrayLayer` is the first array layer accessible to the view.
- `layerCount` is the number of array layers (starting from `baseArrayLayer`) accessible to the view.

The number of mipmap levels and array layers **must** be a subset of the image subresources in the image. If an application wants to use all mip levels or layers in an image after the `baseMipLevel` or `baseArrayLayer`, it **can** set `levelCount` and `layerCount` to the special values `VK_REMAINING_MIP_LEVELS` and `VK_REMAINING_ARRAY_LAYERS` without knowing the exact number of mip levels or layers.

For cube and cube array image views, the layers of the image view starting at `baseArrayLayer` correspond to faces in the order +X, -X, +Y, -Y, +Z, -Z. For cube arrays, each set of six sequential layers is a single cube, so the number of cube maps in a cube map array view is `layerCount / 6`, and image array layer (`baseArrayLayer + i`) is face index (`i mod 6`) of cube `i / 6`. If the number of layers in the view, whether set explicitly in `layerCount` or implied by `VK_REMAINING_ARRAY_LAYERS`, is not a multiple of 6, the last cube map in the array **must** not be accessed.

`aspectMask` **must** be only `VK_IMAGE_ASPECT_COLOR_BIT`, `VK_IMAGE_ASPECT_DEPTH_BIT` or `VK_IMAGE_ASPECT_STENCIL_BIT` if `format` is a color, depth-only or stencil-only format, respectively. If using a depth/stencil format with both depth and stencil components, `aspectMask` **must** include at least one of `VK_IMAGE_ASPECT_DEPTH_BIT` and `VK_IMAGE_ASPECT_STENCIL_BIT`, and **can** include both.

When using an image view of a depth/stencil image to populate a descriptor set (e.g. for sampling in the shader, or for use as an input attachment), the `aspectMask` **must** only include one bit, which selects whether the image view is used for depth reads (i.e. using a floating-point sampler or input attachment in the shader) or stencil reads (i.e. using an unsigned integer sampler or input attachment in the shader). When an image view of a depth/stencil image is used as a depth/stencil framebuffer attachment, the `aspectMask` is ignored and both depth and stencil image subresources are used.

Valid Usage

- VUID-VkImageSubresourceRange-levelCount-01720
If `levelCount` is not `VK_REMAINING_MIP_LEVELS`, it **must** be greater than 0
- VUID-VkImageSubresourceRange-layerCount-01721
If `layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, it **must** be greater than 0

Valid Usage (Implicit)

- VUID-VkImageSubresourceRange-aspectMask-parameter
`aspectMask` **must** be a valid combination of `VkImageAspectFlagBits` values
- VUID-VkImageSubresourceRange-aspectMask-requiredbitmask
`aspectMask` **must** not be 0

Bits which **can** be set in an aspect mask to specify aspects of an image for purposes such as identifying a subresource, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkImageAspectFlagBits {
    VK_IMAGE_ASPECT_COLOR_BIT = 0x00000001,
    VK_IMAGE_ASPECT_DEPTH_BIT = 0x00000002,
    VK_IMAGE_ASPECT_STENCIL_BIT = 0x00000004,
    VK_IMAGE_ASPECT_METADATA_BIT = 0x00000008,
} VkImageAspectFlagBits;
```

- `VK_IMAGE_ASPECT_COLOR_BIT` specifies the color aspect.
- `VK_IMAGE_ASPECT_DEPTH_BIT` specifies the depth aspect.
- `VK_IMAGE_ASPECT_STENCIL_BIT` specifies the stencil aspect.
- `VK_IMAGE_ASPECT_METADATA_BIT` specifies the metadata aspect used for `sparse resource` operations.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkImageAspectFlags;
```

`VkImageAspectFlags` is a bitmask type for setting a mask of zero or more `VkImageAspectFlagBits`.

The `VkComponentMapping` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkComponentMapping {
    VkComponentSwizzle    r;
    VkComponentSwizzle    g;
    VkComponentSwizzle    b;
    VkComponentSwizzle    a;
} VkComponentMapping;
```

- **r** is a [VkComponentSwizzle](#) specifying the component value placed in the R component of the output vector.
- **g** is a [VkComponentSwizzle](#) specifying the component value placed in the G component of the output vector.
- **b** is a [VkComponentSwizzle](#) specifying the component value placed in the B component of the output vector.
- **a** is a [VkComponentSwizzle](#) specifying the component value placed in the A component of the output vector.

Valid Usage (Implicit)

- VUID-VkComponentMapping-r-parameter
r must be a valid [VkComponentSwizzle](#) value
- VUID-VkComponentMapping-g-parameter
g must be a valid [VkComponentSwizzle](#) value
- VUID-VkComponentMapping-b-parameter
b must be a valid [VkComponentSwizzle](#) value
- VUID-VkComponentMapping-a-parameter
a must be a valid [VkComponentSwizzle](#) value

Possible values of the members of [VkComponentMapping](#), specifying the component values placed in each component of the output vector, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkComponentSwizzle {
    VK_COMPONENT_SWIZZLE_IDENTITY = 0,
    VK_COMPONENT_SWIZZLE_ZERO = 1,
    VK_COMPONENT_SWIZZLE_ONE = 2,
    VK_COMPONENT_SWIZZLE_R = 3,
    VK_COMPONENT_SWIZZLE_G = 4,
    VK_COMPONENT_SWIZZLE_B = 5,
    VK_COMPONENT_SWIZZLE_A = 6,
} VkComponentSwizzle;
```

- **VK_COMPONENT_SWIZZLE_IDENTITY** specifies that the component is set to the identity swizzle.

- `VK_COMPONENT_SWIZZLE_ZERO` specifies that the component is set to zero.
- `VK_COMPONENT_SWIZZLE_ONE` specifies that the component is set to either 1 or 1.0, depending on whether the type of the image view format is integer or floating-point respectively, as determined by the [Format Definition](#) section for each `VkFormat`.
- `VK_COMPONENT_SWIZZLE_R` specifies that the component is set to the value of the R component of the image.
- `VK_COMPONENT_SWIZZLE_G` specifies that the component is set to the value of the G component of the image.
- `VK_COMPONENT_SWIZZLE_B` specifies that the component is set to the value of the B component of the image.
- `VK_COMPONENT_SWIZZLE_A` specifies that the component is set to the value of the A component of the image.

Setting the identity swizzle on a component is equivalent to setting the identity mapping on that component. That is:

Table 8. Component Mappings Equivalent To `VK_COMPONENT_SWIZZLE_IDENTITY`

Component	Identity Mapping
<code>components.r</code>	<code>VK_COMPONENT_SWIZZLE_R</code>
<code>components.g</code>	<code>VK_COMPONENT_SWIZZLE_G</code>
<code>components.b</code>	<code>VK_COMPONENT_SWIZZLE_B</code>
<code>components.a</code>	<code>VK_COMPONENT_SWIZZLE_A</code>

To destroy an image view, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyImageView(
    VkDevice          device,
    VkImageView       imageView,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that destroys the image view.
- `imageView` is the image view to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyImageView-imageView-01026
All submitted commands that refer to `imageView` **must** have completed execution
- VUID-vkDestroyImageView-imageView-01027
If `VkAllocationCallbacks` were provided when `imageView` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyImageView-imageView-01028

If no `VkAllocationCallbacks` were provided when `imageView` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- `VOID-vkDestroyImageView-device-parameter`
`device` **must** be a valid `VkDevice` handle
- `VOID-vkDestroyImageView-imageView-parameter`
If `imageView` is not `VK_NULL_HANDLE`, `imageView` **must** be a valid `VkImageView` handle
- `VOID-vkDestroyImageView-pAllocator-parameter`
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- `VOID-vkDestroyImageView-imageView-parent`
If `imageView` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `imageView` **must** be externally synchronized

12.5.1. Image View Format Features

Valid uses of a `VkImageView` **may** depend on the image view's *format features*, defined below. Such constraints are documented in the affected valid usage statement.

- If `VkImageViewCreateInfo::image` was created with `VK_IMAGE_TILING_LINEAR`, then the image view's set of *format features* is the value of `VkFormatProperties::linearTilingFeatures` found by calling `vkGetPhysicalDeviceFormatProperties` on the same `format` as `VkImageViewCreateInfo::format`.
- If `VkImageViewCreateInfo::image` was created with `VK_IMAGE_TILING_OPTIMAL`, then the image view's set of *format features* is the value of `VkFormatProperties::optimalTilingFeatures` found by calling `vkGetPhysicalDeviceFormatProperties` on the same `format` as `VkImageViewCreateInfo::format`.

12.6. Resource Memory Association

Resources are initially created as *virtual allocations* with no backing memory. Device memory is allocated separately (see [Device Memory](#)) and then associated with the resource. This association is done differently for sparse and non-sparse resources.

Resources created with any of the sparse creation flags are considered sparse resources. Resources created without these flags are non-sparse. The details on resource memory association for sparse resources is described in [Sparse Resources](#).

Non-sparse resources **must** be bound completely and contiguously to a single `VkDeviceMemory` object before the resource is passed as a parameter to any of the following operations:

- creating image or buffer views
- updating descriptor sets
- recording commands in a command buffer

Once bound, the memory binding is immutable for the lifetime of the resource.

To determine the memory requirements for a buffer resource, call:

```
// Provided by VK_VERSION_1_0
void vkGetBufferMemoryRequirements(
    VkDevice          device,
    VkBuffer          buffer,
    VkMemoryRequirements* pMemoryRequirements);
```

- `device` is the logical device that owns the buffer.
- `buffer` is the buffer to query.
- `pMemoryRequirements` is a pointer to a `VkMemoryRequirements` structure in which the memory requirements of the buffer object are returned.

Valid Usage (Implicit)

- VUID-vkGetBufferMemoryRequirements-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetBufferMemoryRequirements-buffer-parameter
`buffer` **must** be a valid `VkBuffer` handle
- VUID-vkGetBufferMemoryRequirements-pMemoryRequirements-parameter
`pMemoryRequirements` **must** be a valid pointer to a `VkMemoryRequirements` structure
- VUID-vkGetBufferMemoryRequirements-buffer-parent
`buffer` **must** have been created, allocated, or retrieved from `device`

To determine the memory requirements for an image resource, call:

```
// Provided by VK_VERSION_1_0
void vkGetImageMemoryRequirements(
    VkDevice          device,
    VkImage          image,
    VkMemoryRequirements* pMemoryRequirements);
```

- `device` is the logical device that owns the image.
- `image` is the image to query.

- `pMemoryRequirements` is a pointer to a `VkMemoryRequirements` structure in which the memory requirements of the image object are returned.

Valid Usage

Valid Usage (Implicit)

- VUID-vkGetImageMemoryRequirements-device-parameter `device` **must** be a valid `VkDevice` handle
- VUID-vkGetImageMemoryRequirements-image-parameter `image` **must** be a valid `VkImage` handle
- VUID-vkGetImageMemoryRequirements-pMemoryRequirements-parameter `pMemoryRequirements` **must** be a valid pointer to a `VkMemoryRequirements` structure
- VUID-vkGetImageMemoryRequirements-image-parent `image` **must** have been created, allocated, or retrieved from `device`

The `VkMemoryRequirements` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkMemoryRequirements {
    VkDeviceSize    size;
    VkDeviceSize    alignment;
    uint32_t        memoryTypeBits;
} VkMemoryRequirements;
```

- `size` is the size, in bytes, of the memory allocation **required** for the resource.
- `alignment` is the alignment, in bytes, of the offset within the allocation **required** for the resource.
- `memoryTypeBits` is a bitmask and contains one bit set for every supported memory type for the resource. Bit `i` is set if and only if the memory type `i` in the `VkPhysicalDeviceMemoryProperties` structure for the physical device is supported for the resource.

The implementation guarantees certain properties about the memory requirements returned by `vkGetBufferMemoryRequirements` and `vkGetImageMemoryRequirements`:

- The `memoryTypeBits` member always contains at least one bit set.
- If `buffer` is a `VkBuffer` not created with the `VK_BUFFER_CREATE_SPARSE_BINDING_BIT` or `VK_BUFFER_CREATE_PROTECTED_BIT` bits set, or if `image` is a `linear` image that was not created with the `VK_IMAGE_CREATE_PROTECTED_BIT` bit set, then the `memoryTypeBits` member always contains at least one bit set corresponding to a `VkMemoryType` with a `propertyFlags` that has both the `VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT` bit and the `VK_MEMORY_PROPERTY_HOST_COHERENT_BIT` bit set. In other words, mappable coherent memory **can** always be attached to these objects.

- The `memoryTypeBits` member always contains at least one bit set corresponding to a `VkMemoryType` with a `propertyFlags` that has the `VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT` bit set.
- The `memoryTypeBits` member is identical for all `VkBuffer` objects created with the same value for the `flags` and `usage` members in the `VkBufferCreateInfo` structure passed to `vkCreateBuffer`. Further, if `usage1` and `usage2` of type `VkBufferUsageFlags` are such that the bits set in `usage2` are a subset of the bits set in `usage1`, and they have the same `flags`, then the bits set in `memoryTypeBits` returned for `usage1` **must** be a subset of the bits set in `memoryTypeBits` returned for `usage2`, for all values of `flags`.
- The `alignment` member is a power of two.
- The `alignment` member is identical for all `VkBuffer` objects created with the same combination of values for the `usage` and `flags` members in the `VkBufferCreateInfo` structure passed to `vkCreateBuffer`.
- The `alignment` member satisfies the buffer descriptor offset alignment requirements associated with the `VkBuffer`'s `usage`:
 - If `usage` included `VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT` or `VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT`, `alignment` **must** be an integer multiple of `VkPhysicalDeviceLimits::minTexelBufferOffsetAlignment`.
 - If `usage` included `VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT`, `alignment` **must** be an integer multiple of `VkPhysicalDeviceLimits::minUniformBufferOffsetAlignment`.
 - If `usage` included `VK_BUFFER_USAGE_STORAGE_BUFFER_BIT`, `alignment` **must** be an integer multiple of `VkPhysicalDeviceLimits::minStorageBufferOffsetAlignment`.
- For images created with a color format, the `memoryTypeBits` member is identical for all `VkImage` objects created with the same combination of values for the `tiling` member, the `VK_IMAGE_CREATE_SPARSE_BINDING_BIT` bit of the `flags` member, and the `VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT` of the `usage` member in the `VkImageCreateInfo` structure passed to `vkCreateImage`.
- For images created with a depth/stencil format, the `memoryTypeBits` member is identical for all `VkImage` objects created with the same combination of values for the `format` member, the `tiling` member, the `VK_IMAGE_CREATE_SPARSE_BINDING_BIT` bit of the `flags` member, and the `VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT` of the `usage` member in the `VkImageCreateInfo` structure passed to `vkCreateImage`.
- If the memory requirements are for a `VkImage`, the `memoryTypeBits` member **must** not refer to a `VkMemoryType` with a `propertyFlags` that has the `VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT` bit set if the `image` did not have `VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT` bit set in the `usage` member of the `VkImageCreateInfo` structure passed to `vkCreateImage`.
- If the memory requirements are for a `VkBuffer`, the `memoryTypeBits` member **must** not refer to a `VkMemoryType` with a `propertyFlags` that has the `VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT` bit set.

NOTE

The implication of this requirement is that lazily allocated memory is disallowed for buffers in all cases.

- The `size` member is identical for all `VkBuffer` objects created with the same combination of creation parameters specified in `VkBufferCreateInfo` and its `pNext` chain.

- The **size** member is identical for all **VkImage** objects created with the same combination of creation parameters specified in **VkImageCreateInfo** and its **pNext** chain.

NOTE

This, however, does not imply that they interpret the contents of the bound memory identically with each other.

To attach memory to a buffer object, call:

```
// Provided by VK_VERSION_1_0
VkResult vkBindBufferMemory(
    VkDevice          device,
    VkBuffer           buffer,
    VkDeviceMemory     memory,
    VkDeviceSize       memoryOffset);
```

- **device** is the logical device that owns the buffer and memory.
- **buffer** is the buffer to be attached to memory.
- **memory** is a **VkDeviceMemory** object describing the device memory to attach.
- **memoryOffset** is the start offset of the region of **memory** which is to be bound to the buffer. The number of bytes returned in the **VkMemoryRequirements::size** member in **memory**, starting from **memoryOffset** bytes, will be bound to the specified buffer.

Valid Usage

- VUID-vkBindBufferMemory-buffer-07459
buffer must not have been bound to a memory object
- VUID-vkBindBufferMemory-buffer-01030
buffer must not have been created with any sparse memory binding flags
- VUID-vkBindBufferMemory-memoryOffset-01031
memoryOffset must be less than the size of **memory**
- VUID-vkBindBufferMemory-memory-01035
memory must have been allocated using one of the memory types allowed in the **memoryTypeBits** member of the **VkMemoryRequirements** structure returned from a call to **vkGetBufferMemoryRequirements** with **buffer**
- VUID-vkBindBufferMemory-memoryOffset-01036
memoryOffset must be an integer multiple of the **alignment** member of the **VkMemoryRequirements** structure returned from a call to **vkGetBufferMemoryRequirements** with **buffer**
- VUID-vkBindBufferMemory-size-01037
The **size** member of the **VkMemoryRequirements** structure returned from a call to **vkGetBufferMemoryRequirements** with **buffer must** be less than or equal to the size of **memory** minus **memoryOffset**

Valid Usage (Implicit)

- VUID-vkBindBufferMemory-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkBindBufferMemory-buffer-parameter
buffer **must** be a valid [VkBuffer](#) handle
- VUID-vkBindBufferMemory-memory-parameter
memory **must** be a valid [VkDeviceMemory](#) handle
- VUID-vkBindBufferMemory-buffer-parent
buffer **must** have been created, allocated, or retrieved from **device**
- VUID-vkBindBufferMemory-memory-parent
memory **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **buffer** **must** be externally synchronized

Return Codes

Success

- [VK_SUCCESS](#)

Failure

- [VK_ERROR_OUT_OF_HOST_MEMORY](#)
- [VK_ERROR_OUT_OF_DEVICE_MEMORY](#)

To attach memory to an image object, call:

```
// Provided by VK_VERSION_1_0
VkResult vkBindImageMemory(
    VkDevice          device,
    VkImage           image,
    VkDeviceMemory    memory,
    VkDeviceSize      memoryOffset);
```

- **device** is the logical device that owns the image and memory.
- **image** is the image.
- **memory** is the [VkDeviceMemory](#) object describing the device memory to attach.
- **memoryOffset** is the start offset of the region of **memory** which is to be bound to the image. The number of bytes returned in the [VkMemoryRequirements::size](#) member in **memory**, starting from **memoryOffset** bytes, will be bound to the specified image.

Valid Usage

- VUID-vkBindImageMemory-image-07460
image **must** not have been bound to a memory object
- VUID-vkBindImageMemory-image-01045
image **must** not have been created with any sparse memory binding flags
- VUID-vkBindImageMemory-memoryOffset-01046
memoryOffset **must** be less than the size of **memory**
- VUID-vkBindImageMemory-memory-01047
memory **must** have been allocated using one of the memory types allowed in the **memoryTypeBits** member of the **VkMemoryRequirements** structure returned from a call to **vkGetImageMemoryRequirements** with **image**
- VUID-vkBindImageMemory-memoryOffset-01048
memoryOffset **must** be an integer multiple of the **alignment** member of the **VkMemoryRequirements** structure returned from a call to **vkGetImageMemoryRequirements** with **image**
- VUID-vkBindImageMemory-size-01049
The difference of the size of **memory** and **memoryOffset** **must** be greater than or equal to the **size** member of the **VkMemoryRequirements** structure returned from a call to **vkGetImageMemoryRequirements** with the same **image**

Valid Usage (Implicit)

- VUID-vkBindImageMemory-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkBindImageMemory-image-parameter
image **must** be a valid **VkImage** handle
- VUID-vkBindImageMemory-memory-parameter
memory **must** be a valid **VkDeviceMemory** handle
- VUID-vkBindImageMemory-image-parent
image **must** have been created, allocated, or retrieved from **device**
- VUID-vkBindImageMemory-memory-parent
memory **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **image** **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

Buffer-Image Granularity

The implementation-dependent limit `bufferImageGranularity` specifies a page-like granularity at which linear and non-linear resources **must** be placed in adjacent memory locations to avoid aliasing. Two resources which do not satisfy this granularity requirement are said to [alias](#). `bufferImageGranularity` is specified in bytes, and **must** be a power of two. Implementations which do not impose a granularity restriction **may** report a `bufferImageGranularity` value of one.

NOTE

Despite its name, `bufferImageGranularity` is really a granularity between “linear” and “non-linear” resources.

Given resourceA at the lower memory offset and resourceB at the higher memory offset in the same `VkDeviceMemory` object, where one resource is linear and the other is non-linear (as defined in the [Glossary](#)), and the following:

```
resourceA.end      = resourceA.memoryOffset + resourceA.size - 1
resourceA.endPage  = resourceA.end & ~(bufferImageGranularity-1)
resourceB.start    = resourceB.memoryOffset
resourceB.startPage = resourceB.start & ~(bufferImageGranularity-1)
```

The following property **must** hold:

```
resourceA.endPage < resourceB.startPage
```

That is, the end of the first resource (A) and the beginning of the second resource (B) **must** be on separate “pages” of size `bufferImageGranularity`. `bufferImageGranularity` **may** be different than the physical page size of the memory heap. This restriction is only needed when a linear resource and a non-linear resource are adjacent in memory and will be used simultaneously. The memory ranges of adjacent resources **can** be closer than `bufferImageGranularity`, provided they meet the [alignment](#) requirement for the objects in question.

Sparse block size in bytes and sparse image and buffer memory alignments **must** all be multiples of the `bufferImageGranularity`. Therefore, memory bound to sparse resources naturally satisfies the `bufferImageGranularity`.

12.7. Resource Sharing Mode

Buffer and image objects are created with a *sharing mode* controlling how they **can** be accessed from queues. The supported sharing modes are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSharingMode {
    VK_SHARING_MODE_EXCLUSIVE = 0,
    VK_SHARING_MODE_CONCURRENT = 1,
} VkSharingMode;
```

- **VK_SHARING_MODE_EXCLUSIVE** specifies that access to any range or image subresource of the object will be exclusive to a single queue family at a time.
- **VK_SHARING_MODE_CONCURRENT** specifies that concurrent access to any range or image subresource of the object from multiple queue families is supported.

NOTE **VK_SHARING_MODE_CONCURRENT** may result in lower performance access to the buffer or image than **VK_SHARING_MODE_EXCLUSIVE**.

Ranges of buffers and image subresources of image objects created using **VK_SHARING_MODE_EXCLUSIVE** **must** only be accessed by queues in the queue family that has *ownership* of the resource. Upon creation, such resources are not owned by any queue family; ownership is implicitly acquired upon first use within a queue. Once a resource using **VK_SHARING_MODE_EXCLUSIVE** is owned by some queue family, the application **must** perform a [queue family ownership transfer](#) to make the memory contents of a range or image subresource accessible to a different queue family.

NOTE Images still require a [layout transition](#) from **VK_IMAGE_LAYOUT_UNDEFINED** or **VK_IMAGE_LAYOUT_PREINITIALIZED** before being used on the first queue.

A queue family **can** take ownership of an image subresource or buffer range of a resource created with **VK_SHARING_MODE_EXCLUSIVE**, without an ownership transfer, in the same way as for a resource that was just created; however, taking ownership in this way has the effect that the contents of the image subresource or buffer range are undefined.

Ranges of buffers and image subresources of image objects created using **VK_SHARING_MODE_CONCURRENT** **must** only be accessed by queues from the queue families specified through the **queueFamilyIndexCount** and **pQueueFamilyIndices** members of the corresponding create info structures.

12.8. Memory Aliasing

A range of a **VkDeviceMemory** allocation is *aliased* if it is bound to multiple resources simultaneously, as described below, via [vkBindImageMemory](#), [vkBindBufferMemory](#), or via [sparse memory bindings](#).

Consider two resources, resource_A and resource_B, bound respectively to memory range_A and range_B.

Let paddedRange_A and paddedRange_B be, respectively, range_A and range_B aligned to [bufferImageGranularity](#). If the resources are both linear or both non-linear (as defined in the [Glossary](#)), then the resources *alias* the memory in the intersection of range_A and range_B . If one resource is linear and the other is non-linear, then the resources *alias* the memory in the intersection of paddedRange_A and paddedRange_B .

Applications **can** alias memory, but use of multiple aliases is subject to several constraints.

NOTE

Memory aliasing **can** be useful to reduce the total device memory footprint of an application, if some large resources are used for disjoint periods of time.

When a [non-linear](#), non-[VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT](#) image is bound to an aliased range, all image subresources of the image *overlap* the range. When a linear image is bound to an aliased range, the image subresources that (according to the image's advertised layout) include bytes from the aliased range *overlap* the range. When a [VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT](#) image has sparse image blocks bound to an aliased range, only image subresources including those sparse image blocks *overlap* the range, and when the memory bound to the image's mip tail overlaps an aliased range all image subresources in the mip tail *overlap* the range.

Buffers, and linear image subresources in either the [VK_IMAGE_LAYOUT_PREINITIALIZED](#) or [VK_IMAGE_LAYOUT_GENERAL](#) layouts, are *host-accessible subresources*. That is, the host has a well-defined addressing scheme to interpret the contents, and thus the layout of the data in memory **can** be consistently interpreted across aliases if each of those aliases is a host-accessible subresource. Non-linear images, and linear image subresources in other layouts, are not host-accessible.

If two aliases are both host-accessible, then they interpret the contents of the memory in consistent ways, and data written to one alias **can** be read by the other alias.

Otherwise, the aliases interpret the contents of the memory differently, and writes via one alias make the contents of memory partially or completely undefined to the other alias. If the first alias is a host-accessible subresource, then the bytes affected are those written by the memory operations according to its addressing scheme. If the first alias is not host-accessible, then the bytes affected are those overlapped by the image subresources that were written. If the second alias is a host-accessible subresource, the affected bytes become undefined. If the second alias is not host-accessible, all sparse image blocks (for sparse partially-resident images) or all image subresources (for non-sparse image and fully resident sparse images) that overlap the affected bytes become undefined.

If any image subresources are made undefined due to writes to an alias, then each of those image subresources **must** have its layout transitioned from [VK_IMAGE_LAYOUT_UNDEFINED](#) to a valid layout before it is used, or from [VK_IMAGE_LAYOUT_PREINITIALIZED](#) if the memory has been written by the host. If any sparse blocks of a sparse image have been made undefined, then only the image subresources containing them **must** be transitioned.

Use of an overlapping range by two aliases **must** be separated by a memory dependency using the appropriate [access types](#) if at least one of those uses performs writes, whether the aliases interpret memory consistently or not. If buffer or image memory barriers are used, the scope of the barrier **must** contain the entire range and/or set of image subresources that overlap.

If two aliasing image views are used in the same framebuffer, then the render pass **must** declare the attachments using the `VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT`, and follow the other rules listed in that section.

NOTE

Memory recycled via an application suballocator (i.e. without freeing and reallocating the memory objects) is not substantially different from memory aliasing. However, a suballocator usually waits on a fence before recycling a region of memory, and signaling a fence involves sufficient implicit dependencies to satisfy all the above requirements.

12.8.1. Resource Memory Overlap

Applications **can** safely access resources concurrently via separate device and host operations as long as the accessed memory locations are guaranteed to not overlap, as defined in [Memory Location](#), and the operation, resource, and access are otherwise independently valid.

Some operations have alignment requirements or access ambiguous [memory locations](#), so the semantics of a particular operation **should** be considered when determining the overlap. Such requirements will be described alongside the operation. Operations between host and device when using non-coherent memory are aligned to `nonCoherentAtomSize`, as defined by [vkFlushMappedMemoryRanges](#) and [vkInvalidateMappedMemoryRanges](#).

NOTE

The intent is that buffers (or linear images) can be accessed concurrently, even when they share cache lines, but otherwise do not access the same memory range. The concept of a device cache line size is not exposed in the memory model.

Chapter 13. Samplers

VkSampler objects represent the state of an image sampler which is used by the implementation to read image data and apply filtering and other transformations for the shader.

Samplers are represented by **VkSampler** handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkSampler)
```

To create a sampler object, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateSampler(
    VkDevice                                device,
    const VkSamplerCreateInfo*              pCreateInfo,
    const VkAllocationCallbacks*           pAllocator,
    VkSampler*                              pSampler);
```

- **device** is the logical device that creates the sampler.
- **pCreateInfo** is a pointer to a **VkSamplerCreateInfo** structure specifying the state of the sampler object.
- **pAllocator** controls host memory allocation as described in the **Memory Allocation** chapter.
- **pSampler** is a pointer to a **VkSampler** handle in which the resulting sampler object is returned.

Valid Usage

- VUID-vkCreateSampler-device-09668
device **must** support at least one queue family with one of the **VK_QUEUE_COMPUTE_BIT** or **VK_QUEUE_GRAPHICS_BIT** capabilities
- VUID-vkCreateSampler-maxSamplerAllocationCount-04110
There **must** be less than **VkPhysicalDeviceLimits::maxSamplerAllocationCount** **VkSampler** objects currently created on the device

Valid Usage (Implicit)

- VUID-vkCreateSampler-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkCreateSampler-pCreateInfo-parameter
pCreateInfo **must** be a valid pointer to a valid **VkSamplerCreateInfo** structure
- VUID-vkCreateSampler-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid

[VkAllocationCallbacks](#) structure

- VUID-vkCreateSampler-pSampler-parameter
[pSampler](#) **must** be a valid pointer to a [VkSampler](#) handle

Return Codes

Success

- [VK_SUCCESS](#)

Failure

- [VK_ERROR_OUT_OF_HOST_MEMORY](#)
- [VK_ERROR_OUT_OF_DEVICE_MEMORY](#)

The [VkSamplerCreateInfo](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSamplerCreateInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkSamplerCreateFlags  flags;
    VkFilter              magFilter;
    VkFilter              minFilter;
    VkSamplerMipmapMode   mipmapMode;
    VkSamplerAddressMode  addressModeU;
    VkSamplerAddressMode  addressModeV;
    VkSamplerAddressMode  addressModeW;
    float                 mipLodBias;
    VkBool32              anisotropyEnable;
    float                 maxAnisotropy;
    VkBool32              compareEnable;
    VkCompareOp           compareOp;
    float                 minLod;
    float                 maxLod;
    VkBorderColor         borderColor;
    VkBool32              unnormalizedCoordinates;
} VkSamplerCreateInfo;
```

- [sType](#) is a [VkStructureType](#) value identifying this structure.
- [pNext](#) is `NULL` or a pointer to a structure extending this structure.
- [flags](#) is a bitmask of [VkSamplerCreateFlagBits](#) describing additional parameters of the sampler.
- [magFilter](#) is a [VkFilter](#) value specifying the magnification filter to apply to lookups.
- [minFilter](#) is a [VkFilter](#) value specifying the minification filter to apply to lookups.
- [mipmapMode](#) is a [VkSamplerMipmapMode](#) value specifying the mipmap filter to apply to lookups.
- [addressModeU](#) is a [VkSamplerAddressMode](#) value specifying the addressing mode for U

coordinates outside [0,1).

- `addressModeV` is a `VkSamplerAddressMode` value specifying the addressing mode for V coordinates outside [0,1).
- `addressModeW` is a `VkSamplerAddressMode` value specifying the addressing mode for W coordinates outside [0,1).
- `mipLodBias` is the bias to be added to mipmap LOD calculation and bias provided by image sampling functions in SPIR-V, as described in the [LOD Operation](#) section.
- `anisotropyEnable` is `VK_TRUE` to enable anisotropic filtering, as described in the [Texel Anisotropic Filtering](#) section, or `VK_FALSE` otherwise.
- `maxAnisotropy` is the anisotropy value clamp used by the sampler when `anisotropyEnable` is `VK_TRUE`. If `anisotropyEnable` is `VK_FALSE`, `maxAnisotropy` is ignored.
- `compareEnable` is `VK_TRUE` to enable comparison against a reference value during lookups, or `VK_FALSE` otherwise.
 - Note: Some implementations will default to shader state if this member does not match.
- `compareOp` is a `VkCompareOp` value specifying the comparison operator to apply to fetched data before filtering as described in the [Depth Compare Operation](#) section.
- `minLod` is used to clamp the [minimum of the computed LOD value](#).
- `maxLod` is used to clamp the [maximum of the computed LOD value](#). To avoid clamping the maximum value, set `maxLod` to the constant `VK_LOD_CLAMP_NONE`.
- `borderColor` is a `VkBorderColor` value specifying the predefined border color to use.
- `unnormalizedCoordinates` controls whether to use unnormalized or normalized texel coordinates to address texels of the image. When `unnormalizedCoordinates` is `VK_TRUE`, the range of the image coordinates used to lookup the texel is in the range of zero to the image size in each dimension. When `unnormalizedCoordinates` is `VK_FALSE`, the range of image coordinates is zero to one.

When `unnormalizedCoordinates` is `VK_TRUE`, images the sampler is used with in the shader have the following requirements:

- The `viewType` **must** be either `VK_IMAGE_VIEW_TYPE_1D` or `VK_IMAGE_VIEW_TYPE_2D`.
- The image view **must** have a single layer and a single mip level.

When `unnormalizedCoordinates` is `VK_TRUE`, image built-in functions in the shader that use the sampler have the following requirements:

- The functions **must** not use projection.
- The functions **must** not use offsets.

NOTE

Mapping of OpenGL to Vulkan Filter Modes

`magFilter` values of `VK_FILTER_NEAREST` and `VK_FILTER_LINEAR` directly correspond to `GL_NEAREST` and `GL_LINEAR` magnification filters. `minFilter` and `mipmapMode` combine to correspond to the similarly named OpenGL minification filter of `GL_minFilter_MIPMAP_mipmapMode` (e.g. `minFilter` of `VK_FILTER_LINEAR` and `mipmapMode` of `VK_SAMPLER_MIPMAP_MODE_NEAREST` correspond to `GL_LINEAR_MIPMAP_NEAREST`).

There are no Vulkan filter modes that directly correspond to OpenGL minification filters of `GL_LINEAR` or `GL_NEAREST`, but they **can** be emulated using `VK_SAMPLER_MIPMAP_MODE_NEAREST`, `minLod = 0`, and `maxLod = 0.25`, and using `minFilter = VK_FILTER_LINEAR` or `minFilter = VK_FILTER_NEAREST`, respectively.

Note that using a `maxLod` of zero would cause **magnification** to always be performed, and the `magFilter` to always be used. This is valid, just not an exact match for OpenGL behavior. Clamping the maximum LOD to 0.25 allows the λ value to be non-zero and minification to be performed, while still always rounding down to the base level. If the `minFilter` and `magFilter` are equal, then using a `maxLod` of zero also works.

The maximum number of sampler objects which **can** be simultaneously created on a device is implementation-dependent and specified by the `maxSamplerAllocationCount` member of the `VkPhysicalDeviceLimits` structure.

NOTE

For historical reasons, if `maxSamplerAllocationCount` is exceeded, some implementations may return `VK_ERROR_TOO_MANY_OBJECTS`. Exceeding this limit will result in undefined behavior, and an application should not rely on the use of the returned error code in order to identify when the limit is reached.

Since `VkSampler` is a non-dispatchable handle type, implementations **may** return the same handle for sampler state vectors that are identical. In such cases, all such objects would only count once against the `maxSamplerAllocationCount` limit.

Valid Usage

- VUID-VkSamplerCreateInfo-mipLodBias-01069
The absolute value of `mipLodBias` **must** be less than or equal to `VkPhysicalDeviceLimits::maxSamplerLodBias`
- VUID-VkSamplerCreateInfo-maxLod-01973
`maxLod` **must** be greater than or equal to `minLod`
- VUID-VkSamplerCreateInfo-anisotropyEnable-01070
If the `samplerAnisotropy` feature is not enabled, `anisotropyEnable` **must** be `VK_FALSE`
- VUID-VkSamplerCreateInfo-anisotropyEnable-01071
If `anisotropyEnable` is `VK_TRUE`, `maxAnisotropy` **must** be between `1.0` and `VkPhysicalDeviceLimits::maxSamplerAnisotropy`, inclusive
- VUID-VkSamplerCreateInfo-unnormalizedCoordinates-01072
If `unnormalizedCoordinates` is `VK_TRUE`, `minFilter` and `magFilter` **must** be equal
- VUID-VkSamplerCreateInfo-unnormalizedCoordinates-01073
If `unnormalizedCoordinates` is `VK_TRUE`, `mipmapMode` **must** be `VK_SAMPLER_MIPMAP_MODE_NEAREST`
- VUID-VkSamplerCreateInfo-unnormalizedCoordinates-01074
If `unnormalizedCoordinates` is `VK_TRUE`, `minLod` and `maxLod` **must** be zero
- VUID-VkSamplerCreateInfo-unnormalizedCoordinates-01075
If `unnormalizedCoordinates` is `VK_TRUE`, `addressModeU` and `addressModeV` **must** each be either

VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_EDGE or VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_BORDER

- VUID-VkSamplerCreateInfo-unnormlizedCoordinates-01076
If `unnormlizedCoordinates` is VK_TRUE, `anisotropyEnable` **must** be VK_FALSE
- VUID-VkSamplerCreateInfo-unnormlizedCoordinates-01077
If `unnormlizedCoordinates` is VK_TRUE, `compareEnable` **must** be VK_FALSE
- VUID-VkSamplerCreateInfo-addressModeU-01078
If any of `addressModeU`, `addressModeV` or `addressModeW` are VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_BORDER, `borderColor` **must** be a valid `VkBorderColor` value
- VUID-VkSamplerCreateInfo-addressModeU-01079
If `samplerMirrorClampToEdge` is not enabled, and if the VK_KHR_sampler_mirror_clamp_to_edge extension is not enabled, `addressModeU`, `addressModeV` and `addressModeW` **must** not be VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE
- VUID-VkSamplerCreateInfo-compareEnable-01080
If `compareEnable` is VK_TRUE, `compareOp` **must** be a valid `VkCompareOp` value

Valid Usage (Implicit)

- VUID-VkSamplerCreateInfo-sType-sType
`sType` **must** be VK_STRUCTURE_TYPE_SAMPLER_CREATE_INFO
- VUID-VkSamplerCreateInfo-pNext-pNext
`pNext` **must** be NULL
- VUID-VkSamplerCreateInfo-flags-zeroBitmask
`flags` **must** be 0
- VUID-VkSamplerCreateInfo-magFilter-parameter
`magFilter` **must** be a valid `VkFilter` value
- VUID-VkSamplerCreateInfo-minFilter-parameter
`minFilter` **must** be a valid `VkFilter` value
- VUID-VkSamplerCreateInfo-mipmapMode-parameter
`mipmapMode` **must** be a valid `VkSamplerMipmapMode` value
- VUID-VkSamplerCreateInfo-addressModeU-parameter
`addressModeU` **must** be a valid `VkSamplerAddressMode` value
- VUID-VkSamplerCreateInfo-addressModeV-parameter
`addressModeV` **must** be a valid `VkSamplerAddressMode` value
- VUID-VkSamplerCreateInfo-addressModeW-parameter
`addressModeW` **must** be a valid `VkSamplerAddressMode` value

VK_LOD_CLAMP_NONE is a special constant value used for `VkSamplerCreateInfo::maxLod` to indicate that maximum LOD clamping should not be performed.

```
#define VK_LOD_CLAMP_NONE 1000.0F
```

Bits which **can** be set in `VkSamplerCreateInfo::flags`, specifying additional parameters of a sampler, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSamplerCreateFlagBits {
} VkSamplerCreateFlagBits;
```

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkSamplerCreateFlags;
```

`VkSamplerCreateFlags` is a bitmask type for setting a mask of zero or more `VkSamplerCreateFlagBits`.

Possible values of the `VkSamplerCreateInfo::magFilter` and `minFilter` parameters, specifying filters used for texture lookups, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkFilter {
    VK_FILTER_NEAREST = 0,
    VK_FILTER_LINEAR = 1,
} VkFilter;
```

- `VK_FILTER_NEAREST` specifies nearest filtering.
- `VK_FILTER_LINEAR` specifies linear filtering.

These filters are described in detail in [Texel Filtering](#).

Possible values of the `VkSamplerCreateInfo::mipmapMode`, specifying the mipmap mode used for texture lookups, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSamplerMipmapMode {
    VK_SAMPLER_MIPMAP_MODE_NEAREST = 0,
    VK_SAMPLER_MIPMAP_MODE_LINEAR = 1,
} VkSamplerMipmapMode;
```

- `VK_SAMPLER_MIPMAP_MODE_NEAREST` specifies nearest filtering.
- `VK_SAMPLER_MIPMAP_MODE_LINEAR` specifies linear filtering.

These modes are described in detail in [Texel Filtering](#).

Possible values of the `VkSamplerCreateInfo::addressMode*` parameters, specifying the behavior of sampling with coordinates outside the range [0,1] for the respective u, v, or w coordinate as defined

in the [Wrapping Operation](#) section, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSamplerAddressMode {
    VK_SAMPLER_ADDRESS_MODE_REPEAT = 0,
    VK_SAMPLER_ADDRESS_MODE_MIRRORED_REPEAT = 1,
    VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_EDGE = 2,
    VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_BORDER = 3,
} VkSamplerAddressMode;
```

- `VK_SAMPLER_ADDRESS_MODE_REPEAT` specifies that the repeat wrap mode will be used.
- `VK_SAMPLER_ADDRESS_MODE_MIRRORED_REPEAT` specifies that the mirrored repeat wrap mode will be used.
- `VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_EDGE` specifies that the clamp to edge wrap mode will be used.
- `VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_BORDER` specifies that the clamp to border wrap mode will be used.

Comparison operators compare a *reference* and a *test* value, and return a true (“passed”) or false (“failed”) value depending on the comparison operator chosen. The supported operators are:

```
// Provided by VK_VERSION_1_0
typedef enum VkCompareOp {
    VK_COMPARE_OP_NEVER = 0,
    VK_COMPARE_OP_LESS = 1,
    VK_COMPARE_OP_EQUAL = 2,
    VK_COMPARE_OP_LESS_OR_EQUAL = 3,
    VK_COMPARE_OP_GREATER = 4,
    VK_COMPARE_OP_NOT_EQUAL = 5,
    VK_COMPARE_OP_GREATER_OR_EQUAL = 6,
    VK_COMPARE_OP_ALWAYS = 7,
} VkCompareOp;
```

- `VK_COMPARE_OP_NEVER` specifies that the comparison always evaluates false.
- `VK_COMPARE_OP_LESS` specifies that the comparison evaluates *reference* < *test*.
- `VK_COMPARE_OP_EQUAL` specifies that the comparison evaluates *reference* = *test*.
- `VK_COMPARE_OP_LESS_OR_EQUAL` specifies that the comparison evaluates *reference* ≤ *test*.
- `VK_COMPARE_OP_GREATER` specifies that the comparison evaluates *reference* > *test*.
- `VK_COMPARE_OP_NOT_EQUAL` specifies that the comparison evaluates *reference* ≠ *test*.
- `VK_COMPARE_OP_GREATER_OR_EQUAL` specifies that the comparison evaluates *reference* ≥ *test*.
- `VK_COMPARE_OP_ALWAYS` specifies that the comparison always evaluates true.

Comparison operators are used for:

- The [Depth Compare Operation](#) operator for a sampler, specified by `VkSamplerCreateInfo`

`::compareOp.`

- The stencil comparison operator for the [stencil test](#), specified by `VkStencilOpState::compareOp.`
- The [Depth Comparison](#) operator for the [depth test](#), specified by `VkPipelineDepthStencilStateCreateInfo::depthCompareOp.`

Each such use describes how the *reference* and *test* values for that comparison are determined.

Possible values of `VkSamplerCreateInfo::borderColor`, specifying the border color used for texture lookups, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkBorderColor {
    VK_BORDER_COLOR_FLOAT_TRANSPARENT_BLACK = 0,
    VK_BORDER_COLOR_INT_TRANSPARENT_BLACK = 1,
    VK_BORDER_COLOR_FLOAT_OPAQUE_BLACK = 2,
    VK_BORDER_COLOR_INT_OPAQUE_BLACK = 3,
    VK_BORDER_COLOR_FLOAT_OPAQUE_WHITE = 4,
    VK_BORDER_COLOR_INT_OPAQUE_WHITE = 5,
} VkBorderColor;
```

- `VK_BORDER_COLOR_FLOAT_TRANSPARENT_BLACK` specifies a transparent, floating-point format, black color.
- `VK_BORDER_COLOR_INT_TRANSPARENT_BLACK` specifies a transparent, integer format, black color.
- `VK_BORDER_COLOR_FLOAT_OPAQUE_BLACK` specifies an opaque, floating-point format, black color.
- `VK_BORDER_COLOR_INT_OPAQUE_BLACK` specifies an opaque, integer format, black color.
- `VK_BORDER_COLOR_FLOAT_OPAQUE_WHITE` specifies an opaque, floating-point format, white color.
- `VK_BORDER_COLOR_INT_OPAQUE_WHITE` specifies an opaque, integer format, white color.

These colors are described in detail in [Texel Replacement](#).

To destroy a sampler, call:

```
// Provided by VK_VERSION_1_0
void vkDestroySampler(
    VkDevice          device,
    VkSampler          sampler,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that destroys the sampler.
- `sampler` is the sampler to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroySampler-sampler-01082
All submitted commands that refer to **sampler** **must** have completed execution
- VUID-vkDestroySampler-sampler-01083
If **VkAllocationCallbacks** were provided when **sampler** was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroySampler-sampler-01084
If no **VkAllocationCallbacks** were provided when **sampler** was created, **pAllocator** **must** be **NULL**

Valid Usage (Implicit)

- VUID-vkDestroySampler-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkDestroySampler-sampler-parameter
If **sampler** is not **VK_NULL_HANDLE**, **sampler** **must** be a valid **VkSampler** handle
- VUID-vkDestroySampler-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid **VkAllocationCallbacks** structure
- VUID-vkDestroySampler-sampler-parent
If **sampler** is a valid handle, it **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **sampler** **must** be externally synchronized

Chapter 14. Resource Descriptors

A *descriptor* is an opaque data structure representing a shader resource such as a buffer, buffer view, image view, sampler, or combined image sampler. Descriptors are organized into *descriptor sets*, which are bound during command recording for use in subsequent drawing commands. The arrangement of content in each descriptor set is determined by a *descriptor set layout*, which determines what descriptors can be stored within it. The sequence of descriptor set layouts that **can** be used by a pipeline is specified in a *pipeline layout*. Each pipeline object **can** use up to `maxBoundDescriptorSets` (see [Limits](#)) descriptor sets.

Shaders access resources via variables decorated with a descriptor set and binding number that link them to a descriptor in a descriptor set. The shader interface mapping to bound descriptor sets is described in the [Shader Resource Interface](#) section.

14.1. Descriptor Types

There are a number of different types of descriptor supported by Vulkan, corresponding to different resources or usage. The following sections describe the API definitions of each descriptor type. The mapping of each type to SPIR-V is listed in the [Shader Resource and Descriptor Type Correspondence](#) and [Shader Resource and Storage Class Correspondence](#) tables in the [Shader Interfaces](#) chapter.

14.1.1. Storage Image

A *storage image* (`VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`) is a descriptor type associated with an [image resource](#) via an [image view](#) that load, store, and atomic operations **can** be performed on.

Storage image loads are supported in all shader stages for image views whose [format features](#) contain `VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT`.

Stores to storage images are supported in compute shaders for image views whose [format features](#) contain `VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT`.

Atomic operations on storage images are supported in compute shaders for image views whose [format features](#) contain `VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT`.

When the `fragmentStoresAndAtomics` feature is enabled, stores and atomic operations are also supported for storage images in fragment shaders with the same set of image formats as supported in compute shaders. When the `vertexPipelineStoresAndAtomics` feature is enabled, stores and atomic operations are also supported in vertex, tessellation, and geometry shaders with the same set of image formats as supported in compute shaders.

The image subresources for a storage image **must** be in the `VK_IMAGE_LAYOUT_GENERAL` layout in order to access its data in a shader.

14.1.2. Sampler

A *sampler descriptor* (`VK_DESCRIPTOR_TYPE_SAMPLER`) is a descriptor type associated with a [sampler](#)

object, used to control the behavior of [sampling operations](#) performed on a [sampled image](#).

14.1.3. Sampled Image

A *sampled image* (`VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`) is a descriptor type associated with an [image resource](#) via an [image view](#) that [sampling operations](#) **can** be performed on.

Shaders combine a sampled image variable and a sampler variable to perform sampling operations.

Sampled images are supported in all shader stages for image views whose [format features](#) contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT`.

An image subresources for a sampled image **must** be in one of the following layouts:

- `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`
- `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL`
- `VK_IMAGE_LAYOUT_GENERAL`

14.1.4. Combined Image Sampler

A *combined image sampler* (`VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`) is a single descriptor type associated with both a [sampler](#) and an [image resource](#), combining both a [sampler](#) and [sampled image](#) descriptor into a single descriptor.

The sampler and image in this type of descriptor **can** be used freely with any other samplers and images.

An image subresources for a combined image sampler **must** be in one of the following layouts:

- `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`
- `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL`
- `VK_IMAGE_LAYOUT_GENERAL`

NOTE

On some implementations, it **may** be more efficient to sample from an image using a combination of sampler and sampled image that are stored together in the descriptor set in a combined descriptor.

14.1.5. Uniform Texel Buffer

A *uniform texel buffer* (`VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER`) is a descriptor type associated with a [buffer resource](#) via a [buffer view](#) that [image sampling operations](#) **can** be performed on.

Uniform texel buffers define a tightly-packed 1-dimensional linear array of texels, with texels going through format conversion when read in a shader in the same way as they are for an image.

Load operations from uniform texel buffers are supported in all shader stages for buffer view formats which report [format features](#) support for `VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT`

14.1.6. Storage Texel Buffer

A *storage texel buffer* (`VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER`) is a descriptor type associated with a [buffer resource](#) via a [buffer view](#) that [image load, store, and atomic operations](#) **can** be performed on.

Storage texel buffers define a tightly-packed 1-dimensional linear array of texels, with texels going through format conversion when read in a shader in the same way as they are for an image. Unlike [uniform texel buffers](#), these buffers can also be written to in the same way as for [storage images](#).

Storage texel buffer loads are supported in all shader stages for texel buffer view formats which report [format features](#) support for `VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT`

Stores to storage texel buffers are supported in compute shaders for texel buffer formats which report [format features](#) support for `VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT`

Atomic operations on storage texel buffers are supported in compute shaders for texel buffer formats which report [format features](#) support for `VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT`

When the `fragmentStoresAndAtomics` feature is enabled, stores and atomic operations are also supported for storage texel buffers in fragment shaders with the same set of texel buffer formats as supported in compute shaders. When the `vertexPipelineStoresAndAtomics` feature is enabled, stores and atomic operations are also supported in vertex, tessellation, and geometry shaders with the same set of texel buffer formats as supported in compute shaders.

14.1.7. Storage Buffer

A *storage buffer* (`VK_DESCRIPTOR_TYPE_STORAGE_BUFFER`) is a descriptor type associated with a [buffer resource](#) directly, described in a shader as a structure with various members that load, store, and atomic operations **can** be performed on.

NOTE

Atomic operations **can** only be performed on members of certain types as defined in the [SPIR-V environment appendix](#).

14.1.8. Uniform Buffer

A *uniform buffer* (`VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER`) is a descriptor type associated with a [buffer resource](#) directly, described in a shader as a structure with various members that load operations **can** be performed on.

14.1.9. Dynamic Uniform Buffer

A *dynamic uniform buffer* (`VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC`) is almost identical to a [uniform buffer](#), and differs only in how the offset into the buffer is specified. The base offset calculated by the `VkDescriptorBufferInfo` when initially [updating the descriptor set](#) is added to a [dynamic offset](#) when binding the descriptor set.

14.1.10. Dynamic Storage Buffer

A *dynamic storage buffer* (`VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC`) is almost identical to a *storage buffer*, and differs only in how the offset into the buffer is specified. The base offset calculated by the `VkDescriptorBufferInfo` when initially *updating the descriptor set* is added to a *dynamic offset* when binding the descriptor set.

14.1.11. Input Attachment

An *input attachment* (`VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT`) is a descriptor type associated with an *image resource* via an *image view* that **can** be used for *framebuffer local* load operations in fragment shaders.

All image formats that are supported for color attachments (`VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT`) or depth/stencil attachments (`VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT`) for a given image tiling mode are also supported for input attachments.

An image view used as an input attachment **must** be in one of the following layouts:

- `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`
- `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL`
- `VK_IMAGE_LAYOUT_GENERAL`

14.2. Descriptor Sets

Descriptors are grouped together into descriptor set objects. A descriptor set object is an opaque object containing storage for a set of descriptors, where the types and number of descriptors is defined by a descriptor set layout. The layout object **may** be used to define the association of each descriptor binding with memory or other implementation resources. The layout is used both for determining the resources that need to be associated with the descriptor set, and determining the interface between shader stages and shader resources.

14.2.1. Descriptor Set Layout

A descriptor set layout object is defined by an array of zero or more descriptor bindings. Each individual descriptor binding is specified by a descriptor type, a count (array size) of the number of descriptors in the binding, a set of shader stages that **can** access the binding, and (if using immutable samplers) an array of sampler descriptors.

Descriptor set layout objects are represented by `VkDescriptorSetLayout` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkDescriptorSetLayout)
```

To create descriptor set layout objects, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateDescriptorSetLayout(
    VkDevice device,
    const VkDescriptorSetLayoutCreateInfo* pCreateInfo,
    const VkAllocationCallbacks* pAllocator,
    VkDescriptorSetLayout* pSetLayout);
```

- **device** is the logical device that creates the descriptor set layout.
- **pCreateInfo** is a pointer to a [VkDescriptorSetLayoutCreateInfo](#) structure specifying the state of the descriptor set layout object.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.
- **pSetLayout** is a pointer to a [VkDescriptorSetLayout](#) handle in which the resulting descriptor set layout object is returned.

Valid Usage (Implicit)

- VUID-vkCreateDescriptorSetLayout-device-parameter **device** **must** be a valid [VkDevice](#) handle
- VUID-vkCreateDescriptorSetLayout-pCreateInfo-parameter **pCreateInfo** **must** be a valid pointer to a valid [VkDescriptorSetLayoutCreateInfo](#) structure
- VUID-vkCreateDescriptorSetLayout-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkCreateDescriptorSetLayout-pSetLayout-parameter **pSetLayout** **must** be a valid pointer to a [VkDescriptorSetLayout](#) handle

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_OUT_OF_DEVICE_MEMORY**

Information about the descriptor set layout is passed in a [VkDescriptorSetLayoutCreateInfo](#) structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkDescriptorSetLayoutCreateInfo {
    VkStructureType          sType;
    const void*              pNext;
    VkDescriptorSetLayoutCreateFlags flags;
    uint32_t                 bindingCount;
    const VkDescriptorSetLayoutBinding* pBindings;
} VkDescriptorSetLayoutCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask specifying options for descriptor set layout creation.
- `bindingCount` is the number of elements in `pBindings`.
- `pBindings` is a pointer to an array of `VkDescriptorSetLayoutBinding` structures.

Valid Usage

- VUID-VkDescriptorSetLayoutCreateInfo-binding-00279
The `VkDescriptorSetLayoutBinding::binding` members of the elements of the `pBindings` array **must** each have different values

Valid Usage (Implicit)

- VUID-VkDescriptorSetLayoutCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_CREATE_INFO`
- VUID-VkDescriptorSetLayoutCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkDescriptorSetLayoutCreateInfo-flags-zero-bitmask
`flags` **must** be `0`
- VUID-VkDescriptorSetLayoutCreateInfo-pBindings-parameter
If `bindingCount` is not `0`, `pBindings` **must** be a valid pointer to an array of `bindingCount` valid `VkDescriptorSetLayoutBinding` structures

Bits which **can** be set in `VkDescriptorSetLayoutCreateInfo::flags`, specifying options for descriptor set layout, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkDescriptorSetLayoutCreateFlagBits {
} VkDescriptorSetLayoutCreateFlagBits;
```

NOTE

All bits for this type are defined by extensions, and none of those extensions are

enabled in this build of the specification.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkDescriptorSetLayoutCreateFlags;
```

`VkDescriptorSetLayoutCreateFlags` is a bitmask type for setting a mask of zero or more `VkDescriptorSetLayoutCreateFlagBits`.

The `VkDescriptorSetLayoutBinding` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDescriptorSetLayoutBinding {
    uint32_t          binding;
    VkDescriptorType   descriptorType;
    uint32_t          descriptorCount;
    VkShaderStageFlags stageFlags;
    const VkSampler*   pImmutableSamplers;
} VkDescriptorSetLayoutBinding;
```

- `binding` is the binding number of this entry and corresponds to a resource of the same binding number in the shader stages.
- `descriptorType` is a `VkDescriptorType` specifying which type of resource descriptors are used for this binding.
- `descriptorCount` is the number of descriptors contained in the binding, accessed in a shader as an array. If `descriptorCount` is zero this binding entry is reserved and the resource **must** not be accessed from any stage via this binding within any pipeline using the set layout.
- `stageFlags` member is a bitmask of `VkShaderStageFlagBits` specifying which pipeline shader stages **can** access a resource for this binding. `VK_SHADER_STAGE_ALL` is a shorthand specifying that all defined shader stages, including any additional stages defined by extensions, **can** access the resource.

If a shader stage is not included in `stageFlags`, then a resource **must** not be accessed from that stage via this binding within any pipeline using the set layout. Other than input attachments which are limited to the fragment shader, there are no limitations on what combinations of stages **can** use a descriptor binding, and in particular a binding **can** be used by both graphics stages and the compute stage.

- `pImmutableSamplers` affects initialization of samplers. If `descriptorType` specifies a `VK_DESCRIPTOR_TYPE_SAMPLER` or `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER` type descriptor, then `pImmutableSamplers` **can** be used to initialize a set of *immutable samplers*. Immutable samplers are permanently bound into the set layout and **must** not be changed; updating a `VK_DESCRIPTOR_TYPE_SAMPLER` descriptor with immutable samplers is not allowed and updates to a `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER` descriptor with immutable samplers does not modify the samplers (the image views are updated, but the sampler updates are ignored). If `pImmutableSamplers` is not `NULL`, then it is a pointer to an array of sampler handles that will be copied into the set layout and used for the corresponding binding. Only the sampler handles are

copied; the sampler objects **must** not be destroyed before the final use of the set layout and any descriptor pools and sets created using it. If `pImmutableSamplers` is `NULL`, then the sampler slots are dynamic and sampler handles **must** be bound into descriptor sets using this layout. If `descriptorType` is not one of these descriptor types, then `pImmutableSamplers` is ignored.

The above layout definition allows the descriptor bindings to be specified sparsely such that not all binding numbers between 0 and the maximum binding number need to be specified in the `pBindings` array. Bindings that are not specified have a `descriptorCount` and `stageFlags` of zero, and the value of `descriptorType` is undefined. However, all binding numbers between 0 and the maximum binding number in the `VkDescriptorSetLayoutCreateInfo::pBindings` array **may** consume memory in the descriptor set layout even if not all descriptor bindings are used, though it **should** not consume additional memory from the descriptor pool.

NOTE

The maximum binding number specified **should** be as compact as possible to avoid wasted memory.

Valid Usage

- VUID-VkDescriptorSetLayoutBinding-descriptorType-00282
If `descriptorType` is `VK_DESCRIPTOR_TYPE_SAMPLER` or `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, and `descriptorCount` is not 0 and `pImmutableSamplers` is not `NULL`, `pImmutableSamplers` **must** be a valid pointer to an array of `descriptorCount` valid `VkSampler` handles
- VUID-VkDescriptorSetLayoutBinding-descriptorCount-09465
If `descriptorCount` is not 0, `stageFlags` **must** be `VK_SHADER_STAGE_ALL` or a valid combination of other `VkShaderStageFlagBits` values
- VUID-VkDescriptorSetLayoutBinding-descriptorType-01510
If `descriptorType` is `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT` and `descriptorCount` is not 0, then `stageFlags` **must** be 0 or `VK_SHADER_STAGE_FRAGMENT_BIT`

Valid Usage (Implicit)

- VUID-VkDescriptorSetLayoutBinding-descriptorType-parameter
`descriptorType` **must** be a valid `VkDescriptorType` value

The following examples show a shader snippet using two descriptor sets, and application code that creates corresponding descriptor set layouts.

GLSL Example

```
//  
// binding to a single sampled image descriptor in set 0  
//  
layout (set=0, binding=0) uniform texture2D mySampledImage;  
  
//
```



```

// binding to an array of sampled image descriptors in set 0
//
layout (set=0, binding=1) uniform texture2D myArrayOfSampledImages[12];

//
// binding to a single uniform buffer descriptor in set 1
//
layout (set=1, binding=0) uniform myUniformBuffer
{
    vec4 myElement[32];
};

```

SPIR-V Example

```

...
%1 = OpExtInstImport "GLSL.std.450"
...
OpName %9 "mySampledImage"
OpName %14 "myArrayOfSampledImages"
OpName %18 "myUniformBuffer"
OpMemberName %18 0 "myElement"
OpName %20 ""
OpDecorate %9 DescriptorSet 0
OpDecorate %9 Binding 0
OpDecorate %14 DescriptorSet 0
OpDecorate %14 Binding 1
OpDecorate %17 ArrayStride 16
OpMemberDecorate %18 0 Offset 0
OpDecorate %18 Block
OpDecorate %20 DescriptorSet 1
OpDecorate %20 Binding 0
%2 = OpTypeVoid
%3 = OpTypeFunction %2
%6 = OpTypeFloat 32
%7 = OpTypeImage %6 2D 0 0 0 1 Unknown
%8 = OpTypePointer UniformConstant %7
%9 = OpVariable %8 UniformConstant
%10 = OpTypeInt 32 0
%11 = OpConstant %10 12
%12 = OpTypeArray %7 %11
%13 = OpTypePointer UniformConstant %12
%14 = OpVariable %13 UniformConstant
%15 = OpTypeVector %6 4
%16 = OpConstant %10 32
%17 = OpTypeArray %15 %16
%18 = OpTypeStruct %17
%19 = OpTypePointer Uniform %18
%20 = OpVariable %19 Uniform
...

```

API Example

```
VkResult myResult;

const VkDescriptorSetLayoutBinding myDescriptorSetLayoutBinding[] =
{
    // binding to a single image descriptor
    {
        .binding = 0,
        .descriptorType = VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE,
        .descriptorCount = 1,
        .stageFlags = VK_SHADER_STAGE_FRAGMENT_BIT,
        .pImmutableSamplers = NULL
    },

    // binding to an array of image descriptors
    {
        .binding = 1,
        .descriptorType = VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE,
        .descriptorCount = 12,
        .stageFlags = VK_SHADER_STAGE_FRAGMENT_BIT,
        .pImmutableSamplers = NULL
    },

    // binding to a single uniform buffer descriptor
    {
        .binding = 0,
        .descriptorType = VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER,
        .descriptorCount = 1,
        .stageFlags = VK_SHADER_STAGE_FRAGMENT_BIT,
        .pImmutableSamplers = NULL
    }
};

const VkDescriptorSetLayoutCreateInfo myDescriptorSetLayoutCreateInfo[] =
{
    // Information for first descriptor set with two descriptor bindings
    {
        .sType = VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_CREATE_INFO,
        .pNext = NULL,
        .flags = 0,
        .bindingCount = 2,
        .pBindings = &myDescriptorSetLayoutBinding[0]
    },

    // Information for second descriptor set with one descriptor binding
    {
        .sType = VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_CREATE_INFO,
        .pNext = NULL,
        .flags = 0,
        .bindingCount = 1,
```

```

        .pBindings = &myDescriptorSetLayoutBinding[2]
    }
};

VkDescriptorSetLayout myDescriptorSetLayout[2];

//
// Create first descriptor set layout
//
myResult = vkCreateDescriptorSetLayout(
    myDevice,
    &myDescriptorSetLayoutCreateInfo[0],
    NULL,
    &myDescriptorSetLayout[0]);

//
// Create second descriptor set layout
//
myResult = vkCreateDescriptorSetLayout(
    myDevice,
    &myDescriptorSetLayoutCreateInfo[1],
    NULL,
    &myDescriptorSetLayout[1]);

```

To destroy a descriptor set layout, call:

```

// Provided by VK_VERSION_1_0
void vkDestroyDescriptorSetLayout(
    VkDevice                device,
    VkDescriptorSetLayout    descriptorSetLayout,
    const VkAllocationCallbacks* pAllocator);

```

- **device** is the logical device that destroys the descriptor set layout.
- **descriptorSetLayout** is the descriptor set layout to destroy.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyDescriptorSetLayout-descriptorSetLayout-00284
If **VkAllocationCallbacks** were provided when **descriptorSetLayout** was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyDescriptorSetLayout-descriptorSetLayout-00285
If no **VkAllocationCallbacks** were provided when **descriptorSetLayout** was created, **pAllocator** **must** be **NULL**

Valid Usage (Implicit)

- VUID-vkDestroyDescriptorSetLayout-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkDestroyDescriptorSetLayout-descriptorSetLayout-parameter
If `descriptorSetLayout` is not `VK_NULL_HANDLE`, `descriptorSetLayout` **must** be a valid `VkDescriptorSetLayout` handle
- VUID-vkDestroyDescriptorSetLayout-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkDestroyDescriptorSetLayout-descriptorSetLayout-parent
If `descriptorSetLayout` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `descriptorSetLayout` **must** be externally synchronized

14.2.2. Pipeline Layouts

Access to descriptor sets from a pipeline is accomplished through a *pipeline layout*. Zero or more descriptor set layouts and zero or more push constant ranges are combined to form a pipeline layout object describing the complete set of resources that **can** be accessed by a pipeline. The pipeline layout represents a sequence of descriptor sets with each having a specific layout. This sequence of layouts is used to determine the interface between shader stages and shader resources. Each pipeline is created using a pipeline layout.

Pipeline layout objects are represented by `VkPipelineLayout` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkPipelineLayout)
```

To create a pipeline layout, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreatePipelineLayout(
    VkDevice                                device,
    const VkPipelineLayoutCreateInfo*       pCreateInfo,
    const VkAllocationCallbacks*           pAllocator,
    VkPipelineLayout*                       pPipelineLayout);
```

- `device` is the logical device that creates the pipeline layout.
- `pCreateInfo` is a pointer to a `VkPipelineLayoutCreateInfo` structure specifying the state of the

pipeline layout object.

- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.
- **pPipelineLayout** is a pointer to a [VkPipelineLayout](#) handle in which the resulting pipeline layout object is returned.

Valid Usage (Implicit)

- VUID-vkCreatePipelineLayout-device-parameter **device** **must** be a valid [VkDevice](#) handle
- VUID-vkCreatePipelineLayout-pCreateInfo-parameter **pCreateInfo** **must** be a valid pointer to a valid [VkPipelineLayoutCreateInfo](#) structure
- VUID-vkCreatePipelineLayout-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkCreatePipelineLayout-pPipelineLayout-parameter
pPipelineLayout **must** be a valid pointer to a [VkPipelineLayout](#) handle

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_OUT_OF_DEVICE_MEMORY**

The [VkPipelineLayoutCreateInfo](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineLayoutCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkPipelineLayoutCreateFlags flags;
    uint32_t           setLayoutCount;
    const VkDescriptorSetLayout* pSetLayouts;
    uint32_t           pushConstantRangeCount;
    const VkPushConstantRange* pPushConstantRanges;
} VkPipelineLayoutCreateInfo;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **flags** is a bitmask of [VkPipelineLayoutCreateFlagBits](#) specifying options for pipeline layout creation.

- `setLayoutCount` is the number of descriptor sets included in the pipeline layout.
- `pSetLayouts` is a pointer to an array of `VkDescriptorSetLayout` objects.
- `pushConstantRangeCount` is the number of push constant ranges included in the pipeline layout.
- `pPushConstantRanges` is a pointer to an array of `VkPushConstantRange` structures defining a set of push constant ranges for use in a single pipeline layout. In addition to descriptor set layouts, a pipeline layout also describes how many push constants **can** be accessed by each stage of the pipeline.

NOTE

Push constants represent a high speed path to modify constant data in pipelines that is expected to outperform memory-backed resource updates.

Valid Usage

- VUID-VkPipelineLayoutCreateInfo-setLayoutCount-00286
`setLayoutCount` **must** be less than or equal to `VkPhysicalDeviceLimits::maxBoundDescriptorSets`
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03016
The total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_SAMPLER` and `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER` accessible to any given shader stage across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxPerStageDescriptorSamplers`
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03017
The total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER` and `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` accessible to any given shader stage across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxPerStageDescriptorUniformBuffers`
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03018
The total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER` and `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC` accessible to any given shader stage across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxPerStageDescriptorStorageBuffers`
- VUID-VkPipelineLayoutCreateInfo-descriptorType-06939
The total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, and `VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER`, accessible to any given shader stage across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxPerStageDescriptorSampledImages`
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03020
The total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, and `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` accessible to any given shader stage across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxPerStageDescriptorStorageImages`
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03021
The total number of descriptors in descriptor set layouts with a `descriptorType` of

`VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT` accessible to any given shader stage across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxPerStageDescriptorInputAttachments`

- VUID-VkPipelineLayoutCreateInfo-descriptorType-03028

The total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_SAMPLER` and `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER` accessible across all shader stages and across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxDescriptorSetSamplers`

- VUID-VkPipelineLayoutCreateInfo-descriptorType-03029

The total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER` accessible across all shader stages and across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxDescriptorSetUniformBuffers`

- VUID-VkPipelineLayoutCreateInfo-descriptorType-03030

The total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` accessible across all shader stages and across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxDescriptorSetUniformBuffersDynamic`

- VUID-VkPipelineLayoutCreateInfo-descriptorType-03031

The total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER` accessible across all shader stages and across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxDescriptorSetStorageBuffers`

- VUID-VkPipelineLayoutCreateInfo-descriptorType-03032

The total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC` accessible across all shader stages and across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxDescriptorSetStorageBuffersDynamic`

- VUID-VkPipelineLayoutCreateInfo-descriptorType-03033

The total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, and `VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER` accessible across all shader stages and across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxDescriptorSetSampledImages`

- VUID-VkPipelineLayoutCreateInfo-descriptorType-03034

The total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, and `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` accessible across all shader stages and across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxDescriptorSetStorageImages`

- VUID-VkPipelineLayoutCreateInfo-descriptorType-03035

The total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT` accessible across all shader stages and across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxDescriptorSetInputAttachments`

- VUID-VkPipelineLayoutCreateInfo-pPushConstantRanges-00292
Any two elements of `pPushConstantRanges` **must** not include the same stage in `stageFlags`
- VUID-VkPipelineLayoutCreateInfo-graphicsPipelineLibrary-06753
Elements of `pSetLayouts` **must** be valid `VkDescriptorSetLayout` objects

Valid Usage (Implicit)

- VUID-VkPipelineLayoutCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_LAYOUT_CREATE_INFO`
- VUID-VkPipelineLayoutCreateInfo-flags-zeroBitmask
`flags` **must** be `0`
- VUID-VkPipelineLayoutCreateInfo-pSetLayouts-parameter
If `setLayoutCount` is not `0`, `pSetLayouts` **must** be a valid pointer to an array of `setLayoutCount` valid or `VK_NULL_HANDLE` `VkDescriptorSetLayout` handles
- VUID-VkPipelineLayoutCreateInfo-pPushConstantRanges-parameter
If `pushConstantRangeCount` is not `0`, `pPushConstantRanges` **must** be a valid pointer to an array of `pushConstantRangeCount` valid `VkPushConstantRange` structures

```
typedef enum VkPipelineLayoutCreateFlagBits {
} VkPipelineLayoutCreateFlagBits;
```

All values for this enum are defined by extensions.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineLayoutCreateFlags;
```

`VkPipelineLayoutCreateFlags` is a bitmask type for setting a mask of `VkPipelineLayoutCreateFlagBits`.

The `VkPushConstantRange` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPushConstantRange {
    VkShaderStageFlags    stageFlags;
    uint32_t              offset;
    uint32_t              size;
} VkPushConstantRange;
```

- `stageFlags` is a set of stage flags describing the shader stages that will access a range of push constants. If a particular stage is not included in the range, then accessing members of that range of push constants from the corresponding shader stage will return undefined values.
- `offset` and `size` are the start offset and size, respectively, consumed by the range. Both `offset` and `size` are in units of bytes and **must** be a multiple of 4. The layout of the push constant

variables is specified in the shader.

Valid Usage

- VUID-VkPushConstantRange-offset-00294
`offset` **must** be less than `VkPhysicalDeviceLimits::maxPushConstantsSize`
- VUID-VkPushConstantRange-offset-00295
`offset` **must** be a multiple of 4
- VUID-VkPushConstantRange-size-00296
`size` **must** be greater than 0
- VUID-VkPushConstantRange-size-00297
`size` **must** be a multiple of 4
- VUID-VkPushConstantRange-size-00298
`size` **must** be less than or equal to `VkPhysicalDeviceLimits::maxPushConstantsSize` minus `offset`

Valid Usage (Implicit)

- VUID-VkPushConstantRange-stageFlags-parameter
`stageFlags` **must** be a valid combination of `VkShaderStageFlagBits` values
- VUID-VkPushConstantRange-stageFlags-requiredbitmask
`stageFlags` **must** not be 0

Once created, pipeline layouts are used as part of pipeline creation (see [Pipelines](#)), as part of binding descriptor sets (see [Descriptor Set Binding](#)), and as part of setting push constants (see [Push Constant Updates](#)). Pipeline creation accepts a pipeline layout as input, and the layout **may** be used to map (set, binding, arrayElement) tuples to implementation resources or memory locations within a descriptor set. The assignment of implementation resources depends only on the bindings defined in the descriptor sets that comprise the pipeline layout, and not on any shader source.

All resource variables [statically used](#) in all shaders in a pipeline **must** be declared with a (set, binding, arrayElement) that exists in the corresponding descriptor set layout and is of an appropriate descriptor type and includes the set of shader stages it is used by in `stageFlags`. The pipeline layout **can** include entries that are not used by a particular pipeline. The pipeline layout allows the application to provide a consistent set of bindings across multiple pipeline compiles, which enables those pipelines to be compiled in a way that the implementation **may** cheaply switch pipelines without reprogramming the bindings.

Similarly, the push constant block declared in each shader (if present) **must** only place variables at offsets that are each included in a push constant range with `stageFlags` including the bit corresponding to the shader stage that uses it. The pipeline layout **can** include ranges or portions of ranges that are not used by a particular pipeline.

There is a limit on the total number of resources of each type that **can** be included in bindings in all

descriptor set layouts in a pipeline layout as shown in [Pipeline Layout Resource Limits](#). The “Total Resources Available” column gives the limit on the number of each type of resource that **can** be included in bindings in all descriptor sets in the pipeline layout. Some resource types count against multiple limits. Additionally, there are limits on the total number of each type of resource that **can** be used in any pipeline stage as described in [Shader Resource Limits](#).

Table 9. Pipeline Layout Resource Limits

Total Resources Available	Resource Types
maxDescriptorSetSamplers	sampler
	combined image sampler
maxDescriptorSetSampledImages	sampled image
	combined image sampler
	uniform texel buffer
maxDescriptorSetStorageImages	storage image
	storage texel buffer
maxDescriptorSetUniformBuffers	uniform buffer
	uniform buffer dynamic
maxDescriptorSetUniformBuffersDynamic	uniform buffer dynamic
maxDescriptorSetStorageBuffers	storage buffer
	storage buffer dynamic
maxDescriptorSetStorageBuffersDynamic	storage buffer dynamic
maxDescriptorSetInputAttachments	input attachment

To destroy a pipeline layout, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyPipelineLayout(
    VkDevice          device,
    VkPipelineLayout  pipelineLayout,
    const VkAllocationCallbacks* pAllocator);
```

- **device** is the logical device that destroys the pipeline layout.
- **pipelineLayout** is the pipeline layout to destroy.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyPipelineLayout-pipelineLayout-00299
If **VkAllocationCallbacks** were provided when **pipelineLayout** was created, a compatible set of callbacks **must** be provided here

- VUID-vkDestroyPipelineLayout-pipelineLayout-00300
If no `VkAllocationCallbacks` were provided when `pipelineLayout` was created, `pAllocator` **must** be `NULL`
- VUID-vkDestroyPipelineLayout-pipelineLayout-02004
`pipelineLayout` **must** not have been passed to any `vkCmd*` command for any command buffers that are still in the `recording state` when `vkDestroyPipelineLayout` is called

Valid Usage (Implicit)

- VUID-vkDestroyPipelineLayout-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkDestroyPipelineLayout-pipelineLayout-parameter
If `pipelineLayout` is not `VK_NULL_HANDLE`, `pipelineLayout` **must** be a valid `VkPipelineLayout` handle
- VUID-vkDestroyPipelineLayout-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkDestroyPipelineLayout-pipelineLayout-parent
If `pipelineLayout` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `pipelineLayout` **must** be externally synchronized

Pipeline Layout Compatibility

Two pipeline layouts are defined to be “compatible for `push constants`” if they were created with identical push constant ranges. Two pipeline layouts are defined to be “compatible for set N” if they were created with *identically defined* descriptor set layouts for sets zero through N, and if they were created with identical push constant ranges.

When binding a descriptor set (see [Descriptor Set Binding](#)) to set number N, a previously bound descriptor set bound with lower index M than N is disturbed if the pipeline layouts for set M and N are not compatible for set M. Otherwise, the bound descriptor set in M is not disturbed.

If, additionally, the previously bound descriptor set for set N was bound using a pipeline layout not compatible for set N, then all bindings in sets numbered greater than N are disturbed.

When binding a pipeline, the pipeline **can** correctly access any previously bound descriptor set N if it was bound with compatible pipeline layout for set N, and it was not disturbed.

Layout compatibility means that descriptor sets **can** be bound to a command buffer for use by any pipeline created with a compatible pipeline layout, and without having bound a particular pipeline

first. It also means that descriptor sets **can** remain valid across a pipeline change, and the same resources will be accessible to the newly bound pipeline.

When a descriptor set is disturbed by binding descriptor sets, the disturbed set is considered to contain undefined descriptors bound with the same pipeline layout as the disturbing descriptor set.

Implementor's Note

A consequence of layout compatibility is that when the implementation compiles a pipeline layout and maps pipeline resources to implementation resources, the mechanism for set N **should** only be a function of sets [0..N].

NOTE

Place the least frequently changing descriptor sets near the start of the pipeline layout, and place the descriptor sets representing the most frequently changing resources near the end. When pipelines are switched, only the descriptor set bindings that have been invalidated will need to be updated and the remainder of the descriptor set bindings will remain in place.

The maximum number of descriptor sets that **can** be bound to a pipeline layout is queried from physical device properties (see `maxBoundDescriptorSets` in [Limits](#)).

API Example

```
const VkDescriptorSetLayout layouts[] = { layout1, layout2 };

const VkPushConstantRange ranges[] =
{
    {
        .stageFlags = VK_SHADER_STAGE_VERTEX_BIT,
        .offset = 0,
        .size = 4
    },
    {
        .stageFlags = VK_SHADER_STAGE_FRAGMENT_BIT,
        .offset = 4,
        .size = 4
    },
};

const VkPipelineLayoutCreateInfo createInfo =
{
    .sType = VK_STRUCTURE_TYPE_PIPELINE_LAYOUT_CREATE_INFO,
    .pNext = NULL,
    .flags = 0,
    .setLayoutCount = 2,
    .pSetLayouts = layouts,
    .pushConstantRangeCount = 2,
    .pPushConstantRanges = ranges
};
```

```
VkPipelineLayout myPipelineLayout;
myResult = vkCreatePipelineLayout(
    myDevice,
    &createInfo,
    NULL,
    &myPipelineLayout);
```

14.2.3. Allocation of Descriptor Sets

A *descriptor pool* maintains a pool of descriptors, from which descriptor sets are allocated. Descriptor pools are externally synchronized, meaning that the application **must** not allocate and/or free descriptor sets from the same pool in multiple threads simultaneously.

Descriptor pools are represented by `VkDescriptorPool` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkDescriptorPool)
```

To create a descriptor pool object, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateDescriptorPool(
    VkDevice                                device,
    const VkDescriptorPoolCreateInfo*       pCreateInfo,
    const VkAllocationCallbacks*           pAllocator,
    VkDescriptorPool*                       pDescriptorPool);
```

- `device` is the logical device that creates the descriptor pool.
- `pCreateInfo` is a pointer to a `VkDescriptorPoolCreateInfo` structure specifying the state of the descriptor pool object.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.
- `pDescriptorPool` is a pointer to a `VkDescriptorPool` handle in which the resulting descriptor pool object is returned.

The created descriptor pool is returned in `pDescriptorPool`.

Valid Usage (Implicit)

- VUID-vkCreateDescriptorPool-device-parameter `device` **must** be a valid `VkDevice` handle
- VUID-vkCreateDescriptorPool-pCreateInfo-parameter `pCreateInfo` **must** be a valid pointer to a valid `VkDescriptorPoolCreateInfo` structure
- VUID-vkCreateDescriptorPool-pAllocator-parameter

If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure

- `VUID-vkCreateDescriptorPool-pDescriptorPool-parameter pDescriptorPool` **must** be a valid pointer to a `VkDescriptorPool` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

Additional information about the pool is passed in a `VkDescriptorPoolCreateInfo` structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkDescriptorPoolCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkDescriptorPoolCreateFlags flags;
    uint32_t           maxSets;
    uint32_t           poolSizeCount;
    const VkDescriptorPoolSize* pPoolSizes;
} VkDescriptorPoolCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkDescriptorPoolCreateFlagBits` specifying certain supported operations on the pool.
- `maxSets` is the maximum number of descriptor sets that **can** be allocated from the pool.
- `poolSizeCount` is the number of elements in `pPoolSizes`.
- `pPoolSizes` is a pointer to an array of `VkDescriptorPoolSize` structures, each containing a descriptor type and number of descriptors of that type to be allocated in the pool.

If multiple `VkDescriptorPoolSize` structures containing the same descriptor type appear in the `pPoolSizes` array then the pool will be created with enough storage for the total number of descriptors of each type.

Fragmentation of a descriptor pool is possible and **may** lead to descriptor set allocation failures. A failure due to fragmentation is defined as failing a descriptor set allocation despite the sum of all outstanding descriptor set allocations from the pool plus the requested allocation requiring no more than the total number of descriptors requested at pool creation. Implementations provide certain guarantees of when fragmentation **must** not cause allocation failure, as described below.

If a descriptor pool has not had any descriptor sets freed since it was created or most recently reset then fragmentation **must** not cause an allocation failure (note that this is always the case for a pool created without the `VK_DESCRIPTOR_POOL_CREATE_FREE_DESCRIPTOR_SET_BIT` bit set). Additionally, if all sets allocated from the pool since it was created or most recently reset use the same number of descriptors (of each type) and the requested allocation also uses that same number of descriptors (of each type), then fragmentation **must** not cause an allocation failure.

If an allocation failure occurs due to fragmentation, an application **can** create an additional descriptor pool to perform further descriptor set allocations.

Valid Usage

- VUID-VkDescriptorPoolCreateInfo-descriptorPoolOverallocation-09227
`maxSets` **must** be greater than 0

Valid Usage (Implicit)

- VUID-VkDescriptorPoolCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DESCRIPTOR_POOL_CREATE_INFO`
- VUID-VkDescriptorPoolCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkDescriptorPoolCreateInfo-flags-parameter
`flags` **must** be a valid combination of `VkDescriptorPoolCreateFlagBits` values
- VUID-VkDescriptorPoolCreateInfo-pPoolSizes-parameter
If `poolSizeCount` is not 0, `pPoolSizes` **must** be a valid pointer to an array of `poolSizeCount` valid `VkDescriptorPoolSize` structures

Bits which **can** be set in `VkDescriptorPoolCreateInfo::flags`, enabling operations on a descriptor pool, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkDescriptorPoolCreateFlagBits {
    VK_DESCRIPTOR_POOL_CREATE_FREE_DESCRIPTOR_SET_BIT = 0x00000001,
} VkDescriptorPoolCreateFlagBits;
```

- `VK_DESCRIPTOR_POOL_CREATE_FREE_DESCRIPTOR_SET_BIT` specifies that descriptor sets **can** return their individual allocations to the pool, i.e. all of `vkAllocateDescriptorSets`, `vkFreeDescriptorSets`, and `vkResetDescriptorPool` are allowed. Otherwise, descriptor sets allocated from the pool **must** not be individually freed back to the pool, i.e. only `vkAllocateDescriptorSets` and `vkResetDescriptorPool` are allowed.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkDescriptorPoolCreateFlags;
```

`VkDescriptorPoolCreateFlags` is a bitmask type for setting a mask of zero or more `VkDescriptorPoolCreateFlagBits`.

The `VkDescriptorPoolSize` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDescriptorPoolSize {
    VkDescriptorType    type;
    uint32_t            descriptorCount;
} VkDescriptorPoolSize;
```

- `type` is the type of descriptor.
- `descriptorCount` is the number of descriptors of that type to allocate.

Valid Usage

- VUID-VkDescriptorPoolSize-descriptorCount-00302
`descriptorCount` **must** be greater than 0

Valid Usage (Implicit)

- VUID-VkDescriptorPoolSize-type-parameter
`type` **must** be a valid `VkDescriptorType` value

To destroy a descriptor pool, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyDescriptorPool(
    VkDevice                device,
    VkDescriptorPool         descriptorPool,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that destroys the descriptor pool.
- `descriptorPool` is the descriptor pool to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

When a pool is destroyed, all descriptor sets allocated from the pool are implicitly freed and become invalid. Descriptor sets allocated from a given pool do not need to be freed before destroying that descriptor pool.

Valid Usage

- VUID-vkDestroyDescriptorPool-descriptorPool-00303

All submitted commands that refer to `descriptorPool` (via any allocated descriptor sets) **must** have completed execution

- VUID-vkDestroyDescriptorPool-descriptorPool-00304
If `VkAllocationCallbacks` were provided when `descriptorPool` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyDescriptorPool-descriptorPool-00305
If no `VkAllocationCallbacks` were provided when `descriptorPool` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyDescriptorPool-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkDestroyDescriptorPool-descriptorPool-parameter
If `descriptorPool` is not `VK_NULL_HANDLE`, `descriptorPool` **must** be a valid `VkDescriptorPool` handle
- VUID-vkDestroyDescriptorPool-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkDestroyDescriptorPool-descriptorPool-parent
If `descriptorPool` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `descriptorPool` **must** be externally synchronized

Descriptor sets are allocated from descriptor pool objects, and are represented by `VkDescriptorSet` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkDescriptorSet)
```

To allocate descriptor sets from a descriptor pool, call:

```
// Provided by VK_VERSION_1_0
VkResult vkAllocateDescriptorSets(
    VkDevice device,
    const VkDescriptorSetAllocateInfo* pAllocateInfo,
    VkDescriptorSet* pDescriptorSets);
```

- `device` is the logical device that owns the descriptor pool.

- `pAllocateInfo` is a pointer to a `VkDescriptorSetAllocateInfo` structure describing parameters of the allocation.
- `pDescriptorSets` is a pointer to an array of `VkDescriptorSet` handles in which the resulting descriptor set objects are returned.

The allocated descriptor sets are returned in `pDescriptorSets`.

When a descriptor set is allocated, the initial state is largely uninitialized and all descriptors are undefined, with the exception that samplers with a non-null `pImmutableSamplers` are initialized on allocation. Descriptors also become undefined if the underlying resource or view object is destroyed. Descriptor sets containing undefined descriptors **can** still be bound and used, subject to the following conditions:

- Descriptors that are **statically used** **must** have been populated before the descriptor set is **consumed**.
- Entries that are not used by a pipeline **can** have undefined descriptors.

If an allocation fails due to fragmentation, an indeterminate error is returned with an unspecified error code. Any returned error other than `VK_ERROR_FRAGMENTED_POOL` does not imply its usual meaning: applications **should** assume that the allocation failed due to fragmentation, and create a new descriptor pool.

NOTE

Applications **should** check for a negative return value when allocating new descriptor sets, assume that any error effectively means `VK_ERROR_FRAGMENTED_POOL`, and try to create a new descriptor pool. If `VK_ERROR_FRAGMENTED_POOL` is the actual return value, it adds certainty to that decision.

The reason for this is that `VK_ERROR_FRAGMENTED_POOL` was only added in a later version of the 1.0 specification, and so drivers **may** return other errors if they were written against earlier versions. To ensure full compatibility with earlier patch versions, these other errors are allowed.

Valid Usage (Implicit)

- VUID-vkAllocateDescriptorSets-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkAllocateDescriptorSets-pAllocateInfo-parameter
`pAllocateInfo` **must** be a valid pointer to a valid `VkDescriptorSetAllocateInfo` structure
- VUID-vkAllocateDescriptorSets-pDescriptorSets-parameter
`pDescriptorSets` **must** be a valid pointer to an array of `pAllocateInfo->descriptorSetCount` `VkDescriptorSet` handles
- VUID-vkAllocateDescriptorSets-pAllocateInfo::descriptorSetCount-arraylength
`pAllocateInfo->descriptorSetCount` **must** be greater than 0

Host Synchronization

- Host access to `pAllocateInfo->descriptorPool` **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_FRAGMENTED_POOL`

The `VkDescriptorSetAllocateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDescriptorSetAllocateInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkDescriptorPool      descriptorPool;
    uint32_t             descriptorSetCount;
    const VkDescriptorSetLayout* pSetLayouts;
} VkDescriptorSetAllocateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `descriptorPool` is the pool which the sets will be allocated from.
- `descriptorSetCount` determines the number of descriptor sets to be allocated from the pool.
- `pSetLayouts` is a pointer to an array of descriptor set layouts, with each member specifying how the corresponding descriptor set is allocated.

Valid Usage

- VUID-VkDescriptorSetAllocateInfo-apiVersion-07895
`descriptorSetCount` **must** not be greater than the number of sets that are currently available for allocation in `descriptorPool`
- VUID-VkDescriptorSetAllocateInfo-apiVersion-07896
`descriptorPool` **must** have enough free descriptor capacity remaining to allocate the descriptor sets of the specified layouts

Valid Usage (Implicit)

- VUID-VkDescriptorSetAllocateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DESCRIPTOR_SET_ALLOCATE_INFO`
- VUID-VkDescriptorSetAllocateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkDescriptorSetAllocateInfo-descriptorPool-parameter
`descriptorPool` **must** be a valid `VkDescriptorPool` handle
- VUID-VkDescriptorSetAllocateInfo-pSetLayouts-parameter
`pSetLayouts` **must** be a valid pointer to an array of `descriptorSetCount` valid `VkDescriptorSetLayout` handles
- VUID-VkDescriptorSetAllocateInfo-descriptorSetCount-arraylength
`descriptorSetCount` **must** be greater than `0`
- VUID-VkDescriptorSetAllocateInfo-commonparent
Both of `descriptorPool`, and the elements of `pSetLayouts` **must** have been created, allocated, or retrieved from the same `VkDevice`

To free allocated descriptor sets, call:

```
// Provided by VK_VERSION_1_0
VkResult vkFreeDescriptorSets(
    VkDevice          device,
    VkDescriptorPool   descriptorPool,
    uint32_t           descriptorSetCount,
    const VkDescriptorSet* pDescriptorSets);
```

- `device` is the logical device that owns the descriptor pool.
- `descriptorPool` is the descriptor pool from which the descriptor sets were allocated.
- `descriptorSetCount` is the number of elements in the `pDescriptorSets` array.
- `pDescriptorSets` is a pointer to an array of handles to `VkDescriptorSet` objects.

After calling `vkFreeDescriptorSets`, all descriptor sets in `pDescriptorSets` are invalid.

Valid Usage

- VUID-vkFreeDescriptorSets-pDescriptorSets-00309
All submitted commands that refer to any element of `pDescriptorSets` **must** have completed execution
- VUID-vkFreeDescriptorSets-pDescriptorSets-00310
`pDescriptorSets` **must** be a valid pointer to an array of `descriptorSetCount` `VkDescriptorSet` handles, each element of which **must** either be a valid handle or `VK_NULL_HANDLE`
- VUID-vkFreeDescriptorSets-descriptorPool-00312

`descriptorPool` **must** have been created with the `VK_DESCRIPTOR_POOL_CREATE_FREE_DESCRIPTOR_SET_BIT` flag

Valid Usage (Implicit)

- VUID-vkFreeDescriptorSets-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkFreeDescriptorSets-descriptorPool-parameter
`descriptorPool` **must** be a valid `VkDescriptorPool` handle
- VUID-vkFreeDescriptorSets-descriptorSetCount-arraylength
`descriptorSetCount` **must** be greater than 0
- VUID-vkFreeDescriptorSets-descriptorPool-parent
`descriptorPool` **must** have been created, allocated, or retrieved from `device`
- VUID-vkFreeDescriptorSets-pDescriptorSets-parent
Each element of `pDescriptorSets` that is a valid handle **must** have been created, allocated, or retrieved from `descriptorPool`

Host Synchronization

- Host access to `descriptorPool` **must** be externally synchronized
- Host access to each member of `pDescriptorSets` **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

None

To return all descriptor sets allocated from a given pool to the pool, rather than freeing individual descriptor sets, call:

```
// Provided by VK_VERSION_1_0
VkResult vkResetDescriptorPool(
    VkDevice          device,
    VkDescriptorPool   descriptorPool,
    VkDescriptorPoolResetFlags flags);
```

- `device` is the logical device that owns the descriptor pool.
- `descriptorPool` is the descriptor pool to be reset.

- `flags` is reserved for future use.

Resetting a descriptor pool recycles all of the resources from all of the descriptor sets allocated from the descriptor pool back to the descriptor pool, and the descriptor sets are implicitly freed.

Valid Usage

- VUID-vkResetDescriptorPool-descriptorPool-00313
All uses of `descriptorPool` (via any allocated descriptor sets) **must** have completed execution

Valid Usage (Implicit)

- VUID-vkResetDescriptorPool-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkResetDescriptorPool-descriptorPool-parameter
`descriptorPool` **must** be a valid `VkDescriptorPool` handle
- VUID-vkResetDescriptorPool-flags-zeroBitmask
`flags` **must** be 0
- VUID-vkResetDescriptorPool-descriptorPool-parent
`descriptorPool` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `descriptorPool` **must** be externally synchronized
- Host access to any `VkDescriptorSet` objects allocated from `descriptorPool` **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

None

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkDescriptorPoolResetFlags;
```

`VkDescriptorPoolResetFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

14.2.4. Descriptor Set Updates

Once allocated, descriptor sets **can** be updated with a combination of write and copy operations. To update descriptor sets, call:

```
// Provided by VK_VERSION_1_0
void vkUpdateDescriptorSets(
    VkDevice                                device,
    uint32_t                                descriptorWriteCount,
    const VkWriteDescriptorSet*              pDescriptorWrites,
    uint32_t                                descriptorCopyCount,
    const VkCopyDescriptorSet*               pDescriptorCopies);
```

- **device** is the logical device that updates the descriptor sets.
- **descriptorWriteCount** is the number of elements in the **pDescriptorWrites** array.
- **pDescriptorWrites** is a pointer to an array of **VkWriteDescriptorSet** structures describing the descriptor sets to write to.
- **descriptorCopyCount** is the number of elements in the **pDescriptorCopies** array.
- **pDescriptorCopies** is a pointer to an array of **VkCopyDescriptorSet** structures describing the descriptor sets to copy between.

The operations described by **pDescriptorWrites** are performed first, followed by the operations described by **pDescriptorCopies**. Within each array, the operations are performed in the order they appear in the array.

Each element in the **pDescriptorWrites** array describes an operation updating the descriptor set using descriptors for resources specified in the structure.

Each element in the **pDescriptorCopies** array is a **VkCopyDescriptorSet** structure describing an operation copying descriptors between sets.

If the **dstSet** member of any element of **pDescriptorWrites** or **pDescriptorCopies** is bound, accessed, or modified by any command that was recorded to a command buffer which is currently in the **recording or executable state**, that command buffer becomes **invalid**.

Valid Usage

- VUID-vkUpdateDescriptorSets-pDescriptorWrites-06236
For each element **i** where **pDescriptorWrites[i].descriptorType** is **VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER** or **VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER**, elements of the **pTexelBufferView** member of **pDescriptorWrites[i]** **must** have been created on **device**
- VUID-vkUpdateDescriptorSets-pDescriptorWrites-06237
For each element **i** where **pDescriptorWrites[i].descriptorType** is **VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER**, **VK_DESCRIPTOR_TYPE_STORAGE_BUFFER**, **VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC**, or

VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC, the `buffer` member of any element of the `pBufferInfo` member of `pDescriptorWrites[i]` **must** have been created on `device`

- VUID-vkUpdateDescriptorSets-pDescriptorWrites-06238
For each element `i` where `pDescriptorWrites[i].descriptorType` is `VK_DESCRIPTOR_TYPE_SAMPLER` or `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, and `dstSet` was not allocated with a layout that included immutable samplers for `dstBinding` with `descriptorType`, the `sampler` member of any element of the `pImageInfo` member of `pDescriptorWrites[i]` **must** have been created on `device`
- VUID-vkUpdateDescriptorSets-pDescriptorWrites-06239
For each element `i` where `pDescriptorWrites[i].descriptorType` is `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT`, or `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER` the `imageView` member of any element of `pDescriptorWrites[i]` **must** have been created on `device`
- VUID-vkUpdateDescriptorSets-pDescriptorWrites-06493
For each element `i` where `pDescriptorWrites[i].descriptorType` is `VK_DESCRIPTOR_TYPE_SAMPLER`, `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, or `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT`, `pDescriptorWrites[i].pImageInfo` **must** be a valid pointer to an array of `pDescriptorWrites[i].descriptorCount` valid `VkDescriptorImageInfo` structures
- VUID-vkUpdateDescriptorSets-None-03047
The `dstSet` member of each element of `pDescriptorWrites` or `pDescriptorCopies` **must** not be used by any command that was recorded to a command buffer which is in the `pending state`
- VUID-vkUpdateDescriptorSets-pDescriptorWrites-06993
Host access to `pDescriptorWrites[i].dstSet` and `pDescriptorCopies[i].dstSet` **must** be `externally synchronized`

Valid Usage (Implicit)

- VUID-vkUpdateDescriptorSets-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkUpdateDescriptorSets-pDescriptorWrites-parameter
If `descriptorWriteCount` is not `0`, `pDescriptorWrites` **must** be a valid pointer to an array of `descriptorWriteCount` valid `VkWriteDescriptorSet` structures
- VUID-vkUpdateDescriptorSets-pDescriptorCopies-parameter
If `descriptorCopyCount` is not `0`, `pDescriptorCopies` **must** be a valid pointer to an array of `descriptorCopyCount` valid `VkCopyDescriptorSet` structures

The `VkWriteDescriptorSet` structure is defined as:


```
// Provided by VK_VERSION_1_0
typedef struct VkWriteDescriptorSet {
    VkStructureType      sType;
    const void*          pNext;
    VkDescriptorSet       dstSet;
    uint32_t             dstBinding;
    uint32_t             dstArrayElement;
    uint32_t             descriptorCount;
    VkDescriptorType      descriptorType;
    const VkDescriptorImageInfo* pImageInfo;
    const VkDescriptorBufferInfo* pBufferInfo;
    const VkBufferView*    pTexelBufferView;
} VkWriteDescriptorSet;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `dstSet` is the destination descriptor set to update.
- `dstBinding` is the descriptor binding within that set.
- `dstArrayElement` is the starting element in that array.
- `descriptorCount` is the number of descriptors to update. `descriptorCount` is one of
 - the number of elements in `pImageInfo`
 - the number of elements in `pBufferInfo`
 - the number of elements in `pTexelBufferView`
- `descriptorType` is a `VkDescriptorType` specifying the type of each descriptor in `pImageInfo`, `pBufferInfo`, or `pTexelBufferView`, as described below. It **must** be the same type as the `descriptorType` specified in `VkDescriptorSetLayoutBinding` for `dstSet` at `dstBinding`. The type of the descriptor also controls which array the descriptors are taken from.
- `pImageInfo` is a pointer to an array of `VkDescriptorImageInfo` structures or is ignored, as described below.
- `pBufferInfo` is a pointer to an array of `VkDescriptorBufferInfo` structures or is ignored, as described below.
- `pTexelBufferView` is a pointer to an array of `VkBufferView` handles as described in the [Buffer Views](#) section or is ignored, as described below.

Only one of `pImageInfo`, `pBufferInfo`, or `pTexelBufferView` members is used according to the descriptor type specified in the `descriptorType` member of the containing `VkWriteDescriptorSet` structure, as specified below.

Consecutive Binding Updates

If the `dstBinding` has fewer than `descriptorCount` array elements remaining starting from `dstArrayElement`, then the remainder will be used to update the subsequent binding - `dstBinding+1` starting at array element zero. If a binding has a `descriptorCount` of zero, it is skipped. This behavior applies recursively, with the update affecting consecutive bindings as needed to update all

`descriptorCount` descriptors. Consecutive bindings **must** have identical `VkDescriptorType`, `VkShaderStageFlags`, and immutable samplers references.

Valid Usage

- VUID-VkWriteDescriptorSet-dstBinding-00315
`dstBinding` **must** be less than or equal to the maximum value of `binding` of all `VkDescriptorSetLayoutBinding` structures specified when `dstSet`'s descriptor set layout was created
- VUID-VkWriteDescriptorSet-dstBinding-00316
`dstBinding` **must** be a binding with a non-zero `descriptorCount`
- VUID-VkWriteDescriptorSet-dstBinding-10009
`dstBinding` **must** be a binding with a non-zero `VkDescriptorSetLayoutCreateInfo::bindingCount`
- VUID-VkWriteDescriptorSet-descriptorCount-00317
All consecutive bindings updated via a single `VkWriteDescriptorSet` structure, except those with a `descriptorCount` of zero, **must** have identical `descriptorType` and `stageFlags`
- VUID-VkWriteDescriptorSet-descriptorCount-00318
All consecutive bindings updated via a single `VkWriteDescriptorSet` structure, except those with a `descriptorCount` of zero, **must** all either use immutable samplers or **must** all not use immutable samplers
- VUID-VkWriteDescriptorSet-descriptorType-00319
`descriptorType` **must** match the type of `dstBinding` within `dstSet`
- VUID-VkWriteDescriptorSet-dstSet-00320
`dstSet` **must** be a valid `VkDescriptorSet` handle
- VUID-VkWriteDescriptorSet-dstArrayElement-00321
The sum of `dstArrayElement` and `descriptorCount` **must** be less than or equal to the number of array elements in the descriptor set binding specified by `dstBinding`, and all applicable consecutive bindings
- VUID-VkWriteDescriptorSet-descriptorType-02994
If `descriptorType` is `VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER` or `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER`, each element of `pTexelBufferView` **must** be either a valid `VkBufferView` handle or `VK_NULL_HANDLE`
- VUID-VkWriteDescriptorSet-descriptorType-02995
If `descriptorType` is `VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER` or `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` and the `nullDescriptor` feature is not enabled, each element of `pTexelBufferView` **must** not be `VK_NULL_HANDLE`
- VUID-VkWriteDescriptorSet-descriptorType-00324
If `descriptorType` is `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER`, `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER`, `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC`, or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC`, `pBufferInfo` **must** be a valid pointer to an array of `descriptorCount` valid `VkDescriptorBufferInfo` structures
- VUID-VkWriteDescriptorSet-descriptorType-00325

If `descriptorType` is `VK_DESCRIPTOR_TYPE_SAMPLER` or `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, and `dstSet` was not allocated with a layout that included immutable samplers for `dstBinding` with `descriptorType`, the `sampler` member of each element of `pImageInfo` **must** be a valid `VkSampler` object

- VUID-VkWriteDescriptorSet-descriptorType-02996

If `descriptorType` is `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, or `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, the `imageView` member of each element of `pImageInfo` **must** be either a valid `VkImageView` handle or `VK_NULL_HANDLE`

- VUID-VkWriteDescriptorSet-descriptorType-02997

If `descriptorType` is `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, or `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, and the `nullDescriptor` feature is not enabled, the `imageView` member of each element of `pImageInfo` **must** not be `VK_NULL_HANDLE`

- VUID-VkWriteDescriptorSet-descriptorType-07683

If `descriptorType` is `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT`, the `imageView` member of each element of `pImageInfo` **must** not be `VK_NULL_HANDLE`

- VUID-VkWriteDescriptorSet-descriptorType-00327

If `descriptorType` is `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER` or `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC`, the `offset` member of each element of `pBufferInfo` **must** be a multiple of `VkPhysicalDeviceLimits::minUniformBufferOffsetAlignment`

- VUID-VkWriteDescriptorSet-descriptorType-00328

If `descriptorType` is `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER` or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC`, the `offset` member of each element of `pBufferInfo` **must** be a multiple of `VkPhysicalDeviceLimits::minStorageBufferOffsetAlignment`

- VUID-VkWriteDescriptorSet-descriptorType-00329

If `descriptorType` is `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER`, `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC`, `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER`, or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC`, and the `buffer` member of any element of `pBufferInfo` is the handle of a non-sparse buffer, then that buffer **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-VkWriteDescriptorSet-descriptorType-00330

If `descriptorType` is `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER` or `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC`, the `buffer` member of each element of `pBufferInfo` **must** have been created with `VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT` set

- VUID-VkWriteDescriptorSet-descriptorType-00331

If `descriptorType` is `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER` or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC`, the `buffer` member of each element of `pBufferInfo` **must** have been created with `VK_BUFFER_USAGE_STORAGE_BUFFER_BIT` set

- VUID-VkWriteDescriptorSet-descriptorType-00332

If `descriptorType` is `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER` or `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC`, the `range` member of each element of

pBufferInfo, or the [effective range](#) if range is VK_WHOLE_SIZE, **must** be less than or equal to VkPhysicalDeviceLimits::maxUniformBufferRange

- VUID-VkWriteDescriptorSet-descriptorType-00333

If [descriptorType](#) is VK_DESCRIPTOR_TYPE_STORAGE_BUFFER or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC, the [range](#) member of each element of pBufferInfo, or the [effective range](#) if range is VK_WHOLE_SIZE, **must** be less than or equal to VkPhysicalDeviceLimits::maxStorageBufferRange

- VUID-VkWriteDescriptorSet-descriptorType-08765

If [descriptorType](#) is VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER, the pTexelBufferView [buffer view usage](#) **must** include VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT

- VUID-VkWriteDescriptorSet-descriptorType-08766

If [descriptorType](#) is VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER, the pTexelBufferView [buffer view usage](#) **must** include VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT

- VUID-VkWriteDescriptorSet-descriptorType-00336

If [descriptorType](#) is VK_DESCRIPTOR_TYPE_STORAGE_IMAGE or VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT, the [imageView](#) member of each element of pImageInfo **must** have been created with the [identity swizzle](#)

- VUID-VkWriteDescriptorSet-descriptorType-00337

If [descriptorType](#) is VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE or VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER, the [imageView](#) member of each element of pImageInfo **must** have been created with VK_IMAGE_USAGE_SAMPLED_BIT set

- VUID-VkWriteDescriptorSet-descriptorType-04149

If [descriptorType](#) is VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE the [imageLayout](#) member of each element of pImageInfo **must** be a member of the list given in [Sampled Image](#)

- VUID-VkWriteDescriptorSet-descriptorType-04150

If [descriptorType](#) is VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER the [imageLayout](#) member of each element of pImageInfo **must** be a member of the list given in [Combined Image Sampler](#)

- VUID-VkWriteDescriptorSet-descriptorType-04151

If [descriptorType](#) is VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT the [imageLayout](#) member of each element of pImageInfo **must** be a member of the list given in [Input Attachment](#)

- VUID-VkWriteDescriptorSet-descriptorType-04152

If [descriptorType](#) is VK_DESCRIPTOR_TYPE_STORAGE_IMAGE the [imageLayout](#) member of each element of pImageInfo **must** be a member of the list given in [Storage Image](#)

- VUID-VkWriteDescriptorSet-descriptorType-00338

If [descriptorType](#) is VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT, the [imageView](#) member of each element of pImageInfo **must** have been created with VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT set

- VUID-VkWriteDescriptorSet-descriptorType-00339

If [descriptorType](#) is VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, the [imageView](#) member of each element of pImageInfo **must** have been created with VK_IMAGE_USAGE_STORAGE_BIT set

- VUID-VkWriteDescriptorSet-descriptorType-02752

If [descriptorType](#) is VK_DESCRIPTOR_TYPE_SAMPLER, then [dstSet](#) **must** not have been allocated

with a layout that included immutable samplers for `dstBinding`

Valid Usage (Implicit)

- `VUID-VkWriteDescriptorSet-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_WRITE_DESCRIPTOR_SET`
- `VUID-VkWriteDescriptorSet-pNext-pNext`
`pNext` **must** be `NULL`
- `VUID-VkWriteDescriptorSet-descriptorType-parameter`
`descriptorType` **must** be a valid `VkDescriptorType` value
- `VUID-VkWriteDescriptorSet-descriptorCount-arraylength`
`descriptorCount` **must** be greater than `0`
- `VUID-VkWriteDescriptorSet-commonparent`
Both of `dstSet`, and the elements of `pTexelBufferView` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

The type of descriptors in a descriptor set is specified by `VkWriteDescriptorSet::descriptorType`, which **must** be one of the values:

```
// Provided by VK_VERSION_1_0
typedef enum VkDescriptorType {
    VK_DESCRIPTOR_TYPE_SAMPLER = 0,
    VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER = 1,
    VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE = 2,
    VK_DESCRIPTOR_TYPE_STORAGE_IMAGE = 3,
    VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER = 4,
    VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER = 5,
    VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER = 6,
    VK_DESCRIPTOR_TYPE_STORAGE_BUFFER = 7,
    VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC = 8,
    VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC = 9,
    VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT = 10,
} VkDescriptorType;
```

- `VK_DESCRIPTOR_TYPE_SAMPLER` specifies a `sampler descriptor`.
- `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER` specifies a `combined image sampler descriptor`.
- `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE` specifies a `sampled image descriptor`.
- `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE` specifies a `storage image descriptor`.
- `VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER` specifies a `uniform texel buffer descriptor`.
- `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` specifies a `storage texel buffer descriptor`.
- `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER` specifies a `uniform buffer descriptor`.
- `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER` specifies a `storage buffer descriptor`.

- `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` specifies a [dynamic uniform buffer descriptor](#).
- `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC` specifies a [dynamic storage buffer descriptor](#).
- `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT` specifies an [input attachment descriptor](#).

When a descriptor set is updated via elements of `VkWriteDescriptorSet`, members of `pImageInfo`, `pBufferInfo` and `pTexelBufferView` are only accessed by the implementation when they correspond to descriptor type being defined - otherwise they are ignored. The members accessed are as follows for each descriptor type:

- For `VK_DESCRIPTOR_TYPE_SAMPLER`, only the `sampler` member of each element of `VkWriteDescriptorSet::pImageInfo` is accessed.
- For `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, or `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT`, only the `imageView` and `imageLayout` members of each element of `VkWriteDescriptorSet::pImageInfo` are accessed.
- For `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, all members of each element of `VkWriteDescriptorSet::pImageInfo` are accessed.
- For `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER`, `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER`, `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC`, or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC`, all members of each element of `VkWriteDescriptorSet::pBufferInfo` are accessed.
- For `VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER` or `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER`, each element of `VkWriteDescriptorSet::pTexelBufferView` is accessed.

The `VkDescriptorBufferInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDescriptorBufferInfo {
    VkBuffer      buffer;
    VkDeviceSize  offset;
    VkDeviceSize  range;
} VkDescriptorBufferInfo;
```

- `buffer` is the buffer resource.
- `offset` is the offset in bytes from the start of `buffer`. Access to buffer memory via this descriptor uses addressing that is relative to this starting offset.
- `range` is the size in bytes that is used for this descriptor update, or `VK_WHOLE_SIZE` to use the range from `offset` to the end of the buffer.

NOTE

When setting `range` to `VK_WHOLE_SIZE`, the [effective range](#) **must** not be larger than the maximum range for the descriptor type (`maxUniformBufferRange` or `maxStorageBufferRange`). This means that `VK_WHOLE_SIZE` is not typically useful in the common case where uniform buffer descriptors are suballocated from a buffer that is much larger than `maxUniformBufferRange`.

For `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` and `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC` descriptor types, `offset` is the base offset from which the dynamic offset is applied and `range` is the

static size used for all dynamic offsets.

When `range` is `VK_WHOLE_SIZE` the effective range is calculated at `vkUpdateDescriptorSets` is by taking the size of `buffer` minus the `offset`.

Valid Usage

- VUID-VkDescriptorBufferInfo-offset-00340
`offset` **must** be less than the size of `buffer`
- VUID-VkDescriptorBufferInfo-range-00341
If `range` is not equal to `VK_WHOLE_SIZE`, `range` **must** be greater than 0
- VUID-VkDescriptorBufferInfo-range-00342
If `range` is not equal to `VK_WHOLE_SIZE`, `range` **must** be less than or equal to the size of `buffer` minus `offset`
- VUID-VkDescriptorBufferInfo-buffer-02998
If the `nullDescriptor` feature is not enabled, `buffer` **must** not be `VK_NULL_HANDLE`

Valid Usage (Implicit)

- VUID-VkDescriptorBufferInfo-buffer-parameter
If `buffer` is not `VK_NULL_HANDLE`, `buffer` **must** be a valid `VkBuffer` handle

The `VkDescriptorImageInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDescriptorImageInfo {
    VkSampler      sampler;
    VkImageView    imageView;
    VkImageLayout  imageLayout;
} VkDescriptorImageInfo;
```

- `sampler` is a sampler handle, and is used in descriptor updates for types `VK_DESCRIPTOR_TYPE_SAMPLER` and `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER` if the binding being updated does not use immutable samplers.
- `imageView` is an image view handle, and is used in descriptor updates for types `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, and `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT`.
- `imageLayout` is the layout that the image subresources accessible from `imageView` will be in at the time this descriptor is accessed. `imageLayout` is used in descriptor updates for types `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, and `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT`.

Members of `VkDescriptorImageInfo` that are not used in an update (as described above) are ignored.

Valid Usage

- VUID-VkDescriptorImageInfo-imageView-01976

If `imageView` is created from a depth/stencil image, the `aspectMask` used to create the `imageView` **must** include either `VK_IMAGE_ASPECT_DEPTH_BIT` or `VK_IMAGE_ASPECT_STENCIL_BIT` but not both

- VUID-VkDescriptorImageInfo-imageLayout-09425

If `imageLayout` is `VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL`, then the `aspectMask` used to create `imageView` **must** not include either `VK_IMAGE_ASPECT_DEPTH_BIT` or `VK_IMAGE_ASPECT_STENCIL_BIT`

- VUID-VkDescriptorImageInfo-imageLayout-09426

If `imageLayout` is `VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`, then the `aspectMask` used to create `imageView` **must** not include `VK_IMAGE_ASPECT_COLOR_BIT`

- VUID-VkDescriptorImageInfo-imageLayout-00344

`imageLayout` **must** match the actual `VkImageLayout` of each subresource accessible from `imageView` at the time this descriptor is accessed as defined by the [image layout matching rules](#)

Valid Usage (Implicit)

- VUID-VkDescriptorImageInfo-commonparent

Both of `imageView`, and `sampler` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

The `VkCopyDescriptorSet` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkCopyDescriptorSet {
    VkStructureType    sType;
    const void*        pNext;
    VkDescriptorSet     srcSet;
    uint32_t            srcBinding;
    uint32_t            srcArrayElement;
    VkDescriptorSet     dstSet;
    uint32_t            dstBinding;
    uint32_t            dstArrayElement;
    uint32_t            descriptorCount;
} VkCopyDescriptorSet;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcSet`, `srcBinding`, and `srcArrayElement` are the source set, binding, and array element,

respectively.

- `dstSet`, `dstBinding`, and `dstArrayElement` are the destination set, binding, and array element, respectively.
- `descriptorCount` is the number of descriptors to copy from the source to destination. If `descriptorCount` is greater than the number of remaining array elements in the source or destination binding, those affect consecutive bindings in a manner similar to `VkWriteDescriptorSet` above.

Valid Usage

- VUID-VkCopyDescriptorSet-srcBinding-00345
`srcBinding` **must** be a valid binding within `srcSet`
- VUID-VkCopyDescriptorSet-srcArrayElement-00346
The sum of `srcArrayElement` and `descriptorCount` **must** be less than or equal to the number of array elements in the descriptor set binding specified by `srcBinding`, and all applicable consecutive bindings
- VUID-VkCopyDescriptorSet-dstBinding-00347
`dstBinding` **must** be a valid binding within `dstSet`
- VUID-VkCopyDescriptorSet-dstArrayElement-00348
The sum of `dstArrayElement` and `descriptorCount` **must** be less than or equal to the number of array elements in the descriptor set binding specified by `dstBinding`, and all applicable consecutive bindings
- VUID-VkCopyDescriptorSet-dstBinding-02632
The type of `dstBinding` within `dstSet` **must** be equal to the type of `srcBinding` within `srcSet`
- VUID-VkCopyDescriptorSet-srcSet-00349
If `srcSet` is equal to `dstSet`, then the source and destination ranges of descriptors **must** not overlap, where the ranges **may** include array elements from consecutive bindings
- VUID-VkCopyDescriptorSet-dstBinding-02753
If the descriptor type of the descriptor set binding specified by `dstBinding` is `VK_DESCRIPTOR_TYPE_SAMPLER`, then `dstSet` **must** not have been allocated with a layout that included immutable samplers for `dstBinding`

Valid Usage (Implicit)

- VUID-VkCopyDescriptorSet-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_COPY_DESCRIPTOR_SET`
- VUID-VkCopyDescriptorSet-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkCopyDescriptorSet-srcSet-parameter
`srcSet` **must** be a valid `VkDescriptorSet` handle
- VUID-VkCopyDescriptorSet-dstSet-parameter
`dstSet` **must** be a valid `VkDescriptorSet` handle

- `VUID-VkCopyDescriptorSet-commonparent`
Both of `dstSet`, and `srcSet` **must** have been created, allocated, or retrieved from the same `VkDevice`

14.2.5. Descriptor Set Binding

To bind one or more descriptor sets to a command buffer, call:

```
// Provided by VK_VERSION_1_0
void vkCmdBindDescriptorSets(
    VkCommandBuffer          commandBuffer,
    VkPipelineBindPoint      pipelineBindPoint,
    VkPipelineLayout         layout,
    uint32_t                 firstSet,
    uint32_t                 descriptorSetCount,
    const VkDescriptorSet*   pDescriptorSets,
    uint32_t                 dynamicOffsetCount,
    const uint32_t*          pDynamicOffsets);
```

- `commandBuffer` is the command buffer that the descriptor sets will be bound to.
- `pipelineBindPoint` is a `VkPipelineBindPoint` indicating the type of the pipeline that will use the descriptors. There is a separate set of bind points for each pipeline type, so binding one does not disturb the others.
- `layout` is a `VkPipelineLayout` object used to program the bindings.
- `firstSet` is the set number of the first descriptor set to be bound.
- `descriptorSetCount` is the number of elements in the `pDescriptorSets` array.
- `pDescriptorSets` is a pointer to an array of handles to `VkDescriptorSet` objects describing the descriptor sets to bind to.
- `dynamicOffsetCount` is the number of dynamic offsets in the `pDynamicOffsets` array.
- `pDynamicOffsets` is a pointer to an array of `uint32_t` values specifying dynamic offsets.

`vkCmdBindDescriptorSets` binds descriptor sets `pDescriptorSets[0..descriptorSetCount-1]` to set numbers `[firstSet..firstSet+descriptorSetCount-1]` for subsequent `bound pipeline commands` set by `pipelineBindPoint`. Any bindings that were previously applied via these sets are no longer valid.

Once bound, a descriptor set affects rendering of subsequent commands that interact with the given pipeline type in the command buffer until either a different set is bound to the same set number, or the set is disturbed as described in [Pipeline Layout Compatibility](#).

A compatible descriptor set **must** be bound for all set numbers that any shaders in a pipeline access, at the time that a drawing or dispatching command is recorded to execute using that pipeline. However, if none of the shaders in a pipeline statically use any bindings with a particular set number, then no descriptor set need be bound for that set number, even if the pipeline layout includes a non-trivial descriptor set layout for that set number.

When consuming a descriptor, a descriptor is considered valid if the descriptor is not undefined as described by [descriptor set allocation](#). A descriptor that was disturbed by [Pipeline Layout Compatibility](#), or was never bound by `vkCmdBindDescriptorSets` is not considered valid. If a pipeline accesses a descriptor either statically or dynamically depending on the `VkDescriptorBindingFlagBits`, the consuming descriptor type in the pipeline **must** match the `VkDescriptorType` in `VkDescriptorSetLayoutCreateInfo` for the descriptor to be considered valid.

NOTE

Further validation may be carried out beyond validation for descriptor types, e.g. [Texel Input Validation](#).

If any of the sets being bound include dynamic uniform or storage buffers, then `pDynamicOffsets` includes one element for each array element in each dynamic descriptor type binding in each set. Values are taken from `pDynamicOffsets` in an order such that all entries for set *N* come before set *N*+1; within a set, entries are ordered by the binding numbers in the descriptor set layouts; and within a binding array, elements are in order. `dynamicOffsetCount` **must** equal the total number of dynamic descriptors in the sets being bound.

The effective offset used for dynamic uniform and storage buffer bindings is the sum of the relative offset taken from `pDynamicOffsets`, and the base address of the buffer plus base offset in the descriptor set. The range of the dynamic uniform and storage buffer bindings is the buffer range as specified in the descriptor set.

Each of the `pDescriptorSets` **must** be compatible with the pipeline layout specified by `layout`. The layout used to program the bindings **must** also be compatible with the pipeline used in subsequent [bound pipeline commands](#) with that pipeline type, as defined in the [Pipeline Layout Compatibility](#) section.

The descriptor set contents bound by a call to `vkCmdBindDescriptorSets` **may** be consumed at the following times:

- during host execution of the command, or during shader execution of the resulting draws and dispatches, or any time in between.

Thus, the contents of a descriptor set binding **must** not be altered (overwritten by an update command, or freed) between the first point in time that it **may** be consumed, and when the command completes executing on the queue.

The contents of `pDynamicOffsets` are consumed immediately during execution of `vkCmdBindDescriptorSets`. Once all pending uses have completed, it is legal to update and reuse a descriptor set.

Valid Usage

- VUID-vkCmdBindDescriptorSets-pDescriptorSets-00358
Each element of `pDescriptorSets` **must** have been allocated with a `VkDescriptorSetLayout` that matches (is the same as, or identically defined as) the `VkDescriptorSetLayout` at set *n* in `layout`, where *n* is the sum of `firstSet` and the index into `pDescriptorSets`
- VUID-vkCmdBindDescriptorSets-dynamicOffsetCount-00359

`dynamicOffsetCount` **must** be equal to the total number of dynamic descriptors in `pDescriptorSets`

- VUID-vkCmdBindDescriptorSets-firstSet-00360

The sum of `firstSet` and `descriptorSetCount` **must** be less than or equal to `VkPipelineLayoutCreateInfo::setLayoutCount` provided when `layout` was created

- VUID-vkCmdBindDescriptorSets-pDynamicOffsets-01971

Each element of `pDynamicOffsets` which corresponds to a descriptor binding with type `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` **must** be a multiple of `VkPhysicalDeviceLimits::minUniformBufferOffsetAlignment`

- VUID-vkCmdBindDescriptorSets-pDynamicOffsets-01972

Each element of `pDynamicOffsets` which corresponds to a descriptor binding with type `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC` **must** be a multiple of `VkPhysicalDeviceLimits::minStorageBufferOffsetAlignment`

- VUID-vkCmdBindDescriptorSets-pDescriptorSets-01979

For each dynamic uniform or storage buffer binding in `pDescriptorSets`, the sum of the `effective offset` and the range of the binding **must** be less than or equal to the size of the buffer

- VUID-vkCmdBindDescriptorSets-pDescriptorSets-06715

For each dynamic uniform or storage buffer binding in `pDescriptorSets`, if the range was set with `VK_WHOLE_SIZE` then `pDynamicOffsets` which corresponds to the descriptor binding **must** be 0

- VUID-vkCmdBindDescriptorSets-pDescriptorSets-06563

Each element of `pDescriptorSets` **must** be a valid `VkDescriptorSet`

- VUID-vkCmdBindDescriptorSets-pipelineBindPoint-00361

`pipelineBindPoint` **must** be supported by the `commandBuffer`'s parent `VkCommandPool`'s queue family

Valid Usage (Implicit)

- VUID-vkCmdBindDescriptorSets-commandBuffer-parameter

`commandBuffer` **must** be a valid `VkCommandBuffer` handle

- VUID-vkCmdBindDescriptorSets-pipelineBindPoint-parameter

`pipelineBindPoint` **must** be a valid `VkPipelineBindPoint` value

- VUID-vkCmdBindDescriptorSets-layout-parameter

`layout` **must** be a valid `VkPipelineLayout` handle

- VUID-vkCmdBindDescriptorSets-pDescriptorSets-parameter

`pDescriptorSets` **must** be a valid pointer to an array of `descriptorSetCount` valid or `VK_NULL_HANDLE` `VkDescriptorSet` handles

- VUID-vkCmdBindDescriptorSets-pDynamicOffsets-parameter

If `dynamicOffsetCount` is not 0, `pDynamicOffsets` **must** be a valid pointer to an array of `dynamicOffsetCount` `uint32_t` values

- VUID-vkCmdBindDescriptorSets-commandBuffer-recording

`commandBuffer` **must** be in the `recording` state

- VUID-vkCmdBindDescriptorSets-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, or compute operations
- VUID-vkCmdBindDescriptorSets-descriptorSetCount-arraylength
`descriptorSetCount` **must** be greater than 0
- VUID-vkCmdBindDescriptorSets-commonparent
Each of `commandBuffer`, `layout`, and the elements of `pDescriptorSets` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Graphics Compute	State

14.2.6. Push Constant Updates

As described above in section [Pipeline Layouts](#), the pipeline layout defines shader push constants which are updated via Vulkan commands rather than via writes to memory or copy commands.

NOTE

Push constants represent a high speed path to modify constant data in pipelines that is expected to outperform memory-backed resource updates.

To update push constants, call:

```
// Provided by VK_VERSION_1_0
void vkCmdPushConstants(
    VkCommandBuffer          commandBuffer,
    VkPipelineLayout         layout,
    VkShaderStageFlags       stageFlags,
    uint32_t                 offset,
    uint32_t                 size,
    const void*              pValues);
```

- `commandBuffer` is the command buffer in which the push constant update will be recorded.
- `layout` is the pipeline layout used to program the push constant updates.
- `stageFlags` is a bitmask of `VkShaderStageFlagBits` specifying the shader stages that will use the push constants in the updated range.
- `offset` is the start offset of the push constant range to update, in units of bytes.
- `size` is the size of the push constant range to update, in units of bytes.
- `pValues` is a pointer to an array of `size` bytes containing the new push constant values.

When a command buffer begins recording, all push constant values are undefined.

Push constant values **can** be updated incrementally, causing shader stages in `stageFlags` to read the new data from `pValues` for push constants modified by this command, while still reading the previous data for push constants not modified by this command. When a `bind pipeline command` is issued, the bound pipeline's layout **must** be compatible with the layouts used to set the values of all push constants in the pipeline layout's push constant ranges, as described in [Pipeline Layout Compatibility](#). Binding a pipeline with a layout that is not compatible with the push constant layout does not disturb the push constant values.

NOTE

As `stageFlags` needs to include all flags the relevant push constant ranges were created with, any flags that are not supported by the queue family that the `VkCommandPool` used to allocate `commandBuffer` was created on are ignored.

Valid Usage

- VUID-vkCmdPushConstants-offset-01795
For each byte in the range specified by `offset` and `size` and for each shader stage in `stageFlags`, there **must** be a push constant range in `layout` that includes that byte and that stage
- VUID-vkCmdPushConstants-offset-01796
For each byte in the range specified by `offset` and `size` and for each push constant range that overlaps that byte, `stageFlags` **must** include all stages in that push constant range's `VkPushConstantRange::stageFlags`
- VUID-vkCmdPushConstants-offset-00368
`offset` **must** be a multiple of 4
- VUID-vkCmdPushConstants-size-00369
`size` **must** be a multiple of 4
- VUID-vkCmdPushConstants-offset-00370
`offset` **must** be less than `VkPhysicalDeviceLimits::maxPushConstantsSize`
- VUID-vkCmdPushConstants-size-00371
`size` **must** be less than or equal to `VkPhysicalDeviceLimits::maxPushConstantsSize` minus `offset`

Valid Usage (Implicit)

- VUID-vkCmdPushConstants-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdPushConstants-layout-parameter
`layout` **must** be a valid `VkPipelineLayout` handle
- VUID-vkCmdPushConstants-stageFlags-parameter
`stageFlags` **must** be a valid combination of `VkShaderStageFlagBits` values
- VUID-vkCmdPushConstants-stageFlags-requiredbitmask
`stageFlags` **must** not be `0`
- VUID-vkCmdPushConstants-pValues-parameter
`pValues` **must** be a valid pointer to an array of `size` bytes
- VUID-vkCmdPushConstants-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdPushConstants-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, or compute operations
- VUID-vkCmdPushConstants-size-arraylength
`size` **must** be greater than `0`
- VUID-vkCmdPushConstants-commonparent
Both of `commandBuffer`, and `layout` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Graphics Compute	State

Chapter 15. Shader Interfaces

When a pipeline is created, the set of shaders specified in the corresponding `VkPipelineCreateInfo` structure are implicitly linked at a number of different interfaces.

- [Shader Input and Output Interface](#)
- [Vertex Input Interface](#)
- [Fragment Output Interface](#)
- [Fragment Input Attachment Interface](#)
- [Shader Resource Interface](#)

This chapter describes valid uses for a set of SPIR-V decorations. Any other use of one of these decorations is invalid, with the exception that, when using SPIR-V versions 1.4 and earlier: `Block`, `BufferBlock`, `Offset`, `ArrayStride`, and `MatrixStride` can also decorate types and type members used by variables in the `Private` and `Function` storage classes.

NOTE

In this chapter, there are references to SPIR-V terms such as the `MeshNV` execution model. These terms will appear even in a build of the specification which does not support any extensions. This is as intended, since these terms appear in the unified SPIR-V specification without such qualifiers.

15.1. Shader Input and Output Interfaces

When multiple stages are present in a pipeline, the outputs of one stage form an interface with the inputs of the next stage. When such an interface involves a shader, shader outputs are matched against the inputs of the next stage, and shader inputs are matched against the outputs of the previous stage.

All the variables forming the shader input and output *interfaces* are listed as operands to the `OpEntryPoint` instruction and are declared with the `Input` or `Output` storage classes, respectively, in the SPIR-V module. These generally form the interfaces between consecutive shader stages, regardless of any non-shader stages between the consecutive shader stages.

There are two classes of variables that **can** be matched between shader stages, built-in variables and user-defined variables. Each class has a different set of matching criteria.

`Output` variables of a shader stage have undefined values until the shader writes to them or uses the `Initializer` operand when declaring the variable.

15.1.1. Built-in Interface Block

Shader `built-in` variables meeting the following requirements define the *built-in interface block*. They **must**

- be explicitly declared (there are no implicit built-ins),
- be identified with a `BuiltIn` decoration,

- form object types as described in the [Built-in Variables](#) section, and
- be declared in a block whose top-level members are the built-ins.

There **must** be no more than one built-in interface block per shader per interface .

Built-ins **must** not have any [Location](#) or [Component](#) decorations.

15.1.2. User-defined Variable Interface

The non-built-in variables listed by [OpEntryPoint](#) with the [Input](#) or [Output](#) storage class form the *user-defined variable interface*. These **must** have [numeric type](#) or, recursively, composite types of such types. These variables **must** be identified with a [Location](#) decoration and **can** also be identified with a [Component](#) decoration.

15.1.3. Interface Matching

An output variable, block, or structure member in a given shader stage has an interface match with an input variable, block, or structure member in a subsequent shader stage if they both adhere to the following conditions:

- They have equivalent decorations, other than:
 - one is not decorated with [Component](#) and the other is declared with a [Component](#) of 0
 - [Interpolation decorations](#)
 - [RelaxedPrecision](#) if one is an input variable and the other an output variable
- Their types match as follows:
 - if the input is declared in a tessellation control or geometry shader as an [OpTypeArray](#) with an [Element Type](#) equivalent to the [OpType*](#) declaration of the output, and neither is a structure member; or
 - if in any other case they are declared with an equivalent [OpType*](#) declaration.
- If both are structures and every member has an interface match.

NOTE

The word “structure” above refers to both variables that have an [OpTypeStruct](#) type and interface blocks (which are also declared as [OpTypeStruct](#)).

All input variables and blocks **must** have an interface match in the preceding shader stage, except for built-in variables in fragment shaders. Shaders **can** declare and write to output variables that are not declared or read by the subsequent stage.

The value of an input variable is undefined if the preceding stage does not write to a matching output variable, as described above.

15.1.4. Location Assignment

This section describes [Location](#) assignments for user-defined variables and how many [Location](#) slots are consumed by a given user-variable type. [As mentioned above](#), some inputs and outputs have an additional level of arrayness relative to other shader inputs and outputs. This outer array level is

removed from the type before considering how many **Location** slots the type consumes.

The **Location** value specifies an interface slot comprised of a 32-bit four-component vector conveyed between stages. The **Component** specifies **word components** within these vector **Location** slots. Only types with widths of 32 or 64 are supported in shader interfaces.

Inputs and outputs of the following types consume a single interface **Location**:

- 32-bit scalar and vector types, and
- 64-bit scalar and 2-component vector types.

64-bit three- and four-component vectors consume two consecutive **Location** slots.

If a declared input or output is an array of size n and each element takes m **Location** slots, it will be assigned $m \times n$ consecutive **Location** slots starting with the specified **Location**.

If the declared input or output is an $n \times m$ 32- or 64-bit matrix, it will be assigned multiple **Location** slots starting with the specified **Location**. The number of **Location** slots assigned for each matrix will be the same as for an n -element array of m -component vectors.

An **OpVariable** with a structure type that is not a block **must** be decorated with a **Location**.

When an **OpVariable** with a structure type (either block or non-block) is decorated with a **Location**, the members in the structure type **must** not be decorated with a **Location**. The **OpVariable**'s members are assigned consecutive **Location** slots in declaration order, starting from the first member, which is assigned the **Location** decoration from the **OpVariable**.

When a block-type **OpVariable** is declared without a **Location** decoration, each member in its structure type **must** be decorated with a **Location**. Types nested deeper than the top-level members **must** not have **Location** decorations.

The **Location** slots consumed by block and structure members are determined by applying the rules above in a depth-first traversal of the instantiated members as though the structure or block member were declared as an input or output variable of the same type.

Any two inputs listed as operands on the same **OpEntryPoint** **must** not be assigned the same **Location** slot and **Component** word, either explicitly or implicitly. Any two outputs listed as operands on the same **OpEntryPoint** **must** not be assigned the same **Location** slot and **Component** word, either explicitly or implicitly.

The number of input and output **Location** slots available for a shader input or output interface is limited, and dependent on the shader stage as described in [Shader Input and Output Locations](#). All variables in both the **built-in interface block** and the **user-defined variable interface** count against these limits. Each effective **Location** **must** have a value less than the number of **Location** slots available for the given interface, as specified in the “Locations Available” column in [Shader Input and Output Locations](#).

Table 10. Shader Input and Output Locations

Shader Interface	Locations Available
vertex input	<code>maxVertexInputAttributes</code>
vertex output	<code>maxVertexOutputComponents</code> / 4
tessellation control input	<code>maxTessellationControlPerVertexInputComponents</code> / 4
tessellation control output	<code>maxTessellationControlPerVertexOutputComponents</code> / 4
tessellation evaluation input	<code>maxTessellationEvaluationInputComponents</code> / 4
tessellation evaluation output	<code>maxTessellationEvaluationOutputComponents</code> / 4
geometry input	<code>maxGeometryInputComponents</code> / 4
geometry output	<code>maxGeometryOutputComponents</code> / 4
fragment input	<code>maxFragmentInputComponents</code> / 4
fragment output	<code>maxFragmentOutputAttachments</code>

15.1.5. Component Assignment

The `Component` decoration allows the `Location` to be more finely specified for scalars and vectors, down to the individual `Component` word within a `Location` slot that are consumed. The `Component` word within a `Location` are 0, 1, 2, and 3. A variable or block member starting at `Component` N will consume `Component` words N, N+1, N+2, ... up through its size. For single precision types, it is invalid if this sequence of `Component` words gets larger than 3. A scalar 64-bit type will consume two of these `Component` words in sequence, and a two-component 64-bit vector type will consume all four `Component` words available within a `Location`. A three- or four-component 64-bit vector type **must** not specify a non-zero `Component` decoration. A three-component 64-bit vector type will consume all four `Component` words of the first `Location` and `Component` 0 and 1 of the second `Location`. This leaves `Component` 2 and 3 available for other component-qualified declarations.

A scalar or two-component 64-bit data type **must** not specify a `Component` decoration of 1 or 3. A `Component` decoration **must** not be specified for any type that is not a scalar or vector.

A four-component 64-bit data type will consume all four `Component` words of the first `Location` and all four `Component` words of the second `Location`.

15.2. Vertex Input Interface

When the vertex stage is present in a pipeline, the vertex shader input variables form an interface with the vertex input attributes. The vertex shader input variables are matched by the `Location` and `Component` decorations to the vertex input attributes specified in the `pVertexInputState` member of the `VkGraphicsPipelineCreateInfo` structure.

The vertex shader input variables listed by `OpEntryPoint` with the `Input` storage class form the *vertex input interface*. These variables **must** be identified with a `Location` decoration and **can** also be identified with a `Component` decoration.

For the purposes of interface matching: variables declared without a **Component** decoration are considered to have a **Component** decoration of zero. The number of available vertex input **Location** slots is given by the **maxVertexInputAttributes** member of the **VkPhysicalDeviceLimits** structure.

See [Attribute Location and Component Assignment](#) for details.

All vertex shader inputs declared as above **must** have a corresponding attribute and binding in the pipeline.

15.3. Fragment Output Interface

When the fragment stage is present in a pipeline, the fragment shader outputs form an interface with the output attachments defined by a [render pass instance](#). The fragment shader output variables are matched by the **Location** and **Component** decorations to specified color attachments.

The fragment shader output variables listed by **OpEntryPoint** with the **Output** storage class form the *fragment output interface*. These variables **must** be identified with a **Location** decoration. They **can** also be identified with a **Component** decoration and/or an **Index** decoration. For the purposes of interface matching: variables declared without a **Component** decoration are considered to have a **Component** decoration of zero, and variables declared without an **Index** decoration are considered to have an **Index** decoration of zero.

A fragment shader output variable identified with a **Location** decoration of *i* is associated with the color attachment indicated by **VkSubpassDescription::pColorAttachments[i]**. Values are written to those attachments after passing through the blending unit as described in [Blending](#), if enabled. Locations are consumed as described in [Location Assignment](#). The number of available fragment output **Location** slots is given by the **maxFragmentOutputAttachments** member of the **VkPhysicalDeviceLimits** structure.

When an active fragment shader invocation finishes, the values of all fragment shader outputs are copied out and used as blend inputs or color attachments writes. If the invocation does not set a value for them, the input values to those blending or color attachment writes are undefined.

Components of the output variables are assigned as described in [Component Assignment](#). Output **Component** words identified as 0, 1, 2, and 3 will be directed to the R, G, B, and A inputs to the blending unit, respectively, or to the output attachment if blending is disabled. If two variables are placed within the same **Location**, they **must** have the same underlying type (floating-point or integer). **Component** words which do not correspond to any fragment shader output will also result in undefined values for blending or color attachment writes.

Fragment outputs identified with an **Index** of zero are directed to the first input of the blending unit associated with the corresponding **Location**. Outputs identified with an **Index** of one are directed to the second input of the corresponding blending unit.

There **must** be no output variable which has the same **Location**, **Component**, and **Index** as any other, either explicitly declared or implied.

Output values written by a fragment shader **must** be declared with either **OpTypeFloat** or **OpTypeInt**, and a **Width** of 32. Composites of these types are also permitted. If the color attachment has a signed

or unsigned normalized fixed-point format, color values are assumed to be floating-point and are converted to fixed-point as described in [Conversion From Floating-Point to Normalized Fixed-Point](#); If the color attachment has an integer format, color values are assumed to be integers and converted to the bit-depth of the target. Any value that cannot be represented in the attachment's format is undefined. For any other attachment format no conversion is performed. If the type of the values written by the fragment shader do not match the format of the corresponding color attachment, the resulting values are undefined for those components.

15.4. Fragment Input Attachment Interface

When a fragment stage is present in a pipeline, the fragment shader subpass inputs form an interface with the input attachments of the current subpass. The fragment shader subpass input variables are matched by `InputAttachmentIndex` decorations to the input attachments specified in the `pInputAttachments` array of the `VkSubpassDescription` structure describing the subpass that the fragment shader is executed in.

The fragment shader subpass input variables with the `UniformConstant` storage class and a decoration of `InputAttachmentIndex` that are statically used by `OpEntryPoint` form the *fragment input attachment interface*. These variables **must** be declared with a type of `OpTypeImage`, a `Dim` operand of `SubpassData`, an `Arrayed` operand of 0, and a `Sampled` operand of 2. The `MS` operand of the `OpTypeImage` **must** be 0 if the `samples` field of the corresponding `VkAttachmentDescription` is `VK_SAMPLE_COUNT_1_BIT` and 1 otherwise.

A subpass input variable identified with an `InputAttachmentIndex` decoration of *i* reads from the input attachment indicated by `pInputAttachments[i]` member of `VkSubpassDescription`. If the subpass input variable is declared as an array of size N, it consumes N consecutive input attachments, starting with the index specified. There **must** not be more than one input variable with the same `InputAttachmentIndex` whether explicitly declared or implied by an array declaration per image aspect. A multi-aspect image (e.g. a depth/stencil format) **can** use the same input variable. The number of available input attachment indices is given by the `maxPerStageDescriptorInputAttachments` member of the `VkPhysicalDeviceLimits` structure.

Variables identified with the `InputAttachmentIndex` **must** only be used by a fragment stage. The `numeric format` of the subpass input **must** match the format of the corresponding input attachment, or the values of subpass loads from these variables are undefined. If the framebuffer attachment contains both depth and stencil aspects, the numeric format of the subpass input determines if depth or stencil aspect is accessed by the shader.

See [Input Attachment](#) for more details.

15.4.1. Fragment Input Attachment Compatibility

An input attachment that is statically accessed by a fragment shader **must** be backed by a descriptor that is equivalent to the `VkImageView` in the `VkFramebuffer`, except for `subresourceRange.aspectMask`. The `aspectMask` **must** be equal to the aspect accessed by the shader.

15.5. Shader Resource Interface

When a shader stage accesses buffer or image resources, as described in the [Resource Descriptors](#) section, the shader resource variables **must** be matched with the [pipeline layout](#) that is provided at pipeline creation time.

The set of shader variables that form the *shader resource interface* for a stage are the variables statically used by that stage's [OpEntryPoint](#) with a storage class of [Uniform](#), [UniformConstant](#), or [PushConstant](#). For the fragment shader, this includes the [fragment input attachment interface](#).

The shader resource interface consists of two sub-interfaces: the push constant interface and the descriptor set interface.

15.5.1. Push Constant Interface

The shader variables defined with a storage class of [PushConstant](#) that are statically used by the shader entry points for the pipeline define the *push constant interface*. They **must** be:

- typed as [OpTypeStruct](#),
- identified with a [Block](#) decoration, and
- laid out explicitly using the [Offset](#), [ArrayStride](#), and [MatrixStride](#) decorations as specified in [Offset and Stride Assignment](#).

There **must** be no more than one push constant block statically used per shader entry point.

Each statically used member of a push constant block **must** be placed at an [Offset](#) such that the entire member is entirely contained within the [VkPushConstantRange](#) for each [OpEntryPoint](#) that uses it, and the [stageFlags](#) for that range **must** specify the appropriate [VkShaderStageFlagBits](#) for that stage. The [Offset](#) decoration for any member of a push constant block **must** not cause the space required for that member to extend outside the range [0, [maxPushConstantsSize](#)).

Any member of a push constant block that is declared as an array **must** only be accessed with *dynamically uniform* indices.

15.5.2. Descriptor Set Interface

The *descriptor set interface* is comprised of the shader variables with the storage class of [Uniform](#) or [UniformConstant](#) (including the variables in the [fragment input attachment interface](#)) that are statically used by the shader entry points for the pipeline.

These variables **must** have [DescriptorSet](#) and [Binding](#) decorations specified, which are assigned and matched with the [VkDescriptorSetLayout](#) objects in the pipeline layout as described in [DescriptorSet and Binding Assignment](#).

The [Image Format](#) of an [OpTypeImage](#) declaration **must** not be [Unknown](#), for variables which are used for [OpImageRead](#), [OpImageSparseRead](#), or [OpImageWrite](#) operations, except under the following conditions:

- For [OpImageWrite](#), if the image format is listed in the [storage without format](#) list and if the

`shaderStorageImageWriteWithoutFormat` feature is enabled and the shader module declares the `StorageImageWriteWithoutFormat` capability.

- For `OpImageRead` or `OpImageSparseRead`, if the image format is listed in the [storage without format](#) list and if the `shaderStorageImageReadWithoutFormat` feature is enabled and the shader module declares the `StorageImageReadWithoutFormat` capability.
- For `OpImageRead`, if `Dim` is `SubpassData` (indicating a read from an input attachment).

The **Image Format** of an `OpTypeImage` declaration **must** not be **Unknown**, for variables which are used for `OpAtomic*` operations.

Variables identified with the `Uniform` storage class are used to access transparent buffer backed resources. Such variables **must** be:

- typed as `OpTypeStruct`, or an array of this type,
- identified with a `Block` or `BufferBlock` decoration, and
- laid out explicitly using the `Offset`, `ArrayStride`, and `MatrixStride` decorations as specified in [Offset and Stride Assignment](#).

The `Offset` decoration for any variable in a `Block` **must** not cause the space required for that variable to extend outside the range `[0, maxUniformBufferRange)`. The `Offset` decoration for any variable in a `BufferBlock` **must** not cause the space required for that variable to extend outside the range `[0, maxStorageBufferRange)`.

Variables identified with a storage class of `UniformConstant` and a decoration of `InputAttachmentIndex` **must** be declared as described in [Fragment Input Attachment Interface](#).

SPIR-V variables decorated with a descriptor set and binding that identify a [combined image sampler descriptor](#) **can** have a type of `OpTypeImage`, `OpTypeSampler` (`Sampled=1`), or `OpTypeSampledImage`.

When accessing a resource through such a variable, the resource **must** be selected via compile time constant expressions unless features are enabled to allow dynamically uniform or non-uniform expressions, as described below:

- Storage images (except storage texel buffers and input attachments):
 - Dynamically uniform: `shaderStorageImageArrayDynamicIndexing` and `StorageImageArrayDynamicIndexing`
- Sampled images (except uniform texel buffers), samplers and combined image samplers:
 - Dynamically uniform: `shaderSampledImageArrayDynamicIndexing` and `SampledImageArrayDynamicIndexing`
- Uniform buffers:
 - Dynamically uniform: `shaderUniformBufferArrayDynamicIndexing` and `UniformBufferArrayDynamicIndexing`
- Storage buffers:
 - Dynamically uniform: `shaderStorageBufferArrayDynamicIndexing` and `StorageBufferArrayDynamicIndexing`

Table 11. Shader Resource and Descriptor Type Correspondence

Resource type	Descriptor Type
sampler	VK_DESCRIPTOR_TYPE_SAMPLER or VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER
sampled image	VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE or VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER
storage image	VK_DESCRIPTOR_TYPE_STORAGE_IMAGE
combined image sampler	VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER
uniform texel buffer	VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER
storage texel buffer	VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER
uniform buffer	VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER or VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC
storage buffer	VK_DESCRIPTOR_TYPE_STORAGE_BUFFER or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC
input attachment	VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT

Table 12. Shader Resource and Storage Class Correspondence

Resource type	Storage Class	Type ¹	Decoration(s) ²
sampler	UniformConstant	OpTypeSampler	
sampled image	UniformConstant	OpTypeImage (Sampled=1)	
storage image	UniformConstant	OpTypeImage (Sampled=2)	
combined image sampler	UniformConstant	OpTypeSampledImage OpTypeImage (Sampled=1) OpTypeSampler	
uniform texel buffer	UniformConstant	OpTypeImage (Dim=Buffer, Sampled=1)	
storage texel buffer	UniformConstant	OpTypeImage (Dim=Buffer, Sampled=2)	
uniform buffer	Uniform	OpTypeStruct	Block, Offset, (ArrayStride), (MatrixStride)
storage buffer	Uniform	OpTypeStruct	BufferBlock, Offset, (ArrayStride), (MatrixStride)
input attachment	UniformConstant	OpTypeImage (Dim =SubpassData, Sampled=2)	InputAttachmentIndex

1

Where **OpTypeImage** is referenced, the **Dim** values **Buffer** and **Subpassdata** are only accepted where they are specifically referenced. They do not correspond to resource types where a generic **OpTypeImage** is specified.

In addition to [DescriptorSet](#) and [Binding](#).

15.5.3. DescriptorSet and Binding Assignment

A variable decorated with a [DescriptorSet](#) decoration of *s* and a [Binding](#) decoration of *b* indicates that this variable is associated with the [VkDescriptorSetLayoutBinding](#) that has a [binding](#) equal to *b* in [pSetLayouts\[s\]](#) that was specified in [VkPipelineLayoutCreateInfo](#).

[DescriptorSet](#) decoration values **must** be between zero and [maxBoundDescriptorSets](#) minus one, inclusive. [Binding](#) decoration values **can** be any 32-bit unsigned integer value, as described in [Descriptor Set Layout](#). Each descriptor set has its own binding name space.

If the [Binding](#) decoration is used with an array, the entire array is assigned that binding value. The array **must** be a single-dimensional array and size of the array **must** be no larger than the number of descriptors in the binding. The array **must** not be runtime-sized. The index of each element of the array is referred to as the *arrayElement*. For the purposes of interface matching and descriptor set [operations](#), if a resource variable is not an array, it is treated as if it has an *arrayElement* of zero.

There is a limit on the number of resources of each type that **can** be accessed by a pipeline stage as shown in [Shader Resource Limits](#). The “Resources Per Stage” column gives the limit on the number each type of resource that **can** be statically used for an entry point in any given stage in a pipeline. The “Resource Types” column lists which resource types are counted against the limit. Some resource types count against multiple limits.

The pipeline layout **may** include descriptor sets and bindings which are not referenced by any variables statically used by the entry points for the shader stages in the binding’s [stageFlags](#).

However, if a variable assigned to a given [DescriptorSet](#) and [Binding](#) is statically used by the entry point for a shader stage, the pipeline layout **must** contain a descriptor set layout binding in that descriptor set layout and for that binding number, and that binding’s [stageFlags](#) **must** include the appropriate [VkShaderStageFlagBits](#) for that stage. The variable **must** be of a valid resource type determined by its SPIR-V type and storage class, as defined in [Shader Resource and Storage Class Correspondence](#). The descriptor set layout binding **must** be of a corresponding descriptor type, as defined in [Shader Resource and Descriptor Type Correspondence](#).

There are no limits on the number of shader variables that can have overlapping set and binding values in a shader; but which resources are [statically used](#) has an impact. If any shader variable identifying a resource is [statically used](#) in a shader, then the underlying descriptor bound at the declared set and binding **must support the declared type in the shader** when the shader executes.

NOTE

If multiple shader variables are declared with the same set and binding values, and with the same underlying descriptor type, they can all be statically used within the same shader. However, accesses are not automatically synchronized, and [Aliased](#) decorations should be used to avoid data hazards (see [section 2.18.2 Aliasing in the SPIR-V specification](#)).

If multiple shader variables with the same set and binding values are declared in a

single shader, but with different declared types, where any of those are not supported by the relevant bound descriptor, that shader can only be executed if the variables with the unsupported type are not statically used.

A noteworthy example of using multiple statically-used shader variables sharing the same descriptor set and binding values is a descriptor of type `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER` that has multiple corresponding shader variables in the `UniformConstant` storage class, where some could be `OpTypeImage` (`Sampled=1`), some could be `OpTypeSampler`, and some could be `OpTypeSampledImage`.

Table 13. Shader Resource Limits

Resources per Stage	Resource Types
<code>maxPerStageDescriptorSamplers</code>	sampler
	combined image sampler
<code>maxPerStageDescriptorSampledImages</code>	sampled image
	combined image sampler
	uniform texel buffer
<code>maxPerStageDescriptorStorageImages</code>	storage image
	storage texel buffer
<code>maxPerStageDescriptorUniformBuffers</code>	uniform buffer
	uniform buffer dynamic
<code>maxPerStageDescriptorStorageBuffers</code>	storage buffer
	storage buffer dynamic
<code>maxPerStageDescriptorInputAttachments</code>	input attachment ¹

1

Input attachments **can** only be used in the fragment shader stage

15.5.4. Offset and Stride Assignment

Certain objects **must** be explicitly laid out using the `Offset`, `ArrayStride`, and `MatrixStride`, as described in [SPIR-V explicit layout validation rules](#). All such layouts also **must** conform to the following requirements.

NOTE The numeric order of `Offset` decorations does not need to follow member declaration order.

Alignment Requirements

There are different alignment requirements depending on the specific resources and on the features enabled on the device.

Matrix types are defined in terms of arrays as follows:

- A column-major matrix with C columns and R rows is equivalent to a C element array of vectors with R components.
- A row-major matrix with C columns and R rows is equivalent to an R element array of vectors with C components.

The *scalar alignment* of the type of an **OpTypeStruct** member is defined recursively as follows:

- A scalar of size N has a scalar alignment of N.
- A vector type has a scalar alignment equal to that of its component type.
- An array type has a scalar alignment equal to that of its element type.
- A structure has a scalar alignment equal to the largest scalar alignment of any of its members.
- A matrix type inherits *scalar alignment* from the equivalent array declaration.

The *base alignment* of the type of an **OpTypeStruct** member is defined recursively as follows:

- A scalar has a base alignment equal to its scalar alignment.
- A two-component vector has a base alignment equal to twice its scalar alignment.
- A three- or four-component vector has a base alignment equal to four times its scalar alignment.
- An array has a base alignment equal to the base alignment of its element type.
- A structure has a base alignment equal to the largest base alignment of any of its members. An empty structure has a base alignment equal to the size of the smallest scalar type permitted by the capabilities declared in the SPIR-V module. (e.g., for a 1 byte aligned empty struct in the **StorageBuffer** storage class, **StorageBuffer8BitAccess** or **UniformAndStorageBuffer8BitAccess** **must** be declared in the SPIR-V module.)
- A matrix type inherits *base alignment* from the equivalent array declaration.

The *extended alignment* of the type of an **OpTypeStruct** member is similarly defined as follows:

- A scalar or vector type has an extended alignment equal to its base alignment.
- An array or structure type has an extended alignment equal to the largest extended alignment of any of its members, rounded up to a multiple of 16.
- A matrix type inherits extended alignment from the equivalent array declaration.

Standard Buffer Layout

Every member of an **OpTypeStruct** that is required to be explicitly laid out **must** be aligned according to the first matching rule as follows. If the struct is contained in pointer types of multiple storage classes, it **must** satisfy the requirements for every storage class used to reference it.

1. Any member of an **OpTypeStruct** with a storage class of **Uniform** and a decoration of **Block** **must** be aligned according to its extended alignment.
2. Every other member **must** be aligned according to its base alignment.

The memory layout **must** obey the following rules:

- The **Offset** decoration of any member **must** be a multiple of its alignment.
- Any **ArrayStride** or **MatrixStride** decoration **must** be a multiple of the alignment of the array or matrix as defined above.
- The **Offset** decoration of a member **must** not place it between the end of a structure, an array or a matrix and the next multiple of the alignment of that structure, array or matrix.

NOTE

The **std430 layout** in GLSL satisfies these rules for types using the base alignment.
The **std140 layout** satisfies the rules for types using the extended alignment.

15.6. Built-In Variables

Built-in variables are accessed in shaders by declaring a variable decorated with a **BuiltIn** SPIR-V decoration. The meaning of each **BuiltIn** decoration is as follows. In the remainder of this section, the name of a built-in is used interchangeably with a term equivalent to a variable decorated with that particular built-in. Built-ins that represent integer values **can** be declared as either signed or unsigned 32-bit integers.

As mentioned above, some inputs and outputs have an additional level of arrayness relative to other shader inputs and outputs. This level of arrayness is not included in the type descriptions below, but **must** be included when declaring the built-in.

Any two **Input** storage class **OpVariable** declarations listed as operands on the same **OpEntryPoint** **must** not have the same **BuiltIn** decoration. Any two **Output** storage class **OpVariable** declarations listed as operands on the same **OpEntryPoint** **must** not have the same **BuiltIn** decoration.

ClipDistance

Decorating a variable with the **ClipDistance** built-in decoration will make that variable contain the mechanism for controlling user clipping. **ClipDistance** is an array such that the i^{th} element of the array specifies the clip distance for plane i . A clip distance of 0 means the vertex is on the plane, a positive distance means the vertex is inside the clip half-space, and a negative distance means the vertex is outside the clip half-space.

NOTE

The array variable decorated with **ClipDistance** is explicitly sized by the shader.

NOTE

In the last **pre-rasterization shader stage**, these values will be linearly interpolated across the primitive and the portion of the primitive with interpolated distances less than 0 will be considered outside the clip volume. If **ClipDistance** is then used by a fragment shader, **ClipDistance** contains these linearly interpolated values.

Valid Usage

- VUID-ClipDistance-ClipDistance-04187
The **ClipDistance** decoration **must** be used only within the **MeshEXT**, **MeshNV**, **Vertex**, **Fragment**, **TessellationControl**, **TessellationEvaluation**, or **Geometry Execution Model**
- VUID-ClipDistance-ClipDistance-04188
The variable decorated with **ClipDistance** within the **MeshEXT**, **MeshNV**, or **Vertex Execution**

Model **must** be declared using the **Output Storage Class**

- VUID-ClipDistance-ClipDistance-04189

The variable decorated with **ClipDistance** within the **Fragment Execution Model** **must** be declared using the **Input Storage Class**

- VUID-ClipDistance-ClipDistance-04190

The variable decorated with **ClipDistance** within the **TessellationControl**, **TessellationEvaluation**, or **Geometry Execution Model** **must** not be declared in a **Storage Class** other than **Input** or **Output**

- VUID-ClipDistance-ClipDistance-04191

The variable decorated with **ClipDistance** **must** be declared as an array of 32-bit floating-point values

CullDistance

Decorating a variable with the **CullDistance** built-in decoration will make that variable contain the mechanism for controlling user culling. If any member of this array is assigned a negative value for all vertices belonging to a primitive, then the primitive is discarded before rasterization.

NOTE

In fragment shaders, the values of the **CullDistance** array are linearly interpolated across each primitive.

NOTE

If **CullDistance** decorates an input variable, that variable will contain the corresponding value from the **CullDistance** decorated output variable from the previous shader stage.

Valid Usage

- VUID-CullDistance-CullDistance-04196

The **CullDistance** decoration **must** be used only within the **MeshEXT**, **MeshNV**, **Vertex**, **Fragment**, **TessellationControl**, **TessellationEvaluation**, or **Geometry Execution Model**

- VUID-CullDistance-CullDistance-04197

The variable decorated with **CullDistance** within the **MeshEXT**, **MeshNV** or **Vertex Execution Model** **must** be declared using the **Output Storage Class**

- VUID-CullDistance-CullDistance-04198

The variable decorated with **CullDistance** within the **Fragment Execution Model** **must** be declared using the **Input Storage Class**

- VUID-CullDistance-CullDistance-04199

The variable decorated with **CullDistance** within the **TessellationControl**, **TessellationEvaluation**, or **Geometry Execution Model** **must** not be declared using a **Storage Class** other than **Input** or **Output**

- VUID-CullDistance-CullDistance-04200

The variable decorated with **CullDistance** **must** be declared as an array of 32-bit floating-point values

FragCoord

Decorating a variable with the **FragCoord** built-in decoration will make that variable contain the framebuffer coordinate $(x, y, z, \frac{1}{w})$ of the fragment being processed. The (x,y) coordinate (0,0) is the upper left corner of the upper left pixel in the framebuffer.

When **Sample Shading** is enabled, the x and y components of **FragCoord** reflect the location of one of the samples corresponding to the shader invocation.

Otherwise, the x and y components of **FragCoord** reflect the location of the center of the fragment.

The z component of **FragCoord** is the interpolated depth value of the primitive.

The w component is the interpolated $\frac{1}{w}$.

The **Centroid** interpolation decoration is ignored, but allowed, on **FragCoord**.

Valid Usage

- VUID-FragCoord-FragCoord-04210
The **FragCoord** decoration **must** be used only within the **Fragment Execution Model**
- VUID-FragCoord-FragCoord-04211
The variable decorated with **FragCoord** **must** be declared using the **Input Storage Class**
- VUID-FragCoord-FragCoord-04212
The variable decorated with **FragCoord** **must** be declared as a four-component vector of 32-bit floating-point values

FragDepth

To have a shader supply a fragment-depth value, the shader **must** declare the **DepthReplacing** execution mode. Such a shader's fragment-depth value will come from the variable decorated with the **FragDepth** built-in decoration.

This value will be used for any subsequent depth testing performed by the implementation or writes to the depth attachment. See [fragment shader depth replacement](#) for details.

Valid Usage

- VUID-FragDepth-FragDepth-04213
The **FragDepth** decoration **must** be used only within the **Fragment Execution Model**
- VUID-FragDepth-FragDepth-04214
The variable decorated with **FragDepth** **must** be declared using the **Output Storage Class**
- VUID-FragDepth-FragDepth-04215
The variable decorated with **FragDepth** **must** be declared as a scalar 32-bit floating-point value
- VUID-FragDepth-FragDepth-04216
If the shader dynamically writes to the variable decorated with **FragDepth**, the

DepthReplacing Execution Mode **must** be declared

FrontFacing

Decorating a variable with the **FrontFacing** built-in decoration will make that variable contain whether the fragment is front or back facing. This variable is non-zero if the current fragment is considered to be part of a **front-facing** polygon primitive or of a non-polygon primitive and is zero if the fragment is considered to be part of a back-facing polygon primitive.

Valid Usage

- VUID-FrontFacing-FrontFacing-04229
The **FrontFacing** decoration **must** be used only within the **Fragment Execution Model**
- VUID-FrontFacing-FrontFacing-04230
The variable decorated with **FrontFacing** **must** be declared using the **Input Storage Class**
- VUID-FrontFacing-FrontFacing-04231
The variable decorated with **FrontFacing** **must** be declared as a boolean value

GlobalInvocationId

Decorating a variable with the **GlobalInvocationId** built-in decoration will make that variable contain the location of the current invocation within the global workgroup. Each component is equal to the index of the local workgroup multiplied by the size of the local workgroup plus **LocalInvocationId**.

Valid Usage

- VUID-GlobalInvocationId-GlobalInvocationId-04236
The **GlobalInvocationId** decoration **must** be used only within the **GLCompute**, **MeshEXT**, **TaskEXT**, **MeshNV**, or **TaskNV Execution Model**
- VUID-GlobalInvocationId-GlobalInvocationId-04237
The variable decorated with **GlobalInvocationId** **must** be declared using the **Input Storage Class**
- VUID-GlobalInvocationId-GlobalInvocationId-04238
The variable decorated with **GlobalInvocationId** **must** be declared as a three-component vector of 32-bit integer values

HelperInvocation

Decorating a variable with the **HelperInvocation** built-in decoration will make that variable contain whether the current invocation is a helper invocation. This variable is non-zero if the current fragment being shaded is a helper invocation and zero otherwise. A helper invocation is an invocation of the shader that is produced to satisfy internal requirements such as the generation of derivatives.

NOTE

It is very likely that a helper invocation will have a value of **SampleMask** fragment

shader input value that is zero.

Valid Usage

- VUID-HelperInvocation-HelperInvocation-04239
The **HelperInvocation** decoration **must** be used only within the **Fragment Execution Model**
- VUID-HelperInvocation-HelperInvocation-04240
The variable decorated with **HelperInvocation** **must** be declared using the **Input Storage Class**
- VUID-HelperInvocation-HelperInvocation-04241
The variable decorated with **HelperInvocation** **must** be declared as a boolean value

InvocationId

Decorating a variable with the **InvocationId** built-in decoration will make that variable contain the index of the current shader invocation in a geometry shader, or the index of the output patch vertex in a tessellation control shader.

In a geometry shader, the index of the current shader invocation ranges from zero to the number of **instances** declared in the shader minus one. If the instance count of the geometry shader is one or is not specified, then **InvocationId** will be zero.

Valid Usage

- VUID-InvocationId-InvocationId-04257
The **InvocationId** decoration **must** be used only within the **TessellationControl** or **Geometry Execution Model**
- VUID-InvocationId-InvocationId-04258
The variable decorated with **InvocationId** **must** be declared using the **Input Storage Class**
- VUID-InvocationId-InvocationId-04259
The variable decorated with **InvocationId** **must** be declared as a scalar 32-bit integer value

InstanceIndex

Decorating a variable in a vertex shader with the **InstanceIndex** built-in decoration will make that variable contain the index of the instance that is being processed by the current vertex shader invocation. **InstanceIndex** begins at the **firstInstance** parameter to **vkCmdDraw** or **vkCmdDrawIndexed** or at the **firstInstance** member of a structure consumed by **vkCmdDrawIndirect** or **vkCmdDrawIndexedIndirect**.

Valid Usage

- VUID-InstanceIndex-InstanceIndex-04263
The **InstanceIndex** decoration **must** be used only within the **Vertex Execution Model**

- VUID-InstanceIndex-InstanceIndex-04264
The variable decorated with **InstanceIndex** **must** be declared using the **Input Storage Class**
- VUID-InstanceIndex-InstanceIndex-04265
The variable decorated with **InstanceIndex** **must** be declared as a scalar 32-bit integer value

Layer

Decorating a variable with the **Layer** built-in decoration will make that variable contain the select layer of a multi-layer framebuffer attachment.

In a geometry shader, any variable decorated with **Layer** can be written with the framebuffer layer index to which the primitive produced by that shader will be directed.

If the last active **pre-rasterization shader stage** shader entry point's interface does not include a variable decorated with **Layer**, then the first layer is used. If a **pre-rasterization shader stage** shader entry point's interface includes a variable decorated with **Layer**, it **must** write the same value to **Layer** for all output vertices of a given primitive. If the **Layer** value is less than 0 or greater than or equal to the number of layers in the framebuffer, then primitives **may** still be rasterized, fragment shaders **may** be executed, and the framebuffer values for all layers are undefined.

In a fragment shader, a variable decorated with **Layer** contains the layer index of the primitive that the fragment invocation belongs to.

Valid Usage

- VUID-Layer-Layer-04272
The **Layer** decoration **must** be used only within the **MeshEXT**, **MeshNV**, **Vertex**, **TessellationEvaluation**, **Geometry**, or **Fragment Execution Model**
- VUID-Layer-Layer-04274
The variable decorated with **Layer** within the **MeshEXT**, **MeshNV**, **Vertex**, **TessellationEvaluation**, or **Geometry Execution Model** **must** be declared using the **Output Storage Class**
- VUID-Layer-Layer-04275
The variable decorated with **Layer** within the **Fragment Execution Model** **must** be declared using the **Input Storage Class**
- VUID-Layer-Layer-04276
The variable decorated with **Layer** **must** be declared as a scalar 32-bit integer value
- VUID-Layer-Layer-07039
The variable decorated with **Layer** within the **MeshEXT Execution Model** **must** also be decorated with the **PerPrimitiveEXT** decoration

LocalInvocationId

Decorating a variable with the **LocalInvocationId** built-in decoration will make that variable contain the location of the current compute shader invocation within the local workgroup. Each

component ranges from zero through to the size of the workgroup in that dimension minus one.

NOTE

If the size of the workgroup in a particular dimension is one, then the `LocalInvocationId` in that dimension will be zero. If the workgroup is effectively two-dimensional, then `LocalInvocationId.z` will be zero. If the workgroup is effectively one-dimensional, then both `LocalInvocationId.y` and `LocalInvocationId.z` will be zero.

Valid Usage

- VUID-LocalInvocationId-LocalInvocationId-04281
The `LocalInvocationId` decoration **must** be used only within the `GLCompute`, `MeshEXT`, `TaskEXT`, `MeshNV`, or `TaskNV Execution Model`
- VUID-LocalInvocationId-LocalInvocationId-04282
The variable decorated with `LocalInvocationId` **must** be declared using the `Input Storage Class`
- VUID-LocalInvocationId-LocalInvocationId-04283
The variable decorated with `LocalInvocationId` **must** be declared as a three-component vector of 32-bit integer values

LocalInvocationIndex

Decorating a variable with the `LocalInvocationIndex` built-in decoration will make that variable contain a one-dimensional representation of `LocalInvocationId`. This is computed as:

```
LocalInvocationIndex =  
    LocalInvocationId.z * WorkgroupSize.x * WorkgroupSize.y +  
    LocalInvocationId.y * WorkgroupSize.x +  
    LocalInvocationId.x;
```

Valid Usage

- VUID-LocalInvocationIndex-LocalInvocationIndex-04284
The `LocalInvocationIndex` decoration **must** be used only within the `GLCompute`, `MeshEXT`, `TaskEXT`, `MeshNV`, or `TaskNV Execution Model`
- VUID-LocalInvocationIndex-LocalInvocationIndex-04285
The variable decorated with `LocalInvocationIndex` **must** be declared using the `Input Storage Class`
- VUID-LocalInvocationIndex-LocalInvocationIndex-04286
The variable decorated with `LocalInvocationIndex` **must** be declared as a scalar 32-bit integer value

NumWorkgroups

Decorating a variable with the `NumWorkgroups` built-in decoration will make that variable contain

the number of local workgroups that are part of the dispatch that the invocation belongs to. Each component is equal to the values of the workgroup count parameters passed into the dispatching commands.

Valid Usage

- VUID-NumWorkgroups-NumWorkgroups-04296
The **NumWorkgroups** decoration **must** be used only within the **GLCompute**, **MeshEXT**, or **TaskEXT Execution Model**
- VUID-NumWorkgroups-NumWorkgroups-04297
The variable decorated with **NumWorkgroups** **must** be declared using the **Input Storage Class**
- VUID-NumWorkgroups-NumWorkgroups-04298
The variable decorated with **NumWorkgroups** **must** be declared as a three-component vector of 32-bit integer values

PatchVertices

Decorating a variable with the **PatchVertices** built-in decoration will make that variable contain the number of vertices in the input patch being processed by the shader. In a Tessellation Control Shader, this is the same as the name:patchControlPoints member of **VkPipelineTessellationStateCreateInfo**. In a Tessellation Evaluation Shader, **PatchVertices** is equal to the tessellation control output patch size. When the same shader is used in different pipelines where the patch sizes are configured differently, the value of the **PatchVertices** variable will also differ.

Valid Usage

- VUID-PatchVertices-PatchVertices-04308
The **PatchVertices** decoration **must** be used only within the **TessellationControl** or **TessellationEvaluation Execution Model**
- VUID-PatchVertices-PatchVertices-04309
The variable decorated with **PatchVertices** **must** be declared using the **Input Storage Class**
- VUID-PatchVertices-PatchVertices-04310
The variable decorated with **PatchVertices** **must** be declared as a scalar 32-bit integer value

PointCoord

Decorating a variable with the **PointCoord** built-in decoration will make that variable contain the coordinate of the current fragment within the point being rasterized, normalized to the size of the point with origin in the upper left corner of the point, as described in **Basic Point Rasterization**. If the primitive the fragment shader invocation belongs to is not a point, then the variable decorated with **PointCoord** contains an undefined value.

NOTE Depending on how the point is rasterized, **PointCoord** **may** never reach (0,0) or (1,1).

Valid Usage

- VUID-PointCoord-PointCoord-04311
The **PointCoord** decoration **must** be used only within the **Fragment Execution Model**
- VUID-PointCoord-PointCoord-04312
The variable decorated with **PointCoord** **must** be declared using the **Input Storage Class**
- VUID-PointCoord-PointCoord-04313
The variable decorated with **PointCoord** **must** be declared as a two-component vector of 32-bit floating-point values

PointSize

Decorating a variable with the **PointSize** built-in decoration will make that variable contain the size of point primitives. The value written to the variable decorated with **PointSize** by the last **pre-rasterization shader stage** in the pipeline is used as the framebuffer-space size of points produced by rasterization.

NOTE	When PointSize decorates a variable in the Input Storage Class , it contains the data written to the output variable decorated with PointSize from the previous shader stage.
-------------	--

Valid Usage

- VUID-PointSize-PointSize-04314
The **PointSize** decoration **must** be used only within the **MeshEXT**, **MeshNV**, **Vertex**, **TessellationControl**, **TessellationEvaluation**, or **Geometry Execution Model**
- VUID-PointSize-PointSize-04315
The variable decorated with **PointSize** within the **MeshEXT**, **MeshNV**, or **Vertex Execution Model** **must** be declared using the **Output Storage Class**
- VUID-PointSize-PointSize-04316
The variable decorated with **PointSize** within the **TessellationControl**, **TessellationEvaluation**, or **Geometry Execution Model** **must** not be declared using a **Storage Class** other than **Input** or **Output**
- VUID-PointSize-PointSize-04317
The variable decorated with **PointSize** **must** be declared as a scalar 32-bit floating-point value

Position

Decorating a variable with the **Position** built-in decoration will make that variable contain the position of the current vertex. In the last **pre-rasterization shader stage**, the value of the variable decorated with **Position** is used in subsequent primitive assembly, clipping, and rasterization operations.

NOTE	When Position decorates a variable in the Input Storage Class , it contains the data
-------------	--

written to the output variable decorated with **Position** from the previous shader stage.

Valid Usage

- VUID-Position-Position-04318
The **Position** decoration **must** be used only within the **MeshEXT**, **MeshNV**, **Vertex**, **TessellationControl**, **TessellationEvaluation**, or **Geometry Execution Model**
- VUID-Position-Position-04319
The variable decorated with **Position** within the **MeshEXT**, **MeshNV**, or **Vertex Execution Model** **must** be declared using the **Output Storage Class**
- VUID-Position-Position-04320
The variable decorated with **Position** within the **TessellationControl**, **TessellationEvaluation**, or **Geometry Execution Model** **must** not be declared using a **Storage Class** other than **Input** or **Output**
- VUID-Position-Position-04321
The variable decorated with **Position** **must** be declared as a four-component vector of 32-bit floating-point values

PrimitiveId

Decorating a variable with the **PrimitiveId** built-in decoration will make that variable contain the index of the current primitive.

The index of the first primitive generated by a drawing command is zero, and the index is incremented after every individual point, line, or triangle primitive is processed.

For triangles drawn as points or line segments (see **Polygon Mode**), the primitive index is incremented only once, even if multiple points or lines are eventually drawn.

Variables decorated with **PrimitiveId** are reset to zero between each instance drawn.

Restarting a primitive topology using primitive restart has no effect on the value of variables decorated with **PrimitiveId**.

In tessellation control and tessellation evaluation shaders, it will contain the index of the patch within the current set of rendering primitives that corresponds to the shader invocation.

In a geometry shader, it will contain the number of primitives presented as input to the shader since the current set of rendering primitives was started.

In a fragment shader, it will contain the primitive index written by the geometry shader if a geometry shader is present, or with the value that would have been presented as input to the geometry shader had it been present.

NOTE

When the **PrimitiveId** decoration is applied to an output variable in the geometry shader, the resulting value is seen through the **PrimitiveId** decorated input variable in the fragment shader.

The fragment shader using `PrimitiveId` will need to declare either the `Geometry` or `Tessellation` capability to satisfy the requirement SPIR-V has to use `PrimitiveId`.

Valid Usage

- VUID-PrimitiveId-PrimitiveId-04330
The `PrimitiveId` decoration **must** be used only within the `MeshEXT`, `MeshNV`, `IntersectionKHR`, `AnyHitKHR`, `ClosestHitKHR`, `TessellationControl`, `TessellationEvaluation`, `Geometry`, or `Fragment Execution Model`
- VUID-PrimitiveId-Fragment-04331
If pipeline contains both the `Fragment` and `Geometry Execution Model` and a variable decorated with `PrimitiveId` is read from `Fragment` shader, then the `Geometry` shader **must** write to the output variables decorated with `PrimitiveId` in all execution paths
- VUID-PrimitiveId-Fragment-04332
If pipeline contains both the `Fragment` and `MeshEXT` or `MeshNV Execution Model` and a variable decorated with `PrimitiveId` is read from `Fragment` shader, then the `MeshEXT` or `MeshNV` shader **must** write to the output variables decorated with `PrimitiveId` in all execution paths
- VUID-PrimitiveId-Fragment-04333
If `Fragment Execution Model` contains a variable decorated with `PrimitiveId`, then either the `MeshShadingEXT`, `MeshShadingNV`, `Geometry` or `Tessellation` capability **must** also be declared
- VUID-PrimitiveId-PrimitiveId-04334
The variable decorated with `PrimitiveId` within the `TessellationControl`, `TessellationEvaluation`, `Fragment`, `IntersectionKHR`, `AnyHitKHR`, or `ClosestHitKHR Execution Model` **must** be declared using the `Input Storage Class`
- VUID-PrimitiveId-PrimitiveId-04335
The variable decorated with `PrimitiveId` within the `Geometry Execution Model` **must** be declared using the `Input` or `Output Storage Class`
- VUID-PrimitiveId-PrimitiveId-04336
The variable decorated with `PrimitiveId` within the `MeshEXT` or `MeshNV Execution Model` **must** be declared using the `Output Storage Class`
- VUID-PrimitiveId-PrimitiveId-04337
The variable decorated with `PrimitiveId` **must** be declared as a scalar 32-bit integer value
- VUID-PrimitiveId-PrimitiveId-07040
The variable decorated with `PrimitiveId` within the `MeshEXT Execution Model` **must** also be decorated with the `PerPrimitiveEXT` decoration

SampleId

Decorating a variable with the `SampleId` built-in decoration will make that variable contain the `coverage index` for the current fragment shader invocation. `SampleId` ranges from zero to the number of samples in the framebuffer minus one. If a fragment shader entry point's interface includes an input variable decorated with `SampleId`, `Sample Shading` is considered enabled with a `minSampleShading` value of 1.0.

Valid Usage

- VUID-SampleId-SampleId-04354
The **SampleId** decoration **must** be used only within the **Fragment Execution Model**
- VUID-SampleId-SampleId-04355
The variable decorated with **SampleId** **must** be declared using the **Input Storage Class**
- VUID-SampleId-SampleId-04356
The variable decorated with **SampleId** **must** be declared as a scalar 32-bit integer value

SampleMask

Decorating a variable with the **SampleMask** built-in decoration will make any variable contain the **sample mask** for the current fragment shader invocation.

A variable in the **Input** storage class decorated with **SampleMask** will contain a bitmask of the set of samples covered by the primitive generating the fragment during rasterization. It has a sample bit set if and only if the sample is considered covered for this fragment shader invocation. **SampleMask[]** is an array of integers. Bits are mapped to samples in a manner where bit B of mask M (**SampleMask[M]**) corresponds to sample $32 \times M + B$.

A variable in the **Output** storage class decorated with **SampleMask** is an array of integers forming a bit array in a manner similar to an input variable decorated with **SampleMask**, but where each bit represents coverage as computed by the shader. This computed **SampleMask** is combined with the generated coverage mask in the **multisample coverage** operation.

Variables decorated with **SampleMask** **must** be either an unsized array, or explicitly sized to be no larger than the implementation-dependent maximum sample-mask (as an array of 32-bit elements), determined by the maximum number of samples.

If a fragment shader entry point's interface includes an output variable decorated with **SampleMask**, the sample mask will be undefined for any array elements of any fragment shader invocations that fail to assign a value. If a fragment shader entry point's interface does not include an output variable decorated with **SampleMask**, the sample mask has no effect on the processing of a fragment.

Valid Usage

- VUID-SampleMask-SampleMask-04357
The **SampleMask** decoration **must** be used only within the **Fragment Execution Model**
- VUID-SampleMask-SampleMask-04358
The variable decorated with **SampleMask** **must** be declared using the **Input** or **Output Storage Class**
- VUID-SampleMask-SampleMask-04359
The variable decorated with **SampleMask** **must** be declared as an array of 32-bit integer values

SamplePosition

Decorating a variable with the **SamplePosition** built-in decoration will make that variable contain the sub-pixel position of the sample being shaded. The top left of the pixel is considered to be at coordinate (0,0) and the bottom right of the pixel is considered to be at coordinate (1,1).

If a fragment shader entry point's interface includes an input variable decorated with **SamplePosition**, **Sample Shading** is considered enabled with a **minSampleShading** value of 1.0.

Valid Usage

- VUID-SamplePosition-SamplePosition-04360
The **SamplePosition** decoration **must** be used only within the **Fragment Execution Model**
- VUID-SamplePosition-SamplePosition-04361
The variable decorated with **SamplePosition** **must** be declared using the **Input Storage Class**
- VUID-SamplePosition-SamplePosition-04362
The variable decorated with **SamplePosition** **must** be declared as a two-component vector of 32-bit floating-point values

TessCoord

Decorating a variable with the **TessCoord** built-in decoration will make that variable contain the three-dimensional (u,v,w) barycentric coordinate of the tessellated vertex within the patch. u, v, and w are in the range [0,1] and vary linearly across the primitive being subdivided. For the tessellation modes of **Quads** or **Isolines**, the third component is always zero.

Valid Usage

- VUID-TessCoord-TessCoord-04387
The **TessCoord** decoration **must** be used only within the **TessellationEvaluation Execution Model**
- VUID-TessCoord-TessCoord-04388
The variable decorated with **TessCoord** **must** be declared using the **Input Storage Class**
- VUID-TessCoord-TessCoord-04389
The variable decorated with **TessCoord** **must** be declared as a three-component vector of 32-bit floating-point values

TessLevelOuter

Decorating a variable with the **TessLevelOuter** built-in decoration will make that variable contain the outer tessellation levels for the current patch.

In tessellation control shaders, the variable decorated with **TessLevelOuter** **can** be written to, controlling the tessellation factors for the resulting patch. These values are used by the tessellator to control primitive tessellation and **can** be read by tessellation evaluation shaders.

In tessellation evaluation shaders, the variable decorated with **TessLevelOuter** **can** read the values written by the tessellation control shader.

Valid Usage

- VUID-TessLevelOuter-TessLevelOuter-04390
The **TessLevelOuter** decoration **must** be used only within the **TessellationControl** or **TessellationEvaluation Execution Model**
- VUID-TessLevelOuter-TessLevelOuter-04391
The variable decorated with **TessLevelOuter** within the **TessellationControl Execution Model** **must** be declared using the **Output Storage Class**
- VUID-TessLevelOuter-TessLevelOuter-04392
The variable decorated with **TessLevelOuter** within the **TessellationEvaluation Execution Model** **must** be declared using the **Input Storage Class**
- VUID-TessLevelOuter-TessLevelOuter-04393
The variable decorated with **TessLevelOuter** **must** be declared as an array of size four, containing 32-bit floating-point values

TessLevelInner

Decorating a variable with the **TessLevelInner** built-in decoration will make that variable contain the inner tessellation levels for the current patch.

In tessellation control shaders, the variable decorated with **TessLevelInner** **can** be written to, controlling the tessellation factors for the resulting patch. These values are used by the tessellator to control primitive tessellation and **can** be read by tessellation evaluation shaders.

In tessellation evaluation shaders, the variable decorated with **TessLevelInner** **can** read the values written by the tessellation control shader.

Valid Usage

- VUID-TessLevelInner-TessLevelInner-04394
The **TessLevelInner** decoration **must** be used only within the **TessellationControl** or **TessellationEvaluation Execution Model**
- VUID-TessLevelInner-TessLevelInner-04395
The variable decorated with **TessLevelInner** within the **TessellationControl Execution Model** **must** be declared using the **Output Storage Class**
- VUID-TessLevelInner-TessLevelInner-04396
The variable decorated with **TessLevelInner** within the **TessellationEvaluation Execution Model** **must** be declared using the **Input Storage Class**
- VUID-TessLevelInner-TessLevelInner-04397
The variable decorated with **TessLevelInner** **must** be declared as an array of size two, containing 32-bit floating-point values

VertexIndex

Decorating a variable with the `VertexIndex` built-in decoration will make that variable contain the index of the vertex that is being processed by the current vertex shader invocation. For non-indexed draws, this variable begins at the `firstVertex` parameter to `vkCmdDraw` or the `firstVertex` member of a structure consumed by `vkCmdDrawIndirect` and increments by one for each vertex in the draw. For indexed draws, its value is the content of the index buffer for the vertex plus the `vertexOffset` parameter to `vkCmdDrawIndexed` or the `vertexOffset` member of the structure consumed by `vkCmdDrawIndexedIndirect`.

NOTE `VertexIndex` starts at the same starting value for each instance.

Valid Usage

- VUID-VertexIndex-VertexIndex-04398
The `VertexIndex` decoration **must** be used only within the `Vertex Execution Model`
- VUID-VertexIndex-VertexIndex-04399
The variable decorated with `VertexIndex` **must** be declared using the `Input Storage Class`
- VUID-VertexIndex-VertexIndex-04400
The variable decorated with `VertexIndex` **must** be declared as a scalar 32-bit integer value

ViewportIndex

Decorating a variable with the `ViewportIndex` built-in decoration will make that variable contain the index of the viewport.

In a geometry shader, the variable decorated with `ViewportIndex` can be written to with the viewport index to which the primitive produced by that shader will be directed.

The selected viewport index is used to select the viewport transform and scissor rectangle.

If the last active `pre-rasterization shader stage` shader entry point's interface does not include a variable decorated with `ViewportIndex` then the first viewport is used. If a `pre-rasterization shader stage` shader entry point's interface includes a variable decorated with `ViewportIndex`, it **must** write the same value to `ViewportIndex` for all output vertices of a given primitive.

In a fragment shader, the variable decorated with `ViewportIndex` contains the viewport index of the primitive that the fragment invocation belongs to.

Valid Usage

- VUID-ViewportIndex-ViewportIndex-04404
The `ViewportIndex` decoration **must** be used only within the `MeshEXT`, `MeshNV`, `Vertex`, `TessellationEvaluation`, `Geometry`, or `Fragment Execution Model`
- VUID-ViewportIndex-ViewportIndex-04406
The variable decorated with `ViewportIndex` within the `MeshEXT`, `MeshNV`, `Vertex`, `TessellationEvaluation`, or `Geometry Execution Model` **must** be declared using the `Output Storage Class`

- VUID-ViewportIndex-ViewportIndex-04407
The variable decorated with **ViewportIndex** within the **Fragment Execution Model** **must** be declared using the **Input Storage Class**
- VUID-ViewportIndex-ViewportIndex-04408
The variable decorated with **ViewportIndex** **must** be declared as a scalar 32-bit integer value
- VUID-ViewportIndex-ViewportIndex-07060
The variable decorated with **ViewportIndex** within the **MeshEXT Execution Model** **must** also be decorated with the **PerPrimitiveEXT** decoration

WorkgroupId

Decorating a variable with the **WorkgroupId** built-in decoration will make that variable contain the global workgroup that the current invocation is a member of. Each component ranges from a base value to a base + count value, based on the parameters passed into the dispatching commands.

Valid Usage

- VUID-WorkgroupId-WorkgroupId-04422
The **WorkgroupId** decoration **must** be used only within the **GLCompute**, **MeshEXT**, **TaskEXT**, **MeshNV**, or **TaskNV Execution Model**
- VUID-WorkgroupId-WorkgroupId-04423
The variable decorated with **WorkgroupId** **must** be declared using the **Input Storage Class**
- VUID-WorkgroupId-WorkgroupId-04424
The variable decorated with **WorkgroupId** **must** be declared as a three-component vector of 32-bit integer values

WorkgroupSize

Decorating an object with the **WorkgroupSize** built-in decoration will make that object contain the dimensions of a local workgroup. If an object is decorated with the **WorkgroupSize** decoration, this takes precedence over any **LocalSize** execution mode.

Valid Usage

- VUID-WorkgroupSize-WorkgroupSize-04425
The **WorkgroupSize** decoration **must** be used only within the **GLCompute**, **MeshEXT**, **TaskEXT**, **MeshNV**, or **TaskNV Execution Model**
- VUID-WorkgroupSize-WorkgroupSize-04426
The variable decorated with **WorkgroupSize** **must** be a specialization constant or a constant
- VUID-WorkgroupSize-WorkgroupSize-04427
The variable decorated with **WorkgroupSize** **must** be declared as a three-component vector of 32-bit integer values

Chapter 16. Image Operations

16.1. Image Operations Overview

Vulkan Image Operations are operations performed by those SPIR-V Image Instructions which take an `OpTypeImage` (representing a `VkImageView`) or `OpTypeSampledImage` (representing a (`VkImageView`, `VkSampler`) pair). Read, write, and atomic operations also take texel coordinates as operands, and return a value based on a neighborhood of texture elements (*texels*) within the image. Query operations return properties of the bound image or of the lookup itself. The “Depth” operand of `OpTypeImage` is ignored.

NOTE

Texel is a term which is a combination of the words texture and element. Early interactive computer graphics supported texture operations on textures, a small subset of the image operations on images described here. The discrete samples remain essentially equivalent, however, so we retain the historical term texel to refer to them.

Image Operations include the functionality of the following SPIR-V Image Instructions:

- `OpImageSample*` and `OpImageSparseSample*` read one or more neighboring texels of the image, and `filter` the texel values based on the state of the sampler.
 - Instructions with `ImplicitLod` in the name `determine` the LOD used in the sampling operation based on the coordinates used in neighboring fragments.
 - Instructions with `ExplicitLod` in the name `determine` the LOD used in the sampling operation based on additional coordinates.
 - Instructions with `Proj` in the name apply homogeneous `projection` to the coordinates.
- `OpImageFetch` and `OpImageSparseFetch` return a single texel of the image. No sampler is used.
- `OpImage*Gather` and `OpImageSparse*Gather` read neighboring texels and `return a single component` of each.
- `OpImageRead` (and `OpImageSparseRead`) and `OpImageWrite` read and write, respectively, a texel in the image. No sampler is used.
- `OpImage*Dref*` instructions apply `depth comparison` on the texel values.
- `OpImageSparse*` instructions additionally return a `sparse residency` code.
- `OpImageQuerySize`, `OpImageQuerySizeLod`, `OpImageQueryLevels`, and `OpImageQuerySamples` return properties of the image descriptor that would be accessed. The image itself is not accessed.
- `OpImageQueryLod` returns the LOD parameters that would be used in a sample operation. The actual operation is not performed.

16.1.1. Texel Coordinate Systems

Images are addressed by *texel coordinates*. There are three *texel coordinate systems*:

- normalized texel coordinates [0.0, 1.0]

- unnormalized texel coordinates [0.0, width / height / depth)
- integer texel coordinates [0, width / height / depth)

SPIR-V `OpImageFetch`, `OpImageSparseFetch`, `OpImageRead`, `OpImageSparseRead`, and `OpImageWrite` instructions use integer texel coordinates.

Other image instructions **can** use either normalized or unnormalized texel coordinates (selected by the `unnormalizedCoordinates` state of the sampler used in the instruction), but there are [limitations](#) on what operations, image state, and sampler state is supported. Normalized coordinates are logically [converted](#) to unnormalized as part of image operations, and [certain steps](#) are only performed on normalized coordinates. The array layer coordinate is always treated as unnormalized even when other coordinates are normalized.

Normalized texel coordinates are referred to as (s,t,r,q,a), with the coordinates having the following meanings:

- s: Coordinate in the first dimension of an image.
- t: Coordinate in the second dimension of an image.
- r: Coordinate in the third dimension of an image.
 - (s,t,r) are interpreted as a direction vector for Cube images.
- q: Fourth coordinate, for homogeneous (projective) coordinates.
- a: Coordinate for array layer.

The coordinates are extracted from the SPIR-V operand based on the dimensionality of the image variable and type of instruction. For `Proj` instructions, the components are in order (s, [t,] [r,] q), with t and r being conditionally present based on the `Dim` of the image. For non-`Proj` instructions, the coordinates are (s [t] [r] [a]), with t and r being conditionally present based on the `Dim` of the image and a being conditionally present based on the `Arrayed` property of the image. Projective image instructions are not supported on `Arrayed` images.

Unnormalized texel coordinates are referred to as (u,v,w,a), with the coordinates having the following meanings:

- u: Coordinate in the first dimension of an image.
- v: Coordinate in the second dimension of an image.
- w: Coordinate in the third dimension of an image.
- a: Coordinate for array layer.

Only the u and v coordinates are directly extracted from the SPIR-V operand, because only 1D and 2D (non-`Arrayed`) dimensionalities support unnormalized coordinates. The components are in order (u [v]), with v being conditionally present when the dimensionality is 2D. When normalized coordinates are converted to unnormalized coordinates, all four coordinates are used.

Integer texel coordinates are referred to as (i,j,k,l,n), with the coordinates having the following meanings:

- i: Coordinate in the first dimension of an image.

- j : Coordinate in the second dimension of an image.
- k : Coordinate in the third dimension of an image.
- l : Coordinate for array layer.
- n : Index of the sample within the texel.

They are extracted from the SPIR-V operand in order (i [j] [k] [l] [n]), with j and k conditionally present based on the **Dim** of the image, and l conditionally present based on the **Arrayed** property of the image. n is conditionally present and is taken from the **Sample** image operand.

For all coordinate types, unused coordinates are assigned a value of zero.

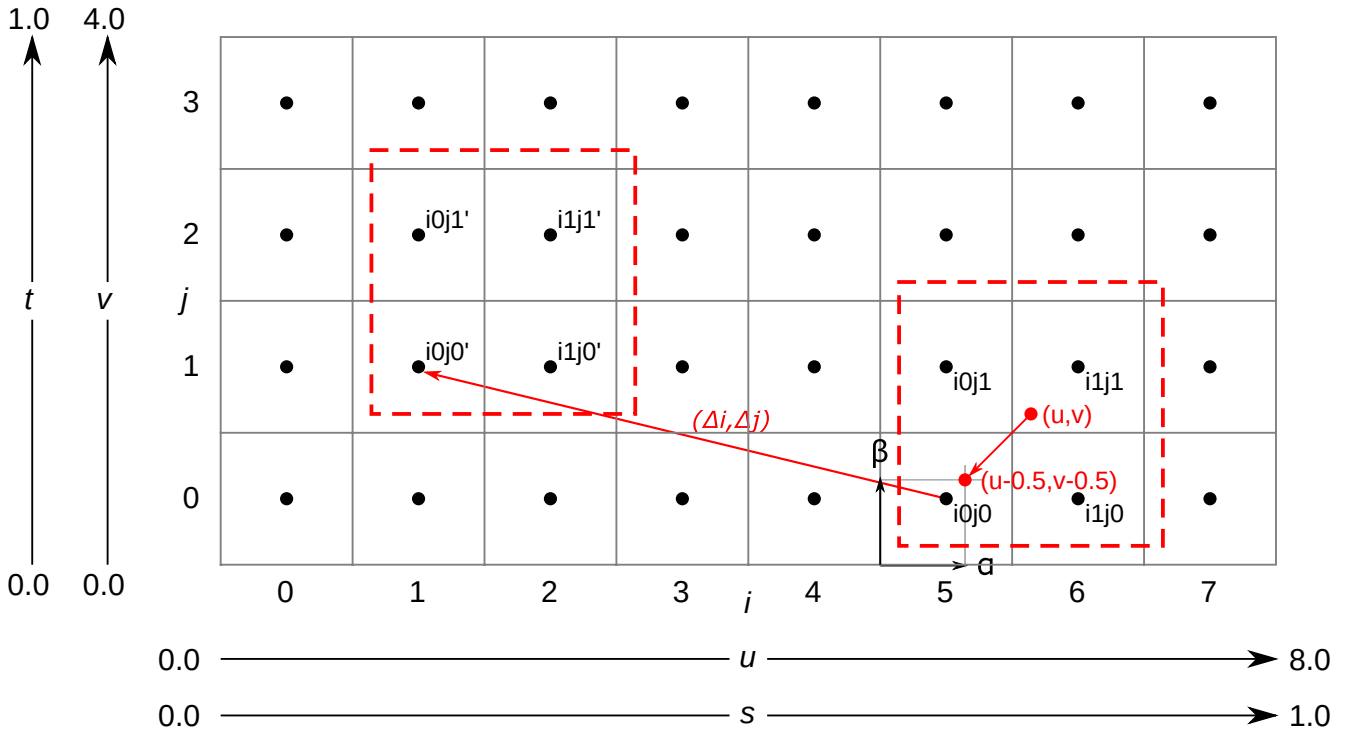


Figure 3. Texel Coordinate Systems, Linear Filtering

The Texel Coordinate Systems - For the example shown of an 8×4 texel two dimensional image.

- Normalized texel coordinates:
 - The s coordinate goes from 0.0 to 1.0.
 - The t coordinate goes from 0.0 to 1.0.
- Unnormalized texel coordinates:
 - The u coordinate within the range 0.0 to 8.0 is within the image, otherwise it is outside the image.
 - The v coordinate within the range 0.0 to 4.0 is within the image, otherwise it is outside the image.
- Integer texel coordinates:
 - The i coordinate within the range 0 to 7 addresses texels within the image, otherwise it is outside the image.
 - The j coordinate within the range 0 to 3 addresses texels within the image, otherwise it is

outside the image.

- Also shown for linear filtering:
 - Given the unnormalized coordinates (u,v) , the four texels selected are i_0j_0 , i_1j_0 , i_0j_1 , and i_1j_1 .
 - The fractions α and β .
 - Given the offset Δ_i and Δ_j , the four texels selected by the offset are $i_0j'_0$, $i_1j'_0$, $i_0j'_1$, and $i_1j'_1$.

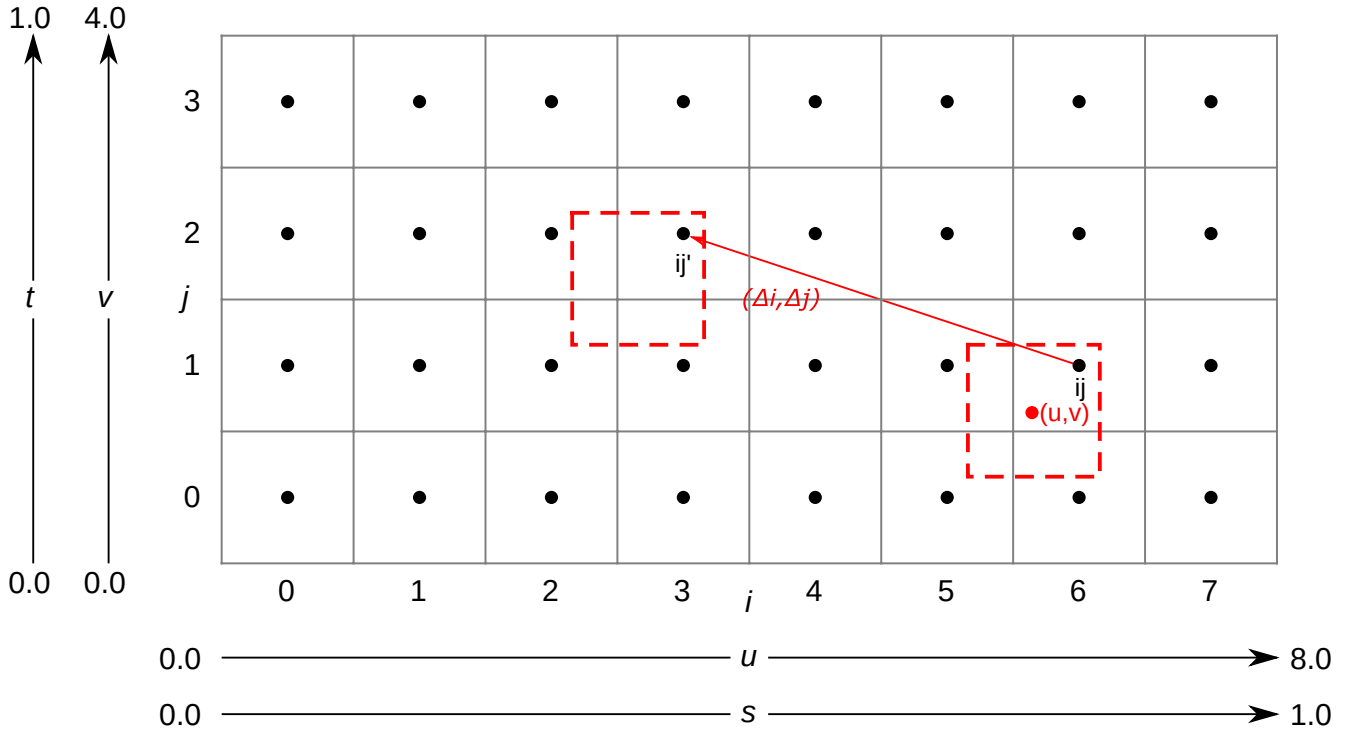


Figure 4. Texel Coordinate Systems, Nearest Filtering

The Texel Coordinate Systems - For the example shown of an 8x4 texel two dimensional image.

- Texel coordinates as above. Also shown for nearest filtering:
 - Given the unnormalized coordinates (u,v) , the texel selected is ij .
 - Given the offset Δ_i and Δ_j , the texel selected by the offset is ij' .

16.2. Conversion Formulas

16.2.1. RGB to Shared Exponent Conversion

An RGB color (red, green, blue) is transformed to a shared exponent color (red_{shared} , $green_{shared}$, $blue_{shared}$, exp_{shared}) as follows:

First, the components (red, green, blue) are clamped to ($red_{clamped}$, $green_{clamped}$, $blue_{clamped}$) as:

$$red_{clamped} = \max(0, \min(sharedexp_{max}, red))$$

$$green_{clamped} = \max(0, \min(sharedexp_{max}, green))$$

$$\text{blue}_{\text{clamped}} = \max(0, \min(\text{sharedexp}_{\text{max}}, \text{blue}))$$

where:

$$\begin{aligned} N &= 9 && \text{number of mantissa bits per component} \\ B &= 15 && \text{exponent bias} \\ E_{\text{max}} &= 31 && \text{maximum possible biased exponent value} \\ \text{sharedexp}_{\text{max}} &= \frac{(2^N - 1)}{2^N} \times 2^{(E_{\text{max}} - B)} \end{aligned}$$

NOTE

NaN, if supported, is handled as in [IEEE 754-2008 minNum\(\)](#) and [maxNum\(\)](#). This results in any NaN being mapped to zero.

The largest clamped component, $\text{max}_{\text{clamped}}$ is determined:

$$\text{max}_{\text{clamped}} = \max(\text{red}_{\text{clamped}}, \text{green}_{\text{clamped}}, \text{blue}_{\text{clamped}})$$

A preliminary shared exponent exp' is computed:

$$\text{exp}' = \begin{cases} \lfloor \log_2(\text{max}_{\text{clamped}}) \rfloor + (B + 1) & \text{for } \text{max}_{\text{clamped}} > 2^{-(B+1)} \\ 0 & \text{for } \text{max}_{\text{clamped}} \leq 2^{-(B+1)} \end{cases}$$

The shared exponent $\text{exp}_{\text{shared}}$ is computed:

$$\begin{aligned} \text{max}_{\text{shared}} &= \lfloor \frac{\text{max}_{\text{clamped}}}{2^{(\text{exp}' - B - N)}} + \frac{1}{2} \rfloor \\ \text{exp}_{\text{shared}} &= \begin{cases} \text{exp}' & \text{for } 0 \leq \text{max}_{\text{shared}} < 2^N \\ \text{exp}' + 1 & \text{for } \text{max}_{\text{shared}} = 2^N \end{cases} \end{aligned}$$

Finally, three integer values in the range 0 to 2^N are computed:

$$\begin{aligned} \text{red}_{\text{shared}} &= \lfloor \frac{\text{red}_{\text{clamped}}}{2^{(\text{exp}_{\text{shared}} - B - N)}} + \frac{1}{2} \rfloor \\ \text{green}_{\text{shared}} &= \lfloor \frac{\text{green}_{\text{clamped}}}{2^{(\text{exp}_{\text{shared}} - B - N)}} + \frac{1}{2} \rfloor \\ \text{blue}_{\text{shared}} &= \lfloor \frac{\text{blue}_{\text{clamped}}}{2^{(\text{exp}_{\text{shared}} - B - N)}} + \frac{1}{2} \rfloor \end{aligned}$$

16.2.2. Shared Exponent to RGB

A shared exponent color ($\text{red}_{\text{shared}}, \text{green}_{\text{shared}}, \text{blue}_{\text{shared}}, \text{exp}_{\text{shared}}$) is transformed to an RGB color (red, green, blue) as follows:

$$red = red_{shared} \times 2^{(exp_{shared} - B - N)}$$

$$green = green_{shared} \times 2^{(exp_{shared} - B - N)}$$

$$blue = blue_{shared} \times 2^{(exp_{shared} - B - N)}$$

where:

N = 9 (number of mantissa bits per component)

B = 15 (exponent bias)

16.3. Texel Input Operations

Texel input instructions are SPIR-V image instructions that read from an image. *Texel input operations* are a set of steps that are performed on state, coordinates, and texel values while processing a texel input instruction, and which are common to some or all texel input instructions. They include the following steps, which are performed in the listed order:

- [Validation operations](#)
 - [Instruction/Sampler/Image validation](#)
 - [Coordinate validation](#)
 - [Sparse validation](#)
- [Format conversion](#)
- [Texel replacement](#)
- [Depth comparison](#)
- [Conversion to RGBA](#)
- [Component swizzle](#)

For texel input instructions involving multiple texels (for sampling or gathering), these steps are applied for each texel that is used in the instruction. Depending on the type of image instruction, other steps are conditionally performed between these steps or involving multiple coordinate or texel values.

16.3.1. Texel Input Validation Operations

Texel input validation operations inspect instruction/image/sampler state or coordinates, and in certain circumstances cause the texel value to be replaced or become undefined. There are a series of validations that the texel undergoes.

Instruction/Sampler/Image View Validation

There are a number of cases where a SPIR-V instruction **can** mismatch with the sampler, the image view, or both, and a number of further cases where the sampler **can** mismatch with the image view. In such cases the value of the texel returned is undefined.

These cases include:

- The sampler `borderColor` is an integer type and the image view `format` is not one of the `VkFormat` integer types or a stencil component of a depth/stencil format.
- The sampler `borderColor` is a float type and the image view `format` is not one of the `VkFormat` float types or a depth component of a depth/stencil format.
- The sampler `borderColor` is one of the opaque black colors (`VK_BORDER_COLOR_FLOAT_OPAQUE_BLACK` or `VK_BORDER_COLOR_INT_OPAQUE_BLACK`) and the image view `VkComponentSwizzle` for any of the `VkComponentMapping` components is not the `identity swizzle`.
- The `VkImageLayout` of any subresource in the image view does not match the `VkDescriptorImageInfo::imageLayout` used to write the image descriptor.
- The SPIR-V Image Format is not `compatible` with the image view's `format`.
- The sampler `unnormalizedCoordinates` is `VK_TRUE` and any of the `limitations of unnormalized coordinates` are violated.
- The SPIR-V instruction is one of the `OpImage*Dref*` instructions and the sampler `compareEnable` is `VK_FALSE`
- The SPIR-V instruction is not one of the `OpImage*Dref*` instructions and the sampler `compareEnable` is `VK_TRUE`
- The SPIR-V instruction is one of the `OpImage*Dref*` instructions and the image view `format` is not one of the depth/stencil formats with a depth component, or the image view aspect is not `VK_IMAGE_ASPECT_DEPTH_BIT`.
- The SPIR-V instruction's image variable's properties are not compatible with the image view:
 - If the image view's `viewType` is one of `VK_IMAGE_VIEW_TYPE_1D_ARRAY`, `VK_IMAGE_VIEW_TYPE_2D_ARRAY`, or `VK_IMAGE_VIEW_TYPE_CUBE_ARRAY` then the instruction **must** have `Arrayed` = 1. Otherwise the instruction **must** have `Arrayed` = 0.
 - If the image was created with `VkImageCreateInfo::samples` equal to `VK_SAMPLE_COUNT_1_BIT`, the instruction **must** have `MS` = 0.
 - If the image was created with `VkImageCreateInfo::samples` not equal to `VK_SAMPLE_COUNT_1_BIT`, the instruction **must** have `MS` = 1.
 - If the `Sampled Type` of the `OpTypeImage` does not match the `SPIR-V Type`.
 - If the `signedness of any read or sample operation` does not match the signedness of the image's format.

If the underlying `VkImage` format has an X component in its format description, undefined values are read from those bits.

NOTE

If the `VkImage` format and `VkImageView` format are the same, these bits will be unused

by format conversion and this will have no effect. However, if the `VkImageView` format is different, then some bits of the result may be undefined. For example, when a `VK_FORMAT_R10X6_UNORM_PACK16` `VkImage` is sampled via a `VK_FORMAT_R16_UNORM` `VkImageView`, the low 6 bits of the value before format conversion are undefined and format conversion may return a range of different values.

NOTE

Some implementations will return undefined values in the case where a sampler uses a `VkSamplerAddressMode` of `VK_SAMPLER_ADDRESS_MODE_MIRRORED_REPEAT`, the sampler is used with operands `Offset`, `ConstOffset`, or `ConstOffsets`, and the value of the offset is larger than or equal to the corresponding width, height, or depth of any accessed image level.

This behavior was not tested prior to Vulkan conformance test suite version 1.3.8.0. Affected implementations will have a conformance test waiver for this issue.

Integer Texel Coordinate Validation

Integer texel coordinates are validated against the size of the image level, and the number of layers and number of samples in the image. For SPIR-V instructions that use integer texel coordinates, this is performed directly on the integer coordinates. For instructions that use normalized or unnormalized texel coordinates, this is performed on the coordinates that result after [conversion](#) to integer texel coordinates.

If the integer texel coordinates do not satisfy all of the conditions

$$0 \leq i < w_s$$

$$0 \leq j < h_s$$

$$0 \leq k < d_s$$

$$0 \leq l < \text{layers}$$

$$0 \leq n < \text{samples}$$

where:

w_s = width of the image level

h_s = height of the image level

d_s = depth of the image level

layers = number of layers in the image

samples = number of samples per texel in the image

then the texel fails integer texel coordinate validation.

There are four cases to consider:

1. Valid Texel Coordinates

- If the texel coordinates pass validation (that is, the coordinates lie within the image),
then the texel value comes from the value in image memory.

2. Border Texel

- If the texel coordinates fail validation, and
- If the read is the result of an image sample instruction or image gather instruction, and
- If the image is not a cube image,

then the texel is a border texel and [texel replacement](#) is performed.

3. Invalid Texel

- If the texel coordinates fail validation, and
- If the read is the result of an image fetch instruction, image read instruction, or atomic instruction,

then the texel is an invalid texel and [texel replacement](#) is performed.

4. Cube Map Edge or Corner

Otherwise the texel coordinates lie beyond the edges or corners of the selected cube map face, and [Cube map edge handling](#) is performed.

Cube Map Edge Handling

If the texel coordinates lie beyond the edges or corners of the selected cube map face (as described in the prior section), the following steps are performed. Note that this does not occur when using `VK_FILTER_NEAREST` filtering within a mip level, since `VK_FILTER_NEAREST` is treated as using `VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_EDGE`.

- Cube Map Edge Texel

- If the texel lies beyond the selected cube map face in either only i or only j , then the coordinates (i,j) and the array layer l are transformed to select the adjacent texel from the appropriate neighboring face.

- Cube Map Corner Texel
 - If the texel lies beyond the selected cube map face in both i and j, then there is no unique neighboring face from which to read that texel. The texel **should** be replaced by the average of the three values of the adjacent texels in each incident face. However, implementations **may** replace the cube map corner texel by other methods. The methods are subject to the constraint that if the three available texels have the same value, the resulting filtered texel **must** have that value.

Sparse Validation

If the texel reads from an unbound region of a sparse image, the texel is a *sparse unbound texel*, and processing continues with [texel replacement](#).

16.3.2. Format Conversion

Texels undergo a format conversion from the [VkFormat](#) of the image view to a vector of either floating-point or signed or unsigned integer components, with the number of components based on the number of components present in the format.

- Color formats have one, two, three, or four components, according to the format.
- Depth/stencil formats are one component. The depth or stencil component is selected by the [aspectMask](#) of the image view.

Each component is converted based on its type and size (as defined in the [Format Definition](#) section for each [VkFormat](#)), using the appropriate equations in [16-Bit Floating-Point Numbers](#), [Unsigned 11-Bit Floating-Point Numbers](#), [Unsigned 10-Bit Floating-Point Numbers](#), [Fixed-Point Data Conversion](#), and [Shared Exponent to RGB](#). Signed integer components smaller than 32 bits are sign-extended.

If the image view format is sRGB, the color components are first converted as if they are UNORM, and then sRGB to linear conversion is applied to the R, G, and B components as described in the “sRGB EOTF” section of the [Khronos Data Format Specification](#). The A component, if present, is unchanged.

If the image view format is block-compressed, then the texel value is first decoded, then converted based on the type and number of components defined by the compressed format.

16.3.3. Texel Replacement

A texel is replaced if it is one (and only one) of:

- a border texel,
- an invalid texel, or
- a sparse unbound texel.

Border texels are replaced with a value based on the image format and the [borderColor](#) of the sampler. The border color is:

Table 14. Border Color B

Sampler borderColor	Corresponding Border Color
VK_BORDER_COLOR_FLOAT_TRANSPARENT_BLACK	$[B_r, B_g, B_b, B_a] = [0.0, 0.0, 0.0, 0.0]$
VK_BORDER_COLOR_FLOAT_OPAQUE_BLACK	$[B_r, B_g, B_b, B_a] = [0.0, 0.0, 0.0, 1.0]$
VK_BORDER_COLOR_FLOAT_OPAQUE_WHITE	$[B_r, B_g, B_b, B_a] = [1.0, 1.0, 1.0, 1.0]$
VK_BORDER_COLOR_INT_TRANSPARENT_BLACK	$[B_r, B_g, B_b, B_a] = [0, 0, 0, 0]$
VK_BORDER_COLOR_INT_OPAQUE_BLACK	$[B_r, B_g, B_b, B_a] = [0, 0, 0, 1]$
VK_BORDER_COLOR_INT_OPAQUE_WHITE	$[B_r, B_g, B_b, B_a] = [1, 1, 1, 1]$

NOTE

The names `VK_BORDER_COLOR*_TRANSPARENT_BLACK`, `VK_BORDER_COLOR*_OPAQUE_BLACK`, and `VK_BORDER_COLOR*_OPAQUE_WHITE` are meant to describe which components are zeros and ones in the vocabulary of compositing, and are not meant to imply that the numerical value of `VK_BORDER_COLOR_INT_OPAQUE_WHITE` is a saturating value for integers.

This is substituted for the texel value by replacing the number of components in the image format

Table 15. Border Texel Components After Replacement

Texel Aspect or Format	Component Assignment
Depth aspect	$D = B_r$
Stencil aspect	$S = B_r$
One component color format	$\text{Color}_r = B_r$
Two component color format	$[\text{Color}_r, \text{Color}_g] = [B_r, B_g]$
Three component color format	$[\text{Color}_r, \text{Color}_g, \text{Color}_b] = [B_r, B_g, B_b]$
Four component color format	$[\text{Color}_r, \text{Color}_g, \text{Color}_b, \text{Color}_a] = [B_r, B_g, B_b, B_a]$

The value returned by a read of an invalid texel is undefined, unless that read operation is from a buffer resource and the `robustBufferAccess` feature is enabled. In that case, an invalid texel is replaced as described by the `robustBufferAccess` feature.

If the `VkPhysicalDeviceSparseProperties::residencyNonResidentStrict` property is `VK_TRUE`, a sparse unbound texel is replaced with 0 or 0.0 values for integer and floating-point components of the image format, respectively.

If `residencyNonResidentStrict` is `VK_FALSE`, the value of the sparse unbound texel is undefined.

16.3.4. Depth Compare Operation

If the image view has a depth/stencil format, the depth component is selected by the `aspectMask`, and the operation is an `OpImage*Dref*` instruction, a depth comparison is performed. The result is 1.0 if the comparison evaluates to true, and 0.0 otherwise. This value replaces the depth component D.

The compare operation is selected by the `VkCompareOp` value set by `VkSamplerCreateInfo`

::compareOp. The reference value from the SPIR-V operand D_{ref} and the texel depth value D_{tex} are used as the *reference* and *test* values, respectively, in that operation.

If the image being sampled has an unsigned normalized fixed-point format, then D_{ref} is clamped to [0,1] before the compare operation.

16.3.5. Conversion to RGBA

The texel is expanded from one, two, or three components to four components based on the image base color:

Table 16. Texel Color After Conversion To RGBA

Texel Aspect or Format	RGBA Color
Depth aspect	$[Color_r, Color_g, Color_b, Color_a] = [D, 0, 0, one]$
Stencil aspect	$[Color_r, Color_g, Color_b, Color_a] = [S, 0, 0, one]$
One component color format	$[Color_r, Color_g, Color_b, Color_a] = [Color_r, 0, 0, one]$
Two component color format	$[Color_r, Color_g, Color_b, Color_a] = [Color_r, Color_g, 0, one]$
Three component color format	$[Color_r, Color_g, Color_b, Color_a] = [Color_r, Color_g, Color_b, one]$
Four component color format	$[Color_r, Color_g, Color_b, Color_a] = [Color_r, Color_g, Color_b, Color_a]$

where one = 1.0f for floating-point formats and depth aspects, and one = 1 for integer formats and stencil aspects.

16.3.6. Component Swizzle

All texel input instructions apply a *swizzle* based on the [VkComponentSwizzle](#) enums in the [components](#) member of the [VkImageViewCreateInfo](#) structure for the image being read.

The swizzle **can** rearrange the components of the texel, or substitute zero or one for any components. It is defined as follows for each color component:

$$Color_{\text{'component}} = \begin{cases} Color_r & \text{for RED swizzle} \\ Color_g & \text{for GREEN swizzle} \\ Color_b & \text{for BLUE swizzle} \\ Color_a & \text{for ALPHA swizzle} \\ 0 & \text{for ZERO swizzle} \\ one & \text{for ONE swizzle} \\ identity & \text{for IDENTITY swizzle} \end{cases}$$

where:

$$one = \begin{cases} 1.0f & \text{for floating-point components} \\ 1 & \text{for integer components} \end{cases}$$

$$identity = \begin{cases} Color_r & \text{for component} = r \\ Color_g & \text{for component} = g \\ Color_b & \text{for component} = b \\ Color_a & \text{for component} = a \end{cases}$$

If the border color is one of the `VK_BORDER_COLOR_*_OPAQUE_BLACK` enums and the `VkComponentSwizzle` is not the `identity swizzle` for all components, the value of the texel after swizzle is undefined.

If the image view has a depth/stencil format and the `VkComponentSwizzle` is `VK_COMPONENT_SWIZZLE_ONE`, the value of the texel after swizzle is undefined.

16.3.7. Sparse Residency

`OpImageSparse*` instructions return a structure which includes a *residency code* indicating whether any texels accessed by the instruction are sparse unbound texels. This code **can** be interpreted by the `OpImageSparseTexelsResident` instruction which converts the residency code to a boolean value.

16.4. Texel Output Operations

Texel output instructions are SPIR-V image instructions that write to an image. *Texel output operations* are a set of steps that are performed on state, coordinates, and texel values while processing a texel output instruction, and which are common to some or all texel output instructions. They include the following steps, which are performed in the listed order:

- [Validation operations](#)
 - [Format validation](#)
 - [Type validation](#)
 - [Coordinate validation](#)
 - [Sparse validation](#)
- [Texel output format conversion](#)

16.4.1. Texel Output Validation Operations

Texel output validation operations inspect instruction/image state or coordinates, and in certain circumstances cause the write to have no effect. There are a series of validations that the texel undergoes.

Texel Format Validation

If the image format of the `OpTypeImage` is not [compatible](#) with the `VkImageView`'s `format`, the write causes the contents of the image's memory to become undefined.

Texel Type Validation

If the **Sampled Type** of the **OpTypeImage** does not match the **SPIR-V Type**, the write causes the value of the texel to become undefined. For integer types, if the **signedness of the access** does not match the signedness of the accessed resource, the write causes the value of the texel to become undefined.

16.4.2. Integer Texel Coordinate Validation

The integer texel coordinates are validated according to the same rules as for texel input **coordinate validation**.

If the texel fails integer texel coordinate validation, then the write has no effect.

16.4.3. Sparse Texel Operation

If the texel attempts to write to an unbound region of a sparse image, the texel is a sparse unbound texel. In such a case, if the **VkPhysicalDeviceSparseProperties::residencyNonResidentStrict** property is **VK_TRUE**, the sparse unbound texel write has no effect. If **residencyNonResidentStrict** is **VK_FALSE**, the write **may** have a side effect that becomes visible to other accesses to unbound texels in any resource, but will not be visible to any device memory allocated by the application.

16.4.4. Texel Output Format Conversion

If the image format is sRGB, a linear to sRGB conversion is applied to the R, G, and B components as described in the “sRGB EOTF” section of the **Khronos Data Format Specification**. The A component, if present, is unchanged.

Texels then undergo a format conversion from the floating-point, signed, or unsigned integer type of the texel data to the **VkFormat** of the image view. If the number of components in the texel data is larger than the number of components in the format, additional components are discarded.

Each component is converted based on its type and size (as defined in the **Format Definition** section for each **VkFormat**). Floating-point outputs are converted as described in **Floating-Point Format Conversions** and **Fixed-Point Data Conversion**. Integer outputs are converted such that their value is preserved. The converted value of any integer that cannot be represented in the target format is undefined.

If the **VkImageView** format has an X component in its format description, undefined values are written to those bits.

If the underlying **VkImage** format has an X component in its format description, undefined values are also written to those bits, even if result format conversion produces a valid value for those bits because the **VkImageView** format is different.

16.5. Normalized Texel Coordinate Operations

If the image sampler instruction provides normalized texel coordinates, some of the following operations are performed.

16.5.1. Projection Operation

For **Proj** image operations, the normalized texel coordinates (s,t,r,q,a) and (if present) the D_{ref} coordinate are transformed as follows:

$$\begin{aligned} s &= \frac{s}{q}, & \text{for 1D, 2D, or 3D image} \\ t &= \frac{t}{q}, & \text{for 2D or 3D image} \\ r &= \frac{r}{q}, & \text{for 3D image} \\ D_{ref} &= \frac{D_{ref}}{q}, & \text{if provided} \end{aligned}$$

16.5.2. Derivative Image Operations

Derivatives are used for LOD selection. These derivatives are either implicit (in an **ImplicitLod** image instruction in a fragment shader) or explicit (provided explicitly by shader to the image instruction in any shader).

For implicit derivatives image instructions, the derivatives of texel coordinates are calculated in the same manner as [derivative operations](#). That is:

$$\begin{aligned} \partial s / \partial x &= dPdx(s), & \partial s / \partial y &= dPdy(s), & \text{for 1D, 2D, Cube, or 3D image} \\ \partial t / \partial x &= dPdx(t), & \partial t / \partial y &= dPdy(t), & \text{for 2D, Cube, or 3D image} \\ \partial r / \partial x &= dPdx(r), & \partial r / \partial y &= dPdy(r), & \text{for Cube or 3D image} \end{aligned}$$

Partial derivatives not defined above for certain image dimensionalities are set to zero.

For explicit LOD image instructions, if the **optional** SPIR-V operand **Grad** is provided, then the operand values are used for the derivatives. The number of components present in each derivative for a given image dimensionality matches the number of partial derivatives computed above.

If the **optional** SPIR-V operand **Lod** is provided, then derivatives are set to zero, the cube map derivative transformation is skipped, and the scale factor operation is skipped. Instead, the floating-point scalar coordinate is directly assigned to λ_{base} as described in [LOD Operation](#).

16.5.3. Cube Map Face Selection and Transformations

For cube map image instructions, the (s,t,r) coordinates are treated as a direction vector (r_x, r_y, r_z). The direction vector is used to select a cube map face. The direction vector is transformed to a per-face texel coordinate system (s_{face}, t_{face}). The direction vector is also used to transform the derivatives to per-face derivatives.

16.5.4. Cube Map Face Selection

The direction vector selects one of the cube map's faces based on the largest magnitude coordinate direction (the major axis direction). Since two or more coordinates **can** have identical magnitude,

the implementation **must** have rules to disambiguate this situation.

The rules **should** have as the first rule that r_z wins over r_y and r_x , and the second rule that r_y wins over r_x . An implementation **may** choose other rules, but the rules **must** be deterministic and depend only on (r_x, r_y, r_z) .

The layer number (corresponding to a cube map face), the coordinate selections for s_c , t_c , r_c , and the selection of derivatives, are determined by the major axis direction as specified in the following two tables.

Table 17. Cube Map Face and Coordinate Selection

Major Axis Direction	Layer Number	Cube Map Face	s_c	t_c	r_c
$+r_x$	0	Positive X	$-r_z$	$-r_y$	r_x
$-r_x$	1	Negative X	$+r_z$	$-r_y$	r_x
$+r_y$	2	Positive Y	$+r_x$	$+r_z$	r_y
$-r_y$	3	Negative Y	$+r_x$	$-r_z$	r_y
$+r_z$	4	Positive Z	$+r_x$	$-r_y$	r_z
$-r_z$	5	Negative Z	$-r_x$	$-r_y$	r_z

Table 18. Cube Map Derivative Selection

Major Axis Direction	$\partial s_c / \partial x$	$\partial s_c / \partial y$	$\partial t_c / \partial x$	$\partial t_c / \partial y$	$\partial r_c / \partial x$	$\partial r_c / \partial y$
$+r_x$	$-\partial r_z / \partial x$	$-\partial r_z / \partial y$	$-\partial r_y / \partial x$	$-\partial r_y / \partial y$	$+\partial r_x / \partial x$	$+\partial r_x / \partial y$
$-r_x$	$+\partial r_z / \partial x$	$+\partial r_z / \partial y$	$-\partial r_y / \partial x$	$-\partial r_y / \partial y$	$-\partial r_x / \partial x$	$-\partial r_x / \partial y$
$+r_y$	$+\partial r_x / \partial x$	$+\partial r_x / \partial y$	$+\partial r_z / \partial x$	$+\partial r_z / \partial y$	$+\partial r_y / \partial x$	$+\partial r_y / \partial y$
$-r_y$	$+\partial r_x / \partial x$	$+\partial r_x / \partial y$	$-\partial r_z / \partial x$	$-\partial r_z / \partial y$	$-\partial r_y / \partial x$	$-\partial r_y / \partial y$
$+r_z$	$+\partial r_x / \partial x$	$+\partial r_x / \partial y$	$-\partial r_y / \partial x$	$-\partial r_y / \partial y$	$+\partial r_z / \partial x$	$+\partial r_z / \partial y$
$-r_z$	$-\partial r_x / \partial x$	$-\partial r_x / \partial y$	$-\partial r_y / \partial x$	$-\partial r_y / \partial y$	$-\partial r_z / \partial x$	$-\partial r_z / \partial y$

16.5.5. Cube Map Coordinate Transformation

$$s_{face} = \frac{1}{2} \times \frac{s_c}{|r_c|} + \frac{1}{2}$$

$$t_{face} = \frac{1}{2} \times \frac{t_c}{|r_c|} + \frac{1}{2}$$

16.5.6. Cube Map Derivative Transformation

The partial derivatives of the [Cube Map Coordinate Transformations](#) can be computed as:

$$\begin{aligned}\frac{\partial s_{face}}{\partial x} &= \frac{\partial}{\partial x} \left(\frac{1}{2} \times \frac{s_c}{|r_c|} + \frac{1}{2} \right) \\ &= \frac{1}{2} \times \frac{\partial}{\partial x} \left(\frac{s_c}{|r_c|} \right) \\ &= \frac{1}{2} \times \left(\frac{|r_c| \times \partial s_c / \partial x - s_c \times \partial r_c / \partial x}{(r_c)^2} \right)\end{aligned}$$

The other derivatives are simplified similarly, resulting in

$$\begin{aligned}\frac{\partial s_{face}}{\partial y} &= \frac{1}{2} \times \left(\frac{|r_c| \times \partial s_c / \partial y - s_c \times \partial r_c / \partial y}{(r_c)^2} \right) \\ \frac{\partial t_{face}}{\partial x} &= \frac{1}{2} \times \left(\frac{|r_c| \times \partial t_c / \partial x - t_c \times \partial r_c / \partial x}{(r_c)^2} \right) \\ \frac{\partial t_{face}}{\partial y} &= \frac{1}{2} \times \left(\frac{|r_c| \times \partial t_c / \partial y - t_c \times \partial r_c / \partial y}{(r_c)^2} \right)\end{aligned}$$

16.5.7. Scale Factor Operation, LOD Operation and Image Level(s) Selection

LOD selection **can** be either explicit (provided explicitly by the image instruction) or implicit (determined from a scale factor calculated from the derivatives). The LOD **must** be computed with `mipmapPrecisionBits` of accuracy.

Scale Factor Operation

The magnitude of the derivatives are calculated by:

$$m_{ux} = |\partial s / \partial x| \times w_{base}$$

$$m_{vx} = |\partial t / \partial x| \times h_{base}$$

$$m_{wx} = |\partial r / \partial x| \times d_{base}$$

$$m_{uy} = |\partial s / \partial y| \times w_{base}$$

$$m_{vy} = |\partial t / \partial y| \times h_{base}$$

$$m_{wy} = |\partial r / \partial y| \times d_{base}$$

where:

$$\partial t / \partial x = \partial t / \partial y = 0 \text{ (for 1D images)}$$

$$\partial r / \partial x = \partial r / \partial y = 0 \text{ (for 1D, 2D or Cube images)}$$

and:

$$w_{\text{base}} = \text{image.w}$$

$$h_{\text{base}} = \text{image.h}$$

$$d_{\text{base}} = \text{image.d}$$

(for the `baseMipLevel`, from the image descriptor).

A point sampled in screen space has an elliptical footprint in texture space. The minimum and maximum scale factors ($\rho_{\min}$, $\rho_{\max}$) **should** be the minor and major axes of this ellipse.

The *scale factors* ρ_x and ρ_y , calculated from the magnitude of the derivatives in x and y, are used to compute the minimum and maximum scale factors.

ρ_x and ρ_y **may** be approximated with functions f_x and f_y , subject to the following constraints:

f_x is continuous and monotonically increasing in each of m_{ux} , m_{vx} , and m_{wx}
 f_y is continuous and monotonically increasing in each of m_{uy} , m_{vy} , and m_{wy}

$$\begin{aligned} \max(|m_{ux}|, |m_{vx}|, |m_{wx}|) &\leq f_x \leq \sqrt{2}(|m_{ux}| + |m_{vx}| + |m_{wx}|) \\ \max(|m_{uy}|, |m_{vy}|, |m_{wy}|) &\leq f_y \leq \sqrt{2}(|m_{uy}| + |m_{vy}| + |m_{wy}|) \end{aligned}$$

The minimum and maximum scale factors ($\rho_{\min}$, $\rho_{\max}$) are determined by:

$$\rho_{\max} = \max(\rho_x, \rho_y)$$

$$\rho_{\min} = \min(\rho_x, \rho_y)$$

The ratio of anisotropy is determined by:

$$\eta = \min(\rho_{\max} / \rho_{\min}, \max_{\text{Aniso}})$$

where:

$\text{sampler.max}_{\text{Aniso}} = \text{maxAnisotropy}$ (from sampler descriptor)

$\text{limits.max}_{\text{Aniso}} = \text{maxSamplerAnisotropy}$ (from physical device limits)

$\text{max}_{\text{Aniso}} = \min(\text{sampler.max}_{\text{Aniso}}, \text{limits.max}_{\text{Aniso}})$

If $\rho_{\max} = \rho_{\min} = 0$, then all the partial derivatives are zero, the fragment's footprint in texel space is a point, and η **should** be treated as 1. If $\rho_{\max} \neq 0$ and $\rho_{\min} = 0$ then all partial derivatives along one axis are zero, the fragment's footprint in texel space is a line segment, and η **should** be treated as $\text{max}_{\text{Aniso}}$. However, anytime the footprint is small in texel space the implementation **may** use a smaller value of η , even when $\rho_{\min}$ is zero or close to zero. If either `VkPhysicalDeviceFeatures::samplerAnisotropy` or `VkSamplerCreateInfo::anisotropyEnable` are `VK_FALSE`, $\text{max}_{\text{Aniso}}$ is set to 1.

If $\eta = 1$, sampling is isotropic. If $\eta > 1$, sampling is anisotropic.

The sampling rate (N) is derived as:

$$N = \lceil \eta \rceil$$

An implementation **may** round N up to the nearest supported sampling rate. An implementation **may** use the value of N as an approximation of η .

LOD Operation

The LOD parameter λ is computed as follows:

$$\lambda_{\text{base}}(x, y) = \begin{cases} \text{shaderOp.Lod} & \text{(from optional SPIR-V operand)} \\ \log_2\left(\frac{\rho_{\max}}{\eta}\right) & \text{otherwise} \end{cases}$$
$$\lambda'(x, y) = \lambda_{\text{base}} + \text{clamp}(\text{sampler.bias} + \text{shaderOp.bias}, -\text{maxSamplerLodBias}, \text{maxSamplerLodBias})$$
$$\lambda = \begin{cases} \text{lod}_{\max}, & \lambda' > \text{lod}_{\max} \\ \lambda', & \text{lod}_{\min} \leq \lambda' \leq \text{lod}_{\max} \\ \text{lod}_{\min}, & \lambda' < \text{lod}_{\min} \\ \text{undefined}, & \text{lod}_{\min} > \text{lod}_{\max} \end{cases}$$

where:

$\text{sampler.bias} = \text{mipLodBias}$ (from sampler descriptor)

$\text{shaderOp.bias} = \begin{cases} \text{Bias} & \text{(from optional SPIR-V operand)} \\ 0 & \text{otherwise} \end{cases}$

$\text{sampler.lod}_{\min} = \text{minLod}$ (from sampler descriptor)

$\text{shaderOp.lod}_{\min} = \begin{cases} \text{MinLod} & \text{(from optional SPIR-V operand)} \\ 0 & \text{otherwise} \end{cases}$

$\text{lod}_{\min} = \max(\text{sampler.lod}_{\min}, \text{shaderOp.lod}_{\min})$

$\text{lod}_{\max} = \text{maxLod}$ (from sampler descriptor)

and maxSamplerLodBias is the value of the `VkPhysicalDeviceLimits` feature `maxSamplerLodBias`.

Image Level(s) Selection

The image level(s) d , d_{hi} , and d_{lo} which texels are read from are determined by an image-level parameter d_l , which is computed based on the LOD parameter, as follows:

$$d_l = \begin{cases} nearest(d'), & \text{mipmapMode is VK_SAMPLER_MIPMAP_MODE_NEAREST} \\ d', & \text{otherwise} \end{cases}$$

where:

$$d' = level_{base} + \text{clamp}(\lambda, 0, q)$$

$$nearest(d') = \begin{cases} [d' + 0.5] - 1, & \text{preferred} \\ [d' + 0.5], & \text{alternative} \end{cases}$$

and:

$$\begin{aligned} level_{base} &= baseMipLevel \\ q &= levelCount - 1 \end{aligned}$$

`baseMipLevel` and `levelCount` are taken from the `subresourceRange` of the image view.

If the sampler's `mipmapMode` is `VK_SAMPLER_MIPMAP_MODE_NEAREST`, then the level selected is $d = d_l$.

If the sampler's `mipmapMode` is `VK_SAMPLER_MIPMAP_MODE_LINEAR`, two neighboring levels are selected:

$$\begin{aligned} d_{hi} &= \lfloor d_l \rfloor \\ d_{lo} &= \min(d_{hi} + 1, level_{base} + q) \\ \delta &= d_l - d_{hi} \end{aligned}$$

δ is the fractional value, quantized to the number of `mipmap precision bits`, used for `linear filtering` between levels.

16.5.8. (s,t,r,q,a) to (u,v,w,a) Transformation

The normalized texel coordinates are scaled by the image level dimensions and the array layer is selected.

This transformation is performed once for each level used in `filtering` (either d , or d_{hi} and d_{lo}).

$$\begin{aligned} u(x, y) &= s(x, y) \times width_{scale} + \Delta_i \\ v(x, y) &= \begin{cases} 0 & \text{for 1D images} \\ t(x, y) \times height_{scale} + \Delta_j & \text{otherwise} \end{cases} \\ w(x, y) &= \begin{cases} 0 & \text{for 2D or Cube images} \\ r(x, y) \times depth_{scale} + \Delta_k & \text{otherwise} \end{cases} \\ a(x, y) &= \begin{cases} a(x, y) & \text{for array images} \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

where:

$$\text{width}_{\text{scale}} = \text{width}_{\text{level}}$$

$$\text{height}_{\text{scale}} = \text{height}_{\text{level}}$$

$$\text{depth}_{\text{scale}} = \text{depth}_{\text{level}}$$

and where $(\Delta_i, \Delta_j, \Delta_k)$ are taken from the image instruction if it includes a **ConstOffset** or **Offset** operand, otherwise they are taken to be zero.

Operations then proceed to Unnormalized Texel Coordinate Operations.

16.6. Unnormalized Texel Coordinate Operations

16.6.1. (u,v,w,a) to (i,j,k,l,n) Transformation and Array Layer Selection

The unnormalized texel coordinates are transformed to integer texel coordinates relative to the selected mipmap level.

The layer index l is computed as:

$$l = \text{clamp}(\text{RNE}(a), 0, \text{layerCount} - 1) + \text{baseArrayLayer}$$

where **layerCount** is the number of layers in the image subresource range of the image view, **baseArrayLayer** is the first layer from the subresource range, and where:

$$\text{RNE}(a) = \begin{cases} \text{roundTiesToEven}(a) & \text{preferred, from IEEE Std 754-2008 Floating-Point Arithmetic} \\ \lfloor a + 0.5 \rfloor & \text{alternative} \end{cases}$$

The sample index n is assigned the value 0.

Nearest filtering (**VK_FILTER_NEAREST**) computes the integer texel coordinates that the unnormalized coordinates lie within:

$$\begin{aligned} i &= \lfloor u + \text{shift} \rfloor \\ j &= \lfloor v + \text{shift} \rfloor \\ k &= \lfloor w + \text{shift} \rfloor \end{aligned}$$

where:

$$\text{shift} = 0.0$$

Linear filtering (**VK_FILTER_LINEAR**) computes a set of neighboring coordinates which bound the

unnormalized coordinates. The integer texel coordinates are combinations of i_0 or i_1 , j_0 or j_1 , k_0 or k_1 , as well as weights α , β , and γ .

$$\begin{aligned}i_0 &= \lfloor u - shift \rfloor \\i_1 &= i_0 + 1 \\j_0 &= \lfloor v - shift \rfloor \\j_1 &= j_0 + 1 \\k_0 &= \lfloor w - shift \rfloor \\k_1 &= k_0 + 1\end{aligned}$$

$$\alpha = \text{frac}(u - shift)$$

$$\beta = \text{frac}(v - shift)$$

$$\gamma = \text{frac}(w - shift)$$

where:

$$shift = 0.5$$

and where:

$$\text{frac}(x) = x - \lfloor x \rfloor$$

where the number of fraction bits retained is specified by `VkPhysicalDeviceLimits::subTexelPrecisionBits`.

16.7. Integer Texel Coordinate Operations

The `OpImageFetch` and `OpImageFetchSparse` SPIR-V instructions **may** supply a LOD from which texels are to be fetched using the optional SPIR-V operand `Lod`. Other integer-coordinate operations **must** not. If the `Lod` is provided then it **must** be an integer.

The image level selected is:

$$d = level_{base} + \begin{cases} Lod & \text{(from optional SPIR-V operand)} \\ 0 & \text{otherwise} \end{cases}$$

If d does not lie in the range $[baseMipLevel, baseMipLevel + levelCount)$ then any values fetched are undefined, and any writes (if supported) are discarded.

16.8. Image Sample Operations

16.8.1. Wrapping Operation

`Cube` images ignore the wrap modes specified in the sampler. Instead, if `VK_FILTER_NEAREST` is used within a mip level then `VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_EDGE` is used, and if `VK_FILTER_LINEAR` is used within a mip level then sampling at the edges is performed as described earlier in the `Cube`

map edge handling section.

The first integer texel coordinate i is transformed based on the `addressModeU` parameter of the sampler.

$$i = \begin{cases} i \bmod size & \text{for repeat} \\ (size - 1) - \text{mirror}((i \bmod (2 \times size)) - size) & \text{for mirrored repeat} \\ \text{clamp}(i, 0, size - 1) & \text{for clamp to edge} \\ \text{clamp}(i, -1, size) & \text{for clamp to border} \\ \text{clamp}(\text{mirror}(i), 0, size - 1) & \text{for mirror clamp to edge} \end{cases}$$

where:

$$\text{mirror}(n) = \begin{cases} n & \text{for } n \geq 0 \\ -(1 + n) & \text{otherwise} \end{cases}$$

j (for 2D and Cube image) and k (for 3D image) are similarly transformed based on the `addressModeV` and `addressModeW` parameters of the sampler, respectively.

16.8.2. Texel Gathering

SPIR-V instructions with `Gather` in the name return a vector derived from 4 texels in the base level of the image view. The rules for the `VK_FILTER_LINEAR` minification filter are applied to identify the four selected texels. Each texel is then converted to an RGBA value according to [conversion to RGBA](#) and then [swizzled](#). A four-component vector is then assembled by taking the component indicated by the `Component` value in the instruction from the swizzled color value of the four texels. If the operation does not use the `ConstOffsets` image operand then the four texels form the 2×2 rectangle used for texture filtering:

$$\begin{aligned} \tau[R] &= \tau_{i_0 j_1}[\text{level}_{base}][\text{comp}] \\ \tau[G] &= \tau_{i_1 j_1}[\text{level}_{base}][\text{comp}] \\ \tau[B] &= \tau_{i_1 j_0}[\text{level}_{base}][\text{comp}] \\ \tau[A] &= \tau_{i_0 j_0}[\text{level}_{base}][\text{comp}] \end{aligned}$$

If the operation does use the `ConstOffsets` image operand then the offsets allow a custom filter to be defined:

$$\begin{aligned} \tau[R] &= \tau_{i_0 j_0 + \Delta_0}[\text{level}_{base}][\text{comp}] \\ \tau[G] &= \tau_{i_0 j_0 + \Delta_1}[\text{level}_{base}][\text{comp}] \\ \tau[B] &= \tau_{i_0 j_0 + \Delta_2}[\text{level}_{base}][\text{comp}] \\ \tau[A] &= \tau_{i_0 j_0 + \Delta_3}[\text{level}_{base}][\text{comp}] \end{aligned}$$

where:

$$\tau[level_{base}][comp] = \begin{cases} \tau[level_{base}][R], & \text{for } comp = 0 \\ \tau[level_{base}][G], & \text{for } comp = 1 \\ \tau[level_{base}][B], & \text{for } comp = 2 \\ \tau[level_{base}][A], & \text{for } comp = 3 \end{cases}$$

comp from SPIR-V operand Component

16.8.3. Texel Filtering

Texel filtering is first performed for each level (either d or d_{hi} and d_{lo}).

If λ is less than or equal to zero, the texture is said to be *magnified*, and the filter mode within a mip level is selected by the **magFilter** in the sampler. If λ is greater than zero, the texture is said to be *minified*, and the filter mode within a mip level is selected by the **minFilter** in the sampler.

Texel Nearest Filtering

Within a mip level, **VK_FILTER_NEAREST** filtering selects a single value using the (i, j, k) texel coordinates, with all texels taken from layer l.

$$\tau[level] = \begin{cases} \tau_{ijk}[level], & \text{for 3D image} \\ \tau_{ij}[level], & \text{for 2D or Cube image} \\ \tau_i[level], & \text{for 1D image} \end{cases}$$

Texel Linear Filtering

Within a mip level, **VK_FILTER_LINEAR** filtering combines 8 (for 3D), 4 (for 2D or Cube), or 2 (for 1D) texel values, together with their linear weights. The linear weights are derived from the fractions computed earlier:

$$\begin{aligned} w_{i_0} &= (1 - \alpha) \\ w_{i_1} &= (\alpha) \\ w_{j_0} &= (1 - \beta) \\ w_{j_1} &= (\beta) \\ w_{k_0} &= (1 - \gamma) \\ w_{k_1} &= (\gamma) \end{aligned}$$

The values of multiple texels, together with their weights, are combined using a weighted average to produce a filtered value:

$$\begin{aligned}\tau_{3D} &= \sum_{k=k_0}^{k_1} \sum_{j=j_0}^{j_1} \sum_{i=i_0}^{i_1} (w_i)(w_j)(w_k)\tau_{ijk} \\ \tau_{2D} &= \sum_{j=j_0}^{j_1} \sum_{i=i_0}^{i_1} (w_i)(w_j)\tau_{ij} \\ \tau_{1D} &= \sum_{i=i_0}^{i_1} (w_i)\tau_i\end{aligned}$$

Texel Mipmap Filtering

`VK_SAMPLER_MIPMAP_MODE_NEAREST` filtering returns the value of a single mipmap level,

$\tau = \tau[d]$.

`VK_SAMPLER_MIPMAP_MODE_LINEAR` filtering combines the values of multiple mipmap levels ($\tau[hi]$ and $\tau[lo]$), together with their linear weights.

The linear weights are derived from the fraction computed earlier:

$$\begin{aligned}w_{hi} &= (1 - \delta) \\ w_{lo} &= (\delta)\end{aligned}$$

The values of multiple mipmap levels together with their linear weights, are combined using a weighted average to produce a final filtered value:

$$\tau = (w_{hi})\tau[hi] + (w_{lo})\tau[lo]$$

Texel Anisotropic Filtering

Anisotropic filtering is enabled by the `anisotropyEnable` in the sampler. When enabled, the image filtering scheme accounts for a degree of anisotropy.

The particular scheme for anisotropic texture filtering is implementation-dependent. Implementations **should** consider the `magFilter`, `minFilter` and `mipmapMode` of the sampler to control the specifics of the anisotropic filtering scheme used. In addition, implementations **should** consider `minLod` and `maxLod` of the sampler.

NOTE

For historical reasons, vendor implementations of anisotropic filtering interpret these sampler parameters in different ways, particularly in corner cases such as `magFilter`, `minFilter` of `NEAREST` or `maxAnisotropy` equal to 1.0. Applications should not expect consistent behavior in such cases, and should use anisotropic filtering only with parameters which are expected to give a quality improvement relative to `LINEAR` filtering.

The following describes one particular approach to implementing anisotropic filtering for the 2D Image case; implementations **may** choose other methods:

Given a `magFilter`, `minFilter` of `VK_FILTER_LINEAR` and a `mipmapMode` of

VK_SAMPLER_MIPMAP_MODE_NEAREST:

Instead of a single isotropic sample, N isotropic samples are sampled within the image footprint of the image level d to approximate an anisotropic filter. The sum $\tau_{2D_{aniso}}$ is defined using the single isotropic $\tau_{2D}(u,v)$ at level d.

$$\tau_{2D_{aniso}} = \frac{1}{N} \sum_{i=1}^N \tau_{2D} \left(u \left(x - \frac{1}{2} + \frac{i}{N+1}, y \right), v \left(x - \frac{1}{2} + \frac{i}{N+1}, y \right) \right), \quad \text{when } \rho_x > \rho_y$$
$$\tau_{2D_{aniso}} = \frac{1}{N} \sum_{i=1}^N \tau_{2D} \left(u \left(x, y - \frac{1}{2} + \frac{i}{N+1} \right), v \left(x, y - \frac{1}{2} + \frac{i}{N+1} \right) \right), \quad \text{when } \rho_y \geq \rho_x$$

16.9. Image Operation Steps

Each step described in this chapter is performed by a subset of the image instructions:

- Texel Input Validation Operations, Format Conversion, Texel Replacement, Conversion to RGBA, and Component Swizzle: Performed by all instructions except **OpImageWrite**.
- Depth Comparison: Performed by **OpImage*Dref** instructions.
- All Texel output operations: Performed by **OpImageWrite**.
- Projection: Performed by all **OpImage*Proj** instructions.
- Derivative Image Operations, Cube Map Operations, Scale Factor Operation, LOD Operation and Image Level(s) Selection, and Texel Anisotropic Filtering: Performed by all **OpImageSample*** and **OpImageSparseSample*** instructions.
- (s,t,r,q,a) to (u,v,w,a) Transformation, Wrapping, and (u,v,w,a) to (i,j,k,l,n) Transformation And Array Layer Selection: Performed by all **OpImageSample**, **OpImageSparseSample**, and **OpImage*Gather** instructions.
- Texel Gathering: Performed by **OpImage*Gather** instructions.
- Texel Filtering: Performed by all **OpImageSample*** and **OpImageSparseSample*** instructions.
- Sparse Residency: Performed by all **OpImageSparse*** instructions.

16.10. Image Query Instructions

16.10.1. Image Property Queries

OpImageQuerySize, **OpImageQuerySizeLod**, **OpImageQueryLevels**, and **OpImageQuerySamples** query properties of the image descriptor that would be accessed by a shader image operation.

OpImageQuerySizeLod returns the size of the image level identified by the **Level of Detail** operand. If that level does not exist in the image, then the value returned is undefined.

16.10.2. LOD Query

OpImageQueryLod returns the Lod parameters that would be used in an image operation with the given image and coordinates. The steps described in this chapter are performed as if for

`OpImageSampleImplicitLod`, up to [Scale Factor Operation](#), [LOD Operation](#) and [Image Level\(s\) Selection](#). The return value is the vector $(\lambda', d_l - \text{level}_{\text{base}})$. These values **may** be subject to implementation-specific maxima and minima for very large, out-of-range values.

Chapter 17. Queries

Queries provide a mechanism to return information about the processing of a sequence of Vulkan commands. Query operations are asynchronous, and as such, their results are not returned immediately. Instead, their results, and their availability status are stored in a [Query Pool](#). The state of these queries **can** be read back on the host, or copied to a buffer object on the device.

The supported query types are [Occlusion Queries](#), [Pipeline Statistics Queries](#), and [Timestamp Queries](#).

17.1. Query Pools

Queries are managed using *query pool* objects. Each query pool is a collection of a specific number of queries of a particular type.

Query pools are represented by `VkQueryPool` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkQueryPool)
```

To create a query pool, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateQueryPool(
    VkDevice                                device,
    const VkQueryPoolCreateInfo*            pCreateInfo,
    const VkAllocationCallbacks*           pAllocator,
    VkQueryPool*                            pQueryPool);
```

- `device` is the logical device that creates the query pool.
- `pCreateInfo` is a pointer to a [VkQueryPoolCreateInfo](#) structure containing the number and type of queries to be managed by the pool.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.
- `pQueryPool` is a pointer to a [VkQueryPool](#) handle in which the resulting query pool object is returned.

Valid Usage

- VUID-vkCreateQueryPool-device-09663
`device` **must** support at least one queue family with one of the `VK_QUEUE_COMPUTE_BIT`, or `VK_QUEUE_GRAPHICS_BIT` capabilities

Valid Usage (Implicit)

- VUID-vkCreateQueryPool-device-parameter
`device` **must** be a valid [VkDevice](#) handle
- VUID-vkCreateQueryPool-pCreateInfo-parameter
`pCreateInfo` **must** be a valid pointer to a valid [VkQueryPoolCreateInfo](#) structure
- VUID-vkCreateQueryPool-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkCreateQueryPool-pQueryPool-parameter
`pQueryPool` **must** be a valid pointer to a [VkQueryPool](#) handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The [VkQueryPoolCreateInfo](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkQueryPoolCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkQueryPoolCreateFlags flags;
    VkQueryType         queryType;
    uint32_t            queryCount;
    VkQueryPipelineStatisticFlags pipelineStatistics;
} VkQueryPoolCreateInfo;
```

- `sType` is a [VkStructureType](#) value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `queryType` is a [VkQueryType](#) value specifying the type of queries managed by the pool.
- `queryCount` is the number of queries managed by the pool.
- `pipelineStatistics` is a bitmask of [VkQueryPipelineStatisticFlagBits](#) specifying which counters will be returned in queries on the new pool, as described below in [Pipeline Statistics Queries](#).

`pipelineStatistics` is ignored if `queryType` is not `VK_QUERY_TYPE_PIPELINE_STATISTICS`.

Valid Usage

- VUID-VkQueryPoolCreateInfo-queryType-00791
If the `pipelineStatisticsQuery` feature is not enabled, `queryType` **must** not be `VK_QUERY_TYPE_PIPELINE_STATISTICS`
- VUID-VkQueryPoolCreateInfo-queryType-00792
If `queryType` is `VK_QUERY_TYPE_PIPELINE_STATISTICS`, `pipelineStatistics` **must** be a valid combination of `VkQueryPipelineStatisticFlagBits` values
- VUID-VkQueryPoolCreateInfo-queryType-09534
If `queryType` is `VK_QUERY_TYPE_PIPELINE_STATISTICS`, `pipelineStatistics` **must** not be zero
- VUID-VkQueryPoolCreateInfo-queryCount-02763
`queryCount` **must** be greater than 0

Valid Usage (Implicit)

- VUID-VkQueryPoolCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_QUERY_POOL_CREATE_INFO`
- VUID-VkQueryPoolCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkQueryPoolCreateInfo-flags-zeroBitmask
`flags` **must** be 0
- VUID-VkQueryPoolCreateInfo-queryType-parameter
`queryType` **must** be a valid `VkQueryType` value

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkQueryPoolCreateFlags;
```

`VkQueryPoolCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

To destroy a query pool, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyQueryPool(
    VkDevice          device,
    VkQueryPool        queryPool,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that destroys the query pool.
- `queryPool` is the query pool to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyQueryPool-queryPool-00793
All submitted commands that refer to **queryPool** **must** have completed execution
- VUID-vkDestroyQueryPool-queryPool-00794
If **VkAllocationCallbacks** were provided when **queryPool** was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyQueryPool-queryPool-00795
If no **VkAllocationCallbacks** were provided when **queryPool** was created, **pAllocator** **must** be **NULL**

NOTE

Applications **can** verify that **queryPool** **can** be destroyed by checking that **vkGetQueryPoolResults()** without the **VK_QUERY_RESULT_PARTIAL_BIT** flag returns **VK_SUCCESS** for all queries that are used in command buffers submitted for execution.

Valid Usage (Implicit)

- VUID-vkDestroyQueryPool-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkDestroyQueryPool-queryPool-parameter
If **queryPool** is not **VK_NULL_HANDLE**, **queryPool** **must** be a valid **VkQueryPool** handle
- VUID-vkDestroyQueryPool-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid **VkAllocationCallbacks** structure
- VUID-vkDestroyQueryPool-queryPool-parent
If **queryPool** is a valid handle, it **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **queryPool** **must** be externally synchronized

Possible values of **VkQueryPoolCreateInfo::queryType**, specifying the type of queries managed by the pool, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkQueryType {
    VK_QUERY_TYPE_OCCLUSION = 0,
    VK_QUERY_TYPE_PIPELINE_STATISTICS = 1,
    VK_QUERY_TYPE_TIMESTAMP = 2,
} VkQueryType;
```

- `VK_QUERY_TYPE_OCCLUSION` specifies an [occlusion query](#).
- `VK_QUERY_TYPE_PIPELINE_STATISTICS` specifies a [pipeline statistics query](#).
- `VK_QUERY_TYPE_TIMESTAMP` specifies a [timestamp query](#).

17.2. Query Operation

The operation of queries is controlled by the commands [vkCmdBeginQuery](#), [vkCmdEndQuery](#), [vkCmdResetQueryPool](#), [vkCmdCopyQueryPoolResults](#), and [vkCmdWriteTimestamp](#).

In order for a `VkCommandBuffer` to record query management commands, the queue family for which its `VkCommandPool` was created **must** support the appropriate type of operations (graphics, compute) suitable for the query type of a given query pool.

Each query in a query pool has a status that is either *unavailable* or *available*, and also has state to store the numerical results of a query operation of the type requested when the query pool was created. Resetting a query via [vkCmdResetQueryPool](#) sets the status to unavailable and makes the numerical results undefined. A query is made available by the operation of [vkCmdEndQuery](#), or [vkCmdWriteTimestamp](#). Both the availability status and numerical results **can** be retrieved by calling either [vkGetQueryPoolResults](#) or [vkCmdCopyQueryPoolResults](#).

After query pool creation, each query is in an uninitialized state and **must** be reset before it is used. Queries **must** also be reset between uses.

To reset a range of queries in a query pool on a queue, call:

```
// Provided by VK_VERSION_1_0
void vkCmdResetQueryPool(
    VkCommandBuffer          commandBuffer,
    VkQueryPool              queryPool,
    uint32_t                 firstQuery,
    uint32_t                 queryCount);
```

- `commandBuffer` is the command buffer into which this command will be recorded.
- `queryPool` is the handle of the query pool managing the queries being reset.
- `firstQuery` is the initial query index to reset.
- `queryCount` is the number of queries to reset.

When executed on a queue, this command sets the status of query indices [`firstQuery`, `firstQuery` +

`queryCount - 1]` to unavailable.

This command defines an execution dependency between other query commands that reference the same query.

The first [synchronization scope](#) includes all commands which reference the queries in `queryPool` indicated by `firstQuery` and `queryCount` that occur earlier in [submission order](#).

The second [synchronization scope](#) includes all commands which reference the queries in `queryPool` indicated by `firstQuery` and `queryCount` that occur later in [submission order](#).

The operation of this command happens after the first scope and happens before the second scope.

Valid Usage

- VUID-vkCmdResetQueryPool-firstQuery-09436
`firstQuery` **must** be less than the number of queries in `queryPool`
- VUID-vkCmdResetQueryPool-firstQuery-09437
The sum of `firstQuery` and `queryCount` **must** be less than or equal to the number of queries in `queryPool`
- VUID-vkCmdResetQueryPool-None-02841
All queries used by the command **must** not be active

Valid Usage (Implicit)

- VUID-vkCmdResetQueryPool-commandBuffer-parameter
`commandBuffer` **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdResetQueryPool-queryPool-parameter
`queryPool` **must** be a valid [VkQueryPool](#) handle
- VUID-vkCmdResetQueryPool-commandBuffer-recording
`commandBuffer` **must** be in the [recording state](#)
- VUID-vkCmdResetQueryPool-commandBuffer-cmdpool
The [VkCommandPool](#) that `commandBuffer` was allocated from **must** support graphics, or compute operations
- VUID-vkCmdResetQueryPool-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdResetQueryPool-commonparent
Both of `commandBuffer`, and `queryPool` **must** have been created, allocated, or retrieved from the same [VkDevice](#)

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized

- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Graphics Compute	Action

Once queries are reset and ready for use, query commands **can** be issued to a command buffer. Occlusion queries and pipeline statistics queries count events - drawn samples and pipeline stage invocations, respectively - resulting from commands that are recorded between a `vkCmdBeginQuery` command and a `vkCmdEndQuery` command within a specified command buffer, effectively scoping a set of drawing and/or dispatching commands. Timestamp queries write timestamps to a query pool.

A query **must** begin and end in the same command buffer, although if it is a primary command buffer, and the `inheritedQueries` feature is enabled, it **can** execute secondary command buffers during the query operation. For a secondary command buffer to be executed while a query is active, it **must** set the `occlusionQueryEnable`, `queryFlags`, and/or `pipelineStatistics` members of `VkCommandBufferInheritanceInfo` to conservative values, as described in the [Command Buffer Recording](#) section. A query **must** either begin and end inside the same subpass of a render pass instance, or **must** both begin and end outside of a render pass instance (i.e. contain entire render pass instances).

To begin a query, call:

```
// Provided by VK_VERSION_1_0
void vkCmdBeginQuery(
    VkCommandBuffer          commandBuffer,
    VkQueryPool              queryPool,
    uint32_t                 query,
    VkQueryControlFlags      flags);
```

- `commandBuffer` is the command buffer into which this command will be recorded.
- `queryPool` is the query pool that will manage the results of the query.
- `query` is the query index within the query pool that will contain the results.
- `flags` is a bitmask of `VkQueryControlFlagBits` specifying constraints on the types of queries that **can** be performed.

If the `queryType` of the pool is `VK_QUERY_TYPE_OCCLUSION` and `flags` contains `VK_QUERY_CONTROL_PRECISE_BIT`, an implementation **must** return a result that matches the actual number of samples passed. This is described in more detail in [Occlusion Queries](#).

After beginning a query, that query is considered *active* within the command buffer it was called in until that same query is ended. Queries active in a primary command buffer when secondary command buffers are executed are considered active for those secondary command buffers.

This command defines an execution dependency between other query commands that reference the same query.

The first [synchronization scope](#) includes all commands which reference the queries in [queryPool](#) indicated by [query](#) that occur earlier in [submission order](#).

The second [synchronization scope](#) includes all commands which reference the queries in [queryPool](#) indicated by [query](#) that occur later in [submission order](#).

The operation of this command happens after the first scope and happens before the second scope.

Valid Usage

- VUID-vkCmdBeginQuery-None-00807
All queries used by the command **must** be *unavailable*
- VUID-vkCmdBeginQuery-queryType-02804
The [queryType](#) used to create [queryPool](#) **must** not be [VK_QUERY_TYPE_TIMESTAMP](#)
- VUID-vkCmdBeginQuery-queryType-00800
If the [occlusionQueryPrecise](#) feature is not enabled, or the [queryType](#) used to create [queryPool](#) was not [VK_QUERY_TYPE_OCCLUSION](#), [flags](#) **must** not contain [VK_QUERY_CONTROL_PRECISE_BIT](#)
- VUID-vkCmdBeginQuery-query-00802
[query](#) **must** be less than the number of queries in [queryPool](#)
- VUID-vkCmdBeginQuery-queryType-00803
If the [queryType](#) used to create [queryPool](#) was [VK_QUERY_TYPE_OCCLUSION](#), the [VkCommandPool](#) that [commandBuffer](#) was allocated from **must** support graphics operations
- VUID-vkCmdBeginQuery-queryType-00804
If the [queryType](#) used to create [queryPool](#) was [VK_QUERY_TYPE_PIPELINE_STATISTICS](#) and any of the [pipelineStatistics](#) indicate graphics operations, the [VkCommandPool](#) that [commandBuffer](#) was allocated from **must** support graphics operations
- VUID-vkCmdBeginQuery-queryType-00805
If the [queryType](#) used to create [queryPool](#) was [VK_QUERY_TYPE_PIPELINE_STATISTICS](#) and any of the [pipelineStatistics](#) indicate compute operations, the [VkCommandPool](#) that [commandBuffer](#) was allocated from **must** support compute operations
- VUID-vkCmdBeginQuery-queryPool-01922
[queryPool](#) **must** have been created with a [queryType](#) that differs from that of any queries that are [active](#) within [commandBuffer](#)

Valid Usage (Implicit)

- VUID-vkCmdBeginQuery-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdBeginQuery-queryPool-parameter
`queryPool` **must** be a valid `VkQueryPool` handle
- VUID-vkCmdBeginQuery-flags-parameter
`flags` **must** be a valid combination of `VkQueryControlFlagBits` values
- VUID-vkCmdBeginQuery-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdBeginQuery-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, or compute operations
- VUID-vkCmdBeginQuery-commonparent
Both of `commandBuffer`, and `queryPool` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Graphics Compute	Action State

Bits which **can** be set in `vkCmdBeginQuery::flags`, specifying constraints on the types of queries that **can** be performed, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkQueryControlFlagBits {
    VK_QUERY_CONTROL_PRECISE_BIT = 0x00000001,
} VkQueryControlFlagBits;
```

- `VK_QUERY_CONTROL_PRECISE_BIT` specifies the precision of `occlusion queries`.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkQueryControlFlags;
```

`VkQueryControlFlags` is a bitmask type for setting a mask of zero or more `VkQueryControlFlagBits`.

To end a query after the set of desired drawing or dispatching commands is executed, call:

```
// Provided by VK_VERSION_1_0
void vkCmdEndQuery(
    VkCommandBuffer          commandBuffer,
    VkQueryPool              queryPool,
    uint32_t                 query);
```

- `commandBuffer` is the command buffer into which this command will be recorded.
- `queryPool` is the query pool that is managing the results of the query.
- `query` is the query index within the query pool where the result is stored.

The command completes the query in `queryPool` identified by `query`, and marks it as available.

This command defines an execution dependency between other query commands that reference the same query.

The first `synchronization scope` includes all commands which reference the queries in `queryPool` indicated by `query` that occur earlier in `submission order`.

The second `synchronization scope` includes only the operation of this command.

Valid Usage

- VUID-vkCmdEndQuery-None-01923
All queries used by the command **must** be `active`
- VUID-vkCmdEndQuery-query-00810
`query` **must** be less than the number of queries in `queryPool`
- VUID-vkCmdEndQuery-None-07007
If called within a subpass of a render pass instance, the corresponding `vkCmdBeginQuery*` command **must** have been called previously within the same subpass

Valid Usage (Implicit)

- VUID-vkCmdEndQuery-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdEndQuery-queryPool-parameter
`queryPool` **must** be a valid `VkQueryPool` handle

- VUID-vkCmdEndQuery-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdEndQuery-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, or compute operations
- VUID-vkCmdEndQuery-commonparent
Both of `commandBuffer`, and `queryPool` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary	Both	Graphics	Action
Secondary		Compute	State

An application **can** retrieve results either by requesting they be written into application-provided memory, or by requesting they be copied into a `VkBuffer`. In either case, the layout in memory is defined as follows:

- The first query's result is written starting at the first byte requested by the command, and each subsequent query's result begins `stride` bytes later.
- Occlusion queries, pipeline statistics queries, and timestamp queries store results in a tightly packed array of unsigned integers, either 32- or 64-bits as requested by the command, storing the numerical results and, if requested, the availability status.
- If `VK_QUERY_RESULT_WITH_AVAILABILITY_BIT` is used, the final element of each query's result is an integer indicating whether the query's result is available, with any non-zero value indicating that it is available.
- Occlusion queries write one integer value - the number of samples passed. Pipeline statistics queries write one integer value for each bit that is enabled in the `pipelineStatistics` when the pool is created, and the statistics values are written in bit order starting from the least significant bit. Timestamp queries write one integer value.
- If more than one query is retrieved and `stride` is not at least as large as the size of the array of values corresponding to a single query, the values written to memory are undefined.

To retrieve status and results for a set of queries, call:

```
// Provided by VK_VERSION_1_0
VkResult vkGetQueryPoolResults(
    VkDevice          device,
    VkQueryPool       queryPool,
    uint32_t          firstQuery,
    uint32_t          queryCount,
    size_t            dataSize,
    void*             pData,
    VkDeviceSize       stride,
    VkQueryResultFlags flags);
```

- **device** is the logical device that owns the query pool.
- **queryPool** is the query pool managing the queries containing the desired results.
- **firstQuery** is the initial query index.
- **queryCount** is the number of queries to read.
- **dataSize** is the size in bytes of the buffer pointed to by **pData**.
- **pData** is a pointer to an application-allocated buffer where the results will be written
- **stride** is the stride in bytes between results for individual queries within **pData**.
- **flags** is a bitmask of **VkQueryResultFlagBits** specifying how and when results are returned.

Any results written for a query are written according to [a layout dependent on the query type](#).

If no bits are set in **flags**, and all requested queries are in the available state, results are written as an array of 32-bit unsigned integer values. Behavior when not all queries are available is described [below](#).

If **VK_QUERY_RESULT_WITH_AVAILABILITY_BIT** is set, results for all queries in **queryPool** identified by **firstQuery** and **queryCount** are copied to **pData**, along with an extra availability value written directly after the results of each query and interpreted as an unsigned integer. A value of zero indicates that the results are not yet available, otherwise the query is complete and results are available. The size of the availability values is 64 bits if **VK_QUERY_RESULT_64_BIT** is set in **flags**. Otherwise, it is 32 bits.

NOTE	If VK_QUERY_RESULT_WITH_AVAILABILITY_BIT is set, the layout of data in the buffer is a <i>(result,availability)</i> pair for each query returned, and stride is the stride between each pair.
-------------	---

Results for any available query written by this command are final and represent the final result of the query. If **VK_QUERY_RESULT_PARTIAL_BIT** is set, then for any query that is unavailable, an intermediate result between zero and the final result value is written for that query. Otherwise, any result written by this command is undefined.

If **VK_QUERY_RESULT_64_BIT** is set, results and, if returned, availability values for all queries are written as an array of 64-bit values. Otherwise, results and availability values are written as an array of 32-bit values. If an unsigned integer query's value overflows the result type, the value **may**

either wrap or saturate.

If `VK_QUERY_RESULT_WAIT_BIT` is set, this command defines an execution dependency with any earlier commands that writes one of the identified queries. The first [synchronization scope](#) includes all instances of `vkCmdEndQuery`, and `vkCmdWriteTimestamp` that reference any query in `queryPool` indicated by `firstQuery` and `queryCount`. The second [synchronization scope](#) includes the host operations of this command.

If `VK_QUERY_RESULT_WAIT_BIT` is not set, `vkGetQueryPoolResults` **may** return `VK_NOT_READY` if there are queries in the unavailable state.

NOTE	Applications must take care to ensure that use of the <code>VK_QUERY_RESULT_WAIT_BIT</code> bit has the desired effect. For example, if a query has been used previously and a command buffer records the commands <code>vkCmdResetQueryPool</code>, <code>vkCmdBeginQuery</code>, and <code>vkCmdEndQuery</code> for that query, then the query will remain in the available state until the <code>vkCmdResetQueryPool</code> command executes on a queue. Applications can use fences or events to ensure that a query has already been reset before checking for its results or availability status. Otherwise, a stale value could be returned from a previous use of the query. The above also applies when <code>VK_QUERY_RESULT_WAIT_BIT</code> is used in combination with <code>VK_QUERY_RESULT_WITH_AVAILABILITY_BIT</code>. In this case, the returned availability status may reflect the result of a previous use of the query unless the <code>vkCmdResetQueryPool</code> command has been executed since the last use of the query.
NOTE	Applications can double-buffer query pool usage, with a pool per frame, and reset queries at the end of the frame in which they are read.

Valid Usage

- VUID-vkGetQueryPoolResults-firstQuery-09436
`firstQuery` **must** be less than the number of queries in `queryPool`
- VUID-vkGetQueryPoolResults-firstQuery-09437
The sum of `firstQuery` and `queryCount` **must** be less than or equal to the number of queries in `queryPool`
- VUID-vkGetQueryPoolResults-queryCount-09438
If `queryCount` is greater than 1, `stride` **must** not be zero
- VUID-vkGetQueryPoolResults-queryType-09439
If the `queryType` used to create `queryPool` was `VK_QUERY_TYPE_TIMESTAMP`, `flags` **must** not contain `VK_QUERY_RESULT_PARTIAL_BIT`
- VUID-vkGetQueryPoolResults-None-09401
All queries used by the command **must** not be uninitialized
- VUID-vkGetQueryPoolResults-flags-02828

If `VK_QUERY_RESULT_64_BIT` is not set in `flags` then `pData` and `stride` **must** be multiples of 4

- VUID-vkGetQueryPoolResults-flags-00815

If `VK_QUERY_RESULT_64_BIT` is set in `flags` then `pData` and `stride` **must** be multiples of 8

- VUID-vkGetQueryPoolResults-stride-08993

If `VK_QUERY_RESULT_WITH_AVAILABILITY_BIT` is set, `stride` **must** be large enough to contain the unsigned integer representing availability in addition to the query result

- VUID-vkGetQueryPoolResults-dataSize-00817

`dataSize` **must** be large enough to contain the result of each query, as described [here](#)

Valid Usage (Implicit)

- VUID-vkGetQueryPoolResults-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetQueryPoolResults-queryPool-parameter
`queryPool` **must** be a valid `VkQueryPool` handle
- VUID-vkGetQueryPoolResults-pData-parameter
`pData` **must** be a valid pointer to an array of `dataSize` bytes
- VUID-vkGetQueryPoolResults-flags-parameter
`flags` **must** be a valid combination of `VkQueryResultFlagBits` values
- VUID-vkGetQueryPoolResults-dataSize-arraylength
`dataSize` **must** be greater than 0
- VUID-vkGetQueryPoolResults-queryPool-parent
`queryPool` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_SUCCESS`
- `VK_NOT_READY`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_DEVICE_LOST`

Bits which **can** be set in `vkGetQueryPoolResults::flags` and `vkCmdCopyQueryPoolResults::flags`, specifying how and when results are returned, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkQueryResultFlagBits {
    VK_QUERY_RESULT_64_BIT = 0x00000001,
    VK_QUERY_RESULT_WAIT_BIT = 0x00000002,
    VK_QUERY_RESULT_WITH_AVAILABILITY_BIT = 0x00000004,
    VK_QUERY_RESULT_PARTIAL_BIT = 0x00000008,
} VkQueryResultFlagBits;
```

- **VK_QUERY_RESULT_64_BIT** specifies the results will be written as an array of 64-bit unsigned integer values. If this bit is not set, the results will be written as an array of 32-bit unsigned integer values.
- **VK_QUERY_RESULT_WAIT_BIT** specifies that Vulkan will wait for each query's status to become available before retrieving its results.
- **VK_QUERY_RESULT_WITH_AVAILABILITY_BIT** specifies that the availability status accompanies the results.
- **VK_QUERY_RESULT_PARTIAL_BIT** specifies that returning partial results is acceptable.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkQueryResultFlags;
```

VkQueryResultFlags is a bitmask type for setting a mask of zero or more **VkQueryResultFlagBits**.

To copy query statuses and numerical results directly to buffer memory, call:

```
// Provided by VK_VERSION_1_0
void vkCmdCopyQueryPoolResults(
    VkCommandBuffer          commandBuffer,
    VkQueryPool              queryPool,
    uint32_t                 firstQuery,
    uint32_t                 queryCount,
    VkBuffer                 dstBuffer,
    VkDeviceSize             dstOffset,
    VkDeviceSize             stride,
    VkQueryResultFlags       flags);
```

- **commandBuffer** is the command buffer into which this command will be recorded.
- **queryPool** is the query pool managing the queries containing the desired results.
- **firstQuery** is the initial query index.
- **queryCount** is the number of queries. **firstQuery** and **queryCount** together define a range of queries.
- **dstBuffer** is a **VkBuffer** object that will receive the results of the copy command.
- **dstOffset** is an offset into **dstBuffer**.
- **stride** is the stride in bytes between results for individual queries within **dstBuffer**. The

required size of the backing memory for `dstBuffer` is determined as described above for `vkGetQueryPoolResults`.

- `flags` is a bitmask of `VkQueryResultFlagBits` specifying how and when results are returned.

Any results written for a query are written according to a [layout dependent on the query type](#).

Results for any query in `queryPool` identified by `firstQuery` and `queryCount` that is available are copied to `dstBuffer`.

If `VK_QUERY_RESULT_WITH_AVAILABILITY_BIT` is set, results for all queries in `queryPool` identified by `firstQuery` and `queryCount` are copied to `dstBuffer`, along with an extra availability value written directly after the results of each query and interpreted as an unsigned integer. A value of zero indicates that the results are not yet available, otherwise the query is complete and results are available.

Results for any available query written by this command are final and represent the final result of the query. If `VK_QUERY_RESULT_PARTIAL_BIT` is set, then for any query that is unavailable, an intermediate result between zero and the final result value is written for that query. Otherwise, any result written by this command is undefined.

If `VK_QUERY_RESULT_64_BIT` is set, results and availability values for all queries are written as an array of 64-bit values. Otherwise, results and availability values are written as an array of 32-bit values. If an unsigned integer query's value overflows the result type, the value **may** either wrap or saturate.

This command defines an execution dependency between other query commands that reference the same query.

The first [synchronization scope](#) includes all commands which reference the queries in `queryPool` indicated by `query` that occur earlier in [submission order](#). If `flags` does not include `VK_QUERY_RESULT_WAIT_BIT`, `vkCmdEndQuery`, and `vkCmdWriteTimestamp` are excluded from this scope.

The second [synchronization scope](#) includes all commands which reference the queries in `queryPool` indicated by `query` that occur later in [submission order](#).

The operation of this command happens after the first scope and happens before the second scope.

`vkCmdCopyQueryPoolResults` is considered to be a transfer operation, and its writes to buffer memory **must** be synchronized using `VK_PIPELINE_STAGE_TRANSFER_BIT` and `VK_ACCESS_TRANSFER_WRITE_BIT` before using the results.

Valid Usage

- VUID-vkCmdCopyQueryPoolResults-firstQuery-09436
`firstQuery` **must** be less than the number of queries in `queryPool`
- VUID-vkCmdCopyQueryPoolResults-firstQuery-09437
The sum of `firstQuery` and `queryCount` **must** be less than or equal to the number of queries in `queryPool`

- VUID-vkCmdCopyQueryPoolResults-queryCount-09438
If **queryCount** is greater than 1, **stride** **must** not be zero
- VUID-vkCmdCopyQueryPoolResults-queryType-09439
If the **queryType** used to create **queryPool** was **VK_QUERY_TYPE_TIMESTAMP**, **flags** **must** not contain **VK_QUERY_RESULT_PARTIAL_BIT**
- VUID-vkCmdCopyQueryPoolResults-None-09402
All queries used by the command **must** not be uninitialized when the command is executed
- VUID-vkCmdCopyQueryPoolResults-dstOffset-00819
dstOffset **must** be less than the size of **dstBuffer**
- VUID-vkCmdCopyQueryPoolResults-flags-00822
If **VK_QUERY_RESULT_64_BIT** is not set in **flags** then **dstOffset** and **stride** **must** be multiples of 4
- VUID-vkCmdCopyQueryPoolResults-flags-00823
If **VK_QUERY_RESULT_64_BIT** is set in **flags** then **dstOffset** and **stride** **must** be multiples of 8
- VUID-vkCmdCopyQueryPoolResults-dstBuffer-00824
dstBuffer **must** have enough storage, from **dstOffset**, to contain the result of each query, as described [here](#)
- VUID-vkCmdCopyQueryPoolResults-dstBuffer-00825
dstBuffer **must** have been created with **VK_BUFFER_USAGE_TRANSFER_DST_BIT** usage flag
- VUID-vkCmdCopyQueryPoolResults-dstBuffer-00826
If **dstBuffer** is non-sparse then it **must** be bound completely and contiguously to a single **VkDeviceMemory** object
- VUID-vkCmdCopyQueryPoolResults-None-07429
All queries used by the command **must** not be active
- VUID-vkCmdCopyQueryPoolResults-None-08752
All queries used by the command **must** have been made *available* by prior executed commands

Valid Usage (Implicit)

- VUID-vkCmdCopyQueryPoolResults-commandBuffer-parameter
commandBuffer **must** be a valid **VkCommandBuffer** handle
- VUID-vkCmdCopyQueryPoolResults-queryPool-parameter
queryPool **must** be a valid **VkQueryPool** handle
- VUID-vkCmdCopyQueryPoolResults-dstBuffer-parameter
dstBuffer **must** be a valid **VkBuffer** handle
- VUID-vkCmdCopyQueryPoolResults-flags-parameter
flags **must** be a valid combination of **VkQueryResultFlagBits** values
- VUID-vkCmdCopyQueryPoolResults-commandBuffer-recording
commandBuffer **must** be in the **recording state**

- VUID-vkCmdCopyQueryPoolResults-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, or compute operations
- VUID-vkCmdCopyQueryPoolResults-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyQueryPoolResults-commonparent
Each of `commandBuffer`, `dstBuffer`, and `queryPool` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Graphics Compute	Action

Rendering operations such as clears, MSAA resolves, attachment load/store operations, and blits **may** count towards the results of queries. This behavior is implementation-dependent and **may** vary depending on the path used within an implementation. For example, some implementations have several types of clears, some of which **may** include vertices and some not.

17.3. Occlusion Queries

Occlusion queries track the number of samples that pass the per-fragment tests for a set of drawing commands. As such, occlusion queries are only available on queue families supporting graphics operations. The application **can** then use these results to inform future rendering decisions. An occlusion query is begun and ended by calling `vkCmdBeginQuery` and `vkCmdEndQuery`, respectively. When an occlusion query begins, the count of passing samples always starts at zero. For each drawing command, the count is incremented as described in [Sample Counting](#). If `flags` does not contain `VK_QUERY_CONTROL_PRECISE_BIT` an implementation **may** generate any non-zero result value for the query if the count of passing samples is non-zero.

NOTE

Not setting `VK_QUERY_CONTROL_PRECISE_BIT` mode **may** be more efficient on some implementations, and **should** be used where it is sufficient to know a boolean result on whether any samples passed the per-fragment tests. In this case, some implementations **may** only return zero or one, indifferent to the actual number of samples passing the per-fragment tests.

Setting `VK_QUERY_CONTROL_PRECISE_BIT` does not guarantee that different implementations return the same number of samples in an occlusion query. Some implementations may kill fragments in the [pre-rasterization shader stage](#), and these killed fragments do not contribute to the final result of the query. It is possible that some implementations generate a zero result value for the query, while others generate a non-zero value.

When an occlusion query finishes, the result for that query is marked as available. The application **can** then either copy the result to a buffer (via `vkCmdCopyQueryPoolResults`) or request it be put into host memory (via `vkGetQueryPoolResults`).

NOTE If occluding geometry is not drawn first, samples **can** pass the depth test, but still not be visible in a final image.

17.4. Pipeline Statistics Queries

Pipeline statistics queries allow the application to sample a specified set of `VkPipeline` counters. These counters are accumulated by Vulkan for a set of either drawing or dispatching commands while a pipeline statistics query is active. As such, pipeline statistics queries are available on queue families supporting either graphics or compute operations. The availability of pipeline statistics queries is indicated by the `pipelineStatisticsQuery` member of the `VkPhysicalDeviceFeatures` object (see `vkGetPhysicalDeviceFeatures` and `vkCreateDevice` for detecting and requesting this query type on a `VkDevice`).

A pipeline statistics query is begun and ended by calling `vkCmdBeginQuery` and `vkCmdEndQuery`, respectively. When a pipeline statistics query begins, all statistics counters are set to zero. While the query is active, the pipeline type determines which set of statistics are available, but these **must** be configured on the query pool when it is created. If a statistic counter is issued on a command buffer that does not support the corresponding operation, or the counter corresponds to a shading stage which is missing from any of the pipelines used while the query is active, the value of that counter is undefined after the query has been made available. At least one statistic counter relevant to the operations supported on the recording command buffer **must** be enabled.

Bits which **can** be set in `VkQueryPoolCreateInfo::pipelineStatistics` for query pools and in `VkCommandBufferInheritanceInfo::pipelineStatistics` for secondary command buffers, individually enabling pipeline statistics counters, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkQueryPipelineStatisticFlagBits {
    VK_QUERY_PIPELINE_STATISTIC_INPUT_ASSEMBLY_VERTICES_BIT = 0x00000001,
    VK_QUERY_PIPELINE_STATISTIC_INPUT_ASSEMBLY_PRIMITIVES_BIT = 0x00000002,
    VK_QUERY_PIPELINE_STATISTIC_VERTEX_SHADER_INVOCATIONS_BIT = 0x00000004,
    VK_QUERY_PIPELINE_STATISTIC_GEOMETRY_SHADER_INVOCATIONS_BIT = 0x00000008,
    VK_QUERY_PIPELINE_STATISTIC_GEOMETRY_SHADER_PRIMITIVES_BIT = 0x00000010,
    VK_QUERY_PIPELINE_STATISTIC_CLIPPING_INVOCATIONS_BIT = 0x00000020,
    VK_QUERY_PIPELINE_STATISTIC_CLIPPING_PRIMITIVES_BIT = 0x00000040,
    VK_QUERY_PIPELINE_STATISTIC_FRAGMENT_SHADER_INVOCATIONS_BIT = 0x00000080,
    VK_QUERY_PIPELINE_STATISTIC_TESSELLATION_CONTROL_SHADER_PATCHES_BIT = 0x00000100,
    VK_QUERY_PIPELINE_STATISTIC_TESSELLATION_EVALUATION_SHADER_INVOCATIONS_BIT =
0x00000200,
    VK_QUERY_PIPELINE_STATISTIC_COMPUTE_SHADER_INVOCATIONS_BIT = 0x00000400,
} VkQueryPipelineStatisticFlagBits;
```

- **VK_QUERY_PIPELINE_STATISTIC_INPUT_ASSEMBLY_VERTICES_BIT** specifies that queries managed by the pool will count the number of vertices processed by the **input assembly** stage. Vertices corresponding to incomplete primitives **may** contribute to the count.
- **VK_QUERY_PIPELINE_STATISTIC_INPUT_ASSEMBLY_PRIMITIVES_BIT** specifies that queries managed by the pool will count the number of primitives processed by the **input assembly** stage. If primitive restart is enabled, restarting the primitive topology has no effect on the count. Incomplete primitives **may** be counted.
- **VK_QUERY_PIPELINE_STATISTIC_VERTEX_SHADER_INVOCATIONS_BIT** specifies that queries managed by the pool will count the number of vertex shader invocations. This counter's value is incremented each time a vertex shader is **invoked**.
- **VK_QUERY_PIPELINE_STATISTIC_GEOMETRY_SHADER_INVOCATIONS_BIT** specifies that queries managed by the pool will count the number of geometry shader invocations. This counter's value is incremented each time a geometry shader is **invoked**. In the case of **instanced geometry shaders**, the geometry shader invocations count is incremented for each separate instanced invocation.
- **VK_QUERY_PIPELINE_STATISTIC_GEOMETRY_SHADER_PRIMITIVES_BIT** specifies that queries managed by the pool will count the number of primitives generated by geometry shader invocations. The counter's value is incremented each time the geometry shader emits a primitive. Restarting primitive topology using the SPIR-V instructions **OpEndPrimitive** or **OpEndStreamPrimitive** has no effect on the geometry shader output primitives count.
- **VK_QUERY_PIPELINE_STATISTIC_CLIPPING_INVOCATIONS_BIT** specifies that queries managed by the pool will count the number of primitives processed by the **Primitive Clipping** stage of the pipeline. The counter's value is incremented each time a primitive reaches the primitive clipping stage.
- **VK_QUERY_PIPELINE_STATISTIC_CLIPPING_PRIMITIVES_BIT** specifies that queries managed by the pool will count the number of primitives output by the **Primitive Clipping** stage of the pipeline. The counter's value is incremented each time a primitive passes the primitive clipping stage. The actual number of primitives output by the primitive clipping stage for a particular input primitive is implementation-dependent but **must** satisfy the following conditions:

- If at least one vertex of the input primitive lies inside the clipping volume, the counter is incremented by one or more.
- Otherwise, the counter is incremented by zero or more.
- `VK_QUERY_PIPELINE_STATISTIC_FRAGMENT_SHADER_INVOCATIONS_BIT` specifies that queries managed by the pool will count the number of fragment shader invocations. The counter's value is incremented each time the fragment shader is [invoked](#).
- `VK_QUERY_PIPELINE_STATISTIC_TESSELLATION_CONTROL_SHADER_PATCHES_BIT` specifies that queries managed by the pool will count the number of patches processed by the tessellation control shader. The counter's value is incremented once for each patch for which a tessellation control shader is [invoked](#).
- `VK_QUERY_PIPELINE_STATISTIC_TESSELLATION_EVALUATION_SHADER_INVOCATIONS_BIT` specifies that queries managed by the pool will count the number of invocations of the tessellation evaluation shader. The counter's value is incremented each time the tessellation evaluation shader is [invoked](#).
- `VK_QUERY_PIPELINE_STATISTIC_COMPUTE_SHADER_INVOCATIONS_BIT` specifies that queries managed by the pool will count the number of compute shader invocations. The counter's value is incremented every time the compute shader is invoked. Implementations **may** skip the execution of certain compute shader invocations or execute additional compute shader invocations for implementation-dependent reasons as long as the results of rendering otherwise remain unchanged.

These values are intended to measure relative statistics on one implementation. Various device architectures will count these values differently. Any or all counters **may** be affected by the issues described in [Query Operation](#).

NOTE

For example, tile-based rendering devices **may** need to replay the scene multiple times, affecting some of the counts.

If a pipeline has `rasterizerDiscardEnable` enabled, implementations **may** discard primitives after the final [pre-rasterization shader stage](#). As a result, if `rasterizerDiscardEnable` is enabled, the clipping input and output primitives counters **may** not be incremented.

When a pipeline statistics query finishes, the result for that query is marked as available. The application **can** copy the result to a buffer (via `vkCmdCopyQueryPoolResults`), or request it be put into host memory (via `vkGetQueryPoolResults`).

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkQueryPipelineStatisticFlags;
```

`VkQueryPipelineStatisticFlags` is a bitmask type for setting a mask of zero or more [VkQueryPipelineStatisticFlagBits](#).

17.5. Timestamp Queries

Timestamps provide applications with a mechanism for timing the execution of commands. A

timestamp is an integer value generated by the `VkPhysicalDevice`. Unlike other queries, timestamps do not operate over a range, and so do not use `vkCmdBeginQuery` or `vkCmdEndQuery`. The mechanism is built around a set of commands that allow the application to tell the `VkPhysicalDevice` to write timestamp values to a `query pool` and then either read timestamp values on the host (using `vkGetQueryPoolResults`) or copy timestamp values to a `VkBuffer` (using `vkCmdCopyQueryPoolResults`). The application **can** then compute differences between timestamps to determine execution time.

The number of valid bits in a timestamp value is determined by the `VkQueueFamilyProperties::timestampValidBits` property of the queue on which the timestamp is written. Timestamps are supported on any queue which reports a non-zero value for `timestampValidBits` via `vkGetPhysicalDeviceQueueFamilyProperties`. If the `timestampComputeAndGraphics` limit is `VK_TRUE`, timestamps are supported by every queue family that supports either graphics or compute operations (see `VkQueueFamilyProperties`).

The number of nanoseconds it takes for a timestamp value to be incremented by 1 **can** be obtained from `VkPhysicalDeviceLimits::timestampPeriod` after a call to `vkGetPhysicalDeviceProperties`.

To request a timestamp and write the value to memory, call:

```
// Provided by VK_VERSION_1_0
void vkCmdWriteTimestamp(
    VkCommandBuffer          commandBuffer,
    VkPipelineStageFlagBits  pipelineStage,
    VkQueryPool              queryPool,
    uint32_t                 query);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pipelineStage` is a `VkPipelineStageFlagBits` value, specifying a stage of the pipeline.
- `queryPool` is the query pool that will manage the timestamp.
- `query` is the query within the query pool that will contain the timestamp.

When `vkCmdWriteTimestamp` is submitted to a queue, it defines an execution dependency on commands that were submitted before it, and writes a timestamp to a query pool.

The first `synchronization scope` includes all commands that occur earlier in `submission order`. The synchronization scope is limited to operations on the pipeline stage specified by `pipelineStage`.

The second `synchronization scope` includes only the timestamp write operation.

NOTE

Implementations may write the timestamp at any stage that is `logically later` than `stage`.

Any timestamp write that `happens-after` another timestamp write in the same submission **must** not have a lower value unless its value overflows the maximum supported integer bit width of the query. If an overflow occurs, the timestamp value **must** wrap back to zero.

NOTE

Comparisons between timestamps should be done between timestamps where they are guaranteed to not decrease. For example, subtracting an older timestamp from a newer one to determine the execution time of a sequence of commands is only a reliable measurement if the two timestamp writes were performed in the same submission.

Valid Usage

- VUID-vkCmdWriteTimestamp-pipelineStage-04074
`pipelineStage` **must** be a [valid stage](#) for the queue family that was used to create the command pool that `commandBuffer` was allocated from
- VUID-vkCmdWriteTimestamp-pipelineStage-04075
If the `geometryShader` feature is not enabled, `pipelineStage` **must** not be `VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT`
- VUID-vkCmdWriteTimestamp-pipelineStage-04076
If the `tessellationShader` feature is not enabled, `pipelineStage` **must** not be `VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT` or `VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT`
- VUID-vkCmdWriteTimestamp-pipelineStage-06490
`pipelineStage` **must** not be `VK_PIPELINE_STAGE_NONE`
- VUID-vkCmdWriteTimestamp-queryPool-01416
`queryPool` **must** have been created with a `queryType` of `VK_QUERY_TYPE_TIMESTAMP`
- VUID-vkCmdWriteTimestamp-timestampValidBits-00829
The command pool's queue family **must** support a non-zero `timestampValidBits`
- VUID-vkCmdWriteTimestamp-query-04904
`query` **must** be less than the number of queries in `queryPool`
- VUID-vkCmdWriteTimestamp-None-00830
All queries used by the command **must** be *unavailable*

Valid Usage (Implicit)

- VUID-vkCmdWriteTimestamp-commandBuffer-parameter
`commandBuffer` **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdWriteTimestamp-pipelineStage-parameter
`pipelineStage` **must** be a valid [VkPipelineStageFlagBits](#) value
- VUID-vkCmdWriteTimestamp-queryPool-parameter
`queryPool` **must** be a valid [VkQueryPool](#) handle
- VUID-vkCmdWriteTimestamp-commandBuffer-recording
`commandBuffer` **must** be in the [recording state](#)
- VUID-vkCmdWriteTimestamp-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support transfer, graphics,

or compute operations

- VUID-vkCmdWriteTimestamp-commonparent

Both of **commandBuffer**, and **queryPool** **must** have been created, allocated, or retrieved from the same **VkDevice**

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the **VkCommandPool** that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Transfer Graphics Compute	Action

Chapter 18. Clear Commands

18.1. Clearing Images Outside a Render Pass Instance

Color and depth/stencil images **can** be cleared outside a render pass instance using [vkCmdClearColorImage](#) or [vkCmdClearDepthStencilImage](#), respectively. These commands are only allowed outside of a render pass instance.

To clear one or more subranges of a color image, call:

```
// Provided by VK_VERSION_1_0
void vkCmdClearColorImage(
    VkCommandBuffer          commandBuffer,
    VkImage                  image,
    VkImageLayout            imageLayout,
    const VkClearColorValue* pColor,
    uint32_t                 rangeCount,
    const VkImageSubresourceRange* pRanges);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `image` is the image to be cleared.
- `imageLayout` specifies the current layout of the image subresource ranges to be cleared, and **must** be `VK_IMAGE_LAYOUT_GENERAL` or `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL`.
- `pColor` is a pointer to a [VkClearColorValue](#) structure containing the values that the image subresource ranges will be cleared to (see [Clear Values](#) below).
- `rangeCount` is the number of image subresource range structures in `pRanges`.
- `pRanges` is a pointer to an array of [VkImageSubresourceRange](#) structures describing a range of mipmap levels, array layers, and aspects to be cleared, as described in [Image Views](#).

Each specified range in `pRanges` is cleared to the value specified by `pColor`.

Valid Usage

- VUID-vkCmdClearColorImage-image-00002
`image` **must** have been created with `VK_IMAGE_USAGE_TRANSFER_DST_BIT` usage flag
- VUID-vkCmdClearColorImage-image-00003
If `image` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdClearColorImage-imageLayout-00004
`imageLayout` **must** specify the layout of the image subresource ranges of `image` specified in `pRanges` at the time this command is executed on a `VkDevice`
- VUID-vkCmdClearColorImage-imageLayout-01394
`imageLayout` **must** be `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL` or `VK_IMAGE_LAYOUT_GENERAL`

- VUID-vkCmdClearColorImage-aspectMask-02498
The `VkImageSubresourceRange::aspectMask` members of the elements of the `pRanges` array **must** each only include `VK_IMAGE_ASPECT_COLOR_BIT`
- VUID-vkCmdClearColorImage-baseMipLevel-01470
The `VkImageSubresourceRange::baseMipLevel` members of the elements of the `pRanges` array **must** each be less than the `mipLevels` specified in `VkImageCreateInfo` when `image` was created
- VUID-vkCmdClearColorImage-pRanges-01692
For each `VkImageSubresourceRange` element of `pRanges`, if the `levelCount` member is not `VK_REMAINING_MIP_LEVELS`, then `baseMipLevel + levelCount` **must** be less than or equal to the `mipLevels` specified in `VkImageCreateInfo` when `image` was created
- VUID-vkCmdClearColorImage-baseArrayLayer-01472
The `VkImageSubresourceRange::baseArrayLayer` members of the elements of the `pRanges` array **must** each be less than the `arrayLayers` specified in `VkImageCreateInfo` when `image` was created
- VUID-vkCmdClearColorImage-pRanges-01693
For each `VkImageSubresourceRange` element of `pRanges`, if the `layerCount` member is not `VK_REMAINING_ARRAY_LAYERS`, then `baseArrayLayer + layerCount` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `image` was created
- VUID-vkCmdClearColorImage-image-00007
`image` **must** not have a compressed or depth/stencil format
- VUID-vkCmdClearColorImage-pColor-04961
`pColor` **must** be a valid pointer to a `VkClearColorValue` union
- VUID-vkCmdClearColorImage-image-09678
If `image`'s format has components other than R and G, it **must** not have a 64-bit component width

Valid Usage (Implicit)

- VUID-vkCmdClearColorImage-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdClearColorImage-image-parameter
`image` **must** be a valid `VkImage` handle
- VUID-vkCmdClearColorImage-imageLayout-parameter
`imageLayout` **must** be a valid `VkImageLayout` value
- VUID-vkCmdClearColorImage-pRanges-parameter
`pRanges` **must** be a valid pointer to an array of `rangeCount` valid `VkImageSubresourceRange` structures
- VUID-vkCmdClearColorImage-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdClearColorImage-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, or

compute operations

- VUID-vkCmdClearColorImage-renderpass

This command **must** only be called outside of a render pass instance

- VUID-vkCmdClearColorImage-rangeCount-arraylength

`rangeCount` **must** be greater than 0

- VUID-vkCmdClearColorImage-commonparent

Both of `commandBuffer`, and `image` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Graphics Compute	Action

To clear one or more subranges of a depth/stencil image, call:

```
// Provided by VK_VERSION_1_0
void vkCmdClearDepthStencilImage(
    VkCommandBuffer          commandBuffer,
    VkImage                  image,
    VkImageLayout             imageLayout,
    const VkClearDepthStencilValue* pDepthStencil,
    uint32_t                 rangeCount,
    const VkImageSubresourceRange* pRanges);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `image` is the image to be cleared.
- `imageLayout` specifies the current layout of the image subresource ranges to be cleared, and **must** be `VK_IMAGE_LAYOUT_GENERAL` or `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL`.
- `pDepthStencil` is a pointer to a `VkClearDepthStencilValue` structure containing the values that the depth and stencil image subresource ranges will be cleared to (see [Clear Values](#) below).
- `rangeCount` is the number of image subresource range structures in `pRanges`.

- `pRanges` is a pointer to an array of `VkImageSubresourceRange` structures describing a range of mipmap levels, array layers, and aspects to be cleared, as described in [Image Views](#).

Valid Usage

- VUID-vkCmdClearDepthStencilImage-pRanges-02659
If the `aspect` member of any element of `pRanges` includes `VK_IMAGE_ASPECT_STENCIL_BIT`, `VK_IMAGE_USAGE_TRANSFER_DST_BIT` **must** have been included in the `VkImageCreateInfo::usage` used to create `image`
- VUID-vkCmdClearDepthStencilImage-pRanges-02660
If the `aspect` member of any element of `pRanges` includes `VK_IMAGE_ASPECT_DEPTH_BIT`, `VK_IMAGE_USAGE_TRANSFER_DST_BIT` **must** have been included in the `VkImageCreateInfo::usage` used to create `image`
- VUID-vkCmdClearDepthStencilImage-image-00010
If `image` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdClearDepthStencilImage-imageLayout-00011
`imageLayout` **must** specify the layout of the image subresource ranges of `image` specified in `pRanges` at the time this command is executed on a `VkDevice`
- VUID-vkCmdClearDepthStencilImage-imageLayout-00012
`imageLayout` **must** be either of `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL` or `VK_IMAGE_LAYOUT_GENERAL`
- VUID-vkCmdClearDepthStencilImage-aspectMask-02824
The `VkImageSubresourceRange::aspectMask` member of each element of the `pRanges` array **must** not include bits other than `VK_IMAGE_ASPECT_DEPTH_BIT` or `VK_IMAGE_ASPECT_STENCIL_BIT`
- VUID-vkCmdClearDepthStencilImage-image-02825
If the `image`'s format does not have a stencil component, then the `VkImageSubresourceRange::aspectMask` member of each element of the `pRanges` array **must** not include the `VK_IMAGE_ASPECT_STENCIL_BIT` bit
- VUID-vkCmdClearDepthStencilImage-image-02826
If the `image`'s format does not have a depth component, then the `VkImageSubresourceRange::aspectMask` member of each element of the `pRanges` array **must** not include the `VK_IMAGE_ASPECT_DEPTH_BIT` bit
- VUID-vkCmdClearDepthStencilImage-baseMipLevel-01474
The `VkImageSubresourceRange::baseMipLevel` members of the elements of the `pRanges` array **must** each be less than the `mipLevels` specified in `VkImageCreateInfo` when `image` was created
- VUID-vkCmdClearDepthStencilImage-pRanges-01694
For each `VkImageSubresourceRange` element of `pRanges`, if the `levelCount` member is not `VK_REMAINING_MIP_LEVELS`, then `baseMipLevel + levelCount` **must** be less than or equal to the `mipLevels` specified in `VkImageCreateInfo` when `image` was created
- VUID-vkCmdClearDepthStencilImage-baseArrayLayer-01476

The `VkImageSubresourceRange::baseArrayLayer` members of the elements of the `pRanges` array **must** each be less than the `arrayLayers` specified in `VkImageCreateInfo` when `image` was created

- VUID-vkCmdClearDepthStencilImage-pRanges-01695
For each `VkImageSubresourceRange` element of `pRanges`, if the `layerCount` member is not `VK_REMAINING_ARRAY_LAYERS`, then `baseArrayLayer + layerCount` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `image` was created
- VUID-vkCmdClearDepthStencilImage-image-00014
`image` **must** have a depth/stencil format

Valid Usage (Implicit)

- VUID-vkCmdClearDepthStencilImage-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdClearDepthStencilImage-image-parameter
`image` **must** be a valid `VkImage` handle
- VUID-vkCmdClearDepthStencilImage-imageLayout-parameter
`imageLayout` **must** be a valid `VkImageLayout` value
- VUID-vkCmdClearDepthStencilImage-pDepthStencil-parameter
`pDepthStencil` **must** be a valid pointer to a valid `VkClearDepthStencilValue` structure
- VUID-vkCmdClearDepthStencilImage-pRanges-parameter
`pRanges` **must** be a valid pointer to an array of `rangeCount` valid `VkImageSubresourceRange` structures
- VUID-vkCmdClearDepthStencilImage-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdClearDepthStencilImage-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdClearDepthStencilImage-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdClearDepthStencilImage-rangeCount-arraylength
`rangeCount` **must** be greater than 0
- VUID-vkCmdClearDepthStencilImage-commonparent
Both of `commandBuffer`, and `image` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Graphics	Action

Clears outside render pass instances are treated as transfer operations for the purposes of memory barriers.

18.2. Clearing Images Inside a Render Pass Instance

To clear one or more regions of color and depth/stencil attachments inside a render pass instance, call:

```
// Provided by VK_VERSION_1_0
void vkCmdClearAttachments(
    VkCommandBuffer          commandBuffer,
    uint32_t                 attachmentCount,
    const VkClearAttachment* pAttachments,
    uint32_t                 rectCount,
    const VkClearRect*       pRects);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `attachmentCount` is the number of entries in the `pAttachments` array.
- `pAttachments` is a pointer to an array of `VkClearAttachment` structures defining the attachments to clear and the clear values to use.
- `rectCount` is the number of entries in the `pRects` array.
- `pRects` is a pointer to an array of `VkClearRect` structures defining regions within each selected attachment to clear.

Unlike other [clear commands](#), `vkCmdClearAttachments` is not a transfer command. It performs its operations in [rasterization order](#). For color attachments, the operations are executed as color attachment writes, by the `VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT` stage. For depth/stencil attachments, the operations are executed as [depth writes](#) and [stencil writes](#) by the `VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT` and `VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT` stages.

`vkCmdClearAttachments` is not affected by the bound pipeline state.

NOTE

It is generally preferable to clear attachments by using the `VK_ATTACHMENT_LOAD_OP_CLEAR` load operation at the start of rendering, as it is more efficient on some implementations.

If any attachment's `aspectMask` to be cleared is not backed by an image view, the clear has no effect

on that aspect.

Valid Usage

- VUID-vkCmdClearAttachments-aspectMask-07884
If the `aspectMask` member of any element of `pAttachments` contains `VK_IMAGE_ASPECT_DEPTH_BIT`, the current subpass instance's depth-stencil attachment **must** be either `VK_ATTACHMENT_UNUSED` or the attachment `format` **must** contain a depth component
- VUID-vkCmdClearAttachments-aspectMask-07885
If the `aspectMask` member of any element of `pAttachments` contains `VK_IMAGE_ASPECT_STENCIL_BIT`, the current subpass instance's depth-stencil attachment **must** be either `VK_ATTACHMENT_UNUSED` or the attachment `format` **must** contain a stencil component
- VUID-vkCmdClearAttachments-aspectMask-07271
If the `aspectMask` member of any element of `pAttachments` contains `VK_IMAGE_ASPECT_COLOR_BIT`, the `colorAttachment` **must** be a valid color attachment index in the current render pass instance
- VUID-vkCmdClearAttachments-rect-02682
The `rect` member of each element of `pRects` **must** have an `extent.width` greater than 0
- VUID-vkCmdClearAttachments-rect-02683
The `rect` member of each element of `pRects` **must** have an `extent.height` greater than 0
- VUID-vkCmdClearAttachments-pRects-00016
The rectangular region specified by each element of `pRects` **must** be contained within the render area of the current render pass instance
- VUID-vkCmdClearAttachments-pRects-06937
The layers specified by each element of `pRects` **must** be contained within every attachment that `pAttachments` refers to, i.e. for each element of `pRects`, `VkClearRect::baseArrayLayer` + `VkClearRect::layerCount` **must** be less than or equal to the number of layers rendered to in the current render pass instance
- VUID-vkCmdClearAttachments-layerCount-01934
The `layerCount` member of each element of `pRects` **must** not be 0
- VUID-vkCmdClearAttachments-None-09679
If the attachment format has components other than R and G, it **must** not have a 64-bit component width

Valid Usage (Implicit)

- VUID-vkCmdClearAttachments-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdClearAttachments-pAttachments-parameter
`pAttachments` **must** be a valid pointer to an array of `attachmentCount` valid `VkClearAttachment` structures

- VUID-vkCmdClearAttachments-pRects-parameter
`pRects` **must** be a valid pointer to an array of `rectCount` `VkClearRect` structures
- VUID-vkCmdClearAttachments-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdClearAttachments-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdClearAttachments-renderpass
This command **must** only be called inside of a render pass instance
- VUID-vkCmdClearAttachments-attachmentCount-arraylength
`attachmentCount` **must** be greater than 0
- VUID-vkCmdClearAttachments-rectCount-arraylength
`rectCount` **must** be greater than 0

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Graphics	Action

The `VkClearRect` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkClearRect {
    VkRect2D    rect;
    uint32_t    baseArrayLayer;
    uint32_t    layerCount;
} VkClearRect;
```

- `rect` is the two-dimensional region to be cleared.
- `baseArrayLayer` is the first layer to be cleared.
- `layerCount` is the number of layers to clear.

The layers [`baseArrayLayer`, `baseArrayLayer` + `layerCount`) counting from the base layer of the

attachment image view are cleared.

The `VkClearAttachment` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkClearAttachment {
    VkImageAspectFlags    aspectMask;
    uint32_t              colorAttachment;
    VkClearColorValue      clearValue;
} VkClearAttachment;
```

- `aspectMask` is a mask selecting the color, depth and/or stencil aspects of the attachment to be cleared.
- `colorAttachment` is only meaningful if `VK_IMAGE_ASPECT_COLOR_BIT` is set in `aspectMask`, in which case it is an index into the currently bound color attachments.
- `clearValue` is the color or depth/stencil value to clear the attachment to, as described in [Clear Values](#) below.

Valid Usage

- VUID-VkClearAttachment-aspectMask-00019
If `aspectMask` includes `VK_IMAGE_ASPECT_COLOR_BIT`, it **must** not include `VK_IMAGE_ASPECT_DEPTH_BIT` or `VK_IMAGE_ASPECT_STENCIL_BIT`
- VUID-VkClearAttachment-aspectMask-00020
`aspectMask` **must** not include `VK_IMAGE_ASPECT_METADATA_BIT`

Valid Usage (Implicit)

- VUID-VkClearAttachment-aspectMask-parameter
`aspectMask` **must** be a valid combination of [VkImageAspectFlagBits](#) values
- VUID-VkClearAttachment-aspectMask-requiredbitmask
`aspectMask` **must** not be 0

18.3. Clear Values

The `VkClearColorValue` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef union VkClearColorValue {
    float        float32[4];
    int32_t       int32[4];
    uint32_t      uint32[4];
} VkClearColorValue;
```

- **float32** are the color clear values when the format of the image or attachment is one of the **numeric formats** with a numeric type that is floating-point. Floating-point values are automatically converted to the format of the image, with the clear value being treated as linear if the image is sRGB.
- **int32** are the color clear values when the format of the image or attachment has a numeric type that is signed integer (**SINT**). Signed integer values are converted to the format of the image by casting to the smaller type (with negative 32-bit values mapping to negative values in the smaller type). If the integer clear value is not representable in the target type (e.g. would overflow in conversion to that type), the clear value is undefined.
- **uint32** are the color clear values when the format of the image or attachment has a numeric type that is unsigned integer (**UINT**). Unsigned integer values are converted to the format of the image by casting to the integer type with fewer bits.

The four array elements of the clear color map to R, G, B, and A components of image formats, in order.

If the image has more than one sample, the same value is written to all samples for any pixels being cleared.

If the image or attachment format has a 64-bit component width, the first 2 array elements of each of the arrays above are reinterpreted as a single 64-bit element for the R component. The next 2 array elements are used in the same way for the G component. In other words, the union behaves as if it had the following additional members:

```
double float64[2];
int64_t int64[2];
uint64_t uint64[2];
```

The **VkClearDepthStencilValue** structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkClearDepthStencilValue {
    float        depth;
    uint32_t      stencil;
} VkClearDepthStencilValue;
```

- **depth** is the clear value for the depth aspect of the depth/stencil attachment. It is a floating-point value which is automatically converted to the attachment's format.

- **stencil** is the clear value for the stencil aspect of the depth/stencil attachment. It is a 32-bit integer value which is converted to the attachment's format by taking the appropriate number of LSBs.

Valid Usage

- VUID-VkClearColorValue-depth-00022
depth **must** be between **0.0** and **1.0**, inclusive

The **VkClearColorValue** union is defined as:

```
// Provided by VK_VERSION_1_0
typedef union VkClearColorValue {
    VkClearColorValue    color;
    VkClearDepthStencilValue depthStencil;
} VkClearColorValue;
```

- **color** specifies the color image clear values to use when clearing a color image or attachment.
- **depthStencil** specifies the depth and stencil clear values to use when clearing a depth/stencil image or attachment.

This union is used where part of the API requires either color or depth/stencil clear values, depending on the attachment, and defines the initial clear values in the [VkRenderPassBeginInfo](#) structure.

18.4. Filling Buffers

To clear buffer data, call:

```
// Provided by VK_VERSION_1_0
void vkCmdFillBuffer(
    VkCommandBuffer    commandBuffer,
    VkBuffer            dstBuffer,
    VkDeviceSize        dstOffset,
    VkDeviceSize        size,
    uint32_t            data);
```

- **commandBuffer** is the command buffer into which the command will be recorded.
- **dstBuffer** is the buffer to be filled.
- **dstOffset** is the byte offset into the buffer at which to start filling, and **must** be a multiple of 4.
- **size** is the number of bytes to fill, and **must** be either a multiple of 4, or **VK_WHOLE_SIZE** to fill the range from **offset** to the end of the buffer. If **VK_WHOLE_SIZE** is used and the remaining size of the buffer is not a multiple of 4, then the nearest smaller multiple is used.
- **data** is the 4-byte word written repeatedly to the buffer to fill **size** bytes of data. The data word

is written to memory according to the host endianness.

`vkCmdFillBuffer` is treated as a “transfer” operation for the purposes of synchronization barriers. The `VK_BUFFER_USAGE_TRANSFER_DST_BIT` **must** be specified in `usage` of `VkBufferCreateInfo` in order for the buffer to be compatible with `vkCmdFillBuffer`.

Valid Usage

- VUID-vkCmdFillBuffer-dstOffset-00024
`dstOffset` **must** be less than the size of `dstBuffer`
- VUID-vkCmdFillBuffer-dstOffset-00025
`dstOffset` **must** be a multiple of 4
- VUID-vkCmdFillBuffer-size-00026
If `size` is not equal to `VK_WHOLE_SIZE`, `size` **must** be greater than 0
- VUID-vkCmdFillBuffer-size-00027
If `size` is not equal to `VK_WHOLE_SIZE`, `size` **must** be less than or equal to the size of `dstBuffer` minus `dstOffset`
- VUID-vkCmdFillBuffer-size-00028
If `size` is not equal to `VK_WHOLE_SIZE`, `size` **must** be a multiple of 4
- VUID-vkCmdFillBuffer-dstBuffer-00029
`dstBuffer` **must** have been created with `VK_BUFFER_USAGE_TRANSFER_DST_BIT` usage flag
- VUID-vkCmdFillBuffer-apiVersion-07894
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics or compute operations
- VUID-vkCmdFillBuffer-dstBuffer-00031
If `dstBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

Valid Usage (Implicit)

- VUID-vkCmdFillBuffer-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdFillBuffer-dstBuffer-parameter
`dstBuffer` **must** be a valid `VkBuffer` handle
- VUID-vkCmdFillBuffer-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdFillBuffer-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics or compute operations
- VUID-vkCmdFillBuffer-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdFillBuffer-commonparent

Both of `commandBuffer`, and `dstBuffer` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Graphics Compute	Action

18.5. Updating Buffers

To update buffer data inline in a command buffer, call:

```
// Provided by VK_VERSION_1_0
void vkCmdUpdateBuffer(
    VkCommandBuffer          commandBuffer,
    VkBuffer                 dstBuffer,
    VkDeviceSize             dstOffset,
    VkDeviceSize             dataSize,
    const void*              pData);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `dstBuffer` is a handle to the buffer to be updated.
- `dstOffset` is the byte offset into the buffer to start updating, and **must** be a multiple of 4.
- `dataSize` is the number of bytes to update, and **must** be a multiple of 4.
- `pData` is a pointer to the source data for the buffer update, and **must** be at least `dataSize` bytes in size.

`dataSize` **must** be less than or equal to 65536 bytes. For larger updates, applications **can** use buffer to buffer `copies`.

NOTE

Buffer updates performed with `vkCmdUpdateBuffer` first copy the data into command buffer memory when the command is recorded (which requires additional storage and may incur an additional allocation), and then copy the data from the command

buffer into `dstBuffer` when the command is executed on a device.

The additional cost of this functionality compared to `buffer to buffer copies` means it is only recommended for very small amounts of data, and is why it is limited to only 65536 bytes.

Applications **can** work around this by issuing multiple `vkCmdUpdateBuffer` commands to different ranges of the same buffer, but it is strongly recommended that they **should** not.

The source data is copied from `pData` to the command buffer when the command is called.

`vkCmdUpdateBuffer` is only allowed outside of a render pass. This command is treated as a “transfer” operation for the purposes of synchronization barriers. The `VK_BUFFER_USAGE_TRANSFER_DST_BIT` **must** be specified in `usage` of `VkBufferCreateInfo` in order for the buffer to be compatible with `vkCmdUpdateBuffer`.

Valid Usage

- VUID-vkCmdUpdateBuffer-dstOffset-00032
`dstOffset` **must** be less than the size of `dstBuffer`
- VUID-vkCmdUpdateBuffer-dataSize-00033
`dataSize` **must** be less than or equal to the size of `dstBuffer` minus `dstOffset`
- VUID-vkCmdUpdateBuffer-dstBuffer-00034
`dstBuffer` **must** have been created with `VK_BUFFER_USAGE_TRANSFER_DST_BIT` usage flag
- VUID-vkCmdUpdateBuffer-dstBuffer-00035
If `dstBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdUpdateBuffer-dstOffset-00036
`dstOffset` **must** be a multiple of 4
- VUID-vkCmdUpdateBuffer-dataSize-00037
`dataSize` **must** be less than or equal to 65536
- VUID-vkCmdUpdateBuffer-dataSize-00038
`dataSize` **must** be a multiple of 4

Valid Usage (Implicit)

- VUID-vkCmdUpdateBuffer-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdUpdateBuffer-dstBuffer-parameter
`dstBuffer` **must** be a valid `VkBuffer` handle
- VUID-vkCmdUpdateBuffer-pData-parameter
`pData` **must** be a valid pointer to an array of `dataSize` bytes

- VUID-vkCmdUpdateBuffer-commandBuffer-recording
`commandBuffer` **must** be in the [recording state](#)
- VUID-vkCmdUpdateBuffer-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support transfer, graphics, or compute operations
- VUID-vkCmdUpdateBuffer-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdUpdateBuffer-dataSize-arraylength
`dataSize` **must** be greater than 0
- VUID-vkCmdUpdateBuffer-commonparent
Both of `commandBuffer`, and `dstBuffer` **must** have been created, allocated, or retrieved from the same [VkDevice](#)

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Transfer Graphics Compute	Action

NOTE

The `pData` parameter was of type `uint32_t*` instead of `void*` prior to version 1.0.19 of the Specification and `VK_HEADER_VERSION` 19 of the [Vulkan Header Files](#). This was a historical anomaly, as the source data may be of other types.

Chapter 19. Copy Commands

An application **can** copy buffer and image data using several methods described in this chapter, depending on the type of data transfer.

All copy commands are treated as “transfer” operations for the purposes of synchronization barriers.

All copy commands that have a source format with an X component in its format description read undefined values from those bits.

All copy commands that have a destination format with an X component in its format description write undefined values to those bits.

19.1. Copying Data Between Buffers

To copy data between buffer objects, call:

```
// Provided by VK_VERSION_1_0
void vkCmdCopyBuffer(
    VkCommandBuffer          commandBuffer,
    VkBuffer                 srcBuffer,
    VkBuffer                 dstBuffer,
    uint32_t                 regionCount,
    const VkBufferCopy*      pRegions);
```

- **commandBuffer** is the command buffer into which the command will be recorded.
- **srcBuffer** is the source buffer.
- **dstBuffer** is the destination buffer.
- **regionCount** is the number of regions to copy.
- **pRegions** is a pointer to an array of **VkBufferCopy** structures specifying the regions to copy.

Each source region specified by **pRegions** is copied from the source buffer to the destination region of the destination buffer. If any of the specified regions in **srcBuffer** overlaps in memory with any of the specified regions in **dstBuffer**, values read from those overlapping regions are undefined.

Valid Usage

- VUID-vkCmdCopyBuffer-srcOffset-00113
The **srcOffset** member of each element of **pRegions** **must** be less than the size of **srcBuffer**
- VUID-vkCmdCopyBuffer-dstOffset-00114
The **dstOffset** member of each element of **pRegions** **must** be less than the size of **dstBuffer**
- VUID-vkCmdCopyBuffer-size-00115
The **size** member of each element of **pRegions** **must** be less than or equal to the size of **srcBuffer** minus **srcOffset**

- VUID-vkCmdCopyBuffer-size-00116
The **size** member of each element of **pRegions** **must** be less than or equal to the size of **dstBuffer** minus **dstOffset**
- VUID-vkCmdCopyBuffer-pRegions-00117
The union of the source regions, and the union of the destination regions, specified by the elements of **pRegions**, **must** not overlap in memory
- VUID-vkCmdCopyBuffer-srcBuffer-00118
srcBuffer **must** have been created with **VK_BUFFER_USAGE_TRANSFER_SRC_BIT** usage flag
- VUID-vkCmdCopyBuffer-srcBuffer-00119
If **srcBuffer** is non-sparse then it **must** be bound completely and contiguously to a single **VkDeviceMemory** object
- VUID-vkCmdCopyBuffer-dstBuffer-00120
dstBuffer **must** have been created with **VK_BUFFER_USAGE_TRANSFER_DST_BIT** usage flag
- VUID-vkCmdCopyBuffer-dstBuffer-00121
If **dstBuffer** is non-sparse then it **must** be bound completely and contiguously to a single **VkDeviceMemory** object

Valid Usage (Implicit)

- VUID-vkCmdCopyBuffer-commandBuffer-parameter
commandBuffer **must** be a valid **VkCommandBuffer** handle
- VUID-vkCmdCopyBuffer-srcBuffer-parameter
srcBuffer **must** be a valid **VkBuffer** handle
- VUID-vkCmdCopyBuffer-dstBuffer-parameter
dstBuffer **must** be a valid **VkBuffer** handle
- VUID-vkCmdCopyBuffer-pRegions-parameter
pRegions **must** be a valid pointer to an array of **regionCount** valid **VkBufferCopy** structures
- VUID-vkCmdCopyBuffer-commandBuffer-recording
commandBuffer **must** be in the **recording state**
- VUID-vkCmdCopyBuffer-commandBuffer-cmdpool
The **VkCommandPool** that **commandBuffer** was allocated from **must** support transfer, graphics, or compute operations
- VUID-vkCmdCopyBuffer-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyBuffer-regionCount-arraylength
regionCount **must** be greater than 0
- VUID-vkCmdCopyBuffer-commonparent
Each of **commandBuffer**, **dstBuffer**, and **srcBuffer** **must** have been created, allocated, or retrieved from the same **VkDevice**

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Transfer Graphics Compute	Action

The `VkBufferCopy` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkBufferCopy {
    VkDeviceSize    srcOffset;
    VkDeviceSize    dstOffset;
    VkDeviceSize    size;
} VkBufferCopy;
```

- `srcOffset` is the starting offset in bytes from the start of `srcBuffer`.
- `dstOffset` is the starting offset in bytes from the start of `dstBuffer`.
- `size` is the number of bytes to copy.

Valid Usage

- VUID-VkBufferCopy-size-01988
The `size` **must** be greater than `0`

19.2. Copying Data Between Images

To copy data between image objects, call:


```
// Provided by VK_VERSION_1_0
void vkCmdCopyImage(
    VkCommandBuffer          commandBuffer,
    VkImage                  srcImage,
    VkImageLayout            srcImageLayout,
    VkImage                  dstImage,
    VkImageLayout            dstImageLayout,
    uint32_t                 regionCount,
    const VkImageCopy*       pRegions);
```

- **commandBuffer** is the command buffer into which the command will be recorded.
- **srcImage** is the source image.
- **srcImageLayout** is the current layout of the source image subresource.
- **dstImage** is the destination image.
- **dstImageLayout** is the current layout of the destination image subresource.
- **regionCount** is the number of regions to copy.
- **pRegions** is a pointer to an array of **VkImageCopy** structures specifying the regions to copy.

Each source region specified by **pRegions** is copied from the source image to the destination region of the destination image. If any of the specified regions in **srcImage** overlaps in memory with any of the specified regions in **dstImage**, values read from those overlapping regions are undefined.

If the format of the destination image has a different **block extent** than the source image (e.g. one is a compressed format), the offset and extent for each of the regions specified is **scaled according to the block extents of each format** to match in size. Copy regions for each image **must** be aligned to a multiple of the texel block extent in each dimension, except at the edges of the image, where region extents **must** match the edge of the image.

Valid Usage

- VUID-vkCmdCopyImage-pRegions-00124
The union of all source regions, and the union of all destination regions, specified by the elements of **pRegions**, **must** not overlap in memory
- VUID-vkCmdCopyImage-srcImageLayout-00128
srcImageLayout **must** specify the layout of the image subresources of **srcImage** specified in **pRegions** at the time this command is executed on a **VkDevice**
- VUID-vkCmdCopyImage-srcImageLayout-01917
srcImageLayout **must** be **VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL**, or **VK_IMAGE_LAYOUT_GENERAL**
- VUID-vkCmdCopyImage-srcImage-09460
If **srcImage** and **dstImage** are the same, and any elements of **pRegions** contains the **srcSubresource** and **dstSubresource** with matching **mipLevel** and overlapping array layers, then the **srcImageLayout** and **dstImageLayout** **must** be **VK_IMAGE_LAYOUT_GENERAL**
- VUID-vkCmdCopyImage-dstImageLayout-00133

`dstImageLayout` **must** specify the layout of the image subresources of `dstImage` specified in `pRegions` at the time this command is executed on a `VkDevice`

- VUID-vkCmdCopyImage-dstImageLayout-01395
`dstImageLayout` **must** be `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL`, or `VK_IMAGE_LAYOUT_GENERAL`
- VUID-vkCmdCopyImage-srcImage-01548
The `VkFormat` of each of `srcImage` and `dstImage` **must** be `size-compatible`
- VUID-vkCmdCopyImage-srcImage-09247
If the `VkFormat` of each of `srcImage` and `dstImage` is a `compressed image format`, the formats **must** have the same texel block extent
- VUID-vkCmdCopyImage-srcImage-00136
The sample count of `srcImage` and `dstImage` **must** match
- VUID-vkCmdCopyImage-srcOffset-01783
The `srcOffset` and `extent` members of each element of `pRegions` **must** respect the image transfer granularity requirements of `commandBuffer`'s command pool's queue family, as described in `VkQueueFamilyProperties`
- VUID-vkCmdCopyImage-dstOffset-01784
The `dstOffset` and `extent` members of each element of `pRegions` **must** respect the image transfer granularity requirements of `commandBuffer`'s command pool's queue family, as described in `VkQueueFamilyProperties`
- VUID-vkCmdCopyImage-apiVersion-07932
If either `srcImage` or `dstImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`, `srcSubresource.baseArrayLayer` and `dstSubresource.baseArrayLayer` **must** both be `0`, and `srcSubresource.layerCount` and `dstSubresource.layerCount` **must** both be `1`
- VUID-vkCmdCopyImage-srcImage-04443
If `srcImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`, `srcSubresource.baseArrayLayer` **must** be `0` and `srcSubresource.layerCount` **must** be `1`
- VUID-vkCmdCopyImage-dstImage-04444
If `dstImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`, `dstSubresource.baseArrayLayer` **must** be `0` and `dstSubresource.layerCount` **must** be `1`
- VUID-vkCmdCopyImage-aspectMask-00142
For each element of `pRegions`, `srcSubresource.aspectMask` **must** specify aspects present in `srcImage`
- VUID-vkCmdCopyImage-aspectMask-00143
For each element of `pRegions`, `dstSubresource.aspectMask` **must** specify aspects present in `dstImage`
- VUID-vkCmdCopyImage-srcOffset-00144
For each element of `pRegions`, `srcOffset.x` and `(extent.width + srcOffset.x)` **must** both be greater than or equal to `0` and less than or equal to the width of the specified `srcSubresource` of `srcImage`
- VUID-vkCmdCopyImage-srcOffset-00145
For each element of `pRegions`, `srcOffset.y` and `(extent.height + srcOffset.y)` **must** both be greater than or equal to `0` and less than or equal to the height of the specified `srcSubresource` of `srcImage`

- VUID-vkCmdCopyImage-srcImage-00146
If `srcImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `srcOffset.y` **must** be `0` and `extent.height` **must** be `1`
- VUID-vkCmdCopyImage-srcOffset-00147
If `srcImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`, `srcOffset.z` and `(extent.depth + srcOffset.z)` **must** both be greater than or equal to `0` and less than or equal to the depth of the specified `srcSubresource` of `srcImage`
- VUID-vkCmdCopyImage-srcImage-01785
If `srcImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `srcOffset.z` **must** be `0` and `extent.depth` **must** be `1`
- VUID-vkCmdCopyImage-dstImage-01786
If `dstImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `dstOffset.z` **must** be `0` and `extent.depth` **must** be `1`
- VUID-vkCmdCopyImage-srcImage-01787
If `srcImage` is of type `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `srcOffset.z` **must** be `0`
- VUID-vkCmdCopyImage-dstImage-01788
If `dstImage` is of type `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `dstOffset.z` **must** be `0`
- VUID-vkCmdCopyImage-apiVersion-07933
If `srcImage` and `dstImage` **must** have the same `VkImageType`
- VUID-vkCmdCopyImage-apiVersion-08969
If `srcImage` or `dstImage` is of type `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `extent.depth` **must** be `1`
- VUID-vkCmdCopyImage-dstOffset-00150
For each element of `pRegions`, `dstOffset.x` and `(extent.width + dstOffset.x)` **must** both be greater than or equal to `0` and less than or equal to the width of the specified `dstSubresource` of `dstImage`
- VUID-vkCmdCopyImage-dstOffset-00151
For each element of `pRegions`, `dstOffset.y` and `(extent.height + dstOffset.y)` **must** both be greater than or equal to `0` and less than or equal to the height of the specified `dstSubresource` of `dstImage`
- VUID-vkCmdCopyImage-dstImage-00152
If `dstImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `dstOffset.y` **must** be `0` and `extent.height` **must** be `1`
- VUID-vkCmdCopyImage-dstOffset-00153
If `dstImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`, `dstOffset.z` and `(extent.depth + dstOffset.z)` **must** both be greater than or equal to `0` and less than or equal to the depth of the specified `dstSubresource` of `dstImage`
- VUID-vkCmdCopyImage-pRegions-07278
For each element of `pRegions`, `srcOffset.x` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `srcImage`
- VUID-vkCmdCopyImage-pRegions-07279

For each element of `pRegions`, `srcOffset.y` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `srcImage`

- VUID-vkCmdCopyImage-pRegions-07280

For each element of `pRegions`, `srcOffset.z` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `srcImage`

- VUID-vkCmdCopyImage-pRegions-07281

For each element of `pRegions`, `dstOffset.x` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `dstImage`

- VUID-vkCmdCopyImage-pRegions-07282

For each element of `pRegions`, `dstOffset.y` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `dstImage`

- VUID-vkCmdCopyImage-pRegions-07283

For each element of `pRegions`, `dstOffset.z` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `dstImage`

- VUID-vkCmdCopyImage-srcImage-01728

For each element of `pRegions`, if the sum of `srcOffset.x` and `extent.width` does not equal the width of the subresource specified by `srcSubresource`, `extent.width` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `srcImage`

- VUID-vkCmdCopyImage-srcImage-01729

For each element of `pRegions`, if the sum of `srcOffset.y` and `extent.height` does not equal the height of the subresource specified by `srcSubresource`, `extent.height` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `srcImage`

- VUID-vkCmdCopyImage-srcImage-01730

For each element of `pRegions`, if the sum of `srcOffset.z` and `extent.depth` does not equal the depth of the subresource specified by `srcSubresource`, `extent.depth` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `srcImage`

- VUID-vkCmdCopyImage-dstImage-01732

For each element of `pRegions`, if the sum of `dstOffset.x` and `extent.width` does not equal the width of the subresource specified by `dstSubresource`, `extent.width` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `dstImage`

- VUID-vkCmdCopyImage-dstImage-01733

For each element of `pRegions`, if the sum of `dstOffset.y` and `extent.height` does not equal the height of the subresource specified by `dstSubresource`, `extent.height` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `dstImage`

- VUID-vkCmdCopyImage-dstImage-01734

For each element of `pRegions`, if the sum of `dstOffset.z` and `extent.depth` does not equal the depth of the subresource specified by `dstSubresource`, `extent.depth` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `dstImage`

- VUID-vkCmdCopyImage-aspect-06662

`VK_IMAGE_USAGE_TRANSFER_SRC_BIT` **must** have been included in the `VkImageCreateInfo::usage` used to create `srcImage`

- VUID-vkCmdCopyImage-aspect-06663

`VK_IMAGE_USAGE_TRANSFER_DST_BIT` **must** have been included in the `VkImageCreateInfo`

::usage used to create dstImage

- VUID-vkCmdCopyImage-srcImage-07966
If srcImage is non-sparse then the image **must** be bound completely and contiguously to a single VkDeviceMemory object
- VUID-vkCmdCopyImage-srcSubresource-07967
The srcSubresource.mipLevel member of each element of pRegions **must** be less than the mipLevels specified in VkImageCreateInfo when srcImage was created
- VUID-vkCmdCopyImage-srcSubresource-07968
srcSubresource.baseArrayLayer + srcSubresource.layerCount of each element of pRegions **must** be less than or equal to the arrayLayers specified in VkImageCreateInfo when srcImage was created
- VUID-vkCmdCopyImage-dstImage-07966
If dstImage is non-sparse then the image **must** be bound completely and contiguously to a single VkDeviceMemory object
- VUID-vkCmdCopyImage-dstSubresource-07967
The dstSubresource.mipLevel member of each element of pRegions **must** be less than the mipLevels specified in VkImageCreateInfo when dstImage was created
- VUID-vkCmdCopyImage-dstSubresource-07968
dstSubresource.baseArrayLayer + dstSubresource.layerCount of each element of pRegions **must** be less than or equal to the arrayLayers specified in VkImageCreateInfo when dstImage was created

Valid Usage (Implicit)

- VUID-vkCmdCopyImage-commandBuffer-parameter
commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdCopyImage-srcImage-parameter
srcImage **must** be a valid VkImage handle
- VUID-vkCmdCopyImage-srcImageLayout-parameter
srcImageLayout **must** be a valid VkImageLayout value
- VUID-vkCmdCopyImage-dstImage-parameter
dstImage **must** be a valid VkImage handle
- VUID-vkCmdCopyImage-dstImageLayout-parameter
dstImageLayout **must** be a valid VkImageLayout value
- VUID-vkCmdCopyImage-pRegions-parameter
pRegions **must** be a valid pointer to an array of regionCount valid VkImageCopy structures
- VUID-vkCmdCopyImage-commandBuffer-recording
commandBuffer **must** be in the recording state
- VUID-vkCmdCopyImage-commandBuffer-cmdpool
The VkCommandPool that commandBuffer was allocated from **must** support transfer, graphics, or compute operations

- VUID-vkCmdCopyImage-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyImage-regionCount-arraylength
regionCount **must** be greater than 0
- VUID-vkCmdCopyImage-commonparent
Each of **commandBuffer**, **dstImage**, and **srcImage** **must** have been created, allocated, or retrieved from the same **VkDevice**

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the **VkCommandPool** that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Transfer Graphics Compute	Action

The **VkImageCopy** structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageCopy {
    VkImageSubresourceLayers    srcSubresource;
    VkOffset3D                  srcOffset;
    VkImageSubresourceLayers    dstSubresource;
    VkOffset3D                  dstOffset;
    VkExtent3D                  extent;
} VkImageCopy;
```

- **srcSubresource** and **dstSubresource** are **VkImageSubresourceLayers** structures specifying the image subresources of the images used for the source and destination image data, respectively.
- **srcOffset** and **dstOffset** select the initial **x**, **y**, and **z** offsets in texels of the sub-regions of the source and destination image data.
- **extent** is the size in texels of the image to copy in **width**, **height** and **depth**.

Valid Usage

- VUID-VkImageCopy-apiVersion-07940
The `aspectMask` member of `srcSubresource` and `dstSubresource` **must** match
- VUID-VkImageCopy-apiVersion-07941
The `layerCount` member of `srcSubresource` and `dstSubresource` **must** match
- VUID-VkImageCopy-extent-06668
`extent.width` **must** not be 0
- VUID-VkImageCopy-extent-06669
`extent.height` **must** not be 0
- VUID-VkImageCopy-extent-06670
`extent.depth` **must** not be 0

Valid Usage (Implicit)

- VUID-VkImageCopy-srcSubresource-parameter
`srcSubresource` **must** be a valid `VkImageSubresourceLayers` structure
- VUID-VkImageCopy-dstSubresource-parameter
`dstSubresource` **must** be a valid `VkImageSubresourceLayers` structure

The `VkImageSubresourceLayers` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageSubresourceLayers {
    VkImageAspectFlags    aspectMask;
    uint32_t              mipLevel;
    uint32_t              baseArrayLayer;
    uint32_t              layerCount;
} VkImageSubresourceLayers;
```

- `aspectMask` is a combination of `VkImageAspectFlagBits`, selecting the color, depth and/or stencil aspects to be copied.
- `mipLevel` is the mipmap level to copy
- `baseArrayLayer` and `layerCount` are the starting layer and number of layers to copy.

Valid Usage

- VUID-VkImageSubresourceLayers-aspectMask-00167
If `aspectMask` contains `VK_IMAGE_ASPECT_COLOR_BIT`, it **must** not contain either of `VK_IMAGE_ASPECT_DEPTH_BIT` or `VK_IMAGE_ASPECT_STENCIL_BIT`
- VUID-VkImageSubresourceLayers-aspectMask-00168
`aspectMask` **must** not contain `VK_IMAGE_ASPECT_METADATA_BIT`
- VUID-VkImageSubresourceLayers-layerCount-09243
`layerCount` **must** not be `VK_REMAINING_ARRAY_LAYERS`

- VUID-VkImageSubresourceLayers-layerCount-01700
If `layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, it **must** be greater than 0

Valid Usage (Implicit)

- VUID-VkImageSubresourceLayers-aspectMask-parameter
`aspectMask` **must** be a valid combination of `VkImageAspectFlagBits` values
- VUID-VkImageSubresourceLayers-aspectMask-requiredbitmask
`aspectMask` **must** not be 0

19.3. Copying Data Between Buffers and Images

Data **can** be copied between buffers and images, enabling applications to load and store data between images and application-defined offsets in buffer memory.

When copying between a buffer and an image, texels in the image and bytes in the buffer are accessed as follows.

Texels at each coordinate (x,y,z,layer) in the image subresource are accessed, where:

x is in the range [`imageOffset.x`, `imageOffset.x` + `imageExtent.width`),

y is in the range [`imageOffset.y`, `imageOffset.y` + `imageExtent.height`),

z is in the range [`imageOffset.z`, `imageOffset.z` + `imageExtent.depth`),

layer is in the range [`imageSubresource.baseArrayLayer`, `imageSubresource.baseArrayLayer` + `imageSubresource.layerCount`)

For each (x,y,z,layer) coordinate in the image, bytes in the buffer are accessed at offsets in the range [`texelOffset`, `texelOffset` + `blockSize`), where:

$$\text{texelOffset} = \text{bufferOffset} + ((x - \text{imageOffset.x}) / \text{blockWidth} \times \text{blockSize}) + ((y - \text{imageOffset.y}) / \text{blockHeight} \times \text{rowExtent}) + ((z - \text{imageOffset.z}) / \text{blockDepth} \times \text{sliceExtent}) + ((\text{layer} - \text{imageSubresource.baseArrayLayer}) \times \text{layerExtent})$$

$$\text{rowExtent} = \lceil \max(\text{bufferRowLength}, \text{imageExtent.width}) / \text{blockWidth} \rceil \times \text{blockSize}$$

$$\text{sliceExtent} = \lceil \max(\text{bufferImageHeight}, \text{imageExtent.height}) / \text{blockHeight} \rceil \times \text{rowExtent}$$

$$\text{layerExtent} = \lceil \text{imageExtent.depth} / \text{blockDepth} \rceil \times \text{sliceExtent}$$

and where blockSize, blockWidth, blockHeight, and blockDepth are the [texel block size and extents](#) of the image's format.

When copying between a buffer and the depth or stencil aspect of an image, data in the buffer is assumed to be laid out as separate planes rather than interleaved. Addressing calculations are thus performed for a different format than the base image, according to the aspect, as described in the following table:

Table 19. Depth/Stencil Aspect Copy Table

Base Format	Depth Aspect Format	Stencil Aspect Format
VK_FORMAT_D16_UNORM	VK_FORMAT_D16_UNORM	-
VK_FORMAT_X8_D24_UNORM_PACK32	VK_FORMAT_X8_D24_UNORM_PACK32	-
VK_FORMAT_D32_SFLOAT	VK_FORMAT_D32_SFLOAT	-
VK_FORMAT_S8_UINT	-	VK_FORMAT_S8_UINT
VK_FORMAT_D16_UNORM_S8_UINT	VK_FORMAT_D16_UNORM	VK_FORMAT_S8_UINT
VK_FORMAT_D24_UNORM_S8_UINT	VK_FORMAT_X8_D24_UNORM_PACK32	VK_FORMAT_S8_UINT
VK_FORMAT_D32_SFLOAT_S8_UINT	VK_FORMAT_D32_SFLOAT	VK_FORMAT_S8_UINT

Each texel block is copied from one resource to the other according to the above addressing equations.

To copy data from a buffer object to an image object, call:

```
// Provided by VK_VERSION_1_0
void vkCmdCopyBufferToImage(
    VkCommandBuffer          commandBuffer,
    VkBuffer                 srcBuffer,
    VkImage                  dstImage,
    VkImageLayout            dstImageLayout,
    uint32_t                 regionCount,
    const VkBufferImageCopy* pRegions);
```

- **commandBuffer** is the command buffer into which the command will be recorded.
- **srcBuffer** is the source buffer.
- **dstImage** is the destination image.
- **dstImageLayout** is the layout of the destination image subresources for the copy.
- **regionCount** is the number of regions to copy.
- **pRegions** is a pointer to an array of [VkBufferImageCopy](#) structures specifying the regions to copy.

Each source region specified by **pRegions** is copied from the source buffer to the destination region of the destination image according to the [addressing calculations](#) for each resource. If any of the

specified regions in `srcBuffer` overlaps in memory with any of the specified regions in `dstImage`, values read from those overlapping regions are undefined. If any region accesses a depth aspect in `dstImage` values copied from `srcBuffer` outside of the range [0,1] will be written as undefined values to the destination image.

Copy regions for the image **must** be aligned to a multiple of the texel block extent in each dimension, except at the edges of the image, where region extents **must** match the edge of the image.

Valid Usage

- VUID-vkCmdCopyBufferToImage-dstImage-07966
If `dstImage` is non-sparse then the image **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdCopyBufferToImage-imageSubresource-07967
The `imageSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `dstImage` was created
- VUID-vkCmdCopyBufferToImage-imageSubresource-07968
`imageSubresource.baseArrayLayer` + `imageSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `dstImage` was created
- VUID-vkCmdCopyBufferToImage-imageSubresource-07970
The image region specified by each element of `pRegions` **must** be contained within the specified `imageSubresource` of `dstImage`
- VUID-vkCmdCopyBufferToImage-imageSubresource-07971
For each element of `pRegions`, `imageOffset.x` and `(imageExtent.width + imageOffset.x)` **must** both be greater than or equal to 0 and less than or equal to the width of the specified `imageSubresource` of `dstImage`
- VUID-vkCmdCopyBufferToImage-imageSubresource-07972
For each element of `pRegions`, `imageOffset.y` and `(imageExtent.height + imageOffset.y)` **must** both be greater than or equal to 0 and less than or equal to the height of the specified `imageSubresource` of `dstImage`
- VUID-vkCmdCopyBufferToImage-dstImage-07973
`dstImage` **must** have a sample count equal to `VK_SAMPLE_COUNT_1_BIT`
- VUID-vkCmdCopyBufferToImage-commandBuffer-07737
If the queue family used to create the `VkCommandPool` which `commandBuffer` was allocated from does not support `VK_QUEUE_GRAPHICS_BIT` or `VK_QUEUE_COMPUTE_BIT`, the `bufferOffset` member of any element of `pRegions` **must** be a multiple of 4
- VUID-vkCmdCopyBufferToImage-imageOffset-07738
The `imageOffset` and `imageExtent` members of each element of `pRegions` **must** respect the image transfer granularity requirements of `commandBuffer`'s command pool's queue family, as described in `VkQueueFamilyProperties`
- VUID-vkCmdCopyBufferToImage-commandBuffer-07739

If the queue family used to create the `VkCommandPool` which `commandBuffer` was allocated from does not support `VK_QUEUE_GRAPHICS_BIT`, for each element of `pRegions`, the `aspectMask` member of `imageSubresource` **must** not be `VK_IMAGE_ASPECT_DEPTH_BIT` or `VK_IMAGE_ASPECT_STENCIL_BIT`

- VUID-vkCmdCopyBufferToImage-pRegions-00171
`srcBuffer` **must** be large enough to contain all buffer locations that are accessed according to [Buffer and Image Addressing](#), for each element of `pRegions`
- VUID-vkCmdCopyBufferToImage-pRegions-00173
The union of all source regions, and the union of all destination regions, specified by the elements of `pRegions`, **must** not overlap in memory
- VUID-vkCmdCopyBufferToImage-srcBuffer-00174
`srcBuffer` **must** have been created with `VK_BUFFER_USAGE_TRANSFER_SRC_BIT` usage flag
- VUID-vkCmdCopyBufferToImage-srcBuffer-00176
If `srcBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdCopyBufferToImage-dstImage-00177
`dstImage` **must** have been created with `VK_IMAGE_USAGE_TRANSFER_DST_BIT` usage flag
- VUID-vkCmdCopyBufferToImage-dstImageLayout-00180
`dstImageLayout` **must** specify the layout of the image subresources of `dstImage` specified in `pRegions` at the time this command is executed on a `VkDevice`
- VUID-vkCmdCopyBufferToImage-dstImageLayout-01396
`dstImageLayout` **must** be `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL`, or `VK_IMAGE_LAYOUT_GENERAL`
- VUID-vkCmdCopyBufferToImage-pRegions-07931
For each element of `pRegions` whose `imageSubresource` contains a depth aspect, the data in `srcBuffer` **must** be in the range [0,1]
- VUID-vkCmdCopyBufferToImage-dstImage-07979
If `dstImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `imageOffset.y` **must** be 0 and `imageExtent.height` **must** be 1
- VUID-vkCmdCopyBufferToImage-imageOffset-09104
For each element of `pRegions`, `imageOffset.z` and `(imageExtent.depth + imageOffset.z)` **must** both be greater than or equal to 0 and less than or equal to the depth of the specified `imageSubresource` of `dstImage`
- VUID-vkCmdCopyBufferToImage-dstImage-07980
If `dstImage` is of type `VK_IMAGE_TYPE_1D` or `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `imageOffset.z` **must** be 0 and `imageExtent.depth` **must** be 1
- VUID-vkCmdCopyBufferToImage-dstImage-07274
For each element of `pRegions`, `imageOffset.x` **must** be a multiple of the [texel block extent width](#) of the `VkFormat` of `dstImage`
- VUID-vkCmdCopyBufferToImage-dstImage-07275
For each element of `pRegions`, `imageOffset.y` **must** be a multiple of the [texel block extent height](#) of the `VkFormat` of `dstImage`

- VUID-vkCmdCopyBufferToImage-dstImage-07276
For each element of `pRegions`, `imageOffset.z` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `dstImage`
- VUID-vkCmdCopyBufferToImage-dstImage-00207
For each element of `pRegions`, if the sum of `imageOffset.x` and `extent.width` does not equal the width of the subresource specified by `imageSubresource`, `extent.width` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `dstImage`
- VUID-vkCmdCopyBufferToImage-dstImage-00208
For each element of `pRegions`, if the sum of `imageOffset.y` and `extent.height` does not equal the height of the subresource specified by `imageSubresource`, `extent.height` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `dstImage`
- VUID-vkCmdCopyBufferToImage-dstImage-00209
For each element of `pRegions`, if the sum of `imageOffset.z` and `extent.depth` does not equal the depth of the subresource specified by `srcSubresource`, `extent.depth` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `dstImage`
- VUID-vkCmdCopyBufferToImage-imageSubresource-09105
For each element of `pRegions`, `imageSubresource.aspectMask` **must** specify aspects present in `dstImage`
- VUID-vkCmdCopyBufferToImage-dstImage-07983
If `dstImage` is of type `VK_IMAGE_TYPE_3D`, for each element of `pRegions`, `imageSubresource.baseArrayLayer` **must** be 0 and `imageSubresource.layerCount` **must** be 1
- VUID-vkCmdCopyBufferToImage-bufferRowLength-09106
For each element of `pRegions`, `bufferRowLength` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `dstImage`
- VUID-vkCmdCopyBufferToImage-bufferImageHeight-09107
For each element of `pRegions`, `bufferImageHeight` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `dstImage`
- VUID-vkCmdCopyBufferToImage-bufferRowLength-09108
For each element of `pRegions`, `bufferRowLength` divided by the `texel block extent width` and then multiplied by the texel block size of `dstImage` **must** be less than or equal to $2^{31}-1$
- VUID-vkCmdCopyBufferToImage-dstImage-07975
If `dstImage` does not have either a depth/stencil format then for each element of `pRegions`, `bufferOffset` **must** be a multiple of the `texel block size`
- VUID-vkCmdCopyBufferToImage-dstImage-07978
If `dstImage` has a depth/stencil format, the `bufferOffset` member of any element of `pRegions` **must** be a multiple of 4

Valid Usage (Implicit)

- VUID-vkCmdCopyBufferToImage-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdCopyBufferToImage-srcBuffer-parameter

`srcBuffer` **must** be a valid `VkBuffer` handle

- VUID-vkCmdCopyBufferToImage-dstImage-parameter
`dstImage` **must** be a valid `VkImage` handle
- VUID-vkCmdCopyBufferToImage-dstImageLayout-parameter
`dstImageLayout` **must** be a valid `VkImageLayout` value
- VUID-vkCmdCopyBufferToImage-pRegions-parameter
`pRegions` **must** be a valid pointer to an array of `regionCount` valid `VkBufferImageCopy` structures
- VUID-vkCmdCopyBufferToImage-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdCopyBufferToImage-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support transfer, graphics, or compute operations
- VUID-vkCmdCopyBufferToImage-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyBufferToImage-regionCount-arraylength
`regionCount` **must** be greater than 0
- VUID-vkCmdCopyBufferToImage-commonparent
Each of `commandBuffer`, `dstImage`, and `srcBuffer` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Transfer Graphics Compute	Action

To copy data from an image object to a buffer object, call:

```
// Provided by VK_VERSION_1_0
void vkCmdCopyImageToBuffer(
    VkCommandBuffer          commandBuffer,
    VkImage                  srcImage,
    VkImageLayout            srcImageLayout,
    VkBuffer                  dstBuffer,
    uint32_t                 regionCount,
    const VkBufferImageCopy* pRegions);
```

- **commandBuffer** is the command buffer into which the command will be recorded.
- **srcImage** is the source image.
- **srcImageLayout** is the layout of the source image subresources for the copy.
- **dstBuffer** is the destination buffer.
- **regionCount** is the number of regions to copy.
- **pRegions** is a pointer to an array of **VkBufferImageCopy** structures specifying the regions to copy.

Each source region specified by **pRegions** is copied from the source image to the destination region of the destination buffer according to the [addressing calculations](#) for each resource. If any of the specified regions in **srcImage** overlaps in memory with any of the specified regions in **dstBuffer**, values read from those overlapping regions are undefined.

Copy regions for the image **must** be aligned to a multiple of the texel block extent in each dimension, except at the edges of the image, where region extents **must** match the edge of the image.

Valid Usage

- VUID-vkCmdCopyImageToBuffer-srcImage-07966
If **srcImage** is non-sparse then the image **must** be bound completely and contiguously to a single **VkDeviceMemory** object
- VUID-vkCmdCopyImageToBuffer-imageSubresource-07967
The **imageSubresource.mipLevel** member of each element of **pRegions** **must** be less than the **mipLevels** specified in **VkImageCreateInfo** when **srcImage** was created
- VUID-vkCmdCopyImageToBuffer-imageSubresource-07968
imageSubresource.baseArrayLayer + **imageSubresource.layerCount** of each element of **pRegions** **must** be less than or equal to the **arrayLayers** specified in **VkImageCreateInfo** when **srcImage** was created
- VUID-vkCmdCopyImageToBuffer-imageSubresource-07970
The image region specified by each element of **pRegions** **must** be contained within the specified **imageSubresource** of **srcImage**
- VUID-vkCmdCopyImageToBuffer-imageSubresource-07971
For each element of **pRegions**, **imageOffset.x** and **(imageExtent.width + imageOffset.x)** **must** both be greater than or equal to **0** and less than or equal to the width of the specified

`imageSubresource` of `srcImage`

- VUID-vkCmdCopyImageToBuffer-imageSubresource-07972

For each element of `pRegions`, `imageOffset.y` and `(imageExtent.height + imageOffset.y)` **must** both be greater than or equal to 0 and less than or equal to the height of the specified `imageSubresource` of `srcImage`

- VUID-vkCmdCopyImageToBuffer-srcImage-07973

`srcImage` **must** have a sample count equal to `VK_SAMPLE_COUNT_1_BIT`

- VUID-vkCmdCopyImageToBuffer-commandBuffer-07746

If the queue family used to create the `VkCommandPool` which `commandBuffer` was allocated from does not support `VK_QUEUE_GRAPHICS_BIT` or `VK_QUEUE_COMPUTE_BIT`, the `bufferOffset` member of any element of `pRegions` **must** be a multiple of 4

- VUID-vkCmdCopyImageToBuffer-imageOffset-07747

The `imageOffset` and `imageExtent` members of each element of `pRegions` **must** respect the image transfer granularity requirements of `commandBuffer`'s command pool's queue family, as described in `VkQueueFamilyProperties`

- VUID-vkCmdCopyImageToBuffer-pRegions-00183

`dstBuffer` **must** be large enough to contain all buffer locations that are accessed according to `Buffer and Image Addressing`, for each element of `pRegions`

- VUID-vkCmdCopyImageToBuffer-pRegions-00184

The union of all source regions, and the union of all destination regions, specified by the elements of `pRegions`, **must** not overlap in memory

- VUID-vkCmdCopyImageToBuffer-srcImage-00186

`srcImage` **must** have been created with `VK_IMAGE_USAGE_TRANSFER_SRC_BIT` usage flag

- VUID-vkCmdCopyImageToBuffer-dstBuffer-00191

`dstBuffer` **must** have been created with `VK_BUFFER_USAGE_TRANSFER_DST_BIT` usage flag

- VUID-vkCmdCopyImageToBuffer-dstBuffer-00192

If `dstBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-vkCmdCopyImageToBuffer-srcImageLayout-00189

`srcImageLayout` **must** specify the layout of the image subresources of `srcImage` specified in `pRegions` at the time this command is executed on a `VkDevice`

- VUID-vkCmdCopyImageToBuffer-srcImageLayout-01397

`srcImageLayout` **must** be `VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL`, or `VK_IMAGE_LAYOUT_GENERAL`

- VUID-vkCmdCopyImageToBuffer-srcImage-07979

If `srcImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `imageOffset.y` **must** be 0 and `imageExtent.height` **must** be 1

- VUID-vkCmdCopyImageToBuffer-imageOffset-09104

For each element of `pRegions`, `imageOffset.z` and `(imageExtent.depth + imageOffset.z)` **must** both be greater than or equal to 0 and less than or equal to the depth of the specified `imageSubresource` of `srcImage`

- VUID-vkCmdCopyImageToBuffer-srcImage-07980

If `srcImage` is of type `VK_IMAGE_TYPE_1D` or `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `imageOffset.z` **must** be 0 and `imageExtent.depth` **must** be 1

- VUID-vkCmdCopyImageToBuffer-srcImage-07274
For each element of `pRegions`, `imageOffset.x` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `srcImage`
- VUID-vkCmdCopyImageToBuffer-srcImage-07275
For each element of `pRegions`, `imageOffset.y` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `srcImage`
- VUID-vkCmdCopyImageToBuffer-srcImage-07276
For each element of `pRegions`, `imageOffset.z` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `srcImage`
- VUID-vkCmdCopyImageToBuffer-srcImage-00207
For each element of `pRegions`, if the sum of `imageOffset.x` and `extent.width` does not equal the width of the subresource specified by `imageSubresource`, `extent.width` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `srcImage`
- VUID-vkCmdCopyImageToBuffer-srcImage-00208
For each element of `pRegions`, if the sum of `imageOffset.y` and `extent.height` does not equal the height of the subresource specified by `imageSubresource`, `extent.height` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `srcImage`
- VUID-vkCmdCopyImageToBuffer-srcImage-00209
For each element of `pRegions`, if the sum of `imageOffset.z` and `extent.depth` does not equal the depth of the subresource specified by `srcSubresource`, `extent.depth` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `srcImage`
- VUID-vkCmdCopyImageToBuffer-imageSubresource-09105
For each element of `pRegions`, `imageSubresource.aspectMask` **must** specify aspects present in `srcImage`
- VUID-vkCmdCopyImageToBuffer-srcImage-07983
If `srcImage` is of type `VK_IMAGE_TYPE_3D`, for each element of `pRegions`, `imageSubresource.baseArrayLayer` **must** be 0 and `imageSubresource.layerCount` **must** be 1
- VUID-vkCmdCopyImageToBuffer-bufferRowLength-09106
For each element of `pRegions`, `bufferRowLength` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `srcImage`
- VUID-vkCmdCopyImageToBuffer-bufferImageHeight-09107
For each element of `pRegions`, `bufferImageHeight` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `srcImage`
- VUID-vkCmdCopyImageToBuffer-bufferRowLength-09108
For each element of `pRegions`, `bufferRowLength` divided by the `texel block extent width` and then multiplied by the texel block size of `srcImage` **must** be less than or equal to $2^{31}-1$
- VUID-vkCmdCopyImageToBuffer-srcImage-07975
If `srcImage` does not have either a depth/stencil format then for each element of `pRegions`, `bufferOffset` **must** be a multiple of the `texel block size`
- VUID-vkCmdCopyImageToBuffer-srcImage-07978

If `srcImage` has a depth/stencil format, the `bufferOffset` member of any element of `pRegions` **must** be a multiple of 4

Valid Usage (Implicit)

- VUID-vkCmdCopyImageToBuffer-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdCopyImageToBuffer-srcImage-parameter
`srcImage` **must** be a valid `VkImage` handle
- VUID-vkCmdCopyImageToBuffer-srcImageLayout-parameter
`srcImageLayout` **must** be a valid `VkImageLayout` value
- VUID-vkCmdCopyImageToBuffer-dstBuffer-parameter
`dstBuffer` **must** be a valid `VkBuffer` handle
- VUID-vkCmdCopyImageToBuffer-pRegions-parameter
`pRegions` **must** be a valid pointer to an array of `regionCount` valid `VkBufferImageCopy` structures
- VUID-vkCmdCopyImageToBuffer-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdCopyImageToBuffer-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support transfer, graphics, or compute operations
- VUID-vkCmdCopyImageToBuffer-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyImageToBuffer-regionCount-arraylength
`regionCount` **must** be greater than 0
- VUID-vkCmdCopyImageToBuffer-commonparent
Each of `commandBuffer`, `dstBuffer`, and `srcImage` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Transfer Graphics Compute	Action

For both [vkCmdCopyBufferToImage](#) and [vkCmdCopyImageToBuffer](#), each element of **pRegions** is a structure defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkBufferImageCopy {
    VkDeviceSize      bufferOffset;
    uint32_t          bufferRowLength;
    uint32_t          bufferImageHeight;
    VkImageSubresourceLayers imageSubresource;
    VkOffset3D         imageOffset;
    VkExtent3D         imageExtent;
} VkBufferImageCopy;
```

- **bufferOffset** is the offset in bytes from the start of the buffer object where the image data is copied from or to.
- **bufferRowLength** and **bufferImageHeight** specify in texels a subregion of a larger two- or three-dimensional image in buffer memory, and control the addressing calculations. If either of these values is zero, that aspect of the buffer memory is considered to be tightly packed according to the **imageExtent**.
- **imageSubresource** is a [VkImageSubresourceLayers](#) used to specify the specific image subresources of the image used for the source or destination image data.
- **imageOffset** selects the initial **x**, **y**, **z** offsets in texels of the sub-region of the source or destination image data.
- **imageExtent** is the size in texels of the image to copy in **width**, **height** and **depth**.

Valid Usage

- VUID-VkBufferImageCopy-bufferRowLength-09101
bufferRowLength **must** be 0, or greater than or equal to the **width** member of **imageExtent**
- VUID-VkBufferImageCopy-bufferImageHeight-09102
bufferImageHeight **must** be 0, or greater than or equal to the **height** member of **imageExtent**
- VUID-VkBufferImageCopy-aspectMask-09103
The **aspectMask** member of **imageSubresource** **must** only have a single bit set
- VUID-VkBufferImageCopy-imageExtent-06659

`imageExtent.width` **must** not be 0

- VUID-VkBufferImageCopy-imageExtent-06660
`imageExtent.height` **must** not be 0
- VUID-VkBufferImageCopy-imageExtent-06661
`imageExtent.depth` **must** not be 0

Valid Usage (Implicit)

- VUID-VkBufferImageCopy-imageSubresource-parameter
`imageSubresource` **must** be a valid `VkImageSubresourceLayers` structure

19.4. Image Copies With Scaling

To copy regions of a source image into a destination image, potentially performing format conversion, arbitrary scaling, and filtering, call:

```
// Provided by VK_VERSION_1_0
void vkCmdBlitImage(
    VkCommandBuffer          commandBuffer,
    VkImage                  srcImage,
    VkImageLayout            srcImageLayout,
    VkImage                  dstImage,
    VkImageLayout            dstImageLayout,
    uint32_t                 regionCount,
    const VkImageBlit*       pRegions,
    VkFilter                  filter);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `srcImage` is the source image.
- `srcImageLayout` is the layout of the source image subresources for the blit.
- `dstImage` is the destination image.
- `dstImageLayout` is the layout of the destination image subresources for the blit.
- `regionCount` is the number of regions to blit.
- `pRegions` is a pointer to an array of `VkImageBlit` structures specifying the regions to blit.
- `filter` is a `VkFilter` specifying the filter to apply if the blits require scaling.

`vkCmdBlitImage` **must** not be used for multisampled source or destination images. Use `vkCmdResolveImage` for this purpose.

As the sizes of the source and destination extents **can** differ in any dimension, texels in the source extent are scaled and filtered to the destination extent. Scaling occurs via the following operations:

- For each destination texel, the integer coordinate of that texel is converted to an unnormalized texture coordinate, using the effective inverse of the equations described in [unnormalized to integer conversion](#):

$$u_{\text{base}} = i + \frac{1}{2}$$

$$v_{\text{base}} = j + \frac{1}{2}$$

$$w_{\text{base}} = k + \frac{1}{2}$$

- These base coordinates are then offset by the first destination offset:

$$u_{\text{offset}} = u_{\text{base}} - x_{\text{dst0}}$$

$$v_{\text{offset}} = v_{\text{base}} - y_{\text{dst0}}$$

$$w_{\text{offset}} = w_{\text{base}} - z_{\text{dst0}}$$

$$a_{\text{offset}} = a - \text{baseArrayCount}_{\text{dst}}$$

- The scale is determined from the source and destination regions, and applied to the offset coordinates:

$$\text{scale}_u = (x_{\text{src1}} - x_{\text{src0}}) / (x_{\text{dst1}} - x_{\text{dst0}})$$

$$\text{scale}_v = (y_{\text{src1}} - y_{\text{src0}}) / (y_{\text{dst1}} - y_{\text{dst0}})$$

$$\text{scale}_w = (z_{\text{src1}} - z_{\text{src0}}) / (z_{\text{dst1}} - z_{\text{dst0}})$$

$$u_{\text{scaled}} = u_{\text{offset}} \times \text{scale}_u$$

$$v_{\text{scaled}} = v_{\text{offset}} \times \text{scale}_v$$

$$w_{\text{scaled}} = w_{\text{offset}} \times \text{scale}_w$$

- Finally the source offset is added to the scaled coordinates, to determine the final unnormalized

coordinates used to sample from `srcImage`:

$$u = u_{\text{scaled}} + x_{\text{src0}}$$

$$v = v_{\text{scaled}} + y_{\text{src0}}$$

$$w = w_{\text{scaled}} + z_{\text{src0}}$$

$$q = \text{mipLevel}$$

$$a = a_{\text{offset}} + \text{baseArrayCount}_{\text{src}}$$

These coordinates are used to sample from the source image, as described in [Image Operations chapter](#), with the filter mode equal to that of `filter`, a mipmap mode of `VK_SAMPLER_MIPMAP_MODE_NEAREST` and an address mode of `VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_EDGE`. Implementations **must** clamp at the edge of the source image, and **may** additionally clamp to the edge of the source region.

NOTE

Due to allowable rounding errors in the generation of the source texture coordinates, it is not always possible to guarantee exactly which source texels will be sampled for a given blit. As rounding errors are implementation-dependent, the exact results of a blitting operation are also implementation-dependent.

Blits are done layer by layer starting with the `baseArrayLayer` member of `srcSubresource` for the source and `dstSubresource` for the destination. `layerCount` layers are blitted to the destination image.

When blitting 3D textures, slices in the destination region bounded by `dstOffsets[0].z` and `dstOffsets[1].z` are sampled from slices in the source region bounded by `srcOffsets[0].z` and `srcOffsets[1].z`. If the `filter` parameter is `VK_FILTER_LINEAR` then the value sampled from the source image is taken by doing linear filtering using the interpolated `z` coordinate represented by `w` in the previous equations. If the `filter` parameter is `VK_FILTER_NEAREST` then the value sampled from the source image is taken from the single nearest slice, with an implementation-dependent arithmetic rounding mode.

The following filtering and conversion rules apply:

- Integer formats **can** only be converted to other integer formats with the same signedness.
- No format conversion is supported between depth/stencil images. The formats **must** match.
- Format conversions on unorm, snorm, scaled and packed float formats of the copied aspect of the image are performed by first converting the pixels to float values.
- For sRGB source formats, nonlinear RGB values are converted to linear representation prior to filtering.

- After filtering, the float values are first clamped and then cast to the destination image format. In case of sRGB destination format, linear RGB values are converted to nonlinear representation before writing the pixel to the image.

Signed and unsigned integers are converted by first clamping to the representable range of the destination format, then casting the value.

Valid Usage

- VUID-vkCmdBlitImage-pRegions-00215
The source region specified by each element of `pRegions` **must** be a region that is contained within `srcImage`
- VUID-vkCmdBlitImage-pRegions-00216
The destination region specified by each element of `pRegions` **must** be a region that is contained within `dstImage`
- VUID-vkCmdBlitImage-pRegions-00217
The union of all destination regions, specified by the elements of `pRegions`, **must** not overlap in memory with any texel that **may** be sampled during the blit operation
- VUID-vkCmdBlitImage-srcImage-01999
The `format features` of `srcImage` **must** contain `VK_FORMAT_FEATURE_BLIT_SRC_BIT`
- VUID-vkCmdBlitImage-srcImage-00219
`srcImage` **must** have been created with `VK_IMAGE_USAGE_TRANSFER_SRC_BIT` usage flag
- VUID-vkCmdBlitImage-srcImage-00220
If `srcImage` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdBlitImage-srcImageLayout-00221
`srcImageLayout` **must** specify the layout of the image subresources of `srcImage` specified in `pRegions` at the time this command is executed on a `VkDevice`
- VUID-vkCmdBlitImage-srcImageLayout-00222
`srcImageLayout` **must** be `VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL` or `VK_IMAGE_LAYOUT_GENERAL`
- VUID-vkCmdBlitImage-srcImage-09459
If `srcImage` and `dstImage` are the same, and an elements of `pRegions` contains the `srcSubresource` and `dstSubresource` with matching `mipLevel` and overlapping array layers, then the `srcImageLayout` and `dstImageLayout` **must** be `VK_IMAGE_LAYOUT_GENERAL`
- VUID-vkCmdBlitImage-dstImage-02000
The `format features` of `dstImage` **must** contain `VK_FORMAT_FEATURE_BLIT_DST_BIT`
- VUID-vkCmdBlitImage-dstImage-00224
`dstImage` **must** have been created with `VK_IMAGE_USAGE_TRANSFER_DST_BIT` usage flag
- VUID-vkCmdBlitImage-dstImage-00225
If `dstImage` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdBlitImage-dstImageLayout-00226
`dstImageLayout` **must** specify the layout of the image subresources of `dstImage` specified in

`pRegions` at the time this command is executed on a `VkDevice`

- VUID-vkCmdBlitImage-dstImageLayout-00227
`dstImageLayout` **must** be `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL` or `VK_IMAGE_LAYOUT_GENERAL`
- VUID-vkCmdBlitImage-srcImage-00229
If either of `srcImage` or `dstImage` was created with a signed integer `VkFormat`, the other **must** also have been created with a signed integer `VkFormat`
- VUID-vkCmdBlitImage-srcImage-00230
If either of `srcImage` or `dstImage` was created with an unsigned integer `VkFormat`, the other **must** also have been created with an unsigned integer `VkFormat`
- VUID-vkCmdBlitImage-srcImage-00231
If either of `srcImage` or `dstImage` was created with a depth/stencil format, the other **must** have exactly the same format
- VUID-vkCmdBlitImage-srcImage-00232
If `srcImage` was created with a depth/stencil format, `filter` **must** be `VK_FILTER_NEAREST`
- VUID-vkCmdBlitImage-srcImage-00233
`srcImage` **must** have been created with a `samples` value of `VK_SAMPLE_COUNT_1_BIT`
- VUID-vkCmdBlitImage-dstImage-00234
`dstImage` **must** have been created with a `samples` value of `VK_SAMPLE_COUNT_1_BIT`
- VUID-vkCmdBlitImage-filter-02001
If `filter` is `VK_FILTER_LINEAR`, then the `format features` of `srcImage` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdBlitImage-srcSubresource-01705
The `srcSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `srcImage` was created
- VUID-vkCmdBlitImage-dstSubresource-01706
The `dstSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `dstImage` was created
- VUID-vkCmdBlitImage-srcSubresource-01707
`srcSubresource.baseArrayLayer` + `srcSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `srcImage` was created
- VUID-vkCmdBlitImage-dstSubresource-01708
`dstSubresource.baseArrayLayer` + `dstSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `dstImage` was created
- VUID-vkCmdBlitImage-srcImage-00240
If either `srcImage` or `dstImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`, `srcSubresource.baseArrayLayer` and `dstSubresource.baseArrayLayer` **must** each be 0, and `srcSubresource.layerCount` and `dstSubresource.layerCount` **must** each be 1
- VUID-vkCmdBlitImage-aspectMask-00241
For each element of `pRegions`, `srcSubresource.aspectMask` **must** specify aspects present in `srcImage`

- VUID-vkCmdBlitImage-aspectMask-00242
For each element of `pRegions`, `dstSubresource.aspectMask` **must** specify aspects present in `dstImage`
- VUID-vkCmdBlitImage-srcOffset-00243
For each element of `pRegions`, `srcOffsets[0].x` and `srcOffsets[1].x` **must** both be greater than or equal to `0` and less than or equal to the width of the specified `srcSubresource` of `srcImage`
- VUID-vkCmdBlitImage-srcOffset-00244
For each element of `pRegions`, `srcOffsets[0].y` and `srcOffsets[1].y` **must** both be greater than or equal to `0` and less than or equal to the height of the specified `srcSubresource` of `srcImage`
- VUID-vkCmdBlitImage-srcImage-00245
If `srcImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `srcOffsets[0].y` **must** be `0` and `srcOffsets[1].y` **must** be `1`
- VUID-vkCmdBlitImage-srcOffset-00246
For each element of `pRegions`, `srcOffsets[0].z` and `srcOffsets[1].z` **must** both be greater than or equal to `0` and less than or equal to the depth of the specified `srcSubresource` of `srcImage`
- VUID-vkCmdBlitImage-srcImage-00247
If `srcImage` is of type `VK_IMAGE_TYPE_1D` or `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `srcOffsets[0].z` **must** be `0` and `srcOffsets[1].z` **must** be `1`
- VUID-vkCmdBlitImage-dstOffset-00248
For each element of `pRegions`, `dstOffsets[0].x` and `dstOffsets[1].x` **must** both be greater than or equal to `0` and less than or equal to the width of the specified `dstSubresource` of `dstImage`
- VUID-vkCmdBlitImage-dstOffset-00249
For each element of `pRegions`, `dstOffsets[0].y` and `dstOffsets[1].y` **must** both be greater than or equal to `0` and less than or equal to the height of the specified `dstSubresource` of `dstImage`
- VUID-vkCmdBlitImage-dstImage-00250
If `dstImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `dstOffsets[0].y` **must** be `0` and `dstOffsets[1].y` **must** be `1`
- VUID-vkCmdBlitImage-dstOffset-00251
For each element of `pRegions`, `dstOffsets[0].z` and `dstOffsets[1].z` **must** both be greater than or equal to `0` and less than or equal to the depth of the specified `dstSubresource` of `dstImage`
- VUID-vkCmdBlitImage-dstImage-00252
If `dstImage` is of type `VK_IMAGE_TYPE_1D` or `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `dstOffsets[0].z` **must** be `0` and `dstOffsets[1].z` **must** be `1`

Valid Usage (Implicit)

- VUID-vkCmdBlitImage-commandBuffer-parameter

`commandBuffer` **must** be a valid `VkCommandBuffer` handle

- VUID-vkCmdBlitImage-srcImage-parameter
`srcImage` **must** be a valid `VkImage` handle
- VUID-vkCmdBlitImage-srcImageLayout-parameter
`srcImageLayout` **must** be a valid `VkImageLayout` value
- VUID-vkCmdBlitImage-dstImage-parameter
`dstImage` **must** be a valid `VkImage` handle
- VUID-vkCmdBlitImage-dstImageLayout-parameter
`dstImageLayout` **must** be a valid `VkImageLayout` value
- VUID-vkCmdBlitImage-pRegions-parameter
`pRegions` **must** be a valid pointer to an array of `regionCount` valid `VkImageBlit` structures
- VUID-vkCmdBlitImage-filter-parameter
`filter` **must** be a valid `VkFilter` value
- VUID-vkCmdBlitImage-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdBlitImage-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdBlitImage-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdBlitImage-regionCount-arraylength
`regionCount` **must** be greater than 0
- VUID-vkCmdBlitImage-commonparent
Each of `commandBuffer`, `dstImage`, and `srcImage` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Graphics	Action

The `VkImageBlit` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageBlit {
    VkImageSubresourceLayers srcSubresource;
    VkOffset3D                srcOffsets[2];
    VkImageSubresourceLayers dstSubresource;
    VkOffset3D                dstOffsets[2];
} VkImageBlit;
```

- **srcSubresource** is the subresource to blit from.
- **srcOffsets** is a pointer to an array of two **VkOffset3D** structures specifying the bounds of the source region within **srcSubresource**.
- **dstSubresource** is the subresource to blit into.
- **dstOffsets** is a pointer to an array of two **VkOffset3D** structures specifying the bounds of the destination region within **dstSubresource**.

For each element of the **pRegions** array, a blit operation is performed for the specified source and destination regions.

Valid Usage

- VUID-VkImageBlit-aspectMask-00238
The **aspectMask** member of **srcSubresource** and **dstSubresource** **must** match
- VUID-VkImageBlit-layerCount-08800
The **layerCount** members of **srcSubresource** or **dstSubresource** **must** match

Valid Usage (Implicit)

- VUID-VkImageBlit-srcSubresource-parameter
srcSubresource **must** be a valid **VkImageSubresourceLayers** structure
- VUID-VkImageBlit-dstSubresource-parameter
dstSubresource **must** be a valid **VkImageSubresourceLayers** structure

19.5. Resolving Multisample Images

To resolve a multisample color image to a non-multisample color image, call:

```
// Provided by VK_VERSION_1_0
void vkCmdResolveImage(
    VkCommandBuffer          commandBuffer,
    VkImage                  srcImage,
    VkImageLayout            srcImageLayout,
    VkImage                  dstImage,
    VkImageLayout            dstImageLayout,
    uint32_t                 regionCount,
    const VkImageResolve*    pRegions);
```

- **commandBuffer** is the command buffer into which the command will be recorded.
- **srcImage** is the source image.
- **srcImageLayout** is the layout of the source image subresources for the resolve.
- **dstImage** is the destination image.
- **dstImageLayout** is the layout of the destination image subresources for the resolve.
- **regionCount** is the number of regions to resolve.
- **pRegions** is a pointer to an array of **VkImageResolve** structures specifying the regions to resolve.

During the resolve the samples corresponding to each pixel location in the source are converted to a single sample before being written to the destination. If the source formats are floating-point or normalized types, the sample values for each pixel are resolved in an implementation-dependent manner. If the source formats are integer types, a single sample's value is selected for each pixel.

srcOffset and **dstOffset** select the initial **x**, **y**, and **z** offsets in texels of the sub-regions of the source and destination image data. **extent** is the size in texels of the source image to resolve in **width**, **height** and **depth**. Each element of **pRegions** **must** be a region that is contained within its corresponding image.

Resolves are done layer by layer starting with **baseArrayLayer** member of **srcSubresource** for the source and **dstSubresource** for the destination. **layerCount** layers are resolved to the destination image.

Valid Usage

- VUID-vkCmdResolveImage-pRegions-00255
The union of all source regions, and the union of all destination regions, specified by the elements of **pRegions**, **must** not overlap in memory
- VUID-vkCmdResolveImage-srcImage-00256
If **srcImage** is non-sparse then it **must** be bound completely and contiguously to a single **VkDeviceMemory** object
- VUID-vkCmdResolveImage-srcImage-00257
srcImage **must** have a sample count equal to any valid sample count value other than **VK_SAMPLE_COUNT_1_BIT**
- VUID-vkCmdResolveImage-dstImage-00258

If `dstImage` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-vkCmdResolveImage-dstImage-00259
`dstImage` **must** have a sample count equal to `VK_SAMPLE_COUNT_1_BIT`
- VUID-vkCmdResolveImage-srcImageLayout-00260
`srcImageLayout` **must** specify the layout of the image subresources of `srcImage` specified in `pRegions` at the time this command is executed on a `VkDevice`
- VUID-vkCmdResolveImage-srcImageLayout-01400
`srcImageLayout` **must** be `VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL` or `VK_IMAGE_LAYOUT_GENERAL`
- VUID-vkCmdResolveImage-dstImageLayout-00262
`dstImageLayout` **must** specify the layout of the image subresources of `dstImage` specified in `pRegions` at the time this command is executed on a `VkDevice`
- VUID-vkCmdResolveImage-dstImageLayout-01401
`dstImageLayout` **must** be `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL` or `VK_IMAGE_LAYOUT_GENERAL`
- VUID-vkCmdResolveImage-dstImage-02003
The `format features` of `dstImage` **must** contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT`
- VUID-vkCmdResolveImage-srcImage-01386
`srcImage` and `dstImage` **must** have been created with the same image format
- VUID-vkCmdResolveImage-srcSubresource-01709
The `srcSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `srcImage` was created
- VUID-vkCmdResolveImage-dstSubresource-01710
The `dstSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `dstImage` was created
- VUID-vkCmdResolveImage-srcSubresource-01711
`srcSubresource.baseArrayLayer` + `srcSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `srcImage` was created
- VUID-vkCmdResolveImage-dstSubresource-01712
`dstSubresource.baseArrayLayer` + `dstSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `dstImage` was created
- VUID-vkCmdResolveImage-srcImage-04446
If `dstImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`, `srcSubresource.layerCount` **must** be 1
- VUID-vkCmdResolveImage-srcImage-04447
If `dstImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`, `dstSubresource.baseArrayLayer` **must** be 0 and `dstSubresource.layerCount` **must** be 1
- VUID-vkCmdResolveImage-srcOffset-00269
For each element of `pRegions`, `srcOffset.x` and `(extent.width + srcOffset.x)` **must** both be greater than or equal to 0 and less than or equal to the width of the specified `srcSubresource` of `srcImage`

- VUID-vkCmdResolveImage-srcOffset-00270
For each element of `pRegions`, `srcOffset.y` and `(extent.height + srcOffset.y)` **must** both be greater than or equal to `0` and less than or equal to the height of the specified `srcSubresource` of `srcImage`
- VUID-vkCmdResolveImage-srcImage-00271
If `srcImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `srcOffset.y` **must** be `0` and `extent.height` **must** be `1`
- VUID-vkCmdResolveImage-srcOffset-00272
For each element of `pRegions`, `srcOffset.z` and `(extent.depth + srcOffset.z)` **must** both be greater than or equal to `0` and less than or equal to the depth of the specified `srcSubresource` of `srcImage`
- VUID-vkCmdResolveImage-srcImage-00273
If `srcImage` is of type `VK_IMAGE_TYPE_1D` or `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `srcOffset.z` **must** be `0` and `extent.depth` **must** be `1`
- VUID-vkCmdResolveImage-dstOffset-00274
For each element of `pRegions`, `dstOffset.x` and `(extent.width + dstOffset.x)` **must** both be greater than or equal to `0` and less than or equal to the width of the specified `dstSubresource` of `dstImage`
- VUID-vkCmdResolveImage-dstOffset-00275
For each element of `pRegions`, `dstOffset.y` and `(extent.height + dstOffset.y)` **must** both be greater than or equal to `0` and less than or equal to the height of the specified `dstSubresource` of `dstImage`
- VUID-vkCmdResolveImage-dstImage-00276
If `dstImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `dstOffset.y` **must** be `0` and `extent.height` **must** be `1`
- VUID-vkCmdResolveImage-dstOffset-00277
For each element of `pRegions`, `dstOffset.z` and `(extent.depth + dstOffset.z)` **must** both be greater than or equal to `0` and less than or equal to the depth of the specified `dstSubresource` of `dstImage`
- VUID-vkCmdResolveImage-dstImage-00278
If `dstImage` is of type `VK_IMAGE_TYPE_1D` or `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `dstOffset.z` **must** be `0` and `extent.depth` **must** be `1`
- VUID-vkCmdResolveImage-srcImage-06762
`srcImage` **must** have been created with `VK_IMAGE_USAGE_TRANSFER_SRC_BIT` usage flag
- VUID-vkCmdResolveImage-dstImage-06764
`dstImage` **must** have been created with `VK_IMAGE_USAGE_TRANSFER_DST_BIT` usage flag

Valid Usage (Implicit)

- VUID-vkCmdResolveImage-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdResolveImage-srcImage-parameter

`srcImage` **must** be a valid `VkImage` handle

- VUID-vkCmdResolveImage-srcImageLayout-parameter
`srcImageLayout` **must** be a valid `VkImageLayout` value
- VUID-vkCmdResolveImage-dstImage-parameter
`dstImage` **must** be a valid `VkImage` handle
- VUID-vkCmdResolveImage-dstImageLayout-parameter
`dstImageLayout` **must** be a valid `VkImageLayout` value
- VUID-vkCmdResolveImage-pRegions-parameter
`pRegions` **must** be a valid pointer to an array of `regionCount` valid `VkImageResolve` structures
- VUID-vkCmdResolveImage-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdResolveImage-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdResolveImage-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdResolveImage-regionCount-arraylength
`regionCount` **must** be greater than 0
- VUID-vkCmdResolveImage-commonparent
Each of `commandBuffer`, `dstImage`, and `srcImage` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Graphics	Action

The `VkImageResolve` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageResolve {
    VkImageSubresourceLayers    srcSubresource;
    VkOffset3D                  srcOffset;
    VkImageSubresourceLayers    dstSubresource;
    VkOffset3D                  dstOffset;
    VkExtent3D                  extent;
} VkImageResolve;
```

- `srcSubresource` and `dstSubresource` are `VkImageSubresourceLayers` structures specifying the image subresources of the images used for the source and destination image data, respectively. Resolve of depth/stencil images is not supported.
- `srcOffset` and `dstOffset` select the initial `x`, `y`, and `z` offsets in texels of the sub-regions of the source and destination image data.
- `extent` is the size in texels of the source image to resolve in `width`, `height` and `depth`.

Valid Usage

- VUID-VkImageResolve-aspectMask-00266
The `aspectMask` member of `srcSubresource` and `dstSubresource` **must** only contain `VK_IMAGE_ASPECT_COLOR_BIT`
- VUID-VkImageResolve-layerCount-08803
The `layerCount` member of `srcSubresource` and `dstSubresource` **must** match

Valid Usage (Implicit)

- VUID-VkImageResolve-srcSubresource-parameter
`srcSubresource` **must** be a valid `VkImageSubresourceLayers` structure
- VUID-VkImageResolve-dstSubresource-parameter
`dstSubresource` **must** be a valid `VkImageSubresourceLayers` structure

Chapter 20. Drawing Commands

Drawing commands (commands with **Draw** in the name) provoke work in a graphics pipeline. Drawing commands are recorded into a command buffer and when executed by a queue, will produce work which executes according to the bound graphics pipeline. A graphics pipeline **must** be bound to a command buffer before any drawing commands are recorded in that command buffer.

Each draw is made up of zero or more vertices and zero or more instances, which are processed by the device and result in the assembly of primitives. Primitives are assembled according to the `pInputAssemblyState` member of the `VkGraphicsPipelineCreateInfo` structure, which is of type `VkPipelineInputAssemblyStateCreateInfo`:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineInputAssemblyStateCreateInfo {
    VkStructureType           sType;
    const void*               pNext;
    VkPipelineInputAssemblyStateCreateFlags flags;
    VkPrimitiveTopology        topology;
    VkBool32                  primitiveRestartEnable;
} VkPipelineInputAssemblyStateCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `topology` is a `VkPrimitiveTopology` defining the primitive topology, as described below.
- `primitiveRestartEnable` controls whether a special vertex index value is treated as restarting the assembly of primitives. This enable only applies to indexed draws (`vkCmdDrawIndexed`, and `vkCmdDrawIndexedIndirect`), and the special index value is either `0xFFFFFFFF` when the `indexType` parameter of `vkCmdBindIndexBuffer` is equal to `VK_INDEX_TYPE_UINT32`, or `0xFFFF` when `indexType` is equal to `VK_INDEX_TYPE_UINT16`. Primitive restart is not allowed for “list” topologies.

Restarting the assembly of primitives discards the most recent index values if those elements formed an incomplete primitive, and restarts the primitive assembly using the subsequent indices, but only assembling the immediately following element through the end of the originally specified elements. The primitive restart index value comparison is performed before adding the `vertexOffset` value to the index value.

Valid Usage

- VUID-VkPipelineInputAssemblyStateCreateInfo-topology-06252
If `topology` is `VK_PRIMITIVE_TOPOLOGY_POINT_LIST`, `VK_PRIMITIVE_TOPOLOGY_LINE_LIST`, `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST`, `VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY`, or `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY`, `primitiveRestartEnable` **must** be `VK_FALSE`

- VUID-VkPipelineInputAssemblyStateCreateInfo-topology-06253
If `topology` is `VK_PRIMITIVE_TOPOLOGY_PATCH_LIST`, `primitiveRestartEnable` **must** be `VK_FALSE`
- VUID-VkPipelineInputAssemblyStateCreateInfo-topology-00429
If the `geometryShader` feature is not enabled, `topology` **must** not be any of `VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY`,
`VK_PRIMITIVE_TOPOLOGY_LINE_STRIP_WITH_ADJACENCY`,
`VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY` or
`VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP_WITH_ADJACENCY`
- VUID-VkPipelineInputAssemblyStateCreateInfo-topology-00430
If the `tessellationShader` feature is not enabled, `topology` **must** not be `VK_PRIMITIVE_TOPOLOGY_PATCH_LIST`

Valid Usage (Implicit)

- VUID-VkPipelineInputAssemblyStateCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_INPUT_ASSEMBLY_STATE_CREATE_INFO`
- VUID-VkPipelineInputAssemblyStateCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkPipelineInputAssemblyStateCreateInfo-flags-zeroBitmask
`flags` **must** be `0`
- VUID-VkPipelineInputAssemblyStateCreateInfo-topology-parameter
`topology` **must** be a valid `VkPrimitiveTopology` value

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineInputAssemblyStateCreateFlags;
```

`VkPipelineInputAssemblyStateCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

20.1. Primitive Topologies

Primitive topology determines how consecutive vertices are organized into primitives, and determines the type of primitive that is used at the beginning of the graphics pipeline. The effective topology for later stages of the pipeline is altered by tessellation or geometry shading (if either is in use) and depends on the execution modes of those shaders.

The primitive topologies defined by `VkPrimitiveTopology` are:

```
// Provided by VK_VERSION_1_0
typedef enum VkPrimitiveTopology {
    VK_PRIMITIVE_TOPOLOGY_POINT_LIST = 0,
    VK_PRIMITIVE_TOPOLOGY_LINE_LIST = 1,
    VK_PRIMITIVE_TOPOLOGY_LINE_STRIP = 2,
    VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST = 3,
    VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP = 4,
    VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN = 5,
    VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY = 6,
    VK_PRIMITIVE_TOPOLOGY_LINE_STRIP_WITH_ADJACENCY = 7,
    VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY = 8,
    VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP_WITH_ADJACENCY = 9,
    VK_PRIMITIVE_TOPOLOGY_PATCH_LIST = 10,
} VkPrimitiveTopology;
```

- `VK_PRIMITIVE_TOPOLOGY_POINT_LIST` specifies a series of [separate point primitives](#).
- `VK_PRIMITIVE_TOPOLOGY_LINE_LIST` specifies a series of [separate line primitives](#).
- `VK_PRIMITIVE_TOPOLOGY_LINE_STRIP` specifies a series of [connected line primitives](#) with consecutive lines sharing a vertex.
- `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST` specifies a series of [separate triangle primitives](#).
- `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP` specifies a series of [connected triangle primitives](#) with consecutive triangles sharing an edge.
- `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN` specifies a series of [connected triangle primitives](#) with all triangles sharing a common vertex.
- `VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY` specifies a series of [separate line primitives with adjacency](#).
- `VK_PRIMITIVE_TOPOLOGY_LINE_STRIP_WITH_ADJACENCY` specifies a series of [connected line primitives with adjacency](#), with consecutive primitives sharing three vertices.
- `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY` specifies a series of [separate triangle primitives with adjacency](#).
- `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP_WITH_ADJACENCY` specifies [connected triangle primitives with adjacency](#), with consecutive triangles sharing an edge.
- `VK_PRIMITIVE_TOPOLOGY_PATCH_LIST` specifies [separate patch primitives](#).

Each primitive topology, and its construction from a list of vertices, is described in detail below with a supporting diagram, according to the following key:

•	Vertex	A point in 3-dimensional space. Positions chosen within the diagrams are arbitrary and for illustration only.
5	Vertex Number	Sequence position of a vertex within the provided vertex data.
	Provoking Vertex	Provoking vertex within the main primitive. The tail is angled towards the relevant primitive. Used in flat shading .

	Primitive Edge	An edge connecting the points of a main primitive.
	Adjacency Edge	Points connected by these lines do not contribute to a main primitive, and are only accessible in a geometry shader .
	Winding Order	The relative order in which vertices are defined within a primitive, used in the facing determination . This ordering has no specific start or end point.

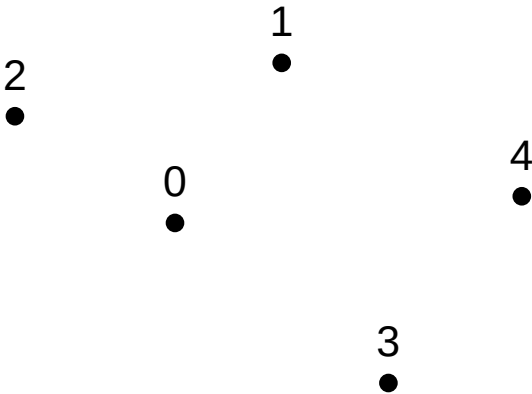
The diagrams are supported with mathematical definitions where the vertices (v) and primitives (p) are numbered starting from 0; v_0 is the first vertex in the provided data and p_0 is the first primitive in the set of primitives defined by the vertices and topology.

20.1.1. Point Lists

When the topology is `VK_PRIMITIVE_TOPOLOGY_POINT_LIST`, each consecutive vertex defines a single point primitive, according to the equation:

$$p_i = \{v_i\}$$

As there is only one vertex, that vertex is the provoking vertex. The number of primitives generated is equal to `vertexCount`.



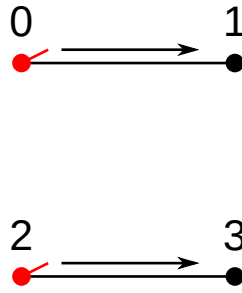
20.1.2. Line Lists

When the primitive topology is `VK_PRIMITIVE_TOPOLOGY_LINE_LIST`, each consecutive pair of vertices defines a single line primitive, according to the equation:

$$p_i = \{v_{2i}, v_{2i+1}\}$$

The number of primitives generated is equal to $\lfloor \text{vertexCount}/2 \rfloor$.

The provoking vertex for p_i is v_{2i} .



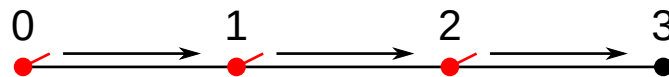
20.1.3. Line Strips

When the primitive topology is `VK_PRIMITIVE_TOPOLOGY_LINE_STRIP`, one line primitive is defined by each vertex and the following vertex, according to the equation:

$$p_i = \{v_i, v_{i+1}\}$$

The number of primitives generated is equal to $\max(0, \text{vertexCount} - 1)$.

The provoking vertex for p_i is v_i .



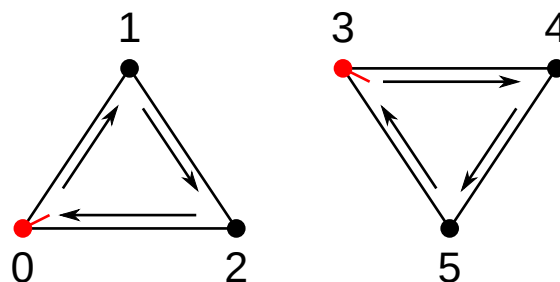
20.1.4. Triangle Lists

When the primitive topology is `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST`, each consecutive set of three vertices defines a single triangle primitive, according to the equation:

$$p_i = \{v_{3i}, v_{3i+1}, v_{3i+2}\}$$

The number of primitives generated is equal to $\lfloor \text{vertexCount} / 3 \rfloor$.

The provoking vertex for p_i is v_{3i} .



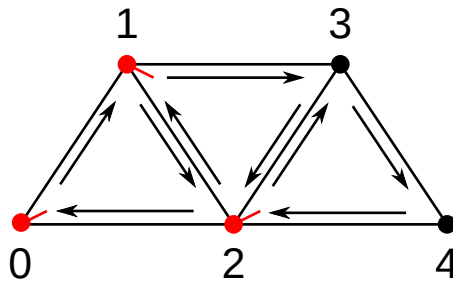
20.1.5. Triangle Strips

When the primitive topology is `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP`, one triangle primitive is defined by each vertex and the two vertices that follow it, according to the equation:

$$p_i = \{v_i, v_{i+(1+i\%2)}, v_{i+(2-i\%2)}\}$$

The number of primitives generated is equal to $\max(0, \text{vertexCount}-2)$.

The provoking vertex for p_i is v_i .



NOTE

The ordering of the vertices in each successive triangle is reversed, so that the winding order is consistent throughout the strip.

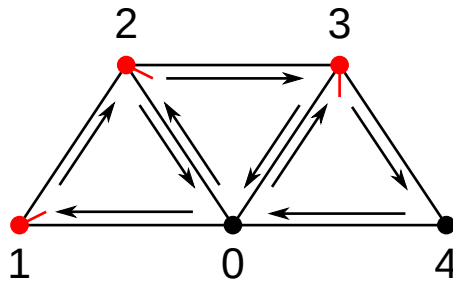
20.1.6. Triangle Fans

When the primitive topology is `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN`, triangle primitives are defined around a shared common vertex, according to the equation:

$$p_i = \{v_{i+1}, v_{i+2}, v_0\}$$

The number of primitives generated is equal to $\max(0, \text{vertexCount}-2)$.

The provoking vertex for p_i is v_{i+1} .



20.1.7. Line Lists With Adjacency

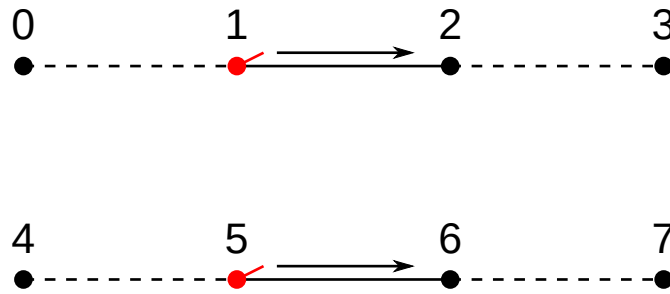
When the primitive topology is `VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY`, each consecutive set of four vertices defines a single line primitive with adjacency, according to the equation:

$$p_i = \{v_{4i}, v_{4i+1}, v_{4i+2}, v_{4i+3}\}$$

A line primitive is described by the second and third vertices of the total primitive, with the remaining two vertices only accessible in a [geometry shader](#).

The number of primitives generated is equal to $\lceil \text{vertexCount}/4 \rceil$.

The provoking vertex for p_i is v_{4i+1} .



20.1.8. Line Strips With Adjacency

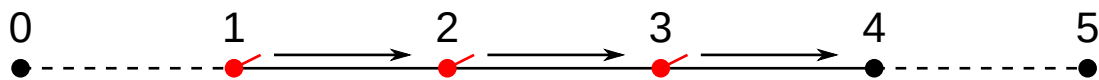
When the primitive topology is `VK_PRIMITIVE_TOPOLOGY_LINE_STRIP_WITH_ADJACENCY`, one line primitive with adjacency is defined by each vertex and the following vertex, according to the equation:

$$p_i = \{v_i, v_{i+1}, v_{i+2}, v_{i+3}\}$$

A line primitive is described by the second and third vertices of the total primitive, with the remaining two vertices only accessible in a [geometry shader](#).

The number of primitives generated is equal to $\max(0, \text{vertexCount} - 3)$.

The provoking vertex for p_i is v_{i+1} .



20.1.9. Triangle Lists With Adjacency

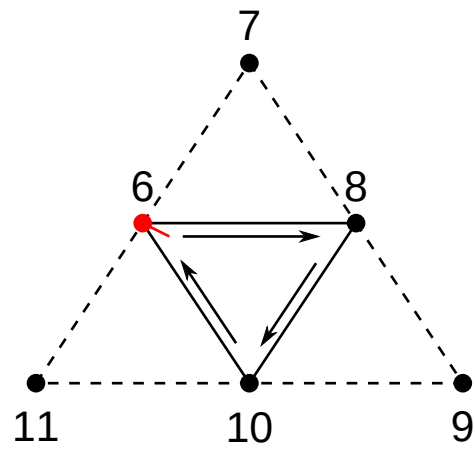
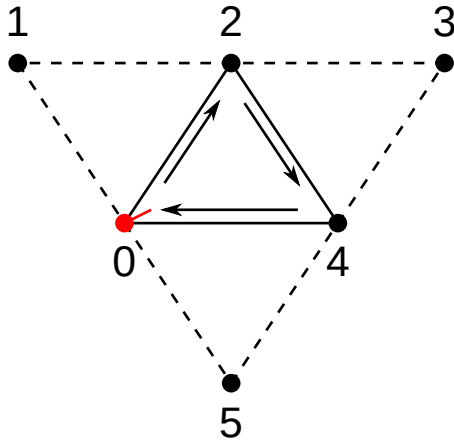
When the primitive topology is `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY`, each consecutive set of six vertices defines a single triangle primitive with adjacency, according to the equations:

$$p_i = \{v_{6i}, v_{6i+1}, v_{6i+2}, v_{6i+3}, v_{6i+4}, v_{6i+5}\}$$

A triangle primitive is described by the first, third, and fifth vertices of the total primitive, with the remaining three vertices only accessible in a [geometry shader](#).

The number of primitives generated is equal to $\lfloor \text{vertexCount} / 6 \rfloor$.

The provoking vertex for p_i is v_{6i} .



20.1.10. Triangle Strips With Adjacency

When the primitive topology is `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP_WITH_ADJACENCY`, one triangle primitive with adjacency is defined by each vertex and the following 5 vertices.

The number of primitives generated, n , is equal to $\lceil \max(0, \text{vertexCount} - 4) / 2 \rceil$.

If $n=1$, the primitive is defined as:

$$p = \{v_0, v_1, v_2, v_5, v_4, v_3\}$$

If $n>1$, the total primitive consists of different vertices according to where it is in the strip:

$$p_i = \{v_{2i}, v_{2i+1}, v_{2i+2}, v_{2i+6}, v_{2i+4}, v_{2i+3}\} \text{ when } i=0$$

$$p_i = \{v_{2i}, v_{2i+3}, v_{2i+4}, v_{2i+6}, v_{2i+2}, v_{2i-2}\} \text{ when } i>0, i<n-1, \text{ and } i\%2=1$$

$$p_i = \{v_{2i}, v_{2i-2}, v_{2i+2}, v_{2i+6}, v_{2i+4}, v_{2i+3}\} \text{ when } i>0, i<n-1, \text{ and } i\%2=0$$

$$p_i = \{v_{2i}, v_{2i+3}, v_{2i+4}, v_{2i+5}, v_{2i+2}, v_{2i-2}\} \text{ when } i=n-1 \text{ and } i\%2=1$$

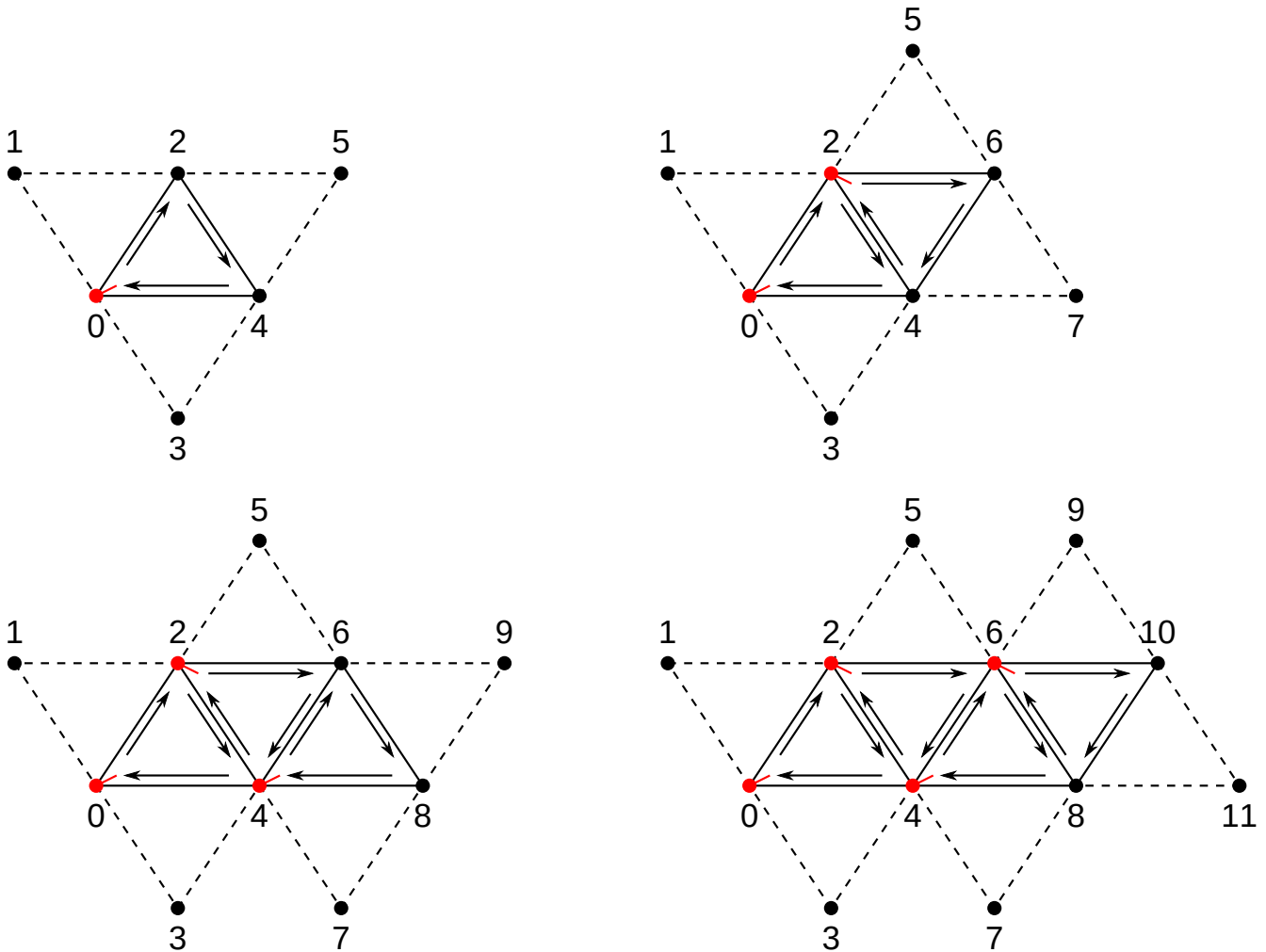
$$p_i = \{v_{2i}, v_{2i-2}, v_{2i+2}, v_{2i+5}, v_{2i+4}, v_{2i+3}\} \text{ when } i=n-1 \text{ and } i\%2=0$$

A triangle primitive is described by the first, third, and fifth vertices of the total primitive in all cases, with the remaining three vertices only accessible in a [geometry shader](#).

NOTE

The ordering of the vertices in each successive triangle is altered so that the winding order is consistent throughout the strip.

The provoking vertex for p_i is always v_{2i} .



20.1.11. Patch Lists

When the primitive topology is `VK_PRIMITIVE_TOPOLOGY_PATCH_LIST`, each consecutive set of m vertices defines a single patch primitive, according to the equation:

$$p_i = \{v_{mi}, v_{mi+1}, \dots, v_{mi+(m-2)}, v_{mi+(m-1)}\}$$

where m is equal to `VkPipelineTessellationStateCreateInfo::patchControlPoints`.

Patch lists are never passed to [vertex post-processing](#), and as such no provoking vertex is defined for patch primitives. The number of primitives generated is equal to $\lceil \text{vertexCount}/m \rceil$.

The vertices comprising a patch have no implied geometry, and are used as inputs to tessellation shaders and the fixed-function tessellator to generate new point, line, or triangle primitives.

20.2. Primitive Order

Primitives generated by [drawing commands](#) progress through the stages of the [graphics pipeline](#) in *primitive order*. Primitive order is initially determined in the following way:

1. Submission order determines the initial ordering
2. For indirect drawing commands, the order in which accessed instances of the

`VkDrawIndirectCommand` are stored in `buffer`, from lower indirect buffer addresses to higher addresses.

3. If a drawing command includes multiple instances, the order in which instances are executed, from lower numbered instances to higher.
4. The order in which primitives are specified by a drawing command:
 - For non-indexed draws, from vertices with a lower numbered `vertexIndex` to a higher numbered `vertexIndex`.
 - For indexed draws, vertices sourced from a lower index buffer addresses to higher addresses.

Within this order implementations further sort primitives:

5. If tessellation shading is active, by an implementation-dependent order of new primitives generated by `tessellation`.
6. If geometry shading is active, by the order new primitives are generated by `geometry shading`.
7. If the `polygon mode` is not `VK_POLYGON_MODE_FILL`, by an implementation-dependent ordering of the new primitives generated within the original primitive.

Primitive order is later used to define `rasterization order`, which determines the order in which fragments output results to a framebuffer.

20.3. Programmable Primitive Shading

Once primitives are assembled, they proceed to the vertex shading stage of the pipeline. If the draw includes multiple instances, then the set of primitives is sent to the vertex shading stage multiple times, once for each instance.

It is implementation-dependent whether vertex shading occurs on vertices that are discarded as part of incomplete primitives, but if it does occur then it operates as if they were vertices in complete primitives and such invocations **can** have side effects.

Vertex shading receives two per-vertex inputs from the primitive assembly stage - the `vertexIndex` and the `instanceIndex`. How these values are generated is defined below, with each command.

Drawing commands fall roughly into two categories:

- Non-indexed drawing commands present a sequential `vertexIndex` to the vertex shader. The sequential index is generated automatically by the device (see [Fixed-Function Vertex Processing](#) for details on both specifying the vertex attributes indexed by `vertexIndex`, as well as binding vertex buffers containing those attributes to a command buffer). These commands are:
 - `vkCmdDraw`
 - `vkCmdDrawIndirect`
- Indexed drawing commands read index values from an *index buffer* and use this to compute the `vertexIndex` value for the vertex shader. These commands are:
 - `vkCmdDrawIndexed`

- [vkCmdDrawIndexedIndirect](#)

To bind an index buffer to a command buffer, call:

```
// Provided by VK_VERSION_1_0
void vkCmdBindIndexBuffer(
    VkCommandBuffer          commandBuffer,
    VkBuffer                 buffer,
    VkDeviceSize             offset,
    VkIndexType              indexType);
```

- **commandBuffer** is the command buffer into which the command is recorded.
- **buffer** is the buffer being bound.
- **offset** is the starting offset in bytes within **buffer** used in index buffer address calculations.
- **indexType** is a [VkIndexType](#) value specifying the size of the indices.

Valid Usage

- VUID-vkCmdBindIndexBuffer-offset-08782
offset must be less than the size of **buffer**
- VUID-vkCmdBindIndexBuffer-offset-08783
The sum of **offset** and the base address of the range of [VkDeviceMemory](#) object that is backing **buffer**, **must** be a multiple of the size of the type indicated by **indexType**
- VUID-vkCmdBindIndexBuffer-buffer-08784
buffer must have been created with the [VK_BUFFER_USAGE_INDEX_BUFFER_BIT](#) flag
- VUID-vkCmdBindIndexBuffer-buffer-08785
If **buffer** is non-sparse then it **must** be bound completely and contiguously to a single [VkDeviceMemory](#) object
- VUID-vkCmdBindIndexBuffer-None-09493
buffer must not be [VK_NULL_HANDLE](#)

Valid Usage (Implicit)

- VUID-vkCmdBindIndexBuffer-commandBuffer-parameter
commandBuffer must be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdBindIndexBuffer-buffer-parameter
If **buffer** is not [VK_NULL_HANDLE](#), **buffer must** be a valid [VkBuffer](#) handle
- VUID-vkCmdBindIndexBuffer-indexType-parameter
indexType must be a valid [VkIndexType](#) value
- VUID-vkCmdBindIndexBuffer-commandBuffer-recording
commandBuffer must be in the [recording state](#)
- VUID-vkCmdBindIndexBuffer-commandBuffer-cmdpool

The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations

- VUID-vkCmdBindIndexBuffer-commonparent
Both of `buffer`, and `commandBuffer` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Graphics	State

Possible values of `vkCmdBindIndexBuffer::indexType`, specifying the size of indices, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkIndexType {
    VK_INDEX_TYPE_UINT16 = 0,
    VK_INDEX_TYPE_UINT32 = 1,
} VkIndexType;
```

- `VK_INDEX_TYPE_UINT16` specifies that indices are 16-bit unsigned integer values.
- `VK_INDEX_TYPE_UINT32` specifies that indices are 32-bit unsigned integer values.

The parameters for each drawing command are specified directly in the command or read from buffer memory, depending on the command. Drawing commands that source their parameters from buffer memory are known as *indirect* drawing commands.

All drawing commands interact with the `robustBufferAccess` feature.

To record a non-indexed draw, call:

```
// Provided by VK_VERSION_1_0
void vkCmdDraw(
    VkCommandBuffer          commandBuffer,
    uint32_t                 vertexCount,
    uint32_t                 instanceCount,
    uint32_t                 firstVertex,
    uint32_t                 firstInstance);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `vertexCount` is the number of vertices to draw.
- `instanceCount` is the number of instances to draw.
- `firstVertex` is the index of the first vertex to draw.
- `firstInstance` is the instance ID of the first instance to draw.

When the command is executed, primitives are assembled using the current primitive topology and `vertexCount` consecutive vertex indices with the first `vertexIndex` value equal to `firstVertex`. The primitives are drawn `instanceCount` times with `instanceIndex` starting with `firstInstance` and increasing sequentially for each instance. The assembled primitives execute the bound graphics pipeline.

Valid Usage

- VUID-vkCmdDraw-magFilter-04553
If a `VkSampler` created with `magFilter` or `minFilter` equal to `VK_FILTER_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdDraw-mipmapMode-04770
If a `VkSampler` created with `mipmapMode` equal to `VK_SAMPLER_MIPMAP_MODE_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdDraw-unnormalizedCoordinates-09635
If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `levelCount` and `layerCount` **must** be 1
- VUID-vkCmdDraw-unnormalizedCoordinates-09636
If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `viewType` **must** be `VK_IMAGE_VIEW_TYPE_1D` or `VK_IMAGE_VIEW_TYPE_2D`
- VUID-vkCmdDraw-aspectMask-06478
If a `VkImageView` is sampled with `depth comparison`, the image view **must** have been created with an `aspectMask` that contains `VK_IMAGE_ASPECT_DEPTH_BIT`

- VUID-vkCmdDraw-None-02691

If a [VkImageView](#) is accessed using atomic operations as a result of this command, then the image view's [format features](#) **must** contain [VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT](#)

- VUID-vkCmdDraw-None-07888

If a [VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER](#) descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's [format features](#) **must** contain [VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT](#)

- VUID-vkCmdDraw-None-08600

For each set n that is statically used by a [bound shader](#), a descriptor set **must** have been bound to n at the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for set n , with the [VkPipelineLayout](#) used to create the current [VkPipeline](#), as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDraw-None-08601

For each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#), as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDraw-None-10068

For each array of resources that is used by a [bound shader](#), the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command

- VUID-vkCmdDraw-None-08114

Descriptors in each bound descriptor set, specified via [vkCmdBindDescriptorSets](#), **must** be valid as described by [descriptor validity](#) if they are statically used by a [bound shader](#)

- VUID-vkCmdDraw-None-08606

A valid pipeline **must** be bound to the pipeline bind point used by this command

- VUID-vkCmdDraw-None-08608

There **must** not have been any calls to dynamic state setting commands for any state specified statically in the [VkPipeline](#) object bound to the pipeline bind point used by this command, since that pipeline was bound

- VUID-vkCmdDraw-None-08609

If the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used to sample from any [VkImage](#) with a [VkImageView](#) of the type [VK_IMAGE_VIEW_TYPE_3D](#), [VK_IMAGE_VIEW_TYPE_CUBE](#), [VK_IMAGE_VIEW_TYPE_1D_ARRAY](#), [VK_IMAGE_VIEW_TYPE_2D_ARRAY](#) or [VK_IMAGE_VIEW_TYPE_CUBE_ARRAY](#), in any shader stage

- VUID-vkCmdDraw-None-08610

If the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions with [ImplicitLod](#), [Dref](#) or [Proj](#) in their name, in any shader stage

- VUID-vkCmdDraw-None-08611

If the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a

[VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions that includes a LOD bias or any offset values, in any shader stage

- VUID-vkCmdDraw-uniformBuffers-06935

If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a uniform buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDraw-storageBuffers-06936

If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a storage buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDraw-viewType-07752

If a [VkImageView](#) is accessed as a result of this command, then the image view's [viewType](#) **must** match the [Dim](#) operand of the [OpTypeImage](#) as described in [Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types](#)

- VUID-vkCmdDraw-format-07753

If a [VkImageView](#) is accessed as a result of this command, then the [numeric type](#) of the image view's [format](#) and the [Sampled Type](#) operand of the [OpTypeImage](#) **must** match

- VUID-vkCmdDraw-OpImageWrite-08795

If a [VkImageView](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have at least as many components as the image view's format

- VUID-vkCmdDraw-OpImageWrite-04469

If a [VkBufferView](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have at least as many components as the buffer view's format

- VUID-vkCmdDraw-None-07288

Any shader invocation executed by this command **must** [terminate](#)

- VUID-vkCmdDraw-None-09600

If a descriptor with type equal to any of [VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE](#), [VK_DESCRIPTOR_TYPE_STORAGE_IMAGE](#), or [VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT](#) is accessed as a result of this command, the image subresource identified by that descriptor **must** be in the image layout identified when the descriptor was written

- VUID-vkCmdDraw-renderPass-02684

The current render pass **must** be [compatible](#) with the [renderPass](#) member of the [VkGraphicsPipelineCreateInfo](#) structure specified when creating the [VkPipeline](#) bound to [VK_PIPELINE_BIND_POINT_GRAPHICS](#)

- VUID-vkCmdDraw-subpass-02685

The subpass index of the current render pass **must** be equal to the [subpass](#) member of the [VkGraphicsPipelineCreateInfo](#) structure specified when creating the [VkPipeline](#) bound to [VK_PIPELINE_BIND_POINT_GRAPHICS](#)

- VUID-vkCmdDraw-None-07748
If any shader statically accesses an input attachment, a valid descriptor **must** be bound to the pipeline via a descriptor set
- VUID-vkCmdDraw-OpTypeImage-07468
If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass
- VUID-vkCmdDraw-None-07469
Input attachment views accessed in a subpass **must** be created with the same `VkFormat` as the corresponding subpass definition, and be created with a `VkImageView` that is compatible with the attachment referenced by the subpass' `pInputAttachments` [`InputAttachmentIndex`] in the currently bound `VkFramebuffer` as specified by [Fragment Input Attachment Compatibility](#)
- VUID-vkCmdDraw-None-06537
Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command
- VUID-vkCmdDraw-None-09000
If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass,

it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDraw-None-09001
If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass,

it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDraw-None-09002
If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass,

it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDraw-None-06539
If any previously recorded command in the current subpass accessed an image subresource used as an attachment in this subpass in any way other than as an attachment, this command **must** not write to that image subresource as an attachment
- VUID-vkCmdDraw-None-06886
If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, `depth writes` **must** be disabled
- VUID-vkCmdDraw-None-06887
If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, `all stencil ops` **must** be `VK_STENCIL_OP_KEEP`
- VUID-vkCmdDraw-None-07831

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VIEWPORT` dynamic state enabled then `vkCmdSetViewport` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07832

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SCISSOR` dynamic state enabled then `vkCmdSetScissor` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07833

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_WIDTH` dynamic state enabled then `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07834

If a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_BIAS` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `depthBiasEnable` is `VK_TRUE`, then `vkCmdSetDepthBounds` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07835

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_BLEND_CONSTANTS` dynamic state enabled then `vkCmdSetBlendConstants` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07836

If a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_BOUNDS` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `depthBoundsTestEnable` is `VK_TRUE`, then `vkCmdSetDepthBounds` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07837

If a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilCompareMask` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07838

If a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilWriteMask` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07839

If a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_REFERENCE` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilReference` **must** have been called and not subsequently `invalidated`

in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-blendEnable-04727

If rasterization is not disabled in the bound graphics pipeline, then for each color attachment in the subpass, if the corresponding image view's [format features](#) do not contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`, then the `blendEnable` member of the corresponding element of the `pAttachments` member of `pColorBlendState` **must** be `VK_FALSE`

- VUID-vkCmdDraw-multisampledRenderToSingleSampled-07284

If rasterization is not disabled in the bound graphics pipeline,

then `rasterizationSamples` for the currently bound graphics pipeline **must** be the same as the current subpass color and/or depth/stencil attachments

- VUID-vkCmdDraw-maxFragmentDualSrcAttachments-09239

If [blending](#) is enabled for any attachment where either the source or destination blend factors for that attachment [use the secondary color input](#), the maximum value of `Location` for any output attachment [statically used](#) in the `Fragment Execution Model` executed by this command **must** be less than `maxFragmentDualSrcAttachments`

- VUID-vkCmdDraw-None-04007

All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or `VK_NULL_HANDLE` buffers bound

- VUID-vkCmdDraw-None-04008

If the `nullDescriptor` feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be `VK_NULL_HANDLE`

- VUID-vkCmdDraw-None-02721

If `robustBufferAccess` is not enabled, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in [Vertex Input Description](#)

Valid Usage (Implicit)

- VUID-vkCmdDraw-commandBuffer-parameter

`commandBuffer` **must** be a valid `VkCommandBuffer` handle

- VUID-vkCmdDraw-commandBuffer-recording

`commandBuffer` **must** be in the [recording state](#)

- VUID-vkCmdDraw-commandBuffer-cmdpool

The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations

- VUID-vkCmdDraw-renderpass

This command **must** only be called inside of a render pass instance

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Graphics	Action

To record an indexed draw, call:

```
// Provided by VK_VERSION_1_0
void vkCmdDrawIndexed(
    VkCommandBuffer          commandBuffer,
    uint32_t                 indexCount,
    uint32_t                 instanceCount,
    uint32_t                 firstIndex,
    int32_t                  vertexOffset,
    uint32_t                 firstInstance);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `indexCount` is the number of vertices to draw.
- `instanceCount` is the number of instances to draw.
- `firstIndex` is the base index within the index buffer.
- `vertexOffset` is the value added to the vertex index before indexing into the vertex buffer.
- `firstInstance` is the instance ID of the first instance to draw.

When the command is executed, primitives are assembled using the current primitive topology and `indexCount` vertices whose indices are retrieved from the index buffer. The index buffer is treated as an array of tightly packed unsigned integers of size defined by the `vkCmdBindIndexBuffer::indexType` parameter with which the buffer was bound.

The first vertex index is at an offset of $\text{firstIndex} \times \text{indexSize} + \text{offset}$ within the bound index buffer, where `offset` is the offset specified by `vkCmdBindIndexBuffer` and `indexSize` is the byte size of the type specified by `indexType`. Subsequent index values are retrieved from consecutive locations in the index buffer. Indices are first compared to the primitive restart value, then zero extended to 32 bits (if the `indexType` is `VK_INDEX_TYPE_UINT16`) and have `vertexOffset` added to them, before being supplied as the `vertexIndex` value.

The primitives are drawn `instanceCount` times with `instanceIndex` starting with `firstInstance` and increasing sequentially for each instance. The assembled primitives execute the bound graphics pipeline.

Valid Usage

- VUID-vkCmdDrawIndexed-magFilter-04553
If a `VkSampler` created with `magFilter` or `minFilter` equal to `VK_FILTER_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdDrawIndexed-mipmapMode-04770
If a `VkSampler` created with `mipmapMode` equal to `VK_SAMPLER_MIPMAP_MODE_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdDrawIndexed-unnormalizedCoordinates-09635
If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `levelCount` and `layerCount` **must** be 1
- VUID-vkCmdDrawIndexed-unnormalizedCoordinates-09636
If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `viewType` **must** be `VK_IMAGE_VIEW_TYPE_1D` or `VK_IMAGE_VIEW_TYPE_2D`
- VUID-vkCmdDrawIndexed-aspectMask-06478
If a `VkImageView` is sampled with `depth comparison`, the image view **must** have been created with an `aspectMask` that contains `VK_IMAGE_ASPECT_DEPTH_BIT`
- VUID-vkCmdDrawIndexed-None-02691
If a `VkImageView` is accessed using atomic operations as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT`
- VUID-vkCmdDrawIndexed-None-07888
If a `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT`
- VUID-vkCmdDrawIndexed-None-08600
For each set n that is statically used by a `bound shader`, a descriptor set **must** have been bound to n at the same pipeline bind point, with a `VkPipelineLayout` that is compatible for set n , with the `VkPipelineLayout` used to create the current `VkPipeline`, as described in [Pipeline Layout Compatibility](#)
- VUID-vkCmdDrawIndexed-None-08601
For each push constant that is statically used by a `bound shader`, a push constant value **must** have been set for the same pipeline bind point, with a `VkPipelineLayout` that is compatible for push constants, with the `VkPipelineLayout` used to create the current

[VkPipeline](#) , as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDrawIndexed-None-10068

For each array of resources that is used by a [bound shader](#), the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command

- VUID-vkCmdDrawIndexed-None-08114

Descriptors in each bound descriptor set, specified via [vkCmdBindDescriptorSets](#), **must** be valid as described by [descriptor validity](#) if they are statically used by a [bound shader](#)

- VUID-vkCmdDrawIndexed-None-08606

A valid pipeline **must** be bound to the pipeline bind point used by this command

- VUID-vkCmdDrawIndexed-None-08608

There **must** not have been any calls to dynamic state setting commands for any state specified statically in the [VkPipeline](#) object bound to the pipeline bind point used by this command, since that pipeline was bound

- VUID-vkCmdDrawIndexed-None-08609

If the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used to sample from any [VkImage](#) with a [VkImageView](#) of the type [VK_IMAGE_VIEW_TYPE_3D](#), [VK_IMAGE_VIEW_TYPE_CUBE](#), [VK_IMAGE_VIEW_TYPE_1D_ARRAY](#), [VK_IMAGE_VIEW_TYPE_2D_ARRAY](#) or [VK_IMAGE_VIEW_TYPE_CUBE_ARRAY](#), in any shader stage

- VUID-vkCmdDrawIndexed-None-08610

If the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions with [ImplicitLod](#), [Dref](#) or [Proj](#) in their name, in any shader stage

- VUID-vkCmdDrawIndexed-None-08611

If the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions that includes a LOD bias or any offset values, in any shader stage

- VUID-vkCmdDrawIndexed-uniformBuffers-06935

If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a uniform buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndexed-storageBuffers-06936

If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a storage buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndexed-viewType-07752

If a [VkImageView](#) is accessed as a result of this command, then the image view's [viewType](#) **must** match the [Dim](#) operand of the [OpTypeImage](#) as described in [Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types](#)

- VUID-vkCmdDrawIndexed-format-07753
If a `VkImageView` is accessed as a result of this command, then the `numeric type` of the image view's `format` and the `Sampled Type` operand of the `OpTypeImage` **must** match
- VUID-vkCmdDrawIndexed-OpImageWrite-08795
If a `VkImageView` is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDrawIndexed-OpImageWrite-04469
If a `VkBufferView` is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDrawIndexed-None-07288
Any shader invocation executed by this command **must** `terminate`
- VUID-vkCmdDrawIndexed-None-09600
If a descriptor with type equal to any of `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, or `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT` is accessed as a result of this command, the image subresource identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDrawIndexed-renderPass-02684
The current render pass **must** be `compatible` with the `renderPass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to `VK_PIPELINE_BIND_POINT_GRAPHICS`
- VUID-vkCmdDrawIndexed-subpass-02685
The subpass index of the current render pass **must** be equal to the `subpass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to `VK_PIPELINE_BIND_POINT_GRAPHICS`
- VUID-vkCmdDrawIndexed-None-07748
If any shader statically accesses an input attachment, a valid descriptor **must** be bound to the pipeline via a descriptor set
- VUID-vkCmdDrawIndexed-OpTypeImage-07468
If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass
- VUID-vkCmdDrawIndexed-None-07469
Input attachment views accessed in a subpass **must** be created with the same `VkFormat` as the corresponding subpass definition, and be created with a `VkImageView` that is compatible with the attachment referenced by the subpass' `pInputAttachments` [`InputAttachmentIndex`] in the currently bound `VkFramebuffer` as specified by `Fragment Input Attachment Compatibility`
- VUID-vkCmdDrawIndexed-None-06537
Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command
- VUID-vkCmdDrawIndexed-None-09000

If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass,

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndexed-None-09001

If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass,

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndexed-None-09002

If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass,

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndexed-None-06539

If any previously recorded command in the current subpass accessed an image subresource used as an attachment in this subpass in any way other than as an attachment, this command **must** not write to that image subresource as an attachment

- VUID-vkCmdDrawIndexed-None-06886

If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, **depth writes must** be disabled

- VUID-vkCmdDrawIndexed-None-06887

If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back **writeMask** are not zero, and stencil test is enabled, **all stencil ops must** be **VK_STENCIL_OP_KEEP**

- VUID-vkCmdDrawIndexed-None-07831

If the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_VIEWPORT** dynamic state enabled then **vkCmdSetViewport must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07832

If the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_SCISSOR** dynamic state enabled then **vkCmdSetScissor must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07833

If the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_LINE_WIDTH** dynamic state enabled then **vkCmdSetLineWidth must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07834

If a graphics pipeline is bound which was created with the **VK_DYNAMIC_STATE_DEPTH_BIAS** dynamic state enabled, the **current value** of **rasterizerDiscardEnable** is **VK_FALSE**, and the **current value** of **depthBiasEnable** is **VK_TRUE**, then **vkCmdSetDepthBounds must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07835

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_BLEND_CONSTANTS` dynamic state enabled then `vkCmdSetBlendConstants` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07836

If a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_BOUNDS` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `depthBoundsTestEnable` is `VK_TRUE`, then `vkCmdSetDepthBounds` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07837

If a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilCompareMask` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07838

If a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilWriteMask` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07839

If a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_REFERENCE` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilReference` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-blendEnable-04727

If rasterization is not disabled in the bound graphics pipeline, then for each color attachment in the subpass, if the corresponding image view's `format features` do not contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`, then the `blendEnable` member of the corresponding element of the `pAttachments` member of `pColorBlendState` **must** be `VK_FALSE`

- VUID-vkCmdDrawIndexed-multisampledRenderToSingleSampled-07284

If rasterization is not disabled in the bound graphics pipeline,

then `rasterizationSamples` for the currently bound graphics pipeline **must** be the same as the current subpass color and/or depth/stencil attachments

- VUID-vkCmdDrawIndexed-maxFragmentDualSrcAttachments-09239

If `blending` is enabled for any attachment where either the source or destination blend factors for that attachment `use the secondary color input`, the maximum value of `Location` for any output attachment `statically used` in the `Fragment Execution Model` executed by this command **must** be less than `maxFragmentDualSrcAttachments`

- VUID-vkCmdDrawIndexed-None-04007
All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or [VK_NULL_HANDLE](#) buffers bound
- VUID-vkCmdDrawIndexed-None-04008
If the [nullDescriptor](#) feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be [VK_NULL_HANDLE](#)
- VUID-vkCmdDrawIndexed-None-02721
If [robustBufferAccess](#) is not enabled, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in [Vertex Input Description](#)
- VUID-vkCmdDrawIndexed-None-07312
A valid index buffer **must** be bound
- VUID-vkCmdDrawIndexed-robustBufferAccess2-08798
If [robustBufferAccess2](#) is not enabled, $(\text{indexSize} \times (\text{firstIndex} + \text{indexCount}) + \text{offset})$ **must** be less than or equal to the size of the bound index buffer, with [indexSize](#) being based on the type specified by [indexType](#), where the index buffer, [indexType](#), and [offset](#) are specified via [vkCmdBindIndexBuffer](#)

Valid Usage (Implicit)

- VUID-vkCmdDrawIndexed-commandBuffer-parameter
[commandBuffer](#) **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdDrawIndexed-commandBuffer-recording
[commandBuffer](#) **must** be in the [recording state](#)
- VUID-vkCmdDrawIndexed-commandBuffer-cmdpool
The [VkCommandPool](#) that [commandBuffer](#) was allocated from **must** support graphics operations
- VUID-vkCmdDrawIndexed-renderpass
This command **must** only be called inside of a render pass instance

Host Synchronization

- Host access to [commandBuffer](#) **must** be externally synchronized
- Host access to the [VkCommandPool](#) that [commandBuffer](#) was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Graphics	Action

To record a non-indexed indirect drawing command, call:

```
// Provided by VK_VERSION_1_0
void vkCmdDrawIndirect(
    VkCommandBuffer          commandBuffer,
    VkBuffer                 buffer,
    VkDeviceSize             offset,
    uint32_t                 drawCount,
    uint32_t                 stride);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `buffer` is the buffer containing draw parameters.
- `offset` is the byte offset into `buffer` where parameters begin.
- `drawCount` is the number of draws to execute, and **can** be zero.
- `stride` is the byte stride between successive sets of draw parameters.

`vkCmdDrawIndirect` behaves similarly to `vkCmdDraw` except that the parameters are read by the device from a buffer during execution. `drawCount` draws are executed by the command, with parameters taken from `buffer` starting at `offset` and increasing by `stride` bytes for each successive draw. The parameters of each draw are encoded in an array of `VkDrawIndirectCommand` structures. If `drawCount` is less than or equal to one, `stride` is ignored.

Valid Usage

- VUID-vkCmdDrawIndirect-magFilter-04553
If a `VkSampler` created with `magFilter` or `minFilter` equal to `VK_FILTER_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdDrawIndirect-mipmapMode-04770
If a `VkSampler` created with `mipmapMode` equal to `VK_SAMPLER_MIPMAP_MODE_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdDrawIndirect-unnormalizedCoordinates-09635
If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a

[VkImageView](#) as a result of this command, then the image view's [levelCount](#) and [layerCount](#) **must** be 1

- VUID-vkCmdDrawIndirect-unnormalizedCoordinates-09636

If a [VkSampler](#) created with [unnormalizedCoordinates](#) equal to [VK_TRUE](#) is used to sample a [VkImageView](#) as a result of this command, then the image view's [viewType](#) **must** be [VK_IMAGE_VIEW_TYPE_1D](#) or [VK_IMAGE_VIEW_TYPE_2D](#)

- VUID-vkCmdDrawIndirect-aspectMask-06478

If a [VkImageView](#) is sampled with [depth comparison](#), the image view **must** have been created with an [aspectMask](#) that contains [VK_IMAGE_ASPECT_DEPTH_BIT](#)

- VUID-vkCmdDrawIndirect-None-02691

If a [VkImageView](#) is accessed using atomic operations as a result of this command, then the image view's [format](#) [features](#) **must** contain [VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT](#)

- VUID-vkCmdDrawIndirect-None-07888

If a [VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER](#) descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's [format](#) [features](#) **must** contain [VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT](#)

- VUID-vkCmdDrawIndirect-None-08600

For each set n that is statically used by a [bound shader](#), a descriptor set **must** have been bound to n at the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for set n , with the [VkPipelineLayout](#) used to create the current [VkPipeline](#), as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDrawIndirect-None-08601

For each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#), as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDrawIndirect-None-10068

For each array of resources that is used by a [bound shader](#), the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command

- VUID-vkCmdDrawIndirect-None-08114

Descriptors in each bound descriptor set, specified via [vkCmdBindDescriptorSets](#), **must** be valid as described by [descriptor validity](#) if they are statically used by a [bound shader](#)

- VUID-vkCmdDrawIndirect-None-08606

A valid pipeline **must** be bound to the pipeline bind point used by this command

- VUID-vkCmdDrawIndirect-None-08608

There **must** not have been any calls to dynamic state setting commands for any state specified statically in the [VkPipeline](#) object bound to the pipeline bind point used by this command, since that pipeline was bound

- VUID-vkCmdDrawIndirect-None-08609

If the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used to

sample from any [VkImage](#) with a [VkImageView](#) of the type [VK_IMAGE_VIEW_TYPE_3D](#), [VK_IMAGE_VIEW_TYPE_CUBE](#), [VK_IMAGE_VIEW_TYPE_1D_ARRAY](#), [VK_IMAGE_VIEW_TYPE_2D_ARRAY](#) or [VK_IMAGE_VIEW_TYPE_CUBE_ARRAY](#), in any shader stage

- VUID-vkCmdDrawIndirect-None-08610

If the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions with [ImplicitLod](#), [Dref](#) or [Proj](#) in their name, in any shader stage

- VUID-vkCmdDrawIndirect-None-08611

If the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions that includes a LOD bias or any offset values, in any shader stage

- VUID-vkCmdDrawIndirect-uniformBuffers-06935

If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a uniform buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndirect-storageBuffers-06936

If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a storage buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndirect-viewType-07752

If a [VkImageView](#) is accessed as a result of this command, then the image view's [viewType](#) **must** match the [Dim](#) operand of the [OpTypeImage](#) as described in [Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types](#)

- VUID-vkCmdDrawIndirect-format-07753

If a [VkImageView](#) is accessed as a result of this command, then the [numeric type](#) of the image view's [format](#) and the [Sampled Type](#) operand of the [OpTypeImage](#) **must** match

- VUID-vkCmdDrawIndirect-OpImageWrite-08795

If a [VkImageView](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have at least as many components as the image view's format

- VUID-vkCmdDrawIndirect-OpImageWrite-04469

If a [VkBufferView](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have at least as many components as the buffer view's format

- VUID-vkCmdDrawIndirect-None-07288

Any shader invocation executed by this command **must** [terminate](#)

- VUID-vkCmdDrawIndirect-None-09600

If a descriptor with type equal to any of [VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE](#), [VK_DESCRIPTOR_TYPE_STORAGE_IMAGE](#), or [VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT](#) is accessed as a result of this command, the image subresource identified by that descriptor **must** be in

the image layout identified when the descriptor was written

- VUID-vkCmdDrawIndirect-renderPass-02684

The current render pass **must** be [compatible](#) with the `renderPass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to `VK_PIPELINE_BIND_POINT_GRAPHICS`

- VUID-vkCmdDrawIndirect-subpass-02685

The subpass index of the current render pass **must** be equal to the `subpass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to `VK_PIPELINE_BIND_POINT_GRAPHICS`

- VUID-vkCmdDrawIndirect-None-07748

If any shader statically accesses an input attachment, a valid descriptor **must** be bound to the pipeline via a descriptor set

- VUID-vkCmdDrawIndirect-OpTypeImage-07468

If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass

- VUID-vkCmdDrawIndirect-None-07469

Input attachment views accessed in a subpass **must** be created with the same `VkFormat` as the corresponding subpass definition, and be created with a `VkImageView` that is compatible with the attachment referenced by the subpass' `pInputAttachments` [`InputAttachmentIndex`] in the currently bound `VkFramebuffer` as specified by [Fragment Input Attachment Compatibility](#)

- VUID-vkCmdDrawIndirect-None-06537

Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command

- VUID-vkCmdDrawIndirect-None-09000

If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass,

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndirect-None-09001

If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass,

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndirect-None-09002

If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass,

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndirect-None-06539

If any previously recorded command in the current subpass accessed an image subresource used as an attachment in this subpass in any way other than as an

attachment, this command **must** not write to that image subresource as an attachment

- VUID-vkCmdDrawIndirect-None-06886

If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, **depth writes must** be disabled

- VUID-vkCmdDrawIndirect-None-06887

If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back **writeMask** are not zero, and stencil test is enabled, **all stencil ops must** be **VK_STENCIL_OP_KEEP**

- VUID-vkCmdDrawIndirect-None-07831

If the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_VIEWPORT** dynamic state enabled then **vkCmdSetViewport must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07832

If the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_SCISSOR** dynamic state enabled then **vkCmdSetScissor must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07833

If the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_LINE_WIDTH** dynamic state enabled then **vkCmdSetLineWidth must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07834

If a graphics pipeline is bound which was created with the **VK_DYNAMIC_STATE_DEPTH_BIAS** dynamic state enabled, the **current value** of **rasterizerDiscardEnable** is **VK_FALSE**, and the **current value** of **depthBiasEnable** is **VK_TRUE**, then **vkCmdSetDepthBounds must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07835

If the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_BLEND_CONSTANTS** dynamic state enabled then **vkCmdSetBlendConstants must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07836

If a graphics pipeline is bound which was created with the **VK_DYNAMIC_STATE_DEPTH_BOUNDS** dynamic state enabled, the **current value** of **rasterizerDiscardEnable** is **VK_FALSE**, and the **current value** of **depthBoundsTestEnable** is **VK_TRUE**, then **vkCmdSetDepthBounds must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07837

If a graphics pipeline is bound which was created with the **VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK** dynamic state enabled, the **current value** of **rasterizerDiscardEnable** is **VK_FALSE**, and the **current value** of **stencilTestEnable** is **VK_TRUE**, then **vkCmdSetStencilCompareMask must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07838

If a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilWriteMask` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07839

If a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_REFERENCE` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilReference` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-blendEnable-04727

If rasterization is not disabled in the bound graphics pipeline, then for each color attachment in the subpass, if the corresponding image view's `format features` do not contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`, then the `blendEnable` member of the corresponding element of the `pAttachments` member of `pColorBlendState` **must** be `VK_FALSE`

- VUID-vkCmdDrawIndirect-multisampledRenderToSingleSampled-07284

If rasterization is not disabled in the bound graphics pipeline,

then `rasterizationSamples` for the currently bound graphics pipeline **must** be the same as the current subpass color and/or depth/stencil attachments

- VUID-vkCmdDrawIndirect-maxFragmentDualSrcAttachments-09239

If `blending` is enabled for any attachment where either the source or destination blend factors for that attachment `use the secondary color input`, the maximum value of `Location` for any output attachment `statically used` in the `Fragment Execution Model` executed by this command **must** be less than `maxFragmentDualSrcAttachments`

- VUID-vkCmdDrawIndirect-None-04007

All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or `VK_NULL_HANDLE` buffers bound

- VUID-vkCmdDrawIndirect-None-04008

If the `nullDescriptor` feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be `VK_NULL_HANDLE`

- VUID-vkCmdDrawIndirect-None-02721

If `robustBufferAccess` is not enabled, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in `Vertex Input Description`

- VUID-vkCmdDrawIndirect-buffer-02708

If `buffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-vkCmdDrawIndirect-buffer-02709

buffer **must** have been created with the `VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT` bit set

- VUID-vkCmdDrawIndirect-offset-02710

offset **must** be a multiple of 4

- VUID-vkCmdDrawIndirect-drawCount-02718

If the `multiDrawIndirect` feature is not enabled, **drawCount** **must** be 0 or 1

- VUID-vkCmdDrawIndirect-drawCount-02719

drawCount **must** be less than or equal to `VkPhysicalDeviceLimits::maxDrawIndirectCount`

- VUID-vkCmdDrawIndirect-drawCount-00476

If **drawCount** is greater than 1, **stride** **must** be a multiple of 4 and **must** be greater than or equal to `sizeof(VkDrawIndirectCommand)`

- VUID-vkCmdDrawIndirect-drawCount-00487

If **drawCount** is equal to 1, $(\text{offset} + \text{sizeof}(\text{VkDrawIndirectCommand}))$ **must** be less than or equal to the size of **buffer**

- VUID-vkCmdDrawIndirect-drawCount-00488

If **drawCount** is greater than 1, $(\text{stride} \times (\text{drawCount} - 1) + \text{offset} + \text{sizeof}(\text{VkDrawIndirectCommand}))$ **must** be less than or equal to the size of **buffer**

Valid Usage (Implicit)

- VUID-vkCmdDrawIndirect-commandBuffer-parameter

commandBuffer **must** be a valid `VkCommandBuffer` handle

- VUID-vkCmdDrawIndirect-buffer-parameter

buffer **must** be a valid `VkBuffer` handle

- VUID-vkCmdDrawIndirect-commandBuffer-recording

commandBuffer **must** be in the `recording state`

- VUID-vkCmdDrawIndirect-commandBuffer-cmdpool

The `VkCommandPool` that **commandBuffer** was allocated from **must** support graphics operations

- VUID-vkCmdDrawIndirect-renderpass

This command **must** only be called inside of a render pass instance

- VUID-vkCmdDrawIndirect-commonparent

Both of **buffer**, and **commandBuffer** **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized

- Host access to the `VkCommandPool` that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Graphics	Action

The `VkDrawIndirectCommand` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDrawIndirectCommand {
    uint32_t    vertexCount;
    uint32_t    instanceCount;
    uint32_t    firstVertex;
    uint32_t    firstInstance;
} VkDrawIndirectCommand;
```

- `vertexCount` is the number of vertices to draw.
- `instanceCount` is the number of instances to draw.
- `firstVertex` is the index of the first vertex to draw.
- `firstInstance` is the instance ID of the first instance to draw.

The members of `VkDrawIndirectCommand` have the same meaning as the similarly named parameters of `vkCmdDraw`.

Valid Usage

- VUID-VkDrawIndirectCommand-firstInstance-00501
If the `drawIndirectFirstInstance` feature is not enabled, `firstInstance` **must** be 0

To record an indexed indirect drawing command, call:

```
// Provided by VK_VERSION_1_0
void vkCmdDrawIndexedIndirect(
    VkCommandBuffer          commandBuffer,
    VkBuffer                  buffer,
    VkDeviceSize              offset,
    uint32_t                  drawCount,
    uint32_t                  stride);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `buffer` is the buffer containing draw parameters.

- `offset` is the byte offset into `buffer` where parameters begin.
- `drawCount` is the number of draws to execute, and **can** be zero.
- `stride` is the byte stride between successive sets of draw parameters.

`vkCmdDrawIndexedIndirect` behaves similarly to `vkCmdDrawIndexed` except that the parameters are read by the device from a buffer during execution. `drawCount` draws are executed by the command, with parameters taken from `buffer` starting at `offset` and increasing by `stride` bytes for each successive draw. The parameters of each draw are encoded in an array of `VkDrawIndexedIndirectCommand` structures. If `drawCount` is less than or equal to one, `stride` is ignored.

Valid Usage

- VUID-vkCmdDrawIndexedIndirect-magFilter-04553
If a `VkSampler` created with `magFilter` or `minFilter` equal to `VK_FILTER_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdDrawIndexedIndirect-mipmapMode-04770
If a `VkSampler` created with `mipmapMode` equal to `VK_SAMPLER_MIPMAP_MODE_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdDrawIndexedIndirect-unnormalizedCoordinates-09635
If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `levelCount` and `layerCount` **must** be 1
- VUID-vkCmdDrawIndexedIndirect-unnormalizedCoordinates-09636
If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `viewType` **must** be `VK_IMAGE_VIEW_TYPE_1D` or `VK_IMAGE_VIEW_TYPE_2D`
- VUID-vkCmdDrawIndexedIndirect-aspectMask-06478
If a `VkImageView` is sampled with `depth comparison`, the image view **must** have been created with an `aspectMask` that contains `VK_IMAGE_ASPECT_DEPTH_BIT`
- VUID-vkCmdDrawIndexedIndirect-None-02691
If a `VkImageView` is accessed using atomic operations as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT`
- VUID-vkCmdDrawIndexedIndirect-None-07888
If a `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT`
- VUID-vkCmdDrawIndexedIndirect-None-08600
For each set n that is statically used by a `bound shader`, a descriptor set **must** have been

bound to n at the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for set n , with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) , as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDrawIndexedIndirect-None-08601

For each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) , as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDrawIndexedIndirect-None-10068

For each array of resources that is used by a [bound shader](#), the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command

- VUID-vkCmdDrawIndexedIndirect-None-08114

Descriptors in each bound descriptor set, specified via [vkCmdBindDescriptorSets](#), **must** be valid as described by [descriptor validity](#) if they are statically used by a [bound shader](#)

- VUID-vkCmdDrawIndexedIndirect-None-08606

A valid pipeline **must** be bound to the pipeline bind point used by this command

- VUID-vkCmdDrawIndexedIndirect-None-08608

There **must** not have been any calls to dynamic state setting commands for any state specified statically in the [VkPipeline](#) object bound to the pipeline bind point used by this command, since that pipeline was bound

- VUID-vkCmdDrawIndexedIndirect-None-08609

If the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used to sample from any [VkImage](#) with a [VkImageView](#) of the type [VK_IMAGE_VIEW_TYPE_3D](#), [VK_IMAGE_VIEW_TYPE_CUBE](#), [VK_IMAGE_VIEW_TYPE_1D_ARRAY](#), [VK_IMAGE_VIEW_TYPE_2D_ARRAY](#) or [VK_IMAGE_VIEW_TYPE_CUBE_ARRAY](#), in any shader stage

- VUID-vkCmdDrawIndexedIndirect-None-08610

If the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions with [ImplicitLod](#), [Dref](#) or [Proj](#) in their name, in any shader stage

- VUID-vkCmdDrawIndexedIndirect-None-08611

If the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions that includes a LOD bias or any offset values, in any shader stage

- VUID-vkCmdDrawIndexedIndirect-uniformBuffers-06935

If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a uniform buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndexedIndirect-storageBuffers-06936

If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this

command accesses a storage buffer, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndexedIndirect-viewType-07752

If a `VkImageView` is accessed as a result of this command, then the image view's `viewType` **must** match the `Dim` operand of the `OpTypeImage` as described in [Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types](#)

- VUID-vkCmdDrawIndexedIndirect-format-07753

If a `VkImageView` is accessed as a result of this command, then the `numeric type` of the image view's `format` and the `Sampled Type` operand of the `OpTypeImage` **must** match

- VUID-vkCmdDrawIndexedIndirect-OpImageWrite-08795

If a `VkImageView` is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the image view's format

- VUID-vkCmdDrawIndexedIndirect-OpImageWrite-04469

If a `VkBufferView` is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the buffer view's format

- VUID-vkCmdDrawIndexedIndirect-None-07288

Any shader invocation executed by this command **must** `terminate`

- VUID-vkCmdDrawIndexedIndirect-None-09600

If a descriptor with type equal to any of `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, or `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT` is accessed as a result of this command, the image subresource identified by that descriptor **must** be in the image layout identified when the descriptor was written

- VUID-vkCmdDrawIndexedIndirect-renderPass-02684

The current render pass **must** be `compatible` with the `renderPass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to `VK_PIPELINE_BIND_POINT_GRAPHICS`

- VUID-vkCmdDrawIndexedIndirect-subpass-02685

The subpass index of the current render pass **must** be equal to the `subpass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to `VK_PIPELINE_BIND_POINT_GRAPHICS`

- VUID-vkCmdDrawIndexedIndirect-None-07748

If any shader statically accesses an input attachment, a valid descriptor **must** be bound to the pipeline via a descriptor set

- VUID-vkCmdDrawIndexedIndirect-OpTypeImage-07468

If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass

- VUID-vkCmdDrawIndexedIndirect-None-07469

Input attachment views accessed in a subpass **must** be created with the same `VkFormat` as the corresponding subpass definition, and be created with a `VkImageView` that is

compatible with the attachment referenced by the subpass' `pInputAttachments` [`InputAttachmentIndex`] in the currently bound `VkFramebuffer` as specified by [Fragment Input Attachment Compatibility](#)

- VUID-vkCmdDrawIndexedIndirect-None-06537
Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command
- VUID-vkCmdDrawIndexedIndirect-None-09000
If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass,

it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndexedIndirect-None-09001
If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass,

it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndexedIndirect-None-09002
If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass,

it **must** not be accessed in any way other than as an attachment by this command
- VUID-vkCmdDrawIndexedIndirect-None-06539
If any previously recorded command in the current subpass accessed an image subresource used as an attachment in this subpass in any way other than as an attachment, this command **must** not write to that image subresource as an attachment
- VUID-vkCmdDrawIndexedIndirect-None-06886
If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, `depth writes` **must** be disabled
- VUID-vkCmdDrawIndexedIndirect-None-06887
If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, `all stencil ops` **must** be `VK_STENCIL_OP_KEEP`
- VUID-vkCmdDrawIndexedIndirect-None-07831
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VIEWPORT` dynamic state enabled then `vkCmdSetViewport` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07832
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SCISSOR` dynamic state enabled then `vkCmdSetScissor` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07833
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_WIDTH` dynamic state enabled then `vkCmdSetLineWidth` **must** have been called and not

subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07834

If a graphics pipeline is bound which was created with the **VK_DYNAMIC_STATE_DEPTH_BIAS** dynamic state enabled, the **current value** of **rasterizerDiscardEnable** is **VK_FALSE**, and the **current value** of **depthBiasEnable** is **VK_TRUE**, then **vkCmdSetDepthBounds** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07835

If the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_BLEND_CONSTANTS** dynamic state enabled then **vkCmdSetBlendConstants** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07836

If a graphics pipeline is bound which was created with the **VK_DYNAMIC_STATE_DEPTH_BOUNDS** dynamic state enabled, the **current value** of **rasterizerDiscardEnable** is **VK_FALSE**, and the **current value** of **depthBoundsTestEnable** is **VK_TRUE**, then **vkCmdSetDepthBounds** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07837

If a graphics pipeline is bound which was created with the **VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK** dynamic state enabled, the **current value** of **rasterizerDiscardEnable** is **VK_FALSE**, and the **current value** of **stencilTestEnable** is **VK_TRUE**, then **vkCmdSetStencilCompareMask** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07838

If a graphics pipeline is bound which was created with the **VK_DYNAMIC_STATE_STENCIL_WRITE_MASK** dynamic state enabled, the **current value** of **rasterizerDiscardEnable** is **VK_FALSE**, and the **current value** of **stencilTestEnable** is **VK_TRUE**, then **vkCmdSetStencilWriteMask** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07839

If a graphics pipeline is bound which was created with the **VK_DYNAMIC_STATE_STENCIL_REFERENCE** dynamic state enabled, the **current value** of **rasterizerDiscardEnable** is **VK_FALSE**, the **current value** of **stencilTestEnable** is **VK_TRUE**, then **vkCmdSetStencilReference** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-blendEnable-04727

If rasterization is not disabled in the bound graphics pipeline, then for each color attachment in the subpass, if the corresponding image view's **format features** do not contain **VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT**, then the **blendEnable** member of the corresponding element of the **pAttachments** member of **pColorBlendState** **must** be **VK_FALSE**

- VUID-vkCmdDrawIndexedIndirect-multisampledRenderToSingleSampled-07284

If rasterization is not disabled in the bound graphics pipeline,

then `rasterizationSamples` for the currently bound graphics pipeline **must** be the same as the current subpass color and/or depth/stencil attachments

- VUID-vkCmdDrawIndexedIndirect-maxFragmentDualSrcAttachments-09239

If `blending` is enabled for any attachment where either the source or destination blend factors for that attachment [use the secondary color input](#), the maximum value of `Location` for any output attachment [statically used](#) in the `Fragment Execution Model` executed by this command **must** be less than `maxFragmentDualSrcAttachments`

- VUID-vkCmdDrawIndexedIndirect-None-04007

All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or `VK_NULL_HANDLE` buffers bound

- VUID-vkCmdDrawIndexedIndirect-None-04008

If the `nullDescriptor` feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be `VK_NULL_HANDLE`

- VUID-vkCmdDrawIndexedIndirect-None-02721

If `robustBufferAccess` is not enabled, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in [Vertex Input Description](#)

- VUID-vkCmdDrawIndexedIndirect-buffer-02708

If `buffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-vkCmdDrawIndexedIndirect-buffer-02709

`buffer` **must** have been created with the `VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT` bit set

- VUID-vkCmdDrawIndexedIndirect-offset-02710

`offset` **must** be a multiple of 4

- VUID-vkCmdDrawIndexedIndirect-drawCount-02718

If the `multiDrawIndirect` feature is not enabled, `drawCount` **must** be 0 or 1

- VUID-vkCmdDrawIndexedIndirect-drawCount-02719

`drawCount` **must** be less than or equal to `VkPhysicalDeviceLimits::maxDrawIndirectCount`

- VUID-vkCmdDrawIndexedIndirect-None-07312

A valid index buffer **must** be bound

- VUID-vkCmdDrawIndexedIndirect-drawCount-00528

If `drawCount` is greater than 1, `stride` **must** be a multiple of 4 and **must** be greater than or equal to `sizeof(VkDrawIndexedIndirectCommand)`

- VUID-vkCmdDrawIndexedIndirect-drawCount-00539

If `drawCount` is equal to 1, `(offset + sizeof(VkDrawIndexedIndirectCommand))` **must** be less than or equal to the size of `buffer`

- VUID-vkCmdDrawIndexedIndirect-drawCount-00540

If `drawCount` is greater than 1, `(stride × (drawCount - 1) + offset + sizeof(VkDrawIndexedIndirectCommand))` **must** be less than or equal to the size of `buffer`

Valid Usage (Implicit)

- VUID-vkCmdDrawIndexedIndirect-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdDrawIndexedIndirect-buffer-parameter
`buffer` **must** be a valid `VkBuffer` handle
- VUID-vkCmdDrawIndexedIndirect-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdDrawIndexedIndirect-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdDrawIndexedIndirect-renderpass
This command **must** only be called inside of a render pass instance
- VUID-vkCmdDrawIndexedIndirect-commonparent
Both of `buffer`, and `commandBuffer` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Graphics	Action

The `VkDrawIndexedIndirectCommand` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDrawIndexedIndirectCommand {
    uint32_t    indexCount;
    uint32_t    instanceCount;
    uint32_t    firstIndex;
    int32_t     vertexOffset;
    uint32_t    firstInstance;
} VkDrawIndexedIndirectCommand;
```

- `indexCount` is the number of vertices to draw.
- `instanceCount` is the number of instances to draw.
- `firstIndex` is the base index within the index buffer.
- `vertexOffset` is the value added to the vertex index before indexing into the vertex buffer.
- `firstInstance` is the instance ID of the first instance to draw.

The members of `VkDrawIndexedIndirectCommand` have the same meaning as the similarly named parameters of `vkCmdDrawIndexed`.

Valid Usage

- VUID-VkDrawIndexedIndirectCommand-robustBufferAccess2-08798
If `robustBufferAccess2` is not enabled, $(\text{indexSize} \times (\text{firstIndex} + \text{indexCount}) + \text{offset})$ **must** be less than or equal to the size of the bound index buffer, with `indexSize` being based on the type specified by `indexType`, where the index buffer, `indexType`, and `offset` are specified via `vkCmdBindIndexBuffer`
- VUID-VkDrawIndexedIndirectCommand-firstInstance-00554
If the `drawIndirectFirstInstance` feature is not enabled, `firstInstance` **must** be 0

Chapter 21. Fixed-Function Vertex Processing

Vertex fetching is controlled via configurable state, as a logically distinct graphics pipeline stage.

21.1. Vertex Attributes

Vertex shaders **can** define input variables, which receive *vertex attribute* data transferred from one or more `VkBuffer`(s) by drawing commands. Vertex shader input variables are bound to buffers via an indirect binding where the vertex shader associates a *vertex input attribute* number with each variable, vertex input attributes are associated to *vertex input bindings* on a per-pipeline basis, and vertex input bindings are associated with specific buffers on a per-draw basis via the `vkCmdBindVertexBuffers` command. Vertex input attribute and vertex input binding descriptions also contain format information controlling how data is extracted from buffer memory and converted to the format expected by the vertex shader.

There are `VkPhysicalDeviceLimits::maxVertexInputAttributes` number of vertex input attributes and `VkPhysicalDeviceLimits::maxVertexInputBindings` number of vertex input bindings (each referred to by zero-based indices), where there are at least as many vertex input attributes as there are vertex input bindings. Applications **can** store multiple vertex input attributes interleaved in a single buffer, and use a single vertex input binding to access those attributes.

In GLSL, vertex shaders associate input variables with a vertex input attribute number using the `location` layout qualifier. The `Component` layout qualifier associates components of a vertex shader input variable with components of a vertex input attribute.

GLSL Example

```
// Assign location M to variableName
layout (location=M, component=2) in vec2 variableName;

// Assign locations [N,N+L) to the array elements of variableNameArray
layout (location=N) in vec4 variableNameArray[L];
```

In SPIR-V, vertex shaders associate input variables with a vertex input attribute number using the `Location` decoration. The `Component` decoration associates components of a vertex shader input variable with components of a vertex input attribute. The `Location` and `Component` decorations are specified via the `OpDecorate` instruction.

SPIR-V Example

```
...
%1 = OpExtInstImport "GLSL.std.450"
...
OpName %9 "variableName"
OpName %15 "variableNameArray"
OpDecorate %18 BuiltIn VertexIndex
```

```

OpDecorate %19 BuiltIn InstanceIndex
OpDecorate %9 Location M
OpDecorate %9 Component 2
OpDecorate %15 Location N
...
%2 = OpTypeVoid
%3 = OpTypeFunction %2
%6 = OpTypeFloat 32
%7 = OpTypeVector %6 2
%8 = OpTypePointer Input %7
%9 = OpVariable %8 Input
%10 = OpTypeVector %6 4
%11 = OpTypeInt 32 0
%12 = OpConstant %11 L
%13 = OpTypeArray %10 %12
%14 = OpTypePointer Input %13
%15 = OpVariable %14 Input
...

```

21.1.1. Attribute Location and Component Assignment

The **Location** decoration specifies which vertex input attribute is used to read and interpret the data that a variable will consume.

When a vertex shader input variable declared using a 16- or 32-bit scalar or vector data type is assigned a **Location**, its value(s) are taken from the components of the input attribute specified with the corresponding `VkVertexInputAttributeDescription::location`. The components used depend on the type of variable and the **Component** decoration specified in the variable declaration, as identified in [Input Attribute Components Accessed By 16-Bit and 32-Bit Input Variables](#). Any 16-bit or 32-bit scalar or vector input will consume a single **Location**. For 16-bit and 32-bit data types, missing components are filled in with default values as described [below](#).

Table 20. Input Attribute Components Accessed By 16-Bit and 32-Bit Input Variables

16-bit or 32-bit data type	Component decoration	Components consumed
scalar	0 or unspecified	(x, o, o, o)
scalar	1	(o, y, o, o)
scalar	2	(o, o, z, o)
scalar	3	(o, o, o, w)
two-component vector	0 or unspecified	(x, y, o, o)
two-component vector	1	(o, y, z, o)
two-component vector	2	(o, o, z, w)
three-component vector	0 or unspecified	(x, y, z, o)
three-component vector	1	(o, y, z, w)

16-bit or 32-bit data type	Component decoration	Components consumed
four-component vector	0 or unspecified	(x, y, z, w)

Components indicated by “o” are available for use by other input variables which are sourced from the same attribute, and if used, are either filled with the corresponding component from the input format (if present), or the default value.

When a vertex shader input variable declared using a 32-bit floating-point matrix type is assigned a **Location** *i*, its values are taken from consecutive input attributes starting with the corresponding **VkVertexInputAttributeDescription::location**. Such matrices are treated as an array of column vectors with values taken from the input attributes identified in **Input Attributes Accessed by 32-Bit Input Matrix Variables**. The **VkVertexInputAttributeDescription::format** **must** be specified with a **VkFormat** that corresponds to the appropriate type of column vector. The **Component** decoration **must** not be used with matrix types.

Table 21. Input Attributes Accessed by 32-Bit Input Matrix Variables

Data type	Column vector type	Locations consumed	Components consumed
mat2	two-component vector	i, i+1	(x, y, o, o), (x, y, o, o)
mat2x3	three-component vector	i, i+1	(x, y, z, o), (x, y, z, o)
mat2x4	four-component vector	i, i+1	(x, y, z, w), (x, y, z, w)
mat3x2	two-component vector	i, i+1, i+2	(x, y, o, o), (x, y, o, o), (x, y, o, o)
mat3	three-component vector	i, i+1, i+2	(x, y, z, o), (x, y, z, o), (x, y, z, o)
mat3x4	four-component vector	i, i+1, i+2	(x, y, z, w), (x, y, z, w), (x, y, z, w)
mat4x2	two-component vector	i, i+1, i+2, i+3	(x, y, o, o), (x, y, o, o), (x, y, o, o), (x, y, o, o)
mat4x3	three-component vector	i, i+1, i+2, i+3	(x, y, z, o), (x, y, z, o), (x, y, z, o), (x, y, z, o)
mat4	four-component vector	i, i+1, i+2, i+3	(x, y, z, w), (x, y, z, w), (x, y, z, w), (x, y, z, w)

Components indicated by “o” are available for use by other input variables which are sourced from the same attribute, and if used, are either filled with the corresponding component from the input (if present), or the default value.

When a vertex shader input variable declared using a scalar or vector 64-bit data type is assigned a **Location** *i*, its values are taken from consecutive input attributes starting with the corresponding **VkVertexInputAttributeDescription::location**. The **Location** slots and **Component** words used depend on the type of variable and the **Component** decoration specified in the variable declaration, as identified in **Input Attribute Locations and Components Accessed by 64-Bit Input Variables**. For 64-

bit data types, no default attribute values are provided. Input variables **must** not use more components than provided by the attribute.

Table 22. Input Attribute Locations and Components Accessed by 64-Bit Input Variables

Input format	Locations consumed	64-bit data type	Location decoration	Component decoration	32-bit components consumed
R64	i	scalar	i	0 or unspecified	(x, y, -, -)
R64G64	i	scalar	i	0 or unspecified	(x, y, o, o)
		scalar	i	2	(o, o, z, w)
		two-component vector	i	0 or unspecified	(x, y, z, w)
R64G64B64	i, i+1	scalar	i	0 or unspecified	(x, y, o, o), (o, o, -, -)
		scalar	i	2	(o, o, z, w), (o, o, -, -)
		scalar	i+1	0 or unspecified	(o, o, o, o), (x, y, -, -)
		two-component vector	i	0 or unspecified	(x, y, z, w), (o, o, -, -)
		three-component vector	i	unspecified	(x, y, z, w), (x, y, -, -)
R64G64B64A64	i, i+1	scalar	i	0 or unspecified	(x, y, o, o), (o, o, o, o)
		scalar	i	2	(o, o, z, w), (o, o, o, o)
		scalar	i+1	0 or unspecified	(o, o, o, o), (x, y, o, o)
		scalar	i+1	2	(o, o, o, o), (o, o, z, w)
		two-component vector	i	0 or unspecified	(x, y, z, w), (o, o, o, o)
		two-component vector	i+1	0 or unspecified	(o, o, o, o), (x, y, z, w)
		three-component vector	i	unspecified	(x, y, z, w), (x, y, o, o)
		four-component vector	i	unspecified	(x, y, z, w), (x, y, z, w)

Components indicated by “o” are available for use by other input variables which are sourced from the same attribute. Components indicated by “-” are not available for input variables as there are

no default values provided for 64-bit data types, and there is no data provided by the input format.

When a vertex shader input variable declared using a 64-bit floating-point matrix type is assigned a **Location** *i*, its values are taken from consecutive input attribute locations. Such matrices are treated as an array of column vectors with values taken from the input attributes as shown in [Input Attribute Locations and Components Accessed by 64-Bit Input Variables](#). Each column vector starts at the **Location** immediately following the last **Location** of the previous column vector. The number of attributes and components assigned to each matrix is determined by the matrix dimensions and ranges from two to eight locations.

When a vertex shader input variable declared using an array type is assigned a location, its values are taken from consecutive input attributes starting with the corresponding **VkVertexInputAttributeDescription::location**. The number of attributes and components assigned to each element are determined according to the data type of the array elements and **Component** decoration (if any) specified in the declaration of the array, as described above. Each element of the array, in order, is assigned to consecutive locations, but all at the same specified component within each location.

Only input variables declared with the data types and component decorations as specified above are supported. Two variables are allowed to share the same **Location** slot only if their **Component** words do not overlap. If multiple variables share the same **Location** slot, they **must** all have the same SPIR-V floating-point component type or all have the same width scalar type components.

21.2. Vertex Input Description

Applications specify vertex input attribute and vertex input binding descriptions as part of graphics pipeline creation by setting the **VkGraphicsPipelineCreateInfo::pVertexInputState** pointer to a **VkPipelineVertexInputStateCreateInfo** structure.

The **VkPipelineVertexInputStateCreateInfo** structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineVertexInputStateCreateInfo {
    VkStructureType          sType;
    const void*              pNext;
    VkPipelineVertexInputStateCreateFlags flags;
    uint32_t                 vertexBindingDescriptionCount;
    const VkVertexInputBindingDescription* pVertexBindingDescriptions;
    uint32_t                 vertexAttributeDescriptionCount;
    const VkVertexInputAttributeDescription* pVertexAttributeDescriptions;
} VkPipelineVertexInputStateCreateInfo;
```

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **flags** is reserved for future use.
- **vertexBindingDescriptionCount** is the number of vertex binding descriptions provided in **pVertexBindingDescriptions**.

- `pVertexBindingDescriptions` is a pointer to an array of `VkVertexInputBindingDescription` structures.
- `vertexAttributeDescriptionCount` is the number of vertex attribute descriptions provided in `pVertexAttributeDescriptions`.
- `pVertexAttributeDescriptions` is a pointer to an array of `VkVertexInputAttributeDescription` structures.

Valid Usage

- VUID-VkPipelineVertexInputStateCreateInfo-vertexBindingDescriptionCount-00613
`vertexBindingDescriptionCount` **must** be less than or equal to `VkPhysicalDeviceLimits::maxVertexInputBindings`
- VUID-VkPipelineVertexInputStateCreateInfo-vertexAttributeDescriptionCount-00614
`vertexAttributeDescriptionCount` **must** be less than or equal to `VkPhysicalDeviceLimits::maxVertexInputAttributes`
- VUID-VkPipelineVertexInputStateCreateInfo-binding-00615
For every `binding` specified by each element of `pVertexAttributeDescriptions`, a `VkVertexInputBindingDescription` **must** exist in `pVertexBindingDescriptions` with the same value of `binding`
- VUID-VkPipelineVertexInputStateCreateInfo-pVertexBindingDescriptions-00616
All elements of `pVertexBindingDescriptions` **must** describe distinct binding numbers
- VUID-VkPipelineVertexInputStateCreateInfo-pVertexAttributeDescriptions-00617
All elements of `pVertexAttributeDescriptions` **must** describe distinct attribute locations

Valid Usage (Implicit)

- VUID-VkPipelineVertexInputStateCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_VERTEX_INPUT_STATE_CREATE_INFO`
- VUID-VkPipelineVertexInputStateCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkPipelineVertexInputStateCreateInfo-flags-zero bitmask
`flags` **must** be `0`
- VUID-VkPipelineVertexInputStateCreateInfo-pVertexBindingDescriptions-parameter
If `vertexBindingDescriptionCount` is not `0`, `pVertexBindingDescriptions` **must** be a valid pointer to an array of `vertexBindingDescriptionCount` valid `VkVertexInputBindingDescription` structures
- VUID-VkPipelineVertexInputStateCreateInfo-pVertexAttributeDescriptions-parameter
If `vertexAttributeDescriptionCount` is not `0`, `pVertexAttributeDescriptions` **must** be a valid pointer to an array of `vertexAttributeDescriptionCount` valid `VkVertexInputAttributeDescription` structures

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineVertexInputStateCreateFlags;
```

`VkPipelineVertexInputStateCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

Each vertex input binding is specified by the `VkVertexInputBindingDescription` structure, defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkVertexInputBindingDescription {
    uint32_t      binding;
    uint32_t      stride;
    VkVertexInputRate inputRate;
} VkVertexInputBindingDescription;
```

- `binding` is the binding number that this structure describes.
- `stride` is the byte stride between consecutive elements within the buffer.
- `inputRate` is a `VkVertexInputRate` value specifying whether vertex attribute addressing is a function of the vertex index or of the instance index.

Valid Usage

- VUID-VkVertexInputBindingDescription-binding-00618
`binding` **must** be less than `VkPhysicalDeviceLimits::maxVertexInputBindings`
- VUID-VkVertexInputBindingDescription-stride-00619
`stride` **must** be less than or equal to `VkPhysicalDeviceLimits::maxVertexInputBindingStride`

Valid Usage (Implicit)

- VUID-VkVertexInputBindingDescription-inputRate-parameter
`inputRate` **must** be a valid `VkVertexInputRate` value

Possible values of `VkVertexInputBindingDescription::inputRate`, specifying the rate at which vertex attributes are pulled from buffers, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkVertexInputRate {
    VK_VERTEX_INPUT_RATE_VERTEX = 0,
    VK_VERTEX_INPUT_RATE_INSTANCE = 1,
} VkVertexInputRate;
```

- `VK_VERTEX_INPUT_RATE_VERTEX` specifies that vertex attribute addressing is a function of the vertex index.

- `VK_VERTEX_INPUT_RATE_INSTANCE` specifies that vertex attribute addressing is a function of the instance index.

Each vertex input attribute is specified by the `VkVertexInputAttributeDescription` structure, defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkVertexInputAttributeDescription {
    uint32_t    location;
    uint32_t    binding;
    VkFormat     format;
    uint32_t    offset;
} VkVertexInputAttributeDescription;
```

- `location` is the shader input location number for this attribute.
- `binding` is the binding number which this attribute takes its data from.
- `format` is the size and type of the vertex attribute data.
- `offset` is a byte offset of this attribute relative to the start of an element in the vertex input binding.

Valid Usage

- VUID-VkVertexInputAttributeDescription-location-00620
`location` **must** be less than `VkPhysicalDeviceLimits::maxVertexInputAttributes`
- VUID-VkVertexInputAttributeDescription-binding-00621
`binding` **must** be less than `VkPhysicalDeviceLimits::maxVertexInputBindings`
- VUID-VkVertexInputAttributeDescription-offset-00622
`offset` **must** be less than or equal to `VkPhysicalDeviceLimits::maxVertexInputAttributeOffset`
- VUID-VkVertexInputAttributeDescription-format-00623
The `format` features of `format` **must** contain `VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT`

Valid Usage (Implicit)

- VUID-VkVertexInputAttributeDescription-format-parameter
`format` **must** be a valid `VkFormat` value

To bind vertex buffers to a command buffer for use in subsequent drawing commands, call:


```
// Provided by VK_VERSION_1_0
void vkCmdBindVertexBuffers(
    VkCommandBuffer                commandBuffer,
    uint32_t                       firstBinding,
    uint32_t                       bindingCount,
    const VkBuffer*                pBuffer,
    const VkDeviceSize*            pOffsets);
```

- **commandBuffer** is the command buffer into which the command is recorded.
- **firstBinding** is the index of the first vertex input binding whose state is updated by the command.
- **bindingCount** is the number of vertex input bindings whose state is updated by the command.
- **pBuffers** is a pointer to an array of buffer handles.
- **pOffsets** is a pointer to an array of buffer offsets.

The values taken from elements *i* of **pBuffers** and **pOffsets** replace the current state for the vertex input binding **firstBinding** + *i*, for *i* in [0, **bindingCount**). The vertex input binding is updated to start at the offset indicated by **pOffsets**[*i*] from the start of the buffer **pBuffers**[*i*]. All vertex input attributes that use each of these bindings will use these updated addresses in their address calculations for subsequent drawing commands.

Valid Usage

- VUID-vkCmdBindVertexBuffers-firstBinding-00624
firstBinding **must** be less than **VkPhysicalDeviceLimits::maxVertexInputBindings**
- VUID-vkCmdBindVertexBuffers-firstBinding-00625
The sum of **firstBinding** and **bindingCount** **must** be less than or equal to **VkPhysicalDeviceLimits::maxVertexInputBindings**
- VUID-vkCmdBindVertexBuffers-pOffsets-00626
All elements of **pOffsets** **must** be less than the size of the corresponding element in **pBuffers**
- VUID-vkCmdBindVertexBuffers-pBuffers-00627
All elements of **pBuffers** **must** have been created with the **VK_BUFFER_USAGE_VERTEX_BUFFER_BIT** flag
- VUID-vkCmdBindVertexBuffers-pBuffers-00628
Each element of **pBuffers** that is non-sparse **must** be bound completely and contiguously to a single **VkDeviceMemory** object
- VUID-vkCmdBindVertexBuffers-pBuffers-04001
If the **nullDescriptor** feature is not enabled, all elements of **pBuffers** **must** not be **VK_NULL_HANDLE**

Valid Usage (Implicit)

- VUID-vkCmdBindVertexBuffers-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdBindVertexBuffers-pBuffers-parameter
`pBuffers` **must** be a valid pointer to an array of `bindingCount` valid or `VK_NULL_HANDLE` `VkBuffer` handles
- VUID-vkCmdBindVertexBuffers-pOffsets-parameter
`pOffsets` **must** be a valid pointer to an array of `bindingCount` `VkDeviceSize` values
- VUID-vkCmdBindVertexBuffers-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdBindVertexBuffers-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdBindVertexBuffers-bindingCount-arraylength
`bindingCount` **must** be greater than 0
- VUID-vkCmdBindVertexBuffers-commonparent
Both of `commandBuffer`, and the elements of `pBuffers` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Graphics	State

21.3. Vertex Input Address Calculation

The address of each attribute for each `vertexIndex` and `instanceIndex` is calculated as follows:

- Let `attribDesc` be the member of `VkPipelineVertexInputStateCreateInfo::pVertexAttributeDescriptions` with `VkVertexInputAttributeDescription::location` equal to the vertex input attribute number.

- Let `bindingDesc` be the member of `VkPipelineVertexInputStateCreateInfo::pVertexBindingDescriptions` with `VkVertexInputAttributeDescription::binding` equal to `attribDesc.binding`.
- Let `vertexIndex` be the index of the vertex within the draw (a value between `firstVertex` and `firstVertex+vertexCount` for `vkCmdDraw`, or a value taken from the index buffer plus `vertexOffset` for `vkCmdDrawIndexed`), and let `instanceIndex` be the instance number of the draw (a value between `firstInstance` and `firstInstance+instanceCount`).
- Let `offset` be an array of offsets into the currently bound vertex buffers specified during `vkCmdBindVertexBuffers` with `pOffsets`.
- Let `stride` be the member of `VkPipelineVertexInputStateCreateInfo::pVertexBindingDescriptions->stride` unless there is dynamic state causing the value to be ignored. In this case the value is set from the last value from one of the following

```
bufferBindingAddress = buffer[binding].baseAddress + offset[binding];

if (bindingDesc.inputRate == VK_VERTEX_INPUT_RATE_VERTEX)
    effectiveVertexOffset = vertexIndex * stride;
else
    effectiveVertexOffset = instanceIndex * stride;

attribAddress = bufferBindingAddress + effectiveVertexOffset + attribDesc.offset;
```

21.3.1. Vertex Input Extraction

For each attribute, raw data is extracted starting at `attribAddress` and is converted from the `VkVertexInputAttributeDescription`'s `format` to either floating-point, unsigned integer, or signed integer based on the `numeric type` of `format`. The numeric type of `format` **must** match the numeric type of the input variable in the shader. The input variable in the shader **must** be declared as a 64-bit data type if and only if `format` is a 64-bit data type. If `format` is a packed format, `attribAddress` **must** be a multiple of the size in bytes of the whole attribute data type as described in [Packed Formats](#). Otherwise, `attribAddress` **must** be a multiple of the size in bytes of the component type indicated by `format` (see [Formats](#)). For attributes that are not 64-bit data types, each component is converted to the format of the input variable based on its type and size (as defined in the [Format Definition](#) section for each `VkFormat`), using the appropriate equations in [16-Bit Floating-Point Numbers](#), [Unsigned 11-Bit Floating-Point Numbers](#), [Unsigned 10-Bit Floating-Point Numbers](#), [Fixed-Point Data Conversion](#), and [Shared Exponent to RGB](#). Signed integer components smaller than 32 bits are sign-extended. Attributes that are not 64-bit data types are expanded to four components in the same way as described in [conversion to RGBA](#). The number of components in the vertex shader input variable need not exactly match the number of components in the format. If the vertex shader has fewer components, the extra components are discarded.

Chapter 22. Tessellation

Tessellation involves three pipeline stages. First, a [tessellation control shader](#) transforms control points of a patch and **can** produce per-patch data. Second, a fixed-function tessellator generates multiple primitives corresponding to a tessellation of the patch in (u,v) or (u,v,w) parameter space. Third, a [tessellation evaluation shader](#) transforms the vertices of the tessellated patch, for example to compute their positions and attributes as part of the tessellated surface. The tessellator is enabled when the pipeline contains both a tessellation control shader and a tessellation evaluation shader.

22.1. Tessellator

If a pipeline includes both tessellation shaders (control and evaluation), the tessellator consumes each input patch (after vertex shading) and produces a new set of independent primitives (points, lines, or triangles). These primitives are logically produced by subdividing a geometric primitive (rectangle or triangle) according to the per-patch outer and inner tessellation levels written by the tessellation control shader. These levels are specified using the [built-in variables](#) `TessLevelOuter` and `TessLevelInner`, respectively. This subdivision is performed in an implementation-dependent manner. If no tessellation shaders are present in the pipeline, the tessellator is disabled and incoming primitives are passed through without modification.

The type of subdivision performed by the tessellator is specified by an `OpExecutionMode` instruction using one of the `Triangles`, `Quads`, or `Isolines` execution modes. This instruction **may** be specified in either the tessellation evaluation or tessellation control shader. Other tessellation-related execution modes **can** also be specified in either the tessellation control or tessellation evaluation shaders.

Any tessellation-related modes specified in both the tessellation control and tessellation evaluation shaders **must** be the same.

Tessellation execution modes include:

- `Triangles`, `Quads`, and `Isolines`. These control the type of subdivision and topology of the output primitives. One mode **must** be set in at least one of the tessellation shader stages.
- `VertexOrderCw` and `VertexOrderCcw`. These control the orientation of triangles generated by the tessellator. One mode **must** be set in at least one of the tessellation shader stages.
- `PointMode`. Controls generation of points rather than triangles or lines. This functionality defaults to disabled, and is enabled if either shader stage includes the execution mode.
- `SpacingEqual`, `SpacingFractionalEven`, and `SpacingFractionalOdd`. Controls the spacing of segments on the edges of tessellated primitives. One mode **must** be set in at least one of the tessellation shader stages.
- `OutputVertices`. Controls the size of the output patch of the tessellation control shader. One value **must** be set in at least one of the tessellation shader stages.

For triangles, the tessellator subdivides a triangle primitive into smaller triangles. For quads, the tessellator subdivides a rectangle primitive into smaller triangles. For isolines, the tessellator subdivides a rectangle primitive into a collection of line segments arranged in strips stretching

across the rectangle in the u dimension (i.e. the coordinates in **TessCoord** are of the form (0,x) through (1,x) for all tessellation evaluation shader invocations that share a line).

Each vertex produced by the tessellator has an associated (u,v,w) or (u,v) position in a normalized parameter space, with parameter values in the range [0,1], as illustrated in figure [Domain parameterization for tessellation primitive modes \(upper-left origin\)](#). The domain space has an upper-left origin.

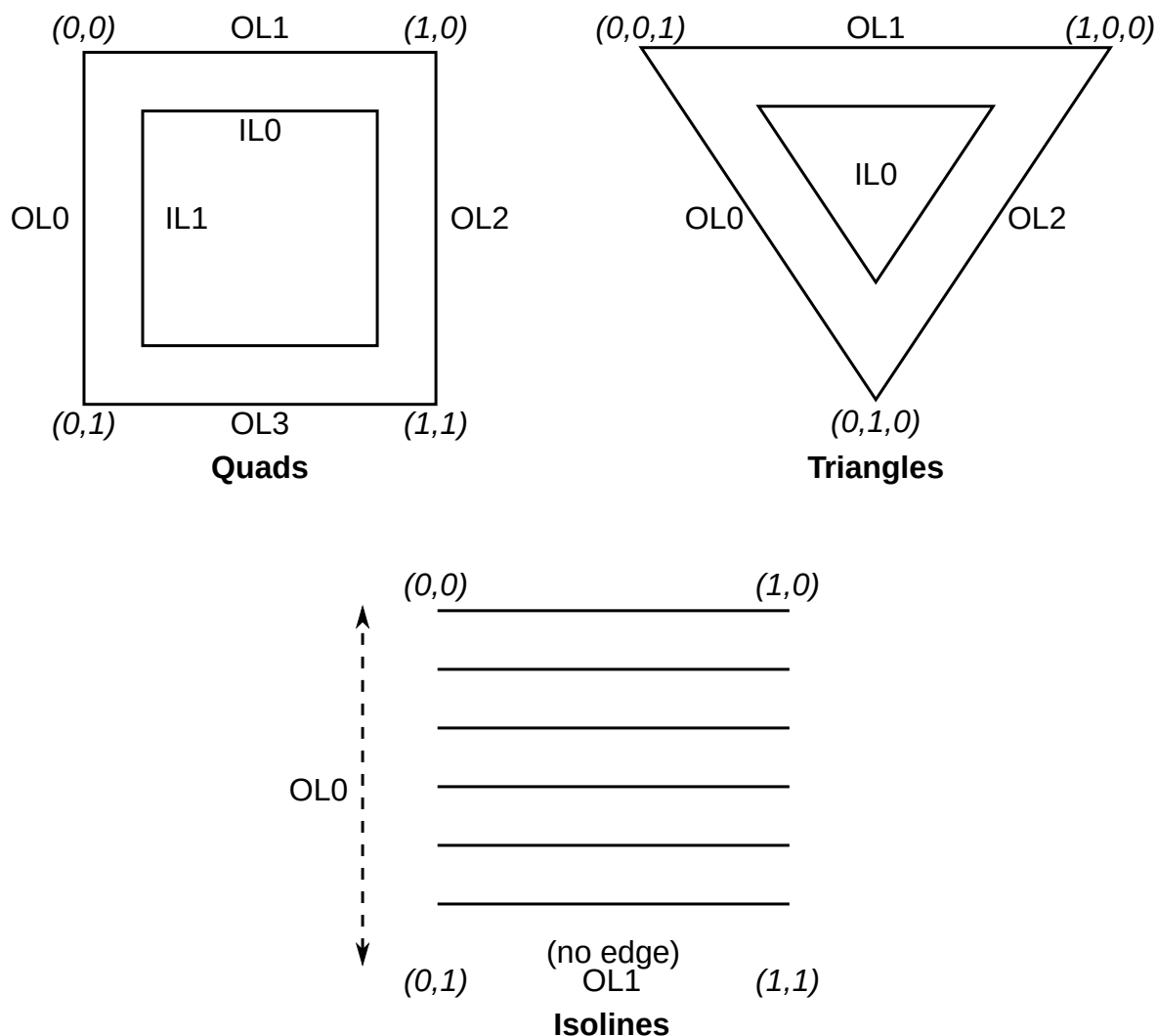


Figure 5. Domain parameterization for tessellation primitive modes (upper-left origin)

Caption

In the domain parameterization diagrams, the coordinates illustrate the value of **TessCoord** at the corners of the domain. The labels on the edges indicate the inner (IL0 and IL1) and outer (OL0 through OL3) tessellation level values used to control the number of subdivisions along each edge of the domain.

For triangles, the vertex's position is a barycentric coordinate (u,v,w), where $u + v + w = 1.0$, and indicates the relative influence of the three vertices of the triangle on the position of the vertex. For quads and isolines, the position is a (u,v) coordinate indicating the relative horizontal and vertical position of the vertex relative to the subdivided rectangle. The subdivision process is explained in more detail in subsequent sections.

22.2. Tessellator Patch Discard

A patch is discarded by the tessellator if any relevant outer tessellation level is less than or equal to zero.

Patches will also be discarded if any relevant outer tessellation level corresponds to a floating-point NaN (not a number) in implementations supporting NaN.

No new primitives are generated and the tessellation evaluation shader is not executed for patches that are discarded. For **Quads**, all four outer levels are relevant. For **Triangles** and **Isolines**, only the first three or two outer levels, respectively, are relevant. Negative inner levels will not cause a patch to be discarded; they will be clamped as described below.

22.3. Tessellator Spacing

Each of the tessellation levels is used to determine the number and spacing of segments used to subdivide a corresponding edge. The method used to derive the number and spacing of segments is specified by an **OpExecutionMode** in the tessellation control or tessellation evaluation shader using one of the identifiers **SpacingEqual**, **SpacingFractionalEven**, or **SpacingFractionalOdd**.

If **SpacingEqual** is used, the floating-point tessellation level is first clamped to $[1, \text{maxLevel}]$, where **maxLevel** is the implementation-dependent maximum tessellation level (**VkPhysicalDeviceLimits::maxTessellationGenerationLevel**). The result is rounded up to the nearest integer n , and the corresponding edge is divided into n segments of equal length in (u,v) space.

If **SpacingFractionalEven** is used, the tessellation level is first clamped to $[2, \text{maxLevel}]$ and then rounded up to the nearest even integer n . If **SpacingFractionalOdd** is used, the tessellation level is clamped to $[1, \text{maxLevel} - 1]$ and then rounded up to the nearest odd integer n . If n is one, the edge will not be subdivided. Otherwise, the corresponding edge will be divided into $n - 2$ segments of equal length, and two additional segments of equal length that are typically shorter than the other segments. The length of the two additional segments relative to the others will decrease monotonically with $n - f$, where f is the clamped floating-point tessellation level. When $n - f$ is zero, the additional segments will have equal length to the other segments. As $n - f$ approaches 2.0, the relative length of the additional segments approaches zero. The two additional segments **must** be placed symmetrically on opposite sides of the subdivided edge. The relative location of these two segments is implementation-dependent, but **must** be identical for any pair of subdivided edges with identical values of f .

When tessellating triangles or quads using **point mode** with fractional odd spacing, the tessellator **may** produce *interior vertices* that are positioned on the edge of the patch if an inner tessellation level is less than or equal to one. Such vertices are considered distinct from vertices produced by subdividing the outer edge of the patch, even if there are pairs of vertices with identical coordinates.

22.4. Tessellation Primitive Ordering

Few guarantees are provided for the relative ordering of primitives produced by tessellation, as they pertain to **primitive order**.

- The output primitives generated from each input primitive are passed to subsequent pipeline stages in an implementation-dependent order.
- All output primitives generated from a given input primitive are passed to subsequent pipeline stages before any output primitives generated from subsequent input primitives.

22.5. Tessellator Vertex Winding Order

When the tessellator produces triangles (in the **Triangles** or **Quads** modes), the orientation of all triangles is specified with an **OpExecutionMode** of **VertexOrderCw** or **VertexOrderCcw** in the tessellation control or tessellation evaluation shaders. If the order is **VertexOrderCw**, the vertices of all generated triangles will have clockwise ordering in (u,v) or (u,v,w) space. If the order is **VertexOrderCcw**, the vertices will have counter-clockwise ordering in that space.

If the tessellation domain has an upper-left origin, the vertices of a triangle have counter-clockwise ordering if

$$a = u_0 v_1 - u_1 v_0 + u_1 v_2 - u_2 v_1 + u_2 v_0 - u_0 v_2$$

is negative, and clockwise ordering if a is positive. u_i and v_i are the u and v coordinates in normalized parameter space of the i th vertex of the triangle.

NOTE

The value a is proportional (with a positive factor) to the signed area of the triangle.

In **Triangles** mode, even though the vertex coordinates have a w value, it does not participate directly in the computation of a , being an affine combination of u and v .

22.6. Triangle Tessellation

If the tessellation primitive mode is **Triangles**, an equilateral triangle is subdivided into a collection of triangles covering the area of the original triangle. First, the original triangle is subdivided into a collection of concentric equilateral triangles. The edges of each of these triangles are subdivided, and the area between each triangle pair is filled by triangles produced by joining the vertices on the subdivided edges. The number of concentric triangles and the number of subdivisions along each triangle except the outermost is derived from the first inner tessellation level. The edges of the outermost triangle are subdivided independently, using the first, second, and third outer tessellation levels to control the number of subdivisions of the $u = 0$ (left), $v = 0$ (bottom), and $w = 0$ (right) edges, respectively. The second inner tessellation level and the fourth outer tessellation level have no effect in this mode.

If the first inner tessellation level and all three outer tessellation levels are exactly one after clamping and rounding, only a single triangle with (u,v,w) coordinates of (0,0,1), (1,0,0), and (0,1,0) is generated. If the inner tessellation level is one and any of the outer tessellation levels is greater than one, the inner tessellation level is treated as though it were originally specified as $1 + \epsilon$ and will result in a two- or three-segment subdivision depending on the tessellation spacing. When used with fractional odd spacing, the three-segment subdivision **may** produce *inner vertices* positioned on the edge of the triangle.

If any tessellation level is greater than one, tessellation begins by producing a set of concentric inner triangles and subdividing their edges. First, the three outer edges are temporarily subdivided using the clamped and rounded first inner tessellation level and the specified tessellation spacing, generating n segments. For the outermost inner triangle, the inner triangle is degenerate — a single point at the center of the triangle — if n is two. Otherwise, for each corner of the outer triangle, an inner triangle corner is produced at the intersection of two lines extended perpendicular to the corner's two adjacent edges running through the vertex of the subdivided outer edge nearest that corner. If n is three, the edges of the inner triangle are not subdivided and it is the final triangle in the set of concentric triangles. Otherwise, each edge of the inner triangle is divided into $n - 2$ segments, with the $n - 1$ vertices of this subdivision produced by intersecting the inner edge with lines perpendicular to the edge running through the $n - 1$ innermost vertices of the subdivision of the outer edge. Once the outermost inner triangle is subdivided, the previous subdivision process repeats itself, using the generated triangle as an outer triangle. This subdivision process is illustrated in [Inner Triangle Tessellation](#).

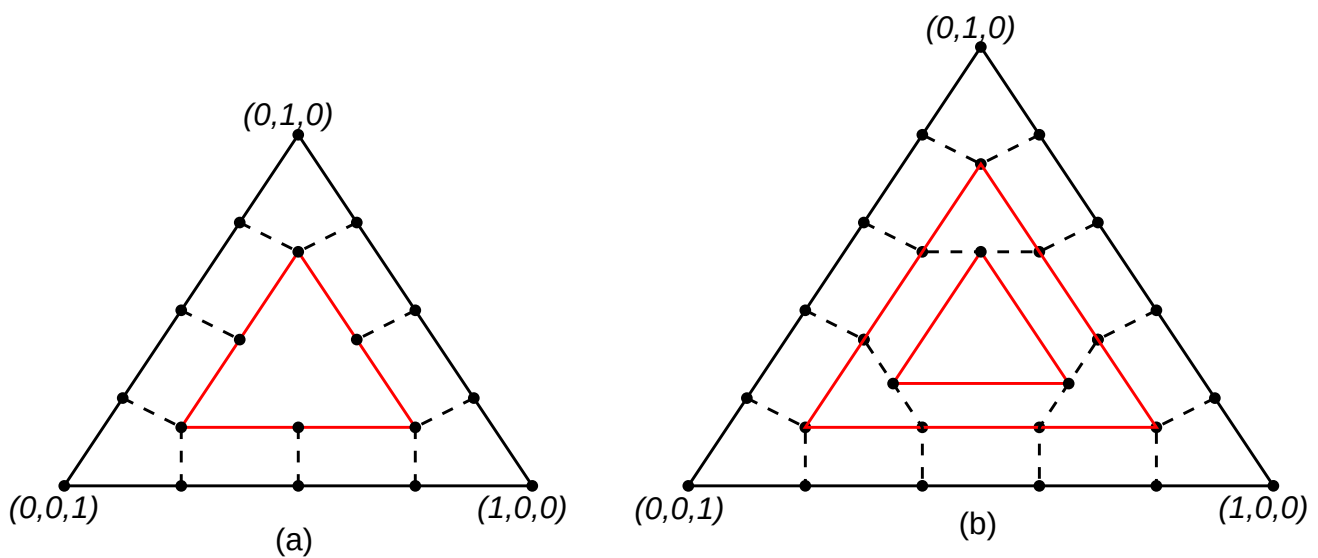


Figure 6. Inner Triangle Tessellation

Caption

In the [Inner Triangle Tessellation](#) diagram, inner tessellation levels of (a) four and (b) five are shown (not to scale). Solid black circles depict vertices along the edges of the concentric triangles. The edges of inner triangles are subdivided by intersecting the edge with segments perpendicular to the edge passing through each inner vertex of the subdivided outer edge. Dotted lines depict edges connecting corresponding vertices on the inner and outer triangle edges.

Once all the concentric triangles are produced and their edges are subdivided, the area between each pair of adjacent inner triangles is filled completely with a set of non-overlapping triangles. In this subdivision, two of the three vertices of each triangle are taken from adjacent vertices on a subdivided edge of one triangle; the third is one of the vertices on the corresponding edge of the other triangle. If the innermost triangle is degenerate (i.e., a point), the triangle containing it is subdivided into six triangles by connecting each of the six vertices on that triangle with the center point. If the innermost triangle is not degenerate, that triangle is added to the set of generated triangles as-is.

After the area corresponding to any inner triangles is filled, the tessellator generates triangles to cover the area between the outermost triangle and the outermost inner triangle. To do this, the temporary subdivision of the outer triangle edge above is discarded. Instead, the $u = 0$, $v = 0$, and $w = 0$ edges are subdivided according to the first, second, and third outer tessellation levels, respectively, and the tessellation spacing. The original subdivision of the first inner triangle is retained. The area between the outer and first inner triangles is completely filled by non-overlapping triangles as described above. If the first (and only) inner triangle is degenerate, a set of triangles is produced by connecting each vertex on the outer triangle edges with the center point.

After all triangles are generated, each vertex in the subdivided triangle is assigned a barycentric (u,v,w) coordinate based on its location relative to the three vertices of the outer triangle.

The algorithm used to subdivide the triangular domain in (u,v,w) space into individual triangles is implementation-dependent. However, the set of triangles produced will completely cover the domain, and no portion of the domain will be covered by multiple triangles.

Output triangles are generated with a topology similar to [triangle lists](#), except that the order in which each triangle is generated, and the order in which the vertices are generated for each triangle, are implementation-dependent. However, the order of vertices in each triangle is consistent across the domain as described in [Tessellator Vertex Winding Order](#).

22.7. Quad Tessellation

If the tessellation primitive mode is **Quads**, a rectangle is subdivided into a collection of triangles covering the area of the original rectangle. First, the original rectangle is subdivided into a regular mesh of rectangles, where the number of rectangles along the $u = 0$ and $u = 1$ (vertical) and $v = 0$ and $v = 1$ (horizontal) edges are derived from the first and second inner tessellation levels, respectively. All rectangles, except those adjacent to one of the outer rectangle edges, are decomposed into triangle pairs. The outermost rectangle edges are subdivided independently, using the first, second, third, and fourth outer tessellation levels to control the number of subdivisions of the $u = 0$ (left), $v = 0$ (bottom), $u = 1$ (right), and $v = 1$ (top) edges, respectively. The area between the inner rectangles of the mesh and the outer rectangle edges are filled by triangles produced by joining the vertices on the subdivided outer edges to the vertices on the edge of the inner rectangle mesh.

If both clamped inner tessellation levels and all four clamped outer tessellation levels are exactly one, only a single triangle pair covering the outer rectangle is generated. Otherwise, if either clamped inner tessellation level is one, that tessellation level is treated as though it was originally specified as $1 + \epsilon$ and will result in a two- or three-segment subdivision depending on the tessellation spacing. When used with fractional odd spacing, the three-segment subdivision **may** produce *inner vertices* positioned on the edge of the rectangle.

If any tessellation level is greater than one, tessellation begins by subdividing the $u = 0$ and $u = 1$ edges of the outer rectangle into m segments using the clamped and rounded first inner tessellation level and the tessellation spacing. The $v = 0$ and $v = 1$ edges are subdivided into n segments using the second inner tessellation level. Each vertex on the $u = 0$ and $v = 0$ edges are joined with the corresponding vertex on the $u = 1$ and $v = 1$ edges to produce a set of vertical and horizontal lines that divide the rectangle into a grid of smaller rectangles. The primitive generator emits a pair of

non-overlapping triangles covering each such rectangle not adjacent to an edge of the outer rectangle. The boundary of the region covered by these triangles forms an inner rectangle, the edges of which are subdivided by the grid vertices that lie on the edge. If either m or n is two, the inner rectangle is degenerate, and one or both of the rectangle's *edges* consist of a single point. This subdivision is illustrated in Figure [Inner Quad Tessellation](#).

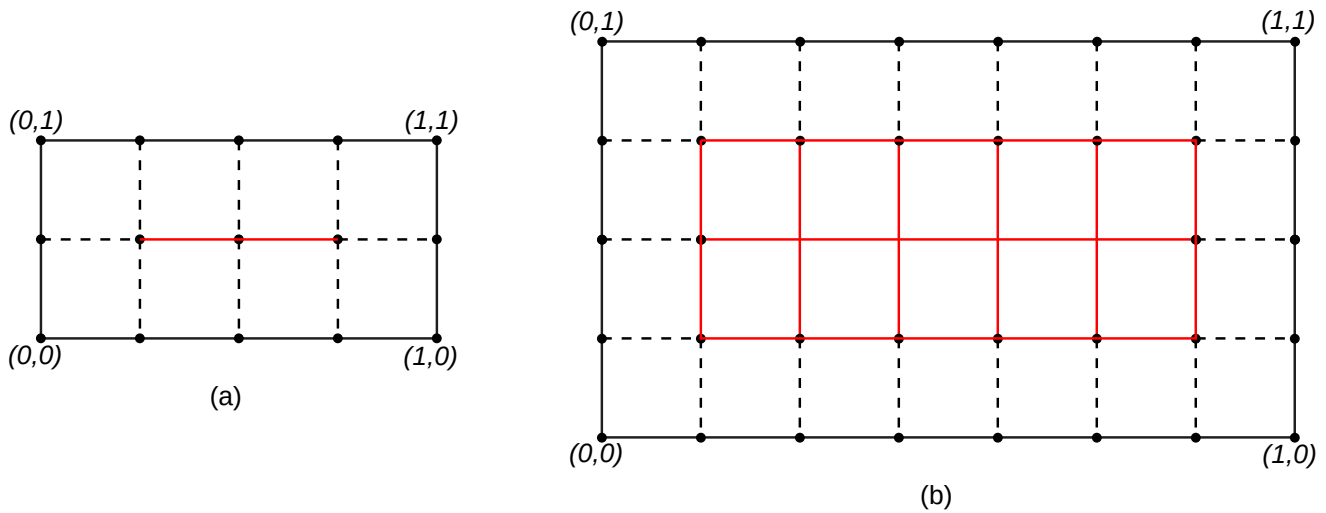


Figure 7. Inner Quad Tessellation

Caption

In the [Inner Quad Tessellation](#) diagram, inner quad tessellation levels of (a) (4,2) and (b) (7,4) are shown. The regions highlighted in red in figure (b) depict the 10 inner rectangles, each of which will be subdivided into two triangles. Solid black circles depict vertices on the boundary of the outer and inner rectangles, where the inner rectangle of figure (a) is degenerate (a single line segment). Dotted lines depict the horizontal and vertical edges connecting corresponding vertices on the inner and outer rectangle edges.

After the area corresponding to the inner rectangle is filled, the tessellator **must** produce triangles to cover the area between the inner and outer rectangles. To do this, the subdivision of the outer rectangle edge above is discarded. Instead, the $u = 0$, $v = 0$, $u = 1$, and $v = 1$ edges are subdivided according to the first, second, third, and fourth outer tessellation levels, respectively, and the tessellation spacing. The original subdivision of the inner rectangle is retained. The area between the outer and inner rectangles is completely filled by non-overlapping triangles. Two of the three vertices of each triangle are adjacent vertices on a subdivided edge of one rectangle; the third is one of the vertices on the corresponding edge of the other rectangle. If either edge of the innermost rectangle is degenerate, the area near the corresponding outer edges is filled by connecting each vertex on the outer edge with the single vertex making up the *inner edge*.

The algorithm used to subdivide the rectangular domain in (u,v) space into individual triangles is implementation-dependent. However, the set of triangles produced will completely cover the domain, and no portion of the domain will be covered by multiple triangles.

Output triangles are generated with a topology similar to [triangle lists](#), except that the order in which each triangle is generated, and the order in which the vertices are generated for each triangle, are implementation-dependent. However, the order of vertices in each triangle is

consistent across the domain as described in [Tessellator Vertex Winding Order](#).

22.8. Isoline Tessellation

If the tessellation primitive mode is **Isolines**, a set of independent horizontal line segments is drawn. The segments are arranged into connected strips called *isolines*, where the vertices of each isoline have a constant *v* coordinate and *u* coordinates covering the full range [0,1]. The number of isolines generated is derived from the first outer tessellation level; the number of segments in each isoline is derived from the second outer tessellation level. Both inner tessellation levels and the third and fourth outer tessellation levels have no effect in this mode.

As with quad tessellation above, isoline tessellation begins with a rectangle. The *u* = 0 and *u* = 1 edges of the rectangle are subdivided according to the first outer tessellation level. For the purposes of this subdivision, the tessellation spacing mode is ignored and treated as `equal_spacing`. An isoline is drawn connecting each vertex on the *u* = 0 rectangle edge to the corresponding vertex on the *u* = 1 rectangle edge, except that no line is drawn between (0,1) and (1,1). If the number of isolines on the subdivided *u* = 0 and *u* = 1 edges is *n*, this process will result in *n* equally spaced lines with constant *v* coordinates of $0, \frac{1}{n}, \frac{2}{n}, \dots, \frac{n-1}{n}$.

Each of the *n* isolines is then subdivided according to the second outer tessellation level and the tessellation spacing, resulting in *m* line segments. Each segment of each line is emitted by the tessellator. These line segments are generated with a topology similar to [line lists](#), except that the order in which each line is generated, and the order in which the vertices are generated for each line segment, are implementation-dependent.

22.9. Tessellation Point Mode

For all primitive modes, the tessellator is capable of generating points instead of lines or triangles. If the tessellation control or tessellation evaluation shader specifies the **OpExecutionMode PointMode**, the primitive generator will generate one point for each distinct vertex produced by tessellation, rather than emitting triangles or lines. Otherwise, the tessellator will produce a collection of line segments or triangles according to the primitive mode. These points are generated with a topology similar to [point lists](#), except the order in which the points are generated for each input primitive is undefined.

22.10. Tessellation Pipeline State

The **pTessellationState** member of **VkGraphicsPipelineCreateInfo** is a pointer to a **VkPipelineTessellationStateCreateInfo** structure.

The **VkPipelineTessellationStateCreateInfo** structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineTessellationStateCreateInfo {
    VkStructureType          sType;
    const void*              pNext;
    VkPipelineTessellationStateCreateFlags flags;
    uint32_t                 patchControlPoints;
} VkPipelineTessellationStateCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `patchControlPoints` is the number of control points per patch.

Valid Usage

- VUID-VkPipelineTessellationStateCreateInfo-patchControlPoints-01214
`patchControlPoints` **must** be greater than zero and less than or equal to `VkPhysicalDeviceLimits::maxTessellationPatchSize`

Valid Usage (Implicit)

- VUID-VkPipelineTessellationStateCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_TESSELLATION_STATE_CREATE_INFO`
- VUID-VkPipelineTessellationStateCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkPipelineTessellationStateCreateInfo-flags-zero bitmask
`flags` **must** be 0

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineTessellationStateCreateFlags;
```

`VkPipelineTessellationStateCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

Chapter 23. Geometry Shading

The geometry shader operates on a group of vertices and their associated data assembled from a single input primitive, and emits zero or more output primitives and the group of vertices and their associated data required for each output primitive. Geometry shading is enabled when a geometry shader is included in the pipeline.

23.1. Geometry Shader Input Primitives

Each geometry shader invocation has access to all vertices in the primitive (and their associated data), which are presented to the shader as an array of inputs.

The input primitive type expected by the geometry shader is specified with an `OpExecutionMode` instruction in the geometry shader, and **must** match the incoming primitive type specified by either the pipeline's `primitive topology` if tessellation is inactive, or the `tessellation mode` if tessellation is active, as follows:

- An input primitive type of `InputPoints` **must** only be used with a pipeline topology of `VK_PRIMITIVE_TOPOLOGY_POINT_LIST`, or with a tessellation shader specifying `PointMode`. The input arrays always contain one element, as described by the `point list topology` or `tessellation in point mode`.
- An input primitive type of `InputLines` **must** only be used with a pipeline topology of `VK_PRIMITIVE_TOPOLOGY_LINE_LIST` or `VK_PRIMITIVE_TOPOLOGY_LINE_STRIP`, or with a tessellation shader specifying `IsoLines` that does not specify `PointMode`. The input arrays always contain two elements, as described by the `line list topology` or `line strip topology`, or by `isoline tessellation`.
- An input primitive type of `InputLinesAdjacency` **must** only be used when tessellation is inactive, with a pipeline topology of `VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY` or `VK_PRIMITIVE_TOPOLOGY_LINE_STRIP_WITH_ADJACENCY`. The input arrays always contain four elements, as described by the `line list with adjacency topology` or `line strip with adjacency topology`.
- An input primitive type of `Triangles` **must** only be used with a pipeline topology of `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST`, `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP`, or `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN`; or with a tessellation shader specifying `Quads` or `Triangles` that does not specify `PointMode`. The input arrays always contain three elements, as described by the `triangle list topology`, `triangle strip topology`, or `triangle fan topology`, or by `triangle` or `quad tessellation`. Vertices **may** be in a different absolute order than specified by the topology, but **must** adhere to the specified winding order.
- An input primitive type of `InputTrianglesAdjacency` **must** only be used when tessellation is inactive, with a pipeline topology of `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY` or `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP_WITH_ADJACENCY`. The input arrays always contain six elements, as described by the `triangle list with adjacency topology` or `triangle strip with adjacency topology`. Vertices **may** be in a different absolute order than specified by the topology, but **must** adhere to the specified winding order, and the vertices making up the main primitive **must** still occur at the first, third, and fifth index.

23.2. Geometry Shader Output Primitives

A geometry shader generates primitives in one of three output modes: points, line strips, or triangle strips. The primitive mode is specified in the shader using an `OpExecutionMode` instruction with the `OutputPoints`, `OutputLineStrip` or `OutputTriangleStrip` modes, respectively. Each geometry shader **must** include exactly one output primitive mode.

The vertices output by the geometry shader are assembled into points, lines, or triangles based on the output primitive type and the resulting primitives are then further processed as described in [Rasterization](#). If the number of vertices emitted by the geometry shader is not sufficient to produce a single primitive, vertices corresponding to incomplete primitives are not processed by subsequent pipeline stages. The number of vertices output by the geometry shader is limited to a maximum count specified in the shader.

The maximum output vertex count is specified in the shader using an `OpExecutionMode` instruction with the mode set to `OutputVertices` and the maximum number of vertices that will be produced by the geometry shader specified as a literal. Each geometry shader **must** specify a maximum output vertex count.

23.3. Multiple Invocations of Geometry Shaders

Geometry shaders **can** be invoked more than one time for each input primitive. This is known as *geometry shader instancing* and is requested by including an `OpExecutionMode` instruction with `mode` specified as `Invocations` and the number of invocations specified as an integer literal.

In this mode, the geometry shader will execute at least *n* times for each input primitive, where *n* is the number of invocations specified in the `OpExecutionMode` instruction. The instance number is available to each invocation as a built-in input using `InvocationId`.

23.4. Geometry Shader Primitive Ordering

Limited guarantees are provided for the relative ordering of primitives produced by a geometry shader, as they pertain to [primitive order](#).

- For instanced geometry shaders, the output primitives generated from each input primitive are passed to subsequent pipeline stages using the invocation number to order the primitives, from least to greatest.
- All output primitives generated from a given input primitive are passed to subsequent pipeline stages before any output primitives generated from subsequent input primitives.

Chapter 24. Fixed-Function Vertex Post-Processing

After [pre-rasterization shader stages](#), the following fixed-function operations are applied to vertices of the resulting primitives:

- Flat shading (see [Flat Shading](#)).
- Primitive clipping, including application-defined half-spaces (see [Primitive Clipping](#)).
- Shader output attribute clipping (see [Clipping Shader Outputs](#)).
- Perspective division on clip coordinates (see [Coordinate Transformations](#)).
- Viewport mapping, including depth range scaling (see [Controlling the Viewport](#)).
- Front face determination for polygon primitives (see [Basic Polygon Rasterization](#)).

Next, rasterization is performed on primitives as described in chapter [Rasterization](#).

24.1. Flat Shading

Flat shading a vertex output attribute means to assign all vertices of the primitive the same value for that output. The output values assigned are those of the *provoking vertex* of the primitive. Flat shading is applied to those vertex attributes that [match](#) fragment input attributes which are decorated as [Flat](#).

If neither [geometry](#) nor [tessellation shading](#) is active, the provoking vertex is determined by the [primitive topology](#) defined by `VkPipelineInputAssemblyStateCreateInfo:topology` used to execute the [drawing command](#).

If [geometry shading](#) is active, the provoking vertex is determined by the [primitive topology](#) defined by the [OutputPoints](#), [OutputLineStrip](#), or [OutputTriangleStrip](#) execution mode.

If [tessellation shading](#) is active but [geometry shading](#) is not, the provoking vertex **may** be any of the vertices in each primitive.

24.2. Primitive Clipping

Primitives are culled against the *cull volume* and then clipped to the *clip volume*. In clip coordinates, the *view volume* is defined by:

$$\begin{aligned} -w_c &\leq x_c \leq w_c \\ -w_c &\leq y_c \leq w_c \\ z_m &\leq z_c \leq w_c \end{aligned}$$

where z_m is equal to zero.

This view volume **can** be further restricted by as many as `VkPhysicalDeviceLimits::maxClipDistances` application-defined half-spaces.

The cull volume is the intersection of up to `VkPhysicalDeviceLimits::maxCullDistances` application-defined half-spaces (if no application-defined cull half-spaces are enabled, culling against the cull volume is skipped).

A shader **must** write a single cull distance for each enabled cull half-space to elements of the `CullDistance` array. If the cull distance for any enabled cull half-space is negative for all of the vertices of the primitive under consideration, the primitive is discarded. Otherwise the primitive is clipped against the clip volume as defined below.

The clip volume is the intersection of up to `VkPhysicalDeviceLimits::maxClipDistances` application-defined half-spaces with the view volume (if no application-defined clip half-spaces are enabled, the clip volume is the view volume).

A shader **must** write a single clip distance for each enabled clip half-space to elements of the `ClipDistance` array. Clip half-space i is then given by the set of points satisfying the inequality

$$c_i(\mathbf{P}) \geq 0$$

where $c_i(\mathbf{P})$ is the clip distance i at point $\mathbf{P}$. For point primitives, $c_i(\mathbf{P})$ is simply the clip distance for the vertex in question. For line and triangle primitives, per-vertex clip distances are interpolated using a weighted mean, with weights derived according to the algorithms described in sections [Basic Line Segment Rasterization](#) and [Basic Polygon Rasterization](#), using the perspective interpolation equations.

The number of application-defined clip and cull half-spaces that are enabled is determined by the explicit size of the built-in arrays `ClipDistance` and `CullDistance`, respectively, declared as an output in the interface of the entry point of the final shader stage before clipping.

Depth clamping is enabled or disabled via the `depthClampEnable` enable of the `VkPipelineRasterizationStateCreateInfo` structure. Depth clipping is disabled when `depthClampEnable` is `VK_TRUE`.

When depth clipping is disabled, the plane equation

$$Z_m \leq Z_c \leq W_c$$

(see the clip volume definition above) is ignored by view volume clipping (effectively, there is no near or far plane clipping).

If the primitive under consideration is a point or line segment, then clipping passes it unchanged if its vertices lie entirely within the clip volume.

If a point's vertex lies outside of the clip volume, the entire primitive **may** be discarded.

If either of a line segment's vertices lie outside of the clip volume, the line segment **may** be clipped, with new vertex coordinates computed for each vertex that lies outside the clip volume. A clipped line segment endpoint lies on both the original line segment and the boundary of the clip volume.

This clipping produces a value, $0 \leq t \leq 1$, for each clipped vertex. If the coordinates of a clipped

vertex are $\mathbf{P}$ and the unclipped line segment's vertex coordinates are $\mathbf{P}_1$ and $\mathbf{P}_2$, then t satisfies the following equation

$$\mathbf{P} = t \mathbf{P}_1 + (1-t) \mathbf{P}_2.$$

t is used to clip vertex output attributes as described in [Clipping Shader Outputs](#).

If the primitive is a polygon, it passes unchanged if every one of its edges lies entirely inside the clip volume, and is either clipped or discarded otherwise. If the edges of the polygon intersect the boundary of the clip volume, the intersecting edges are reconnected by new edges that lie along the boundary of the clip volume - in some cases requiring the introduction of new vertices into a polygon.

If a polygon intersects an edge of the clip volume's boundary, the clipped polygon **must** include a point on this boundary edge.

Primitives rendered with application-defined half-spaces **must** satisfy a complementarity criterion. Suppose a series of primitives is drawn where each vertex i has a single specified clip distance d_i (or a number of similarly specified clip distances, if multiple half-spaces are enabled). Next, suppose that the same series of primitives are drawn again with each such clip distance replaced by $-d_i$ (and the graphics pipeline is otherwise the same). In this case, primitives **must** not be missing any pixels, and pixels **must** not be drawn twice in regions where those primitives are cut by the clip planes.

24.3. Clipping Shader Outputs

Next, vertex output attributes are clipped. The output values associated with a vertex that lies within the clip volume are unaffected by clipping. If a primitive is clipped, however, the output values assigned to vertices produced by clipping are clipped.

Let the output values assigned to the two vertices $\mathbf{P}_1$ and $\mathbf{P}_2$ of an unclipped edge be $\mathbf{c}_1$ and $\mathbf{c}_2$. The value of t (see [Primitive Clipping](#)) for a clipped point $\mathbf{P}$ is used to obtain the output value associated with $\mathbf{P}$ as

$$\mathbf{c} = t \mathbf{c}_1 + (1-t) \mathbf{c}_2.$$

(Multiplying an output value by a scalar means multiplying each of x , y , z , and w by the scalar.)

Since this computation is performed in clip space before division by w_c , clipped output values are perspective-correct.

Polygon clipping creates a clipped vertex along an edge of the clip volume's boundary. This situation is handled by noting that polygon clipping proceeds by clipping against one half-space at a time. Output value clipping is done in the same way, so that clipped points always occur at the intersection of polygon edges (possibly already clipped) with the clip volume's boundary.

For vertex output attributes whose matching fragment input attributes are decorated with **NoPerspective**, the value of t used to obtain the output value associated with $\mathbf{P}$ will be adjusted to produce results that vary linearly in framebuffer space.

Output attributes of integer or unsigned integer type **must** always be flat shaded. Flat shaded attributes are constant over the primitive being rasterized (see [Basic Line Segment Rasterization](#) and [Basic Polygon Rasterization](#)), and no interpolation is performed. The output value **c** is taken from either **c₁** or **c₂**, since flat shading has already occurred and the two values are identical.

24.4. Coordinate Transformations

Clip coordinates for a vertex result from shader execution, which yields a vertex coordinate **Position**.

Perspective division on clip coordinates yields *normalized device coordinates*, followed by a *viewport* transformation (see [Controlling the Viewport](#)) to convert these coordinates into *framebuffer coordinates*.

If a vertex in clip coordinates has a position given by

$$\begin{pmatrix} x_c \\ y_c \\ z_c \\ w_c \end{pmatrix}$$

then the vertex's normalized device coordinates are

$$\begin{pmatrix} x_d \\ y_d \\ z_d \end{pmatrix} = \begin{pmatrix} \frac{x_c}{w_c} \\ \frac{y_c}{w_c} \\ \frac{z_c}{w_c} \end{pmatrix}$$

24.5. Controlling the Viewport

The viewport transformation is determined by the selected viewport's width and height in pixels, **p_x** and **p_y**, respectively, and its center (**o_x**, **o_y**) (also in pixels), as well as its depth range min and max determining a depth range scale value **p_z** and a depth range bias value **o_z** (defined below). The vertex's framebuffer coordinates (**x_f**, **y_f**, **z_f**) are given by

$$x_f = (p_x / 2) x_d + o_x$$

$$y_f = (p_y / 2) y_d + o_y$$

$$z_f = p_z \times z_d + o_z$$

Multiple viewports are available, numbered zero up to **VkPhysicalDeviceLimits::maxViewports** minus one. The number of viewports used by a pipeline is controlled by the **viewportCount** member of the **VkPipelineViewportStateCreateInfo** structure used in pipeline creation.

x_f and y_f have limited precision, where the number of fractional bits retained is specified by `VkPhysicalDeviceLimits::subPixelPrecisionBits`.

The `VkPipelineViewportStateCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineViewportStateCreateInfo {
    VkStructureType          sType;
    const void*              pNext;
    VkPipelineViewportStateCreateFlags flags;
    uint32_t                 viewportCount;
    const VkViewport*        pViewports;
    uint32_t                 scissorCount;
    const VkRect2D*          pScissors;
} VkPipelineViewportStateCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `viewportCount` is the number of viewports used by the pipeline.
- `pViewports` is a pointer to an array of `VkViewport` structures, defining the viewport transforms. If the viewport state is dynamic, this member is ignored.
- `scissorCount` is the number of `scissors` and **must** match the number of viewports.
- `pScissors` is a pointer to an array of `VkRect2D` structures defining the rectangular bounds of the scissor for the corresponding viewport. If the scissor state is dynamic, this member is ignored.

Valid Usage

- VUID-VkPipelineViewportStateCreateInfo-viewportCount-01216
If the `multiViewport` feature is not enabled, `viewportCount` **must** not be greater than 1
- VUID-VkPipelineViewportStateCreateInfo-scissorCount-01217
If the `multiViewport` feature is not enabled, `scissorCount` **must** not be greater than 1
- VUID-VkPipelineViewportStateCreateInfo-viewportCount-01218
`viewportCount` **must** be less than or equal to `VkPhysicalDeviceLimits::maxViewports`
- VUID-VkPipelineViewportStateCreateInfo-scissorCount-01219
`scissorCount` **must** be less than or equal to `VkPhysicalDeviceLimits::maxViewports`
- VUID-VkPipelineViewportStateCreateInfo-x-02821
The `x` and `y` members of `offset` member of any element of `pScissors` **must** be greater than or equal to 0
- VUID-VkPipelineViewportStateCreateInfo-offset-02822
Evaluation of (`offset.x` + `extent.width`) **must** not cause a signed integer addition overflow for any element of `pScissors`
- VUID-VkPipelineViewportStateCreateInfo-offset-02823

Evaluation of `(offset.y + extent.height)` **must** not cause a signed integer addition overflow for any element of `pScissors`

- VUID-VkPipelineViewportStateCreateInfo-scissorCount-04134
If `scissorCount` and `viewportCount` are both not dynamic, then `scissorCount` and `viewportCount` **must** be identical
- VUID-VkPipelineViewportStateCreateInfo-viewportCount-04135
`viewportCount` **must** be greater than 0
- VUID-VkPipelineViewportStateCreateInfo-scissorCount-04136
`scissorCount` **must** be greater than 0

Valid Usage (Implicit)

- VUID-VkPipelineViewportStateCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_VIEWPORT_STATE_CREATE_INFO`
- VUID-VkPipelineViewportStateCreateInfo-pNext-pNext
`pNext` **must** be NULL
- VUID-VkPipelineViewportStateCreateInfo-flags-zerobitmask
`flags` **must** be 0

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineViewportStateCreateFlags;
```

`VkPipelineViewportStateCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

If a geometry shader is active and has an output variable decorated with `ViewportIndex`, the viewport transformation uses the viewport corresponding to the value assigned to `ViewportIndex` taken from an implementation-dependent vertex of each primitive. If `ViewportIndex` is outside the range zero to `viewportCount` minus one for a primitive, or if the geometry shader did not assign a value to `ViewportIndex` for all vertices of a primitive due to flow control, the values resulting from the viewport transformation of the vertices of such primitives are undefined. If no geometry shader is active, or if the geometry shader does not have an output decorated with `ViewportIndex`, the viewport numbered zero is used by the viewport transformation.

A single vertex **can** be used in more than one individual primitive, in primitives such as `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP`. In this case, the viewport transformation is applied separately for each primitive.

To [dynamically set](#) the viewport transformation parameters, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetViewport(
    VkCommandBuffer                commandBuffer,
    uint32_t                       firstViewport,
    uint32_t                       viewportCount,
    const VkViewport*              pViewports);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `firstViewport` is the index of the first viewport whose parameters are updated by the command.
- `viewportCount` is the number of viewports whose parameters are updated by the command.
- `pViewports` is a pointer to an array of `VkViewport` structures specifying viewport parameters.

This command sets the viewport transformation parameters state for subsequent drawing commands when the graphics pipeline is created with `VK_DYNAMIC_STATE_VIEWPORT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineViewportStateCreateInfo::pViewports` values used to create the currently active pipeline.

The viewport parameters taken from element `i` of `pViewports` replace the current state for the viewport index `firstViewport + i`, for `i` in `[0, viewportCount)`.

Valid Usage

- VUID-vkCmdSetViewport-firstViewport-01223
The sum of `firstViewport` and `viewportCount` **must** be between `1` and `VkPhysicalDeviceLimits::maxViewports`, inclusive
- VUID-vkCmdSetViewport-firstViewport-01224
If the `multiViewport` feature is not enabled, `firstViewport` **must** be `0`
- VUID-vkCmdSetViewport-viewportCount-01225
If the `multiViewport` feature is not enabled, `viewportCount` **must** be `1`

Valid Usage (Implicit)

- VUID-vkCmdSetViewport-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetViewport-pViewports-parameter
`pViewports` **must** be a valid pointer to an array of `viewportCount` valid `VkViewport` structures
- VUID-vkCmdSetViewport-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetViewport-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations

- VUID-vkCmdSetViewport-viewportCount-arraylength
`viewportCount` **must** be greater than 0

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Graphics	State

Both `VkPipelineViewportStateCreateInfo` and `vkCmdSetViewport` use `VkViewport` to set the viewport transformation parameters.

The `VkViewport` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkViewport {
    float    x;
    float    y;
    float    width;
    float    height;
    float    minDepth;
    float    maxDepth;
} VkViewport;
```

- `x` and `y` are the viewport's upper left corner (x,y).
- `width` and `height` are the viewport's width and height, respectively.
- `minDepth` and `maxDepth` are the depth range for the viewport.

NOTE Despite their names, `minDepth` **can** be less than, equal to, or greater than `maxDepth`.

The framebuffer depth coordinate `zf` **may** be represented using either a fixed-point or floating-point representation. However, a floating-point representation **must** be used if the depth/stencil attachment has a floating-point depth component. If an m-bit fixed-point representation is used, we assume that it represents each value $\frac{k}{2^m-1}$, where $k \in \{0, 1, \dots, 2^m-1\}$, as k (e.g. 1.0 is represented in binary as a string of all ones).

The viewport parameters shown in the above equations are found from these values as

$$o_x = x + \text{width} / 2$$

$$o_y = y + \text{height} / 2$$

$$o_z = \text{minDepth}$$

$$p_x = \text{width}$$

$$p_y = \text{height}$$

$$p_z = \text{maxDepth} - \text{minDepth}$$

The width and height of the [implementation-dependent maximum viewport dimensions](#) **must** be greater than or equal to the width and height of the largest image which **can** be created and attached to a framebuffer.

The floating-point viewport bounds are represented with an [implementation-dependent precision](#).

Valid Usage

- VUID-VkViewport-width-01770
width must be greater than **0.0**
- VUID-VkViewport-width-01771
width must be less than or equal to `VkPhysicalDeviceLimits::maxViewportDimensions[0]`
- VUID-VkViewport-apiVersion-07917
If the `VK_KHR_maintenance1` extension is not enabled, the `VK_AMD_negative_viewport_height` extension is not enabled, and `VkPhysicalDeviceProperties::apiVersion` is less than Vulkan 1.1, **height must** be greater than **0.0**
- VUID-VkViewport-height-01773
The absolute value of **height must** be less than or equal to `VkPhysicalDeviceLimits::maxViewportDimensions[1]`
- VUID-VkViewport-x-01774
x must be greater than or equal to `viewportBoundsRange[0]`
- VUID-VkViewport-x-01232
(**x + width must** be less than or equal to `viewportBoundsRange[1]`
- VUID-VkViewport-y-01775
y must be greater than or equal to `viewportBoundsRange[0]`

- VUID-VkViewport-y-01233
(y + height) **must** be less than or equal to viewportBoundsRange[1]
- VUID-VkViewport-minDepth-01234
minDepth **must** be between 0.0 and 1.0, inclusive
- VUID-VkViewport-maxDepth-01235
maxDepth **must** be between 0.0 and 1.0, inclusive

Chapter 25. Rasterization

Rasterization is the process by which a primitive is converted to a two-dimensional image. Each discrete location of this image contains associated data such as depth, color, or other attributes.

Rasterizing a primitive begins by determining which squares of an integer grid in framebuffer coordinates are occupied by the primitive, and assigning one or more depth values to each such square. This process is described below for points, lines, and polygons.

A grid square, including its (x,y) framebuffer coordinates, z (depth), and associated data added by fragment shaders, is called a fragment. A fragment is located by its upper left corner, which lies on integer grid coordinates.

Rasterization operations also refer to a fragment's sample locations, which are offset by fractional values from its upper left corner. The rasterization rules for points, lines, and triangles involve testing whether each sample location is inside the primitive. Fragments need not actually be square, and rasterization rules are not affected by the aspect ratio of fragments. Display of non-square grids, however, will cause rasterized points and line segments to appear fatter in one direction than the other.

We assume that fragments are square, since it simplifies antialiasing and texturing. After rasterization, fragments are processed by [fragment operations](#).

Several factors affect rasterization, including the members of [VkPipelineRasterizationStateCreateInfo](#) and [VkPipelineMultisampleStateCreateInfo](#).

The [VkPipelineRasterizationStateCreateInfo](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineRasterizationStateCreateInfo {
    VkStructureType           sType;
    const void*               pNext;
    VkPipelineRasterizationStateCreateFlags flags;
    VkBool32                  depthClampEnable;
    VkBool32                  rasterizerDiscardEnable;
    VkPolygonMode              polygonMode;
    VkCullModeFlags            cullMode;
    VkFrontFace                frontFace;
    VkBool32                  depthBiasEnable;
    float                      depthBiasConstantFactor;
    float                      depthBiasClamp;
    float                      depthBiasSlopeFactor;
    float                      lineWidth;
} VkPipelineRasterizationStateCreateInfo;
```

- `sType` is a [VkStructureType](#) value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.

- **depthClampEnable** controls whether to clamp the fragment's depth values as described in [Depth Test](#). Enabling depth clamp will also disable clipping primitives to the z planes of the frustum as described in [Primitive Clipping](#).
- **rasterizerDiscardEnable** controls whether primitives are discarded immediately before the rasterization stage.
- **polygonMode** is the triangle rendering mode. See [VkPolygonMode](#).
- **cullMode** is the triangle facing direction used for primitive culling. See [VkCullModeFlagBits](#).
- **frontFace** is a [VkFrontFace](#) value specifying the front-facing triangle orientation to be used for culling.
- **depthBiasEnable** controls whether to bias fragment depth values.
- **depthBiasConstantFactor** is a scalar factor controlling the constant depth value added to each fragment.
- **depthBiasClamp** is the maximum (or minimum) depth bias of a fragment.
- **depthBiasSlopeFactor** is a scalar factor applied to a fragment's slope in depth bias calculations.
- **lineWidth** is the width of rasterized line segments.

Valid Usage

- VUID-VkPipelineRasterizationStateCreateInfo-depthClampEnable-00782
If the **depthClamp** feature is not enabled, **depthClampEnable** **must** be **VK_FALSE**
- VUID-VkPipelineRasterizationStateCreateInfo-polygonMode-01507
If the **fillModeNonSolid** feature is not enabled, **polygonMode** **must** be **VK_POLYGON_MODE_FILL**

Valid Usage (Implicit)

- VUID-VkPipelineRasterizationStateCreateInfo-sType-sType
sType **must** be **VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_STATE_CREATE_INFO**
- VUID-VkPipelineRasterizationStateCreateInfo-pNext-pNext
pNext **must** be **NULL**
- VUID-VkPipelineRasterizationStateCreateInfo-flags-zeroBitmask
flags **must** be **0**
- VUID-VkPipelineRasterizationStateCreateInfo-polygonMode-parameter
polygonMode **must** be a valid [VkPolygonMode](#) value
- VUID-VkPipelineRasterizationStateCreateInfo-cullMode-parameter
cullMode **must** be a valid combination of [VkCullModeFlagBits](#) values
- VUID-VkPipelineRasterizationStateCreateInfo-frontFace-parameter
frontFace **must** be a valid [VkFrontFace](#) value

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineRasterizationStateCreateFlags;
```

`VkPipelineRasterizationStateCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

The `VkPipelineMultisampleStateCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineMultisampleStateCreateInfo {
    VkStructureType           sType;
    const void*               pNext;
    VkPipelineMultisampleStateCreateFlags flags;
    VkSampleCountFlagBits     rasterizationSamples;
    VkBool32                  sampleShadingEnable;
    float                     minSampleShading;
    const VkSampleMask*       pSampleMask;
    VkBool32                  alphaToCoverageEnable;
    VkBool32                  alphaToOneEnable;
} VkPipelineMultisampleStateCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `rasterizationSamples` is a `VkSampleCountFlagBits` value specifying the number of samples used in rasterization.
- `sampleShadingEnable` **can** be used to enable [Sample Shading](#).
- `minSampleShading` specifies a minimum fraction of sample shading if `sampleShadingEnable` is `VK_TRUE`.
- `pSampleMask` is a pointer to an array of `VkSampleMask` values used in the [sample mask test](#).
- `alphaToCoverageEnable` controls whether a temporary coverage value is generated based on the alpha component of the fragment's first color output as specified in the [Multisample Coverage](#) section.
- `alphaToOneEnable` controls whether the alpha component of the fragment's first color output is replaced with one as described in [Multisample Coverage](#).

Each bit in the sample mask is associated with a unique [sample index](#) as defined for the [coverage mask](#). Each bit `b` for mask word `w` in the sample mask corresponds to sample index `i`, where $i = 32 \times w + b$. `pSampleMask` has a length equal to $\lceil \text{rasterizationSamples} / 32 \rceil$ words.

If `pSampleMask` is `NULL`, it is treated as if the mask has all bits set to 1.

Valid Usage

- VUID-VkPipelineMultisampleStateCreateInfo-sampleShadingEnable-00784
If the `sampleRateShading` feature is not enabled, `sampleShadingEnable` **must** be `VK_FALSE`
- VUID-VkPipelineMultisampleStateCreateInfo-alphaToOneEnable-00785
If the `alphaToOne` feature is not enabled, `alphaToOneEnable` **must** be `VK_FALSE`
- VUID-VkPipelineMultisampleStateCreateInfo-minSampleShading-00786
`minSampleShading` **must** be in the range [0,1]

Valid Usage (Implicit)

- VUID-VkPipelineMultisampleStateCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_MULTISAMPLE_STATE_CREATE_INFO`
- VUID-VkPipelineMultisampleStateCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkPipelineMultisampleStateCreateInfo-flags-zerobitmask
`flags` **must** be 0
- VUID-VkPipelineMultisampleStateCreateInfo-rasterizationSamples-parameter
`rasterizationSamples` **must** be a valid `VkSampleCountFlagBits` value
- VUID-VkPipelineMultisampleStateCreateInfo-pSampleMask-parameter
If `pSampleMask` is not `NULL`, `pSampleMask` **must** be a valid pointer to an array of $\lceil \frac{\text{rasterizationSamples}}{32} \rceil$ `VkSampleMask` values

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineMultisampleStateCreateFlags;
```

`VkPipelineMultisampleStateCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

The elements of the sample mask array are of type `VkSampleMask`, each representing 32 bits of coverage information:

```
// Provided by VK_VERSION_1_0
typedef uint32_t VkSampleMask;
```

Rasterization only generates fragments which cover one or more pixels inside the framebuffer. Pixels outside the framebuffer are never considered covered in the fragment. Fragments which would be produced by application of any of the primitive rasterization rules described below but which lie outside the framebuffer are not produced, nor are they processed by any later stage of the pipeline, including any of the [fragment operations](#).

Surviving fragments are processed by fragment shaders. Fragment shaders determine associated

data for fragments, and **can** also modify or replace their assigned depth values.

25.1. Discarding Primitives Before Rasterization

Primitives are discarded before rasterization if the `rasterizerDiscardEnable` member of `VkPipelineRasterizationStateCreateInfo` is enabled. When enabled, primitives are discarded after they are processed by the last active shader stage in the pipeline before rasterization.

25.2. Rasterization Order

Within a subpass of a `render pass instance`, for a given (x,y,layer,sample) sample location, the following operations are guaranteed to execute in *rasterization order*, for each separate primitive that includes that sample location:

1. `Fragment operations`, in the order defined
2. `Blending`, `logic operations`, and color writes

Execution of these operations for each primitive in a subpass occurs in `primitive order`.

25.3. Multisampling

Multisampling is a mechanism to antialias all Vulkan primitives: points, lines, and polygons. The technique is to sample all primitives multiple times at each pixel. Each sample in each framebuffer attachment has storage for a color, depth, and/or stencil value, such that per-fragment operations apply to each sample independently. The color sample values **can** be later *resolved* to a single color (see `Resolving Multisample Images` and the `Render Pass` chapter for more details on how to resolve multisample images to non-multisample images).

Vulkan defines rasterization rules for single-sample modes in a way that is equivalent to a multisample mode with a single sample in the center of each fragment.

Each fragment includes a `coverage mask` with a single bit for each sample in the fragment, and a number of depth values and associated data for each sample.

It is understood that each pixel has `rasterizationSamples` locations associated with it. These locations are exact positions, rather than regions or areas, and each is referred to as a sample point. The sample points associated with a pixel **must** be located inside or on the boundary of the unit square that is considered to bound the pixel. Furthermore, the relative locations of sample points **may** be identical for each pixel in the framebuffer, or they **may** differ.

If the current pipeline includes a fragment shader with one or more variables in its interface decorated with `Sample` and `Input`, the data associated with those variables will be assigned independently for each sample. The values for each sample **must** be evaluated at the location of the sample. The data associated with any other variables not decorated with `Sample` and `Input` need not be evaluated independently for each sample.

A *coverage mask* is generated for each fragment, based on which samples within that fragment are determined to be within the area of the primitive that generated the fragment.

Single pixel fragments have one set of samples. Each set of samples has a number of samples determined by `VkPipelineMultisampleStateCreateInfo::rasterizationSamples`. Each sample in a set is assigned a unique *sample index* i in the range $[0, \text{rasterizationSamples})$.

Each sample in a fragment is also assigned a unique *coverage index* j in the range $[0, n \times \text{rasterizationSamples})$, where n is the number of sets in the fragment. If the fragment contains a single set of samples, the *coverage index* is always equal to the *sample index*.

The coverage mask includes B bits packed into W words, defined as:

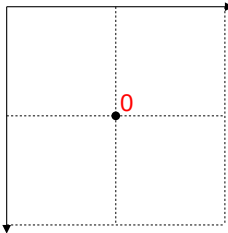
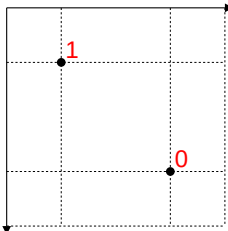
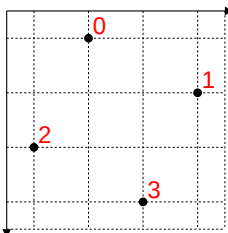
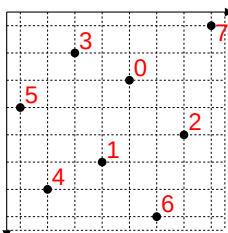
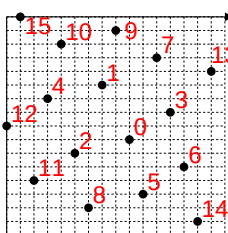
$$B = n \times \text{rasterizationSamples}$$

$$W = \lceil B/32 \rceil$$

Bit b in coverage mask word w is `1` if the sample with coverage index $j = 32 \times w + b$ is covered, and `0` otherwise.

If the `standardSampleLocations` member of `VkPhysicalDeviceLimits` is `VK_TRUE`, then the sample counts `VK_SAMPLE_COUNT_1_BIT`, `VK_SAMPLE_COUNT_2_BIT`, `VK_SAMPLE_COUNT_4_BIT`, `VK_SAMPLE_COUNT_8_BIT`, and `VK_SAMPLE_COUNT_16_BIT` have sample locations as listed in the following table, with the i th entry in the table corresponding to sample index i . `VK_SAMPLE_COUNT_32_BIT` and `VK_SAMPLE_COUNT_64_BIT` do not have standard sample locations. Locations are defined relative to an origin in the upper left corner of the fragment.

Table 23. Standard Sample Locations

Sample count	Sample Locations	
VK_SAMPLE_COUNT_1_BIT	(0.5,0.5)	
VK_SAMPLE_COUNT_2_BIT	(0.75,0.75) (0.25,0.25)	
VK_SAMPLE_COUNT_4_BIT	(0.375, 0.125) (0.875, 0.375) (0.125, 0.625) (0.625, 0.875)	
VK_SAMPLE_COUNT_8_BIT	(0.5625, 0.3125) (0.4375, 0.6875) (0.8125, 0.5625) (0.3125, 0.1875) (0.1875, 0.8125) (0.0625, 0.4375) (0.6875, 0.9375) (0.9375, 0.0625)	
VK_SAMPLE_COUNT_16_BIT	(0.5625, 0.5625) (0.4375, 0.3125) (0.3125, 0.625) (0.75, 0.4375) (0.1875, 0.375) (0.625, 0.8125) (0.8125, 0.6875) (0.6875, 0.1875) (0.375, 0.875) (0.5, 0.0625) (0.25, 0.125) (0.125, 0.75) (0.0, 0.5) (0.9375, 0.25) (0.875, 0.9375) (0.0625, 0.0)	

25.4. Sample Shading

Sample shading **can** be used to specify a minimum number of unique samples to process for each fragment. If sample shading is enabled, an implementation **must** invoke the fragment shader at least $\max(\text{VkPipelineMultisampleStateCreateInfo::minSampleShading} \times \text{VkPipelineMultisampleStateCreateInfo::rasterizationSamples}, 1)$ times per fragment. If `VkPipelineMultisampleStateCreateInfo::sampleShadingEnable` is `VK_TRUE`, sample shading is enabled.

If a fragment shader entry point **statically uses** an input variable decorated with a `BuiltIn` of `SampleId` or `SamplePosition`, sample shading is enabled and a value of `1.0` is used instead of `minSampleShading`. If a fragment shader entry point **statically uses** an input variable decorated with `Sample`, sample shading **may** be enabled and a value of `1.0` will be used instead of `minSampleShading` if it is.

NOTE

If a shader decorates an input variable with `Sample` and that value meaningfully impacts the output of a shader, sample shading will be enabled to ensure that the input is in fact interpolated per-sample. This is inherent to the specification and not spelled out here - if an application simply declares such a variable it is implementation-defined whether sample shading is enabled or not. It is possible to see the effects of this by using atomics in the shader or using a pipeline statistics query to query the number of fragment invocations, even if the shader itself does not use any per-sample variables.

If there are fewer fragment invocations than **covered samples**, implementations **may** include those samples in fragment shader invocations in any manner as long as covered samples are all shaded at least once, and each invocation that is not a **helper invocation** covers at least one sample.

25.5. Points

A point is drawn by generating a set of fragments in the shape of a square centered around the vertex of the point. Each vertex has an associated point size controlling the width/height of that square. The point size is taken from the (potentially clipped) shader built-in `PointSize` written by:

- the geometry shader, if active;
- the tessellation evaluation shader, if active and no geometry shader is active;
- the vertex shader, otherwise

and clamped to the implementation-dependent point size range `[pointSizeRange[0], pointSizeRange[1]]`. The value written to `PointSize` **must** be greater than zero.

Not all point sizes need be supported, but the size `1.0` **must** be supported. The range of supported sizes and the size of evenly-spaced gradations within that range are implementation-dependent. The range and gradations are obtained from the `pointSizeRange` and `pointSizeGranularity` members of `VkPhysicalDeviceLimits`. If, for instance, the size range is from `0.1` to `2.0` and the gradation size is `0.1`, then the sizes `0.1`, `0.2`, ..., `1.9`, `2.0` are supported. Additional point sizes **may** also be supported. There is no requirement that these sizes be equally spaced. If an unsupported size is requested, the nearest supported size is used instead.

25.5.1. Basic Point Rasterization

Point rasterization produces a fragment for each fragment area group of framebuffer pixels with one or more sample points that intersect a region centered at the point's (x_p, y_p) . This region is a square with side equal to the current point size. Coverage bits that correspond to sample points that intersect the region are 1, other coverage bits are 0. All fragments produced in rasterizing a point are assigned the same associated data, which are those of the vertex corresponding to the point. However, the fragment shader built-in `PointCoord` contains point sprite texture coordinates. The s and t point sprite texture coordinates vary from zero to one across the point horizontally left-to-right and vertically top-to-bottom, respectively. The following formulas are used to evaluate s and t :

$$s = \frac{1}{2} + \frac{(x_p - x_f)}{\text{size}}$$
$$t = \frac{1}{2} + \frac{(y_p - y_f)}{\text{size}}$$

where size is the point's size; (x_p, y_p) is the location at which the point sprite coordinates are evaluated - this **may** be the framebuffer coordinates of the fragment center, or the location of a sample; and (x_f, y_f) is the exact, unrounded framebuffer coordinate of the vertex for the point.

25.6. Line Segments

To [dynamically set](#) the line width, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetLineWidth(
    VkCommandBuffer          commandBuffer,
    float                    lineWidth);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `lineWidth` is the width of rasterized line segments.

This command sets the line width for subsequent drawing commands when the graphics pipeline is created with `VK_DYNAMIC_STATE_LINE_WIDTH` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineRasterizationStateCreateInfo::lineWidth` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetLineWidth-lineWidth-00788
If the `wideLines` feature is not enabled, `lineWidth` **must** be `1.0`

Valid Usage (Implicit)

- VUID-vkCmdSetLineWidth-commandBuffer-parameter

`commandBuffer` **must** be a valid `VkCommandBuffer` handle

- VUID-vkCmdSetLineWidth-commandBuffer-recording

`commandBuffer` **must** be in the `recording` state

- VUID-vkCmdSetLineWidth-commandBuffer-cmdpool

The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Graphics	State

Not all line widths need be supported for line segment rasterization, but width 1.0 antialiased segments **must** be provided. The range and gradations are obtained from the `lineWidthRange` and `lineWidthGranularity` members of `VkPhysicalDeviceLimits`. If, for instance, the size range is from 0.1 to 2.0 and the gradation size is 0.1, then the sizes 0.1, 0.2, ..., 1.9, 2.0 are supported. Additional line widths **may** also be supported. There is no requirement that these widths be equally spaced. If an unsupported width is requested, the nearest supported width is used instead.

25.6.1. Basic Line Segment Rasterization

Rasterized line segments produce fragments which intersect a rectangle centered on the line segment. Two of the edges are parallel to the specified line segment; each is at a distance of one-half the current width from that segment in directions perpendicular to the direction of the line. The other two edges pass through the line endpoints and are perpendicular to the direction of the specified line segment. Coverage bits that correspond to sample points that intersect the rectangle are 1, other coverage bits are 0.

Next we specify how the data associated with each rasterized fragment are obtained. Let $\mathbf{p}_r = (x_d, y_d)$ be the framebuffer coordinates at which associated data are evaluated. This **may** be the center of a fragment or the location of a sample within the fragment. When `rasterizationSamples` is `VK_SAMPLE_COUNT_1_BIT`, the fragment center **must** be used. Let $\mathbf{p}_a = (x_a, y_a)$ and $\mathbf{p}_b = (x_b, y_b)$ be initial and final endpoints of the line segment, respectively. Set

$$t = \frac{(\mathbf{p}_r - \mathbf{p}_a) \cdot (\mathbf{p}_b - \mathbf{p}_a)}{\|\mathbf{p}_b - \mathbf{p}_a\|^2}$$

(Note that $t = 0$ at $\mathbf{p}_a$ and $t = 1$ at $\mathbf{p}_b$. Also note that this calculation projects the vector from $\mathbf{p}_a$ to $\mathbf{p}_r$ onto the line, and thus computes the normalized distance of the fragment along the line.)

If `strictLines` is `VK_TRUE`, line segments are rasterized using perspective or linear interpolation.

Perspective interpolation for a line segment interpolates two values in a manner that is correct when taking the perspective of the viewport into consideration, by way of the line segment's clip coordinates. An interpolated value f can be determined by

$$f = \frac{(1-t)f_a/w_a + tf_b/w_b}{(1-t)/w_a + t/w_b}$$

where f_a and f_b are the data associated with the starting and ending endpoints of the segment, respectively; w_a and w_b are the clip w coordinates of the starting and ending endpoints of the segment, respectively.

Linear interpolation for a line segment directly interpolates two values, and an interpolated value f can be determined by

$$f = (1-t)f_a + tf_b$$

where f_a and f_b are the data associated with the starting and ending endpoints of the segment, respectively.

The clip coordinate w for a sample is determined using perspective interpolation. The depth value z for a sample is determined using linear interpolation. Interpolation of fragment shader input values are determined by [Interpolation decorations](#).

The above description documents the preferred method of line rasterization, and **must** be used when the implementation advertises the `strictLines` limit in `VkPhysicalDeviceLimits` as `VK_TRUE`.

When `strictLines` is `VK_FALSE`, the edges of the lines are generated as a parallelogram surrounding the original line. The major axis is chosen by noting the axis in which there is the greatest distance between the line start and end points. If the difference is equal in both directions then the X axis is chosen as the major axis. Edges 2 and 3 are aligned to the minor axis and are centered on the endpoints of the line as in [Non strict lines](#), and each is `LineWidth` long. Edges 0 and 1 are parallel to the line and connect the endpoints of edges 2 and 3. Coverage bits that correspond to sample points that intersect the parallelogram are 1, other coverage bits are 0.

Samples that fall exactly on the edge of the parallelogram follow the polygon rasterization rules.

Interpolation occurs as if the parallelogram was decomposed into two triangles where each pair of vertices at each end of the line has identical attributes.

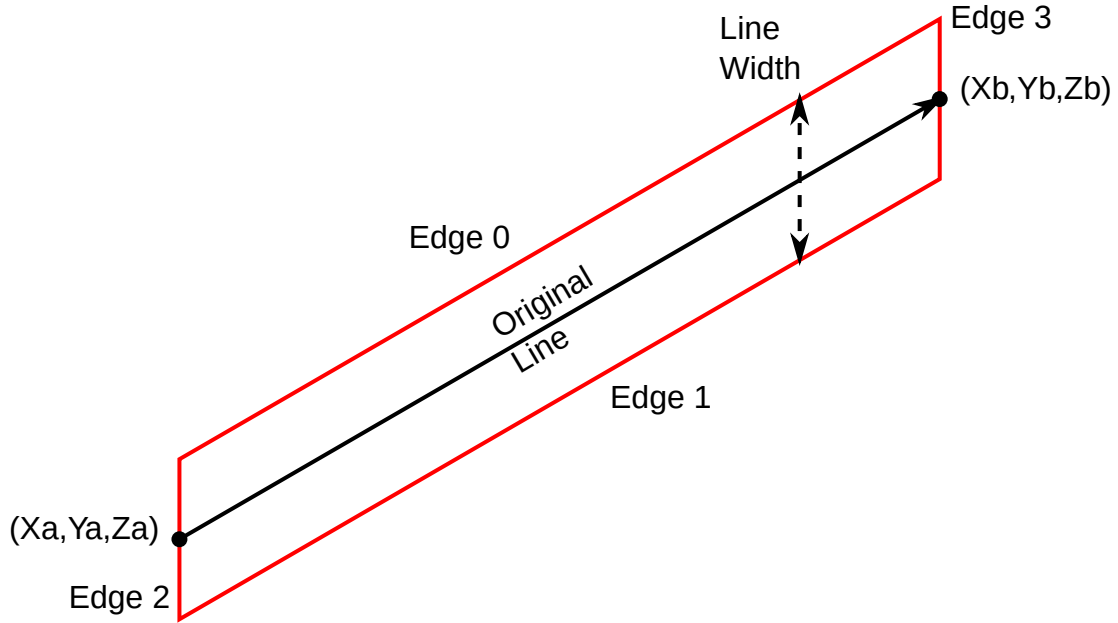


Figure 8. Non strict lines

Only when `strictLines` is `VK_FALSE` implementations **may** deviate from the non-strict line algorithm described above in the following ways:

- Implementations **may** instead interpolate each fragment according to the formula in [Basic Line Segment Rasterization](#) using the original line segment endpoints.
- Rasterization of non-antialiased non-strict line segments **may** be performed using the rules defined in [Bresenham Line Segment Rasterization](#).

25.6.2. Bresenham Line Segment Rasterization

Non-strict lines **may** also follow these rasterization rules for non-antialiased lines.

Line segment rasterization begins by characterizing the segment as either *x-major* or *y-major*. *x-major* line segments have slope in the closed interval $[-1,1]$; all other line segments are *y-major* (slope is determined by the segment's endpoints). We specify rasterization only for *x-major* segments except in cases where the modifications for *y-major* segments are not self-evident.

Ideally, Vulkan uses a *diamond-exit* rule to determine those fragments that are produced by rasterizing a line segment. For each fragment f with center at framebuffer coordinates x_f and y_f , define a diamond-shaped region that is the intersection of four half planes:

$$R_f = \{(x, y) \mid |x - x_f| + |y - y_f| < \frac{1}{2}\}$$

Essentially, a line segment starting at p_a and ending at p_b produces those fragments f for which the segment intersects R_f , except if p_b is contained in R_f .

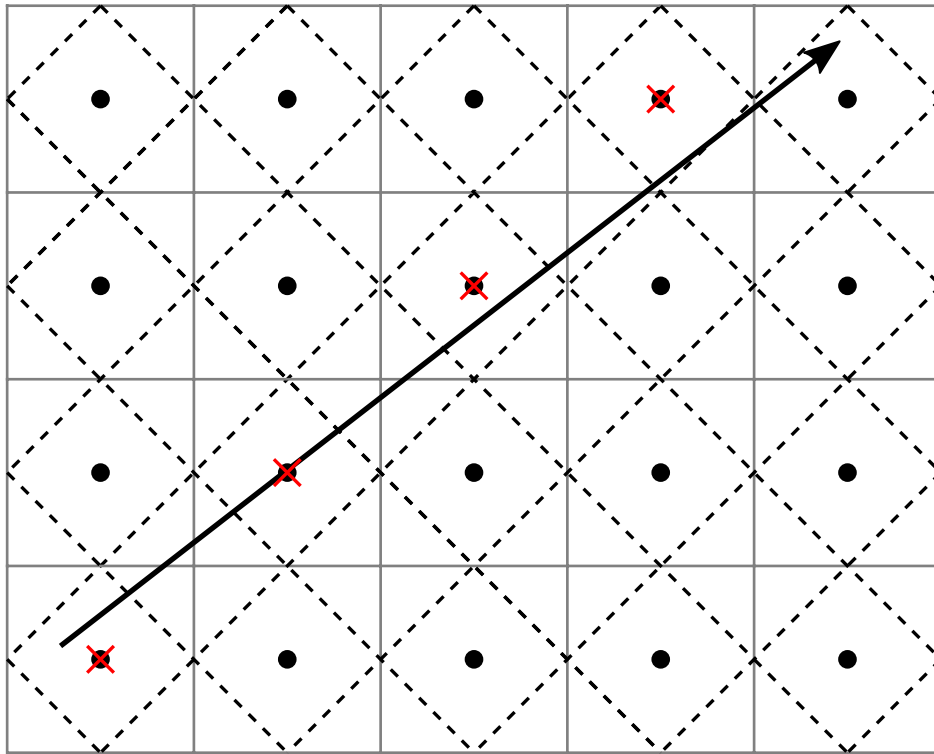


Figure 9. Visualization of Bresenham's algorithm

To avoid difficulties when an endpoint lies on a boundary of R_f we (in principle) perturb the supplied endpoints by a tiny amount. Let p_a and p_b have framebuffer coordinates (x_a, y_a) and (x_b, y_b) , respectively. Obtain the perturbed endpoints p_a' given by $(x_a, y_a) - (\epsilon, \epsilon^2)$ and p_b' given by $(x_b, y_b) - (\epsilon, \epsilon^2)$. Rasterizing the line segment starting at p_a and ending at p_b produces those fragments f for which the segment starting at p_a' and ending on p_b' intersects R_f , except if p_b' is contained in R_f . ϵ is chosen to be so small that rasterizing the line segment produces the same fragments when δ is substituted for ϵ for any $0 < \delta \leq \epsilon$.

When p_a and p_b lie on fragment centers, this characterization of fragments reduces to Bresenham's algorithm with one modification: lines produced in this description are "half-open", meaning that the final fragment (corresponding to p_b) is not drawn. This means that when rasterizing a series of connected line segments, shared endpoints will be produced only once rather than twice (as would occur with Bresenham's algorithm).

Implementations **may** use other line segment rasterization algorithms, subject to the following rules:

- The coordinates of a fragment produced by the algorithm **must** not deviate by more than one unit in either x or y framebuffer coordinates from a corresponding fragment produced by the diamond-exit rule.
- The total number of fragments produced by the algorithm **must** not differ from that produced by the diamond-exit rule by more than one.
- For an x -major line, two fragments that lie in the same framebuffer-coordinate column **must** not be produced (for a y -major line, two fragments that lie in the same framebuffer-coordinate row **must** not be produced).

- If two line segments share a common endpoint, and both segments are either x-major (both left-to-right or both right-to-left) or y-major (both bottom-to-top or both top-to-bottom), then rasterizing both segments **must** not produce duplicate fragments. Fragments also **must** not be omitted so as to interrupt continuity of the connected segments.

The actual width w of Bresenham lines is determined by rounding the line width to the nearest integer, clamping it to the implementation-dependent `LineWidthRange` (with both values rounded to the nearest integer), then clamping it to be no less than 1.

Bresenham line segments of width other than one are rasterized by offsetting them in the minor direction (for an x-major line, the minor direction is y , and for a y-major line, the minor direction is x) and producing a row or column of fragments in the minor direction. If the line segment has endpoints given by (x_0, y_0) and (x_1, y_1) in framebuffer coordinates, the segment with endpoints $(x_0, y_0 - \frac{w-1}{2})$ and $(x_1, y_1 - \frac{w-1}{2})$ is rasterized, but instead of a single fragment, a column of fragments of height w (a row of fragments of length w for a y-major segment) is produced at each x (y for y-major) location. The lowest fragment of this column is the fragment that would be produced by rasterizing the segment of width 1 with the modified coordinates.

The preferred method of attribute interpolation for a wide line is to generate the same attribute values for all fragments in the row or column described above, as if the adjusted line was used for interpolation and those values replicated to the other fragments, except for `FragCoord` which is interpolated as usual. Implementations **may** instead interpolate each fragment according to the formula in [Basic Line Segment Rasterization](#), using the original line segment endpoints.

When Bresenham lines are being rasterized, sample locations **may** all be treated as being at the pixel center (this **may** affect attribute and depth interpolation).

NOTE

The sample locations described above are **not** used for determining coverage, they are only used for things like attribute interpolation. The rasterization rules that determine coverage are defined in terms of whether the line intersects **pixels**, as opposed to the point sampling rules used for other primitive types. So these rules are independent of the sample locations. One consequence of this is that Bresenham lines cover the same pixels regardless of the number of rasterization samples, and cover all samples in those pixels (unless masked out or killed).

25.7. Polygons

A polygon results from the decomposition of a triangle strip, triangle fan or a series of independent triangles. Like points and line segments, polygon rasterization is controlled by several variables in the `VkPipelineRasterizationStateCreateInfo` structure.

25.7.1. Basic Polygon Rasterization

The first step of polygon rasterization is to determine whether the triangle is *back-facing* or *front-facing*. This determination is made based on the sign of the (clipped or unclipped) polygon's area computed in framebuffer coordinates. One way to compute this area is:

$$a = -\frac{1}{2} \sum_{i=0}^{n-1} x_f^i y_f^{i \oplus 1} - x_f^{i \oplus 1} y_f^i$$

where x_f^i and y_f^i are the x and y framebuffer coordinates of the i th vertex of the n -vertex polygon (vertices are numbered starting at zero for the purposes of this computation) and $i \oplus 1$ is $(i + 1) \bmod n$.

The interpretation of the sign of a is determined by the [VkPipelineRasterizationStateCreateInfo::frontFace](#) property of the currently active pipeline. Possible values are:

```
// Provided by VK_VERSION_1_0
typedef enum VkFrontFace {
    VK_FRONT_FACE_COUNTER_CLOCKWISE = 0,
    VK_FRONT_FACE_CLOCKWISE = 1,
} VkFrontFace;
```

- [VK_FRONT_FACE_COUNTER_CLOCKWISE](#) specifies that a triangle with positive area is considered front-facing.
- [VK_FRONT_FACE_CLOCKWISE](#) specifies that a triangle with negative area is considered front-facing.

Any triangle which is not front-facing is back-facing, including zero-area triangles.

Once the orientation of triangles is determined, they are culled according to the [VkPipelineRasterizationStateCreateInfo::cullMode](#) property of the currently active pipeline. Possible values are:

```
// Provided by VK_VERSION_1_0
typedef enum VkCullModeFlagBits {
    VK_CULL_MODE_NONE = 0,
    VK_CULL_MODE_FRONT_BIT = 0x00000001,
    VK_CULL_MODE_BACK_BIT = 0x00000002,
    VK_CULL_MODE_FRONT_AND_BACK = 0x00000003,
} VkCullModeFlagBits;
```

- [VK_CULL_MODE_NONE](#) specifies that no triangles are discarded
- [VK_CULL_MODE_FRONT_BIT](#) specifies that front-facing triangles are discarded
- [VK_CULL_MODE_BACK_BIT](#) specifies that back-facing triangles are discarded
- [VK_CULL_MODE_FRONT_AND_BACK](#) specifies that all triangles are discarded.

Following culling, fragments are produced for any triangles which have not been discarded.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkCullModeFlags;
```

[VkCullModeFlags](#) is a bitmask type for setting a mask of zero or more [VkCullModeFlagBits](#).

The rule for determining which fragments are produced by polygon rasterization is called *point sampling*. The two-dimensional projection obtained by taking the x and y framebuffer coordinates of the polygon's vertices is formed. Fragments are produced for any fragment area groups of pixels for which any sample points lie inside of this polygon. Coverage bits that correspond to sample points that satisfy the point sampling criteria are 1, other coverage bits are 0. Special treatment is given to a sample whose sample location lies on a polygon edge. In such a case, if two polygons lie on either side of a common edge (with identical endpoints) on which a sample point lies, then exactly one of the polygons **must** result in a covered sample for that fragment during rasterization. As for the data associated with each fragment produced by rasterizing a polygon, we begin by specifying how these values are produced for fragments in a triangle.

Barycentric coordinates are a set of three numbers, a, b, and c, each in the range [0,1], with $a + b + c = 1$. These coordinates uniquely specify any point p within the triangle or on the triangle's boundary as

$$p = a p_a + b p_b + c p_c$$

where p_a , p_b , and p_c are the vertices of the triangle. a, b, and c are determined by:

$$a = \frac{A(pp_b p_c)}{A(p_a p_b p_c)}, \quad b = \frac{A(pp_a p_c)}{A(p_a p_b p_c)}, \quad c = \frac{A(pp_a p_b)}{A(p_a p_b p_c)},$$

where $A(lmn)$ denotes the area in framebuffer coordinates of the triangle with vertices l, m, and n.

Denote an associated datum at p_a , p_b , or p_c as f_a , f_b , or f_c , respectively.

Perspective interpolation for a triangle interpolates three values in a manner that is correct when taking the perspective of the viewport into consideration, by way of the triangle's clip coordinates. An interpolated value f can be determined by

$$f = \frac{a f_a / w_a + b f_b / w_b + c f_c / w_c}{a / w_a + b / w_b + c / w_c}$$

where w_a , w_b , and w_c are the clip w coordinates of p_a , p_b , and p_c , respectively. a, b, and c are the barycentric coordinates of the location at which the data are produced.

Linear interpolation for a triangle directly interpolates three values, and an interpolated value f can be determined by

$$f = a f_a + b f_b + c f_c$$

where f_a , f_b , and f_c are the data associated with p_a , p_b , and p_c , respectively.

The clip coordinate w for a sample is determined using perspective interpolation. The depth value z for a sample is determined using linear interpolation. Interpolation of fragment shader input values are determined by [Interpolation decorations](#).

For a polygon with more than three edges, such as are produced by clipping a triangle, a convex combination of the values of the datum at the polygon's vertices **must** be used to obtain the value

assigned to each fragment produced by the rasterization algorithm. That is, it **must** be the case that at every fragment

$$f = \sum_{i=1}^n a_i f_i$$

where n is the number of vertices in the polygon and f_i is the value of f at vertex i . For each i , $0 \leq a_i \leq 1$ and $\sum_{i=1}^n a_i = 1$. The values of a_i **may** differ from fragment to fragment, but at vertex i , $a_i = 1$ and $a_j = 0$ for $j \neq i$.

NOTE

One algorithm that achieves the required behavior is to triangulate a polygon (without adding any vertices) and then treat each triangle individually as already discussed. A scan-line rasterizer that linearly interpolates data along each edge and then linearly interpolates data across each horizontal span from edge to edge also satisfies the restrictions (in this case the numerator and denominator of [perspective interpolation](#) are iterated independently, and a division is performed for each fragment).

25.7.2. Polygon Mode

Possible values of the `VkPipelineRasterizationStateCreateInfo::polygonMode` property of the currently active pipeline, specifying the method of rasterization for polygons, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkPolygonMode {
    VK_POLYGON_MODE_FILL = 0,
    VK_POLYGON_MODE_LINE = 1,
    VK_POLYGON_MODE_POINT = 2,
} VkPolygonMode;
```

- `VK_POLYGON_MODE_POINT` specifies that polygon vertices are drawn as points.
- `VK_POLYGON_MODE_LINE` specifies that polygon edges are drawn as line segments.
- `VK_POLYGON_MODE_FILL` specifies that polygons are rendered using the polygon rasterization rules in this section.

These modes affect only the final rasterization of polygons: in particular, a polygon's vertices are shaded and the polygon is clipped and possibly culled before these modes are applied.

The point size of the final rasterization of polygons when `polygons mode` is `VK_POLYGON_MODE_POINT` is implementation-dependent, and the point size **may** either be `PointSize` or 1.0.

25.7.3. Depth Bias

The depth values of all fragments generated by the rasterization of a polygon **can** be biased (offset) by a single depth bias value o that is computed for that polygon.

Depth Bias Enable

The depth bias computation is enabled by the [VkPipelineRasterizationStateCreateInfo::depthBiasEnable](#) value used to create the currently active pipeline. If the depth bias enable is [VK_FALSE](#), no bias is applied and the fragment's depth values are unchanged.

Depth Bias Computation

The depth bias depends on three parameters:

- [depthBiasSlopeFactor](#) scales the maximum depth slope m of the polygon
- [depthBiasConstantFactor](#) scales the parameter r of the depth attachment
- the scaled terms are summed to produce a value which is then clamped to a minimum or maximum value specified by [depthBiasClamp](#)

[depthBiasSlopeFactor](#), [depthBiasConstantFactor](#), and [depthBiasClamp](#) can each be positive, negative, or zero. These parameters are set as described for [vkCmdSetDepthBias](#) below.

The maximum depth slope m of a triangle is

$$m = \sqrt{\left(\frac{\partial z_f}{\partial x_f}\right)^2 + \left(\frac{\partial z_f}{\partial y_f}\right)^2}$$

where (x_f, y_f, z_f) is a point on the triangle. m may be approximated as

$$m = \max\left(\left|\frac{\partial z_f}{\partial x_f}\right|, \left|\frac{\partial z_f}{\partial y_f}\right|\right).$$

r is the minimum resolvable difference that depends on the depth attachment representation. It is the smallest difference in framebuffer coordinate z values that is guaranteed to remain distinct throughout polygon rasterization and in the depth attachment. All pairs of fragments generated by the rasterization of two polygons with otherwise identical vertices, but z_f values that differ by r , will have distinct depth values.

For fixed-point depth attachment representations, r is constant throughout the range of the entire depth attachment. Its value is implementation-dependent but **must** be at most

$$r = 2 \times 2^{-n}$$

where n is the number of bits used for the depth aspect.

For floating-point depth attachment, there is no single minimum resolvable difference. In this case, the minimum resolvable difference for a given polygon is dependent on the maximum exponent, e , in the range of z values spanned by the primitive. If n is the number of bits in the floating-point mantissa, the minimum resolvable difference, r , for the given primitive is defined as

$$r = 2^{e-n}$$

If no depth attachment is present, r is undefined.

The bias value o for a polygon is

$$o = \text{dbclamp}(m \times \text{depthBiasSlopeFactor} + r \times \text{depthBiasConstantFactor})$$

where $\text{dbclamp}(x) = \begin{cases} x & \text{depthBiasClamp} = 0 \text{ or NaN} \\ \min(x, \text{depthBiasClamp}) & \text{depthBiasClamp} > 0 \\ \max(x, \text{depthBiasClamp}) & \text{depthBiasClamp} < 0 \end{cases}$

m is computed as described above. If the depth attachment uses a fixed-point representation, m is a function of depth values in the range $[0,1]$, and o is applied to depth values in the same range.

Depth bias is applied to triangle topology primitives received by the rasterizer regardless of [polygon mode](#). Depth bias **may** also be applied to line and point topology primitives received by the rasterizer.

To [dynamically set](#) the depth bias parameters, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetDepthBias(
    VkCommandBuffer          commandBuffer,
    float                    depthBiasConstantFactor,
    float                    depthBiasClamp,
    float                    depthBiasSlopeFactor);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `depthBiasConstantFactor` is a scalar factor controlling the constant depth value added to each fragment.
- `depthBiasClamp` is the maximum (or minimum) depth bias of a fragment.
- `depthBiasSlopeFactor` is a scalar factor applied to a fragment's slope in depth bias calculations.

This command sets the depth bias parameters for subsequent drawing commands when the graphics pipeline is created with `VK_DYNAMIC_STATE_DEPTH_BIAS` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the corresponding `VkPipelineRasterizationStateCreateInfo::depthBiasConstantFactor`, `depthBiasClamp`, and `depthBiasSlopeFactor` values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetDepthBias-depthBiasClamp-00790
If the `depthBiasClamp` feature is not enabled, `depthBiasClamp` **must** be `0.0`

Valid Usage (Implicit)

- VUID-vkCmdSetDepthBias-commandBuffer-parameter

`commandBuffer` **must** be a valid `VkCommandBuffer` handle

- VUID-vkCmdSetDepthBias-commandBuffer-recording

`commandBuffer` **must** be in the `recording` state

- VUID-vkCmdSetDepthBias-commandBuffer-cmdpool

The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Graphics	State

Chapter 26. Fragment Operations

Fragments produced by rasterization go through a number of operations to determine whether or how values produced by fragment shading are written to the framebuffer.

The following fragment operations adhere to [rasterization order](#), and are typically performed in this order:

1. [Scissor test](#)
2. [Sample mask test](#)
3. Certain [Fragment shading](#) operations:
 - [Sample Mask Accesses](#)
 - [Depth Replacement](#)
4. [Multisample coverage](#)
5. [Depth bounds test](#)
6. [Stencil test](#)
7. [Depth test](#)
8. [Sample counting](#)
9. [Coverage reduction](#)

The [coverage mask](#) generated by rasterization describes the initial coverage of each sample covered by the fragment. Fragment operations will update the coverage mask to add or subtract coverage where appropriate. If a fragment operation results in all bits of the coverage mask being 0, the fragment is discarded, and no further operations are performed. Fragments can also be programmatically discarded in a fragment shader by executing one of

- [OpKill](#).

When one of the fragment operations in this chapter is described as “replacing” a fragment shader output, that output is replaced unconditionally, even if no fragment shader previously wrote to that output.

If there is a [fragment shader](#) and it declares the [EarlyFragmentTests](#) execution mode, [fragment shading](#) and [multisample coverage](#) operations **should** instead be performed after [sample counting](#), and [sample mask test](#) **may** instead be performed after [sample counting](#).

For a pipeline with the following properties:

- a fragment shader is specified
- the fragment shader does not write to storage resources;
- the fragment shader specifies the [DepthReplacing](#) execution mode; and
- either
 - the fragment shader specifies the [DepthUnchanged](#) execution mode;

- the fragment shader specifies the `DepthLess` execution mode and the pipeline uses a `VkPipelineDepthStencilStateCreateInfo::depthCompareOp` of `VK_COMPARE_OP_GREATER` or `VK_COMPARE_OP_GREATER_OR_EQUAL`; or
- the fragment shader specifies the `DepthGreater` execution mode and the pipeline uses a `VkPipelineDepthStencilStateCreateInfo::depthCompareOp` of `VK_COMPARE_OP_LESS` or `VK_COMPARE_OP_LESS_OR_EQUAL`

the implementation **may** perform `depth bounds test` before `fragment shading` and perform an additional `depth test` immediately after that using the interpolated depth value generated by rasterization.

Once all fragment operations have completed, fragment shader outputs for covered color attachment samples pass through `framebuffer operations`.

26.1. Scissor Test

The scissor test compares the framebuffer coordinates (x_f, y_f) of each sample covered by a fragment against a *scissor rectangle* at the index equal to the fragment's `ViewportIndex`.

Each scissor rectangle is defined by a `VkRect2D`. These values are either set by the `VkPipelineViewportStateCreateInfo` structure during pipeline creation, or dynamically by the `vkCmdSetScissor` command.

A given sample is considered inside a scissor rectangle if x_f is in the range [`VkRect2D::offset.x`, `VkRect2D::offset.x` + `VkRect2D::extent.x`), and y_f is in the range [`VkRect2D::offset.y`, `VkRect2D::offset.y` + `VkRect2D::extent.y`). Samples with coordinates outside the scissor rectangle at the corresponding `ViewportIndex` will have their coverage set to 0.

To `dynamically set` the scissor rectangles, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetScissor(
    VkCommandBuffer          commandBuffer,
    uint32_t                 firstScissor,
    uint32_t                 scissorCount,
    const VkRect2D*          pScissors);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `firstScissor` is the index of the first scissor whose state is updated by the command.
- `scissorCount` is the number of scissors whose rectangles are updated by the command.
- `pScissors` is a pointer to an array of `VkRect2D` structures defining scissor rectangles.

The scissor rectangles taken from element i of `pScissors` replace the current state for the scissor index `firstScissor` + i , for i in $[0, \text{scissorCount})$.

This command sets the scissor rectangles for subsequent drawing commands when the graphics pipeline is created with `VK_DYNAMIC_STATE_SCISSOR` set in `VkPipelineDynamicStateCreateInfo`

::pDynamicStates. Otherwise, this state is specified by the [VkPipelineViewportStateCreateInfo::pScissors](#) values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetScissor-firstScissor-00592
The sum of `firstScissor` and `scissorCount` **must** be between `1` and `VkPhysicalDeviceLimits::maxViewports`, inclusive
- VUID-vkCmdSetScissor-firstScissor-00593
If the `multiViewport` feature is not enabled, `firstScissor` **must** be `0`
- VUID-vkCmdSetScissor-scissorCount-00594
If the `multiViewport` feature is not enabled, `scissorCount` **must** be `1`
- VUID-vkCmdSetScissor-x-00595
The `x` and `y` members of `offset` member of any element of `pScissors` **must** be greater than or equal to `0`
- VUID-vkCmdSetScissor-offset-00596
Evaluation of $(\text{offset.x} + \text{extent.width})$ **must** not cause a signed integer addition overflow for any element of `pScissors`
- VUID-vkCmdSetScissor-offset-00597
Evaluation of $(\text{offset.y} + \text{extent.height})$ **must** not cause a signed integer addition overflow for any element of `pScissors`

Valid Usage (Implicit)

- VUID-vkCmdSetScissor-commandBuffer-parameter
`commandBuffer` **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdSetScissor-pScissors-parameter
`pScissors` **must** be a valid pointer to an array of `scissorCount` [VkRect2D](#) structures
- VUID-vkCmdSetScissor-commandBuffer-recording
`commandBuffer` **must** be in the [recording state](#)
- VUID-vkCmdSetScissor-commandBuffer-cmdpool
The [VkCommandPool](#) that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetScissor-scissorCount-arraylength
`scissorCount` **must** be greater than `0`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the [VkCommandPool](#) that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Graphics	State

26.2. Sample Mask Test

The sample mask test compares the [coverage mask](#) for a fragment with the *sample mask* defined by `VkPipelineMultisampleStateCreateInfo::pSampleMask`.

Each bit of the coverage mask is associated with a sample index as described in the [rasterization chapter](#). If the bit in `VkPipelineMultisampleStateCreateInfo::pSampleMask` which is associated with that same sample index is set to 0, the coverage mask bit is set to 0.

26.3. Fragment Shading

[Fragment shaders](#) are invoked for each fragment, or as [helper invocations](#).

Most operations in the fragment shader are not performed in [rasterization order](#), with exceptions called out in the following sections.

For fragment shaders invoked by fragments, the following rules apply:

- A fragment shader **must** not be executed if a [fragment operation](#) that executes before fragment shading discards the fragment.
- A fragment shader **may** not be executed if:
 - An implementation determines that another fragment shader, invoked by a subsequent primitive in [primitive order](#), overwrites all results computed by the shader (including writes to storage resources).
 - Any other [fragment operation](#) discards the fragment, and the shader does not write to any storage resources.
 - If a fragment shader statically computes the same values for different framebuffer locations, and does not write to any storage resources, multiple fragments **may** be shaded by one fragment shader invocation. This **may** affect `VK_QUERY_PIPELINE_STATISTIC_FRAGMENT_SHADER_INVOCATIONS_BIT` results, but **must** otherwise not be visible behavior to applications.
- Otherwise, at least one fragment shader **must** be executed.
 - If [sample shading](#) is enabled and multiple invocations per fragment are **required**, additional invocations **must** be executed as specified.
 - Each covered sample **must** be included in at least one fragment shader invocation.

If no fragment shader is included in the pipeline, no fragment shader is executed, and undefined values **may** be written to all color attachment outputs during this fragment operation.

NOTE

Multiple fragment shader invocations may be executed for the same fragment for any number of implementation-dependent reasons. When there is more than one fragment shader invocation per fragment, the association of samples to invocations is implementation-dependent. Stores and atomics performed by these additional invocations have the normal effect.

26.3.1. Sample Mask

Reading from the `SampleMask` built-in in the `Input` storage class will return the coverage mask for the current fragment as calculated by fragment operations that executed prior to fragment shading.

If `sample shading` is enabled, fragment shaders will only see values of `1` for samples being shaded - other bits will be `0`.

Each bit of the coverage mask is associated with a sample index as described in the [rasterization chapter](#). If the bit in `SampleMask` which is associated with that same sample index is set to `0`, that coverage mask bit is set to `0`.

Values written to the `SampleMask` built-in in the `Output` storage class will be used by the `multisample coverage` operation, with the same encoding as the input built-in.

26.3.2. Depth Replacement

Writing to the `FragDepth` built-in will replace the fragment's calculated depth values for each sample in the input `SampleMask`. `Depth testing` performed after the fragment shader for this fragment will use this new value as z_f .

26.4. Multisample Coverage

If a fragment shader is active and its entry point's interface includes a built-in output variable decorated with `SampleMask`, the coverage mask is ANDed with the bits of the `SampleMask` built-in to generate a new coverage mask. If `sample shading` is enabled, bits written to `SampleMask` corresponding to samples that are not being shaded by the fragment shader invocation are ignored. If no fragment shader is active, or if the active fragment shader does not include `SampleMask` in its interface, the coverage mask is not modified.

Next, the fragment alpha value and coverage mask are modified based on the `alphaToCoverageEnable` and `alphaToOneEnable` members of the `VkPipelineMultisampleStateCreateInfo` structure.

All alpha values in this section refer only to the alpha component of the fragment shader output that has a `Location` and `Index` decoration of zero (see the [Fragment Output Interface](#) section). If that shader output has an integer or unsigned integer type, then these operations are skipped.

If `alphaToCoverageEnable` is enabled, a temporary coverage mask is generated where each bit is determined by the fragment's alpha value, which is ANDed with the fragment coverage mask.

No specific algorithm is specified for converting the alpha value to a temporary coverage mask. It is intended that the number of 1's in this value be proportional to the alpha value (clamped to [0,1]), with all 1's corresponding to a value of 1.0 and all 0's corresponding to 0.0. The algorithm **may** be different at different framebuffer coordinates.

NOTE

Using different algorithms at different framebuffer coordinates **may** help to avoid artifacts caused by regular coverage sample locations.

Finally, if `alphaToOneEnable` is enabled, each alpha value is replaced by the maximum representable alpha value for fixed-point color attachments, or by 1.0 for floating-point attachments. Otherwise, the alpha values are not changed.

26.5. Depth and Stencil Operations

Pipeline state controlling the `depth bounds tests`, `stencil test`, and `depth test` is specified through the members of the `VkPipelineDepthStencilStateCreateInfo` structure.

The `VkPipelineDepthStencilStateCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineDepthStencilStateCreateInfo {
    VkStructureType           sType;
    const void*               pNext;
    VkPipelineDepthStencilStateCreateFlags flags;
    VkBool32                  depthTestEnable;
    VkBool32                  depthWriteEnable;
    VkCompareOp               depthCompareOp;
    VkBool32                  depthBoundsTestEnable;
    VkBool32                  stencilTestEnable;
    VkStencilOpState           front;
    VkStencilOpState           back;
    float                     minDepthBounds;
    float                     maxDepthBounds;
} VkPipelineDepthStencilStateCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `depthTestEnable` controls whether `depth testing` is enabled.
- `depthWriteEnable` controls whether `depth writes` are enabled when `depthTestEnable` is `VK_TRUE`. Depth writes are always disabled when `depthTestEnable` is `VK_FALSE`.
- `depthCompareOp` is a `VkCompareOp` value specifying the comparison operator to use in the `Depth Comparison` step of the `depth test`.
- `depthBoundsTestEnable` controls whether `depth bounds testing` is enabled.
- `stencilTestEnable` controls whether `stencil testing` is enabled.

- `front` and `back` are `VkStencilOpState` values controlling the corresponding parameters of the `stencil test`.
- `minDepthBounds` is the minimum depth bound used in the `depth bounds test`.
- `maxDepthBounds` is the maximum depth bound used in the `depth bounds test`.

Valid Usage

- VUID-VkPipelineDepthStencilStateCreateInfo-depthBoundsTestEnable-00598
If the `depthBounds` feature is not enabled, `depthBoundsTestEnable` **must** be `VK_FALSE`

Valid Usage (Implicit)

- VUID-VkPipelineDepthStencilStateCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_DEPTH_STENCIL_STATE_CREATE_INFO`
- VUID-VkPipelineDepthStencilStateCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkPipelineDepthStencilStateCreateInfo-flags-zeroBitmask
`flags` **must** be `0`
- VUID-VkPipelineDepthStencilStateCreateInfo-depthCompareOp-parameter
`depthCompareOp` **must** be a valid `VkCompareOp` value
- VUID-VkPipelineDepthStencilStateCreateInfo-front-parameter
`front` **must** be a valid `VkStencilOpState` structure
- VUID-VkPipelineDepthStencilStateCreateInfo-back-parameter
`back` **must** be a valid `VkStencilOpState` structure

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineDepthStencilStateCreateFlags;
```

`VkPipelineDepthStencilStateCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

26.6. Depth Bounds Test

The depth bounds test compares the depth value z_a in the depth/stencil attachment at each sample's framebuffer coordinates (x_f, y_f) and `sample index` i against a set of *depth bounds*.

The depth bounds are determined by two floating-point values defining a minimum (`minDepthBounds`) and maximum (`maxDepthBounds`) depth value. These values are either set by the `VkPipelineDepthStencilStateCreateInfo` structure during pipeline creation, or dynamically by `vkCmdSetDepthBounds`.

A given sample is considered within the depth bounds if z_a is in the range [`minDepthBounds`

,[maxDepthBounds](#)]. Samples with depth attachment values outside of the depth bounds will have their coverage set to [0](#).

If the depth bounds test is disabled, or if there is no depth attachment, the coverage mask is unmodified by this operation.

To [dynamically set](#) the depth bounds range, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetDepthBounds(
    VkCommandBuffer          commandBuffer,
    float                    minDepthBounds,
    float                    maxDepthBounds);
```

- [commandBuffer](#) is the command buffer into which the command will be recorded.
- [minDepthBounds](#) is the minimum depth bound.
- [maxDepthBounds](#) is the maximum depth bound.

This command sets the depth bounds range for subsequent drawing commands when the graphics pipeline is created with [VK_DYNAMIC_STATE_DEPTH_BOUNDS](#) set in [VkPipelineDynamicStateCreateInfo::pDynamicStates](#). Otherwise, this state is specified by the [VkPipelineDepthStencilStateCreateInfo::minDepthBounds](#) and [VkPipelineDepthStencilStateCreateInfo::maxDepthBounds](#) values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetDepthBounds-minDepthBounds-00600
[minDepthBounds](#) **must** be between [0.0](#) and [1.0](#), inclusive
- VUID-vkCmdSetDepthBounds-maxDepthBounds-00601
[maxDepthBounds](#) **must** be between [0.0](#) and [1.0](#), inclusive

Valid Usage (Implicit)

- VUID-vkCmdSetDepthBounds-commandBuffer-parameter
[commandBuffer](#) **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdSetDepthBounds-commandBuffer-recording
[commandBuffer](#) **must** be in the [recording state](#)
- VUID-vkCmdSetDepthBounds-commandBuffer-cmdpool
The [VkCommandPool](#) that [commandBuffer](#) was allocated from **must** support graphics operations

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Graphics	State

26.7. Stencil Test

The stencil test compares the stencil attachment value s_a in the depth/stencil attachment at each sample's framebuffer coordinates (x_f, y_f) and `sample index` i against a *stencil reference value*.

If the stencil test is not enabled, as specified by `VkPipelineDepthStencilStateCreateInfo::stencilTestEnable`, or if there is no stencil attachment, the coverage mask is unmodified by this operation.

The stencil test is controlled by one of two sets of stencil-related state, the front stencil state and the back stencil state. Stencil tests and writes use the back stencil state when processing fragments generated by `back-facing polygons`, and the front stencil state when processing fragments generated by `front-facing polygons` or any other primitives.

The comparison operation performed is determined by the `VkCompareOp` value set by `VkStencilOpState::compareOp` during pipeline creation.

The compare mask s_c and stencil reference value s_r of the front or the back stencil state set determine arguments of the comparison operation. s_c is set by the `VkPipelineDepthStencilStateCreateInfo` structure during pipeline creation, or by the `vkCmdSetStencilCompareMask` command. s_r is set by `VkPipelineDepthStencilStateCreateInfo` or by `vkCmdSetStencilReference`.

s_r and s_a are each independently combined with s_c using a bitwise **AND** operation to create masked reference and attachment values s'_r and s'_a . s'_r and s'_a are used as the *reference* and *test* values, respectively, in the operation specified by the `VkCompareOp`.

If the comparison evaluates to false, the coverage for the sample is set to `0`.

A new stencil value s_g is generated according to a stencil operation defined by `VkStencilOp` parameters set by `VkPipelineDepthStencilStateCreateInfo`. If the stencil test fails, `failOp` defines the stencil operation used. If the stencil test passes however, the stencil op used is based on the `depth`

test - if it passes, `VkPipelineDepthStencilStateCreateInfo::passOp` is used, otherwise `VkPipelineDepthStencilStateCreateInfo::depthFailOp` is used.

The stencil attachment value s_a is then updated with the generated stencil value s_g according to the write mask s_w defined by `writeMask` in `VkPipelineDepthStencilStateCreateInfo::front` and `VkPipelineDepthStencilStateCreateInfo::back` as:

$$s_a = (s_a \& \neg s_w) \mid (s_g \& s_w)$$

The `VkStencilOpState` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkStencilOpState {
    VkStencilOp    failOp;
    VkStencilOp    passOp;
    VkStencilOp    depthFailOp;
    VkCompareOp    compareOp;
    uint32_t       compareMask;
    uint32_t       writeMask;
    uint32_t       reference;
} VkStencilOpState;
```

- `failOp` is a `VkStencilOp` value specifying the action performed on samples that fail the stencil test.
- `passOp` is a `VkStencilOp` value specifying the action performed on samples that pass both the depth and stencil tests.
- `depthFailOp` is a `VkStencilOp` value specifying the action performed on samples that pass the stencil test and fail the depth test.
- `compareOp` is a `VkCompareOp` value specifying the comparison operator used in the stencil test.
- `compareMask` selects the bits of the unsigned integer stencil values participating in the stencil test.
- `writeMask` selects the bits of the unsigned integer stencil values updated by the stencil test in the stencil framebuffer attachment.
- `reference` is an integer stencil reference value that is used in the unsigned stencil comparison.

Valid Usage (Implicit)

- VUID-VkStencilOpState-failOp-parameter `failOp` must be a valid `VkStencilOp` value
- VUID-VkStencilOpState-passOp-parameter `passOp` must be a valid `VkStencilOp` value
- VUID-VkStencilOpState-depthFailOp-parameter `depthFailOp` must be a valid `VkStencilOp` value
- VUID-VkStencilOpState-compareOp-parameter

`compareOp` **must** be a valid `VkCompareOp` value

To [dynamically set](#) the stencil compare mask, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetStencilCompareMask(
    VkCommandBuffer          commandBuffer,
    VkStencilFaceFlags       faceMask,
    uint32_t                 compareMask);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `faceMask` is a bitmask of `VkStencilFaceFlagBits` specifying the set of stencil state for which to update the compare mask.
- `compareMask` is the new value to use as the stencil compare mask.

This command sets the stencil compare mask for subsequent drawing commands when the graphics pipeline is created with `VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkStencilOpState::compareMask` value used to create the currently active pipeline, for both front and back faces.

Valid Usage (Implicit)

- VUID-vkCmdSetStencilCompareMask-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetStencilCompareMask-faceMask-parameter
`faceMask` **must** be a valid combination of `VkStencilFaceFlagBits` values
- VUID-vkCmdSetStencilCompareMask-faceMask-requiredbitmask
`faceMask` **must** not be 0
- VUID-vkCmdSetStencilCompareMask-commandBuffer-recording
`commandBuffer` **must** be in the [recording state](#)
- VUID-vkCmdSetStencilCompareMask-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Graphics	State

`VkStencilFaceFlagBits` values are:

```
// Provided by VK_VERSION_1_0
typedef enum VkStencilFaceFlagBits {
    VK_STENCIL_FACE_FRONT_BIT = 0x00000001,
    VK_STENCIL_FACE_BACK_BIT = 0x00000002,
    VK_STENCIL_FACE_FRONT_AND_BACK = 0x00000003,
    // VK_STENCIL_FACE_FRONT_AND_BACK is a deprecated alias
    VK_STENCIL_FACE_FRONT_AND_BACK = VK_STENCIL_FACE_FRONT_AND_BACK,
} VkStencilFaceFlagBits;
```

- `VK_STENCIL_FACE_FRONT_BIT` specifies that only the front set of stencil state is updated.
- `VK_STENCIL_FACE_BACK_BIT` specifies that only the back set of stencil state is updated.
- `VK_STENCIL_FACE_FRONT_AND_BACK` is the combination of `VK_STENCIL_FACE_FRONT_BIT` and `VK_STENCIL_FACE_BACK_BIT`, and specifies that both sets of stencil state are updated.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkStencilFaceFlags;
```

`VkStencilFaceFlags` is a bitmask type for setting a mask of zero or more `VkStencilFaceFlagBits`.

To [dynamically set](#) the stencil write mask, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetStencilWriteMask(
    VkCommandBuffer          commandBuffer,
    VkStencilFaceFlags        faceMask,
    uint32_t                  writeMask);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `faceMask` is a bitmask of `VkStencilFaceFlagBits` specifying the set of stencil state for which to update the write mask, as described above for `vkCmdSetStencilCompareMask`.
- `writeMask` is the new value to use as the stencil write mask.

This command sets the stencil write mask for subsequent drawing commands when the graphics pipeline is created with `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK` set in

`VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `writeMask` value used to create the currently active pipeline, for both `VkPipelineDepthStencilStateCreateInfo::front` and `VkPipelineDepthStencilStateCreateInfo::back` faces.

Valid Usage (Implicit)

- VUID-vkCmdSetStencilWriteMask-commandBuffer-parameter `commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetStencilWriteMask-faceMask-parameter `faceMask` **must** be a valid combination of `VkStencilFaceFlagBits` values
- VUID-vkCmdSetStencilWriteMask-faceMask-requiredbitmask `faceMask` **must** not be 0
- VUID-vkCmdSetStencilWriteMask-commandBuffer-recording `commandBuffer` **must** be in the `recording` state
- VUID-vkCmdSetStencilWriteMask-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Graphics	State

To [dynamically set](#) the stencil reference value, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetStencilReference(
    VkCommandBuffer          commandBuffer,
    VkStencilFaceFlags       faceMask,
    uint32_t                 reference);
```

- `commandBuffer` is the command buffer into which the command will be recorded.

- **faceMask** is a bitmask of **VkStencilFaceFlagBits** specifying the set of stencil state for which to update the reference value, as described above for **vkCmdSetStencilCompareMask**.
- **reference** is the new value to use as the stencil reference value.

This command sets the stencil reference value for subsequent drawing commands when the graphics pipeline is created with **VK_DYNAMIC_STATE_STENCIL_REFERENCE** set in **VkPipelineDynamicStateCreateInfo::pDynamicStates**. Otherwise, this state is specified by the **VkPipelineDepthStencilStateCreateInfo::reference** value used to create the currently active pipeline, for both front and back faces.

Valid Usage (Implicit)

- VUID-vkCmdSetStencilReference-commandBuffer-parameter **commandBuffer** **must** be a valid **VkCommandBuffer** handle
- VUID-vkCmdSetStencilReference-faceMask-parameter **faceMask** **must** be a valid combination of **VkStencilFaceFlagBits** values
- VUID-vkCmdSetStencilReference-faceMask-requiredbitmask **faceMask** **must** not be 0
- VUID-vkCmdSetStencilReference-commandBuffer-recording **commandBuffer** **must** be in the **recording** state
- VUID-vkCmdSetStencilReference-commandBuffer-cmdpool
The **VkCommandPool** that **commandBuffer** was allocated from **must** support graphics operations

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the **VkCommandPool** that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Graphics	State

Possible values of the **failOp**, **passOp**, and **depthFailOp** members of **VkStencilOpState**, specifying what happens to the stored stencil value if this or certain subsequent tests fail or pass, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkStencilOp {
    VK_STENCIL_OP_KEEP = 0,
    VK_STENCIL_OP_ZERO = 1,
    VK_STENCIL_OP_REPLACE = 2,
    VK_STENCIL_OP_INCREMENT_AND_CLAMP = 3,
    VK_STENCIL_OP_DECREMENT_AND_CLAMP = 4,
    VK_STENCIL_OP_INVERT = 5,
    VK_STENCIL_OP_INCREMENT_AND_WRAP = 6,
    VK_STENCIL_OP_DECREMENT_AND_WRAP = 7,
} VkStencilOp;
```

- **VK_STENCIL_OP_KEEP** keeps the current value.
- **VK_STENCIL_OP_ZERO** sets the value to 0.
- **VK_STENCIL_OP_REPLACE** sets the value to **reference**.
- **VK_STENCIL_OP_INCREMENT_AND_CLAMP** increments the current value and clamps to the maximum representable unsigned value.
- **VK_STENCIL_OP_DECREMENT_AND_CLAMP** decrements the current value and clamps to 0.
- **VK_STENCIL_OP_INVERT** bitwise-inverts the current value.
- **VK_STENCIL_OP_INCREMENT_AND_WRAP** increments the current value and wraps to 0 when the maximum value would have been exceeded.
- **VK_STENCIL_OP_DECREMENT_AND_WRAP** decrements the current value and wraps to the maximum possible value when the value would go below 0.

For purposes of increment and decrement, the stencil bits are considered as an unsigned integer.

26.8. Depth Test

The depth test compares the depth value z_a in the depth/stencil attachment at each sample's framebuffer coordinates (x_f, y_f) and **sample index** i against the sample's depth value z_f . If there is no depth attachment then the depth test is skipped.

The depth test occurs in three stages, as detailed in the following sections.

26.8.1. Depth Clamping and Range Adjustment

If **VkPipelineRasterizationStateCreateInfo::depthClampEnable** is enabled, z_f is clamped to $[z_{\min}, z_{\max}]$, where $z_{\min} = \min(n, f)$, $z_{\max} = \max(n, f)$, and n and f are the **minDepth** and **maxDepth** depth range values of the viewport used by this fragment, respectively.

Following depth clamping:

- If z_f is not in the range $[z_{\min}, z_{\max}]$, then z_f is undefined following this step.

26.8.2. Depth Comparison

If the depth test is not enabled, as specified by [VkPipelineDepthStencilStateCreateInfo::depthTestEnable](#), then this step is skipped.

The comparison operation performed is determined by the [VkCompareOp](#) value set by [VkPipelineDepthStencilStateCreateInfo::depthCompareOp](#) during pipeline creation. z_f and z_a are used as the *reference* and *test* values, respectively, in the operation specified by the [VkCompareOp](#).

If the comparison evaluates to false, the coverage for the sample is set to 0.

26.8.3. Depth Attachment Writes

If depth writes are enabled, as specified by [VkPipelineDepthStencilStateCreateInfo::depthWriteEnable](#), and the comparison evaluated to true, the depth attachment value z_a is set to the sample's depth value z_f . If there is no depth attachment, no value is written.

26.9. Sample Counting

Occlusion queries use query pool entries to track the number of samples that pass all the per-fragment tests. The mechanism of collecting an occlusion query value is described in [Occlusion Queries](#).

The occlusion query sample counter increments by one for each sample with a coverage value of 1 in each fragment that survives all the per-fragment tests, including scissor, sample mask, alpha to coverage, stencil, and depth tests.

26.10. Coverage Reduction

Coverage reduction takes the coverage information for a fragment and converts that to a boolean coverage value for each color sample in each pixel covered by the fragment.

26.10.1. Pixel Coverage

Coverage for each pixel is first extracted from the total fragment coverage mask. This consists of [rasterizationSamples](#) unique coverage samples for each pixel in the fragment area, each with a unique [sample index](#). If the fragment only contains a single pixel, coverage for the pixel is equivalent to the fragment coverage.

26.10.2. Color Sample Coverage

Once pixel coverage is determined, coverage for each individual color sample corresponding to that pixel is determined.

The number of [rasterizationSamples](#) is identical to the number of samples in the color attachments. A color sample is covered if the pixel coverage sample with the same [sample index](#) i is covered.

Chapter 27. The Framebuffer

27.1. Blending

Blending combines the incoming *source* fragment's R, G, B, and A values with the *destination* R, G, B, and A values of each sample stored in the framebuffer at the fragment's (x_f, y_f) location. Blending is performed for each color sample covered by the fragment, rather than just once for each fragment.

Source and destination values are combined according to the [blend operation](#), quadruplets of source and destination weighting factors determined by the [blend factors](#), and a [blend constant](#), to obtain a new set of R, G, B, and A values, as described below.

Blending is computed and applied separately to each color attachment used by the subpass, with separate controls for each attachment.

Prior to performing the blend operation, signed and unsigned normalized fixed-point color components undergo an implied conversion to floating-point as specified by [Conversion from Normalized Fixed-Point to Floating-Point](#). Blending computations are treated as if carried out in floating-point, and basic blend operations are performed with a precision and dynamic range no lower than that used to represent destination components.

NOTE

Blending is only defined for floating-point, UNORM, SNORM, and sRGB formats. Within those formats, the implementation may only support blending on some subset of them. Which formats support blending is indicated by `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`.

The pipeline blend state is included in the `VkPipelineColorBlendStateCreateInfo` structure during graphics pipeline creation:

The `VkPipelineColorBlendStateCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineColorBlendStateCreateInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkPipelineColorBlendStateCreateFlags flags;
    VkBool32             logicOpEnable;
    VkLogicOp            logicOp;
    uint32_t             attachmentCount;
    const VkPipelineColorBlendAttachmentState* pAttachments;
    float                blendConstants[4];
} VkPipelineColorBlendStateCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.

- `logicOpEnable` controls whether to apply [Logical Operations](#).
- `logicOp` selects which logical operation to apply.
- `attachmentCount` is the number of [VkPipelineColorBlendAttachmentState](#) elements in `pAttachments`.
- `pAttachments` is a pointer to an array of [VkPipelineColorBlendAttachmentState](#) structures defining blend state for each color attachment.
- `blendConstants` is a pointer to an array of four values used as the R, G, B, and A components of the blend constant that are used in blending, depending on the [blend factor](#).

Valid Usage

- VUID-VkPipelineColorBlendStateCreateInfo-pAttachments-00605
If the `independentBlend` feature is not enabled, all elements of `pAttachments` **must** be identical
- VUID-VkPipelineColorBlendStateCreateInfo-logicOpEnable-00606
If the `logicOp` feature is not enabled, `logicOpEnable` **must** be `VK_FALSE`
- VUID-VkPipelineColorBlendStateCreateInfo-logicOpEnable-00607
If `logicOpEnable` is `VK_TRUE`, `logicOp` **must** be a valid [VkLogicOp](#) value
- VUID-VkPipelineColorBlendStateCreateInfo-pAttachments-07353
If `attachmentCount` is not `0` `pAttachments` **must** be a valid pointer to an array of `attachmentCount` valid [VkPipelineColorBlendAttachmentState](#) structures

Valid Usage (Implicit)

- VUID-VkPipelineColorBlendStateCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_COLOR_BLEND_STATE_CREATE_INFO`
- VUID-VkPipelineColorBlendStateCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkPipelineColorBlendStateCreateInfo-flags-zeroBitmask
`flags` **must** be `0`
- VUID-VkPipelineColorBlendStateCreateInfo-pAttachments-parameter
If `attachmentCount` is not `0`, and `pAttachments` is not `NULL`, `pAttachments` **must** be a valid pointer to an array of `attachmentCount` valid [VkPipelineColorBlendAttachmentState](#) structures

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineColorBlendStateCreateFlags;
```

`VkPipelineColorBlendStateCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

The `VkPipelineColorBlendAttachmentState` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineColorBlendAttachmentState {
    VkBool32          blendEnable;
    VkBlendFactor      srcColorBlendFactor;
    VkBlendFactor      dstColorBlendFactor;
    VkBlendOp          colorBlendOp;
    VkBlendFactor      srcAlphaBlendFactor;
    VkBlendFactor      dstAlphaBlendFactor;
    VkBlendOp          alphaBlendOp;
    VkColorComponentFlags colorWriteMask;
} VkPipelineColorBlendAttachmentState;
```

- `blendEnable` controls whether blending is enabled for the corresponding color attachment. If blending is not enabled, the source fragment's color for that attachment is passed through unmodified.
- `srcColorBlendFactor` selects which blend factor is used to determine the source factors (S_r, S_g, S_b).
- `dstColorBlendFactor` selects which blend factor is used to determine the destination factors (D_r, D_g, D_b).
- `colorBlendOp` selects which blend operation is used to calculate the RGB values to write to the color attachment.
- `srcAlphaBlendFactor` selects which blend factor is used to determine the source factor S_a .
- `dstAlphaBlendFactor` selects which blend factor is used to determine the destination factor D_a .
- `alphaBlendOp` selects which blend operation is used to calculate the alpha values to write to the color attachment.
- `colorWriteMask` is a bitmask of `VkColorComponentFlagBits` specifying which of the R, G, B, and/or A components are enabled for writing, as described for the [Color Write Mask](#).

Valid Usage

- VUID-VkPipelineColorBlendAttachmentState-srcColorBlendFactor-00608
If the `dualSrcBlend` feature is not enabled, `srcColorBlendFactor` **must** not be `VK_BLEND_FACTOR_SRC1_COLOR`, `VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR`, `VK_BLEND_FACTOR_SRC1_ALPHA`, or `VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA`
- VUID-VkPipelineColorBlendAttachmentState-dstColorBlendFactor-00609
If the `dualSrcBlend` feature is not enabled, `dstColorBlendFactor` **must** not be `VK_BLEND_FACTOR_SRC1_COLOR`, `VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR`, `VK_BLEND_FACTOR_SRC1_ALPHA`, or `VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA`
- VUID-VkPipelineColorBlendAttachmentState-srcAlphaBlendFactor-00610
If the `dualSrcBlend` feature is not enabled, `srcAlphaBlendFactor` **must** not be `VK_BLEND_FACTOR_SRC1_COLOR`, `VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR`, `VK_BLEND_FACTOR_SRC1_ALPHA`, or `VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA`

- VUID-VkPipelineColorBlendAttachmentState-dstAlphaBlendFactor-00611

If the `dualSrcBlend` feature is not enabled, `dstAlphaBlendFactor` **must** not be `VK_BLEND_FACTOR_SRC1_COLOR`, `VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR`, `VK_BLEND_FACTOR_SRC1_ALPHA`, or `VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA`

Valid Usage (Implicit)

- VUID-VkPipelineColorBlendAttachmentState-srcColorBlendFactor-parameter
`srcColorBlendFactor` **must** be a valid `VkBlendFactor` value
- VUID-VkPipelineColorBlendAttachmentState-dstColorBlendFactor-parameter
`dstColorBlendFactor` **must** be a valid `VkBlendFactor` value
- VUID-VkPipelineColorBlendAttachmentState-colorBlendOp-parameter
`colorBlendOp` **must** be a valid `VkBlendOp` value
- VUID-VkPipelineColorBlendAttachmentState-srcAlphaBlendFactor-parameter
`srcAlphaBlendFactor` **must** be a valid `VkBlendFactor` value
- VUID-VkPipelineColorBlendAttachmentState-dstAlphaBlendFactor-parameter
`dstAlphaBlendFactor` **must** be a valid `VkBlendFactor` value
- VUID-VkPipelineColorBlendAttachmentState-alphaBlendOp-parameter
`alphaBlendOp` **must** be a valid `VkBlendOp` value
- VUID-VkPipelineColorBlendAttachmentState-colorWriteMask-parameter
`colorWriteMask` **must** be a valid combination of `VkColorComponentFlagBits` values

27.1.1. Blend Factors

The source and destination color and alpha blending factors are selected from the enum:


```
// Provided by VK_VERSION_1_0
typedef enum VkBlendFactor {
    VK_BLEND_FACTOR_ZERO = 0,
    VK_BLEND_FACTOR_ONE = 1,
    VK_BLEND_FACTOR_SRC_COLOR = 2,
    VK_BLEND_FACTOR_ONE_MINUS_SRC_COLOR = 3,
    VK_BLEND_FACTOR_DST_COLOR = 4,
    VK_BLEND_FACTOR_ONE_MINUS_DST_COLOR = 5,
    VK_BLEND_FACTOR_SRC_ALPHA = 6,
    VK_BLEND_FACTOR_ONE_MINUS_SRC_ALPHA = 7,
    VK_BLEND_FACTOR_DST_ALPHA = 8,
    VK_BLEND_FACTOR_ONE_MINUS_DST_ALPHA = 9,
    VK_BLEND_FACTOR_CONSTANT_COLOR = 10,
    VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR = 11,
    VK_BLEND_FACTOR_CONSTANT_ALPHA = 12,
    VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA = 13,
    VK_BLEND_FACTOR_SRC_ALPHA_SATURATE = 14,
    VK_BLEND_FACTOR_SRC1_COLOR = 15,
    VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR = 16,
    VK_BLEND_FACTOR_SRC1_ALPHA = 17,
    VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA = 18,
} VkBlendFactor;
```

The semantics of the enum values are described in the table below:

Table 24. Blend Factors

VkBlendFactor	RGB Blend Factors (S_r, S_g, S_b) or (D_r, D_g, D_b)	Alpha Blend Factor (S_a or D_a)
VK_BLEND_FACTOR_ZERO	(0,0,0)	0
VK_BLEND_FACTOR_ONE	(1,1,1)	1
VK_BLEND_FACTOR_SRC_COLOR	(R_{s0}, G_{s0}, B_{s0})	A_{s0}
VK_BLEND_FACTOR_ONE_MINUS_SRC_COLOR	($1-R_{s0}, 1-G_{s0}, 1-B_{s0}$)	$1-A_{s0}$
VK_BLEND_FACTOR_DST_COLOR	(R_d, G_d, B_d)	A_d
VK_BLEND_FACTOR_ONE_MINUS_DST_COLOR	($1-R_d, 1-G_d, 1-B_d$)	$1-A_d$
VK_BLEND_FACTOR_SRC_ALPHA	(A_{s0}, A_{s0}, A_{s0})	A_{s0}
VK_BLEND_FACTOR_ONE_MINUS_SRC_ALPHA	($1-A_{s0}, 1-A_{s0}, 1-A_{s0}$)	$1-A_{s0}$
VK_BLEND_FACTOR_DST_ALPHA	(A_d, A_d, A_d)	A_d
VK_BLEND_FACTOR_ONE_MINUS_DST_ALPHA	($1-A_d, 1-A_d, 1-A_d$)	$1-A_d$
VK_BLEND_FACTOR_CONSTANT_COLOR	(R_c, G_c, B_c)	A_c
VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR	($1-R_c, 1-G_c, 1-B_c$)	$1-A_c$
VK_BLEND_FACTOR_CONSTANT_ALPHA	(A_c, A_c, A_c)	A_c

VkBlendFactor	RGB Blend Factors (S_r, S_g, S_b) or (D_r, D_g, D_b)	Alpha Blend Factor (S_a or D_a)
VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA	$(1-A_c, 1-A_c, 1-A_c)$	$1-A_c$
VK_BLEND_FACTOR_SRC_ALPHA_SATURATE	$(f, f, f); f = \min(A_{s0}, 1-A_d)$	1
VK_BLEND_FACTOR_SRC1_COLOR	(R_{s1}, G_{s1}, B_{s1})	A_{s1}
VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR	$(1-R_{s1}, 1-G_{s1}, 1-B_{s1})$	$1-A_{s1}$
VK_BLEND_FACTOR_SRC1_ALPHA	(A_{s1}, A_{s1}, A_{s1})	A_{s1}
VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA	$(1-A_{s1}, 1-A_{s1}, 1-A_{s1})$	$1-A_{s1}$

In this table, the following conventions are used:

- R_{s0}, G_{s0}, B_{s0} and A_{s0} represent the first source color R, G, B, and A components, respectively, for the fragment output location corresponding to the color attachment being blended.
- R_{s1}, G_{s1}, B_{s1} and A_{s1} represent the second source color R, G, B, and A components, respectively, used in dual source blending modes, for the fragment output location corresponding to the color attachment being blended.
- R_d, G_d, B_d and A_d represent the R, G, B, and A components of the destination color. That is, the color currently in the corresponding color attachment for this fragment/sample.
- R_c, G_c, B_c and A_c represent the blend constant R, G, B, and A components, respectively.

To [dynamically set and change](#) the blend constants, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetBlendConstants(
    VkCommandBuffer          commandBuffer,
    const float               blendConstants[4]);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `blendConstants` is a pointer to an array of four values specifying the R_c , G_c , B_c , and A_c components of the blend constant color used in blending, depending on the [blend factor](#).

This command sets blend constants for subsequent drawing commands when the graphics pipeline is created with `VK_DYNAMIC_STATE_BLEND_CONSTANTS` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineColorBlendStateCreateInfo::blendConstants` values used to create the currently active pipeline.

Valid Usage (Implicit)

- VUID-vkCmdSetBlendConstants-commandBuffer-parameter `commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetBlendConstants-commandBuffer-recording

`commandBuffer` **must** be in the `recording` state

- VUID-vkCmdSetBlendConstants-commandBuffer-cmdpool

The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Graphics	State

27.1.2. Dual-Source Blending

Blend factors that use the secondary color input ($R_{s1}, G_{s1}, B_{s1}, A_{s1}$) (`VK_BLEND_FACTOR_SRC1_COLOR`, `VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR`, `VK_BLEND_FACTOR_SRC1_ALPHA`, and `VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA`) **may** consume implementation resources that could otherwise be used for rendering to multiple color attachments. Therefore, the number of color attachments that **can** be used in a framebuffer **may** be lower when using dual-source blending.

Dual-source blending is only supported if the `dualSrcBlend` feature is enabled.

The maximum number of color attachments that **can** be used in a subpass when using dual-source blending functions is implementation-dependent and is reported as the `maxFragmentDualSrcAttachments` member of `VkPhysicalDeviceLimits`.

Color outputs **can** be bound to the first and second inputs of the blender using the `Index` decoration, as described in [Fragment Output Interface](#). If the second color input to the blender is not written in the shader, or if no output is bound to the second input of a blender, the value of the second input is undefined.

27.1.3. Blend Operations

Once the source and destination blend factors have been selected, they along with the source and destination components are passed to the blending operations. RGB and alpha components **can** use different operations. Possible values of `VkBlendOp`, specifying the operations, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkBlendOp {
    VK_BLEND_OP_ADD = 0,
    VK_BLEND_OP_SUBTRACT = 1,
    VK_BLEND_OP_REVERSE_SUBTRACT = 2,
    VK_BLEND_OP_MIN = 3,
    VK_BLEND_OP_MAX = 4,
} VkBlendOp;
```

The semantics of the basic blend operations are described in the table below:

Table 25. Basic Blend Operations

VkBlendOp	RGB Components	Alpha Component
VK_BLEND_OP_ADD	$R = R_{s0} \times S_r + R_d \times D_r$ $G = G_{s0} \times S_g + G_d \times D_g$ $B = B_{s0} \times S_b + B_d \times D_b$	$A = A_{s0} \times S_a + A_d \times D_a$
VK_BLEND_OP_SUBTRACT	$R = R_{s0} \times S_r - R_d \times D_r$ $G = G_{s0} \times S_g - G_d \times D_g$ $B = B_{s0} \times S_b - B_d \times D_b$	$A = A_{s0} \times S_a - A_d \times D_a$
VK_BLEND_OP_REVERSE_SUBTRACT	$R = R_d \times D_r - R_{s0} \times S_r$ $G = G_d \times D_g - G_{s0} \times S_g$ $B = B_d \times D_b - B_{s0} \times S_b$	$A = A_d \times D_a - A_{s0} \times S_a$
VK_BLEND_OP_MIN	$R = \min(R_{s0}, R_d)$ $G = \min(G_{s0}, G_d)$ $B = \min(B_{s0}, B_d)$	$A = \min(A_{s0}, A_d)$
VK_BLEND_OP_MAX	$R = \max(R_{s0}, R_d)$ $G = \max(G_{s0}, G_d)$ $B = \max(B_{s0}, B_d)$	$A = \max(A_{s0}, A_d)$

In this table, the following conventions are used:

- R_{s0} , G_{s0} , B_{s0} and A_{s0} represent the first source color R, G, B, and A components, respectively.
- R_d , G_d , B_d and A_d represent the R, G, B, and A components of the destination color. That is, the color currently in the corresponding color attachment for this fragment/sample.
- S_r , S_g , S_b and S_a represent the source blend factor R, G, B, and A components, respectively.
- D_r , D_g , D_b and D_a represent the destination blend factor R, G, B, and A components, respectively.

The blending operation produces a new set of values R, G, B and A, which are written to the framebuffer attachment. If blending is not enabled for this attachment, then R, G, B and A are assigned R_{s0} , G_{s0} , B_{s0} and A_{s0} , respectively.

If the color attachment is fixed-point, the components of the source and destination values and blend factors are each clamped to [0,1] or [-1,1] respectively for an unsigned normalized or signed normalized color attachment prior to evaluating the blend operations. If the color attachment is floating-point, no clamping occurs.

If the [numeric format](#) of a framebuffer attachment uses sRGB encoding, the R, G, and B destination color values (after conversion from fixed-point to floating-point) are considered to be encoded for the sRGB color space and hence are linearized prior to their use in blending. Each R, G, and B component is converted from nonlinear to linear as described in the “sRGB EOTF” section of the [Khronos Data Format Specification](#). If the format is not sRGB, no linearization is performed.

If the [numeric format](#) of a framebuffer attachment uses sRGB encoding, then the final R, G and B values are converted into the nonlinear sRGB representation before being written to the framebuffer attachment as described in the “sRGB EOTF⁻¹” section of the Khronos Data Format

Specification.

If the [numeric format](#) of a framebuffer color attachment is not sRGB encoded then the resulting c_s values for R, G and B are unmodified. The value of A is never sRGB encoded. That is, the alpha component is always stored in memory as linear.

If the framebuffer color attachment is `VK_ATTACHMENT_UNUSED`, no writes are performed through that attachment. Writes are not performed to framebuffer color attachments greater than or equal to the `VkSubpassDescription::colorAttachmentCount` value.

27.2. Logical Operations

The application **can** enable a *logical operation* between the fragment's color values and the existing value in the framebuffer attachment. This logical operation is applied prior to updating the framebuffer attachment. Logical operations are applied only for signed and unsigned integer and normalized integer framebuffers. Logical operations are not applied to floating-point or sRGB format color attachments.

Logical operations are controlled by the `logicOpEnable` and `logicOp` members of `VkPipelineColorBlendStateCreateInfo`. If `logicOpEnable` is `VK_TRUE`, then a logical operation selected by `logicOp` is applied between each color attachment and the fragment's corresponding output value, and blending of all attachments is treated as if it were disabled. Any attachments using color formats for which logical operations are not supported simply pass through the color values unmodified. The logical operation is applied independently for each of the red, green, blue, and alpha components. The `logicOp` is selected from the following operations:

```
// Provided by VK_VERSION_1_0
typedef enum VkLogicOp {
    VK_LOGIC_OP_CLEAR = 0,
    VK_LOGIC_OP_AND = 1,
    VK_LOGIC_OP_AND_REVERSE = 2,
    VK_LOGIC_OP_COPY = 3,
    VK_LOGIC_OP_AND_INVERTED = 4,
    VK_LOGIC_OP_NO_OP = 5,
    VK_LOGIC_OP_XOR = 6,
    VK_LOGIC_OP_OR = 7,
    VK_LOGIC_OP_NOR = 8,
    VK_LOGIC_OP_EQUIVALENT = 9,
    VK_LOGIC_OP_INVERT = 10,
    VK_LOGIC_OP_OR_REVERSE = 11,
    VK_LOGIC_OP_COPY_INVERTED = 12,
    VK_LOGIC_OP_OR_INVERTED = 13,
    VK_LOGIC_OP_NAND = 14,
    VK_LOGIC_OP_SET = 15,
} VkLogicOp;
```

The logical operations supported by Vulkan are summarized in the following table in which

- $\neg$ is bitwise invert,
- $\square$ is bitwise and,
- $\sqcup$ is bitwise or,
- $\oplus$ is bitwise exclusive or,
- s is the fragment's R_{s0} , G_{s0} , B_{s0} or A_{s0} component value for the fragment output corresponding to the color attachment being updated, and
- d is the color attachment's R, G, B or A component value:

Table 26. Logical Operations

Mode	Operation
VK_LOGIC_OP_CLEAR	0
VK_LOGIC_OP_AND	$s \square d$
VK_LOGIC_OP_AND_REVERSE	$s \square \neg d$
VK_LOGIC_OP_COPY	s
VK_LOGIC_OP_AND_INVERTED	$\neg s \square d$
VK_LOGIC_OP_NO_OP	d
VK_LOGIC_OP_XOR	$s \oplus d$
VK_LOGIC_OP_OR	$s \sqcup d$
VK_LOGIC_OP_NOR	$\neg (s \sqcup d)$
VK_LOGIC_OP_EQUIVALENT	$\neg (s \square d)$
VK_LOGIC_OP_INVERT	$\neg d$
VK_LOGIC_OP_OR_REVERSE	$s \sqcup \neg d$
VK_LOGIC_OP_COPY_INVERTED	$\neg s$
VK_LOGIC_OP_OR_INVERTED	$\neg s \sqcup d$
VK_LOGIC_OP_NAND	$\neg (s \square d)$
VK_LOGIC_OP_SET	all 1s

The result of the logical operation is then written to the color attachment as controlled by the component write mask, described in [Blend Operations](#).

27.3. Color Write Mask

Bits which **can** be set in `VkPipelineColorBlendAttachmentState::colorWriteMask`, determining whether the final color values R, G, B and A are written to the framebuffer attachment, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkColorComponentFlagBits {
    VK_COLOR_COMPONENT_R_BIT = 0x00000001,
    VK_COLOR_COMPONENT_G_BIT = 0x00000002,
    VK_COLOR_COMPONENT_B_BIT = 0x00000004,
    VK_COLOR_COMPONENT_A_BIT = 0x00000008,
} VkColorComponentFlagBits;
```

- **VK_COLOR_COMPONENT_R_BIT** specifies that the R value is written to the color attachment for the appropriate sample. Otherwise, the value in memory is unmodified.
- **VK_COLOR_COMPONENT_G_BIT** specifies that the G value is written to the color attachment for the appropriate sample. Otherwise, the value in memory is unmodified.
- **VK_COLOR_COMPONENT_B_BIT** specifies that the B value is written to the color attachment for the appropriate sample. Otherwise, the value in memory is unmodified.
- **VK_COLOR_COMPONENT_A_BIT** specifies that the A value is written to the color attachment for the appropriate sample. Otherwise, the value in memory is unmodified.

The color write mask operation is applied regardless of whether blending is enabled.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkColorComponentFlags;
```

VkColorComponentFlags is a bitmask type for setting a mask of zero or more **VkColorComponentFlagBits**.

Chapter 28. Dispatching Commands

Dispatching commands (commands with **Dispatch** in the name) provoke work in a compute pipeline. Dispatching commands are recorded into a command buffer and when executed by a queue, will produce work which executes according to the bound compute pipeline. A compute pipeline **must** be bound to a command buffer before any dispatching commands are recorded in that command buffer.

To record a dispatch, call:

```
// Provided by VK_VERSION_1_0
void vkCmdDispatch(
    VkCommandBuffer          commandBuffer,
    uint32_t                 groupCountX,
    uint32_t                 groupCountY,
    uint32_t                 groupCountZ);
```

- **commandBuffer** is the command buffer into which the command will be recorded.
- **groupCountX** is the number of local workgroups to dispatch in the X dimension.
- **groupCountY** is the number of local workgroups to dispatch in the Y dimension.
- **groupCountZ** is the number of local workgroups to dispatch in the Z dimension.

When the command is executed, a global workgroup consisting of **groupCountX** × **groupCountY** × **groupCountZ** local workgroups is assembled.

Valid Usage

- VUID-vkCmdDispatch-magFilter-04553
If a **VkSampler** created with **magFilter** or **minFilter** equal to **VK_FILTER_LINEAR**, and **compareEnable** equal to **VK_FALSE** is used to sample a **VkImageView** as a result of this command, then the image view's **format features** **must** contain **VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT**
- VUID-vkCmdDispatch-mipmapMode-04770
If a **VkSampler** created with **mipmapMode** equal to **VK_SAMPLER_MIPMAP_MODE_LINEAR**, and **compareEnable** equal to **VK_FALSE** is used to sample a **VkImageView** as a result of this command, then the image view's **format features** **must** contain **VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT**
- VUID-vkCmdDispatch-unnormalizedCoordinates-09635
If a **VkSampler** created with **unnormalizedCoordinates** equal to **VK_TRUE** is used to sample a **VkImageView** as a result of this command, then the image view's **levelCount** and **layerCount** **must** be 1
- VUID-vkCmdDispatch-unnormalizedCoordinates-09636
If a **VkSampler** created with **unnormalizedCoordinates** equal to **VK_TRUE** is used to sample a **VkImageView** as a result of this command, then the image view's **viewType** **must** be

VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D

- VUID-vkCmdDispatch-aspectMask-06478

If a [VkImageView](#) is sampled with [depth comparison](#), the image view **must** have been created with an [aspectMask](#) that contains [VK_IMAGE_ASPECT_DEPTH_BIT](#)

- VUID-vkCmdDispatch-None-02691

If a [VkImageView](#) is accessed using atomic operations as a result of this command, then the image view's [format](#) [features](#) **must** contain [VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT](#)

- VUID-vkCmdDispatch-None-07888

If a [VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER](#) descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's [format](#) [features](#) **must** contain [VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT](#)

- VUID-vkCmdDispatch-None-08600

For each set n that is statically used by a [bound shader](#), a descriptor set **must** have been bound to n at the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for set n , with the [VkPipelineLayout](#) used to create the current [VkPipeline](#), as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDispatch-None-08601

For each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#), as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDispatch-None-10068

For each array of resources that is used by a [bound shader](#), the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command

- VUID-vkCmdDispatch-None-08114

Descriptors in each bound descriptor set, specified via [vkCmdBindDescriptorSets](#), **must** be valid as described by [descriptor validity](#) if they are statically used by a [bound shader](#)

- VUID-vkCmdDispatch-None-08606

A valid pipeline **must** be bound to the pipeline bind point used by this command

- VUID-vkCmdDispatch-None-08608

There **must** not have been any calls to dynamic state setting commands for any state specified statically in the [VkPipeline](#) object bound to the pipeline bind point used by this command, since that pipeline was bound

- VUID-vkCmdDispatch-None-08609

If the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used to sample from any [VkImage](#) with a [VkImageView](#) of the type [VK_IMAGE_VIEW_TYPE_3D](#), [VK_IMAGE_VIEW_TYPE_CUBE](#), [VK_IMAGE_VIEW_TYPE_1D_ARRAY](#), [VK_IMAGE_VIEW_TYPE_2D_ARRAY](#) or [VK_IMAGE_VIEW_TYPE_CUBE_ARRAY](#), in any shader stage

- VUID-vkCmdDispatch-None-08610

If the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a

VkSampler object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V **OpImageSample*** or **OpImageSparseSample*** instructions with **ImplicitLod**, **Dref** or **Proj** in their name, in any shader stage

- VUID-vkCmdDispatch-None-08611

If the **VkPipeline** object bound to the pipeline bind point used by this command accesses a **VkSampler** object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V **OpImageSample*** or **OpImageSparseSample*** instructions that includes a LOD bias or any offset values, in any shader stage

- VUID-vkCmdDispatch-uniformBuffers-06935

If any stage of the **VkPipeline** object bound to the pipeline bind point used by this command accesses a uniform buffer, and the **robustBufferAccess** feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDispatch-storageBuffers-06936

If any stage of the **VkPipeline** object bound to the pipeline bind point used by this command accesses a storage buffer, and the **robustBufferAccess** feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDispatch-viewType-07752

If a **VkImageView** is accessed as a result of this command, then the image view's **viewType** **must** match the **Dim** operand of the **OpTypeImage** as described in [Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types](#)

- VUID-vkCmdDispatch-format-07753

If a **VkImageView** is accessed as a result of this command, then the **numeric type** of the image view's **format** and the **Sampled Type** operand of the **OpTypeImage** **must** match

- VUID-vkCmdDispatch-OpImageWrite-08795

If a **VkImageView** is accessed using **OpImageWrite** as a result of this command, then the **Type** of the **Texel** operand of that instruction **must** have at least as many components as the image view's format

- VUID-vkCmdDispatch-OpImageWrite-04469

If a **VkBufferView** is accessed using **OpImageWrite** as a result of this command, then the **Type** of the **Texel** operand of that instruction **must** have at least as many components as the buffer view's format

- VUID-vkCmdDispatch-None-07288

Any shader invocation executed by this command **must terminate**

- VUID-vkCmdDispatch-None-09600

If a descriptor with type equal to any of **VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE**, **VK_DESCRIPTOR_TYPE_STORAGE_IMAGE**, or **VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT** is accessed as a result of this command, the image subresource identified by that descriptor **must** be in the image layout identified when the descriptor was written

- VUID-vkCmdDispatch-groupCountX-00386

groupCountX **must** be less than or equal to **VkPhysicalDeviceLimits::maxComputeWorkGroupCount[0]**

- VUID-vkCmdDispatch-groupCountY-00387
`groupCountY` **must** be less than or equal to `VkPhysicalDeviceLimits::maxComputeWorkGroupCount[1]`
- VUID-vkCmdDispatch-groupCountZ-00388
`groupCountZ` **must** be less than or equal to `VkPhysicalDeviceLimits::maxComputeWorkGroupCount[2]`

Valid Usage (Implicit)

- VUID-vkCmdDispatch-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdDispatch-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdDispatch-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support compute operations
- VUID-vkCmdDispatch-renderpass
This command **must** only be called outside of a render pass instance

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Compute	Action

To record an indirect dispatching command, call:

```
// Provided by VK_VERSION_1_0
void vkCmdDispatchIndirect(
    VkCommandBuffer          commandBuffer,
    VkBuffer                 buffer,
    VkDeviceSize             offset);
```

- `commandBuffer` is the command buffer into which the command will be recorded.

- `buffer` is the buffer containing dispatch parameters.
- `offset` is the byte offset into `buffer` where parameters begin.

`vkCmdDispatchIndirect` behaves similarly to `vkCmdDispatch` except that the parameters are read by the device from a buffer during execution. The parameters of the dispatch are encoded in a `VkDispatchIndirectCommand` structure taken from `buffer` starting at `offset`.

Valid Usage

- VUID-vkCmdDispatchIndirect-magFilter-04553
If a `VkSampler` created with `magFilter` or `minFilter` equal to `VK_FILTER_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdDispatchIndirect-mipmapMode-04770
If a `VkSampler` created with `mipmapMode` equal to `VK_SAMPLER_MIPMAP_MODE_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdDispatchIndirect-unnormalizedCoordinates-09635
If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `levelCount` and `layerCount` **must** be 1
- VUID-vkCmdDispatchIndirect-unnormalizedCoordinates-09636
If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `viewType` **must** be `VK_IMAGE_VIEW_TYPE_1D` or `VK_IMAGE_VIEW_TYPE_2D`
- VUID-vkCmdDispatchIndirect-aspectMask-06478
If a `VkImageView` is sampled with `depth comparison`, the image view **must** have been created with an `aspectMask` that contains `VK_IMAGE_ASPECT_DEPTH_BIT`
- VUID-vkCmdDispatchIndirect-None-02691
If a `VkImageView` is accessed using atomic operations as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT`
- VUID-vkCmdDispatchIndirect-None-07888
If a `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT`
- VUID-vkCmdDispatchIndirect-None-08600
For each set n that is statically used by a `bound shader`, a descriptor set **must** have been bound to n at the same pipeline bind point, with a `VkPipelineLayout` that is compatible for set n , with the `VkPipelineLayout` used to create the current `VkPipeline`, as described in [Pipeline Layout Compatibility](#)
- VUID-vkCmdDispatchIndirect-None-08601

For each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) , as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDispatchIndirect-None-10068

For each array of resources that is used by a [bound shader](#), the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command

- VUID-vkCmdDispatchIndirect-None-08114

Descriptors in each bound descriptor set, specified via [vkCmdBindDescriptorSets](#), **must** be valid as described by [descriptor validity](#) if they are statically used by a [bound shader](#)

- VUID-vkCmdDispatchIndirect-None-08606

A valid pipeline **must** be bound to the pipeline bind point used by this command

- VUID-vkCmdDispatchIndirect-None-08608

There **must** not have been any calls to dynamic state setting commands for any state specified statically in the [VkPipeline](#) object bound to the pipeline bind point used by this command, since that pipeline was bound

- VUID-vkCmdDispatchIndirect-None-08609

If the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used to sample from any [VkImage](#) with a [VkImageView](#) of the type [VK_IMAGE_VIEW_TYPE_3D](#), [VK_IMAGE_VIEW_TYPE_CUBE](#), [VK_IMAGE_VIEW_TYPE_1D_ARRAY](#), [VK_IMAGE_VIEW_TYPE_2D_ARRAY](#) or [VK_IMAGE_VIEW_TYPE_CUBE_ARRAY](#), in any shader stage

- VUID-vkCmdDispatchIndirect-None-08610

If the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions with [ImplicitLod](#), [Dref](#) or [Proj](#) in their name, in any shader stage

- VUID-vkCmdDispatchIndirect-None-08611

If the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions that includes a LOD bias or any offset values, in any shader stage

- VUID-vkCmdDispatchIndirect-uniformBuffers-06935

If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a uniform buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDispatchIndirect-storageBuffers-06936

If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a storage buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDispatchIndirect-viewType-07752

If a `VkImageView` is accessed as a result of this command, then the image view's `viewType` **must** match the `Dim` operand of the `OpTypeImage` as described in [Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types](#)

- VUID-vkCmdDispatchIndirect-format-07753

If a `VkImageView` is accessed as a result of this command, then the `numeric type` of the image view's `format` and the `Sampled Type` operand of the `OpTypeImage` **must** match

- VUID-vkCmdDispatchIndirect-OpImageWrite-08795

If a `VkImageView` is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the image view's `format`

- VUID-vkCmdDispatchIndirect-OpImageWrite-04469

If a `VkBufferView` is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the buffer view's `format`

- VUID-vkCmdDispatchIndirect-None-07288

Any shader invocation executed by this command **must** `terminate`

- VUID-vkCmdDispatchIndirect-None-09600

If a descriptor with type equal to any of `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, or `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT` is accessed as a result of this command, the image subresource identified by that descriptor **must** be in the image layout identified when the descriptor was written

- VUID-vkCmdDispatchIndirect-buffer-02708

If `buffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-vkCmdDispatchIndirect-buffer-02709

`buffer` **must** have been created with the `VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT` bit set

- VUID-vkCmdDispatchIndirect-offset-02710

`offset` **must** be a multiple of 4

- VUID-vkCmdDispatchIndirect-offset-00407

The sum of `offset` and the size of `VkDispatchIndirectCommand` **must** be less than or equal to the size of `buffer`

Valid Usage (Implicit)

- VUID-vkCmdDispatchIndirect-commandBuffer-parameter

`commandBuffer` **must** be a valid `VkCommandBuffer` handle

- VUID-vkCmdDispatchIndirect-buffer-parameter

`buffer` **must** be a valid `VkBuffer` handle

- VUID-vkCmdDispatchIndirect-commandBuffer-recording

`commandBuffer` **must** be in the `recording state`

- VUID-vkCmdDispatchIndirect-commandBuffer-cmdpool

The `VkCommandPool` that `commandBuffer` was allocated from **must** support compute

operations

- VUID-vkCmdDispatchIndirect-renderpass

This command **must** only be called outside of a render pass instance

- VUID-vkCmdDispatchIndirect-commonparent

Both of **buffer**, and **commandBuffer** **must** have been created, allocated, or retrieved from the same **VkDevice**

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the **VkCommandPool** that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Compute	Action

The **VkDispatchIndirectCommand** structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDispatchIndirectCommand {
    uint32_t    x;
    uint32_t    y;
    uint32_t    z;
} VkDispatchIndirectCommand;
```

- **x** is the number of local workgroups to dispatch in the X dimension.
- **y** is the number of local workgroups to dispatch in the Y dimension.
- **z** is the number of local workgroups to dispatch in the Z dimension.

The members of **VkDispatchIndirectCommand** have the same meaning as the corresponding parameters of **vkCmdDispatch**.

Valid Usage

- VUID-VkDispatchIndirectCommand-x-00417
x **must** be less than or equal to **VkPhysicalDeviceLimits::maxComputeWorkGroupCount[0]**
- VUID-VkDispatchIndirectCommand-y-00418

y must be less than or equal to `VkPhysicalDeviceLimits::maxComputeWorkGroupCount[1]`

- VUID-VkDispatchIndirectCommand-z-00419

z must be less than or equal to `VkPhysicalDeviceLimits::maxComputeWorkGroupCount[2]`

Chapter 29. Sparse Resources

As documented in [Resource Memory Association](#), `VkBuffer` and `VkImage` resources in Vulkan **must** be bound completely and contiguously to a single `VkDeviceMemory` object. This binding **must** be done before the resource is used, and the binding is immutable for the lifetime of the resource.

Sparse resources relax these restrictions and provide these additional features:

- Sparse resources **can** be bound non-contiguously to one or more `VkDeviceMemory` allocations.
- Sparse resources **can** be re-bound to different memory allocations over the lifetime of the resource.
- Sparse resources **can** have descriptors generated and used orthogonally with memory binding commands.

29.1. Sparse Resource Features

Sparse resources have several features that **must** be enabled explicitly at resource creation time. The features are enabled by including bits in the `flags` parameter of `VkImageCreateInfo` or `VkBufferCreateInfo`. Each feature also has one or more corresponding feature enables specified in `VkPhysicalDeviceFeatures`.

- The `sparseBinding` feature is the base, and provides the following capabilities:
 - Resources **can** be bound at some defined (sparse block) granularity.
 - The entire resource **must** be bound to memory before use regardless of regions actually accessed.
 - No specific mapping of image region to memory offset is defined, i.e. the location that each texel corresponds to in memory is implementation-dependent.
 - Sparse buffers have a well-defined mapping of buffer range to memory range, where an offset into a range of the buffer that is bound to a single contiguous range of memory corresponds to an identical offset within that range of memory.
 - Requested via the `VK_IMAGE_CREATE_SPARSE_BINDING_BIT` and `VK_BUFFER_CREATE_SPARSE_BINDING_BIT` bits.
 - A sparse image created using `VK_IMAGE_CREATE_SPARSE_BINDING_BIT` (but not `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`) supports all formats that non-sparse usage supports, and supports both `VK_IMAGE_TILING_OPTIMAL` and `VK_IMAGE_TILING_LINEAR` tiling.
- *Sparse Residency* builds on (and requires) the `sparseBinding` feature. It includes the following capabilities:
 - Resources do not have to be completely bound to memory before use on the device.
 - Images have a prescribed sparse image block layout, allowing specific rectangular regions of the image to be bound to specific offsets in memory allocations.
 - Consistency of access to unbound regions of the resource is defined by the absence or presence of `VkPhysicalDeviceSparseProperties::residencyNonResidentStrict`. If this property is present, accesses to unbound regions of the resource are well defined and behave as if the

data bound is populated with all zeros; writes are discarded. When this property is absent, accesses are considered safe, but reads will return undefined values.

- Requested via the `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` and `VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT` bits.
- Sparse residency support is advertised on a finer grain via the following features:
 - The `sparseResidencyBuffer` feature provides support for creating `VkBuffer` objects with the `VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT`.
 - The `sparseResidencyImage2D` feature provides support for creating 2D single-sampled `VkImage` objects with `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.
 - The `sparseResidencyImage3D` feature provides support for creating 3D `VkImage` objects with `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.
 - The `sparseResidency2Samples` feature provides support for creating 2D `VkImage` objects with 2 samples and `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.
 - The `sparseResidency4Samples` feature provides support for creating 2D `VkImage` objects with 4 samples and `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.
 - The `sparseResidency8Samples` feature provides support for creating 2D `VkImage` objects with 8 samples and `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.
 - The `sparseResidency16Samples` feature provides support for creating 2D `VkImage` objects with 16 samples and `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.

Implementations supporting `sparseResidencyImage2D` are only **required** to support sparse 2D, single-sampled images. Support for sparse 3D and MSAA images is **optional** and **can** be enabled via `sparseResidencyImage3D`, `sparseResidency2Samples`, `sparseResidency4Samples`, `sparseResidency8Samples`, and `sparseResidency16Samples`.

- A sparse image created using `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` supports all non-compressed color formats with power-of-two element size that non-sparse usage supports. Additional formats **may** also be supported and **can** be queried via `vkGetPhysicalDeviceSparseImageFormatProperties`. `VK_IMAGE_TILING_LINEAR` tiling is not supported.
- The `sparseResidencyAliased` feature provides the following capability that **can** be enabled per resource:

Allows physical memory ranges to be shared between multiple locations in the same sparse resource or between multiple sparse resources, with each binding of a memory location observing a consistent interpretation of the memory contents.

See [Sparse Memory Aliasing](#) for more information.

29.2. Sparse Buffers and Fully-Resident Images

Both `VkBuffer` and `VkImage` objects created with the `VK_IMAGE_CREATE_SPARSE_BINDING_BIT` or `VK_BUFFER_CREATE_SPARSE_BINDING_BIT` bits **can** be thought of as a linear region of address space. In the `VkImage` case if `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` is not used, this linear region is entirely

opaque, meaning that there is no application-visible mapping between texel location and memory offset.

Unless `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` or `VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT` are also used, the entire resource **must** be bound to one or more `VkDeviceMemory` objects before use.

29.2.1. Sparse Buffer and Fully-Resident Image Block Size

The sparse block size in bytes for sparse buffers and fully-resident images is reported as `VkMemoryRequirements::alignment`. `alignment` represents both the memory alignment requirement and the binding granularity (in bytes) for sparse resources.

29.3. Sparse Partially-Resident Buffers

`VkBuffer` objects created with the `VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT` bit allow the buffer to be made only partially resident. Partially resident `VkBuffer` objects are allocated and bound identically to `VkBuffer` objects using only the `VK_BUFFER_CREATE_SPARSE_BINDING_BIT` feature. The only difference is the ability for some regions of the buffer to be unbound during device use.

29.4. Sparse Partially-Resident Images

`VkImage` objects created with the `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` bit allow specific rectangular regions of the image called sparse image blocks to be bound to specific ranges of memory. This allows the application to manage residency at either image subresource or sparse image block granularity. Each image subresource (outside of the [mip tail](#)) starts on a sparse block boundary and has dimensions that are integer multiples of the corresponding dimensions of the sparse image block.

NOTE

Applications **can** use these types of images to control LOD based on total memory consumption. If memory pressure becomes an issue the application **can** unbind and disable specific mipmap levels of images without having to recreate resources or modify texel data of unaffected levels.

The application **can** also use this functionality to access subregions of the image in a “megatexture” fashion. The application **can** create a large image and only populate the region of the image that is currently being used in the scene.

29.4.1. Accessing Unbound Regions

The following member of `VkPhysicalDeviceSparseProperties` affects how data in unbound regions of sparse resources are handled by the implementation:

- `residencyNonResidentStrict`

If this property is not present, reads of unbound regions of the image will return undefined values. Both reads and writes are still considered *safe* and will not affect other resources or populated regions of the image.

If this property is present, all reads of unbound regions of the image will behave as if the region was bound to memory populated with all zeros; writes will be discarded.

[Image operations](#) performed on unbound memory **may** still alter some component values in the natural way for those accesses, e.g. substituting a value of one for alpha in formats that do not have an alpha component.

Example: Reading the alpha component of an unbacked `VK_FORMAT_R8_UNORM` image will return a value of 1.0f.

See [Physical Device Enumeration](#) for instructions for retrieving physical device properties.

Implementor's Note

For implementations that **cannot** natively handle access to unbound regions of a resource, the implementation **may** allocate and bind memory to the unbound regions. Reads and writes to unbound regions will access the implementation-managed memory instead.

Given that the values resulting from reads of unbound regions are undefined in this scenario, implementations **may** use the same physical memory for all unbound regions of multiple resources within the same process.

29.4.2. Mip Tail Regions

Sparse images created using `VK_IMAGE_CREATE_SPARSE_BINDING_BIT` (without also using `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`) have no specific mapping of image region or image subresource to memory offset defined, so the entire image **can** be thought of as a linear opaque address region. However, images created with `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` do have a prescribed sparse image block layout, and hence each image subresource **must** start on a sparse block boundary. Within each array layer, the set of mip levels that have a smaller size than the sparse block size in bytes are grouped together into a *mip tail region*.

If the `VK_SPARSE_IMAGE_FORMAT_ALIGNED_MIP_SIZE_BIT` flag is present in the `flags` member of `VkSparseImageFormatProperties`, for the image's `format`, then any mip level which has dimensions that are not integer multiples of the corresponding dimensions of the sparse image block, and all subsequent mip levels, are also included in the mip tail region.

The following member of `VkPhysicalDeviceSparseProperties` **may** affect how the implementation places mip levels in the mip tail region:

- `residencyAlignedMipSize`

Each mip tail region is bound to memory as an opaque region (i.e. **must** be bound using a `VkSparseImageOpaqueMemoryBindInfo` structure) and **may** be of a size greater than or equal to the sparse block size in bytes. This size is guaranteed to be an integer multiple of the sparse block size in bytes.

An implementation **may** choose to allow each array-layer's mip tail region to be bound to memory independently or require that all array-layer's mip tail regions be treated as one. This is dictated by `VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT` in `VkSparseImageMemoryRequirements::flags`.

The following diagrams depict how `VK_SPARSE_IMAGE_FORMAT_ALIGNED_MIP_SIZE_BIT` and `VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT` alter memory usage and requirements.

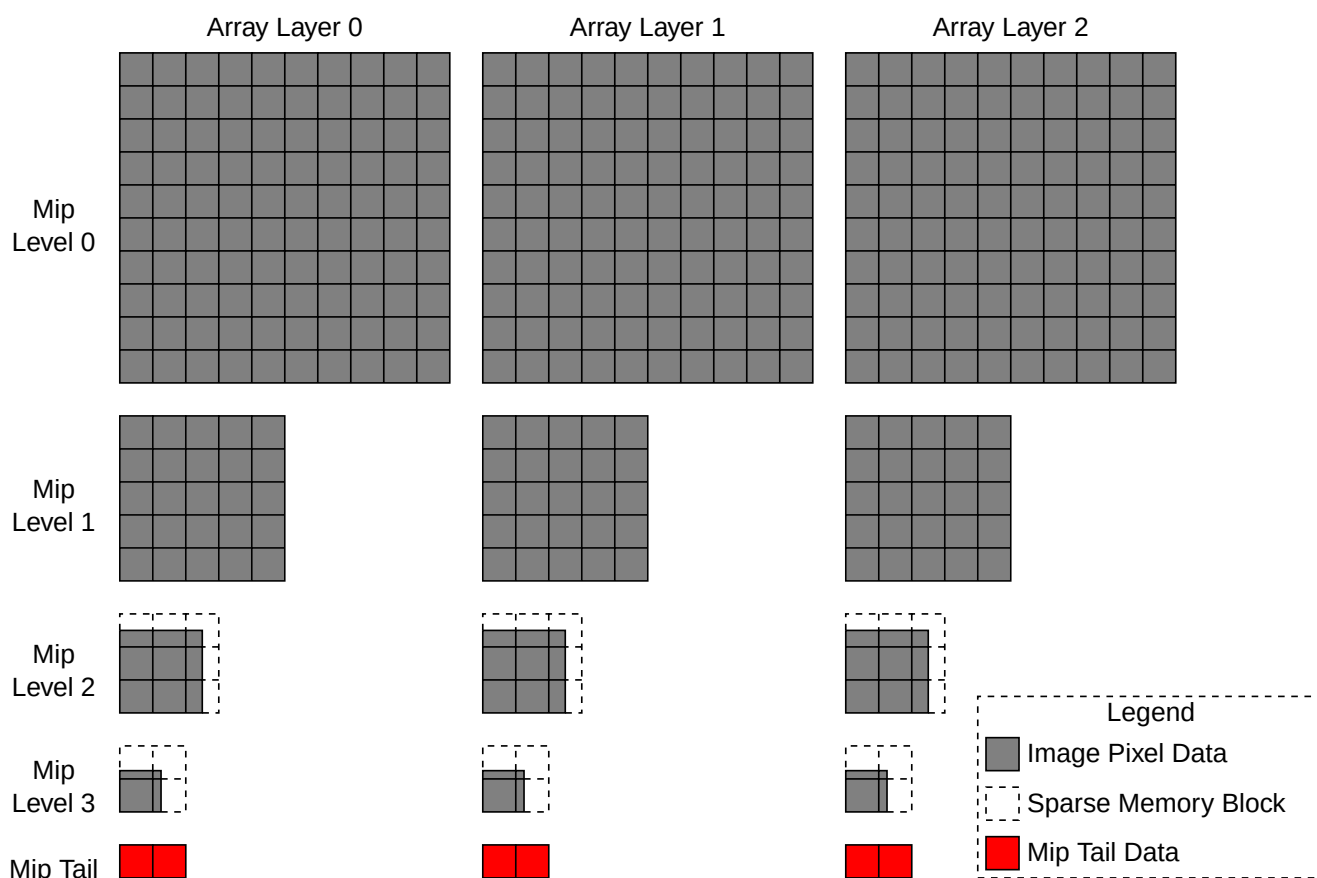


Figure 10. Sparse Image

In the absence of `VK_SPARSE_IMAGE_FORMAT_ALIGNED_MIP_SIZE_BIT` and `VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT`, each array layer contains a mip tail region containing texel data for all mip levels smaller than the sparse image block in any dimension.

Mip levels that are as large or larger than a sparse image block in all dimensions **can** be bound individually. Right-edges and bottom-edges of each level are allowed to have partially used sparse blocks. Any bound partially-used-sparse-blocks **must** still have their full sparse block size in bytes allocated in memory.

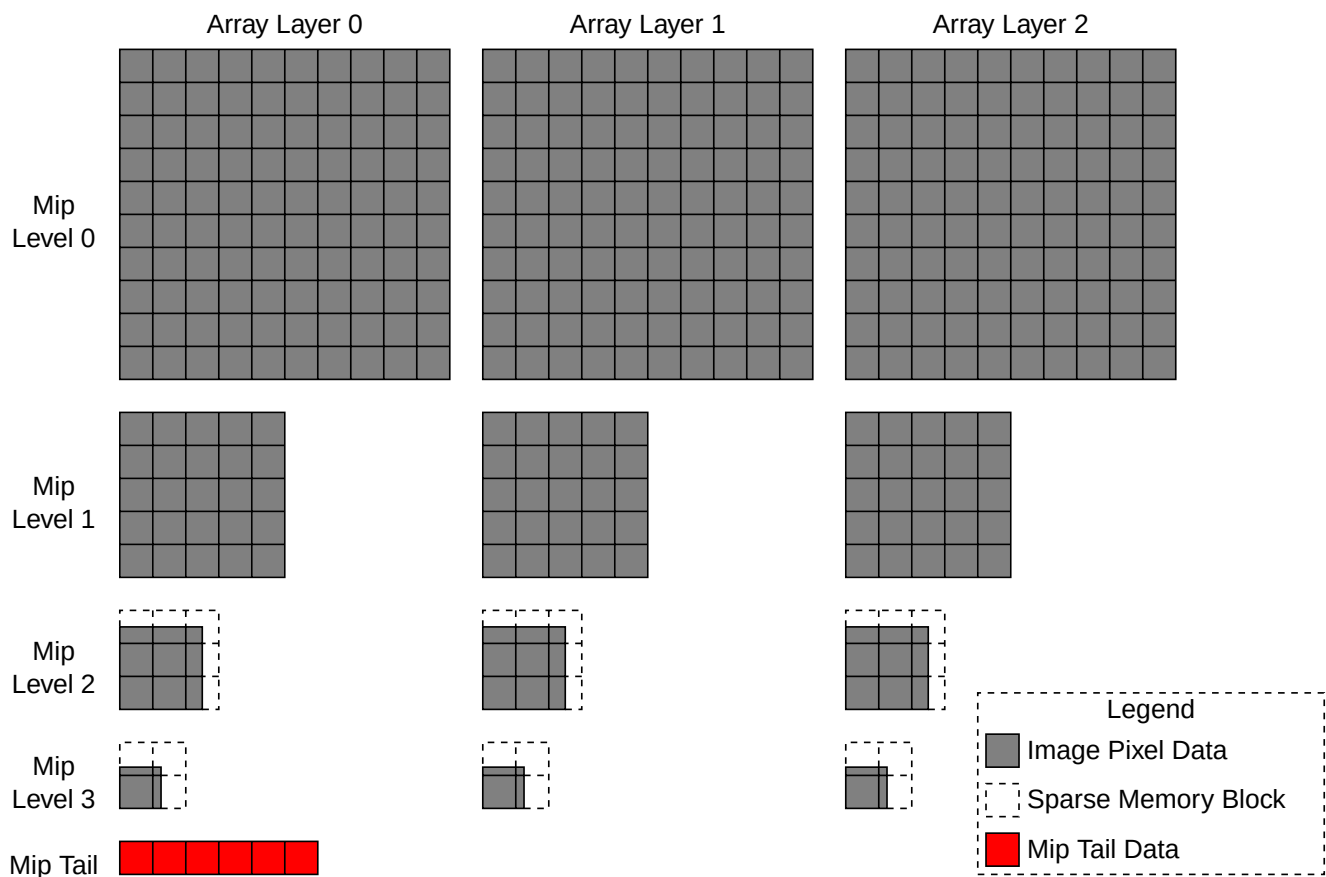


Figure 11. Sparse Image with Single Mip Tail

When `VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT` is present all array layers will share a single mip tail region.

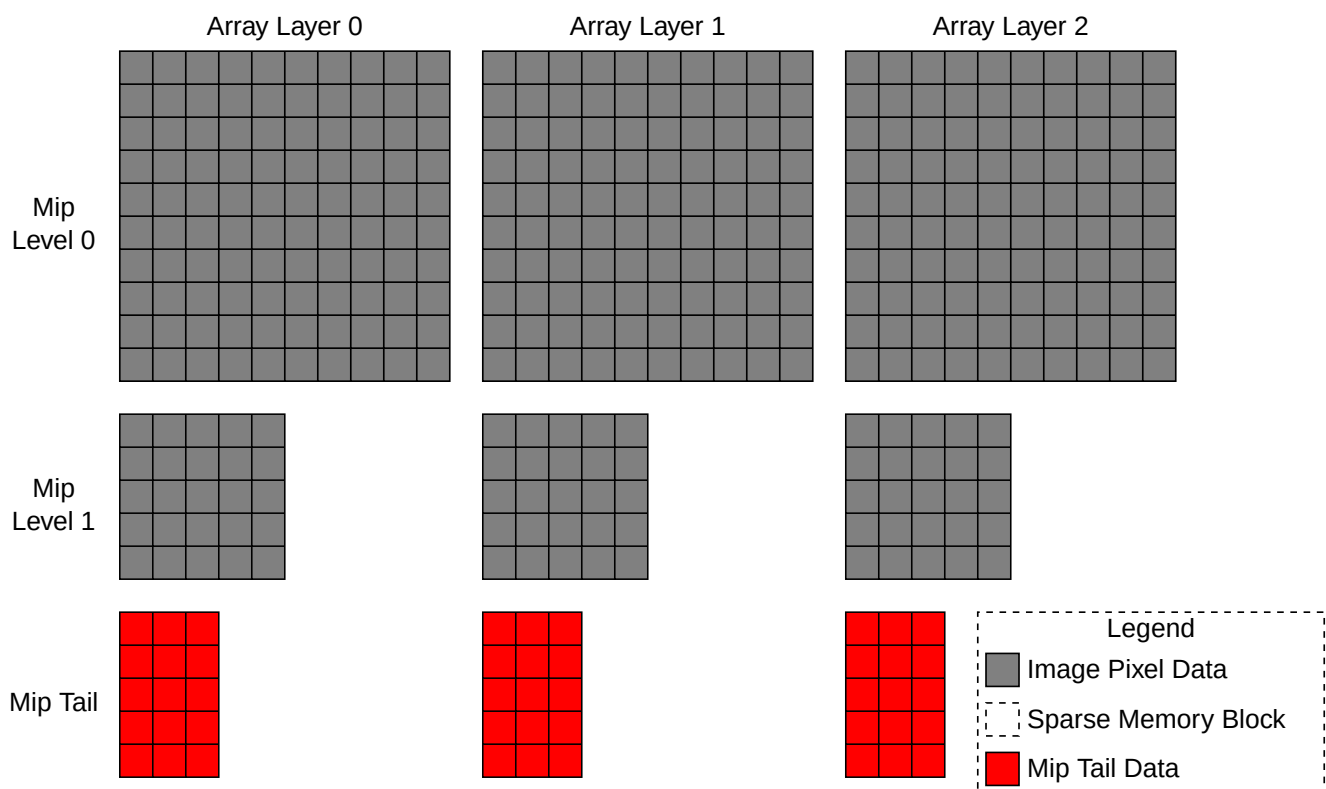


Figure 12. Sparse Image with Aligned Mip Size

NOTE The mip tail regions are presented here in 2D arrays simply for figure size reasons.

Each mip tail is logically a single array of sparse blocks with an implementation-dependent mapping of texels or compressed texel blocks to sparse blocks.

When `VK_SPARSE_IMAGE_FORMAT_ALIGNED_MIP_SIZE_BIT` is present the first mip level that would contain partially used sparse blocks begins the mip tail region. This level and all subsequent levels are placed in the mip tail. Only the first N mip levels whose dimensions are an exact multiple of the sparse image block dimensions **can** be bound and unbound on a sparse block basis.

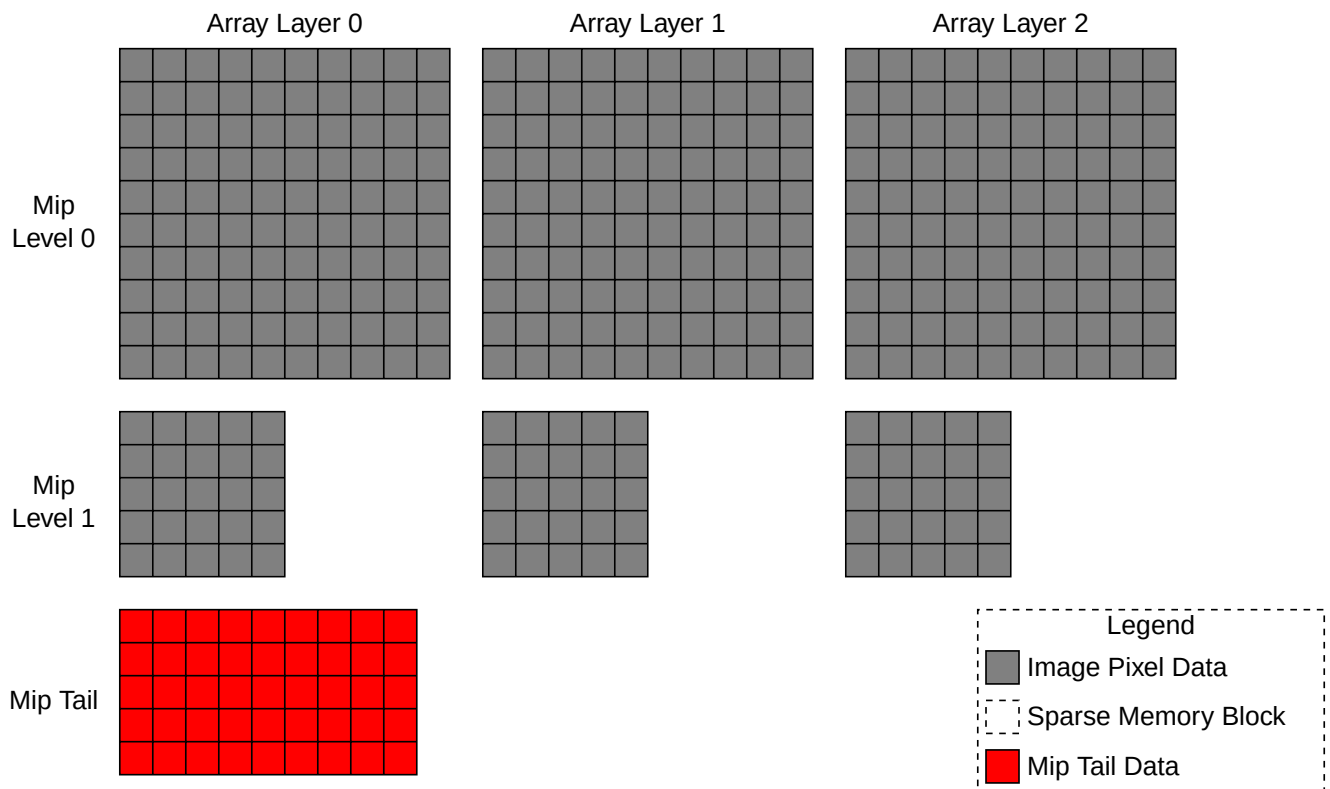


Figure 13. Sparse Image with Aligned Mip Size and Single Mip Tail

NOTE

The mip tail region is presented here in a 2D array simply for figure size reasons. It is logically a single array of sparse blocks with an implementation-dependent mapping of texels or compressed texel blocks to sparse blocks.

When both `VK_SPARSE_IMAGE_FORMAT_ALIGNED_MIP_SIZE_BIT` and `VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT` are present the constraints from each of these flags are in effect.

29.4.3. Standard Sparse Image Block Shapes

Standard sparse image block shapes define a standard set of dimensions for sparse image blocks that depend on the format of the image. Layout of texels or compressed texel blocks within a sparse image block is implementation-dependent. All currently defined standard sparse image block shapes are 64 KB in size.

For block-compressed formats (e.g. `VK_FORMAT_BC5_UNORM_BLOCK`), the texel size is the size of the compressed texel block (e.g. 128-bit for `BC5`) thus the dimensions of the standard sparse image block shapes apply in terms of compressed texel blocks.

NOTE

For block-compressed formats, the dimensions of a sparse image block in terms of texels **can** be calculated by multiplying the sparse image block dimensions by the compressed texel block dimensions.

Table 27. Standard Sparse Image Block Shapes (Single Sample)

TEXEL SIZE (bits)	Block Shape (2D)	Block Shape (3D)
8-Bit	$256 \times 256 \times 1$	$64 \times 32 \times 32$
16-Bit	$256 \times 128 \times 1$	$32 \times 32 \times 32$
32-Bit	$128 \times 128 \times 1$	$32 \times 32 \times 16$
64-Bit	$128 \times 64 \times 1$	$32 \times 16 \times 16$
128-Bit	$64 \times 64 \times 1$	$16 \times 16 \times 16$

Table 28. Standard Sparse Image Block Shapes (MSAA)

TEXEL SIZE (bits)	Block Shape (2X)	Block Shape (4X)	Block Shape (8X)	Block Shape (16X)
8-Bit	$128 \times 256 \times 1$	$128 \times 128 \times 1$	$64 \times 128 \times 1$	$64 \times 64 \times 1$
16-Bit	$128 \times 128 \times 1$	$128 \times 64 \times 1$	$64 \times 64 \times 1$	$64 \times 32 \times 1$
32-Bit	$64 \times 128 \times 1$	$64 \times 64 \times 1$	$32 \times 64 \times 1$	$32 \times 32 \times 1$
64-Bit	$64 \times 64 \times 1$	$64 \times 32 \times 1$	$32 \times 32 \times 1$	$32 \times 16 \times 1$
128-Bit	$32 \times 64 \times 1$	$32 \times 32 \times 1$	$16 \times 32 \times 1$	$16 \times 16 \times 1$

Implementations that support the standard sparse image block shape for all formats listed in the [Standard Sparse Image Block Shapes \(Single Sample\)](#) and [Standard Sparse Image Block Shapes \(MSAA\)](#) tables **may** advertise the following `VkPhysicalDeviceSparseProperties`:

- `residencyStandard2DBlockShape`
- `residencyStandard2DMultisampleBlockShape`
- `residencyStandard3DBlockShape`

Reporting each of these features does *not* imply that all possible image types are supported as sparse. Instead, this indicates that no supported sparse image of the corresponding type will use custom sparse image block dimensions for any formats that have a corresponding standard sparse image block shape.

29.4.4. Custom Sparse Image Block Shapes

An implementation that does not support a standard image block shape for a particular sparse partially-resident image **may** choose to support a custom sparse image block shape for it instead. The dimensions of such a custom sparse image block shape are reported in `VkSparseImageFormatProperties::imageGranularity`. As with standard sparse image block shapes, the size in bytes of the custom sparse image block shape will be reported in `VkMemoryRequirements::alignment`.

Custom sparse image block dimensions are reported through `vkGetPhysicalDeviceSparseImageFormatProperties` and `vkGetImageSparseMemoryRequirements`.

An implementation **must** not support both the standard sparse image block shape and a custom

sparse image block shape for the same image. The standard sparse image block shape **must** be used if it is supported.

29.4.5. Multiple Aspects

Partially resident images are allowed to report separate sparse properties for different aspects of the image. One example is for depth/stencil images where the implementation separates the depth and stencil data into separate planes. Another reason for multiple aspects is to allow the application to manage memory allocation for implementation-private *metadata* associated with the image. See the figure below:

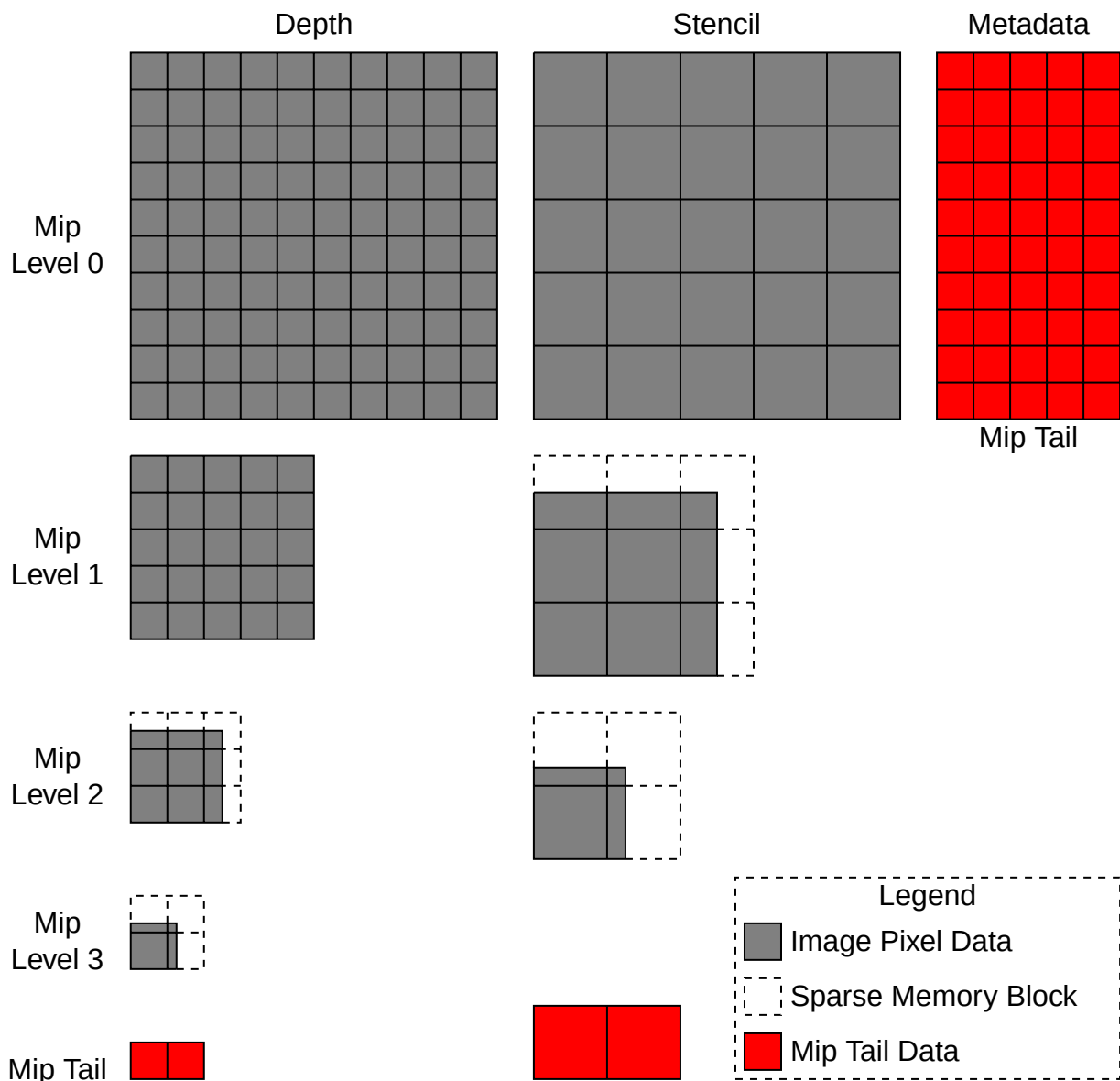


Figure 14. Multiple Aspect Sparse Image

NOTE

The mip tail regions are presented here in 2D arrays simply for figure size reasons. Each mip tail is logically a single array of sparse blocks with an implementation-dependent mapping of texels or compressed texel blocks to sparse blocks.

In the figure above the depth, stencil, and metadata aspects all have unique sparse properties. The per-texel stencil data is $\frac{1}{4}$ the size of the depth data, hence the stencil sparse blocks include $4 \times$ the number of texels. The sparse block size in bytes for all of the aspects is identical and defined by

VkMemoryRequirements::alignment.

Metadata

The metadata aspect of an image has the following constraints:

- All metadata is reported in the mip tail region of the metadata aspect.
- All metadata **must** be bound prior to device use of the sparse image.

29.5. Sparse Memory Aliasing

By default sparse resources have the same aliasing rules as non-sparse resources. See [Memory Aliasing](#) for more information.

VkDevice objects that have the `sparseResidencyAliased` feature enabled are able to use the `VK_BUFFER_CREATE_SPARSE_ALIASED_BIT` and `VK_IMAGE_CREATE_SPARSE_ALIASED_BIT` flags for resource creation. These flags allow resources to access physical memory bound into multiple locations within one or more sparse resources in a *data consistent* fashion. This means that reading physical memory from multiple aliased locations will return the same value.

Care **must** be taken when performing a write operation to aliased physical memory. Memory dependencies **must** be used to separate writes to one alias from reads or writes to another alias. Writes to aliased memory that are not properly guarded against accesses to different aliases will have undefined results for all accesses to the aliased memory.

Applications that wish to make use of data consistent sparse memory aliasing **must** abide by the following guidelines:

- All sparse resources that are bound to aliased physical memory **must** be created with the `VK_BUFFER_CREATE_SPARSE_ALIASED_BIT` / `VK_IMAGE_CREATE_SPARSE_ALIASED_BIT` flag.
- All resources that access aliased physical memory **must** interpret the memory in the same way. This implies the following:
 - Buffers and images **cannot** alias the same physical memory in a data consistent fashion. The physical memory ranges **must** be used exclusively by buffers or used exclusively by images for data consistency to be guaranteed.
 - Memory in sparse image mip tail regions **cannot** access aliased memory in a data consistent fashion.
 - Sparse images that alias the same physical memory **must** have compatible formats and be using the same sparse image block shape in order to access aliased memory in a data consistent fashion.

Failure to follow any of the above guidelines will require the application to abide by the normal, non-sparse resource [aliasing rules](#). In this case memory **cannot** be accessed in a data consistent fashion.

NOTE

Enabling sparse resource memory aliasing **can** be a way to lower physical memory use, but it **may** reduce performance on some implementations. An application

developer **can** test on their target HW and balance the memory / performance trade-offs measured.

29.6. Sparse Resource Implementation Guidelines (Informative)

This section is Informative. It is included to aid in implementors' understanding of sparse resources.

Device Virtual Address

The basic **sparseBinding** feature allows the resource to reserve its own device virtual address range at resource creation time rather than relying on a bind operation to set this. Without any other creation flags, no other constraints are relaxed compared to normal resources. All pages **must** be bound to physical memory before the device accesses the resource.

The **sparseResidency** features allow sparse resources to be used even when not all pages are bound to memory. Implementations that support access to unbound pages without causing a fault **may** support **residencyNonResidentStrict**.

Not faulting on access to unbound pages is not enough to support **residencyNonResidentStrict**. An implementation **must** also guarantee that reads after writes to unbound regions of the resource always return data for the read as if the memory contains zeros. Depending on any caching hierarchy of the implementation this **may** not always be possible.

Any implementation that does not fault, but does not guarantee correct read values **must** not support **residencyNonResidentStrict**.

Any implementation that **cannot** access unbound pages without causing a fault will require the implementation to bind the entire device virtual address range to physical memory. Any pages that the application does not bind to memory **may** be bound to one (or more) "placeholder" physical page(s) allocated by the implementation. Given the following properties:

- A process **must** not access memory from another process
- Reads return undefined values

It is sufficient for each host process to allocate these placeholder pages and use them for all resources in that process. Implementations **may** allocate more often (per instance, per device, or per resource).

Binding Memory

The byte size reported in **VkMemoryRequirements::size** **must** be greater than or equal to the amount of physical memory **required** to fully populate the resource. Some implementations require "holes" in the device virtual address range that are never accessed. These holes **may** be included in the **size** reported for the resource.

Including or not including the device virtual address holes in the resource size will alter how

the implementation provides support for `VkSparseImageOpaqueMemoryBindInfo`. This operation **must** be supported for all sparse images, even ones created with `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.

- If the holes are included in the size, this bind function becomes very easy. In most cases the `resourceOffset` is simply a device virtual address offset and the implementation can easily determine what device virtual address to bind. The cost is that the application **may** allocate more physical memory for the resource than it needs.
- If the holes are not included in the size, the application **can** allocate less physical memory than otherwise for the resource. However, in this case the implementation **must** account for the holes when mapping `resourceOffset` to the actual device virtual address intended to be mapped.

If the application always uses `VkSparseImageMemoryBindInfo` to bind memory for the non-tail mip levels, any holes that are present in the resource size **may** never be bound.

NOTE

Since `VkSparseImageMemoryBindInfo` uses texel locations to determine which device virtual addresses to bind, it is impossible to bind device virtual address holes with this operation.

Binding Metadata Memory

All metadata for sparse images have their own sparse properties and are embedded in the mip tail region for said properties. See the [Multiaspect](#) section for details.

Given that metadata is in a mip tail region, and the mip tail region **must** be reported as contiguous (either globally or per-array-layer), some implementations will have to resort to complicated offset → device virtual address mapping for handling `VkSparseImageOpaqueMemoryBindInfo`.

To make this easier on the implementation, the `VK_SPARSE_MEMORY_BIND_METADATA_BIT` explicitly specifies when metadata is bound with `VkSparseImageOpaqueMemoryBindInfo`. When this flag is not present, the `resourceOffset` **may** be treated as a strict device virtual address offset.

When `VK_SPARSE_MEMORY_BIND_METADATA_BIT` is present, the `resourceOffset` **must** have been derived explicitly from the `imageMipTailOffset` in the sparse resource properties returned for the metadata aspect. By manipulating the value returned for `imageMipTailOffset`, the `resourceOffset` does not have to correlate directly to a device virtual address offset, and **may** instead be whatever value makes it easiest for the implementation to derive the correct device virtual address.

29.7. Sparse Resource API

The APIs related to sparse resources are grouped into the following categories:

- [Physical Device Features](#)
- [Physical Device Sparse Properties](#)

- [Sparse Image Format Properties](#)
- [Sparse Resource Creation](#)
- [Sparse Resource Memory Requirements](#)
- [Binding Resource Memory](#)

29.7.1. Physical Device Features

Some sparse-resource related features are reported and enabled in `VkPhysicalDeviceFeatures`. These features **must** be supported and enabled on the `VkDevice` object before applications **can** use them. See [Physical Device Features](#) for information on how to get and set enabled device features, and for more detailed explanations of these features.

Sparse Physical Device Features

- `sparseBinding`: Support for creating `VkBuffer` and `VkImage` objects with the `VK_BUFFER_CREATE_SPARSE_BINDING_BIT` and `VK_IMAGE_CREATE_SPARSE_BINDING_BIT` flags, respectively.
- `sparseResidencyBuffer`: Support for creating `VkBuffer` objects with the `VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT` flag.
- `sparseResidencyImage2D`: Support for creating 2D single-sampled `VkImage` objects with `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.
- `sparseResidencyImage3D`: Support for creating 3D `VkImage` objects with `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.
- `sparseResidency2Samples`: Support for creating 2D `VkImage` objects with 2 samples and `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.
- `sparseResidency4Samples`: Support for creating 2D `VkImage` objects with 4 samples and `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.
- `sparseResidency8Samples`: Support for creating 2D `VkImage` objects with 8 samples and `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.
- `sparseResidency16Samples`: Support for creating 2D `VkImage` objects with 16 samples and `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.
- `sparseResidencyAliased`: Support for creating `VkBuffer` and `VkImage` objects with the `VK_BUFFER_CREATE_SPARSE_ALIASED_BIT` and `VK_IMAGE_CREATE_SPARSE_ALIASED_BIT` flags, respectively.

29.7.2. Physical Device Sparse Properties

Some features of the implementation are not possible to disable, and are reported to allow applications to alter their sparse resource usage accordingly. These read-only capabilities are reported in the `VkPhysicalDeviceProperties::sparseProperties` member, which is a `VkPhysicalDeviceSparseProperties` structure.

The `VkPhysicalDeviceSparseProperties` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPhysicalDeviceSparseProperties {
    VkBool32    residencyStandard2DBlockShape;
    VkBool32    residencyStandard2DMultisampleBlockShape;
    VkBool32    residencyStandard3DBlockShape;
    VkBool32    residencyAlignedMipSize;
    VkBool32    residencyNonResidentStrict;
} VkPhysicalDeviceSparseProperties;
```

- **residencyStandard2DBlockShape** is **VK_TRUE** if the physical device will access all single-sample 2D sparse resources using the standard sparse image block shapes (based on image format), as described in the [Standard Sparse Image Block Shapes \(Single Sample\)](#) table. If this property is not supported the value returned in the **imageGranularity** member of the **VkSparseImageFormatProperties** structure for single-sample 2D images is not **required** to match the standard sparse image block dimensions listed in the table.
- **residencyStandard2DMultisampleBlockShape** is **VK_TRUE** if the physical device will access all multisample 2D sparse resources using the standard sparse image block shapes (based on image format), as described in the [Standard Sparse Image Block Shapes \(MSAA\)](#) table. If this property is not supported, the value returned in the **imageGranularity** member of the **VkSparseImageFormatProperties** structure for multisample 2D images is not **required** to match the standard sparse image block dimensions listed in the table.
- **residencyStandard3DBlockShape** is **VK_TRUE** if the physical device will access all 3D sparse resources using the standard sparse image block shapes (based on image format), as described in the [Standard Sparse Image Block Shapes \(Single Sample\)](#) table. If this property is not supported, the value returned in the **imageGranularity** member of the **VkSparseImageFormatProperties** structure for 3D images is not **required** to match the standard sparse image block dimensions listed in the table.
- **residencyAlignedMipSize** is **VK_TRUE** if images with mip level dimensions that are not integer multiples of the corresponding dimensions of the sparse image block **may** be placed in the mip tail. If this property is not reported, only mip levels with dimensions smaller than the **imageGranularity** member of the **VkSparseImageFormatProperties** structure will be placed in the mip tail. If this property is reported the implementation is allowed to return **VK_SPARSE_IMAGE_FORMAT_ALIGNED_MIP_SIZE_BIT** in the **flags** member of **VkSparseImageFormatProperties**, indicating that mip level dimensions that are not integer multiples of the corresponding dimensions of the sparse image block will be placed in the mip tail.
- **residencyNonResidentStrict** specifies whether the physical device **can** consistently access non-resident regions of a resource. If this property is **VK_TRUE**, access to non-resident regions of resources will be guaranteed to return values as if the resource was populated with 0; writes to non-resident regions will be discarded.

29.7.3. Sparse Image Format Properties

Given that certain aspects of sparse image support, including the sparse image block dimensions, **may** be implementation-dependent, **vkGetPhysicalDeviceSparseImageFormatProperties** **can** be used to query for sparse image format properties prior to resource creation. This command is used

to check whether a given set of sparse image parameters is supported and what the sparse image block shape will be.

Sparse Image Format Properties API

The `VkSparseImageFormatProperties` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSparseImageFormatProperties {
    VkImageAspectFlags    aspectMask;
    VkExtent3D            imageGranularity;
    VkSparseImageFormatFlags flags;
} VkSparseImageFormatProperties;
```

- `aspectMask` is a bitmask `VkImageAspectFlagBits` specifying which aspects of the image the properties apply to.
- `imageGranularity` is the width, height, and depth of the sparse image block in texels or compressed texel blocks.
- `flags` is a bitmask of `VkSparseImageFormatFlagBits` specifying additional information about the sparse resource.

Bits which **may** be set in `VkSparseImageFormatProperties::flags`, specifying additional information about the sparse resource, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSparseImageFormatFlagBits {
    VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT = 0x00000001,
    VK_SPARSE_IMAGE_FORMAT_ALIGNED_MIP_SIZE_BIT = 0x00000002,
    VK_SPARSE_IMAGE_FORMAT_NONSTANDARD_BLOCK_SIZE_BIT = 0x00000004,
} VkSparseImageFormatFlagBits;
```

- `VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT` specifies that the image uses a single mip tail region for all array layers.
- `VK_SPARSE_IMAGE_FORMAT_ALIGNED_MIP_SIZE_BIT` specifies that the first mip level whose dimensions are not integer multiples of the corresponding dimensions of the sparse image block begins the mip tail region.
- `VK_SPARSE_IMAGE_FORMAT_NONSTANDARD_BLOCK_SIZE_BIT` specifies that the image uses non-standard sparse image block dimensions, and the `imageGranularity` values do not match the standard sparse image block dimensions for the given format.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkSparseImageFormatFlags;
```

`VkSparseImageFormatFlags` is a bitmask type for setting a mask of zero or more `VkSparseImageFormatFlagBits`.

`vkGetPhysicalDeviceSparseImageFormatProperties` returns an array of `VkSparseImageFormatProperties`. Each element describes properties for one set of image aspects that are bound simultaneously for a `VkImage` created with the provided image creation parameters. This is usually one element for each aspect in the image, but for interleaved depth/stencil images there is only one element describing the combined aspects.

```
// Provided by VK_VERSION_1_0
void vkGetPhysicalDeviceSparseImageFormatProperties(
    VkPhysicalDevice    physicalDevice,
    VkFormat            format,
    VkImageType         type,
    VkSampleCountFlagBits samples,
    VkImageUsageFlags   usage,
    VkImageTiling       tiling,
    uint32_t*           pPropertyCount,
    VkSparseImageFormatProperties* pProperties);
```

- `physicalDevice` is the physical device from which to query the sparse image format properties.
- `format` is the image format.
- `type` is the dimensionality of the image.
- `samples` is a `VkSampleCountFlagBits` value specifying the number of samples per texel.
- `usage` is a bitmask describing the intended usage of the image.
- `tiling` is the tiling arrangement of the texel blocks in memory.
- `pPropertyCount` is a pointer to an integer related to the number of sparse format properties available or queried, as described below.
- `pProperties` is either `NULL` or a pointer to an array of `VkSparseImageFormatProperties` structures.

If `pProperties` is `NULL`, then the number of sparse format properties available is returned in `pPropertyCount`. Otherwise, `pPropertyCount` **must** point to a variable set by the application to the number of elements in the `pProperties` array, and on return the variable is overwritten with the number of structures actually written to `pProperties`. If `pPropertyCount` is less than the number of sparse format properties available, at most `pPropertyCount` structures will be written.

If `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` is not supported for the given arguments, `pPropertyCount` will be zero upon return, and no data will be written to `pProperties`.

Multiple aspects are returned for depth/stencil images that are implemented as separate planes by the implementation. The depth and stencil data planes each have unique `VkSparseImageFormatProperties` data.

Depth/stencil images with depth and stencil data interleaved into a single plane will return a single `VkSparseImageFormatProperties` structure with the `aspectMask` set to `VK_IMAGE_ASPECT_DEPTH_BIT | VK_IMAGE_ASPECT_STENCIL_BIT`.

Valid Usage

- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-samples-01094
`samples` **must** be a valid `VkSampleCountFlagBits` value that is set in `VkImageFormatProperties::sampleCounts` returned by `vkGetPhysicalDeviceImageFormatProperties` with `format`, `type`, `tiling`, and `usage` equal to those in this command

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-physicalDevice-parameter
`physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-format-parameter
`format` **must** be a valid `VkFormat` value
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-type-parameter
`type` **must** be a valid `VkImageType` value
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-samples-parameter
`samples` **must** be a valid `VkSampleCountFlagBits` value
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-usage-parameter
`usage` **must** be a valid combination of `VkImageUsageFlagBits` values
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-usage-requiredbitmask
`usage` **must** not be `0`
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-tiling-parameter
`tiling` **must** be a valid `VkImageTiling` value
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-pPropertyCount-parameter
`pPropertyCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-pProperties-parameter
If the value referenced by `pPropertyCount` is not `0`, and `pProperties` is not `NULL`, `pProperties` **must** be a valid pointer to an array of `pPropertyCount` `VkSparseImageFormatProperties` structures

29.7.4. Sparse Resource Creation

Sparse resources require that one or more sparse feature flags be specified (as part of the `VkPhysicalDeviceFeatures` structure described previously in the [Physical Device Features](#) section) when calling `vkCreateDevice`. When the appropriate device features are enabled, the `VK_BUFFER_CREATE_SPARSE_*` and `VK_IMAGE_CREATE_SPARSE_*` flags **can** be used. See [vkCreateBuffer](#) and [vkCreateImage](#) for details of the resource creation APIs.

NOTE

Specifying `VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT` or `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` requires specifying `VK_BUFFER_CREATE_SPARSE_BINDING_BIT` or `VK_IMAGE_CREATE_SPARSE_BINDING_BIT`,

respectively, as well. This means that resources **must** be created with the appropriate `*_SPARSE_BINDING_BIT` to be used with the sparse binding command (`vkQueueBindSparse`).

29.7.5. Sparse Resource Memory Requirements

Sparse resources have specific memory requirements related to binding sparse memory. These memory requirements are reported differently for `VkBuffer` objects and `VkImage` objects.

Buffer and Fully-Resident Images

Buffers (both fully and partially resident) and fully-resident images **can** be bound to memory using only the data from `VkMemoryRequirements`. For all sparse resources the `VkMemoryRequirements::alignment` member specifies both the binding granularity in bytes and the **required** alignment of `VkDeviceMemory`.

Partially Resident Images

Partially resident images have a different method for binding memory. As with buffers and fully resident images, the `VkMemoryRequirements::alignment` field specifies the binding granularity in bytes for the image.

Requesting sparse memory requirements for `VkImage` objects using `vkGetImageSparseMemoryRequirements` will return an array of one or more `VkSparseImageMemoryRequirements` structures. Each structure describes the sparse memory requirements for a group of aspects of the image.

The sparse image **must** have been created using the `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` flag to retrieve valid sparse image memory requirements.

Sparse Image Memory Requirements

The `VkSparseImageMemoryRequirements` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSparseImageMemoryRequirements {
    VkSparseImageFormatProperties    formatProperties;
    uint32_t                        imageMipTailFirstLod;
    VkDeviceSize                    imageMipTailSize;
    VkDeviceSize                    imageMipTailOffset;
    VkDeviceSize                    imageMipTailStride;
} VkSparseImageMemoryRequirements;
```

- `formatProperties` is a `VkSparseImageFormatProperties` structure specifying properties of the image format.
- `imageMipTailFirstLod` is the first mip level at which image subresources are included in the mip tail region.
- `imageMipTailSize` is the memory size (in bytes) of the mip tail region. If `formatProperties.flags`

contains `VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT`, this is the size of the whole mip tail, otherwise this is the size of the mip tail of a single array layer. This value is guaranteed to be a multiple of the sparse block size in bytes.

- `imageMipTailOffset` is the opaque memory offset used with `VkSparseImageOpaqueMemoryBindInfo` to bind the mip tail region(s).
- `imageMipTailStride` is the offset stride between each array-layer's mip tail, if `formatProperties.flags` does not contain `VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT` (otherwise the value is undefined).

To query sparse memory requirements for an image, call:

```
// Provided by VK_VERSION_1_0
void vkGetImageSparseMemoryRequirements(
    VkDevice          device,
    VkImage           image,
    uint32_t*         pSparseMemoryRequirementCount,
    VkSparseImageMemoryRequirements* pSparseMemoryRequirements);
```

- `device` is the logical device that owns the image.
- `image` is the `VkImage` object to get the memory requirements for.
- `pSparseMemoryRequirementCount` is a pointer to an integer related to the number of sparse memory requirements available or queried, as described below.
- `pSparseMemoryRequirements` is either `NULL` or a pointer to an array of `VkSparseImageMemoryRequirements` structures.

If `pSparseMemoryRequirements` is `NULL`, then the number of sparse memory requirements available is returned in `pSparseMemoryRequirementCount`. Otherwise, `pSparseMemoryRequirementCount` **must** point to a variable set by the application to the number of elements in the `pSparseMemoryRequirements` array, and on return the variable is overwritten with the number of structures actually written to `pSparseMemoryRequirements`. If `pSparseMemoryRequirementCount` is less than the number of sparse memory requirements available, at most `pSparseMemoryRequirementCount` structures will be written.

If the image was not created with `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` then `pSparseMemoryRequirementCount` will be zero and `pSparseMemoryRequirements` will not be written to.

NOTE

It is legal for an implementation to report a larger value in `VkMemoryRequirements::size` than would be obtained by adding together memory sizes for all `VkSparseImageMemoryRequirements` returned by `vkGetImageSparseMemoryRequirements`. This **may** occur when the implementation requires unused padding in the address range describing the resource.

Valid Usage (Implicit)

- VUID-vkGetImageSparseMemoryRequirements-device-parameter `device` **must** be a valid `VkDevice` handle

- VUID-vkGetImageSparseMemoryRequirements-image-parameter
`image` **must** be a valid `VkImage` handle
- VUID-vkGetImageSparseMemoryRequirements-pSparseMemoryRequirementCount-parameter
`pSparseMemoryRequirementCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetImageSparseMemoryRequirements-pSparseMemoryRequirements-parameter
If the value referenced by `pSparseMemoryRequirementCount` is not `0`, and `pSparseMemoryRequirements` is not `NULL`, `pSparseMemoryRequirements` **must** be a valid pointer to an array of `pSparseMemoryRequirementCount` `VkSparseImageMemoryRequirements` structures
- VUID-vkGetImageSparseMemoryRequirements-image-parent
`image` **must** have been created, allocated, or retrieved from `device`

29.7.6. Binding Resource Memory

Non-sparse resources are backed by a single physical allocation prior to device use (via `vkBindImageMemory` or `vkBindBufferMemory`), and their backing **must** not be changed. On the other hand, sparse resources **can** be bound to memory non-contiguously and these bindings **can** be altered during the lifetime of the resource.

NOTE

It is important to note that freeing a `VkDeviceMemory` object with `vkFreeMemory` will not cause resources (or resource regions) bound to the memory object to become unbound. Applications **must** not access resources bound to memory that has been freed.

Sparse memory bindings execute on a queue that includes the `VK_QUEUE_SPARSE_BINDING_BIT` bit. Applications **must** use `synchronization primitives` to guarantee that other queues do not access ranges of memory concurrently with a binding change. Applications **can** access other ranges of the same resource while a bind operation is executing.

NOTE

Implementations **must** provide a guarantee that simultaneously binding sparse blocks while another queue accesses those same sparse blocks via a sparse resource **must** not access memory owned by another process or otherwise corrupt the system.

While some implementations **may** include `VK_QUEUE_SPARSE_BINDING_BIT` support in queue families that also include graphics and compute support, other implementations **may** only expose a `VK_QUEUE_SPARSE_BINDING_BIT`-only queue family. In either case, applications **must** use `synchronization primitives` to explicitly request any ordering dependencies between sparse memory binding operations and other graphics/compute/transfer operations, as sparse binding operations are not automatically ordered against command buffer execution, even within a single queue.

When binding memory explicitly for the `VK_IMAGE_ASPECT_METADATA_BIT` the application **must** use the `VK_SPARSE_MEMORY_BIND_METADATA_BIT` in the `VkSparseMemoryBind::flags` field when binding memory. Binding memory for metadata is done the same way as binding memory for the mip tail, with the

addition of the `VK_SPARSE_MEMORY_BIND_METADATA_BIT` flag.

Binding the mip tail for any aspect **must** only be performed using `VkSparseImageOpaqueMemoryBindInfo`. If `formatProperties.flags` contains `VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT`, then it **can** be bound with a single `VkSparseMemoryBind` structure, with `resourceOffset = imageMipTailOffset` and `size = imageMipTailSize`.

If `formatProperties.flags` does not contain `VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT` then the offset for the mip tail in each array layer is given as:

```
arrayMipTailOffset = imageMipTailOffset + arrayLayer * imageMipTailStride;
```

and the mip tail **can** be bound with `layerCount` `VkSparseMemoryBind` structures, each using `size = imageMipTailSize` and `resourceOffset = arrayMipTailOffset` as defined above.

Sparse memory binding is handled by the following APIs and related data structures.

Sparse Memory Binding Functions

The `VkSparseMemoryBind` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSparseMemoryBind {
    VkDeviceSize      resourceOffset;
    VkDeviceSize      size;
    VkDeviceMemory     memory;
    VkDeviceSize      memoryOffset;
    VkSparseMemoryBindFlags flags;
} VkSparseMemoryBind;
```

- `resourceOffset` is the offset into the resource.
- `size` is the size of the memory region to be bound.
- `memory` is the `VkDeviceMemory` object that the range of the resource is bound to. If `memory` is `VK_NULL_HANDLE`, the range is unbound.
- `memoryOffset` is the offset into the `VkDeviceMemory` object to bind the resource range to. If `memory` is `VK_NULL_HANDLE`, this value is ignored.
- `flags` is a bitmask of `VkSparseMemoryBindFlagBits` specifying usage of the binding operation.

The *binding range* [`resourceOffset`, `resourceOffset + size`) has different constraints based on `flags`. If `flags` contains `VK_SPARSE_MEMORY_BIND_METADATA_BIT`, the binding range **must** be within the mip tail region of the metadata aspect. This metadata region is defined by:

```
metadataRegion = [base, base + imageMipTailSize)
```


$\text{base} = \text{imageMipTailOffset} + \text{imageMipTailStride} \times n$

and `imageMipTailOffset`, `imageMipTailSize`, and `imageMipTailStride` values are from the `VkSparseImageMemoryRequirements` corresponding to the metadata aspect of the image, and `n` is a valid array layer index for the image,

`imageMipTailStride` is considered to be zero for aspects where `VkSparseImageMemoryRequirements::formatProperties.flags` contains `VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT`.

If `flags` does not contain `VK_SPARSE_MEMORY_BIND_METADATA_BIT`, the binding range **must** be within the range `[0, VkMemoryRequirements::size)`.

Valid Usage

- VUID-VkSparseMemoryBind-memory-01096
If `memory` is not `VK_NULL_HANDLE`, `memory` and `memoryOffset` **must** match the memory requirements of the resource, as described in section [Resource Memory Association](#)
- VUID-VkSparseMemoryBind-resourceOffset-09491
If the resource being bound is a `VkBuffer`, `resourceOffset`, `memoryOffset` and `size` **must** be an integer multiple of the `alignment` of the `VkMemoryRequirements` structure returned from a call to `vkGetBufferMemoryRequirements` with the buffer resource
- VUID-VkSparseMemoryBind-resourceOffset-09492
If the resource being bound is a `VkImage`, `resourceOffset` and `memoryOffset` **must** be an integer multiple of the `alignment` of the `VkMemoryRequirements` structure returned from a call to `vkGetImageMemoryRequirements` with the image resource
- VUID-VkSparseMemoryBind-memory-01097
If `memory` is not `VK_NULL_HANDLE`, `memory` **must** not have been created with a memory type that reports `VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT` bit set
- VUID-VkSparseMemoryBind-size-01098
`size` **must** be greater than 0
- VUID-VkSparseMemoryBind-resourceOffset-01099
`resourceOffset` **must** be less than the size of the resource
- VUID-VkSparseMemoryBind-size-01100
`size` **must** be less than or equal to the size of the resource minus `resourceOffset`
- VUID-VkSparseMemoryBind-memoryOffset-01101
`memoryOffset` **must** be less than the size of `memory`
- VUID-VkSparseMemoryBind-size-01102
`size` **must** be less than or equal to the size of `memory` minus `memoryOffset`

Valid Usage (Implicit)

- VUID-VkSparseMemoryBind-memory-parameter
If `memory` is not `VK_NULL_HANDLE`, `memory` **must** be a valid `VkDeviceMemory` handle

- VUID-VkSparseMemoryBind-flags-parameter
flags must be a valid combination of [VkSparseMemoryBindFlagBits](#) values

Bits which **can** be set in [VkSparseMemoryBind::flags](#), specifying usage of a sparse memory binding operation, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSparseMemoryBindFlagBits {
    VK_SPARSE_MEMORY_BIND_METADATA_BIT = 0x00000001,
} VkSparseMemoryBindFlagBits;
```

- [VK_SPARSE_MEMORY_BIND_METADATA_BIT](#) specifies that the memory being bound is only for the metadata aspect.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkSparseMemoryBindFlags;
```

[VkSparseMemoryBindFlags](#) is a bitmask type for setting a mask of zero or more [VkSparseMemoryBindFlagBits](#).

Memory is bound to [VkBuffer](#) objects created with the [VK_BUFFER_CREATE_SPARSE_BINDING_BIT](#) flag using the following structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkSparseBufferMemoryBindInfo {
    VkBuffer          buffer;
    uint32_t          bindCount;
    const VkSparseMemoryBind* pBinds;
} VkSparseBufferMemoryBindInfo;
```

- **buffer** is the [VkBuffer](#) object to be bound.
- **bindCount** is the number of [VkSparseMemoryBind](#) structures in the **pBinds** array.
- **pBinds** is a pointer to an array of [VkSparseMemoryBind](#) structures.

Valid Usage (Implicit)

- VUID-VkSparseBufferMemoryBindInfo-buffer-parameter
buffer must be a valid [VkBuffer](#) handle
- VUID-VkSparseBufferMemoryBindInfo-pBinds-parameter
pBinds must be a valid pointer to an array of **bindCount** valid [VkSparseMemoryBind](#) structures
- VUID-VkSparseBufferMemoryBindInfo-bindCount-arraylength
bindCount must be greater than 0

Memory is bound to opaque regions of `VkImage` objects created with the `VK_IMAGE_CREATE_SPARSE_BINDING_BIT` flag using the following structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkSparseImageOpaqueMemoryBindInfo {
    VkImage          image;
    uint32_t         bindCount;
    const VkSparseMemoryBind* pBinds;
} VkSparseImageOpaqueMemoryBindInfo;
```

- `image` is the `VkImage` object to be bound.
- `bindCount` is the number of `VkSparseMemoryBind` structures in the `pBinds` array.
- `pBinds` is a pointer to an array of `VkSparseMemoryBind` structures.

NOTE

This structure is normally used to bind memory to fully-resident sparse images or for mip tail regions of partially resident images. However, it **can** also be used to bind memory for the entire binding range of partially resident images.

If the `pBinds[i].flags` of an element `i` of `pBinds` does not contain `VK_SPARSE_MEMORY_BIND_METADATA_BIT`, the `resourceOffset` is in the range `[0, VkMemoryRequirements::size)`. This range includes data from all aspects of the image, including metadata. For most implementations this will probably mean that the `resourceOffset` is a simple device address offset within the resource. It is possible for an application to bind a range of memory that includes both resource data and metadata. However, the application would not know what part of the image the memory is used for, or if any range is being used for metadata.

If the `pBinds[i].flags` of an element `i` of `pBinds` contains `VK_SPARSE_MEMORY_BIND_METADATA_BIT`, the binding range specified **must** be within the mip tail region of the metadata aspect. In this case the `resourceOffset` is not **required** to be a simple device address offset within the resource. However, it is defined to be within `[imageMipTailOffset, imageMipTailOffset + imageMipTailSize)` for the metadata aspect. See `VkSparseMemoryBind` for the full constraints on binding region with this flag present.

Valid Usage

- VUID-VkSparseImageOpaqueMemoryBindInfo-pBinds-01103

If the `flags` member of any element of `pBinds` contains `VK_SPARSE_MEMORY_BIND_METADATA_BIT`, the binding range defined **must** be within the mip tail region of the metadata aspect of `image`

Valid Usage (Implicit)

- VUID-VkSparseImageOpaqueMemoryBindInfo-image-parameter

image must be a valid [VkImage](#) handle

- VUID-VkSparseImageOpaqueMemoryBindInfo-pBinds-parameter
pBinds must be a valid pointer to an array of **bindCount** valid [VkSparseMemoryBind](#) structures
- VUID-VkSparseImageOpaqueMemoryBindInfo-bindCount-arraylength
bindCount must be greater than 0

Memory **can** be bound to sparse image blocks of [VkImage](#) objects created with the [VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT](#) flag using the following structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkSparseImageMemoryBindInfo {
    VkImage          image;
    uint32_t         bindCount;
    const VkSparseImageMemoryBind* pBinds;
} VkSparseImageMemoryBindInfo;
```

- **image** is the [VkImage](#) object to be bound
- **bindCount** is the number of [VkSparseImageMemoryBind](#) structures in **pBinds** array
- **pBinds** is a pointer to an array of [VkSparseImageMemoryBind](#) structures

Valid Usage

- VUID-VkSparseImageMemoryBindInfo-subresource-01722
The **subresource.mipLevel** member of each element of **pBinds** must be less than the **mipLevels** specified in [VkImageCreateInfo](#) when **image** was created
- VUID-VkSparseImageMemoryBindInfo-subresource-01723
The **subresource.arrayLayer** member of each element of **pBinds** must be less than the **arrayLayers** specified in [VkImageCreateInfo](#) when **image** was created
- VUID-VkSparseImageMemoryBindInfo-subresource-01106
The **subresource.aspectMask** member of each element of **pBinds** must be valid for the **format** specified in [VkImageCreateInfo](#) when **image** was created
- VUID-VkSparseImageMemoryBindInfo-image-02901
image must have been created with [VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT](#) set

Valid Usage (Implicit)

- VUID-VkSparseImageMemoryBindInfo-image-parameter
image must be a valid [VkImage](#) handle
- VUID-VkSparseImageMemoryBindInfo-pBinds-parameter
pBinds must be a valid pointer to an array of **bindCount** valid [VkSparseImageMemoryBind](#) structures

- VUID-VkSparseImageMemoryBindInfo-bindCount-arraylength
bindCount **must** be greater than 0

The **VkSparseImageMemoryBind** structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSparseImageMemoryBind {
    VkImageSubresource      subresource;
    VkOffset3D              offset;
    VkExtent3D              extent;
    VkDeviceMemory          memory;
    VkDeviceSize            memoryOffset;
    VkSparseMemoryBindFlags flags;
} VkSparseImageMemoryBind;
```

- **subresource** is the image *aspect* and region of interest in the image.
- **offset** are the coordinates of the first texel within the image subresource to bind.
- **extent** is the size in texels of the region within the image subresource to bind. The extent **must** be a multiple of the sparse image block dimensions, except when binding sparse image blocks along the edge of an image subresource it **can** instead be such that any coordinate of **offset** + **extent** equals the corresponding dimensions of the image subresource.
- **memory** is the **VkDeviceMemory** object that the sparse image blocks of the image are bound to. If **memory** is **VK_NULL_HANDLE**, the sparse image blocks are unbound.
- **memoryOffset** is an offset into **VkDeviceMemory** object. If **memory** is **VK_NULL_HANDLE**, this value is ignored.
- **flags** are sparse memory binding flags.

Valid Usage

- VUID-VkSparseImageMemoryBind-memory-01104
If the **sparseResidencyAliased** feature is not enabled, and if any other resources are bound to ranges of **memory**, the range of **memory** being bound **must** not overlap with those bound ranges
- VUID-VkSparseImageMemoryBind-memory-01105
memory and **memoryOffset** **must** match the memory requirements of the calling command's **image**, as described in section [Resource Memory Association](#)
- VUID-VkSparseImageMemoryBind-offset-01107
offset.x **must** be a multiple of the sparse image block width (**VkSparseImageFormatProperties::imageGranularity.width**) of the image
- VUID-VkSparseImageMemoryBind-extent-09388
extent.width **must** be greater than 0
- VUID-VkSparseImageMemoryBind-extent-01108
extent.width **must** either be a multiple of the sparse image block width of the image, or

else (`extent.width + offset.x`) **must** equal the width of the image subresource

- VUID-VkSparseImageMemoryBind-offset-01109
`offset.y` **must** be a multiple of the sparse image block height (`VkSparseImageFormatProperties::imageGranularity.height`) of the image
- VUID-VkSparseImageMemoryBind-extent-09389
`extent.height` **must** be greater than 0
- VUID-VkSparseImageMemoryBind-extent-01110
`extent.height` **must** either be a multiple of the sparse image block height of the image, or else (`extent.height + offset.y`) **must** equal the height of the image subresource
- VUID-VkSparseImageMemoryBind-offset-01111
`offset.z` **must** be a multiple of the sparse image block depth (`VkSparseImageFormatProperties::imageGranularity.depth`) of the image
- VUID-VkSparseImageMemoryBind-extent-09390
`extent.depth` **must** be greater than 0
- VUID-VkSparseImageMemoryBind-extent-01112
`extent.depth` **must** either be a multiple of the sparse image block depth of the image, or else (`extent.depth + offset.z`) **must** equal the depth of the image subresource

Valid Usage (Implicit)

- VUID-VkSparseImageMemoryBind-subresource-parameter
`subresource` **must** be a valid `VkImageSubresource` structure
- VUID-VkSparseImageMemoryBind-memory-parameter
If `memory` is not `VK_NULL_HANDLE`, `memory` **must** be a valid `VkDeviceMemory` handle
- VUID-VkSparseImageMemoryBind-flags-parameter
`flags` **must** be a valid combination of `VkSparseMemoryBindFlagBits` values

To submit sparse binding operations to a queue, call:

```
// Provided by VK_VERSION_1_0
VkResult vkQueueBindSparse(
    VkQueue          queue,
    uint32_t         bindInfoCount,
    const VkBindSparseInfo* pBindInfo,
    VkFence          fence);
```

- `queue` is the queue that the sparse binding operations will be submitted to.
- `bindInfoCount` is the number of elements in the `pBindInfo` array.
- `pBindInfo` is a pointer to an array of `VkBindSparseInfo` structures, each specifying a sparse binding submission batch.
- `fence` is an **optional** handle to a fence to be signaled. If `fence` is not `VK_NULL_HANDLE`, it

defines a [fence signal operation](#).

`vkQueueBindSparse` is a [queue submission command](#), with each batch defined by an element of `pBindInfo` as a `VkBindSparseInfo` structure. Batches begin execution in the order they appear in `pBindInfo`, but **may** complete out of order.

Within a batch, a given range of a resource **must** not be bound more than once. Across batches, if a range is to be bound to one allocation and offset and then to another allocation and offset, then the application **must** guarantee (usually using semaphores) that the binding operations are executed in the correct order, as well as to order binding operations against the execution of command buffer submissions.

As no operation to `vkQueueBindSparse` causes any pipeline stage to access memory, synchronization primitives used in this command effectively only define execution dependencies.

Additional information about fence and semaphore operation is described in [the synchronization chapter](#).

Valid Usage

- VUID-vkQueueBindSparse-fence-01113
If `fence` is not `VK_NULL_HANDLE`, `fence` **must** be unsignaled
- VUID-vkQueueBindSparse-fence-01114
If `fence` is not `VK_NULL_HANDLE`, `fence` **must** not be associated with any other queue command that has not yet completed execution on that queue
- VUID-vkQueueBindSparse-pSignalSemaphores-01115
Each element of the `pSignalSemaphores` member of each element of `pBindInfo` **must** be unsignaled when the semaphore signal operation it defines is executed on the device
- VUID-vkQueueBindSparse-pWaitSemaphores-01116
When a semaphore wait operation referring to a binary semaphore defined by any element of the `pWaitSemaphores` member of any element of `pBindInfo` executes on `queue`, there **must** be no other queues waiting on the same semaphore
- VUID-vkQueueBindSparse-pWaitSemaphores-03245
All elements of the `pWaitSemaphores` member of all elements of `pBindInfo` referring to a semaphore **must** reference a semaphore signal operation that has been submitted for execution and any [semaphore signal operations](#) on which it depends **must** have also been submitted for execution

Valid Usage (Implicit)

- VUID-vkQueueBindSparse-queue-parameter
`queue` **must** be a valid `VkQueue` handle
- VUID-vkQueueBindSparse-pBindInfo-parameter
If `bindInfoCount` is not 0, `pBindInfo` **must** be a valid pointer to an array of `bindInfoCount` valid `VkBindSparseInfo` structures

- VUID-vkQueueBindSparse-fence-parameter
If **fence** is not [VK_NULL_HANDLE](#), **fence** **must** be a valid [VkFence](#) handle
- VUID-vkQueueBindSparse-queue
The **queue** **must** support sparse binding operations
- VUID-vkQueueBindSparse-commonparent
Both of **fence**, and **queue** that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same [VkDevice](#)

Host Synchronization

- Host access to **queue** **must** be externally synchronized
- Host access to **fence** **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Supported Queue Types	Command Type
-	-	SPARSE_BINDING	-

Return Codes

Success

- [VK_SUCCESS](#)

Failure

- [VK_ERROR_OUT_OF_HOST_MEMORY](#)
- [VK_ERROR_OUT_OF_DEVICE_MEMORY](#)
- [VK_ERROR_DEVICE_LOST](#)

The [VkBindSparseInfo](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkBindSparseInfo {
    VkStructureType          sType;
    const void*              pNext;
    uint32_t                 waitSemaphoreCount;
    const VkSemaphore*       pWaitSemaphores;
    uint32_t                 bufferBindCount;
    const VkSparseBufferMemoryBindInfo* pBufferBinds;
    uint32_t                 imageOpaqueBindCount;
    const VkSparseImageOpaqueMemoryBindInfo* pImageOpaqueBinds;
    uint32_t                 imageBindCount;
    const VkSparseImageMemoryBindInfo* pImageBinds;
    uint32_t                 signalSemaphoreCount;
    const VkSemaphore*       pSignalSemaphores;
} VkBindSparseInfo;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **waitSemaphoreCount** is the number of semaphores upon which to wait before executing the sparse binding operations for the batch.
- **pWaitSemaphores** is a pointer to an array of semaphores upon which to wait on before the sparse binding operations for this batch begin execution. If semaphores to wait on are provided, they define a [semaphore wait operation](#).
- **bufferBindCount** is the number of sparse buffer bindings to perform in the batch.
- **pBufferBinds** is a pointer to an array of [VkSparseBufferMemoryBindInfo](#) structures.
- **imageOpaqueBindCount** is the number of opaque sparse image bindings to perform.
- **pImageOpaqueBinds** is a pointer to an array of [VkSparseImageOpaqueMemoryBindInfo](#) structures, indicating opaque sparse image bindings to perform.
- **imageBindCount** is the number of sparse image bindings to perform.
- **pImageBinds** is a pointer to an array of [VkSparseImageMemoryBindInfo](#) structures, indicating sparse image bindings to perform.
- **signalSemaphoreCount** is the number of semaphores to be signaled once the sparse binding operations specified by the structure have completed execution.
- **pSignalSemaphores** is a pointer to an array of semaphores which will be signaled when the sparse binding operations for this batch have completed execution. If semaphores to be signaled are provided, they define a [semaphore signal operation](#).

Valid Usage (Implicit)

- VUID-VkBindSparseInfo-sType-sType
sType **must** be `VK_STRUCTURE_TYPE_BIND_SPARSE_INFO`
- VUID-VkBindSparseInfo-pNext-pNext
pNext **must** be `NULL`

- VUID-VkBindSparseInfo-pWaitSemaphores-parameter
If `waitSemaphoreCount` is not 0, `pWaitSemaphores` **must** be a valid pointer to an array of `waitSemaphoreCount` valid `VkSemaphore` handles
- VUID-VkBindSparseInfo-pBufferBinds-parameter
If `bufferBindCount` is not 0, `pBufferBinds` **must** be a valid pointer to an array of `bufferBindCount` valid `VkSparseBufferMemoryBindInfo` structures
- VUID-VkBindSparseInfo-pImageOpaqueBinds-parameter
If `imageOpaqueBindCount` is not 0, `pImageOpaqueBinds` **must** be a valid pointer to an array of `imageOpaqueBindCount` valid `VkSparseImageOpaqueMemoryBindInfo` structures
- VUID-VkBindSparseInfo-pImageBinds-parameter
If `imageBindCount` is not 0, `pImageBinds` **must** be a valid pointer to an array of `imageBindCount` valid `VkSparseImageMemoryBindInfo` structures
- VUID-VkBindSparseInfo-pSignalSemaphores-parameter
If `signalSemaphoreCount` is not 0, `pSignalSemaphores` **must** be a valid pointer to an array of `signalSemaphoreCount` valid `VkSemaphore` handles
- VUID-VkBindSparseInfo-commonparent
Both of the elements of `pSignalSemaphores`, and the elements of `pWaitSemaphores` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

Chapter 30. Extending Vulkan

New functionality **may** be added to Vulkan via either new extensions or new versions of the core, or new versions of an extension in some cases.

This chapter describes how Vulkan is versioned, how compatibility is affected between different versions, and compatibility rules that are followed by the Vulkan Working Group.

30.1. Instance and Device Functionality

Commands that enumerate instance properties, or that accept a [VkInstance](#) object as a parameter, are considered instance-level functionality.

Commands that dispatch from a [VkDevice](#) object or a child object of a [VkDevice](#), or take any of them as a parameter, are considered device-level functionality. Types defined by a [device extension](#) are also considered device-level functionality.

Commands that dispatch from [VkPhysicalDevice](#), or accept a [VkPhysicalDevice](#) object as a parameter, are considered either instance-level or device-level functionality depending if the functionality is specified by an [instance extension](#) or [device extension](#) respectively.

30.2. Core Versions

The Vulkan Specification is regularly updated with bug fixes and clarifications. Occasionally new functionality is added to the core and at some point it is expected that there will be a desire to perform a large, breaking change to the API. In order to indicate to developers how and when these changes are made to the specification, and to provide a way to identify each set of changes, the Vulkan API maintains a version number.

30.2.1. Version Numbers

The Vulkan version number comprises four parts indicating the variant, major, minor and patch version of the Vulkan API Specification.

The *variant* indicates the variant of the Vulkan API supported by the implementation. This is always 0 for the Vulkan API.

NOTE

A non-zero variant indicates the API is a variant of the Vulkan API and applications will typically need to be modified to run against it. The variant field was a later addition to the version number, added in version 1.2.175 of the Specification. As Vulkan uses variant 0, this change is fully backwards compatible with the previous version number format for Vulkan implementations. New version number macros have been added for this change and the old macros deprecated. For existing applications using the older format and macros, an implementation with non-zero variant will decode as a very high Vulkan version. The high version number should be detectable by applications performing suitable version checking.

The *major version* indicates a significant change in the API, which will encompass a wholly new version of the specification.

The *minor version* indicates the incorporation of new functionality into the core specification.

The *patch version* indicates bug fixes, clarifications, and language improvements have been incorporated into the specification.

Compatibility guarantees made about versions of the API sharing any of the same version numbers are documented in [Core Versions](#)

The version number is used in several places in the API. In each such use, the version numbers are packed into a 32-bit integer as follows:

- The variant is a 3-bit integer packed into bits 31-29.
- The major version is a 7-bit integer packed into bits 28-22.
- The minor version number is a 10-bit integer packed into bits 21-12.
- The patch version number is a 12-bit integer packed into bits 11-0.

VK_API_VERSION_VARIANT extracts the API variant number from a packed version number:

```
// Provided by VK_VERSION_1_0
#define VK_API_VERSION_VARIANT(version) ((uint32_t)(version) >> 29U)
```

VK_API_VERSION_MAJOR extracts the API major version number from a packed version number:

```
// Provided by VK_VERSION_1_0
#define VK_API_VERSION_MAJOR(version) (((uint32_t)(version) >> 22U) & 0x7FU)
```

VK_VERSION_MAJOR extracts the API major version number from a packed version number:

```
// Provided by VK_VERSION_1_0
// VK_VERSION_MAJOR is deprecated, but no reason was given in the API XML
// DEPRECATED: This define is deprecated. VK_API_VERSION_MAJOR should be used instead.
#define VK_VERSION_MAJOR(version) ((uint32_t)(version) >> 22U)
```

VK_API_VERSION_MINOR extracts the API minor version number from a packed version number:

```
// Provided by VK_VERSION_1_0
#define VK_API_VERSION_MINOR(version) (((uint32_t)(version) >> 12U) & 0x3FFU)
```

VK_VERSION_MINOR extracts the API minor version number from a packed version number:

```
// Provided by VK_VERSION_1_0
// VK_VERSION_MINOR is deprecated, but no reason was given in the API XML
// DEPRECATED: This define is deprecated. VK_API_VERSION_MINOR should be used instead.
#define VK_VERSION_MINOR(version) (((uint32_t)(version) >> 12U) & 0x3FFU)
```

VK_API_VERSION_PATCH extracts the API patch version number from a packed version number:

```
// Provided by VK_VERSION_1_0
#define VK_API_VERSION_PATCH(version) ((uint32_t)(version) & 0xFFFU)
```

VK_VERSION_PATCH extracts the API patch version number from a packed version number:

```
// Provided by VK_VERSION_1_0
// VK_VERSION_PATCH is deprecated, but no reason was given in the API XML
// DEPRECATED: This define is deprecated. VK_API_VERSION_PATCH should be used instead.
#define VK_VERSION_PATCH(version) ((uint32_t)(version) & 0xFFFU)
```

VK_MAKE_API_VERSION constructs an API version number.

```
// Provided by VK_VERSION_1_0
#define VK_MAKE_API_VERSION(variant, major, minor, patch) \
    (((uint32_t)(variant)) << 29U) | (((uint32_t)(major)) << 22U) | \
    (((uint32_t)(minor)) << 12U) | ((uint32_t)(patch)))
```

- **variant** is the variant number.
- **major** is the major version number.
- **minor** is the minor version number.
- **patch** is the patch version number.

VK_MAKE_VERSION constructs an API version number.

```
// Provided by VK_VERSION_1_0
// VK_MAKE_VERSION is deprecated, but no reason was given in the API XML
// DEPRECATED: This define is deprecated. VK_MAKE_API_VERSION should be used instead.
#define VK_MAKE_VERSION(major, minor, patch) \
    (((uint32_t)(major)) << 22U) | (((uint32_t)(minor)) << 12U) | \
    ((uint32_t)(patch)))
```

- **major** is the major version number.
- **minor** is the minor version number.
- **patch** is the patch version number.

VK_API_VERSION_1_0 returns the API version number for Vulkan 1.0.0.

```
// Provided by VK_VERSION_1_0
// Vulkan 1.0 version number
#define VK_API_VERSION_1_0 VK_MAKE_API_VERSION(0, 1, 0, 0)// Patch version should
always be set to 0
```

30.2.2. Querying Version Support

NOTE

In Vulkan 1.0, there is no mechanism to detect the separate versions of [instance-level](#) and [device-level functionality](#) supported. However, the [vkEnumerateInstanceVersion](#) command was added in Vulkan 1.1 to determine the supported version of instance-level functionality - querying for this function via [vkGetInstanceProcAddr](#) will return [NULL](#) on implementations that only support Vulkan 1.0 functionality. For more information on this, please refer to the Vulkan 1.1 specification.

The version of device-level functionality can be queried by calling [vkGetPhysicalDeviceProperties](#) and is returned in [VkPhysicalDeviceProperties::apiVersion](#), encoded as described in [Version Numbers](#).

30.3. Layers

When a layer is enabled, it inserts itself into the call chain for Vulkan commands the layer is interested in. Layers **can** be used for a variety of tasks that extend the base behavior of Vulkan beyond what is required by the specification - such as call logging, tracing, validation, or providing additional extensions.

NOTE

For example, an implementation is not expected to check that the value of enums used by the application fall within allowed ranges. Instead, a validation layer would do those checks and flag issues. This avoids a performance penalty during production use of the application because those layers would not be enabled in production.

NOTE

Vulkan layers **may** wrap object handles (i.e. return a different handle value to the application than that generated by the implementation). This is generally discouraged, as it increases the probability of incompatibilities with new extensions. The validation layers wrap handles in order to track the proper use and destruction of each object. See the [“Architecture of the Vulkan Loader Interfaces”](#) document for additional information.

To query the available layers, call:

```
// Provided by VK_VERSION_1_0
VkResult vkEnumerateInstanceLayerProperties(
    uint32_t*                                pPropertyCount,
    VkLayerProperties*                        pProperties);
```

- `pPropertyCount` is a pointer to an integer related to the number of layer properties available or queried, as described below.
- `pProperties` is either `NULL` or a pointer to an array of `VkLayerProperties` structures.

If `pProperties` is `NULL`, then the number of layer properties available is returned in `pPropertyCount`. Otherwise, `pPropertyCount` **must** point to a variable set by the application to the number of elements in the `pProperties` array, and on return the variable is overwritten with the number of structures actually written to `pProperties`. If `pPropertyCount` is less than the number of layer properties available, at most `pPropertyCount` structures will be written, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that not all the available properties were returned.

The list of available layers may change at any time due to actions outside of the Vulkan implementation, so two calls to `vkEnumerateInstanceLayerProperties` with the same parameters **may** return different results, or retrieve different `pPropertyCount` values or `pProperties` contents. Once an instance has been created, the layers enabled for that instance will continue to be enabled and valid for the lifetime of that instance, even if some of them become unavailable for future instances.

Valid Usage (Implicit)

- VUID-vkEnumerateInstanceLayerProperties-pPropertyCount-parameter
`pPropertyCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkEnumerateInstanceLayerProperties-pProperties-parameter
If the value referenced by `pPropertyCount` is not `0`, and `pProperties` is not `NULL`, `pProperties` **must** be a valid pointer to an array of `pPropertyCount` `VkLayerProperties` structures

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkLayerProperties` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkLayerProperties {
    char        layerName[VK_MAX_EXTENSION_NAME_SIZE];
    uint32_t    specVersion;
    uint32_t    implementationVersion;
    char        description[VK_MAX_DESCRIPTION_SIZE];
} VkLayerProperties;
```

- `layerName` is an array of `VK_MAX_EXTENSION_NAME_SIZE` `char` containing a null-terminated UTF-8 string which is the name of the layer. Use this name in the `ppEnabledLayerNames` array passed in the `VkInstanceCreateInfo` structure to enable this layer for an instance.
- `specVersion` is the Vulkan version the layer was written to, encoded as described in [Version Numbers](#).
- `implementationVersion` is the version of this layer. It is an integer, increasing with backward compatible changes.
- `description` is an array of `VK_MAX_DESCRIPTION_SIZE` `char` containing a null-terminated UTF-8 string which provides additional details that **can** be used by the application to identify the layer.

`VK_MAX_EXTENSION_NAME_SIZE` is the length in `char` values of an array containing a layer or extension name string, as returned in `VkLayerProperties::layerName`, `VkExtensionProperties::extensionName`, and other queries.

```
#define VK_MAX_EXTENSION_NAME_SIZE    256U
```

`VK_MAX_DESCRIPTION_SIZE` is the length in `char` values of an array containing a string with additional descriptive information about a query, as returned in `VkLayerProperties::description` and other queries.

```
#define VK_MAX_DESCRIPTION_SIZE    256U
```

To enable a layer, the name of the layer **should** be added to the `ppEnabledLayerNames` member of `VkInstanceCreateInfo` when creating a `VkInstance`.

Loader implementations **may** provide mechanisms outside the Vulkan API for enabling specific layers. Layers enabled through such a mechanism are *implicitly enabled*, while layers enabled by including the layer name in the `ppEnabledLayerNames` member of `VkInstanceCreateInfo` are *explicitly enabled*. Implicitly enabled layers are loaded before explicitly enabled layers, such that implicitly enabled layers are closer to the application, and explicitly enabled layers are closer to the driver. Except where otherwise specified, implicitly enabled and explicitly enabled layers differ only in the way they are enabled, and the order in which they are loaded. Explicitly enabling a layer that is implicitly enabled results in this layer being loaded as an implicitly enabled layer; it has no additional effect.

30.3.1. Device Layer Deprecation

Previous versions of this specification distinguished between instance and device layers. Instance layers were only able to intercept commands that operate on `VkInstance` and `VkPhysicalDevice`, except they were not able to intercept `vkCreateDevice`. Device layers were enabled for individual devices when they were created, and could only intercept commands operating on that device or its child objects.

Device-only layers are now deprecated, and this specification no longer distinguishes between instance and device layers. Layers are enabled during instance creation, and are able to intercept all commands operating on that instance or any of its child objects. At the time of deprecation there were no known device-only layers and no compelling reason to create one.

In order to maintain compatibility with implementations released prior to device-layer deprecation, applications **should** still enumerate and enable device layers. The behavior of `vkEnumerateDeviceLayerProperties` and valid usage of the `ppEnabledLayerNames` member of `VkDeviceCreateInfo` maximizes compatibility with applications written to work with the previous requirements.

To enumerate device layers, call:

```
// Provided by VK_VERSION_1_0
VkResult vkEnumerateDeviceLayerProperties(
    VkPhysicalDevice      physicalDevice,
    uint32_t*             pPropertyCount,
    VkLayerProperties*     pProperties);
```

- `physicalDevice` is the physical device that will be queried.
- `pPropertyCount` is a pointer to an integer related to the number of layer properties available or queried.
- `pProperties` is either `NULL` or a pointer to an array of `VkLayerProperties` structures.

If `pProperties` is `NULL`, then the number of layer properties available is returned in `pPropertyCount`. Otherwise, `pPropertyCount` **must** point to a variable set by the application to the number of elements in the `pProperties` array, and on return the variable is overwritten with the number of structures actually written to `pProperties`. If `pPropertyCount` is less than the number of layer properties available, at most `pPropertyCount` structures will be written, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that not all the available properties were returned.

The list of layers enumerated by `vkEnumerateDeviceLayerProperties` **must** be exactly the sequence of layers enabled for the instance. The members of `VkLayerProperties` for each enumerated layer **must** be the same as the properties when the layer was enumerated by `vkEnumerateInstanceLayerProperties`.

NOTE

Due to platform details on Android, `vkEnumerateDeviceLayerProperties` may be called with `physicalDevice` equal to `NULL` during layer discovery. This behavior will only be observed by layer implementations, and not the underlying Vulkan driver.

Valid Usage (Implicit)

- `VUID-vkEnumerateDeviceLayerProperties-physicalDevice-parameter`
`physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- `VUID-vkEnumerateDeviceLayerProperties-pPropertyCount-parameter`
`pPropertyCount` **must** be a valid pointer to a `uint32_t` value
- `VUID-vkEnumerateDeviceLayerProperties-pProperties-parameter`
If the value referenced by `pPropertyCount` is not 0, and `pProperties` is not `NULL`, `pProperties` **must** be a valid pointer to an array of `pPropertyCount` `VkLayerProperties` structures

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `ppEnabledLayerNames` and `enabledLayerCount` members of `VkDeviceCreateInfo` are deprecated and their values **must** be ignored by implementations. However, for compatibility, only an empty list of layers or a list that exactly matches the sequence enabled at instance creation time are valid, and validation layers **should** issue diagnostics for other cases.

Regardless of the enabled layer list provided in `VkDeviceCreateInfo`, the sequence of layers active for a device will be exactly the sequence of layers enabled when the parent instance was created.

30.4. Extensions

Extensions **may** define new Vulkan commands, structures, and enumerants. For compilation purposes, the interfaces defined by registered extensions, including new structures and enumerants as well as function pointer types for new commands, are defined in the Khronos-supplied `vulkan_core.h` together with the core API. However, commands defined by extensions **may** not be available for static linking - in which case function pointers to these commands **should** be queried at runtime as described in [Command Function Pointers](#). Extensions **may** be provided by layers as well as by a Vulkan implementation.

Because extensions **may** extend or change the behavior of the Vulkan API, extension authors **should** add support for their extensions to the Khronos validation layers. This is especially important for new commands whose parameters have been wrapped by the validation layers. See the [“Architecture of the Vulkan Loader Interfaces”](#) document for additional information.

NOTE

To enable an instance extension, the name of the extension **can** be added to the

`ppEnabledExtensionNames` member of `VkInstanceCreateInfo` when creating a `VkInstance`.

To enable a device extension, the name of the extension **can** be added to the `ppEnabledExtensionNames` member of `VkDeviceCreateInfo` when creating a `VkDevice`.

Enabling an extension (with no further use of that extension) does not change the behavior of functionality exposed by the core Vulkan API or any other extension, other than making valid the use of the commands, enums and structures defined by that extension.

Valid Usage sections for individual commands and structures do not currently contain which extensions have to be enabled in order to make their use valid, although they might do so in the future. It is defined only in the [Valid Usage for Extensions](#) section.

30.4.1. Instance Extensions

Instance extensions add new [instance-level functionality](#) to the API, outside of the core specification.

To query the available instance extensions, call:

```
// Provided by VK_VERSION_1_0
VkResult vkEnumerateInstanceExtensionProperties(
    const char*          pLayerName,
    uint32_t*            pPropertyCount,
    VkExtensionProperties* pProperties);
```

- `pLayerName` is either `NULL` or a pointer to a null-terminated UTF-8 string naming the layer to retrieve extensions from.
- `pPropertyCount` is a pointer to an integer related to the number of extension properties available or queried, as described below.
- `pProperties` is either `NULL` or a pointer to an array of `VkExtensionProperties` structures.

When `pLayerName` parameter is `NULL`, only extensions provided by the Vulkan implementation or by implicitly enabled layers are returned. When `pLayerName` is the name of a layer, the instance extensions provided by that layer are returned.

If `pProperties` is `NULL`, then the number of extensions properties available is returned in `pPropertyCount`. Otherwise, `pPropertyCount` **must** point to a variable set by the application to the number of elements in the `pProperties` array, and on return the variable is overwritten with the number of structures actually written to `pProperties`. If `pPropertyCount` is less than the number of extension properties available, at most `pPropertyCount` structures will be written, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that not all the available properties were returned.

Because the list of available layers may change externally between calls to

`vkEnumerateInstanceExtensionProperties`, two calls may retrieve different results if a `pLayerName` is available in one call but not in another. The extensions supported by a layer may also change between two calls, e.g. if the layer implementation is replaced by a different version between those calls.

Implementations **must** not advertise any pair of extensions that cannot be enabled together due to behavioral differences, or any extension that cannot be enabled against the advertised version.

Valid Usage (Implicit)

- VUID-vkEnumerateInstanceExtensionProperties-pLayerName-parameter
If `pLayerName` is not `NULL`, `pLayerName` **must** be a null-terminated UTF-8 string
- VUID-vkEnumerateInstanceExtensionProperties-pPropertyCount-parameter
`pPropertyCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkEnumerateInstanceExtensionProperties-pProperties-parameter
If the value referenced by `pPropertyCount` is not `0`, and `pProperties` is not `NULL`, `pProperties` **must** be a valid pointer to an array of `pPropertyCount` `VkExtensionProperties` structures

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_LAYER_NOT_PRESENT`

30.4.2. Device Extensions

Device extensions add new [device-level functionality](#) to the API, outside of the core specification.

To query the extensions available to a given physical device, call:

```
// Provided by VK_VERSION_1_0
VkResult vkEnumerateDeviceExtensionProperties(
    VkPhysicalDevice      physicalDevice,
    const char*           pLayerName,
    uint32_t*             pPropertyCount,
    VkExtensionProperties* pProperties);
```

- `physicalDevice` is the physical device that will be queried.

- `pLayerName` is either `NULL` or a pointer to a null-terminated UTF-8 string naming the layer to retrieve extensions from.
- `pPropertyCount` is a pointer to an integer related to the number of extension properties available or queried, and is treated in the same fashion as the `vkEnumerateInstanceExtensionProperties::pPropertyCount` parameter.
- `pProperties` is either `NULL` or a pointer to an array of `VkExtensionProperties` structures.

When `pLayerName` parameter is `NULL`, only extensions provided by the Vulkan implementation or by implicitly enabled layers are returned. When `pLayerName` is the name of a layer, the device extensions provided by that layer are returned.

Implementations **must** not advertise any pair of extensions that cannot be enabled together due to behavioral differences, or any extension that cannot be enabled against the advertised version.

NOTE

Due to platform details on Android, `vkEnumerateDeviceExtensionProperties` may be called with `physicalDevice` equal to `NULL` during layer discovery. This behavior will only be observed by layer implementations, and not the underlying Vulkan driver.

Valid Usage (Implicit)

- VUID-vkEnumerateDeviceExtensionProperties-physicalDevice-parameter `physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkEnumerateDeviceExtensionProperties-pLayerName-parameter
If `pLayerName` is not `NULL`, `pLayerName` **must** be a null-terminated UTF-8 string
- VUID-vkEnumerateDeviceExtensionProperties-pPropertyCount-parameter
`pPropertyCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkEnumerateDeviceExtensionProperties-pProperties-parameter
If the value referenced by `pPropertyCount` is not `0`, and `pProperties` is not `NULL`, `pProperties` **must** be a valid pointer to an array of `pPropertyCount` `VkExtensionProperties` structures

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_LAYER_NOT_PRESENT`

The `VkExtensionProperties` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkExtensionProperties {
    char        extensionName[VK_MAX_EXTENSION_NAME_SIZE];
    uint32_t    specVersion;
} VkExtensionProperties;
```

- `extensionName` is an array of `VK_MAX_EXTENSION_NAME_SIZE` `char` containing a null-terminated UTF-8 string which is the name of the extension.
- `specVersion` is the version of this extension. It is an integer, incremented with backward compatible changes.

Accessing Device-Level Functionality From a [VkPhysicalDevice](#)

Some device extensions also add support for physical-device-level functionality. Physical-device-level functionality **can** be used, if the required extension is supported as advertised by [vkEnumerateDeviceExtensionProperties](#) for a given [VkPhysicalDevice](#).

Accessing Device-Level Functionality From a [VkDevice](#)

For commands that are dispatched from a [VkDevice](#), or from a child object of a [VkDevice](#), device extensions **must** be enabled in [vkCreateDevice](#).

30.5. Extension Dependencies

Some extensions are dependent on other extensions, or on specific core API versions, to function. To enable extensions with dependencies, any *required extensions* **must** also be enabled through the same API mechanisms when creating an instance with [vkCreateInstance](#) or a device with [vkCreateDevice](#). Each extension which has such dependencies documents them in the [appendix summarizing that extension](#).

If an extension is supported (as queried by [vkEnumerateInstanceExtensionProperties](#) or [vkEnumerateDeviceExtensionProperties](#)), then *required extensions* of that extension **must** also be supported for the same instance or physical device.

Any device extension that has an instance extension dependency that is not enabled by [vkCreateInstance](#) is considered to be unsupported, hence it **must** not be returned by [vkEnumerateDeviceExtensionProperties](#) for any [VkPhysicalDevice](#) child of the instance. Instance extensions do not have dependencies on device extensions.

If a required extension has been [promoted](#) to another extension or to a core API version, then as a *general* rule, the dependency is also satisfied by the promoted extension or core version. This will be true so long as any features required by the original extension are also required or enabled by the promoted extension or core version. However, in some cases an extension is promoted while making some of its features optional in the promoted extension or core version. In this case, the dependency **may** not be satisfied. The only way to be certain is to look at the descriptions of the original dependency and the promoted version in the [Layers & Extensions](#) and [Core Revisions](#) appendices.

NOTE

There is metadata in `vk.xml` describing some aspects of promotion, especially `requires`, `promotedto` and `deprecatedby` attributes of `<extension>` tags. However, the metadata does not yet fully describe this scenario. In the future, we may extend the XML schema to describe the full set of extensions and versions satisfying a dependency. As discussed in more detail for [Promotion](#) below, when an extension is promoted it does not mean that a mechanical substitution of an extension API by the corresponding promoted API will work in exactly the same fashion; be supported at runtime; or even exist.

30.6. Compatibility Guarantees (Informative)

This section is marked as informal as there is no binding responsibility on implementations of the Vulkan API - these guarantees are however a contract between the Vulkan Working Group and developers using this Specification.

30.6.1. Core Versions

Each of the [major](#), [minor](#), and [patch versions](#) of the Vulkan specification provide different compatibility guarantees.

Patch Versions

A difference in the patch version indicates that a set of bug fixes or clarifications have been made to the Specification. Informative enums returned by Vulkan commands that will not affect the runtime behavior of a valid application may be added in a patch version (e.g. [VkVendorId](#)).

The specification's patch version is strictly increasing for a given major version of the specification; any change to a specification as described above will result in the patch version being increased by 1. Patch versions are applied to all minor versions, even if a given minor version is not affected by the provoking change.

Specifications with different patch versions but the same major and minor version are *fully compatible* with each other - such that a valid application written against one will work with an implementation of another.

NOTE

If a patch version includes a bug fix or clarification that could have a significant impact on developer expectations, these will be highlighted in the change log. Generally the Vulkan Working Group tries to avoid these kinds of changes, instead fixing them in either an extension or core version.

Minor Versions

Changes in the minor version of the specification indicate that new functionality has been added to the core specification. This will usually include new interfaces in the header, and **may** also include behavior changes and bug fixes. Core functionality **may** be deprecated in a minor version, but will not be obsoleted or removed.

The specification's minor version is strictly increasing for a given major version of the

specification; any change to a specification as described above will result in the minor version being increased by 1. Changes that can be accommodated in a patch version will not increase the minor version.

Specifications with a lower minor version are *backwards compatible* with an implementation of a specification with a higher minor version for core functionality and extensions issued with the KHR vendor tag. Vendor and multi-vendor extensions are not guaranteed to remain functional across minor versions, though in general they are with few exceptions - see [Obsolescence](#) for more information.

Major Versions

A difference in the major version of specifications indicates a large set of changes which will likely include interface changes, behavioral changes, removal of [deprecated functionality](#), and the modification, addition, or replacement of other functionality.

The specification's major version is monotonically increasing; any change to the specification as described above will result in the major version being increased. Changes that can be accommodated in a patch or minor version will not increase the major version.

The Vulkan Working Group intends to only issue a new major version of the Specification in order to realize significant improvements to the Vulkan API that will necessarily require breaking compatibility.

A new major version will likely include a wholly new version of the specification to be issued - which could include an overhaul of the versioning semantics for the minor and patch versions. The patch and minor versions of a specification are therefore not meaningful across major versions. If a major version of the specification includes similar versioning semantics, it is expected that the patch and the minor version will be reset to 0 for that major version.

30.6.2. Extensions

A KHR extension **must** be able to be enabled alongside any other KHR extension, and for any minor or patch version of the core Specification beyond the minimum version it requires. A multi-vendor extension **should** be able to be enabled alongside any KHR extension or other multi-vendor extension, and for any minor or patch version of the core Specification beyond the minimum version it requires. A vendor extension **should** be able to be enabled alongside any KHR extension, multi-vendor extension, or other vendor extension from the same vendor, and for any minor or patch version of the core Specification beyond the minimum version it requires. A vendor extension **may** be able to be enabled alongside vendor extensions from another vendor.

The one other exception to this is if a vendor or multi-vendor extension is [made obsolete](#) by either a core version or another extension, which will be highlighted in the [extension appendix](#).

Promotion

Extensions, or features of an extension, **may** be promoted to a new [core version of the API](#), or a newer extension which an equal or greater number of implementors are in favor of.

When extension functionality is promoted, minor changes **may** be introduced, limited to the

following:

- Naming
- Non-intrusive parameter changes
- [Feature advertisement/enablement](#)
- Combining structure parameters into larger structures
- Author ID suffixes changed or removed

NOTE

If extension functionality is promoted, there is no guarantee of direct compatibility, however it should require little effort to port code from the original feature to the promoted one.

The Vulkan Working Group endeavors to ensure that larger changes are marked as either [deprecated](#) or [obsoleted](#) as appropriate, and can do so retroactively if necessary.

Extensions that are promoted are listed as being promoted in their extension appendices, with reference to where they were promoted to.

When an extension is promoted, any backwards compatibility aliases which exist in the extension will **not** be promoted.

NOTE

As a hypothetical example, if the `VK_KHR_surface` extension were promoted to part of a future core version, the `VK_COLOR_SPACE_SRGB_NONLINEAR_KHR` token defined by that extension would be promoted to `VK_COLOR_SPACE_SRGB_NONLINEAR`. However, the `VK_COLORSPACE_SRGB_NONLINEAR_KHR` token aliases `VK_COLOR_SPACE_SRGB_NONLINEAR_KHR`. The `VK_COLORSPACE_SRGB_NONLINEAR_KHR` would not be promoted, because it is a backwards compatibility alias that exists only due to a naming mistake when the extension was initially published.

Deprecation

Extensions **may** be marked as deprecated when the intended use cases either become irrelevant or can be solved in other ways. Generally, a new feature will become available to solve the use case in another extension or core version of the API, but it is not guaranteed.

NOTE

Features that are intended to replace deprecated functionality have no guarantees of compatibility, and applications may require drastic modification in order to make use of the new features.

Extensions that are deprecated are listed as being deprecated in their extension appendices, with an explanation of the deprecation and any features that are relevant.

Obsoletion

Occasionally, an extension will be marked as obsolete if a new version of the core API or a new extension is fundamentally incompatible with it. An obsoleted extension **must** not be used with the

extension or core version that obsoleted it.

Extensions that are obsoleted are listed as being obsoleted in their extension appendices, with reference to what they were obsoleted by.

Aliases

When an extension is promoted or deprecated by a newer feature, some or all of its functionality **may** be replicated into the newer feature. Rather than duplication of all the documentation and definitions, the specification instead identifies the identical commands and types as *aliases* of one another. Each alias is mentioned together with the definition it aliases, with the older aliases marked as “equivalents”. Each alias of the same command has identical behavior, and each alias of the same type has identical meaning - they can be used interchangeably in an application with no compatibility issues.

NOTE

For promoted types, the aliased extension type is semantically identical to the new core type. The C99 headers simply `typedef` the older aliases to the promoted types.

For promoted command aliases, however, there are two separate command definitions, due to the fact that the C99 ABI has no way to alias command definitions without resorting to macros. Calling either command will produce identical behavior within the bounds of the specification, and should still invoke the same path in the implementation. Debug tools may use separate commands with different debug behavior; to write the appropriate command name to an output log, for instance.

Special Use Extensions

Some extensions exist only to support a specific purpose or specific class of application. These are referred to as “special use extensions”. Use of these extensions in applications not meeting the special use criteria is not recommended.

Special use cases are restricted, and only those defined below are used to describe extensions:

Table 29. Extension Special Use Cases

Special Use	XML Tag	Full Description
CAD support	cadsupport	Extension is intended to support specialized functionality used by CAD/CAM applications.
D3D support	d3demulation	Extension is intended to support D3D emulation layers, and applications ported from D3D, by adding functionality specific to D3D.
Developer tools	devtools	Extension is intended to support developer tools such as capture-replay libraries.
Debugging tools	debugging	Extension is intended for use by applications when debugging.

Special Use	XML Tag	Full Description
OpenGL / ES support	glemulation	Extension is intended to support OpenGL and/or OpenGL ES emulation layers, and applications ported from those APIs, by adding functionality specific to those APIs.

Special use extensions are identified in the metadata for each such extension in the [Layers & Extensions](#) appendix, using the name in the “Special Use” column above.

Special use extensions are also identified in `vk.xml` with the short name in “XML Tag” column above, as described in the “API Extensions (`extension` tag)” section of the [registry schema documentation](#).

Chapter 31. Features

Features advertise additional functionality which **can** be enabled in the API. If a feature is *supported* by a physical device, it **must** be *enabled* when creating a [VkDevice](#) in order to use its associated functionality with that [VkDevice](#). If a feature is *unsupported*, that functionality **cannot** be used with that [VkDevice](#).

For convenience, new core versions of Vulkan **may** introduce new unified feature structures for features promoted from extensions. At the same time, the extension's original feature structure (if any) is also promoted to the core API, and is an alias of the extension's structure. This results in multiple names for the same feature: in the original extension's feature structure and the promoted structure alias, in the unified feature structure. When a feature was implicitly supported and enabled in the extension, but an explicit name was added during promotion, then the extension itself acts as an alias for the feature as listed in the table below.

All aliases of the same feature in the core API **must** be reported consistently: either all **must** be reported as supported, or none of them. When a promoted extension is available, any corresponding feature aliases **must** be supported.

Table 30. Extension Feature Aliases

Extension	Feature(s)
-----------	------------

To query supported features, call:

```
// Provided by VK_VERSION_1_0
void vkGetPhysicalDeviceFeatures(
    VkPhysicalDevice          physicalDevice,
    VkPhysicalDeviceFeatures* pFeatures);
```

- [physicalDevice](#) is the physical device from which to query the supported features.
- [pFeatures](#) is a pointer to a [VkPhysicalDeviceFeatures](#) structure in which the physical device features are returned. For each feature, a value of [VK_TRUE](#) specifies that the feature is supported on this physical device, and [VK_FALSE](#) specifies that the feature is not supported.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceFeatures-physicalDevice-parameter
[physicalDevice](#) **must** be a valid [VkPhysicalDevice](#) handle
- VUID-vkGetPhysicalDeviceFeatures-pFeatures-parameter
[pFeatures](#) **must** be a valid pointer to a [VkPhysicalDeviceFeatures](#) structure

Fine-grained features used by a logical device **must** be enabled at [VkDevice](#) creation time. If a feature is enabled that the physical device does not support, [VkDevice](#) creation will fail and return [VK_ERROR_FEATURE_NOT_PRESENT](#).

The fine-grained features are enabled by passing a pointer to the `VkPhysicalDeviceFeatures` structure via the `pEnabledFeatures` member of the `VkDeviceCreateInfo` structure that is passed into the `vkCreateDevice` call. If a member of `pEnabledFeatures` is `VK_TRUE` or `VK_FALSE`, then the device will be created with the indicated feature enabled or disabled, respectively.

If an application wishes to enable all features supported by a device, it **can** simply pass in the `VkPhysicalDeviceFeatures` structure that was previously returned by `vkGetPhysicalDeviceFeatures`. To disable an individual feature, the application **can** set the desired member to `VK_FALSE` in the same structure. Setting `pEnabledFeatures` to `NULL` is equivalent to setting all members of the structure to `VK_FALSE`.

NOTE

Some features, such as `robustBufferAccess`, **may** incur a runtime performance cost. Application writers **should** carefully consider the implications of enabling all supported features.

The `VkPhysicalDeviceFeatures` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPhysicalDeviceFeatures {
    VkBool32    robustBufferAccess;
    VkBool32    fullDrawIndexUint32;
    VkBool32    imageCubeArray;
    VkBool32    independentBlend;
    VkBool32    geometryShader;
    VkBool32    tessellationShader;
    VkBool32    sampleRateShading;
    VkBool32    dualSrcBlend;
    VkBool32    logicOp;
    VkBool32    multiDrawIndirect;
    VkBool32    drawIndirectFirstInstance;
    VkBool32    depthClamp;
    VkBool32    depthBiasClamp;
    VkBool32    fillModeNonSolid;
    VkBool32    depthBounds;
    VkBool32    wideLines;
    VkBool32    largePoints;
    VkBool32    alphaToOne;
    VkBool32    multiViewport;
    VkBool32    samplerAnisotropy;
    VkBool32    textureCompressionETC2;
    VkBool32    textureCompressionASTC_LDR;
    VkBool32    textureCompressionBC;
    VkBool32    occlusionQueryPrecise;
    VkBool32    pipelineStatisticsQuery;
    VkBool32    vertexPipelineStoresAndAtomics;
    VkBool32    fragmentStoresAndAtomics;
    VkBool32    shaderTessellationAndGeometryPointSize;
    VkBool32    shaderImageGatherExtended;
    VkBool32    shaderStorageImageExtendedFormats;
    VkBool32    shaderStorageImageMultisample;
```

```

VkBool32    shaderStorageImageReadWithoutFormat;
VkBool32    shaderStorageImageWriteWithoutFormat;
VkBool32    shaderUniformBufferArrayDynamicIndexing;
VkBool32    shaderSampledImageArrayDynamicIndexing;
VkBool32    shaderStorageBufferArrayDynamicIndexing;
VkBool32    shaderStorageImageArrayDynamicIndexing;
VkBool32    shaderClipDistance;
VkBool32    shaderCullDistance;
VkBool32    shaderFloat64;
VkBool32    shaderInt64;
VkBool32    shaderInt16;
VkBool32    shaderResourceResidency;
VkBool32    shaderResourceMinLod;
VkBool32    sparseBinding;
VkBool32    sparseResidencyBuffer;
VkBool32    sparseResidencyImage2D;
VkBool32    sparseResidencyImage3D;
VkBool32    sparseResidency2Samples;
VkBool32    sparseResidency4Samples;
VkBool32    sparseResidency8Samples;
VkBool32    sparseResidency16Samples;
VkBool32    sparseResidencyAliased;
VkBool32    variableMultisampleRate;
VkBool32    inheritedQueries;
} VkPhysicalDeviceFeatures;

```

This structure describes the following features:

- **robustBufferAccess** specifies that accesses to buffers are bounds-checked against the range of the buffer descriptor (as determined by **VkDescriptorBufferInfo::range**, **VkBufferViewCreateInfo::range**, or the size of the buffer). Out of bounds accesses **must** not cause application termination, and the effects of shader loads, stores, and atomics **must** conform to an implementation-dependent behavior as described below.
 - A buffer access is considered to be out of bounds if any of the following are true:
 - The pointer was formed by **OpImageTexelPointer** and the coordinate is less than zero or greater than or equal to the number of whole elements in the bound range.
 - The pointer was not formed by **OpImageTexelPointer** and the object pointed to is not wholly contained within the bound range.

NOTE

If a SPIR-V **OpLoad** instruction loads a structure and the tail end of the structure is out of bounds, then all members of the structure are considered out of bounds even if the members at the end are not statically used.

- If any buffer access is determined to be out of bounds, then any other access of the same type (load, store, or atomic) to the same buffer that accesses an address less than 16 bytes away from the out of bounds address **may** also be considered out of bounds.

- If the access is a load that reads from the same memory locations as a prior store in the same shader invocation, with no other intervening accesses to the same memory locations in that shader invocation, then the result of the load **may** be the value stored by the store instruction, even if the access is out of bounds. If the load is **volatile**, then an out of bounds load **must** return the appropriate out of bounds value.
- Out-of-bounds buffer loads will return any of the following values:
 - Values from anywhere within the memory range(s) bound to the buffer (possibly including bytes of memory past the end of the buffer, up to the end of the bound range).
 - Zero values, or (0,0,0,x) vectors for vector reads where x is a valid value represented in the type of the vector components and **may** be any of:
 - 0, 1, or the maximum representable positive integer value, for signed or unsigned integer components
 - 0.0 or 1.0, for floating-point components
- Out-of-bounds writes **may** modify values within the memory range(s) bound to the buffer, but **must** not modify any other memory.
- Out-of-bounds atomics **may** modify values within the memory range(s) bound to the buffer, but **must** not modify any other memory, and return an undefined value.
- Vertex input attributes are considered out of bounds if the offset of the attribute in the bound vertex buffer range plus the size of the attribute is greater than either:
 - `vertexBufferSize`, if `bindingStride == 0`; or
 - `(vertexBufferSize - (vertexBufferSize % bindingStride))`

where `vertexBufferSize` is the byte size of the memory range bound to the vertex buffer binding and `bindingStride` is the byte stride of the corresponding vertex input binding. Further, if any vertex input attribute using a specific vertex input binding is out of bounds, then all vertex input attributes using that vertex input binding for that vertex shader invocation are considered out of bounds.

- If a vertex input attribute is out of bounds, it will be assigned one of the following values:
 - Values from anywhere within the memory range(s) bound to the buffer, converted according to the format of the attribute.
 - Zero values, format converted according to the format of the attribute.
 - Zero values, or (0,0,0,x) vectors, as described above.
- If `robustBufferAccess` is not enabled, applications **must** not perform out of bounds accesses .
- `fullDrawIndexUint32` specifies the full 32-bit range of indices is supported for indexed draw calls when using a `VkIndexType` of `VK_INDEX_TYPE_UINT32`. `maxDrawIndexedIndexValue` is the maximum index value that **may** be used (aside from the primitive restart index, which is always $2^{32}-1$ when the `VkIndexType` is `VK_INDEX_TYPE_UINT32`). If this feature is supported, `maxDrawIndexedIndexValue` **must** be $2^{32}-1$; otherwise it **must** be no smaller than $2^{24}-1$. See `maxDrawIndexedIndexValue`.
- `imageCubeArray` specifies whether image views with a `VkImageViewType` of

VK_IMAGE_VIEW_TYPE_CUBE_ARRAY **can** be created, and that the corresponding `SampledCubeArray` and `ImageCubeArray` SPIR-V capabilities **can** be used in shader code.

- `independentBlend` specifies whether the `VkPipelineColorBlendAttachmentState` settings are controlled independently per-attachment. If this feature is not enabled, the `VkPipelineColorBlendAttachmentState` settings for all color attachments **must** be identical. Otherwise, a different `VkPipelineColorBlendAttachmentState` **can** be provided for each bound color attachment.
- `geometryShader` specifies whether geometry shaders are supported. If this feature is not enabled, the `VK_SHADER_STAGE_GEOMETRY_BIT` and `VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT` enum values **must** not be used. This also specifies whether shader modules **can** declare the `Geometry` capability.
- `tessellationShader` specifies whether tessellation control and evaluation shaders are supported. If this feature is not enabled, the `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT`, `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`, `VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT`, `VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT`, and `VK_STRUCTURE_TYPE_PIPELINE_TESSELLATION_STATE_CREATE_INFO` enum values **must** not be used. This also specifies whether shader modules **can** declare the `Tessellation` capability.
- `sampleRateShading` specifies whether `Sample Shading` and multisample interpolation are supported. If this feature is not enabled, the `sampleShadingEnable` member of the `VkPipelineMultisampleStateCreateInfo` structure **must** be `VK_FALSE` and the `minSampleShading` member is ignored. This also specifies whether shader modules **can** declare the `SampleRateShading` capability.
- `dualSrcBlend` specifies whether blend operations which take two sources are supported. If this feature is not enabled, the `VK_BLEND_FACTOR_SRC1_COLOR`, `VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR`, `VK_BLEND_FACTOR_SRC1_ALPHA`, and `VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA` enum values **must** not be used as source or destination blending factors. See [Dual-Source Blending](#).
- `logicOp` specifies whether logic operations are supported. If this feature is not enabled, the `logicOpEnable` member of the `VkPipelineColorBlendStateCreateInfo` structure **must** be `VK_FALSE`, and the `logicOp` member is ignored.
- `multiDrawIndirect` specifies whether multiple draw indirect is supported. If this feature is not enabled, the `drawCount` parameter to the `vkCmdDrawIndirect` and `vkCmdDrawIndexedIndirect` commands **must** be 0 or 1. The `maxDrawIndirectCount` member of the `VkPhysicalDeviceLimits` structure **must** also be 1 if this feature is not supported. See [maxDrawIndirectCount](#).
- `drawIndirectFirstInstance` specifies whether indirect drawing calls support the `firstInstance` parameter. If this feature is not enabled, the `firstInstance` member of all `VkDrawIndirectCommand` and `VkDrawIndexedIndirectCommand` structures that are provided to the `vkCmdDrawIndirect` and `vkCmdDrawIndexedIndirect` commands **must** be 0.
- `depthClamp` specifies whether depth clamping is supported. If this feature is not enabled, the `depthClampEnable` member of the `VkPipelineRasterizationStateCreateInfo` structure **must** be `VK_FALSE`. Otherwise, setting `depthClampEnable` to `VK_TRUE` will enable depth clamping.
- `depthBiasClamp` specifies whether depth bias clamping is supported. If this feature is not enabled, the `depthBiasClamp` member of the `VkPipelineRasterizationStateCreateInfo` structure **must** be 0.0 unless the `VK_DYNAMIC_STATE_DEPTH_BIAS` dynamic state is enabled, in which case the `depthBiasClamp` parameter to `vkCmdSetDepthBias` **must** be 0.0.

- `fillModeNonSolid` specifies whether point and wireframe fill modes are supported. If this feature is not enabled, the `VK_POLYGON_MODE_POINT` and `VK_POLYGON_MODE_LINE` enum values **must** not be used.
- `depthBounds` specifies whether depth bounds tests are supported. If this feature is not enabled, the `depthBoundsTestEnable` member of the `VkPipelineDepthStencilStateCreateInfo` structure **must** be `VK_FALSE` unless the `VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE` dynamic state is enabled, in which case the `depthBoundsTestEnable` parameter to `vkCmdSetDepthBoundsTestEnable` **must** be `VK_FALSE`. When `depthBoundsTestEnable` is `VK_FALSE`, the `minDepthBounds` and `maxDepthBounds` members of the `VkPipelineDepthStencilStateCreateInfo` structure are ignored.
- `wideLines` specifies whether lines with width other than 1.0 are supported. If this feature is not enabled, the `lineWidth` member of the `VkPipelineRasterizationStateCreateInfo` structure **must** be 1.0 unless the `VK_DYNAMIC_STATE_LINE_WIDTH` dynamic state is enabled, in which case the `lineWidth` parameter to `vkCmdSetLineWidth` **must** be 1.0. When this feature is supported, the range and granularity of supported line widths are indicated by the `lineWidthRange` and `lineWidthGranularity` members of the `VkPhysicalDeviceLimits` structure, respectively.
- `largePoints` specifies whether points with size greater than 1.0 are supported. If this feature is not enabled, only a point size of 1.0 written by a shader is supported. The range and granularity of supported point sizes are indicated by the `pointSizeRange` and `pointSizeGranularity` members of the `VkPhysicalDeviceLimits` structure, respectively.
- `alphaToOne` specifies whether the implementation is able to replace the alpha value of the fragment shader color output in the `Multisample Coverage` fragment operation. If this feature is not enabled, then the `alphaToOneEnable` member of the `VkPipelineMultisampleStateCreateInfo` structure **must** be `VK_FALSE`. Otherwise setting `alphaToOneEnable` to `VK_TRUE` will enable alpha-to-one behavior.
- `multiViewport` specifies whether more than one viewport is supported. If this feature is not enabled:
 - The `viewportCount` and `scissorCount` members of the `VkPipelineViewportStateCreateInfo` structure **must** be 1.
 - The `firstViewport` and `viewportCount` parameters to the `vkCmdSetViewport` command **must** be 0 and 1, respectively.
 - The `firstScissor` and `scissorCount` parameters to the `vkCmdSetScissor` command **must** be 0 and 1, respectively.
- `samplerAnisotropy` specifies whether anisotropic filtering is supported. If this feature is not enabled, the `anisotropyEnable` member of the `VkSamplerCreateInfo` structure **must** be `VK_FALSE`.
- `textureCompressionETC2` specifies whether all of the ETC2 and EAC compressed texture formats are supported. If this feature is enabled, then the `VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT`, `VK_FORMAT_FEATURE_BLIT_SRC_BIT` and `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT` features **must** be supported in `optimalTilingFeatures` for the following formats:
 - `VK_FORMAT_ETC2_R8G8B8_UNORM_BLOCK`
 - `VK_FORMAT_ETC2_R8G8B8_SRGB_BLOCK`
 - `VK_FORMAT_ETC2_R8G8B8A1_UNORM_BLOCK`
 - `VK_FORMAT_ETC2_R8G8B8A1_SRGB_BLOCK`

- `VK_FORMAT_ETC2_R8G8B8A8_UNORM_BLOCK`
- `VK_FORMAT_ETC2_R8G8B8A8_SRGB_BLOCK`
- `VK_FORMAT_EAC_R11_UNORM_BLOCK`
- `VK_FORMAT_EAC_R11_SNORM_BLOCK`
- `VK_FORMAT_EAC_R11G11_UNORM_BLOCK`
- `VK_FORMAT_EAC_R11G11_SNORM_BLOCK`

To query for additional properties, or if the feature is not enabled, `vkGetPhysicalDeviceFormatProperties` and `vkGetPhysicalDeviceImageFormatProperties` can be used to check for supported properties of individual formats as normal.

- `textureCompressionASTC_LDR` specifies whether all of the ASTC LDR compressed texture formats are supported. If this feature is enabled, then the `VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT`, `VK_FORMAT_FEATURE_BLIT_SRC_BIT` and `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT` features **must** be supported in `optimalTilingFeatures` for the following formats:

- `VK_FORMAT_ASTC_4x4_UNORM_BLOCK`
- `VK_FORMAT_ASTC_4x4_SRGB_BLOCK`
- `VK_FORMAT_ASTC_5x4_UNORM_BLOCK`
- `VK_FORMAT_ASTC_5x4_SRGB_BLOCK`
- `VK_FORMAT_ASTC_5x5_UNORM_BLOCK`
- `VK_FORMAT_ASTC_5x5_SRGB_BLOCK`
- `VK_FORMAT_ASTC_6x5_UNORM_BLOCK`
- `VK_FORMAT_ASTC_6x5_SRGB_BLOCK`
- `VK_FORMAT_ASTC_6x6_UNORM_BLOCK`
- `VK_FORMAT_ASTC_6x6_SRGB_BLOCK`
- `VK_FORMAT_ASTC_8x5_UNORM_BLOCK`
- `VK_FORMAT_ASTC_8x5_SRGB_BLOCK`
- `VK_FORMAT_ASTC_8x6_UNORM_BLOCK`
- `VK_FORMAT_ASTC_8x6_SRGB_BLOCK`
- `VK_FORMAT_ASTC_8x8_UNORM_BLOCK`
- `VK_FORMAT_ASTC_8x8_SRGB_BLOCK`
- `VK_FORMAT_ASTC_10x5_UNORM_BLOCK`
- `VK_FORMAT_ASTC_10x5_SRGB_BLOCK`
- `VK_FORMAT_ASTC_10x6_UNORM_BLOCK`
- `VK_FORMAT_ASTC_10x6_SRGB_BLOCK`
- `VK_FORMAT_ASTC_10x8_UNORM_BLOCK`
- `VK_FORMAT_ASTC_10x8_SRGB_BLOCK`

- `VK_FORMAT_ASTC_10x10_UNORM_BLOCK`
- `VK_FORMAT_ASTC_10x10_SRGB_BLOCK`
- `VK_FORMAT_ASTC_12x10_UNORM_BLOCK`
- `VK_FORMAT_ASTC_12x10_SRGB_BLOCK`
- `VK_FORMAT_ASTC_12x12_UNORM_BLOCK`
- `VK_FORMAT_ASTC_12x12_SRGB_BLOCK`

To query for additional properties, or if the feature is not enabled, `vkGetPhysicalDeviceFormatProperties` and `vkGetPhysicalDeviceImageFormatProperties` **can** be used to check for supported properties of individual formats as normal.

- `textureCompressionBC` specifies whether all of the BC compressed texture formats are supported. If this feature is enabled, then the `VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT`, `VK_FORMAT_FEATURE_BLIT_SRC_BIT` and `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT` features **must** be supported in `optimalTilingFeatures` for the following formats:

- `VK_FORMAT_BC1_RGB_UNORM_BLOCK`
- `VK_FORMAT_BC1_RGB_SRGB_BLOCK`
- `VK_FORMAT_BC1_RGBA_UNORM_BLOCK`
- `VK_FORMAT_BC1_RGBA_SRGB_BLOCK`
- `VK_FORMAT_BC2_UNORM_BLOCK`
- `VK_FORMAT_BC2_SRGB_BLOCK`
- `VK_FORMAT_BC3_UNORM_BLOCK`
- `VK_FORMAT_BC3_SRGB_BLOCK`
- `VK_FORMAT_BC4_UNORM_BLOCK`
- `VK_FORMAT_BC4_SNORM_BLOCK`
- `VK_FORMAT_BC5_UNORM_BLOCK`
- `VK_FORMAT_BC5_SNORM_BLOCK`
- `VK_FORMAT_BC6H_UFLOAT_BLOCK`
- `VK_FORMAT_BC6H_SFLOAT_BLOCK`
- `VK_FORMAT_BC7_UNORM_BLOCK`
- `VK_FORMAT_BC7_SRGB_BLOCK`

To query for additional properties, or if the feature is not enabled, `vkGetPhysicalDeviceFormatProperties` and `vkGetPhysicalDeviceImageFormatProperties` **can** be used to check for supported properties of individual formats as normal.

- `occlusionQueryPrecise` specifies whether occlusion queries returning actual sample counts are supported. Occlusion queries are created in a `VkQueryPool` by specifying the `queryType` of `VK_QUERY_TYPE_OCCLUSION` in the `VkQueryPoolCreateInfo` structure which is passed to `vkCreateQueryPool`. If this feature is enabled, queries of this type **can** enable `VK_QUERY_CONTROL_PRECISE_BIT` in the `flags` parameter to `vkCmdBeginQuery`. If this feature is not

supported, the implementation supports only boolean occlusion queries. When any samples are passed, boolean queries will return a non-zero result value, otherwise a result value of zero is returned. When this feature is enabled and `VK_QUERY_CONTROL_PRECISE_BIT` is set, occlusion queries will report the actual number of samples passed.

- `pipelineStatisticsQuery` specifies whether the pipeline statistics queries are supported. If this feature is not enabled, queries of type `VK_QUERY_TYPE_PIPELINE_STATISTICS` **cannot** be created, and none of the `VkQueryPipelineStatisticFlagBits` bits **can** be set in the `pipelineStatistics` member of the `VkQueryPoolCreateInfo` structure.
- `vertexPipelineStoresAndAtomics` specifies whether storage buffers and images support stores and atomic operations in the vertex, tessellation, and geometry shader stages. If this feature is not enabled, all storage image, storage texel buffer, and storage buffer variables used by these stages in shader modules **must** be decorated with the `NonWritable` decoration (or the `readonly` memory qualifier in GLSL).
- `fragmentStoresAndAtomics` specifies whether storage buffers and images support stores and atomic operations in the fragment shader stage. If this feature is not enabled, all storage image, storage texel buffer, and storage buffer variables used by the fragment stage in shader modules **must** be decorated with the `NonWritable` decoration (or the `readonly` memory qualifier in GLSL).
- `shaderTessellationAndGeometryPointSize` specifies whether the `PointSize` built-in decoration is available in the tessellation control, tessellation evaluation, and geometry shader stages. If this feature is not enabled, members decorated with the `PointSize` built-in decoration **must** not be read from or written to and all points written from a tessellation or geometry shader will have a size of 1.0. This also specifies whether shader modules **can** declare the `TessellationPointSize` capability for tessellation control and evaluation shaders, or if the shader modules **can** declare the `GeometryPointSize` capability for geometry shaders. An implementation supporting this feature **must** also support one or both of the `tessellationShader` or `geometryShader` features.
- `shaderImageGatherExtended` specifies whether the extended set of image gather instructions are available in shader code. If this feature is not enabled, the `OpImage*Gather` instructions do not support the `Offset` and `ConstOffsets` operands. This also specifies whether shader modules **can** declare the `ImageGatherExtended` capability.
- `shaderStorageImageExtendedFormats` specifies whether all the “storage image extended formats” below are supported; if this feature is supported, then the `VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT` **must** be supported in `optimalTilingFeatures` for the following formats:
 - `VK_FORMAT_R16G16_SFLOAT`
 - `VK_FORMAT_B10G11R11_UFLOAT_PACK32`
 - `VK_FORMAT_R16_SFLOAT`
 - `VK_FORMAT_R16G16B16A16_UNORM`
 - `VK_FORMAT_A2B10G10R10_UNORM_PACK32`
 - `VK_FORMAT_R16G16_UNORM`
 - `VK_FORMAT_R8G8_UNORM`
 - `VK_FORMAT_R16_UNORM`
 - `VK_FORMAT_R8_UNORM`

- `VK_FORMAT_R16G16B16A16_SNORM`
- `VK_FORMAT_R16G16_SNORM`
- `VK_FORMAT_R8G8_SNORM`
- `VK_FORMAT_R16_SNORM`
- `VK_FORMAT_R8_SNORM`
- `VK_FORMAT_R16G16_SINT`
- `VK_FORMAT_R8G8_SINT`
- `VK_FORMAT_R16_SINT`
- `VK_FORMAT_R8_SINT`
- `VK_FORMAT_A2B10G10R10_UINT_PACK32`
- `VK_FORMAT_R16G16_UINT`
- `VK_FORMAT_R8G8_UINT`
- `VK_FORMAT_R16_UINT`
- `VK_FORMAT_R8_UINT`

NOTE

`shaderStorageImageExtendedFormats` feature only adds a guarantee of format support, which is specified for the whole physical device. Therefore enabling or disabling the feature via `vkCreateDevice` has no practical effect.

To query for additional properties, or if the feature is not supported, `vkGetPhysicalDeviceFormatProperties` and `vkGetPhysicalDeviceImageFormatProperties` **can** be used to check for supported properties of individual formats, as usual rules allow.

`VK_FORMAT_R32G32_UINT`, `VK_FORMAT_R32G32_SINT`, and `VK_FORMAT_R32G32_SFLOAT` from `StorageImageExtendedFormats` SPIR-V capability, are already covered by core Vulkan **mandatory format support**.

- `shaderStorageImageMultisample` specifies whether multisampled storage images are supported. If this feature is not enabled, images that are created with a `usage` that includes `VK_IMAGE_USAGE_STORAGE_BIT` **must** be created with `samples` equal to `VK_SAMPLE_COUNT_1_BIT`. This also specifies whether shader modules **can** declare the `StorageImageMultisample` and `ImageMSArray` capabilities.
- `shaderStorageImageReadWithoutFormat` specifies whether storage images and storage texel buffers require a format qualifier to be specified when reading.
- `shaderStorageImageWriteWithoutFormat` specifies whether storage images and storage texel buffers require a format qualifier to be specified when writing.
- `shaderUniformBufferArrayDynamicIndexing` specifies whether arrays of uniform buffers **can** be indexed by integer expressions that are dynamically uniform within the invocation group in shader code. If this feature is not enabled, resources with a descriptor type of `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER` or `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` **must** be indexed only by constant integral expressions when aggregated into arrays in shader code. This

also specifies whether shader modules **can** declare the `UniformBufferArrayDynamicIndexing` capability.

- `shaderSampledImageArrayDynamicIndexing` specifies whether arrays of samplers or sampled images **can** be indexed by integer expressions that are dynamically uniform within the invocation group in shader code. If this feature is not enabled, resources with a descriptor type of `VK_DESCRIPTOR_TYPE_SAMPLER`, `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, or `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE` **must** be indexed only by constant integral expressions when aggregated into arrays in shader code. This also specifies whether shader modules **can** declare the `SampledImageArrayDynamicIndexing` capability.
- `shaderStorageBufferArrayDynamicIndexing` specifies whether arrays of storage buffers **can** be indexed by integer expressions that are dynamically uniform within the invocation group in shader code. If this feature is not enabled, resources with a descriptor type of `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER` or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC` **must** be indexed only by constant integral expressions when aggregated into arrays in shader code. This also specifies whether shader modules **can** declare the `StorageBufferArrayDynamicIndexing` capability.
- `shaderStorageImageArrayDynamicIndexing` specifies whether arrays of storage images **can** be indexed by integer expressions that are dynamically uniform within the invocation group in shader code. If this feature is not enabled, resources with a descriptor type of `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE` **must** be indexed only by constant integral expressions when aggregated into arrays in shader code. This also specifies whether shader modules **can** declare the `StorageImageArrayDynamicIndexing` capability.
- `shaderClipDistance` specifies whether clip distances are supported in shader code. If this feature is not enabled, any members decorated with the `ClipDistance` built-in decoration **must** not be read from or written to in shader modules. This also specifies whether shader modules **can** declare the `ClipDistance` capability.
- `shaderCullDistance` specifies whether cull distances are supported in shader code. If this feature is not enabled, any members decorated with the `CullDistance` built-in decoration **must** not be read from or written to in shader modules. This also specifies whether shader modules **can** declare the `CullDistance` capability.
- `shaderFloat64` specifies whether 64-bit floats (doubles) are supported in shader code. If this feature is not enabled, 64-bit floating-point types **must** not be used in shader code. This also specifies whether shader modules **can** declare the `Float64` capability. Declaring and using 64-bit floats is enabled for all storage classes that SPIR-V allows with the `Float64` capability.
- `shaderInt64` specifies whether 64-bit integers (signed and unsigned) are supported in shader code. If this feature is not enabled, 64-bit integer types **must** not be used in shader code. This also specifies whether shader modules **can** declare the `Int64` capability. Declaring and using 64-bit integers is enabled for all storage classes that SPIR-V allows with the `Int64` capability.
- `shaderInt16` specifies whether 16-bit integers (signed and unsigned) are supported in shader code. If this feature is not enabled, 16-bit integer types **must** not be used in shader code. This also specifies whether shader modules **can** declare the `Int16` capability. However, this only enables a subset of the storage classes that SPIR-V allows for the `Int16` SPIR-V capability: Declaring and using 16-bit integers in the `Private`, `Workgroup`, and `Function` storage classes is enabled, while declaring them in the interface storage classes (e.g., `UniformConstant`, `Uniform`, `StorageBuffer`, `Input`, `Output`, and `PushConstant`) is not enabled.

- **shaderResourceResidency** specifies whether image operations that return resource residency information are supported in shader code. If this feature is not enabled, the **OpImageSparse*** instructions **must** not be used in shader code. This also specifies whether shader modules **can** declare the **SparseResidency** capability. The feature requires at least one of the **sparseResidency*** features to be supported.
- **shaderResourceMinLod** specifies whether image operations specifying the minimum resource LOD are supported in shader code. If this feature is not enabled, the **MinLod** image operand **must** not be used in shader code. This also specifies whether shader modules **can** declare the **MinLod** capability.
- **sparseBinding** specifies whether resource memory **can** be managed at opaque sparse block level instead of at the object level. If this feature is not enabled, resource memory **must** be bound only on a per-object basis using the **vkBindBufferMemory** and **vkBindImageMemory** commands. In this case, buffers and images **must** not be created with **VK_BUFFER_CREATE_SPARSE_BINDING_BIT** and **VK_IMAGE_CREATE_SPARSE_BINDING_BIT** set in the **flags** member of the **VkBufferCreateInfo** and **VkImageCreateInfo** structures, respectively. Otherwise resource memory **can** be managed as described in [Sparse Resource Features](#).
- **sparseResidencyBuffer** specifies whether the device **can** access partially resident buffers. If this feature is not enabled, buffers **must** not be created with **VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT** set in the **flags** member of the **VkBufferCreateInfo** structure.
- **sparseResidencyImage2D** specifies whether the device **can** access partially resident 2D images with 1 sample per pixel. If this feature is not enabled, images with an **imageType** of **VK_IMAGE_TYPE_2D** and **samples** set to **VK_SAMPLE_COUNT_1_BIT** **must** not be created with **VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT** set in the **flags** member of the **VkImageCreateInfo** structure.
- **sparseResidencyImage3D** specifies whether the device **can** access partially resident 3D images. If this feature is not enabled, images with an **imageType** of **VK_IMAGE_TYPE_3D** **must** not be created with **VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT** set in the **flags** member of the **VkImageCreateInfo** structure.
- **sparseResidency2Samples** specifies whether the physical device **can** access partially resident 2D images with 2 samples per pixel. If this feature is not enabled, images with an **imageType** of **VK_IMAGE_TYPE_2D** and **samples** set to **VK_SAMPLE_COUNT_2_BIT** **must** not be created with **VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT** set in the **flags** member of the **VkImageCreateInfo** structure.
- **sparseResidency4Samples** specifies whether the physical device **can** access partially resident 2D images with 4 samples per pixel. If this feature is not enabled, images with an **imageType** of **VK_IMAGE_TYPE_2D** and **samples** set to **VK_SAMPLE_COUNT_4_BIT** **must** not be created with **VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT** set in the **flags** member of the **VkImageCreateInfo** structure.
- **sparseResidency8Samples** specifies whether the physical device **can** access partially resident 2D images with 8 samples per pixel. If this feature is not enabled, images with an **imageType** of **VK_IMAGE_TYPE_2D** and **samples** set to **VK_SAMPLE_COUNT_8_BIT** **must** not be created with **VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT** set in the **flags** member of the **VkImageCreateInfo** structure.
- **sparseResidency16Samples** specifies whether the physical device **can** access partially resident 2D

images with 16 samples per pixel. If this feature is not enabled, images with an `imageType` of `VK_IMAGE_TYPE_2D` and `samples` set to `VK_SAMPLE_COUNT_16_BIT` **must** not be created with `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` set in the `flags` member of the `VkImageCreateInfo` structure.

- `sparseResidencyAliased` specifies whether the physical device **can** correctly access data aliased into multiple locations. If this feature is not enabled, the `VK_BUFFER_CREATE_SPARSE_ALIASED_BIT` and `VK_IMAGE_CREATE_SPARSE_ALIASED_BIT` enum values **must** not be used in `flags` members of the `VkBufferCreateInfo` and `VkImageCreateInfo` structures, respectively.
- `variableMultisampleRate` specifies whether all pipelines that will be bound to a command buffer during a subpass which uses no attachments **must** have the same value for `VkPipelineMultisampleStateCreateInfo::rasterizationSamples`. If set to `VK_TRUE`, the implementation supports variable multisample rates in a subpass which uses no attachments. If set to `VK_FALSE`, then all pipelines bound in such a subpass **must** have the same multisample rate. This has no effect in situations where a subpass uses any attachments.
- `inheritedQueries` specifies whether a secondary command buffer **may** be executed while a query is active.

`nullDescriptor` support requires the `VK_EXT_robustness2` extension.

31.1. Feature Requirements

All Vulkan graphics implementations **must** support the following features:

- If Vulkan 1.0 is supported, `robustBufferAccess` **must** be supported

All other features defined in the Specification are **optional**.

Chapter 32. Limits

Limits are implementation-dependent minimums, maximums, and other device characteristics that an application **may** need to be aware of.

The `VkPhysicalDeviceLimits` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPhysicalDeviceLimits {
    uint32_t          maxImageDimension1D;
    uint32_t          maxImageDimension2D;
    uint32_t          maxImageDimension3D;
    uint32_t          maxImageDimensionCube;
    uint32_t          maxImageArrayLayers;
    uint32_t          maxTexelBufferElements;
    uint32_t          maxUniformBufferRange;
    uint32_t          maxStorageBufferRange;
    uint32_t          maxPushConstantsSize;
    uint32_t          maxMemoryAllocationCount;
    uint32_t          maxSamplerAllocationCount;
    VkDeviceSize      bufferImageGranularity;
    VkDeviceSize      sparseAddressSpaceSize;
    uint32_t          maxBoundDescriptorSets;
    uint32_t          maxPerStageDescriptorSamplers;
    uint32_t          maxPerStageDescriptorUniformBuffers;
    uint32_t          maxPerStageDescriptorStorageBuffers;
    uint32_t          maxPerStageDescriptorSampledImages;
    uint32_t          maxPerStageDescriptorStorageImages;
    uint32_t          maxPerStageDescriptorInputAttachments;
    uint32_t          maxPerStageResources;
    uint32_t          maxDescriptorSetSamplers;
    uint32_t          maxDescriptorSetUniformBuffers;
    uint32_t          maxDescriptorSetUniformBuffersDynamic;
    uint32_t          maxDescriptorSetStorageBuffers;
    uint32_t          maxDescriptorSetStorageBuffersDynamic;
    uint32_t          maxDescriptorSetSampledImages;
    uint32_t          maxDescriptorSetStorageImages;
    uint32_t          maxDescriptorSetInputAttachments;
    uint32_t          maxVertexInputAttributes;
    uint32_t          maxVertexInputBindings;
    uint32_t          maxVertexInputAttributeOffset;
    uint32_t          maxVertexInputBindingStride;
    uint32_t          maxVertexOutputComponents;
    uint32_t          maxTessellationGenerationLevel;
    uint32_t          maxTessellationPatchSize;
    uint32_t          maxTessellationControlPerVertexInputComponents;
    uint32_t          maxTessellationControlPerVertexOutputComponents;
    uint32_t          maxTessellationControlPerPatchOutputComponents;
    uint32_t          maxTessellationControlTotalOutputComponents;
    uint32_t          maxTessellationEvaluationInputComponents;
```


uint32_t	maxTessellationEvaluationOutputComponents;
uint32_t	maxGeometryShaderInvocations;
uint32_t	maxGeometryInputComponents;
uint32_t	maxGeometryOutputComponents;
uint32_t	maxGeometryOutputVertices;
uint32_t	maxGeometryTotalOutputComponents;
uint32_t	maxFragmentInputComponents;
uint32_t	maxFragmentOutputAttachments;
uint32_t	maxFragmentDualSrcAttachments;
uint32_t	maxFragmentCombinedOutputResources;
uint32_t	maxComputeSharedMemorySize;
uint32_t	maxComputeWorkGroupCount[3];
uint32_t	maxComputeWorkGroupInvocations;
uint32_t	maxComputeWorkGroupSize[3];
uint32_t	subPixelPrecisionBits;
uint32_t	subTexelPrecisionBits;
uint32_t	mipmapPrecisionBits;
uint32_t	maxDrawIndexedIndexValue;
uint32_t	maxDrawIndirectCount;
float	maxSamplerLodBias;
float	maxSamplerAnisotropy;
uint32_t	maxViewports;
uint32_t	maxViewportDimensions[2];
float	viewportBoundsRange[2];
uint32_t	viewportSubPixelBits;
size_t	minMemoryMapAlignment;
VkDeviceSize	minTexelBufferOffsetAlignment;
VkDeviceSize	minUniformBufferOffsetAlignment;
VkDeviceSize	minStorageBufferOffsetAlignment;
int32_t	minTexelOffset;
uint32_t	maxTexelOffset;
int32_t	minTexelGatherOffset;
uint32_t	maxTexelGatherOffset;
float	minInterpolationOffset;
float	maxInterpolationOffset;
uint32_t	subPixelInterpolationOffsetBits;
uint32_t	maxFramebufferWidth;
uint32_t	maxFramebufferHeight;
uint32_t	maxFramebufferLayers;
VkSampleCountFlags	framebufferColorSampleCounts;
VkSampleCountFlags	framebufferDepthSampleCounts;
VkSampleCountFlags	framebufferStencilSampleCounts;
VkSampleCountFlags	framebufferNoAttachmentsSampleCounts;
uint32_t	maxColorAttachments;
VkSampleCountFlags	sampledImageColorSampleCounts;
VkSampleCountFlags	sampledImageIntegerSampleCounts;
VkSampleCountFlags	sampledImageDepthSampleCounts;
VkSampleCountFlags	sampledImageStencilSampleCounts;
VkSampleCountFlags	storageImageSampleCounts;
uint32_t	maxSampleMaskWords;
VkBool32	timestampComputeAndGraphics;

```

float          timestampPeriod;
uint32_t       maxClipDistances;
uint32_t       maxCullDistances;
uint32_t       maxCombinedClipAndCullDistances;
uint32_t       discreteQueuePriorities;
float          pointSizeRange[2];
float          lineWidthRange[2];
float          pointSizeGranularity;
float          lineWidthGranularity;
VkBool32       strictLines;
VkBool32       standardSampleLocations;
VkDeviceSize   optimalBufferCopyOffsetAlignment;
VkDeviceSize   optimalBufferCopyRowPitchAlignment;
VkDeviceSize   nonCoherentAtomSize;
} VkPhysicalDeviceLimits;

```

The `VkPhysicalDeviceLimits` are properties of the physical device. These are available in the `limits` member of the `VkPhysicalDeviceProperties` structure which is returned from `vkGetPhysicalDeviceProperties`.

- `maxImageDimension1D` is the largest dimension (`width`) that is guaranteed to be supported for all images created with an `imageType` of `VK_IMAGE_TYPE_1D`. Some combinations of image parameters (format, usage, etc.) **may** allow support for larger dimensions, which **can** be queried using `vkGetPhysicalDeviceImageFormatProperties`.
- `maxImageDimension2D` is the largest dimension (`width` or `height`) that is guaranteed to be supported for all images created with an `imageType` of `VK_IMAGE_TYPE_2D` and without `VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT` set in `flags`. Some combinations of image parameters (format, usage, etc.) **may** allow support for larger dimensions, which **can** be queried using `vkGetPhysicalDeviceImageFormatProperties`.
- `maxImageDimension3D` is the largest dimension (`width`, `height`, or `depth`) that is guaranteed to be supported for all images created with an `imageType` of `VK_IMAGE_TYPE_3D`. Some combinations of image parameters (format, usage, etc.) **may** allow support for larger dimensions, which **can** be queried using `vkGetPhysicalDeviceImageFormatProperties`.
- `maxImageDimensionCube` is the largest dimension (`width` or `height`) that is guaranteed to be supported for all images created with an `imageType` of `VK_IMAGE_TYPE_2D` and with `VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT` set in `flags`. Some combinations of image parameters (format, usage, etc.) **may** allow support for larger dimensions, which **can** be queried using `vkGetPhysicalDeviceImageFormatProperties`.
- `maxImageArrayLayers` is the maximum number of layers (`arrayLayers`) for an image.
- `maxTexelBufferElements` is the maximum number of addressable texels for a buffer view created on a buffer which was created with the `VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT` or `VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT` set in the `usage` member of the `VkBufferCreateInfo` structure.
- `maxUniformBufferRange` is the maximum value that **can** be specified in the `range` member of a `VkDescriptorBufferInfo` structure passed to `vkUpdateDescriptorSets` for descriptors of type `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER` or `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC`.

- `maxStorageBufferRange` is the maximum value that **can** be specified in the `range` member of a `VkDescriptorBufferInfo` structure passed to `vkUpdateDescriptorSets` for descriptors of type `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER` or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC`.
- `maxPushConstantsSize` is the maximum size, in bytes, of the pool of push constant memory. For each of the push constant ranges indicated by the `pPushConstantRanges` member of the `VkPipelineLayoutCreateInfo` structure, `(offset + size)` **must** be less than or equal to this limit.
- `maxMemoryAllocationCount` is the maximum number of device memory allocations, as created by `vkAllocateMemory`, which **can** simultaneously exist.
- `maxSamplerAllocationCount` is the maximum number of sampler objects, as created by `vkCreateSampler`, which **can** simultaneously exist on a device.
- `bufferImageGranularity` is the granularity, in bytes, at which buffer or linear image resources, and optimal image resources **can** be bound to adjacent offsets in the same `VkDeviceMemory` object without aliasing. See [Buffer-Image Granularity](#) for more details.
- `sparseAddressSpaceSize` is the total amount of address space available, in bytes, for sparse memory resources. This is an upper bound on the sum of the sizes of all sparse resources, regardless of whether any memory is bound to them.
- `maxBoundDescriptorSets` is the maximum number of descriptor sets that **can** be simultaneously used by a pipeline. All `DescriptorSet` decorations in shader modules **must** have a value less than `maxBoundDescriptorSets`. See [Descriptor Sets](#).
- `maxPerStageDescriptorSamplers` is the maximum number of samplers that **can** be accessible to a single shader stage in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_SAMPLER` or `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER` count against this limit. A descriptor is accessible to a shader stage when the `stageFlags` member of the `VkDescriptorSetLayoutBinding` structure has the bit for that shader stage set. See [Sampler](#) and [Combined Image Sampler](#).
- `maxPerStageDescriptorUniformBuffers` is the maximum number of uniform buffers that **can** be accessible to a single shader stage in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER` or `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` count against this limit. A descriptor is accessible to a shader stage when the `stageFlags` member of the `VkDescriptorSetLayoutBinding` structure has the bit for that shader stage set. See [Uniform Buffer](#) and [Dynamic Uniform Buffer](#).
- `maxPerStageDescriptorStorageBuffers` is the maximum number of storage buffers that **can** be accessible to a single shader stage in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER` or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC` count against this limit. A descriptor is accessible to a pipeline shader stage when the `stageFlags` member of the `VkDescriptorSetLayoutBinding` structure has the bit for that shader stage set. See [Storage Buffer](#) and [Dynamic Storage Buffer](#).
- `maxPerStageDescriptorSampledImages` is the maximum number of sampled images that **can** be accessible to a single shader stage in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, or `VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER` count against this limit. A descriptor is accessible to a pipeline shader stage when the `stageFlags` member of the `VkDescriptorSetLayoutBinding` structure has the bit for that shader stage set. See [Combined Image Sampler](#), [Sampled Image](#), and [Uniform Texel Buffer](#).

- `maxPerStageDescriptorStorageImages` is the maximum number of storage images that **can** be accessible to a single shader stage in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, or `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` count against this limit. A descriptor is accessible to a pipeline shader stage when the `stageFlags` member of the `VkDescriptorSetLayoutBinding` structure has the bit for that shader stage set. See [Storage Image](#), and [Storage Texel Buffer](#).
- `maxPerStageDescriptorInputAttachments` is the maximum number of input attachments that **can** be accessible to a single shader stage in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT` count against this limit. A descriptor is accessible to a pipeline shader stage when the `stageFlags` member of the `VkDescriptorSetLayoutBinding` structure has the bit for that shader stage set. These are only supported for the fragment stage. See [Input Attachment](#).
- `maxPerStageResources` is the maximum number of resources that **can** be accessible to a single shader stage in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, `VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER`, `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER`, `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER`, `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER`, `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC`, `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC`, or `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT` count against this limit. For the fragment shader stage the framebuffer color attachments also count against this limit.
- `maxDescriptorSetSamplers` is the maximum number of samplers that **can** be included in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_SAMPLER` or `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER` count against this limit. See [Sampler](#) and [Combined Image Sampler](#).
- `maxDescriptorSetUniformBuffers` is the maximum number of uniform buffers that **can** be included in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER` or `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` count against this limit. See [Uniform Buffer](#) and [Dynamic Uniform Buffer](#).
- `maxDescriptorSetUniformBuffersDynamic` is the maximum number of dynamic uniform buffers that **can** be included in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` count against this limit. See [Dynamic Uniform Buffer](#).
- `maxDescriptorSetStorageBuffers` is the maximum number of storage buffers that **can** be included in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER` or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC` count against this limit. See [Storage Buffer](#) and [Dynamic Storage Buffer](#).
- `maxDescriptorSetStorageBuffersDynamic` is the maximum number of dynamic storage buffers that **can** be included in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC` count against this limit. See [Dynamic Storage Buffer](#).
- `maxDescriptorSetSampledImages` is the maximum number of sampled images that **can** be included in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, or `VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER` count against this limit. See [Combined Image Sampler](#), [Sampled Image](#), and [Uniform Texel Buffer](#).

- `maxDescriptorSetStorageImages` is the maximum number of storage images that **can** be included in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, or `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` count against this limit. See [Storage Image](#), and [Storage Texel Buffer](#).
- `maxDescriptorSetInputAttachments` is the maximum number of input attachments that **can** be included in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT` count against this limit. See [Input Attachment](#).
- `maxVertexInputAttributes` is the maximum number of vertex input attributes that **can** be specified for a graphics pipeline. These are described in the array of `VkVertexInputAttributeDescription` structures that are provided at graphics pipeline creation time via the `pVertexAttributeDescriptions` member of the `VkPipelineVertexInputStateCreateInfo` structure. See [Vertex Attributes](#) and [Vertex Input Description](#).
- `maxVertexInputBindings` is the maximum number of vertex buffers that **can** be specified for providing vertex attributes to a graphics pipeline. These are described in the array of `VkVertexInputBindingDescription` structures that are provided at graphics pipeline creation time via the `pVertexBindingDescriptions` member of the `VkPipelineVertexInputStateCreateInfo` structure. The `binding` member of `VkVertexInputBindingDescription` **must** be less than this limit. See [Vertex Input Description](#).
- `maxVertexInputAttributeOffset` is the maximum vertex input attribute offset that **can** be added to the vertex input binding stride. The `offset` member of the `VkVertexInputAttributeDescription` structure **must** be less than or equal to this limit. See [Vertex Input Description](#).
- `maxVertexInputBindingStride` is the maximum vertex input binding stride that **can** be specified in a vertex input binding. The `stride` member of the `VkVertexInputBindingDescription` structure **must** be less than or equal to this limit. See [Vertex Input Description](#).
- `maxVertexOutputComponents` is the maximum number of components of output variables which **can** be output by a vertex shader. See [Vertex Shaders](#).
- `maxTessellationGenerationLevel` is the maximum tessellation generation level supported by the fixed-function tessellation primitive generator. See [Tessellation](#).
- `maxTessellationPatchSize` is the maximum patch size, in vertices, of patches that **can** be processed by the tessellation control shader and tessellation primitive generator. The `patchControlPoints` member of the `VkPipelineTessellationStateCreateInfo` structure specified at pipeline creation time and the value provided in the `OutputVertices` execution mode of shader modules **must** be less than or equal to this limit. See [Tessellation](#).
- `maxTessellationControlPerVertexInputComponents` is the maximum number of components of input variables which **can** be provided as per-vertex inputs to the tessellation control shader stage.
- `maxTessellationControlPerVertexOutputComponents` is the maximum number of components of per-vertex output variables which **can** be output from the tessellation control shader stage.
- `maxTessellationControlPerPatchOutputComponents` is the maximum number of components of per-patch output variables which **can** be output from the tessellation control shader stage.
- `maxTessellationControlTotalOutputComponents` is the maximum total number of components of per-vertex and per-patch output variables which **can** be output from the tessellation control shader stage.

- `maxTessellationEvaluationInputComponents` is the maximum number of components of input variables which **can** be provided as per-vertex inputs to the tessellation evaluation shader stage.
- `maxTessellationEvaluationOutputComponents` is the maximum number of components of per-vertex output variables which **can** be output from the tessellation evaluation shader stage.
- `maxGeometryShaderInvocations` is the maximum invocation count supported for instanced geometry shaders. The value provided in the `Invocations` execution mode of shader modules **must** be less than or equal to this limit. See [Geometry Shading](#).
- `maxGeometryInputComponents` is the maximum number of components of input variables which **can** be provided as inputs to the geometry shader stage.
- `maxGeometryOutputComponents` is the maximum number of components of output variables which **can** be output from the geometry shader stage.
- `maxGeometryOutputVertices` is the maximum number of vertices which **can** be emitted by any geometry shader.
- `maxGeometryTotalOutputComponents` is the maximum total number of components of output variables, across all emitted vertices, which **can** be output from the geometry shader stage.
- `maxFragmentInputComponents` is the maximum number of components of input variables which **can** be provided as inputs to the fragment shader stage.
- `maxFragmentOutputAttachments` is the maximum number of output attachments which **can** be written to by the fragment shader stage.
- `maxFragmentDualSrcAttachments` is the maximum number of output attachments which **can** be written to by the fragment shader stage when blending is enabled and one of the dual source blend modes is in use. See [Dual-Source Blending](#) and `dualSrcBlend`.
- `maxFragmentCombinedOutputResources` is the total number of storage buffers, storage images, and output `Location` decorated color attachments (described in [Fragment Output Interface](#)) which **can** be used in the fragment shader stage.
- `maxComputeSharedMemorySize` is the maximum total storage size, in bytes, available for variables declared with the `Workgroup` storage class in shader modules (or with the `shared` storage qualifier in GLSL) in the compute shader stage.
- `maxComputeWorkGroupCount[3]` is the maximum number of local workgroups that **can** be dispatched by a single dispatching command. These three values represent the maximum number of local workgroups for the X, Y, and Z dimensions, respectively. The workgroup count parameters to the dispatching commands **must** be less than or equal to the corresponding limit. See [Dispatching Commands](#).
- `maxComputeWorkGroupInvocations` is the maximum total number of compute shader invocations in a single local workgroup. The product of the X, Y, and Z sizes, as specified by the `LocalSize` execution mode in shader modules or by the object decorated by the `WorkgroupSize` decoration, **must** be less than or equal to this limit.
- `maxComputeWorkGroupSize[3]` is the maximum size of a local compute workgroup, per dimension. These three values represent the maximum local workgroup size in the X, Y, and Z dimensions, respectively. The `x`, `y`, and `z` sizes, as specified by the `LocalSize` execution mode or by the object decorated by the `WorkgroupSize` decoration in shader modules, **must** be less than or equal to the

corresponding limit.

- `subPixelPrecisionBits` is the number of bits of subpixel precision in framebuffer coordinates x_f and y_f . See [Rasterization](#).
- `subTexelPrecisionBits` is the number of bits of precision in the division along an axis of an image used for minification and magnification filters. $2^{\text{subTexelPrecisionBits}}$ is the actual number of divisions along each axis of the image represented. Sub-texel values calculated during image sampling will snap to these locations when generating the filtered results.
- `mipmapPrecisionBits` is the number of bits of division that the LOD calculation for mipmap fetching get snapped to when determining the contribution from each mip level to the mip filtered results. $2^{\text{mipmapPrecisionBits}}$ is the actual number of divisions.
- `maxDrawIndexedIndexValue` is the maximum index value that **can** be used for indexed draw calls when using 32-bit indices. This excludes the primitive restart index value of 0xFFFFFFFF. See [fullDrawIndexUint32](#).
- `maxDrawIndirectCount` is the maximum draw count that is supported for indirect drawing calls. See [multiDrawIndirect](#).
- `maxSamplerLodBias` is the maximum absolute sampler LOD bias. The sum of the `mipLodBias` member of the [VkSamplerCreateInfo](#) structure and the `Bias` operand of image sampling operations in shader modules (or 0 if no `Bias` operand is provided to an image sampling operation) are clamped to the range $[-\text{maxSamplerLodBias}, +\text{maxSamplerLodBias}]$. See [\[samplers-mipLodBias\]](#).
- `maxSamplerAnisotropy` is the maximum degree of sampler anisotropy. The maximum degree of anisotropic filtering used for an image sampling operation is the minimum of the `maxAnisotropy` member of the [VkSamplerCreateInfo](#) structure and this limit. See [\[samplers-maxAnisotropy\]](#).
- `maxViewports` is the maximum number of active viewports. The `viewportCount` member of the [VkPipelineViewportStateCreateInfo](#) structure that is provided at pipeline creation **must** be less than or equal to this limit.
- `maxViewportDimensions[2]` are the maximum viewport dimensions in the X (width) and Y (height) dimensions, respectively. The maximum viewport dimensions **must** be greater than or equal to the largest image which **can** be created and used as a framebuffer attachment. See [Controlling the Viewport](#).
- `viewportBoundsRange[2]` is the [minimum, maximum] range that the corners of a viewport **must** be contained in. This range **must** be at least $[-2 \times \text{size}, 2 \times \text{size} - 1]$, where `size` = $\max(\text{maxViewportDimensions}[0], \text{maxViewportDimensions}[1])$. See [Controlling the Viewport](#).

NOTE

The intent of the `viewportBoundsRange` limit is to allow a maximum sized viewport to be arbitrarily shifted relative to the output target as long as at least some portion intersects. This would give a bounds limit of $[-\text{size} + 1, 2 \times \text{size} - 1]$ which would allow all possible non-empty-set intersections of the output target and the viewport. Since these numbers are typically powers of two, picking the signed number range using the smallest possible number of bits ends up with the specified range.

- `viewportSubPixelBits` is the number of bits of subpixel precision for viewport bounds. The subpixel precision that floating-point viewport bounds are interpreted at is given by this limit.

- `minMemoryMapAlignment` is the minimum **required** alignment, in bytes, of host visible memory allocations within the host address space. When mapping a memory allocation with `vkMapMemory`, subtracting `offset` bytes from the returned pointer will always produce an integer multiple of this limit. See [Host Access to Device Memory Objects](#). The value **must** be a power of two.
- `minTexelBufferOffsetAlignment` is the minimum **required** alignment, in bytes, for the `offset` member of the `VkBufferViewCreateInfo` structure for texel buffers. The value **must** be a power of two. `VkBufferViewCreateInfo::offset` **must** be a multiple of this value.
- `minUniformBufferOffsetAlignment` is the minimum **required** alignment, in bytes, for the `offset` member of the `VkDescriptorBufferInfo` structure for uniform buffers. When a descriptor of type `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER` or `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` is updated, the `offset` **must** be an integer multiple of this limit. Similarly, dynamic offsets for uniform buffers **must** be multiples of this limit. The value **must** be a power of two.
- `minStorageBufferOffsetAlignment` is the minimum **required** alignment, in bytes, for the `offset` member of the `VkDescriptorBufferInfo` structure for storage buffers. When a descriptor of type `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER` or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC` is updated, the `offset` **must** be an integer multiple of this limit. Similarly, dynamic offsets for storage buffers **must** be multiples of this limit. The value **must** be a power of two.
- `minTexelOffset` is the minimum offset value for the `ConstOffset` image operand of any of the `OpImageSample*` or `OpImageFetch*` image instructions.
- `maxTexelOffset` is the maximum offset value for the `ConstOffset` image operand of any of the `OpImageSample*` or `OpImageFetch*` image instructions.
- `minTexelGatherOffset` is the minimum offset value for the `Offset`, `ConstOffset`, or `ConstOffsets` image operands of any of the `OpImage*Gather` image instructions.
- `maxTexelGatherOffset` is the maximum offset value for the `Offset`, `ConstOffset`, or `ConstOffsets` image operands of any of the `OpImage*Gather` image instructions.
- `minInterpolationOffset` is the base minimum (inclusive) negative offset value for the `Offset` operand of the `InterpolateAtOffset` extended instruction.
- `maxInterpolationOffset` is the base maximum (inclusive) positive offset value for the `Offset` operand of the `InterpolateAtOffset` extended instruction.
- `subPixelInterpolationOffsetBits` is the number of fractional bits that the `x` and `y` offsets to the `InterpolateAtOffset` extended instruction **may** be rounded to as fixed-point values.
- `maxFramebufferWidth` is the maximum width for a framebuffer. The `width` member of the `VkFramebufferCreateInfo` structure **must** be less than or equal to this limit.
- `maxFramebufferHeight` is the maximum height for a framebuffer. The `height` member of the `VkFramebufferCreateInfo` structure **must** be less than or equal to this limit.
- `maxFramebufferLayers` is the maximum layer count for a layered framebuffer. The `layers` member of the `VkFramebufferCreateInfo` structure **must** be less than or equal to this limit.
- `framebufferColorSampleCounts` is a bitmask¹ of `VkSampleCountFlagBits` indicating the color sample counts that are supported for all framebuffer color attachments with floating- or fixed-point formats. There is no limit specifying the color sample counts that are supported for all color attachments with integer formats.

- `framebufferDepthSampleCounts` is a bitmask¹ of `VkSampleCountFlagBits` indicating the supported depth sample counts for all framebuffer depth/stencil attachments, when the format includes a depth component.
- `framebufferStencilSampleCounts` is a bitmask¹ of `VkSampleCountFlagBits` indicating the supported stencil sample counts for all framebuffer depth/stencil attachments, when the format includes a stencil component.
- `framebufferNoAttachmentsSampleCounts` is a bitmask¹ of `VkSampleCountFlagBits` indicating the supported sample counts for a [subpass which uses no attachments](#).
- `maxColorAttachments` is the maximum number of color attachments that **can** be used by a subpass in a render pass. The `colorAttachmentCount` member of the `VkSubpassDescription` structure **must** be less than or equal to this limit.
- `sampledImageColorSampleCounts` is a bitmask¹ of `VkSampleCountFlagBits` indicating the sample counts supported for all 2D images created with `VK_IMAGE_TILING_OPTIMAL`, `usage` containing `VK_IMAGE_USAGE_SAMPLED_BIT`, and a non-integer color format.
- `sampledImageIntegerSampleCounts` is a bitmask¹ of `VkSampleCountFlagBits` indicating the sample counts supported for all 2D images created with `VK_IMAGE_TILING_OPTIMAL`, `usage` containing `VK_IMAGE_USAGE_SAMPLED_BIT`, and an integer color format.
- `sampledImageDepthSampleCounts` is a bitmask¹ of `VkSampleCountFlagBits` indicating the sample counts supported for all 2D images created with `VK_IMAGE_TILING_OPTIMAL`, `usage` containing `VK_IMAGE_USAGE_SAMPLED_BIT`, and a depth format.
- `sampledImageStencilSampleCounts` is a bitmask¹ of `VkSampleCountFlagBits` indicating the sample counts supported for all 2D images created with `VK_IMAGE_TILING_OPTIMAL`, `usage` containing `VK_IMAGE_USAGE_SAMPLED_BIT`, and a stencil format.
- `storageImageSampleCounts` is a bitmask¹ of `VkSampleCountFlagBits` indicating the sample counts supported for all 2D images created with `VK_IMAGE_TILING_OPTIMAL`, and `usage` containing `VK_IMAGE_USAGE_STORAGE_BIT`.
- `maxSampleMaskWords` is the maximum number of array elements of a variable decorated with the `SampleMask` built-in decoration.
- `timestampComputeAndGraphics` specifies support for timestamps on all graphics and compute queues. If this limit is `VK_TRUE`, all queues that advertise the `VK_QUEUE_GRAPHICS_BIT` or `VK_QUEUE_COMPUTE_BIT` in the `VkQueueFamilyProperties::queueFlags` support `VkQueueFamilyProperties::timestampValidBits` of at least 36. See [Timestamp Queries](#).
- `timestampPeriod` is the number of nanoseconds **required** for a timestamp query to be incremented by 1. See [Timestamp Queries](#).
- `maxClipDistances` is the maximum number of clip distances that **can** be used in a single shader stage. The size of any array declared with the `ClipDistance` built-in decoration in a shader module **must** be less than or equal to this limit.
- `maxCullDistances` is the maximum number of cull distances that **can** be used in a single shader stage. The size of any array declared with the `CullDistance` built-in decoration in a shader module **must** be less than or equal to this limit.
- `maxCombinedClipAndCullDistances` is the maximum combined number of clip and cull distances that **can** be used in a single shader stage. The sum of the sizes of all arrays declared with the

`ClipDistance` and `CullDistance` built-in decoration used by a single shader stage in a shader module **must** be less than or equal to this limit.

- `discreteQueuePriorities` is the number of discrete priorities that **can** be assigned to a queue based on the value of each member of `VkDeviceQueueCreateInfo::pQueuePriorities`. This **must** be at least 2, and levels **must** be spread evenly over the range, with at least one level at 1.0, and another at 0.0. See [Queue Priority](#).
- `pointSizeRange[2]` is the range `[minimum,maximum]` of supported sizes for points. Values written to variables decorated with the `PointSize` built-in decoration are clamped to this range.
- `lineWidthRange[2]` is the range `[minimum,maximum]` of supported widths for lines. Values specified by the `lineWidth` member of the `VkPipelineRasterizationStateCreateInfo` or the `lineWidth` parameter to `vkCmdSetLineWidth` are clamped to this range.
- `pointSizeGranularity` is the granularity of supported point sizes. Not all point sizes in the range defined by `pointSizeRange` are supported. This limit specifies the granularity (or increment) between successive supported point sizes.
- `lineWidthGranularity` is the granularity of supported line widths. Not all line widths in the range defined by `lineWidthRange` are supported. This limit specifies the granularity (or increment) between successive supported line widths.
- `strictLines` specifies whether lines are rasterized according to the preferred method of rasterization. If set to `VK_FALSE`, lines **may** be rasterized under a relaxed set of rules. If set to `VK_TRUE`, lines are rasterized as per the strict definition. See [Basic Line Segment Rasterization](#).
- `standardSampleLocations` specifies whether rasterization uses the standard sample locations as documented in [Multisampling](#). If set to `VK_TRUE`, the implementation uses the documented sample locations. If set to `VK_FALSE`, the implementation **may** use different sample locations.
- `optimalBufferCopyOffsetAlignment` is the optimal buffer offset alignment in bytes for `vkCmdCopyBufferToImage` and `vkCmdCopyImageToBuffer`. The per texel alignment requirements are enforced, but applications **should** use the optimal alignment for optimal performance and power use. The value **must** be a power of two.
- `optimalBufferCopyRowPitchAlignment` is the optimal buffer row pitch alignment in bytes for `vkCmdCopyBufferToImage` and `vkCmdCopyImageToBuffer`. Row pitch is the number of bytes between texels with the same X coordinate in adjacent rows (Y coordinates differ by one). The per texel alignment requirements are enforced, but applications **should** use the optimal alignment for optimal performance and power use. The value **must** be a power of two.
- `nonCoherentAtomSize` is the size and alignment in bytes that bounds concurrent access to [host-mapped device memory](#). The value **must** be a power of two.

1

For all bitmasks of `VkSampleCountFlagBits`, the sample count limits defined above represent the minimum supported sample counts for each image type. Individual images **may** support additional sample counts, which are queried using `vkGetPhysicalDeviceImageFormatProperties` as described in [Supported Sample Counts](#).

Bits which **may** be set in the sample count limits returned by `VkPhysicalDeviceLimits`, as well as in other queries and structures representing image sample counts, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSampleCountFlagBits {
    VK_SAMPLE_COUNT_1_BIT = 0x00000001,
    VK_SAMPLE_COUNT_2_BIT = 0x00000002,
    VK_SAMPLE_COUNT_4_BIT = 0x00000004,
    VK_SAMPLE_COUNT_8_BIT = 0x00000008,
    VK_SAMPLE_COUNT_16_BIT = 0x00000010,
    VK_SAMPLE_COUNT_32_BIT = 0x00000020,
    VK_SAMPLE_COUNT_64_BIT = 0x00000040,
} VkSampleCountFlagBits;
```

- `VK_SAMPLE_COUNT_1_BIT` specifies an image with one sample per pixel.
- `VK_SAMPLE_COUNT_2_BIT` specifies an image with 2 samples per pixel.
- `VK_SAMPLE_COUNT_4_BIT` specifies an image with 4 samples per pixel.
- `VK_SAMPLE_COUNT_8_BIT` specifies an image with 8 samples per pixel.
- `VK_SAMPLE_COUNT_16_BIT` specifies an image with 16 samples per pixel.
- `VK_SAMPLE_COUNT_32_BIT` specifies an image with 32 samples per pixel.
- `VK_SAMPLE_COUNT_64_BIT` specifies an image with 64 samples per pixel.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkSampleCountFlags;
```

`VkSampleCountFlags` is a bitmask type for setting a mask of zero or more `VkSampleCountFlagBits`.

32.1. Limit Requirements

The following table specifies the **required** minimum/maximum for all Vulkan graphics implementations. Where a limit corresponds to a fine-grained device feature which is **optional**, the feature name is listed with two **required** limits, one when the feature is supported and one when it is not supported. If an implementation supports a feature, the limits reported are the same whether or not the feature is enabled.

Table 31. Required Limit Types

Type	Limit	Feature
<code>uint32_t</code>	<code>maxImageDimension1D</code>	-
<code>uint32_t</code>	<code>maxImageDimension2D</code>	-
<code>uint32_t</code>	<code>maxImageDimension3D</code>	-
<code>uint32_t</code>	<code>maxImageDimensionCube</code>	-
<code>uint32_t</code>	<code>maxImageArrayLayers</code>	-
<code>uint32_t</code>	<code>maxTexelBufferElements</code>	-
<code>uint32_t</code>	<code>maxUniformBufferRange</code>	-

Type	Limit	Feature
uint32_t	maxStorageBufferRange	-
uint32_t	maxPushConstantsSize	-
uint32_t	maxMemoryAllocationCount	-
uint32_t	maxSamplerAllocationCount	-
VkDeviceSize	bufferImageGranularity	-
VkDeviceSize	sparseAddressSpaceSize	sparseBinding
uint32_t	maxBoundDescriptorSets	-
uint32_t	maxPerStageDescriptorSamplers	-
uint32_t	maxPerStageDescriptorUniformBuffers	-
uint32_t	maxPerStageDescriptorStorageBuffers	-
uint32_t	maxPerStageDescriptorSampledImages	-
uint32_t	maxPerStageDescriptorStorageImages	-
uint32_t	maxPerStageDescriptorInputAttachments	-
uint32_t	maxPerStageResources	-
uint32_t	maxDescriptorSetSamplers	-
uint32_t	maxDescriptorSetUniformBuffers	-
uint32_t	maxDescriptorSetUniformBuffersDynamic	-
uint32_t	maxDescriptorSetStorageBuffers	-
uint32_t	maxDescriptorSetStorageBuffersDynamic	-
uint32_t	maxDescriptorSetSampledImages	-
uint32_t	maxDescriptorSetStorageImages	-
uint32_t	maxDescriptorSetInputAttachments	-
uint32_t	maxVertexInputAttributes	-
uint32_t	maxVertexInputBindings	-
uint32_t	maxVertexInputAttributeOffset	-
uint32_t	maxVertexInputBindingStride	-
uint32_t	maxVertexOutputComponents	-
uint32_t	maxTessellationGenerationLevel	tessellationShader
uint32_t	maxTessellationPatchSize	tessellationShader
uint32_t	maxTessellationControlPerVertexInputComponents	tessellationShader
uint32_t	maxTessellationControlPerVertexOutputComponents	tessellationShaders
uint32_t	maxTessellationControlPerPatchOutputComponents	tessellationShader
uint32_t	maxTessellationControlTotalOutputComponents	tessellationShader
uint32_t	maxTessellationEvaluationInputComponents	tessellationShader

Type	Limit	Feature
uint32_t	maxTessellationEvaluationOutputComponents	tessellationShader
uint32_t	maxGeometryShaderInvocations	geometryShader
uint32_t	maxGeometryInputComponents	geometryShader
uint32_t	maxGeometryOutputComponents	geometryShader
uint32_t	maxGeometryOutputVertices	geometryShader
uint32_t	maxGeometryTotalOutputComponents	geometryShader
uint32_t	maxFragmentInputComponents	-
uint32_t	maxFragmentOutputAttachments	-
uint32_t	maxFragmentDualSrcAttachments	dualSrcBlend
uint32_t	maxFragmentCombinedOutputResources	-
uint32_t	maxComputeSharedMemorySize	-
3 × uint32_t	maxComputeWorkGroupCount	-
uint32_t	maxComputeWorkGroupInvocations	-
3 × uint32_t	maxComputeWorkGroupSize	-
uint32_t	subPixelPrecisionBits	-
uint32_t	subTexelPrecisionBits	-
uint32_t	mipmapPrecisionBits	-
uint32_t	maxDrawIndexedIndexValue	fullDrawIndexUint32
uint32_t	maxDrawIndirectCount	multiDrawIndirect
float	maxSamplerLodBias	-
float	maxSamplerAnisotropy	samplerAnisotropy
uint32_t	maxViewports	multiViewport
2 × uint32_t	maxViewportDimensions	-
2 × float	viewportBoundsRange	-
uint32_t	viewportSubPixelBits	-
size_t	minMemoryMapAlignment	-
VkDeviceSize	minTexelBufferOffsetAlignment	-
VkDeviceSize	minUniformBufferOffsetAlignment	-
VkDeviceSize	minStorageBufferOffsetAlignment	-
int32_t	minTexelOffset	-
uint32_t	maxTexelOffset	-
int32_t	minTexelGatherOffset	shaderImageGatherExtended
uint32_t	maxTexelGatherOffset	shaderImageGatherExtended
float	minInterpolationOffset	sampleRateShading
float	maxInterpolationOffset	sampleRateShading

Type	Limit	Feature
uint32_t	subPixelInterpolationOffsetBits	sampleRateShading
uint32_t	maxFramebufferWidth	-
uint32_t	maxFramebufferHeight	-
uint32_t	maxFramebufferLayers	-
VkSampleCountFlags	framebufferColorSampleCounts	-
VkSampleCountFlags	framebufferDepthSampleCounts	-
VkSampleCountFlags	framebufferStencilSampleCounts	-
VkSampleCountFlags	framebufferNoAttachmentsSampleCounts	-
uint32_t	maxColorAttachments	-
VkSampleCountFlags	sampledImageColorSampleCounts	-
VkSampleCountFlags	sampledImageIntegerSampleCounts	-
VkSampleCountFlags	sampledImageDepthSampleCounts	-
VkSampleCountFlags	sampledImageStencilSampleCounts	-
VkSampleCountFlags	storageImageSampleCounts	shaderStorageImageMultisample
uint32_t	maxSampleMaskWords	-
VkBool32	timestampComputeAndGraphics	-
float	timestampPeriod	-
uint32_t	maxClipDistances	shaderClipDistance
uint32_t	maxCullDistances	shaderCullDistance
uint32_t	maxCombinedClipAndCullDistances	shaderCullDistance
uint32_t	discreteQueuePriorities	-
2 × float	pointSizeRange	largePoints
2 × float	lineWidthRange	wideLines
float	pointSizeGranularity	largePoints
float	lineWidthGranularity	wideLines
VkBool32	strictLines	-
VkBool32	standardSampleLocations	-
VkDeviceSize	optimalBufferCopyOffsetAlignment	-

Type	Limit	Feature
VkDeviceSize	optimalBufferCopyRowPitchAlignment	-
VkDeviceSize	nonCoherentAtomSize	-

Table 32. Required Limits

Limit	Unsupport ed Limit	Supported Limit	Limit Type ¹
maxImageDimension1D	-	4096	min
maxImageDimension2D	-	4096	min
maxImageDimension3D	-	256	min
maxImageDimensionCube	-	4096	min
maxImageArrayLayers	-	256	min
maxTexelBufferElements	-	65536	min
maxUniformBufferRange	-	16384	min
maxStorageBufferRange	-	2 ²⁷	min
maxPushConstantsSize	-	128	min
maxMemoryAllocationCount	-	4096	min
maxSamplerAllocationCount	-	4000	min
bufferImageGranularity	-	131072	max
sparseAddressSpaceSize	0	2 ³¹	min
maxBoundDescriptorSets	-	4	min
maxPerStageDescriptorSamplers	-	16	min
maxPerStageDescriptorUniformBuffers	-	12	min
maxPerStageDescriptorStorageBuffers	-	4	min
maxPerStageDescriptorSampledImages	-	16	min
maxPerStageDescriptorStorageImages	-	4	min
maxPerStageDescriptorInputAttachments	-	4	min
maxPerStageResources	-	128 ²	min
maxDescriptorSetSamplers	-	96 ⁸	min, $n \times$ PerStage
maxDescriptorSetUniformBuffers	-	72 ⁸	min, $n \times$ PerStage
maxDescriptorSetUniformBuffersDynamic	-	8	min
maxDescriptorSetStorageBuffers	-	24 ⁸	min, $n \times$ PerStage
maxDescriptorSetStorageBuffersDynamic	-	4	min

Limit	Unsupport ed Limit	Supported Limit	Limit Type ¹
maxDescriptorSetSampledImages	-	96 ⁸	min, $n \times$ PerStage
maxDescriptorSetStorageImages	-	24 ⁸	min, $n \times$ PerStage
maxDescriptorSetInputAttachments	-	4	min
maxVertexInputAttributes	-	16	min
maxVertexInputBindings	-	16	min
maxVertexInputAttributeOffset	-	2047	min
maxVertexInputBindingStride	-	2048	min
maxVertexOutputComponents	-	64	min
maxTessellationGenerationLevel	0	64	min
maxTessellationPatchSize	0	32	min
maxTessellationControlPerVertexInputComponents	0	64	min
maxTessellationControlPerVertexOutputComponents	0	64	min
maxTessellationControlPerPatchOutputComponents	0	120	min
maxTessellationControlTotalOutputComponents	0	2048	min
maxTessellationEvaluationInputComponents	0	64	min
maxTessellationEvaluationOutputComponents	0	64	min
maxGeometryShaderInvocations	0	32	min
maxGeometryInputComponents	0	64	min
maxGeometryOutputComponents	0	64	min
maxGeometryOutputVertices	0	256	min
maxGeometryTotalOutputComponents	0	1024	min
maxFragmentInputComponents	-	64	min
maxFragmentOutputAttachments	-	4	min
maxFragmentDualSrcAttachments	0	1	min
maxFragmentCombinedOutputResources	-	4	min
maxComputeSharedMemorySize	-	16384	min
maxComputeWorkGroupCount	-	(65535,65535,65535)	min
maxComputeWorkGroupInvocations	-	128	min
maxComputeWorkGroupSize	-	(128,128,64)	min
subPixelPrecisionBits	-	4	min

Limit	Unsupport ed Limit	Supported Limit	Limit Type ¹
subTexelPrecisionBits	-	4	min
mipmapPrecisionBits	-	4	min
maxDrawIndexedIndexValue	$2^{24}-1$	$2^{32}-1$	min
maxDrawIndirectCount	1	$2^{16}-1$	min
maxSamplerLodBias	-	2	min
maxSamplerAnisotropy	1	16	min
maxViewports	1	16	min
maxViewportDimensions	-	$(4096,4096)^3$	min
viewportBoundsRange	-	$(-8192,8191)^4$	(max,min)
viewportSubPixelBits	-	0	min
minMemoryMapAlignment	-	64	min
minTexelBufferOffsetAlignment	-	256	max
minUniformBufferOffsetAlignment	-	256	max
minStorageBufferOffsetAlignment	-	256	max
minTexelOffset	-	-8	max
maxTexelOffset	-	7	min
minTexelGatherOffset	0	-8	max
maxTexelGatherOffset	0	7	min
minInterpolationOffset	0.0	-0.5^5	max
maxInterpolationOffset	0.0	$0.5 - (1 \text{ ULP})^5$	min
subPixelInterpolationOffsetBits	0	4^5	min
maxFramebufferWidth	-	4096	min
maxFramebufferHeight	-	4096	min
maxFramebufferLayers	-	256	min
framebufferColorSampleCounts	-	(VK_SAMPLE_COUNT_1_BIT VK_SAMPLE_COUNT_4_BIT)	min
framebufferDepthSampleCounts	-	(VK_SAMPLE_COUNT_1_BIT VK_SAMPLE_COUNT_4_BIT)	min

Limit	Unsupport ed Limit	Supported Limit	Limit Type ¹
framebufferStencilSampleCounts	-	(VK_SAMPLE_COUNT_1_BIT VK_SAMPLE_COUNT_4_BIT)	min
framebufferNoAttachmentsSampleCounts	-	(VK_SAMPLE_COUNT_1_BIT VK_SAMPLE_COUNT_4_BIT)	min
maxColorAttachments	-	4	min
sampledImageColorSampleCounts	-	(VK_SAMPLE_COUNT_1_BIT VK_SAMPLE_COUNT_4_BIT)	min
sampledImageIntegerSampleCounts	-	VK_SAMPLE_COUNT_1_BIT	min
sampledImageDepthSampleCounts	-	(VK_SAMPLE_COUNT_1_BIT VK_SAMPLE_COUNT_4_BIT)	min
sampledImageStencilSampleCounts	-	(VK_SAMPLE_COUNT_1_BIT VK_SAMPLE_COUNT_4_BIT)	min
storageImageSampleCounts	VK_SAMPLE_COUNT_1_BIT	(VK_SAMPLE_COUNT_1_BIT VK_SAMPLE_COUNT_4_BIT)	min
maxSampleMaskWords	-	1	min
timestampComputeAndGraphics	-	-	implementation-dependent
timestampPeriod	-	-	duration
maxClipDistances	0	8	min
maxCullDistances	0	8	min
maxCombinedClipAndCullDistances	0	8	min
discreteQueuePriorities	-	2	min
pointSizeRange	(1.0,1.0)	(1.0,64.0 - ULP) ⁶	(max,min)
lineWidthRange	(1.0,1.0)	(1.0,8.0 - ULP) ⁷	(max,min)
pointSizeGranularity	0.0	1.0 ⁶	max, fixed point increment

Limit	Unsupport ed Limit	Supported Limit	Limit Type ¹
lineWidthGranularity	0.0	1.0 ⁷	max, fixed point increment
strictLines	-	-	implementatio n-dependent
standardSampleLocations	-	-	implementatio n-dependent
optimalBufferCopyOffsetAlignment	-	-	recommendati on
optimalBufferCopyRowPitchAlignment	-	-	recommendati on
nonCoherentAtomSize	-	256	max

1

The **Limit Type** column specifies the limit is either the minimum limit all implementations **must** support, the maximum limit all implementations **must** support, or the exact value all implementations **must** support. For bitmasks a minimum limit is the least bits all implementations **must** set, but they **may** have additional bits set beyond this minimum.

2

The `maxPerStageResources` **must** be at least the smallest of the following:

- the sum of the `maxPerStageDescriptorUniformBuffers`, `maxPerStageDescriptorStorageBuffers`, `maxPerStageDescriptorSampledImages`, `maxPerStageDescriptorStorageImages`, `maxPerStageDescriptorInputAttachments`, `maxColorAttachments` limits, or
- 128.

It **may** not be possible to reach this limit in every stage.

3

See `maxViewportDimensions` for the **required** relationship to other limits.

4

See `viewportBoundsRange` for the **required** relationship to other limits.

5

The values `minInterpolationOffset` and `maxInterpolationOffset` describe the closed interval of supported interpolation offsets: `[minInterpolationOffset, maxInterpolationOffset]`. The ULP is determined by `subPixelInterpolationOffsetBits`. If `subPixelInterpolationOffsetBits` is 4, this provides increments of $(1/2^4) = 0.0625$, and thus the range of supported interpolation offsets would be `[-0.5, 0.4375]`.

6

The point size ULP is determined by `pointSizeGranularity`. If the `pointSizeGranularity` is 0.125, the range of supported point sizes **must** be at least [1.0, 63.875].

7

The line width ULP is determined by `lineWidthGranularity`. If the `lineWidthGranularity` is 0.0625, the range of supported line widths **must** be at least [1.0, 7.9375].

8

The minimum `maxDescriptorSet*` limit is n times the corresponding *specification* minimum `maxPerStageDescriptor*` limit, where n is the number of shader stages supported by the `VkPhysicalDevice`. If all shader stages are supported, $n = 6$ (vertex, tessellation control, tessellation evaluation, geometry, fragment, compute).

Chapter 33. Formats

Supported buffer and image formats **may** vary across implementations. A minimum set of format features are guaranteed, but others **must** be explicitly queried before use to ensure they are supported by the implementation.

The features for the set of formats ([VkFormat](#)) supported by the implementation are queried individually using the [vkGetPhysicalDeviceFormatProperties](#) command.

33.1. Format Definition

The following image formats **can** be passed to, and **may** be returned from Vulkan commands. The memory required to store each format is discussed with that format, and also summarized in the [Representation and Texel Block Size](#) section and the [Compatible formats](#) table.

```
// Provided by VK_VERSION_1_0
typedef enum VkFormat {
    VK_FORMAT_UNDEFINED = 0,
    VK_FORMAT_R4G4_UNORM_PACK8 = 1,
    VK_FORMAT_R4G4B4A4_UNORM_PACK16 = 2,
    VK_FORMAT_B4G4R4A4_UNORM_PACK16 = 3,
    VK_FORMAT_R5G6B5_UNORM_PACK16 = 4,
    VK_FORMAT_B5G6R5_UNORM_PACK16 = 5,
    VK_FORMAT_R5G5B5A1_UNORM_PACK16 = 6,
    VK_FORMAT_B5G5R5A1_UNORM_PACK16 = 7,
    VK_FORMAT_A1R5G5B5_UNORM_PACK16 = 8,
    VK_FORMAT_R8_UNORM = 9,
    VK_FORMAT_R8_SNORM = 10,
    VK_FORMAT_R8_USCALED = 11,
    VK_FORMAT_R8_SSCALED = 12,
    VK_FORMAT_R8_UINT = 13,
    VK_FORMAT_R8_SINT = 14,
    VK_FORMAT_R8_SRGB = 15,
    VK_FORMAT_R8G8_UNORM = 16,
    VK_FORMAT_R8G8_SNORM = 17,
    VK_FORMAT_R8G8_USCALED = 18,
    VK_FORMAT_R8G8_SSCALED = 19,
    VK_FORMAT_R8G8_UINT = 20,
    VK_FORMAT_R8G8_SINT = 21,
    VK_FORMAT_R8G8_SRGB = 22,
    VK_FORMAT_R8G8B8_UNORM = 23,
    VK_FORMAT_R8G8B8_SNORM = 24,
    VK_FORMAT_R8G8B8_USCALED = 25,
    VK_FORMAT_R8G8B8_SSCALED = 26,
    VK_FORMAT_R8G8B8_UINT = 27,
    VK_FORMAT_R8G8B8_SINT = 28,
    VK_FORMAT_R8G8B8_SRGB = 29,
    VK_FORMAT_B8G8R8_UNORM = 30,
    VK_FORMAT_B8G8R8_SNORM = 31,
```

```
VK_FORMAT_B8G8R8_USCALED = 32,  
VK_FORMAT_B8G8R8_SSCALED = 33,  
VK_FORMAT_B8G8R8_UINT = 34,  
VK_FORMAT_B8G8R8_SINT = 35,  
VK_FORMAT_B8G8R8_SRGB = 36,  
VK_FORMAT_R8G8B8A8_UNORM = 37,  
VK_FORMAT_R8G8B8A8_SNORM = 38,  
VK_FORMAT_R8G8B8A8_USCALED = 39,  
VK_FORMAT_R8G8B8A8_SSCALED = 40,  
VK_FORMAT_R8G8B8A8_UINT = 41,  
VK_FORMAT_R8G8B8A8_SINT = 42,  
VK_FORMAT_R8G8B8A8_SRGB = 43,  
VK_FORMAT_B8G8R8A8_UNORM = 44,  
VK_FORMAT_B8G8R8A8_SNORM = 45,  
VK_FORMAT_B8G8R8A8_USCALED = 46,  
VK_FORMAT_B8G8R8A8_SSCALED = 47,  
VK_FORMAT_B8G8R8A8_UINT = 48,  
VK_FORMAT_B8G8R8A8_SINT = 49,  
VK_FORMAT_B8G8R8A8_SRGB = 50,  
VK_FORMAT_A8B8G8R8_UNORM_PACK32 = 51,  
VK_FORMAT_A8B8G8R8_SNORM_PACK32 = 52,  
VK_FORMAT_A8B8G8R8_USCALED_PACK32 = 53,  
VK_FORMAT_A8B8G8R8_SSCALED_PACK32 = 54,  
VK_FORMAT_A8B8G8R8_UINT_PACK32 = 55,  
VK_FORMAT_A8B8G8R8_SINT_PACK32 = 56,  
VK_FORMAT_A8B8G8R8_SRGB_PACK32 = 57,  
VK_FORMAT_A2R10G10B10_UNORM_PACK32 = 58,  
VK_FORMAT_A2R10G10B10_SNORM_PACK32 = 59,  
VK_FORMAT_A2R10G10B10_USCALED_PACK32 = 60,  
VK_FORMAT_A2R10G10B10_SSCALED_PACK32 = 61,  
VK_FORMAT_A2R10G10B10_UINT_PACK32 = 62,  
VK_FORMAT_A2R10G10B10_SINT_PACK32 = 63,  
VK_FORMAT_A2B10G10R10_UNORM_PACK32 = 64,  
VK_FORMAT_A2B10G10R10_SNORM_PACK32 = 65,  
VK_FORMAT_A2B10G10R10_USCALED_PACK32 = 66,  
VK_FORMAT_A2B10G10R10_SSCALED_PACK32 = 67,  
VK_FORMAT_A2B10G10R10_UINT_PACK32 = 68,  
VK_FORMAT_A2B10G10R10_SINT_PACK32 = 69,  
VK_FORMAT_R16_UNORM = 70,  
VK_FORMAT_R16_SNORM = 71,  
VK_FORMAT_R16_USCALED = 72,  
VK_FORMAT_R16_SSCALED = 73,  
VK_FORMAT_R16_UINT = 74,  
VK_FORMAT_R16_SINT = 75,  
VK_FORMAT_R16_SFLOAT = 76,  
VK_FORMAT_R16G16_UNORM = 77,  
VK_FORMAT_R16G16_SNORM = 78,  
VK_FORMAT_R16G16_USCALED = 79,  
VK_FORMAT_R16G16_SSCALED = 80,  
VK_FORMAT_R16G16_UINT = 81,  
VK_FORMAT_R16G16_SINT = 82,
```

```
VK_FORMAT_R16G16_SFLOAT = 83,  
VK_FORMAT_R16G16B16_UNORM = 84,  
VK_FORMAT_R16G16B16_SNORM = 85,  
VK_FORMAT_R16G16B16_USCALED = 86,  
VK_FORMAT_R16G16B16_SSCALED = 87,  
VK_FORMAT_R16G16B16_UINT = 88,  
VK_FORMAT_R16G16B16_SINT = 89,  
VK_FORMAT_R16G16B16_SFLOAT = 90,  
VK_FORMAT_R16G16B16A16_UNORM = 91,  
VK_FORMAT_R16G16B16A16_SNORM = 92,  
VK_FORMAT_R16G16B16A16_USCALED = 93,  
VK_FORMAT_R16G16B16A16_SSCALED = 94,  
VK_FORMAT_R16G16B16A16_UINT = 95,  
VK_FORMAT_R16G16B16A16_SINT = 96,  
VK_FORMAT_R16G16B16A16_SFLOAT = 97,  
VK_FORMAT_R32_UINT = 98,  
VK_FORMAT_R32_SINT = 99,  
VK_FORMAT_R32_SFLOAT = 100,  
VK_FORMAT_R32G32_UINT = 101,  
VK_FORMAT_R32G32_SINT = 102,  
VK_FORMAT_R32G32_SFLOAT = 103,  
VK_FORMAT_R32G32B32_UINT = 104,  
VK_FORMAT_R32G32B32_SINT = 105,  
VK_FORMAT_R32G32B32_SFLOAT = 106,  
VK_FORMAT_R32G32B32A32_UINT = 107,  
VK_FORMAT_R32G32B32A32_SINT = 108,  
VK_FORMAT_R32G32B32A32_SFLOAT = 109,  
VK_FORMAT_R64_UINT = 110,  
VK_FORMAT_R64_SINT = 111,  
VK_FORMAT_R64_SFLOAT = 112,  
VK_FORMAT_R64G64_UINT = 113,  
VK_FORMAT_R64G64_SINT = 114,  
VK_FORMAT_R64G64_SFLOAT = 115,  
VK_FORMAT_R64G64B64_UINT = 116,  
VK_FORMAT_R64G64B64_SINT = 117,  
VK_FORMAT_R64G64B64_SFLOAT = 118,  
VK_FORMAT_R64G64B64A64_UINT = 119,  
VK_FORMAT_R64G64B64A64_SINT = 120,  
VK_FORMAT_R64G64B64A64_SFLOAT = 121,  
VK_FORMAT_B10G11R11_UFLOAT_PACK32 = 122,  
VK_FORMAT_E5B9G9R9_UFLOAT_PACK32 = 123,  
VK_FORMAT_D16_UNORM = 124,  
VK_FORMAT_X8_D24_UNORM_PACK32 = 125,  
VK_FORMAT_D32_SFLOAT = 126,  
VK_FORMAT_S8_UINT = 127,  
VK_FORMAT_D16_UNORM_S8_UINT = 128,  
VK_FORMAT_D24_UNORM_S8_UINT = 129,  
VK_FORMAT_D32_SFLOAT_S8_UINT = 130,  
VK_FORMAT_BC1_RGB_UNORM_BLOCK = 131,  
VK_FORMAT_BC1_RGB_SRGB_BLOCK = 132,  
VK_FORMAT_BC1_RGBA_UNORM_BLOCK = 133,
```

```
VK_FORMAT_BC1_RGBA_SRGB_BLOCK = 134,  
VK_FORMAT_BC2_UNORM_BLOCK = 135,  
VK_FORMAT_BC2_SRGB_BLOCK = 136,  
VK_FORMAT_BC3_UNORM_BLOCK = 137,  
VK_FORMAT_BC3_SRGB_BLOCK = 138,  
VK_FORMAT_BC4_UNORM_BLOCK = 139,  
VK_FORMAT_BC4_SNORM_BLOCK = 140,  
VK_FORMAT_BC5_UNORM_BLOCK = 141,  
VK_FORMAT_BC5_SNORM_BLOCK = 142,  
VK_FORMAT_BC6H_UFLOAT_BLOCK = 143,  
VK_FORMAT_BC6H_SFLOAT_BLOCK = 144,  
VK_FORMAT_BC7_UNORM_BLOCK = 145,  
VK_FORMAT_BC7_SRGB_BLOCK = 146,  
VK_FORMAT_ETC2_R8G8B8_UNORM_BLOCK = 147,  
VK_FORMAT_ETC2_R8G8B8_SRGB_BLOCK = 148,  
VK_FORMAT_ETC2_R8G8B8A1_UNORM_BLOCK = 149,  
VK_FORMAT_ETC2_R8G8B8A1_SRGB_BLOCK = 150,  
VK_FORMAT_ETC2_R8G8B8A8_UNORM_BLOCK = 151,  
VK_FORMAT_ETC2_R8G8B8A8_SRGB_BLOCK = 152,  
VK_FORMAT_EAC_R11_UNORM_BLOCK = 153,  
VK_FORMAT_EAC_R11_SNORM_BLOCK = 154,  
VK_FORMAT_EAC_R11G11_UNORM_BLOCK = 155,  
VK_FORMAT_EAC_R11G11_SNORM_BLOCK = 156,  
VK_FORMAT_ASTC_4x4_UNORM_BLOCK = 157,  
VK_FORMAT_ASTC_4x4_SRGB_BLOCK = 158,  
VK_FORMAT_ASTC_5x4_UNORM_BLOCK = 159,  
VK_FORMAT_ASTC_5x4_SRGB_BLOCK = 160,  
VK_FORMAT_ASTC_5x5_UNORM_BLOCK = 161,  
VK_FORMAT_ASTC_5x5_SRGB_BLOCK = 162,  
VK_FORMAT_ASTC_6x5_UNORM_BLOCK = 163,  
VK_FORMAT_ASTC_6x5_SRGB_BLOCK = 164,  
VK_FORMAT_ASTC_6x6_UNORM_BLOCK = 165,  
VK_FORMAT_ASTC_6x6_SRGB_BLOCK = 166,  
VK_FORMAT_ASTC_8x5_UNORM_BLOCK = 167,  
VK_FORMAT_ASTC_8x5_SRGB_BLOCK = 168,  
VK_FORMAT_ASTC_8x6_UNORM_BLOCK = 169,  
VK_FORMAT_ASTC_8x6_SRGB_BLOCK = 170,  
VK_FORMAT_ASTC_8x8_UNORM_BLOCK = 171,  
VK_FORMAT_ASTC_8x8_SRGB_BLOCK = 172,  
VK_FORMAT_ASTC_10x5_UNORM_BLOCK = 173,  
VK_FORMAT_ASTC_10x5_SRGB_BLOCK = 174,  
VK_FORMAT_ASTC_10x6_UNORM_BLOCK = 175,  
VK_FORMAT_ASTC_10x6_SRGB_BLOCK = 176,  
VK_FORMAT_ASTC_10x8_UNORM_BLOCK = 177,  
VK_FORMAT_ASTC_10x8_SRGB_BLOCK = 178,  
VK_FORMAT_ASTC_10x10_UNORM_BLOCK = 179,  
VK_FORMAT_ASTC_10x10_SRGB_BLOCK = 180,  
VK_FORMAT_ASTC_12x10_UNORM_BLOCK = 181,  
VK_FORMAT_ASTC_12x10_SRGB_BLOCK = 182,  
VK_FORMAT_ASTC_12x12_UNORM_BLOCK = 183,  
VK_FORMAT_ASTC_12x12_SRGB_BLOCK = 184,
```



```
} VkFormat;
```

- **VK_FORMAT_UNDEFINED** specifies that the format is not specified.
- **VK_FORMAT_R4G4_UNORM_PACK8** specifies a two-component, 8-bit packed unsigned normalized format that has a 4-bit R component in bits 4..7, and a 4-bit G component in bits 0..3.
- **VK_FORMAT_R4G4B4A4_UNORM_PACK16** specifies a four-component, 16-bit packed unsigned normalized format that has a 4-bit R component in bits 12..15, a 4-bit G component in bits 8..11, a 4-bit B component in bits 4..7, and a 4-bit A component in bits 0..3.
- **VK_FORMAT_B4G4R4A4_UNORM_PACK16** specifies a four-component, 16-bit packed unsigned normalized format that has a 4-bit B component in bits 12..15, a 4-bit G component in bits 8..11, a 4-bit R component in bits 4..7, and a 4-bit A component in bits 0..3.
- **VK_FORMAT_R5G6B5_UNORM_PACK16** specifies a three-component, 16-bit packed unsigned normalized format that has a 5-bit R component in bits 11..15, a 6-bit G component in bits 5..10, and a 5-bit B component in bits 0..4.
- **VK_FORMAT_B5G6R5_UNORM_PACK16** specifies a three-component, 16-bit packed unsigned normalized format that has a 5-bit B component in bits 11..15, a 6-bit G component in bits 5..10, and a 5-bit R component in bits 0..4.
- **VK_FORMAT_R5G5B5A1_UNORM_PACK16** specifies a four-component, 16-bit packed unsigned normalized format that has a 5-bit R component in bits 11..15, a 5-bit G component in bits 6..10, a 5-bit B component in bits 1..5, and a 1-bit A component in bit 0.
- **VK_FORMAT_B5G5R5A1_UNORM_PACK16** specifies a four-component, 16-bit packed unsigned normalized format that has a 5-bit B component in bits 11..15, a 5-bit G component in bits 6..10, a 5-bit R component in bits 1..5, and a 1-bit A component in bit 0.
- **VK_FORMAT_A1R5G5B5_UNORM_PACK16** specifies a four-component, 16-bit packed unsigned normalized format that has a 1-bit A component in bit 15, a 5-bit R component in bits 10..14, a 5-bit G component in bits 5..9, and a 5-bit B component in bits 0..4.
- **VK_FORMAT_R8_UNORM** specifies a one-component, 8-bit unsigned normalized format that has a single 8-bit R component.
- **VK_FORMAT_R8_SNORM** specifies a one-component, 8-bit signed normalized format that has a single 8-bit R component.
- **VK_FORMAT_R8_USCALED** specifies a one-component, 8-bit unsigned scaled integer format that has a single 8-bit R component.
- **VK_FORMAT_R8_SSCALED** specifies a one-component, 8-bit signed scaled integer format that has a single 8-bit R component.
- **VK_FORMAT_R8_UINT** specifies a one-component, 8-bit unsigned integer format that has a single 8-bit R component.
- **VK_FORMAT_R8_SINT** specifies a one-component, 8-bit signed integer format that has a single 8-bit R component.
- **VK_FORMAT_R8_SRGB** specifies a one-component, 8-bit unsigned normalized format that has a single 8-bit R component stored with sRGB nonlinear encoding.
- **VK_FORMAT_R8G8_UNORM** specifies a two-component, 16-bit unsigned normalized format that has an

8-bit R component in byte 0, and an 8-bit G component in byte 1.

- **VK_FORMAT_R8G8_SNORM** specifies a two-component, 16-bit signed normalized format that has an 8-bit R component in byte 0, and an 8-bit G component in byte 1.
- **VK_FORMAT_R8G8_USCALED** specifies a two-component, 16-bit unsigned scaled integer format that has an 8-bit R component in byte 0, and an 8-bit G component in byte 1.
- **VK_FORMAT_R8G8_SSCALED** specifies a two-component, 16-bit signed scaled integer format that has an 8-bit R component in byte 0, and an 8-bit G component in byte 1.
- **VK_FORMAT_R8G8_UINT** specifies a two-component, 16-bit unsigned integer format that has an 8-bit R component in byte 0, and an 8-bit G component in byte 1.
- **VK_FORMAT_R8G8_SINT** specifies a two-component, 16-bit signed integer format that has an 8-bit R component in byte 0, and an 8-bit G component in byte 1.
- **VK_FORMAT_R8G8_SRGB** specifies a two-component, 16-bit unsigned normalized format that has an 8-bit R component stored with sRGB nonlinear encoding in byte 0, and an 8-bit G component stored with sRGB nonlinear encoding in byte 1.
- **VK_FORMAT_R8G8B8_UNORM** specifies a three-component, 24-bit unsigned normalized format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, and an 8-bit B component in byte 2.
- **VK_FORMAT_R8G8B8_SNORM** specifies a three-component, 24-bit signed normalized format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, and an 8-bit B component in byte 2.
- **VK_FORMAT_R8G8B8_USCALED** specifies a three-component, 24-bit unsigned scaled format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, and an 8-bit B component in byte 2.
- **VK_FORMAT_R8G8B8_SSCALED** specifies a three-component, 24-bit signed scaled format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, and an 8-bit B component in byte 2.
- **VK_FORMAT_R8G8B8_UINT** specifies a three-component, 24-bit unsigned integer format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, and an 8-bit B component in byte 2.
- **VK_FORMAT_R8G8B8_SINT** specifies a three-component, 24-bit signed integer format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, and an 8-bit B component in byte 2.
- **VK_FORMAT_R8G8B8_SRGB** specifies a three-component, 24-bit unsigned normalized format that has an 8-bit R component stored with sRGB nonlinear encoding in byte 0, an 8-bit G component stored with sRGB nonlinear encoding in byte 1, and an 8-bit B component stored with sRGB nonlinear encoding in byte 2.
- **VK_FORMAT_B8G8R8_UNORM** specifies a three-component, 24-bit unsigned normalized format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, and an 8-bit R component in byte 2.
- **VK_FORMAT_B8G8R8_SNORM** specifies a three-component, 24-bit signed normalized format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, and an 8-bit R component in byte 2.
- **VK_FORMAT_B8G8R8_USCALED** specifies a three-component, 24-bit unsigned scaled format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, and an 8-bit R component in byte 2.
- **VK_FORMAT_B8G8R8_SSCALED** specifies a three-component, 24-bit signed scaled format that has an 8-

bit B component in byte 0, an 8-bit G component in byte 1, and an 8-bit R component in byte 2.

- **VK_FORMAT_B8G8R8_UINT** specifies a three-component, 24-bit unsigned integer format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, and an 8-bit R component in byte 2.
- **VK_FORMAT_B8G8R8_SINT** specifies a three-component, 24-bit signed integer format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, and an 8-bit R component in byte 2.
- **VK_FORMAT_B8G8R8_SRGB** specifies a three-component, 24-bit unsigned normalized format that has an 8-bit B component stored with sRGB nonlinear encoding in byte 0, an 8-bit G component stored with sRGB nonlinear encoding in byte 1, and an 8-bit R component stored with sRGB nonlinear encoding in byte 2.
- **VK_FORMAT_R8G8B8A8_UNORM** specifies a four-component, 32-bit unsigned normalized format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, an 8-bit B component in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_R8G8B8A8_SNORM** specifies a four-component, 32-bit signed normalized format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, an 8-bit B component in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_R8G8B8A8_USCALED** specifies a four-component, 32-bit unsigned scaled format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, an 8-bit B component in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_R8G8B8A8_SSCALED** specifies a four-component, 32-bit signed scaled format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, an 8-bit B component in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_R8G8B8A8_UINT** specifies a four-component, 32-bit unsigned integer format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, an 8-bit B component in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_R8G8B8A8_SINT** specifies a four-component, 32-bit signed integer format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, an 8-bit B component in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_R8G8B8A8_SRGB** specifies a four-component, 32-bit unsigned normalized format that has an 8-bit R component stored with sRGB nonlinear encoding in byte 0, an 8-bit G component stored with sRGB nonlinear encoding in byte 1, an 8-bit B component stored with sRGB nonlinear encoding in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_B8G8R8A8_UNORM** specifies a four-component, 32-bit unsigned normalized format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, an 8-bit R component in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_B8G8R8A8_SNORM** specifies a four-component, 32-bit signed normalized format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, an 8-bit R component in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_B8G8R8A8_USCALED** specifies a four-component, 32-bit unsigned scaled format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, an 8-bit R component in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_B8G8R8A8_SSCALED** specifies a four-component, 32-bit signed scaled format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, an 8-bit R component in byte 2, and

an 8-bit A component in byte 3.

- **VK_FORMAT_B8G8R8A8_UINT** specifies a four-component, 32-bit unsigned integer format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, an 8-bit R component in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_B8G8R8A8_SINT** specifies a four-component, 32-bit signed integer format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, an 8-bit R component in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_B8G8R8A8_SRGB** specifies a four-component, 32-bit unsigned normalized format that has an 8-bit B component stored with sRGB nonlinear encoding in byte 0, an 8-bit G component stored with sRGB nonlinear encoding in byte 1, an 8-bit R component stored with sRGB nonlinear encoding in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_A8B8G8R8_UNORM_PACK32** specifies a four-component, 32-bit packed unsigned normalized format that has an 8-bit A component in bits 24..31, an 8-bit B component in bits 16..23, an 8-bit G component in bits 8..15, and an 8-bit R component in bits 0..7.
- **VK_FORMAT_A8B8G8R8_SNORM_PACK32** specifies a four-component, 32-bit packed signed normalized format that has an 8-bit A component in bits 24..31, an 8-bit B component in bits 16..23, an 8-bit G component in bits 8..15, and an 8-bit R component in bits 0..7.
- **VK_FORMAT_A8B8G8R8_USCALED_PACK32** specifies a four-component, 32-bit packed unsigned scaled integer format that has an 8-bit A component in bits 24..31, an 8-bit B component in bits 16..23, an 8-bit G component in bits 8..15, and an 8-bit R component in bits 0..7.
- **VK_FORMAT_A8B8G8R8_SSCALED_PACK32** specifies a four-component, 32-bit packed signed scaled integer format that has an 8-bit A component in bits 24..31, an 8-bit B component in bits 16..23, an 8-bit G component in bits 8..15, and an 8-bit R component in bits 0..7.
- **VK_FORMAT_A8B8G8R8_UINT_PACK32** specifies a four-component, 32-bit packed unsigned integer format that has an 8-bit A component in bits 24..31, an 8-bit B component in bits 16..23, an 8-bit G component in bits 8..15, and an 8-bit R component in bits 0..7.
- **VK_FORMAT_A8B8G8R8_SINT_PACK32** specifies a four-component, 32-bit packed signed integer format that has an 8-bit A component in bits 24..31, an 8-bit B component in bits 16..23, an 8-bit G component in bits 8..15, and an 8-bit R component in bits 0..7.
- **VK_FORMAT_A8B8G8R8_SRGB_PACK32** specifies a four-component, 32-bit packed unsigned normalized format that has an 8-bit A component in bits 24..31, an 8-bit B component stored with sRGB nonlinear encoding in bits 16..23, an 8-bit G component stored with sRGB nonlinear encoding in bits 8..15, and an 8-bit R component stored with sRGB nonlinear encoding in bits 0..7.
- **VK_FORMAT_A2R10G10B10_UNORM_PACK32** specifies a four-component, 32-bit packed unsigned normalized format that has a 2-bit A component in bits 30..31, a 10-bit R component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit B component in bits 0..9.
- **VK_FORMAT_A2R10G10B10_SNORM_PACK32** specifies a four-component, 32-bit packed signed normalized format that has a 2-bit A component in bits 30..31, a 10-bit R component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit B component in bits 0..9.
- **VK_FORMAT_A2R10G10B10_USCALED_PACK32** specifies a four-component, 32-bit packed unsigned scaled integer format that has a 2-bit A component in bits 30..31, a 10-bit R component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit B component in bits 0..9.

- `VK_FORMAT_A2R10G10B10_SSCALED_PACK32` specifies a four-component, 32-bit packed signed scaled integer format that has a 2-bit A component in bits 30..31, a 10-bit R component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit B component in bits 0..9.
- `VK_FORMAT_A2R10G10B10_UINT_PACK32` specifies a four-component, 32-bit packed unsigned integer format that has a 2-bit A component in bits 30..31, a 10-bit R component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit B component in bits 0..9.
- `VK_FORMAT_A2R10G10B10_SINT_PACK32` specifies a four-component, 32-bit packed signed integer format that has a 2-bit A component in bits 30..31, a 10-bit R component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit B component in bits 0..9.
- `VK_FORMAT_A2B10G10R10_UNORM_PACK32` specifies a four-component, 32-bit packed unsigned normalized format that has a 2-bit A component in bits 30..31, a 10-bit B component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit R component in bits 0..9.
- `VK_FORMAT_A2B10G10R10_SNORM_PACK32` specifies a four-component, 32-bit packed signed normalized format that has a 2-bit A component in bits 30..31, a 10-bit B component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit R component in bits 0..9.
- `VK_FORMAT_A2B10G10R10_USCALED_PACK32` specifies a four-component, 32-bit packed unsigned scaled integer format that has a 2-bit A component in bits 30..31, a 10-bit B component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit R component in bits 0..9.
- `VK_FORMAT_A2B10G10R10_SSCALED_PACK32` specifies a four-component, 32-bit packed signed scaled integer format that has a 2-bit A component in bits 30..31, a 10-bit B component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit R component in bits 0..9.
- `VK_FORMAT_A2B10G10R10_UINT_PACK32` specifies a four-component, 32-bit packed unsigned integer format that has a 2-bit A component in bits 30..31, a 10-bit B component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit R component in bits 0..9.
- `VK_FORMAT_A2B10G10R10_SINT_PACK32` specifies a four-component, 32-bit packed signed integer format that has a 2-bit A component in bits 30..31, a 10-bit B component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit R component in bits 0..9.
- `VK_FORMAT_R16_UNORM` specifies a one-component, 16-bit unsigned normalized format that has a single 16-bit R component.
- `VK_FORMAT_R16_SNORM` specifies a one-component, 16-bit signed normalized format that has a single 16-bit R component.
- `VK_FORMAT_R16_USCALED` specifies a one-component, 16-bit unsigned scaled integer format that has a single 16-bit R component.
- `VK_FORMAT_R16_SSCALED` specifies a one-component, 16-bit signed scaled integer format that has a single 16-bit R component.
- `VK_FORMAT_R16_UINT` specifies a one-component, 16-bit unsigned integer format that has a single 16-bit R component.
- `VK_FORMAT_R16_SINT` specifies a one-component, 16-bit signed integer format that has a single 16-bit R component.
- `VK_FORMAT_R16_SFLOAT` specifies a one-component, 16-bit signed floating-point format that has a single 16-bit R component.

- **VK_FORMAT_R16G16_UNORM** specifies a two-component, 32-bit unsigned normalized format that has a 16-bit R component in bytes 0..1, and a 16-bit G component in bytes 2..3.
- **VK_FORMAT_R16G16_SNORM** specifies a two-component, 32-bit signed normalized format that has a 16-bit R component in bytes 0..1, and a 16-bit G component in bytes 2..3.
- **VK_FORMAT_R16G16_USCALED** specifies a two-component, 32-bit unsigned scaled integer format that has a 16-bit R component in bytes 0..1, and a 16-bit G component in bytes 2..3.
- **VK_FORMAT_R16G16_SSCALED** specifies a two-component, 32-bit signed scaled integer format that has a 16-bit R component in bytes 0..1, and a 16-bit G component in bytes 2..3.
- **VK_FORMAT_R16G16_UINT** specifies a two-component, 32-bit unsigned integer format that has a 16-bit R component in bytes 0..1, and a 16-bit G component in bytes 2..3.
- **VK_FORMAT_R16G16_SINT** specifies a two-component, 32-bit signed integer format that has a 16-bit R component in bytes 0..1, and a 16-bit G component in bytes 2..3.
- **VK_FORMAT_R16G16_SFLOAT** specifies a two-component, 32-bit signed floating-point format that has a 16-bit R component in bytes 0..1, and a 16-bit G component in bytes 2..3.
- **VK_FORMAT_R16G16B16_UNORM** specifies a three-component, 48-bit unsigned normalized format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, and a 16-bit B component in bytes 4..5.
- **VK_FORMAT_R16G16B16_SNORM** specifies a three-component, 48-bit signed normalized format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, and a 16-bit B component in bytes 4..5.
- **VK_FORMAT_R16G16B16_USCALED** specifies a three-component, 48-bit unsigned scaled integer format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, and a 16-bit B component in bytes 4..5.
- **VK_FORMAT_R16G16B16_SSCALED** specifies a three-component, 48-bit signed scaled integer format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, and a 16-bit B component in bytes 4..5.
- **VK_FORMAT_R16G16B16_UINT** specifies a three-component, 48-bit unsigned integer format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, and a 16-bit B component in bytes 4..5.
- **VK_FORMAT_R16G16B16_SINT** specifies a three-component, 48-bit signed integer format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, and a 16-bit B component in bytes 4..5.
- **VK_FORMAT_R16G16B16_SFLOAT** specifies a three-component, 48-bit signed floating-point format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, and a 16-bit B component in bytes 4..5.
- **VK_FORMAT_R16G16B16A16_UNORM** specifies a four-component, 64-bit unsigned normalized format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, a 16-bit B component in bytes 4..5, and a 16-bit A component in bytes 6..7.
- **VK_FORMAT_R16G16B16A16_SNORM** specifies a four-component, 64-bit signed normalized format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, a 16-bit B component in bytes 4..5, and a 16-bit A component in bytes 6..7.

- **VK_FORMAT_R16G16B16A16_USCALED** specifies a four-component, 64-bit unsigned scaled integer format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, a 16-bit B component in bytes 4..5, and a 16-bit A component in bytes 6..7.
- **VK_FORMAT_R16G16B16A16_SSCALED** specifies a four-component, 64-bit signed scaled integer format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, a 16-bit B component in bytes 4..5, and a 16-bit A component in bytes 6..7.
- **VK_FORMAT_R16G16B16A16_UINT** specifies a four-component, 64-bit unsigned integer format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, a 16-bit B component in bytes 4..5, and a 16-bit A component in bytes 6..7.
- **VK_FORMAT_R16G16B16A16_SINT** specifies a four-component, 64-bit signed integer format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, a 16-bit B component in bytes 4..5, and a 16-bit A component in bytes 6..7.
- **VK_FORMAT_R16G16B16A16_SFLOAT** specifies a four-component, 64-bit signed floating-point format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, a 16-bit B component in bytes 4..5, and a 16-bit A component in bytes 6..7.
- **VK_FORMAT_R32_UINT** specifies a one-component, 32-bit unsigned integer format that has a single 32-bit R component.
- **VK_FORMAT_R32_SINT** specifies a one-component, 32-bit signed integer format that has a single 32-bit R component.
- **VK_FORMAT_R32_SFLOAT** specifies a one-component, 32-bit signed floating-point format that has a single 32-bit R component.
- **VK_FORMAT_R32G32_UINT** specifies a two-component, 64-bit unsigned integer format that has a 32-bit R component in bytes 0..3, and a 32-bit G component in bytes 4..7.
- **VK_FORMAT_R32G32_SINT** specifies a two-component, 64-bit signed integer format that has a 32-bit R component in bytes 0..3, and a 32-bit G component in bytes 4..7.
- **VK_FORMAT_R32G32_SFLOAT** specifies a two-component, 64-bit signed floating-point format that has a 32-bit R component in bytes 0..3, and a 32-bit G component in bytes 4..7.
- **VK_FORMAT_R32G32B32_UINT** specifies a three-component, 96-bit unsigned integer format that has a 32-bit R component in bytes 0..3, a 32-bit G component in bytes 4..7, and a 32-bit B component in bytes 8..11.
- **VK_FORMAT_R32G32B32_SINT** specifies a three-component, 96-bit signed integer format that has a 32-bit R component in bytes 0..3, a 32-bit G component in bytes 4..7, and a 32-bit B component in bytes 8..11.
- **VK_FORMAT_R32G32B32_SFLOAT** specifies a three-component, 96-bit signed floating-point format that has a 32-bit R component in bytes 0..3, a 32-bit G component in bytes 4..7, and a 32-bit B component in bytes 8..11.
- **VK_FORMAT_R32G32B32A32_UINT** specifies a four-component, 128-bit unsigned integer format that has a 32-bit R component in bytes 0..3, a 32-bit G component in bytes 4..7, a 32-bit B component in bytes 8..11, and a 32-bit A component in bytes 12..15.
- **VK_FORMAT_R32G32B32A32_SINT** specifies a four-component, 128-bit signed integer format that has a 32-bit R component in bytes 0..3, a 32-bit G component in bytes 4..7, a 32-bit B component in bytes 8..11, and a 32-bit A component in bytes 12..15.

- **VK_FORMAT_R32G32B32A32_SFLOAT** specifies a four-component, 128-bit signed floating-point format that has a 32-bit R component in bytes 0..3, a 32-bit G component in bytes 4..7, a 32-bit B component in bytes 8..11, and a 32-bit A component in bytes 12..15.
- **VK_FORMAT_R64_UINT** specifies a one-component, 64-bit unsigned integer format that has a single 64-bit R component.
- **VK_FORMAT_R64_SINT** specifies a one-component, 64-bit signed integer format that has a single 64-bit R component.
- **VK_FORMAT_R64_SFLOAT** specifies a one-component, 64-bit signed floating-point format that has a single 64-bit R component.
- **VK_FORMAT_R64G64_UINT** specifies a two-component, 128-bit unsigned integer format that has a 64-bit R component in bytes 0..7, and a 64-bit G component in bytes 8..15.
- **VK_FORMAT_R64G64_SINT** specifies a two-component, 128-bit signed integer format that has a 64-bit R component in bytes 0..7, and a 64-bit G component in bytes 8..15.
- **VK_FORMAT_R64G64_SFLOAT** specifies a two-component, 128-bit signed floating-point format that has a 64-bit R component in bytes 0..7, and a 64-bit G component in bytes 8..15.
- **VK_FORMAT_R64G64B64_UINT** specifies a three-component, 192-bit unsigned integer format that has a 64-bit R component in bytes 0..7, a 64-bit G component in bytes 8..15, and a 64-bit B component in bytes 16..23.
- **VK_FORMAT_R64G64B64_SINT** specifies a three-component, 192-bit signed integer format that has a 64-bit R component in bytes 0..7, a 64-bit G component in bytes 8..15, and a 64-bit B component in bytes 16..23.
- **VK_FORMAT_R64G64B64_SFLOAT** specifies a three-component, 192-bit signed floating-point format that has a 64-bit R component in bytes 0..7, a 64-bit G component in bytes 8..15, and a 64-bit B component in bytes 16..23.
- **VK_FORMAT_R64G64B64A64_UINT** specifies a four-component, 256-bit unsigned integer format that has a 64-bit R component in bytes 0..7, a 64-bit G component in bytes 8..15, a 64-bit B component in bytes 16..23, and a 64-bit A component in bytes 24..31.
- **VK_FORMAT_R64G64B64A64_SINT** specifies a four-component, 256-bit signed integer format that has a 64-bit R component in bytes 0..7, a 64-bit G component in bytes 8..15, a 64-bit B component in bytes 16..23, and a 64-bit A component in bytes 24..31.
- **VK_FORMAT_R64G64B64A64_SFLOAT** specifies a four-component, 256-bit signed floating-point format that has a 64-bit R component in bytes 0..7, a 64-bit G component in bytes 8..15, a 64-bit B component in bytes 16..23, and a 64-bit A component in bytes 24..31.
- **VK_FORMAT_B10G11R11_UFLOAT_PACK32** specifies a three-component, 32-bit packed unsigned floating-point format that has a 10-bit B component in bits 22..31, an 11-bit G component in bits 11..21, an 11-bit R component in bits 0..10. See [Unsigned 10-Bit Floating-Point Numbers](#) and [Unsigned 11-Bit Floating-Point Numbers](#).
- **VK_FORMAT_E5B9G9R9_UFLOAT_PACK32** specifies a three-component, 32-bit packed unsigned floating-point format that has a 5-bit shared exponent in bits 27..31, a 9-bit B component mantissa in bits 18..26, a 9-bit G component mantissa in bits 9..17, and a 9-bit R component mantissa in bits 0..8.
- **VK_FORMAT_D16_UNORM** specifies a one-component, 16-bit unsigned normalized format that has a single 16-bit depth component.

- `VK_FORMAT_X8_D24_UNORM_PACK32` specifies a two-component, 32-bit format that has 24 unsigned normalized bits in the depth component and, **optionally**, 8 bits that are unused.
- `VK_FORMAT_D32_SFLOAT` specifies a one-component, 32-bit signed floating-point format that has 32 bits in the depth component.
- `VK_FORMAT_S8_UINT` specifies a one-component, 8-bit unsigned integer format that has 8 bits in the stencil component.
- `VK_FORMAT_D16_UNORM_S8_UINT` specifies a two-component, 24-bit format that has 16 unsigned normalized bits in the depth component and 8 unsigned integer bits in the stencil component.
- `VK_FORMAT_D24_UNORM_S8_UINT` specifies a two-component, 32-bit packed format that has 8 unsigned integer bits in the stencil component, and 24 unsigned normalized bits in the depth component.
- `VK_FORMAT_D32_SFLOAT_S8_UINT` specifies a two-component format that has 32 signed float bits in the depth component and 8 unsigned integer bits in the stencil component. There are **optionally** 24 bits that are unused.
- `VK_FORMAT_BC1_RGB_UNORM_BLOCK` specifies a three-component, block-compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGB texel data. This format has no alpha and is considered opaque.
- `VK_FORMAT_BC1_RGB_SRGB_BLOCK` specifies a three-component, block-compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGB texel data with sRGB nonlinear encoding. This format has no alpha and is considered opaque.
- `VK_FORMAT_BC1_RGBA_UNORM_BLOCK` specifies a four-component, block-compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGB texel data, and provides 1 bit of alpha.
- `VK_FORMAT_BC1_RGBA_SRGB_BLOCK` specifies a four-component, block-compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGB texel data with sRGB nonlinear encoding, and provides 1 bit of alpha.
- `VK_FORMAT_BC2_UNORM_BLOCK` specifies a four-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data with the first 64 bits encoding alpha values followed by 64 bits encoding RGB values.
- `VK_FORMAT_BC2_SRGB_BLOCK` specifies a four-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data with the first 64 bits encoding alpha values followed by 64 bits encoding RGB values with sRGB nonlinear encoding.
- `VK_FORMAT_BC3_UNORM_BLOCK` specifies a four-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data with the first 64 bits encoding alpha values followed by 64 bits encoding RGB values.
- `VK_FORMAT_BC3_SRGB_BLOCK` specifies a four-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data with the first 64 bits encoding alpha values followed by 64 bits encoding RGB values with sRGB nonlinear encoding.
- `VK_FORMAT_BC4_UNORM_BLOCK` specifies a one-component, block-compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized red texel data.

- **VK_FORMAT_BC4_SNORM_BLOCK** specifies a one-component, block-compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of signed normalized red texel data.
- **VK_FORMAT_BC5_UNORM_BLOCK** specifies a two-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RG texel data with the first 64 bits encoding red values followed by 64 bits encoding green values.
- **VK_FORMAT_BC5_SNORM_BLOCK** specifies a two-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of signed normalized RG texel data with the first 64 bits encoding red values followed by 64 bits encoding green values.
- **VK_FORMAT_BC6H_UFLOAT_BLOCK** specifies a three-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned floating-point RGB texel data.
- **VK_FORMAT_BC6H_SFLOAT_BLOCK** specifies a three-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of signed floating-point RGB texel data.
- **VK_FORMAT_BC7_UNORM_BLOCK** specifies a four-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_BC7_SRGB_BLOCK** specifies a four-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- **VK_FORMAT_ETC2_R8G8B8_UNORM_BLOCK** specifies a three-component, ETC2 compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGB texel data. This format has no alpha and is considered opaque.
- **VK_FORMAT_ETC2_R8G8B8_SRGB_BLOCK** specifies a three-component, ETC2 compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGB texel data with sRGB nonlinear encoding. This format has no alpha and is considered opaque.
- **VK_FORMAT_ETC2_R8G8B8A1_UNORM_BLOCK** specifies a four-component, ETC2 compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGB texel data, and provides 1 bit of alpha.
- **VK_FORMAT_ETC2_R8G8B8A1_SRGB_BLOCK** specifies a four-component, ETC2 compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGB texel data with sRGB nonlinear encoding, and provides 1 bit of alpha.
- **VK_FORMAT_ETC2_R8G8B8A8_UNORM_BLOCK** specifies a four-component, ETC2 compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data with the first 64 bits encoding alpha values followed by 64 bits encoding RGB values.
- **VK_FORMAT_ETC2_R8G8B8A8_SRGB_BLOCK** specifies a four-component, ETC2 compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data with the first 64 bits encoding alpha values followed by 64 bits encoding RGB values with sRGB nonlinear encoding applied.
- **VK_FORMAT_EAC_R11_UNORM_BLOCK** specifies a one-component, ETC2 compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized red texel data.
- **VK_FORMAT_EAC_R11_SNORM_BLOCK** specifies a one-component, ETC2 compressed format where each

64-bit compressed texel block encodes a 4×4 rectangle of signed normalized red texel data.

- **VK_FORMAT_EAC_R11G11_UNORM_BLOCK** specifies a two-component, ETC2 compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RG texel data with the first 64 bits encoding red values followed by 64 bits encoding green values.
- **VK_FORMAT_EAC_R11G11_SNORM_BLOCK** specifies a two-component, ETC2 compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of signed normalized RG texel data with the first 64 bits encoding red values followed by 64 bits encoding green values.
- **VK_FORMAT_ASTC_4x4_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_4x4_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- **VK_FORMAT_ASTC_5x4_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 5×4 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_5x4_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 5×4 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- **VK_FORMAT_ASTC_5x5_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 5×5 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_5x5_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 5×5 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- **VK_FORMAT_ASTC_6x5_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 6×5 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_6x5_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 6×5 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- **VK_FORMAT_ASTC_6x6_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 6×6 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_6x6_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 6×6 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- **VK_FORMAT_ASTC_8x5_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes an 8×5 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_8x5_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes an 8×5 rectangle of unsigned normalized RGBA

texel data with sRGB nonlinear encoding applied to the RGB components.

- **VK_FORMAT_ASTC_8x6_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes an 8×6 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_8x6_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes an 8×6 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- **VK_FORMAT_ASTC_8x8_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes an 8×8 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_8x8_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes an 8×8 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- **VK_FORMAT_ASTC_10x5_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×5 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_10x5_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×5 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- **VK_FORMAT_ASTC_10x6_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×6 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_10x6_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×6 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- **VK_FORMAT_ASTC_10x8_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×8 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_10x8_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×8 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- **VK_FORMAT_ASTC_10x10_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×10 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_10x10_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×10 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- **VK_FORMAT_ASTC_12x10_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 12×10 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_12x10_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 12×10 rectangle of unsigned normalized RGBA texel data.

texel data with sRGB nonlinear encoding applied to the RGB components.

- `VK_FORMAT_ASTC_12x12_UNORM_BLOCK` specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 12×12 rectangle of unsigned normalized RGBA texel data.
- `VK_FORMAT_ASTC_12x12_SRGB_BLOCK` specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 12×12 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.

33.1.1. Multi-planar Format Image Aspect

When using `VkImageAspectFlagBits` to select a plane of a multi-planar format, the following are the valid options:

- Two planes
 - `VK_IMAGE_ASPECT_PLANE_0_BIT`
 - `VK_IMAGE_ASPECT_PLANE_1_BIT`
- Three planes
 - `VK_IMAGE_ASPECT_PLANE_0_BIT`
 - `VK_IMAGE_ASPECT_PLANE_1_BIT`
 - `VK_IMAGE_ASPECT_PLANE_2_BIT`

33.1.2. Packed Formats

For the purposes of address alignment when accessing buffer memory containing vertex attribute or texel data, the following formats are considered *packed* - components of the texels or attributes are stored in bitfields packed into one or more 8-, 16-, or 32-bit fundamental data type.

- **Packed into 8-bit data types:**
 - `VK_FORMAT_R4G4_UNORM_PACK8`
- **Packed into 16-bit data types:**
 - `VK_FORMAT_R4G4B4A4_UNORM_PACK16`
 - `VK_FORMAT_B4G4R4A4_UNORM_PACK16`
 - `VK_FORMAT_R5G6B5_UNORM_PACK16`
 - `VK_FORMAT_B5G6R5_UNORM_PACK16`
 - `VK_FORMAT_R5G5B5A1_UNORM_PACK16`
 - `VK_FORMAT_B5G5R5A1_UNORM_PACK16`
 - `VK_FORMAT_A1R5G5B5_UNORM_PACK16`
- **Packed into 32-bit data types:**
 - `VK_FORMAT_A8B8G8R8_UNORM_PACK32`
 - `VK_FORMAT_A8B8G8R8_SNORM_PACK32`

- `VK_FORMAT_A8B8G8R8_USCALED_PACK32`
- `VK_FORMAT_A8B8G8R8_SSCALED_PACK32`
- `VK_FORMAT_A8B8G8R8_UINT_PACK32`
- `VK_FORMAT_A8B8G8R8_SINT_PACK32`
- `VK_FORMAT_A8B8G8R8_SRGB_PACK32`
- `VK_FORMAT_A2R10G10B10_UNORM_PACK32`
- `VK_FORMAT_A2R10G10B10_SNORM_PACK32`
- `VK_FORMAT_A2R10G10B10_USCALED_PACK32`
- `VK_FORMAT_A2R10G10B10_SSCALED_PACK32`
- `VK_FORMAT_A2R10G10B10_UINT_PACK32`
- `VK_FORMAT_A2R10G10B10_SINT_PACK32`
- `VK_FORMAT_A2B10G10R10_UNORM_PACK32`
- `VK_FORMAT_A2B10G10R10_SNORM_PACK32`
- `VK_FORMAT_A2B10G10R10_USCALED_PACK32`
- `VK_FORMAT_A2B10G10R10_SSCALED_PACK32`
- `VK_FORMAT_A2B10G10R10_UINT_PACK32`
- `VK_FORMAT_A2B10G10R10_SINT_PACK32`
- `VK_FORMAT_B10G11R11_UFLOAT_PACK32`
- `VK_FORMAT_E5B9G9R9_UFLOAT_PACK32`
- `VK_FORMAT_X8_D24_UNORM_PACK32`

33.1.3. Identification of Formats

A “format” is represented by a single enum value. The name of a format is usually built up by using the following pattern:

```
VK_FORMAT_{component-format|compression-scheme}_{numeric-format}
```

The component-format indicates either the size of the R, G, B, and A components (if they are present) in the case of a color format, or the size of the depth (D) and stencil (S) components (if they are present) in the case of a depth/stencil format (see below). An X indicates a component that is unused, but **may** be present for padding.

Table 33. Interpretation of Numeric Format

Numeric format	Type-Declaration instructions	Numeric type	Description
UNORM	OpTypeFloat	floating-point	The components are unsigned normalized values in the range [0,1]
SNORM	OpTypeFloat	floating-point	The components are signed normalized values in the range [-1,1]
USCALED	OpTypeFloat	floating-point	The components are unsigned integer values that get converted to floating-point in the range $[0, 2^n - 1]$
SSCALED	OpTypeFloat	floating-point	The components are signed integer values that get converted to floating-point in the range $[-2^{n-1}, 2^{n-1} - 1]$
UINT	OpTypeInt	unsigned integer	The components are unsigned integer values in the range $[0, 2^n - 1]$
SINT	OpTypeInt	signed integer	The components are signed integer values in the range $[-2^{n-1}, 2^{n-1} - 1]$
UFLOAT	OpTypeFloat	floating-point	The components are unsigned floating-point numbers (used by packed, shared exponent, and some compressed formats)
SFLOAT	OpTypeFloat	floating-point	The components are signed floating-point numbers
SRGB	OpTypeFloat	floating-point	The R, G, and B components are unsigned normalized values that represent values using sRGB nonlinear encoding, while the A component (if one exists) is a regular unsigned normalized value

n is the number of bits in the component.

The suffix **_PACKnn** indicates that the format is packed into an underlying type with **nn** bits.

The suffix **_BLOCK** indicates that the format is a block-compressed format, with the representation of multiple pixels encoded interdependently within a region.

Table 34. Interpretation of Compression Scheme

Compression scheme	Description
BC	Block Compression. See Block-Compressed Image Formats .
ETC2	Ericsson Texture Compression. See ETC Compressed Image Formats .
EAC	ETC2 Alpha Compression. See ETC Compressed Image Formats .
ASTC	Adaptive Scalable Texture Compression (LDR Profile). See ASTC Compressed Image Formats .

33.1.4. Representation and Texel Block Size

Color formats **must** be represented in memory in exactly the form indicated by the format's name. This means that promoting one format to another with more bits per component and/or additional components **must** not occur for color formats. Depth/stencil formats have more relaxed requirements as discussed [below](#).

Each format has a *texel block size*, the number of bytes used to store one *texel block* (a single addressable element of an uncompressed image, or a single compressed block of a compressed image). The texel block size for each format is shown in the [Compatible formats](#) table.

The representation of non-packed formats is that the first component specified in the name of the format is in the lowest memory addresses and the last component specified is in the highest memory addresses. See [Byte mappings for non-packed/compressed color formats](#). The in-memory ordering of bytes within a component is determined by the host endianness.

Table 35. Byte Mappings for Non-Packed/Compressed Color Formats

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	← Byte		
R																VK_FORMAT_R8_*		
R	G															VK_FORMAT_R8G8_*		
R	G	B														VK_FORMAT_R8G8B8_*		
B	G	R														VK_FORMAT_B8G8R8_*		
R	G	B	A													VK_FORMAT_R8G8B8A8_*		
B	G	R	A													VK_FORMAT_B8G8R8A8_*		
R																VK_FORMAT_R16_*		
R		G															VK_FORMAT_R16G16_*	
R		G		B													VK_FORMAT_R16G16B16_*	
R		G		B		A												VK_FORMAT_R16G16B16A16_*
R																	VK_FORMAT_R32_*	
R			G															VK_FORMAT_R32G32_*
R			G				B										VK_FORMAT_R32G32B32_*	
R			G				B				A							VK_FORMAT_R32G32B32A32_*
R																	VK_FORMAT_R64_*	
R							G											VK_FORMAT_R64G64_*
VK_FORMAT_R64G64B64_* as VK_FORMAT_R64G64_* but with B in bytes 16-23																		
VK_FORMAT_R64G64B64A64_* as VK_FORMAT_R64G64B64_* but with A in bytes 24-31																		

Packed formats store multiple components within one underlying type. The bit representation is that the first component specified in the name of the format is in the most-significant bits and the last component specified is in the least-significant bits of the underlying type. The in-memory ordering of bytes comprising the underlying type is determined by the host endianness.

Table 36. Bit Mappings for Packed 8-Bit Formats

Bit							
7	6	5	4	3	2	1	0
VK_FORMAT_R4G4_UNORM_PACK8							
R				G			
3	2	1	0	3	2	1	0

Table 37. Bit Mappings for Packed 16-Bit Formats

Bit																
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
VK_FORMAT_R4G4B4A4_UNORM_PACK16																
R				G				B				A				
3	2	1	0	3	2	1	0	3	2	1	0	3	2	1	0	
VK_FORMAT_B4G4R4A4_UNORM_PACK16																
B				G				R				A				
3	2	1	0	3	2	1	0	3	2	1	0	3	2	1	0	
VK_FORMAT_R5G6B5_UNORM_PACK16																
R					G						B					
4	3	2	1	0	5	4	3	2	1	0	4	3	2	1	0	
VK_FORMAT_B5G6R5_UNORM_PACK16																
B					G						R					
4	3	2	1	0	5	4	3	2	1	0	4	3	2	1	0	
VK_FORMAT_R5G5B5A1_UNORM_PACK16																
R					G						B					A
4	3	2	1	0	4	3	2	1	0	4	3	2	1	0	0	
VK_FORMAT_B5G5R5A1_UNORM_PACK16																
B					G						R					A
4	3	2	1	0	4	3	2	1	0	4	3	2	1	0	0	
VK_FORMAT_A1R5G5B5_UNORM_PACK16																
A	R					G					B					
0	4	3	2	1	0	4	3	2	1	0	4	3	2	1	0	

Table 38. Bit Mappings for Packed 32-Bit Formats

Bit																															
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VK_FORMAT_A8B8G8R8*_PACK32																															
A								B								G								R							
7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
VK_FORMAT_A2R10G10B10*_PACK32																															
A		R										G										B									

Bit																																
1	0	9	8	7	6	5	4	3	2	1	0	9	8	7	6	5	4	3	2	1	0	9	8	7	6	5	4	3	2	1	0	
VK_FORMAT_A2B10G10R10_*_PACK32																																
A		B								G										R												
1	0	9	8	7	6	5	4	3	2	1	0	9	8	7	6	5	4	3	2	1	0	9	8	7	6	5	4	3	2	1	0	
VK_FORMAT_B10G11R11_UFLOAT_PACK32																																
B										G										R												
9	8	7	6	5	4	3	2	1	0	10	9	8	7	6	5	4	3	2	1	0	10	9	8	7	6	5	4	3	2	1	0	
VK_FORMAT_E5B9G9R9_UFLOAT_PACK32																																
E					B								G										R									
4	3	2	1	0	8	7	6	5	4	3	2	1	0	8	7	6	5	4	3	2	1	0	8	7	6	5	4	3	2	1	0	
VK_FORMAT_X8_D24_UNORM_PACK32																																
X									D																							
7	6	5	4	3	2	1	0	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	

33.1.5. Depth/Stencil Formats

Depth/stencil formats are considered opaque and need not be stored in the exact number of bits per texel or component ordering indicated by the format enum. However, implementations **must** not substitute a different depth or stencil precision than is described in the format (e.g. D16 **must** not be implemented as D24 or D32).

33.1.6. Format Compatibility Classes

Uncompressed color formats are *compatible* with each other if they occupy the same number of bits per texel block . Compressed color formats are compatible with each other if the only difference between them is the [numeric format](#) of the uncompressed pixels. Each depth/stencil format is only compatible with itself. In the [following](#) table, all the formats in the same row are compatible. Each format has a defined *texel block extent* specifying how many texels each texel block represents in each dimension.

Table 39. Compatible Formats

Class, Texel Block Size, Texel Block Extent, # Texels/Block	Formats
8-bit Block size 1 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R4G4_UNORM_PACK8, VK_FORMAT_R8_UNORM, VK_FORMAT_R8_SNORM, VK_FORMAT_R8_USCALED, VK_FORMAT_R8_SSCALED, VK_FORMAT_R8_UINT, VK_FORMAT_R8_SINT, VK_FORMAT_R8_SRGB

Class, Texel Block Size, Texel Block Extent, # Texels/Block	Formats
16-bit Block size 2 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R4G4B4A4_UNORM_PACK16, VK_FORMAT_B4G4R4A4_UNORM_PACK16, VK_FORMAT_R5G6B5_UNORM_PACK16, VK_FORMAT_B5G6R5_UNORM_PACK16, VK_FORMAT_R5G5B5A1_UNORM_PACK16, VK_FORMAT_B5G5R5A1_UNORM_PACK16, VK_FORMAT_A1R5G5B5_UNORM_PACK16, VK_FORMAT_R8G8_UNORM, VK_FORMAT_R8G8_SNORM, VK_FORMAT_R8G8_USCALED, VK_FORMAT_R8G8_SSCALED, VK_FORMAT_R8G8_UINT, VK_FORMAT_R8G8_SINT, VK_FORMAT_R8G8_SRGB, VK_FORMAT_R16_UNORM, VK_FORMAT_R16_SNORM, VK_FORMAT_R16_USCALED, VK_FORMAT_R16_SSCALED, VK_FORMAT_R16_UINT, VK_FORMAT_R16_SINT, VK_FORMAT_R16_SFLOAT
24-bit Block size 3 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R8G8B8_UNORM, VK_FORMAT_R8G8B8_SNORM, VK_FORMAT_R8G8B8_USCALED, VK_FORMAT_R8G8B8_SSCALED, VK_FORMAT_R8G8B8_UINT, VK_FORMAT_R8G8B8_SINT, VK_FORMAT_R8G8B8_SRGB, VK_FORMAT_B8G8R8_UNORM, VK_FORMAT_B8G8R8_SNORM, VK_FORMAT_B8G8R8_USCALED, VK_FORMAT_B8G8R8_SSCALED, VK_FORMAT_B8G8R8_UINT, VK_FORMAT_B8G8R8_SINT, VK_FORMAT_B8G8R8_SRGB

Class, Texel Block Size, Texel Block Extent, # Texels/Block	Formats
32-bit Block size 4 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R8G8B8A8_UNORM, VK_FORMAT_R8G8B8A8_SNORM, VK_FORMAT_R8G8B8A8_USCALED, VK_FORMAT_R8G8B8A8_SSCALED, VK_FORMAT_R8G8B8A8_UINT, VK_FORMAT_R8G8B8A8_SINT, VK_FORMAT_R8G8B8A8_SRGB, VK_FORMAT_B8G8R8A8_UNORM, VK_FORMAT_B8G8R8A8_SNORM, VK_FORMAT_B8G8R8A8_USCALED, VK_FORMAT_B8G8R8A8_SSCALED, VK_FORMAT_B8G8R8A8_UINT, VK_FORMAT_B8G8R8A8_SINT, VK_FORMAT_B8G8R8A8_SRGB, VK_FORMAT_A8B8G8R8_UNORM_PACK32, VK_FORMAT_A8B8G8R8_SNORM_PACK32, VK_FORMAT_A8B8G8R8_USCALED_PACK32, VK_FORMAT_A8B8G8R8_SSCALED_PACK32, VK_FORMAT_A8B8G8R8_UINT_PACK32, VK_FORMAT_A8B8G8R8_SINT_PACK32, VK_FORMAT_A8B8G8R8_SRGB_PACK32, VK_FORMAT_A2R10G10B10_UNORM_PACK32, VK_FORMAT_A2R10G10B10_SNORM_PACK32, VK_FORMAT_A2R10G10B10_USCALED_PACK32, VK_FORMAT_A2R10G10B10_SSCALED_PACK32, VK_FORMAT_A2R10G10B10_UINT_PACK32, VK_FORMAT_A2R10G10B10_SINT_PACK32, VK_FORMAT_A2B10G10R10_UNORM_PACK32, VK_FORMAT_A2B10G10R10_SNORM_PACK32, VK_FORMAT_A2B10G10R10_USCALED_PACK32, VK_FORMAT_A2B10G10R10_SSCALED_PACK32, VK_FORMAT_A2B10G10R10_UINT_PACK32, VK_FORMAT_A2B10G10R10_SINT_PACK32, VK_FORMAT_R16G16_UNORM, VK_FORMAT_R16G16_SNORM, VK_FORMAT_R16G16_USCALED, VK_FORMAT_R16G16_SSCALED, VK_FORMAT_R16G16_UINT, VK_FORMAT_R16G16_SINT, VK_FORMAT_R16G16_SFLOAT, VK_FORMAT_R32_UINT, VK_FORMAT_R32_SINT, VK_FORMAT_R32_SFLOAT, VK_FORMAT_B10G11R11_UFLOAT_PACK32, VK_FORMAT_E5B9G9R9_UFLOAT_PACK32

Class, Texel Block Size, Texel Block Extent, # Texels/Block	Formats
48-bit Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R16G16B16_UNORM, VK_FORMAT_R16G16B16_SNORM, VK_FORMAT_R16G16B16_USCALED, VK_FORMAT_R16G16B16_SSCALED, VK_FORMAT_R16G16B16_UINT, VK_FORMAT_R16G16B16_SINT, VK_FORMAT_R16G16B16_SFLOAT
64-bit Block size 8 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R16G16B16A16_UNORM, VK_FORMAT_R16G16B16A16_SNORM, VK_FORMAT_R16G16B16A16_USCALED, VK_FORMAT_R16G16B16A16_SSCALED, VK_FORMAT_R16G16B16A16_UINT, VK_FORMAT_R16G16B16A16_SINT, VK_FORMAT_R16G16B16A16_SFLOAT, VK_FORMAT_R32G32_UINT, VK_FORMAT_R32G32_SINT, VK_FORMAT_R32G32_SFLOAT, VK_FORMAT_R64_UINT, VK_FORMAT_R64_SINT, VK_FORMAT_R64_SFLOAT
96-bit Block size 12 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R32G32B32_UINT, VK_FORMAT_R32G32B32_SINT, VK_FORMAT_R32G32B32_SFLOAT
128-bit Block size 16 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R32G32B32A32_UINT, VK_FORMAT_R32G32B32A32_SINT, VK_FORMAT_R32G32B32A32_SFLOAT, VK_FORMAT_R64G64_UINT, VK_FORMAT_R64G64_SINT, VK_FORMAT_R64G64_SFLOAT
192-bit Block size 24 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R64G64B64_UINT, VK_FORMAT_R64G64B64_SINT, VK_FORMAT_R64G64B64_SFLOAT
256-bit Block size 32 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R64G64B64A64_UINT, VK_FORMAT_R64G64B64A64_SINT, VK_FORMAT_R64G64B64A64_SFLOAT

Class, Texel Block Size, Texel Block Extent, # Texels/Block	Formats
D16 Block size 2 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_D16_UNORM
D24 Block size 4 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_X8_D24_UNORM_PACK32
D32 Block size 4 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_D32_SFLOAT
S8 Block size 1 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_S8_UINT
D16S8 Block size 3 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_D16_UNORM_S8_UINT
D24S8 Block size 4 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_D24_UNORM_S8_UINT
D32S8 Block size 5 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_D32_SFLOAT_S8_UINT
BC1_RGB Block size 8 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_BC1_RGB_UNORM_BLOCK, VK_FORMAT_BC1_RGB_SRGB_BLOCK
BC1_RGBA Block size 8 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_BC1_RGBA_UNORM_BLOCK, VK_FORMAT_BC1_RGBA_SRGB_BLOCK
BC2 Block size 16 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_BC2_UNORM_BLOCK, VK_FORMAT_BC2_SRGB_BLOCK

Class, Texel Block Size, Texel Block Extent, # Texels/Block	Formats
BC3 Block size 16 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_BC3_UNORM_BLOCK, VK_FORMAT_BC3_SRGB_BLOCK
BC4 Block size 8 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_BC4_UNORM_BLOCK, VK_FORMAT_BC4_SNORM_BLOCK
BC5 Block size 16 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_BC5_UNORM_BLOCK, VK_FORMAT_BC5_SNORM_BLOCK
BC6H Block size 16 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_BC6H_UFLOAT_BLOCK, VK_FORMAT_BC6H_SFLOAT_BLOCK
BC7 Block size 16 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_BC7_UNORM_BLOCK, VK_FORMAT_BC7_SRGB_BLOCK
ETC2_RGB Block size 8 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_ETC2_R8G8B8_UNORM_BLOCK, VK_FORMAT_ETC2_R8G8B8_SRGB_BLOCK
ETC2_RGBA Block size 8 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_ETC2_R8G8B8A1_UNORM_BLOCK, VK_FORMAT_ETC2_R8G8B8A1_SRGB_BLOCK
ETC2_EAC_RGBA Block size 16 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_ETC2_R8G8B8A8_UNORM_BLOCK, VK_FORMAT_ETC2_R8G8B8A8_SRGB_BLOCK
EAC_R Block size 8 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_EAC_R11_UNORM_BLOCK, VK_FORMAT_EAC_R11_SNORM_BLOCK
EAC_RG Block size 16 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_EAC_R11G11_UNORM_BLOCK, VK_FORMAT_EAC_R11G11_SNORM_BLOCK

Class, Texel Block Size, Texel Block Extent, # Texels/Block	Formats
ASTC_4x4 Block size 16 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_ASTC_4x4_UNORM_BLOCK, VK_FORMAT_ASTC_4x4_SRGB_BLOCK
ASTC_5x4 Block size 16 byte 5x4x1 block extent 20 texel/block	VK_FORMAT_ASTC_5x4_UNORM_BLOCK, VK_FORMAT_ASTC_5x4_SRGB_BLOCK
ASTC_5x5 Block size 16 byte 5x5x1 block extent 25 texel/block	VK_FORMAT_ASTC_5x5_UNORM_BLOCK, VK_FORMAT_ASTC_5x5_SRGB_BLOCK
ASTC_6x5 Block size 16 byte 6x5x1 block extent 30 texel/block	VK_FORMAT_ASTC_6x5_UNORM_BLOCK, VK_FORMAT_ASTC_6x5_SRGB_BLOCK
ASTC_6x6 Block size 16 byte 6x6x1 block extent 36 texel/block	VK_FORMAT_ASTC_6x6_UNORM_BLOCK, VK_FORMAT_ASTC_6x6_SRGB_BLOCK
ASTC_8x5 Block size 16 byte 8x5x1 block extent 40 texel/block	VK_FORMAT_ASTC_8x5_UNORM_BLOCK, VK_FORMAT_ASTC_8x5_SRGB_BLOCK
ASTC_8x6 Block size 16 byte 8x6x1 block extent 48 texel/block	VK_FORMAT_ASTC_8x6_UNORM_BLOCK, VK_FORMAT_ASTC_8x6_SRGB_BLOCK
ASTC_8x8 Block size 16 byte 8x8x1 block extent 64 texel/block	VK_FORMAT_ASTC_8x8_UNORM_BLOCK, VK_FORMAT_ASTC_8x8_SRGB_BLOCK
ASTC_10x5 Block size 16 byte 10x5x1 block extent 50 texel/block	VK_FORMAT_ASTC_10x5_UNORM_BLOCK, VK_FORMAT_ASTC_10x5_SRGB_BLOCK
ASTC_10x6 Block size 16 byte 10x6x1 block extent 60 texel/block	VK_FORMAT_ASTC_10x6_UNORM_BLOCK, VK_FORMAT_ASTC_10x6_SRGB_BLOCK

Class, Texel Block Size, Texel Block Extent, # Texels/Block	Formats
ASTC_10x8 Block size 16 byte 10x8x1 block extent 80 texel/block	VK_FORMAT_ASTC_10x8_UNORM_BLOCK, VK_FORMAT_ASTC_10x8_SRGB_BLOCK
ASTC_10x10 Block size 16 byte 10x10x1 block extent 100 texel/block	VK_FORMAT_ASTC_10x10_UNORM_BLOCK, VK_FORMAT_ASTC_10x10_SRGB_BLOCK
ASTC_12x10 Block size 16 byte 12x10x1 block extent 120 texel/block	VK_FORMAT_ASTC_12x10_UNORM_BLOCK, VK_FORMAT_ASTC_12x10_SRGB_BLOCK
ASTC_12x12 Block size 16 byte 12x12x1 block extent 144 texel/block	VK_FORMAT_ASTC_12x12_UNORM_BLOCK, VK_FORMAT_ASTC_12x12_SRGB_BLOCK

Size Compatibility

Color formats with the same texel block size are considered *size-compatible*. If two size-compatible formats have different block extents (i.e. for compressed formats), then an image with size $A \times B \times C$ in one format with a block extent of $a \times b \times c$ can be represented as an image with size $X \times Y \times Z$ in the other format with block extent $x \times y \times z$ at the ratio between the block extents for each format, where

$$\lceil A/a \rceil = \lceil X/x \rceil$$

$$\lceil B/b \rceil = \lceil Y/y \rceil$$

$$\lceil C/c \rceil = \lceil Z/z \rceil$$

NOTE

For example, a 7x3 image in the `VK_FORMAT_ASTC_8x5_UNORM_BLOCK` format can be represented as a 1x1 `VK_FORMAT_R64G64_UINT` image.

Copy operations are able to copy between size-compatible formats in different resources to enable manipulation of data in different formats. The extent used in these copy operations always matches the source image, and is resized to the expectations of the block extents noted above for the destination image.

33.2. Format Properties

To query supported format features which are properties of the physical device, call:

```
// Provided by VK_VERSION_1_0
void vkGetPhysicalDeviceFormatProperties(
    VkPhysicalDevice    physicalDevice,
    VkFormat            format,
    VkFormatProperties* pFormatProperties);
```

- `physicalDevice` is the physical device from which to query the format properties.
- `format` is the format whose properties are queried.
- `pFormatProperties` is a pointer to a `VkFormatProperties` structure in which physical device properties for `format` are returned.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceFormatProperties-physicalDevice-parameter `physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceFormatProperties-format-parameter `format` **must** be a valid `VkFormat` value
- VUID-vkGetPhysicalDeviceFormatProperties-pFormatProperties-parameter `pFormatProperties` **must** be a valid pointer to a `VkFormatProperties` structure

The `VkFormatProperties` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkFormatProperties {
    VkFormatFeatureFlags    linearTilingFeatures;
    VkFormatFeatureFlags    optimalTilingFeatures;
    VkFormatFeatureFlags    bufferFeatures;
} VkFormatProperties;
```

- `linearTilingFeatures` is a bitmask of `VkFormatFeatureFlagBits` specifying features supported by images created with a `tiling` parameter of `VK_IMAGE_TILING_LINEAR`.
- `optimalTilingFeatures` is a bitmask of `VkFormatFeatureFlagBits` specifying features supported by images created with a `tiling` parameter of `VK_IMAGE_TILING_OPTIMAL`.
- `bufferFeatures` is a bitmask of `VkFormatFeatureFlagBits` specifying features supported by buffers.

NOTE

If no format feature flags are supported, then the only possible use would be image transfers - which alone are not useful. As such, if no format feature flags are supported, the format itself is not supported, and images of that format cannot be

created.

If **format** is a block-compressed format, then **bufferFeatures** **must** not support any features for the format.

Bits which **can** be set in the **VkFormatProperties** features **linearTilingFeatures**, **optimalTilingFeatures**, and **bufferFeatures** are:

```
// Provided by VK_VERSION_1_0
typedef enum VkFormatFeatureFlagBits {
    VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT = 0x00000001,
    VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT = 0x00000002,
    VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT = 0x00000004,
    VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT = 0x00000008,
    VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT = 0x00000010,
    VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT = 0x00000020,
    VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT = 0x00000040,
    VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT = 0x00000080,
    VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT = 0x00000100,
    VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT = 0x00000200,
    VK_FORMAT_FEATURE_BLIT_SRC_BIT = 0x00000400,
    VK_FORMAT_FEATURE_BLIT_DST_BIT = 0x00000800,
    VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT = 0x00001000,
} VkFormatFeatureFlagBits;
```

These values **may** be set in **linearTilingFeatures** and **optimalTilingFeatures**, specifying that the features are supported by **images** or **image views** created with the queried **vkGetPhysicalDeviceFormatProperties::format**:

- **VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT** specifies that an image view **can** be **sampled from**.
- **VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT** specifies that an image view **can** be used as a **storage image**.
- **VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT** specifies that an image view **can** be used as storage image that supports atomic operations.
- **VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT** specifies that an image view **can** be used as a framebuffer color attachment and as an input attachment.
- **VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT** specifies that an image view **can** be used as a framebuffer color attachment that supports blending.
- **VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT** specifies that an image view **can** be used as a framebuffer depth/stencil attachment and as an input attachment.
- **VK_FORMAT_FEATURE_BLIT_SRC_BIT** specifies that an image **can** be used as **srcImage** for the **vkCmdBlitImage** command.
- **VK_FORMAT_FEATURE_BLIT_DST_BIT** specifies that an image **can** be used as **dstImage** for the **vkCmdBlitImage** command.
- **VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT** specifies that if

`VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT` is also set, an image view **can** be used with a sampler that has either of `magFilter` or `minFilter` set to `VK_FILTER_LINEAR`, or `mipmapMode` set to `VK_SAMPLER_MIPMAP_MODE_LINEAR`. If `VK_FORMAT_FEATURE_BLIT_SRC_BIT` is also set, an image can be used as the `srcImage` to `vkCmdBlitImage` with a `filter` of `VK_FILTER_LINEAR`. This bit **must** only be exposed for formats that also support the `VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT` or `VK_FORMAT_FEATURE_BLIT_SRC_BIT`.

If the format being queried is a depth/stencil format, this bit only specifies that the depth aspect (not the stencil aspect) of an image of this format supports linear filtering, and that linear filtering of the depth aspect is supported whether depth compare is enabled in the sampler or not. Where depth comparison is supported it **may** be linear filtered whether this bit is present or not, but where this bit is not present the filtered value **may** be computed in an implementation-dependent manner which differs from the normal rules of linear filtering. The resulting value **must** be in the range [0,1] and **should** be proportional to, or a weighted average of, the number of comparison passes or failures.

The following bits **may** be set in `bufferFeatures`, specifying that the features are supported by `buffers` or `buffer views` created with the queried `vkGetPhysicalDeviceFormatProperties::format`:

- `VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT` specifies that the format **can** be used to create a buffer view that **can** be bound to a `VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER` descriptor.
- `VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT` specifies that the format **can** be used to create a buffer view that **can** be bound to a `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` descriptor.
- `VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT` specifies that atomic operations are supported on `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` with this format.
- `VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT` specifies that the format **can** be used as a vertex attribute format (`VkVertexInputAttributeDescription::format`).

NOTE	<code>VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT</code> and <code>VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT</code> are only intended to be advertised for single-component formats, since SPIR-V atomic operations require a scalar type.
------	---

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkFormatFeatureFlags;
```

`VkFormatFeatureFlags` is a bitmask type for setting a mask of zero or more `VkFormatFeatureFlagBits`.

33.2.1. Potential Format Features

Some `valid usage conditions` depend on the format features supported by a `VkImage` whose `VkImageTiling` is unknown. In such cases the exact `VkFormatFeatureFlagBits` supported by the `VkImage` cannot be determined, so the valid usage conditions are expressed in terms of the *potential format features* of the `VkImage` format.

The *potential format features* of a `VkFormat` are defined as follows:

- The union of `VkFormatFeatureFlagBits` supported when the `VkImageTiling` is `VK_IMAGE_TILING_OPTIMAL` or `VK_IMAGE_TILING_LINEAR`

33.3. Required Format Support

Implementations **must** support at least the following set of features on the listed formats. For images, these features **must** be supported for every `VkImageType` (including arrayed and cube variants) unless otherwise noted. These features are supported on existing formats without needing to advertise an extension or needing to explicitly enable them. Support for additional functionality beyond the requirements listed here is queried using the `vkGetPhysicalDeviceFormatProperties` command.

NOTE

Unless otherwise excluded below, the required formats are supported for all `VkImageCreateFlags` values as long as those flag values are otherwise allowed.

The following tables show which feature bits **must** be supported for each format.

Table 40. Key for Format Feature Tables

□	This feature must be supported on the named format
†	This feature must be supported on at least some of the named formats, with more information in the table where the symbol appears
‡	This feature must be supported with some caveats or preconditions, with more information in the table where the symbol appears
§	This feature must be supported with some caveats or preconditions, with more information in the table where the symbol appears

Table 41. Feature Bits in `optimalTilingFeatures`

<code>VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT</code>
<code>VK_FORMAT_FEATURE_BLIT_SRC_BIT</code>
<code>VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT</code>
<code>VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT</code>
<code>VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT</code>
<code>VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT</code>
<code>VK_FORMAT_FEATURE_BLIT_DST_BIT</code>
<code>VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT</code>
<code>VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT</code>

Table 42. Feature Bits in `bufferFeatures`

<code>VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT</code>
<code>VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT</code>

VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT
VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT

Table 43. Mandatory Format Support: Sub-Byte Components

	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT												
	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT												
	VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT												
	VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT												
	VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT												
	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT												
	VK_FORMAT_FEATURE_BLIT_DST_BIT												
	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT												
	VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT												
	VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT												
	VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT												
	VK_FORMAT_FEATURE_BLIT_SRC_BIT												
	VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT												
Format													
VK_FORMAT_UNDEFINED													
VK_FORMAT_R4G4_UNORM_PACK8													
VK_FORMAT_R4G4B4A4_UNORM_PACK16													
VK_FORMAT_B4G4R4A4_UNORM_PACK16	1	1	1										
VK_FORMAT_R5G6B5_UNORM_PACK16	1	1	1			1	1	1					
VK_FORMAT_B5G6R5_UNORM_PACK16													
VK_FORMAT_R5G5B5A1_UNORM_PACK16													
VK_FORMAT_B5G5R5A1_UNORM_PACK16													
VK_FORMAT_A1R5G5B5_UNORM_PACK16	1	1	1			1	1	1					

Table 44. Mandatory Format Support: 1-3 Byte-Sized Components

[illegible]

VK_FORMAT_B8G8R8_USCALED													
VK_FORMAT_B8G8R8_SSCALED													
VK_FORMAT_B8G8R8_UINT													
VK_FORMAT_B8G8R8_SINT													
VK_FORMAT_B8G8R8_SRGB													

Format features marked with ‡ **must** be supported for `optimalTilingFeatures` if the `VkPhysicalDevice` supports the `shaderStorageImageExtendedFormats` feature.

Table 45. Mandatory Format Support: 4 Byte-Sized Components

VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT													0			
VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT																
VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT											0					
VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT																
VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT									0							
VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT																
VK_FORMAT_FEATURE_BLIT_DST_BIT																
VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT																
VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT					0											
VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT																
VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT																
VK_FORMAT_FEATURE_BLIT_SRC_BIT			0													
VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT																
Format																
VK_FORMAT_R8G8B8A8_UNORM	0	0					0	0					0	0	0	
VK_FORMAT_R8G8B8A8_SNORM	0	0					0	0						0	0	0
VK_FORMAT_R8G8B8A8_USCALED																
VK_FORMAT_R8G8B8A8_SSCALED																
VK_FORMAT_R8G8B8A8_UINT	0	0				0		0	0			0	0	0		
VK_FORMAT_R8G8B8A8_SINT	0	0				0		0	0			0	0	0		
VK_FORMAT_R8G8B8A8_SRGB	0	0	0			0	0	0								
VK_FORMAT_B8G8R8A8_UNORM	0	0	0			0	0	0		0	0					
VK_FORMAT_B8G8R8A8_SNORM																
VK_FORMAT_B8G8R8A8_USCALED																
VK_FORMAT_B8G8R8A8_SSCALED																
VK_FORMAT_B8G8R8A8_UINT																
VK_FORMAT_B8G8R8A8_SINT																
VK_FORMAT_B8G8R8A8_SRGB	0	0	0			0	0	0								
VK_FORMAT_A8B8G8R8_UNORM_PACK32	0	0	0			0	0	0		0	0	0				
VK_FORMAT_A8B8G8R8_SNORM_PACK32	0	0	0							0	0	0				
VK_FORMAT_A8B8G8R8_USCALED_PACK32																
VK_FORMAT_A8B8G8R8_SSCALED_PACK32																
VK_FORMAT_A8B8G8R8_UINT_PACK32	0	0				0	0			0	0	0				
VK_FORMAT_A8B8G8R8_SINT_PACK32	0	0				0	0			0	0	0				
VK_FORMAT_A8B8G8R8_SRGB_PACK32	0	0	0			0	0	0								

Table 46. Mandatory Format Support: 10- and 12-Bit Components

	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT												
	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT												
	VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT												
	VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT												
	VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT												
	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT												
	VK_FORMAT_FEATURE_BLIT_DST_BIT												
	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT												
	VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT												
	VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT												
	VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT												
	VK_FORMAT_FEATURE_BLIT_SRC_BIT												
	VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT												
Format													
VK_FORMAT_A2R10G10B10_UNORM_PACK32													
VK_FORMAT_A2R10G10B10_SNORM_PACK32													
VK_FORMAT_A2R10G10B10_USCALED_PACK32													
VK_FORMAT_A2R10G10B10_SSCALED_PACK32													
VK_FORMAT_A2R10G10B10_UINT_PACK32													
VK_FORMAT_A2R10G10B10_SINT_PACK32													
VK_FORMAT_A2B10G10R10_UNORM_PACK32	␣	␣	␣	‡		␣	␣	␣		␣	␣		
VK_FORMAT_A2B10G10R10_SNORM_PACK32													
VK_FORMAT_A2B10G10R10_USCALED_PACK32													
VK_FORMAT_A2B10G10R10_SSCALED_PACK32													
VK_FORMAT_A2B10G10R10_UINT_PACK32	␣	␣		‡		␣	␣				␣		
VK_FORMAT_A2B10G10R10_SINT_PACK32													
Format features marked with ‡ must be supported for optimalTilingFeatures if the VkPhysicalDevice supports the shaderStorageImageExtendedFormats feature.													

Table 47. Mandatory Format Support: 16-bit Components

	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT											
	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT											
	VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT											
	VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT											
	VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT											
	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT											
	VK_FORMAT_FEATURE_BLIT_DST_BIT											
	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT											
	VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT											
	VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT											
	VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT											
	VK_FORMAT_FEATURE_BLIT_SRC_BIT											
	VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT											
Format												
VK_FORMAT_R16_UNORM				‡					□			
VK_FORMAT_R16_SNORM				‡					□			
VK_FORMAT_R16_USCALED												
VK_FORMAT_R16_SSCALED												
VK_FORMAT_R16_UINT	□	□		‡		□	□			□	□	
VK_FORMAT_R16_SINT	□	□		‡		□	□			□	□	
VK_FORMAT_R16_SFLOAT	□	□	□	‡		□	□	□		□	□	
VK_FORMAT_R16G16_UNORM				‡						□		
VK_FORMAT_R16G16_SNORM				‡						□		
VK_FORMAT_R16G16_USCALED												
VK_FORMAT_R16G16_SSCALED												
VK_FORMAT_R16G16_UINT	□	□		‡		□	□			□	□	
VK_FORMAT_R16G16_SINT	□	□		‡		□	□			□	□	
VK_FORMAT_R16G16_SFLOAT	□	□	□	‡	§	□	□	□		□	□	§
VK_FORMAT_R16G16B16_UNORM												
VK_FORMAT_R16G16B16_SNORM												
VK_FORMAT_R16G16B16_USCALED												
VK_FORMAT_R16G16B16_SSCALED												
VK_FORMAT_R16G16B16_UINT												
VK_FORMAT_R16G16B16_SINT												
VK_FORMAT_R16G16B16_SFLOAT												
VK_FORMAT_R16G16B16A16_UNORM				‡						□		
VK_FORMAT_R16G16B16A16_SNORM				‡						□		

VK_FORMAT_R16G16B16A16_USCALED													
VK_FORMAT_R16G16B16A16_SSCALED													
VK_FORMAT_R16G16B16A16_UINT	□	□		□		□	□			□	□	□	
VK_FORMAT_R16G16B16A16_SINT	□	□		□		□	□			□	□	□	
VK_FORMAT_R16G16B16A16_SFLOAT	□	□	□	□	§	□	□	□		□	□	□	§

Format features marked with § **must** be supported for `optimalTilingFeatures` if the `VkPhysicalDevice` supports the `shaderStorageImageExtendedFormats` feature.

Table 48. Mandatory Format Support: 32-bit Components

VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT													
VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT													0
VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT												0	
VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT											0		
VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT										0			
VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT									0				
VK_FORMAT_FEATURE_BLIT_DST_BIT								0					
VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT							0						
VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT						0							
VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT					0								
VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT				0									
VK_FORMAT_FEATURE_BLIT_SRC_BIT			0										
VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT		0											
Format	0												
VK_FORMAT_R32_UINT	0	0		0	0	0	0			0	0	0	0
VK_FORMAT_R32_SINT	0	0		0	0	0	0			0	0	0	0
VK_FORMAT_R32_SFLOAT	0	0		0		0	0			0	0	0	
VK_FORMAT_R32G32_UINT	0	0		0		0	0			0	0	0	
VK_FORMAT_R32G32_SINT	0	0		0		0	0			0	0	0	
VK_FORMAT_R32G32_SFLOAT	0	0		0		0	0			0	0	0	
VK_FORMAT_R32G32B32_UINT										0			
VK_FORMAT_R32G32B32_SINT										0			
VK_FORMAT_R32G32B32_SFLOAT										0			
VK_FORMAT_R32G32B32A32_UINT	0	0		0		0	0			0	0	0	
VK_FORMAT_R32G32B32A32_SINT	0	0		0		0	0			0	0	0	
VK_FORMAT_R32G32B32A32_SFLOAT	0	0		0		0	0			0	0	0	

Table 49. Mandatory Format Support: 64-bit/uneven Components

	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT												
	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT												
	VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT												
	VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT												
	VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT												
	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT												
	VK_FORMAT_FEATURE_BLIT_DST_BIT												
	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT												
	VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT												
	VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT												
	VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT												
	VK_FORMAT_FEATURE_BLIT_SRC_BIT												
	VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT												
Format													
VK_FORMAT_R64_UINT													
VK_FORMAT_R64_SINT													
VK_FORMAT_R64_SFLOAT													
VK_FORMAT_R64G64_UINT													
VK_FORMAT_R64G64_SINT													
VK_FORMAT_R64G64_SFLOAT													
VK_FORMAT_R64G64B64_UINT													
VK_FORMAT_R64G64B64_SINT													
VK_FORMAT_R64G64B64_SFLOAT													
VK_FORMAT_R64G64B64A64_UINT													
VK_FORMAT_R64G64B64A64_SINT													
VK_FORMAT_R64G64B64A64_SFLOAT													
VK_FORMAT_B10G11R11_UFLOAT_PACK32	‡	‡	‡	‡							‡		
VK_FORMAT_E5B9G9R9_UFLOAT_PACK32	‡	‡	‡										
Format features marked with ‡ must be supported for optimalTilingFeatures if the VkPhysicalDevice supports the shaderStorageImageExtendedFormats feature.													

Table 50. Mandatory Format Support: Depth/Stencil With `VkImageType VK_IMAGE_TYPE_2D`

VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT													
VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT													
VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT													
VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT													
VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT													
VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT													
VK_FORMAT_FEATURE_BLIT_DST_BIT													
VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT													
VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT													
VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT													
VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT													
VK_FORMAT_FEATURE_BLIT_SRC_BIT													
VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT													
Format													
VK_FORMAT_D16_UNORM													
VK_FORMAT_X8_D24_UNORM_PACK32													
VK_FORMAT_D32_SFLOAT													
VK_FORMAT_S8_UINT													
VK_FORMAT_D16_UNORM_S8_UINT													
VK_FORMAT_D24_UNORM_S8_UINT													
VK_FORMAT_D32_SFLOAT_S8_UINT													
VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT feature must be supported for at least one of VK_FORMAT_X8_D24_UNORM_PACK32 and VK_FORMAT_D32_SFLOAT, and must be supported for at least one of VK_FORMAT_D24_UNORM_S8_UINT and VK_FORMAT_D32_SFLOAT_S8_UINT.													
bufferFeatures must not support any features for these formats													

Table 51. Mandatory Format Support: BC Compressed Formats With `VkImageType VK_IMAGE_TYPE_2D` and `VK_IMAGE_TYPE_3D`

[illegible]

Table 52. Mandatory Format Support: ETC2 and EAC Compressed Formats With VkImageType VK_IMAGE_TYPE_2D

	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT											
	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT											
	VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT											
	VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT											
	VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT											
	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT											
	VK_FORMAT_FEATURE_BLIT_DST_BIT											
	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT											
	VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT											
	VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT											
	VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT											
	VK_FORMAT_FEATURE_BLIT_SRC_BIT											
	VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT											
Format												
VK_FORMAT_ETC2_R8G8B8_UNORM_BLOCK	†	†	†									
VK_FORMAT_ETC2_R8G8B8_SRGB_BLOCK	†	†	†									
VK_FORMAT_ETC2_R8G8B8A1_UNORM_BLOCK	†	†	†									
VK_FORMAT_ETC2_R8G8B8A1_SRGB_BLOCK	†	†	†									
VK_FORMAT_ETC2_R8G8B8A8_UNORM_BLOCK	†	†	†									
VK_FORMAT_ETC2_R8G8B8A8_SRGB_BLOCK	†	†	†									
VK_FORMAT_EAC_R11_UNORM_BLOCK	†	†	†									
VK_FORMAT_EAC_R11_SNORM_BLOCK	†	†	†									
VK_FORMAT_EAC_R11G11_UNORM_BLOCK	†	†	†									
VK_FORMAT_EAC_R11G11_SNORM_BLOCK	†	†	†									

The VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT, VK_FORMAT_FEATURE_BLIT_SRC_BIT and VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT features **must** be supported in optimalTilingFeatures for all the formats in at least one of: this table, [Mandatory Format Support: BC Compressed Formats With VkImageType VK_IMAGE_TYPE_2D and VK_IMAGE_TYPE_3D](#), or [Mandatory Format Support: ASTC LDR Compressed Formats With VkImageType VK_IMAGE_TYPE_2D](#).

Table 53. Mandatory Format Support: ASTC LDR Compressed Formats With `VkImageType VK_IMAGE_TYPE_2D`

VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT													
VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT													
VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT													
VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT													
VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT													
VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT													
VK_FORMAT_FEATURE_BLIT_DST_BIT													
VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT													
VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT													
VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT													
VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT													
VK_FORMAT_FEATURE_BLIT_SRC_BIT													
VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT													
Format													
VK_FORMAT_ASTC_4x4_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_4x4_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_5x4_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_5x4_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_5x5_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_5x5_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_6x5_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_6x5_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_6x6_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_6x6_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_8x5_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_8x5_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_8x6_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_8x6_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_8x8_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_8x8_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_10x5_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_10x5_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_10x6_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_10x6_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_10x8_UNORM_BLOCK	†	†	†										

VK_FORMAT_ASTC_10x8_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_10x10_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_10x10_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_12x10_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_12x10_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_12x12_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_12x12_SRGB_BLOCK	†	†	†										

The `VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT`, `VK_FORMAT_FEATURE_BLIT_SRC_BIT` and `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT` features **must** be supported in `optimalTilingFeatures` for all the formats in at least one of: this table, [Mandatory Format Support: BC Compressed Formats With `VkImageType` `VK\_IMAGE\_TYPE\_2D` and `VK\_IMAGE\_TYPE\_3D`](#), or [Mandatory Format Support: ETC2 and EAC Compressed Formats With `VkImageType` `VK\_IMAGE\_TYPE\_2D`](#).

33.3.1. Formats Without Shader Storage Format

The device-level features for using a storage image or a storage texel buffer with an image format of `Unknown`, `shaderStorageImageReadWithoutFormat` and `shaderStorageImageWriteWithoutFormat`, only apply to the following formats:

- `VK_FORMAT_R8G8B8A8_UNORM`
- `VK_FORMAT_R8G8B8A8_SNORM`
- `VK_FORMAT_R8G8B8A8_UINT`
- `VK_FORMAT_R8G8B8A8_SINT`
- `VK_FORMAT_R32_UINT`
- `VK_FORMAT_R32_SINT`
- `VK_FORMAT_R32_SFLOAT`
- `VK_FORMAT_R32G32_UINT`
- `VK_FORMAT_R32G32_SINT`
- `VK_FORMAT_R32G32_SFLOAT`
- `VK_FORMAT_R32G32B32A32_UINT`
- `VK_FORMAT_R32G32B32A32_SINT`
- `VK_FORMAT_R32G32B32A32_SFLOAT`
- `VK_FORMAT_R16G16B16A16_UINT`
- `VK_FORMAT_R16G16B16A16_SINT`
- `VK_FORMAT_R16G16B16A16_SFLOAT`
- `VK_FORMAT_R16G16_SFLOAT`
- `VK_FORMAT_B10G11R11_UFLOAT_PACK32`
- `VK_FORMAT_R16_SFLOAT`

- `VK_FORMAT_R16G16B16A16_UNORM`
- `VK_FORMAT_A2B10G10R10_UNORM_PACK32`
- `VK_FORMAT_R16G16_UNORM`
- `VK_FORMAT_R8G8_UNORM`
- `VK_FORMAT_R16_UNORM`
- `VK_FORMAT_R8_UNORM`
- `VK_FORMAT_R16G16B16A16_SNORM`
- `VK_FORMAT_R16G16_SNORM`
- `VK_FORMAT_R8G8_SNORM`
- `VK_FORMAT_R16_SNORM`
- `VK_FORMAT_R8_SNORM`
- `VK_FORMAT_R16G16_SINT`
- `VK_FORMAT_R8G8_SINT`
- `VK_FORMAT_R16_SINT`
- `VK_FORMAT_R8_SINT`
- `VK_FORMAT_A2B10G10R10_UINT_PACK32`
- `VK_FORMAT_R16G16_UINT`
- `VK_FORMAT_R8G8_UINT`
- `VK_FORMAT_R16_UINT`
- `VK_FORMAT_R8_UINT`

NOTE

This list of formats is the union of required storage formats from [Required Format Support](#) section and formats listed in `shaderStorageImageExtendedFormats`.

33.3.2. Format Feature Dependent Usage Flags

Certain resource usage flags depend on support for the corresponding format feature flag for the format in question. The following tables list the `VkBufferUsageFlagBits` and `VkImageUsageFlagBits` that have such dependencies, and the format feature flags they depend on. Additional restrictions, including, but not limited to, further required format feature flags specific to the particular use of the resource **may** apply, as described in the respective sections of this specification.

Table 54. Format Feature Dependent Buffer Usage Flags

Buffer usage flag	Required format feature flag
<code>VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT</code>	<code>VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT</code>
<code>VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT</code>	<code>VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT</code>
<code>VK_BUFFER_USAGE_VERTEX_BUFFER_BIT</code>	<code>VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT</code>

Table 55. Format Feature Dependent Image Usage Flags

Image usage flag	Required format feature flag
VK_IMAGE_USAGE_SAMPLED_BIT	VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT
VK_IMAGE_USAGE_STORAGE_BIT	VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT
VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT
VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT	VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT
VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT or VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT

Chapter 34. Additional Capabilities

This chapter describes additional capabilities beyond the minimum capabilities described in the [Limits](#) and [Formats](#) chapters, including:

- [Additional Image Capabilities](#)

34.1. Additional Image Capabilities

Additional image capabilities, such as larger dimensions or additional sample counts for certain image types, or additional capabilities for *linear* tiling format images, are described in this section.

To query additional capabilities specific to image types, call:

```
// Provided by VK_VERSION_1_0
VkResult vkGetPhysicalDeviceImageFormatProperties(
    VkPhysicalDevice    physicalDevice,
    VkFormat            format,
    VkImageType         type,
    VkImageTiling       tiling,
    VkImageUsageFlags   usage,
    VkImageCreateFlags   flags,
    VkImageFormatProperties* pImageFormatProperties);
```

- `physicalDevice` is the physical device from which to query the image capabilities.
- `format` is a [VkFormat](#) value specifying the image format, corresponding to [VkImageCreateInfo::format](#).
- `type` is a [VkImageType](#) value specifying the image type, corresponding to [VkImageCreateInfo::imageType](#).
- `tiling` is a [VkImageTiling](#) value specifying the image tiling, corresponding to [VkImageCreateInfo::tiling](#).
- `usage` is a bitmask of [VkImageUsageFlagBits](#) specifying the intended usage of the image, corresponding to [VkImageCreateInfo::usage](#).
- `flags` is a bitmask of [VkImageCreateFlagBits](#) specifying additional parameters of the image, corresponding to [VkImageCreateInfo::flags](#).
- `pImageFormatProperties` is a pointer to a [VkImageFormatProperties](#) structure in which capabilities are returned.

The `format`, `type`, `tiling`, `usage`, and `flags` parameters correspond to parameters that would be consumed by [vkCreateImage](#) (as members of [VkImageCreateInfo](#)).

If `format` is not a supported image format, or if the combination of `format`, `type`, `tiling`, `usage`, and `flags` is not supported for images, then [vkGetPhysicalDeviceImageFormatProperties](#) returns [VK_ERROR_FORMAT_NOT_SUPPORTED](#).

The limitations on an image format that are reported by `vkGetPhysicalDeviceImageFormatProperties` have the following property: if `usage1` and `usage2` of type `VkImageUsageFlags` are such that the bits set in `usage1` are a subset of the bits set in `usage2`, and `flags1` and `flags2` of type `VkImageCreateFlags` are such that the bits set in `flags1` are a subset of the bits set in `flags2`, then the limitations for `usage1` and `flags1` **must** be no more strict than the limitations for `usage2` and `flags2`, for all values of `format`, `type`, and `tiling`.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceImageFormatProperties-physicalDevice-parameter `physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceImageFormatProperties-format-parameter `format` **must** be a valid `VkFormat` value
- VUID-vkGetPhysicalDeviceImageFormatProperties-type-parameter `type` **must** be a valid `VkImageType` value
- VUID-vkGetPhysicalDeviceImageFormatProperties-tiling-parameter `tiling` **must** be a valid `VkImageTiling` value
- VUID-vkGetPhysicalDeviceImageFormatProperties-usage-parameter `usage` **must** be a valid combination of `VkImageUsageFlagBits` values
- VUID-vkGetPhysicalDeviceImageFormatProperties-usage-requiredbitmask `usage` **must** not be 0
- VUID-vkGetPhysicalDeviceImageFormatProperties-flags-parameter `flags` **must** be a valid combination of `VkImageCreateFlagBits` values
- VUID-vkGetPhysicalDeviceImageFormatProperties-pImageFormatProperties-parameter `pImageFormatProperties` **must** be a valid pointer to a `VkImageFormatProperties` structure

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_FORMAT_NOT_SUPPORTED`

The `VkImageFormatProperties` structure is defined as:


```
// Provided by VK_VERSION_1_0
typedef struct VkImageFormatProperties {
    VkExtent3D          maxExtent;
    uint32_t            maxMipLevels;
    uint32_t            maxArrayLayers;
    VkSampleCountFlags  sampleCounts;
    VkDeviceSize        maxResourceSize;
} VkImageFormatProperties;
```

- **maxExtent** are the maximum image dimensions. See the [Allowed Extent Values](#) section below for how these values are constrained by **type**.
- **maxMipLevels** is the maximum number of mipmap levels. **maxMipLevels** **must** be equal to the number of levels in the complete mipmap chain based on the **maxExtent.width**, **maxExtent.height**, and **maxExtent.depth**, except when one of the following conditions is true, in which case it **may** instead be 1:
 - **vkGetPhysicalDeviceImageFormatProperties::tiling** was **VK_IMAGE_TILING_LINEAR**
- **maxArrayLayers** is the maximum number of array layers. **maxArrayLayers** **must** be no less than **VkPhysicalDeviceLimits::maxImageArrayLayers**, except when one of the following conditions is true, in which case it **may** instead be 1:
 - **tiling** is **VK_IMAGE_TILING_LINEAR**
 - **tiling** is **VK_IMAGE_TILING_OPTIMAL** and **type** is **VK_IMAGE_TYPE_3D**
- **sampleCounts** is a bitmask of **VkSampleCountFlagBits** specifying all the supported sample counts for this image as described [below](#).
- **maxResourceSize** is an upper bound on the total image size in bytes, inclusive of all image subresources. Implementations **may** have an address space limit on total size of a resource, which is advertised by this property. **maxResourceSize** **must** be at least 2^{31} .

NOTE

There is no mechanism to query the size of an image before creating it, to compare that size against **maxResourceSize**. If an application attempts to create an image that exceeds this limit, the creation will fail and **vkCreateImage** will return **VK_ERROR_OUT_OF_DEVICE_MEMORY**. While the advertised limit **must** be at least 2^{31} , it **may** not be possible to create an image that approaches that size, particularly for **VK_IMAGE_TYPE_1D**.

If the combination of parameters to **vkGetPhysicalDeviceImageFormatProperties** is not supported by the implementation for use in **vkCreateImage**, then all members of **VkImageFormatProperties** will be filled with zero.

NOTE

Filling **VkImageFormatProperties** with zero for unsupported formats is an exception to the usual rule that output structures have undefined contents on error. This exception was unintentional, but is preserved for backwards compatibility.

34.1.1. Supported Sample Counts

`vkGetPhysicalDeviceImageFormatProperties` returns a bitmask of `VkSampleCountFlagBits` in `sampleCounts` specifying the supported sample counts for the image parameters.

`sampleCounts` will be `VK_SAMPLE_COUNT_1_BIT` if at least one of the following conditions is true:

- `tiling` is `VK_IMAGE_TILING_LINEAR`
- `type` is not `VK_IMAGE_TYPE_2D`
- `flags` contains `VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT`
- Neither the `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT` flag nor the `VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT` flag in `VkFormatProperties::optimalTilingFeatures` returned by `vkGetPhysicalDeviceFormatProperties` is set

Otherwise, the bits set in `sampleCounts` will be the sample counts supported for the specified values of `usage` and `format`. For each bit set in `usage`, the supported sample counts relate to the limits in `VkPhysicalDeviceLimits` as follows:

- If `usage` includes `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT` and `format` is a floating- or fixed-point color format, a superset of `VkPhysicalDeviceLimits::framebufferColorSampleCounts`
- If `usage` includes `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`, and `format` includes a depth component, a superset of `VkPhysicalDeviceLimits::framebufferDepthSampleCounts`
- If `usage` includes `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`, and `format` includes a stencil component, a superset of `VkPhysicalDeviceLimits::framebufferStencilSampleCounts`
- If `usage` includes `VK_IMAGE_USAGE_SAMPLED_BIT`, and `format` includes a color component, a superset of `VkPhysicalDeviceLimits::sampledImageColorSampleCounts`
- If `usage` includes `VK_IMAGE_USAGE_SAMPLED_BIT`, and `format` includes a depth component, a superset of `VkPhysicalDeviceLimits::sampledImageDepthSampleCounts`
- If `usage` includes `VK_IMAGE_USAGE_SAMPLED_BIT`, and `format` is an integer format, a superset of `VkPhysicalDeviceLimits::sampledImageIntegerSampleCounts`
- If `usage` includes `VK_IMAGE_USAGE_STORAGE_BIT`, a superset of `VkPhysicalDeviceLimits::storageImageSampleCounts`

If multiple bits are set in `usage`, `sampleCounts` will be the intersection of the per-usage values described above.

If none of the bits described above are set in `usage`, then there is no corresponding limit in `VkPhysicalDeviceLimits`. In this case, `sampleCounts` **must** include at least `VK_SAMPLE_COUNT_1_BIT`.

34.1.2. Allowed Extent Values Based on Image Type

Implementations **may** support extent values larger than the `required minimum/maximum values` for certain types of images. `VkImageFormatProperties::maxExtent` for each type is subject to the constraints below.

NOTE	Implementations must support images with dimensions up to the <code>required</code>
-------------	--

minimum/maximum values for all types of images. It follows that the query for additional capabilities **must** return extent values that are at least as large as the required values.

For `VK_IMAGE_TYPE_1D`:

- `maxExtent.width` $\geq$ `VkPhysicalDeviceLimits::maxImageDimension1D`
- `maxExtent.height` = 1
- `maxExtent.depth` = 1

For `VK_IMAGE_TYPE_2D` when `flags` does not contain `VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT`:

- `maxExtent.width` $\geq$ `VkPhysicalDeviceLimits::maxImageDimension2D`
- `maxExtent.height` $\geq$ `VkPhysicalDeviceLimits::maxImageDimension2D`
- `maxExtent.depth` = 1

For `VK_IMAGE_TYPE_2D` when `flags` contains `VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT`:

- `maxExtent.width` $\geq$ `VkPhysicalDeviceLimits::maxImageDimensionCube`
- `maxExtent.height` $\geq$ `VkPhysicalDeviceLimits::maxImageDimensionCube`
- `maxExtent.depth` = 1

For `VK_IMAGE_TYPE_3D`:

- `maxExtent.width` $\geq$ `VkPhysicalDeviceLimits::maxImageDimension3D`
- `maxExtent.height` $\geq$ `VkPhysicalDeviceLimits::maxImageDimension3D`
- `maxExtent.depth` $\geq$ `VkPhysicalDeviceLimits::maxImageDimension3D`

Chapter 35. Debugging

To aid developers in tracking down errors in the application’s use of Vulkan, particularly in combination with an external debugger or profiler, *debugging extensions* may be available.

The `VkObjectType` enumeration defines values, each of which corresponds to a specific Vulkan handle type. These values **can** be used to associate debug information with a particular type of object through one or more extensions.

```
// Provided by VK_VERSION_1_0
typedef enum VkObjectType {
    VK_OBJECT_TYPE_UNKNOWN = 0,
    VK_OBJECT_TYPE_INSTANCE = 1,
    VK_OBJECT_TYPE_PHYSICAL_DEVICE = 2,
    VK_OBJECT_TYPE_DEVICE = 3,
    VK_OBJECT_TYPE_QUEUE = 4,
    VK_OBJECT_TYPE_SEMAPHORE = 5,
    VK_OBJECT_TYPE_COMMAND_BUFFER = 6,
    VK_OBJECT_TYPE_FENCE = 7,
    VK_OBJECT_TYPE_DEVICE_MEMORY = 8,
    VK_OBJECT_TYPE_BUFFER = 9,
    VK_OBJECT_TYPE_IMAGE = 10,
    VK_OBJECT_TYPE_EVENT = 11,
    VK_OBJECT_TYPE_QUERY_POOL = 12,
    VK_OBJECT_TYPE_BUFFER_VIEW = 13,
    VK_OBJECT_TYPE_IMAGE_VIEW = 14,
    VK_OBJECT_TYPE_SHADER_MODULE = 15,
    VK_OBJECT_TYPE_PIPELINE_CACHE = 16,
    VK_OBJECT_TYPE_PIPELINE_LAYOUT = 17,
    VK_OBJECT_TYPE_RENDER_PASS = 18,
    VK_OBJECT_TYPE_PIPELINE = 19,
    VK_OBJECT_TYPE_DESCRIPTOR_SET_LAYOUT = 20,
    VK_OBJECT_TYPE_SAMPLER = 21,
    VK_OBJECT_TYPE_DESCRIPTOR_POOL = 22,
    VK_OBJECT_TYPE_DESCRIPTOR_SET = 23,
    VK_OBJECT_TYPE_FRAMEBUFFER = 24,
    VK_OBJECT_TYPE_COMMAND_POOL = 25,
} VkObjectType;
```

Table 56. `VkObjectType` and Vulkan Handle Relationship

<code>VkObjectType</code>	Vulkan Handle Type
<code>VK_OBJECT_TYPE_UNKNOWN</code>	Unknown/Undefined Handle
<code>VK_OBJECT_TYPE_INSTANCE</code>	<code>VkInstance</code>
<code>VK_OBJECT_TYPE_PHYSICAL_DEVICE</code>	<code>VkPhysicalDevice</code>
<code>VK_OBJECT_TYPE_DEVICE</code>	<code>VkDevice</code>
<code>VK_OBJECT_TYPE_QUEUE</code>	<code>VkQueue</code>

VkObjectType	Vulkan Handle Type
VK_OBJECT_TYPE_SEMAPHORE	VkSemaphore
VK_OBJECT_TYPE_COMMAND_BUFFER	VkCommandBuffer
VK_OBJECT_TYPE_FENCE	VkFence
VK_OBJECT_TYPE_DEVICE_MEMORY	VkDeviceMemory
VK_OBJECT_TYPE_BUFFER	VkBuffer
VK_OBJECT_TYPE_IMAGE	VkImage
VK_OBJECT_TYPE_EVENT	VkEvent
VK_OBJECT_TYPE_QUERY_POOL	VkQueryPool
VK_OBJECT_TYPE_BUFFER_VIEW	VkBufferView
VK_OBJECT_TYPE_IMAGE_VIEW	VkImageView
VK_OBJECT_TYPE_SHADER_MODULE	VkShaderModule
VK_OBJECT_TYPE_PIPELINE_CACHE	VkPipelineCache
VK_OBJECT_TYPE_PIPELINE_LAYOUT	VkPipelineLayout
VK_OBJECT_TYPE_RENDER_PASS	VkRenderPass
VK_OBJECT_TYPE_PIPELINE	VkPipeline
VK_OBJECT_TYPE_DESCRIPTOR_SET_LAYOUT	VkDescriptorSetLayout
VK_OBJECT_TYPE_SAMPLER	VkSampler
VK_OBJECT_TYPE_DESCRIPTOR_POOL	VkDescriptorPool
VK_OBJECT_TYPE_DESCRIPTOR_SET	VkDescriptorSet
VK_OBJECT_TYPE_FRAMEBUFFER	VkFramebuffer
VK_OBJECT_TYPE_COMMAND_POOL	VkCommandPool

If this Specification was generated with any such extensions included, they will be described in the remainder of this chapter.

Appendix A: Vulkan Environment for SPIR-V

Shaders for Vulkan are defined by the [Khronos SPIR-V Specification](#) as well as the [Khronos SPIR-V Extended Instructions for GLSL](#) Specification. This appendix defines additional SPIR-V requirements applying to Vulkan shaders.

Versions and Formats

A Vulkan 1.0 implementation **must** support the 1.0 version of SPIR-V and the 1.0 version of the SPIR-V Extended Instructions for GLSL.

A SPIR-V module passed into [vkCreateShaderModule](#) is interpreted as a series of 32-bit words in host endianness, with literal strings packed as described in section 2.2 of the SPIR-V Specification. The first few words of the SPIR-V module **must** be a magic number and a SPIR-V version number, as described in section 2.3 of the SPIR-V Specification.

Capabilities

The [table below](#) lists the set of SPIR-V capabilities that **may** be supported in Vulkan implementations. The application **must** not use any of these capabilities in SPIR-V passed to [vkCreateShaderModule](#) unless one of the following conditions is met for the [VkDevice](#) specified in the [device](#) parameter of [vkCreateShaderModule](#):

- The corresponding field in the table is blank.
- Any corresponding Vulkan feature is enabled.
- Any corresponding Vulkan extension is enabled.
- Any corresponding Vulkan property is supported.
- The corresponding core version is supported (as returned by [VkPhysicalDeviceProperties::apiVersion](#)).

Table 57. List of SPIR-V Capabilities and Corresponding Vulkan Features, Extensions, or Core Version

SPIR-V OpCapability	Vulkan feature, extension, or core version
Matrix	VK_VERSION_1_0
Shader	VK_VERSION_1_0
InputAttachment	VK_VERSION_1_0
Sampled1D	VK_VERSION_1_0
Image1D	VK_VERSION_1_0

SPIR-V OpCapability Vulkan feature, extension, or core version
SampledBuffer VK_VERSION_1_0
ImageBuffer VK_VERSION_1_0
ImageQuery VK_VERSION_1_0
DerivativeControl VK_VERSION_1_0
Geometry VkPhysicalDeviceFeatures::geometryShader
Tessellation VkPhysicalDeviceFeatures::tessellationShader
Float64 VkPhysicalDeviceFeatures::shaderFloat64
Int64 VkPhysicalDeviceFeatures::shaderInt64
Int16 VkPhysicalDeviceFeatures::shaderInt16
TessellationPointSize VkPhysicalDeviceFeatures::shaderTessellationAndGeometryPointSize
GeometryPointSize VkPhysicalDeviceFeatures::shaderTessellationAndGeometryPointSize
ImageGatherExtended VkPhysicalDeviceFeatures::shaderImageGatherExtended
StorageImageMultisample VkPhysicalDeviceFeatures::shaderStorageImageMultisample
UniformBufferArrayDynamicIndexing VkPhysicalDeviceFeatures::shaderUniformBufferArrayDynamicIndexing
SampledImageArrayDynamicIndexing VkPhysicalDeviceFeatures::shaderSampledImageArrayDynamicIndexing
StorageBufferArrayDynamicIndexing VkPhysicalDeviceFeatures::shaderStorageBufferArrayDynamicIndexing
StorageImageArrayDynamicIndexing VkPhysicalDeviceFeatures::shaderStorageImageArrayDynamicIndexing
ClipDistance VkPhysicalDeviceFeatures::shaderClipDistance
CullDistance VkPhysicalDeviceFeatures::shaderCullDistance

SPIR-V OpCapability	Vulkan feature, extension, or core version
ImageCubeArray	VkPhysicalDeviceFeatures::imageCubeArray
SampleRateShading	VkPhysicalDeviceFeatures::sampleRateShading
SparseResidency	VkPhysicalDeviceFeatures::shaderResourceResidency
MinLod	VkPhysicalDeviceFeatures::shaderResourceMinLod
SampledCubeArray	VkPhysicalDeviceFeatures::imageCubeArray
ImageMSArray	VkPhysicalDeviceFeatures::shaderStorageImageMultisample
StorageImageExtendedFormats	VK_VERSION_1_0
InterpolationFunction	VkPhysicalDeviceFeatures::sampleRateShading
StorageImageReadWithoutFormat	VkPhysicalDeviceFeatures::shaderStorageImageReadWithoutFormat
StorageImageWriteWithoutFormat	VkPhysicalDeviceFeatures::shaderStorageImageWriteWithoutFormat
MultiViewport	VkPhysicalDeviceFeatures::multiViewport

The application **must** not pass a SPIR-V module containing any of the following to `vkCreateShaderModule`:

- any `OpCapability` not listed above,
- an unsupported capability, or
- a capability which corresponds to a Vulkan feature or extension which has not been enabled.

SPIR-V Extensions

The following table lists SPIR-V extensions that implementations **may** support. The application **must** not pass a SPIR-V module to `vkCreateShaderModule` that uses the following SPIR-V extensions unless one of the following conditions is met for the `VkDevice` specified in the `device` parameter of `vkCreateShaderModule`:

- Any corresponding Vulkan extension is enabled.
- The corresponding core version is supported (as returned by `VkPhysicalDeviceProperties::apiVersion`).

Table 58. List of SPIR-V Extensions and Corresponding Vulkan Extensions or Core Version

SPIR-V OpExtension	Vulkan extension or core version
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Validation Rules Within a Module

A SPIR-V module passed to `vkCreateShaderModule` **must** conform to the following rules:

Standalone SPIR-V Validation

The following rules **can** be validated with only the SPIR-V module itself. They do not depend on knowledge of the implementation and its capabilities or knowledge of runtime information, such as enabled features.

Valid Usage

- VUID-StandaloneSpirv-None-04633
Every entry point **must** have no return value and accept no arguments
- VUID-StandaloneSpirv-None-04634
The static function-call graph for an entry point **must** not contain cycles; that is, static recursion is not allowed
- VUID-StandaloneSpirv-None-04635
The `Logical` or `PhysicalStorageBuffer64` addressing model **must** be selected
- VUID-StandaloneSpirv-None-04636
`Scope` for execution **must** be limited to `Workgroup` or `Subgroup`
- VUID-StandaloneSpirv-None-04637
If the `Scope` for execution is `Workgroup`, then it **must** only be used in the task, mesh, tessellation control, or compute `Execution Model`
- VUID-StandaloneSpirv-None-04638
`Scope` for memory **must** be limited to `Device`, `QueueFamily`, `Workgroup`, `ShaderCallKHR`, `Subgroup`, or `Invocation`
- VUID-StandaloneSpirv-ExecutionModel-07320
If the `Execution Model` is `TessellationControl`, and the `MemoryModel` is `GLSL450`, the `Scope` for memory **must** not be `Workgroup`
- VUID-StandaloneSpirv-None-07321
If the `Scope` for memory is `Workgroup`, then it **must** only be used in the task, mesh, tessellation control, or compute `Execution Model`
- VUID-StandaloneSpirv-None-04640
If the `Scope` for memory is `ShaderCallKHR`, then it **must** only be used in ray generation, intersection, closest hit, any-hit, miss, and callable `Execution Model`
- VUID-StandaloneSpirv-None-04641
If the `Scope` for memory is `Invocation`, then memory semantics **must** be `None`
- VUID-StandaloneSpirv-None-04642

Scope for group operations **must** be limited to Subgroup

- VUID-StandaloneSpirv-SubgroupVoteKHR-07951
If none of the SubgroupVoteKHR, GroupNonUniform, or SubgroupBallotKHR capabilities are declared, Scope for memory **must** not be Subgroup
- VUID-StandaloneSpirv-None-04643
Storage Class **must** be limited to UniformConstant, Input, Uniform, Output, Workgroup, Private, Function, PushConstant, Image, StorageBuffer, RayPayloadKHR, IncomingRayPayloadKHR, HitAttributeKHR, CallableDataKHR, IncomingCallableDataKHR, ShaderRecordBufferKHR, PhysicalStorageBuffer, or TileImageEXT
- VUID-StandaloneSpirv-None-04644
If the Storage Class is Output, then it **must** not be used in the GLCompute, RayGenerationKHR, IntersectionKHR, AnyHitKHR, ClosestHitKHR, MissKHR, or CallableKHR Execution Model
- VUID-StandaloneSpirv-None-04645
If the Storage Class is Workgroup, then it **must** only be used in the task, mesh, or compute Execution Model
- VUID-StandaloneSpirv-None-08720
If the Storage Class is TileImageEXT, then it **must** only be used in the fragment execution model
- VUID-StandaloneSpirv-OpAtomicStore-04730
OpAtomicStore **must** not use Acquire, AcquireRelease, or SequentiallyConsistent memory semantics
- VUID-StandaloneSpirv-OpAtomicLoad-04731
OpAtomicLoad **must** not use Release, AcquireRelease, or SequentiallyConsistent memory semantics
- VUID-StandaloneSpirv-OpMemoryBarrier-04732
OpMemoryBarrier **must** use one of Acquire, Release, AcquireRelease, or SequentiallyConsistent memory semantics
- VUID-StandaloneSpirv-OpMemoryBarrier-04733
OpMemoryBarrier **must** include at least one Storage Class
- VUID-StandaloneSpirv-OpControlBarrier-04650
If the semantics for OpControlBarrier includes one of Acquire, Release, AcquireRelease, or SequentiallyConsistent memory semantics, then it **must** include at least one Storage Class
- VUID-StandaloneSpirv-OpVariable-04651
Any OpVariable with an Initializer operand **must** have Output, Private, Function, or Workgroup as its Storage Class operand
- VUID-StandaloneSpirv-OpVariable-04734
Any OpVariable with an Initializer operand and Workgroup as its Storage Class operand **must** use OpConstantNull as the initializer
- VUID-StandaloneSpirv-OpReadClockKHR-04652
Scope for OpReadClockKHR **must** be limited to Subgroup or Device
- VUID-StandaloneSpirv-OriginLowerLeft-04653
The OriginLowerLeft Execution Mode **must** not be used; fragment entry points **must** declare

OriginUpperLeft

- VUID-StandaloneSpirv-PixelCenterInteger-04654
The **PixelCenterInteger Execution Mode** **must** not be used (pixels are always centered at half-integer coordinates)
- VUID-StandaloneSpirv-UniformConstant-04655
Any variable in the **UniformConstant Storage Class** **must** be typed as either **OpTypeImage**, **OpTypeSampler**, **OpTypeSampledImage**, **OpTypeAccelerationStructureKHR**, or an array of one of these types
- VUID-StandaloneSpirv-Uniform-06807
Any variable in the **Uniform** or **StorageBuffer Storage Class** **must** be typed as **OpTypeStruct** or an array of this type
- VUID-StandaloneSpirv-PushConstant-06808
Any variable in the **PushConstant Storage Class** **must** be typed as **OpTypeStruct**
- VUID-StandaloneSpirv-OpTypeImage-04656
OpTypeImage **must** declare a scalar 32-bit float, 64-bit integer, or 32-bit integer type for the “Sampled Type” (**RelaxedPrecision** **can** be applied to a sampling instruction and to the variable holding the result of a sampling instruction)
- VUID-StandaloneSpirv-OpTypeImage-04657
OpTypeImage **must** have a “Sampled” operand of 1 (sampled image) or 2 (storage image)
- VUID-StandaloneSpirv-OpTypeSampledImage-06671
OpTypeSampledImage **must** have a **OpTypeImage** with a “Sampled” operand of 1 (sampled image)
- VUID-StandaloneSpirv-Image-04965
The **SPIR-V Type** of the **Image Format** operand of an **OpTypeImage** **must** match the **Sampled Type**, as defined in **Image Format and Type Matching**
- VUID-StandaloneSpirv-OpImageTexelPointer-04658
If an **OpImageTexelPointer** is used in an atomic operation, the image type of the **image** parameter to **OpImageTexelPointer** **must** have an image format of **R64i**, **R64ui**, **R32f**, **R32i**, or **R32ui**
- VUID-StandaloneSpirv-OpImageQuerySizeLod-04659
OpImageQuerySizeLod, **OpImageQueryLod**, and **OpImageQueryLevels** **must** only consume an “Image” operand whose type has its “Sampled” operand set to 1
- VUID-StandaloneSpirv-OpTypeImage-09638
An **OpTypeImage** **must** not have a “Dim” operand of **Rect**
- VUID-StandaloneSpirv-OpTypeImage-06214
An **OpTypeImage** with a “Dim” operand of **SubpassData** **must** have an “Arrayed” operand of 0 (non-arrayed) and a “Sampled” operand of 2 (storage image)
- VUID-StandaloneSpirv-SubpassData-04660
The (u,v) coordinates used for a **SubpassData** **must** be the <id> of a constant vector (0,0)
- VUID-StandaloneSpirv-OpTypeImage-06924
Objects of types **OpTypeImage**, **OpTypeSampler**, **OpTypeSampledImage**, **OpTypeAccelerationStructureKHR**, and arrays of these types **must** not be stored to or

modified

- VUID-StandaloneSpirv-Uniform-06925
Any variable in the **Uniform Storage Class** decorated as **Block** **must** not be stored to or modified
- VUID-StandaloneSpirv-Offset-04663
Image operand **Offset** **must** only be used with **OpImage*Gather** instructions
- VUID-StandaloneSpirv-Offset-04865
Any image instruction which uses an **Offset**, **ConstOffset**, or **ConstOffsets** image operand, **must** only consume a “Sampled Image” operand whose type has its “Sampled” operand set to 1
- VUID-StandaloneSpirv-OpImageGather-04664
The “Component” operand of **OpImageGather**, and **OpImageSparseGather** **must** be the <id> of a constant instruction
- VUID-StandaloneSpirv-OpImage-04777
OpImage*Dref* instructions **must** not consume an image whose **Dim** is 3D
- VUID-StandaloneSpirv-None-04667
Structure types **must** not contain opaque types
- VUID-StandaloneSpirv-BuiltIn-04668
Any **BuiltIn** decoration not listed in **Built-In Variables** **must** not be used
- VUID-StandaloneSpirv-OpEntryPoint-09658
For a given **OpEntryPoint**, any **BuiltIn** decoration **must** not be used more than once by the **Input** interface
- VUID-StandaloneSpirv-OpEntryPoint-09659
For a given **OpEntryPoint**, any **BuiltIn** decoration **must** not be used more than once by the **Output** interface
- VUID-StandaloneSpirv-Location-06672
The **Location** or **Component** decorations **must** only be used with the **Input**, **Output**, **RayPayloadKHR**, **IncomingRayPayloadKHR**, **HitAttributeKHR**, **HitObjectAttributeNV**, **CallableDataKHR**, **IncomingCallableDataKHR**, or **ShaderRecordBufferKHR** storage classes
- VUID-StandaloneSpirv-Location-04915
The **Location** or **Component** decorations **must** not be used with **BuiltIn**
- VUID-StandaloneSpirv-Location-04916
The **Location** decorations **must** be used on **user-defined variables**
- VUID-StandaloneSpirv-Location-04917
If a **user-defined variable** is not a pointer to a **Block** decorated **OpTypeStruct**, then the **OpVariable** **must** have a **Location** decoration
- VUID-StandaloneSpirv-Location-04918
If a **user-defined variable** has a **Location** decoration, and the variable is a pointer to a **OpTypeStruct**, then the members of that structure **must** not have **Location** decorations
- VUID-StandaloneSpirv-Location-04919
If a **user-defined variable** does not have a **Location** decoration, and the variable is a pointer to a **Block** decorated **OpTypeStruct**, then each member of the struct **must** have a

Location decoration

- VUID-StandaloneSpirv-Component-04920
The **Component** decoration value **must** not be greater than 3
- VUID-StandaloneSpirv-Component-04921
If the **Component** decoration is used on an **OpVariable** that has a **OpTypeVector** type with a **Component Type** with a **Width** that is less than or equal to 32, the sum of its **Component Count** and the **Component** decoration value **must** be less than or equal to 4
- VUID-StandaloneSpirv-Component-04922
If the **Component** decoration is used on an **OpVariable** that has a **OpTypeVector** type with a **Component Type** with a **Width** that is equal to 64, the sum of two times its **Component Count** and the **Component** decoration value **must** be less than or equal to 4
- VUID-StandaloneSpirv-Component-04923
The **Component** decorations value **must** not be 1 or 3 for scalar or two-component 64-bit data types
- VUID-StandaloneSpirv-Component-04924
The **Component** decorations **must** not be used with any type that is not a scalar or vector, or an array of such a type
- VUID-StandaloneSpirv-Component-07703
The **Component** decorations **must** not be used for a 64-bit vector type with more than two components
- VUID-StandaloneSpirv-Input-09557
The pointers of any **Input** or **Output Interface user-defined variables** **must** not contain any **PhysicalStorageBuffer Storage Class** pointers
- VUID-StandaloneSpirv-GLSLShared-04669
The **GLSLShared** and **GLSLPacked** decorations **must** not be used
- VUID-StandaloneSpirv-Flat-04670
The **Flat**, **NoPerspective**, **Sample**, and **Centroid** decorations **must** only be used on variables with the **Output** or **Input Storage Class**
- VUID-StandaloneSpirv-Flat-06201
The **Flat**, **NoPerspective**, **Sample**, and **Centroid** decorations **must** not be used on variables with the **Output** storage class in a fragment shader
- VUID-StandaloneSpirv-Flat-06202
The **Flat**, **NoPerspective**, **Sample**, and **Centroid** decorations **must** not be used on variables with the **Input** storage class in a vertex shader
- VUID-StandaloneSpirv-PerVertexKHR-06777
The **PerVertexKHR** decoration **must** only be used on variables with the **Input Storage Class** in a fragment shader
- VUID-StandaloneSpirv-Flat-04744
Any variable with integer or double-precision floating-point type and with **Input Storage Class** in a fragment shader, **must** be decorated **Flat**
- VUID-StandaloneSpirv-ViewportRelativeNV-04672
The **ViewportRelativeNV** decoration **must** only be used on a variable decorated with **Layer**

in the vertex, tessellation evaluation, or geometry shader stages

- VUID-StandaloneSpirv-ViewportRelativeNV-04673

The **ViewportRelativeNV** decoration **must** not be used unless a variable decorated with one of **ViewportIndex** or **ViewportMaskNV** is also statically used by the same **OpEntryPoint**

- VUID-StandaloneSpirv-ViewportMaskNV-04674

The **ViewportMaskNV** and **ViewportIndex** decorations **must** not both be statically used by one or more **OpEntryPoint**'s that form the **pre-rasterization shader stages** of a graphics pipeline

- VUID-StandaloneSpirv-FPRoundingMode-04675

Rounding modes other than round-to-nearest-even and round-towards-zero **must** not be used for the **FPRoundingMode** decoration

- VUID-StandaloneSpirv-Invariant-04677

Variables decorated with **Invariant** and variables with structure types that have any members decorated with **Invariant** **must** be in the **Output** or **Input Storage Class**, **Invariant** used on an **Input Storage Class** variable or structure member has no effect

- VUID-StandaloneSpirv-VulkanMemoryModel-04678

If the **VulkanMemoryModel** capability is not declared, the **Volatile** decoration **must** be used on any variable declaration that includes one of the **SMIDNV**, **WarpIDNV**, **SubgroupSize**, **SubgroupLocalInvocationId**, **SubgroupEqMask**, **SubgroupGeMask**, **SubgroupGtMask**, **SubgroupLeMask**, or **SubgroupLtMask BuiltIn** decorations when used in the ray generation, closest hit, miss, intersection, or callable shaders, or with the **RayTmaxKHR BuiltIn** decoration when used in an intersection shader

- VUID-StandaloneSpirv-VulkanMemoryModel-04679

If the **VulkanMemoryModel** capability is declared, the **OpLoad** instruction **must** use the **Volatile** memory semantics when it accesses into any variable that includes one of the **SMIDNV**, **WarpIDNV**, **SubgroupSize**, **SubgroupLocalInvocationId**, **SubgroupEqMask**, **SubgroupGeMask**, **SubgroupGtMask**, **SubgroupLeMask**, or **SubgroupLtMask BuiltIn** decorations when used in the ray generation, closest hit, miss, intersection, or callable shaders, or with the **RayTmaxKHR BuiltIn** decoration when used in an intersection shader

- VUID-StandaloneSpirv-OpTypeRuntimeArray-04680

OpTypeRuntimeArray **must** only be used for:

- the last member of a **Block**-decorated **OpTypeStruct** in **StorageBuffer** or **PhysicalStorageBuffer** storage **Storage Class**
- **BufferBlock**-decorated **OpTypeStruct** in the **Uniform** storage **Storage Class**
- the outermost dimension of an arrayed variable in the **StorageBuffer**, **Uniform**, or **UniformConstant** storage **Storage Class**
- variables in the **NodePayloadAMD** storage **Storage Class** when the **CoalescingAMD** **Execution Mode** is specified

- VUID-StandaloneSpirv-Function-04681

A type *T* that is an array sized with a specialization constant **must** neither be, nor be contained in, the type *T2* of a variable *V*, unless either: a) *T* is equal to *T2*, b) *V* is declared in the **Function**, or **Private Storage Class**, c) *V* is a non-Block variable in the **Workgroup Storage Class**, or d) *V* is an interface variable with an additional level of arrayness, **as described in interface matching**, and *T* is the member type of the array type *T2*

- VUID-StandaloneSpirv-OpControlBarrier-04682
If **OpControlBarrier** is used in ray generation, intersection, any-hit, closest hit, miss, fragment, vertex, tessellation evaluation, or geometry shaders, the execution Scope **must** be **Subgroup**
- VUID-StandaloneSpirv-LocalSize-06426
For each compute shader entry point, either a **LocalSize** or **LocalSizeId Execution Mode**, or an object decorated with the **WorkgroupSize** decoration **must** be specified
- VUID-StandaloneSpirv-DerivativeGroupQuadsNV-04684
For compute shaders using the **DerivativeGroupQuadsNV** execution mode, the first two dimensions of the local workgroup size **must** be a multiple of two
- VUID-StandaloneSpirv-DerivativeGroupLinearNV-04778
For compute shaders using the **DerivativeGroupLinearNV** execution mode, the product of the dimensions of the local workgroup size **must** be a multiple of four
- VUID-StandaloneSpirv-DerivativeGroupQuadsKHR-10151
For compute, mesh, or task shaders using the **DerivativeGroupQuadsKHR** execution mode, the first two dimensions of the local workgroup size **must** be a multiple of two
- VUID-StandaloneSpirv-DerivativeGroupLinearKHR-10152
For compute, mesh, or task shaders using the **DerivativeGroupLinearKHR** execution mode, the product of the dimensions of the local workgroup size **must** be a multiple of four
- VUID-StandaloneSpirv-OpGroupNonUniformBallotBitCount-04685
If **OpGroupNonUniformBallotBitCount** is used, the group operation **must** be limited to **Reduce**, **InclusiveScan**, or **ExclusiveScan**
- VUID-StandaloneSpirv-None-04686
The *Pointer* operand of all atomic instructions **must** have a **Storage Class** limited to **Uniform**, **Workgroup**, **Image**, **StorageBuffer**, **PhysicalStorageBuffer**, or **TaskPayloadWorkgroupEXT**
- VUID-StandaloneSpirv-Offset-04687
Output variables or block members decorated with **Offset** that have a 64-bit type, or a composite type containing a 64-bit type, **must** specify an **Offset** value aligned to a 8 byte boundary
- VUID-StandaloneSpirv-Offset-04689
The size of any output block containing any member decorated with **Offset** that is a 64-bit type **must** be a multiple of 8
- VUID-StandaloneSpirv-Offset-04690
The first member of an output block specifying a **Offset** decoration **must** specify a **Offset** value that is aligned to an 8 byte boundary if that block contains any member decorated with **Offset** and is a 64-bit type
- VUID-StandaloneSpirv-Offset-04691
Output variables or block members decorated with **Offset** that have a 32-bit type, or a composite type contains a 32-bit type, **must** specify an **Offset** value aligned to a 4 byte boundary
- VUID-StandaloneSpirv-Offset-04692
Output variables, blocks or block members decorated with **Offset** **must** only contain base types that have components that are either 32-bit or 64-bit in size

- VUID-StandaloneSpirv-Offset-04716
Only variables or block members in the output interface decorated with **Offset** **can** be captured for transform feedback, and those variables or block members **must** also be decorated with **XfbBuffer** and **XfbStride**, or inherit **XfbBuffer** and **XfbStride** decorations from a block containing them
- VUID-StandaloneSpirv-XfbBuffer-04693
All variables or block members in the output interface of the entry point being compiled decorated with a specific **XfbBuffer** value **must** all be decorated with identical **XfbStride** values
- VUID-StandaloneSpirv-Stream-04694
If any variables or block members in the output interface of the entry point being compiled are decorated with **Stream**, then all variables belonging to the same **XfbBuffer** **must** specify the same **Stream** value
- VUID-StandaloneSpirv-XfbBuffer-04696
For any two variables or block members in the output interface of the entry point being compiled with the same **XfbBuffer** value, the ranges determined by the **Offset** decoration and the size of the type **must** not overlap
- VUID-StandaloneSpirv-XfbBuffer-04697
All block members in the output interface of the entry point being compiled that are in the same block and have a declared or inherited **XfbBuffer** decoration **must** specify the same **XfbBuffer** value
- VUID-StandaloneSpirv-RayPayloadKHR-04698
RayPayloadKHR Storage Class **must** only be used in ray generation, closest hit or miss shaders
- VUID-StandaloneSpirv-IncomingRayPayloadKHR-04699
IncomingRayPayloadKHR Storage Class **must** only be used in closest hit, any-hit, or miss shaders
- VUID-StandaloneSpirv-IncomingRayPayloadKHR-04700
There **must** be at most one variable with the **IncomingRayPayloadKHR Storage Class** in the input interface of an entry point
- VUID-StandaloneSpirv-HitAttributeKHR-04701
HitAttributeKHR Storage Class **must** only be used in intersection, any-hit, or closest hit shaders
- VUID-StandaloneSpirv-HitAttributeKHR-04702
There **must** be at most one variable with the **HitAttributeKHR Storage Class** in the input interface of an entry point
- VUID-StandaloneSpirv-HitAttributeKHR-04703
A variable with **HitAttributeKHR Storage Class** **must** only be written to in an intersection shader
- VUID-StandaloneSpirv-CallableDataKHR-04704
CallableDataKHR Storage Class **must** only be used in ray generation, closest hit, miss, and callable shaders
- VUID-StandaloneSpirv-IncomingCallableDataKHR-04705

IncomingCallableDataKHR Storage Class **must** only be used in callable shaders

- VUID-StandaloneSpirv-IncomingCallableDataKHR-04706

There **must** be at most one variable with the **IncomingCallableDataKHR Storage Class** in the input interface of an entry point

- VUID-StandaloneSpirv-ShaderRecordBufferKHR-07119

ShaderRecordBufferKHR Storage Class **must** only be used in ray generation, intersection, any-hit, closest hit, callable, or miss shaders

- VUID-StandaloneSpirv-Base-07650

The **Base** operand of **OpPtrAccessChain** **must** have a storage class of **Workgroup**, **StorageBuffer**, or **PhysicalStorageBuffer**

- VUID-StandaloneSpirv-Base-07651

If the **Base** operand of **OpPtrAccessChain** has a **Workgroup Storage Class**, then the **VariablePointers** capability **must** be declared

- VUID-StandaloneSpirv-Base-07652

If the **Base** operand of **OpPtrAccessChain** has a **StorageBuffer Storage Class**, then the **VariablePointers** or **VariablePointersStorageBuffer** capability **must** be declared

- VUID-StandaloneSpirv-PhysicalStorageBuffer64-04708

If the **PhysicalStorageBuffer64** addressing model is enabled, all instructions that support memory access operands and that use a physical pointer **must** include the **Aligned** operand

- VUID-StandaloneSpirv-PhysicalStorageBuffer64-04709

If the **PhysicalStorageBuffer64** addressing model is enabled, any access chain instruction that accesses into a **RowMajor** matrix **must** only be used as the **Pointer** operand to **OpLoad** or **OpStore**

- VUID-StandaloneSpirv-PhysicalStorageBuffer64-04710

If the **PhysicalStorageBuffer64** addressing model is enabled, **OpConvertUToPtr** and **OpConvertPtrToU** **must** use an integer type whose **Width** is 64

- VUID-StandaloneSpirv-OpTypeForwardPointer-04711

OpTypeForwardPointer **must** have a **Storage Class** of **PhysicalStorageBuffer**

- VUID-StandaloneSpirv-None-04745

All block members in a variable with a **Storage Class** of **PushConstant** declared as an array **must** only be accessed by dynamically uniform indices

- VUID-StandaloneSpirv-OpVariable-06673

There **must** not be more than one **OpVariable** in the **PushConstant Storage Class** listed in the **Interface** for each **OpEntryPoint**

- VUID-StandaloneSpirv-OpEntryPoint-06674

Each **OpEntryPoint** **must** not statically use more than one **OpVariable** in the **PushConstant Storage Class**

- VUID-StandaloneSpirv-OpEntryPoint-08721

Each **OpEntryPoint** **must** not have more than one **Input** variable assigned the same **Component** word inside a **Location** slot, either explicitly or implicitly

- VUID-StandaloneSpirv-OpEntryPoint-08722

Each **OpEntryPoint** **must** not have more than one **Output** variable assigned the same **Component** word inside a **Location** slot, either explicitly or implicitly

- VUID-StandaloneSpirv-Result-04780
The **Result Type** operand of any **OpImageRead** or **OpImageSparseRead** instruction **must** be a vector of four components
- VUID-StandaloneSpirv-Base-04781
The **Base** operand of any **OpBitCount**, **OpBitReverse**, **OpBitFieldInsert**, **OpBitFieldSExtract**, or **OpBitFieldUExtract** instruction **must** be a 32-bit integer scalar or a vector of 32-bit integers
- VUID-StandaloneSpirv-PushConstant-06675
Any variable in the **PushConstant** or **StorageBuffer** storage class **must** be decorated as **Block**
- VUID-StandaloneSpirv-Uniform-06676
Any variable in the **Uniform Storage Class** **must** be decorated as **Block** or **BufferBlock**
- VUID-StandaloneSpirv-UniformConstant-06677
Any variable in the **UniformConstant**, **StorageBuffer**, or **Uniform Storage Class** **must** be decorated with **DescriptorSet** and **Binding**
- VUID-StandaloneSpirv-InputAttachmentIndex-06678
Variables decorated with **InputAttachmentIndex** **must** be in the **UniformConstant Storage Class**
- VUID-StandaloneSpirv-DescriptorSet-06491
If a variable is decorated by **DescriptorSet** or **Binding**, the **Storage Class** **must** correspond to an entry in [Shader Resource and Storage Class Correspondence](#)
- VUID-StandaloneSpirv-Input-06778
Variables with a **Storage Class** of **Input** in a fragment shader stage that are decorated with **PerVertexKHR** **must** be declared as arrays
- VUID-StandaloneSpirv-MeshEXT-07102
The module **must** not contain both an entry point that uses the **TaskEXT** or **MeshEXT Execution Model** and an entry point that uses the **TaskNV** or **MeshNV Execution Model**
- VUID-StandaloneSpirv-MeshEXT-07106
In mesh shaders using the **MeshEXT Execution Model** **OpSetMeshOutputsEXT** **must** be called before any outputs are written
- VUID-StandaloneSpirv-MeshEXT-07107
In mesh shaders using the **MeshEXT Execution Model** all variables declared as output **must** not be read from
- VUID-StandaloneSpirv-MeshEXT-07108
In mesh shaders using the **MeshEXT Execution Model** for **OpSetMeshOutputsEXT** instructions, the “Vertex Count” and “Primitive Count” operands **must** not depend on **ViewIndex**
- VUID-StandaloneSpirv-MeshEXT-07109
In mesh shaders using the **MeshEXT Execution Model** variables decorated with **PrimitivePointIndicesEXT**, **PrimitiveLineIndicesEXT**, or **PrimitiveTriangleIndicesEXT** declared as an array **must** not be accessed by indices that depend on **ViewIndex**
- VUID-StandaloneSpirv-MeshEXT-07110

In mesh shaders using the **MeshEXT Execution Model** any values stored in variables decorated with **PrimitivePointIndicesEXT**, **PrimitiveLineIndicesEXT**, or **PrimitiveTriangleIndicesEXT** **must** not depend on **ViewIndex**

- VUID-StandaloneSpirv-MeshEXT-07111

In mesh shaders using the **MeshEXT Execution Model** variables in workgroup or private **Storage Class** declared as or containing a composite type **must** not be accessed by indices that depend on **ViewIndex**

- VUID-StandaloneSpirv-MeshEXT-07330

In mesh shaders using the **MeshEXT Execution Model** the **OutputVertices Execution Mode** **must** be greater than 0

- VUID-StandaloneSpirv-MeshEXT-07331

In mesh shaders using the **MeshEXT Execution Model** the **OutputPrimitivesEXT Execution Mode** **must** be greater than 0

- VUID-StandaloneSpirv-Input-07290

Variables with a **Storage Class** of **Input** or **Output** and a type of **OpTypeBool** **must** be decorated with the **BuiltIn** decoration

- VUID-StandaloneSpirv-TileImageEXT-08723

The tile image variable declarations **must** obey the constraints on the **TileImageEXT Storage Class** and the **Location** decoration described in **Fragment Tile Image Interface**

- VUID-StandaloneSpirv-None-08724

The **TileImageEXT Storage Class** **must** only be used for declaring tile image variables

- VUID-StandaloneSpirv-Pointer-08973

The **Storage Class** of the **Pointer** operand to **OpCooperativeMatrixLoadKHR** or **OpCooperativeMatrixStoreKHR** **must** be limited to **Workgroup**, **StorageBuffer**, or **PhysicalStorageBuffer**

- VUID-StandaloneSpirv-UniformBufferArrayDynamicIndexing-10127

If the **UniformBufferArrayDynamicIndexing** capability is not declared, and an instruction accesses memory through a uniform buffer, the uniform buffer through which that memory is accessed **must** be determined by **constant integral expressions**

- VUID-StandaloneSpirv-SampledImageArrayDynamicIndexing-10128

If the **SampledImageArrayDynamicIndexing** capability is not declared, and an instruction accesses memory through a sampled image or sampler, the sampled image or sampler through which that memory is accessed **must** be determined by **constant integral expressions**

- VUID-StandaloneSpirv-StorageBufferArrayDynamicIndexing-10129

If the **StorageBufferArrayDynamicIndexing** capability is not declared, and an instruction accesses memory through a storage buffer, the storage buffer through which that memory is accessed **must** be determined by **constant integral expressions**

- VUID-StandaloneSpirv-StorageImageArrayDynamicIndexing-10130

If the **StorageImageArrayDynamicIndexing** capability is not declared, and an instruction accesses memory through a storage image, the storage image through which that memory is accessed **must** be determined by **constant integral expressions**

- VUID-StandaloneSpirv-InputAttachmentArrayDynamicIndexing-10131

If the `InputAttachmentArrayDynamicIndexing` capability is not declared, and an instruction accesses memory through an input attachment, the input attachment through which that memory is accessed **must** be determined by [constant integral expressions](#)

- VUID-StandaloneSpirv-UniformTexelBufferArrayDynamicIndexing-10132

If the `UniformTexelBufferArrayDynamicIndexing` capability is not declared, and an instruction accesses memory through a uniform texel buffer, the uniform texel buffer through which that memory is accessed **must** be determined by [constant integral expressions](#)

- VUID-StandaloneSpirv-StorageTexelBufferArrayDynamicIndexing-10133

If the `StorageTexelBufferArrayDynamicIndexing` capability is not declared, and an instruction accesses memory through a storage texel buffer, the storage texel buffer through which that memory is accessed **must** be determined by [constant integral expressions](#)

Runtime SPIR-V Validation

The following rules **must** be validated at runtime. These rules depend on knowledge of the implementation and its capabilities and knowledge of runtime information, such as enabled features.

Valid Usage

- VUID-RuntimeSpirv-OpTypeImage-06269

If `shaderStorageImageWriteWithoutFormat` is not enabled, any variable created with a “Type” of `OpTypeImage` that has a “Sampled” operand of 2 and an “Image Format” operand of `Unknown` **must** be decorated with `NonWritable`

- VUID-RuntimeSpirv-OpTypeImage-06270

If `shaderStorageImageReadWithoutFormat` is not enabled, any variable created with a “Type” of `OpTypeImage` that has a “Sampled” operand of 2 and an “Image Format” operand of `Unknown` **must** be decorated with `NonReadable`

- VUID-RuntimeSpirv-None-09558

Any variable created with a “Type” of `OpTypeImage` that has a “Dim” operand of `SubpassData` **must** be decorated with `InputAttachmentIndex`

- VUID-RuntimeSpirv-OpImageWrite-07112

`OpImageWrite` to any `Image` whose `Image Format` is not `Unknown` **must** have the `Texel` operand contain at least as many components as the corresponding `VkFormat` as given in the [SPIR-V Image Format compatibility table](#)

- VUID-RuntimeSpirv-Location-06272

The sum of `Location` and the number of locations the variable it decorates consumes **must** be less than or equal to the value for the matching `Execution Model` defined in [Shader Input and Output Locations](#)

- VUID-RuntimeSpirv-Location-06428

The maximum number of storage buffers, storage images, and output `Location` decorated color attachments written to in the `Fragment Execution Model` **must** be less than or equal to

maxFragmentCombinedOutputResources

- VUID-RuntimeSpirv-DescriptorSet-06323

DescriptorSet and **Binding** decorations **must** obey the constraints on **Storage Class**, type, and descriptor type described in [DescriptorSet and Binding Assignment](#)

- VUID-RuntimeSpirv-NonWritable-06340

If **fragmentStoresAndAtomics** is not enabled, then all storage image, storage texel buffer, and storage buffer variables in the fragment stage **must** be decorated with the **NonWritable** decoration

- VUID-RuntimeSpirv-NonWritable-06341

If **vertexPipelineStoresAndAtomics** is not enabled, then all storage image, storage texel buffer, and storage buffer variables in the vertex, tessellation, and geometry stages **must** be decorated with the **NonWritable** decoration

- VUID-RuntimeSpirv-None-06342

If **subgroupQuadOperationsInAllStages** is **VK_FALSE**, then **quad subgroup operations** **must** not be used except for in fragment and compute stages

- VUID-RuntimeSpirv-Offset-06344

The first element of the **Offset** operand of **InterpolateAtOffset** **must** be greater than or equal to:

$$\text{frag}_{\text{width}} \times \text{minInterpolationOffset}$$

where $\text{frag}_{\text{width}}$ is the width of the current fragment in pixels

- VUID-RuntimeSpirv-Offset-06345

The first element of the **Offset** operand of **InterpolateAtOffset** **must** be less than or equal to

$$\text{frag}_{\text{width}} \times (\text{maxInterpolationOffset} + \text{ULP}) - \text{ULP}$$

where $\text{frag}_{\text{width}}$ is the width of the current fragment in pixels and $\text{ULP} = 1 / 2^{\text{subPixelInterpolationOffsetBits}}$

- VUID-RuntimeSpirv-Offset-06346

The second element of the **Offset** operand of **InterpolateAtOffset** **must** be greater than or equal to

$$\text{frag}_{\text{height}} \times \text{minInterpolationOffset}$$

where $\text{frag}_{\text{height}}$ is the height of the current fragment in pixels

- VUID-RuntimeSpirv-Offset-06347

The second element of the **Offset** operand of **InterpolateAtOffset** **must** be less than or equal to

$$\text{frag}_{\text{height}} \times (\text{maxInterpolationOffset} + \text{ULP}) - \text{ULP}$$

where $\text{frag}_{\text{height}}$ is the height of the current fragment in pixels and $\text{ULP} = 1 / 2^{\text{subPixelInterpolationOffsetBits}}$

- VUID-RuntimeSpirv-x-06429

In compute shaders using the **GLCompute Execution Model** the **x** size in **LocalSize** or **LocalSizeId** **must** be less than or equal to **VkPhysicalDeviceLimits::maxComputeWorkGroupSize[0]**

- VUID-RuntimeSpirv-y-06430

In compute shaders using the **GLCompute Execution Model** the **y** size in **LocalSize** or **LocalSizeId** **must** be less than or equal to **VkPhysicalDeviceLimits**

::maxComputeWorkGroupSize[1]

- VUID-RuntimeSpirv-z-06431

In compute shaders using the **GLCompute Execution Model** the **z** size in **LocalSize** or **LocalSizeId** **must** be less than or equal to **VkPhysicalDeviceLimits::maxComputeWorkGroupSize[2]**

- VUID-RuntimeSpirv-x-06432

In compute shaders using the **GLCompute Execution Model** the product of **x** size, **y** size, and **z** size in **LocalSize** or **LocalSizeId** **must** be less than or equal to **VkPhysicalDeviceLimits::maxComputeWorkGroupInvocations**

- VUID-RuntimeSpirv-LocalSizeId-06433

The **Execution Mode LocalSizeId** **must** not be used

- VUID-RuntimeSpirv-OpTypeVector-06816

Any **OpTypeVector** output interface variables **must** not have a higher **Component Count** than a matching **OpTypeVector** input interface variable

- VUID-RuntimeSpirv-OpEntryPoint-08743

Any **user-defined variables** shared between the **OpEntryPoint** of two shader stages, and declared with **Input** as its **Storage Class** for the subsequent shader stage, **must** have all **Location** slots and **Component** words declared in the preceding shader stage's **OpEntryPoint** with **Output** as the **Storage Class**

- VUID-RuntimeSpirv-OpEntryPoint-07754

Any **user-defined variables** between the **OpEntryPoint** of two shader stages **must** have the same type and width for each **Component**

- VUID-RuntimeSpirv-OpVariable-08746

Any **OpVariable**, **Block**-decorated **OpTypeStruct**, or **Block**-decorated **OpTypeStruct** members shared between the **OpEntryPoint** of two shader stages **must** have matching decorations as defined in **interface matching**

- VUID-RuntimeSpirv-Workgroup-06530

The sum of size in bytes for variables and **padding** in the **Workgroup Storage Class** in the **GLCompute Execution Model** **must** be less than or equal to **maxComputeSharedMemorySize**

- VUID-RuntimeSpirv-OpVariable-06373

Any **OpVariable** with **Workgroup** as its **Storage Class** **must** not have an **Initializer** operand

- VUID-RuntimeSpirv-OpImage-06376

If an **OpImage*Gather** operation has an image operand of **Offset**, **ConstOffset**, or **ConstOffsets** the offset value **must** be greater than or equal to **minTexelGatherOffset**

- VUID-RuntimeSpirv-OpImage-06377

If an **OpImage*Gather** operation has an image operand of **Offset**, **ConstOffset**, or **ConstOffsets** the offset value **must** be less than or equal to **maxTexelGatherOffset**

- VUID-RuntimeSpirv-OpImageSample-06435

If an **OpImageSample*** or **OpImageFetch*** operation has an image operand of **ConstOffset** then the offset value **must** be greater than or equal to **minTexelOffset**

- VUID-RuntimeSpirv-OpImageSample-06436

If an **OpImageSample*** or **OpImageFetch*** operation has an image operand of **ConstOffset** then the offset value **must** be less than or equal to **maxTexelOffset**

- VUID-RuntimeSpirv-samples-08725
If an `OpTypeImage` has an `MS` operand 0, its bound image **must** have been created with `VkImageCreateInfo::samples` as `VK_SAMPLE_COUNT_1_BIT`
- VUID-RuntimeSpirv-samples-08726
If an `OpTypeImage` has an `MS` operand 1, its bound image **must** not have been created with `VkImageCreateInfo::samples` as `VK_SAMPLE_COUNT_1_BIT`
- VUID-RuntimeSpirv-OpEntryPoint-08727
Each `OpEntryPoint` **must** not have more than one variable decorated with `InputAttachmentIndex` per image aspect of the attachment image bound to it, either explicitly or implicitly as described by [input attachment interface](#)
- VUID-RuntimeSpirv-MeshEXT-09218
In mesh shaders using the `MeshEXT` or `MeshNV Execution Model` and the `OutputPoints Execution Mode`, if the number of output points is greater than 0, a `PointSize` decorated variable **must** be written to for each output point
- VUID-RuntimeSpirv-protectedNoFault-09645
If `protectedNoFault` is not supported, the `Storage Class` of the `PhysicalStorageBuffer` **must** not be used if the buffer being accessed is [protected](#)

Precision and Operation of SPIR-V Instructions

The following rules apply to half, single, and double-precision floating-point instructions:

- Positive and negative infinities and positive and negative zeros are generated as dictated by [IEEE 754](#), but subject to the precisions allowed in the [table below](#).
- Signaling NaNs are not required to be generated and exceptions are never raised. Signaling NaN **may** be converted to quiet NaNs values by any floating-point instruction.
- The set of operations `OpPhi`, `OpSelect`, `OpFunctionCall`, `OpReturnValue`, `OpVectorExtractDynamic`, `OpVectorInsertDynamic`, `OpVectorShuffle`, `OpCompositeConstruct`, `OpCompositeExtract`, `OpCompositeInsert`, `OpTranspose`, `OpCopyObject`, `OpCopyLogical`, `OpCopyMemory`, `OpAtomicLoad`, `OpAtomicStore`, `OpAtomicExchange`, `OpStore`, and `OpLoad` are referred to as *bit-preserving operations*.
- By default, the implementation **may** perform optimizations on half, single, or double-precision floating-point instructions that ignore sign of a zero, or assume that arguments and results are not NaNs or infinities.
- All bit-preserving operations and the following instructions **must** not flush denormalized values: `OpConstant`, `OpConstantComposite`, `OpSpecConstant`, `OpSpecConstantComposite`, and `OpBitcast`.
- Any denormalized value input into a shader or potentially generated by any instruction in a shader (except those listed above) **may** be flushed to 0.
- The rounding mode **cannot** be set, and results will be [correctly rounded](#), as described below.
- NaNs **may** not be generated. Instructions that operate on a NaN **may** not result in a NaN.

The precision of double-precision instructions is at least that of single precision.

The precision of individual operations is defined in [Precision of Individual Operations](#). Subject to

the constraints below, however, implementations **may** reorder or combine operations, resulting in expressions exhibiting different precisions than might be expected from the constituent operations.

Evaluation of Expressions

Implementations **may** rearrange floating-point operations using any of the mathematical properties governing the expressions in precise arithmetic, even where the floating-point operations do not share these properties. This includes, but is not limited to, associativity and distributivity, and **may** involve a different number of rounding steps than would occur if the operations were not rearranged. In shaders that use the **SignedZeroInfNanPreserve Execution Mode** the values **must** be preserved if they are generated after any rearrangement but the **Execution Mode** does not change which rearrangements are valid. This rearrangement **can** be prevented for particular operations by using the **NoContraction** decoration.

Precision of Individual Operations

The precision of individual operations is defined either in terms of rounding (correctly rounded), as an error bound in ULP, or as inherited from a formula as follows:

Correct Result

Operations that are described as returning the "correct result" will return the infinitely precise result which, due to the nature of the operation, will not need rounding.

Correctly Rounded

Operations described as "correctly rounded" will return the infinitely precise result, x , rounded so as to be representable in floating-point. The rounding mode used is not defined but **must** obey the following rules. If x is exactly representable then x will be returned. Otherwise, either the floating-point value closest to and no less than x or the value closest to and no greater than x will be returned.

ULP

Where an error bound of n ULP (units in the last place) is given, for an operation with infinitely precise result x the value returned **must** be in the range $[x - n \times \text{ulp}(x), x + n \times \text{ulp}(x)]$. The function $\text{ulp}(x)$ is defined as follows:

If there exist non-equal, finite floating-point numbers a and b such that $a \leq x \leq b$ then $\text{ulp}(x)$ is the minimum possible distance between such numbers, $\text{ulp}(x) = \min_{a, b} |b - a|$. If such numbers do not exist then $\text{ulp}(x)$ is defined to be the difference between the two non-equal, finite floating-point numbers nearest to x .

Where the range of allowed return values includes any value of magnitude larger than that of the largest representable finite floating-point number, operations **may**, additionally, return either an infinity of the appropriate sign or the finite number with the largest magnitude of the appropriate sign. If the infinitely precise result of the operation is not mathematically defined then the value returned is undefined.

Inherited From ...

Where an operation's precision is described as being inherited from a formula, the result returned **must** be at least as accurate as the result of computing an approximation to x using a formula equivalent to the given formula applied to the supplied inputs. Specifically, the formula given may be transformed using the mathematical associativity, commutativity and distributivity of the operators involved to yield an equivalent formula. The SPIR-V precision rules, when applied to each such formula and the given input values, define a range of permitted values. If NaN is one of the permitted values then the operation may return any result, otherwise let the largest permitted value in any of the ranges be $F_{\max}$ and the smallest be $F_{\min}$. The operation **must** return a value in the range $[x - E, x + E]$ where $E = \max(|x - F_{\min}|, |x - F_{\max}|)$.

For single precision (32 bit) instructions, precisions are **required** to be at least as follows:

Table 59. Precision of Core SPIR-V Instructions

Instruction	Single precision, unless decorated with RelaxedPrecision	Half precision
OpFNegate	Correct result.	
OpFAdd	Correctly rounded.	
OpFSub	Correctly rounded.	
OpFMul, OpVectorTimesScalar, OpMatrixTimesScalar	Correctly rounded.	
OpMatrixTimesVector	Inherited from $\sum_{j=0}^{n-1} M_{ij} \times v_j$.	
OpVectorTimesMatrix	Inherited from $\sum_{j=0}^{n-1} v_j \times M_{ji}$.	
OpMatrixTimesMatrix	Inherited from $\sum_{k=0}^{n-1} M_{ik} \times N_{kj}$.	
OpOuterProduct	Correctly rounded.	
OpDot(x, y)	Inherited from $\sum_{i=0}^{n-1} x_i \times y_i$.	
OpIsNan, OpIsInf	Correct result.	
OpFOrdEqual, OpFUnordEqual	Correct result.	
OpFOrdNotEqual, OpFUnordNotEqual	Correct result.	
OpFOrdLessThan, OpFUnordLessThan	Correct result.	
OpFOrdGreaterThan, OpFUnordGreaterThan	Correct result.	
OpFOrdLessThanEqual, OpFUnordLessThanEqual	Correct result.	
OpFOrdGreaterThanEqual, OpFUnordGreaterThanEqual	Correct result.	
OpFDiv(x,y)	2.5 ULP for $ y = 0$ or $ y $ in the range $[2^{-126}, 2^{126}]$.	2.5 ULP for $ y = 0$ or $ y $ in the range $[2^{-14}, 2^{14}]$.
OpFRem(x,y)	Inherited from $x - y \times \text{trunc}(x/y)$.	
OpFMod(x,y)	Inherited from $x - y \times \text{floor}(x/y)$.	

Instruction	Single precision, unless decorated with RelaxedPrecision	Half precision
OpQuantizeToF16	Correctly rounded.	
conversions between types	Correctly rounded.	

NOTE

The OpFRem and OpFMod instructions use cheap approximations of remainder, and the error can be large due to the discontinuity in trunc() and floor(). This can produce mathematically unexpected results in some cases, such as FMod(x,x) computing x rather than 0, and can also cause the result to have a different sign than the infinitely precise result.

Table 60. Precision of GLSL.std.450 Instructions

Instruction	Single precision, unless decorated with RelaxedPrecision	Half precision
fma()	Inherited from OpFMul followed by OpFAdd.	
exp(x), exp2(x)	$3 + 2 \times x $ ULP.	$1 + 2 \times x $ ULP.
log(), log2()	3 ULP outside the range [0.5, 2.0]. Absolute error $< 2^{-21}$ inside the range [0.5, 2.0].	3 ULP outside the range [0.5, 2.0]. Absolute error $< 2^{-7}$ inside the range [0.5, 2.0].
pow(x, y)	Inherited from exp2($y \times \log_2(x)$).	
sqrt()	Inherited from 1.0 / inversesqrt().	
inversesqrt()	2 ULP.	
radians(x)	Inherited from $x \times C_{\pi_180}$, where C_{π_180} is a correctly rounded approximation to $\frac{\pi}{180}$.	
degrees(x)	Inherited from $x \times C_{180_pi}$, where C_{180_pi} is a correctly rounded approximation to $\frac{180}{\pi}$.	
sin()	Absolute error $\leq 2^{-11}$ inside the range $[-\pi, \pi]$.	Absolute error $\leq 2^{-7}$ inside the range $[-\pi, \pi]$.
cos()	Absolute error $\leq 2^{-11}$ inside the range $[-\pi, \pi]$.	Absolute error $\leq 2^{-7}$ inside the range $[-\pi, \pi]$.
tan()	Inherited from $\frac{\sin()}{\cos()}$.	
asin(x)	Inherited from atan2(x, sqrt(1.0 - x × x)).	
acos(x)	Inherited from atan2(sqrt(1.0 - x × x), x).	
atan(), atan2()	4096 ULP	5 ULP.
sinh(x)	Inherited from $(\exp(x) - \exp(-x)) \times 0.5$.	
cosh(x)	Inherited from $(\exp(x) + \exp(-x)) \times 0.5$.	
tanh()	Inherited from $\frac{\sinh()}{\cosh()}$.	

Instruction	Single precision, unless decorated with RelaxedPrecision	Half precision
<code>asinh(x)</code>	Inherited from $\log(x + \text{sqrt}(x \times x + 1.0))$.	
<code>acosh(x)</code>	Inherited from $\log(x + \text{sqrt}(x \times x - 1.0))$.	
<code>atanh(x)</code>	Inherited from $\log(\frac{1.0+x}{1.0-x}) \times 0.5$.	
<code>frexp()</code>	Correct result.	
<code>ldexp()</code>	Correctly rounded.	
<code>length(x)</code>	Inherited from $\text{sqrt}(\text{dot}(x, x))$.	
<code>distance(x, y)</code>	Inherited from $\text{length}(x - y)$.	
<code>cross()</code>	Inherited from <code>OpFSub(OpFMul, OpFMul)</code> .	
<code>normalize(x)</code>	Inherited from $x \times \text{inversesqrt}(\text{dot}(x, x))$.	
<code>faceforward(N, I, NRef)</code>	Inherited from $\text{dot}(N\text{Ref}, I) < 0.0 ? N : -N$.	
<code>reflect(x, y)</code>	Inherited from $x - 2.0 \times \text{dot}(y, x) \times y$.	
<code>refract(I, N, eta)</code>	Inherited from $k < 0.0 ? 0.0 : \text{eta} \times I - (\text{eta} \times \text{dot}(N, I) + \text{sqrt}(k)) \times N$, where $k = 1 - \text{eta} \times \text{eta} \times (1.0 - \text{dot}(N, I) \times \text{dot}(N, I))$.	
<code>round</code>	Correctly rounded.	
<code>roundEven</code>	Correctly rounded.	
<code>trunc</code>	Correctly rounded.	
<code>fabs</code>	Correct result.	
<code>fsign</code>	Correct result.	
<code>floor</code>	Correctly rounded.	
<code>ceil</code>	Correctly rounded.	
<code>fract</code>	Correctly rounded.	
<code>modf</code>	Correctly rounded.	
<code>fmin</code>	Correct result.	
<code>fmax</code>	Correct result.	
<code>fclamp</code>	Correct result.	
<code>fmix(x, y, a)</code>	Inherited from $x \times (1.0 - a) + y \times a$.	
<code>step</code>	Correctly rounded.	
<code>smoothStep(edge0, edge1, x)</code>	Inherited from $t \times t \times (3.0 - 2.0 \times t)$, where $t = \text{clamp}(\frac{x - \text{edge0}}{\text{edge1} - \text{edge0}}, 0.0, 1.0)$.	
<code>nmin</code>	Correct result.	
<code>nmax</code>	Correct result.	
<code>nclamp</code>	Correct result.	

Instruction	Single precision, unless decorated with RelaxedPrecision	Half precision
packHalf2x16	Correctly rounded.	

GLSL.std.450 extended instructions specifically defined in terms of the above instructions inherit the above errors. GLSL.std.450 extended instructions not listed above and not defined in terms of the above have undefined precision.

For the **OpSRem** and **OpSMod** instructions, if either operand is negative the result is undefined.

NOTE

While the **OpSRem** and **OpSMod** instructions are supported by the Vulkan environment, they require non-negative values and thus do not enable additional functionality beyond what **OpUMod** provides.

Signedness of SPIR-V Image Accesses

SPIR-V associates a signedness with all integer image accesses. This is required in certain parts of the SPIR-V and the Vulkan image access pipeline to ensure defined results. The signedness is determined from a combination of the access instruction's **Image Operands** and the underlying image's **Sampled Type** as follows:

1. If the instruction's **Image Operands** contains the **SignExtend** operand then the access is signed.
2. If the instruction's **Image Operands** contains the **ZeroExtend** operand then the access is unsigned.
3. Otherwise, the image accesses signedness matches that of the **Sampled Type** of the **OpTypeImage** being accessed.

Image Format and Type Matching

When specifying the **Image Format** of an **OpTypeImage**, the converted bit width and type, as shown in the table below, **must** match the **Sampled Type**. The signedness **must** match the **signedness of any access** to the image.

NOTE

Formatted accesses are always converted from a shader readable type to the resource's format or vice versa via **Format Conversion** for reads and **Texel Output Format Conversion** for writes. As such, the bit width and format below do not necessarily match 1:1 with what might be expected for some formats.

For a given **Image Format**, the **Sampled Type** **must** be the type described in the *Type* column of the below table, with its **Literal Width** set to that in the *Bit Width* column. Every access that is made to the image **must** have a signedness equal to that in the *Signedness* column (where applicable).

Image Format	Type-Declaration instructions	Bit Width	Signedness
Unknown	Any	Any	Any

Image Format	Type-Declaration instructions	Bit Width	Signedness
Rgba32f	OpTypeFloat	32	N/A
Rg32f			
R32f			
Rgba16f			
Rg16f			
R16f			
Rgba16			
Rg16			
R16			
Rgba16Snorm			
Rg16Snorm			
R16Snorm			
Rgb10A2			
R11fG11fB10f			
Rgba8			
Rg8			
R8			
Rgba8Snorm			
Rg8Snorm			
R8Snorm			

Image Format	Type-Declaration instructions	Bit Width	Signedness
Rgba32i	OpTypeInt	32	1
Rg32i			
R32i			
Rgba16i			
Rg16i			
R16i			
Rgba8i			
Rg8i			
R8i			
Rgba32ui			0
Rg32ui			
R32ui			
Rgba16ui			
Rg16ui			
R16ui			
Rgb10a2ui			
Rgba8ui			
Rg8ui			
R8ui			
R64i	OpTypeInt	64	1
R64ui			0

The *SPIR-V Type* is defined by an instruction in SPIR-V, declared with the Type-Declaration Instruction, Bit Width, and Signedness from above.

Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types

SPIR-V **Image Dim** values are compatible with **VkImageView viewType** values as defined below:

Table 61. SPIR-V and Vulkan ImageView Dimension Compatibility

SPIR-V Image Dim	Compatible Vulkan ImageView viewType
1D	VK_IMAGE_VIEW_TYPE_1D, VK_IMAGE_VIEW_TYPE_1D_ARRAY
2D	VK_IMAGE_VIEW_TYPE_2D, VK_IMAGE_VIEW_TYPE_2D_ARRAY
3D	VK_IMAGE_VIEW_TYPE_3D
Cube	VK_IMAGE_VIEW_TYPE_CUBE, VK_IMAGE_VIEW_TYPE_CUBE_ARRAY

Compatibility Between SPIR-V Image Formats and Vulkan Formats

SPIR-V **Image Format** values are compatible with **VkFormat** values as defined below:

Table 62. SPIR-V and Vulkan Image Format Compatibility

SPIR-V Image Format	Compatible Vulkan Format
Unknown	Any
R8	VK_FORMAT_R8_UNORM
R8Snorm	VK_FORMAT_R8_SNORM
R8ui	VK_FORMAT_R8_UINT
R8i	VK_FORMAT_R8_SINT
Rg8	VK_FORMAT_R8G8_UNORM
Rg8Snorm	VK_FORMAT_R8G8_SNORM
Rg8ui	VK_FORMAT_R8G8_UINT
Rg8i	VK_FORMAT_R8G8_SINT
Rgba8	VK_FORMAT_R8G8B8A8_UNORM
Rgba8Snorm	VK_FORMAT_R8G8B8A8_SNORM
Rgba8ui	VK_FORMAT_R8G8B8A8_UINT
Rgba8i	VK_FORMAT_R8G8B8A8_SINT
Rgb10A2	VK_FORMAT_A2B10G10R10_UNORM_PACK32
Rgb10a2ui	VK_FORMAT_A2B10G10R10_UINT_PACK32
R16	VK_FORMAT_R16_UNORM
R16Snorm	VK_FORMAT_R16_SNORM
R16ui	VK_FORMAT_R16_UINT
R16i	VK_FORMAT_R16_SINT
R16f	VK_FORMAT_R16_SFLOAT
Rg16	VK_FORMAT_R16G16_UNORM
Rg16Snorm	VK_FORMAT_R16G16_SNORM
Rg16ui	VK_FORMAT_R16G16_UINT
Rg16i	VK_FORMAT_R16G16_SINT
Rg16f	VK_FORMAT_R16G16_SFLOAT
Rgba16	VK_FORMAT_R16G16B16A16_UNORM
Rgba16Snorm	VK_FORMAT_R16G16B16A16_SNORM
Rgba16ui	VK_FORMAT_R16G16B16A16_UINT
Rgba16i	VK_FORMAT_R16G16B16A16_SINT
Rgba16f	VK_FORMAT_R16G16B16A16_SFLOAT
R32ui	VK_FORMAT_R32_UINT
R32i	VK_FORMAT_R32_SINT
R32f	VK_FORMAT_R32_SFLOAT

SPIR-V Image Format	Compatible Vulkan Format
Rg32ui	VK_FORMAT_R32G32_UINT
Rg32i	VK_FORMAT_R32G32_SINT
Rg32f	VK_FORMAT_R32G32_SFLOAT
Rgba32ui	VK_FORMAT_R32G32B32A32_UINT
Rgba32i	VK_FORMAT_R32G32B32A32_SINT
Rgba32f	VK_FORMAT_R32G32B32A32_SFLOAT
R64ui	VK_FORMAT_R64_UINT
R64i	VK_FORMAT_R64_SINT
R11fG11fB10f	VK_FORMAT_B10G11R11_UFLOAT_PACK32

Appendix B: Compressed Image Formats

The compressed texture formats used by Vulkan are described in the specifically identified sections of the [Khronos Data Format Specification](#), version 1.3.

Unless otherwise described, the quantities encoded in these compressed formats are treated as normalized, unsigned values.

Those formats listed as sRGB-encoded have in-memory representations of R, G and B components which are nonlinearly-encoded as R', G', and B'; any alpha component is unchanged. As part of filtering, the nonlinear R', G', and B' values are converted to linear R, G, and B components; any alpha component is unchanged. The conversion between linear and nonlinear encoding is performed as described in the “KHR_DF_TRANSFER_SRGB” section of the Khronos Data Format Specification.

Block-Compressed Image Formats

BC1, BC2 and BC3 formats are described in “S3TC Compressed Texture Image Formats” chapter of the [Khronos Data Format Specification](#). BC4 and BC5 are described in the “RGTC Compressed Texture Image Formats” chapter. BC6H and BC7 are described in the “BPTC Compressed Texture Image Formats” chapter.

Table 63. Mapping of Vulkan BC Formats to Descriptions

VkFormat	Khronos Data Format Specification description
Formats described in the “S3TC Compressed Texture Image Formats” chapter	
VK_FORMAT_BC1_RGB_UNORM_BLOCK	BC1 with no alpha
VK_FORMAT_BC1_RGB_SRGB_BLOCK	BC1 with no alpha, sRGB-encoded
VK_FORMAT_BC1_RGBA_UNORM_BLOCK	BC1 with alpha
VK_FORMAT_BC1_RGBA_SRGB_BLOCK	BC1 with alpha, sRGB-encoded
VK_FORMAT_BC2_UNORM_BLOCK	BC2
VK_FORMAT_BC2_SRGB_BLOCK	BC2, sRGB-encoded
VK_FORMAT_BC3_UNORM_BLOCK	BC3
VK_FORMAT_BC3_SRGB_BLOCK	BC3, sRGB-encoded
Formats described in the “RGTC Compressed Texture Image Formats” chapter	
VK_FORMAT_BC4_UNORM_BLOCK	BC4 unsigned
VK_FORMAT_BC4_SNORM_BLOCK	BC4 signed
VK_FORMAT_BC5_UNORM_BLOCK	BC5 unsigned
VK_FORMAT_BC5_SNORM_BLOCK	BC5 signed
Formats described in the “BPTC Compressed Texture Image Formats” chapter	
VK_FORMAT_BC6H_UFLOAT_BLOCK	BC6H (unsigned version)
VK_FORMAT_BC6H_SFLOAT_BLOCK	BC6H (signed version)
VK_FORMAT_BC7_UNORM_BLOCK	BC7
VK_FORMAT_BC7_SRGB_BLOCK	BC7, sRGB-encoded

ETC Compressed Image Formats

The following formats are described in the “ETC2 Compressed Texture Image Formats” chapter of the [Khronos Data Format Specification](#).

Table 64. Mapping of Vulkan ETC Formats to Descriptions

VkFormat	Khronos Data Format Specification description
VK_FORMAT_ETC2_R8G8B8_UNORM_BLOCK	RGB ETC2
VK_FORMAT_ETC2_R8G8B8_SRGB_BLOCK	RGB ETC2 with sRGB encoding
VK_FORMAT_ETC2_R8G8B8A1_UNORM_BLOCK	RGB ETC2 with punch-through alpha
VK_FORMAT_ETC2_R8G8B8A1_SRGB_BLOCK	RGB ETC2 with punch-through alpha and sRGB
VK_FORMAT_ETC2_R8G8B8A8_UNORM_BLOCK	RGBA ETC2
VK_FORMAT_ETC2_R8G8B8A8_SRGB_BLOCK	RGBA ETC2 with sRGB encoding
VK_FORMAT_EAC_R11_UNORM_BLOCK	Unsigned R11 EAC
VK_FORMAT_EAC_R11_SNORM_BLOCK	Signed R11 EAC
VK_FORMAT_EAC_R11G11_UNORM_BLOCK	Unsigned RG11 EAC
VK_FORMAT_EAC_R11G11_SNORM_BLOCK	Signed RG11 EAC

ASTC Compressed Image Formats

ASTC formats are described in the “ASTC Compressed Texture Image Formats” chapter of the [Khronos Data Format Specification](#).

Table 65. Mapping of Vulkan ASTC Formats to Descriptions

VkFormat	Compressed texel block dimensions	Requested mode
VK_FORMAT_ASTC_4x4_UNORM_BLOCK	4 × 4	Linear LDR
VK_FORMAT_ASTC_4x4_SRGB_BLOCK	4 × 4	sRGB
VK_FORMAT_ASTC_5x4_UNORM_BLOCK	5 × 4	Linear LDR
VK_FORMAT_ASTC_5x4_SRGB_BLOCK	5 × 4	sRGB
VK_FORMAT_ASTC_5x5_UNORM_BLOCK	5 × 5	Linear LDR
VK_FORMAT_ASTC_5x5_SRGB_BLOCK	5 × 5	sRGB
VK_FORMAT_ASTC_6x5_UNORM_BLOCK	6 × 5	Linear LDR
VK_FORMAT_ASTC_6x5_SRGB_BLOCK	6 × 5	sRGB
VK_FORMAT_ASTC_6x6_UNORM_BLOCK	6 × 6	Linear LDR
VK_FORMAT_ASTC_6x6_SRGB_BLOCK	6 × 6	sRGB
VK_FORMAT_ASTC_8x5_UNORM_BLOCK	8 × 5	Linear LDR
VK_FORMAT_ASTC_8x5_SRGB_BLOCK	8 × 5	sRGB
VK_FORMAT_ASTC_8x6_UNORM_BLOCK	8 × 6	Linear LDR
VK_FORMAT_ASTC_8x6_SRGB_BLOCK	8 × 6	sRGB
VK_FORMAT_ASTC_8x8_UNORM_BLOCK	8 × 8	Linear LDR
VK_FORMAT_ASTC_8x8_SRGB_BLOCK	8 × 8	sRGB
VK_FORMAT_ASTC_10x5_UNORM_BLOCK	10 × 5	Linear LDR
VK_FORMAT_ASTC_10x5_SRGB_BLOCK	10 × 5	sRGB
VK_FORMAT_ASTC_10x6_UNORM_BLOCK	10 × 6	Linear LDR
VK_FORMAT_ASTC_10x6_SRGB_BLOCK	10 × 6	sRGB
VK_FORMAT_ASTC_10x8_UNORM_BLOCK	10 × 8	Linear LDR
VK_FORMAT_ASTC_10x8_SRGB_BLOCK	10 × 8	sRGB
VK_FORMAT_ASTC_10x10_UNORM_BLOCK	10 × 10	Linear LDR
VK_FORMAT_ASTC_10x10_SRGB_BLOCK	10 × 10	sRGB
VK_FORMAT_ASTC_12x10_UNORM_BLOCK	12 × 10	Linear LDR
VK_FORMAT_ASTC_12x10_SRGB_BLOCK	12 × 10	sRGB
VK_FORMAT_ASTC_12x12_UNORM_BLOCK	12 × 12	Linear LDR

VkFormat	Compressed texel block dimensions	Requested mode
VK_FORMAT_ASTC_12x12_SRGB_BLOCK	12 × 12	sRGB

ASTC textures containing any HDR blocks **should** not be passed into the API using an sRGB or UNORM texture format.

NOTE

An HDR block in a texture passed using a LDR UNORM format will return the appropriate ASTC error color if the implementation supports only the ASTC LDR profile, but may result in either the error color or a decompressed HDR color if the implementation supports HDR decoding.

The ASTC decode mode is decode_float16.

Note that an implementation **may** use HDR mode when linear LDR mode is requested.

Appendix C: Core Revisions (Informative)

New minor versions of the Vulkan API are defined periodically by the Khronos Vulkan Working Group. These consist of some amount of additional functionality added to the core API, potentially including both new functionality and functionality [promoted](#) from extensions.

Vulkan Version 1.0

Vulkan Version 1.0 was the initial release of the Vulkan API.

New Macros

- [VK_API_VERSION](#)
- [VK_API_VERSION_1_0](#)
- [VK_API_VERSION_MAJOR](#)
- [VK_API_VERSION_MINOR](#)
- [VK_API_VERSION_PATCH](#)
- [VK_API_VERSION_VARIANT](#)
- [VK_DEFINE_HANDLE](#)
- [VK_DEFINE_NON_DISPATCHABLE_HANDLE](#)
- [VK_HEADER_VERSION](#)
- [VK_HEADER_VERSION_COMPLETE](#)
- [VK_MAKE_API_VERSION](#)
- [VK_MAKE_VERSION](#)
- [VK_NULL_HANDLE](#)
- [VK_USE_64_BIT_PTR_DEFINES](#)
- [VK_VERSION_MAJOR](#)
- [VK_VERSION_MINOR](#)
- [VK_VERSION_PATCH](#)

New Base Types

- [VkBool32](#)
- [VkDeviceAddress](#)
- [VkDeviceSize](#)
- [VkFlags](#)
- [VkSampleMask](#)

New Object Types

- [VkBuffer](#)
- [VkBufferView](#)
- [VkCommandBuffer](#)
- [VkCommandPool](#)
- [VkDescriptorPool](#)
- [VkDescriptorSet](#)
- [VkDescriptorSetLayout](#)
- [VkDevice](#)
- [VkDeviceMemory](#)
- [VkEvent](#)
- [VkFence](#)
- [VkFramebuffer](#)
- [VkImage](#)
- [VkImageView](#)
- [VkInstance](#)
- [VkPhysicalDevice](#)
- [VkPipeline](#)
- [VkPipelineCache](#)
- [VkPipelineLayout](#)
- [VkQueryPool](#)
- [VkQueue](#)
- [VkRenderPass](#)
- [VkSampler](#)
- [VkSemaphore](#)
- [VkShaderModule](#)

New Commands

- [vkAllocateCommandBuffers](#)
- [vkAllocateDescriptorSets](#)
- [vkAllocateMemory](#)
- [vkBeginCommandBuffer](#)
- [vkBindBufferMemory](#)
- [vkBindImageMemory](#)

- [vkCmdBeginQuery](#)
- [vkCmdBeginRenderPass](#)
- [vkCmdBindDescriptorSets](#)
- [vkCmdBindIndexBuffer](#)
- [vkCmdBindPipeline](#)
- [vkCmdBindVertexBuffers](#)
- [vkCmdBlitImage](#)
- [vkCmdClearAttachments](#)
- [vkCmdClearColorImage](#)
- [vkCmdClearDepthStencilImage](#)
- [vkCmdCopyBuffer](#)
- [vkCmdCopyBufferToImage](#)
- [vkCmdCopyImage](#)
- [vkCmdCopyImageToBuffer](#)
- [vkCmdCopyQueryPoolResults](#)
- [vkCmdDispatch](#)
- [vkCmdDispatchIndirect](#)
- [vkCmdDraw](#)
- [vkCmdDrawIndexed](#)
- [vkCmdDrawIndexedIndirect](#)
- [vkCmdDrawIndirect](#)
- [vkCmdEndQuery](#)
- [vkCmdEndRenderPass](#)
- [vkCmdExecuteCommands](#)
- [vkCmdFillBuffer](#)
- [vkCmdNextSubpass](#)
- [vkCmdPipelineBarrier](#)
- [vkCmdPushConstants](#)
- [vkCmdResetEvent](#)
- [vkCmdResetQueryPool](#)
- [vkCmdResolveImage](#)
- [vkCmdSetBlendConstants](#)
- [vkCmdSetDepthBias](#)
- [vkCmdSetDepthBounds](#)
- [vkCmdSetEvent](#)

- [vkCmdSetLineWidth](#)
- [vkCmdSetScissor](#)
- [vkCmdSetStencilCompareMask](#)
- [vkCmdSetStencilReference](#)
- [vkCmdSetStencilWriteMask](#)
- [vkCmdSetViewport](#)
- [vkCmdUpdateBuffer](#)
- [vkCmdWaitEvents](#)
- [vkCmdWriteTimestamp](#)
- [vkCreateBuffer](#)
- [vkCreateBufferView](#)
- [vkCreateCommandPool](#)
- [vkCreateComputePipelines](#)
- [vkCreateDescriptorPool](#)
- [vkCreateDescriptorSetLayout](#)
- [vkCreateDevice](#)
- [vkCreateEvent](#)
- [vkCreateFence](#)
- [vkCreateFramebuffer](#)
- [vkCreateGraphicsPipelines](#)
- [vkCreateImage](#)
- [vkCreateImageView](#)
- [vkCreateInstance](#)
- [vkCreatePipelineCache](#)
- [vkCreatePipelineLayout](#)
- [vkCreateQueryPool](#)
- [vkCreateRenderPass](#)
- [vkCreateSampler](#)
- [vkCreateSemaphore](#)
- [vkCreateShaderModule](#)
- [vkDestroyBuffer](#)
- [vkDestroyBufferView](#)
- [vkDestroyCommandPool](#)
- [vkDestroyDescriptorPool](#)
- [vkDestroyDescriptorSetLayout](#)

- [vkDestroyDevice](#)
- [vkDestroyEvent](#)
- [vkDestroyFence](#)
- [vkDestroyFramebuffer](#)
- [vkDestroyImage](#)
- [vkDestroyImageView](#)
- [vkDestroyInstance](#)
- [vkDestroyPipeline](#)
- [vkDestroyPipelineCache](#)
- [vkDestroyPipelineLayout](#)
- [vkDestroyQueryPool](#)
- [vkDestroyRenderPass](#)
- [vkDestroySampler](#)
- [vkDestroySemaphore](#)
- [vkDestroyShaderModule](#)
- [vkDeviceWaitIdle](#)
- [vkEndCommandBuffer](#)
- [vkEnumerateDeviceExtensionProperties](#)
- [vkEnumerateDeviceLayerProperties](#)
- [vkEnumerateInstanceExtensionProperties](#)
- [vkEnumerateInstanceLayerProperties](#)
- [vkEnumeratePhysicalDevices](#)
- [vkFlushMappedMemoryRanges](#)
- [vkFreeCommandBuffers](#)
- [vkFreeDescriptorSets](#)
- [vkFreeMemory](#)
- [vkGetBufferMemoryRequirements](#)
- [vkGetDeviceMemoryCommitment](#)
- [vkGetDeviceProcAddr](#)
- [vkGetDeviceQueue](#)
- [vkGetEventStatus](#)
- [vkGetFenceStatus](#)
- [vkGetImageMemoryRequirements](#)
- [vkGetImageSparseMemoryRequirements](#)
- [vkGetImageSubresourceLayout](#)

- [vkGetInstanceProcAddr](#)
- [vkGetPhysicalDeviceFeatures](#)
- [vkGetPhysicalDeviceFormatProperties](#)
- [vkGetPhysicalDeviceImageFormatProperties](#)
- [vkGetPhysicalDeviceMemoryProperties](#)
- [vkGetPhysicalDeviceProperties](#)
- [vkGetPhysicalDeviceQueueFamilyProperties](#)
- [vkGetPhysicalDeviceSparseImageFormatProperties](#)
- [vkGetPipelineCacheData](#)
- [vkGetQueryPoolResults](#)
- [vkGetRenderAreaGranularity](#)
- [vkInvalidateMappedMemoryRanges](#)
- [vkMapMemory](#)
- [vkMergePipelineCaches](#)
- [vkQueueBindSparse](#)
- [vkQueueSubmit](#)
- [vkQueueWaitIdle](#)
- [vkResetCommandBuffer](#)
- [vkResetCommandPool](#)
- [vkResetDescriptorPool](#)
- [vkResetEvent](#)
- [vkResetFences](#)
- [vkSetEvent](#)
- [vkUnmapMemory](#)
- [vkUpdateDescriptorSets](#)
- [vkWaitForFences](#)

New Structures

- [VkAllocationCallbacks](#)
- [VkApplicationInfo](#)
- [VkAttachmentDescription](#)
- [VkAttachmentReference](#)
- [VkBaseInStructure](#)
- [VkBaseOutStructure](#)
- [VkBindSparseInfo](#)

- [VkBufferCopy](#)
- [VkBufferCreateInfo](#)
- [VkBufferImageCopy](#)
- [VkBufferMemoryBarrier](#)
- [VkBufferViewCreateInfo](#)
- [VkClearAttachment](#)
- [VkClearDepthStencilValue](#)
- [VkClearRect](#)
- [VkCommandBufferAllocateInfo](#)
- [VkCommandBufferBeginInfo](#)
- [VkCommandBufferInheritanceInfo](#)
- [VkCommandPoolCreateInfo](#)
- [VkComponentMapping](#)
- [VkComputePipelineCreateInfo](#)
- [VkCopyDescriptorSet](#)
- [VkDescriptorBufferInfo](#)
- [VkDescriptorImageInfo](#)
- [VkDescriptorPoolCreateInfo](#)
- [VkDescriptorPoolSize](#)
- [VkDescriptorSetAllocateInfo](#)
- [VkDescriptorSetLayoutBinding](#)
- [VkDescriptorSetLayoutCreateInfo](#)
- [VkDeviceCreateInfo](#)
- [VkDeviceQueueCreateInfo](#)
- [VkDispatchIndirectCommand](#)
- [VkDrawIndexedIndirectCommand](#)
- [VkDrawIndirectCommand](#)
- [VkEventCreateInfo](#)
- [VkExtensionProperties](#)
- [VkExtent2D](#)
- [VkExtent3D](#)
- [VkFenceCreateInfo](#)
- [VkFormatProperties](#)
- [VkFramebufferCreateInfo](#)
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- [VkImageBlit](#)
- [VkImageCopy](#)
- [VkImageCreateInfo](#)
- [VkImageFormatProperties](#)
- [VkImageMemoryBarrier](#)
- [VkImageResolve](#)
- [VkImageSubresource](#)
- [VkImageSubresourceLayers](#)
- [VkImageSubresourceRange](#)
- [VkImageViewCreateInfo](#)
- [VkInstanceCreateInfo](#)
- [VkLayerProperties](#)
- [VkMappedMemoryRange](#)
- [VkMemoryAllocateInfo](#)
- [VkMemoryBarrier](#)
- [VkMemoryHeap](#)
- [VkMemoryRequirements](#)
- [VkMemoryType](#)
- [VkOffset2D](#)
- [VkOffset3D](#)
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- [VkPhysicalDeviceLimits](#)
- [VkPhysicalDeviceMemoryProperties](#)
- [VkPhysicalDeviceProperties](#)
- [VkPhysicalDeviceSparseProperties](#)
- [VkPipelineCacheCreateInfo](#)
- [VkPipelineCacheHeaderVersionOne](#)
- [VkPipelineColorBlendAttachmentState](#)
- [VkPipelineColorBlendStateCreateInfo](#)
- [VkPipelineDepthStencilStateCreateInfo](#)
- [VkPipelineDynamicStateCreateInfo](#)
- [VkPipelineInputAssemblyStateCreateInfo](#)
- [VkPipelineMultisampleStateCreateInfo](#)
- [VkPipelineRasterizationStateCreateInfo](#)
- [VkPipelineShaderStageCreateInfo](#)

- [VkPipelineTessellationStateCreateInfo](#)
- [VkPipelineVertexInputStateCreateInfo](#)
- [VkPipelineViewportStateCreateInfo](#)
- [VkPushConstantRange](#)
- [VkQueryPoolCreateInfo](#)
- [VkQueueFamilyProperties](#)
- [VkRect2D](#)
- [VkRenderPassBeginInfo](#)
- [VkRenderPassCreateInfo](#)
- [VkSamplerCreateInfo](#)
- [VkSemaphoreCreateInfo](#)
- [VkSparseBufferMemoryBindInfo](#)
- [VkSparseImageFormatProperties](#)
- [VkSparseImageMemoryBind](#)
- [VkSparseImageMemoryBindInfo](#)
- [VkSparseImageMemoryRequirements](#)
- [VkSparseImageOpaqueMemoryBindInfo](#)
- [VkSparseMemoryBind](#)
- [VkSpecializationInfo](#)
- [VkSpecializationMapEntry](#)
- [VkStencilOpState](#)
- [VkSubmitInfo](#)
- [VkSubpassDependency](#)
- [VkSubpassDescription](#)
- [VkSubresourceLayout](#)
- [VkVertexInputAttributeDescription](#)
- [VkVertexInputBindingDescription](#)
- [VkViewport](#)
- [VkWriteDescriptorSet](#)
- Extending [VkBindDescriptorSetsInfoKHR](#), [VkPushConstantsInfoKHR](#), [VkPushDescriptorSetInfoKHR](#), [VkPushDescriptorSetWithTemplateInfoKHR](#), [VkSetDescriptorBufferOffsetsInfoEXT](#), [VkBindDescriptorBufferEmbeddedSamplersInfoEXT](#), [VkIndirectCommandsLayoutCreateInfoEXT](#):
 - [VkPipelineLayoutCreateInfo](#)
- Extending [VkPipelineShaderStageCreateInfo](#):
 - [VkShaderModuleCreateInfo](#)

New Unions

- [VkClearColorValue](#)
- [VkClearValue](#)

New Function Pointers

- [PFN_vkAllocationFunction](#)
- [PFN_vkFreeFunction](#)
- [PFN_vkInternalAllocationNotification](#)
- [PFN_vkInternalFreeNotification](#)
- [PFN_vkReallocationFunction](#)
- [PFN_vkVoidFunction](#)

New Enums

- [VkAccessFlagBits](#)
- [VkAttachmentDescriptionFlagBits](#)
- [VkAttachmentLoadOp](#)
- [VkAttachmentStoreOp](#)
- [VkBlendFactor](#)
- [VkBlendOp](#)
- [VkBorderColor](#)
- [VkBufferCreateFlagBits](#)
- [VkBufferUsageFlagBits](#)
- [VkColorComponentFlagBits](#)
- [VkCommandBufferLevel](#)
- [VkCommandBufferResetFlagBits](#)
- [VkCommandBufferUsageFlagBits](#)
- [VkCommandPoolCreateFlagBits](#)
- [VkCommandPoolResetFlagBits](#)
- [VkCompareOp](#)
- [VkComponentSwizzle](#)
- [VkCullModeFlagBits](#)
- [VkDependencyFlagBits](#)
- [VkDescriptorPoolCreateFlagBits](#)
- [VkDescriptorSetLayoutCreateFlagBits](#)
- [VkDescriptorType](#)

- [VkDynamicState](#)
- [VkEventCreateFlagBits](#)
- [VkFenceCreateFlagBits](#)
- [VkFilter](#)
- [VkFormat](#)
- [VkFormatFeatureFlagBits](#)
- [VkFramebufferCreateFlagBits](#)
- [VkFrontFace](#)
- [VkImageAspectFlagBits](#)
- [VkImageCreateFlagBits](#)
- [VkImageLayout](#)
- [VkImageTiling](#)
- [VkImageType](#)
- [VkImageUsageFlagBits](#)
- [VkImageViewCreateFlagBits](#)
- [VkImageViewType](#)
- [VkIndexType](#)
- [VkInstanceCreateFlagBits](#)
- [VkInternalAllocationType](#)
- [VkLogicOp](#)
- [VkMemoryHeapFlagBits](#)
- [VkMemoryMapFlagBits](#)
- [VkMemoryPropertyFlagBits](#)
- [VkObjectType](#)
- [VkPhysicalDeviceType](#)
- [VkPipelineBindPoint](#)
- [VkPipelineCacheHeaderVersion](#)
- [VkPipelineCreateFlagBits](#)
- [VkPipelineShaderStageCreateFlagBits](#)
- [VkPipelineStageFlagBits](#)
- [VkPolygonMode](#)
- [VkPrimitiveTopology](#)
- [VkQueryControlFlagBits](#)
- [VkQueryPipelineStatisticFlagBits](#)
- [VkQueryResultFlagBits](#)

- [VkQueryType](#)
- [VkQueueFlagBits](#)
- [VkRenderPassCreateFlagBits](#)
- [VkResult](#)
- [VkSampleCountFlagBits](#)
- [VkSamplerAddressMode](#)
- [VkSamplerCreateFlagBits](#)
- [VkSamplerMipmapMode](#)
- [VkShaderStageFlagBits](#)
- [VkSharingMode](#)
- [VkSparseImageFormatFlagBits](#)
- [VkSparseMemoryBindFlagBits](#)
- [VkStencilFaceFlagBits](#)
- [VkStencilOp](#)
- [VkStructureType](#)
- [VkSubpassContents](#)
- [VkSubpassDescriptionFlagBits](#)
- [VkSystemAllocationScope](#)
- [VkVendorId](#)
- [VkVertexInputRate](#)

New Bitmasks

- [VkAccessFlags](#)
- [VkAttachmentDescriptionFlags](#)
- [VkBufferCreateFlags](#)
- [VkBufferUsageFlags](#)
- [VkBufferViewCreateFlags](#)
- [VkColorComponentFlags](#)
- [VkCommandBufferResetFlags](#)
- [VkCommandBufferUsageFlags](#)
- [VkCommandPoolCreateFlags](#)
- [VkCommandPoolResetFlags](#)
- [VkCullModeFlags](#)
- [VkDependencyFlags](#)
- [VkDescriptorPoolCreateFlags](#)

- [VkDescriptorPoolResetFlags](#)
- [VkDescriptorSetLayoutCreateFlags](#)
- [VkDeviceCreateFlags](#)
- [VkDeviceQueueCreateFlags](#)
- [VkEventCreateFlags](#)
- [VkFenceCreateFlags](#)
- [VkFormatFeatureFlags](#)
- [VkFramebufferCreateFlags](#)
- [VkImageAspectFlags](#)
- [VkImageCreateFlags](#)
- [VkImageUsageFlags](#)
- [VkImageViewCreateFlags](#)
- [VkInstanceCreateFlags](#)
- [VkMemoryHeapFlags](#)
- [VkMemoryMapFlags](#)
- [VkMemoryPropertyFlags](#)
- [VkPipelineCacheCreateFlags](#)
- [VkPipelineColorBlendStateCreateFlags](#)
- [VkPipelineCreateFlags](#)
- [VkPipelineDepthStencilStateCreateFlags](#)
- [VkPipelineDynamicStateCreateFlags](#)
- [VkPipelineInputAssemblyStateCreateFlags](#)
- [VkPipelineLayoutCreateFlags](#)
- [VkPipelineMultisampleStateCreateFlags](#)
- [VkPipelineRasterizationStateCreateFlags](#)
- [VkPipelineShaderStageCreateFlags](#)
- [VkPipelineStageFlags](#)
- [VkPipelineTessellationStateCreateFlags](#)
- [VkPipelineVertexInputStateCreateFlags](#)
- [VkPipelineViewportStateCreateFlags](#)
- [VkQueryControlFlags](#)
- [VkQueryPipelineStatisticFlags](#)
- [VkQueryPoolCreateFlags](#)
- [VkQueryResultFlags](#)
- [VkQueueFlags](#)

- [VkRenderPassCreateFlags](#)
- [VkSampleCountFlags](#)
- [VkSamplerCreateFlags](#)
- [VkSemaphoreCreateFlags](#)
- [VkShaderModuleCreateFlags](#)
- [VkShaderStageFlags](#)
- [VkSparseImageFormatFlags](#)
- [VkSparseMemoryBindFlags](#)
- [VkStencilFaceFlags](#)
- [VkSubpassDescriptionFlags](#)

New Headers

- [vk_platform](#)

New Enum Constants

- [VK_ATTACHMENT_UNUSED](#)
- [VK_FALSE](#)
- [VK_LOD_CLAMP_NONE](#)
- [VK_MAX_DESCRIPTION_SIZE](#)
- [VK_MAX_EXTENSION_NAME_SIZE](#)
- [VK_MAX_MEMORY_HEAPS](#)
- [VK_MAX_MEMORY_TYPES](#)
- [VK_MAX_PHYSICAL_DEVICE_NAME_SIZE](#)
- [VK_QUEUE_FAMILY_IGNORED](#)
- [VK_REMAINING_ARRAY_LAYERS](#)
- [VK_REMAINING_MIP_LEVELS](#)
- [VK_SUBPASS_EXTERNAL](#)
- [VK_TRUE](#)
- [VK_UUID_SIZE](#)
- [VK_WHOLE_SIZE](#)

Appendix D: Layers & Extensions

(Informative)

Extensions to the Vulkan API **can** be defined by authors, groups of authors, and the Khronos Vulkan Working Group. In order not to compromise the readability of the Vulkan Specification, the core Specification does not incorporate most extensions. The online Registry of extensions is available at URL

<https://registry.khronos.org/vulkan/>

and allows generating versions of the Specification incorporating different extensions.

Authors creating extensions and layers **must** follow the mandatory procedures described in the [Vulkan Documentation and Extensions](#) document when creating extensions and layers.

The remainder of this appendix documents a set of extensions chosen when this document was built. Versions of the Specification published in the Registry include:

- Core API + mandatory extensions required of all Vulkan implementations.
- Core API + all registered and published Khronos (**KHR**) extensions.
- Core API + all registered and published extensions.

Extensions are grouped as Khronos **KHR**, multivendor **EXT**, and then alphabetically by author ID. Within each group, extensions are listed in alphabetical order by their name.

Extension Dependencies

Extensions which have dependencies on specific core versions or on other extensions will list such dependencies.

For core versions, the specified version **must** be supported at runtime. All extensions implicitly require support for Vulkan 1.0.

For a device extension, use of any device-level functionality defined by that extension requires that any extensions that extension depends on be enabled.

For any extension, use of any instance-level functionality defined by that extension requires only that any extensions that extension depends on be supported at runtime.

Extension Interactions

Some extensions define APIs which are only supported when other extensions or core versions are supported at runtime. Such interactions are noted as “API Interactions”.

List of Extensions

Appendix E: API Boilerplate

This appendix defines Vulkan API features that are infrastructure required for a complete functional description of Vulkan, but do not logically belong elsewhere in the Specification.

Vulkan Header Files

Vulkan is defined as an API in the C99 language. Khronos provides a corresponding set of header files for applications using the API, which may be used in either C or C++ code. The interface descriptions in the specification are the same as the interfaces defined in these header files, and both are derived from the `vk.xml` XML API Registry, which is the canonical machine-readable description of the Vulkan API. The Registry, scripts used for processing it into various forms, and documentation of the registry schema are available as described at <https://registry.khronos.org/vulkan/#apiregistry>.

Language bindings for other languages can be defined using the information in the Specification and the Registry. Khronos does not provide any such bindings, but third-party developers have created some additional bindings.

Vulkan Combined API Header `vulkan.h` (Informative)

Applications normally will include the header `vulkan.h`. In turn, `vulkan.h` always includes the following headers:

- `vk_platform.h`, defining platform-specific macros and headers.
- `vulkan_core.h`, defining APIs for the Vulkan core and all registered extensions *other* than `window system-specific` and `provisional` extensions, which are included in separate header files.

In addition, specific preprocessor macros defined at the time `vulkan.h` is included cause header files for the corresponding window system-specific and provisional interfaces to be included, as described below.

Vulkan Platform-Specific Header `vk_platform.h` (Informative)

Platform-specific macros and interfaces are defined in `vk_platform.h`. These macros are used to control platform-dependent behavior, and their exact definitions are under the control of specific platforms and Vulkan implementations.

Platform-Specific Calling Conventions

On many platforms the following macros are empty strings, causing platform- and compiler-specific default calling conventions to be used.

`VKAPI_ATTR` is a macro placed before the return type in Vulkan API function declarations. This macro controls calling conventions for C++11 and GCC/Clang-style compilers.

`VKAPI_CALL` is a macro placed after the return type in Vulkan API function declarations. This macro controls calling conventions for MSVC-style compilers.

VKAPI_PTR is a macro placed between the '(' and '*' in Vulkan API function pointer declarations. This macro also controls calling conventions, and typically has the same definition as **VKAPI_ATTR** or **VKAPI_CALL**, depending on the compiler.

With these macros, a Vulkan function declaration takes the form of:

```
VKAPI_ATTR <return_type> VKAPI_CALL <command_name>(<command_parameters>);
```

Additionally, a Vulkan function pointer type declaration takes the form of:

```
typedef <return_type> (VKAPI_PTR *PFN_<command_name>)(<command_parameters>);
```

Platform-Specific Header Control

If the **VK_NO_STDINT_H** macro is defined by the application at compile time, extended integer types used by the Vulkan API, such as **uint8_t**, **must** also be defined by the application. Otherwise, the Vulkan headers will not compile. If **VK_NO_STDINT_H** is not defined, the system **<stdint.h>** is used to define these types. There is a fallback path when Microsoft Visual Studio version 2008 and earlier versions are detected at compile time.

If the **VK_NO_STDDEF_H** macro is defined by the application at compile time, **size_t**, **must** also be defined by the application. Otherwise, the Vulkan headers will not compile. If **VK_NO_STDDEF_H** is not defined, the system **<stddef.h>** is used to define this type.

Vulkan Core API Header **vulkan_core.h**

Applications that do not make use of window system-specific extensions may simply include **vulkan_core.h** instead of **vulkan.h**, although there is usually no reason to do so. In addition to the Vulkan API, **vulkan_core.h** also defines a small number of C preprocessor macros that are described below.

Vulkan Header File Version Number

VK_HEADER_VERSION is the version number of the **vulkan_core.h** header. This value is kept synchronized with the patch version of the released Specification.

```
// Provided by VK_VERSION_1_0
// Version of this file
#define VK_HEADER_VERSION 300
```

VK_HEADER_VERSION_COMPLETE is the complete version number of the **vulkan_core.h** header, comprising the major, minor, and patch versions. The major/minor values are kept synchronized with the complete version of the released Specification. This value is intended for use by automated tools to identify exactly which version of the header was used during their generation.

Applications should not use this value as their **VkApplicationInfo::apiVersion**. Instead applications

should explicitly select a specific fixed major/minor API version using, for example, one of the `VK_API_VERSION_*` values.

```
// Provided by VK_VERSION_1_0
// Complete version of this file
#define VK_HEADER_VERSION_COMPLETE VK_MAKE_API_VERSION(0, 1, 3, VK_HEADER_VERSION)
```

`VK_API_VERSION` is now commented out of `vulkan_core.h` and **cannot** be used.

```
// Provided by VK_VERSION_1_0
// DEPRECATED: This define has been removed. Specific version defines (e.g.
VK_API_VERSION_1_0), or the VK_MAKE_VERSION macro, should be used instead.
// #define VK_API_VERSION VK_MAKE_API_VERSION(0, 1, 0, 0) // Patch version should
always be set to 0
```

Vulkan Handle Macros

`VK_DEFINE_HANDLE` defines a [dispatchable handle](#) type.

```
// Provided by VK_VERSION_1_0

#define VK_DEFINE_HANDLE(object) typedef struct object##_T* object;
```

- `object` is the name of the resulting C type.

The only dispatchable handle types are those related to device and instance management, such as [VkDevice](#).

`VK_DEFINE_NON_DISPATCHABLE_HANDLE` defines a [non-dispatchable handle](#) type.

```
// Provided by VK_VERSION_1_0

#ifndef VK_DEFINE_NON_DISPATCHABLE_HANDLE
    #if (VK_USE_64_BIT_PTR_DEFINES==1)
        #define VK_DEFINE_NON_DISPATCHABLE_HANDLE(object) typedef struct object##_T
    *object;
    #else
        #define VK_DEFINE_NON_DISPATCHABLE_HANDLE(object) typedef uint64_t object;
    #endif
#endif
```

- `object` is the name of the resulting C type.

Most Vulkan handle types, such as [VkBuffer](#), are non-dispatchable.

NOTE The `vulkan_core.h` header allows the `VK_DEFINE_NON_DISPATCHABLE_HANDLE`

and `VK_NULL_HANDLE` definitions to be overridden by the application. If `VK_DEFINE_NON_DISPATCHABLE_HANDLE` is already defined when `vulkan_core.h` is compiled, the default definitions for `VK_DEFINE_NON_DISPATCHABLE_HANDLE` and `VK_NULL_HANDLE` are skipped. This allows the application to define a binary-compatible custom handle which **may** provide more type-safety or other features needed by the application. Applications **must** not define handles in a way that is not binary compatible - where binary compatibility is platform dependent.

`VK_NULL_HANDLE` is a reserved value representing a non-valid object handle. It may be passed to and returned from Vulkan commands only when [specifically allowed](#).

```
// Provided by VK_VERSION_1_0

#ifndef VK_DEFINE_NON_DISPATCHABLE_HANDLE
    #if (VK_USE_64_BIT_PTR_DEFINES==1)
        #if (defined(__cplusplus) && (__cplusplus >= 201103L)) || (defined(_MSVC_LANG)
&& (_MSVC_LANG >= 201103L))
            #define VK_NULL_HANDLE nullptr
        #else
            #define VK_NULL_HANDLE ((void*)0)
        #endif
    #else
        #define VK_NULL_HANDLE 0ULL
    #endif
#endif
#ifndef VK_NULL_HANDLE
    #define VK_NULL_HANDLE 0
#endif
```

`VK_USE_64_BIT_PTR_DEFINES` defines whether the default non-dispatchable handles are declared using either a 64-bit pointer type or a 64-bit unsigned integer type.

`VK_USE_64_BIT_PTR_DEFINES` is set to '1' to use a 64-bit pointer type or any other value to use a 64-bit unsigned integer type.

```
// Provided by VK_VERSION_1_0

#ifndef VK_USE_64_BIT_PTR_DEFINES
    #if defined(__LP64__) || defined(WIN64) || (defined(__x86_64__) &&
!defined(__ILP32__) ) || defined(_M_X64) || defined(__ia64) || defined (_M_IA64) ||
defined(__aarch64__) || defined(__powerpc64__) || (defined(__riscv) && __riscv_xlen ==
64)
        #define VK_USE_64_BIT_PTR_DEFINES 1
    #else
        #define VK_USE_64_BIT_PTR_DEFINES 0
    #endif
#endif
```

NOTE

The `vulkan_core.h` header allows the `VK_USE_64_BIT_PTR_DEFINES` definition to be overridden by the application. This allows the application to select either a 64-bit pointer type or a 64-bit unsigned integer type for non-dispatchable handles in the case where the predefined preprocessor check does not identify the desired configuration.

NOTE

This macro was introduced starting with the Vulkan 1.2.174 headers, and its availability can be checked at compile time by requiring `VK_HEADER_VERSION >= 174`.

It is not available if you are using older headers, such as may be shipped with an older Vulkan SDK. Developers requiring this functionality may wish to include a copy of the current Vulkan headers with their project in this case.

Window System-Specific Header Control (Informative)

To use a Vulkan extension supporting a platform-specific window system, header files for that window system **must** be included at compile time, or platform-specific types **must** be forward-declared. The Vulkan header files are unable to determine whether or not an external header is available at compile time, so platform-specific extensions are provided in separate headers from the core API and platform-independent extensions, allowing applications to decide which ones they need to be defined and how the external headers are included.

Extensions dependent on particular sets of platform headers, or that forward-declare platform-specific types, are declared in a header named for that platform. Before including these platform-specific Vulkan headers, applications **must** include both `vulkan_core.h` and any external native headers the platform extensions depend on.

As a convenience for applications that do not need the flexibility of separate platform-specific Vulkan headers, `vulkan.h` includes `vulkan_core.h`, and then conditionally includes platform-specific Vulkan headers and the external headers they depend on. Applications control which platform-specific headers are included by #defining macros before including `vulkan.h`.

The correspondence between platform-specific extensions, external headers they require, the platform-specific header which declares them, and the preprocessor macros which enable inclusion by `vulkan.h` are shown in the [following table](#).

Table 66. Window System Extensions and Headers

Extension Name	Window System Name	Platform-specific Header	Required External Headers	Controlling <code>vulkan.h</code> Macro
<code>VK_KHR_android_surface</code>	Android	<code>vulkan_android.h</code>	None	<code>VK_USE_PLATFORM_ANDROID_KHR</code>
<code>VK_KHR_wayland_surface</code>	Wayland	<code>vulkan_wayland.h</code>	<code><wayland-client.h></code>	<code>VK_USE_PLATFORM_WAYLAND_KHR</code>

Extension Name	Window System Name	Platform-specific Header	Required External Headers	Controlling <code>vulkan.h</code> Macro
VK_KHR_win32_surface , VK_KHR_external_memory_win32 , VK_KHR_win32_keyed_mutex , VK_KHR_external_semaphore_win32 , VK_KHR_external_fence_win32 , VK_NV_external_memory_win32 , VK_NV_win32_keyed_mutex	Microsoft Windows	vulkan_win32.h	<code><windows.h></code>	<code>VK_USE_PLATFORM_WIN32_KHR</code>
VK_KHR_xcb_surface	X11 Xcb	vulkan_xcb.h	<code><xcb/xcb.h></code>	<code>VK_USE_PLATFORM_XCB_KHR</code>
VK_KHR_xlib_surface	X11 Xlib	vulkan_xlib.h	<code><X11/Xlib.h></code>	<code>VK_USE_PLATFORM_XLIB_KHR</code>
VK_EXT_directfb_surface	DirectFB	vulkan_directfb.h	<code><directfb/directfb.h></code>	<code>VK_USE_PLATFORM_DIRECTFB_EXT</code>
VK_EXT_acquire_xlib_display	X11 XRandR	vulkan_xlib_xrandr.h	<code><X11/Xlib.h></code> , <code><X11/extensions/Xrandr.h></code>	<code>VK_USE_PLATFORM_XLIB_XRANDR_EXT</code>
VK_GGP_stream_descriptor_surface , VK_GGP_frame_token	Google Games Platform	vulkan_ggp.h	<code><ggp_c/vulkan_types.h></code>	<code>VK_USE_PLATFORM_GGP</code>
VK_MVK_ios_surface	iOS	vulkan_ios.h	None	<code>VK_USE_PLATFORM_IOS_MVK</code>
VK_MVK_macos_surface	macOS	vulkan_macos.h	None	<code>VK_USE_PLATFORM_MACOS_MVK</code>
VK_NN_vi_surface	VI	vulkan_vi.h	None	<code>VK_USE_PLATFORM_VI_NN</code>
VK_FUCHSIA_imagepipe_surface	Fuchsia	vulkan_fuchsia.h	<code><zircon/types.h></code>	<code>VK_USE_PLATFORM_FUCHSIA</code>
VK_EXT_metal_surface	Metal on CoreAnimation	vulkan_metal.h	None	<code>VK_USE_PLATFORM_METAL_EXT</code>
VK_QNX_screen_surface	QNX Screen	vulkan_screen.h	<code><screen/screen.h></code>	<code>VK_USE_PLATFORM_SCREEN_QNX</code>

NOTE

This section describes the purpose of the headers independently of the specific underlying functionality of the window system extensions themselves. Each extension name will only link to a description of that extension when viewing a specification built with that extension included.

Provisional Extension Header Control (Informative)

Provisional extensions **should** not be used in production applications. The functionality defined by such extensions **may** change in ways that break backwards compatibility between revisions, and before final release of a non-provisional version of that extension.

Provisional extensions are defined in a separate *provisional header*, `vulkan_beta.h`, allowing applications to decide whether or not to include them. The mechanism is similar to [window system-specific headers](#): before including `vulkan_beta.h`, applications **must** include `vulkan_core.h`.

NOTE

Sometimes a provisional extension will include a subset of its interfaces in `vulkan_core.h`. This may occur if the provisional extension is promoted from an existing vendor or EXT extension and some of the existing interfaces are defined as aliases of the provisional extension interfaces. All other interfaces of that provisional extension which are not aliased will be included in `vulkan_beta.h`.

As a convenience for applications, `vulkan.h` conditionally includes `vulkan_beta.h`. Applications **can** control inclusion of `vulkan_beta.h` by #defining the macro `VK_ENABLE_BETA_EXTENSIONS` before including `vulkan.h`.

NOTE

Starting in version 1.2.171 of the Specification, all provisional enumerants are protected by the macro `VK_ENABLE_BETA_EXTENSIONS`. Applications needing to use provisional extensions must always define this macro, even if they are explicitly including `vulkan_beta.h`. This is a minor change to behavior, affecting only provisional extensions.

NOTE

This section describes the purpose of the provisional header independently of the specific provisional extensions which are contained in that header at any given time. The extension appendices for provisional extensions note their provisional status, and link back to this section for more information. Provisional extensions are intended to provide early access for bleeding-edge developers, with the understanding that extension interfaces may change in response to developer feedback. Provisional extensions are very likely to eventually be updated and released as non-provisional extensions, but there is no guarantee this will happen, or how long it will take if it does happen.

Video Std Headers

Performing video coding operations usually involves the application having to provide various parameters, data structures, or other syntax elements specific to the particular video compression standard used, and the associated semantics are covered by the specification of those.

The interface descriptions of these are available in the header files derived from the `video.xml` XML file, which is the canonical machine-readable description of data structures and enumerations that are associated with the externally-provided video compression standards.

Video Std Header Name	Description	Header File	Related Extensions
<code>vulkan_video_codecs_common</code>	Codec-independent common definitions	<code><vk_video/vulkan_video_codecs_common.h></code>	-
<code>vulkan_video_codec_h264std</code>	ITU-T H.264 common definitions	<code><vk_video/vulkan_video_codec_h264std.h></code>	VK_KHR_video_decode_h264, VK_KHR_video_encode_h264
<code>vulkan_video_codec_h264std_decode</code>	ITU-T H.264 decode-specific definitions	<code><vk_video/vulkan_video_codec_h264std_decode.h></code>	VK_KHR_video_decode_h264
<code>vulkan_video_codec_h264std_encode</code>	ITU-T H.264 encode-specific definitions	<code><vk_video/vulkan_video_codec_h264std_encode.h></code>	VK_KHR_video_encode_h264
<code>vulkan_video_codec_h265std</code>	ITU-T H.265 common definitions	<code><vk_video/vulkan_video_codec_h265std.h></code>	VK_KHR_video_decode_h265, VK_KHR_video_encode_h265
<code>vulkan_video_codec_h265std_decode</code>	ITU-T H.265 decode-specific definitions	<code><vk_video/vulkan_video_codec_h265std_decode.h></code>	VK_KHR_video_decode_h265
<code>vulkan_video_codec_h265std_encode</code>	ITU-T H.265 encode-specific definitions	<code><vk_video/vulkan_video_codec_h265std_encode.h></code>	VK_KHR_video_encode_h265
<code>vulkan_video_codec_av1std</code>	AV1 common definitions	<code><vk_video/vulkan_video_codec_av1std.h></code>	VK_KHR_video_decode_av1
<code>vulkan_video_codec_av1std_decode</code>	AV1 decode-specific definitions	<code><vk_video/vulkan_video_codec_av1std_decode.h></code>	VK_KHR_video_decode_av1

Appendix F: Invariance

The Vulkan specification is not pixel exact. It therefore does not guarantee an exact match between images produced by different Vulkan implementations. However, the specification does specify exact matches, in some cases, for images produced by the same implementation. The purpose of this appendix is to identify and provide justification for those cases that require exact matches.

Repeatability

The obvious and most fundamental case is repeated issuance of a series of Vulkan commands. For any given Vulkan and framebuffer state vector, and for any Vulkan command, the resulting Vulkan and framebuffer state **must** be identical whenever the command is executed on that initial Vulkan and framebuffer state. This repeatability requirement does not apply when using shaders containing side effects (image and buffer variable stores and atomic operations), because these memory operations are not guaranteed to be processed in a defined order.

One purpose of repeatability is avoidance of visual artifacts when a double-buffered scene is redrawn. If rendering is not repeatable, swapping between two buffers rendered with the same command sequence **may** result in visible changes in the image. Such false motion is distracting to the viewer. Another reason for repeatability is testability.

Repeatability, while important, is a weak requirement. Given only repeatability as a requirement, two scenes rendered with one (small) polygon changed in position might differ at every pixel. Such a difference, while within the law of repeatability, is certainly not within its spirit. Additional invariance rules are desirable to ensure useful operation.

Multi-Pass Algorithms

Invariance is necessary for a whole set of useful multi-pass algorithms. Such algorithms render multiple times, each time with a different Vulkan mode vector, to eventually produce a result in the framebuffer. Examples of these algorithms include:

- “Erasing” a primitive from the framebuffer by redrawing it, either in a different color or using the XOR logical operation.
- Using stencil operations to compute capping planes.

Invariance Rules

For a given Vulkan device:

Rule 1 *For any given Vulkan and framebuffer state vector, and for any given Vulkan command, the resulting Vulkan and framebuffer state **must** be identical each time the command is executed on that initial Vulkan and framebuffer state.*

Rule 2 *Changes to the following state values have no side effects (the use of any other state value is not affected by the change):*

Required:

- *Color and depth/stencil attachment contents*
- *Scissor parameters (other than enable)*
- *Write masks (color, depth, stencil)*
- *Clear values (color, depth, stencil)*

Strongly suggested:

- *Stencil parameters (other than enable)*
- *Depth test parameters (other than enable)*
- *Blend parameters (other than enable)*
- *Logical operation parameters (other than enable)*

Corollary 1 *Fragment generation is invariant with respect to the state values listed in Rule 2.*

Rule 3 *The arithmetic of each per-fragment operation is invariant except with respect to parameters that directly control it.*

Corollary 2 *Images rendered into different color attachments of the same framebuffer, either simultaneously or separately using the same command sequence, are pixel identical.*

Rule 4 *Identical pipelines will produce the same result when run multiple times with the same input. The wording “Identical pipelines” means [VkPipeline](#) objects that have been created with identical SPIR-V binaries and identical state, which are then used by commands executed using the same Vulkan state vector. Invariance is relaxed for shaders with side effects, such as performing stores or atomics.*

Rule 5 *All fragment shaders that either conditionally or unconditionally assign [FragCoord.z](#) to [FragDepth](#) are depth-invariant with respect to each other, for those fragments where the assignment to [FragDepth](#) actually is done.*

If a sequence of Vulkan commands specifies primitives to be rendered with shaders containing side effects (image and buffer variable stores and atomic operations), invariance rules are relaxed. In particular, rule 1, corollary 2, and rule 4 do not apply in the presence of shader side effects.

The following weaker versions of rules 1 and 4 apply to Vulkan commands involving shader side effects:

Rule 6 *For any given Vulkan and framebuffer state vector, and for any given Vulkan command, the contents of any framebuffer state not directly or indirectly affected by results of shader image or buffer variable stores or atomic operations **must** be identical each time the command is executed on that initial Vulkan and framebuffer state.*

Rule 7 *Identical pipelines will produce the same result when run multiple times with the same input as long as:*

- *shader invocations do not use image atomic operations;*

- *no framebuffer memory is written to more than once by image stores, unless all such stores write the same value; and*
- *no shader invocation, or other operation performed to process the sequence of commands, reads memory written to by an image store.*

NOTE

The OpenGL specification has the following invariance rule: Consider a primitive p' obtained by translating a primitive p through an offset (x, y) in window coordinates, where x and y are integers. As long as neither p' nor p is clipped, it **must** be the case that each fragment f' produced from p' is identical to a corresponding fragment f from p except that the center of f' is offset by (x, y) from the center of f .

This rule does not apply to Vulkan and is an intentional difference from OpenGL.

When any sequence of Vulkan commands triggers shader invocations that perform image stores or atomic operations, and subsequent Vulkan commands read the memory written by those shader invocations, these operations **must** be explicitly synchronized.

Tessellation Invariance

When using a pipeline containing tessellation evaluation shaders, the fixed-function tessellation primitive generator consumes the input patch specified by an application and emits a new set of primitives. The following invariance rules are intended to provide repeatability guarantees. Additionally, they are intended to allow an application with a carefully crafted tessellation evaluation shader to ensure that the sets of triangles generated for two adjacent patches have identical vertices along shared patch edges, avoiding “cracks” caused by minor differences in the positions of vertices along shared edges.

Rule 1 *When processing two patches with identical outer and inner tessellation levels, the tessellation primitive generator will emit an identical set of point, line, or triangle primitives as long as the pipeline used to process the patch primitives has tessellation evaluation shaders specifying the same tessellation mode, spacing, vertex order, and point mode decorations. Two sets of primitives are considered identical if and only if they contain the same number and type of primitives and the generated tessellation coordinates for the vertex numbered m of the primitive numbered n are identical for all values of m and n .*

Rule 2 *The set of vertices generated along the outer edge of the subdivided primitive in triangle and quad tessellation, and the tessellation coordinates of each, depend only on the corresponding outer tessellation level and the spacing decorations in the tessellation shaders of the pipeline.*

Rule 3 *The set of vertices generated when subdividing any outer primitive edge is always symmetric. For triangle tessellation, if the subdivision generates a vertex with tessellation coordinates of the form $(0, x, 1-x)$, $(x, 0, 1-x)$, or $(x, 1-x, 0)$, it will also generate a vertex with coordinates of exactly $(0, 1-x, x)$, $(1-x, 0, x)$, or $(1-x, x, 0)$, respectively. For quad tessellation, if the subdivision generates a vertex with coordinates of $(x, 0)$ or $(0, x)$, it will also generate a vertex with coordinates of exactly $(1-x, 0)$ or $(0, 1-x)$, respectively. For isoline tessellation, if it generates vertices at $(0, x)$ and $(1, x)$ where x is not zero, it will also generate vertices at exactly $(0, 1-x)$ and $(1, 1-x)$, respectively.*

Rule 4 *The set of vertices generated when subdividing outer edges in triangular and quad tessellation*

must be independent of the specific edge subdivided, given identical outer tessellation levels and spacing. For example, if vertices at $(x, 1 - x, 0)$ and $(1 - x, x, 0)$ are generated when subdividing the $w = 0$ edge in triangular tessellation, vertices **must** be generated at $(x, 0, 1 - x)$ and $(1 - x, 0, x)$ when subdividing an otherwise identical $v = 0$ edge. For quad tessellation, if vertices at $(x, 0)$ and $(1 - x, 0)$ are generated when subdividing the $v = 0$ edge, vertices **must** be generated at $(0, x)$ and $(0, 1 - x)$ when subdividing an otherwise identical $u = 0$ edge.

Rule 5 When processing two patches that are identical in all respects enumerated in rule 1 except for vertex order, the set of triangles generated for triangle and quad tessellation **must** be identical except for vertex and triangle order. For each triangle $n1$ produced by processing the first patch, there **must** be a triangle $n2$ produced when processing the second patch each of whose vertices has the same tessellation coordinates as one of the vertices in $n1$.

Rule 6 When processing two patches that are identical in all respects enumerated in rule 1 other than matching outer tessellation levels and/or vertex order, the set of interior triangles generated for triangle and quad tessellation **must** be identical in all respects except for vertex and triangle order. For each interior triangle $n1$ produced by processing the first patch, there **must** be a triangle $n2$ produced when processing the second patch each of whose vertices has the same tessellation coordinates as one of the vertices in $n1$. A triangle produced by the tessellator is considered an interior triangle if none of its vertices lie on an outer edge of the subdivided primitive.

Rule 7 For quad and triangle tessellation, the set of triangles connecting an inner and outer edge depends only on the inner and outer tessellation levels corresponding to that edge and the spacing decorations.

Rule 8 The value of all defined components of **TessCoord** will be in the range $[0, 1]$. Additionally, for any defined component x of **TessCoord**, the results of computing $1.0 - x$ in a tessellation evaluation shader will be exact. If any floating-point values in the range $[0, 1]$ fail to satisfy this property, such values **must** not be used as tessellation coordinate components.

Appendix G: Lexicon

This appendix defines terms, abbreviations, and API prefixes used in the Specification.

Glossary

The terms defined in this section are used consistently throughout the Specification and may be used with or without capitalization.

Accessible (Descriptor Binding)

A descriptor binding is accessible to a shader stage if that stage is included in the `stageFlags` of the descriptor binding. Descriptors using that binding **can** only be used by stages in which they are accessible.

Acquire Operation (Resource)

An operation that acquires ownership of an image subresource or buffer range.

Adjacent Vertex

A vertex in an adjacency primitive topology that is not part of a given primitive, but is accessible in geometry shaders.

Alias (API type/command)

An identical definition of another API type/command with the same behavior but a different name.

Aliased Range (Memory)

A range of a device memory allocation that is bound to multiple resources simultaneously.

Allocation Scope

An association of a host memory allocation to a parent object or command, where the allocation's lifetime ends before or at the same time as the parent object is freed or destroyed, or during the parent command.

API command

Any command defined in the Vulkan specification. These entry points all have a `vk` prefix.

Aspect (Image)

Some image types contain multiple kinds (called “aspects”) of data for each pixel, where each aspect is used in a particular way by the pipeline and **may** be stored differently or separately from other aspects. For example, the color components of an image format make up the color aspect of the image, and **can** be used as a framebuffer color attachment. Some operations, like depth testing, operate only on specific aspects of an image.

Attachment (Render Pass)

A zero-based integer index name used in render pass creation to refer to a framebuffer attachment that is accessed by one or more subpasses. The index also refers to an attachment description which includes information about the properties of the image view that will later be

attached.

Availability Operation

An operation that causes the values generated by specified memory write accesses to become available for future access.

Available

A state of values written to memory that allows them to be made visible.

Back-Facing

See Facingness.

Batch

A single structure submitted to a queue as part of a [queue submission command](#), describing a set of queue operations to execute.

Backwards Compatibility

A given version of the API is backwards compatible with an earlier version if an application, relying only on valid behavior and functionality defined by the earlier specification, is able to correctly run against each version without any modification. This assumes no active attempt by that application to not run when it detects a different version.

Binary Semaphore

A semaphore with a boolean payload indicating whether the semaphore is signaled or unsignaled. Represented by a [VkSemaphore](#) object .

Binding (Memory)

An association established between a range of a resource object and a range of a memory object. These associations determine the memory locations affected by operations performed on elements of a resource object. Memory bindings are established using the [vkBindBufferMemory](#) command for non-sparse buffer objects, using the [vkBindImageMemory](#) command for non-sparse image objects , and using the [vkQueueBindSparse](#) command for sparse resources .

Blend Constant

Four floating-point (RGBA) values used as an input to blending.

Blending

Arithmetic operations between a fragment color value and a value in a color attachment that produce a final color value to be written to the attachment.

Buffer

A resource that represents a linear array of data in device memory. Represented by a [VkBuffer](#) object.

Buffer View

An object that represents a range of a specific buffer, and state controlling how the contents are interpreted. Represented by a [VkBufferView](#) object.

Built-In Variable

A variable decorated in a shader, where the decoration makes the variable take values provided by the execution environment or values that are generated by fixed-function pipeline stages.

Built-In Interface Block

A block defined in a shader containing only variables decorated with built-in decorations, and is used to match against other shader stages.

Clip Coordinates

The homogeneous coordinate space in which vertex positions (**Position** decoration) are written by [pre-rasterization shader stages](#).

Clip Distance

A built-in output from [pre-rasterization shader stages](#) defining a clip half-space against which the primitive is clipped.

Clip Volume

The intersection of the view volume with all clip half-spaces.

Color Attachment

A subpass attachment point, or image view, that is the target of fragment color outputs and blending.

Color Renderable Format

A [VkFormat](#) where **VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT** is set in one of the following, depending on the image's tiling:

- [VkFormatProperties::linearTilingFeatures](#)
- [VkFormatProperties::optimalTilingFeatures](#)

Combined Image Sampler

A descriptor type that includes both a sampled image and a sampler.

Command Buffer

An object that records commands to be submitted to a queue. Represented by a [VkCommandBuffer](#) object.

Command Pool

An object that command buffer memory is allocated from, and that owns that memory. Command pools aid multithreaded performance by enabling different threads to use different allocators, without internal synchronization on each use. Represented by a [VkCommandPool](#) object.

Compatible Allocator

When allocators are compatible, allocations from each allocator **can** be freed by the other allocator.

Compatible Image Formats

When formats are compatible, images created with one of the formats **can** have image views created from it using any of the compatible formats. Also see *Size-Compatible Image Formats*.

Compatible Queues

Queues within a queue family. Compatible queues have identical properties.

Complete Mipmap Chain

The entire set of mip levels that can be provided for an image, from the largest application-specified mip level size down to the *minimum mip level size*. See [Image Mip Level Sizing](#).

Component (Format)

A distinct part of a format. Color components are represented with **R**, **G**, **B**, and **A**. Depth and stencil components are represented with **D** and **S**. Formats **can** have multiple instances of the same component. Some formats have other notations such as **E** or **X** which are not considered a component of the format.

Compressed Texel Block

An element of an image having a block-compressed format, comprising a rectangular block of texel values that are encoded as a single value in memory. Compressed texel blocks of a particular block-compressed format have a corresponding width, height, and depth defining the dimensions of these elements in units of texels, and a size in bytes of the encoding in memory.

Constant Integral Expressions

A SPIR-V constant instruction whose type is **OpTypeInt**. See *Constant Instruction* in section 2.2.1 “Instructions” of the [Khronos SPIR-V Specification](#).

Coverage Index

The index of a sample in the coverage mask.

Coverage Mask

A bitfield associated with a fragment representing the samples that were determined to be covered based on the result of rasterization, and then subsequently modified by fragment operations or the fragment shader.

Cull Distance

A built-in output from [pre-rasterization shader stages](#) defining a cull half-space where the primitive is rejected if all vertices have a negative value for the same cull distance.

Cull Volume

The intersection of the view volume with all cull half-spaces.

Decoration (SPIR-V)

Auxiliary information such as built-in variables, stream numbers, invariance, interpolation type, relaxed precision, etc., added to variables or structure-type members through decorations.

Deprecated (feature)

A feature is deprecated if it is no longer recommended as the correct or best way to achieve its

intended purpose.

Depth/Stencil Attachment

A subpass attachment point, or image view, that is the target of depth and/or stencil test operations and writes.

Depth/Stencil Format

A [VkFormat](#) that includes depth and/or stencil components.

Depth/Stencil Image (or ImageView)

A [VkImage](#) (or [VkImageView](#)) with a depth/stencil format.

Derivative Group

A set of fragment shader invocations that cooperate to compute derivatives, including implicit derivatives for sampled image operations.

Descriptor

Information about a resource or resource view written into a descriptor set that is used to access the resource or view from a shader.

Descriptor Binding

An entry in a descriptor set layout corresponding to zero or more descriptors of a single descriptor type in a set. Defined by a [VkDescriptorSetLayoutBinding](#) structure.

Descriptor Pool

An object that descriptor sets are allocated from, and that owns the storage of those descriptor sets. Descriptor pools aid multithreaded performance by enabling different threads to use different allocators, without internal synchronization on each use. Represented by a [VkDescriptorPool](#) object.

Descriptor Set

An object that resource descriptors are written into via the API, and that **can** be bound to a command buffer such that the descriptors contained within it **can** be accessed from shaders. Represented by a [VkDescriptorSet](#) object.

Descriptor Set Layout

An object defining the set of resources (types and counts) and their relative arrangement (in the binding namespace) within a descriptor set. Used when allocating descriptor sets and when creating pipeline layouts. Represented by a [VkDescriptorSetLayout](#) object.

Device

The processor(s) and execution environment that perform tasks requested by the application via the Vulkan API.

Device Memory

Memory accessible to the device. Represented by a [VkDeviceMemory](#) object.

Device-Level Command

Any command that is dispatched from a logical device, or from a child object of a logical device.

Device-Level Functionality

All device-level commands and objects, and their structures, enumerated types, and enumerants. Additionally, physical-device-level functionality defined by a [device extension](#) is also considered device-level functionality.

Device-Level Object

Logical device objects and their child objects. For example, [VkDevice](#), [VkQueue](#), and [VkCommandBuffer](#) objects are device-level objects.

Device-Local Memory

Memory that is connected to the device, and **may** be more performant for device access than host-local memory.

Direct Drawing Commands

Drawing commands that take all their parameters as direct arguments to the command (and not sourced via structures in buffer memory as the *indirect drawing commands*). Includes [vkCmdDraw](#), and [vkCmdDrawIndexed](#).

Dispatchable Command

A non-global command. The first argument to each dispatchable command is a dispatchable handle type.

Dispatchable Handle

A handle of a pointer handle type which **may** be used by layers as part of intercepting API commands.

Dispatching Commands

Commands that provoke work using a compute pipeline. Includes [vkCmdDispatch](#) and [vkCmdDispatchIndirect](#).

Drawing Commands

Commands that provoke work using a graphics pipeline. Includes [vkCmdDraw](#), [vkCmdDrawIndexed](#), [vkCmdDrawIndirect](#), and [vkCmdDrawIndexedIndirect](#).

Duration (Command)

The *duration* of a Vulkan command refers to the interval between calling the command and its return to the caller.

Dynamic Storage Buffer

A storage buffer whose offset is specified each time the storage buffer is bound to a command buffer via a descriptor set.

Dynamic Uniform Buffer

A uniform buffer whose offset is specified each time the uniform buffer is bound to a command buffer via a descriptor set.

Dynamically Uniform

See *Dynamically Uniform* in section 2.2 “Terms” of the [Khronos SPIR-V Specification](#).

Element

Arrays are composed of multiple elements, where each element exists at a unique index within that array. Used primarily to describe data passed to or returned from the Vulkan API.

Explicitly-Enabled Layer

A layer enabled by the application by adding it to the enabled layer list in [vkCreateInstance](#) or [vkCreateDevice](#).

Event

A synchronization primitive that is signaled when execution of previous commands completes through a specified set of pipeline stages. Events can be waited on by the device and polled by the host. Represented by a [VkEvent](#) object.

Executable State (Command Buffer)

A command buffer that has ended recording commands and **can** be executed. See also Initial State and Recording State.

Execution Dependency

A dependency that guarantees that certain pipeline stages’ work for a first set of commands has completed execution before certain pipeline stages’ work for a second set of commands begins execution. This is accomplished via pipeline barriers, subpass dependencies, events, or implicit ordering operations.

Execution Dependency Chain

A sequence of execution dependencies that transitively act as a single execution dependency.

Extension Scope

The set of objects and commands that **can** be affected by an extension. Extensions are either device scope or instance scope.

Extending Structure

A structure type which may appear in the [pNext chain](#) of another structure, extending the functionality of the other structure. Extending structures may be defined by either core API versions or extensions.

External synchronization

A type of synchronization **required** of the application, where parameters defined to be externally synchronized **must** not be used simultaneously in multiple threads.

Facingness (Polygon)

A classification of a polygon as either front-facing or back-facing, depending on the orientation (winding order) of its vertices.

Facingness (Fragment)

A fragment is either front-facing or back-facing, depending on the primitive it was generated

from. If the primitive was a polygon (regardless of polygon mode), the fragment inherits the facingness of the polygon. All other fragments are front-facing.

Fence

A synchronization primitive that is signaled when a set of batches or sparse binding operations complete execution on a queue. Fences **can** be waited on by the host. Represented by a [VkFence](#) object.

Flat Shading

A property of a vertex attribute that causes the value from a single vertex (the provoking vertex) to be used for all vertices in a primitive, and for interpolation of that attribute to return that single value unaltered.

Format Features

A set of features from [VkFormatFeatureFlagBits](#) that a [VkFormat](#) is capable of using for various commands. The list is determined by factors such as [VkImageTiling](#).

Fragment

A rectangular framebuffer region with associated data produced by [rasterization](#) and processed by [fragment operations](#) including the fragment shader.

Fragment Area

The width and height, in pixels, of a fragment.

Fragment Input Attachment Interface

Variables with [UniformConstant](#) storage class and a decoration of [InputAttachmentIndex](#) that are statically used by a fragment shader's entry point, which receive values from input attachments.

Fragment Output Interface

A fragment shader entry point's variables with [Output](#) storage class, which output to color and/or depth/stencil attachments.

Framebuffer

A collection of image views and a set of dimensions that, in conjunction with a render pass, define the inputs and outputs used by drawing commands. Represented by a [VkFramebuffer](#) object.

Framebuffer Attachment

One of the image views used in a framebuffer.

Framebuffer Coordinates

A coordinate system in which adjacent pixels' coordinates differ by 1 in x and/or y, with (0,0) in the upper left corner and pixel centers at half-integers.

Framebuffer-Space

Operating with respect to framebuffer coordinates.

Framebuffer-Local

A framebuffer-local dependency guarantees that only for a single framebuffer region, the first set of operations happens-before the second set of operations.

Framebuffer-Global

A framebuffer-global dependency guarantees that for all framebuffer regions, the first set of operations happens-before the second set of operations.

Framebuffer Region

A framebuffer region is a set of sample (x, y, layer, sample) coordinates that is a subset of the entire framebuffer.

Front-Facing

See Facingness.

Full Compatibility

A given version of the API is fully compatible with another version if an application, relying only on valid behavior and functionality defined by either of those specifications, is able to correctly run against each version without any modification. This assumes no active attempt by that application to not run when it detects a different version.

Global Command

A Vulkan command for which the first argument is not a dispatchable handle type.

Global Workgroup

A collection of local workgroups dispatched by a single dispatching command.

Handle

An opaque integer or pointer value used to refer to a Vulkan object. Each object type has a unique handle type.

Happen-after, happens-after

A transitive, irreflexive and antisymmetric ordering relation between operations. An execution dependency with a source of **A** and a destination of **B** enforces that **B** happens-after **A**. The inverse relation of happens-before.

Happen-before, happens-before

A transitive, irreflexive and antisymmetric ordering relation between operations. An execution dependency with a source of **A** and a destination of **B** enforces that **A** happens-before **B**. The inverse relation of happens-after.

Helper Invocation

A fragment shader invocation that is created solely for the purposes of evaluating derivatives for use in non-helper fragment shader invocations, and which does not have side effects.

Host

The processor(s) and execution environment that the application runs on, and that the Vulkan API is exposed on.

Host Mapped Device Memory

Device memory that is mapped for host access using [vkMapMemory](#).

Host Memory

Memory not accessible to the device, used to store implementation data structures.

Host-Accessible Subresource

A buffer, or a linear image subresource in either the [VK_IMAGE_LAYOUT_PREINITIALIZED](#) or [VK_IMAGE_LAYOUT_GENERAL](#) layout. Host-accessible subresources have a well-defined addressing scheme which can be used by the host.

Host-Local Memory

Memory that is not local to the device, and **may** be less performant for device access than device-local memory.

Host-Visible Memory

Device memory that **can** be mapped on the host and **can** be read and written by the host.

ICD

Installable Client Driver. An ICD is represented as a [VkPhysicalDevice](#).

Identically Defined Objects

Objects of the same type where all arguments to their creation or allocation functions, with the exception of [pAllocator](#), are

1. Vulkan handles which refer to the same object or
2. identical scalar or enumeration values or
3. Host pointers which point to an array of values or structures which also satisfy these three constraints.

Image

A resource that represents a multi-dimensional formatted interpretation of device memory. Represented by a [VkImage](#) object.

Image Subresource

A specific mipmap level, layer, and set of aspects of an image.

Image Subresource Range

A set of image subresources that are contiguous mipmap levels and layers.

Image View

An object that represents an image subresource range of a specific image, and state controlling how the contents are interpreted. Represented by a [VkImageView](#) object.

Immutable Sampler

A sampler descriptor provided at descriptor set layout creation time for a specific binding. This sampler is then used for that binding in all descriptor sets allocated with the layout, and it **cannot** be changed.

Implicitly-Enabled Layer

A layer enabled by a loader-defined mechanism outside the Vulkan API, rather than explicitly by the application during instance or device creation.

Index Buffer

A buffer bound via [vkCmdBindIndexBuffer](#) which is the source of index values used to fetch vertex attributes for a [vkCmdDrawIndexed](#) or [vkCmdDrawIndexedIndirect](#) command.

Indexed Drawing Commands

Drawing commands which use an *index buffer* as the source of index values used to fetch vertex attributes for a drawing command. Includes [vkCmdDrawIndexed](#), and [vkCmdDrawIndexedIndirect](#).

Indirect Commands

Drawing or dispatching commands that source some of their parameters from structures in buffer memory. Includes [vkCmdDrawIndirect](#), [vkCmdDrawIndexedIndirect](#), and [vkCmdDispatchIndirect](#).

Indirect Drawing Commands

Drawing commands that source some of their parameters from structures in buffer memory. Includes [vkCmdDrawIndirect](#), and [vkCmdDrawIndexedIndirect](#).

Initial State (Command Buffer)

A command buffer that has not begun recording commands. See also Recording State and Executable State.

Input Attachment

A descriptor type that represents an image view, and supports unfiltered read-only access in a shader, only at the fragment's location in the view.

Instance

The top-level Vulkan object, which represents the application's connection to the implementation. Represented by a [VkInstance](#) object.

Instance-Level Command

Any command that is dispatched from an instance, or from a child object of an instance, except for physical devices and their children.

Instance-Level Functionality

All instance-level commands and objects, and their structures, enumerated types, and enumerants.

Instance-Level Object

High-level Vulkan objects, which are not physical devices, nor children of physical devices. For example, [VkInstance](#) is an instance-level object.

Internal Synchronization

A type of synchronization **required** of the implementation, where parameters not defined to be

externally synchronized **may** require internal mutexing to avoid multithreaded race conditions.

Invocation (Shader)

A single execution of an entry point in a SPIR-V module. For example, a single vertex's execution of a vertex shader or a single fragment's execution of a fragment shader.

Invocation Group

A set of shader invocations that are executed in parallel and that **must** execute the same control flow path in order for control flow to be considered dynamically uniform.

Linear Resource

A resource is *linear* if it is one of the following:

- a [VkBuffer](#)
- a [VkImage](#) created with `VK_IMAGE_TILING_LINEAR`

A resource is *non-linear* if it is one of the following:

- a [VkImage](#) created with `VK_IMAGE_TILING_OPTIMAL`

Local Workgroup

A collection of compute shader invocations invoked by a single dispatching command, which share data via `WorkgroupLocal` variables and can synchronize with each other.

Logical Device

An object that represents the application's interface to the physical device. The logical device is the parent of most Vulkan objects. Represented by a [VkDevice](#) object.

Logical Operation

Bitwise operations between a fragment color value and a value in a color attachment, that produce a final color value to be written to the attachment.

Lost Device

A state that a logical device **may** be in as a result of unrecoverable implementation errors, or other exceptional conditions.

Mappable

See Host-Visible Memory.

Memory Dependency

A memory dependency is an execution dependency which includes availability and visibility operations such that:

- The first set of operations happens-before the availability operation
- The availability operation happens-before the visibility operation
- The visibility operation happens-before the second set of operations

Memory Domain

A memory domain is an abstract place to which memory writes are made available by availability operations and memory domain operations. The memory domains correspond to the set of agents that the write **can** then be made visible to. The memory domains are *host*, *device*, *shader*, *workgroup instance* (for workgroup instance there is a unique domain for each compute workgroup) and *subgroup instance* (for subgroup instance there is a unique domain for each subgroup).

Memory Domain Operation

An operation that makes the writes that are available to one memory domain available to another memory domain.

Memory Heap

A region of memory from which device memory allocations **can** be made.

Memory Type

An index used to select a set of memory properties (e.g. mappable, cached) for a device memory allocation.

Minimum Mip Level Size

The smallest size that is permitted for a mip level. For conventional images this is 1x1x1. See [Image Mip Level Sizing](#).

Mip Tail Region

The set of mipmap levels of a sparse residency texture that are too small to fill a sparse block, and that **must** all be bound to memory collectively and opaquely.

Non-Dispatchable Handle

A handle of an integer handle type. Handle values **may** not be unique, even for two objects of the same type.

Non-Indexed Drawing Commands

Drawing commands for which the vertex attributes are sourced in linear order from the vertex input attributes for a drawing command (i.e. they do not use an *index buffer*). Includes [vkCmdDraw](#), and [vkCmdDrawIndirect](#).

Normalized

A value that is interpreted as being in the range [0,1] as a result of being implicitly divided by some other value.

Normalized Device Coordinates

A coordinate space after perspective division is applied to clip coordinates, and before the viewport transformation converts them to framebuffer coordinates.

Obsolete (feature)

A feature is obsolete if it can no longer be used.

Overlapped Range (Aliased Range)

The aliased range of a device memory allocation that intersects a given image subresource of an image or range of a buffer.

Ownership (Resource)

If an entity (e.g. a queue family) has ownership of a resource, access to that resource is well-defined for access by that entity.

Packed Format

A format whose components are stored as a single texel block in memory, with their relative locations defined within that element.

Physical Device

An object that represents a single device in the system. Represented by a [VkPhysicalDevice](#) object.

Physical-Device-Level Command

Any command that is dispatched from a physical device.

Physical-Device-Level Functionality

All physical-device-level commands and objects, and their structures, enumerated types, and enumerants.

Physical-Device-Level Object

Physical device objects. For example, [VkPhysicalDevice](#) is a physical-device-level object.

Pipeline

An object controlling how graphics or compute work is executed on the device. A pipeline includes one or more shaders, as well as state controlling any non-programmable stages of the pipeline. Represented by a [VkPipeline](#) object.

Pipeline Barrier

An execution and/or memory dependency recorded as an explicit command in a command buffer, that forms a dependency between the previous and subsequent commands.

Pipeline Cache

An object that **can** be used to collect and retrieve information from pipelines as they are created, and **can** be populated with previously retrieved information in order to accelerate pipeline creation. Represented by a [VkPipelineCache](#) object.

Pipeline Layout

An object defining the set of resources (via a collection of descriptor set layouts) and push constants used by pipelines that are created using the layout. Used when creating a pipeline and when binding descriptor sets and setting push constant values. Represented by a [VkPipelineLayout](#) object.

Pipeline Stage

A logically independent execution unit that performs some of the operations defined by an

action command.

pNext Chain

A set of structures [chained together](#) through their **pNext** members.

Point Sampling (Rasterization)

A rule that determines whether a fragment sample location is covered by a polygon primitive by testing whether the sample location is in the interior of the polygon in framebuffer-space, or on the boundary of the polygon according to the tie-breaking rules.

Potential Format Features

The union of all [VkFormatFeatureFlagBits](#) that the implementation supports for a specified [VkFormat](#), over all supported image tilings.

Pre-rasterization

Operations that execute before [rasterization](#), and any state associated with those operations.

Preserve Attachment

One of a list of attachments in a subpass description that is not read or written by the subpass, but that is read or written on earlier and later subpasses and whose contents **must** be preserved through this subpass.

Primary Command Buffer

A command buffer that **can** execute secondary command buffers, and **can** be submitted directly to a queue.

Primitive Topology

State controlling how vertices are assembled into primitives, e.g. as lists of triangles, strips of lines, etc.

Promoted (feature)

A feature from an older extension is considered promoted if it is made available as part of a new core version or newer extension with wider support.

Provisional

A feature is released provisionally in order to get wider feedback on the functionality before it is finalized. Provisional features may change in ways that break backwards compatibility, and thus are not recommended for use in production applications.

Provoking Vertex

The vertex in a primitive from which flat shaded attribute values are taken. This is generally the “first” vertex in the primitive, and depends on the primitive topology.

Push Constants

A small bank of values writable via the API and accessible in shaders. Push constants allow the application to set values used in shaders without creating buffers or modifying and binding descriptor sets for each update.

Push Constant Interface

The set of variables with `PushConstant` storage class that are statically used by a shader entry point, and which receive values from push constant commands.

Query Pool

An object containing a number of query entries and their associated state and results. Represented by a `VkQueryPool` object.

Queue

An object that executes command buffers and sparse binding operations on a device. Represented by a `VkQueue` object.

Queue Family

A set of queues that have common properties and support the same functionality, as advertised in `VkQueueFamilyProperties`.

Queue Operation

A unit of work to be executed by a specific queue on a device, submitted via a `queue submission command`. Each queue submission command details the specific queue operations that occur as a result of calling that command. Queue operations typically include work that is specific to each command, and synchronization tasks.

Queue Submission

Zero or more batches and an optional fence to be signaled, passed to a command for execution on a queue. See the [Devices and Queues chapter](#) for more information.

Recording State (Command Buffer)

A command buffer that is ready to record commands. See also Initial State and Executable State.

Release Operation (Resource)

An operation that releases ownership of an image subresource or buffer range.

Render Pass

An object that represents a set of framebuffer attachments and phases of rendering using those attachments. Represented by a `VkRenderPass` object.

Render Pass Instance

A use of a render pass in a command buffer.

Required Extensions

Extensions that **must** be enabled alongside extensions dependent on them (see [Extension Dependencies](#)).

Reset (Command Buffer)

Resetting a command buffer discards any previously recorded commands and puts a command buffer in the initial state.

Residency Code

An integer value returned by sparse image instructions, indicating whether any sparse unbound texels were accessed.

Resolve Attachment

A subpass attachment point, or image view, that is the target of a multisample resolve operation from the corresponding color attachment at the end of the subpass.

Sample Index

The index of a sample within a [single set of samples](#).

Sample Shading

Invoking the fragment shader multiple times per fragment, with the covered samples partitioned among the invocations.

Sampled Image

A descriptor type that represents an image view, and supports filtered (sampled) and unfiltered read-only access in a shader.

Sampler

An object containing state controlling how sampled image data is sampled (or filtered) when accessed in a shader. Also a descriptor type describing the object. Represented by a [VkSampler](#) object.

Secondary Command Buffer

A command buffer that **can** be executed by a primary command buffer, and **must** not be submitted directly to a queue.

Self-Dependency

A subpass dependency from a subpass to itself, i.e. with [srcSubpass](#) equal to [dstSubpass](#). A self-dependency is not automatically performed during a render pass instance, rather a subset of it **can** be performed via [vkCmdPipelineBarrier](#) during the subpass.

Semaphore

A synchronization primitive that supports signal and wait operations, and **can** be used to synchronize operations within a queue or across queues. Represented by a [VkSemaphore](#) object.

Shader

Instructions selected (via an entry point) from a shader module, which are executed in a shader stage.

Shader Code

A stream of instructions used to describe the operation of a shader.

Shader Module

A collection of shader code, potentially including several functions and entry points, that is used to create shaders in pipelines. Represented by a [VkShaderModule](#) object.

Shader Stage

A stage of the graphics or compute pipeline that executes shader code.

Side Effect

A store to memory or atomic operation on memory from a shader invocation.

Size-Compatible Image Formats

When a compressed image format and an uncompressed image format are size-compatible, it means that the texel block size of the uncompressed format **must** equal the texel block size of the compressed format.

Sparse Block

An element of a sparse resource that can be independently bound to memory. Sparse blocks of a particular sparse resource have a corresponding size in bytes that they use in the bound memory.

Sparse Image Block

A sparse block in a sparse partially-resident image. In addition to the sparse block size in bytes, sparse image blocks have a corresponding width, height, and depth defining the dimensions of these elements in units of texels or compressed texel blocks, the latter being used in case of sparse images having a block-compressed format.

Sparse Unbound Texel

A texel read from a region of a sparse texture that does not have memory bound to it.

Static Use

An object in a shader is statically used by a shader entry point if any function in the entry point's call tree contains an instruction using the object. A reference in the entry point's interface list does not constitute a static use. Static use is used to constrain the set of descriptors used by a shader entry point.

Storage Buffer

A descriptor type that represents a buffer, and supports reads, writes, and atomics in a shader.

Storage Image

A descriptor type that represents an image view, and supports unfiltered loads, stores, and atomics in a shader.

Storage Texel Buffer

A descriptor type that represents a buffer view, and supports unfiltered, formatted reads, writes, and atomics in a shader.

Subpass

A phase of rendering within a render pass, that reads and writes a subset of the attachments.

Subpass Dependency

An execution and/or memory dependency between two subpasses described as part of render pass creation, and automatically performed between subpasses in a render pass instance. A

subpass dependency limits the overlap of execution of the pair of subpasses, and **can** provide guarantees of memory coherence between accesses in the subpasses.

Subpass Description

Lists of attachment indices for input attachments, color attachments, depth/stencil attachment, resolve attachments, and preserve attachments used by the subpass in a render pass.

Subset (Self-Dependency)

A subset of a self-dependency is a pipeline barrier performed during the subpass of the self-dependency, and whose stage masks and access masks each contain a subset of the bits set in the identically named mask in the self-dependency.

Texel Block

A single addressable element of an image with an uncompressed [VkFormat](#), or a single compressed block of an image with a compressed [VkFormat](#).

Texel Block Size

The size (in bytes) used to store a texel block of a compressed or uncompressed image.

Texel Coordinate System

One of three coordinate systems (normalized, unnormalized, integer) defining how texel coordinates are interpreted in an image or a specific mipmap level of an image.

Uniform Texel Buffer

A descriptor type that represents a buffer view, and supports unfiltered, formatted, read-only access in a shader.

Uniform Buffer

A descriptor type that represents a buffer, and supports read-only access in a shader.

Units in the Last Place (ULP)

A measure of floating-point error loosely defined as the smallest representable step in a floating-point format near a given value. For the precise definition see [Precision and Operation of SPIR-V instructions](#) or Jean-Michel Muller, “On the definition of $ulp(x)$ ”, RR-5504, INRIA. Other sources may also use the term “unit of least precision”.

Unnormalized

A value that is interpreted according to its conventional interpretation, and is not normalized.

User-Defined Variable Interface

A shader entry point’s variables with [Input](#) or [Output](#) storage class that are not built-in variables.

Vertex Input Attribute

A graphics pipeline resource that produces input values for the vertex shader by reading data from a vertex input binding and converting it to the attribute’s format.

Variable-Sized Descriptor Binding

A descriptor binding whose size will be specified when a descriptor set is allocated using this

layout.

Vertex Input Binding

A graphics pipeline resource that is bound to a buffer and includes state that affects addressing calculations within that buffer.

Vertex Input Interface

A vertex shader entry point's variables with **Input** storage class, which receive values from vertex input attributes.

View Volume

A subspace in homogeneous coordinates, corresponding to post-projection x and y values between -1 and +1, and z values between 0 and +1.

Viewport Transformation

A transformation from normalized device coordinates to framebuffer coordinates, based on a viewport rectangle and depth range.

Visibility Operation

An operation that causes available values to become visible to specified memory accesses.

Visible

A state of values written to memory that allows them to be accessed by a set of operations.

Common Abbreviations

The abbreviations and acronyms defined in this section are sometimes used in the Specification and the API where they are considered clear and commonplace.

Src

Source

Dst

Destination

Min

Minimum

Max

Maximum

Rect

Rectangle

Info

Information

LOD

Level of Detail

Log

Logarithm

ID

Identifier

UUID

Universally Unique Identifier

Op

Operation

R

Red color component

G

Green color component

B

Blue color component

A

Alpha color component

RTZ

Round towards zero

RTE

Round to nearest even

Prefixes

Prefixes are used in the API to denote specific semantic meaning of Vulkan names, or as a label to avoid name clashes, and are explained here:

VK/Vk/vk

Vulkan namespace

All types, commands, enumerants and defines in this specification are prefixed with these two characters.

PFN/pfn

Function Pointer

Denotes that a type is a function pointer, or that a variable is of a pointer type.

p

Pointer

Variable is a pointer.

vkCmd

Commands that record commands in command buffers

These API commands do not result in immediate processing on the device. Instead, they record the requested action in a command buffer for execution when the command buffer is submitted to a queue.

s

Structure

Used to denote the `VK_STRUCTURE_TYPE*` member of each structure in `sType`

Appendix H: Credits (Informative)

Vulkan 1.3 is the result of contributions from many people and companies participating in the Khronos Vulkan Working Group, as well as input from the Vulkan Advisory Panel.

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